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McDonald's Blood Flow in Arteries

THEORETICAL, EXPERIMENTAL
AND CLINICAL PRINCIPLES
SIXTH EDITION

Wilmer W Nichols
Michael F O'Rourke
Charalambos Vlachopoulos

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McDonald's Blood Flow in Arteries Theoretical, Experimental and Clinical Principles

Sixth edition

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To the memory of friends and colleagues

Donald A McDonald

(Born June 3, 1917; died in Birmingham, AL May 24, 1973)

John R Womersley

(Born June 20, 1907; died in Columbus, OH March 7, 1958)

Michael G Taylor

(Born December 30; 1926; died in Sydney, AU January 10, 2006)

William R Milnor

(Born May 4, 1920; died in Baltimore, MD January 3, 2008)

Raymond P Kelly

(Born December 5, 1956; died in Sydney, AU October 5, 2000)

As an animal Body consists not only of a wonderful texture of solid Parts, but also of large proportion of Fluids, which are continually circulating and flowing, thro' an inimitable Embroidery of Blood-Vessels, and other inconceivably minute Canals: and as the healthy State of an Animal principally consists, in the maintaining of a due *Equilibrium* between those Solids and Fluids; it has ever since the important Discovery of the Circulation of the Blood, been looked upon as a Matter well worth enquiring into, to find the Force and Velocity with which these Fluids are impelled; as a likely means to give a considerable Insight into Animal Oeconomy.

S. Hales

...when you can measure what you are speaking about, and express it in numbers, you know something about it; but when you cannot measure it, when you cannot express it in numbers, your knowledge is of a meagre and unsatisfactory kind; it may be the beginning of knowledge, but you have scarcely, in your thoughts, advanced to the stage of Science whatever the matter may be.

Kelvin

Science like life feeds on its own decay. New facts burn old rules; then newly developed concepts bind old and new together into a reconciling law.

William James

Die Methode ist Alles

Carl Ludwig

Contents

Foreword by Michael Taylor	ix
Foreword by David A. Kass and Myron L. Weisfeldt	x
Preface	xi
List of Abbreviations	xiii
1 Introduction	1
2 The nature of flow of a liquid	13
3 Properties of the arterial wall: theory	55
4 Properties of the arterial wall: practice	77
5 Endothelial function	111
6 General principles for measuring arterial waves	129
7 Pulsatile pressure–flow relations	165
8 Ultrasound	177
9 Wave reflections	195
10 Contours of pressure and flow waves in arteries	225
11 Principles of recording and analysis of arterial waveforms	255
12 Vascular impedance	273
13 Aortic input impedance as ventricular load	311
14 Coupling of the left ventricle with the systemic circulation: implications to cardiac failure	321
15 Cardiac failure: clinical implications	345
16 The pulmonary circulation	357
17 The coronary circulation	375
18 Special circulations	397
19 Aging	411
20 Hypertension	447
21 Interpretation of blood pressure in epidemiological studies and clinical trials	469
22 Arterial biomarkers	485
23 Atherosclerosis	499
24 Specific arterial disease	507
25 Generalized and metabolic disease	523
26 Therapeutic strategies	535
27 Exercise	553
28 Central arterial pressure	569
29 Lifestyle and environment	579

30	Pressure pulse waveform analysis	595
	Bibliography	639
	Index	743

Foreword by Michael Taylor

It is very pleasant to be invited to provide a few words of introduction to the fifth edition of McDonald's "Blood Flow in Arteries", for it takes me back fifty years, to the first time I met Donald McDonald and John Womersley in January 1955 at St. Bartholomew's hospital in London. Womersley was an applied mathematician at the National Physical Laboratory, and his daughter was a medical student at Barts. She had told him about McDonald's studies on pulsatile arterial blood flow and he became intrigued. By 1955 he and McDonald had been working

together very fruitfully for some time, but no one then would have predicted how far their work would lead. This book first appeared in 1960, and now is going into its fifth edition. Over time its character has evolved, with gradually increasing clinical emphasis, but essentially it remains a tribute to the imagination and foresight of Donald McDonald.

Michael Taylor
Sydney 2004

Foreword by David A. Kass and Myron L. Weisfeldt

With each heart beat, both ventricles eject blood into their respective vascular systems – but do so for only a fraction of the time. The rest of the cycle is allocated to muscle relaxation and refilling of the chambers so the process can start all over again. This fact belies a profound yet often ignored feature of the circulation, namely that, that it is pulsatile. The cyclical nature of flow exiting each ventricle results in pulse pressures that both define the mechanical forces experienced by vascular tissues, and the loads imposed on each ventricle. This hemodynamic behavior is conserved across mammalian species – indeed, the pulse and mean pressure at the base of the brain is remarkably similar between mouse, dog, human, or even giraffe, whose heart has to work a bit extra to provide it. Perhaps not surprisingly, organ physiology has developed to take full advantage of the pulsatile nature of flow and pressure, with vascular signaling that is particularly responsive to vessel stretch and phasic shear stress, and distributed compliances that buffer the pulse as required. As is now well recognized, loss of such buffering capacity and attendant augmentation of the pulse pressure is common in older individuals. Rather than being a ‘normal’ feature of aging, this change is associated with substantial morbidity and mortality. For all the sophisticated molecular biology being applied to identify risks for cardiovascular disease, the simple difference between systolic and diastolic blood pressure is an amazingly potent predictor. Yet for most physicians in medical practice, including those who specialize in cardiovascular disease, we primarily measure and treat means – pressure, flow, resistance – rather than the properties that result in greater pulsatility. While any first-year medical student can define pulse pressure, the properties that give rise to it are traditionally more difficult to measure and can require sophisticated mathematics. As our ability to selectively target arterial properties that lead to excessive pulsatility remain limited, one might reasonably ask, why bother trying?

The answer to this last question and to the many others regarding large artery properties and the pulsatile nature of blood flow are to be found in the newly revised sixth edition of the unparalleled volume, *McDonald's Blood Flow in Arteries*. The authors, all seminal leaders in this field, have updated prior chapters and added several new ones to enhance the discussion in several critical areas. As

before, the book provides a lucid and comprehensive presentation of the history of blood flow and pressure measurements, and the physics and mathematics required to assess pulsatile loading – a.k.a. impedance. They describe newer methodologies that have translated what was once a complex and rather arcane research tool into something more accessible and easily measured in out-patient clinics. In a new chapter on central arterial pressure, methods for its estimation from non-invasively recorded data are explained, and importantly, evidence that such estimation provides better assessment of cardiovascular risk and the benefits of pharmacologic therapy are discussed. Other chapters deal with the impact of pulsatile loading on the heart, blood vessel signaling and biology, and how the normally distensible arterial system in both systemic and pulmonary circulations modulates this pulsatility. There are a number of chapters exploring how vascular pulsatility becomes modified by aging and by disease. A new chapter highlights the impact of common environmental and lifestyle factors as well. Other updated chapters highlight the physiology and pathophysiology of pulmonary and coronary vascular circuits, and in a chapter, this analysis is now extended to cerebral and renal circulations.

The new edition provides an excellent reference for those already engaged in the field, and for those embarking anew as researchers, students, and clinicians. The measurement and understanding of pulsatile circulatory dynamics and its implications to the body are important for clarifying the pathophysiology of many organ diseases, and central for charting the course for better and more appropriate therapies to treat them. This continues to be particularly true in the elderly, a population in which pulsatile features of the circulation are often prominent. The volume is readable, comprehensive, authoritative and remarkably revealing to the casual and expert reader. The subject has always been important, and volumes like this bring to light just why, and will hopefully stimulate new research and disease therapies based upon these insights.

David A. Kass and Myron L. Weisfeldt
Johns Hopkins Medical Institutions
Baltimore, MD, April 25, 2011

Preface

The first edition of *Blood Flow in Arteries* appeared in 1960 as a monograph for the British Physiological Society and introduced a new approach to the study of arterial hemodynamics. It was the product of six years' collaboration between an inventive physiologist, Donald McDonald, a talented applied mathematician, John Womersley, and a young medical graduate, Michael Taylor, at St Bartholomew's Hospital in London – the site of William Harvey's revolutionary work just over 300 years before.

As befits arterial pressure and flow, the first edition concentrated on pulsatile phenomena, and brought to its analysis techniques that were established in engineering science, but had hitherto not been applied comprehensively to a biological system. The major advance was to consider the arterial system as in a 'steady state' of oscillation so that waveforms could be analyzed in the frequency domain – as a series of integral harmonics – and whose relationship with other pulsations could also be considered in the frequency domain. The approach broke completely with the traditional approach of analyzing arterial pulses in the time domain – as a fluctuation over a cardiac cycle between systolic and diastolic values, with peak and nadir of the pressure pulse (systolic and diastolic pressure) considered to have particular physiological and clinical significance.

The new approach considered any pulse waveform as having a mean value, and a fluctuation around this mean, and with the fluctuations capable of being expressed as a series of harmonic components. This new approach stepped further by considering pressure/flow relationships to be near linear, so that the relationships between individual harmonic components of other pulses (of flow at the same site as pressure, or of pressure at another site) could be expressed as a function of frequency. This approach had the advantage of separating pulsatile from steady flow phenomenon and was entirely consistent with the method used then (and now) of relating mean pressure to mean flow for calculation of peripheral resistance. But the new approach, championed by McDonald in his 1960 monograph, was that the arterial system could be considered as a linear system – that any harmonic component of a pressure wave could be related to the same harmonic of another pressure or flow wave recorded at the same time, and to no other harmonic. This was revolutionary (even though linearity was assumed in calculation of resistance from mean pressure and flow). It was, however, soundly based on Womersley's classic mathematical studies on the equations of motion as these applied to blood flow in arteries. Womersley had shown

that the non-linear terms were indeed sufficiently small (in relation to measuring errors) that they could be neglected to a first approximation.

The 1960 monograph ran entirely counter to the prevailing approaches to arterial physiology, particularly as practiced in the United States, but was sufficiently appealing for both Womersley and McDonald to be later attracted to institutions in the USA, and Taylor back to Australia. John Womersley died in Columbus, Ohio as the original monograph was being completed.

Donald McDonald wrote the second edition of *Blood Flow in Arteries* while Professor of Physiology at the University of Alabama in Birmingham. He died when the book was in galley proof stage. The book was completed by his US colleagues Bill Milnor (of Johns Hopkins) and Wilmer Nichols (of the University of Florida). This edition extended the earlier work and introduced some clinical applications.

The third to fifth editions of *Blood Flow in Arteries* were written together by Wilmer Nichols (a pupil of Donald McDonald) and Michael O'Rourke (a pupil of Michael Taylor). At a time when arterial disease was exploding as a clinical problem in developed and developing countries, the book came more and more to consider clinical implications of the original hemodynamic discoveries and to show that the same principles could be expanded and applied to clinical practice – both with regard to the arteries that are affected by disease, and the organs, especially the heart, that are secondarily affected by changes in pressure and flow fluctuations. The authors, both protégés of McDonald, were anxious to retain the attractive features of the original 1960 monograph – emphasis on mechanisms with strong scientific base, embrace of new technology, a focus on practical applications, a comprehensive coordinated style, a disrespect of cant, and a challenge to established authority.

In this sixth edition, we attempt to endorse the same approach to problems of blood flow in arteries as we see them today. The main problems are in the clinical arena and relate to pressure as much as to flow, and to the heart, which generates pulsatile pressure and flow, as well as all organs which require flow so as to live and function normally. Our emphasis remains on arterial pulsations resulting from ventricular ejection, not on the steady component of flow and pressure, and on mechanisms, i.e. on theoretical principles, physiological principles, and clinical principles. This book cannot be a text on arterial disease. It is intended to help those dealing with arterial function, aging, and disease, when they wish to understand underlying principles.

‘Principles’ and ‘reasons’ underlie all good practice. ‘Evidence’ follows. Emphasis on principles may explain why earlier editions of this book have been so popular and so successful in anticipating future developments in clinical practice. As with the first edition, this sixth edition will prove contentious, since it provides a challenge to clinicians and epidemiologists who have seen just one way to measure arterial pressure and cardiac load with the cuff sphygmomanometer. If it succeeds, it will have continued the same approach and traditions, not only of the first edition of ‘McDonald’, but of Harvey’s treatise as well.

This sixth edition appears 50 years after the first. As befits a mature publication which has introduced a new approach to clinical issues related to aging and disease, it is appropriate that the new steps be summarized in the Introduction, then expanded in the text that follows. Scientific advances carry little benefit unless they can be applied. Application requires simplification and approximation. This was done by Einthoven with the electrocardiogram (ECG), by analysis on the basis of an equilateral triangle in the frontal plane of the body. In this sixth edition, which may complete the stewardship of *Blood Flow in Arteries* by the first two authors, we expand

McDonald’s concept of linearity into ‘Fifteen Commandments’ (or general principles) which may guide further advance in the clinical arena. These are summarized in Chapter 1.

We cannot adequately thank those we should in extending the work of illustrious pioneers. We have encountered the same frustrations and setbacks as have they, yet have received generous support from colleagues around the world. We are especially grateful to Arnold Hoeks and Robert Reneman from the Cardiovascular Research Institute, Maastricht, The Netherlands for their assistance with Chapter 5 (Endothelial function) and their contribution of Chapter 8 (Ultrasound). Our younger colleagues are extending this work and have our admiration and respect, particularly Ray Kelly, whose premature death robbed this field and this book of a worthy heir. Above all we thank our dear wives, Arlene, Margaret, and Antonia, who have contributed more than anyone could know.

Wilmer W Nichols
Michael F O’Rourke
Charalambos Vlachopoulos

List of Abbreviations

AASI	arterial stiffness index	EDRF	endothelium-derived relaxing factor
ABI	ankle-brachial index	EECP	enhanced external counterpulsation
AC	alternating current	EMF	electromagnetic flowmeter
ACCT	Anglo-Cardiff Collaborative Trial	eNOS	enzyme nitric oxide synthase
ACE	angiotensin-converting enzyme	EPC	endothelial progenitor cells
ACEI	angiotensin-converting enzyme inhibitor	ESPVR	end-systolic pressure-volume relationship
ACEI	angiotensin-converting enzyme inhibitors	ESRD	end-stage renal disease
ACEI	angiotensin-converting enzymes inhibitors	FAI	flow augmentation index
ACh	acetylcholine	FF	form factor
AGE	advanced glycation end product	FFR	fractional flow reserve
AGE	advanced glycosylation end-products	FMD	Flow-mediated vasodilation
AHA	American Heart Association	FMD	flow-mediated dilation
AIx	augmentation index	HRT	hormone replacement therapy
ARB	angiotensin receptor blocker	iCFR	Invasive CFR
ARB	angiotensin receptor blockers	ICG	impedance cardiography
ARB	angiotensin receptor type 1 blocker	IMT	intima-media thickness
ARIC	Atherosclerosis Risk in Communities	IMT	intima-medial thickness
ASCOT	Anglo-Scandinavian Cardiac Outcomes Trial	ISH	International Society of Hypertension
BA	brachial artery	ISH	isolated systolic hypertension
BHS	Bogalusa Heart Study	IVUS	intravascular ultrasonic
BLSA	Baltimore Longitudinal Study of Aging	IVUS	intravascular ultrasound
BP	blood pressure	IVUS	invasive intravascular ultrasound
CAC	coronary artery calcium	L-NMMA	L-monomethyl arginine
CAD	coronary artery disease	LBBB	left ventricular bundle branch block
CAFE	Conduit Artery Function Evaluation	LDA	laser Doppler anemometry
CC	compliance coefficient	LDF	laser Doppler flowmetry
CCA	common carotid artery	LDL	low-density lipoprotein
CCB	calcium channel blocker	LV	left ventricular
CCB	calcium channel blockers	MAP	mean aortic pressure
CCTA	coronary CT angiography	MCP	monocyte chemoattractant protein
CDC	Center for Disease Control	MESA	Multi-Ethnic Study of Atherosclerosis
CFA	common femoral artery	MI	myocardial infarction
CFR	coronary flow reserve	MMP	matrix metalloproteinase
cGMP	cyclic guanosine monophosphate	MP	microparticles
CHD	coronary heart disease	MRI	magnetic resonance imaging
CO	cardiac output	NO	nitric oxide
CPTE	chronic pulmonary thromboembolism	NTG	nitroglycerin
CRP	C-reactive protein	NYHA	New York Heart Association
CT	computed tomography	OCT	optical coherence tomography
CUPS	Chennai Urban Population Study	P-V	pressure-volume
CV	cardiovascular	PA	pulmonary arterial
CVD	cardiovascular disease	PAD	peripheral arterial disease
CW	continuous wave	PAT	pulse amplitude tonometry
DC	direct current	PDA	patent ductus arteriosus
DC	distensibility coefficient	PDE5	phosphodiesterase 5
DHEA	dehydroepiandrosterone	PDE5	phosphodiesterase type 5
DPTI	diastolic pressure-time index	PDE5	phosphodiesterase-5
ECG	echocardiogram	PET	positron emission tomography
ECG	electrocardiogram	PI	pulsatility index
EDRF	endothelial-derived relaxing factor	PP	pulse pressure

PPH	primary pulmonary hypertension	SHM	simple harmonic motion
PRAM	pressure recording analytical method	SIH	secondary intracranial hypertension
PRF	pulse repetition frequency	SMC	smooth muscle cells
PRU	peripheral resistance unit	SNR	signal-to-noise ratio
PWA	pulse waveform analysis	SPTI	systolic pressure–time index
PWV	pulse wave velocity	SRES	shear rate estimating system
QCA	quantitative coronary angiography	SV	stroke volume
RBC	red blood cell	TEE	transesophageal echocardiography
REASON	pREterax in regression of Arterial Stiffness in a contrOLled double-bliNd	TGF- β	transforming growth factor- β
RF	radiofrequency	TTI	tension–time index
RR	relative risk	US	ultrasound
rtPA	recombinant tissue plasminogen activator	VSD	ventricular septal defect
SAC	systemic arterial bed	WBV	whole blood viscosity
SFA	superficial femoral artery	WSR	wall shear rate
SHEP	Systolic Hypertension in the Elderly Project	WSS	wall shear stress

Introduction

In the beginning	3	Analysis in the frequency and time domains	10
The arterial system in steady-state oscillation	8	Application to clinical problems	10

The most obvious feature of ‘blood flow in arteries’ is that it is pulsatile. Therefore, the field of survey is the character of pulsatile flow and the physical laws governing it. This means that the survey must also comprehend the form of the pressure wave of the pulse, in relation both to the flow wave and to the means by which the pulse wave (of flow, as well as pressure) is propagated over the arterial system.

The ability to measure and interpret variations of pressure and flow in humans depends on an understanding of physiologic principles and is based on a heritage well over 100 years old. Studies of pressure preceded those of flow, since reliable tools were available for pressure measurement 100 years ago, but for flow some 40 years less. Relevant studies of the circulation and of underlying physical principles, however, stretch back over 400 years through Harvey and Newton to Galileo.

Pulsatile pressure is recorded routinely in human cardiovascular diagnostic laboratories, operating rooms, and critical care units in the present generation, and pulsatile flow patterns are now used invasively and non-invasively for the quantification of cardiac and vascular shunts and of valvular and vascular stenoses. These techniques have assumed practical importance in routine cardiologic and vascular diagnostic, investigative, and monitoring procedures.

Earlier editions of this book did not anticipate the full clinical implications and potential application of non-invasive pressure and flow measurements in human studies. In particular, the development of diagnostic ultrasound, applanation tonometry, magnetic resonance imaging (MRI), and computed tomography (CT) were not foreseen. These issues are covered in separate chapters. New to the sixth edition are chapters on central blood pressure, change in pulsatile cerebral and renal blood flow with aging and relevance to syndromes of dementia and kidney function, on physical biomarkers which may

anticipate and help interpret results of forthcoming trials and studies, including Framingham and on lifestyle and the environment.

Earlier editions of this book referred to particular conferences, books and articles that had advanced the field. Such reference is no longer appropriate, since arterial hemodynamics is now a mainstream topic in the literature, including the most august experimental and clinical journals, which deal in cardiovascular function and disease. Implications of central aortic pressure measurement rated discussion in Highlights of the Year in *Journal of the American College of Cardiology* 2009 (DeMaria *et al.*, 2010). An award for the best clinical research paper for 2007 was presented to Peter Lacy for the CAFE study (Williams *et al.*, 2006) by the American Heart Association, while Eugene Braunwald in his keynote address to the American Society of Hypertension’s annual meeting in 2010 stressed the historical studies of Frederick Mahomed on the arterial pressure waveform (Braunwald, 2010). The most eminent general medical journals, such as the *New England Journal of Medicine* and the *Lancet*, however, continue to prefer the older, established approaches to arterial pressure and flow measurement and analysis, with scant reference (except in correspondence) to phenomena such as wave reflection and pressure wave amplification in the arterial tree.

This sixth edition of *Blood Flow in Arteries* seeks to justify its place as a reference in the field of pulsatile flow for those who might wish direction to greater detail elsewhere, but who still want to retain orientation, perspective, and balance in what was, is, and always will be a particularly complex and challenging multidisciplinary area. Our challenge, originally met by McDonald alone in 1960, is to achieve this in a sound but interesting and provocative way.

While our principal interest is in pulsatile flow, we must also deal with pulsatile pressure and diameter as they are related to flow. We will be attempting to redress the

problem pointed out by Jarisch in 1928, and quoted again by Neil in 1983, that:

The development of our knowledge of the circulation has been bedevilled by the fact that the measurement of blood flow is so complicated whereas that of blood pressure is so easy: hence the blood pressure manometer has exerted an almost hypnotic influence, though bodily organs don't need pressure but flow.

Sir James Mackenzie (1917), the father of modern cardiology, was even more scathing in his criticism of those who accepted uncritically the numbers provided by the cuff sphygmomanometer: 'we get now a definite and precise figure which most people imagine to be more scientific'.

There is still preoccupation with the sphygmomanometer in the diagnosis and management of hypertension. At the turn of the century, attention was deflected from diastolic to systolic pressure as the major index of risk in hypertension (Black, 1999; He and Whelton, 1999; Staessen *et al.*, 2000b) and to the importance of large arteries both in determining systolic and pulse pressure and as the target of damage when pressure is high (Laurent *et al.*, 2001b; Boutouyrie *et al.*, 2002). This has led to a new approach to hypertension (O'Rourke, 1976; O'Rourke, 1983a; Nichols *et al.*, 1986b; Ting *et al.*, 1986; Safar, 1987; O'Brien and Staessen, 1999; Safar, 2001; Beevers *et al.*, 2001a, Beevers *et al.*, 2001b, Beevers *et al.*, 2001c), which has been aided by new methods for determining arterial pressure, flow, and diameter non-invasively (Avolio *et al.*, 1988; Kelly *et al.*, 1988a; Kelly *et al.*, 1988b; Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; Kelly *et al.*, 1989c; Kelly *et al.*, 1989d; Kelly *et al.*, 1989e; Kelly *et al.*, 1989f; Pedersen *et al.*, 1993; Pauca *et al.*, 2001; Schofield *et al.*, 2002a; das Chagas de Azevedo *et al.*, 2005; Fesler *et al.*, 2007; Chugh and Bakris, 2007; Giannattasio *et al.*, 2008; Wolak *et al.*, 2008).

The 1990s saw confirmation of the view that systolic blood pressure is of greater importance than diastolic (at least in people over age 60) as a risk factor for cardiovascular disease (Rutan *et al.*, 1989; Folkow and Svanborg, 1993; Franklin and Weber, 1994; Franklin *et al.*, 1997) and saw further development of tonometric and ultrasonic methods for more precise assessment of arterial pressure, flow, and diameter changes (Kelly *et al.*, 1990a; Kelly *et al.*, 1990b; Kelly and Fitchett, 1992; Länne *et al.*, 1992a; Länne *et al.*, 1992b). The landmark SHEP publication (1991) showed that treatment of elevated systolic pressure reduced cardiovascular mortality and morbidity, even in people whose diastolic pressure was normal or low. Tonometric and other methods for recording pressure have been used to identify the boost to pressure in central arteries caused by the early return of wave reflection and have linked such boost to measures of left ventricular hypertrophy and performance. There has been new interest in pulse pressure (systolic minus diastolic) as a marker of arterial stiffness, and a prognostic indicator in

older people (O'Rourke and Frohlich, 1999; Dart and Kingwell, 2001; Safar, 2001; Van Bortel *et al.*, 2001a; Domanski *et al.*, 2002; Franklin *et al.*, 2009; Roman *et al.*, 2009). This issue remains contentious, since older, larger studies have not shown the same superiority of pulse pressure over systolic or diastolic pressure (MacMahon *et al.*, 1990; Prospective Studies Collaboration, 2002; Mosley *et al.*, 2007). An explanation arises when one considers the relative youth of subjects in the trials of entry (when amplification of the pulse is considerable), and the different criteria used for measuring systolic and (particularly) diastolic pressure. As discussed in Chapter 28, this issue can be clarified in the future through measurement of central rather than brachial pressure, and with due consideration of aortic aging change (Benetos *et al.*, 2010; Mitchell *et al.*, 2010; Vlachopoulos *et al.*, 2010a; Vlachopoulos *et al.*, 2010b; Wang *et al.*, 2010; Weber *et al.*, 2010). Interest in pulse pressure is moving on to amplitude and contour of the pressure pulse, and in the central arteries. Ultrasonic methods have been further refined for measurement of arterial caliber (Pedersen *et al.*, 1993; Astrand *et al.*, 2005), arterial caliber change (Sonesson *et al.*, 1994; Reneman *et al.*, 2005), wall thickness (Pearson *et al.*, 1994; Astrand *et al.*, 2005), and flow velocity profiles (Yamamoto *et al.*, 1994; Tortoli *et al.*, 2003), using both invasive and non-invasive approaches (see Chapters 6 and 8) (Dewey, 2009; Hermeling *et al.*, 2009; Joly *et al.*, 2009; Muiesan and Paini, 2009).

Over the past 25 years, there has been a surge of interest in the vascular endothelium, on substances produced by the endothelium that influence components of the blood within the lumen, and on the tone of smooth muscle in the vessel wall. This subject has commanded the attention of some of the best minds in molecular biology, has been widely covered in the literature, and was the subject of the Nobel Prize Award in 1998 (Moncada *et al.*, 1991; Griffith *et al.*, 1994; Rubanyi, 1996; Cardillo and Panza, 1998). Findings give new insight into such subjects as hemostasis and clot formation (Verhamme and Hoylaerts, 2006), arterial spasm (Kawano and Ogawa, 2005; Yasue *et al.*, 2008; Miwa *et al.*, 2009) and variations of arterial tone (Fields and Makhoul, 1998), and atherogenesis (Dobarro *et al.*, 2009; Herrera *et al.*, 2010; Napoli and Ignarro, 2009) and cellular proliferation after angioplasty (Torella *et al.*, 2004; Finn *et al.*, 2010; Libby *et al.*, 2010).

Over the past 20 years also, there has been acceptance of plaque disruption as the initial event causing coronary thrombosis and myocardial infarction, and on the value of urgent stenting, and of thrombolysis, used early, for dissolution of the clot and limitation of infarction size, not only in the heart, but in the brain as well (Wang *et al.*, 2006; Ay *et al.*, 2008; Braunwald, 2009; Libby *et al.*, 2009; Sanz *et al.*, 2009a). All this leads on to more precision in the approach to understanding how plaques develop in the first place, the influence of inflammation, and why they disrupt. Attention is directed at disrupting forces

within the arterial wall and shearing stresses on the blood–vascular interface. Attention is also directed at the factors which can limit the benefits of angioplasty and stenting – particularly intimal proliferation with bare metal stents and late thrombosis with drug eluting stents (Htay and Liu, 2005; Mehdizadeh *et al.*, 2009; Nakazawa *et al.*, 2009; Nakazawa *et al.*, 2010).

IN THE BEGINNING ...

The sudden swing from ‘natural philosophy’ into modern science is most vividly characterized in the person of Galileo Galilei (1564–1642). From 1598 to 1610, he was Professor of Mathematics at the University of Padua, Padova, Italy. During this period, William Harvey (1578–1657) was studying medicine there, and although Harvey (1628) never mentioned Galileo in his book *Exercitatio Anatomica de Motu Cordis et Sanguinis in Animalibus* (Anatomical Essay on the Motion of the Heart and Blood in Animals), it is difficult to believe that he was not influenced by the new quantitative teaching and reasoning in his approach to the problem of the circulation of blood. Others were prepared to accept passage of blood through the lungs into the left atrium and through small systemic blood vessels into the right atrium; it was Harvey who stressed the magnitude of such flow. His challenge to the existing order attracted many detractors, especially in France, and he defended his views with colourful language (Fig. 1.1). Modern cardiovascular physiology and biophysics began with Harvey and *Exercitatio Anatomica de Motu Cordis*. However, Harvey typified another characteristic of many of the later pioneers in this field, i.e. interest, competence, and often prominence in more than one field of endeavor. Perhaps a commitment to science and a catholicity of interest accounted for the unusual multidisciplinary achievements of these men. Galileo achieved prominence as a mathematician and as an astronomer, Harvey as anatomist, physiologist and physician (indeed as personal physician to King Charles I).

Perhaps the greatest mathematician and physicist of all time was Sir Isaac Newton (1642–1727). Newton’s monumental work *Principia Mathematica* contained as an addended proposition the concept of fluid viscosity; this has now become central to the consideration of arterial blood flow, pressure–flow relationships, and vascular resistance. Newton left Cambridge in 1696 to take on and excel at an administrative task as Warden of the Royal Mint in London and to gain office, ultimately as president, of the Royal Society (Manuel, 1968). It was in this capacity that he authorized publication of a book by another Cambridge graduate, the Reverend Stephen Hales (1677–1761), entitled *Haemastaticks* (1733).

Although Galen (130–200 AD) palpated the arterial pulse and classified it in terms of strength, rate, and rhythm, Hales was the first to measure arterial pressure in an animal, to document the response of arterial pressure

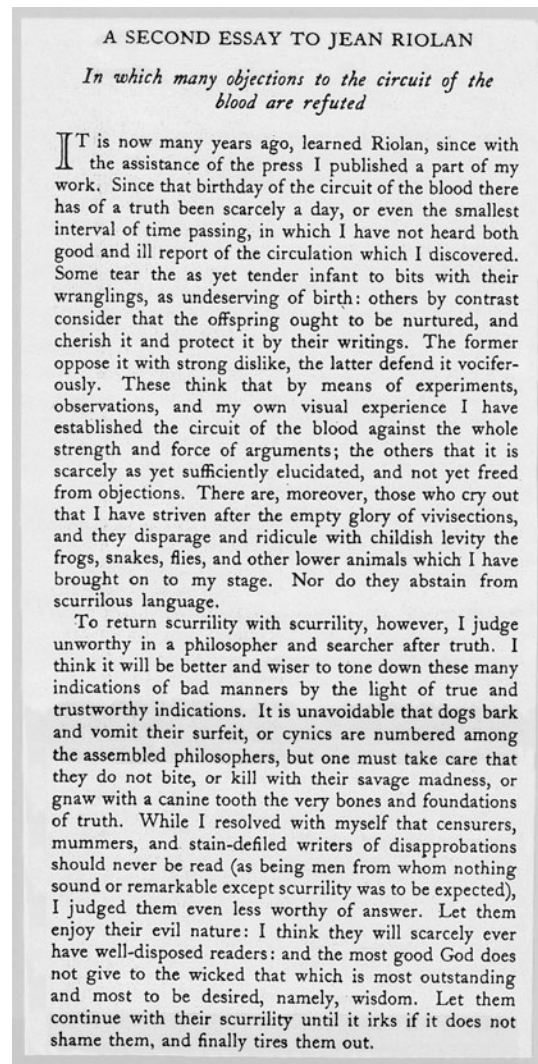


Figure 1.1 Introductory paragraphs of William Harvey's second letter to the anatomist Jean Riolan of Paris. Translated from the original Latin by KJ Franklin of St Bartholomew's Hospital, London

to blood loss, to formulate the concept of peripheral resistance, and to show in an elegantly simple experiment that the greatest resistance to blood flow resides in the tiny blood vessels that are invisible to the naked eye. Hales also likened the arterial system to the inverted, air-filled dome of the contemporary fire engine that smoothed the oscillations due to intermittent pumping so that flow of water through the fire hose nozzle was almost perfectly continuous (Fig. 1.2). Hales saw the arterial system acting as a cushion (or buffer) and the arterioles as resistance to the intermittent pumping of the heart, so that flow through the tissues was virtually devoid of cardiac pulsations. In the German translation of *Haemastaticks*, the fire engine compression chamber was referred to as *Windkessel* – a term which has stuck to the description of the functional cushioning role of the arterial system.

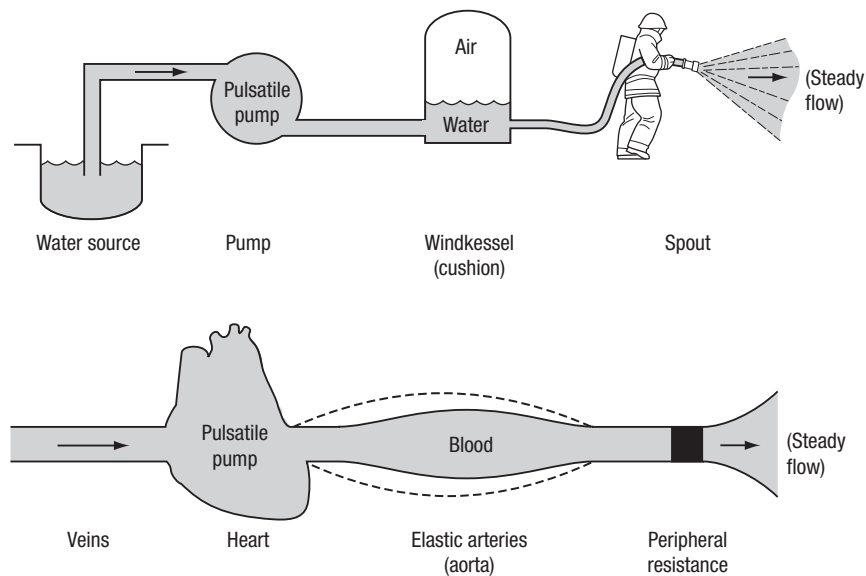


Figure 1.2 The *Windkessel* model of the arterial system. The arterial tree is likened to a medieval fire cart, with distensibility represented by the inverted air-filled chamber, which changes pulsatile flow from the intermittently acting pump, and with peripheral resistance represented by the fire hose nozzle. Reproduced from Westerhof *et al.* (2009)

The eighteenth century was further notable for the great development of the theoretic mathematical treatment of fluid dynamics. While this was based on the work of Newton, it usually dealt with an ‘ideal’ fluid in which Newton’s *defectus lubricatis* or viscosity was ignored. The leader in this field was Leonhart Euler (1707–83), who is described as the most prolific mathematician that ever lived; he established the general equations of fluid motion virtually as they are used today. Euler was a close friend of Daniel Bernoulli (1700–82), who is remembered for his equation describing pressure change with interconversion of potential to kinetic energy.

Daniel Bernoulli was a member of the illustrious Bernoulli family, which included at least three generations of famous and productive mathematicians and physicists (Hopkins, 1991). Daniel, the second son of Jacob’s brother Johann (Fye, 2001), who was a trained physician, learned mathematics from his older brother Nicolaus III. He became Professor of Mathematics in 1725 at St Petersburg, Russia where he did most of the work on his principal opus, *Hydrodynamica*. Life there irked him so greatly that he returned at the first opportunity to Basel, where he became Professor of Anatomy and Botany, and finally of Physics. His mathematical work included calculus, differential equations, probability, the theory of vibrating strings and many other problems in applied mathematics. Daniel Bernoulli has been called the founder of mathematical physics.

The nineteenth century saw further advances in the theory and application of arterial hemodynamics, principally in Germany, England, and France, and again often by young achievers in many disciplines. One of the most prominent was Poiseuille (1799–1869), whose research

in cardiovascular physiology began when he was still a medical student (Hopkins, 1991). Poiseuille was fundamentally a physician with training in physics. His major contribution was to establish the relationship between flow, pressure gradient, and dimensions of a capillary tube; Poiseuille showed experimentally that flow was related to the fourth power of the tube’s internal diameter (1846). Poiseuille’s equation is now expressed as:

$$P_1 - P_2 = \frac{\text{Flow} \times \text{viscosity} \times \text{length} \times 8}{\pi \times (\text{radius})^4}$$

but the precise derivation of this from theoretic principles was attributed to Hagenbach (1860). Poiseuille expressed (viscosity/ π) as a constant. His measurements of relatively low venous pressure in animals (Poiseuille, 1830) suggested to him that a marked drop in pressure must occur somewhere beyond the high pressure that he had measured in arteries down to approximately 2 mm in diameter (Poiseuille, 1828), thus confirming Hales’ contention that the greatest component of peripheral resistance resides in the smallest blood vessels (Fig. 1.3) (Poiseuille, 1828). This (earlier) work explains Poiseuille’s interest in tiny capillary tubes for determining mean pressure–flow relationships.

A man of similar background to Poiseuille as physicist and physician, but who covered an even greater range, was Thomas Young (1773–1829). Young’s scientific exploits are legendary (Hopkins, 1991). His most famous researches in biophysics were concerned with human vision, particularly with the perception of color. The theory that human vision is based on three primary colors – red, green, and violet – is commemorated today as the Young–Helmholtz theory. In fact, his work on light went

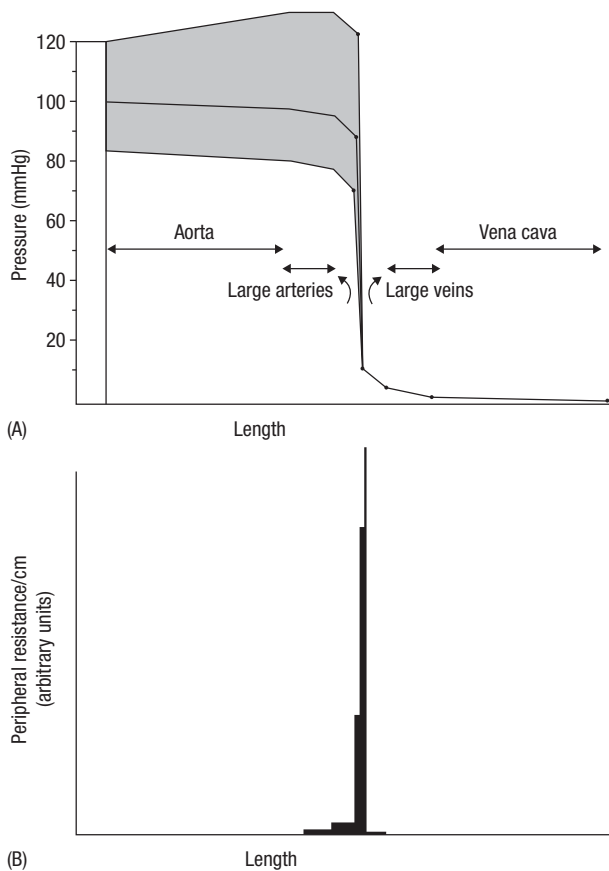


Figure 1.3 (A) Change in pressure between left ventricle and right atrium along the whole vascular pathway. Mean pressure falls precipitously in the arterioles. Pressure wave amplitude is greater in peripheral than in central arteries as a consequence of wave reflection, but falls precipitously in prearteriolar vessels. (B) Change in resistance per unit length along the vascular pathway. Reproduced from O'Rourke (1982b)

much deeper, and that on interference gratings led to the acceptance of the wave nature of light (Williams, 1969). However, the other great body of biophysical work done by Young was into the nature of elasticity, in particular the relationship between the elastic properties of arteries and the velocity of propagation of the arterial pulse (Young, 1808; Young, 1809; Laird, 1980). His work on the nature of elasticity led to the concept of the elastic modulus to which his name is still attached. In determining wave velocity, he derived the formula (given as eqn 3.49) that Bramwell and Hill (1922a) attributed to AV Hill and which other writers (e.g. Bergel and Schultz, 1971) attributed to Otto Frank (1865–1944). This preceded the work of Moens (1878) and of Korteweg (1878) (eqn 3.51) into the relationship between the arterial pulse wave velocity and Young's elastic modulus of the arterial wall. It is finally of interest to add that Thomas Young created an independent reputation in successfully deciphering Egyptian hieroglyphics after the discovery of the Rosetta Stone.

The reference to Egypt creates a link with Fourier (1768–1830), who was one of the band of savants that Napoleon took with him to Egypt and who were stranded there for several years. Fourier's work on the analysis of periodic functions was apparently carried out here and is published in his monograph on heat. Fourier never contributed directly to hemodynamics, but his name occurs so frequently as originator of the Fourier (or harmonic) series that it can hardly be omitted from a historical survey.

The next work of interest in the field of arterial distensibility and wave velocity was the result of collaboration in Germany between the brothers WE and EH Weber (physicist and physician, respectively) and published in a monograph entitled *Wellenlehre* in 1825. This established many of the properties of propagated and reflected waves. This work by the Webers on wave propagation was soon followed by the experimental work of Moens in Leiden, published in his short monograph of 1878. The mathematical analyses of Korteweg and Resal on his data give the equation (written as eqn 3.51) which remains the most useful relationship between pulse wave velocity and arterial properties.

The work of Young, Weber, and other physicians had important clinical implications that were well appreciated at the time. These became more important with the description by Bright (1827) of a relationship between hardening of the arteries, albuminuria, cardiac hypertrophy, and risk of death from stroke and cardiac failure. This work was extended by many, including Marey (1863) and Mahomed (1872) who first described changes in contour of the arterial pressure pulse with age and in hypertension, changes that were forgotten with the introduction of the cuff sphygmomanometer, then rediscovered 100 years later (see Chapter 18) (Braunwald, 2010). Like Poiseuille, Mahomed's research contributions began as a medical student when he developed and described the first quantitative sphygmograph (Mahomed, 1872). Using this, he was the first to detail the natural history of 'essential' hypertension (Mahomed, 1874) and to separate 'essential' from renal hypertension (see Chapter 19 and O'Rourke's (1992), Swales' (1996a), Lewis' (1997), and Cameron and Hicks' (1996) historical tributes to this great pioneer).

The implications of arterial stiffening with respect to cardiac load were, it seems, appreciated at this time, together with the relentless stiffening that occurs in humans with age. Roy (1880) pointed out that 'only in the case of young children do we find that the elasticity of arteries is so perfectly adapted to the requirements of the organism as it is in the case of the lower animals'. The clinical achievements of the later nineteenth century were all the more remarkable when one appreciates that there were no accurate means of measuring arterial blood pressure. The cuff of Riva-Rocci was not introduced until 1896 and the use of Korotkov's sounds not until 1905. Prior to this, high arterial blood pressure was assessed

on the basis of the force required to compress the radial artery and on the amplitude and contour of the arterial pressure pulse. A variety of sphygmographs had been introduced by Marey (1860), Mahomed (1872), Mackenzie (1902), and others, but these could not be calibrated in absolute terms and their dynamic frequency response was poor. Although Torricelli (1608–1647), Galileo's successor as Professor of Mathematics at the Academia in Florence, was the first to use mercury in a glass tube to measure (atmospheric) pressure, Poiseuille (1828) introduced the mercury manometer (U-tube) to measure absolute values of intra-arterial blood pressure. Ludwig (1847) added a float and pointer, enabling pressure to be recorded on a smoked drum. The inertia of such systems was, of course, considerable, so that only mean pressure with slight fluctuations could be obtained. It was known that these small fluctuations occurred simultaneously with cardiac contraction, but their magnitude was too low to represent systolic and diastolic blood pressures. One of Ludwig's pupils, Adolph Fick (Acierno, 2000), was the first to devise a manometer that responded rapidly enough to record systolic and diastolic pressures relatively accurately (1864).

The outstanding difference between the records obtained with Ludwig's mercury manometer and Fick's C-spring recorder was the magnitude of the fluctuations (or pulsations). How large they should be was unknown, and renewed efforts were made to increase the frequency response of the recording system. In 1883, Fick introduced his straight-spring manometer and recorded the first arterial pressure wave that showed a slight hint of a dicrotic notch, although Landois had already demonstrated the presence of a true dicrotic notch in 1874 (Luciani, 1911). He placed a small needle into a dog artery and directed the jet of blood onto the rotating drum of a kymograph. There, clearly displayed, was the dicrotic notch, along with systolic and diastolic pressures. Landois called his record a 'hemautograph'. The development of reliable manometers that could accurately measure pulsatile blood pressure had to await Otto Frank's great work of 1903.

The French physiologist/physician EJ Marey made important contributions not only to non-invasive recordings of arterial pressure waves (Marey, 1863), but also to direct measurements of arterial blood flow. Marey's textbook of 1881 (see Snellen, 1980) includes a complete chapter on the velocity of blood flow in arteries. The technique used was that of a double Pitot tube, which is based on the Bernoulli principle and is used in modern aircraft to measure air speed. (A defective pitot tube is thought to have contributed to the fatal crash of an Air France A330 into the South Atlantic Ocean in 2008.) The Marey tracings look surprisingly like those that we accept as normal today, clearly showing the back flow phase of flow in the carotid and femoral arteries. As with Mahomed's observations, these findings took another 100 years to be rediscovered and accepted (see Chapter 9). Marey's

special interest was in biological motion – of birds in flight and of running horses. His rapid-sequence photographs were the forerunner of modern cinematography (and coronary angiography). As with others previously mentioned, he was a master of many fields. Carl Ludwig appears to have been the first to introduce a continuously recording blood flowmeter. This was an ingenious though awkward device in which blood, flowing through an exteriorized tube, pushed before it a column of oil. As with his pressure recordings, this provided better registration of mean than of pulsatile flow. Adolph Fick is known for the principle enunciated by him that related blood flow through an organ to the arteriovenous concentration gradient of a substance and the amount of the substance taken up or given off by that organ (Fick, 1870). This principle became the standard for measurement of cardiac output in humans following the first human cardiac catheterization by Forssmann (on himself) in 1929 and its popularization by Cournand, Richards, and others in the early 1940s, and the developments which have followed (Mueller and Sanborn, 1995; Ryan, 2002; Braunwald, 2009).

The early twentieth century was completely dominated by Otto Frank and his pupils in Munich. Frank had been highly trained in physics and mathematics and apparently did not turn to physiology until in his late twenties. Frank's first major contribution was to cardiac contraction (in 1895), and it is for this that he is best remembered today. Frank was the first to describe the influence of ventricular dimension on ventricular contraction; this phenomenon, clarified by Starling (Patterson *et al.*, 1914; Starling, 1918), is usually referred to as the Frank–Starling mechanism (Konhilas *et al.*, 2002) or the 'law of the heart' (Katz, 2002). From this early work of Frank sprang the need to produce a manometer that could measure pulsatile blood pressure accurately (Frank, 1903). The manometer that emerged, as with Einthoven's electrocardiogram introduced at the same time (Einthoven, 1903; Fisch, 2000), was based on a very sound physical analysis of the requirements of such an instrument; the analysis remains valid to this day.

Frank passed from cardiac dynamics to manometry, to the measurement of arterial pressure waves, and on to analysis of arterial behavior (Frank, 1905). Frank's penultimate goal was the understanding of arterial properties and arterial function, so that (ultimately) flow could be predicted from pressure change. Beginning with accurate arterial pressure waves, he sought a theoretic basis for arterial function. The framework of his approach was that of manometer theory – that the arterial system behaved like a manometer with a 'lumped' distensible section or *Windkessel* and 'lumped' peripheral resistance. Such analysis ignored wave propagation and wave reflection (McDonald and Taylor, 1959). Frank had the uneasy task of reconciling this theory with the obvious secondary pressure oscillations or diastolic waves that were apparent on his pressure tracings from peripheral arteries. He did

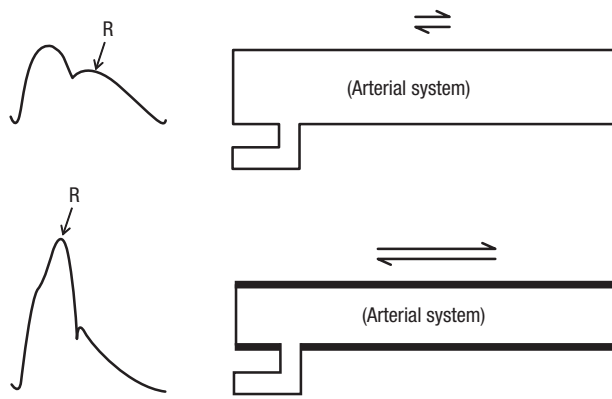


Figure 1.4 Tubular models of the arterial system with the heart (at left) and conduit arteries and lumped peripheral resistance (at right). Coronary arteries are represented by the 'hook'. Pressure waves in the ascending aorta are represented at left. The top model represents the normal youthful arterial tree. The bottom model represents the stiffened arterial tree of an older person with increased aortic pulse wave velocity. Reflection (R) is represented by the diastolic wave in the top case and by the late systolic wave in the bottom case. Reproduced from O'Rourke *et al.* (1987)

so with the awkward compromise of accepting at the same time the infinite wave travel of a *Windkessel* and the finite wave travel of an elastic liquid-filled tube (Fig. 1.4). The solution was more Chinese (related to paradoxical logic) than Germanic (related to Aristotelian logic), more feminine than masculine, utterly uncharacteristic of the man as history portrays him personally. Frank (1899) viewed the arterial pressure pulse as having a *Grundform* or basic pattern upon which was superimposed a *Grundschiwingung* or damped oscillation created by wave travel and reflection generating a type of resonance. A similar approach was later introduced by Cohn and colleagues (Cohn and Finkelstein, 1992; Cohn *et al.*, 1995) and by Davies *et al.* (2007) for calculating the components of arterial compliance. Assuming wave reflection as being from a closed-end reflecting site (as is usually accepted today), Frank proceeded to calculate the length of the *Windkessel* and then went on to estimate flow. The compromise was brilliant. Frank's contributions have been criticized by many (including ourselves) as contradictory and illogical. Viewed from a farther distance in time, and allowing for the primitive state of knowledge at that time and the need to attempt some reconciliation of apparently conflicting observations so that the field might advance, it is clear that one should be far more generous. One's criticisms are more appropriately directed at the unwieldy mathematical edifices that have been created about the initial simple analog model by those followers of Frank who have overlooked the conflicting basic assumptions that were so apparent to him. Such contributions are still seen appearing anew in the literature today (Davies *et al.*, 2007; Cameron, 2009; Manisty *et al.*, 2009). This subject was

well reviewed by Apéria (1940) and by McDonald and Taylor (1959). It was alluded to also by Birkhoff (1960), who, quoting Sir Cyril Hinshelwood, said 'fluid dynamists in the nineteenth century were divided into hydraulic engineers who observed what could not be explained and mathematicians who explained what could not be observed', and added, 'it is my impression that many survivors of both species are still with us'. Some prosper in clinical departments where physical and physiological scrutiny is lacking or dismissed. Good clinical researchers can be 'hoodwinked' (Sharman *et al.*, 2009b).

The First World War destroyed cardiovascular research in Europe. The classic article by Patterson *et al.* (1914) on 'The Law of the Heart' only just made publication, as the sole item in the last issue of the *Journal of Physiology* to be published in five years. Piper was killed in the war. Starling was directed to practical matters of military importance. Frank's work was terminated by the war and never recommenced. His laboratory was destroyed by a bomb shortly after his death at the end of the Second World War. In cardiovascular physiology, the period since the First World War saw the emergence of and ultimate domination by research scientists in the United States. Early pioneers include Katz, Wiggers (a student of Frank), and Hamilton, all of whom wrote influential textbooks and reviews before and after the Second World War. In England, Bramwell and Hill (1922b) described cardiac function in relation to hydraulic load and showed how arterial stiffness (measured as pulse wave velocity (PWV)) was an important component of this load, and how PWV increased in humans with age (Fig. 1.4).

The electromagnetic flowmeter was introduced independently by Wetterer (1937) in Germany (1937) and by Kolin (1936) in the United States. Interference due to the electric potentials created by the heart were overcome, and Deppe and Wetterer (1940) used it successfully on the ascending aorta; its use there and in other arteries in animals became commonplace. Gregg and colleagues (Gregg and Green, 1940; Gregg *et al.*, 1948; Gregg and Fisher, 1963) recorded accurate flow waves in the coronary and other arteries of dogs using this instrument. The period between the world wars had seen little advancement in cardiovascular medicine apart from acceptance of the cuff sphygmomanometer and the electrocardiograph into clinical practice. This period contrasts with that which followed 1944. One can still argue, as did Mackenzie in 1917, that the sphygmomanometer proved a disservice to cardiovascular medicine. While it enabled pressure to be recorded reasonably accurately, quickly, and non-invasively in individual subjects, it concentrated attention on 'the systolic' and 'the diastolic' pressure, inducing an aura of complacency of mind. Clinicians, and indeed researchers, came to consider that systolic and diastolic pressure had some particular biological significance instead of estimating just the highest and the lowest point of a pressure wave in the brachial artery at the time of measurement. Such a view takes no account

of pressure wave amplification between central and peripheral arteries; this renders both systolic and pulse pressure different in different arteries and susceptible to the various influences that alter amplification. The view also encourages the simplistic notion referred to previously, first voiced posthumously in Mackenzie's text but attributable to James Orr, his literary executor, that systolic pressure is a manifestation of cardiac strength and diastolic pressure a manifestation of arteriolar tone (and resistance). This is the basis of the myth that still persists in many quarters that diastolic pressure is the appropriate gauge of elevated arteriolar tone in systemic hypertension (O'Rourke, 1983a; Fisher, 1985; Nichols and Edwards, 2001; Arguedas *et al.*, 2009).

The clinical scene since 1944 has not yet been able to overcome the shackles of restrictive thought that surround concepts of systolic and diastolic pressure, but has exploded in other directions. The introduction of cardiac catheterization by Cournand and Ranges (1941) led to surgical correction of cardiac defects and to the introduction of other invasive and non-invasive techniques into the clinical arena. Intravascular pressure is now recorded routinely not only in the diagnostic laboratory, but in the operating room and critical care area as well. Cardiac output can be recorded by a conventional Fick technique (Acierno, 2000), or by indicator dilution or thermodilution at the same time. Micromanometers have been introduced, permitting the measurement not only of arterial pressure waves and their differentials but also of cardiovascular sound (Murgu and Millar, 1972). Electromagnetic velocity sensors have been miniaturized and mounted on cardiac catheters for the measurement of phasic blood flow velocity (Mills and Shillingford, 1967; Nichols *et al.*, 1980a). Catheter-mounted ultrasonic devices were developed and used for the measurement of flow velocity waveforms (Hartley and Cole, 1974a; Hartley and Cole, 1974b; Bertolet *et al.*, 1996; von Mering *et al.*, 2004; Pepine *et al.*, 2007) and for the visualization of arterial wall structure (Nishimura *et al.*, 1991; Reneman *et al.*, 2006; Nicholls *et al.*, 2008; Bayturan *et al.*, 2009). Ultrasonic techniques have become office procedures for visualizing the chambers of the heart and the blood vessels, and for Doppler non-invasive flow velocity recording. Possibilities for other imaging techniques, such as magnetic resonance imaging and computed tomography, abound and are presently giving new insight into the development of cerebral microvascular disease (Batesman, 2002b; Henry-Feugeas *et al.*, 2005; O'Rourke and Hashimoto, 2007; Lin *et al.*, 2007; Kang *et al.*, 2009). The disease on which they can be used has become not only the most common in the Western world, but correctable to boot (Gibbons *et al.*, 2009; Lauer, 2009; Matter *et al.*, 2009; Wu *et al.*, 2009; Achenbach and Raggi, 2010; Mark *et al.*, 2010).

With the introduction of new methods for measuring phasic blood flow, and with the availability of existing techniques for measuring phasic pressure, a need arose

for more sophisticated analysis of the recorded data. Until 1955, there had been little attempt to do anything other than analyze the waves as they were displayed (i.e. in the time domain). This approach had been explored almost *ad nauseam* for pressure and showed little prospect of giving new information on flow or on the relationship between pressure and flow waves. Physiologists (as well as clinicians) of the time had been content to describe waves in terms of their obvious features, using such descriptive terms as 'foot', 'peak', 'anacrotic wave', and 'dicrotic notch'. This is clearly unsatisfactory from the analytic point of view, however familiar it may be in conveying information of a sort from one person to another. However, the terms used were confusing and descriptions often conflicting (O'Rourke, 1971; O'Rourke, 1982c). Neither the American Heart Association, the American College of Cardiology, nor the European Society of Cardiology have yet been able to provide a consensus document on description and terminology of that most fundamental of all signs, the arterial pulse! Yet methods of description and of analysis have been available and routinely used in the physical sciences. These, of course, utilize the fundamental language of science – mathematics. This subject was addressed by DA McDonald with the mathematician John Womersley in a series of papers published in 1955 and afterwards (McDonald, 1955; leading up to the review by McDonald and Taylor (1959) and the first edition of this book in 1960. The most fundamental and important advance was the demonstration that the arterial system could be viewed as being in a steady state of oscillation, so could validly be analyzed in the frequency domain (Fig. 1.5).

THE ARTERIAL SYSTEM IN STEADY-STATE OSCILLATION

In an attempt to apply mathematical techniques to the circulation, it is obvious that one must be able to describe the phenomena studied in such a way that numerical values can be attached to them. In terms of physical and mathematical practice, a wave is regarded either as an isolated phenomenon or as a repetitive one. That is, it is a transient or it is a periodic oscillation. As the mathematical description of a periodic oscillation is simpler than that of a transient one, it was considered reasonable to suppose that it was the preferred approach, provided that it was valid. Regularity of the heart beat is one of its most characteristic features, and in terms of the length of an individual pulse this regularity is normally maintained for a very long period of time. This is indeed a condition of steady-state oscillation. In these circumstances, any wave that is repeated regularly can be represented by a Fourier series; that is, the wave shape can be described as the sum of a set of sinusoidal waves whose frequencies are

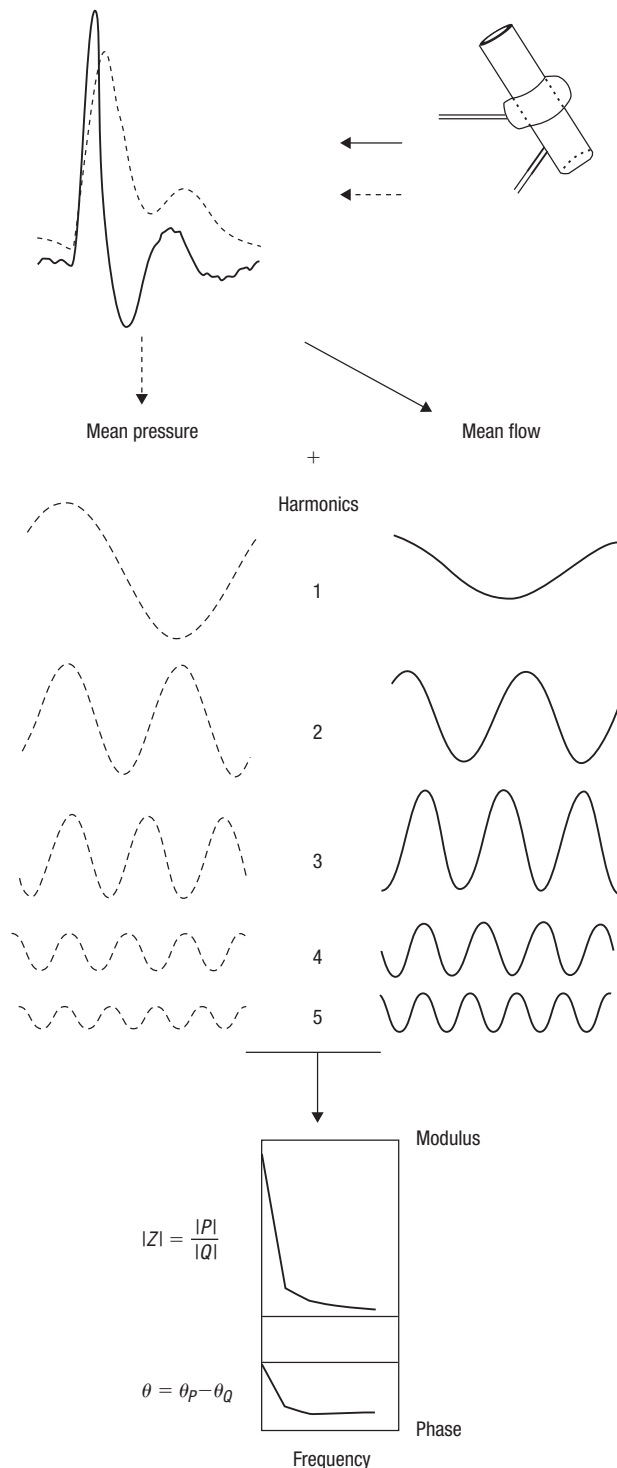


Figure 1.5 Use of Fourier analysis to determine arterial input impedance. Pressure (dotted line) and flow (solid line) measured in a peripheral artery are decomposed into mean values and a series of harmonic (sine) waves at multiples of heart rate frequency. Only the first to fifth are shown. Impedance modulus (top line of lower panel) is the modulus of pressure (P) divided by modulus of flow (Q) at the different frequencies, and phase (bottom line of lower panel) is the delay between pressure and flow harmonics. Reproduced from O'Rourke and Taylor (1966)

all integral multiples of the frequency of repetition of the wave.

The Fourier series and the method of performing Fourier analysis are described in Chapter 11. It is familiar to most people in terms of the harmonic analysis of musical sounds, a connection that is emphasized by the use of phrases such as 'harmonic components' or 'over-tones'. It is widely used in the physical sciences of acoustic, electrical, and aeronautical engineering. In physiology, it is generally accepted as a basis for determining the frequency response of manometers and for recording arterial pressure pulses (see Chapter 6). It is, therefore, logical to extend its use to the flow pulse and the physical relation between pulsatile pressure and flow. It has the advantage of dealing with simple harmonic motion, which, mathematically, is the most tractable form of oscillatory motion, the compound synthesized curve or wave being the simple sum of the component harmonic terms. This property of summation of its component terms is one reason for its outstanding usefulness in describing curves. From this is derived the important practical property of the series that the curve described by a certain number of terms is the best approximation possible for that number of terms. Computing further terms will only improve the fit where it is faulty and does not require any alteration in the lower harmonics.

In applying Fourier analysis to the circulation, certain approximations have to be made. In the first place, there are irregularities in the pulse frequency. As it is a requirement for the Fourier series that the system should be in a 'steady state', any alteration in frequency must be allowed to settle down, just as the conditions due to the starting up in a system must have disappeared before a steady state is attained. The rapidity with which a steady state is attained depends largely on the damping in the system which is being driven; in the arterial system of a dog, early evidence suggested that this is attained within 1.0 s and probably within 0.5 s (i.e. once the pulse has been regular for at most two or three beats). Later evidence (O'Rourke and Taylor, 1966) indicated that a steady state could be achieved well within 0.5 s, so that individual waves could be regarded as part of a steady-state oscillation, even though the heart was beating irregularly.

A more complex difficulty with respect to application of Fourier analysis arises over the question of non-linearity. The circulation is not a linear system, so that, even though a Fourier series may represent a pressure pulse or a flow pulse, one cannot, strictly speaking, say that one harmonic term of pressure is related exclusively to the corresponding harmonic term of flow. The harmonic components must interact with each other. This non-linear effect was calculated by Womersley (1957b) with respect to the motion of blood and found to be small enough to be regarded as negligible to a first-order approximation. Later experimental work has confirmed Womersley's predictions again and again (Dick *et al.*, 1968; Nichols *et al.*, 1977a; Noble, 1979b; O'Rourke,

1982a; O'Rourke, 1982b; Milnor, 1989). This issue is no longer a subject of contention, though it could re-emerge as more accurate measuring and analytical techniques are introduced.

The physical reality of the Fourier representation of impulses having complex shape has often been debated. The apparent artificiality of the statement that the only frequencies represented are integral multiples of the fundamental frequency suggests to the skeptic that this is merely a mathematic device. Yet it can easily be seen that if frequencies other than integral multiples were present, they would not repeat at the fundamental interval, so successive waves would not be of the same form, which denies the condition for applying the analysis. Put another way, this means that the cyclical period to be analyzed must be one in which the whole pattern is exactly repeated.

Further discussion on Fourier analysis, frequency spectrum analysis (Taylor, 1966a; Taylor, 1966b; Taylor, 1966c; Taylor, 1966d), and other analyses of wave contour is presented in Chapter 11. Womersley, McDonald, and Taylor established the validity and suitability of applying Fourier analysis to circulatory phenomena even if they were not the first to propose its use (Apéria, 1940; Porjé, 1946; Randall, 1958; McDonald and Taylor, 1959). This approach has been utilized by many over the last 50 years and forms the basis for the work presented in this sixth edition of *Blood Flow in Arteries*, as it did in the first edition.

ANALYSIS IN THE FREQUENCY AND TIME DOMAINS

The concept of the arterial system as being in steady-state oscillation is stressed here because it is foreign to most physiologists and physicians, and because it has been so helpful to us in studies of the arterial system. It has, however, led to the more general concept of frequency spectrum analysis (Taylor, 1965), which, while it does in theory describe the pulse (or better still a series of pulses) as a transient, expresses results in the frequency domain just as does Fourier analysis, while having the additional advantage of covering the whole frequency range between the individual harmonic frequencies of a single pulse. The full potential is realized when applied to data obtained with the heart beating irregularly, as with random electrical stimuli applied to the heart or when the heart rhythm has converted to atrial fibrillation (Taylor, 1966d). Frequency spectrum analysis and other analytical techniques, including the method of characteristics and controversial use of wave intensity analysis, are described more fully in Chapter 11.

It goes without saying that the results of one method of analysis must be reconcilable with the results of another method when applied to the same data. It should be possible to reach the same conclusions with any analytic

method. This was a problem at the time that the first edition of this book was published in 1960, when the field of arterial hemodynamics was polarized into those who could only see the pulse as a transient (see Hamilton, 1962; Spencer and Denison, 1963), and those like McDonald who could only see the pulse as a series of harmonics. It is possible to draw these two approaches together and link them with others (see Avolio *et al.*, 2009a; O'Rourke, 2009a; and other papers in a special issue of *Medical, Biological, Engineering and Computing*).

In a book such as this, emphasis must be given to details, theoretic and experimental, of blood flow and pressure-flow relationships in different segments of the arterial tree, but the overall perspective of the circulation and its purpose must be retained. Described in its simplest form, the circulation consists of a pump, the heart, which forces blood periodically and rhythmically into a branching system of elastic tubes. The pulsations so generated travel centrifugally and are partially reflected at points of discontinuity (or change in impedance) to travel backwards again (see Chapter 10), while pulsations themselves are normally damped by the time they reach the smallest branches, the capillaries, which are in intimate contact with the cells of the tissues. The blood then returns in a more or less steady stream to the right side of the heart with secondary pulsations imposed in veins by skeletal muscular activity and by the heart itself. The circulation is, of course, far more complicated than this, and the arterial system markedly so, with varied geometric patterns of branching, non-uniform elasticity of arteries, non-linear elastic wall properties, and anomalous viscosity of blood in smaller vessels. Solutions to these problems are sought by detailed descriptions of individual parts of the arterial system, using analytic techniques that are foreign to many and often difficult to comprehend. However, once arterial hemodynamics has been dissected into its component parts, one can synthesize the whole from analysis of arterial pressure and flow waves in single arteries; one then finds that the analysis of the whole complex system of the vascular tree proves to be a great deal simpler than a first look suggests. This book has been written in the hope that such a view – of cohesion, of harmony, and of simplicity arising from complexity – can be transmitted to men and women entering the field, with the expectation that young and flexible minds will enlarge and extend the field and provide further clarification. In this spirit, we provide a summary list of what we see has evolved over the 50 years of this book's publication.

APPLICATION TO CLINICAL PROBLEMS

Application requires simplification. On the basis of what is presented throughout this book, and on the presumption that recordings are accurate, we offer the following as present art, and a tentative foundation for future work.

- To a first approximation, a linear relationship may be assumed between harmonic components of pressure and of flow waves, such that the relationship between pressure and flow at the same point may be described as vascular impedance, and the relationship between pressure waves at different points can be described as a transfer function, i.e. in terms of modulus and phase plotted against frequency.
- Fatigue and fracture of non-living components of the arterial wall can be explained and predicted from conventional engineering theory on strain $\times$ frequency curves – i.e. from the relationship between degree of stretch at each cycle and the cumulative number of cycles. This explains degeneration of the aorta and proximal elastic arteries with age, the progressive increase in aortic systolic and pulse pressure, or aortic diameter and increased left ventricular (LV) load and predisposition to hypertrophy and ischemia with age.
- Other things being equal, wave reflection in the arterial tree can be estimated from the secondary boost to arterial pressure after the peak of flow velocity in the ascending aorta.
- The relationship between velocity of LV contraction and afterload is such that increased load or impairment of contractility reduces aortic flow velocity after the early peak of flow velocity in the ascending aorta.
- The secondary boost or augmentation of pressure in late systole can only be interpreted in terms of wave reflection when the aortic flow wave is of almost normal contour – i.e. when the heart contracts normally and functions as a flow source – with output essentially independent of afterload.
- The major component of LV afterload is wave reflection from peripheral sites, mainly in the lower body. Wave reflection can increase pressure or decrease flow or cause a mixture of both. Pressure increase dominates when the heart acts as a flow source. Flow decrease dominates when contraction is weak and the heart acts as a pressure source.
- Drugs which reduce wave reflection have a favorable effect on the heart when reflection returns during systole – i.e. in adults from the third decade on. When the heart is beating strongly and normally, wave reflection decrease is manifest as fall in aortic and LV late systolic pressure (an antihypertensive effect). When the heart contracts weakly, wave reflection decrease is manifest as increase in aortic flow in late systole, with increased LV emptying (an antiheart failure effect). Hence, such drugs are effective in treating both hypertension and heart failure.
- Drugs which decrease muscular tone have a far greater effect on peripheral muscular arteries than on central elastic arteries, and so a far greater effect on wave reflection than on stiffness of the aorta and other elastic central arteries.
- Stiffening of the aorta with age is due to the fracture of elastin fibers with transfer of stress to more rigid components of the wall. The aorta becomes less able to cushion flow pulsations; these extend peripherally into the smallest arteries of vasodilated organs, principally the brain and kidney. Consequent high pulsatile shear stress and radial stress in the small vessels causes damage with obstruction of lumen and disruption of the wall. Consequences include dementia and renal failure.
- The century-old method for assessment of blood pressure (the brachial cuff) is an inadequate method for prognosis and gauge of therapy, and needs to be supplemented by other methods for precise description of arterial stresses and LV load.
- Measurement of aortic pulse wave velocity as an index of stiffness is a robust predictor of cardiovascular events.
- Measurement of aortic central pressure by validated methods adds substantially to use of cuff brachial pressure for predicting cardiovascular events.
- Central aortic pressure cannot reliably be estimated by applanation tonometry applied to the brachial artery at the elbow.
- Subtle effects of gender and race cannot reliably be estimated from cuff sphygmomanometry, but can be uncovered non-invasively from the pressure waveform measured at the wrist or in the neck.
- Entrainment between heart rate and step rate can explain some benefits and hazards of long distance running – the rhythm of running to which athletes aspire, and cardiac arrest from extreme ischemia.

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The nature of flow of a liquid

Steady pressure–flow relations (Poiseuille's equation)	14	Inlet length	41
Viscous properties of whole blood	19	Non-laminar liquid flow	48
Bernoulli's equation	26	The measurement of turbulence	51
Lateral and impact pressure	27	Flow disturbances in the region of cardiac valves	52
Torricelli's equation	28	Flow disturbances in large arteries	52
The effect of change in the size of the vascular bed	29	Turbulence as a cause of post-stenotic dilatation and aneurysms	53
The measurement of peripheral resistance	31	The potential role of turbulence in atherogenesis	54
The dimensions of the arterial tree	35		
The velocity profile in pulsatile flow	37		

It is convenient in mechanics to classify matter into solids and fluids. A solid substance has a definite size and shape and can be either perfectly elastic or plastic (see Chapter 3). A fluid is a substance that can flow and includes both liquids and gases. Gases are easily compressed, whereas liquids are almost incompressible. This text deals almost exclusively with liquids and the physical laws that govern their flow.

In considering the vascular system with its contained blood, we are naturally concerned with the physical laws governing the flow of liquids through solid tubes. The simplest example is that of a straight, uniform, rigid tube with a steady (or constant) rate of non-turbulent liquid flow through it. To maintain such a steady flow, there must be a constant head of pressure applied to the liquid because of its viscosity or 'internal friction'. Steady flow in a rigid cylindrical tube is described by the well-known 'Poiseuille equation', which states that the pressure drop (or gradient) along the tube is directly proportional to the length of the tube, the rate of flow through the tube, and the viscosity of the liquid, and inversely proportional to the fourth power of the internal radius. If dye is injected into the tube, the liquid in the axis of the tube moves much faster than that near the wall, and the front of the dye assumes a paraboloid shape. This occurs because the particles of liquid are flowing in a series of laminae parallel to the sides of the tube and the liquid in contact with the wall is stationary and each successive lamina is slipping against the viscous friction of the lamina outside it. When flow occurs in such parallel laminae, it is called

'laminar' (Kaufmann, 1963). If two tubes are joined to form a trunk and dye is injected into only one of them, there is no mixing transversely across the tube, and the two streams remain distinct and are seen to flow side by side in the main trunk; this phenomenon is called 'streamlining' (McDonald, 1974; Nichols and O'Rourke, 1990; Nichols and O'Rourke, 1998). Hence, laminar flow is often called 'streamline flow'. Alternatively, it may be called 'Poiseuille-type flow' because it obeys the Poiseuille equation (Caro *et al.*, 1978; Milnor, 1989; Fung, 1997; Wood, 1999).

If the rate of flow through a tube is continuously increased, there comes a point when the resistance to flow increases quite sharply and the Poiseuille relation no longer applies. Dye injected at this stage into the stream shows the liquid mixing across the tube and the particles of dye, and hence of the liquid, are no longer moving regularly in the line of flow, but are following more or less random paths across the entire cross section of the tube in addition to their main movement along the tube. The flow is then said to be 'turbulent' (see below). The pressure–flow relationships of turbulent flow are not predictable with precision, therefore, it is important to know whether the flow being studied is laminar or turbulent. It should be emphasized, however, that this classical distinction of flow types is only correctly defined for steady flow of a Newtonian liquid in straight rigid cylindrical tubes; there are intermediate stages of instability in the liquid that become important in the irregular flow systems of the living animal.

STEADY PRESSURE–FLOW RELATIONS (POISEUILLE'S EQUATION)

The first work dealing with the steady flow of a Newtonian liquid in cylindrical tubes was that of Girard (1813) using brass tubes of 2.0–3.0 mm diameter. He obtained the relationship

$$Q = \frac{KD^3(P_1 - P_2)}{L}, \quad 2.1$$

where Q is the volume flow per unit time, K is a constant, D the internal diameter and $(P_1 - P_2)$ the pressure drop (or difference) along the length, L , of the tube. Thus he observed that the flow varied directly with the cube of the internal radius. Ten years later, Navier (1785–1836) made the first deductions of the theoretic equations for flow of viscous liquids in cylindrical tubes, but obtained the incorrect result, already apparently confirmed by Girard's experimental work, that the flow was proportional to the cube of the radius.

The work of Navier (1827) was not completely successful and was later (Poisson, 1831) developed by Poisson (1781–1840), but both used assumptions of intermolecular forces that are now regarded as quite unrealistic. Stokes (1819–1903) reworked the entire problem (Stokes, 1845), and the resulting equations are known as the Navier–Stokes equations (see Chapter 4), which are regarded as the basic equations for the flow of viscous liquids. The viscous Navier–Stokes equations can be used to describe the flow and the linearly viscoelastic membrane equations to model the mechanical properties of the arterial wall.

The first published experimental work indicating that liquid flow is proportional to the fourth power of the radius was due to Hagen in 1839. He used brass tubes of a similar size to those of Girard; however, the results were not very accurate. The exponent of the radius derived from his results was actually 4.12, and he assumed that the real value was probably 4.0. Poiseuille (1799–1869) published his first results in 1842, although there was not a full paper published until 1846 (Poiseuille, 1846). Due to Hagen's priority of publication, some references use the name 'Hagen–Poiseuille equation'. Poiseuille's work, however, was much more detailed and precise (Joly, 1968), and, therefore, it is fair to name it Poiseuille's equation.

Poiseuille's first important work, in 1828, was the introduction of the mercury filled U-tube manometer to measure blood pressure (in mmHg), a technique that was quickly adopted by other scientists and is still in use today. Curiously enough, his early experiments with this device led him to believe that pressure remained constant as blood moved through the larger arteries. Measurement of the extremely small decrease in mean arterial pressure (1.0–2.0 mmHg) that actually occurs along the aorta and out to the smallest arteries exceeded the limits of

resolution of his measuring device, and he concluded that 'a molecule of blood moved with the same pressure during its entire course through the arterial system' (Poiseuille, 1828). His measurements of relatively low pressure in the veins (Poiseuille, 1830), however, suggested to him that a marked drop in pressure must occur somewhere beyond the small arteries and this led him to study liquid flow in very small tubes. Herrick (1942) gives a most interesting account of the physiologic observations that Poiseuille made regarding the circulation before he turned to his classic work on the flow of liquids.

Poiseuille may be regarded as fortunate in two respects. In the first place, although he had training in physics, he was also a physician who wanted to apply the results of his investigations to the understanding of the blood circulation. Hence, Poiseuille worked with glass tubes of capillary size where his predecessors had been engineers and worked with much larger tubes. Minute tubes maintain laminar flow more easily and also greatly facilitate accurate measurement. In the second place, he was deflected from his original intention of using blood as a test liquid because no satisfactory means of rendering it incoagulable was known, and he was compelled to confine his investigation to water. Blood flowing in capillaries shows anomalous viscous properties that would have introduced great complications in these pioneer studies.

Poiseuille used rigid uniform capillary tubes varying in internal diameter from 0.03 to 0.14 mm because of his conclusion that tubes of such small size provided the greatest resistance to blood flow through the circulation. His results were expressed by the equation

$$Q = \frac{KD^4(P_1 - P_2)}{L}, \quad 2.2$$

where, as above, Q , $(P_1 - P_2)$, and L are the volume flow, pressure drop, and length of the tube, respectively; D is the tube internal diameter and K is a constant.

The value of K was determined under various conditions and shown to fall with decreasing temperature. This constant is clearly a measure of the viscosity, but by purely experimental work it is not possible to define it other than empirically.

The form of Poiseuille's equation with which we are familiar is, in fact, dependent on the theoretic solution of the problem. Navier's early work on the equations of motion for viscous liquids were amplified and corrected by Stokes in the 1840s, and the Navier–Stokes equations are the general solution of this problem. Stokes, however, did not tackle the particular case of liquid flow in a tube. The solution of this case was made independently by Wiedemann in 1856 and Hagenbach in 1860, who both produced the result that

$$Q = \frac{\pi R^4(P_1 - P_2)}{8\mu L}, \quad 2.3$$

where μ is the dynamic viscosity of the liquid and R the internal radius of the tube. It can thus be seen that Poiseuille’s constant K is $\pi/128\mu$ (as $R^4 = D^4/16$), or $\mu = \pi/128K$.

Hagenbach calculated μ from Poiseuille’s data and obtained the result in modern units of 0.013084 poise at 10°C (the modern value is 0.013077 poise) (see Bingham and Jackson, 1918; Barr, 1931).

The method of derivation of the solution used by Hagenbach was a simple one, and it seems odd that it should have taken so long to produce. Compared with the advances made in pure mathematics by the middle of the nineteenth century, this is a very elementary problem (in applied mathematics) and furthermore a problem that had been subjected to experimental investigation for over 40 years.

The basic assumption made for this solution, essentially that made by Newton (1642–1727), is that every particle of liquid is moving parallel to the axis of the tube with a constant velocity V , and the force opposing the flow over unit area is proportional to the viscosity and the velocity gradient of the liquid. In a cylindrical tube, the particles traveling at the same velocity will be symmetrically arranged as cylindrical laminae.

In a tube of constant internal radius R , let us consider a cylindrical unit of liquid length L and radius r . The viscous force retarding its motion will be the area of its surface ($2\pi rL$) multiplied by its viscosity (μ) and the velocity gradient across the tube (dV/dr):

$$F_{\text{visc}} = 2\pi r\mu L \left(\frac{dV}{dr} \right).$$

The force exerted by the pressure on the end of the cylinder of liquid is the pressure multiplied by the cross-sectional area, minus the force on the distal end, so that

$$F(P) = \pi r^2(P_1 - P_2),$$

where P_1 and P_2 are the pressures at the ends of the length (L) of the cylinder. The pressure difference per unit length $(P_1 - P_2)/L$ is known as the ‘pressure gradient’. These forces are equal and opposite, so that

$$\pi r^2(P_1 - P_2) = -2\pi r\mu L \left(\frac{dV}{dr} \right)$$

or

$$r(P_1 - P_2) = -2\mu L \left(\frac{dV}{dr} \right).$$

Hence, the velocity gradient is

$$\frac{dV}{dr} = -\frac{r(P_1 - P_2)}{2\mu L}.$$

By integration, we find that the velocity at radius r is

$$V = -\frac{r^2(P_1 - P_2)}{4\mu L} + C. \quad 2.4$$

To determine the value of the constant of integration C , it is necessary to make a further assumption of the boundary conditions that the lamina in contact with the wall is at rest (an assumption made by Newton); that is, when $r = R$, $V = 0$. With substitution of these values into eqn 2.4, we see that

$$C = \frac{R^2(P_1 - P_2)}{4\mu L}, \quad 2.5$$

so that

$$\begin{aligned} V &= -\frac{r^2(P_1 - P_2)}{4\mu L} + \frac{R^2(P_1 - P_2)}{4\mu L} \\ &= \frac{(P_1 - P_2)}{4\mu L} (R^2 - r^2). \end{aligned} \quad 2.6$$

This is the equation for a parabola, where $V = 0$ when $r = R$ and is a maximum when $r = 0$, at the axis of the tube (Fig. 2.1).

To obtain the volume of flow, it is necessary to determine the volume of the paraboloid which has this parabola as its profile. That is, we must determine the volume of the solid of revolution of this parabola. This volume is

$$Q = \int_0^R 2\pi V r \, dr$$

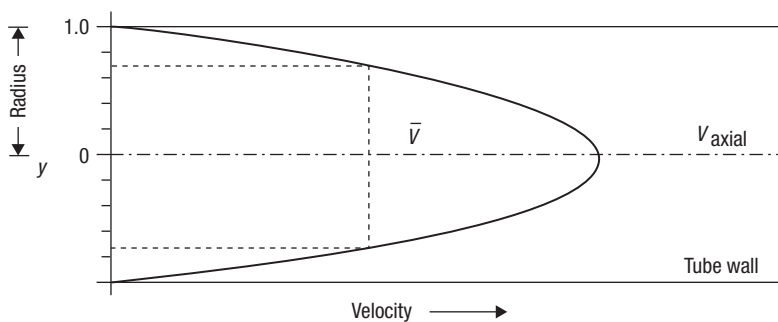


Figure 2.1 The velocity profile in steady laminar flow. This is a parabola (see eqn 2.6). The actual equation of the curve shown is $\bar{V} = 10(1 - y^2)$, where $y = r/R$, the ratio of the radius r of the lamina considered to the radius R of the tube. The average velocity V across the tube can be shown to be half the axial velocity (see eqns 2.8 and 2.9)

or, substituting the value for V above (eqn 2.6),

$$Q = \frac{2\pi(P_1 - P_2)}{4\mu L} \int_0^R r(R^2 - r^2) dr \quad 2.7$$

$$= \frac{\pi R^4(P_1 - P_2)}{8\mu L}.$$

This is commonly referred to as the Poiseuille equation. It is important to note that volume flow is directly related to the constant $\pi/8$ and the fourth power of the radius, therefore, flow increases exponentially with increases in internal radius.

The average velocity $\bar{V}$ of the liquid across the tube is obtained by dividing the volume flow Q by the cross-sectional area πR^2 , so that from eqn 2.7

$$\bar{V} = \frac{R^2(P_1 - P_2)}{8\mu L}. \quad 2.8$$

The maximum velocity is at the axis of the tube, so that from eqn 2.6, by substituting $r=0$, we obtain

$$V_{\text{axial}} = \frac{R^2(P_1 - P_2)}{4\mu L}. \quad 2.9$$

Thus the average velocity is half the axial velocity (Fig. 2.1). The lamina of the liquid flowing at this velocity is the one of radius r , such that $(R^2 - r^2) = R^2/2$, or $r = 0.707R$. (The term 'average' velocity is used in the sense of the average across the tube at any moment, and the term 'mean' velocity for the mean of the time-dependent variation throughout a cycle when the flow is pulsatile.)

The simple derivation of Poiseuille's equation given above is often treated as the standard one (Kaufmann, 1963; Caro *et al.*, 1978; Milnor, 1989; Fung, 1997), but is unsatisfactory in terms of the infinitesimal calculus because it postulates a 'cylinder' of finite size, and is physically unreal because this is considered as if it were a solid with a viscous drag only on its surface. A better method, apparently first due to Lamb about 1890 (Lamb, 1932), is to treat a thin cylindrical liquid shell of thickness ∂r , inner radius r , and length L ; its axis z is coincident with the axis of the tube, and the velocity V is everywhere parallel to the axis, and a function of the distance r from this axis (see Fung (1997)). The normal pressure on the end is now

$$F_p = (P_1 - P_2)2\pi r \partial r. \quad 2.10$$

The retarding force on the inner surface will be written as before, but we are concerned with the differences between the two surfaces, which will be the rate of change across the thickness, or

$$F_{\text{visc}} = -\frac{\partial}{\partial r} \left(\mu \frac{\partial V}{\partial r} 2\pi r L \right) \partial r. \quad 2.11$$

Equating these, we obtain

$$\frac{\partial}{\partial r} r \left(\frac{\partial V}{\partial r} \right) = -\frac{r(P_1 - P_2)}{\mu L}. \quad 2.12$$

Integrating once, we obtain

$$\frac{\partial V}{\partial r} = -\frac{r(P_1 - P_2)}{2\mu L} + \frac{A}{r}, \quad 2.13$$

where A is the constant of integration. Flow is symmetrical at the axis, so $\partial V/\partial r = 0$ when $r = 0$; hence $A = 0$, and

$$\frac{\partial V}{\partial r} = -\frac{r(P_1 - P_2)}{2\mu L}.$$

Alternatively, Lamb (1932) writes the solution of eqn 2.12 as

$$V = -\frac{r^2(P_1 - P_2)}{4\mu L} + A \ln r + B \quad 2.14$$

and points out that, since the velocity of the liquid must be finite at the axis, then we must have $A = 0$. The solution for $B = 0$, due to zero velocity at the wall, is as for C in eqn 2.5. Equation 2.12 may also be written in the form

$$r \left(\frac{\partial^2 V}{\partial r^2} \right) + \frac{\partial V}{\partial r} + \frac{r(P_1 - P_2)}{\mu L} = 0 \quad 2.15$$

or

$$\frac{\partial^2 V}{\partial r^2} + \frac{1}{r} \frac{\partial V}{\partial r} + \frac{(P_1 - P_2)}{\mu L} = 0, \quad 2.16$$

and we shall later find this form in Womersley's derivations in Chapter 7. (In the later usage, the symbol w is commonly used in place of V .)

To evaluate Poiseuille's equation (2.7), all units must be consistent. Thus, in the centimeter-gram-second (c.g.s.) system, Q is in cm^3/s and R and L are in centimeters, while P is in dyne/cm^2 . The conversion of P from mmHg, so commonly used in medicine and physiology, is accomplished by multiplying by one-tenth to convert to cmHg and then multiplying by the density of mercury, $13.6 \text{ g}/\text{cm}^3$ and the gravitational constant g (approximately $980 \text{ cm}/\text{s}^2$); that is, $1.0 \text{ mmHg} = 1.33 \times 10^3 \text{ dyne}/\text{cm}^2$.

The c.g.s. unit of viscosity is the poise and is named in honor of Poiseuille. It is the force required to move a unit area through a liquid to create a unit velocity gradient, and is thus defined as $1.0 \text{ dyne-s}/\text{cm}^2$ or $1.0 \text{ g}/\text{cm-s}$. For convenience, the centipoise is often used. The viscosity of water is very close to 0.01 poise (1.0 centipoise) at 20°C ; more precisely, it is 1.0 centipoise at 20.2°C (Bingham and Jackson, 1918). The value taken for blood with a hematocrit of 45 percent in most calculations in this book is 0.04 poise (4.0 centipoise) at 37°C .

The applicability of Poiseuille's equation to the circulation

The conditions under which Poiseuille's equation applies precisely are implicit in the method by which it is derived theoretically. In view of its importance in the hydrodynamics of the circulation, these conditions should be considered in more detail. They are as follows:

1. **The liquid is homogeneous and its viscosity is the same at all rates of shear.** Blood is, of course, a suspension of particles, but, in tubes in which the internal diameter is large compared with the size of the RBCs, it behaves as a Newtonian liquid. In tubes with an internal radius less than 0.5 mm, changes in apparent viscosity occur. This is important in the study of liquid flow in small vessels and is discussed more fully below. In the larger arteries and veins, however, blood may be considered as a homogenous liquid with a viscosity that is independent of the velocity gradient.
2. **The liquid does not slip at the wall.** This was the assumption that velocity is zero when $r=R$, which was made in evaluating the constant of integration in the equations above. As Poiseuille's equation would not be valid if this were not true, the equation may be used as a test for the assumption; it is held to be universally true for liquids (Kaufmann, 1963; Fung, 1997). Even when gas flows over a solid surface, there appears to be no appreciable slip under normal conditions, although it may have to be considered with a rarefied gas. These conditions are derived from the detailed review by Goldstein (1938). This point is of some importance as it has occasionally been suggested that some of the anomalous flow properties of blood vessels may be due to non-wettable properties of their endothelial lining with a consequent slip at the wall. Even if the endothelium was shown to be non-wettable, the conclusion, that slipping would result, is unjustifiable. For example, Poiseuille's equation holds for the flow of mercury in a glass tube so that there must be zero velocity at the wall. The appearance in flowing blood of a narrow but cell-free zone at the walls of narrow tubes has led some workers to treat this lower viscosity zone as a region of 'slip'. This can be convenient as a mathematic device, but should not be taken as a physical reality.
3. **The flow is laminar; that is, the liquid is at all points moving parallel to the walls of the tube.** At rates of liquid flow above a critical value this is no longer true and the flow is turbulent. Turbulent flow may occur in the largest blood vessels, but as the flow here is also pulsatile,

condition (4) below is not satisfied, so that Poiseuille's equation cannot be applied. The evidence available on the occurrence of turbulent flow in the circulation is reviewed below, where it will be seen that laminar flow is almost certainly present in all the vessels where flow is sufficiently steady to consider applying Poiseuille's equation.

4. **The rate of flow is 'steady' and is not subjected to acceleration or deceleration.** If the velocity is altered, the pressure gradient is partly utilized in communicating kinetic energy to the liquid, and the equations do not apply. As the flow in all large arteries and the intrathoracic veins (Brecher, 1956) is markedly pulsatile, it is clear that Poiseuille's equation cannot be applied in these vessels. The velocity profiles and the pressure–flow relationship of flow in these circumstances is considered below.
5. **The tube is long compared with the region being studied.** Close to the inlet (or entrance) of a tube, flow has not yet become established with the paraboloid velocity profile characteristic of laminar flow. The distance required to establish the steady form of flow is known as the 'inlet (or entrance) length', and here Poiseuille's equation does not apply. Within the inlet length, the assumption made in the derivation that there are no accelerations along the axis of the tube is not true. It is, therefore, a special case of the steady flow condition discussed in (4) above. In viscometers where a flow from a reservoir through a narrow tube is measured, a correction for this effect must be applied (Newman and Searle, 1957). In the circulation, there is never this condition of a stationary reservoir leading to steady laminar flow, so that this correction does not apply. This special problem of the inlet length with pulsating flow in relation to the proximal aorta is discussed below.
6. **The tube is cylindrical in shape.** Two stipulations are contained in this assumption: the cross-section of the tube is circular, and the walls of the tube are parallel. Most arteries of the systemic circulation are circular in cross-section, but many veins and the pulmonary arteries tend to be elliptical rather than circular vessels, and their relatively thin, compliant walls make their cross-section subject to the effects of gravity and external pressure from surrounding tissues. The external jugular vein, for example, which is visible as a rounded structure in the neck in many human subjects when they are supine, collapses to a much flatter cross-section when the subject sits upright. The requirement of parallel walls is probably never exactly met in blood vessels because individual arteries taper (i.e. become narrower) as they progress toward the periphery

(see Table 2.3); in the veins, the process is reversed. Because of this tapering, velocity changes constantly as a function of distance along the vessel axis. In a typical artery, with steady blood flow, average velocity increases as the artery narrows, and the kinetic energy of the blood in motion increases accordingly. This tends to reduce the pressure distally, as pressure energy is converted to kinetic energy (Bernoulli effect, see below). The artery thus exhibits an extra pressure gradient due to tapering as well as the gradient associated with losses due to blood viscosity. Observations on the canine aorta and coronary arteries indicate an angle of taper of 1.0–2.0 degrees (Ling *et al.*, 1973; Atabek *et al.*, 1975). The results of Ling and Atabek suggest that this small degree of narrowing has a definite effect on the mean pressure gradient and a lesser influence on the pulsatile components of pressure and flow. They found that the experimentally measured mean pressure gradient in the aorta (Ling *et al.*, 1973) and coronary arteries (Atabek *et al.*, 1975) was from three to six times the value calculated from Womersley's equation for a rigid tube. In human proximal left coronary artery tree regions where atherosclerosis frequently occurs, low wall pressure gradient appears. At distal segments, wall pressure gradient increases substantially due to increased velocity resulting from increased vessel tapering. Low wall pressure gradient occurs at bifurcations in regions opposite the apexes, which are anatomic sites, predisposed for atherosclerotic development. Endothelial cells respond to the combined effects of locally low wall pressure gradient and low wall shear stress gradient and provide a mechanism promoting atherosclerosis (Giannoglou *et al.*, 2005). Because of greater tapering, the apparent pressure gradient in the main pulmonary artery may be much greater (O'Rourke, 1968; Milnor *et al.*, 1969). Strictly speaking, the assumption of a cylindrical vessel also implies that no branches leave the segment under consideration, and this requirement is met for only short distances in most vessels. The distance between arterial side branches varies widely, but it is rare to find an arterial segment more than 3.0 or 4.0 cm in length without grossly visible branches; microscopic branches appear at shorter intervals. Tapering and side branches in exactly the right proportions can combine to produce constant velocity along an artery, and such a system can be hemodynamically equivalent to a cylinder. The thoracic aorta is an example; the drainage through intercostal branches compensating very closely for the progressive tapering of the main vessel. Whether the same is true in other large arteries is not known.

7. **The tube is rigid; the diameter does not vary with the internal pressure.** Blood vessels are viscoelastic structures, and their diameter is a function of transmural pressure. Under these conditions, the flow will not be solely determined by the pressure gradient. The change in diameter with pressure is much less in the small arteries and arterioles and is less predictable because the wall is largely composed of smooth muscle cells. These commonly constrict in response to an increase in transmural pressure. It is only in such small vessels that the flow is sufficiently steady to apply Poiseuille's equation (eqn 2.7). This equation assumes that the radius of the vessel is constant along its length. If this is not the case, such as when the vessel walls are deformable, then the flow of blood inside the vessel will not be fully developed, and eqn 2.7 will not strictly be valid (Middleman, 1995). Volumetric flow rate in such a system is significantly less than that predicted by eqn 2.7. In a model for steady laminar flow through a deformable gel-coated tapered channel, Yang *et al.* (2000) demonstrated that the mechanism of the flow rate reduction is due to kinetic energy end effects caused by tapering of the channel, rather than to a turbulent layer caused by oscillations in the channel wall.

It has been assumed that the laws of Poiseuille apply also to the flow of blood through the capillaries of the body. Ewald (1877) questioned this conclusion, but Lewy (1897) found that Poiseuille's law holds as long as no sedimentation occurs. duPre Denning and Watson (1900), from studies of the effect of bore size on blood flow, concluded that blood containing a large number of RBCs (erythrocytes) did not obey Poiseuille's law if the tube diameter was less than 300 microns (μm); that is, the viscosity was apparently a function of radius for smaller capillaries. They noted an increase in viscosity with a decrease in tube diameter.

Fahraeus and Lindquist (1931) also studied blood flow quantitatively in glass capillary tubes and found the marginal layer to be most prominent in tubes with diameters of less than 300 μm . They noted that the 'apparent' viscosity of blood as calculated from Poiseuille's equation decreased with a decrease in tube diameter, in contradiction of duPre Denning and Watson. This phenomenon, known as the Fahraeus–Lindquist effect, was confirmed *in vitro* by Stadler *et al.* (1990), who demonstrated that the decrease of apparent blood viscosity continues down to diameters just less than 10 μm and then increases (Fig. 2.2). Whitaker and Winton (1933) studied blood flow in the hind limb of a dog and suggested that the Fahraeus–Lindquist effect explained the lower apparent viscosity they observed in vessels compared with glass tubes.

Fahraeus and Lindquist concluded that Poiseuille's equation applies under a wide range of conditions in

physiology. For example, it appears to apply to whole blood suspensions under two conditions: (1) the suspension is flowing in tubes with diameters greater than 300 μm ; and (2) the flow velocity is sufficiently great that the asymptotic apparent viscosity applies throughout the vessel. When there are deviations from the asymptotes, deviations from Poiseuille's equation occur. Many studies have been made of the deviations of blood flow from Poiseuille's equation, including those of Zamir (1976), Mayrovitz (1987), and Yang *et al.* (2000).

The assumptions underlying the Poiseuille equation that are discussed above can be applied only approximately, at best, in the living circulation. Using the steady-flow criterion alone, the equation is only directly applied, in the vascular tree, to vessels which are so small, numerous, and inaccessible that even their dimension can only be roughly measured, so that no hydrodynamic sophistication is possible (Lee, 1977). In vessels the size of arterioles, where flow is still pulsatile (and diameter measurements are more accurate), several studies (e.g. Mayrovitz, 1987) have been published that suggest flow is proportional to the third (rather than the fourth) power of the radius, an observation first reported by Girard in brass tubes in 1813 (Girard, 1813).

Baez *et al.* (1960) found that the diameter of capillaries and other minute vessels varied very little with changes of the internal pressure in the range of 20–100 mmHg. So rigidly did the capillaries (which have no smooth muscle) appear to behave that Fung *et al.* (1966) were led to postulate that the extracellular fluid has a gel-like structure that effectively strengthens the wall. Recent studies have also shown that the diameter of terminal arterioles changes very little with large changes in internal pressure

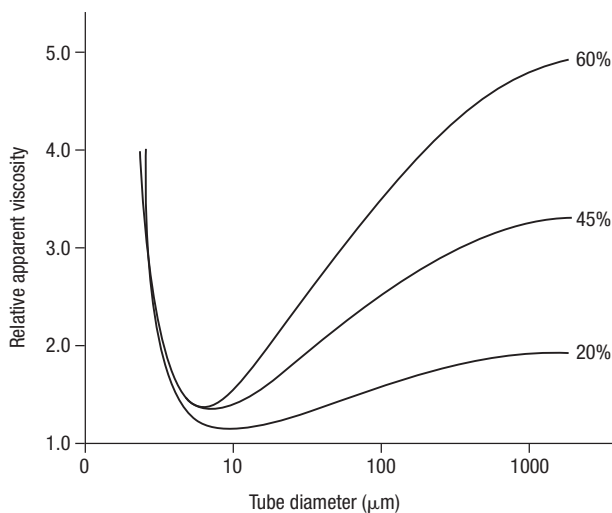


Figure 2.2 The variation of viscosity with tube diameter, illustrating the Fahraeus–Lindquist effect. Blood with different hematocrits (20, 45, and 60 percent) was perfused through glass tubes with internal diameters between 3.3 and 1978 μm . Blood viscosity increases along with tube diameter above about 9.0 μm and also varies with changes in hematocrit (see Fig. 2.6)

(Meyer *et al.*, 1988). Within the limits of accuracy of determination of such vessel dimensions under experimental conditions, it would appear that changes in internal pressure do not produce significant changes in the size of very small vessels. The changes in arterial diameter with pulsatile pressure are discussed in Chapters 4 and 6.

VISCOUS PROPERTIES OF WHOLE BLOOD

It is well known that compared to *in vitro* determinations, blood behaves as a less viscous fluid under *in vivo* flow conditions (Fig. 2.3). The experiments of Whittaker and Winton (1933) were among the first to study the *in vivo* effects of altered blood rheology, and experimental studies have provided additional evidence for the complexity of *in vivo* hemodynamics–hemorheology relationships (Baskurt, 2008). Careful studies indicate that the impact of a given blood rheology alteration is determined by the properties of the experimental model and method used to modify blood rheology. In addition, vascular control mechanisms may play a major role in the resulting hemodynamic effects of a hemorheological alteration: (1) a response simply related to metabolic autoregulation in which there is a compensatory vasodilation due to altered *in vivo* blood flow and organ/tissue hypoxia; (2) modulation of endothelial function (e.g. nitric oxide (NO) production) via altering wall shear stress, thereby leading to changes of vascular properties. The *in vivo* effects of

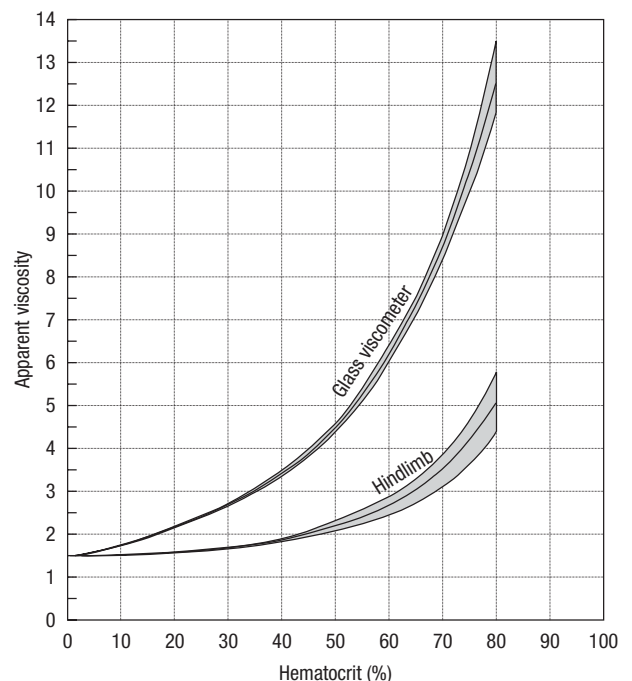


Figure 2.3 Effect of hematocrit alterations on apparent viscosity of blood in the dog hind limb and a glass tube viscometer. Reproduced from Whittaker and Winton (1933)

altered RBC aggregation have been investigated in both humans and various experimental models (Shin *et al.*, 2004; Lipowsky, 2005; Baskurt and Meiselman, 2007; Lee *et al.*, 2007; Christy *et al.*, 2010). Shin *et al.* (2004) employed a novel technique to quantify aggregation characteristics by analysis of light transmission in a slit flow system. Other recent studies have used the popular Myrenne-aggregometer (Lee *et al.*, 2007; Christy *et al.*, 2010), and have provided important data on RBC aggregation and neutrophil function in an aging population. These data indicate that both the magnitude of the hemodynamic effect and the direction of the alteration depend on the intensity of RBC aggregation. Using the same novel technique, RBC aggregation has been shown to be an important determinant of endothelial function through its effects on RBC axial distribution and wall shear stress (Baskurt, 2008).

In Poiseuille, or laminar, flow, concentric cylindrical layers of liquid undergo shearing or slide over each other. The force applied to the liquid layer is termed the 'wall shear stress' and the velocity gradient obtained between adjacent layers of the liquid is the 'shear rate' (Pearson, 2001). Resistance to flow arises from friction between the adjacent layers of the liquid; this frictional flow resistance is the liquid's viscosity. 'Viscosity' is defined as the ratio of shear stress and shear rate (Samijo *et al.*, 1997). Therefore, shear stress is the product of viscosity and shear rate (Samijo *et al.*, 1997). In general, the measurement of whole blood and plasma viscosity requires mechanical tests, such as the capillary tube viscometer (Çinar *et al.*, 2001), where liquid is sheared by flow past the stationary inner wall of a capillary tube and the cone-and-plate viscometer (Eckmann *et al.*, 2000; De Backer *et al.*, 2002), where the test liquid is sheared between two surfaces, one fixed and one moving, with the torque being related to viscosity. The falling-ball, oscillating and concentric cylinder, and porous bed viscometers have also been used to measure viscosity (Crowley *et al.*, 1991; Reinhart *et al.*, 1998). A suitable system for the measurement of blood or plasma viscosity must have several features for clinical applications. These include rapid, reproducible, and precise measurements, small blood or plasma sample volume, simulate *in vivo* time-varying flow conditions using oscillatory flow in a tube, precise thermal control, simple operation, and minimal exposure of operator to blood-borne pathogens. The porous bed viscometer is a convenient means to measure whole blood viscosity in humans and it should be useful as a screening test for clinical and epidemiologic studies (Crowley *et al.*, 1991). In a later study, Crowley *et al.* (1994) found strong positive correlations between blood viscosity and packed cell volume, total cholesterol, triglycerides, and low density lipoprotein cholesterol and a negative correlation between blood viscosity and high density lipoprotein cholesterol.

Reference to Poiseuille's equation (2.3) indicates that an increase in viscosity is associated with a decrease in

blood flow and an increase in blood pressure and vascular resistance, while a decrease in viscosity has the opposite effect. Whole blood viscosity is mainly determined by the concentration of RBCs, whereas the shear rate at the vessel wall is a function of plasma viscosity (Gustafsson *et al.*, 1981). Also, whole blood viscosity is higher in the brachial than the common carotid artery (Dammers *et al.*, 2003). The effect of alterations in hematocrit are most pronounced in the myocardium and brain in terms of blood flow index and are different from liver, kidney, and intestine (Baskurt, 2008).

Early investigators conceptualized blood as a viscous liquid, assuming that the viscosity controls its flow properties (Lowe, 1988). But blood is not a liquid in the ordinary sense; it is a liquidized suspension of elastic cells and, therefore, should be treated as a viscoelastic substance. The viscoelasticity reflects the cumulative effects of many blood parameters, such as plasma viscosity, RBC deformability, platelet aggregation, and hematocrit. Blood viscosity (or viscoelasticity) is mainly determined by the concentration of RBCs, whereas shear rate at the vessel wall is a function of plasma viscosity (Gustafsson *et al.*, 1981). Viscoelasticity and flow velocity of blood determine shear stress at the vessel wall, which stimulates the endothelium to produce vasoactive substances, such as NO (Bugá *et al.*, 1991) prostacyclin (Frangos *et al.*, 1985), and endothelin (Malek and Izumo, 1992). By modifying the production of vasoactive mediators by endothelial cells, shear stress regulates arterial diameter (Furchgott, 1983; Pohl *et al.*, 1986), while shear stress, and radial and circumferential strain are determinants of endothelial genes' expression (Dekker *et al.*, 2006). Extensive research on blood viscoelasticity and the factors affecting it has provided a firm foundation for the increasing interest in viscoelasticity among researchers in both physiology and clinical medicine. Changes in blood viscoelasticity are associated with such pathologies as diabetes (Isogai *et al.*, 1984), hypertension (Letcher *et al.*, 1981), coronary artery disease (Junker *et al.*, 1998; Koenig *et al.*, 1998; De Backer *et al.*, 2002), myocardial infarction (Fisman and Smilovitch, 1997), ischemic stroke (Turczynski *et al.*, 2002), cardiovascular mortality (Brown *et al.*, 2001), cancer (von Tempelhoff *et al.*, 2000), all-cause mortality (Koenig *et al.*, 2000), and changes in endothelial function in both animals (Tsai *et al.*, 1998) and humans (Melkumyants *et al.*, 1989; Bertuglia, 2001; Giannattasio *et al.*, 2002).

A Newtonian liquid is, by definition, one in which the coefficient of viscosity is constant at all rates of shear (McDonald, 1974; Fung, 1997; Pearson, 2001). This is closely approximated by most homogeneous liquids, but suspensions of particles show deviations from it. As might be expected, such anomalous behavior becomes most apparent as the particle size becomes appreciably large in comparison with the dimensions of the channel in which it is flowing (see Fig. 2.2). Blood is essentially a suspension of erythrocytes, leukocytes, and platelets in plasma and

shows anomalous viscous properties. Excellent reviews of this subject are given by Whitmore (1968), Schmid-Schonbein (1976), Dintenfass and Seaman (1981), Dintenfass (1985), Liepsch *et al.* (1994), Pearson (2001), Reinhart (2001), Schuff-Werner and Holdt (2002), and Baskurt (2008).

The anomalous viscous properties of blood are of two kinds:

1. At low shear rates, the apparent viscosity increases markedly. Indeed, it has been shown almost certainly that flow may cease in the presence of a measurable stress, implying that there is a 'yield stress' (McDonald, 1974; Morris *et al.*, 1987). However, recent model studies (Secomb *et al.*, 1986; Skalak *et al.*, 1986) suggest that no critical minimum pressure or 'yield stress' is required to maintain flow in a uniform tube (Secomb, 1987).
2. In small tubes, the apparent viscosity at higher rates of shear is smaller than it is in larger tubes. This progressive diminution with tube size begins to be detectable with tubes of less than 1.0 mm internal diameter (i.e. about 100 times the major diameter of the RBC) and becomes marked in tubes of the order of 100–200 μm diameter (Fahraeus and Lindquist, 1931; Reinke *et al.*, 1987). Relatively little detailed quantitative work has been done on glass tubes smaller than 40 μm internal diameter because of technical difficulties, but more recent studies used glass tubes with internal diameters of 29 μm or less (Barbee and Cokelet, 1971; Reinke *et al.*, 1987). With tube sizes of 10 μm or smaller, the deformation of the cells will introduce more complicating factors (Gaehtgens and Papenfuss, 1979).

These two types of anomalies may be referred to as the 'low shear' and 'high shear' effects. They have been the object of intensive research for many years, and the literature on the subject is very large. This interest is in part due to the challenge of the very nature of viscosity, which is, in turn, allied to our inability to predict the properties of liquids in terms of their molecular properties. On the other hand, the hydrodynamic description of the circulation clearly requires a measurement of the viscosity of the blood as accurately as it may be achieved, especially since flow resistance is determined by the rheological behavior of blood flowing through microvessels (Pries *et al.*, 1994).

The brief review given here will emphasize the quantitation of anomalies that may introduce errors into the hydrodynamic analyses of blood flow that assumes Newtonian viscosity of blood. The 'high shear' or 'small tube' effect, mentioned above, clearly applies in a study of the small vessels. In applying a physical equation such as that of Poiseuille (eqn 2.7) to a vessel of the microcirculation, one is faced not only with viscosity changes, but also with the fact that some of the parameters of the equation

cannot be measured accurately in the living vessel. Even the dimensions are inaccessible except in a few regions, such as the microcirculation of the mesentery (Pries *et al.*, 1994), the tenuissimus muscle (Meyer *et al.*, 1988), the hamster cremaster muscle (Kiani *et al.*, 1993; Kurjiaka and Segal, 1995; de Wit *et al.*, 1997), the hamster skinfold model (Tsai *et al.*, 1998), the bat wing (Davis, 1988), the rabbit omentum (Schmid-Schonbein and Zweifach, 1975), and the rat spinotrapezius muscle (Warnke and Skalak, 1990; Warnke and Skalak, 1992). Furthermore, the radius of small vessels normally appears to vary continuously (Meyer *et al.*, 1988). Nevertheless, some of the anomalies of blood viscosity are large and may introduce systematic errors into physical analysis of blood flow. For example, in microvessels with diameters less than about 100 μm , the intravascular hematocrit is always less than the inflow hematocrit, since the average erythrocyte velocity is higher than the mean blood flow velocity (Barbee and Cokelet, 1971; Albrecht *et al.*, 1979). This phenomenon was first described by Fahraeus in 1928.

Recent reviews of the rheological properties of blood that are recommended for reading are by Rosenson and Lowe (1998), Sotiriou and Cheng (2000), Pearson (2001), Reinhart (2001), Schuff-Werner and Holdt (2002), Raddino *et al.* (2002), Lipowsky (2005), and Baskurt and Meiselman (2007) and older reviews by Wayland (1967), Cokelet (1972), and Chien (1979). The more detailed reviews by Bayliss (Bayliss, 1960; Bayliss, 1962) have long been standard references. The monograph by Whitmore (1968) provides an excellent coverage of the basic principles of blood and plasma viscosity. The review of Merrill (1969) is particularly thorough in considering the low shear behavior of blood to which he and his group made a big contribution; the review by Johnson (1969), on the other hand, is more physiologically oriented towards work done in living capillaries. The recent text by Fung (1997) reviews both *in vitro* and *in vivo* blood rheology studies, and the reviews by Dintenfass and Seaman (1981), Dintenfass (1985), and Raddino *et al.* (2002) address the importance of blood viscosity in cardiovascular disease.

Wall shear stress and shear rate

The coefficient of viscosity μ in a flowing liquid was defined earlier as the constant of proportionality between the stress applied τ and the velocity gradient perpendicular to the direction of shear, or rate of shear, dV/dr , of the liquid laminae (see eqn 2.17) according to Poiseuille's theory of flow (O'Rourke and Nichols, 2005; Reneman *et al.*, 2006; Katritsis *et al.*, 2007; Efstathopoulos *et al.*, 2008a; Efstathopoulos *et al.*, 2008b). If the viscosity is measured in a concentric cylinder (Couette) or a cone-and-plate viscometer, the rate of shear is approximately constant throughout the liquid and is measured directly, as is the applied stress. Results on changes in viscosity with changing rate of shear τ are displayed by plotting

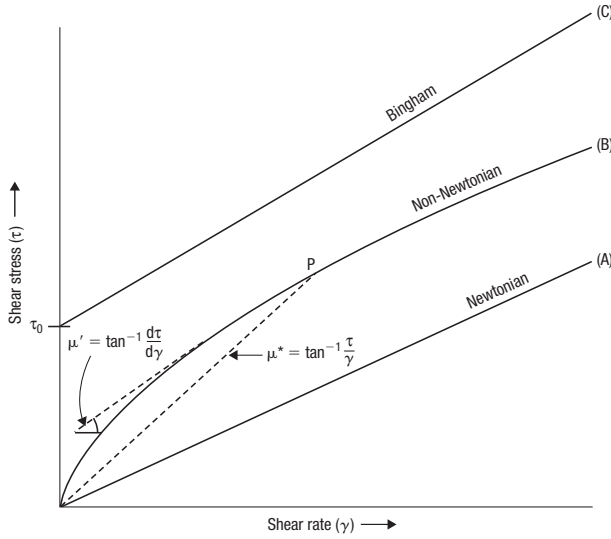


Figure 2.4 Types of flow behavior. Shear stress is the ordinate and shear rate the abscissa. (A) A Newtonian liquid: the relation between stress and shear is linear from the origin. (B) A non-Newtonian liquid with asymptotic Newtonian behavior. Beyond point P , at higher shear rates, the relationship is linear; below that it is non-linear, but the curve passes through the origin. The 'differential' viscosity, μ' , is defined by the slope of the curve $\mu' = \tan^{-1} d\tau/d\gamma$. The 'generalized' viscosity, μ^* , is the ratio of the applied stress to the shear rate; thus, at point P , $\mu^* = \tan^{-1} \tau/\gamma$. Wayland (1967) calls this the 'secant' viscosity. (C) A Bingham body: when stress is applied, there is no flow until it exceeds a certain value τ_0 . This value of τ_0 is called the 'yield stress'

stress τ against the rate of shear (Fig. 2.4). Shear stress is in pressure units, dyne/cm² or Pascal, i.e. N/m² (1 N/m² = 10 dyne/cm²) and viscosity is measured in poise (1 poise = 1 dyne-sec/cm²) or in Pascal-sec.

Most of the work on this subject has been done in straight cylindrical tubes, a situation analogous to that of blood flow in arteries. In order to be able to compare results from the two methods directly, it is necessary to define shear stress and rate of shear in terms of the pressure gradient, the rate of flow, and the tube radius. From eqn 2.4, we can write

$$\tau = \mu \left(\frac{dV}{dr} \right) = - \frac{r(P_1 - P_2)}{2L}. \quad 2.17$$

It is usual in these formulations to use ΔP as the pressure gradient, so that $\Delta P = (P_1 - P_2)/L$, and (ignoring the sign, which is irrelevant in this context) we obtain

$$\tau = r\Delta P/2, \quad 2.18$$

where r is the radius of a given lamina of liquid. The radius r is zero at the axis, so that the stress is also zero. The maximum stress is at the wall where $r=R$, and

$$\tau = R\Delta P/2. \quad 2.19$$

It is this maximum value of stress at the wall that is used as the characteristic shearing stress of liquid flow in a tube.

The rate of shear also varies from zero at the axis to a maximum at the wall ($r=R$). Its maximum value can be derived from the volume flow Q , or the velocity of flow V :

$$\frac{Q}{\pi R^2} = V = \frac{R^2 \Delta P}{8\mu}. \quad 2.20$$

Combining this with eqn 2.4 and eliminating the viscosity term, we obtain

$$\gamma = \frac{dV}{dr} = \frac{4V}{R} = \frac{4Q}{\pi R^3}. \quad 2.21$$

These derivations applied to flow in a tube are dependent on the equations that describe the parabolic velocity profile found in a Newtonian liquid. As will be seen below, variations of viscosity with rate of shear cause a deviation from this form, as will oscillatory flow, so that the evaluation of the rate of shear and stress at the wall in this way is only approximate (Reneman *et al.*, 2006). Nevertheless, in plotting the results graphically, the values calculated in this way are the clearest way of demonstrating regions of deviation when liquid flow behavior in tubes of different size are being compared. In recent years *in vivo* measurements of wall shear stress in humans have become possible using ultrasound (Reneman *et al.*, 2006) and phase contrast magnetic resonance imaging (MRI) (Oshinski *et al.*, 2006; Pantos *et al.*, 2007a; Sui *et al.*, 2008). Reneman and colleagues prefer the ultrasound technique because of its better spatial, and especially, temporal resolution (Reneman and Hoeks, 2008).

When shear stress is plotted against shear rate, the resulting line will express the behavior of the viscosity. Newtonian viscosity will give a straight line passing through the origin, as in Fig. 2.4A. The behavior of blood, however, is represented by Fig. 2.4B in which, at higher shear rates, the line is essentially straight and curves around to the origin at low shear rates. Another type of flow is of a material represented by Fig. 2.4C, where there is no flow up to a stress of τ_0 and flows thereafter; the stress is called 'yield stress'. This type of material is called a Bingham body (after Bingham, who first coined the term 'rheology'). Such a curve initially represents the behavior of a solid for stresses below the yield stress, and thereafter the curve represents the behavior of a liquid.

The effect of hematocrit on the flow properties of human blood was examined at shear rates up to 700/s by Brooks *et al.* (1970). An illustration of their findings is shown in Fig. 2.5. Up to hematocrits of approximately 12 percent, whole blood (i.e. erythrocytes suspended in plasma) was found to be Newtonian at all rates of shear (i.e. independent of the shear rate). As the hematocrit is increased (to about 70 percent in the figure), the viscosity increases and reaches an asymptotic value above

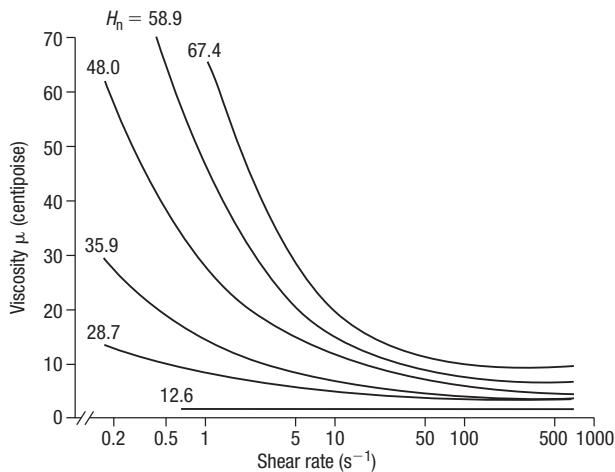


Figure 2.5 Relationship between viscosity and rate of shear for human erythrocytes suspended in their own plasma at 25°C for various volume concentrations H_n of the erythrocytes (hematocrit $\times 0.96$). Reproduced from Brooks *et al.* (1970)

approximately 100/s. At shear rates below 100/s, the viscosity increases markedly with increase in hematocrit (Whittaker and Winton, 1933; Whitmore, 1968; Brooks *et al.*, 1970; Liepsch *et al.*, 1994). This non-Newtonian behavior of blood at low rates of shear is partially due to the behavior of the rouleaux (Caro *et al.*, 1978). It has been observed that the average length of the rouleaux in blood decreases as the shear rate is increased until the erythrocytes are present individually, and such a change is associated with a reduction in viscosity. Also, the viscoelastic properties (Pfafferott *et al.*, 1982; Pfafferott *et al.*, 1985) and aggregation (Christy *et al.*, 2010) of RBCs change with age. At constant media viscosity and shear stress, young cells deform more than old ones and aggregation increases. Blood flow to the tissues depends upon the physical properties of the blood vessels, the pumping capacity of the heart and the viscoelastic properties of the blood (i.e. cell structure and plasma viscosity). Hydrodynamic principles are different for non-Newtonian liquids, such as blood, as compared to Newtonian liquids, such as water. Structural viscosity of flowing blood is dependent upon the diameter of the vessel. In large vessels, the concentration and rigidity of cellular components, mainly RBCs, contribute to the flow-determining structural whole blood viscosity, whereas in arterioles and capillaries, the viscosity of the plasma primarily determines flow (Schuff-Werner and Holt, 2002). Elevated plasma viscosity in extreme hemodilution increases perivascular NO concentration and microvascular perfusion (Tsai *et al.*, 2005).

Viscosity of plasma

It has been established that plasma (blood from which all cellular elements have been removed) behaves like a

Newtonian viscous liquid. Careful tests have been made in concentric cylinder viscometers (Cokelet *et al.*, 1963) and in capillary tubes (Merrill *et al.*, 1965) over a range of shear rates from 0.1 to over 1200/s, and no significant departure from linearity was found. A coefficient of viscosity of about 1.2 centipoise for plasma has been reported (Chien *et al.*, 1966; Gregerson *et al.*, 1967; Chien *et al.*, 1971). Other authors have reported plasma to be non-Newtonian in varying degrees, but only when an apparatus is used that allows plasma–air interfaces to occur; it appears that the abnormal results are due to a denatured protein layer at the interface (Wayland, 1967). As plasma is a suspension of fibrinogen, immunoglobulins, albumin, and lipoproteins in an electrolyte solution, it might be expected to show some deviation from the behavior of a pure liquid. As Wayland (1967) pointed out, however, the longest dimension of any of the particles is the length of the fibrinogen molecule (about 50 nm). Even in a capillary of 5.0 μm bore, this particle dimension would only be 1.0 percent of the lumen. As noted above, deviations of viscosity are not seen until the particle size is a much larger fraction of the tube diameter. In certain pathologic states, there is some suggestive evidence that measurable non-Newtonian behavior may be found in plasma. Modification of plasma, such as changes in the concentration of fibrinogen, lipoproteins, plasma proteins, and clinically introduced blood volume expanders, can have major effects on blood viscoelasticity (Barasch *et al.*, 1995; Eckmann *et al.*, 2000; Otto *et al.*, 2000; Ercan *et al.*, 2002). For example, changing the plasma composition by addition of blood volume expanders can affect aggregation and deformability of the cells, resulting in a shear rate-dependent viscoelasticity that deviates from that of normal blood. During the last 20 years, fibrinogen has been associated with the development of atherosclerotic plaque (Munkvad *et al.*, 1990) and the formation of thrombus at the atheroma site (Wilhelmsen *et al.*, 1984). Furthermore, the importance of fibrinogen as an independent risk factor for atherosclerosis has been recognized (Yarnell *et al.*, 1991; Kesmarky *et al.*, 1998; Koenig, 2003; Lominadze *et al.*, 2010). Also, large-scale studies such as the Northwick Park Heart (Meade *et al.*, 1986), Framingham (Kannel *et al.*, 1987), Edinburgh Artery (Lowe *et al.*, 1997), Caerphilly and Speedwell (Yarnell *et al.*, 1991), PRIME (Scarabin *et al.*, 2003), and Strong Heart (Palmieri *et al.*, 2003b; De Marco *et al.*, 2009) studies demonstrated that elevated plasma fibrinogen (and plasma viscosity) levels were associated with increased incidence of coronary heart disease (CHD), stroke, and hypertension. In meta-analysis, an association between even modest increases (10 percent) in fibrinogen and future CHD end points was found (Koenig, 2003). Fibrinogen levels were also associated with unstable and stable coronary artery disease, and coronary complications after interventions. The reasons, however, why fibrinogen is elevated in cardiovascular disease and in atherosclerosis in general, are only incompletely understood; but all cells

involved in the atherogenic process are able to produce cytokines which induce an acute phase reaction, and thus increase fibrinogen plasma levels. The potential pathophysiological mechanisms by which elevated fibrinogen levels mediate cardiovascular risk are manifold. It forms the substrate for thrombin and is converted to fibrin in the final step in the coagulation cascade; it is essential for platelet aggregation; it modulates endothelial function; it promotes smooth muscle cell proliferation and migration; it interacts with the binding of plasmin with its receptor, and finally it represents a major acute phase protein. Whether or not fibrinogen is causally involved in atherothrombogenesis still remains to be determined, and even though other unsolved issues await conclusive answers, fibrinogen and plasma viscosity have emerged as important additional markers of cardiovascular risk (Junker *et al.*, 1998; Lominadze *et al.*, 2010). It is known, however, that plasma fibrinogen levels (and plasma viscosity) are associated with cardiovascular risk factors, such as age, hypercholesterolemia, diabetes, hypertension, body mass index, and smoking in both men and women (Barasch *et al.*, 1995; Otto *et al.*, 2000; Ercan *et al.*, 2002; Pinon *et al.*, 2002; De Marco *et al.*, 2009).

Both low high-density (Stamos and Rosenson, 1999) and high low-density (Rosenson and Lowe, 1998) lipoprotein levels are associated with elevated plasma viscosity. In patients with hyperlipoproteinemia, the risk can be reduced by aggressive lipid-lowering therapy (Braunwald, 1997a; Braunwald, 1997b). Low-density lipoprotein (LDL) immunophoresis is an excellent therapeutic option in patients with familial hypercholesterolemia: long-term LDL apheresis treatment is known to reduce plasma LDL-cholesterol and plasma viscosity (Otto *et al.*, 2002). Agents such as statins and gemfibrozil that lower LDL and triglycerides, respectively, improve cardiovascular risk and may lower plasma viscosity (Stein and Rosenson, 1998; Banyai *et al.*, 2001). C-reactive protein, a sensitive indicator of inflammation and cardiovascular risk, is positively associated with plasma fibrinogen and viscosity (Mendall *et al.*, 2000) and is significantly reduced by pravastatin (Ridker *et al.*, 1999). Also, hormone replacement therapy improves cardiovascular risk by lowering plasma viscosity in postmenopausal women (Rosenson *et al.*, 1998).

Viscosity of blood

As erythrocytes are progressively added to plasma, the viscosity increases. Marked non-Newtonian properties are not seen until the concentration of cells exceeds 10 percent (Chien *et al.*, 1966). If blood viscosity is measured in a tube with an internal radius of about 0.5 mm (500 μm) or larger, and shear rates are not less than 200–300/s, viscosity will be effectively independent of tube size but will vary with erythrocyte concentration.

Erythrocyte concentration is expressed as the hematocrit value, which is the percentage volume of the blood in which the erythrocytes have sedimented by centrifugation; this packed cell volume includes a small volume of plasma ‘trapped’ between the cells, but this is usually neglected unless specifically stated, when it is usually called the ‘true’ hematocrit. The difference is small but not negligible; a hematocrit of 45 percent represents a true cell volume of about 41 percent.

The variation of viscosity with the hematocrit is associated with alterations in the pressure–flow relationship. For a given blood pressure, flow decreases as hematocrit and viscosity increase (Fig. 2.6). The hematocrit values shown in Fig. 2.6 are based on those of Whittaker and Winton (1933) and Çinar *et al.* (1999), but they are in reasonable agreement with those derived from other such authors as Whitmore (1968) and Amin and Sivs (1985). For practical purposes, the viscosity may be regarded as varying linearly with erythrocyte concentration from 0 percent (plasma) to a hematocrit of about 45 percent. This covers most experimental or clinical conditions (Dintenfass, 1976; Lowe *et al.*, 1981). The viscosity rises and flow decreases much more rapidly with hematocrit values above this. In tubes with diameters from 5.0 to 15 μm , apparent blood viscosity is only slightly higher than that

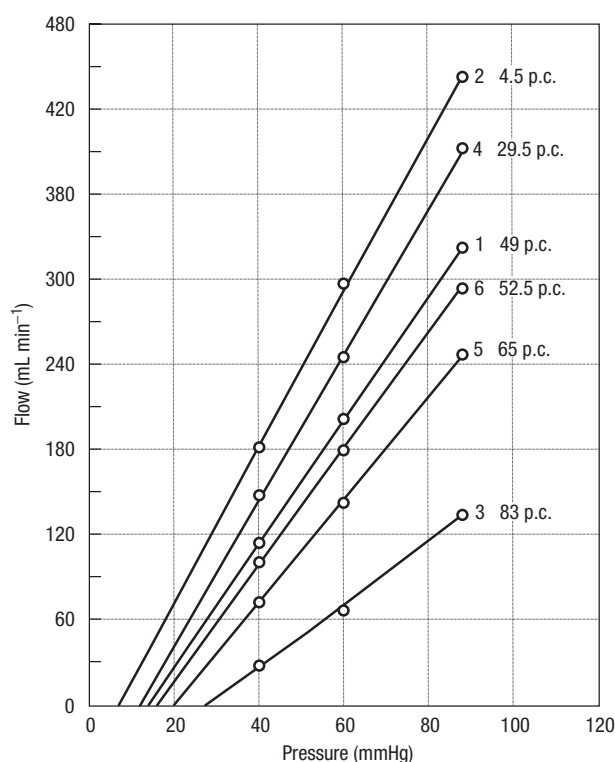


Figure 2.6 Pressure–flow curves in a dog hind limb with hematocrit varying from 4.5 to 83 percent as shown. Each curve represents the mean of values obtained during raising and lowering of the blood pressure. Reproduced from Whittaker and Winton (1933)

of the suspending plasma and does not exhibit a strong dependence on hematocrit (Cokelet, 1976).

Whole blood, unlike plasma, behaves in a non-Newtonian fashion. That is, its viscosity is markedly dependent on the prevailing shear rate. At high shear rates, above approximately 100/s, whole blood achieves its minimum viscosity value (Stuart and Kenny, 1980). In this situation, the RBCs maximally deform and align themselves to produce the minimum resistance to flow. With progressive lowering of the shear rate below this level, RBC aggregation occurs increasingly and the viscosity rises exponentially. Thus, in measuring blood viscosity, it is important to fix the shear rate at which the value of viscosity is being determined. If blood is stationary, a force, termed the 'yield stress value', is required to initiate flow (Pearson, 2001).

The single most important determinant of whole blood viscosity is the RBC content of the blood. In the earliest experiments using capillary viscometers, it was observed that blood viscosity did not increase significantly until the hematocrit was above 55 percent (Wells and Merrill, 1961). However, with the use of rotational viscometers, where the shear rate can be fixed, it can be shown that there is a linear relationship between the logarithm of blood viscosity and hematocrit at each shear rate. In addition, the relationship between blood viscosity and hematocrit becomes steeper as the shear rate is reduced (Fig. 2.7) (Pearson, 2001). Thus, in the physiological range of hematocrit from 40 to 50 percent, there is a 25 percent rise in viscosity at a shear rate of 230/s, but a nearly 80 percent rise at a shear rate of 0.77/s.

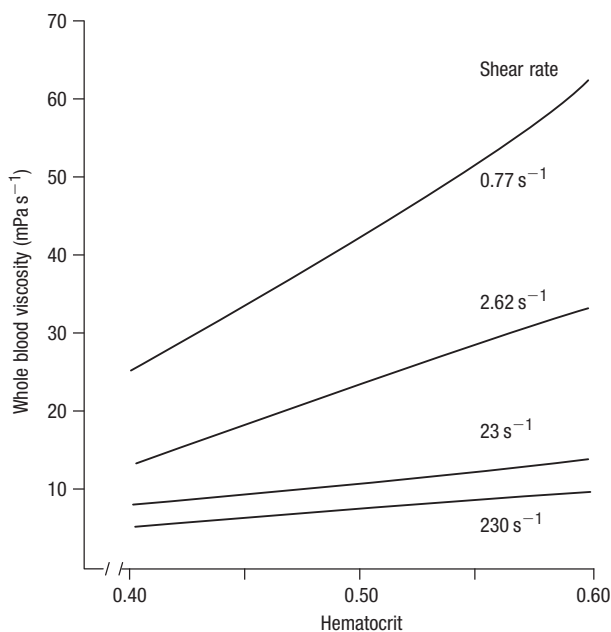


Figure 2.7 The relationship of whole blood viscosity (mPa s^{-1}) measured *in vitro* to hematocrit at a range of shear rates (inverse seconds). From Pearson (1987)

The yield stress value reflects the attractive forces between RBCs in stationary conditions (Merrill *et al.*, 1963). The value for yield stress depends on the plasma protein concentration, notably globulin and fibrinogen at any given hematocrit, but the hematocrit is the dominant influence on the yield stress value. There are different methods of measuring yield stress. Thus, there is no precise agreement on the hematocrit effect, but best-fit relationships between the two parameters have been given by the logarithm of yield stress against hematocrit or the cubed root of yield stress against hematocrit (Pearson, 1987).

Since one of the main objectives of blood flow is to maintain oxygen delivery to the tissues, the high hemoglobin value in erythrocytoses (polycythemia), and hence the high oxygen-carrying capacity of the blood, mitigates the influence of increased hematocrit and blood viscosity in reducing blood flow (see Fig. 2.6) and, thus, reducing oxygen delivery to the tissue. This situation is quite dissimilar to all other forms of hyperviscosity, where patients are generally anemic. Finally, when considering blood flow *in vivo*, one has to appreciate that it is highly dynamic and that disturbances of the normal cardiovascular physiology, such as heart failure and vessel disease, can lead to significant deviations from the normal physiology (Pearson, 2001). For example, cerebral blood flow rates have been extensively measured both at presentation and following treatment to lower the hematocrit to normal. For each patient and in each group of erythrocytosis, a negative correlation was found between cerebral blood flow and hematocrit (Thomas *et al.*, 1977). The typical findings are shown in Fig. 2.8.

Results from the Edinburgh Artery Study showed a close linear relationship between whole blood viscosity (and its major determinants) and carotid artery intima-media thickness in men 60–80 years old. These findings suggest that whole blood viscosity and its major determinants are associated not only with incident cardiovascular events, but also with the early stages of atherosclerosis (Lowe *et al.*, 2000; Gorodeski, 2002). Indeed, elevated blood viscosity is associated with high low-density and low high-density lipoproteins (Rosenson and Lowe, 1998; Stamos and Rosenson, 1999), therefore, improvement of the lipid profile should reduce blood viscosity. Lowe *et al.* (2000) found that after one year, lipoprotein reduction by pravastatin was associated with significant reductions in both plasma and blood viscosity. Also, other investigators found that blood and plasma viscosity were significantly elevated in patients with cardiac 'syndrome X' (see Chapter 17) (Turczynski *et al.*, 1999). In animal studies, a decrease in blood viscosity evoked by hemodilution at a constant flow rate led to arterial vasoconstriction, whereas an increase in viscosity due to hemoconcentration caused arterial vasodilation. These blood viscosity-induced changes in arterial diameter depended on an intact endothelium (Melkumyants *et al.*, 1989).

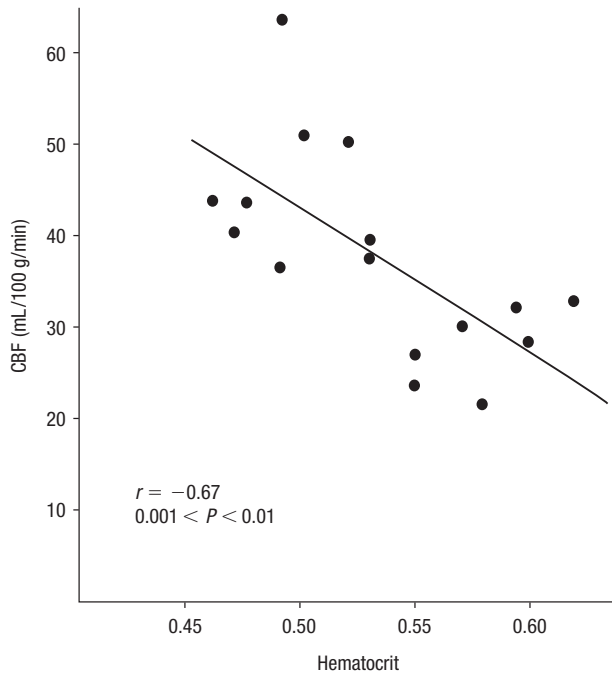


Figure 2.8 The inverse relationship between cerebral blood flow (CBF) and hematocrit in patients with polycythemia vera and idiopathic erythrocytosis (some patients had been phlebotomized to achieve a lower hematocrit). From Pearson (1987)

BERNOULLI'S EQUATION

Up to this point, we have concerned ourselves with the behavior of viscous liquids. Daniel Bernoulli (1700–82) developed the theoretic behavior of ‘ideal’ liquids, ignoring the forces of viscosity. One theorem (or equation) he developed for steady flow is important in the hydrodynamics of the circulation and will be considered briefly here. This equation has been discussed in some detail in its application to aortic stenosis by Pasipoularides *et al.* (1984) and in relation to intraventricular pressure gradients by Pasipoularides (1990). The equation has also been used recently to estimate pulmonary artery systolic and diastolic blood pressure (Lanzarini *et al.*, 2002; Selimovic *et al.*, 2007). In general, Bernoulli’s equation is used to determine liquid velocities by means of pressure measurements and may be deduced from the principle of the conservation of energy (Newman and Searle, 1957; Wood, 1999; Badeer, 2001). If we imagine a section CE of a tube (Fig. 2.9) of changing diameter from C to D and from D to E with a flow of liquid passing through it, then the amount of liquid passing into the section of the tube at C must be equal to that passing the point D and out at E. If the cross-sections of the tube at C, D, and E are respectively A_1 , A_2 , and A_3 , and the velocities are V_1 , V_2 , and V_3 , then the volume of liquid entering per unit time (Q) is A_1V_1 and the equation of continuity gives

$$Q = A_1V_1 = A_2V_2 = A_3V_3. \quad 2.22$$

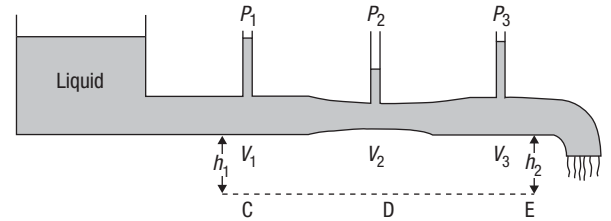


Figure 2.9 Diagram to illustrate Bernoulli’s principle. The cross-sectional area of the rigid tube decreases from A_1 at site C to A_2 at site D. The cross-sectional area and velocity at site E are the same as those at site C. Lateral pressures P_1 , P_2 , and P_3 are indicated by the height of the fluid in the small vertical tubes at each cross-section. Since $A_1 > A_2$, $V_1 < V_2$, and if the tube is horizontal $P_1 > P_2$ and $P_2 < P_3$. Since there is flow along the tube, P_1 is slightly greater than P_3

If the pressures at C, D, and E are P_1 , P_2 , and P_3 , the work done on the fluid entering the tube per unit time (W_p) at C is $P_1A_1V_1$, at D it is $P_2A_2V_2$ and at E it is $P_3A_3V_3$. Liquid in motion also has kinetic energy (W_k) that depends on its mass m and the square of its velocity. Mass in turn is the product of density ρ and the volume flow, so that

$$m = \rho A_1 V_1,$$

and the kinetic energy at the point of entry C is

$$W_k = \frac{1}{2} V_1^2 \rho A_1 V_1.$$

If there is a difference in height between the two sites C and D, there will be a net external gravitational force (W_g) acting on the mass of liquid; for heights h_1 and h_2 this will be $g\rho A_1 V_1 h_1$ and $g\rho A_1 V_1 h_2$, respectively.

The total energy (W_t) at the entry will be equal to the energy at D, so that

$$\begin{aligned} W_t &= P_1 A_1 V_1 + \rho A_1 V_1 \left(\frac{1}{2} V_1^2 + g h_1 \right) \\ &= P_2 A_2 V_2 + \rho A_2 V_2 \left(\frac{1}{2} V_2^2 + g h_2 \right), \end{aligned}$$

which can be reduced to

$$\begin{aligned} (P_1 - P_2) + \frac{1}{2} \rho (V_1^2 - V_2^2) \\ + \rho g (h_1 - h_2) = 0. \end{aligned} \quad 2.23$$

Equation 2.23 is called Bernoulli’s equation for steady, non-viscous, incompressible liquid flow. The energy thus consists of three parts: the pressure energy W_p , the kinetic energy W_k , and the potential energy in a gravitational field W_g .

If the tube in Fig. 2.9 is positioned so that $h_1 = h_2$, then $(h_1 - h_2) = 0$, and

$$P_1 = P_2 + \frac{1}{2} \rho (V_2^2 - V_1^2); \quad 2.24$$

and since $V_1 < V_2$, it follows that $P_1 > P_2$.

It can thus be seen that if the velocity of a liquid is increased in a horizontal tube, the lateral pressure must be correspondingly decreased. This effect is particularly apparent at a constriction in a vessel (point D) and is applied in devices such as the Pitot and orifice (or Venturi) meters that have been used for measuring blood flow. Bernoulli's equation can also be applied to unsteady flows. The expression has been derived for the general case by considering the fluctuations of flow as variables in respect of position along a streamline in respect to time (Kaufmann, 1963), but the problem can be approached more generally through the Navier–Stokes equations. The unsteady Bernoulli equation accurately predicted the measured pressure gradient across a stenotic aortic valve in dogs (Bermejo *et al.*, 2002).

LATERAL AND IMPACT PRESSURE

Kinetic energy must be taken into account when pressures are measured through fluid-filled catheters inserted into arteries (O'Rourke, 1982a; O'Rourke, 1982b; Badeer, 2001). If the catheter opening is oriented parallel to the bloodstream so that the flow 'impacts' on the catheter tip, the blood velocity at this point falls to zero (a stagnation point) and the kinetic energy is converted into pressure energy (stagnation pressure). The 'impact' or end-on pressure is therefore higher than the 'lateral' pressure measured with the catheter opening perpendicular to the bloodstream by an amount equivalent to the converted kinetic energy. The difference between impact and lateral pressure in an artery is therefore totally dependent on the

square of the linear flow velocity and can be calculated from the Bernoulli equation above (O'Rourke, 1982b). The greatest differences between impact and lateral pressure occur when the flow velocity is high during exercise or when the artery is narrowed, for example, by an electromagnetic flow transducer. Figure 2.10 shows the difference between the impact and lateral pressure waves measured in the dog ascending aorta when the vessel is narrowed by the placement of a rigid cuff-type flow probe (O'Rourke, 1968). The presence of the flow probe caused no perceptible change in the contours of the flow and lateral pressure waves, but considerable alteration in the contour and amplitude of the impact pressure wave. If the pressure is measured with the catheter facing downstream, the impact pressure will be lower than the lateral pressure by an amount equal to the kinetic energy pressure.

For most physiologic purposes, lateral pressures are the ones that should be measured, according to Milnor (1989); however, the question arises of which pressure (lateral or impact) should be used in impedance calculations. This subject was addressed by Burton (1965) in relation to peripheral resistance calculations. He suggested that the total energy expressed as impact pressure be used in the calculation. The subject is also discussed in detail by O'Rourke (O'Rourke, 1982a; O'Rourke, 1982b), where he suggests that impact pressure be used in the calculation of vascular impedance since this pressure also includes the kinetic energy imparted to the blood at high flow velocity. In practice, the difference between 'impact' and 'lateral' pressure is small unless the artery is compressed externally or narrowed by disease.

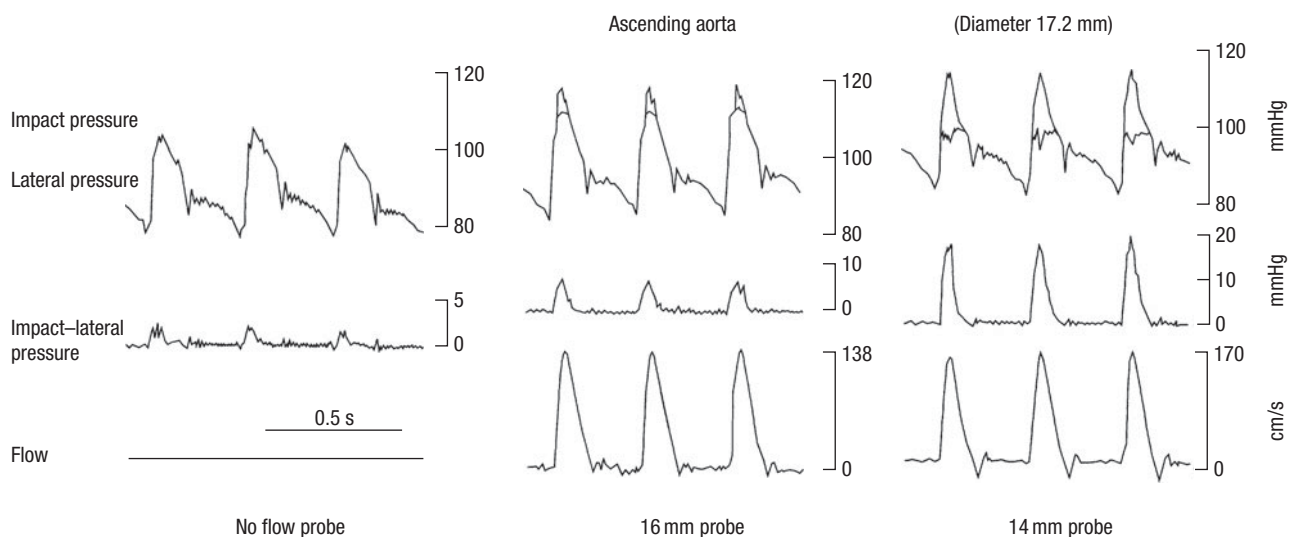


Figure 2.10 Measurements of 'lateral pressure' and 'impact pressure' (upper traces) through liquid-filled catheters inserted into the ascending aorta of a dog. When the catheter opening is parallel to the direction of flow, only the laterally directed pressure is communicated to the pressure transducer. When the catheter opening faces upstream, the blood that 'impacts' upon the opening loses some (or all) of its kinetic energy, which is converted into pressure, making the measured 'impact pressure' higher than the 'lateral pressure' (middle trace). Blood flow velocity is measured with a cuff-type flow probe (lower trace). With a 14 mm probe, the aorta (diameter 17.2 mm) is moderately narrowed and velocity is increased (note the difference in scale). Reproduced from O'Rourke (1968)

TORRICELLI'S EQUATION

Consider a reservoir filled with liquid whose level is maintained constant by an appropriate supply line. Some distance h_2 below the fluid level (h_1), there is an opening or orifice with well-rounded corners, small in comparison with the cross-sectional area of the reservoir, through which a steady flow of liquid exits (see Fig. 2.11A). If P_1 and V_1 are the pressure and velocity at the level surface in the reservoir, and P_2 and V_2 are those at the orifice, we have from Bernoulli's equation (eqn 2.23)

$$V_2^2 - V_1^2 = \frac{2}{\rho}(P_1 - P_2) + 2g(h_1 - h_2). \quad 2.25$$

By use of the continuity equation above, V_1 can be expressed in terms of V_2 , since $V_1 = V_2 A_2 / A_1$, where A_1 and A_2 are the cross-sectional areas of the tank and the orifice, respectively. In general, A_2 is small compared with A_1 , so that V_1 is small compared with V_2 , and V_1^2 in eqn 2.25 can be neglected. If, furthermore, both the liquid level and the flow out of the orifice are open to the atmosphere, and if we neglect the difference in atmospheric pressure between the two levels, $P_1 = P_2$ and eqn 2.25 becomes

$$V_2^2 = 2g(h_1 - h_2)$$

or

$$V = [2g(h_1 - h_2)]^{1/2}. \quad 2.26$$

Equation 2.26 gives the speed with which a liquid escapes from a vessel through an orifice and is known as Torricelli's (1608–47) equation.

The volume flow rate Q corresponding to the velocity in eqn 2.26 is

$$Q = \pi R^2 V = AV = A[2g(h_1 - h_2)]^{1/2}. \quad 2.27$$

Experiments have shown that the actual exit velocity is somewhat smaller than the theoretic velocity owing to the effect of friction, which causes a certain loss in total

energy. One can account for this by introducing a correction factor η and writing eqn 2.27 as

$$Q = \eta A[2g(h_1 - h_2)]^{1/2}. \quad 2.28a$$

Another phenomenon that is closely connected with the flow from a tank is the narrowing or constriction of the jet emerging from the tank. For, if one does not use an opening with well-rounded contours as in Fig. 2.11A, but permits the liquid to flow through a sharp-edged orifice in the vessel wall (Fig. 2.11B), the streamlines converging from inside the vessel toward the orifice cannot suddenly turn into the horizontal direction, but do so gradually and cause the jet to contract; its cross-section A_c is smaller than the cross-section A of the orifice. The ratio $\psi = A_c/A$ is called the 'contraction coefficient'. Hence, the volume flow rate is

$$Q = VA_c = \eta\psi A[2g(h_1 - h_2)]^{1/2}. \quad 2.28b$$

In general, $\eta\psi$ can be determined only empirically.

Gorlin and Gorlin (1951) used eqn 2.28b to calculate the area of stenotic cardiac valves and central circulatory shunts:

$$A = \frac{Q}{44.3\pi\psi[(P_1 - P_2)]^{1/2}} \quad 2.29$$

where $\eta\psi$ is the 'discharge coefficient' (an empirical constant), and $(P_1 - P_2)$ is the pressure gradient across the orifice and $44.3 = \sqrt{2g^{1/2}} = \sqrt{2 \times 980^{1/2}}$. When using eqn 2.29 to calculate valve area, it must be kept in mind that flow across the atrioventricular valves occurs during diastole, and flow across the semilunar valves occurs during systole. The empirical constant for the tricuspid, pulmonic, and aortic valves, as well as for a patent ductus arteriosus or ventricular septal defect, is assumed to be 1.0. The empirical constant for the mitral valve was originally reported by Gorlin and Gorlin to be 0.7, but in a later report (Cohen and Gorlin, 1972) a value of 0.85 was considered more accurate. Simplified versions of the Gorlin equation have also been reported (Cannon *et al.*, 1985).

Possibly the most important clinical use of the above equations (or modifications of them) employs Doppler

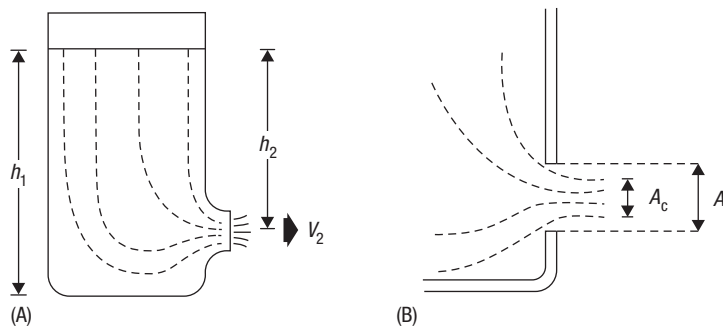


Figure 2.11 Diagram to illustrate Torricelli's principle. (A) Liquid flow from a reservoir through an orifice with well-rounded corners. (B) Liquid flow through an orifice with sharp edges. The cross-sectional area A_c of the stream is smaller than that of the orifice, A

echocardiography to calculate the pressure gradient across a narrowed part of the cardiovascular system (Hatle, 1984; Topol, 2007). The accuracy and validity of this non-invasive technique have been verified in numerous laboratories (Hatle *et al.*, 1978; Stramm and Martin, 1983; Teirstein *et al.*, 1985). The equations are used extensively to estimate the area of (Zoghbi *et al.*, 1986; Poh *et al.*, 2008) and pressure gradient across normal (Firstenberg *et al.*, 2000) and stenotic mitral (Geibel *et al.*, 1991; Devlin *et al.*, 2001), aortic (Barker *et al.*, 2002), and prosthetic valves (Peteiro *et al.*, 1998).

THE EFFECT OF CHANGE IN THE SIZE OF THE VASCULAR BED

We have already considered the relations between the pressure gradient, rate of flow, and internal lumen for steady flow of liquid in a straight tube. Changes in the lumen of arteries, however, are virtually all associated with the occurrence of curves and branches. These branches are, individually, narrower than the parent trunk, but the total cross-sectional area nearly always increases with successive branching. It is therefore of interest to see how a change in the size and shape of the channel due to the presence of curves and branches will affect the rate of flow, the Reynolds number (eqn 2.57), and the pressure gradient. While this can be derived simply for steady flow, it is of somewhat academic interest in a consideration of the larger arteries, because pulsatile flow has quite different pressure-flow relationships. Using analogous electrical terms, we are here calculating the changes in resistance in the vascular network, whereas we really need to know the changes in impedance, which is considered in Chapter 10. The latter problem is more difficult, so that it is worth considering the simplest conditions first. These conditions apply in general throughout the venous system and (approximately) in small arteries.

Effects of alterations in regional arterial geometry on liquid flow have been studied extensively using Murray's law (Murray, 1926) and Dean's equation (Dean, 1928). In 1926, the physiologist Cecil Murray published a theoretical explanation for the relationship between the radius of an artery immediately upstream from a branch point (parent artery) and the radii of arteries immediately downstream (daughter arteries). In its simplest form, i.e. when an artery of radius R_k branches into n arteries of radius R_{k+1} , the relationship termed Murray's law states that the radius of the parent artery cubed equals the sum of the radii of daughter branches cubed. Murray's derivation of this relationship assumed that there is an energy requirement for producing the blood contained in a vessel that is proportional to the volume of blood in a vessel and that there is an energy requirement for pumping blood through a vessel that is given by Poiseuille's law for flow in a straight tube. When the radius of a branching artery is increased, the cost of blood in the artery increases, but the

cost of pumping blood through the artery decreases. While Murray did not suggest a mechanism for the regulation of the radius of an artery, other scientists have. A recent hypothesis is that shear force at the inner surface triggers circumferential growth if the force is above a threshold (Barakat *et al.*, 2005). This is an attractive hypothesis because high values of shear in fluids can damage or destroy cells (Painter *et al.*, 2006). Furthermore, turbulence, which occurs above critical values of shear in fluids, is associated with atherosclerosis of arterial walls. Analysis of blood flow in the entire coronary arterial tree and its relationship with arterial branch lengths and lumen volume using Murray's law has been reported recently (Zhou *et al.*, 1999; Mittal *et al.*, 2005; Molloy and Wong, 2007). Take a main vessel, cross-sectional area A_1 , that divides into n branches of equal size and each of cross-sectional area A_2 (Fig. 2.12). If the total vascular bed changes by a factor d , so that $dA_1 = nA_2$, we can describe the behavior of the bed in terms of the ratio d without considering actual cross-sectional areas. Let the pressure gradient, $(P - P')/L$ or ΔP , in the main trunk be ΔP_1 and that in the branches ΔP_2 , similarly Q_1 and Q_2 the respective volume rates of flow, R_1 and R_2 the radii, and V_1 and V_2 the average linear velocities of flow. The only determining condition is that the rate of flow into and out of the system must be equal, or Q_1 and nQ_2 . Since the rate of flow is the mean velocity times the cross-sectional area of the tube,

$$Q_1 = V_1 \pi R_1^2 = n V_2 \pi R_2^2$$

or

$$\frac{V_1}{V_2} = \frac{n R_2^2}{R_1^2} = d; \quad 2.30$$

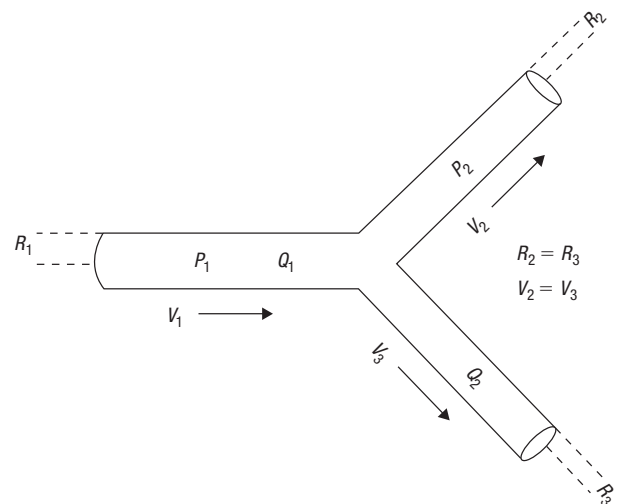


Figure 2.12 A junction with two branches, of equal radii (R_2 , R_3) and velocity (V_2 , V_3). If the total cross-sectional area of the two branches is greater than that of the parent vessel, V_2 and V_3 are less than V_1

that is to say, the mean velocity of flow in the branches will be less by a factor of d whatever the number of branches. Furthermore, the Reynolds number in the branches will be less than that in the main artery.

In a similar manner, from Poiseuille's equation, we may compare the pressure gradients in the main trunk and the branches:

$$Q_1 = \frac{\pi R_1^4 \Delta P_1}{8\mu} = nQ_2 = \frac{n\pi R_2^4 \Delta P_2}{8\mu},$$

so that

$$\frac{\Delta P_1}{\Delta P_2} = \frac{nR_2^4}{R_1^4}$$

and, as

$$\begin{aligned} R_2^2/R_1^2 &= d/n, \\ \Delta P_1/\Delta P_2 &= d^2 n, \end{aligned} \quad 2.31$$

from which we see the interesting result that for steady flow the pressure gradient remains the same only if $d = n^{1/2}$ and increases when $d < n^{1/2}$. As the minimum value of n is 2.0, the pressure gradient will be greater in the branches of a bifurcation unless the cross-sectional area of the vascular bed increases by a factor of more than 1.414. The average value for d that is often quoted is 1.26 at a major arterial bifurcation (Milsum and Roberge, 1973); this appears to originate from the work of Blum (1919) and was used by Hess (1927) and Zamir (1977). In this situation, we can see that the velocity in the branches (eqn 2.30) will be $1/1.26$, or 0.8 , of that in the parent trunk. The pressure gradient, on the other hand, will increase by the factor $2/(1.26)^2$ or 1.26 .

In a steady-flow system, the relation between pressure gradient and liquid flow is commonly expressed as the resistance (see the following section); if the resistance is expressed as the ratio of pressure gradient and volume flow (see eqn 2.34), $\Delta P/Q$, and is indicated by R_s , then $R_{s1} = \Delta P_1/Q_1$, and from eqn 2.31

$$\frac{R_{s1}}{R_{s2}} = \frac{\Delta P_1 Q_2}{\Delta P_2 Q_1} = \frac{d^2}{n} \frac{1}{n} = \frac{d^2}{n^2}. \quad 2.32$$

If the resistance is expressed in terms of the average flow velocity and indicated by r_s , we have, $r_{s1} = \Delta P_1/V_1$ and from eqns 2.26 and 2.27 we obtain

$$\frac{r_{s1}}{r_{s2}} = \frac{\Delta P_1}{\Delta P_2} = \frac{V_2}{V_1} = \frac{d^2}{n} \frac{1}{n} = \frac{d^2}{n}. \quad 2.33$$

Therefore, the resistance will remain unchanged in an individual branch beyond a subdivision only if the total cross-sectional area of the branches is larger than that of the parent trunk by a factor that is the number

of branches – which is a minimum of two. This applies equally whether the resistance is measured in relation to volume flow or to flow velocity. As no one has observed an increase in the size of the vascular bed by as large a factor as two at any bifurcation, it follows that the vascular resistance measured in each branch will always increase.

If, however, the vascular resistance of all branches is taken together, the volume flow is the same on either side of the junction and the corresponding resistance will vary as the pressure gradient (eqn 2.33). As pointed out above, this means that at a bifurcation, the cross-sectional area must increase by a factor of $2.0^{1/2}$ for the resistance (in terms of volume flow) to remain unchanged.

It is interesting to note that, in all the relations above, the only one that is determined by the total cross-sectional area of the bed alone is the velocity of flow. The measurement of the mean blood flow velocity on either side of a vascular subdivision could therefore be used as a measure of the change in the size of the vascular bed in the living animal. The dimensions of the vessel in the living animal are the subject of considerable controversy and are discussed below. In a more general way, the changes in mean velocity can be used to indicate where the main changes in size of the bed occur. For example, in the dog, the mean blood flow velocity in the proximal aorta is usually 15–20 cm/s and in the femoral artery about 10–12 cm/s. This implies that the vascular bed measured in arteries of the same order as the femoral has increased by about 50 percent compared with the aorta. In the capillaries, however, where the mean blood flow velocity is about 0.05 cm/s, there must be an increase of the vascular bed to 300–400 times that of the aorta. Such generalizations are, of course, only approximate as they imply an equal degree of arteriolar constriction in all the main vascular beds so that blood flow is not being shunted preferentially through any one of them.

The consideration of the division of flow between unequal branches cannot be satisfied in such a general way because there are a different set of pressure–flow relations for each. As before, $Q_1 = Q_2 + Q_3$, and from Poiseuille's equation we see that

$$\frac{Q_2}{Q_3} = \frac{\Delta P_2 R_2^4}{\Delta P_3 R_3^4}.$$

ΔP is the pressure gradient (the fall in pressure per unit length), and unless we know the length and fall in pressure gradient in both channels, the flow down each branch will fall to the same capillary pressure, $\Delta P_2 = \Delta P_3$, but this makes the further assumption that the overall lengths of the two branches are the same. If ΔP_2 and ΔP_3 are the same, the relative flows in the branches will, of course, be proportional to the fourth power of their radii, R_2 and R_3 . In recent years, the fast development of new methods in microcirculatory research has made it possible to determine a large number of rheologically and functionally

relevant parameters in individual microvessels down to the capillary level. In many studies, it was found that at bifurcations the hematocrit in a side branch was lower than in the parent vessel. This phenomenon has been described as plasma skimming by Krogh (1922) and has been extensively studied *in vitro* (Pries *et al.*, 1981) and *in vivo*. In these studies, a number of variables have been identified which influence the disparity between blood and RBC volume flow distribution at a branch point. The most relevant of these variables are the distribution of volume flow, the tube to particle diameter ratio, the average hematocrit and the hematocrit profile in the feeding vessel. The branches' angles between the segments constituting a bifurcation proved to be of minor importance. It has been found that plasma skimming takes place not only in the microcirculation as previously discussed, but also in the larger arterial branches that carry blood to the organs, such as the brain (Mchedlishvili and Varazashvili, 1982).

The presence of atherosclerotic plaques has been shown to be closely related to arterial geometry (Caro, 1994). The carotid artery bifurcation, the coronary arteries, the infrarenal aorta and the arteries supplying the lower extremities may be markedly diseased, while other vessels are spared. Due to these findings, local hemodynamic factors are considered to play an important role in the initiation and development of atherosclerosis (Chapter 21). Studies on postmortem human arteries (Zarins *et al.*, 1983) and in experimental animals (Sawchuk *et al.*, 1994) showed positive correlation between plaque thickness and the presence of low shear stress, departure of unidirectional flow and regions of flow separation and recirculation.

Comparison between flow velocity measurements and plaque distributions in human carotid artery specimens revealed that at the outer wall of the carotid sinus, where wall shear stress and mean flow velocities are low (Lee *et al.*, 2001), plaque thickness is maximal, whereas at the flow divider wall, where wall shear stress and velocities are high, plaque thickness is minimal (Zarins *et al.*, 1983). At the side walls, where circumferential flow velocities (secondary flow) are present, plaque thickness is intermediate. Cross-sectional axial velocity profiles were calculated and measured for the midsinus and endsinus internal carotid artery by Botnar *et al.* (2000a; 2000b). At both locations, a skewed velocity profile with slow velocities at the outer vessel wall, medium velocities at the side walls, and high velocities at the flow divider (inner) wall were observed. Various flow quantification techniques, such as laser Doppler anemometry (LDA) (Ku and Giddens, 1987; Liepsch *et al.*, 1994), ultrasound (US) Doppler (Hatle and Angelson, 1990; Lee *et al.*, 2001) or MRI (Botnar *et al.*, 2000a; Botnar *et al.*, 2000b; Ford *et al.*, 2008) are commonly used to study arterial hemodynamics. The choice of one of these techniques depends on the specific application and the demand on spatial and temporal resolution.

THE MEASUREMENT OF PERIPHERAL RESISTANCE

The application of Poiseuille's equation (eqn 2.7) to any set of arteries, in the way it was used in the preceding section, requires knowledge of the dimensions of those arteries. When considering the vascular system of the body, or of one of its regions, we do not have this information. The formula can, however, be simplified by writing

$$P_1 - P_2 = R_s Q = r_s V; \quad 2.34$$

and from analogy with Ohm's law, the terms R_s or r_s , are called the fluid (or vascular) resistances because the pressure gradient is analogous with the potential difference and the blood flow with the current. It is usual to calculate the resistance in terms of volume flow, but, as was seen in the previous section, it is often more illuminating to compare the velocity of flow in various vessels, so we will also consider the 'velocity resistance'. Pursuing the analogy, the Poiseuille equation only applies to steady flow, the equivalent of direct current. In arterial channels where the flow is pulsatile, this might be thought to be inapplicable. Pulsatile arterial flow, however, always has a steady-flow component – the mean flow – and it is possible to apply the direct current concept to this mean flow. The corresponding pressure–flow relation for the oscillatory flow is, using alternating current theory, called the 'vascular impedance' (see Chapter 12). In the broader implications of this analogy, it is often more useful to take velocity as the analogy so, at risk of confusion, it is here also applied to resistance; the convenience of this measure is more apparent in comparing impedance at different sites of the same animal, at the same sites of different animals, and the relationship between impedance and wave velocity.

From Poiseuille's equation, it can be seen that the resistance in eqn 2.34 is expressed by

$$R_s = 8\mu L / \pi R^4 \quad 2.35$$

and

$$r_s = 8\mu L / R^2. \quad 2.36$$

The length of any vascular channel is virtually constant for anatomical reasons, and the mean viscosity of the blood can also be regarded as constant (although the viscosity does vary with the radius in small vessels) (Pries *et al.*, 1994). The vascular resistance (i.e. opposition to the flow of blood) is thus very largely determined by the radius of the vessels and the systemic hematocrit (Fig. 2.6) (Pries *et al.*, 1994). In the complete vascular bed, the total resistance may be regarded as the resistance of, respectively, the arteries, the arterioles, the capillaries, and the veins in series. As the resistance is proportional to the

drop in mean pressure, it is apparent that the resistance of the arterioles constitutes the largest proportion of the whole (Zweifach and Lipowsky, 1984). Thus, the mean arterial pressure is normally about 100 mmHg and has fallen very little in the smallest arteries in which it has been measured (Poiseuille, 1828; Pauca *et al.*, 1992; Pauca *et al.*, 2001). In the capillaries, it is generally agreed to be about 30–34 mmHg at the proximal (arterial) end and 12–15 mmHg at the distal (venous) end. Most of the fall (up to 60 mmHg) beyond the conduit arteries will occur within the 5 mm or so immediately proximal to the capillaries, in arterioles less than 200 μm in diameter (Fig. 2.13) (Zweifach, 1974; Zweifach and Lipowsky, 1977; Zweifach and Lipowsky, 1984). The pressure in the large veins will only be a few mmHg, so that, of a total pressure drop of 100 mmHg, up to 60 percent may occur in the arterioles, about 15 percent both in the capillaries and veins (totalling 30 percent) and about 10 percent in the large (about 2.0 percent) and small arteries (about 8.0 percent) (see Fig. 1.3). The total peripheral vascular resistance is thus dominated by the caliber of the arterioles, but the other components of the vascular bed are by no means negligible (Zweifach and Lipowsky, 1977; Zweifach and Lipowsky, 1984). The fact that the arterioles can be actively altered in size by smooth muscle cell activation has focused special attention on their contribution to the peripheral vascular resistance. The control of arteriolar diameter reflects the sum of multiple inputs to vascular smooth muscle cells. In skeletal muscle, activation of sympathetic nerves results in vasoconstriction (Kurjiaka and Segal, 1995). In contrast, increasing the metabolic activity of muscle fibers (e.g. during exercise) induces vasodilation. Second order arterioles (resting diameter, 15–54 μm) can decrease their diameter by as much as 80 percent during vasoconstriction and can increase their diameter by as much as 60 percent during

vasodilatation (Fig. 2.14) (Kurjiaka and Segal, 1995). These changes in arteriolar diameter can cause a 33-fold change in vascular resistance. Indeed, under controlled conditions, the measurement of the resistance is a useful and simple way of studying vasomotor activity (Honig, 1979). Nevertheless, it is regrettable that the term ‘peripheral resistance’ has so often been used as if it were synonymous with ‘arteriolar resistance’, and it is unfortunate, too, that many believe that resistance is constant at different levels of pressure gradient (and of flow) so that a change in resistance always signifies a corresponding change in vascular caliber.

The value of the concept of peripheral, or vascular, resistance has been questioned by some authors. The basis of most of their comments has been due partly to the somewhat slipshod way in which approximations have been introduced without assessment of the consequent errors, and partly to verbal shortcuts, such as regarding the arteriolar vessels as the whole of the peripheral resistance. In addition, it has been criticized as being an artificial, or abstract, idea, useful only for manipulating experimental data. To some extent, this is true, but only to the extent that the concept of electrical resistance is abstract – for it also only expresses the relationship between two measured variables: current and potential drop in a circuit. Similarly, the measurement of changes in the resistance of a circuit can only give the net, or overall, effect and cannot distinguish changes of individual components.

The measurements that need to be made are basically the same in all cases – the mean pressure at the beginning and the end of the vascular bed to be studied and the total blood flow through the bed. This means that instead of eqn 2.34, the total peripheral resistance of the systemic circulation is calculated as the difference between the mean aortic pressure (MAP) and the mean right atrial pressure, divided by the cardiac output.

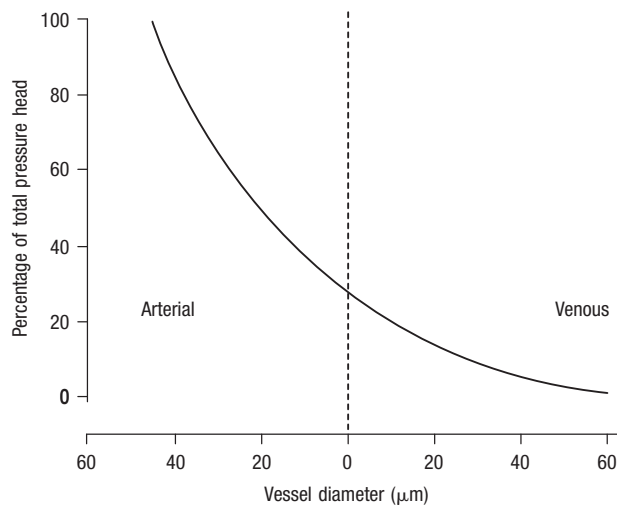


Figure 2.13 Graph illustrating pressure distributions measured in the cat mesentery. Data from Zweifach (1974) and Zweifach and Lipowsky (1977)

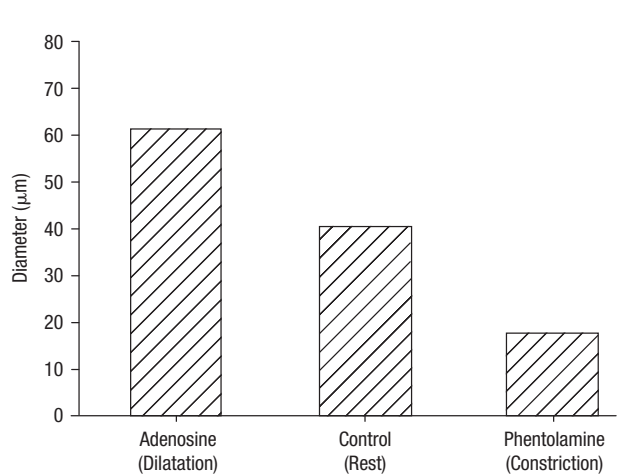


Figure 2.14 Diagram showing the active change in diameter of second-order arterioles during vasoconstriction with phentolamine and vasodilatation with topical adenosine. Data from Kurjiaka and Segal (1995)

As the pressure in the great veins is very small in comparison with the MAP, it is usually taken as zero, and the formula for resistance is then written as

$$R_s = \frac{\text{MAP}}{\text{Cardiac output}}. \quad 2.37$$

In smaller regions of the systemic bed, the approximation of taking the venous pressure as zero may introduce appreciable errors (Milnor, 1989). Furthermore, in the pulmonary circulation, the arterial pressure is much lower than aortic pressure and left atrial pressure higher than right atrial pressure (see Chapter 16); it is then very inaccurate to estimate pulmonary vascular resistance without measuring left atrial pressure. In the presence of heart failure, venous pressure must be included for systematic resistance to be calculated accurately.

The units in which the peripheral resistance is expressed have varied with different authors. The peripheral resistance unit (PRU) introduced by HD Green is in the empirical units of mmHg/mL/min; in practice, this works out in numbers of convenient dimensions in regional vascular beds, but as with all arbitrary units, is difficult to convert or correlate with other dimensions. In standard physical units (centimeter-gram-second), the corresponding units are dyne-s/cm⁵ if volume flow rate is used, or dyne-s/cm³ if blood flow velocity is used.

Taking some average numbers for the human subject, we can calculate some representative values.

1. **Total peripheral resistance of the systemic circulation.** If the cardiac output (CO) is 6.6 L/min (110 cm³/s), the MAP 95 mmHg and the mean velocity of flow 16 cm/s, then

$$R_s = \frac{95}{6600} = 0.014 \text{ PRU}$$

or

$$R_s = \frac{9.5 \times 13.6 \times 980}{110} = 1151 \text{ dyne-s/cm}^5.$$

A useful conversion to remember is

$$R_s = \frac{\text{MAP (mmHg)}}{\text{CO (L/min)}} \times 80 = \frac{95}{6.6} \times 80 = 1151 \text{ dyne-s/cm}^5.$$

If mean blood flow velocity is used to calculate resistance, we have

$$r_s = \frac{9.5 \times 13.6 \times 980}{16} = 7914 \text{ dyne-s/cm}^3.$$

2. **Peripheral resistance of the femoral vascular bed.** If volume flow is 225 cm³/min (3.75 cm³/s) and the mean velocity is 12 cm/s, then

$$R_s = \frac{95}{225} = 0.42 \text{ PRU}$$

or

$$R_s = \frac{9.5 \times 13.6 \times 980}{3.75} = 3.38 \times 10^4 \text{ dyne-s/cm}^5$$

or

$$r_s = 10.6 \times 10^3 \text{ dyne-s/cm}^3.$$

3. **Vascular resistance of a single capillary.** Taking the following values for a typical capillary, we may obtain comparative values for the smallest vessels: pressure drop 15 mmHg, mean velocity 0.5 mm/s, and a diameter of 10 μm.

$$R_s = 5 \times 10^{11} \text{ dyne-s/cm}^5$$

and

$$r_s = 4 \times 10^5 \text{ dyne-s/cm}^3.$$

(From the same values, it may be computed that there must be a total of 2×10^9 capillaries of this size and that their total surface area is 32 m². The femoral artery values taken above are scaled up proportionately from values observed in the dog.)

Normal values of systemic and pulmonary vascular resistance for man and dog are given in Table 2.1. Body weight and age are specified in the table because resistances normally vary with these factors (Nichols *et al.*, 1985a; Milnor, 1989).

The contribution of the various grades of vessel to the peripheral resistance has been studied by a few workers (Zweifach and Lipowsky, 1977; Zweifach and Lipowsky, 1984; Pries *et al.*, 1994). The estimate given above is very approximate, and the pressure drop across various sections, including the arteries, will vary considerably with vasomotor activity. It is also of interest to know whether a change in mean pressure can alter the peripheral resistance by causing a passive change in the caliber of the small vessels. Read *et al.* (1958) measured the total peripheral resistance in dogs that were being perfused by a pump, so that the total flow was accurately known, and found that a rise in venous pressure consistently caused a fall in the resistance. At its greatest value, there was about 20 percent change in R_s for an increase of 20 mmHg in venous pressure and a fall of about 60 percent with a rise in venous pressure to 60 mmHg. This is thought to be

Table 2.1 Normal values of total vascular resistance at rest (dyne-s/cm⁵)

Vascular bed	Man (73 kg; age 50)		Dog (20 kg; age 2.5)	
	Mean	SD	Mean	SD
Systemic	1344 (1268)	183 (161)	3140 (3018)	570 (546)
Pulmonary	194 (88)	34 (17)	480 (320)	53 (34)

Values in parentheses were calculated by eqn 2.34 using aortic and right atrial pressures and main pulmonary artery and left atrial (or wedge) pressures. Other values are input resistance calculated without using atrial pressures.

mainly due to a dilatation of the small venous channels, since the arterial pressure increased a corresponding amount. In the vascular bed of a leg that was being per-fused, Phillips *et al.* (1955) measured the resistance when both arterial and venous pressures were changed so that the pressure gradient remained the same but the trans-mural pressure (i.e. the distending pressure) was altered. They found evidence that an increase of transmural pressure caused a fall in resistance. In a later paper, Levy (1958) found that this change was much more marked when arterial pressures were changed than when the changes were on the venous side. The change in arteriolar caliber and local resistance with a change in distending pressure is probably due to the myogenic response (Hwa and Bevan, 1986), often referred to as the Bayliss phenomenon (Baez, 1968).

The term ‘myogenic’ is used to describe an adjustment in smooth muscle of the contractile machinery that is set in motion by mechanical forces acting directly on the muscle cell. The response is presumably activated *in situ* by the variable wall tension set up by the prevailing transmural pressure (Burrows and Johnson, 1978). Myo-genic responses occur even after the removal of neurogenic

and blood-borne humoral influences. A basic implication of the myogenic concept is that pressure is the fulcrum around which local arteriolar adjustments are made. This is illustrated in Fig. 2.15, in which blood flow was stopped in mesenteric arterioles while static pressure was elevated through the vascular bed (Johnson and Intaglietta, 1976). Under these conditions, a maintained contraction (reduction in diameter) of the arterioles was observed.

Measurement of peripheral resistance is not, therefore, a very good analytical method for distinguishing the behavior of individual areas of a vascular bed. For separating the effects of changes in flow or vasomotor activity, however, it is simple and direct. As an example, some results of Leusen *et al.* (1954) may be used. In a series of dogs, a fall in the pressure of their isolated carotid sinuses caused an increase in mean arterial blood pressure from 125 to 194 mmHg (55.2 percent). At the same time, the cardiac output increased from 3.3 to 4.31 L/min (30.3 percent) and the total peripheral resistance from 3140 to 3750 dyne-s/cm⁵ (19.4 percent). It is clear that the increase in cardiac output played a greater part in increasing the blood pressure than did vasoconstriction.

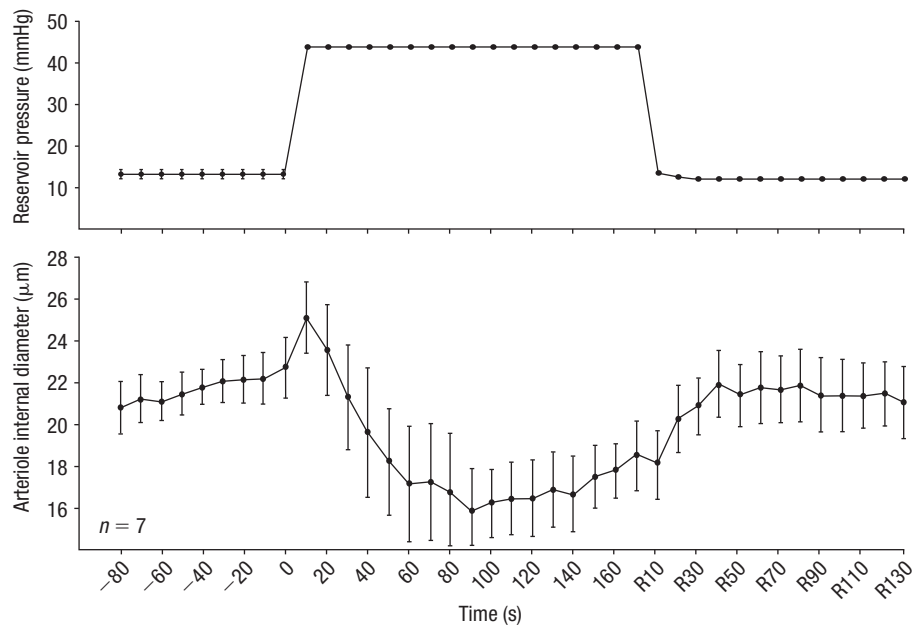


Figure 2.15 Change in arteriolar internal diameter with a step change in static intravascular pressure. The studies were performed with blood flow arrested in isolated cat mesenteric preparation and arterial and venous pressure elevated simultaneously. Reproduced from Johnson and Intaglietta (1976)

However, after a hemorrhage of 10 percent of their estimated blood volume, the corresponding increases were arterial pressure 72.5 percent, cardiac output 26.3 percent, and peripheral resistance 38.5 percent. The rise in pressure was then due more to vasomotor activity than to the increase in cardiac output. These values alone, however, cannot give any information on which areas of the vascular bed have constricted. From general knowledge, it is assumed that it will largely be in the arterioles, but venous constriction may well play a part in the response to sinus hypotension (Zweifach and Lipowsky, 1984). Such constriction would cause an increase in cardiac output but would also increase the total resistance, although there is little information from which we can estimate the proportion of the total increase that this would be. In some hypertensive patients, the elevation in blood pressure is caused by an increase in cardiac output. These patients have a peripheral resistance within the normal range (Nichols *et al.*, 1986b).

THE DIMENSIONS OF THE ARTERIAL TREE

Nothing demonstrates the difference between applying hydrodynamics to a man-made system and to the living array of arteries so dramatically as the uncertainty about the size of the lumen of blood vessels, the thickness of their walls and, in the smaller vessels, their length. In inanimate tubes, these dimensions are known with great precision. The problems arise, in part, from the range in size of individuals of a species, and from the fact that functioning arteries are elastic tubes subject to a distending pressure. The first factor requires the measurement of as many vessels as possible in the same individual, which is most easily done after death; this, however, removes the distending pressure, with a consequent diminution in size. As the elastic properties of the wall are non-uniformly distributed, this, in turn, will alter the ratios of the cross-sectional areas at each point of branching. The use of controlled-setting plastics (or resins) (which has provided a means whereby internal casts of an arterial tree can be made) has improved the situation. Ideally, these plastics can be injected after death at a normal arterial pressure that is held steady until the casting material is set and measurements of the internal bore of individual vessels can be made on the complete cast. For all the smaller branches down to the capillaries, their small size and dispersal within the tissues has made us rely on measurements using some method of magnification (Wiedemann, 1974; Pries *et al.*, 1994; Kurjiaka and Segal, 1995). Radiology with injected contrast media has been used by some workers (e.g. Reynolds *et al.*, 1952; Arndt *et al.*, 1971). Techniques for measuring vessel diameter *in situ* are discussed in Chapter 6.

In view of the complexity of the vascular network, it is necessary to be selective. The overall pattern of branching with considerations of total cross-sectional area, pressure

drop, resistance, and flow rates was considered in detail by Hess (1927), largely based on the extensive German literature up to that time. The total cross-sectional area has long been known to increase greatly by the time the level of the capillaries has been reached; this was shown above in relation to the fall in flow velocity and the rise in resistance in a number of parallel channels (see Chapter 1). A detailed numerical survey into the cross-sectional area data down to the smallest vessels (Table 2.2) has been made by Iberall (1967) and is described by Milnor (1989) (Table 2.2).

The data on aortic and major arterial dimensions that are considered here are from the excellent paper of Patel *et al.* (1963b) and from unpublished results from McDonald's laboratory. These measurements were made on casts of the canine arterial tree made immediately post-mortem. They can only be regarded as guides because the cast technique involves errors that are difficult to allow for. The resin that McDonald's group used (Batson No. 17) is subject to a linear shrinkage of at least 10 percent on setting; Patel and co-workers used 'Jeltrate', which sets into a rubbery material in which contraction is very small. McDonald *et al.* (1974) injected the liquid resin, which flows easily through the capillaries, at a sustained pressure of 100 mmHg. Once setting begins, because of shrinkage, the effective distending pressure is not known; in the earlier experiments, the injection was not well controlled and the distending pressure was almost certainly less than 100 mmHg. Patel *et al.* (1963b) do not mention the injection pressure they used but say that the cast measurements were adjusted in the light of the external diameters determined in the living animal.

Figure 2.16 is a diagrammatic scale drawing of a large part of the arterial tree; the dimensions of selected vessels of the dog are given in Table 2.3. The relative sizes of other vessels may thus be determined approximately. The values for the radius are all rather smaller than those of Patel *et al.* (1963b), also given in the table. This difference may be due, in part, to the smaller size of McDonald's (1974) dogs and in part to the adjustments of size to the living artery mentioned above. It certainly indicates that there is a considerable individual variation in these laboratory animals.

The aorta tapers from its origin at the left ventricle to its termination at the iliac bifurcation (Patel *et al.*, 1963b; Iberall, 1967; Ling *et al.*, 1973; McDonald, 1974; Langewouters, 1982; Latham *et al.*, 1985; Latham *et al.*, 1987). The cross-sectional area (A) change in the dog aorta can be expressed by the following exponential function (Caro *et al.*, 1978):

$$A = A_0 e^{-(kx/r_0)},$$

where A_0 and r_0 are the area and radius at the upstream site, x is the distance from the upstream site, and k is the 'taper factor', which normally lies between 0.02 and 0.05. The degree of tapering seen in the dog is illustrated in

Table 2.2 Model of vascular dimensions in a 20-kg dog (reproduced from Milnor (1989))

Class number	Vessels	Mean diameter (mm)	Number of vessels	Mean length (mm)	Total cross-section (cm ²)	Total blood volume length	
						(mL)	(%)
Systemic							
1	Aorta	(19–4.5)	1		(2.8–0.2)	60	11
2	Arteries	4.000	40	150.0	5.0	75	
3	Arteries	1.300	500	45.0	6.6	30	
4	Arteries	0.450	6000	13.5	9.5	13	
5	Arteries	0.150	110 000	4.0	19.4	8	
6	Arterioles	0.050	2.8 10 ⁶	1.2	55.0	7	5
7	Capillaries	0.008	2.7 10 ⁹	0.65	1357.0	88	
8	Venules	0.100	1.0 10 ⁷	1.6	785.4	126	
9	Veins	0.280	660 000	4.8	406.4	196	
10	Veins	0.700	40 000	13.5	154.0	208	
11	Veins	1.800	2100	45.0	53.4	240	67
12	Veins	4.500	110	150.0	17.5	263	
13	Venae cavae	(5–14)	2		(0.2–1.5)	92	
						1406	
Pulmonary							
1	Main artery	1.600	1	28.0	2.0	6	3
2	Arteries	4.000	20	10.0	2.5	25	
3	Arteries	1.000	1550	14.0	12.2	17	
4	Arterioles	0.100	1.5 10 ⁶	0.7	120.0	8	4
5	Capillaries	0.008	2.7 10 ⁹	0.5	1357.0	68	
6	Venules	0.110	2.0 10 ⁶	0.7	190.0	13	
7	Veins	1.100	1650	14.0	15.7	22	5
8	Veins	4.200	25	100.0		35	
9	Main veins	8.000	4	30.0		6	
						200	
Heart							
	Atria		2			30	5
	Ventricles		2			54	
						84	
						1690	100

Fig. 2.17 (Patel *et al.*, 1963b; Fry *et al.*, 1964). The measurements in the figure were obtained from postmortem canine aortic casts; the value of k in this example is 0.033. Although the aorta itself tapers, the total cross-sectional area of the aorta and its major branches summed together changes very little (see Table 2.2 above). However, as the arterial tree branches beyond its major branches, the total cross-sectional area increases progressively to the capillary bed (Fig. 2.18). These comments apply to experimental animals. In humans, with aging, taper is affected by degeneration and dilation of proximal elastic arteries, especially the aorta (see Chapter 18).

One approach for a mathematic model to deal with the multiplicity of branches along the length of the aorta that has been suggested at various times is to make it analogous to a tube that is leaking along its length; this is a problem for which the analysis is known, but the word ‘leak’ is used in its everyday sense of being small in proportion to the main flow. A consideration of the relative

flow through branches shows that this is misleading. The values for man are more dependable than are available for the dog, as these mean flow measurements through the principal organs are standard clinical diagnostic procedures. Representative values are given, for humans at rest, in Table 2.4. It can be seen that, of the flow leaving the ventricle (cardiac output), approximately 25 percent flows to the head and upper limbs (and the heart), and approximately 50 percent flows out through the large abdominal branches. The flow through the termination of the aorta is only some 25 percent of the total cardiac output. These values are, percentage-wise, comparable for the dog, although the flow to the hind legs is probably proportionately less in the dog, in which the weight of the muscle there is a smaller fraction of the body weight than is the muscle in the legs of habitually erect humans. The consideration of these flows illustrates more dramatically the functional role of the branches than do the dimensions of the branches.

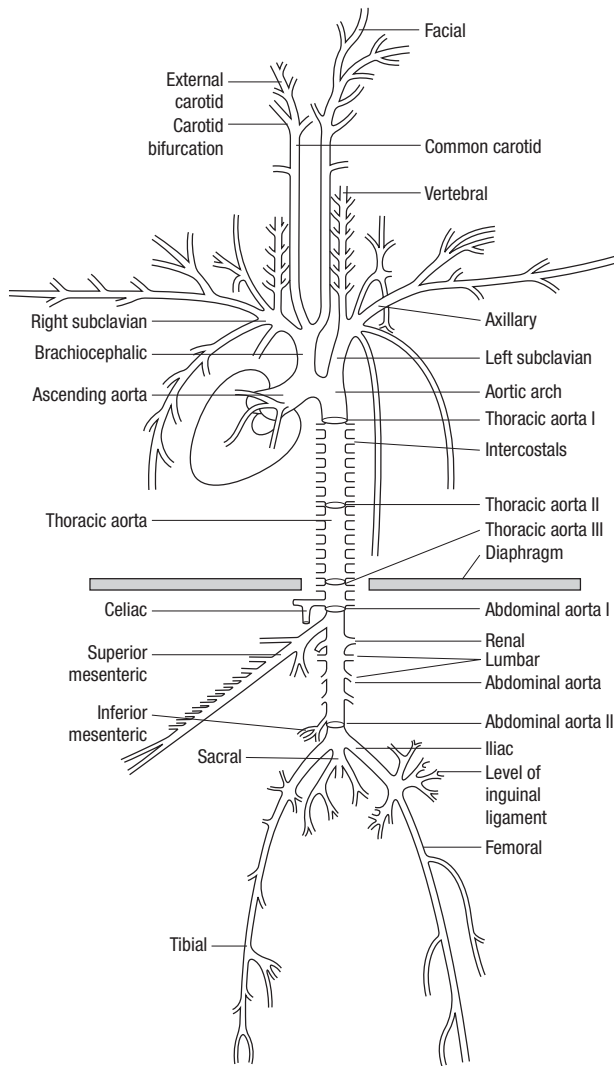


Figure 2.16 Diagrammatic representation of the major branches of the arterial tree of the adult dog (20 kg). The drawing is to scale, but the scale of the diameters is double the scale in lengths. The measurements are from plastic casts of the vessels; the numerical values refer to the central set of numbers in Table 2.3. The rings marked thoracic aorta I, etc., show the levels at which the diameters and cross-sectional areas were measured.

A corresponding set of values of flows within various regions in the dog is also given by Green (1950). He also prepared a table of a set of values for the dimensions, number of branches, etc., that is a standard reference. The only revision that might be desirable today would be to incorporate the measurements of the aorta and large arteries that have been reviewed above. For the smaller values, reliance has long been made on measurements of vessels throughout the mesenteric bed by Mall (1888), although he also studied the vessels of the stomach, adrenal glands, spleen, and liver (Mall, 1905). Attention was focused on Mall's early data by Schleier (1918) by an analysis of the hemodynamic factors implicit in his sets of dimensions of the branching system. Schleier's table was introduced into the textbook contribution of Bazett

(1941). A most valuable paper by Iberall (1967) has drawn our attention to Mall's neglected work and also more recent work by Suwa *et al.* (1963) in Japan. This latter work, in humans, analyzed the geometry of the circulations of the kidney, intestine, femoral bed, pancreas, heart, cerebral cortex, and basal ganglia of the brain. It should be emphasized, however, that under normal living conditions the internal diameter of arterioles and capillaries varies widely under vasoconstrictor and dilator influences (Kurjiaka and Segal, 1995). From these values and the simple relations between total cross-sectional area and number of branches arising from each parent branch, which were discussed above, the changes in resistance and mean velocity can be easily derived (Pries *et al.*, 1994). It is found that the resistance (and hence, by implication, the impedance) increases very rapidly over the very short distance between the tertiary arterial branches and the capillaries. It is on such considerations that it is postulated that the large rise of impedance in these vessels is the main site of wave reflection, although wave reflections are influenced by the tapering anatomy of the aorta and large arteries, especially, with aging in humans (see Chapters 10, 12, and 18) (Segers and Verdonck, 2000).

From the physiologic point of view, the capillaries may be thought of as the functional *raison d'être* of the circulation, where the transport of oxygen, and other substances required by the tissue cells, and the removal of metabolic products, necessary to maintain the life of the cell of the body, takes place. This brief summary of the anatomic array of tubes supplying these capillary beds is given to put in perspective the role of the arteries in the distribution of blood, from the pump (the heart) to the tissues of the body; the role of the veins in returning the blood to the heart is equally important but for present purposes may be regarded as being arrayed in the same way as the arterial tree. This is a book on blood flow in arteries and cannot hope to deal with the complex and intriguing functions of the microcirculation.

THE VELOCITY PROFILE IN PULSATILE FLOW

As blood flow in arteries is not steady, one of the basic assumptions of deriving the Poiseuille equation is violated. Thus, it is to be anticipated that the velocity profile will not be of the same parabolic form that is found in steady laminar flow (Tortoli, 1996).

The equation for the motion of a viscous liquid in laminar flow in a tube of circular cross-section, radius R , was derived earlier for steady flow. In its general form for an incompressible liquid, we can write (Womersley, 1955a; Womersley, 1955b; Womersley, 1957a)

$$\frac{\partial^2 w}{\partial r^2} + \frac{1}{r} \frac{\partial w}{\partial r} + \frac{1}{\mu} \frac{\partial P}{\partial z} = \frac{\rho}{\mu} \frac{\partial w}{\partial t} = \frac{1}{\nu} \frac{\partial w}{\partial t}. \quad 2.38$$

Table 2.3 Average internal radius and cross-sectional area values for three sets of data measured in dogs. The data at the left are taken from Patel *et al.* (1963b) and are based on external measurements made with casts of the arteries. The other two sets of data are from McDonald (1974)

Artery	Radius (cm)	Area (cm ²)	Radius (cm)	Area (cm ²)	Radius (cm)	Area (cm ²)
Asc. aorta	1.01	3.21	0.75	1.77	0.73	1.77
Thor. A I	0.82	2.11	0.55	0.95	0.48	0.78
Thor. A II	0.66	1.38	0.48	0.72	0.40	0.52
Thor. A III	0.61	1.19	0.45	0.64	0.39	0.51
Abd. A I	0.45	0.64	0.40	0.50	0.40	0.52
Abd. A II	0.44	0.60	0.35	0.38	0.35	0.39
Brach. ceph.	0.53	0.88	0.40	0.50	0.38	0.49
L. subcl.	0.44	0.60	0.35	0.38	0.35	0.41
Celiac	0.29	0.27	0.23	0.17	0.25	0.21
Sup. mes.	0.34	0.35	0.23	0.17	0.23	0.18
Renal (2)	0.19	0.12	0.15	0.07	0.19	0.12
Iliac (2)	0.23	0.17	0.24	0.18	0.23	0.13
Sacral	0.23	0.17	0.20	0.13	0.18	0.10
Int. cost. (20)	0.09	0.029	0.06	0.01	0.06	0.01
Inf. mes.	–	–	0.10	0.03	0.12	0.05
Lumber (8)	–	–	0.10	0.03	0.08	0.02
Comm. carot	–	–	0.20	0.13	0.25	0.20
Vertebral (2)	–	–	0.12	0.05	–	–
R. subcl.	–	–	0.28	0.25	–	–
Axillary	–	–	0.20	0.13	–	–
Femoral	–	–	0.15	0.07	–	–

The location of the thoracic (Thor.) and abdominal (Abd.) aortic measurements are indicated in Fig. 2.16. Asc., ascending; L. and R. subcl., left and right subclavian; Sup. and Inf. mes., superior and inferior mesenteric; Int. cost, intercostals; Comm. carot., common carotid.

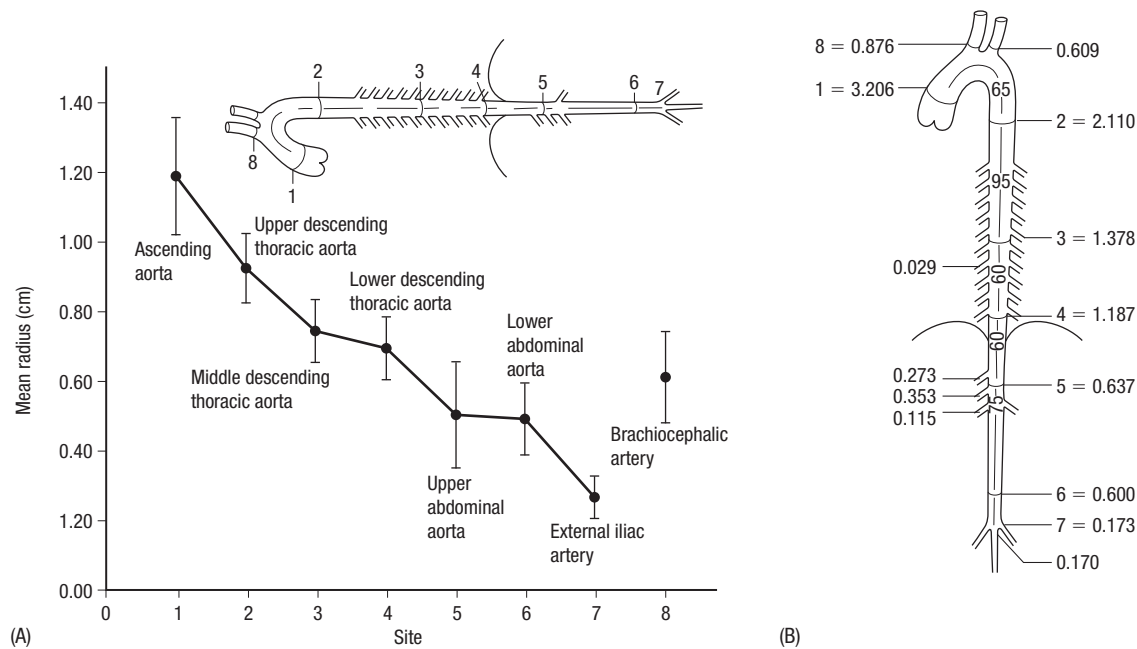


Figure 2.17 (A) Taper in the aorta: the mean external radius of the aorta at various sites measured at physiological pressure in dogs. Reproduced from Fry *et al.* (1964). (B) Internal cross-sectional area (cm²) at various sites in the aorta and its main branches. Reproduced from Patel *et al.* (1963b)

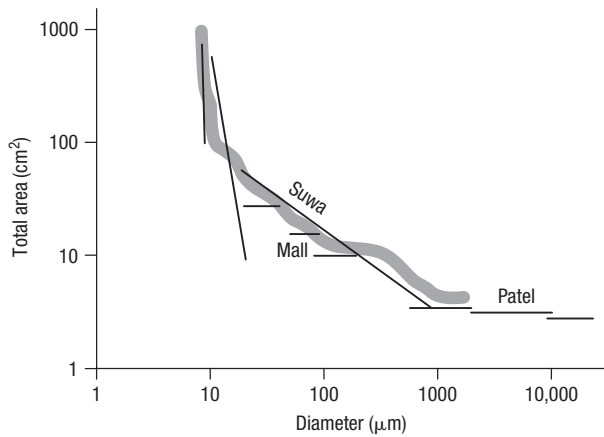


Figure 2.18 The distribution of cross-sectional area of the total vascular bed based on the values taken from Iberall (1967). The names inserted on the graph indicate the areas on the curve which are taken from the estimates of the authors quoted; thus the large vessel values are from Patel *et al.* (1963b) and the others from a series of papers by Mall and Suwa, cited by Iberall

Table 2.4 Distribution of blood flow in humans at rest

Circulation	Blood flow	
	(mL/min)	(% total)
Splanchnic (liver, intestines, spleen)	1400	24
Renal	1100	19
Cerebral	750	13
Coronary	250	4
Skeletal muscle	1200	21
Skin	500	9
Other organs	600	10
	5800	100

	Flow into various regions	
	(mL/min)	(% total)
Upper body		
Brain and heart	1000	
Muscle and skin (est.)	500	
	1500	26
Trunk: (4300 mL/min enters descending aorta, 74%)		
Liver, intestines and spleen	1400	
Kidneys	1100	
Muscle and skin (est.)	300	
	2800	48
Terminal aorta		
Pelvic organs and legs	1500	26

Following common convention, the axis of the tube is taken as the z axis and the velocity in the direction of that axis is w (the velocities in the x and y axes for a rigid tube will both be zero).

For steady flow, the velocity does not vary with time, so that $\partial w / \partial t = 0$; the pressure gradient $\partial P / \partial z$ can then be measured over a finite distance and written as

$$(P_1 - P_2) / L = \partial P / \partial z.$$

The Newtonian coefficient of viscosity, as stated above, is μ and the density of the liquid is ρ . The kinematic viscosity ν is μ / ρ .

For simplicity, the form of the pressure gradient will be taken as a simple harmonic motion and written in complex form (A^* is the complex conjugate of A).

$$\frac{\partial P}{\partial z} = A^* e^{i\omega t} \quad 2.39$$

where $\omega = 2\pi f$ is the angular frequency in radians per second of the oscillatory motion, with f the frequency in Hertz (cycles/s). The solution for this case is especially important because, with the aid of the Fourier series (see Chapter 11), any periodic function, such as the arterial pulse, can be represented.

With this substitution, we rewrite eqn 2.38 in the form

$$\frac{\partial^2 w}{\partial r^2} + \frac{1}{r} \frac{\partial w}{\partial r} - \frac{1}{\nu} \frac{\partial w}{\partial t} = -\frac{A^*}{\mu} e^{i\omega t}. \quad 2.40$$

This may be simplified mathematically by the substitution $w = u e^{i\omega t}$, where, by the cancellation of $e^{i\omega t}$ throughout the equation we obtain

$$\frac{d^2 u}{dr^2} + \frac{1}{r} \frac{du}{dr} - \frac{i\omega}{\nu} u = -\frac{A^*}{\mu}. \quad 2.41$$

This is a form of Bessel's equation (Pipes, 1958), and its solution, appropriate to the boundary conditions, is

$$u = \frac{A^*}{i\omega\rho} \left\{ 1 - \frac{J_0[r\sqrt{(\omega/\nu)}i^{3/2}]}{J_0[R\sqrt{(\omega/\nu)}i^{3/2}]} \right\} \quad 2.42$$

where an expression of the form $J_0(xi^{3/2})$ is a Bessel (1784–1846) function of the first kind of order zero and complex argument (Pipes, 1958). They are tabulated by Dwight (1961); it is his tables that have been used in the profiles calculated in this chapter.

The quantity $R(\omega/\nu)^{1/2}$ is a non-dimensional parameter that characterizes kinematic similarities in the liquid motion. It is written as the symbol α :

$$\alpha = R(\omega/\nu)^{1/2}. \quad 2.43$$

The radius is also made non-dimensional by substituting the fractional radius, $y = r/R$. The solution for the velocity

w then converts eqn 2.42 to

$$w = \frac{A^* R^2}{i\mu\alpha^2} \left\{ 1 - \frac{J_0(\alpha y i^{3/2})}{J_0(\alpha i^{3/2})} \right\} e^{i\omega t}. \quad 2.44$$

This gives the velocity of motion of the lamina of liquid at a fraction of the radius, y , from the axis of the tube. The corresponding volume flow is derived by integrating across the tube as was done for Poiseuille flow; the oscillatory volume flow is considered in Chapter 7. Although the terminology and derivation of Womersley (Womersley, 1955a; Womersley, 1957a) is followed here, the same, or essentially similar, solutions have previously been achieved by other authors, notably Witzig (1914), Apéria (1940), Iberall (1950), Lambossy (1952a), and Morgan and Kiely (1954). Witzig's work was in a doctoral thesis and remained unknown until cited by Lambossy (1952a); it is remarkable in that he actually obtained velocity profiles (at values of α of 4.0 and 10), like those illustrated in this chapter, in reference to arterial flow. Lambossy also calculated profiles over a range of values of α relevant to the circulation, while Morgan and Ferrante (1955) were more concerned with the limiting cases; Apéria (1940) and Iberall (1950) only dealt with the general mathematical solution.

The numerical solution of eqn 2.44 is concerned with complex quantities and may be made by separation into real and imaginary parts (the real and imaginary parts of a Bessel function of the first kind are tabulated as Ber and Bei functions) – as did Witzig (1914) and later Lambossy (1952a). Alternatively, they may be expressed as modulus and phase, which Womersley regarded as more tractable. This involves the substitutions

$$\begin{aligned} J_0(\alpha y^{3/2}) &= M_0(y) e^{i\omega_0(y)} \\ J_0(\alpha i^{3/2}) &= M_0 e^{i\omega_0}. \end{aligned} \quad 2.45$$

If the real part of the pressure gradient, $A^* e^{i\omega t}$ is also written in modulus and phase form as $M \cos(\omega t - \phi)$, then

$$\begin{aligned} w &= \frac{M}{\omega\rho} \sin(\omega t - \phi) \\ &\quad - \frac{M_0(y)}{M_0} \sin(\omega t - \phi - \delta_0), \end{aligned} \quad 2.46$$

where $\delta_0 = \theta_0 - \theta_0(y)$.

It can be seen that, for the lamina at the wall, $r = R$ and $y = 1$, so that $M_0(y) = M_0$ and $\delta_0 = 0$, the expression in the square brackets is thus zero (i.e. the flow velocity at the wall is zero). This is the same boundary condition that was discussed in relation to Poiseuille's equation.

For the purposes of calculation, Womersley compressed the equation further by writing

$$h_0 = \frac{M_0(y)}{M_0} \quad 2.47$$

and introduced the terms M'_0 and ε_0 by the following definitions:

$$M'_0 = (1 + h_0^2 - 2h_0 \cos \delta_0)^{1/2} \quad 2.48$$

and

$$\tan \varepsilon_0 = \frac{h_0 \sin \delta_0}{1 - h_0 \cos \delta_0}, \quad 2.49$$

so that

$$w = \frac{M}{\omega\rho} M'_0 \sin(\omega t - \phi + \varepsilon_0); \quad 2.50$$

or, since $\alpha^2 = R^2(\omega\rho/\mu)$ (eqn 2.43), eqn 2.50 may be written as

$$w = \frac{MR^2}{\mu} \frac{M'_0}{\alpha^2} \sin(\omega t - \phi + \varepsilon_0). \quad 2.51$$

In this form, it can be seen that for steady flow (eqn 2.6) the term M'_0/α^2 is analogous to $(1 - y^2)/4$, where $y^2 = r^2/R^2$.

(The zero subscript in M'_0/α^2 indicates that only Bessel functions of zero order are involved and is modified when functions of a higher order are involved – see Chapter 7.)

The forms of the velocity profile created by a pressure which oscillates sinusoidally were discussed in some detail by Hale *et al.* (1955). Some examples calculated from eqn 2.51 are shown in Fig. 2.19. The pressure gradient was assumed to be of the form $M \cos(\omega t - \phi)$, the standard expression for one harmonic component of a Fourier series (see Chapter 7). In this example, $M = 1$ and $\phi = 0$, so that the gradient is $\cos \omega t$. The angular frequency ωt is respectively in the ratio 1, 2, 3, and 4, which gives a values in the ratios of the square roots (i.e. $\sqrt{1}$, $\sqrt{2}$, $\sqrt{3}$, and 2). The actual α values used were those taken from an experiment on the femoral artery of a dog in which the pulse rate was 2.8 Hz. Only half a cycle is illustrated (at intervals of 15°) because with simple harmonic motion the second half is the inverted form of the first half – as shown by comparing the profiles at 0° and 180° .

It will be seen from the illustrations that, even at the lowest frequency shown, a true parabolic profile is not formed at any time. There is a phase lag between the applied pressure and the movement of the liquid; being a cosine function, the amplitude of the gradient is a maximum at 0° , while the maximum for the total flow integrated across the tube is at about 60° in example A, and at about 77° in example D. The laminae that move first are those nearest the wall, and flow successively involves the laminae towards the axis of the tube. At the wall is a lamina of zero velocity; the laminae near the wall always have a low velocity owing to the effect of viscosity. Hence they have a low momentum and reverse easily when the gradient reverses. As we move towards the axis of the tube, the momentum becomes progressively higher relative to

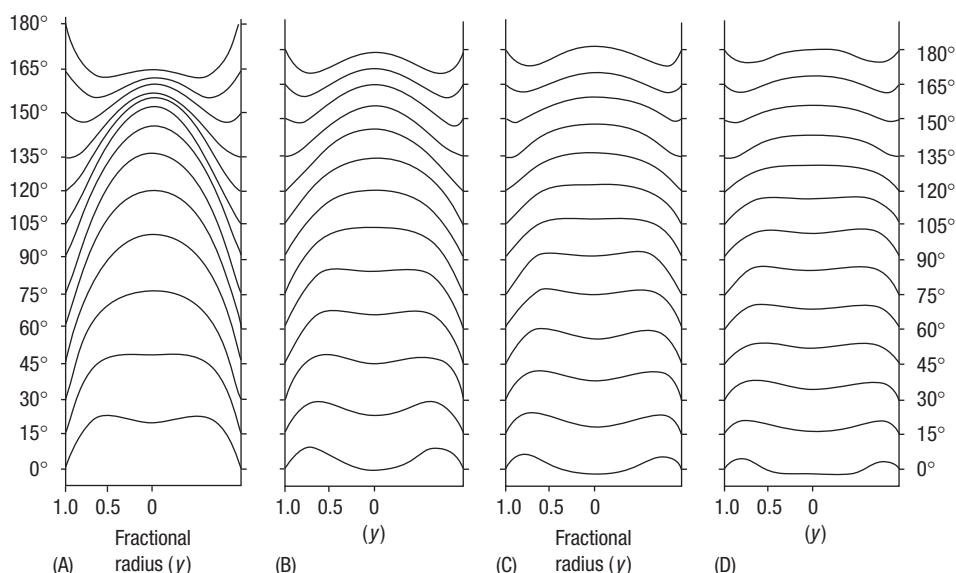


Figure 2.19 (A) The velocity profiles, at intervals of 15° , of the flow resulting from a sinusoidal pressure gradient ($\cos \omega t$) in a tube. In this case, $\alpha = 3.34$, corresponding to the fundamental harmonic of the flow curve illustrated in Fig. 2.21. Note that reversal of flow starts in the laminae near the wall. As this is harmonic motion, only half a cycle is illustrated as the remainder will be the same in form but opposite sign, for example, compare 180° and 0° . (B) A similar set of profiles for harmonic motion of double the frequency of A ($\alpha = 4.72$). The amplitude and phase of the pressure are the same here (and in C and D) as in A. The effects of the larger α are thus seen to be a flattening of the profile of the central region, a reduction of amplitude of the flow and the rate of reversal of flow increases close to the wall. (C) The third harmonic with $\alpha = 5.78$. The effects of the higher frequency noted in B are here further accentuated. (D) The fourth harmonic ($\alpha = 6.67$) again shows the same effect. The rapidly varying part of the flow lies between $y = 0.8$ and $y = 1.0$, and the central mass of the fluid reciprocates almost like a solid core

the viscous drag, so that there is a greater lag between the pressure gradient and the movement of the liquid in the center of the tube. As the frequency increases, α increases and the velocity profile becomes very flattened. An increase of diameter without a change of frequency will also cause an increase in α and produce a similar alteration in the profile. The liquid in the central portion of the tube is virtually unsheared, and the significant velocity gradients are only found in the layers near the wall. Thus, the liquid begins to behave rather like a solid mass sliding inside a thin layer of viscous liquid surrounding it. In order to visualize the velocity profiles in an artery, it is necessary to sum the profiles of the main harmonic components, with their appropriate amplitudes and phases, together with a parabolic profile representing the steady-flow component (i.e. the mean forward flow). Such a summation has been carried out in Fig. 2.20 for a typical cycle for the femoral artery from the four harmonics illustrated in Fig. 2.19. It can be seen that in the fast systolic rush, all the harmonics are most nearly in phase and create a profile that approaches the form of a parabola. Then, reversal of flow following reversal of the pressure gradient begins in the peripheral laminae and progressively involves those towards the axis. The oscillations of certain selected laminae throughout the cycle in relation to the pressure gradient are shown in Fig. 2.21. Thus, at 135° of the cycle, the average velocity in the artery is approximately zero; the axial laminae are still

flowing forward when the mural laminae are flowing backward. During back flow, the harmonics are considerably out of phase with one another, and the profile, in contrast to that in forward flow, is very much flattened. Indeed, the maximum retrograde velocity does not occur at the axis but in the laminae with a fractional radius of between 0.3 and 0.4. From Fig. 2.20, it can be seen that, even in a vessel as small as the femoral artery, the point of flow reversal in the most peripheral lamina ($y = 0.95$) is about 25° later than pressure gradient reversal; in the axial lamina ($y = 0.0$), reversal occurs about 40° later still.

Finally, the viscous behavior of blood due to the fact that it is a suspension of cells in plasma was discussed earlier. As the laminae very close to the wall have a lower concentration of cells, the viscosity in this region is lower so that the shear is greater than it would be if the viscosity were uniform throughout. The effect of this on the profile is to make it steeper at the wall than it would otherwise be.

INLET LENGTH

In the consideration of the basic assumptions of the Poiseuille equation, it was pointed out that, in a tube leading from a reservoir, the pressure-flow relation was different in the proximal part of the tube. The region between the reservoir and the point where a constant flow regime is 'established' is called the 'inlet' (or sometimes

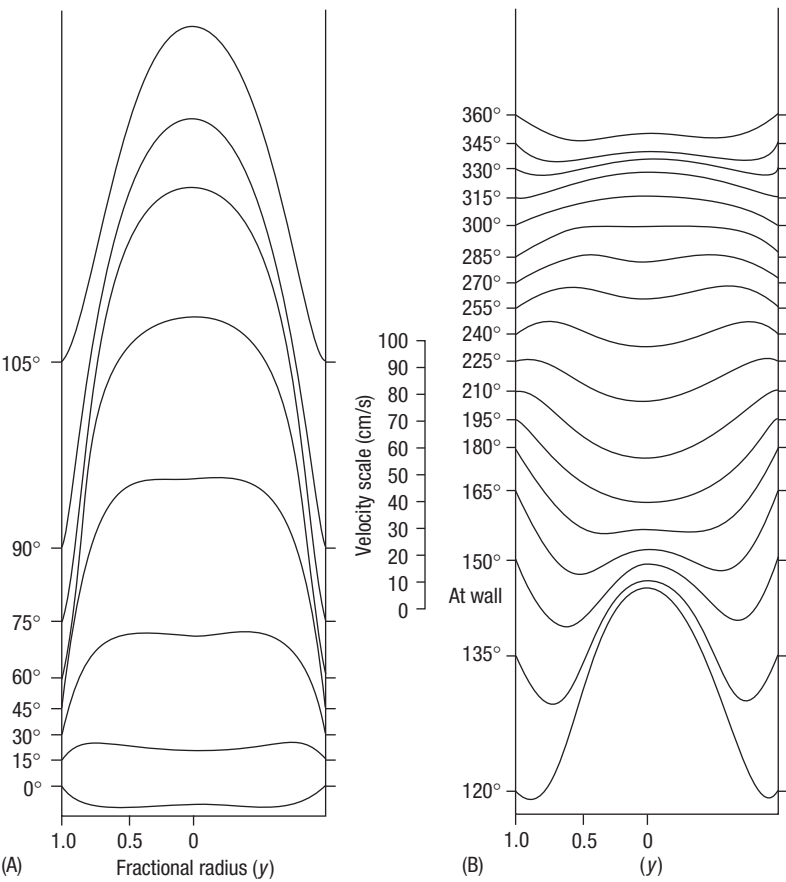


Figure 2.20 Velocity profiles calculated from the measured pressure gradient in the femoral artery of the dog. The first four harmonic components with the same values of α as in Fig. 2.19 are summed together with a parabola (axial velocity 30 cm/s) representing the steady forward flow. The maximum forward velocity occurs in the axis because here the harmonic components are in phase, but the maximum backward velocity lies between $y=0.3$ and 0.4 at 180° . The reversal of flow beginning near the wall is clearly seen. The variation in velocity of certain of these laminae is also plotted in Fig. 2.21 in a similar way to that of ordinary average velocity flow curves. Average velocity is used for the average of all laminae across the tube at any moment; mean velocity is reserved here for the mean value throughout the cycle, i.e. the steady, or constant, term of the Fourier series. The pulse frequency in this case was 2.8 per second, so that the cycle length was approximately 360° . One degree of arc is thus approximately 1.0 ms

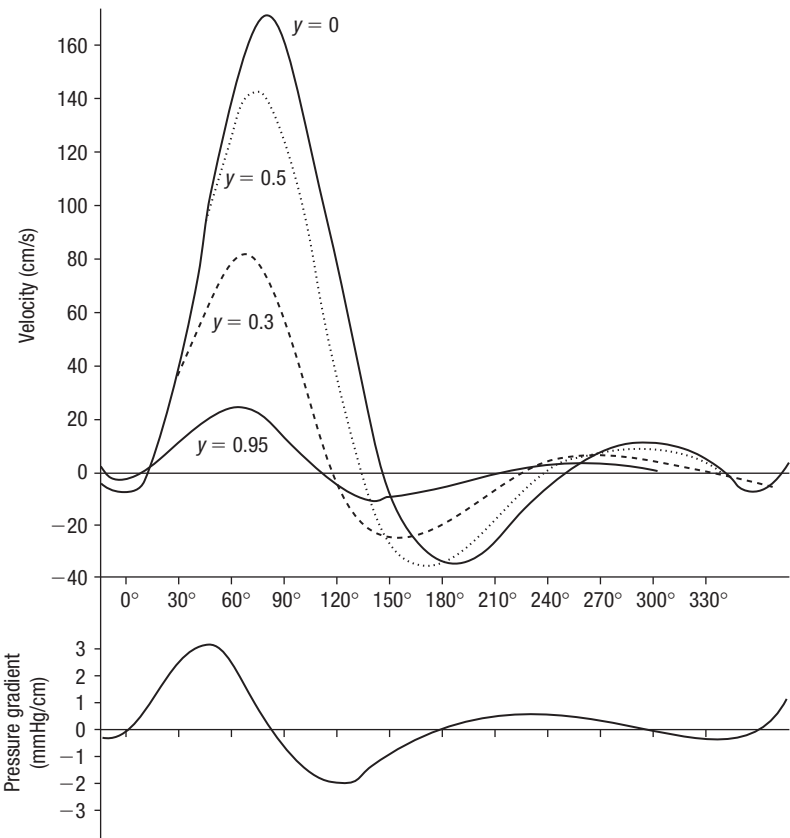


Figure 2.21 The velocity of pulsatile flow in the dog femoral artery displayed in terms of individual laminae in the stream corresponding to the profiles shown in Fig. 2.20. The position of a lamina is defined by $y=r/R$. Thus, $y=0$ is in the axis, $y=0.5$ is midway between the axis and the wall, and $y=0.95$ is close to the wall. The measured pressure gradient from which the curves were computed is displayed below. It can be seen that the flow of liquid near the wall follows the pressure gradient most closely and that the phase lag increases to a maximum at the axis. The peak mean forward velocity was 105 cm/s at 75° and the peak mean backward velocity was 25 cm/s at 165°

'entrance') length. When considering the circulation, there is a clear analogy with the outflow into the aortic tube from the ventricular reservoir. Strictly speaking, a similar situation occurs when flow is from a larger artery into a smaller branch. The conditions within this region near the entrance may be visualized in this way. At the orifice of the reservoir, there is initially the same velocity all the way across. Once inside the tube, the layer of liquid immediately in contact with the wall will become stationary. The laminae close to the tube wall begin to slide on it subject to the force of viscosity and form a boundary layer. The bulk of the liquid in the center of the tube will move as a mass, affected very little by the forces of viscosity, and will have a flat velocity profile. As flow proceeds down the tube, however, the boundary layer will grow in thickness as the viscous drag involves more and more of the liquid. Finally, the boundary layer comes to occupy the whole of the tube and steady viscous flow is established. The velocity profile is then parabolic. The stages of development of the profile are shown in Fig. 2.22. If the Reynolds number is above the critical value, the established flow will be turbulent, but up to that point (i.e. within the inlet) one cannot, strictly speaking, define the flow as either laminar or turbulent (Han *et al.*, 2001).

This conceptual description of the manner in which flow becomes established in the inlet region is essentially the same as that for determining the form of the profile in oscillatory flow. The differences only arise from the fact that the standard concept in calculating inlet length is an initially flat profile across the tube at the reservoir with only forward flow, whereas with continuously oscillatory flow the profile is never completely flat. From this comparison, we think of the oscillatory flow profile as though it were that of a steady flow that has not had time to become completely established. We should, therefore, expect the inlet length for oscillatory flow to be shorter than for steady flow; indeed, the higher the value of α , the shorter one would expect the oscillatory inlet length to be.

The effect of inlet length conditions will be to cause a flattening of the central region of the velocity profile. This may be, in practical applications, of less importance than the change in pressure-flow relations. The pressure gradient in the inlet length is higher for a given rate of flow than when the same flow is established.

Formulae for the calculation of the inlet length for steady flow are given in all standard hydrodynamic texts. They are all of the form, for an inlet length L , when the established flow is laminar,

$$L = \frac{2k\rho\bar{V}r^2}{\mu} = \frac{2k\bar{V}r^2}{\nu}, \quad 2.52$$

where r is radius, $\bar{V}$ is mean velocity, $\nu = \mu/\rho$ is the kinematic viscosity, and k is a derived constant. It is, therefore, somewhat confusing to find a range of values for k in various sources. Goldstein (1938) gives $k = 0.057$. Prandtl and Tietjens (1934), however, say that this is a lowest limit, which is too short, and recommended an experimentally determined value of $k = 0.01$; the highest value quoted by Chang and Atabek (1961) is $k = 0.13$. These last authors use $k = 0.08$, and as they are the only authorities to have determined it for oscillatory flow, their value is used here.

The inlet length for the steady flow component in arteries is therefore taken as

$$L = 4.2 \bar{V} r^2, \quad 2.53$$

since $\nu = 0.038 \text{ cm}^2/\text{s}$.

Some numerical examples will indicate the inlet lengths in the arterial tree, as applied to the steady component of flow.

1. **Dog aorta.** Radius 0.75 cm; mean flow velocity 20 cm/s. The inlet length is 47 cm; this is approximately the length of the aorta.
2. **Human aorta.** Radius 1.4 cm; mean flow velocity 16 cm/s. The inlet length is 132 cm. This greatly exceeds the lengths to which the dimension used would apply but shows that inlet length conditions should be considered as being operative throughout all major arteries in humans.

If the flow of the liquid in the reservoir is turbulent, the inlet length is much shorter. The formula given by Goldstein (1938) is

$$L = 3.74r(r\bar{V})^{1/4}. \quad 2.54$$

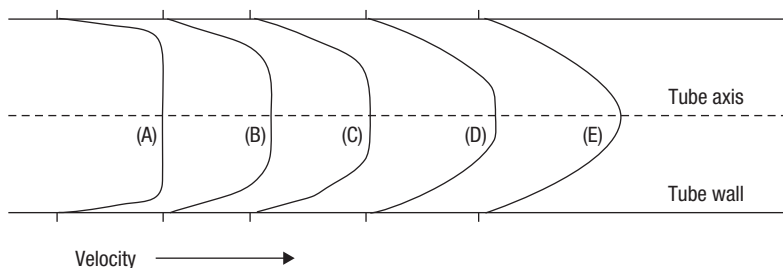


Figure 2.22 Successive velocity profiles within the 'inlet length' between two parallel walls leading from a reservoir – it is assumed that the pattern in a cylindrical tube would be similar. (A) Near the inlet, the shear in the liquid is confined to a boundary layer near the wall, and there is no velocity gradient in the center of the channel. (B), (C) and (D) The boundary layer increases in thickness progressively, with a corresponding diminution of the unsheared region. (E) Finally, the parabolic profile of steady laminar flow is established

Calculating the two examples above with this formula, we obtain: (1) dog aorta, $L = 3.3$ cm, or shorter than the ascending aorta; (2) human aorta, $L = 11.4$ cm, or within the arch of the aorta.

Flow out of the heart is predominantly laminar and the blood in the heart, although perhaps well stirred, is not really turbulent. The inlet lengths calculated from eqn 2.54 are, therefore, manifestly too short. On the other hand, the heart is not the very large, still reservoir of hydrodynamic experiments, so that we can equally argue that the lengths calculated from eqn 2.54 are unrealistically long. From the flow patterns recorded near the valves, it appears that the outflow tract of the ventricles should be considered as part of the inlet.

When the entrance velocity is oscillatory or pulsatile, the inlet length will also fluctuate. This phenomenon has been studied by Chang and Atabek (1961) and Atabek *et al.* (1964), who found that the ratio of 'pulsatile' inlet length to steady inlet length varied by 36.5 percent at a value of $\alpha = 4.0$, appropriate for the iliac artery of a dog. The variability is even greater in larger arteries with higher values of α and blood flow velocity. Florio and Mueller (1968) extended the work of Atabek *et al.* (1964) and also showed that the inlet phenomenon can be regarded as the superposition of a developing steady flow and a fully developed oscillatory flow. Using the values given above for the dog aorta, the velocity profile would be expected to show more or less complete development to a parabolic profile at lengths of about 47 cm using steady flow criterion and over a range of 30–70 cm under oscillatory conditions. Deviation from a flat velocity profile should be expected before this, and it is not unreasonable that in

the descending aorta of dogs we do in fact find the velocity profile deviating from its shape at the inlet (Fig. 2.23). Indeed, studies in the rabbit mesentery by Tangelder *et al.* (1986) show truly parabolic velocity profiles in the arterioles (Fig. 2.24).

Experimental measurements of the velocity profile

The technique used for recording the velocity profile by Atabek *et al.* (1964) in their models involved the pulsed electrolytic release of a line of minute bubbles from a thin wire run across a diameter. The successive form taken up by this line marker could then be photographed. This technique clearly cannot be used in the living animal. The same limitation applied to the method of recording the rates of shear in bentonite solutions used by Attinger *et al.* (1966b). They recorded profiles that were skewed, but this was almost certainly due to bends in the tube causing secondary flows – an effect difficult to avoid. Müller (1954) used a Pitot tube that could be traversed across the lumen of a tube; with this device he verified the form of the profile for $\alpha = 6.43$ calculated by Lambossy (Lambossy, 1952a; Lambossy, 1952b). This instrument was reported by von Deschanden *et al.* (1956) to have recorded a flat profile in the thoracic aorta. Freis and Heath (1964) traversed thermistors across the ascending aorta and reported a flat velocity front, but their results were not plotted as profiles.

The problem with any instrument introduced into the stream is to avoid distortion of the natural profile.

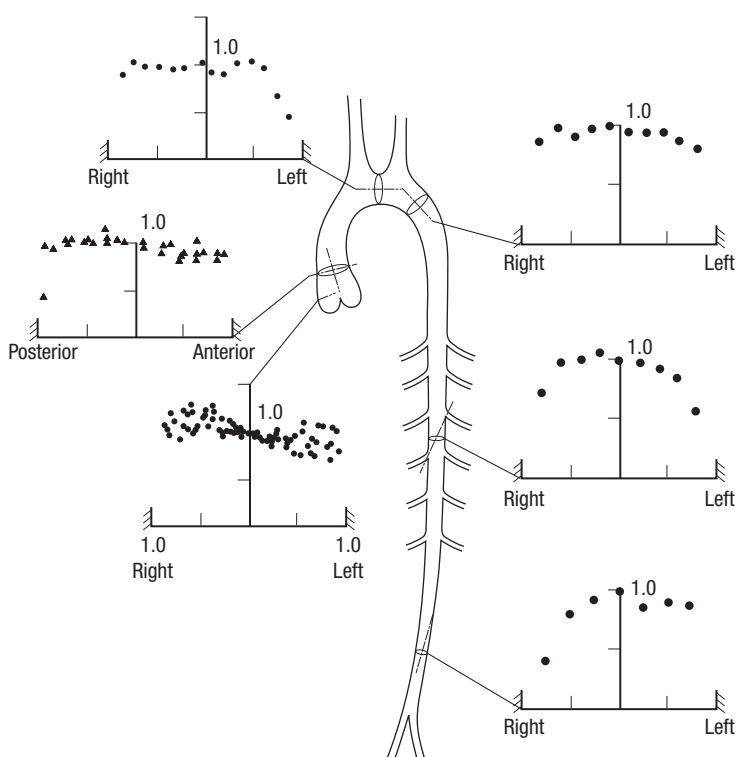


Figure 2.23 Velocity profiles in the aorta of the dog. For each graph, the mean velocity at each site on the vessel diameter is expressed relative to the centerline mean velocity. Reproduced from Schultz *et al.* (1969)

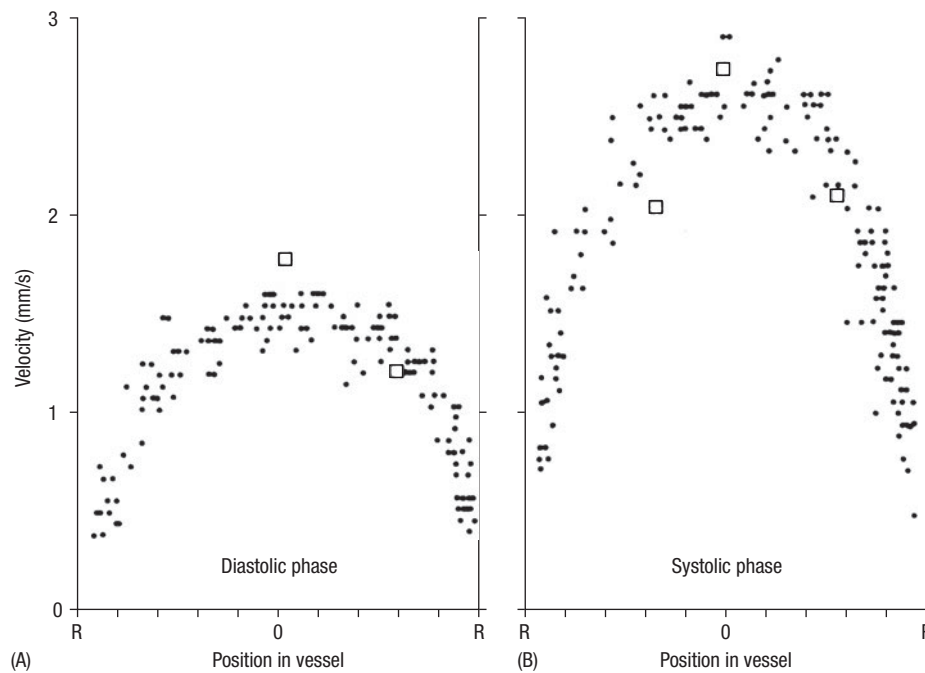


Figure 2.24 Velocity profiles of blood platelets in an arteriole (diameter $25\text{ }\mu\text{m}$) of the mesentery of a rabbit at (A) low (diastolic phase) and (B) high (systolic phase) velocity. Each dot represents the velocity of one platelet. An open square indicates the velocity of a leukocyte. Fluorescent dye was used for labeling and visualization was accomplished by intravital fluorescence video microscopy, using flashed illumination. Reproduced from Tangelder *et al.* (1986)

The fine probe used by Schultz *et al.* (1969) in their hot-film anemometer has provided a detailed account of the profiles in both dogs and humans. Reuben *et al.* (1970), Seed and Wood (1970), Tunstall-Pedoe (1970), Falsetti *et al.* (1972), and Paulsen and Hasenkam (1983) used a similar technique and recorded similar profiles in the thoracic aorta of dogs and the main pulmonary artery of dogs and humans. Figure 2.23 shows velocity profiles at various sites recorded in the aorta of the dog. While the profile is essentially flat across the vessel where measurements could be made, a profile in the thoracic aorta shows a central convexity that bears more similarity to that anticipated for established flow. Thus, although the calculations made above indicate that a fully developed flow pattern will take a longer distance to become established, the evidence of these profiles suggests that the tapering conformation of these arteries will create shorter inlet length conditions than might be expected. In addition, the analysis of the profile in the inlet length shows that a flat profile in the proximal aorta is consistent with an essentially laminar flow that the records of Schultz *et al.* (1969) also show to exist commonly. It is not, therefore, to be taken as evidence of turbulence even though turbulent flow would also have a flat profile. Although there are geometric differences in humans when compared with the dog, velocity profiles in the larger arteries are similar. This holds true for both the aorta (Fig. 2.25) and pulmonary artery (Fig. 2.26). Fluid mechanics, including velocity profiles, of large arteries and at bifurcations are discussed

in detail by Caro *et al.* (1978), Giddens *et al.* (1985), Henry and Lucas (1994), Yamamoto *et al.* (1994), and Kornet *et al.* (1999).

Sets of velocity profiles have been recorded in the aorta by Farthing and Peronneau (1979) in dogs and Hagl *et al.* (1978) in humans using the pulsed Doppler velocimeter described by Peronneau and Leger (1969) and Peronneau *et al.* (Peronneau *et al.*, 1970; Peronneau *et al.*, 1977). With this technique, the profile can be measured in a matter of milliseconds, and it can thus make a virtually continuous traverse, which gives it an advantage over the heated-film technique where only certain spots in the traverse can be made. Tortoli (1996) has developed a Doppler ultrasound multigate system capable of providing about 50 full profiles per second. The pulsed Doppler can also measure velocity right up to the wall. The velocity profiles obtained with the pulsed Doppler technique are very similar to those obtained with the heated film anemometer. Other recent methods used to measure velocity profiles are intravascular (Henry and Lucas, 1994) and extravascular ultrasound (Segadal and Matre 1987; Yamamoto *et al.*, 1994; Samijo *et al.*, 1997; Sloth *et al.*, 1997; Kornet *et al.*, 1998; Kornet *et al.*, 1999; Kornet *et al.*, 2000; Thury *et al.*, 2002; Tortoli *et al.*, 2002; Dammers *et al.*, 2003; Tortoli *et al.*, 2003), laser Doppler (Bharadvaj *et al.*, 1982a), and magnetic resonance imaging (Fogel *et al.*, 1997; Botnar *et al.*, 2000; Han *et al.*, 2001; Glor *et al.*, 2002; Ford *et al.*, 2008; Sui *et al.*, 2008). The high-frequency pulsed Doppler velocimeter used by Yamamoto

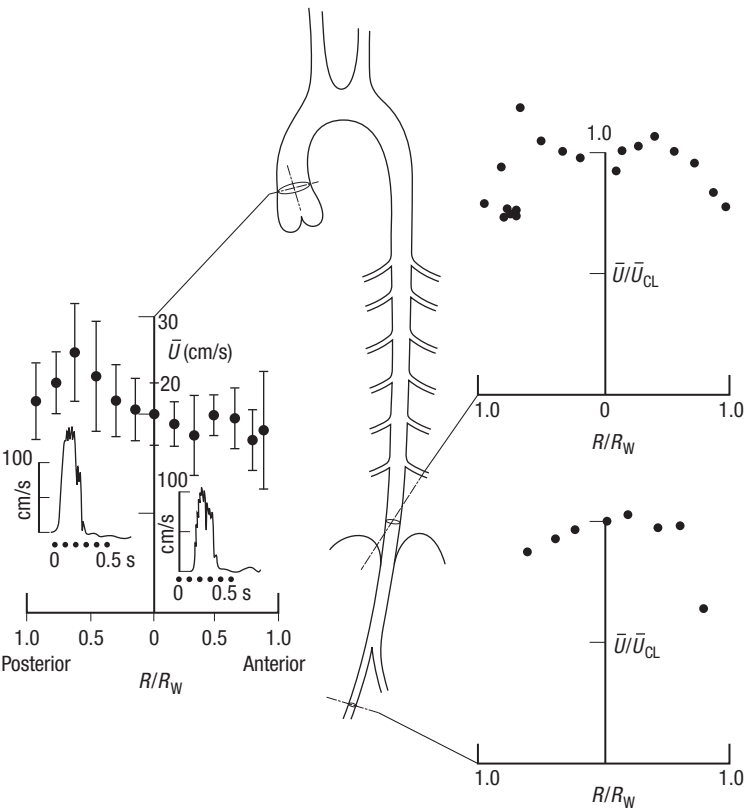


Figure 2.25 Velocity distribution in the ascending aorta, lower thoracic aorta, and femoral artery of humans. Data obtained from three patients. Reproduced from Bergel (1972)

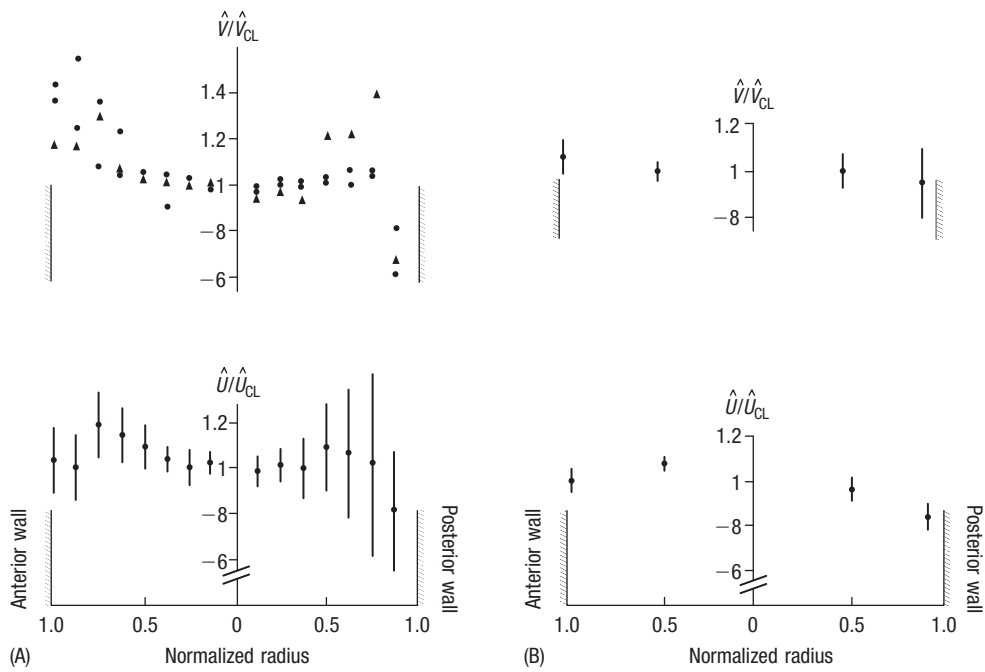


Figure 2.26 Velocity profiles across the transverse axis of the main pulmonary artery of (left) dogs and (right) humans obtained with a thin-film anemometer. Reproduced from Reuben *et al.* (1970)

et al. (1994) was developed at the Kawasaki Medical School in collaboration with Fujitsu Laboratory Co. and has been described in detail by Ogasawara (1984) and Kajiya *et al.* (1986). This velocimeter measures the

Doppler shift from 80 channels in real time with sample volumes of about $0.8 \text{ mm}^2 \times 0.2 \text{ mm}$ (radius, 0.5 mm; thickness, 0.2 mm). The high-pass filter has a cut-off frequency of 375 kHz. The shear rate estimating system

(SRES) developed by Reneman's group in Maastricht (Samijo *et al.*, 1997) has been used to measure shear rate, shear stress, and velocity profiles in the common carotid (Kornet *et al.*, 1998), brachial (Dammers *et al.*, 2003), and femoral arteries (Kornet *et al.*, 1999; Kornet *et al.*, 2000) of humans. Results from these studies indicate that mean shear stress is higher in the elastic carotid artery than in the muscular brachial artery. Furthermore, the velocity profile is flatter in the carotid than the femoral artery and both are influenced by wave reflections originating from the periphery and the flow divider. The multigate Doppler processing system developed by Tortoli *et al.* (Tortoli, 1996; Tortoli *et al.*, 2003) can be used in conjunction with an esophageal probe to obtain velocity profiles in the human aorta (Tortoli *et al.*, 2002). In this study, the velocity profiles in the upper descending thoracic aorta were found to be almost flat throughout the entire cardiac cycle (Fig. 2.27), while those in the ascending aorta are skewed toward the outer wall (see Fig. 2.25 and Segadal and Matre (1987)). Fogel *et al.* (1997), using MRI, found that surgically reconstructed aortas display disturbed flow, and the velocity profiles across the ascending aortic diameter are more varied than those in aortas without reconstruction and they are skewed anteriorly. Ford *et al.* (2008) and Sui *et al.* (2008) used MRI to measure velocity profiles and wall shear stress in the common carotid artery. When averaged over the cardiac cycle, roughly 60 percent of the velocity profiles were classified as skewed and mean wall shear stress was 0.85 N/m^2 .

The paper by Ling *et al.* (1968) is of particular interest in that they attempted to measure the actual shear stress at the wall of the thoracic aorta. In a group of very interesting papers, Fry (Fry, 1968; Fry, 1969a; Fry, 1969b) studied the acute yield stress of the vascular endothelium subjected to the shearing stress of blood flow in the aorta. In order to increase the stress of a localized region of the endothelium, he introduced a plastic plug with a restricted channel machined on one side into the thoracic aorta of a dog. The linear velocity of flow would thus be greatly increased on one side and the adjacent endothelium consequently subjected to an increased shear. Damage to the endothelial cells was assessed histologically from cell configuration and also by the abnormal uptake of dyes. The shear stress was calculated from the relevant

hydrodynamic equations. It was estimated that the acute yield stress was 379 dyne/cm^2 .

While estimates of the shear stress with normal flows can be calculated, the heated-film probe creates the possibility of a more direct measurement. The range of peak values that Ling *et al.* (1968) obtained was $80\text{--}160 \text{ dyne/cm}^2$ (assuming a blood viscosity of 0.04 poise). From this, it would appear that the critical level for normal endothelium is not very much above the shear stresses that are normally attained. The measured normal shear stresses must be regarded as tentative because the heated probe should be precisely flush with the internal surface – a condition very difficult to achieve.

Nevertheless, this work provides the first definite relationship between hydrodynamic stresses in the arterial wall and the genesis of arterial disease. (Fry also found that where an intra-arterial catheter made contact with the wall, abnormal staining reactions appeared in the endothelial cells.) Even an increase in the permeability of the endothelial cells, without gross structural damage, might well facilitate the accommodation of lipids in the arterial wall, which is the dominant feature of atherosclerosis. It has long been suspected that mechanical and hydrodynamic factors (notably turbulence) are concerned in the genesis of atherosclerosis, but this has hitherto been largely speculative (Davies *et al.*, 1986). Duguid and Robertson (1957) discussed the role of radial stresses in the arterial wall, and the whole subject is considered more fully by Mitchell and Schwartz (1965), Stehbens (1975), Patel and Vaishnav (1980), Stone *et al.* (2003), Reneman *et al.* (2006), and Katritsis *et al.* (2007). The implication is, from the foregoing, that atheromatous lesions should occur predominantly in regions of high shear stress; however, there are regions of predicted high shear stress that are spared atherosclerosis, and more recent experiments suggest that the occurrence of low shear stresses in regions of laminar flow separation correlates with lesion development (Caro *et al.*, 1971; Zarins *et al.*, 1983; Ku *et al.*, 1985a; Ku *et al.*, 1985b; Caro, 1994; Stone *et al.*, 2003). Areas of low shear stress and secondary flows occur at arterial branches and bifurcations (Sabbah *et al.*, 1984) where the velocity profile is skewed toward the flow divider (Fig. 2.28). A normal sinus bulb, as shown in the figure, is expected to have flow separation, secondary flow patterns, low mean

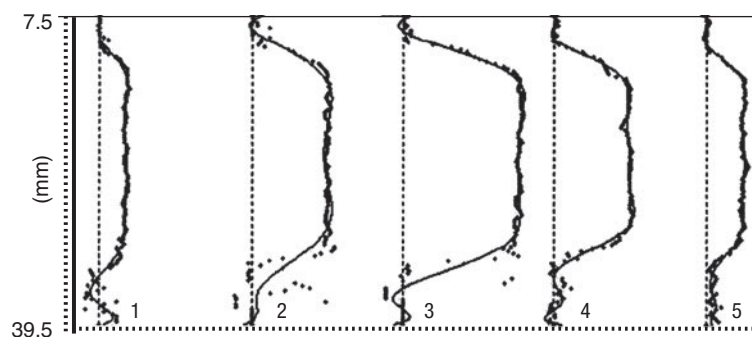


Figure 2.27 Velocity profiles at various times in systole detected at a location 7–10 cm down from the aortic arch in a patient with no cardiovascular disease. From Tortoli *et al.* (2002)

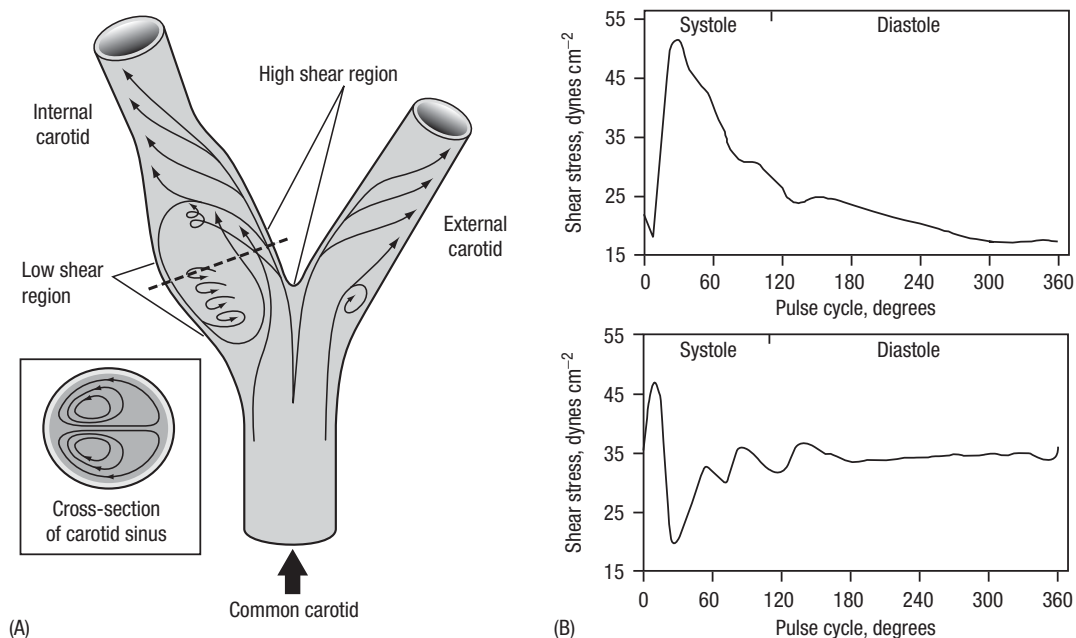


Figure 2.28 Representation of flow patterns in a detailed model of a human carotid artery (A), and the associated patterns of blood flow and endothelial shear stress (B). Areas of normal flow with physiologic, vasculoprotective shear stress are seen at the inner aspect of the bifurcation of the artery, with the measured shear stress throughout a cardiac cycle shown in the top panel of B. Areas of low, reversed, and disturbed flow are seen at the waist of the bifurcation and the measured shear stress throughout the cardiac cycle are shown in the bottom panel of B. Reproduced from Ku *et al.* (1985a); Ku *et al.* (1985b)

wall shear stresses along the outer sinus wall and oscillating wall shear stress along this wall. Morphologic evidence demonstrates that the outer sinus wall is a strongly preferred site for early atheromatous lesions (Zarins *et al.*, 1983; Giddens *et al.*, 1985a; Ku *et al.*, 1985a; Ku *et al.*, 1985b; Giddens *et al.*, 1993).

Wall shear stress influences vascular biology in many ways, including affecting the dimensions of blood vessels and the morphology, mechanics, and metabolism of the endothelial cells (Furchgott, 1983; Pohl *et al.*, 1986; Lansman, 1988; Liepsch, 1990; Koller and Huang, 1999) and endothelial gene expression (Dekker *et al.*, 2006). The mechanism whereby chronic wall shear stress influences the occurrence of atherosclerosis has not been established. However, it has been suggested that it involves regulation of the release by the endothelium of NO and other vasoactive mediators (Henderson, 1991).

Nitric oxide is produced by the endothelium, and its release is influenced by many factors, including hydrodynamic wall shear stress (see Chapter 5) (Melkumyants and Balashov, 1990; Henderson, 1991). It has a short half-life in the circulation and is thought, therefore, to exert its effects mainly in the immediate vicinity of its site of release. It is a relaxant of vascular smooth muscle, and the predilection of atherosclerosis for low shear regions could result from a low release of the substance in these regions, with a consequent elevation of local arterial wall smooth muscle tone (Henderson, 1991). Wall shear stress, directly measured from the blood flow velocity profile, is a determinant of local vasodilation, by contrast to central

velocity. Although wall shear stress acutely regulates smooth muscle tone and diameter, it does not acutely regulate local arterial stiffness (Boutouyrie *et al.*, 2004a).

Blood vessel walls are poroelastic, and their porosity is subject to modification by physiologic mechanical stresses, including those generated by wall smooth muscle tone (Caro and Lever, 1983). A mechanism proposed to account for the preferred occurrence of atherosclerosis in low shear regions in arteries involves a local reduction of medial porosity and permeability, resulting from a low local release of NO, and an increased tendency for macromolecules, including LDL, to accumulate locally within the intima. These conditions could also favor the local oxidation of LDL and thus further increase the likelihood of the development of atherosclerosis (Henderson, 1991; Libby, 2002).

The modulation of NO release by wall shear stress could also influence the development of atherosclerosis via another mechanism. Nitric oxide inhibits both platelet aggregation and adhesion, and some theories for atherogenesis implicate the intravascular aggregation of platelets and their adhesion to the vessel wall (Ross, 1986). Therefore, in this scheme, too, the lesions would develop preferentially in regions that experience low wall shear stress (see Chapter 21).

NON-LAMINAR LIQUID FLOW

It was observed both by Hagen and by Poiseuille that the law relating steady flow and the pressure gradient in a

tube ceased to be kept when there was a high-velocity rate of flow. This was rightly attributed to the breakdown of laminar flow and the onset of turbulent flow. They observed the appearance of turbulence, but it was not until the work of Osborne Reynolds (1842–1912) in 1883 and subsequently, that the conditions determining the transition from laminar to turbulent flow were described precisely (Rott, 1990). Reynolds injected a filament of dye into the axial stream of a cylindrical tube in which all irregularities at the inlet and of the wall had been eliminated. At low velocities of flow, the motion of the liquid, as shown by the dye, was smooth and regular. Any disturbance introduced into the liquid was soon damped out. At higher velocities of flow, however, the liquid became more sensitive to disturbances. At a critical point, a stage was reached when smooth flow could no longer be maintained and 'the motion of the liquid became wildly irregular and the tube appeared to be filled with interlacing and constantly varying streams, crossing and recrossing the pipe' (Lamb, 1932).

The critical point was found to be dependent on the diameter of the tube, the mean velocity of the flow, and the density and viscosity of the liquid. This was expressed as a dimensionless quantity known as the Reynolds number R_e , which (Schlichting, 1968; Reynolds, 1974), when applied to liquid flow in a circular tube, is

$$R_e = \frac{\bar{V} D \rho}{\mu} = 26 \bar{V} D \quad 2.55$$

where $\bar{V}$ is the mean velocity of flow (cm/s), D is the diameter of the tube (cm), ρ is the density (g/cm³) of the liquid, and μ its dynamic viscosity (dyne-s/cm²). The fraction ρ/μ is known as the kinematic viscosity, written as ν . The c.g.s. unit of kinematic viscosity is the stokes. Reynolds' formula is thus often expressed as

$$R_e = \frac{\bar{V} D}{\nu} = 16.56 \frac{(\text{volume flow})}{\text{radius}}. \quad 2.56$$

It will be seen not only that an increase in mean velocity causes turbulence, but also that at a given velocity of flow, turbulence will occur in large tubes before it will in small ones.

The fact that the characteristic property of the liquid concerned in determining the stability of laminar flow is the kinematic viscosity gives rise to some apparently surprising results. Thus, a gas-like air would appear to be more easily disturbed than blood, or even water. Its absolute viscosity is, of course, much less than that of water (at 20°C, the absolute viscosity of air is 0.0181 centipoise and water is 1.0 centipoise), but because the density of air is proportionately lower, its kinematic viscosity is actually some ten times as great as that of water at 20°C (the kinematic viscosity of air is 10.5×10^{-2} and water is 1.007×10^{-2} stokes). Furthermore, as the viscosity of gases increases with temperature

while that of liquids decreases, the difference is greater at body temperature. The kinematic viscosity of blood at body temperature is about 3.8×10^{-2} stokes at 38°C and that of water is 0.686×10^{-2} stokes, while that of air (at 40°C) is 16.9×10^{-2} stokes (Goldstein, 1938). Thus, at body temperature, a steady laminar stream of air in a bronchus, for example, would be stable at a velocity four times greater than that of blood in a blood vessel of comparable size. This also serves to emphasize that, in terms of hydrodynamics, both blood and water are thought of as liquids of low viscosity.

A certain amount of confusion arises in physiologic literature from the use of radius instead of diameter in this equation, especially as quoted values are often compared without reference to different ways of computing the values. The general formula for Reynolds number for a channel of any shape is

$$R_e = 4m\bar{V}/\nu \quad 2.57$$

where m is the 'mean hydraulic depth', which is defined as the cross-sectional area divided by the perimeter (Goldstein, 1938). Thus, for a circle

$$m = \text{radius}/2 \quad \text{and} \quad 4m = \text{diameter}.$$

Hence, eqn 2.57 is used throughout in the present work in order to conform to standard hydrodynamic practice, and the values of other investigators are converted, where necessary, to comparable values.

The critical value of the Reynolds number is usually stated as about 2000 (Schlichting, 1968). This, however, is an experimentally determined value and is highly dependent on the conditions of the experiment. Reynolds called it the 'lower critical value'. With very carefully controlled flow from an undisturbed reservoir into a tube, Reynolds himself was able to maintain laminar flow up to a value of 12 000 and much higher values have been attained.

The standard Reynolds number of 2000 is obtained when there are disturbed conditions in the reservoir. With such conditions, it is difficult to cause turbulence in a regular tube at values below 2000. Other disturbances are found, however, at lower values. The period of laminar flow has been described by Schiller (see Goldstein (1938)) as falling into three regimes. The first is the period of completely undisturbed flow with the streamline of liquid running parallel to the sides of the tube. These streams begin to exhibit a wave motion at a higher Reynolds number, which is described as the second regime, and these oscillations get larger and vortices start forming at the border and are carried down the tube. The third regime is that with vortex formation, and as the vortices become larger and more frequent, this merges into a condition of turbulence. The stage at which these conditions of disturbed flow, heralds of the approaching breakdown of the flow pattern appear, depends on the

degree of disturbance in the reservoir or on roughness at the entry of the tube.

The mere evaluation of the Reynolds number is not, of itself, a proof of the existence of laminar or turbulent flow. The best way to demonstrate turbulence is to measure the pressure gradient–flow relationship and demonstrate that this deviates from that of laminar flow. Whereas, with unvarying laminar flow, the pressure drop per unit length (pressure gradient) varies linearly with the rate of flow, when turbulence is present it varies approximately with a higher power of the rate of flow ($\Delta P \propto Q^2$). In a pressure gradient–flow diagram, this shows as a discontinuity (Fig. 2.29). In other words, a greater pressure gradient is required to maintain turbulent flow than is required for laminar flow. The conditions for turbulent flow have not been precisely determined because it is not possible to give a precise description of the motion of the liquid. The liquid particles, in addition to flowing along the line of the tube, pursue random pathways in other directions, and small eddies or vortices are formed. Thus, extra energy is required to maintain the increased movement of liquid that does not directly contribute to the flow; that is, an increased pressure gradient is required to maintain it. Whereas in laminar flow $Q \propto \Delta P$ when the flow becomes turbulent, the relation is $Q^n \propto \Delta P$ where n is greater than 1.0, and increases to 2.0 with increasing Reynolds number (Goldstein, 1938). Where measurement of the pressure gradient and flow is not practicable, the type of flow may be observed from the distribution of injected dye, as Reynolds did in his pioneer experiments.

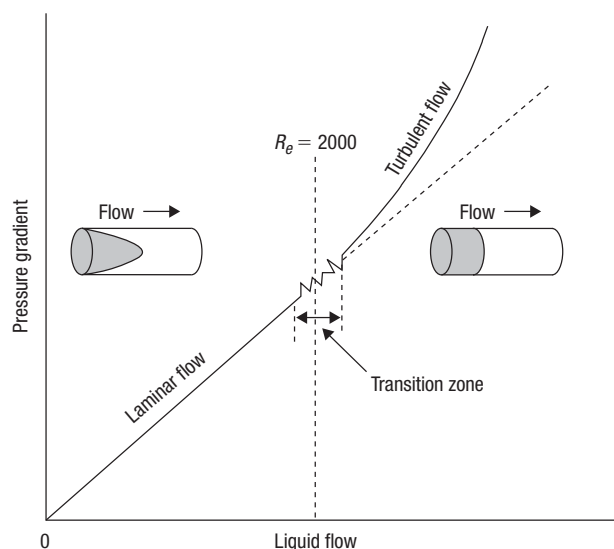


Figure 2.29 Variation of volume flow rate with the pressure gradient for steady flow in a long, straight circular tube to indicate the changing relationship as flow rate increases. In this example, below a Reynolds number (R_e) of 2000, flow is linearly related to the pressure gradient (i.e. laminar) and the velocity profile is parabolic; above an R_e of 2000, the flow is approximately equal to the square root of the pressure gradient, is turbulent, and exhibits a flat velocity profile

Whereas such dye will be distributed in a paraboloid velocity profile in steady laminar flow, in turbulent flow the dye is seen to be mixed across the stream and the velocity profile is much more nearly the same (i.e. flat) across the tube, although there will still be a region of shear near the wall (Fig. 2.29). The extent to which this region of shear extends and the width of the 'flat' front varies with the degree of turbulence. For although the distinction between laminar flow and turbulent flow is relatively abrupt, turbulence, in the sense of disorder of motion of the liquid, will continue to increase with increasing Reynolds number for some considerable time. This is shown by the changing relation of the pressure gradient and flow.

The transition from turbulence to laminar flow with conditions of decreasing Reynolds number is even more difficult to define than the onset of turbulence. In general, it may be said that, in a given tube, the restoration of laminar flow will occur at a lower Reynolds number than that at which the onset of turbulence occurs. As variation in the rate of flow is found in all blood vessels in which the Reynolds number is in the critical region, this point is of considerable interest. As stated above, disturbances generated in truly laminar flow will be damped out, and this is commonly used to define turbulence as the condition of steady flow where a disturbance, once started, will increase in size until it involves the entire body of the liquid. The consideration of turbulence in pulsating flow such as is found in arteries and the larger veins is, therefore, somewhat difficult. If a disturbance in the flow is short lasting, one cannot know whether its transient character is due to a regime of instability that is too brief to produce turbulence throughout the liquid, or whether the flow is inherently stable and hence has damped out the disturbance.

Helps and McDonald (1954) observed that vortex rings were formed in the inferior vena cava when flow reversed, but they were damped out when forward flow was re-established. They thought that this could not be considered as true turbulence. Their reasons were that the Reynolds number was well below the critical value throughout the cycle and, furthermore, that the disturbance occurred at a time when flow velocity was minimal. In addition, the vortex rings did not involve all the blood in the vein. Helps and McDonald suggested that some term such as 'disturbed flow' should be used for conditions such as this, which are neither clearly laminar nor clearly turbulent. On the other hand, McDonald (1952) using high-speed cinematography, showed in the rabbit aorta that, while laminar flow with a parabola-like profile could be seen in early systole, the dye pattern was disrupted completely during the peak systolic ejection and the flow appeared to be turbulent. A laminar flow pattern was re-established in diastole, although it was impossible to determine at what point it started, and streams could not be clearly seen until well into diastole when the flow velocity was very low. In this case, it may be more

justifiable to call the flow 'turbulent' because the disturbance spread right across the artery and was initiated at a time of maximal flow velocity. The 'turbulent' phase was very brief, however, and probably did not persist for more than 50 ms. Furthermore, Evans (1955) has pointed out that the dye appears to be disposed in the form of helical vortices and is not randomly distributed, but he regards this as a usual phenomenon in early turbulence. The Reynolds number in this case, calculated from the peak flow velocity, was at most 1000 and thus considerably below the usually accepted critical value of 2000. Hale *et al.* (1955), however, suggested that the pulsating flow in the rabbit aorta might reach an 'effective Reynolds number', calculated at the maximal rate of shear of 2000, so that the latter discrepancy might be unimportant.

The first example described above (the inferior vena cava) may, therefore, illustrate one group of causes of disturbed flow. This type of disturbed flow is due to certain conditions, in this case probably a rapid reversal of flow, that create large vortices which are damped out by the inherent stability of the flow. The second example (the rabbit aorta) would then fall into a second group where a very rapid acceleration may produce a period of true instability that is short lived because the high velocity is only maintained for a very short time.

The practical point that arises from this discussion is that the mixing of streams of blood in a vessel, as evidenced by sampling of oxygen content or the dispersion of a dye, may be due to causes other than the existence of turbulence. For much experimental work, the most important fact to establish is the presence of adequate mixing, so it may be argued that the cause of such mixing is of little interest. In the study of the hemodynamics of the circulation, however, it is desirable to know whether the pressure-flow relationship is essentially that of laminar flow, albeit altered somewhat by the creation of eddies, or whether there is fully established turbulence where the pressure-flow relationship may be considerably different.

THE MEASUREMENT OF TURBULENCE

Hot-film anemometry

Blood velocity disturbances both in laboratory animals and in humans can be measured with a constant-temperature anemometer in association with hot-film sensors mounted on rigid L-shaped probes or at the tip of cardiac catheters (see Chapter 6). As blood flows past a heated sensor, it cools it, causing the electrical resistance to change (Ling *et al.*, 1968; Seed and Wood, 1970; Clark, 1974). The voltage required to maintain a constant temperature is related to the blood flow velocity near the tip. Thus, the anemometer system detects the velocity in the region of the sensor. Because of its high-frequency

response, such a system is sensitive to the rapid and random fluctuations of velocity that occur in turbulence. Hot-film probes in water have measured fluctuations of velocity at frequencies as high as 1000 Hz (Clark, 1974; Walburn *et al.*, 1980).

The hot-film sensor is essentially a conducting film on a ceramic substrate. Its cone shape is particularly valuable in preventing fibrin accumulation on the tip of the sensor. A disadvantage of most hot-film probes is that reversed flow cannot be measured. However, a second film, when placed close to the heated film, can be used for detecting flow reversal (Seed and Wood, 1970).

Rigid, usually L-shaped, hot-film probes have been used during thoracotomy in patients and animals (Schultz *et al.*, 1969; Seed and Wood, 1971; Seed and Thomas, 1972; Nerem *et al.*, 1974), but invasion of the artery near the site of the measurement is necessary to insert the probe. In spite of this disadvantage, such probes are useful because the characteristics of velocity can be determined at various positions across the vessel lumen, and the profile of velocity (characterization of velocity across the cross-section) can be measured.

Hot-film probes that can be introduced through catheters during cardiac catheterization enhance accessibility to the great vessels. The accuracy of the measurement of axial velocity, however, may be somewhat diminished if the sensor is at an angle to the axial stream of flow. The catheter-tip method, nevertheless, provides a means to determine the magnitude of flow disturbances in the great vessels (Stein and Sabbah, 1976; Stein *et al.*, 1979a; Stein *et al.*, 1979b).

Laser Doppler anemometry

The development of the laser has made it possible to use the Doppler effect as an optical method for measuring velocity at a localized point in liquids. In LDA, laser light is scattered by the particles suspended in the fluid (Woo and Yoganathan, 1985; Liepsch *et al.*, 1994). The scattered light undergoes a frequency shift, which must be interpreted by electro-optic techniques to obtain information concerning the velocity of the particles. This method for measuring velocity requires that the fluid be translucent and contain particles that scatter light. Because of the requirement of a translucent medium, measurements relevant to the cardiovascular system with LDA have primarily been obtained *in vitro*. However, laser Doppler velocity measurements using an optical fiber as a light guide have been accomplished in the coronary arteries of dogs (Kajiya, 1983).

Doppler ultrasound for turbulence

Doppler ultrasound permits a non-invasive impression of turbulence in patients (Iooss *et al.*, 2002). The

measurements do not give results as rigorous as those of hot-film anemometry or LDA, but useful information may be obtained (see Chapters 6 and 8). Most studies that employ Doppler ultrasound use broadening of the Doppler spectrum as evidence of turbulence (Casty and Giddens, 1984; Cloutier *et al.*, 2001). This method lacks precision because Doppler spectral broadening may arise from sources other than turbulence (i.e. from Doppler ambiguity; see George and Lumley (1973)). Doppler ambiguity includes the primary effects of transit time broadening, gradients in mean velocity across the sample volume and turbulent fluctuations within the sample volume, along with higher-order effects of Brownian motion and source broadening. Satisfactory measurements of turbulence intensities have, nevertheless, been shown *in vitro* with Doppler ultrasound measurements (Giddens and Khalifa, 1982). Many clinical applications, such as the evaluation of carotid stenosis, involve real-time spectral displays of either continuous wave or pulsed Doppler signals (Baskett *et al.*, 1977). For an excellent review of the development of ultrasound techniques to measure blood flow and turbulence, see Veyrat (2001), Rasulo *et al.* (2008), and Bonello *et al.* (2009).

FLOW DISTURBANCES IN THE REGION OF CARDIAC VALVES

In patients with a normally functioning aortic valve and normal cardiac output, blood flow disturbances (measured with a hot-film anemometer probe) developed at peak velocity, were maintained throughout ejection and persisted into the deceleration phase (Stein and Sabbah, 1976; Stein *et al.*, 1979a). Even though blood flow disturbances were present near the aortic valve, they were minimal or absent in the mid-ascending aorta and always absent in the proximal aortic arch. Peak ascending aortic blood flow velocity ranged between 71 and 143 cm/s.

Disturbed flow was also observed in the region of normally functioning pulmonary valves (Stein *et al.*, 1979a). Flow disturbances, however, were lower in the main pulmonary artery than in the ascending aorta and sometimes barely discernible. One factor that may have contributed to the higher turbulence in the aorta may be the higher velocity in the aorta. Peak velocity in the pulmonary artery ranged from 27 to 78 cm/s, whereas in the aorta it ranged from 71 to 143 cm/s (Stein *et al.*, 1979a). Even though volumetric flow is identical in the pulmonary artery and aorta, maximal velocity is lower in the pulmonary artery owing to several factors. Among these are the larger diameter of the main pulmonary artery (Greenfield and Griggs, 1963) and pulmonary valve (Davies *et al.*, 1975), and a longer systolic ejection period within the pulmonary artery (Murgu *et al.*, 1975). In addition, the greater distensibility of the pulmonary artery may have a damping effect on turbulence that is independent of velocity (Stein *et al.*, 1980).

In patients with aortic valvular disease, disturbed or turbulent blood flow is markedly increased compared with patients with normal valves (Stein and Sabbah, 1976). In patients with pure aortic insufficiency, the intensity of turbulence was approximately three times greater than in patients with a normal aortic valve (Stein and Sabbah, 1976). In these patients, flow disturbances may be attributed to an increased forward flow across the valve productive of high velocities during ejection. Peak aortic velocities in the range of 231–256 cm/s were recorded in these patients, as were prominent levels of turbulence extended to the mid-ascending aorta and proximal aortic arch. In patients with aortic stenosis, the level of turbulence was approximately ten times that observed in normal subjects (Stein and Sabbah, 1976). Peak velocities ranged from 308 to 459 cm/s (Stein and Sabbah, 1976). Disturbances occurred before peak velocities were reached and extended throughout systole. Further clinical implications are dealt with in Chapter 8.

FLOW DISTURBANCES IN LARGE ARTERIES

Coulter and Pappenheimer (1949) studied the steady flow of blood through tubes of 0.126 cm internal radius, i.e. similar in dimensions to the femoral artery of the dog. By the orthodox method of recording the discontinuity in the pressure gradient–flow graph, they recorded the onset of turbulence at an average Reynolds number of 1600. This is sufficiently close to the accepted value for the critical value for other liquids to indicate that blood does not behave anomalously in this respect.

The effect of pulsation on the stability of laminar flow is clearly of considerable interest. There seems little doubt that oscillatory flow will become unstable at Reynolds numbers that are less than 2000, even when these are calculated from the maximum mean flow velocities. In the rabbit aorta quoted above, for example, the Reynolds number was less than 1000 when derived in the usual way. Simple model experiments that McDonald did, and also those reported by Evans (1955), demonstrate that rapid alterations in the rate of flow will readily cause a break-up of the pattern of injected dye. Hale *et al.* (1955) considered this question quantitatively and made the assumption that the critical Reynolds number was simply related to the maximum rate of shear in the liquid. By calculating this for oscillating flows, they derived the equivalent steady-flow velocity that would give the same rate of shear. The ratio of the equivalent mean velocity to the actual measured mean velocity when multiplied by the Reynolds number would give an ‘effective Reynolds number’. This ratio was dependent on the value of α . The ratio remained at 1.0 for a value of α up to 2.0. When α was greater than 3.0, it increased linearly so that at $\alpha = 4.0$ the ratio was 1.2, and at $\alpha = 10$ the ratio was 2.8. The only experimental evidence in support of the assumption that instability depends on rate of shear was based on a

comparison of flow patterns in the rabbit aorta and the dog femoral artery. The maximal Reynolds number calculated in the normal way was about 1000 for each vessel, yet high-speed cine films demonstrated turbulence in the aorta of the rabbit but no turbulence in the femoral artery of the dog. The pulse rate of the rabbit is, however, faster than that of the dog, and allowing for this difference in the value of α , the effective Reynolds number in the rabbit came close to 2000, whereas in the dog it was only 1300. The frequency of the pulse rate would thus be an important factor in determining the stability of arterial flow.

Since the development of techniques for measuring high-fidelity pulsatile blood flow velocity in intact animals (see above), numerous studies have reported data from a variety of animals, including humans. Seed and Wood (1971) and Nerem *et al.* (1974) confirmed the existence of disturbed flow in the ascending aorta and descending thoracic aorta of dogs and horses. In human patients undergoing diagnostic cardiac catheterization, Stein *et al.* (1979b) showed no flow disturbances in the abdominal aorta and common iliac arteries. The average peak velocity recorded in the aorta in the region of the renal artery was 28 cm/s. The velocity was only somewhat lower in the proximal portion of the common iliac artery (24 cm/s). Higher values for peak flow velocity are common with Doppler flow techniques (see Chapter 8). Using the earlier data, the Reynolds number at the aortic bifurcation was 730 (range, 400–1100). In the common iliac artery, the average Reynolds number ranged between 390 and 620.

The predisposition of the coronary and carotid arteries to develop atherosclerosis has led many investigators to assess the nature of flow in these vessels. A proper determination of flow disturbances in the coronary arteries with hot-film probes presents major technical difficulties. Nevertheless, an attempt was made to measure blood velocity in the coronary arteries of anesthetized horses (Nerem *et al.*, 1976). In both the left anterior descending coronary artery and the left circumflex coronary artery, flow was generally free of disturbances. Peak velocities in the range of 9–83 cm/s and Reynolds numbers of 188–1671 were reported at various distances from the coronary ostium. Similar observations were reported using a pulsed ultrasound Doppler velocity meter in ponies (Wells *et al.*, 1974).

Hemodynamic factors that tend to affect the magnitude of blood flow disturbances include velocity, viscosity, and density of the blood. For example, an increased blood velocity or a reduction of blood viscosity would increase the intensity of turbulence. The pulsatile nature of blood flow may also lead to flow instability. The effects of unsteady flow on the transition from laminar to turbulent flow have been evaluated by several investigators (Hale *et al.*, 1955; Parker, 1977), but an understanding of the stability of unsteady flow, even in tube flow, is still poor (Parker, 1977; Chahed *et al.*, 1991). Lesions that produce a partial obstruction to flow or an irregularity of the vessel

wall would augment flow disturbances. The normal aortic and pulmonary valves also cause flow disturbances, because the valvular leaflets act as natural projections into the stream of flow (Sabbah and Stein, 1979). Fluttering of the free margin of the leaflets may also contribute to the disturbances.

Several anatomic and hematologic factors affect the character of blood flow in the cardiovascular system. Anatomic factors that tend to diminish disturbances of flow at branches of vessels include branch to trunk area ratios less than 1.0 (Walburn *et al.*, 1979) and acute angles of branching (Roach, 1977b). Arterial tapering (Walburn and Stein, 1981) and vessel distensibility (Stein *et al.*, 1980) also diminish disturbances. Hematological factors that minimize disturbances include a normal concentration of RBCs and deformability of the RBCs (Stein *et al.*, 1975; Sabbah and Stein, 1976).

TURBULENCE AS A CAUSE OF POST-STENOTIC DILATATION AND ANEURYSMS

Turbulent blood flow has been suggested to have a direct effect on the arterial wall. Post-stenotic dilatation was attributed to progressive weakening of the wall by turbulence (Roach, 1963). The association of turbulence with a post-stenotic dilatation is likely because it commonly occurs with aortic or pulmonary stenosis (Kjellberg *et al.*, 1955) and may be found distal to aortic coarctations (Zaroff *et al.*, 1959; Kieffer *et al.*, 1961) where turbulence is known to occur. The mechanism by which turbulence may contribute to the development of a post-stenotic dilatation is conjectural. Several factors, when combined, may produce the dilatation (Holman, 1955). The conversion of the high kinetic energy of the swiftly moving stream into high potential energy or lateral pressure was postulated to be one factor in the development of a post-stenotic dilatation. Shocks of impacts of alternating high and low pressure may contribute to arterial distension. The lower velocity due to the widening of the vessel, according to the Bernoulli equation, would itself increase lateral pressure and cause even greater dilatation (Holman, 1955). Gow *et al.* (1992) studied the effects of chronic high frequency (50 Hz) vibrations on the rabbit aorta and concluded that vibration does not cause post-stenotic dilatation and suggested that its cause is likely to involve the vascular smooth muscle stimulated by the effect of turbulent flow on the endothelium and NO activity. Various experimental models have been used to study the effect of turbulence on wall shear stress and velocity profiles in aneurysms of the aorta and cerebral arteries. Khanafer *et al.* (2007) found that turbulence increases pressure and shear stress in an aortic aneurysm under resting and exercise flow conditions and maximum turbulence-induced shear stress occurred at the distal end of the aneurysm. Yamaguchi *et al.* (2008) found that

hemodynamics in large aneurysms of the anterior communicating artery contribute to thrombus formation.

THE POTENTIAL ROLE OF TURBULENCE IN ATHEROGENESIS

Atherosclerosis and its complications represent the leading cause of morbidity and mortality in the industrialized, as well as in the developing, countries. Classical cardiovascular risk factors have been identified over the past decades leading to recommendations for lifestyle modifications and to the development of efficient and well-tolerated drug regimens aimed at reducing the occurrence of cardiovascular complications. The endothelium due to its position in the circulation is the first organ being exposed to circulating noxious elements and solutes, as well as to the mechanical aggressions generated by heartbeats and pulsating blood flow (Hayoz and Mazzolai, 2007). Identification of disturbed or turbulent blood flow in the ascending aorta or pulmonary artery, although it offers explanations for mechanisms of cardiac murmurs (Stein, 1981), post-stenotic dilatation (Roach, 1963), and the continuing disease process of aortic valvular stenosis (Stein *et al.*, 1977), probably has little, if any, bearing on atherogenesis. Absence of turbulence or disturbed flow in the peripheral circulation negates the potential role of turbulence in the initiation of atherosclerosis. Such

blood flow disturbances, however, once initiated, may lead to a progression of atherosclerosis (Gimbrone, 1999; Gimbrone *et al.*, 2000; Libby, 2002; de Nigris *et al.*, 2003; Li *et al.*, 2007; Wong *et al.*, 2009). Irrespective of the initial cause of atherosclerosis, once raised intimal plaques create partial obstructions to flow, turbulence downstream from such obstructions is likely.

Several mechanisms by which the presence of turbulence within the arterial tree may lead to the progression of atherosclerosis have been proposed. High-frequency pressure fluctuations may have a damaging effect on the arterial wall (Holman, 1955). The acceleration of atherosclerosis may also result from the production of localized areas of increased lateral wall pressure (local hypertension) (Holman, 1955). More recently, Davies *et al.* (1986) showed that exposure of endothelial cells to turbulent flow stimulated endothelial deoxyribonucleic acid (DNA) synthesis. This implied that such blood flow characteristics may be a major determinant of a hemodynamically induced endothelial cell turnover or endothelial stripping (Li *et al.*, 2007). Turbulence may also contribute to platelet activation through high shear stresses (Stein *et al.*, 1982; Walburn *et al.*, 1985) and the interaction of blood elements with the arterial wall (Jorgensen, 1978). Platelet deposits on the endothelial surface, according to the encrustation theory of atherogenesis, may participate in the disease process (Murphy *et al.*, 1962; Mustard *et al.*, 1962).

Properties of the arterial wall: theory

Mechanical properties of an elastic body	55	The velocity of the 'foot' and other points of the pressure wave	69
The behavior of an elastic tube under a distending pressure	59	The phase velocity of harmonic components of waves	70
Definition of compliance, distensibility, and stiffness	61	The use of transient excitation to measure wave velocity	71
The dynamic behavior of elastic materials	61	The effect of arterial distension on wave velocity	73
Wave propagation in an elastic tube	64	Attenuation of propagated waves in viscoelastic tubes and arteries	74
Pulse wave velocity measurement in viscoelastic arteries	66		

In consideration of the pressure–flow relationships of oscillatory flow in Chapters 2 and 7, the artery is treated primarily as a cylindrical tube of constant (or non-varying) diameter. In reality, an artery is a viscoelastic tube consisting of three layers (Fig. 3.1) (Gasser *et al.*, 2006) whose diameter varies with a pulsating pressure which is generated by a pulsating flow of blood (Sonesson *et al.*, 1994); in addition, the arterial system will propagate pressure, flow and diameter waves, generated by the ejection of a viscous liquid (blood) from the ventricle, at a certain velocity, which is largely determined by the elastic properties of the arterial wall (Bramwell and Hill, 1922a; Bramwell and Hill, 1922b; Hardung, 1962; Cox, 1968; McDonald, 1968b; Nichols and McDonald, 1972; Farrar *et al.*, 1978; Milnor and Bertram, 1978; Corte *et al.*, 1979; Kelly and Fitchett, 1992; Saeki *et al.*, 1995; Laurent *et al.*, 2006). The model considered, therefore, was inconsistent in that the artery was treated partly as a straight rigid tube (its diameter was invariant) and partly as an elastic tube (the velocity of the pulse wave was given a realistic value corresponding to that found *in vivo*). For, in a rigid tube filled with an incompressible liquid, the wave velocity would be infinitely high, and filled with a real liquid it would be the velocity of sound in blood – about 1500 m/s.

Analyses of the behavior of pulsatile liquid flows in various types of elastic tubes are considered in Chapters 2, 7, and 9. There it is seen that many of the physical conditions with which we are concerned, such as the viscoelastic properties of blood, the damping of propagated waves, and the presence of reflected waves, change the velocity of propagation of the pulse wave. The degree of this change can only be measured experimentally if the 'natural' velocity (i.e. the pulse wave velocity along the

arteries being studied in relation to the properties of the wall alone) is known. To study the hydrodynamics (or hemodynamics) of the arterial system, a knowledge of the elastic properties of the arterial wall, from which one can predict the 'natural' velocity, is of fundamental importance. Indeed, knowledge of the viscoelastic properties of the blood vessels has long been recognized as playing an essential role in cardiovascular behavior.

This chapter reviews the essential concepts of large artery rheology, including methods of quantitation and physical definitions on which studies of elasticity should be based. The chapter is divided into two parts: the first part deals with the theory of arterial mechanics and the second part with pulse wave velocity and attenuation. Excellent reviews of these topics have been published by Bergel (1960), Gow (Gow, 1972; Gow, 1980), Caro *et al.* (1978), Dobrin (Dobrin, 1978; Dobrin, 1983), Cox (1979), Langewouters (1982), O'Rourke (1982a), Milnor (1989), and more recently by Fung (1997), Levy (1999), Dobrin (2000), Nichols and Edwards (2001), Cheng *et al.* (2002c), O'Rourke *et al.* (2002), Safar and Benetos (2003), Dao *et al.* (2005), Nichols (2005), Laurent *et al.* (2006), Vlachopoulos *et al.* (2006c), McEniery *et al.* (2007), Atkinson (2008), Boutouyrie (2008), Nichols *et al.* (2008), O'Rourke and Hashimoto (2008), Warriner *et al.* (2008), Nichols and Epstein (2009), Boutouyrie *et al.* (2010), and Reusz *et al.* (2010).

MECHANICAL PROPERTIES OF AN ELASTIC BODY

No solid body is completely rigid. When forces act on a solid body without displacing it, various parts of the body

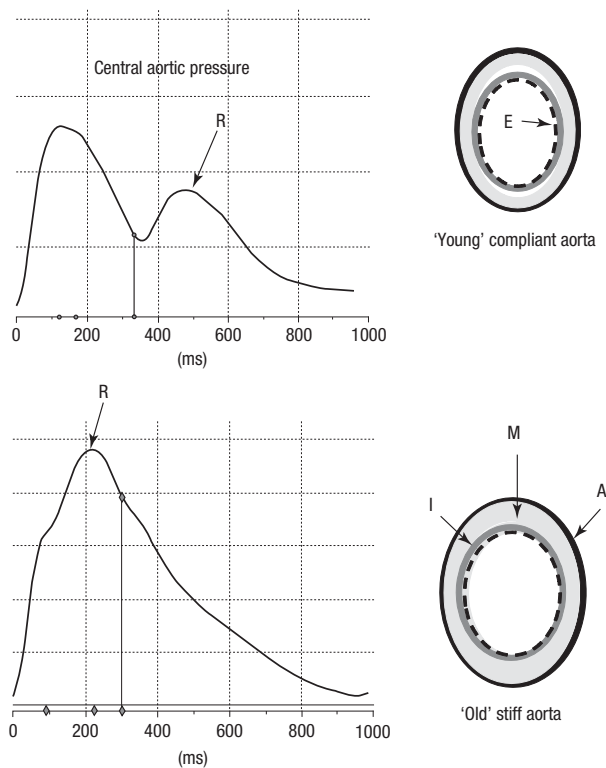


Figure 3.1 Typical central aortic pressure waveforms, showing the reflected (R) wave, from a person with a 'young' compliant aorta (top) compared to one with an 'old' stiff aorta (bottom). The aorta is composed of three layers: intima (I), media (M), adventitia (A). I is the innermost layer consisting of a single layer of endothelial (E) cells, a thin basement membrane and a subendothelial layer. The subendothelial layer mainly comprises thinly dispersed smooth muscle cells and bundles of collagen fibrils. M is composed of smooth muscle cells, a network of elastic and collagen fibrils and elastic laminae which separate M into a number of transversely isotropic fiber-reinforced units. A is the outermost layer surrounded by loose connective tissue. The primary constituents of A are thick bundles of collagen fibrils arranged in helical structures. Reproduced from Covic *et al.*, (2004) and Gasser *et al.*, (2006)

move relative to one another. If the body regains exactly its original form when the force is removed, it is said to be 'perfectly elastic'. If the body retains the deformation, it is said to be 'plastic'.

The theory of elasticity deals with the relations between the forces applied to a body and its consequent change or deformation. The force per unit area that produces the deformation is called the 'stress'; the deformation, described as the ratio of the deformation to its original form, is called the 'strain'. Because it is a ratio, strain is dimensionless. The ability to withstand a stress is a property that distinguishes a solid from a liquid, as a liquid when stressed will undergo a viscous flow. Nevertheless, a larger number of substances exhibit properties appropriate to both an elastic solid and a viscous liquid,

and the deformation suffered by such a material will depend on both the magnitude of the stress and the rate at which it is applied. Such substances are called 'visco-elastic', and it is to this large class that the arterial wall belongs. When deforming stresses are applied slowly, however, the viscous behavior may be neglected, so we will first outline the purely elastic properties of substances.

No substance is perfectly elastic when very large forces are applied to it, but for the small deformations that are considered in such materials as steel wire, which are the main study of elastic theory, the deformation is proportional to the force applied. This proportionality was first described by Robert Hooke (1635–1703) in 1676 in the famous phrase *Ut tensio sic vis* (as the (ex)tension so the force) and is known as Hooke's law. The point at which Hooke's law ceases to apply is known as the 'elastic limit', and when the solid has deformed beyond this point it cannot regain its original form and acquires a permanent set. With larger loads still, the 'yield point' is reached when the deformation continues to increase without further load and usually leads rapidly to breakage. The classical theory of elasticity stems from behavior below the elastic limit and bases itself throughout on two assumptions – that the deformations are infinitesimal and that the structure of the material is continuous and uniform, or homogeneous. The treatment below follows the excellent summary by Bergel (1960) and Dobrin (Dobrin, 1978; Dobrin, 1983) very closely and is very similar to that given in the earlier editions of *Blood Flow in Arteries*. It will be seen that neither of the two assumptions mentioned above applies well to the arterial wall. In the first place, the wall is easily extensible and is more analogous to substances such as rubber that are often classed as elastomers and undergo large deformations. In the second place, the main elastic components of the arterial wall are collagen and elastin, which are fibrous in nature, have different elastic modulus and different attachment to smooth muscle and are supported in a liquid, or semi-liquid, matrix of water and mucoproteins. The arterial wall is, therefore, far from homogeneous. Nevertheless, the majority of analyses of arterial mechanics are based on classical theory so that it is necessary to consider its main elements.

Strain

When a body of length x_0 is extended to a length x_1 (Fig. 3.2), the relative increase in length $(x_1 - x_0)/x_0$ is the 'longitudinal strain' (Bergel, 1960; Patel and Fry, 1966; Dobrin, 1978; Dobrin, 1983; Cinthio *et al.*, 2006; Zhang *et al.*, 2007a; Zhang *et al.*, 2007b). An increase in length is a positive, a decrease a negative, strain. When there is a change in volume, there is a 'compressive strain'; when there is an angular deformation (a displacement of two points in parallel planes in a direction parallel to those planes), we refer to a 'shear

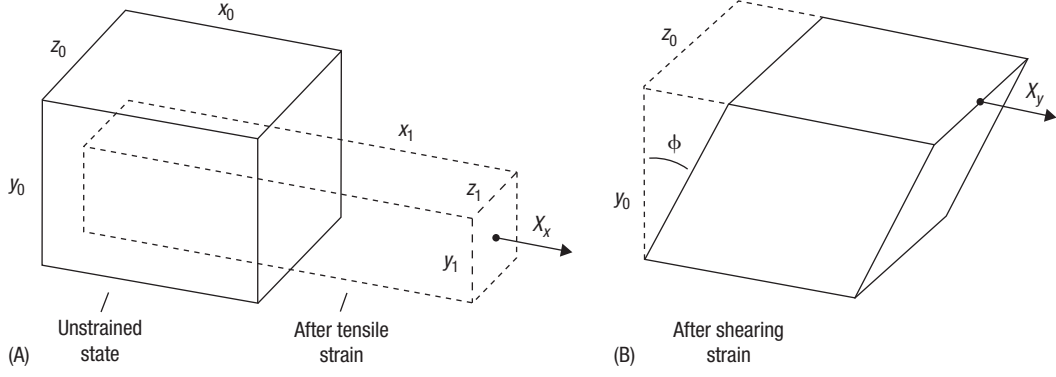


Figure 3.2 Longitudinal (or tensile) strains. (A) An elastic solid of dimensions x_0 , y_0 , z_0 in the unstrained state before deformation (solid lines) and after a positive tensile strain ϵ_{xx} resulting from the tensile stress X_x (broken lines). Strains in the y -axis and z -axis are ϵ_{yy} and ϵ_{zz} , respectively, and are both negative. (B) Shearing strain produced by the stress X_y . One measure of strain is the angle of shear, ϕ . Adapted from Milnor (1989)

strain' (Cinthio *et al.*, 2006). Considering only longitudinal strain, it is obvious that a substance that is stretched simultaneously gets thinner in the transverse direction (Learoyd and Taylor, 1966; Patel *et al.*, 1969; Cox, 1975b) (Fig. 3.2). Expressed mathematically, the 'tensile', or longitudinal strain ϵ , is given by

$$\epsilon_{xx} = \frac{x_1 - x_0}{x_0}; \quad 3.1$$

while in the width and breadth, or y and z directions, the strains are given by

$$\epsilon_{yy} = \frac{y_1 - y_0}{y_0} = -\sigma_{yx}\epsilon_{xx} \quad 3.2$$

and

$$\epsilon_{zz} = \frac{z_1 - z_0}{z_0} = -\sigma_{zx}\epsilon_{xx} \quad 3.3$$

The symbol σ , the ratio of transverse to longitudinal strain, is called the 'Poisson ratio'. This ratio is a characteristic property of the material and for small strains it is constant, but it cannot necessarily be assumed that $\sigma_{yz} = \sigma_{zx}$. Similarly, primary extensions in the y and z directions give four more Poisson ratios, making a total of six; these may, or may not, be of the same value, depending on the nature of the material.

If we consider an infinitesimal cube within the substance of an object that has faces parallel to the axes x , y , and z , the strain at any face can be resolved into three components in directions at right angles to each other. These will be one tensile and two shearing strains, which may be written as ϵ_{xx} , ϵ_{xy} , and ϵ_{xz} , where ϵ_{xx} indicates a strain separating two points on the x -axis in the direction of the x -axis, and ϵ_{xy} a strain separating these points on the y -axis, etc. We thus have nine components of strain at a point:

$$\epsilon_{xx}, \epsilon_{xy}, \epsilon_{xz}, \epsilon_{yx}, \epsilon_{yy}, \epsilon_{yz}, \epsilon_{zx}, \epsilon_{zy}, \epsilon_{zz}.$$

It may be shown (Love, 1944) that the strains of the form ϵ_{xy} and ϵ_{yx} are identical, so they may be eliminated and six independent components of strain remain:

$$\epsilon_{xx}, \epsilon_{xy}, \epsilon_{yy}, \epsilon_{yz}, \epsilon_{zz}, \epsilon_{zx}.$$

The corresponding six independent Poisson ratios are

$$\sigma_{yx}, \sigma_{zx}, \sigma_{xy}, \sigma_{zy}, \sigma_{xz}, \sigma_{yz},$$

and the value of σ can be shown to fall in the range $-1.0 < \sigma < 0.5$. Although it is theoretically possible, no isotropic material is known to possess a negative Poisson ratio, so that the effective range is 0.0–0.5. For small extensions with a ratio of 0.5, the volume of the solid remains the same when it is stretched – an isovolumetric deformation. The value for rubber and rubber-like substances is usually close to this value (rubber, 0.48) (Newman and Searle, 1957). For most metals it lies in the range 0.25–0.4, so that they undergo an increase of volume in stretching; when $\sigma = 0.0$, the increase in volume will be proportional to the change in length. The value of σ for the arterial wall is almost certainly close to 0.5; the evidence is considered below.

Reconsidering the cube of dimensions x_0 , y_0 , and z_0 subjected to tensile strain ϵ_{xx} , after deformation the dimensions are

$$\begin{aligned} x_1 &= x_0(1 + \epsilon_{xx}) \\ y_1 &= y_0(1 - \sigma_{yx}\epsilon_{xx}) \\ z_1 &= z_0(1 - \sigma_{zx}\epsilon_{xx}). \end{aligned} \quad 3.4$$

The new volume V_1 is $x_1y_1z_1$ compared with the original volume V_0 , which was $x_0y_0z_0$. The change in volume (ΔV) is thus

$$\begin{aligned} \Delta V &= V_1 - V_0 \\ &= x_0(1 + \epsilon_{xx})y_0(1 - \sigma_{yx}\epsilon_{xx}) \\ &\quad z_0(1 - \sigma_{zx}\epsilon_{xx}) - x_0y_0z_0; \end{aligned} \quad 3.5$$

and if ε_{xx} is small so that its powers may be neglected,

$$\begin{aligned}\Delta V &= \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz} \\ &= \varepsilon_{xx}(1 - \sigma_{yx} - \sigma_{zx}).\end{aligned}\quad 3.6$$

If the material has the same elastic properties in all these axes, it is said to be 'isotropic', and all the values of σ will be the same, so we may then rewrite eqn 3.6 as

$$\Delta V = \varepsilon_{xx}(1 - 2\sigma), \quad 3.7$$

which demonstrates that, for small strains, a value of $\sigma = 0.5$ leads to no change in volume. Furthermore, even if the elastic properties in each direction are not the same, but are isovolumetric under strain, the average values of σ must be equal to 0.5. This is because the average overall value of σ must be 0.5; thus if some are less than 0.5, others will need to be greater than 0.5 or negative. These considerations need to be borne in mind when assessing experimental work that has studied the arterial wall as an 'anisotropic' material – a difficult procedure with biological tissues – which is discussed below.

Stress

Stress is the intensity of force F acting across a given plane in a body and, if it is evenly distributed over the area A , the stress is F/A . The units of stress are thus force per unit area (Halliday and Resnick, 1965; McDonald, 1974; Milnor, 1989; Nichols and Nicolini, 1994). Numerous reviews have appeared in the literature since the discovery that arterial wall shear stress is an important determinant of endothelial function (see Chapter 5). Tension or stress in the arterial wall is related to arterial pressure (Laplace) and stretch of the wall (increased diameter) (Dobrin, 1999; Zulliger *et al.*, 2002), while wall shear stress is related to blood flow (Poiseuille) (drag on the endothelium) (Pries *et al.* 2005; Reneman *et al.*, 2006; Katritsis *et al.*, 2007) (see below). Increased shear (pressure) stress leads to aneurysm and rupture, while decreased circumferential (flow) stress leads to endothelial dysfunction and atherosclerosis.

The stress on a point in a plane may be resolved into those normal to the axis (tensile or compressive) and tangential to the axis (shearing stresses) (Fig. 3.2). The components along the three axes are designated by subscripts (thus X_x, X_y, X_z, Y_x etc.), where the capital letters indicate the direction of the stress component. X_y and Y_x must be equal or a rotational resultant of force would exist. We are therefore left with six independent components of stress:

$$X_x, X_y, Y_x, Y_z, Z_z, Z_x. \quad 3.8$$

The relationship between stress and strain

Hooke's law states that, within certain limits, strain is proportional to stress. This may be expanded into a

generalized statement of Hooke's law: 'Each of the six independent components of strain may be expressed as a linear function of the six components of stress, and vice versa.' With six stress components and six strain components, it can be seen that this will lead to 36 constants of proportionality; of these, 15 can be shown to be inter-related, leaving some 21 constants to be considered when the material is anisotropic.

If, however, the material is isotropic and has the same elastic behavior in each of the three axes, the number of constants of proportionality is reduced to two. For a material that is macroscopically inhomogeneous, as is the arterial wall, this is an uncertain assumption, although much experimental evidence indicates that under *in vivo* conditions this is approximately true. It can be seen that the advantages, in simplicity, for making such an assumption, if possible, are very great.

The relationship between stress and strain is expressed as an 'elastic modulus' (Fig. 3.3). As strain is a ratio, and therefore dimensionless, all these moduli will have the dimensions of stress (i.e. force per unit area, or dyne/cm²).

The modulus in the longitudinal direction is called 'Young's modulus' in honor of the pioneer work of Thomas Young (Young, 1808; Young, 1809) and is designated by E or Y (in this text, E is used). Thus,

$$\begin{aligned}E &= \frac{\text{Longitudinal force per unit area}}{\text{Extension per unit length}} \\ &= \frac{X_x}{\varepsilon_{xx}}.\end{aligned}\quad 3.9$$

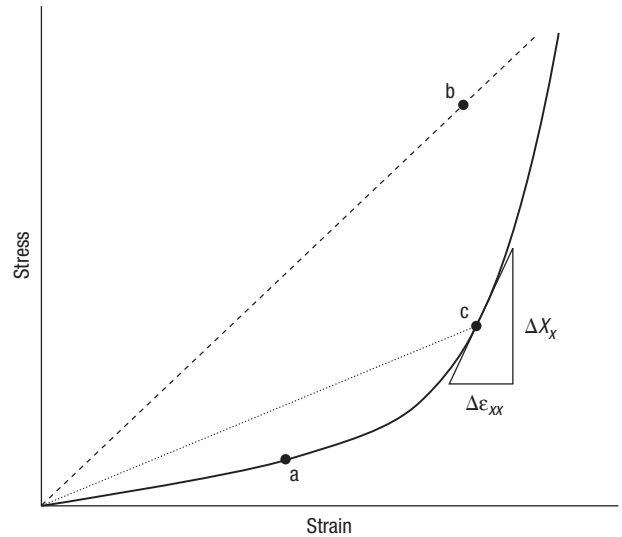


Figure 3.3 Linear (broken line) and non-linear (solid line) relations between stress (X_x) and strain (ε_{xx}) in the x -direction. The ratio stress/strain is Young's (or the elastic) modulus. This is applicable at points a and b because the curves are linear and pass through the origin. At point c, the curve is non-linear; therefore, an 'incremented' elastic modulus (which is the slope of the curve at point c) may be calculated. Reproduced from Dobrin (1978)

The 'shear modulus', or modulus of rigidity η , is the ratio of shear stress to angular strain, i.e.

$$\eta = \frac{X_y}{\epsilon_{xy}} \quad 3.10$$

(μ is the common symbol for shear modulus in standard texts, but it is not used here since it is used for viscosity of a liquid).

The 'bulk modulus' B is the ratio of compressive stress to strain, that is, to the relative change in volume. Taking a cube under compression, the mean stress is

$$-(X_x + Y_y + Z_z)/3. \quad 3.11$$

This is commonly considered as a pressure P that will exert the same force in all directions (Pascal's law) so that

$$P = X_x = Y_y = Z_z. \quad 3.12$$

The change in volume (ΔV) is given by eqn 3.6, and volumetric strain is $\Delta V/V_0$, so that

$$B = -\frac{PV_0}{\Delta V} = \frac{(X_x + Y_y + Z_z)V_0}{3(\epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz})}. \quad 3.13$$

In addition, a generalized longitudinal modulus (χ) is defined as

$$\chi = \frac{\text{Longitudinal load per unit area}}{\text{Longitudinal strain}},$$

so that, with the Poisson ratio, we have five elastic constants that are defined. In an isotropic material, only two need to be evaluated independently, and all five may be expressed in terms of any two of them, thus,

$$E = 2(1 + \sigma)\eta = \frac{9B\eta}{(3B + \eta)} \quad 3.14$$

$$\eta = \frac{E}{2(1 + \sigma)} \quad 3.15$$

$$B = \frac{E}{3(1 - 2\sigma)} = \frac{2(1 + \sigma)\eta}{3(1 - 2\sigma)} \quad 3.16$$

$$\sigma = \frac{3B - 2\eta}{2(3B + \eta)} \quad 3.17$$

$$\chi = \frac{E\sigma}{(1 + \sigma)(1 - 2\sigma)}. \quad 3.18$$

In the treatment that follows, the two elastic constants that have been used to characterize the wall are Young's modulus E and the Poisson ratio σ .

From these, we derive certain results, when σ is approximately 0.5, that are often quoted without stating this condition (McDonald, 1974). For example, $E = 3\eta$, and it will be seen from eqn 3.16 that as σ approaches 0.5, the bulk modulus must become very large. Carew *et al.* (1968) found that the arterial wall is virtually incompressible, with a bulk modulus of approximately 4.4×10^9 dyne/cm². McDonald (1974) reasoned that the arterial wall contains about 70 percent water and its bulk modulus would therefore be approximately 2.2×10^{10} dyne/cm²; and since Young's modulus of the arterial wall is in the range $2.0\text{--}8.0 \times 10^6$ dyne/cm², Poisson's ratio would be approximately 0.5.

THE BEHAVIOR OF AN ELASTIC TUBE UNDER A DISTENDING PRESSURE

The increase in length of an elastic material is proportional to the tension (or force) applied to it. In the case of a tube of cylindrical cross-section, the applied force will be the circumferential tension in the wall due to the lengthening caused by a distending pressure P . For a very thin-walled cylindrical shell of inside radius R , this tension T is given by the relation commonly referred to as the law of Laplace (Pierre S de Laplace, 1749–1827), i.e.

$$T = PR \quad 3.19$$

The dimensions of T will be dyne/cm, and it is often called the longitudinal (or wall) stress or tension (Peterson *et al.*, 1960).

If the wall has a thickness h , the circumferential stress τ is given by

$$\tau = PR/h \quad 3.20$$

(often known as Lamé's equation). If the outside radius is used in the equation, it is called Barlow's equation.

It will be seen from eqns 3.19 and 3.20 that the tension or stress will not vary linearly with the distending pressure, so that even though the elastic properties of the wall are Hookean, the pressure–volume relation will not be linear. To avoid this confusion, tension–volume or tension–circumference plots should be made to study non-Hookean properties of the wall.

If $T = PR$, a small increase in P or δP will cause an increase in tension, δT , so that we obtain

$$\begin{aligned} (T + \delta T) &= (P + \delta P)(R + \delta R) \\ &= PR + P\delta P + P\delta R + \delta R\delta P; \end{aligned}$$

and if δR and δP are small, then

$$\delta T = R\delta P + P\delta R. \quad 3.21$$

The strain in the circumferential length of the wall is $2\pi\delta R/2\pi R$. So that from eqn 3.9 we have

$$\delta T = EA \frac{2\pi\delta R}{2\pi R} = EA \frac{\delta R}{R}. \quad 3.22$$

If the length of the tube is L and it is isotropic, the length will remain constant and the cross-section of the wall will be $A = hL$; from eqns 3.21 and 3.22 we have

$$R\delta P + P\delta R = EhL \frac{\delta R}{R}, \quad 3.23$$

and at the limiting value

$$\frac{dR}{dP} = R \left(\frac{EhL}{R} - P \right)^{-1}. \quad 3.24$$

In terms of volume,

$$\frac{dV}{dP} = \frac{dV}{dR} \frac{dR}{dP} = 2\pi R \frac{dR}{dP}, \quad 3.25$$

so that

$$\frac{dV}{dP} = \frac{2\pi R^2}{(EhL/R) - P} = \frac{2\pi R^3}{(EhL - PR)}. \quad 3.26$$

Thus, it is clear that if the elastic modulus remains constant, the rate of increase of volume with pressure will increase until, at the point where $EhL/R = P$, it will become infinite and the tube will burst (or rupture). As will be seen below, however, on account of progressive recruitment of stiffer collagenous fibers, the elastic modulus of blood vessels increases as the circumferential strain increases, and this enables arteries and veins to remain stable over a wide range of pressures. They may reach their 'yield strength', but seldom reach their 'ultimate strength'. The shape of the resultant curve is more complex than these relations suggest, because with increasing distension the wall gets thinner and the ratio h/R becomes smaller. Furthermore, although the length L remains constant with a vessel *in situ*, this is not true for an isolated segment over the whole pressure range unless the vessel is held at a constant length. If the tension–circumference graph is linear, the elastic behavior obeys Hooke's law and Young's modulus is the slope of the curve (see Fig. 3.3). If the curve is concave towards the tension (or stress) axis, this indicates an increase in Young's modulus with tension; this is characteristic of the arterial wall and is discussed below.

The ratio of wall thickness to radius of an artery under physiologic pressure lies in the range 0.06–0.16 (Bergel, 1961a; Bergel, 1961b; Learoyd and Taylor, 1966; Gow and Taylor, 1968; Milnor and Bertram, 1978; Langewouters, 1982; Dobrin, 1983). While this may be approximated quite well by the 'thin-wall' relations above, for accurate measurements a better relationship must be derived. Each element in the wall of a pressurized tube is subject to stresses in the longitudinal, radial, and circumferential (tangential) directions. The strains can be considered additive if the first strain does not alter the second stress, and so on. If the strains are small and the tube is isotropic, Love (1944) showed that they may be resolved as follows

when the internal radius is R_i with an intravascular pressure P_i and the outer radius R_o with an extravascular pressure P_o ; the stresses are

$$S_\theta = \left(\frac{P_i R_i^2 - P_o R_o^2}{R_o^2 - R_i^2} \right) + \left(\frac{P_i - P_o}{R_o^2 - R_i^2} \right) \left(\frac{R_o^2 R_i^2}{R^2} \right) \quad 3.27$$

$$S_r = \left(\frac{P_i R_i^2 - P_o R_o^2}{R_o^2 - R_i^2} \right) - \left(\frac{P_i - P_o}{R_o^2 - R_i^2} \right) \left(\frac{R_o^2 R_i^2}{R^2} \right) \quad 3.28$$

$$S_z = \frac{\chi}{(\chi + \eta)} \left(\frac{P_i R_i^2 - P_o R_o^2}{R_o^2 - R_i^2} \right) + \frac{\varepsilon(3\chi + 2\eta)\eta}{(\chi + \eta)}, \quad 3.29$$

where S_θ is the circumferential stress, S_r is the radial stress, S_z is the longitudinal stress and ε is the longitudinal extension. R is any chosen radius between R_i and R_o . If the tube is closed at each end by a flat plate so that the forces of the plates will be balanced by the longitudinal tension, then

$$\pi(R_o^2 - R_i^2)S_z = \pi(R_i^2 P_i - R_o^2 P_o), \quad 3.30$$

which gives for the longitudinal extension, ε ,

$$\varepsilon = \frac{1}{(\varepsilon\chi + 2\eta)} \left(\frac{P_i R_i^2 - P_o R_o^2}{R_o^2 - R_i^2} \right); \quad 3.31$$

and substituting for the values of χ and η in eqns 3.15 and 3.18, we obtain

$$\varepsilon = \frac{(1 - 2\sigma)}{2E} \left(\frac{P_i R_i^2 - P_o R_o^2}{R_o^2 - R_i^2} \right). \quad 3.32$$

It can be seen that for moderately small distensions of a closed isotropic tube, the longitudinal extension will be zero if Poisson's ratio σ is 0.5.

Considering distention, if the radial displacement of a shell of radius R is ΔR , then

$$\Delta R = NR + G/R, \quad 3.33$$

where

$$N = \frac{(P_i R_i^2 - P_o R_o^2)}{2(\chi + \eta)(R_o^2 - R_i^2)} - \frac{\chi\varepsilon}{2(\chi + \eta)} \quad 3.34a$$

and

$$G = \frac{(P_i - P_o)R_o^2 R_i^2}{2\eta(R_o^2 - R_i^2)}, \quad 3.34b$$

and from eqns 3.15 and 3.18 we have

$$2(\chi + \eta) = 2 \left[\frac{E\sigma}{(1 + \sigma)(1 - 2\sigma)} + \frac{E}{2(1 + \sigma)} \right] \\ = \frac{E}{(1 + \sigma)(1 - 2\sigma)}. \quad 3.35$$

On substituting eqns 3.35, 3.34a, and 3.34b into eqn 3.33 with the conditions $P_i = \Delta P$, $P_o = 0$, $\varepsilon = 0$ and $R = R_o$ (i.e. taking the displacement at the outer wall), we have

$$\Delta R = \frac{\Delta P R_i^2 R_o (1 + \sigma)(1 - 2\sigma)}{E(R_o^2 - R_i^2)} \\ + \frac{\Delta P R_o^2 R_i^2 2(1 + \sigma)}{2E(R_o^2 - R_i^2) R_o} \quad 3.36 \\ = \frac{\Delta P R_i^2 2(1 - \sigma^2) R_o}{E(R_o^2 - R_i^2)}$$

and

$$E = \frac{\Delta P R_i^2 2(1 - \sigma^2) R_o}{(R_o^2 - R_i^2) \Delta R}, \quad 3.37$$

or when $\sigma = 0.5$

$$E = \frac{1.5 \Delta P R_i^2 R_o}{(R_o^2 - R_i^2) \Delta R} \quad 3.38$$

In practice, eqn 3.38 is used to calculate the incremental elastic modulus (E_{inc}).

DEFINITION OF COMPLIANCE, DISTENSIBILITY, AND STIFFNESS

In an isolated arterial segment filled with a liquid, ‘compliance’ is defined as a change in volume (ΔV) for a given change in pressure (ΔP), i.e. $\Delta V/\Delta P$. ‘Distensibility’ is compliance divided by the initial volume (V). If the segment is tethered, as it is in the intact circulation and the flow (Q) is pulsatile, then compliance (and distensibility) can be estimated as a change in radius, diameter, flow, or cross-sectional area for a given change in pulse pressure (Honda *et al.*, 1994; Stefanadis *et al.*, 1995b; Resnick *et al.*, 1997; van den Bergmortel *et al.*, 1998; Zanchi *et al.*, 1998; Arnett *et al.*, 1999; Nagai *et al.*, 1999; Cheng *et al.*, 2002; Lalande *et al.*, 2002; O’Rourke *et al.*, 2002; Vulliemoz *et al.*, 2002; Danias *et al.*, 2003; Safar and Benetos, 2003). Arterial stiffness or elastance (elastic modulus) are the inverse of compliance and distensibility. Some studies have used ‘stroke volume/pulse pressure’ as a measure of arterial compliance (Ferguson *et al.*, 1984; Chemla *et al.*, 1998; de Simone *et al.*, 1999; Saba *et al.*, 1999; Celentano *et al.*, 2001; de Simone *et al.*, 2002; Haluska *et al.*, 2010) and others have used ‘pulse pressure/stroke volume’ as a

measure of arterial elastance or stiffness (Bella *et al.*, 2002; Pavlopoulos and Nihoyannopoulos, 2009). Some authors have used stroke volume index in the calculation (Kööbi *et al.*, 2003; Palmieri *et al.*, 2003). Although the ratio of pulse pressure and stroke volume has been used by several investigators in the past as a measure of arterial stiffness, this method is considered a very crude approximation and should be avoided in future research (Van Bortel *et al.*, 2002). When using the above equations to measure elastic properties of the arterial wall, it must be remembered that the two variables used in the calculation must be measured at the same site because of wave shape changes during transmission.

THE DYNAMIC BEHAVIOR OF ELASTIC MATERIALS

The properties of a ‘perfectly’ elastic body are independent of the rate at which a stress is applied. In many materials, including all living tissues, the time factor is important: when they are extended rapidly, they are stiffer than when they are extended slowly. The analogy for a perfectly elastic solid is that of a spring and its extension, which is proportional to the force that is applied and remains constant as long as the force is maintained. On the other hand, the force applied to a liquid causes flow, the rate of which is proportional to the force and which is constant when the force is constant. Living tissues show both of these properties and are classified as viscoelastic.

While the elastic behavior is characterized by a spring, with an elastic modulus (or spring constant) E , the viscous flow behavior is characterized by a ‘dashpot’ which is similar to a shock absorber on an automobile. This is a cylinder containing a liquid of viscosity μ , in which there is a movable vane connected to the spring (Fig. 3.4). The response of the spring to a tension T is given by

$$\Delta L/L = T/E, \quad 3.39$$

while that of a dashpot is

$$dL/dt = T/\mu. \quad 3.40$$

If a spring and a dashpot are connected in series, we have what is known as a Maxwell model (or element) (Fig. 3.4, left) (Hodis and Zamir, 2008). On suddenly extending such an element, only the spring will stretch and the initial tension T_0 will be $(\Delta L/L)E$; the dashpot will, however, begin to move and the tension will begin to decay exponentially to zero according to the formula

$$T = T_0 e^{-Et/\mu}. \quad 3.41$$

This phenomenon is known as ‘stress relaxation’ (Zatzman *et al.*, 1954; Craiem and Armentano, 2008) or delayed compliance (Fig. 3.5), and if the analogy with a Maxwell

model is adequate, the material cannot maintain a stress and is classified as an elastic liquid (Seidel and Murphy, 1976).

At a time $t = \mu/E$, the tension has fallen to $1/e$ of its original value, and this is referred to as the 'relaxation time'. If a second spring (elastic modulus E_2) is placed in parallel with the dashpot (Fig. 3.4, right), the tension will not decay to zero and its final value will be

$$T = \frac{\Delta L}{L} \left(\frac{E_1 E_2}{E_1 + E_2} \right). \quad 3.42$$

If a spring and a dashpot alone are placed in parallel, it is known as a Voigt model (Fig. 3.4, middle) (Craiem and Armentano, 2007; Craiem *et al.*, 2008). Each unit undergoes the same extension, but the whole cannot respond instantaneously and is said to have a retarded elastic response to an applied force as it will slowly extend to its final length. This phenomenon is commonly called 'creep'. The extension in response to an applied force is given by

$$\frac{\Delta L}{L} = \frac{T}{E} (1 - e^{Et/\mu}), \quad 3.43$$

and in this case the quantity μ/E is called the 'retardation time' (i.e. the time taken to achieve $1/e$ of its final length).

On removing the load, the converse phenomenon occurs, so that the shortening returns on a different curve to that of the lengthening. The two curves thus form a curve which is known as a 'hysteresis loop'; the area within the loop represents the energy that has been dissipated by the viscous element.

A combination of a Voigt and a Maxwell model (Fig. 3.4, right) is a St Venant model, and this is the simplest one that has been used to analyze the viscoelastic behavior of living materials. This type of response has been pursued in considerable detail by Apter (Apter, 1966; Apter, 1967). The decay in force in stress relaxation is, however, never a simple exponential (Fig. 3.5) and it is necessary to postulate a spectrum of such elements.

As the dynamic behavior of the arterial wall is the periodic strain imposed by the pulse wave, most workers have preferred to analyze the response of the wall to a stress in the form of a simple harmonic motion. In this case, the viscous elements will cause a phase lag of angle ϕ between the applied stress and the resultant strain.

The first formulation to analyze this in physiology was by Hardung (Hardung, 1952; Hardung, 1953; Hardung, 1962), who expressed the elastic modulus E' in complex form, as has subsequently been adopted by others (Bergel, 1961b; Learoyd and Taylor, 1966; Gow and Taylor, 1968; Apter *et al.*, 1970; Gow *et al.*, 1974; Newman *et al.*, 1975;

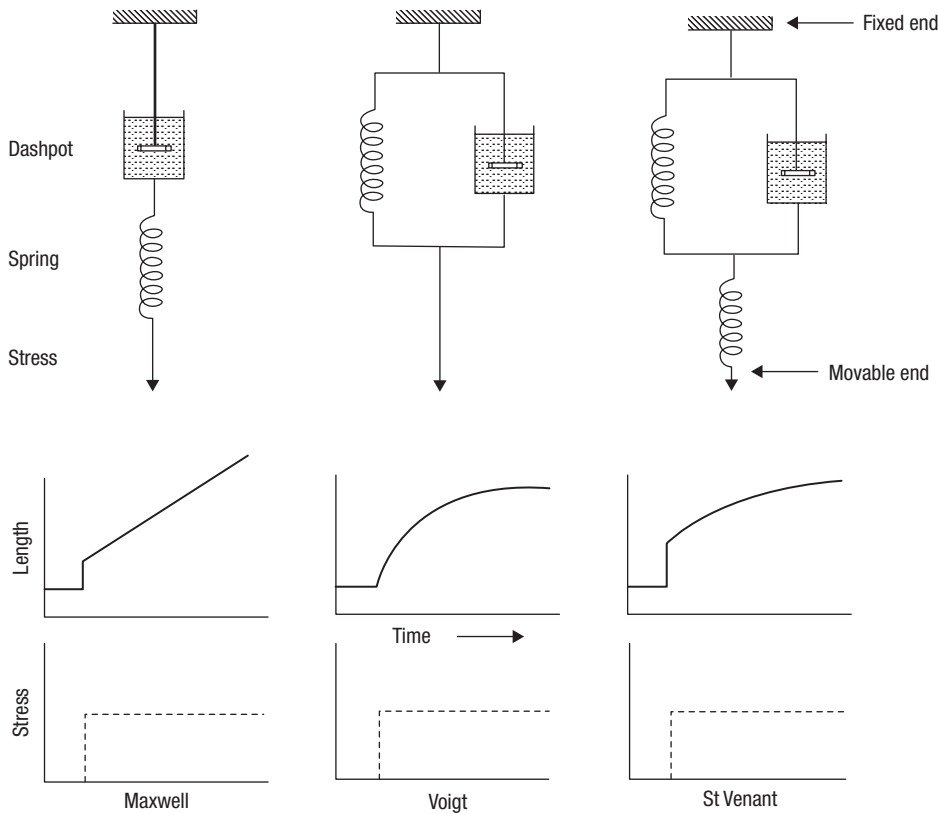


Figure 3.4 Viscoelastic behavior of three standard mechanical models (Maxwell, Voigt, St Venant). Springs represent purely elastic, and dashpots purely viscous, elements. The length response of each model to an imposed step function of stress is shown. Adapted from Milnor (1989)

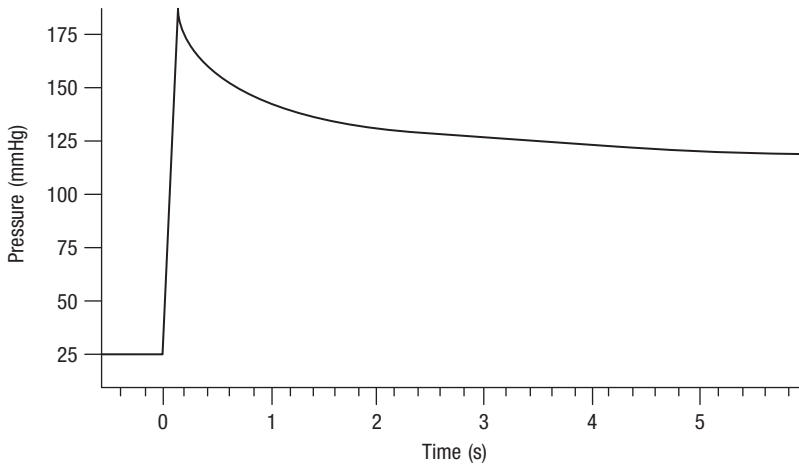


Figure 3.5 Stress relaxation in a dog carotid artery *in vitro*. At time zero, the volume of the arterial segment was suddenly increased, then held constant. Reproduced from Zatzman *et al.* (1954)

Langewouters *et al.*, 1985):

$$E' = E_{\text{dyn}} + i\mu\omega, \quad 3.44$$

where the real part E_{dyn} is given by

$$E_{\text{dyn}} = \frac{\Delta P}{\Delta l} \frac{l_m}{q_m} \cos \phi \quad 3.45$$

and the imaginary part by

$$\mu\omega = \frac{\Delta P}{\Delta l} \frac{l_m}{q_m} \sin \phi; \quad 3.46$$

l_m and q_m are the average length and cross-section of the specimen. Some investigators used strips or rings, while others used arterial segments, where l would represent circumference; it should be mentioned that studies using different specimens may give different results (Cox, 1983a). ΔP is the amplitude of the applied stress and $\Delta l/l_m$ the strain; ω is the angular frequency; ϕ and μ are the phase lag and viscosity as stated above.

The magnitude of the dynamic elastic modulus is given by

$$|E'| = \sqrt{(E_{\text{dyn}} + \mu\omega^2)}. \quad 3.47$$

The viscoelastic properties of the arterial wall have also been represented by a different mathematical formulation by Womersley (1957b). He wrote both the elastic modulus and the Poisson ratio in complex form. This formulation is easier to handle in the wave equations, but is more difficult to visualize in a simple physical setting. Its relationship with the formulation of eqn 3.44 was discussed by Taylor (1959b) and McDonald and Gessner (1968) and has been used to construct one-dimensional and viscoelastic models of the systemic arterial tree to study blood flow and wave propagation (Azer and Peskin, 2007; Bessems *et al.*, 2008; Warriner *et al.*, 2008). Pressure and flow waveforms obtained in the aorta, lower limb arteries, and cerebral arteries using the Witzig-Womersley

model are displayed in Figs 3.6 and 3.7 (Reymond *et al.*, 2009).

The dynamic elastic modulus, which would be expected to rise continuously with frequency (and does so with most viscoelastic substances, such as plastic tubing) in the arterial wall, only increases markedly up to a frequency below 2.0 Hz and thereafter remains virtually constant. Equally, the viscous term μ is highly frequency-dependent, so that the term $\mu\omega$ remains virtually constant with increasing frequency; therefore the ‘viscosity’ must be regarded as something of a mathematical abstraction, and no arrangement of springs and viscous dashpots has yet been found which mimics the dynamic elastic behavior of the arterial wall.

Another way of analyzing the behavior of the arterial wall is to treat it as an elastomer, a class of substances that covers the long molecular chain polymers. Elastomers are grouped (Stacy *et al.*, 1955) by the properties they have in common. Elastin is a representative ideal protein elastomer (Urry *et al.*, 2002):

1. They have a low Young’s modulus, of the order of 10^7 dyne/cm². Most other solids have moduli of the order of 10^{10} to 10^{11} dyne/cm². They thus include all highly extensible rubber-like substances.
2. They can be stretched to great lengths before their breaking point or ultimate strength is reached.
3. Their stress–strain curve is usually sigmoid, or S-shaped.

The theory of large deformations in elastomers in terms of the coiling and uncurling of their long molecular chains using thermodynamic postulates has been quite successful in studying rubber elasticity, and on the assumption that these properties are shared by soft body tissues, elastomeric theory has been applied in physiology by King (1950) and King and Lawton (1950). A review of their analysis was given by Stacy *et al.* (1955). More recently, the mechanical characterization of polyurethane elastomer for biomedical applications has been reported (Kanyanta and Ivankovic, 2010).

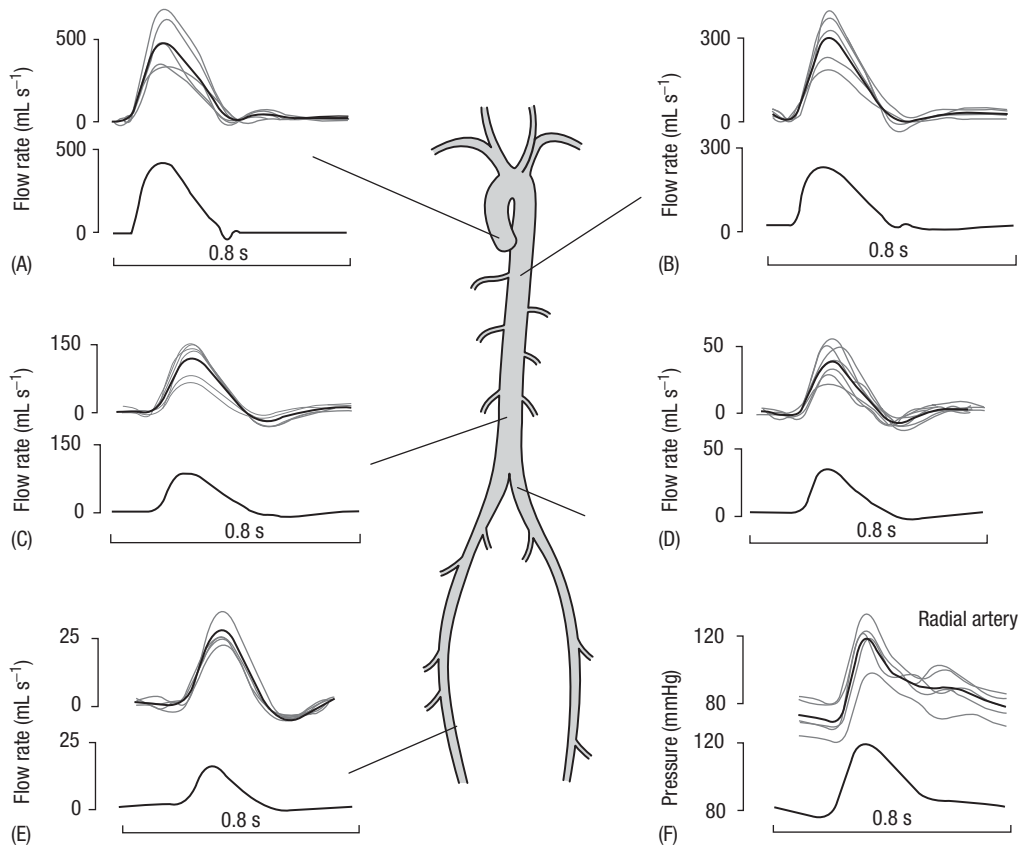


Figure 3.6 A, ascending aorta; B, thoracic aorta; C, abdominal aorta; D, common iliac; E, femoral artery; F, radial artery. Model results (bottom) compared with *in vivo* measurements of flow and pressure waves (top) at various systemic arteries' locations. Thick line represents the averaged waveform. Flow waves were measured using magnetic resonance imaging (MRI) or ultrasound and pressure waves were measured using applanation tonometry. Reproduced from Reymond *et al.* (2009)

WAVE PROPAGATION IN AN ELASTIC TUBE

The elastic property of the wall of a tube plays an important part in determining the velocity of propagation of a pulse wave. The study of this problem is closely related to that of the analysis of the velocity, c_0 , of sound waves in air, which was first studied by Isaac Newton. His equation was

$$c_0 = \sqrt{B/\rho}, \quad 3.48$$

where B was the bulk modulus and ρ the density. The analogy between the velocity of wave propagation in a compressible fluid (e.g. air) and that in an incompressible fluid (i.e. a liquid) in an elastic tube is very similar. Therefore, by substitution of eqn 3.13, the expression for the bulk modulus, into eqn 3.48, we obtain

$$c_0 = \sqrt{VdP/\rho dV}, \quad 3.49$$

which is the form of the equation derived by Frank (1920) and Bramwell and Hill (1922a) from the Moens–Korteweg equation (eqn 3.51). The above derivation from Newton's

equation was, however, first made by Thomas Young in his Croonian lecture of 1809. Further analyses were made by EH Weber (1850) and later in detail by Résal (1876), Korteweg (1878), and Witzig (1914). A stimulating historical review of this subject was published by Lambossy (1950) and experimental validation of a time-domain-based wave propagation model of blood flow in visco-elastic vessels was recently published (Bessemers *et al.*, 2007; Bessemers *et al.*, 2008).

A simple derivation linking Young's treatment with the later work may be made from eqn 3.49 when expanding dV/dP as in eqn 3.25. We may write, with L constant,

$$\begin{aligned} c_0 &= \sqrt{\pi R^2 dP / \rho 2\pi R dR} \\ &= \sqrt{RdP / \rho 2dR}, \end{aligned} \quad 3.50$$

and this form is very useful in calculating the equivalent wave velocity from measurement of the radial dilatation of arteries *in situ* (Aars, 1971a; Aars, 1971b), *in vivo* (Gozna *et al.*, 1974; Merillon *et al.*, 1978; Meinders and Hoeks, 2004), or *in vitro* (Learoyd and Taylor, 1966; Langewouters *et al.*, 1984).

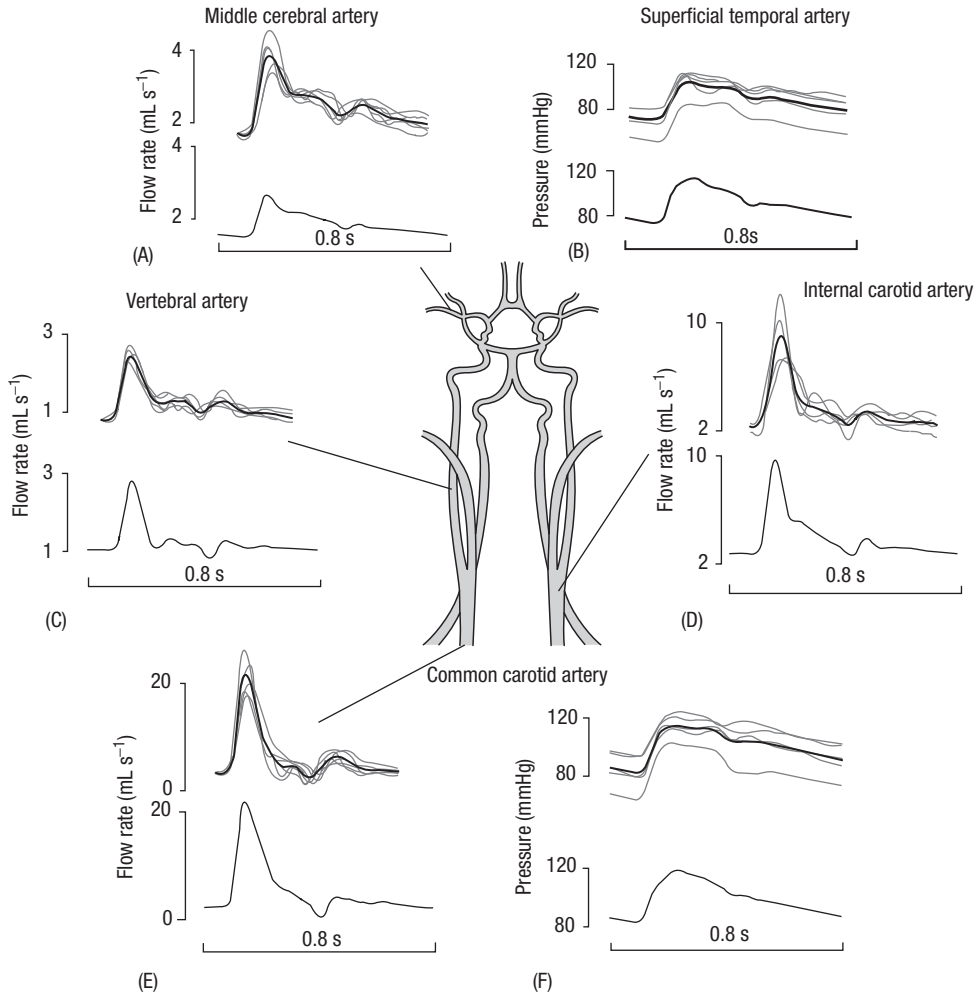


Figure 3.7 Model results (bottom) compared with *in vivo* measurements of flow and pressure waves (top panels) at various cerebral artery locations. Thick line represents the averaged waveform. Blood flow was measured with color-coded duplex ultrasound. Pressure was measured with applanation tonometry in the superficial temporal artery (B) and common carotid (E and F). A, middle cerebral artery; C, vertebral artery. Reproduced from Reymond *et al.* (2009)

Taking eqn 3.49 a step further and substituting the derived value for dV/dP of eqn 3.26 for a unit length of circumference, L , and taking the mean value of P as an arbitrary zero, we obtain, with E as the Young's modulus in the circumferential direction, and h the wall thickness,

$$c_0 = \sqrt{Eh/2R\rho}. \quad 3.51$$

This is known as the Moens–Korteweg equation. It was used with the addition of a constant outside the square-root term by Moens (1878) in his experimental studies of the velocity of a pressure wave in a rubber tube filled with water. He used a value of 0.9 for this arbitrary constant.

The assumptions underlying the Moens–Korteweg equation are that the tube has a thin wall (i.e. $h/2R$ is small) and is filled with an ‘ideal’ incompressible inviscid liquid. The first two assumptions are quite reasonable; the value of $h/2R$ is normally less than 0.1 and, as was seen earlier, the bulk modulus of water is between 1.03 and

1.04 times greater than the elastic modulus of the wall. In larger arteries, the effect of viscosity is also small, but it will greatly retard the velocity in the smaller arteries.

The errors due to assuming that the wall is ‘thin’ have been explored by Bergel (1960). In the Moens–Korteweg equation, the wall is assumed to be so thin that there is no appreciable change in wall thickness. In a wall of finite thickness, Poisson's ratio must be incorporated. The full correction equation by Bergel (1960) is given as

$$\left(\frac{c_t}{c_0}\right)^2 = \frac{(2 - \gamma)}{[2 - 2\gamma(1 - \sigma - 2\sigma^2) + \gamma^2(1 - \sigma - 2\sigma^2) - 2\sigma^2]}, \quad 3.52$$

where c_t is the ‘true’ derived velocity and c_0 is the velocity calculated from eqn 3.51; γ is the ratio h/R_{outer} . Taking a Poisson ratio of 0.5 results in the conclusion that the true derived velocity is higher than c_0 by a factor of $(4/3)^{1/2} = 1.15$ even if the wall is very thin. Only when the Poisson

ratio is equal to zero does the discrepancy disappear. In practice, an effective simplification of eqn 3.52 in terms of eqn 3.51 is

$$c_t = \sqrt{Eh/2\rho R(1 - \sigma^2)}. \quad 3.53$$

The difference between Moens–Korteweg and ‘corrected’ wave velocities was found by Bergel (1961b) to lie between 16 and 24 percent; this discrepancy was at that time not taken to be of great account because the measurement error of the pulse wave velocity was considerable. Using the technique of measuring wave-front velocities with a delay line (McDonald, 1968b), however, Nichols and McDonald (1972) were able to reduce the standard error of their measurements to ± 3.0 percent, so that these refinements of relating the elastic constants to the wave velocity have now become relevant.

In general, calculation of the wave velocity from the elastic modulus has utilized the Moens–Korteweg equation, eqn 3.51 or eqn 3.53; the latter is more accurate when viscoelastic effects are allowed for. Various other modifications have been used but have not been proved to be more accurate. Opatowski (1967) made a theoretic analysis which he considered showed that the definitions for the elastic modulus overestimate the true value. Nevertheless, the Moens–Korteweg and Bramwell–Hill equations are still used to calculate arterial stiffness and pulse wave velocity (Cantini *et al.*, 2001; Westerhof *et al.*, 2006; Hermeling *et al.*, 2010; Taviani *et al.*, 2010).

PULSE WAVE VELOCITY MEASUREMENT IN VISCOELASTIC ARTERIES

The velocity (c_0) of travel (or pulse wave velocity, PWV) of flow velocity, pressure, and diameter waves generated by ventricular ejection has been proclaimed as an important parameter in all the analyses of behavior of the arterial tree and PWV (carotid-femoral and brachial-ankle) has recently been proposed as a marker of cardiovascular risk (Blacher *et al.*, 1999a; Blacher *et al.*, 1999b; Asmar *et al.*, 2001a; Asmar *et al.*, 2001b; Laurent *et al.*, 2001a; Meaume *et al.*, 2001; Shoji *et al.*, 2001; Boutouyrie *et al.*, 2002; Cruickshank *et al.*, 2002; Lebrun *et al.*, 2002; London and Cohn, 2002; Safar *et al.*, 2002a; Blacher *et al.*, 2003; Laurent *et al.*, 2003; Ohnishi *et al.*, 2003; Chirinos *et al.*, 2005; Pannier *et al.*, 2005; Shokawa *et al.*, 2005; Sutton-Tyrrell *et al.*, 2005; Mattace-Raso *et al.*, 2006; Willum-Hansen *et al.*, 2006; Cernes *et al.*, 2008; Deloach and Townsend, 2008; Ikonomidis *et al.*, 2008; Yu *et al.*, 2008; Mitchell, 2009a; Mitchell, 2009b; Mitchell *et al.*, 2010; McEniery *et al.*, 2010a; McEniery *et al.*, 2010b; Tsuchikura *et al.*, 2010; Turin *et al.*, 2010; Vlachopoulos *et al.*, 2010d) and non-invasive assessment of elastic artery stiffness has recently entered the European Society of Hypertension/European Society of Cardiology guidelines for the management of hypertension (Mancia *et al.*,

2007a). c_0 (PWV) is considered the ‘gold standard’ for assessment of arterial elastance (inverse of distensibility), being directly related to direct measurement of arterial stiffness (or elastic modulus, see eqn 3.51) and is highly reproducible (Wilkinson *et al.*, 1998; Sutton-Tyrrell *et al.*, 2001a; Naidu *et al.*, 2005; Grotenhuis *et al.*, 2009; Hickson *et al.*, 2009b; Lee and Park, 2009; Sigrist *et al.*, 2010). Numerous excellent reviews have been published recently (Mackenzie *et al.*, 2002; Schiffrin, 2004; Kass, 2005; Khoshdel *et al.*, 2006; Hamilton *et al.*, 2007; McEniery *et al.*, 2007; Boutouyrie, 2008; Fernhall and Agiovlasitis, 2008; Graham *et al.*, 2008; Nichols *et al.*, 2008; O’Rourke and Hashimoto, 2008; Cecelja and Chowienczyk, 2009; Tomiyama and Yamashina, 2010), including the expert consensus document on arterial stiffness by Laurent *et al.* (2006) and the article from the Reference values for Arterial Stiffness Collaboration (2010). At first thought, PWV is one of the easiest measurements to make. Nevertheless, there is some uncertainty in the results that have been obtained from the various methods. Although the value desired can equally well be obtained from the travel (or transit) time between two measuring sites of the flow (or flow velocity) or diameter wave (or an artificial perturbation) as from the pressure wave over a given distance (see Fig. 3.8), it is technically much easier to measure pressure than flow or diameter, and most of the older previous studies only considered propagation velocity of the invasively measured pressure wave. However, some recent studies have used the blood flow velocity wave to measure PWV in humans (Avolio *et al.*, 1983; Avolio *et al.*, 1985a; Avolio *et al.*, 1986; Lehmann *et al.*, 1993; Sutton-Tyrrell *et al.*, 2001b; Baguet *et al.*, 2003; Blacher *et al.*, 2003; Bonapace *et al.*, 2003; Pannier *et al.*, 2005; Sutton-Tyrrell *et al.*, 2005; Tillin *et al.*, 2007; Jiang *et al.*, 2008; Redheuil *et al.*, 2010) and animals (Hartley *et al.*, 2005; Williams *et al.*, 2007). The methods (ultrasound and magnetic resonance imaging (MRI)) used in these studies are very attractive since they are totally non-invasive. Several methods have been proposed to determine PWV using velocity-encoded MRI, including transit-time (Redheuil *et al.*, 2010), flow-area (Vulliemoz *et al.*, 2002), and cross-correlation (Fielden *et al.*, 2008) methods. The reproducibility and comparison of these different techniques were reported recently (Ibrahim *et al.*, 2010). While each of the methods studied had its own advantage and disadvantage, at high field strength, the transit-time and cross-correlation methods resulted in closer and more reproducible aortic PWV measurements, and the associated image processing required less user interaction, than in the flow-area method. One can obtain very precise values of PWV over short distances (< 10 cm) with the transit-time MRI method (Redheuil *et al.*, 2010), even though time intervals between sampling are 16–32 ms and time of transit < 16 ms. The method requires measurement of the flow pulse at two sites over multiple cycles so that distinct ensemble-averaged flow pulses can be generated. Unfortunately, some values of

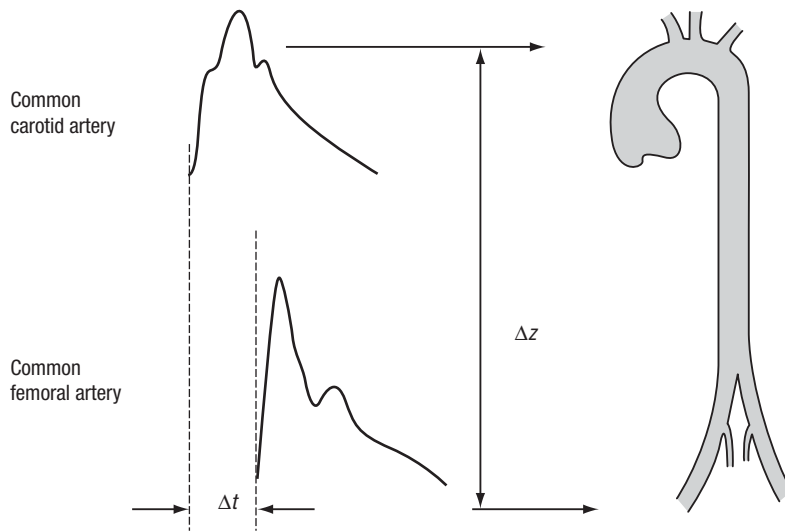


Figure 3.8 Measurement of carotid-femoral pulse wave velocity using simultaneous non-invasive measurements of pressure in the respective arteries using the foot to foot method. The delay time (Δt) between the two pressures is the transmission time and when divided by distance (Δz) between the two pressure recording sites gives pulse wave velocity (c_0 or pulse wave velocity (PWV) in cm/s). Reproduced from Laurent *et al.* (2006)

MRI-determined aortic PWV have not considered this issue, and have been reported (and editorialized) without the error being recognized (need for ensemble-averaging) (O'Rourke *et al.*, 2010a). MRI also permits correct measurement of distance in three dimensions, so that elongation of the aorta with age (which is far greater than dilation with age) can be shown (Sugawara *et al.*, 2008). The MRI techniques, used appropriately, give lower values of ascending aortic PWV in youth, and ascending aortic PWV change with age which are much greater (Redheuil *et al.*, 2010) (about four-fold from age 20 to 80 years) than measured over long distances – from aortic root to aortic bifurcation (Cruickshank *et al.*, 2002) or to femoral artery (Avolio *et al.*, 1983; Avolio *et al.*, 1985a; Nichols and O'Rourke, 2005), as measured by Doppler flow velocity or arterial tonometry. Also, several recent studies have used non-invasive measurement of peripheral artery pressure or displacement waves (e.g. carotid-femoral, carotid-radial, and carotid-dorsalis pedis) and semi-automatic devices to obtain PWV in humans. These include the Complior (Asmar *et al.*, 1995a; Asmar *et al.*, 2001a; Asmar *et al.*, 2001b; Pannier *et al.*, 2001; Smulyan *et al.*, 2001; Kidawa *et al.*, 2003; Millasseau *et al.*, 2005; Khoshdel *et al.*, 2006; Protogerou *et al.*, 2006; Ueda *et al.*, 2008; Ali *et al.*, 2009; Joly *et al.*, 2009; Polónia *et al.*, 2009; Scuteri *et al.*, 2009; Strozecki *et al.*, 2010), the SphygmoCor (Wilkinson *et al.*, 1999; Yasmin and Brown, 1999; Kelly *et al.*, 2001; Millasseau *et al.*, 2005; Heffernan *et al.*, 2006; Li *et al.*, 2006; Qasem and Avolio, 2008; Rajzer *et al.*, 2008; Polónia *et al.*, 2009; Weber *et al.*, 2009; Adkisson *et al.*, 2010; McEniery *et al.*, 2010a; McEniery *et al.*, 2010b), the PulsePen (Alecú *et al.*, 2006; Giannattasio *et al.*, 2008; Joly *et al.*, 2009; Benetos *et al.*, 2010; Salvi *et al.*, 2010), the Vicorder (Hickson *et al.*, 2009b; van Leeuwen-Segarceanu *et al.*, 2010), the PeriScope (Naidu *et al.*, 2005), the Form/ABI (Tomiyaama *et al.*, 2005a; Tomiyaama *et al.*, 2005b; Xie *et al.*, 2009; Tomiyaama *et al.*, 2010; Tsuchikura *et al.*, 2010) and the VaSera (Shirai *et al.*, 2006; Kubozono *et al.*, 2007). Several reviews have been published on methodological

issues (O'Rourke *et al.*, 2002; Pannier *et al.*, 2002; Laurent *et al.*, 2006). The Complior device measures the time delay between two simultaneously measured pressure waves (Fig. 3.8), while the SphygmoCor and the PulsePen measure the time delay between two sequentially measured pressure waves using the electrocardiogram (ECG) as a reference (Fig. 3.9). Distance between the two recording sites must be inserted into each device and is usually measured with calipers or a tape measure applied to the external surface of the body. Aortic PWVs obtained using these devices are somewhat higher than those obtained directly from the artery because the externally measured distance between the two sites is greater than that measured internally (Asmar, 1999; Patel *et al.*, 2003). The difference in distance gives an internally (or invasively) measured aortic PWV that is at least 1.2 m/s less than the externally (or non-invasively) measured carotid-femoral value. Some investigators use the total distance between the carotid and femoral sites (Asmar, 1999; Asmar *et al.*, 2001a; Asmar *et al.*, 2001b; Pannier *et al.*, 2005; Sutton-Tyrrell *et al.*, 2005), while others subtract the distance between the carotid site and the suprasternal notch from the carotid-femoral distance (Avolio *et al.*, 1983; van der Heijden-Spek *et al.*, 2000) and others measure flow velocity in the aortic arch (suprasternal notch) and use the distance from this site to the femoral artery (Avolio *et al.*, 1983; Avolio *et al.*, 1985a). These conditions and recommendations are discussed by van Bortel *et al.* (2002) and Sugawara *et al.* (2010a, 2010b). For the most accurate method, Weber *et al.* (2009) recommend subtracting carotid-suprasternal notch distance from suprasternal-femoral distance if signals (pressure, velocity, or displacement) are recorded in the carotid and femoral arteries. Also, PWV can be measured automatically using an oscillometric technique. The Vicorder system measures aortic (or carotid-femoral) PWV, while the PeriScope, the Form/ABI, and the VaSera systems measure brachial-ankle PWV. All the above methods must measure a pulse displacement signal at two sites of the arterial tree a known distance apart and calculate the

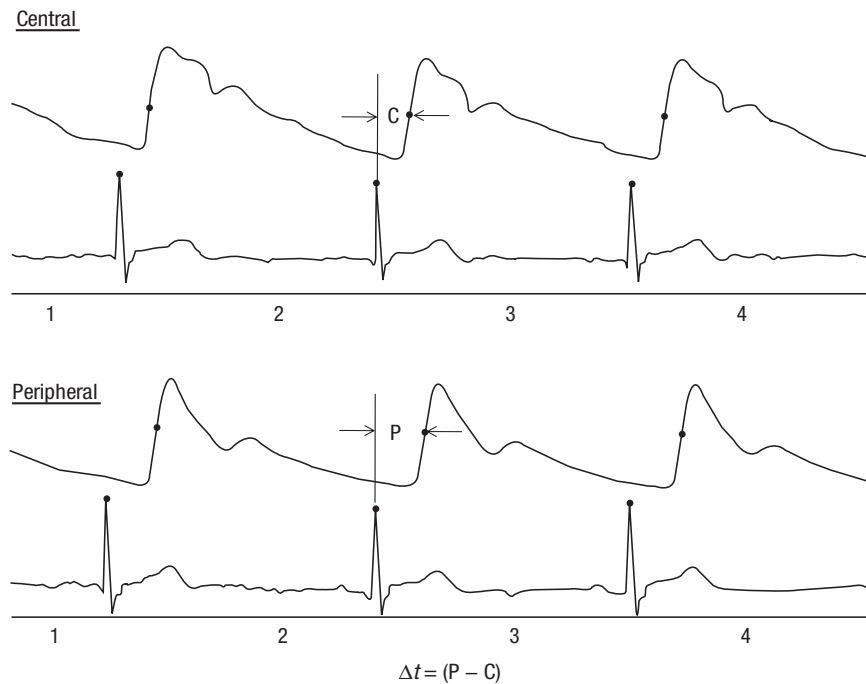


Figure 3.9 Non-invasively sequential recordings of central and peripheral pressure waveforms using applanation tonometry. The SphygmoCor system calculates pulse wave velocity from the time delay between maximum dp/dt of the two pressures using the electrocardiogram (ECG) as a reference

travel (transit) time between them. To eliminate some of the problems associated with the ‘two site’ method for obtaining arterial PWV, some investigators have recently reported results using a ‘one site’ method to calculate local PWV (Harada *et al.*, 2002; Hermeling *et al.*, 2007; Hermeling *et al.*, 2008; Antonini-Canterin *et al.*, 2009; Hermeling *et al.*, 2009; Hermeling *et al.*, 2010; Taviani *et al.*, 2010). These methods use either the water-hammer, Moens–Korteweg (eqn 3.51) or Bramwell–Hill (eqn 3.49) equation and require measurement of two variables such as pressure and blood flow velocity (Doppler ultrasound) or pressure and diameter (ultrasound wall track system) (Hoeks *et al.*, 1990; Hoeks *et al.*, 1992; Kool *et al.*, 1994; Paini *et al.*, 2006; Giannattasio *et al.*, 2008). Qasem and Avolio (2008) proposed a method for determining PWV from waveform decomposition of the central aortic pressure wave measured non-invasively. Other investigators have used MRI (Rogers *et al.*, 2001; Stevanov *et al.*, 2001; Danias *et al.*, 2003; Redheuil *et al.*, 2010) and tissue Doppler imaging (Eriksson *et al.*, 2002) to obtain segmental aortic PWV. MRI could be complementary to the other methods because it can measure diameter and distension of deeper arteries like the aorta (see above). Also, a new oscillometric device (the Arteriograph) was recently introduced to estimate aortic PWV and arterial stiffness (Baulmann *et al.*, 2008; Rajzer *et al.*, 2008; Jatoi *et al.*, 2009; Nemcsik *et al.*, 2009; Trachet *et al.*, 2010). All of these new methods are non-invasive and individual comparisons have been reported (Millasseau *et al.*, 2005; Baulmann *et al.*, 2008; Rajzer *et al.*, 2008; Salvi *et al.*,

2008; Hickson *et al.*, 2009b; Jatoi *et al.*, 2009; Joly *et al.*, 2009; Nemcsik *et al.*, 2009). The main experimental problem associated with PWV measurements using the ‘two site’ method centers around the fact that pressure and flow waves change dramatically in shape as they travel (see Fig. 10.1) and thus raises difficulties in establishing a single value that is definitive for the entire wave. Also, basic experimental studies (Langewouters *et al.*, 1984; Armentano *et al.*, 1991) have shown that the elastic behavior of the arterial system is nonlinear or pressure dependent. PWV and, therefore, arterial stiffness is directly or indirectly influenced by all factors influencing blood pressure (Alfie *et al.*, 2005; Kim *et al.*, 2007). Therefore, before starting hemodynamic measurements, subjects should rest for some time in a quiet room (at room temperature) in an attempt to get cardiovascular function and vasomotor tone as close to basal resting conditions as possible (Van Bortel *et al.*, 2002). This resting period largely differs between investigators from 5 up to 30 minutes (Wilkinson *et al.*, 1998; McVeigh *et al.*, 1999). Study subjects should rest for at least 10 minutes (in general) to minimize deviation from basal resting condition (van der Heijden-Spek *et al.*, 2000). For most methods/devices, a supine position of the subject is obligatory, however, measurements can be made in either arm. With some devices, measurements can also be taken in the sitting position. As blood pressure differs somewhat between the sitting and the supine position, it is likely that PWV will also differ depending on the subject’s position. Like blood pressure, arterial diameter and stiffness show a

diurnal variation. Although Kool *et al.* (1991) observed an increase in arterial diameter and stiffness during sleep, the changes were probably related to changes in blood pressure (Marchiando and Elston, 2003; Kim *et al.*, 2007; Kollias *et al.*, 2009). Indeed, several investigators have derived an arterial stiffness index (AASI) from ambulatory blood pressure measurements (Dolan *et al.*, 2006; Hansen *et al.*, 2006; Li *et al.*, 2006; Andreadis *et al.*, 2010; Li *et al.*, 2010).

After a meal, systemic vascular resistance decreases accompanied with an increase in heart rate and cardiac output (Waalder *et al.*, 1991). In a substantial number of subjects and particularly in elderly subjects, blood pressure decreases after a meal (Imai *et al.*, 1998) and may alter PWV. These hemodynamic changes last for up to 2 hours in the case of a small meal and longer after a large meal (Waalder *et al.*, 1991). Therefore, for repeated measurements in particular, measurements should be performed at the same time of the day, and not within 3 hours after a meal. Subjects should not drink beverages containing caffeine within 3 hours before assessments (Vlachopoulos *et al.*, 2003c; Riksen *et al.*, 2009). Smoking increases arterial stiffness and PWV acutely and lasts for 1.5 to 2.5 hours (Failla *et al.*, 1997), therefore, it is recommended that the study subject refrain from smoking for at least 3 hours before assessments. Chronic exposure (active and passive) to cigarette smoke also causes an increase in arterial stiffness (Malayeri *et al.*, 2008; Xie *et al.*, 2009; Mayer *et al.*, 2010; Tomiyama *et al.*, 2010).

THE VELOCITY OF THE 'FOOT' AND OTHER POINTS OF THE PRESSURE WAVE

The method that has been most commonly used in the past to measure PWV has measured the time of travel of the 'foot' of the wave (pressure, flow velocity, or diameter) over a known distance. The 'foot' is defined as the point, at the end of diastole (Fig. 3.8), when the steep rise of the wavefront begins. In essence, the principle involved is that of determining a point of identity on the traveling wave and using its velocity as characteristic of the whole wave. In view of the various factors that cause the form of the wave to change, the identity of features of the wave cannot be taken as self-evident.

By analogy with the interaction of a transient wave with reflected waves that it creates, the early part of the wave will be little affected by the reflections (see Chapter 9). Therefore, it is reasonable to treat the early wavefront as a region that maintains its identity in the propagated wave (Nichols and McDonald, 1972; McDonald, 1974; O'Rourke, 1982b; Milnor, 1989; Hartley *et al.*, 1997; Mitchell *et al.*, 1997b; Wang *et al.*, 2000; Fitch *et al.*, 2001). The 'foot' of the wavefront is, at first sight, the easiest point to recognize; it is, however, clear-cut only when the wave is recorded at a relatively slow time base. When the sweep speed is increased, as is necessary for recording over

short distances, it becomes much more difficult to define with any precision (see Fig. 10.6). Various techniques have been used to avoid this difficulty. Laszt and Muller (Laszt and Muller 1952a; Laszt and Muller, 1952b; Laszt and Muller, 1952c) defined a point, representing the foot, by extrapolating the wavefront downward and measuring from the intersection of this line with a straight-line extrapolation of the last part of the diastolic curve. Anliker *et al.* (Anliker *et al.*, 1968a; Anliker *et al.*, 1968b) used a similar technique for the superimposed sine waves that they used. Frank (1905) suggested measuring the velocity of a point at one-fifth of the height of the rising limb. Hermeling *et al.* (2009) used the dicrotic notch (or incisura) identified by the second heart sound as an alternative time-reference point to measure local PWV in the common carotid artery by M-mode ultrasonography. In a later article, they suggested that assessment of arterial stiffness should discriminate between systolic and diastolic pressure ranges (Hermeling *et al.*, 2010). Kapal *et al.* (1951) had previously measured the velocities of four 'corresponding points' on the rising limb of the pressure wave and found that they were not appreciably different from each other, nor from the velocity of the foot. They thus confirmed the concept that the rising limb, or wavefront, retains its identity reasonably well. A similar technique was used by Pruett *et al.* (1988) to study the relationship between PWV and arterial blood pressure.

This concept was utilized by McDonald (1968b) and Nichols and McDonald (1972) in developing a technique that used a double-beam cathode ray oscilloscope equipped with an adjustable delay time between the two beams (or pressure waves). Using a double-lumen arterial catheter, with a 5.0 cm interval between the recording points, introduced through a peripheral artery, they recorded the wave velocity at successive 5.0 cm intervals along the length of the aorta of the dog and at slightly larger intervals in the more peripheral vessels. Initially, two pressure waves separated by the transmission time would be seen; after carefully monitoring the two waves in terms of amplitude, the proximal pressure recording was then delayed until the wavefronts were superimposed. The time of delay then represents the time of travel of the wavefront over the known distance. From this, the 'wavefront velocity' is easily calculated. The precision of the measurement depends on the rate of sweep, and this was used at the highest rate that would allow all the wavefront to be retained on the screen. The proportion of the wavefront that can be precisely superimposed is about 75 percent of the wave amplitude in the thoracic aorta; in the femoral or more distal arteries closer to major reflecting sites, the proportion is usually between 30 and 50 percent. The findings of Kapal *et al.* (1951) were thus confirmed. Other points of possible identity can also be checked. Another method of measuring the PWV which is closely linked to that described above was used by Fry and colleagues (Greenfield and Fry, 1962; Patel *et al.*, 1969). In this method, the time derivative of the pulse wave was

recorded and the transit time of the peak of the derivative measured. As this point of maximum rate of rise in pressure is usually about half-way up the wavefront (see Fig. 3.9), or somewhat earlier, this falls within the area that can always be matched in the method of McDonald (1968b) and is thus essentially the same measure. This technique could be easily adapted to the delay-line method and might well be better than using the actual wavefront as it would provide a steeper slope for matching.

THE PHASE VELOCITY OF HARMONIC COMPONENTS OF WAVES

When studies of the velocities of the harmonic components were first undertaken, it was thought that this offered promise of an accurate measure of PWV. It was then found that the measured, or apparent, phase velocity showed great variations with frequency. These variations were shown to be due to the presence of reflected waves, which are discussed in Chapter 9. The pattern of the phase velocities that was found follows that of the impedance modulus very closely; indeed, because these two values are closely related (see Chapter 12 and the discussion below), we can attribute any major discrepancy between the two distributions to measurement error in one or the other. Especially where the apparent phase velocity is high, this error is most likely to be in the phase velocity because the phase shift is small and the phase discrimination using analog-to-digital converters and magnetic tape recording is rarely less than ± 1.0 degrees. In most situations, the apparent phase velocity is high for components of the pulse wave below 2.0 Hz. In the ascending aorta of dogs and young adults, the phase velocity fluctuates very little. In a peripheral artery such as the femoral, the fluctuations (like that of impedance moduli) are still marked above 2.0 Hz, but, nevertheless, have usually become very small above 8.0–10 Hz. The disappearance of reflection effects at these higher frequencies is due to two causes:

1. The attenuation of the reflected wave becomes progressively greater with frequency, and the interaction with the reflected wave, therefore, becomes much less.
2. As the phase velocity is measured over a finite interval, this produces some averaging of the changes, and, as the frequency increases, the measuring interval forms a progressively greater fraction of a wavelength and the averaging effect becomes dominant.

Relationship of the apparent phase velocity to the wavefront velocity

The fluctuations of apparent phase velocity that are caused by wave reflections were shown by Taylor (Taylor,

1957a; Taylor, 1957b) to be around the velocity that is characterized by the elastic properties of the tube. The relatively steady value of phase velocity that is attained in the higher frequency components was, in fact, found to be in good agreement with wave velocities measured from the foot of the wave. McDonald (1968b) and more recently Mookerjee *et al.* (2010) also found that there was good agreement with the wavefront velocity. Furthermore, he found that the mean value of the apparent phase velocity above 2.0–2.5 Hz (which were the slowest rates of the fundamental frequency in this series) was very close to the wavefront velocity that was measured in the aorta. Figure 3.10 shows the frequency spectrum of apparent phase velocities in a dog experiment where the heart had been slowed by vagal stimulation. The mean of all velocity values is shown by the broken line and is substantially higher than the mean of the values above 2.0 Hz, shown by the solid line. This latter value was virtually identical with the wavefront velocity; for this situation, the matching of the mean value of the phase velocities above 2.0 Hz is always extremely close to the wavefront velocity.

McDonald and Taylor (1959) concerned themselves with the steady value that the apparent phase velocities attained at the higher frequencies and argued that the similarity of this value to the velocity of the foot was due to the fact that the sharp inflection there would be largely determined by the higher-frequency components. They assumed that the viscoelastic properties of the arterial wall would cause a progressive increase in the dynamic elastic modulus (and hence the wave velocity) with frequency, as with all classical viscoelastic materials. Thus, the foot-to-foot wave velocity would represent the characteristic wave

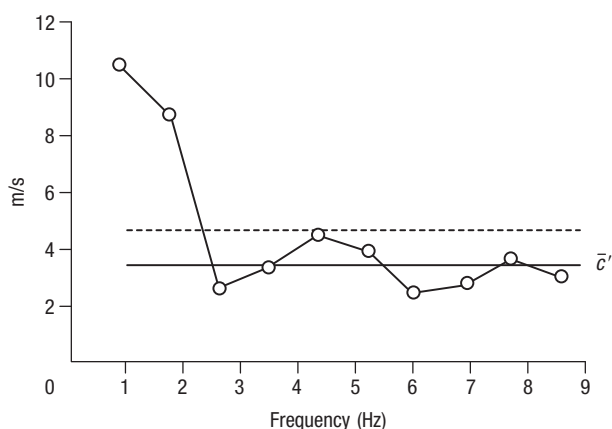


Figure 3.10 The distribution of phase velocities of the pulse wave in the dog ascending aorta when the heart rate had been slowed to below 1.0 Hz by vagal stimulation. It can be seen that the velocity is elevated in the lowest two harmonics, but that the variation above 2.0 Hz is quite small. The mean value of all the harmonics is shown by the broken line and is significantly higher than the mean value of the harmonics above 2.0 Hz (solid line). This latter value coincides with the wavefront velocity

velocity of the higher-frequency components, but not the velocity of the main low-frequency terms. The work of Bergel (Bergel, 1960; Bergel, 1961b), however, showed that the dynamic elastic modulus remains virtually unchanged at frequencies above 2.0 Hz, and probably above 1.0 Hz. Similar observations were made by Callaghan *et al.* (1984) and Li *et al.* (1981) regarding the PWV. The value of the foot, or the wavefront, velocity is thus, on the evidence presented above, a good measure of the characteristic velocity of all the harmonic components of the pulse wave. Alternatively, when a Fourier analysis is being made, the mean value of the apparent phase velocities above the minimum representing the quarter-wavelength frequency is also a satisfactory method. This approach was taken, for example, by Milnor *et al.* (1969) to measure the pulmonary artery PWV.

Relationship of the apparent phase velocity spectra to the input impedance moduli spectra

In discussing the relationship of oscillatory pressure to flow in Chapter 7, it is seen that the derivative of pressure in respect of space, dP/dz , can be written as

$$-\frac{dP}{dz} = \frac{1}{c'} \frac{dP}{dt}, \quad 3.54$$

where c' is the apparent phase velocity (i.e. $c' = \omega \Delta z / \Delta \phi$).

Writing an harmonic component of the pressure P as $|P| \cos(\omega t - \phi)$, we see that

$$\frac{dP}{dz} = -\frac{\omega}{c'} |P| \sin(\omega t - \phi), \quad 3.55$$

and the equation for flow becomes

$$Q = \frac{\pi R^2}{\rho c'} |P| M'_{10} \sin(\omega t - \phi - \pi/2 + \varepsilon_{10}). \quad 3.56$$

The modulus of the impedance is then

$$|Z| = \frac{|P|}{|Q|} = \frac{\rho c'}{\pi R^2 M'_{10}}. \quad 3.57$$

For a given artery, the ratio $\rho/\pi R^2$ changes very little, while the modulus of the Bessel function, M'_{10} , is dependent on the value of α and does not change greatly in the larger arteries as it is asymptotically approaching its limit of 1.0. Therefore, the apparent phase velocity will vary fairly closely with the modulus of the input impedance in large vessels where the value of M'_{10} changes relatively little with frequency. This comparison is seen in Fig. 3.11, which shows data measured in the ascending aorta of a dog during vagal stimulation to slow the heart rate.

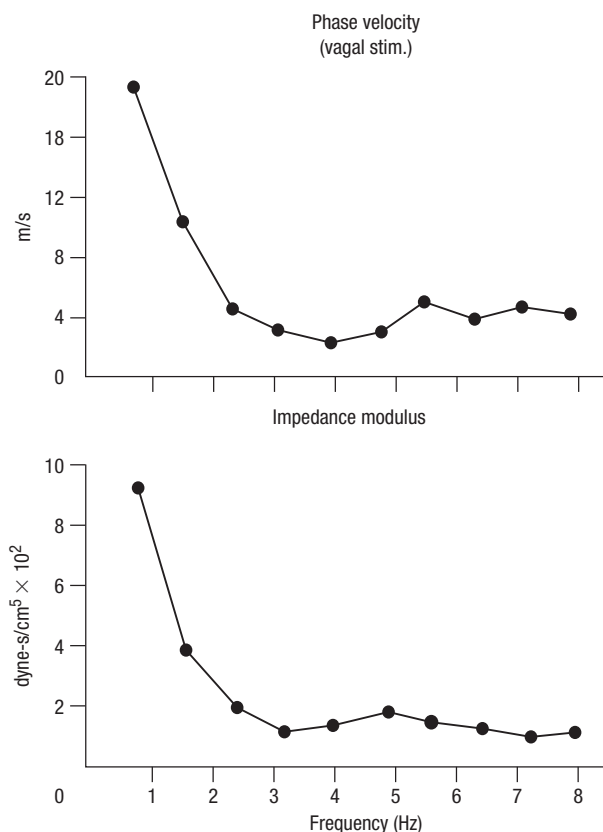


Figure 3.11 Relationship between apparent phase velocity versus frequency measured in the dog ascending aorta and aortic input impedance moduli during vagal stimulation to slow the heart

THE USE OF TRANSIENT EXCITATION TO MEASURE WAVE VELOCITY

All the methods of measuring PWV discussed above use the pressure, flow velocity, or diameter (or distension) wave created by systolic ejection. The presence of wave reflection has posed a problem in interpreting these results. To avoid this problem, Landowne (1957) developed a method in which a transient spike of pressure could be superimposed on the pulse waveform by tapping the artery sharply with a solenoid-operated mechanism. The transit time of the spike between two points in the arteries of the arm was measured and its velocity of propagation determined. In this experiment, the spike was created in the brachial artery (of patients) and the pressure measured at one position in that artery and another in the radial artery. A similar method was used by Arts *et al.* (1979) to measure PWV in the coronary artery of dogs. The velocities that Landowne (1957) recorded were, however, much higher than those recorded by the more usual methods. McDonald and Taylor (1959) assumed that this was due to an increase of elastic stiffness at the high frequencies that constituted the brief transient. This explanation has proved to be untenable in the light of the

work of Bergel (1961b) and others – particularly of Anliker and his group, who made a detailed study of the propagation velocity of transient waves and trains of waves at a high frequency.

The technique used by Anliker *et al.* (1968b) used short trains of some five sinusoidal oscillations (Fig. 3.12). From the Fourier integral of a train of this length, it was shown that approximately 90 percent or more of the energy was represented by the frequency of the oscillation. These oscillations were usually imposed with a mechanical oscillator applied to the arterial wall. The non-axisymmetric component that this produced was found to be attenuated very rapidly, so that if the proximal pressure measurement was made at some 5.0–6.0 cm distant, this wave was found to be virtually axisymmetrical. The advantages of this technique were:

1. The superimposed waves were of small amplitude so that they could be assumed to be propagated linearly.
2. Short trains of high-frequency oscillations are attenuated to such a degree that no reflected waves are created.
3. As the excitation was very brief compared with the length of the cardiac cycle, the propagation velocity could be tested at various parts of the cycle. This last facility was especially used to test the effect of changes in distending pressure on the wave velocity.

There is some suggestion that PWV and, therefore, arterial compliance (or stiffness) is affected by pacing induced

heart rate (Mangoni *et al.*, 1996; Sa Cunha *et al.*, 1997; Stefanadis *et al.*, 1997; Lantelme *et al.*, 2002; Albaladejo *et al.*, 2003; Millasseau *et al.*, 2005; Su *et al.*, 2007). Albaladejo *et al.* (2003) suggest that a relationship between aortic PWV and heart rate exists for males, but not for females, while other studies found an increase in carotid-femoral (Lantelme *et al.*, 2002; Millasseau *et al.*, 2005) and ankle-brachial (Su *et al.*, 2007) PWV in mixed groups of males and females. Local invasive measurements of descending thoracic aortic diameter decreased and distensibility increased during pacing-induced tachycardia (Stefanadis *et al.*, 1997). One common occurrence in all the studies was an increase in arterial diastolic pressure and a decrease in pulse pressure. PWV previously measured using high-frequency vibrations imposed on the pressure wave (see Fig. 3.12) (Anliker *et al.*, 1968a), which has no relationship to heart rate, provides values similar to those obtained using the foot-to-foot method to calculate PWV. These considerations indicate that there should be no effect of heart rate on measured values of aortic PWV. The method used in the studies referred to (Lantelme *et al.*, 2002; Albaladejo *et al.*, 2003), and in others where a relationship between PWV and heart rate seemed possible, was different to that used in older studies. This method employed the Complior device (Asmar *et al.*, 1995a; Asmar, 1999), which did not use the wave foot for identification of pulse arrival, but rather a proprietary algorithm to identify the whole of the rising limb of the pulse wave. In the initial studies (Asmar *et al.*, 1995a; Asmar, 1999), there was no suggestion of heart rate dependency for PWV. However, it must be noted that

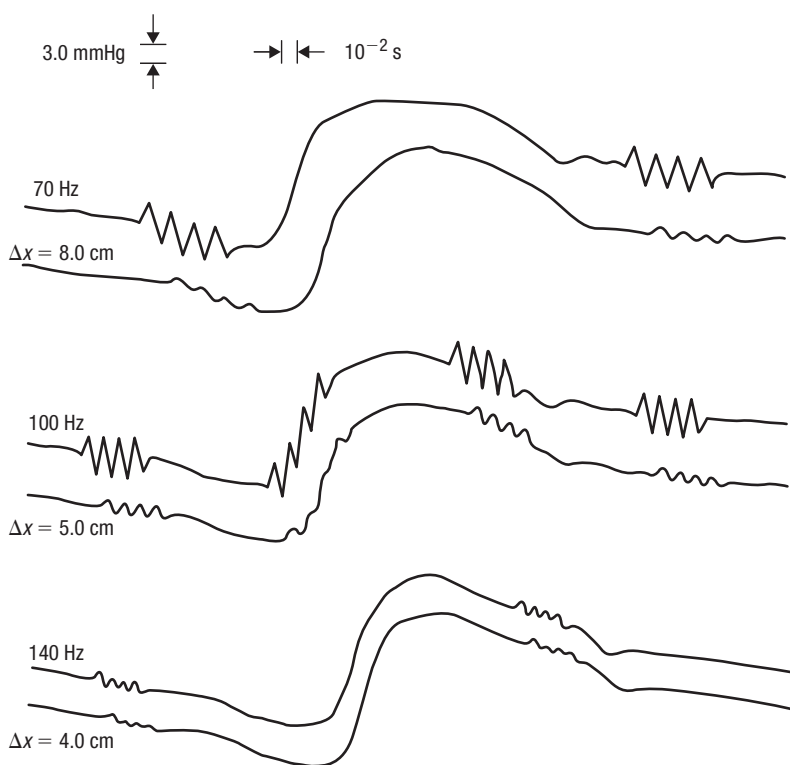


Figure 3.12 An illustration of the technique used by Anliker and colleagues to measure the velocity of radial waves. An external vibrator imposed short trains of high-frequency waves on the intact vessel and the resultant pressure was recorded at two points a known distance apart. The velocity of travel can thus be calculated from the transmission time. From the downstream recording, it can be seen how rapidly these waves are attenuated so that there is no possibility of reflected waves being created and interacting with the centrifugal wave. The frequencies used are far higher than those that can be detected in the natural waves, but as it was also shown that the velocity of pressure waves is virtually independent of frequency, they may be taken as a measure of the velocity of the normal pulse wave. Reproduced from Anliker *et al.*, (1968a)

such studies were less comprehensive than those undertaken more recently (Albaladejo *et al.*, 2003; Millasseau *et al.*, 2005), which have shown dependency on heart rate. The study by Millasseau *et al.*, used both the Complior and the SphygmoCor to measure carotid-femoral PWV.

A positive relationship between spontaneous heart rate (sinus rhythm) and PWV assessed by the conventional method (foot-to-foot) in an epidemiological study was reported by Sa Cunha *et al.* (1997). Also, Mangoni *et al.* (1996) found a positive relationship between arterial stiffness and paced heart rate in rats, while Yasmin and Brown (1999), Wilkinson *et al.* (2002), and Cameron *et al.* (1998) found no relationship between heart rate and PWV in humans. Refinement of the Anliker technique by Hermeling *et al.* (Hermeling *et al.*, 2007; Hermeling *et al.*, 2008; Hermeling *et al.*, 2009; Hermeling *et al.*, 2010) should help clarify this issue.

THE EFFECT OF ARTERIAL DISTENSION ON WAVE VELOCITY

The fact that an increase in mean arterial pressure increases the PWV was appreciated by Frank (Frank, 1905; Frank, 1920) and later studied in detail by Bramwell and Hill (1922b). They used excised arterial segments that they filled with mercury to slow the wave velocity and so allow more accurate measurements of the transit time. They

concluded that the foot-to-foot wave velocity increased proportionately with the diastolic pressure. This concept was extended in another brief paper (Bramwell and Hill, 1923) to assuming that each portion of the wave traveled at the characteristic velocity associated with the pressure at that point. It is apparent by inspection of the changing form of the pulse wave (see Fig. 10.1) that points of equal pressure on the rising and falling phases of the pulse do not necessarily have the same velocity, while the incisura and peak dP/dT travel at the same velocity as the foot of the wave, although at a considerably higher pressure. McDonald and Taylor (1959) and more recently Kim *et al.* (2007) found that while the foot-to-foot velocity correlated well with the mean arterial pressure (and with the minimum diastolic pressure), the velocity of the point of maximum pressure correlated very poorly with it. The fallacy inherent in many of these concepts is the assumption that arbitrary points selected on two successive curves have a specific identity.

By using brief transient excitation, Anliker *et al.* (1968a) were able to study the velocity of a discrete wave at the various pressures of the successive phases of the cardiac cycle. They also varied the mean pressure by vagal slowing or arrest of the heart and by brief periods of aortic occlusion either proximal or distal to the thoracic aorta where all their measurements were made. A typical distribution of the velocity of the transient waves with pressure is shown in Fig. 3.13. This shows a change

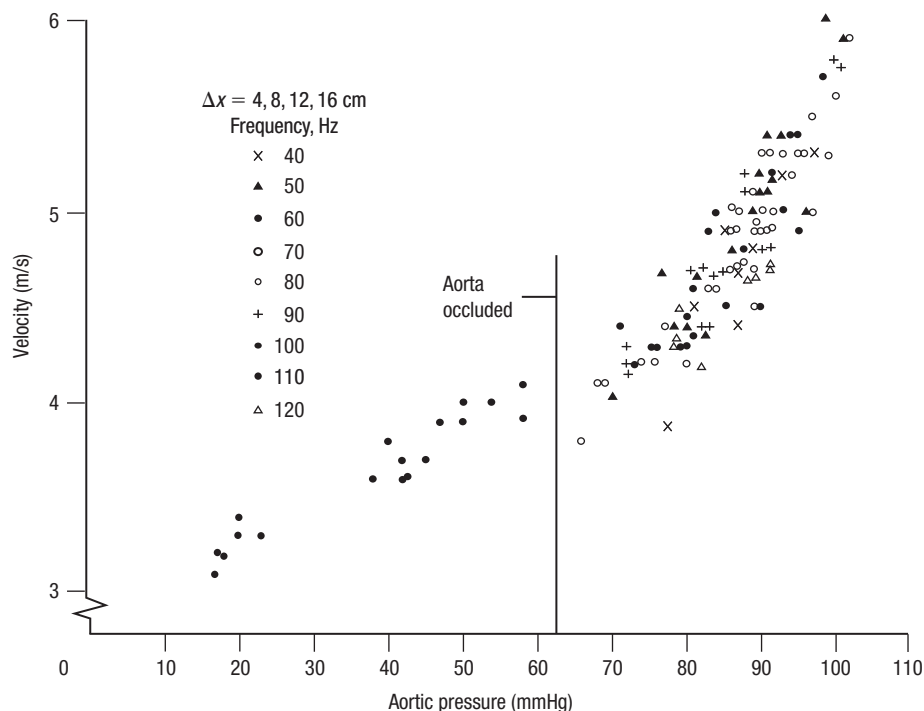


Figure 3.13 The velocity of small imposed transient waves at different transmural pressures. The lowest pressures are recorded during partial aortic occlusion proximal to the site of measurement; to the right of the vertical line, the velocities are taken to be those of the pressures during diastole and systole. As pressure increases, the arterial wall becomes stiffer and pulse wave velocity increases. The increase in wave velocity is more pronounced above pressures of about 75 mmHg. Reproduced from Anliker *et al.* (1968a)

in velocity from about 3.7 m/s at 30 mmHg to 6.0 m/s at 100 mmHg. Fitch *et al.* (2001) observed an increase in aortic pulse velocity from about 4.7 m/s at 110 mmHg to 7.8 m/s at 190 mmHg in rats. This type of behavior is to be expected from the non-linear elastic properties of arteries that was discussed above.

Hughes *et al.* (1979) showed that the modulus of elasticity E of the aorta increases exponentially with increasing intravascular pressure P . The relationship is of the form $E = E_0 e^{\zeta P}$, where E_0 is the zero-pressure modulus, ζ is a constant that depends on the particular vessel (typically $0.016 < \zeta < 0.018$), and $e = 2.71828$. For the dog ascending aorta, Hughes *et al.* reported $E = 667e^{0.017P}$, and for the descending thoracic aorta, the value given was $E = 687e^{0.016P}$.

Substituting $E = E_0 e^{\zeta P}$ into the Moens–Korteweg equation (eqn 3.51) yields

$$c_0 = \sqrt{\frac{hE_0 e^{\zeta P}}{2R\rho}} \quad 3.58$$

Therefore, from Figs 3.13 and 3.14 and eqn 3.58, it is clear that aortic PWV increases exponentially with an increase in arterial pressure.

When the PWV is measured from the mean apparent phase velocities, this value is derived from the whole wave and will only be affected by the mean blood flow velocity, which is less than 5.0 percent of the PWV. Such values are shown at various mean arterial pressures in Fig. 3.14. These velocities were measured in the ascending aorta of the dog (Nichols and McDonald, 1972). It can be seen that the velocity remains unchanged at about 4.0 m/s from a

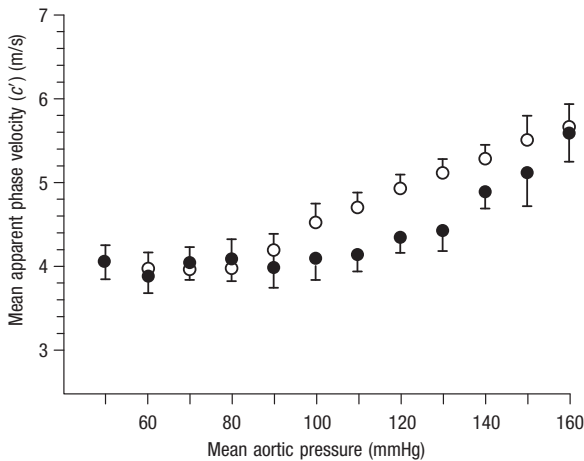


Figure 3.14 Mean apparent phase velocities measured over a wide pressure range (barostat) in the ascending aorta of a group of dogs before (closed circles) and during intravenous infusion of norepinephrine (open circles). Smooth muscle contraction in peripheral arteries increases aortic pressure and elastic modulus causing an increase in wave velocity at pressures above 100 mmHg. Reproduced from Nichols and McDonald (1972)

mean pressure of 60–110 mmHg. A comparable set of values is shown that were measured during continuous intravenous infusion of norepinephrine. The wave velocity is virtually unchanged up to a pressure of 100 mmHg, but thereafter there is an appreciable increase in PWV, presumably attributable to the fact that contraction of smooth muscle cells in peripheral arteries increases aortic pressure and transfers stress from elastin to collagen which causes stiffening of the aortic wall. This dependence of PWV on arterial blood pressure was also observed by Pruett *et al.* (1988).

ATTENUATION OF PROPAGATED WAVES IN VISCOELASTIC TUBES AND ARTERIES

When a pressure wave is propagated in a reflectionless, viscoelastic tube, it is progressively attenuated and its amplitude diminishes exponentially as it travels.

A traveling pressure wave in a reflectionless, long viscoelastic tube model may be expressed mathematically as

$$P_z = P_0 e^{i\omega(t-z/c)} = P_0 e^{i\omega t} e^{-i\omega z/c}, \quad 3.59$$

where $P_0 e^{i\omega t}$ is the pressure at the origin and P_z is the pressure at a distance z downstream; c is the complex wave velocity defined as

$$1/c = (b - ia)/\omega, \quad 3.60$$

where a is the attenuation constant and b the phase constant. In electrical transmission line theory, a complex propagation constant γ is defined as

$$\gamma = i\omega/c = a + ib, \quad 3.61$$

and on substitution, eqn 3.59 becomes

$$P_z = P_0 e^{i\omega t} e^{-\gamma z}, \quad 3.62$$

which can be written as

$$P_z = P_0 e^{-az} e^{i(\omega t - bz)}. \quad 3.63$$

Since the system in question has no reflections, these same equations also apply to the propagation of flow waves. The phase velocity c_1 of the traveling wave is given by

$$c_1 = \omega/b, \quad 3.64$$

so that the velocity of propagation of a wave can be found from the imaginary part of γ , while the attenuation per unit length is given by the real part. The amplitude of an oscillation after traveling a distance z is thus

$$|P_z| = |P_0| e^{-az}, \quad 3.65$$

and after traveling one wavelength, λ , is

$$\begin{aligned} |P_z| &= |P_0|e^{-a\lambda} \\ &= |P_0|e^{-ac/f} \\ &= |P_0|e^{-2\pi a/b}. \end{aligned} \quad 3.66$$

The measured arterial pulse is made up of a forward-traveling wave and a backward-traveling (or reflected) wave (see Chapter 9), so that simple comparison of upstream and downstream pulse amplitude is invalid. The forward wave is attenuated as it travels toward the periphery and the reflected wave is attenuated as it travels toward the heart. With the high-frequency trains of waves used by Anliker and his group, the condition of wave propagation in one direction only is met, and they determined their attenuation coefficients in this manner. To standardize these terms, the attenuation coefficient and its complement, the transmission coefficient, are usually expressed per wavelength (i.e. $e^{-a\lambda}$ and $e^{-\pi a/b}$, respectively).

For pressure waves in the thoracic aorta, the value of $a\lambda$ (termed K in their papers) was between 0.7 and 1.0, and showed no significant variation in the frequency range 40–200 Hz. In the carotid artery, the value of K was between 0.8 and 1.5 for pressure waves.

As the transmission per wavelength in a typical viscoelastic medium such as a rubber tube is markedly dependent on frequency, it is necessary, before using these values in analyzing the propagation and characteristics of the natural pulse wave, to measure the attenuation over the frequency range of the significant harmonic components of the pulse. This is more difficult than for the high-frequency range because of the presence of reflected waves. To measure the attenuation *in vivo* in these circumstances requires the measurement of the pressure at three equally spaced points (Taylor, 1959a; McDonald and Gessner, 1968; Li *et al.*, 1981), the measurement of two pressures, one flow and one diameter (Cox, 1971), or the measurement of two flows and two pressures (Milnor and Nichols, 1975; Milnor and Bertram, 1978); pulse transmission characteristics have also been derived from two pressure recordings (Busse *et al.*, 1979a). The principle of the three-pressure method is outlined below.

If the distance between pressures P_1 and P_2 and P_2 and P_3 are both z centimeters, and pressure P_2 is the sum of a centripetal (or incident) wave (P_i) and a centrifugal (or reflected) wave (P_r) (see Chapter 9), then

$$P_2 = P_i + P_r \quad 3.67$$

we can write

$$P_1 = P_i e^{\gamma z} + P_r e^{-\gamma z} \quad 3.68$$

and

$$P_3 = P_i e^{-\gamma z} + P_r e^{\gamma z}. \quad 3.69$$

Therefore, it follows that

$$\cosh \gamma z = \frac{P_1 + P_3}{2P_2} \quad 3.70$$

or

$$\gamma = \frac{1}{z} \cosh^{-1} \left(\frac{P_1 + P_3}{2P_2} \right) \quad 3.71$$

Experimentally, the difficulties of the method are centered around the placement and precise matching of three manometer systems. Rarely can one position the manometer so that the proximal and distal systems are precisely equidistant from the center one; provided the difference between the two intervals is small, this can be overcome by using an iterative computing technique, as was done by McDonald and Gessner (1968). The relatively small attenuation per centimeter in the lower frequency terms makes it necessary to use as long an interval as possible, and those authors found that a suitable long overall interval of 15–18 cm without the intervening of major branching could only be achieved in the thoracic aorta and common carotid arteries; even so, there was considerable scatter in the results. The method presupposes that the propagation constant is unchanged throughout the length studied (i.e. its elastic properties and internal diameter remain approximately constant). The multisensor catheter used by Latham *et al.* (Latham *et al.*, 1985; Latham *et al.*, 1987) seems ideal for such studies since the pressure sensors are placed at equal intervals along the catheter.

The methods used to measure attenuation of propagated waves were compared in arteries and viscoelastic tubes in the presence of reflections (Ursino *et al.*, 1994; Bertram and She, 1995; Bertram *et al.*, 1997; Bertram *et al.*, 1999). The results from these studies suggest that a full satisfactory method of measuring arterial attenuation has yet to be devised (Bertram *et al.*, 1999).

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Properties of the arterial wall: practice

Arterial wall structure and geometry	81	The role of smooth muscle cells in the arterial wall	97
Measurement of the elastic parameters of the arterial wall	83	Attenuation of propagated waves	99

The function of the systemic arterial system is to deliver blood at high pressure and in a continuous stream to peripheral vascular beds. From a simplistic perspective, it can be separated into three anatomical regions to serve the left ventricle as a pulsatile pump and the tissues that need a supply of blood. Each region has a distinct and separate function: (1) The large arteries, especially the elastic arteries (aorta, brachiocephalic, carotid, etc.), serve predominately as a cushioning reservoir or 'Windkessel' that stores blood during systole and expels it to the tissue during diastole. (2) The long muscular arteries act predominately as conduits, distributing blood to the extremities; these arteries, also, actively modify wave propagation by changing smooth muscle tone and diameter with little change in mean arterial blood pressure. (3) The arterioles, by changing their caliber, alter peripheral resistance and, therefore, aid in the maintenance of mean arterial blood pressure, and the delivery of a steady or continuous flow of blood to the organs and tissues according to their need. Major changes in the central elastic arteries occur over long periods of time, while acute alterations in wall properties (e.g. with rise or fall in distending pressure) are passive (Nichols and Edwards, 2001; Giannattasio and Mancina, 2002). Changes in the muscular arteries and arterioles most often occur acutely and alterations in wall properties are active (Boutouyrie *et al.*, 2000; Nichols and Edwards, 2001; Safar *et al.*, 2003).

For technical reasons, it is much easier to measure the properties (physical and mechanical) of excised perfused arterial segments or rings than it is to measure them in the living animal, and a large amount of data has been obtained using isolated *in vitro* preparations (e.g. Bergel, 1960; Bergel, 1961a; Learoyd and Taylor, 1966; Dobrin and Rovick, 1969; Gow *et al.*, 1974; Langewouters, 1982; Cox, 1983a; Cox, 1983b; Langewouters *et al.*, 1984; Dobrin, 1986; Dobrin, 1999; Paniagua *et al.*, 2000; Paniagua *et al.*,

2001; Laurant *et al.*, 2004; Tahvanainen *et al.*, 2006; Nadasy *et al.*, 2010). It is therefore necessary to consider the validity of observations made in this way when applying them to an artery *in vivo* in the living animal. In the common carotid artery (CCA) of normotensive and hypertensive rats, diameter and distensibility are less *in vitro* than *in vivo* (Zanchi *et al.*, 1998); even prolonged immobilization of the human radial artery causes distensibility to decrease (Giannattasio *et al.*, 1998).

From general principles, the obvious reason for doubting the importance of results obtained from isolated preparations is that a post-mortem change, such as is usually seen in biological tissues, has to occur (Zanchi *et al.*, 1998). However, the truly elastic constituents of the vessel wall are the collagen and elastin fibers that form a closely meshed matrix (Angouras *et al.*, 2000; Masson *et al.*, 2008; Curci, 2009; Danpinid *et al.*, 2010; Timmins *et al.*, 2010). These are chemically very stable – in fact, two of the most stable substances known. However, impairment of blood supply to the arterial wall leads to abnormal morphology of both elastin and collagen fibers of the outer media, resulting in increased stiffness under a wide range of stresses (Angouras *et al.*, 2000). Also, injury to the internal arterial surface (denudation) causes increases in elastin and collagen that appear within about 7 days (Nili *et al.*, 2002) and repeated pulsations over an extended period of time cause alterations in medial elastin that adversely effect its function (Avolio *et al.*, 1998) and can lead to aneurysm disease (Curci, 2009). Elastin loses its functionality in cerebral arteries with aging, leading to stiffer less compliant arteries. However, the area fraction of elastin remains fairly the same. The loss of functionality may thus be attributed to fragmentation and structural reorganization of elastin occurring with age and continued pulsations (Fonck *et al.*, 2009). There is also a smooth muscle component, and, in the absence of an

adequate oxygen supply, this will not maintain a contraction; in other circumstances, it may undergo rigor mortis and go into spasm. In both cases, the tension in the vessel wall will change and thus alter the apparent elastic modulus (or stiffness) (Nili *et al.*, 2002). It has been argued that since a contractile tissue, like smooth muscle, is not essentially elastic, the tissue may be properly studied without the added variations in muscle tension. This is incorrect. The effect of smooth muscle on the arterial wall is discussed separately below.

The important confusion in applying measurements on isolated arterial segments to their function under physiologic conditions arises from the dimensional changes that occur when they are excised. These changes will be considered under three headings: (1) the length of the segment; (2) the thickness of the wall; and (3) the initial dimensions before applying stress.

The length of an arterial segment

Arteries in the body are naturally under a condition of longitudinal tension. This is demonstrated by the fact that if any artery is severed, the cut ends retract to some extent from each other; when it is dissected free from the surrounding tissues, the artery further shortens by a much greater amount. Although this retraction of excised arteries was studied by Fuchs (1900) at the beginning of the twentieth century, the significance of the phenomenon to the study of elastic measurements in excised arteries does not seem to have been appreciated until the work of Bergel (1960), McDonald (1974), and Learoyd and Taylor (1966). Average values for the relative shortening in terms of *in vivo* length in various arteries found by Fuchs (1900) and Bergel (1960) ranged from 21 to 42 percent, with the greatest retraction occurring in the smaller arteries. These values are very similar to those found in normal human arteries by Learoyd and Taylor (1966). In young subjects (under 35 years of age), the amount of retraction was 22 percent in the upper thoracic aorta and 39 percent in the femoral artery. In older subjects (over 35 years of age), retraction was 10–15 percent less than in younger subjects (Fig. 4.1).

A retraction of 40 percent means that a 5.0 cm length in the body will shorten by 2.0 cm when excised. Before the tissue connections are removed, the corresponding shortening is found to be only about 0.3–0.5 cm. This observation, together with the finding that there is very little, if any, longitudinal movement of the artery during passage of the pulse wave, led to the concept called ‘tethering’ of arteries. This longitudinal restraint has been studied quantitatively by Patel and Fry (1966); their data and the implications of tethering in the analysis of wave propagation are discussed by McDonald (1974) and Milnor (1989). It does, however, raise the conceptual problem of the limits of the functional arterial wall when considering excised arteries. Clearly, the effective

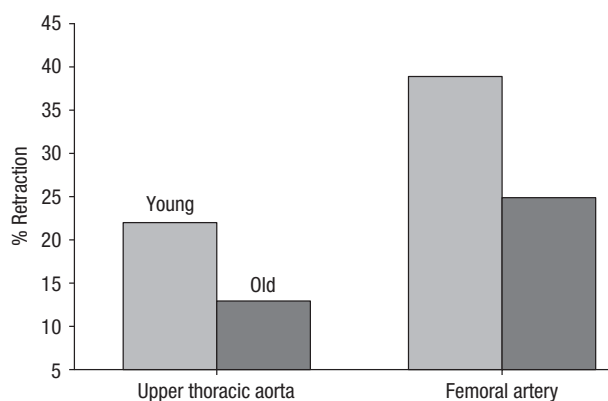


Figure 4.1 Percentage retraction of upper thoracic aorta and femoral artery segments from ‘young’ (11–20 years of age) and ‘old’ (36–50 years of age) subjects. Data from Learoyd and Taylor (1966)

longitudinal elasticity, at least, will be markedly altered by removing this longitudinal restraint from the outside of the artery. Its implications in measuring the thickness and the composition of the ‘true’ wall are discussed below.

When the ends of an excised and retracted artery are blocked and the segment is subjected to an internal pressure, it lengthens. This was studied by Fenn (1957). As an isotropic tube of Poisson ratio 0.5 will remain at a constant length when inflated (eqn 3.32), Fenn concluded that the arterial wall was either markedly anisotropic, with an elastic modulus that was much less in the longitudinal than in the circumferential direction, or that the Poisson ratio was much less than 0.5. As evidence discussed below indicates that the Poisson ratio is indeed very close to 0.5, the artery must be anisotropic. There is, in fact, no question that the excised retracted artery is markedly anisotropic (Dobrin, 1986). For example, McDonald (1974) found that the longitudinal elastic modulus of the excised dog femoral artery was about 2.0×10^6 dyne/cm² for up to 90 percent of its *in vivo* length, while at a distending pressure of 100 mmHg the circumferential elastic modulus was about 10×10^6 dyne/cm².

The lengthening that accompanies distention is not, however, sufficient to account for all natural lengths of an artery in the body. At a distending pressure of 100 mmHg, Bergel (1960) found that the length of the excised segment was about 90 percent of its *in vivo* length and that further increases in pressure produced little extra extension, and the arteries never reached their natural length. There is, however, some variability between arteries. The thoracic aorta will often regain its *in vivo* length, and at high pressures (e.g. 200 mmHg) may even exceed it; in the abdominal aorta, McDonald (1974) and Bergel (1960) found that the natural length is never reached and a considerable additional force is needed to pull it out to its *in vivo* length. Arteries in their natural position must, therefore, always be subject to a longitudinal stress. The reason for this is a matter for speculation, and it is

postulated that it is due to different growth rates between the artery and the surrounding tissues.

In order to study the circumferential elastic modulus of an excised artery, it is necessary to measure its length in the body and then extend and hold it at this length during distension; these measurements will then be relevant to the physiologic situation. Similarly, measurements of the longitudinal elastic modulus should be made for small extensions close to the natural length. Values derived from the force–length curve at initial lengths markedly different from the *in vivo* length will be very different from those at the natural length. Where the elastic modulus or distensibility has been measured by the change in volume on distension of the unrestrained segment, the inaccuracy is very marked. To eliminate longitudinal retraction and minimize the problems associated with measurements on excised arterial segments, some investigators (Frohlich *et al.*, 1992a) have assessed arterial distensibility (static measurement) from pressure–volume curves obtained in intact vessels ligated at two sites a given distance apart. Small catheters inserted into the lumen of the segment are used to measure intra-arterial pressure and volume infusion.

Another type of arterial sample that has often been used is the arterial strip. If one of these is cut in the longitudinal direction, the same considerations apply as for the whole segment. It is more usual, however, to cut the strip in a closely pitched helix as the majority of the muscle cells are orientated in this direction. In this case, it is virtually impossible to estimate the natural length, and data from such preparations must be treated with caution (McDonald, 1974; Cox, 1983a). With arterial rings, there can also be no control of the width of the strip. The free edges in all strips also lead to minor distortions (called ‘edge effects’ in the elasticity literature). Also, care must be taken during preparation so that endothelial cells are not disrupted. Problems associated with isolated arterial rings were realized by Furchgott and Zawadzki (1980) in their early studies on endothelial function. Recent studies have demonstrated that arterioles with diameters about 200 μm taken from the gluteal region of humans can be used to study vessel properties and endothelial function (Paniagua *et al.*, 2000; Paniagua *et al.*, 2001). When subjected to progressive increases in flow, vessel diameter increased in relation to the increased flow rate (or shear stress). This response is dependent upon the presence and integrity of the endothelium. In normal humans, the maximum increase in diameter of the brachial artery following release of a 5-minute occlusion (reactive hyperemia) occurs at about 60 s, while the maximum increase in flow occurs at about 15 s (see Chapter 5) (Stadler *et al.*, 1996; Stadler *et al.*, 1998).

Before 1960, there had been only a limited number of studies of the dynamic elastic properties of arteries *in situ*, although Alexander (1954) had shown that this was feasible. Since the article by Peterson *et al.* (1960), in which a differential transformer was used to measure the radial

pulsation caused by the pulse wave, there have been several studies of this nature using improved techniques to measure pulsatile arterial diameter (see below and Chapter 6). Isolated segments maintained *in situ* were also used by Patel *et al.* (Patel *et al.*, 1969; Patel *et al.*, 1973), Patel and Vaishnev (1972), Zanchi *et al.* (1998), Hwang *et al.* (2007), and Colleran *et al.* (2008) in a series of articles, and this method has the advantage of maintaining the initial length and the tissue attachments. Since 1980, most studies of arterial properties have used ultrasonic techniques to measure arterial diameter. In large animals, ultrasonic crystals are attached to the external surface of the artery and are used to measure pulsatile diameter. This invasive measurement, along with pressure at the same site, has been used in both acute (White *et al.*, 1994) and chronic (Armentano *et al.*, 1991; Barra *et al.*, 1993; Armentano *et al.*, 1995) preparations to study arterial wall mechanics. In human (Stale *et al.*, 1990; Länne *et al.*, 1992a; Länne *et al.*, 1992b; Asmar *et al.*, 1993; Kool *et al.*, 1993; Pedersen *et al.*, 1993; Sonesson *et al.*, 1993; Sonesson *et al.*, 1994; Giannattasio *et al.*, 1995; Weber *et al.*, 1996; Giannattasio *et al.*, 1998; Liang *et al.*, 1998; Stadler *et al.*, 1998; Arnett *et al.*, 1999; Bank *et al.*, 1999) and small animal (Lacolley *et al.*, 1995; Bezie *et al.*, 1998; Zanchi *et al.*, 1998) central and peripheral arteries, ultrasonic (B-mode, M-mode, and echo-tracking) techniques are used to measure pulsatile arterial diameter non-invasively (see Chapter 3). In the carotid artery, ultrasound is used to obtain indices of wall properties. In the aorta and cerebral arteries, the technique is used primarily to detect aneurysms and in peripheral muscular arteries it is used to measure endothelial function.

Also, estimation of local dynamic aortic elastic properties in humans can be obtained non-invasively using magnetic resonance imaging (MRI) and cardiac-gated computed tomography (CT) (Honda *et al.*, 1994; Hundley *et al.*, 2001; Lalande *et al.*, 2002; Rerkpattanapipat *et al.*, 2002; Vulliemoz *et al.*, 2002; Figueroa *et al.*, 2009; Morrison *et al.*, 2009; Redheuil *et al.*, 2010). When these methods are used to calculate compliance and distensibility, arterial pressure (or flow) and diameter (or cross-sectional area) must be measured at the same site, or errors will be introduced in the calculation (O’Rourke *et al.*, 2002). Pulsatile peripheral artery pressure and flow velocity can be measured non-invasively using applanation tonometry and ultrasound, respectively. These tonometric, ultrasonic, MRI, and CT methods are discussed in Chapters 3, 5, 6, 8, and 30.

The thickness of the arterial wall

Nowhere are the difficulties of applying precise elastic theory to the arterial wall more apparent than in the problem of determining the thickness of the wall. Yet, in ordinary physical studies, this would be regarded as very elementary. The imprecision with which one can define

the limits of the outer wall in relation to the surrounding tissues has already been mentioned. In addition, in isolated arterial segment studies, the retraction of the segment will cause the arterial wall to thicken. The radial collapse due to removal of the distending pressure will also create wall thickening. A combination of these factors, plus fixation artifacts, makes estimates of wall thickness from ordinary histologic sections almost useless. However, an article by Fischer *et al.* (1991) reported values of wall thickness (1.05 mm) for the dog descending thoracic aorta in fixed segments that were similar to those calculated *in vivo* using eqn 4.1 below. Estimation of the *in vivo* thickness is based on the assumption that the volume of the arterial wall does not change on extension. This was shown to be the case by Lawton (1954) on arterial strips and, with more confidence, by Carew *et al.* (1968) on arteries *in situ*.

The method of measurement originally used by Dobrin and Rovick (1969) was to determine the volume V of the specimen from its weight W and its density ρ (i.e. from the weight in air and in water). The dimensions of the *in vivo* external (or outside) radius R_o and the length L allow one to calculate the internal radius R_i , and thus the wall thickness $h = (R_o - R_i)$:

$$\begin{aligned} V &= \frac{W}{\rho} = L\pi(R_o^2 - R_i^2) \\ R_i &= \sqrt{R_o^2 - \frac{V}{L\pi}} \\ R_o &= \sqrt{R_i^2 + \frac{V}{L\pi}} \end{aligned} \quad 4.1$$

This method has since been used by several different investigators (e.g. Cox, 1983b; Dobrin and Canfield, 1984; Hajdu *et al.*, 1990; Armentano *et al.*, 1991; Barra *et al.*, 1993; Armentano *et al.*, 1995). The density of the wall is usually taken to be 1.06–1.066 g/mL (i.e. very close to that of blood, which is 1.05–1.055 g/mL). In humans with no cardiovascular risk factors, measurements of CCA intima-media thickness (IMT) using high-resolution B-mode ultrasound showed an increase from 0.59 to 0.95 mm in males and from 0.52 to 0.93 mm in females over the age range 20–79 years (Fig. 4.2) (Jarauta *et al.*, 2010). Furthermore, IMT was found to be significantly higher in hypertensive individuals (Sharma *et al.*, 2009). IMT is also used to estimate arterial mechanical properties (Reneman *et al.*, 2005; Kawasaki *et al.*, 2009) and as a marker of atherosclerosis (Pignoli and Longo, 1986; Yamasaki *et al.*, 1994; de Groot *et al.*, 2009) (see Chapter 23). Also, results from the Chennai Urban Population Study (CUPS) showed that arterial stiffness and endothelial function are correlated with IMT (Ravikumar *et al.*, 2002). The new ultrasound method recently introduced by Rossi *et al.* (2010) provides accurate robust estimates of CCA internal and external diameter and IMT waveforms (see Chapter 8). Peterson *et al.* (1960) obtained a mean value of 0.15 for h/R_o , as did

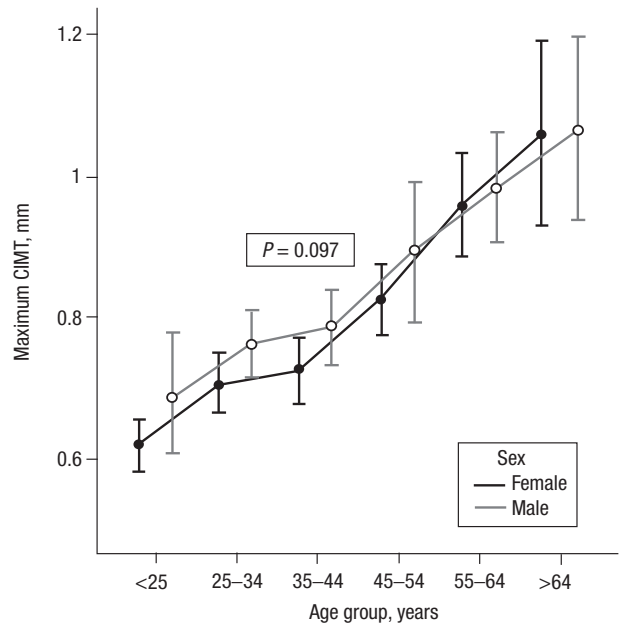


Figure 4.2 Comparison of the mean maximum common carotid artery intima-media thickness (CIMT) among groups of normal humans according to age measured using B-mode ultrasound. Reproduced from Jarauta *et al.* (2010)

Hurthle (1920), while Patel *et al.* (1969) recorded mean values from 0.10 to 0.123 in the thoracic aorta. These values are in the range of those obtained by Bergel (Bergel, 1961a; Bergel, 1961b), Gow and Taylor (1968), and Milnor and Bertram (1978) for various arteries, including the aorta, carotid, iliac, and femoral, and are in close agreement with those reported by Barra *et al.* (1993) in conscious dogs and Pearson *et al.* (1994) in humans. Pearson *et al.* used transesophageal echocardiography (TEE) to assess the effects of age, gender, and hypertension on descending thoracic aortic wall size, thickness, and stiffness. Aortic wall thickness averaged 1.1 mm in diastole in both males and females (mean age 45 years) and decreased to 0.9 at end-systole. The ratio for h/R_o averaged 0.12 and was significantly higher in females than in males (0.124 versus 0.106). Barra *et al.* measured R_o using ultrasonic crystals and calculated R_i for the dog descending thoracic aorta using eqn 4.1 above. The ratio for h/R_o calculated using their data was 0.126 and decreased to 0.116 when the arterial pressure was increased with phenylephrine because wall thickness decreased while radius increased. McDonald (1974) found no significant difference in the ratio for h/R_o in any artery from the carotid to the saphenous of dogs. These results are similar to those found by Learoyd and Taylor (1966) and Fonck *et al.* (2009) in isolated arteries from young human subjects; the ratio for h/R_o was 0.15 in the left posterior cerebral artery (Fonck *et al.*, 2009). In older subjects, however, the ratio increased progressively from the thoracic aorta to the femoral artery and was 0.21 in the left posterior cerebral artery; the ratio for h/R_o is lower for the pulmonary artery

(0.087, Cox (1984); 0.045, Pollack *et al.* (1968)). The discrepancies between the estimates of arterial wall thickness in various studies can obviously lead to discrepancies in the calculated elastic modulus; however, they probably stem from individual variations in the assessment of the outer limit of the arterial wall. This is the decision in answer to the question, ‘what is artery and what is surrounding connective tissue?’ Another relevant question in humans can be, ‘what is the load-bearing part of the artery and what is the intimal component of hyperplastic endothelial cells?’

The ‘initial’ dimensions of the artery: the incremental elastic modulus

In defining Young’s modulus (eqn 3.9) as the ratio of stress (pressure) to strain (dimension), the length of the initial condition was symbolized by x_0 (eqn 3.1). In a tube, the length x_0 is the circumference, or $2\pi R$. In classical elastic theory, R would be the dimension (or radius) where there is no stress. This would indicate the analogy with the artery when there is zero distending pressure; the length of a segment x_0 *in situ* we have already seen to be subjected to longitudinal stress. Similarly, an artery that is cut open longitudinally will open up, indicating that there is a residual circumferential stress, even though it is small (Bergel, 1960). Furthermore, it can be seen from Fig. 4.3 that the stress–strain (i.e. incremental elastic modulus–radius) curve is markedly concave towards the stress

axis (the ordinate). If, therefore, the value of R is always taken as that at zero pressure, the slope of the line deviates greatly from that of the curve and we are dealing with large strains greatly in excess of the very small strains postulated in Hooke’s law. If, however, the stress is increasing by a series of small steps, the strain at each step can be regarded as the ratio of the increment of extension to the length pertaining before the last increment of stress was added. This is the measure commonly used when large extensions are possible in rubber-like material; it is called the ‘incremental elastic modulus’ (E_{inc}) and can be measured non-invasively in intact human arteries using real-time B-mode ultrasound (London *et al.*, 2004a; Aggoun *et al.*, 2008). Because the limit as the increments get very small is the slope of the tangent to the curve, it may also be referred to as the ‘tangential elastic modulus’. As the radial pulsation with each arterial pulse is only a small percentage of the mean radius (see below), this would appear to be a very realistic manner in which to describe the elastic behavior at any given pressure (Bezie *et al.*, 1998). It is to be noted that a statement of the value of the elastic modulus needs to be accompanied by the mean stress (or pressure) at which it was measured. In spontaneously hypertensive rats, E_{inc} was found to be 17×10^6 dyne/cm² at a mean pressure of 172 mmHg compared to 6.3×10^6 dyne/cm² at a mean pressure of 87 mmHg in normotensives (Bezie *et al.*, 1998). In young children, E_{inc} was found to be 14×10^5 dyne/cm² at a mean pressure of 73 mmHg (Aggoun *et al.*, 2008). The derivation of the incremental elastic modulus is illustrated in Fig. 3.3. If the modulus is defined by the strain referred to that at zero distending pressure, this should properly be called the ‘secant elastic modulus’.

ARTERIAL WALL STRUCTURE AND GEOMETRY

Arteries can be roughly subdivided into two types: elastic and muscular, although there are arteries that are intermediate between both types. Elastic arteries have relatively large diameters and are located close to the heart, while muscular arteries are located at the periphery (except in the case of coronary arteries). For a full interpretation of the elastic behavior of the arterial wall, it is necessary to know the composition of the elastic material and how that material is arranged or oriented in the wall (Gasser *et al.*, 2006; Timmins *et al.*, 2010). The predominant elastic materials of the arterial wall are collagen and elastin. They are fibrous in nature, although elastin tends to occur in more or less continuous sheets. Another major constituent of the muscular artery wall is smooth muscle cells (SMC) (Zulliger *et al.*, 2004; Fonck *et al.*, 2007) which, while contributing to the tension in the wall, cannot properly be regarded as a true elastic material (McDonald, 1974). Arterial SMC and fibroblasts continually produce and degrade matrix constituents in preferred modes and at altered rates in response to changing mechanical

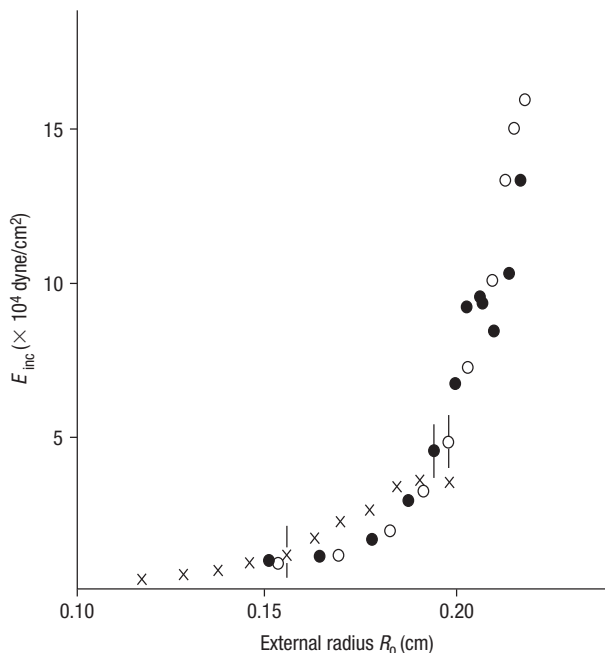


Figure 4.3 Plot of incremental circumferential elastic modulus (stress) versus external radius (strain) for a dog femoral artery. Three successive experiments (different points for each experiment) are superimposed

stimuli. Smooth muscle cells similarly exhibit vasoactive changes that contribute to the control of overall structure, function, and mechanical behavior. Constrained mixture models provide a useful framework in which to quantify arterial growth and remodeling, for they can account for cell-mediated changes in individual structurally significant constituents. The level of smooth muscle activation or tone may change in response to biomechanical stimuli such as flow or pressure (Bevan and Laher, 1991), hormonal stimuli, neural stimuli, and drugs. Furthermore, smooth muscle tone has been proposed to be a very important parameter in arterial remodeling (Zulliger *et al.*, 2004). Collagen and elastin are complexed with an amorphous substance formed of mucoprotein. This substance is not itself elastic, but contributes to the elastic and viscous properties of the artery.

The arterial wall is divided into three concentric layers or regions (see Fig. 3.1): the tunics intima, media, and adventitia (Gasser *et al.*, 2006). Demarcation between the tunics intima and media is provided by the internal elastic lamina. This is a complex structure that consists of a fenestrated membrane of elastin lined on the intimal side by a coarse fibrous network. The tunica intima consists of the vascular endothelium, which is normally, in youth, a single layer of cells, and a thin layer of elastin and collagen fibers that anchors it to the internal elastic lamina (Chapter 5). The outer elastic lamina demarcates the tunica adventitia. The adventitia is a region of collagen and some elastin tissues that merges with the surrounding connective tissue consisting of nerves, small blood vessels, and fibroblasts. The tunica media forms the major part of the wall and is the determinant of the vessel's mechanical properties. The intermediate layers have a fibrous structure, the fibers running circularly or in a tight helix. Between these layers lie smooth muscle, cells mostly parallel to the elastin, although some are oriented longitudinally. The structure of the tunica media has been subjected to detailed study by Wolinsky and Glagov (1964), who described it as being made up of an orderly array of lamellar units. Extensibilities of the lamellae are equal across the media, suggesting equal stiffness of the lamellae (Dobrin, 1999). Because of the interlocked structure of elastin, smooth muscle, and collagen in the media, these findings suggest that although the media is histologically heterogeneous, it acts mechanically as a homogeneous material. At physiologic distending pressures, in experimental animals and young humans, elastin and collagen fibers and SMC are precisely oriented and form well-defined layers. Relatively thick elastin bands are seen as concentric plates, while finer elastin fibers form networks between them. Collagen fibers are dispersed in the interstices and are arranged circumferentially. They are laid down in the wall with some slackness (accordion fashion) and are initially not stretched. This arrangement provides a safety net to prevent vessel bursting at high transmural pressures. Although the finer elastin fibers connect with the elastin lamellae, no attachments between

collagen and elastin fibers can be demonstrated. Anatomic (Walmsley, 1984; Bezie *et al.*, 1998) and physiologic (Attinger, 1968; Dobrin and Doyle, 1970; Bagshaw and Attinger, 1972) evidence indicates that the SMC extend circumferentially between adjacent lamellae among the fine elastin and collagen fibers, and connect to the latter. The first analysis of the three-dimensional state of residual stress and stretch in an artery wall consisting of the three layers, modeled as a circular cylindrical tube, was reported recently (Holzapfel and Ogden, 2010a). The analysis is based on experimental results on human aortas with non-atherosclerotic intimal thickening documented in a recent paper by Holzapfel *et al.* (2007). The intima was included in the analysis because it has significant thickness and load-bearing capacity, unlike in a young, healthy human aorta. The mathematical model takes into account bending and stretching in both the circumferential and axial directions in each layer of the wall (Holzapfel and Ogden, 2010b). The relative amounts of the various components of the arterial wall have been measured in a number of dog (Harkness *et al.*, 1957; Fischer and Llauro, 1966; Cox, 1978) and human studies (Faber and Moller-Hou, 1952; Cleary, 1963). Results showed that, as in most soft living tissue, water comprises most of the wet weight of the wall (approximately 70 percent). Of the dry weight, about 50 percent is made up of elastin and collagen; the remainder consists of smooth muscle and non-fibrous matrix. The distribution of elastin and collagen differs strikingly between the central and the peripheral arteries. In the proximal aorta, elastin is the dominant component, while in the distal abdominal aorta the content reverses, and in peripheral arteries collagen dominates (Fig. 4.4). Harkness *et al.* (1957) found that in the thoracic aorta, elastin formed 60 percent of the total fibrous element and collagen 40 percent. In the extrathoracic arteries, the composition reversed, with elastin forming 30 percent and collagen 70 percent. The transition occurred rapidly over the distal 5.0 cm of the thoracic aorta above the diaphragm and over a similar distance in the branches leaving the arch of the aorta. Thereafter, no significant change in composition could be found in any artery.

A similar study was reported by Apter *et al.* (1966) for the aorta, in which chemical analyses were compared with histologic estimates. This produced divergent results between the two methods used for estimating collagen, but both are substantially in agreement with Harkness *et al.* (1957). Fischer and Llauro (1966) also estimated arterial collagen and elastin content. Their studies showed the same distinction in content between the ascending aorta (the only intrathoracic vessel they studied) and the extrathoracic arteries. Cleary (1963) also confirmed the findings of Harkness *et al.* (1957) in dogs and found that a similar distribution occurred in rabbits. In humans, he found similar values for the thoracic aorta, with a high elastin content, but made the interesting observation that the transition to the low-elastin type of artery did not occur until the distal abdominal aorta was reached. The

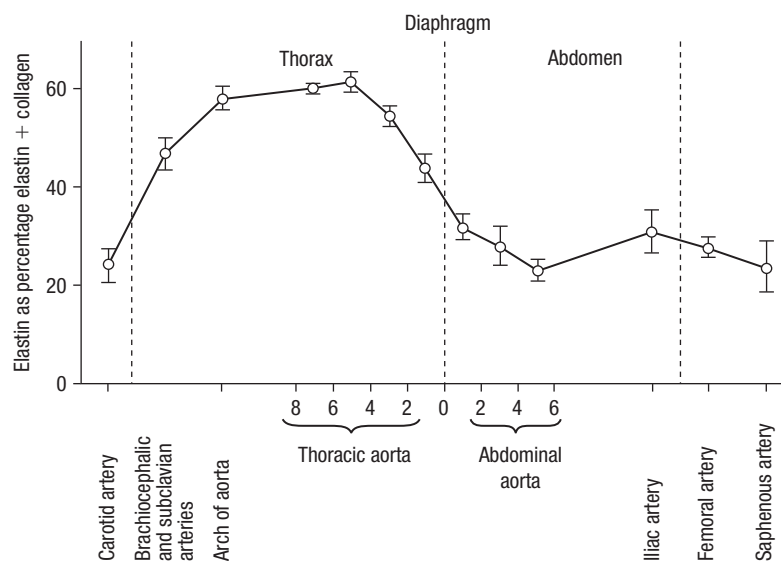


Figure 4.4 The chemical composition of the fibrous elements of the arterial wall of a dog. The vertical axis represents elastin expressed as a percentage of total elastin and collagen. Note that elastin greatly predominates in the thoracic aorta, while outside the thorax the reverse is the case. The transition from one type to the other occurs over about 5.0 cm, and there is no significant difference in composition between the extrathoracic arteries even though the elastic modulus steadily increases with increasing distance from the heart. Reproduced from Harkness *et al.* (1957)

elastic modulus of collagen (1000×10^6 dyne/cm²) is much higher than that of elastin (3×10^6 dyne/cm²) (Dobrin and Canfield, 1984; Armentano *et al.*, 1991; Barra *et al.*, 1993; Armentano *et al.*, 1995), so that as the distance from the heart increases, the arteries become stiffer and the elastic modulus (Learoyd and Taylor, 1966; Gow, 1972) and pulse wave velocity (Nichols and McDonald, 1972; Latham *et al.*, 1985; Latham *et al.*, 1987) increase (Fig. 4.5). This relation is verified by the Moens–Korteweg equation (eqn 3.51).

Dobrin and Canfield (1984) reported results suggesting that in the intact dog carotid artery, subjected to physiologic levels of strain, elastin bears a portion of both circumferential and longitudinal load and provides for structural integrity and for the compliance of the artery at low pressure, whereas collagen bears a portion of only circumferential load and gives the tensile resistance required at high pressures (Fonck *et al.*, 2007). Most theoretical models used to characterize the mechanical behavior of the arterial wall have assumed a linear elastic theory (Gow and Taylor, 1968; Cox, 1978; Vatner *et al.*, 1984). Since the stress–strain relationship of the arterial wall is non-linear, its elastic response cannot be represented by a single elastic modulus. Using a simple model in which elastin and collagen are arranged in parallel and in which the collagen fiber recruitment occurs as transmural pressure increases, Armentano *et al.* (Armentano *et al.*, 1991; Armentano *et al.*, 1995) divided the single elastic modulus into two elastic moduli, one representing the stiffness due to elastin fibers at low levels of pressure and the other representing the stiffness due to collagen fibers at high levels of pressure (Fig. 4.6). Since at low and normal levels of pressure the resistance to stretch is due to elastin alone (Roach and Burton, 1957), the first portion of the pressure–diameter relationship exhibits a linear behavior and can be characterized by a linear mathematical model. At high levels of pressure, the pressure–diameter relationship becomes curvilinear with a

steeper slope, thus indicating the participation of collagen behavior. The transition (or break point) from elastin to collagen fiber recruitment in the dog descending thoracic aorta occurs at a pressure of 110–125 mmHg (Fig. 4.7).

MEASUREMENT OF THE ELASTIC PARAMETERS OF THE ARTERIAL WALL

The circumferential elastic modulus

Measurement of elastic parameters of intact arteries usually falls into two main groups: those relating the radial pulsation (change in diameter or area) of the artery to the pulsatile change in distending pressure, and those which derive the elastic modulus from pulse wave velocity (Laurent *et al.*, 2006; Masson *et al.*, 2008; Nemes *et al.*, 2008) by means of the water hammer (eqn 4.4), Moens–Korteweg (eqn 3.51) and Bramwell–Hill equations (eqn 3.49) (see Chapter 3). This latter relationship was useful in the past in checking the accuracy of values obtained by the first general method when the performance of diameter gauges was in question. The checking of one method against the other in the same animal was clearly desirable. However, since the introduction and use of non-restraining and non-invasive ultrasonic, MRI, and CT techniques to measure arterial diameter, it is not necessary to check their accuracy against another method (see Chapter 6).

Bergel (Bergel, 1960; Bergel, 1961a; Bergel, 1961b) made a large series of measurements on isolated segments of dog arteries. The vessels were held in air at their *in vivo* length and the pressure within was recorded while the volume was changed. The external diameter was recorded by directing a collimated light beam across the vessel on to a photoelectric tube. Changes in the diameter of the

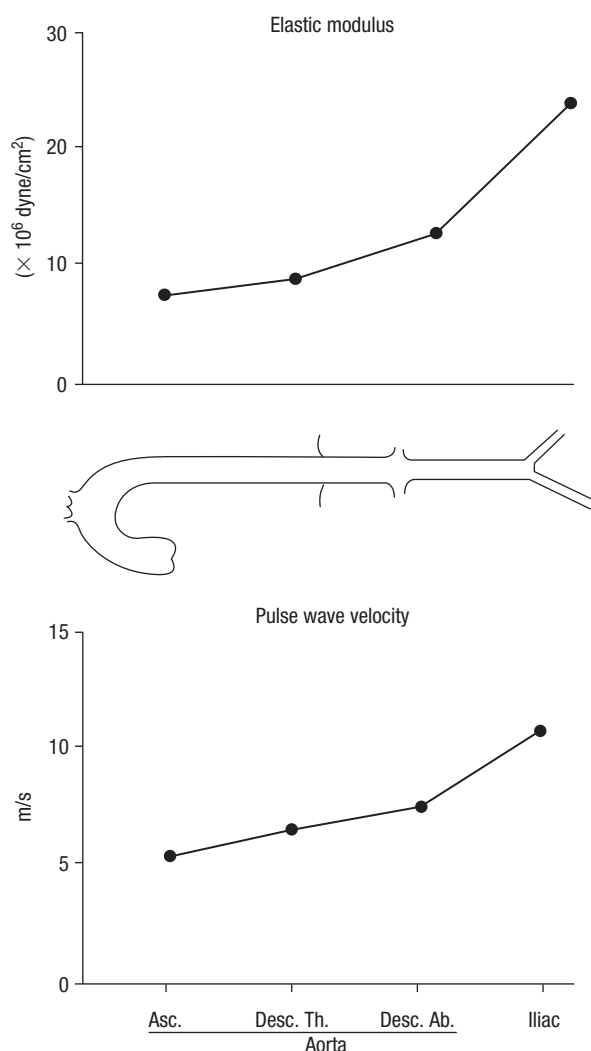


Figure 4.5 Elastic modulus and pulse wave velocity measurements in the ascending aorta, descending thoracic aorta, abdominal aorta, and iliac artery of a young normal subject. Both elastic modulus and pulse wave velocity increase as the distance from the heart increases, owing to decreased elastin and increased collagen content. Reproduced from Nichols *et al.* (1987)

segment caused a variation in the light falling on the tube. The use of a light beam ultrasonography and MRI to measure wall movements has the great advantage that they impose no restraint on the arterial wall (see Chapter 6). The Bergel method is, however, impossible to apply *in situ*, and difficult to use in a bath of oxygenated solution to maintain the functional activity of the muscle cells. However, the relatively new ultrasound, MRI, and CT methods can be applied in intact animals and humans (Honda *et al.*, 1994; Bezie *et al.*, 1998; Giannattasio *et al.*, 1998; Figueroa *et al.*, 2009; Joly *et al.*, 2009; Rossi *et al.*, 2010).

Bergel performed two sets of experiments. The first involved static (Bergel, 1961a), or maintained, distensions,

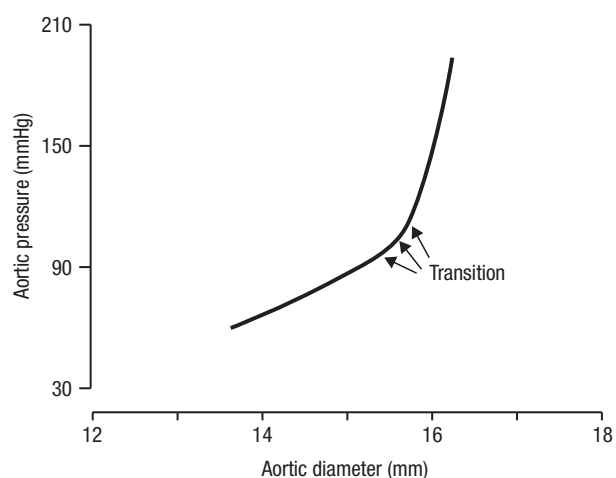


Figure 4.6 Pressure-diameter relation obtained in the descending thoracic aorta of a conscious dog; ultrasonic crystals were used to measure aortic diameter. Aortic pressure and diameter were varied by nitroglycerin and angiotensin administration. At low levels of arterial blood pressure, wall stress is supported by elastin fibers, while at higher levels of pressure, wall stress is supported by collagen fibers. In this example, transition occurs at a pressure of 110–125 mmHg. Reproduced from Armentano *et al.* (1991)

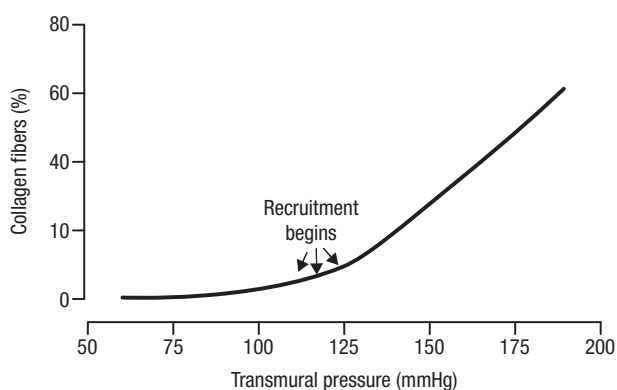


Figure 4.7 Recruitment of collagen fibers as a function of transmural pressure in the descending thoracic aorta of a conscious dog. Recruitment begins at an arterial pressure of about 115 mmHg. Reproduced from Armentano *et al.* (1991)

each held for 2.0 minutes to allow for the completion of stress relaxation. The first few inflations after the artery had been removed from the body for some time showed marked hysteresis loops (i.e. the descending pressure limb was markedly divergent from the curve for increasing pressure). After these initial cycles, there was very little hysteresis of the curves obtained in slow steps.

The average values for the incremental elastic modulus (E_{inc} , eqn 3.38) at various pressures are displayed in Fig. 4.8. It will be seen that the points for the abdominal aorta, femoral and carotid arteries fall close together in these static measurements. The thoracic aorta is markedly less

stiff in the pressure range 80–180 mmHg, but the arteries become indistinguishable at 60 mmHg; the thoracic aorta actually appears stiffer than the carotid above 220 mmHg. These relations are altered with age in the human arterial tree (see Chapter 19). An increase in stiffness with age is virtually confined to the aorta and central, predominantly elastic arteries; there is little or no consistent change in stiffness of the peripheral muscular arteries with age (McEniery *et al.*, 2007). Central elastic artery stiffness is less than peripheral muscular artery stiffness in youth. However, in middle-age central aortic stiffness equals and then exceeds peripheral muscular artery stiffness in old age. This difference in stiffness between elastic and muscular arteries has been referred to as a reversal of the arterial-stiffness gradient (Mitchell *et al.*, 2004a). Findings

of increased stiffness and collagen with age in the central, but not the peripheral arteries of humans are consistent with the theory previously proposed that fracture of elastic load-bearing elements of the arterial wall causes the wall to stretch, so that stress is transferred to the stiff collagenous elements in the wall, just as when an artery is acutely distended by a greater blood pressure within (Fig. 4.9) (Stewart *et al.*, 2006; Kim *et al.*, 2007). These age-related changes in arterial wall properties indicate that left ventricular afterload is confined to the large proximal elastic arteries during the aging process. Bergel (Bergel, 1960; Bergel, 1961b) used the same apparatus mentioned above to determine the dynamic viscoelastic properties of arteries. Instead of incremental volume changes, a sinusoidal pump oscillated the pressure inside the segments and the variations in diameter were monitored continuously. The mean pressure was held at 100 mmHg. The ratio of diameter to pressure was analyzed into real and imaginary parts and calculated as E_{dyn} (the dynamic elastic modulus) and $\mu\omega$ as in eqns 3.44, 3.45, 3.46, and 3.47. The lowest frequency measured was 2.0 Hz, and it was found that there was no significant rise in elastic modulus up to 20 Hz. The phase lag was between 0.1 and 0.15 radians, with the result that the term $\mu\omega$ remained virtually constant over the measured frequency range. This means that the viscosity term μ falls inversely with frequency; this is unlike any known viscous material and indicates that the simple conceptual model of a single spring and dashpot should not be regarded as a realistic representation. In an ‘orthodox’ viscoelastic substance, both E_{dyn} and $\mu\omega$ rise steadily with frequency.

Similar investigations were made by the same techniques on canine and human coronary arteries by Gow and Hadfield (1979) and Gow *et al.* (1974), and on other human arteries by Learoyd and Taylor (1966). One interesting point of distinction between the human values and the values for the dog is that in the group of young humans (under 35 years), the elasticity of the carotid artery was the same as that of the aorta, while that of the femoral artery was much higher, whereas in the dog the value for the carotid artery was very similar to that for the femoral. The groups of subjects were not very large, but it is possible that this difference may be connected with the normal upright stance of humans.

Learoyd and Taylor (1966) also studied the effect of change of length of the specimen beyond its natural length

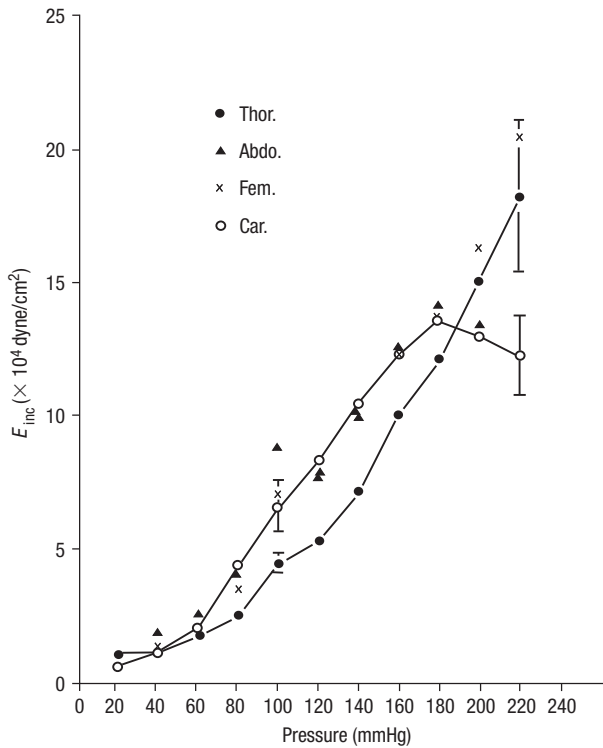


Figure 4.8 The average incremental static elastic modulus for dog arteries from four different sites (thoracic aorta, closed circles; abdominal aorta, closed triangles; femoral artery, crosses; carotid artery, open circles)

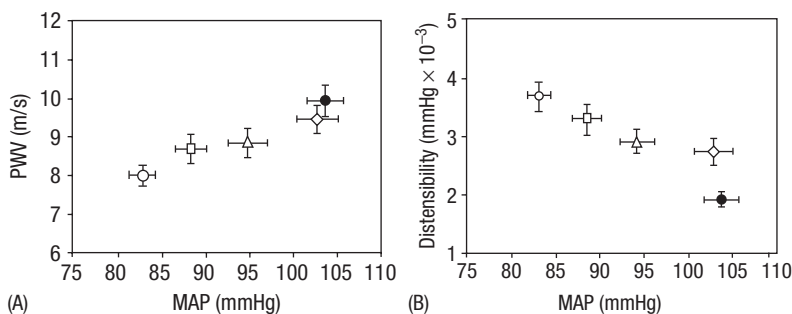


Figure 4.9 Carotid-femoral pulse wave velocity (PWV) (A) and carotid distensibility (B) in normotensive subjects during infusion of saline (closed circle); Ang II, 75 ng min⁻¹ (open square); Ang II, 150 ng min⁻¹ (open triangle); and Ang II, 300 ng min⁻¹ (open diamond). Reproduced from Stewart *et al.* (2006)

on the circumferential elastic modulus. When the modulus was plotted against the external radius, the effect of lengthening was to increase the circumferential elastic modulus markedly. However, when the elastic modulus was plotted against internal pressure, there was little change, and indeed there was a tendency for a decrease in elastic modulus. This had also been observed by Bergel (1960). The reason for this effect is that, at any given distending pressure, an increase in length will cause a decrease in radius. As the tension in the wall is a product of pressure and radius, the initial tension will be reduced, thus reducing the elastic modulus. This suggests that some complexity in considering the non-linear static elastic behavior of arteries would be avoided if the stress ordinate were to be plotted as tension.

The dynamic elastic behavior of human arteries studied by Gow and Hadfield (1979) and Learoyd and Taylor (1966) was qualitatively similar to that reported by Bergel (1961b) and Gow *et al.* (1974) in the dog. The values of the elastic modulus obtained in humans were all distinctly greater than those in the dog (Nagai *et al.*, 1999). For excellent reviews on the biomechanics and fiber orientation in the arterial wall see Gasser *et al.* (2006) and Kassab (2006).

The viscoelastic properties of arterial strips were studied by Lawton (1955), Cox (1978), and Katsuda *et al.* (2004). Qualitatively, the results are very similar to those quoted above, and Hardung showed that the rise in modulus from the static value to the dynamic value is complete at a frequency of 1.0 Hz and almost so by 0.5 Hz. Remington also did extensive work using arterial strips (e.g. Remington (1957)). Numerous specimens, including strips, rings, and cylindrical segments, have been used to study the mechanical properties of arteries (Dobrin and Rovick, 1969; Cox, 1974; Gow *et al.*, 1974; Cox, 1982; Dobrin and Canfield, 1984; Dobrin, 1986; Dobrin, 1999). Improved methodologies have been developed using ring segments and intact cylindrical segments of arteries and employed for studies of active and passive mechanics (Hajdu *et al.*, 1990; Frohlich *et al.*, 1992a). Cox (1983a) compared the arterial wall mechanics derived using these two techniques and found some significant differences. He concluded that these differences could be related to the intrinsic difference in specimen geometry that exists in the two preparations. Maximum values of active force development were not found to be significantly different between the two preparations. Therefore, ring segments or cylindrical segments can be used in studies of active force development/muscle length relationships. However, data on passive mechanics derived from ring segments do not accurately represent mechanical properties of intact cylindrical segments.

Frohlich and colleagues studied the changes in cardiovascular mass, left ventricular pumping ability (Frank-Starling curves), and aortic distensibility after chronic ingestion of angiotensin-converting enzyme (ACE) inhibitors or calcium antagonists (Frohlich *et al.*, 1992a) in

normotensive and spontaneously hypertensive rats. As stated above, aortic distensibility was obtained by ligating a segment of the intact vessel proximal and distal to the carotid artery and constructing pressure–volume curves. All agents tested were vasodilators (or smooth muscle relaxants); they improved aortic distensibility (i.e. a rightward shift of the pressure–volume curve; an example is shown in Fig. 4.10), decreased left ventricular mass (in hypertensive rats), and improved left ventricular pumping ability. Reduction in elastance (or stiffness) with vasodilators in the elastic aorta of rats may be exaggerated because of the small caliber and large proportion of SMC. In larger human elastic arteries, vasoactive drugs have minimal direct (or active) effects on vessel wall properties (Nichols and Edwards, 2001; Weber *et al.*, 2003), however, elastic artery wall properties will change with a change in distending pressure caused by the drug's action on muscular arteries (see Fig. 4.9). Hajdu *et al.* (1990) studied the effects of aging on mechanics and composition of cerebral arterioles in rats (see below). Pressure and diameter were measured in pial arterioles through an open skull preparation (Baumbach *et al.*, 1988).

Radial pulsation and indices of arterial stiffness

Intact living arteries have been studied by several investigators using various methods to measure radial pulsation (i.e. pressure, flow, diameter, and area) and calculate indices of arterial stiffness (or elastance) and its

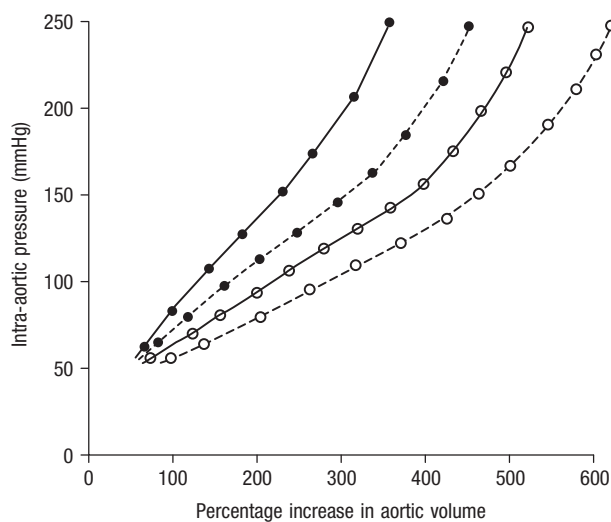


Figure 4.10 Pressure–volume curves obtained from the aortic arch of normotensive (open circles) and spontaneously hypertensive (closed circles) rats before (solid lines) and after (broken lines) a 3-week treatment with the calcium antagonist nifedipine. Nifedipine treatment improved aortic distensibility (i.e. decreased the slope of the pressure–volume curve, indicating decreased stiffness) in both normotensive and hypertensive rats. Data from Frohlich *et al.* (1992a)

inverse (compliance or distensibility). The early studies on arterial properties were performed in anesthetized dogs. Patel and Fry (Patel and Fry, 1966) (see Chapter 6) published a series of articles using a strain gauge caliper designed by Mallos (1962). Peterson *et al.* (1960) used a differential transformer, a device that moves a metal core in a coil and changes the inductance. Gow (1966), Gow and Hadfield (1979), Gow *et al.* (1974), and Gow and Taylor (1968) introduced an electrical caliper that uses the same principle and is extremely lightweight. The development of reliable miniature ultrasonic crystals that can be attached to the external surface of the artery has made possible continuous measurement of arterial diameter in both anesthetized (White *et al.*, 1994) and conscious animals (Latson *et al.*, 1988; Armentano *et al.*, 1991; Barra *et al.*, 1993; Armentano *et al.*, 1995). Arndt *et al.* (1971), Leitz and Arndt (1968), Gozna *et al.* (1974), Merillon *et al.* (1978), and Stefanadis *et al.* (1987) used x-ray angiography to record carotid artery and aortic diameters in cats and humans, and Fleischmann *et al.* (2001) used contrast-enhanced helical CT scans (Rubin *et al.*, 1998) in humans to establish the relationship between mean segmental aortic diameter and age, sex, and body surface area. Baumbach *et al.* (1988) and Hajdu *et al.* (1990) used a Leitz microscope attached to a closed-circuit video system to measure pial arteriolar diameter in rats. A similar method was used by Paniagua *et al.* (Paniagua *et al.*, 2000; Paniagua *et al.*, 2001) to measure the diameter of arterioles taken from the gluteal region of humans.

Due to its high resolution and non-invasiveness, vascular ultrasound is the modality of choice for measurement of vascular physiology *in vivo*, especially in awake humans. Non-invasive ultrasound methods for measurement of arterial diameter (or distension) include B-mode (Arndt *et al.*, 1968; Levenson *et al.*, 1981; Safar *et al.*, 1981a; Safar *et al.*, 1981b; Simon *et al.*, 1983a; Simon *et al.*, 1983b; Safar *et al.*, 1986a; Safar *et al.*, 1986b; Levenson *et al.*, 1987a; Stefanadis *et al.*, 1990; Celermajer *et al.*, 1992; Celermajer *et al.*, 1993a; Pedersen *et al.*, 1993; Celermajer *et al.*, 1996; Bonyhay *et al.*, 1997; McGrath *et al.*, 1998; Stadler *et al.*, 1998; van Popele *et al.*, 2000a; Dinunno *et al.*, 2001; Myers *et al.*, 2002), M-mode (MacSweeney *et al.*, 1992; Alessandri *et al.*, 1995; Schmidt-Trucksäss *et al.*, 2003) and echo tracking (Länne *et al.*, 1992a; Länne *et al.*, 1992b; Sonesson *et al.*, 1993; Länne *et al.*, 1994; Laurent *et al.*, 1994; Armentano *et al.*, 1995; Weber *et al.*, 1996; Ahlgren *et al.*, 1997; Liang *et al.*, 1998; Arnett *et al.*, 1999; Bank *et al.*, 1999; Liao *et al.*, 1999; Lenard *et al.*, 2000; Giannattasio and Mancina, 2002; Studinger *et al.*, 2003). Of these, echo tracking is the most frequently applied method (see Chapter 3). These ultrasonic methods for measuring arterial diameter have been reviewed by Stadler *et al.* (Stadler *et al.*, 1996a; Stadler *et al.*, 1996b; Stadler *et al.*, 1997), Reneman and Hoeks (2000), Pannier *et al.* (2002), and Laurent *et al.* (2006). They can be used to measure diameter changes in the aorta (ascending, arch, and descending) and carotid, brachial, radial, iliac,

and femoral arteries and are applicable to measurements of endothelial function (flow-mediated dilation) and arterial compliance and distensibility. Ultrasonic methods have been used to measure arterial diameter in awake humans (adult and children), the human fetus (Brodzski *et al.*, 1998; Fujita *et al.*, 2002; Giannattasio *et al.*, 2008) and small animals (Lacolley *et al.*, 1995). Also, TEE allows a comprehensive evaluation of the size, morphology, and distensibility of the thoracic aorta (Pearson *et al.*, 1994; Tomochika *et al.*, 1996; Meissner *et al.*, 1999; Agmon *et al.*, 2003).

As mentioned above, the most recent method used to measure arterial pulsations and local (or segmental) dynamic elastic properties and pulse wave velocity is MRI (Honda *et al.*, 1994; Grotenhuis *et al.*, 2009; Ibrahim *et al.*, 2010; Markl *et al.*, 2010; Redheuil *et al.*, 2010). The method has been used to measure thoracic aortic diameter and determine compliance and distensibility in normal subjects (Vullemoz *et al.*, 2002; Potthast *et al.*, 2010; Redheuil *et al.*, 2010) and in patients with ischemic heart disease (Bogren *et al.*, 1989), heart failure (Hundley *et al.*, 2001; Rerkpattanapipat *et al.*, 2002), Marfan syndrome (Groenink *et al.*, 1998; Lalande *et al.*, 2002; Nollen *et al.*, 2004), aortic atherosclerosis (Harloff *et al.*, 2010), and essential hypertension (Resnick *et al.*, 1997). Furthermore, a recent large clinical trial (the Multi-Ethnic Study of Atherosclerosis (MESA)) using MRI reported an adverse relation of proximal aortic wall thickness and distensibility to cardiovascular risk factors (Malayeri *et al.*, 2008). El-Barhoun *et al.* (2009) measured Circle of Willis artery diameters; basilar artery diameter showed an increase with age, but the other arteries did not. These non-invasive methods are ideal for measuring arterial diameter and are discussed in more detail in Chapters 3, 6, and 8. Intravascular devices have also been developed for measuring arterial diameter in large animals (Hughes *et al.*, 1985) and humans (Bank *et al.*, 1995; Stefanadis *et al.*, 1995b). With all mechanical devices, especially if they have a spring in them, it is difficult to determine whether the gauge introduces a restraint on the arterial wall. Discrepancies between the values of elastic modulus reported by different workers have been attributed to this factor (McDonald, 1974). Patel *et al.* (1964) argued that the higher values of elastic modulus reported by Bergel were due to an absence of blood supply in the excised arteries. Indeed, recent results have shown that removal of the vasa vasorum leads to an acute decrease in distensibility of the ascending aorta (Stefanadis *et al.*, 1995a; Angouras *et al.*, 2000). Even gentle dissection of an artery can lower its distensibility. The values of wave velocity from Patel *et al.*'s results were always considerably higher than those measured *in vivo*; the elastic modulus is also lower in *in vivo* preparations (Gow and Hadfield, 1979; Zanchi *et al.*, 1998). Later, the strain gauge used by Patel was replaced with a lighter one, with a resulting decrease in the elastic modulus recorded (Patel *et al.*, 1969). These results are very similar to those obtained by Bergel (1961b) and Gow and Taylor (1968).

Gow and Taylor (1968) also attempted to measure the viscoelastic components of arteries *in situ* by subjecting the pressure and diameter curves to harmonic analysis. For the lower harmonics, they recorded a small phase lag of radial movement behind the pressure, as Learoyd and Taylor (1966) and Bergel (1961b) had found in excised specimens and as Armentano *et al.* (1995) had found in the intact aorta. The behavior of the small higher harmonics, however, became quite unpredictable. They attributed this to the non-linear behavior of the arterial wall causing interaction between the harmonic terms, and they demonstrated that calculated small non-linearities would produce similar effects. The variations in pattern between individual animals were, however, marked, and it would be difficult to give a generalized quantitative measure of the non-linear terms.

Arterial stiffness (or distensibility) is emerging as the most important determinant of increased systolic and pulse blood pressure in our aging community, and therefore, the root cause of a host of cardiovascular complications and events, including left ventricular hypertrophy and cardiac failure, aneurysm formation and rupture, and a major contributor to atherosclerotic and small artery disease and thus to dementia, stroke, myocardial infarction, and renal failure.

Although appreciated for years (Roy, 1880; Bramwell and Hill, 1922a; Bramwell and Hill, 1922b), it is only in recent times, after acceptance of the ill effects of systolic and pulse pressure in the elderly (He and Whelton, 1999), that serious attention has been directed at precise measurement of arterial stiffness. The issue, although superficially simple, is complex, as older treatises on the subject will attest (Bergel, 1961a; Bergel, 1961b; Gow, 1980).

A host of indices have been introduced to quantify arterial stiffness. As is usual when multiple indices exist, no one has proved superior, and all have problems in measurement and interpretation. The subject was addressed at a satellite symposium of the 1994 International Society of Hypertension (ISH) meeting in Sydney, and a list of indices for measurement of arterial stiffness was put forward, and agreed to as an interim measure (Table 4.1) (Cheng *et al.*, 2002; Mackenzie *et al.*, 2002; O'Rourke *et al.*, 2002; Oliver and Webb, 2003). These measurements were recently embraced by an expert consensus document on arterial stiffness (Laurent *et al.*, 2006). Methods and devices for measuring these indices have also been reported (Pannier *et al.*, 2002), as well as terminology for describing them (Gosling and Budge, 2003) and their clinical applications (Laurent *et al.*, 2006; O'Rourke and Franklin, 2006; O'Rourke, 2009a) (see Chapter 3). There are reservations associated with many of these indices because of one or more of the following conditions: (1) use of inappropriate arterial models (e.g. Windkessel); (2) assumption of cardiac output values obtained using methods that have been poorly validated; (3) relate proximal diameter change to pressure change at a distant site; (4) influenced by heart rate or myocardial contractility; and (5) neglect of wave reflections. General recommendations are as follows (O'Rourke *et al.*, 2002):

1. As a generic term, stiffness is preferable, especially to compliance and distensibility, which is more frequently used to describe adherence with therapy or advice, or adherence to protocol.

Table 4.1 Definitions and units of various indices of arterial stiffness

Index	Equation	Units (cgs)
Arterial distensibility	$(D_s - D_d)/(P_s - P_d)D_d$	cm ² /dyne
Arterial compliance	$(D_s - D_d)/(P_s - P_d)$	cm ³ /dyne
Volume elastic modulus	$[(P_s - P_d)/(V_s - V_d)]/V_d$	dyne/cm ²
Peterson's elastic modulus	$[(P_s - P_d)D_d]/(D_s - D_d)$	dyne/cm ²
Young's elastic modulus	$[(P_s - P_d)D_d]/(D_s - D_d)h$	dyne/cm
Pulse wave velocity	$(z_1 - z_2)/(t_1 - t_2)$	cm/s
Pressure augmentation	$(P_s - P_i)$	dyne/cm ²
Augmentation index	$[(P_s - P_i)/(P_s - P_d)] \times 100$	Percent
Characteristic impedance	$(P_i - P_d)/\text{Peak flow (or velocity)}$	dyne-s/cm ⁵ or dyne-s/cm ³
Stiffness index (β)	$[D_d \ln(P_s/P_d)]/(D_s - D_d)$	Non-dimensional
Large artery elasticity index (or capacitive compliance, C_1)	Relation between arterial volume fall and pressure decline during exponential diastolic pressure decay	cm ⁵ /dyne
Small artery elasticity index (or oscillatory compliance, C_2)	Relation between oscillating arterial volume change and oscillating pressure change around exponential diastolic pressure decay	cm ⁵ /dyne

P_i , pressure; D , diameter; V , volume; h , wall thickness; $(z_1 - z_2)$, distance between measuring sites; $(t_1 - t_2)$, travel time of pulse; d , diastole; i , inflection point; $\ln$, natural logarithm; s , systole.

2. For regional measurements, diameter and pressure or flow velocity and pressure should be measured at the same point. When this cannot be done, note should be made of possible confounders, such as age, drugs, or heart rate.
3. For regional and other measurements, note should be made of distending pressure, and comparisons made at the same distending pressure. If technical correction for distending pressure is not possible, validated statistical 'adjustment' should be carried out and clearly described.
4. For indices that give absolute values, absolute initial size should always be given.
5. For global measures of arterial mechanics, analysis of the arterial pressure waveform is often performed. Although various techniques (and underlying models) are used, they should all take into account forward (or incident) and reflected (or backward) waves as important components of the measured waveform.
6. A preference is given to pure physical measures, especially pulse wave velocity.
7. Reference values for all arterial properties must be given as a function of age.
8. Indices should be comprehensively applicable under different conditions.

A direct comparison of the values for the elastic modulus (or arterial stiffness) from all published accounts is not possible because in most cases the wall thickness was not recorded, so that the 'pressure-strain' elastic modulus, E_p , introduced by Peterson *et al.* (1960) has been substituted (Imura *et al.*, 1986; Nagai *et al.*, 1999):

$$E_p = D_d \frac{(P_s - P_d)}{(D_s - D_d)} \quad 4.2$$

where $(P_s - P_d)$ and $(D_s - D_d)$ are pulse pressure and pulse diameter, respectively, and D_d is the end-diastolic diameter. If static values are desired, $(P_s - P_d)$ and $(D_s - D_d)$ are quasistatic changes in pressure and diameter, respectively, and D_d is the mean diameter corresponding to the mean pressure. The inverse of E_p has been referred to as arterial compliance by Gosling (1976) and more recently as arterial distensibility by O'Rourke *et al.* (2002). By rearranging eqn 4.2, it is seen that both pulse pressure $(P_s - P_d)$ and systolic pressure (P_s) are directly and positively related to arterial elastic modulus (or stiffness) and diastolic pressure (P_d) is directly but inversely (or negatively) related to arterial elastance, i.e. the stiffer the artery the higher the systolic and pulse pressure and the lower the diastolic blood pressure (Burt *et al.*, 1995; Nichols and Edwards, 2001). This equation, along with eqns 4.3 and 4.4, is more closely associated with arterial stiffness when measured in central elastic arteries and disregarding the influence of wave reflections.

Numerous articles have used peripheral pulse pressure as an index of arterial stiffness and found a close relationship with cardiovascular events (Liao *et al.*, 1999; Safar, 2000; Vaccarino *et al.*, 2000; Dart *et al.*, 2001; Franklin *et al.*, 2001; Lu *et al.*, 2001; Domanski *et al.*, 2002; Benjo *et al.*, 2007; Anderson *et al.*, 2008; Yasuno *et al.*, 2010), including cardiovascular mortality (Benetos *et al.*, 1997; Alderman *et al.*, 1998; Safar *et al.*, 2002b; Thomas *et al.*, 2008); mortality is related to pulse pressure even in normotensive individuals. Plots of hazard ratios for new-onset diabetes considering both systolic and diastolic pressure revealed that a higher pulse pressure with a lower diastolic pressure, indicating that the increased pulse pressure was largely due to increased arterial stiffness, was strongly associated with the risk of new-onset diabetes (Yasuno *et al.*, 2010). Furthermore, data from the International VERapamil SR-trandolapril Study (INVEST) arterial diastolic blood pressure exhibited a J- or U-shaped relationship with primary outcome (Bangalore *et al.*, 2009). Also, central pulse pressure is better than peripheral pulse pressure as a measure of aortic stiffness and predictor of certain cardiovascular risk factors (Waddell *et al.*, 2001; Safar *et al.*, 2002b; Jankowski *et al.*, 2004; Chirinos *et al.*, 2005; Agabiti-Rosei *et al.*, 2007; Roman *et al.*, 2007; McEniery *et al.*, 2008; Pini *et al.*, 2008; Ghiadoni *et al.*, 2009; Roman *et al.*, 2009; McEniery *et al.*, 2010a; McEniery *et al.*, 2010b; Roman *et al.*, 2010; Vlachopoulos *et al.*, 2010d).

E_p has been reported in normal humans and in patients with various cardiovascular diseases using both invasive and non-invasive techniques (for references see Table 4.2 and O'Rourke *et al.* (2002); Roman *et al.* (2005); Carallo *et al.* (2006)). Several large epidemiological studies are currently evaluating the natural history of elastic artery stiffness including the Atherosclerosis Risk in Communities (ARIC) study (Arnett *et al.*, 1999; Liao *et al.*, 1999), the Baltimore Longitudinal Study of Aging (BLSA) (Nagai *et al.*, 1999) and the Bogalusa Heart Study (BHS) (Urbina *et al.*, 2004; Toprak *et al.*, 2009) using eqn 4.2. These three studies obtained E_p in the CCA from non-invasive measurements of diameter by ultrasonography, but made the mistake of using pressure measured in the brachial artery by sphygmomanometry. The ARIC study included 15 800 males and females, ages 45 to 64 years, with a brachial blood pressure of 118/71 mmHg and E_p of 1.18×10^6 dyne/cm² (Table 4.2). The BLSA study included 100 volunteers including 47 males (25–90 years) and 63 females (28–87 years). E_p was 0.89×10^6 dyne/cm² in the entire group, but was higher in males (1.01×10^6 dyne/cm²) than in females (0.80×10^6 dyne/cm²); E_p was directly and linearly related to aortic pulse wave velocity. The BHS included 786 young adults (age range 24–43 years, average 36; 42 percent male, 70 percent white). Widened pulse pressure (47 ± 11 versus 37 ± 7 mmHg), the presence of diabetes mellitus, increased body mass index, and elevated E_p (0.60 versus 0.44×10^6 dyne/cm²) were associated with concentric left ventricular

Table 4.2 Pressure–strain elastic modulus (E_p) of arteries

Species	Age (years)	Artery	Pressure (mmHg)	D_d (mm)	$D_s - D_d$ (mm)	E_p (dynes/cm ²) 3×10^6	Reference
Human	40–50	Ascending aorta	124/80	32.3	2.2	0.88	Stefanadis <i>et al.</i> (1990)
Human	58	Ascending aorta	125/74	28	2.8	0.78	Patrianakos <i>et al.</i> (2006)
Human	47	Descending aorta	125/78	21.1	1.8	0.76	Pearson <i>et al.</i> (1994)
Human	43	Descending aorta	123/75	18.2	1.8	0.68	Pearson <i>et al.</i> (1994)
Human	19–35	Abdominal aorta	116/64	15.5	1.5	0.70	Länne <i>et al.</i> (1992a)
Human	25	Abdominal aorta	117/70	15.6	1.4	0.69	Länne <i>et al.</i> (1992a)
Human	27	Abdominal aorta	120/76	13.2	1.59	0.52	Sonesson <i>et al.</i> (1993)
Human	56	Common carotid	119/70	5.5	0.40	0.89	Nagai <i>et al.</i> (1999)
Human	45–64	Common carotid	118/71	7.73	0.41	1.18	Arnett <i>et al.</i> (1999)
Human	20–39	Abdominal aorta	120/69	11.7	1.08	0.74	Kawasaki <i>et al.</i> (1987)
		Common carotid	117/70	7.1	0.64	0.70	
		Brachial	116/69	3.6	0.24	0.94	
		Femoral	116/68	8.1	0.45	1.15	
Conscious dog		High descending aorta (angiotensin)	131/80 202/84	15.2 15.2	1.74 2.43	0.59 2.1	Armentano <i>et al.</i> (1991)
Conscious dog		High descending aorta (phenylephrine)	125/73 193/126	15.5 16.37	1.75 1.25	0.62 1.17	Armentano <i>et al.</i> (1995)
Conscious dog		High descending aorta (phenylephrine)	127/74 193/126	15.7 16.4	1.6 1.2	0.69 1.22	Barra <i>et al.</i> (1993)
Anesthetized sheep		Main pulmonary artery	19/11	22.6	1.3	0.23	Santana <i>et al.</i> (2005)
Sheep		Increased pulmonary pressure	29/15	23.6	1.1	0.47	

D_d , end diastolic diameter; D_s , peak systolic diameter.

hypertrophy compared with normal geometry. Aortic PWV, the ‘gold standard’ for measuring arterial stiffness, has also been used in several clinical trials as a marker of cardiovascular risk (see Chapter 3). These include the Caerphilly Prospective Study (McEniery *et al.*, 2010a; McEniery *et al.*, 2010b), the Framingham Heart Study (Mitchell *et al.*, 2010), the Chronic Renal Insufficiency Cohort study (CRIC) (Townsend *et al.*, 2010), the Rotterdam study (Sijpe *et al.*, 2009), the Bogalusa Heart Study (Toprak *et al.*, 2009), and the Hawaii–Los Angeles–Hiroshima Study (Shokawa *et al.*, 2005).

Another equation proposed to characterize the elastic properties of arteries supposedly independent of distending pressure was introduced by Kawasaki *et al.* (1987) and has been used in humans by several investigators, including Hirai *et al.* (1989), Hirata *et al.* (1991), Roman *et al.* (1992), Länne *et al.* (1992a), Länne *et al.* (1992b), Sonesson *et al.* (1993), Pearson *et al.* (1994), Gatzka *et al.* (1998), Liao *et al.* (1999), Roman *et al.* (2005), Armentano *et al.* (2006), Patrianakos *et al.* (2006), Ou *et al.* (2008a; 2008b), Morovic *et al.* (2009), Sanz *et al.* (2009a, 2009b), Shroff *et al.* (2009), Urbina *et al.* (2009b), Bradley *et al.* (2010), and Dengo *et al.* (2010) (Table 4.3). These studies used either ultrasound or magnetic resonance imaging to measure arterial diameter noninvasively. The β -stiffness index has also been measured in excised isolated human carotid (Armentano *et al.*, 2006) and

coronary arterial segments (Tajaddini *et al.*, 2005). This stiffness index β , is given by

$$\beta = \frac{D_d \ln \left(\frac{P_s}{P_d} \right)}{(D_s - D_d)}. \quad 4.3$$

Characteristic impedance (Z_o), which is directly and linearly related to pulse wave velocity (c_o), is another valuable index of arterial stiffness. It is the relationship between pulsatile pressure and pulsatile flow measured at the same site in an artery in the absence of wave reflections (see Chapter 12). Z_o and c_o are almost identical when flow is expressed as cm/s and pressure as dyne/cm². Most investigators, however, use volume flow rate in the calculation and report impedance in units of dyne-s/cm⁵. To minimize the influence of reflections (oscillations in the spectra), ascending aortic Z_o can be estimated as the arithmetic mean of the higher frequency input impedance moduli > 2.0 Hz (see Table 4.4) (Nichols *et al.*, 1977a). Some investigators average input impedance moduli for harmonics 3.0 through 10 (Segers *et al.*, 2007a) or frequencies 2.0 to 16 Hz (Lucas *et al.*, 1988). These methods of calculating Z_o require measurement of pressure and flow waves with systems that have a flat frequency response > 15 Hz and the use of Fourier analysis, so as to determine input impedance spectra (Chapter 12)

Table 4.3 Stiffness index (β) of arteries

Species	Condition	Age (years)	Artery	Pressure (mmHg)	β (non-dimensional)	Reference
Human	Healthy	29	Ascending aorta	112/75	5.9	Hirata <i>et al.</i> (1991)
	Marfan	26		126/81	10.9	
Human	Healthy	58	Ascending aorta	125/74	5.5	Patrianakos <i>et al.</i> (2006)
	ESRD	61		138/70	8.9	
Human	Healthy	62	Aortic arch	130/77	9.0	Gatzka <i>et al.</i> (1998)
	CAD	63		133/74	16.0	
Human	Healthy male	47	Descending aorta	125/78	5.71	Pearson <i>et al.</i> (1994)
	Female	43		123/75	5.1	
	Hypertensives	47		150/92	13.88	
Human	Healthy	6–81	Abdominal aorta	Normotensive	4.3–9.8	Kawasaki <i>et al.</i> (1987)
Human	Healthy	29	Abdominal aorta	112/75	3.9	Hirata <i>et al.</i> (1991)
	Marfan	26		126/81	7.1	
Human	Healthy	60	Abdominal aorta	126/74	8.58	Hirai <i>et al.</i> (1989)
	CAD	55–59		128/79–140/83	12.4–22.4	
Human	Healthy	6–81	Carotid	Normotensive	4.3–11.3	Kawasaki <i>et al.</i> (1987)
Human	Healthy	56	Carotid	113/69	10.3	Liao <i>et al.</i> (1999)
Human	Healthy	60	Carotid	124/74	9.17	Hirai <i>et al.</i> (1989)
	CAD	55–59		128/79–140/83	9.4–13.2	
Human	Hypertensives	56	Carotid		11.87	Liao <i>et al.</i> (1999)
Human	Lean	18	Carotid	110/66	9.6	Urbina <i>et al.</i> (2009b)
	Obese	18		119/69	11.2	
	Type 2 diabetes	19		125/73	10.8	
Human	Healthy	6–81	Brachial	Normotensive	8.6–13.7	Kawasaki <i>et al.</i> (1987)
Human	Healthy	6–81	Femoral	Normotensive	9.4–15.3	Kawasaki <i>et al.</i> (1987)

(O'Rourke and Taylor, 1966a). An alternative method for calculating Z_o , and one that is much simpler, is to relate peak flow to the change in pressure using the so-called 'water hammer equation', i.e

$$c_o = \Delta P / \rho \Delta V \quad 4.4$$

where ΔP is change in pressure, ρ is blood density, c_o is pulse wave velocity, and ΔV is change in blood flow velocity. This equation is often referred to as the 'Joukowsky equation' and its derivation and validation were published by Joukowsky (1898) in a comprehensive study of pressure waves in water supply lines. The same equation had earlier been derived and validated, through experiments in water-filled rubber tubes, by Kries (1883) in a study of the pulse. By making a few reasonable approximations, we can write

$$Z_o = \Delta P_Q / \Delta Q = (P_i - P_d) / \text{peak flow} \quad 4.5$$

where P_d is end diastolic pressure and P_i is the inflection point on the pressure curve and marks the initial upstroke of the reflected wave, and where peak flow occurs (ΔQ , Fig. 4.11a) (Dujardin and Stone, 1981; Dujardin *et al.*, 1982; Lucas *et al.*, 1988). This method of estimating Z_o is based upon the assumptions that blood density is constant and the early part (up to peak flow) of the pressure and flow waves are not influenced by wave reflections. Evaluation of the method in both humans ($n=40$, $r=0.92$) and dogs ($n=56$, $r=0.94$) indicates that it is very reliable compared

to the Fourier analysis method of obtaining Z_o (Fig. 4.11b). Some investigators (Mitchell *et al.*, 2001; Mitchell *et al.*, 2002; Mitchell *et al.*, 2003; Mitchell *et al.*, 2007a; Mitchell *et al.*, 2007b; Segers *et al.*, 2007a, 2007b; Sweitzer *et al.*, 2007; Chirinos *et al.*, 2009a; Chirinos *et al.*, 2009b) make the mistake of using carotid pressure (calibrated against brachial cuff pressure) and aortic flow to calculate Z_o . This mistake introduces an error in the calculation causing abnormally high values of Z_o (see Table 4.4). Chemla *et al.* (2008) have proposed a new equation based upon eqn 4.3 that uses triangulation of the aortic flow wave (Westerhof *et al.*, 2006) obtained from stroke volume (SV):

$$Z_o = [(P_i - P_d) \times ED] / (SV \times \Delta t) \quad 4.6$$

where ED is systolic ejection duration and Δt is pulse wave travel time (Chapter 9). Other equations, similar to the water hammer equation, have also been used to calculate Z_o (Patrianakos *et al.*, 2006; Segers *et al.*, 2006). In the above studies, aortic blood flow velocity waves were obtained with both Doppler ultrasound and MRI.

The above equations for estimating arterial elastance (or stiffness) are used to obtain information in a particular artery at a particular site. In 1986, Liu *et al.* introduced an equation for estimating total arterial compliance of a systemic arterial bed (SAC).

$$\text{SAC} = \frac{A_d}{R(P_{es} - P_d)} \quad 4.7$$

Table 4.4 Characteristic impedance (Z_0) of arteries

Species	Condition	Age (years)	Artery	Pressure (mmHg)	Z_0 (dyne-s/cm ⁵)	Reference
Human	Healthy	34	Ascending aorta	117/77	47	Murgo <i>et al.</i> (1980a, 1980b)
Human	Healthy (majority)	38	Ascending aorta		73	Merillon <i>et al.</i> (1976)
Human	Healthy (majority)	47	Ascending aorta	100 (mean)	97	O'Rourke and Avolio (1980)
Human	Healthy (majority)	42	Ascending aorta	121/78	94	Ting <i>et al.</i> (1986)
Human	Healthy	57	Ascending aorta	126/73	184	Mitchell <i>et al.</i> (2001)
	Heart failure	61		128/66	225	
	Hypertensive	61		163/84	237	Mitchell <i>et al.</i> (2002)
	Healthy (female)	56		114/64	185	Mitchell <i>et al.</i> (2003)
	Hypertensive	61		167/83	268	
	Healthy male	60		125/69	159	
	Hypertensive	60		162/88	208	
	Healthy	28	Ascending aorta	108/65	136	Sweitzer <i>et al.</i> (2007)
	Type 1 diabetes	27		113/59	179	
Human	Healthy	41	Ascending aorta	114/70	106	Chirinos <i>et al.</i> (2009c)
Human	Hypertensive	54	Ascending aorta	134/62	160	
Human	Hypertensives	35	Ascending aorta	168/99	146	Ting <i>et al.</i> (1986)
Human	CAD	56	Ascending aorta	126/71	136	Kelly and Fitchett (1992)
Human	Hypertensives	32	Ascending aorta		81	Merillon <i>et al.</i> (1976)
Human	Heart failure	21–70	Ascending aorta	89 (mean)	138	Binkley <i>et al.</i> (1990)
Human	Healthy	32	Ascending aorta	114/78	53	Nichols <i>et al.</i> (1977a)
Human	CAD	48	Ascending aorta	109/68	95	Nichols <i>et al.</i> (1977a)
Human	Hypertensives	48	Ascending aorta	161/86	202	Nichols <i>et al.</i> (1977a)
Human	Heart failure	47	Ascending aorta	90 (mean)	131	Pepine <i>et al.</i> (1978)
Human	Healthy	19–62	Ascending aorta	108/77–137/80	33–85	Nichols <i>et al.</i> (1985a, 1985b)
Human	Healthy	52	Ascending aorta	126/74	72	Nichols <i>et al.</i> (1993)
Human	Hypertensives	54	Ascending aorta	180/110	110	Nichols <i>et al.</i> (1993)
Human	Heart failure	58	Ascending aorta	127/72	80	Kromer <i>et al.</i> (1992)
Human	Heart failure	41	Ascending aorta	92 (mean)	93	Laskey <i>et al.</i> (1985)
Dog	Normal		Ascending aorta	110/80	121	Eaton <i>et al.</i> (1993)
	Heart failure			94/66	186	

where R is the total peripheral resistance or steady component of left ventricular load calculated as (mean arterial pressure)/(cardiac output) and A_d is area under the diastolic portion of the ascending aortic pressure curve. P_{es} is end-systolic aortic pressure (pressure at the dicrotic notch) and P_d is end-diastolic aortic pressure (Fig. 4.12). This equation can be applied non-invasively by measuring carotid artery pressure by tonometry and aortic blood flow by a Doppler flow velocimeter placed on the suprasternal notch or magnetic resonance imaging, however, the use of aortic pressure (measured directly or derived) instead of carotid pressure would improve the accuracy of the method. This method has been used in normal human volunteers (Liang *et al.*, 1998), premenopausal (Nestel *et al.*, 1997a), and post-menopausal (McGrath *et al.*, 1998) women, obese and hypertensive subjects (Nestel *et al.*, 1997b; Teede *et al.*, 2006), patients with aortic valve stenosis (Antonini-Canterin *et al.*, 2009), patients with age-related macular degeneration (CHARM) (McCarty *et al.*, 2008) and individuals undergoing exercise training (Cameron and Dart, 1994; Maeda *et al.*, 2008). In 50 healthy volunteers (29–70 years, average age

46.5) with blood pressure 121/71 mmHg, Liang *et al.* reported an average value of 0.67 mL/mmHg. The age-related change is shown in Fig. 4.13A. SAC was reduced in obesity and post-menopausal women and increased with exercise training, fish oil, and soy isoflavones (Nestel *et al.*, 1997a; Nestel *et al.*, 1997b; Maeda *et al.*, 2008); in hypertensive subjects SAC was reduced and was not altered with soy isoflavones (Teede *et al.*, 2006). SAC was increased in subjects with age-related macular degeneration (McCarty *et al.*, 2008), but was not changed in patients with aortic valve stenosis even though aortic augmentation index was elevated (Antonini-Canterin *et al.*, 2009). Systemic arterial compliance as an index of arterial stiffness has shown considerable variability and has not been very helpful in studying aging change. It is surprising that the above reported differences have been seen, but not confirmed, with the use of other methods that have better reproducibility.

The list of arterial stiffness indices in Table 4.1 was further examined and general recommendations made at a consensus conference convened in 2001 by ME Safar and E Frohlich. Recommendations of this representative group

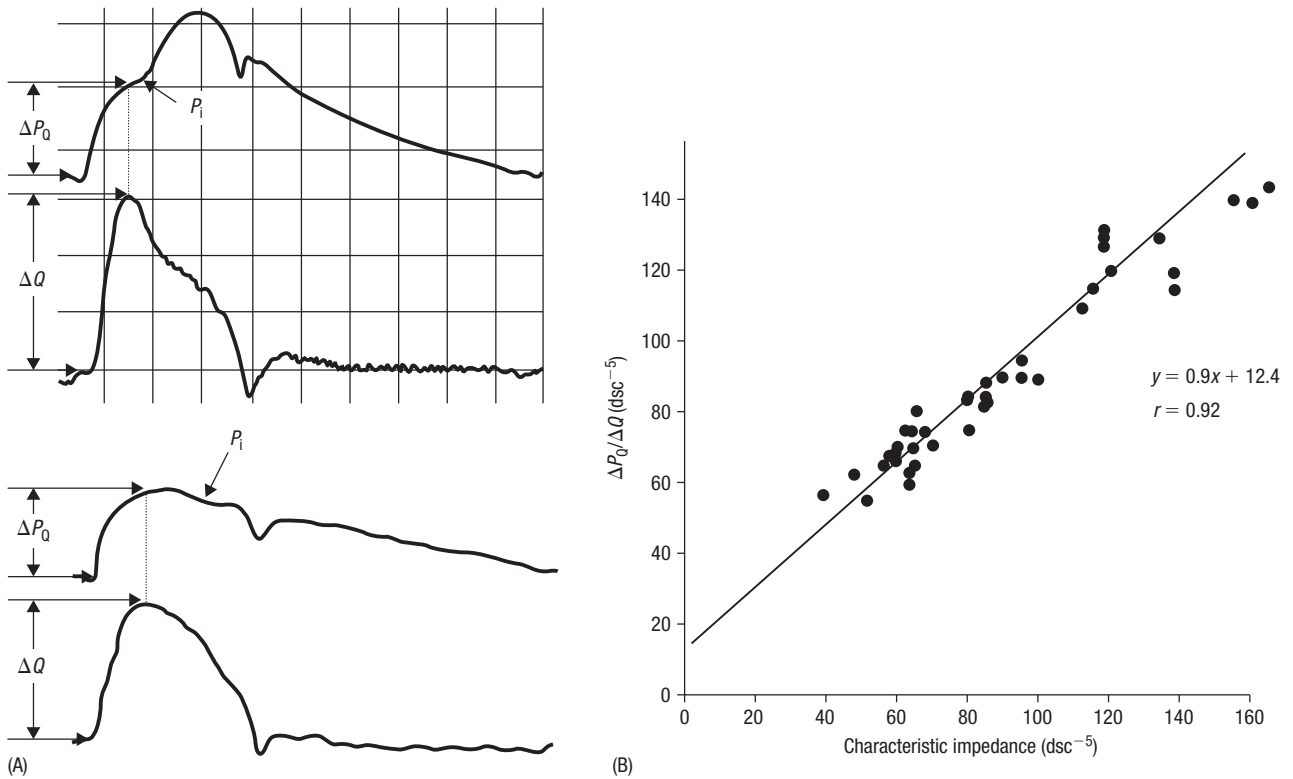


Figure 4.11 (A) Pressure and flow waves measured in the ascending aorta of two humans (middle-aged, top and young, bottom) and the method used to calculate characteristic impedance ($Z_0 = \Delta P_0 / \Delta Q$). P_d is end-diastolic pressure and P_i is the inflection point. (B) Relationship between Z_0 calculated from pressure and flow waves in the time domain and Z_0 obtained from the high frequency moduli (> 2 Hz) of the impedance spectra in the frequency domain

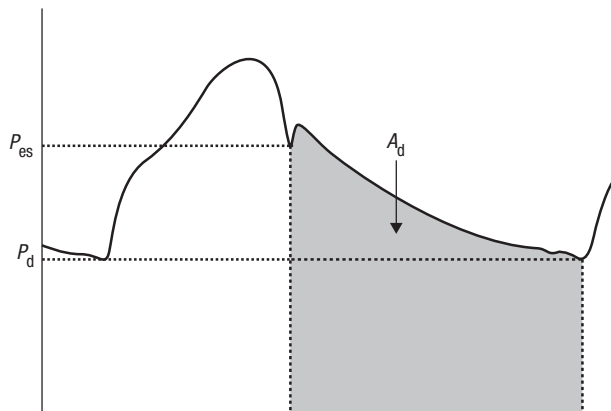


Figure 4.12 Illustration of method used to calculate total systemic arterial compliance (SAC). A_d , area under the diastolic portion of the pressure curve, P_{es} , end-systolic pressure, and P_d , end-diastolic aortic pressure. Modified from Liang *et al.* (1998)

were published in a series of articles commissioned by the late Larry Resnick (Laurent *et al.*, 2002; London and Cohn, 2002; O'Rourke *et al.*, 2002; Pannier *et al.*, 2002). The subject was further addressed by the ESH/ESC Guideline Committee on Hypertension. It was suggested

that measures of aortic PWV and pulse wave analysis be used as indices of arterial stiffness and target organ damage (Zanchetti *et al.*, 2003). Furthermore, arterial stiffness has been recognized as integral to the diagnosis and treatment of hypertension (Mancia *et al.*, 2007a).

The value of the above indices in clinical practice depends on their ability to predict cardiovascular events independently of other factors, such as age, blood pressure, etc. The series of workshops held in Paris by Michael Safar has focused on this issue over a period of 25 years. At this time (2010), the consensus view is that aortic PWV is the best index of aortic stiffness (Vlachopoulos *et al.*, 2010a; Mitchell *et al.*, 2010b; McEniery *et al.*, 2011) – confirming earlier work on normal subjects at different ages, and patients with hypertension, diabetes, and renal failure, and the consensus views by Laurent *et al.* (2006). Also, an emerging consensus views central aortic systolic and pulse pressure, augmentation, and amplification as predictive of cardiovascular events (Benetos *et al.*, 2010; Vlachopoulos *et al.*, 2010a; Vlachopoulos *et al.*, 2010b; Wang *et al.*, 2010; Weber *et al.*, 2010; Roman *et al.*, 2010). Studies that have failed to support this view appear to have serious methodological problems (O'Rourke and Adji, 2010) as discussed later in chapters on aging and hypertension.

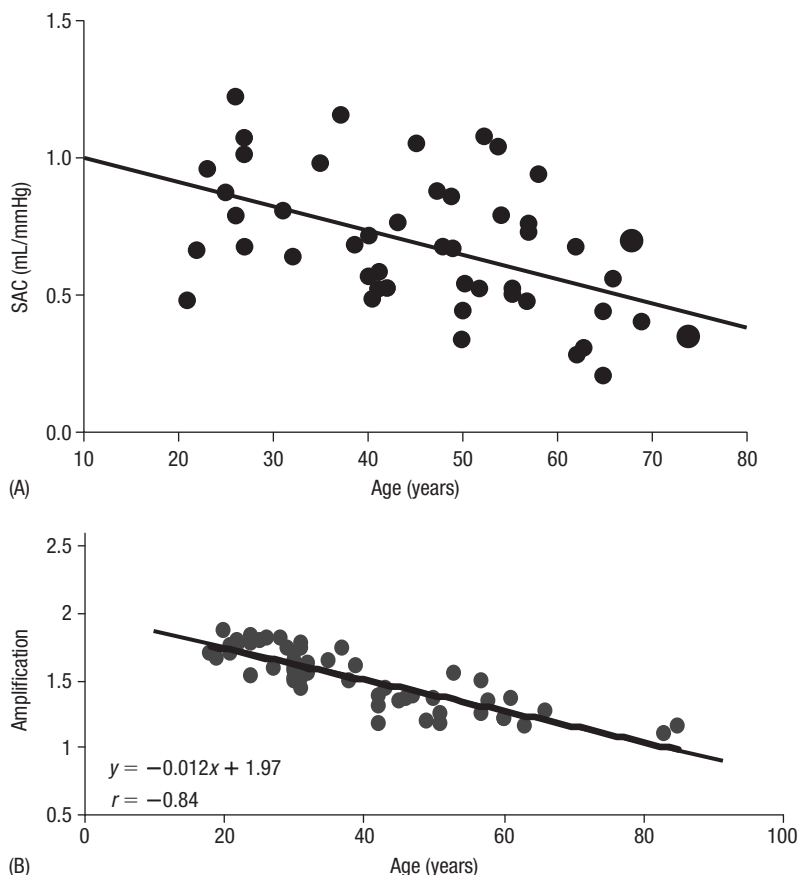


Figure 4.13 (A) Age-related changes in total system arterial compliance (SAC) in healthy humans (20 males and 30 females). From Liang *et al.* (1998). (B) Age-related changes in pulse pressure amplification (brachial pulse pressure/central aortic pulse pressure) in a different group of healthy humans (22 males and 31 females)

Arterial disease

Chronic arterial diseases, such as that seen in Marfan syndrome (Jeremy *et al.*, 1994; Groenink *et al.*, 1998; Lalande *et al.*, 2002; Segers *et al.*, 2006), Pompe's disease (Nemes *et al.*, 2007), Kawasaki's disease (Liu *et al.*, 2009), Raynaud's (secondary) disease (Cheng *et al.*, 2003a), and abdominal aortic aneurysm (Cheng *et al.*, 2003b) can affect the arterial media and increase arterial stiffness, as can hypertension (Blacher *et al.*, 1999a; Boutouyrie *et al.*, 2002), end-stage renal disease (Blacher *et al.*, 1999b; Blacher *et al.*, 2003), diabetes mellitus (Westerbacka *et al.*, 2001; Cruickshank *et al.*, 2002; Haller *et al.*, 2004; Sweitzer *et al.*, 2007; McEniery *et al.*, 2008; Wadwa *et al.*, 2010), obesity (Sutton-Tyrrell *et al.*, 2001b; Danias *et al.*, 2003; Wildman *et al.*, 2003; Chirinos *et al.*, 2009a; Chirinos *et al.*, 2009b), and hypercholesterolemia (Wilkinson *et al.*, 2002; McEniery *et al.*, 2008). These diseases are presented in detail in Chapters 15, 20, and 24. Atherosclerosis, which is covered in Chapter 23, principally affects the non-load bearing arterial intima and usually causes adverse effects through narrowing an artery and restricting blood flow. However, changes in the arterial intima with disease can interfere with the release of vasoactive substances to the media, with resultant changes in muscle tone, and thus in arterial caliber and stiffness (Lind *et al.*, 2009).

Diseases related directly to the heart, such as coronary artery disease (Fig. 4.14) (Nichols *et al.*, 1977a; Stefanadis

et al., 1987; Hirai *et al.*, 1989; Dart *et al.*, 1991; Hirai *et al.*, 1989; Kelly and Fitchett, 1992; Cameron *et al.*, 1996; Gatzka *et al.*, 1998; Waddell *et al.*, 2001; Medley *et al.*, 2002; Weber *et al.*, 2004), myocardial ischemia and infarction (Bogren *et al.*, 1989; Hirai *et al.*, 1989; Madhavan *et al.*, 1994; Mitchell *et al.*, 1997b), and chronic heart failure (Pepine *et al.*, 1978; Pepine *et al.*, 1979; Yin *et al.*, 1983b; Brin and Yin, 1984; Merillon *et al.*, 1984; Laskey *et al.*, 1985; Laskey and Kussmaul, 1987; Nichols and Pepine, 1992; Duprez *et al.*, 1998; Khan *et al.*, 1999; Hundley *et al.*, 2001; Mitchell *et al.*, 2001; Rerkpattanapipat *et al.*, 2002) resulting from increased central arterial stiffness which eventually progresses to severe left ventricular dysfunction and reduced arterial pressure cause indices of arterial stiffness and wave reflection to decrease (Denardo *et al.*, 2010). The increase in left ventricular external load may result from one or more of a variety of chronic events, including increased arterial wall retention of sodium and sympathetic nervous stimulation and decreased endothelial function. This may be accompanied by prominent neurohumoral activation with increased circulatory plasma catecholamines and renin–angiotensin–aldosterone concentrations (Levine *et al.*, 1982). Increased stiffness of the central elastic arteries (Mancia *et al.*, 1998; Giannattasio and Mancia, 2002; Laurent *et al.*, 2002) is the primary cause of increased systolic and pulse pressure with advancing age and in patients with hypertension (O'Rourke, 2003; Safar

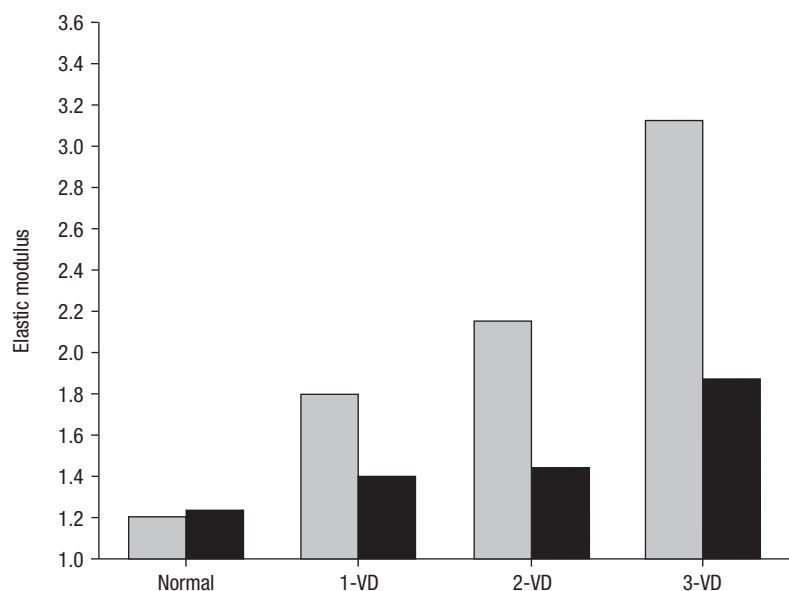


Figure 4.14 Bar graphs of pressure-strain elastic modulus ($\text{dyne/cm}^2 \times 10^6$) of abdominal aorta (light stippled) and carotid artery (dark stippled) in normal subjects and in patients with 1, 2, and 3 coronary vessel disease (VD) and myocardial infarction. Age and arterial blood pressure were similar in the various groups. Elastic modulus increased (distensibility decreased) in both vessels (more in the aorta than the carotid) as the severity of coronary heart disease increased. Data from Hirai *et al.* (1989)

and Benetos, 2003), and is due to degeneration and hyperplasia of the arterial wall (see Chapters 19 and 20). Diastolic pressure decreases as arterial stiffness increases and becomes obvious after age 55–60 years and in ‘isolated systolic hypertension’. As arterial stiffness increases, transmission velocity of both forward and backward (or reflected) traveling waves increases which causes the reflected wave to arrive earlier in the central aorta and augments pressure in late systole and decreases left ventricular ejection during deceleration (Chapter 9). These changes in arterial wall properties cause an increase in left ventricular afterload, hypertrophy (Levy, 1999), and myocardial oxygen consumption and a decrease in myocardial perfusion pressure which may induce an imbalance in the supply/demand ratio, especially in hypertrophied hearts with significant coronary artery disease (Chapter 17). An increase in systolic pressure increases arterial wall circumferential stress which promotes local fatigue and development of atherosclerosis (Chapter 23). Also, acute intake of caffeine and cigarette smoking (acute and chronic) cause an acute increase in arterial stiffness which is associated with an increase in arterial blood pressure (Mahmud and Feely, 2003b; Vlachopoulos *et al.*, 2003c). These deleterious modifications in arterial wall properties, especially of the large elastic arteries, measurement techniques, definitions, and reference values are reviewed in depth by Cheng *et al.* (2002), Mackenzie *et al.* (2002), O’Rourke *et al.* (2002), Pannier *et al.* (2002), Oliver and Webb (2003), Khoshdel *et al.* (2006), Laurent *et al.* (2006), and Reference Values for Arterial Stiffness Collaboration (2010).

The ultrasonic echo and the MRI techniques used in measuring the pulsatile changes in diameter of the aorta and peripheral arteries are of particular interest. These techniques share with optical methods the advantage that no restraint is imposed on the arterial wall, and they can be used non-invasively, which makes them attractive for

use in humans. Arndt *et al.* (1968) reported an average pulsatile change of 14.3 percent of the diastolic radius of the carotid artery with a pulse pressure of 46 mmHg (124/78 mmHg) in nine subjects. In terms of mean radius (radial pulsation), this is ± 7.0 percent and is considerably larger than that recorded in corresponding dog arteries. It does, however, correspond well with the wave velocity of 640 cm/s that they recorded. The calculated value of E_p was $0.46 \times 10^6 \text{ dyne/cm}^2$. These values are in striking contrast to those measured by Greenfield *et al.* (1964), who recorded only about one-tenth of the pulsatile change in diameter with a corresponding E_p of $4.3 \times 10^6 \text{ dyne/cm}^2$ and are somewhat less than those calculated by Kawasaki *et al.* (1987) and Arnett *et al.* (1999) which were $0.70 \times 10^6 \text{ dyne/cm}^2$ and $1.18 \times 10^6 \text{ dyne/cm}^2$, respectively (Table 4.2). Arndt *et al.* (1971) attempted to explain the discrepancy as a result of exposing the artery to attach the gauge, and as mentioned above, dissection of an artery lowers its compliance and inserting sutures in the wall may produce some local spasm of the vascular smooth muscle. Armentano *et al.* (Armentano *et al.*, 1991; Armentano *et al.*, 1995) and Barra *et al.* (1993) measured diameter high in the descending thoracic aorta of conscious dogs using ultrasonic crystals and calculated a radial pulsation of ± 4.8 to ± 5.4 percent; aortic pressure ranged from 125/73 to 132/80 mmHg (Table 4.5). Smooth muscle activation with phenylephrine increased aortic pressure (to 193/126 mmHg) and diameter (from 17.3/15.7 to 17.6/16.4 mm) and decreased radial pulsation (± 3.5 percent). Also, it appears as though radial pulsation is less (± 2.2 percent) in anesthetized open-chest dogs (Kohnno *et al.*, 1987; White *et al.*, 1994) than in conscious dogs.

From the above discussion, it is clear that large differences in the values of radial pulsation and elastic modulus have been reported (see also Milnor (1989)). These differences are due to the type of preparation, the

Table 4.5 Values of radial pulsation of arteries

Species	Artery	Pressure (mmHg)	R_m (mm)	Radial pulsation (%)	Reference
Human (40–50 years)	Ascending aorta	124/80	16.4	± 3.3	Stefanadis <i>et al.</i> (1990)
Human (male, av. 47 years)	Descending thoracic aorta	132/77	10.9	± 3.9	Stefanadis <i>et al.</i> (1997)
Human (male, av. 47 years)	Descending thoracic aorta	125/78	10.8	± 3.8	Pearson <i>et al.</i> (1994)
Human (female, av. 43 years)	Descending thoracic aorta	123/75	9.4	± 4.7	Pearson <i>et al.</i> (1994)
Human (19–35 years)	Abdominal aorta	116/64	8.1	± 4.6	Länne <i>et al.</i> (1992a)
Human (male, av. 25 years)	Abdominal aorta	117/70	8.1	± 4.3	Länne <i>et al.</i> (1992b)
Human (female, av. 26 years)	Abdominal aorta	120/76	6.6	± 5.7	Sonesson <i>et al.</i> (1993)
Conscious dog	High descending thoracic aorta	127/74	8.2	± 4.8	Barra <i>et al.</i> (1993)
'Phenylephrine'	High descending thoracic aorta	193/126	8.6	± 3.5	Barra <i>et al.</i> (1993)
Conscious dog	High descending thoracic aorta	125/73	8.3	± 5.3	Armentano <i>et al.</i> (1995)
Conscious dog	High descending thoracic aorta	132/80	7.97	± 5.1	Fischer <i>et al.</i> (1991)
Anesthetized dog	Thoracic aorta	91/72	7.65	± 2.2	Kohno <i>et al.</i> (1987)
Anesthetized dog	Abdominal aorta	110/60	3.1	± 2.0	White <i>et al.</i> (1994)

R_m , mean radius.

level of intraluminal pressure (see Figs 4.9 and 4.15), the technique used to measure arterial diameter and the age of the experimental animal or human (see below). To minimize the errors introduced in the measurements of arterial wall properties, recent publications have used *in vivo* techniques that require minimal (ultrasonic crystals) or no (echo and MRI) contact with the arterial wall. Results from some of these studies are presented in Table 4.2.

The values recorded in the aorta for the radial pulsation with the pulse pressure wave may seem to be small in terms of the apparently obvious expansion seen by the naked eye. It should be remembered, however, that considerable lateral movement of arteries often occurs with each heart beat. This is especially marked in arteries running in a curved course as, for example, in mesenteric vessels. The actual radial movement of arteries is small and requires very precise methods for its measurement. The various methods used to measure arterial diameter are discussed in Chapters 6 and 9.

The longitudinal elastic modulus

Lawton and Greene (1956) made measurements of longitudinal movement of the dog abdominal aorta by cinematographic observation of beads sutured to the surface of the vessel. The movements were very small, and they often recorded a shortening length during the systolic rise in pressure. This was confirmed in a more detailed study by Patel *et al.* (1961); in the thoracic aorta, however, a small lengthening of the vessel was found. The average value was 0.011 percent per mmHg pressure (mean pulse pressure 38 mmHg). The effect in the abdominal aorta was coupled to the movements of the thoracic aorta rather than being due to the pressure rise in the vessel because it persisted after the aorta was occluded at the diaphragm.

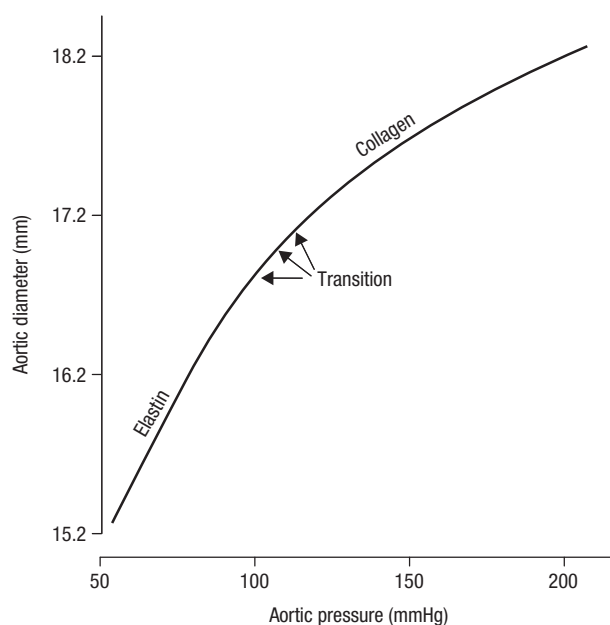


Figure 4.15 Pressure–diameter curve of the human abdominal aorta. Recordings were taken both at rest and during isometric exercise. Note the transition point at 90–110 mmHg indicating a stiffer aortic wall above than below this point. Data from Länne *et al.* (1992a)

Respiratory movements of the diaphragm also caused a small longitudinal movement in the aorta; during inspiration, the thoracic aorta lengthens by 1.5–2.0 mm and the abdominal aorta shortens. A small movement of the ascending aorta towards the heart and a headward movement in the arch of the aorta were also recorded.

The results of studies in these and other arteries of the dog were published by Patel *et al.* (1964). The longitudinal modulus of the thoracic aorta was measured in a detailed study of the anisotropic elastic properties by Patel *et al.* (1969). They used a segment of the aorta *in situ* that was

isolated by inserting plugs into each end and attaching it to a force gauge (Janicki and Patel, 1968), which also allowed applied extensions of the artery to be made. At a mean pressure of 116 mmHg, the average longitudinal elastic modulus was 9.8×10^6 dyne/cm². When released from the longitudinal tissue attachments, the modulus decreased by 32 percent. This reduces it to a value that is similar to the corresponding circumferential elastic modulus of 7.3×10^6 dyne/cm². This is designated transverse anisotropy, and it was also found by Tickner and Sacks (1967) in excised specimens of the thoracic aorta and femoral artery of dogs, and in human brachial, superior mesenteric and external arteries at internal pressures of 100–120 mmHg when extended to the *in vivo* length. The stress–strain diagrams of the excised femoral artery published by Attinger (1968) are also essentially the same up to strains of between 1.5 and 1.6 of the excised dimensions. At greater strains, the artery becomes considerably stiffer in the circumferential than in the longitudinal direction. Anliker *et al.* (Anliker *et al.*, 1968a; Anliker *et al.*, 1968b) generated axial waves in the carotid artery of the dog *in situ*. While the axial velocity is about three times greater than the velocity of the pressure wave, the calculated difference would be a factor of five. They therefore concluded that the longitudinal elastic modulus in this artery is smaller than the circumferential elastic modulus. It is possible that, because of the range of movements of the head, the longitudinal restraint of the surrounding tissues is less in the carotid than in other arteries. Otherwise, *in situ* the tethering of the arteries in the longitudinal direction is usually greater than in the circumferential direction, as found by Patel *et al.* (1969), at usual physiologic pressures. At much higher pressures, they found the situation reversed. When the Poisson ratio is 0.5, then from the relationship in eqn 3.10, the shear modulus will have a value of one-third that of the circumferential elastic modulus, and these are the values usually assumed to be true. This modulus is of little practical value, since Patel and Fry (1964) showed that the shear strains in the arterial wall are negligible as an artery is distended, and hence the structure must be elastically symmetric.

THE ROLE OF SMOOTH MUSCLE CELLS IN THE ARTERIAL WALL

A complete account of the physiology of vascular smooth muscle will not be attempted here; many excellent reviews are available (e.g. Somlyo and Somlyo, 1968; Gow, 1972; Dobrin, 1978; Bohr *et al.*, 1980; Furchgott, 1983; Vanhoutte *et al.*, 1986; Shadwick, 1999; Wagenseil and Mecham, 2009).

Since the discovery of endothelium-derived relaxing factor (EDRF or nitric oxide (NO)) by Furchgott and Zawadzki (1980), it has become apparent that the endothelium plays a major role in modulation of vascular smooth muscle contraction and relaxation (Chapter 5)

(Furchgott, 1983; Vanhoutte *et al.*, 1986). Therefore, results from some earlier experiments on isolated specimens may be incorrect.

In arteries, vascular SMC and fibroblasts continually produce and degrade matrix constituents in preferred modes and at altered rates in response to changing mechanical stimuli. SMC similarly exhibit vasoactive changes that contribute to the control of overall structure, function, and mechanical behavior (Zulliger *et al.*, 2002; Zulliger *et al.*, 2004). A major function of the vascular SMC in medium to large vessels is to synthesize and organize the unique extracellular matrix responsible for the mechanical properties of the wall. Unlike cells in the smaller muscular arteries and arterioles, the SMC of the large elastic conduit arteries contribute much less to the passive mechanical properties of the vessel wall (Faury *et al.*, 2003). SMC exhibit a predominantly circumferential orientation, and their contraction is mainly in the circumferential direction (Herlihy, 1980; Wagenseil and Mecham, 2009). Furthermore, when exposed to sudden changes in blood pressure, SMC are stretched circumferentially and their contractile apparatus is activated, leading to an autonomous contraction known as the myogenic response (Seow, 2000). Even when pressure is augmented slowly in *ex vivo* preparation, SMC tone increases as a result of the myogenic mechanism (Fridez *et al.*, 2001). The contribution of smooth muscle to the functional behavior of the larger conduit arteries has been a controversial topic, but recent results from numerous studies, using new non-invasive methods to measure pulsatile arterial diameter and pressure, have shed new light on the topic (Weber *et al.*, 1996; Bank and Kaiser, 1998; Bank *et al.*, 1999). While in the smallest arteries and the arterioles, contraction of the vascular smooth muscle causes large changes in the vascular lumen, with corresponding changes in the peripheral resistance, sympathetic stimulation or increases in circulatory vasoactive substances like norepinephrine was thought to produce insignificant changes in the size of the larger elastic arteries (Faury *et al.*, 2003). This was attributed to the fact that the wall tension due to the distending blood pressure is already very high in a large artery, and it did not seem likely that the tension the vascular SMC in the wall could generate would be sufficient to alter lumen diameter significantly. This was the point of view taken by McDonald (1974), but the value for the maximal tension developed by vascular SMC in his argument was, from later evidence, much too low. The values he used were from Burton (1954), who calculated a Young's modulus for maximally contracted smooth muscle of about 1.0×10^5 dyne/cm², which is more than an order of magnitude less than that of the aortic wall (Table 4.2).

The assessment of the role of contracted vascular smooth muscular tension may be more clearly stated in numerical terms. If we take, for example, the thoracic aorta of a dog with a mean radius of 0.825 cm and a mean internal pressure of 98 mmHg, from Laplace's law the

linear tension in the wall is $P \times R$; that is, $130\,634 \times 0.825$ (10.8×10^4 dyne/cm). Assuming a wall thickness of 0.0923 cm (Barra *et al.*, 1993), the wall stress will be 11.68×10^5 dyne/cm². The earlier values suggested that the maximally contracted smooth muscle would not be able to exert a force of much more than 1.0×10^4 dyne/cm² and would clearly make little material difference. The actual tension that the smooth muscle can exert is now known to be far greater than this (McDonald, 1974; Barra *et al.*, 1993; Armentano *et al.*, 1995; Bank *et al.*, 1995). Somlyo and Somlyo (1968) concluded that the active contractile force that can be developed by vascular smooth muscle is between 1.5 and 2.5×10^5 dyne/cm². This is much less than the value of 2.73×10^6 dyne/cm² calculated from the experimental data on the carotid artery of the dog by Dobrin and Rovick (1969). Lundholm and Mohme-Lundholm (1966) made measurements from which a value of 2.62×10^5 dyne/cm² was obtained for the vascular smooth muscle of the mesenteric artery, taking a value of 75 percent of the cross-sectional area as being due to muscle. The assessment of the effective cross-section of the arterial wall that is due to smooth muscle is somewhat imprecise; it is commonly estimated from the total area of smooth muscle in histologic section, but Dobrin and Rovick (1969) derived it from estimates of the intracellular space, and Apter (1966) estimated it from the protein content. The amount of smooth muscle in the thoracic aorta has been estimated at between 25 and 35 percent; if we take one-third as the value, the tension at maximal contraction will be $8.0 - 9.0 \times 10^5$ dyne/cm² in the aortic wall. If the elastic tension is 5.5×10^5 dyne/cm², it is clearly possible to double the tension in the wall even when the muscle is contracting less than maximally. With arteries of small radius, the effect will be even greater because the initial wall tension will be correspondingly less. In large arteries such as the human aorta, the elastic tension developed by SMC is most likely reduced.

The controversy over the constricting effect of SMC appears to have arisen from the fact that as circulating norepinephrine, for example, will increase the peripheral resistance owing to the concomitant arteriolar constriction, there is an increase in blood pressure that counteracts the constrictor effect in the artery. It is only in recent years that due emphasis has been placed on separating these two effects. A pressure-stabilizing device, or 'barostat', was used by Nichols and McDonald (1972) to keep a constant arterial pressure. Using a Pieper intra-arterial gauge (Pieper and Paul, 1969), a diminution of aortic diameter with vasoconstrictor substances was measured. Using a similar device, Gerova and Gero (1967) also showed that sympathetic stimulation can significantly reduce the diameter of the dog femoral artery, but when the pressure was allowed to rise this effect was abolished. In the left circumflex coronary artery, Vatner *et al.* (1980), using ultrasonic crystals to measure vessel diameter, found that, during vasoconstriction with methoxamine administration, the diameter decreased even though arterial

pressure increased. Sympathetic stimulation also causes a significant reduction in large artery diameter (Pagani *et al.*, 1975). Exciting arterial smooth muscle can decrease arterial diameter by 20–50 percent as reported for the dog iliac (Cox, 1976), carotid (Dobrin and Rovick, 1969), femoral (Gerova and Gero, 1967; Gow, 1970), coronary (Vatner *et al.*, 1980) and pulmonary (Ingram *et al.*, 1970) arteries, the cat descending thoracic aorta (Pagani *et al.*, 1975), the rabbit ear artery (Speden, 1973; Speden, 1975), and the human radial (Weber *et al.*, 1996) and brachial arteries (Bank *et al.*, 1995; Bank *et al.*, 1996; Bank *et al.*, 1999).

It is clear that the contractile tension (or stress) of vascular smooth muscle is capable of decreasing the diameter of even the largest muscular arteries. The smooth muscle, however, does not appear to be able to contract beyond 25–30 percent below its resting length (Lundholm and Mohme-Lundholm, 1966), which limits the constriction that can be achieved. It is interesting that most arteries exhibit their greatest capacity to actively constrict at transmural pressures comparable with those to which they are exposed *in vivo* (Dobrin, 1983).

The effect of arterial smooth muscle contraction on the elastic modulus has been interpreted in totally opposite ways (see also Chapter 26). Wiggers and Wegria (1938) and later Alexander (1954) and Gow (Gow, 1972; Gow, 1980) found that the distensibility of large arteries was increased (i.e. stiffness decreased) by smooth muscle contraction, and this would normally be interpreted as a reduction of the elastic modulus. The addition of a tensile force in parallel with the elastic fibers would be expected *a priori* to make the wall stiffer (i.e. to increase the elastic modulus). This was found by Gerova and Gero (1967), Cox (Cox, 1976; Cox, 1982), Barra *et al.* (1993), Armentano *et al.* (1995), and Bank *et al.* (1995) and, in the 'barostat' experiments described above, we found that vasoconstrictor agents caused a rise in the wave velocity (Fig. 3.14) and elastic modulus of the aorta (Nichols and McDonald, 1972), as did Bargainer (1967) in the pulmonary artery. The difference in interpretation is undoubtedly due to the conditions under which the measurements were made (McDonald, 1974; Cox, 1982).

The earlier workers tended to use distensibility as the ratio of the actual radius or volume rise to the accompanying rise in pressure. Thus, Burton (1965) defines an 'absolute distensibility' as $\Delta R/\Delta P$. The strain properly defined is the ratio of the increment ΔR to the initial radius R_0 . Under these conditions, the value of the distensibility is arbitrary if the starting dimensions are unknown. In published work where the elastic tension and strain are recorded, there seems little question that a greater increase of tension is required to distend an artery at any given arterial size when the smooth muscle is stimulated to contract. This defines an increase in the incremental elastic modulus and represents the physiologic condition when the artery is *in situ* and the muscle is largely contracting isometrically (Gerova and Gero, 1967).

When excised specimens are used with a starting condition of zero distending pressure, the application of norepinephrine will cause a marked shortening (or reduction) of the circumference. Again, a given increase in circumferential length will therefore represent a greater fraction of the initial length (i.e. a larger strain), and the modulus calculated in this way will have decreased. This is apparent from the tension–length diagrams published by Attinger (1968) for the excised femoral artery, which increase in slope after the application of norepinephrine, so indicate an increased elastic modulus. The curves are then recalculated in terms of the reduced initial length and are termed ‘corrected’. Under these conditions, the modulus would be reduced.

The divergence of interpretation has been clearly discussed by Dobrin and Rovick (1969) in terms of their experimental data on the dog carotid artery, and in the reviews by Gow (1980) and Cox (1982); this is also discussed in Chapter 26. When the value of the elastic modulus is plotted against the transmural pressure, the modulus is reduced after norepinephrine. In the isolated dog carotid artery segment, when stress is plotted against strain, the elastic modulus is increased with smooth muscle contraction (i.e. the slope of the stress–strain curve is increased) compared with a control artery treated with potassium cyanide to eliminate all muscle contraction (Fig. 4.16) (Dobrin and Rovick, 1969). Activation of SMC with phenylephrine also increased the elastic modulus of the intact dog thoracic aorta somewhat (Fig. 4.17) (Armentano *et al.*, 1995). In both these studies, at very high strains, the elastic modulus of the control artery exceeds that for the artery with contracting smooth muscle; this may be interpreted as being due to the muscle fibers sustaining the stress rather than the collagen fibers,

whose high elastic modulus largely determines the elastic properties of an artery at large distensions. For a discussion (and references) concerning the effects of SMC relaxation on arterial properties, see below and Chapters 9 and 26.

The incremental elastic modulus at a given arterial diameter which increases with smooth muscle contraction would seem to be the most desirable value to use in functional terms, for it is constant with an increase in the velocity of wave propagation that we have seen also occurs under these conditions.

ATTENUATION OF PROPAGATED WAVES

The first predictions of attenuation of propagated waves were made by Bergel (1961b) from measurement of the complex elastic properties of excised vessels. This technique was also used by Learoyd and Taylor (1966) and extended to the study of arteries *in vivo* by Gow and Taylor (1968). The results from the three-point pressure method in the thoracic aorta are therefore compared (Fig. 4.18) with those obtained from isolated vessels and also the results obtained by Anliker *et al.* (Anliker *et al.*, 1968a; Anliker *et al.* 1968b) with a view to establishing norms for future studies on other arteries.

The values obtained in the excised thoracic aorta (maintained in oxygenated Krebs solution to preserve the viability of the SMC component) by McDonald and Gessner (1968) are displayed in Fig. 4.19. They are compared with the values obtained by Bergel and also with the attenuation to be expected due to the viscosity of the blood alone when the wall is expected to be purely elastic, as in the Womersley formulation discussed in Chapter 2.

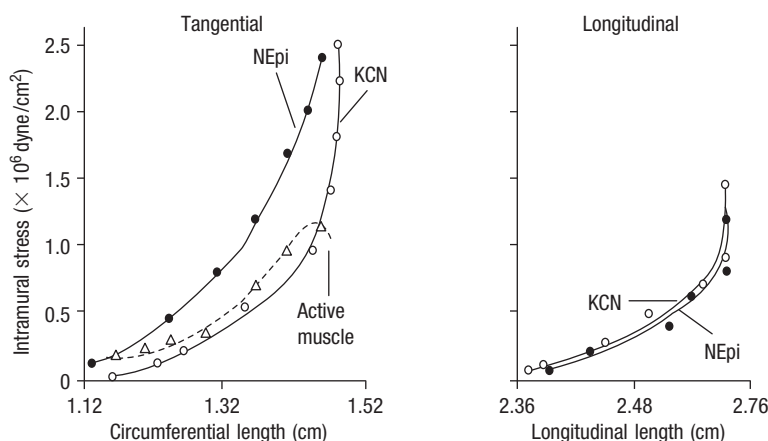


Figure 4.16 The stress–strain diagrams for the dog carotid artery comparing the vessel when smooth muscle tone is abolished by potassium cyanide (KCN curve) and when it is in a strong state of contraction due to norepinephrine (NEpi curve). The broken line is calculated from the difference between the two curves at the left to indicate the force developed by the muscle. It is seen that there is no muscular force developed in the longitudinal direction. The apparent elastic modulus is increased with muscle contraction when the stress is calculated against the strain. If it is calculated against the pressure, the modulus appears to decrease with muscle contraction because the contracting muscle reduces the circumference and reduces the strain. This apparently paradoxical effect of muscular contraction reducing the stiffness of the wall is to be explained by the elastin and collagen fibrous elements being in a meshwork and the stress falling progressively on the much stiffer collagen fibers as the strain is increased. Reproduced from Dobrin and Rovick (1969)

The curves representing Bergel's results are compatible with those of McDonald and Gessner, although the latter show a higher attenuation, because the wave velocity determined by Bergel was higher (so that 10 cm was a

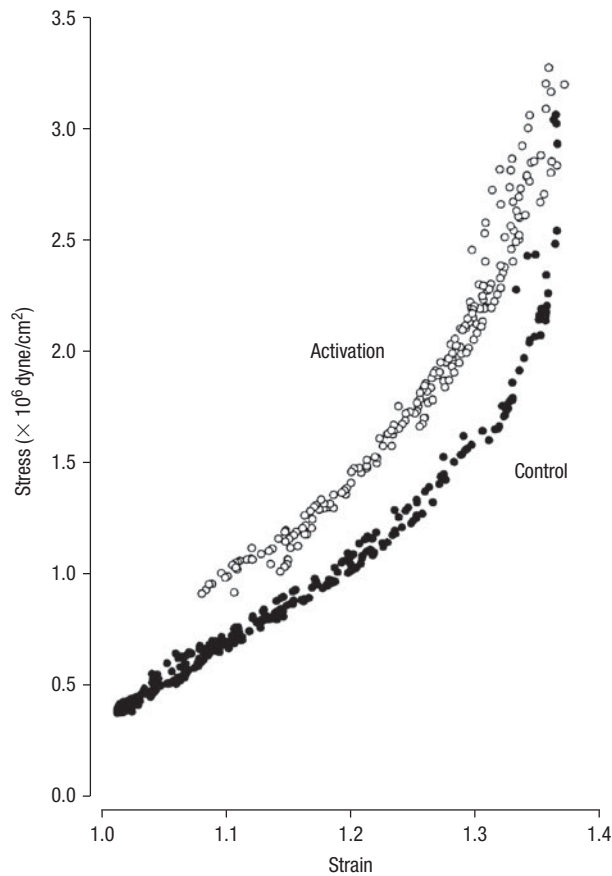


Figure 4.17 Stress-strain relation obtained in the descending thoracic aorta of a conscious dog during control (closed circles) and during vascular smooth muscle contraction with phenylephrine infusion (open circles). Aortic pressure and diameter (ultrasonic crystals) were varied by aortic occlusion. The difference between the two curves represents the force developed by smooth muscle contraction. Reproduced from Armentano *et al.* (1995)

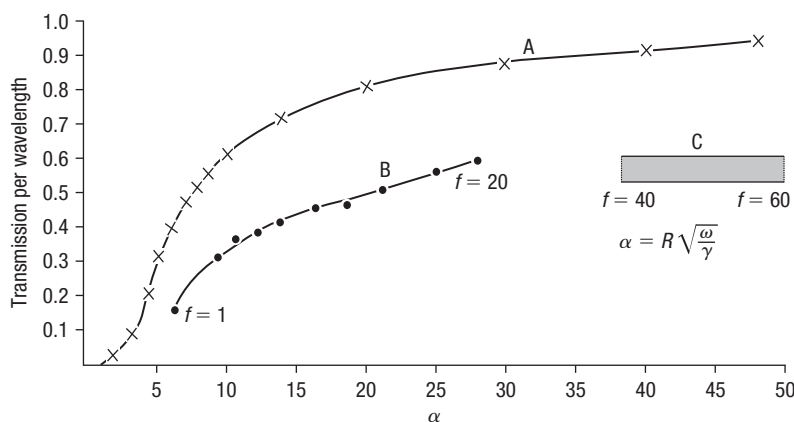


Figure 4.18 Demonstration of the attenuating properties of the arterial wall over a wide range of values of α . All values are for the dog thoracic aorta. (A) The theoretic transmission (according to Womersley (1957b) equations) of the wave when damping is only due to the viscosity of blood. (B) The transmission in the aorta measured by McDonald and Gessner (1968). (C) The range of values recorded by Anliker *et al.* (1968a) for the frequency range 40–60 Hz. It is clear that graphs B and C could be reasonably extrapolated together. This again shows that the attenuation due to the viscosity of the blood contributes only about 25–30 percent of the damping of the traveling wave. The data show that transmission per wavelength reaches a plateau of 0.5–0.6 at $\alpha = 30$

smaller fraction of a wavelength). The value of the attenuation term per wavelength was also compatible with those determined by Anliker *et al.* (1968b). This is most simply demonstrated by the calculation of the transmission per wavelength for the thoracic aorta which is displayed graphically against the value of α in Fig. 4.18 (the graph only includes the high-frequency values up to 60 Hz, as it is known that they remain constant thereafter). In Fig. 4.19, the measured attenuation in the artery is much higher (i.e. the transmission is much lower) than that due to the viscosity of the blood alone, and the major component is due to the 'viscous' component of the wall.

In the carotid and femoral arteries, the attenuation per wavelength is higher, but, because the wave velocity is also higher, the attenuation per 10 cm (Fig. 4.20) is comparable to that in the thoracic aorta. Again, the larger values in the data of McDonald and Gessner over those of Bergel in the perfused vessel were due, in part, to the fact that in the former case the wave velocity was about 7.0 m/s, while in Bergel's it was somewhat above 9.0 m/s. In the *in vivo* carotid artery (Fig. 4.20A), the attenuation varied from values close to those in the perfused isolated artery to values that were considerably higher, although the wave velocity was probably the same. The data on the vessel in the living animal are more limited than in the perfused artery because only the frequency components of the pressure wave could be used, and once they became small in the higher harmonics the results were indeterminable. Values for the transmission per wavelength as shown in the figure emphasize how much less the transmission is when compared to Womersley's model. The transmission characteristics in the femoral artery derived using the three-pressure method are consistent with the values reported by Milnor and Bertram (1978) and Milnor and Nichols (1975) using two pressures and two flows (Fig. 4.20A). Giving vasoconstrictor drugs tended to increase the attenuation. The findings all tend to support Bergel's conclusion that the viscous element in the wall is largely represented by the smooth muscle component. In Bergel's experiments, the vessel was filled with saline and

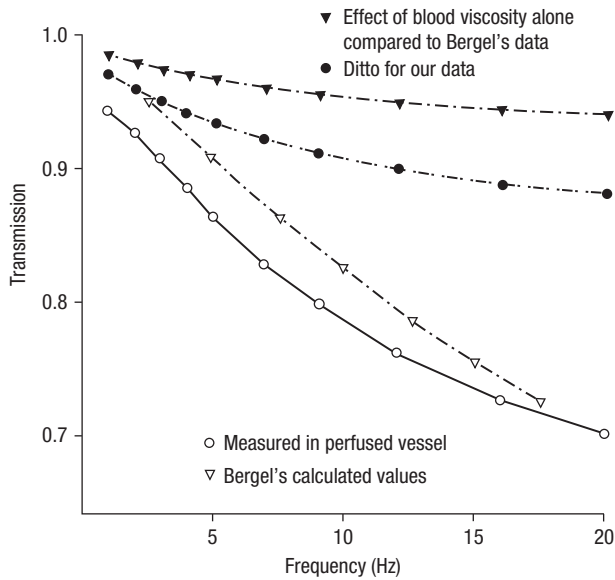


Figure 4.19 The transmission/10 cm versus frequency of the pressure wave in the perfused thoracic aorta. Open circles are from McDonald and Gessner (1968); open triangles are from Bergel (1961b). The upper two lines give the corresponding values from Womersley (1957b) equations incorporating a perfect elastic wall. Two curves are necessary because wave velocity was much lower in the experiments of McDonald and Gessner compared with Bergel's, and hence 10 cm was a longer fraction of the wavelength

suspended in air so that the smooth muscle was presumed to be completely relaxed. In the experiments of McDonald and Gessner (1968), Li *et al.* (1981), Milnor and Bertram (1978), and Milnor and Nichols (1975), the wall was oxygenated even in the isolated arteries, and the somewhat greater attenuation is probably due to the more active state of the SMC.

The data on the dynamic elastic modulus of human arteries recorded by Learoyd and Taylor (1966) are comparable to that of Bergel in dogs, and the calculated attenuation is comparable to the values discussed above.

The attenuation in pulmonary vessels has not been measured in any comparable way. The transmission of pressure waves from the main pulmonary artery to a lobar vein was studied by Caro and McDonald (1961) in the perfused rabbit lung. At 3.0–5.0 Hz (the frequency of the minimum for the input impedance), the mean pressure oscillation created by the sinusoidal pump was 2.52 cmH₂O; in a lobar vein, the amplitude was 0.62 cmH₂O. This indicates a transmission of almost 25 percent. In the experiment with the lowest peripheral resistance, the transmission was approximately 45 percent. Maloney *et al.* (1968) performed somewhat similar experiments in the perfused dog lung, but studied lower frequencies; they were also concerned with changes in alveolar pressure. They found that oscillatory flow transmission was approximately 75 percent at 0.03 Hz, approximately

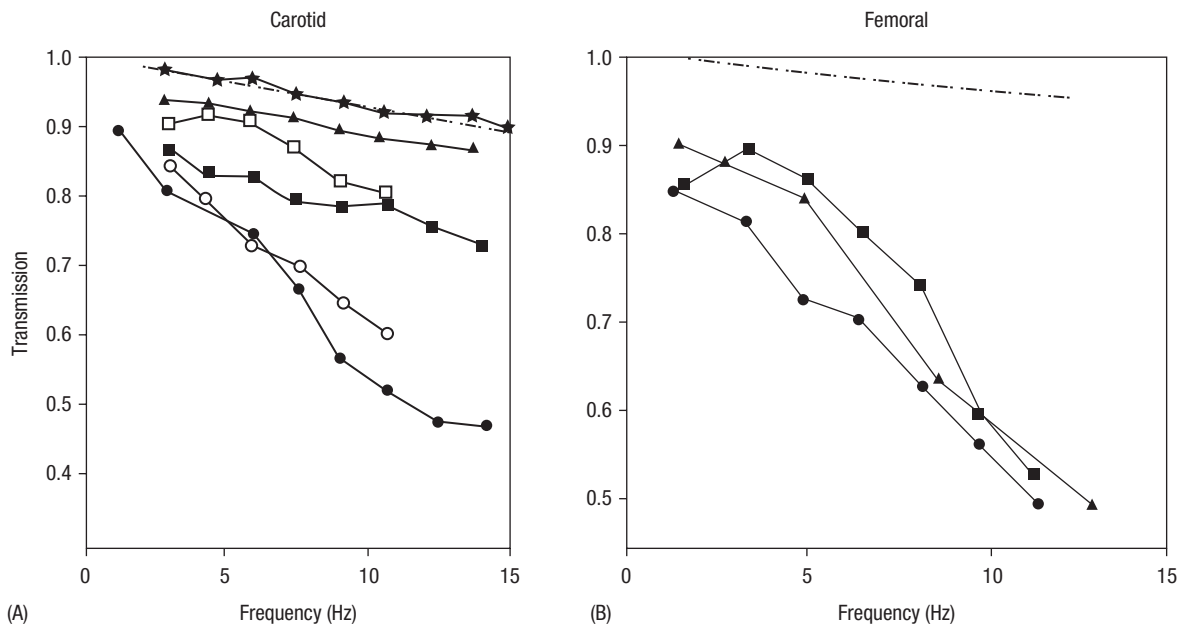


Figure 4.20 Comparison of the transmission/10 cm versus frequency of the arterial pressure wave calculated using various published models. (A) Carotid artery, --- calculated from Womersley (1957b) theory; ▲ calculated from Bergel (1961b) data, *in vitro*; ■ McDonald and Gessner (1968), perfused, *in vitro*; □ McDonald and Gessner (1968), *in vivo*; ○ McDonald and Gessner (1968), *in vitro*; * Wetterer and Kenner (1968), *in vivo*; • Li *et al.* (1981), *in vivo*. (B) Femoral artery: --- Womersley (1957b) model; ■ Milnor and Nichols (1975), *in vivo*; ▲ Milnor and Bertram (1978), *in vivo*; • Li *et al.* (1981), *in vivo*. Reproduced from Li *et al.* (1981)

50 percent at 0.1 Hz, and approximately 30 percent at 1.0 Hz. They also found the same values for transmission from vein to artery, although Caro *et al.* (1967a) had found that transmission was markedly asymmetric under certain conditions. Caro *et al.* (1967b) also studied the transmission of pressure waves between the pulmonary artery and a lobar vein in human patients. The forward transmission was of the order of 30–40 percent in patients with normal pressures. They also found asymmetric transmission, with a reduction in the backward direction.

Pressure wave transmission characteristics, especially in humans, are commonly derived as a transmission (or amplification) ratio of the distal and proximal pulse pressures (Wilkinson *et al.*, 2001; Laurent *et al.*, 2003; Mahmud and Feely, 2003b; Waring *et al.*, 2003; McEniery *et al.*, 2005a; McEniery *et al.*, 2005b; Protogerou *et al.*, 2007b; Casey *et al.*, 2008b; Edwards *et al.*, 2008; Nijdam *et al.*, 2008; Safar *et al.*, 2008; Protogerou *et al.*, 2009a; Protogerou *et al.*, 2009b; Segers *et al.*, 2009; Benetos *et al.*, 2010; Lieber *et al.*, 2010; Mitchell *et al.*, 2010; Polónia *et al.*, 2010; Salvi *et al.*, 2010b; Sumiya *et al.*, 2010); for an excellent recent review of the role of pulse pressure amplification in arterial hypertension, see Avolio (2009). Since the pressure pulse increases as it travels to the periphery because of increased stiffness and wave reflections, the transmission ratio from the ascending aorta to the femoral arteries is about 2.0 (or amplification almost 100 percent) in the dog (McDonald, 1974) and kangaroo (Avolio *et al.*, 1985b). This amplification of the pressure wave is somewhat less marked in humans (O'Rourke *et al.*, 1968; O'Rourke, 1982b; Nichols *et al.*, 1993; Nichols, 2005), amounting to about 1.73 (or 74 percent amplification) in young adults (age 20 years) and decreasing steadily with age, reaching a value just greater than 1.0 (about 5.0 percent amplification) at age 80 years (see Fig. 4.13B). These values are in line with those published recently by Wilkinson *et al.* (Wilkinson *et al.*, 2001;

Wilkinson *et al.*, 2002) in normotensives and higher than those published by Laurent *et al.* (2003) for aged-matched hypertensives (males = 1.28; females = 1.18). The reason for this species difference is not completely clear, but it is probably due to a lower apparent reflection coefficient in humans. O'Rourke *et al.* (1968) suggested that the difference may result from the greater degree of dispersion of reflecting sites in humans, because the lower limbs (legs) are longer. In young human adults, the pressure wave changes shape dramatically during transmission from the heart to the periphery, while in the elderly and hypertensive individuals with degeneration of the elastic arteries, the pressure wave in the ascending aorta is very similar to the pressure wave recorded in a peripheral artery (Fig. 4.21). The pressure falls almost exponentially from its systolic peak and there is no secondary wave in any artery; pressure moduli remain relatively constant along the aorta. The reason for the disappearance of pressure amplification in older humans is not entirely clear, but it is likely due to elastic artery degeneration, decreased distensibility and increased elastic artery PWV, and amplitude and timing of wave reflection from the lower body (see Chapters 9 and 19) (Nichols and Edwards, 2001; Nichols and O'Rourke, 2009). Pulse pressure amplification is inversely or negatively related to established markers of cardiovascular disease (Nijdam *et al.*, 2008) especially in the elderly (Benetos *et al.*, 2010; Salvi *et al.*, 2010b). Acute catecholamine administration or sympathetic stimulation of the SMC causes vasoconstriction and increases wave reflection amplitude in systole and thus reduces pulse pressure amplification (Edwards *et al.*, 2008; Nichols and Epstein, 2009); vasodilation has the opposite effect (Protogerou *et al.*, 2007b; Safar *et al.*, 2008). The newer anti-hypertensive drugs (angiotensin-converting enzymes inhibitors (ACEI), angiotensin receptor blockers (ARB), and dihydropyridine

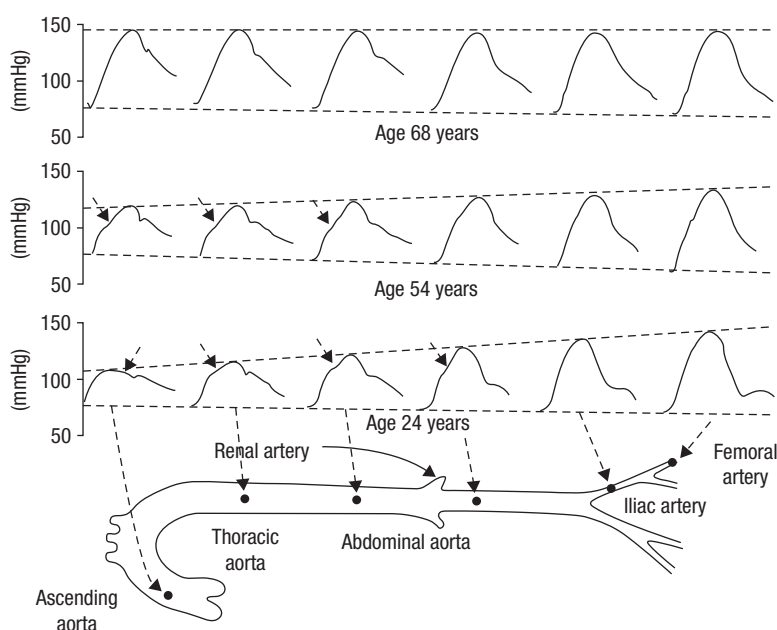


Figure 4.21 Pressure waves recorded along the arterial tree from the proximal ascending aorta to the femoral artery in three adult humans aged 24, 54, and 68 years. In the oldest subject, there is little amplification in the pressure wave during transmission; however, in the youngest subject, the amplitude of the pressure wave increases approximately 60 percent during transmission. Reproduced from Nichols *et al.* (1993)

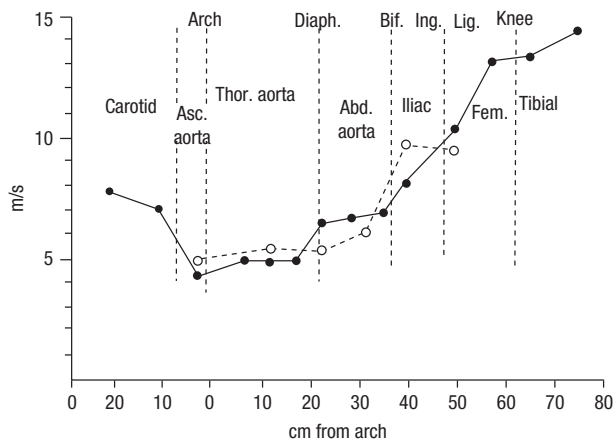


Figure 4.22 A diagram to show the progressive increase in wavefront velocity of the pulse wave with increasing distance from the heart in the dog. The mean pressure was 130–160 cmH₂O for all readings. It will be seen that in the ascending aorta the wave velocity is 4.0 m/s, rises slightly to about 4.7 m/s in the thoracic aorta, and finally attains 13–15 m/s in the femoral and tibial arteries. The rise towards the head is also marked, but is ultimately less elevated because the distances are shorter. Reproduced from Nichols and McDonald (1972). Pulse wave velocities recorded by Latham *et al.* (1987) in young adult humans are indicated by the broken line

calcium channel blockers (CCB)), as well as nitrates, have a more beneficial effect on pulse pressure amplification than the older drugs (diuretics and β -blockers) (Protogerou *et al.*, 2009a; Protogerou *et al.*, 2009b). These modifications in pulse pressure amplification are due to alterations in muscular artery stiffness and reflected wave amplitude and can occur with or without a change in elastic artery stiffness or PWV.

The distribution of arterial wave velocities in the dog (McDonald, 1968b) are illustrated in Fig. 4.22; the velocity over each interval is represented by a point at the middle of the interval. The point representing the ascending aorta was added from a later paper by Nichols and McDonald (1972); repeated measurements along the aorta were found to be in close agreement with those illustrated. The values are in general agreement with those of Dow and Hamilton (1939), who also measured the wave velocities (foot-to-foot with intra-arterial catheters over intervals of 5.0 cm). Their paper was, for many years, the only clear evidence of a progressive and marked rise in wave velocity with increasing distance from the heart because the series of measurements were made in the same dog. Laszt and Muller (Laszt and Muller, 1952a; Laszt and Muller, 1952b; Laszt and Muller, 1952c), in a series of papers, made careful recordings of the normal patterns of pressure waves throughout the arterial system and measured the PWV over intervals of 10–15 cm. A summary of these findings, together with values from several other studies, is given in Table 4.6. Values of the pulse wave velocity obtained in various arteries of humans are also given in Fig. 4.23. Latham *et al.* (1985) utilized a specially designed

catheter with six micromanometers equally spaced at 10 cm intervals to measure PWV and pressure wave contours along the human aorta (age 42 ± 5 years); a similar study was also reported in primates by Latham *et al.* (1987). Some values of pulse wave velocity collected in humans, dogs, rats, and kangaroos are given in Table 4.6 and Figs 4.22, 4.23, and 4.24; Fig. 4.24 shows the age-related PWV change in elastic and muscular arteries.

Most of the recent PWV studies in humans have employed non-invasive techniques to measure pulsatile pressure, blood flow velocity, or diameter (see Chapter 3) (Avolio *et al.*, 1983; Levenson *et al.*, 1984a; Levenson *et al.*, 1984b; Avolio *et al.*, 1985a; Avolio *et al.*, 1986; Toto-Moukouo *et al.*, 1986; Levenson *et al.*, 1987a; Levenson *et al.*, 1987b; Safar *et al.*, 1988; Lehmann *et al.*, 1993; Wilkinson *et al.*, 1999; Yasmin and Brown, 1999; Asmar *et al.*, 2001a; Asmar *et al.*, 2001b; Kelly *et al.*, 2001; Pannier *et al.*, 2001; Rogers *et al.*, 2001; Smulyan *et al.*, 2001; Stevanov *et al.*, 2001; Sutton-Tyrrell *et al.*, 2001b; Harada *et al.*, 2002; Vulliemoz *et al.*, 2002; Wilkinson *et al.*, 2002; Blacher *et al.*, 2003; Bonapace *et al.*, 2003; Danias *et al.*, 2003; Ichihara *et al.*, 2003; Kidawa *et al.*, 2003; Kidawa *et al.*, 2003; Vlachopoulos *et al.*, 2003c; Mitchell *et al.*, 2004a; Greenwald, 2007). These studies show that hypertension, advancing age, cigarette smoking, diabetes (type 1 and 2), end-stage renal disease, cardiac syndrome X, heart failure, coronary artery disease, and obesity cause an increase in central elastic artery stiffness (and PWV), while a low-salt diet (Avolio *et al.*, 1986), enhanced external counter-pulsation and exercise conditioning (light, moderate, or vigorous intensity) (Vaitkevicius *et al.*, 1993; Edwards *et al.*, 2004a; Edwards *et al.*, 2004b; Nichols *et al.*, 2006; Gando *et al.*, 2010) cause a decrease. Acute bouts of exercise and vasodilators decrease muscular artery stiffness and PWV (but have little active effect on central elastic artery stiffness (Chapter 26) (Sugawara *et al.*, 2003). Early studies of arterial mechanical properties dealt mostly with normal arteries of experimental animals studied *in vitro* (arterial segments), or *in vivo* under standard controlled conditions. Complexities abound here and relate to non-linear pressure–diameter relationships, effect of wall viscosity and calculation of the elastic modulus from measurements of wall thickness and intimal length or vessel diameter.

When applied to human subjects, these matters are elevated to complexity at another level of magnitude. Representative of problems is the issue of gender, hormones, and menopause. Claims for differences in arterial stiffness had not been supported by large, more detailed studies between males and females (Avolio *et al.*, 1983; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b; Mitchell *et al.*, 2004a) in the earlier editions of this text, however, recent studies have appeared showing gender-related differences in arterial geometry and wave reflections (McEniery *et al.*, 2005a; McEniery *et al.*, 2005b; Alecu *et al.*, 2006; Vermeersch *et al.*, 2008; Cernes *et al.*, 2010; Lieber *et al.*, 2010). Other problems related to the effects of aging, of disease, and of drug therapy, and to

Table 4.6 Pulse wave velocity in arteries

Artery	Species	Wave velocity (cm/s)	Reference
Carotid-femoral artery	Human	730	Van der Heijden-Spek <i>et al.</i> (2000) The Reference Values for Arterial Stiffness' Collaboration (2010)
	Human (< 30 yrs)	620	
	Human (30–39)	650	
	Human (40–49)	720	
	Human (50–59)	830	
	Human (60–69)	1030	
Aorta	Human (≥ 70)	1090	Cantini <i>et al.</i> (2001) Hartley <i>et al.</i> (2000) Fitch <i>et al.</i> (2006)
	Rat	511	
	Mouse	380	
Ascending aorta	Mouse	340	Luchsinger <i>et al.</i> (1964) Merillon <i>et al.</i> (1976) Merillon <i>et al.</i> (1980) Latham <i>et al.</i> (1985) Nichols and McDonald (1972)
	Human	520	
	Human	387	
	Human	545	
	Human	440	
	Dog	350–472	
Thoracic aorta	Human	550–650	Wezler and Boger (1939) Luchsinger <i>et al.</i> (1964) Latham <i>et al.</i> (1985) Dow and Hamilton (1939) Laszt and Müller (1952a) McDonald (1968b)
	Human	400	
	Human	530	
	Dog	400–700	
	Dog	480–550	
	Dog	410–480	
Abdominal aorta	Dog	400–500	Anliker <i>et al.</i> (1968b) Luchsinger <i>et al.</i> (1964) Latham <i>et al.</i> (1985) Dow and Hamilton (1939) Laszt and Müller (1952a) McDonald (1968b)
	Human	500	
	Human	570	
	Dog	600–800	
	Dog	550–850	
	Dog	600–740	
Iliac	Dog	660	Li <i>et al.</i> (1981) Latham <i>et al.</i> (1985) McDonald (1968b)
	Human	880	
	Dog	700–800	
Femoral	Dog	765	Li <i>et al.</i> (1981) Kapal <i>et al.</i> (1951) Dow and Hamilton (1939) Laszt and Müller (1952a) McDonald (1968b) Cox (1971)
	Human	800	
	Dog	800–1200	
	Dog	850–1300	
	Dog	830–1030	
	Dog	1185	
Popliteal	Dog	878	Li <i>et al.</i> (1981)
Tibial	Dog	1220–1310	McDonald (1968b)
Carotid	Dog	1040–1430	McDonald (1968b)
	Dog	610–780	McDonald (1968b)
	Dog	850	Busse <i>et al.</i> (1979b)
Radial	Dog	804	Li <i>et al.</i> (1981)
	Human	730	Levenson <i>et al.</i> (1987a)
Coronary	Dog	860	Arts <i>et al.</i> (1979)
Pulmonary	Human	182	Caro and Harrison (1962)
	Human	168	Milnor <i>et al.</i> (1969)
	Dog	255	Attinger (1963)
	Dog	275	Bargainer (1967)
Ascending aorta to femoral	Human	767	Avolio <i>et al.</i> (1985a)
	Dog	619	Latson <i>et al.</i> (1988)
	Kangaroo	496	Avolio <i>et al.</i> (1985b)

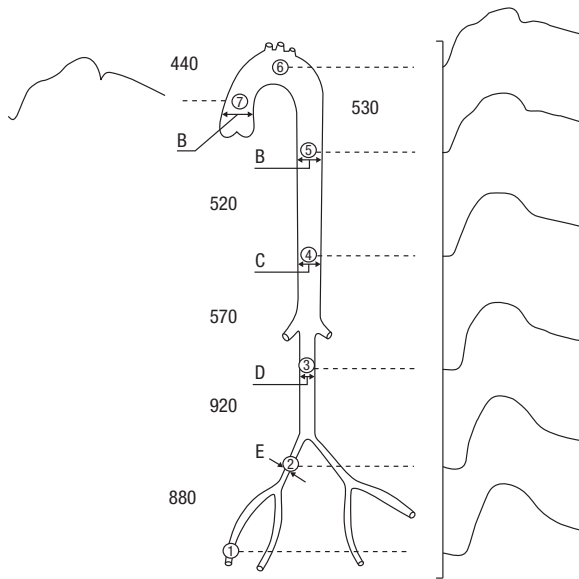


Figure 4.23 Schema showing the location of the micromanometers (separated by 10 cm intervals) along the human aorta. Measurements in the ascending aorta were obtained with a different catheter with micromanometers separated by 5.0 cm. Pulse wave velocity values (cm/s) measured over each interval are shown. Reproduced from Latham *et al.* (1985)

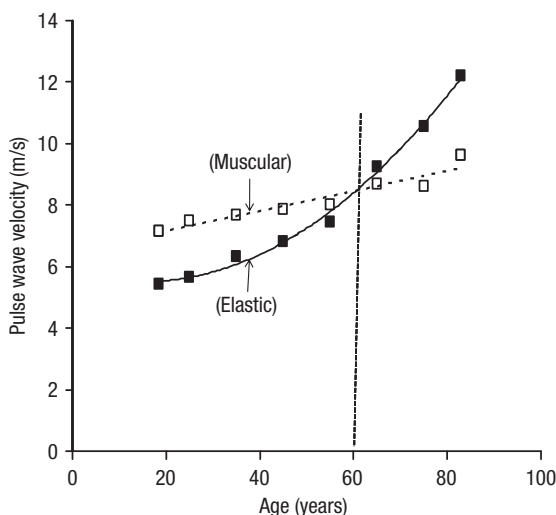


Figure 4.24 Age-related changes in carotid-femoral and brachial artery pulse wave velocity (PWV) in healthy subjects. Data from the Anglo-Cardiff Collaborative Trial (ACCT) study population (McEniery *et al.*, 2005a and McEniery *et al.*, 2005b). Note the relatively linear relationship for muscular artery PWV (dotted line and open squares), compared with the curved relationship for elastic artery PWV (solid line and closed squares) which changes much more, especially in subjects > 50 years of age (McEniery *et al.*, 2007). The broken vertical line represents the age at which the arterial-stiffness gradient reverses (Nichols and O'Rourke, 2009)

the less accurate methods that were previously available for measuring pressure, flow velocity, and arterial dimensions. Problems also relate to the different effects of aging, of disease, and of drugs at different sites in the arterial tree,

and often to the wish of clinicians to simplify the subject by accepting the simplest (Windkessel) model of the arterial tree. The problems are magnified by the fact that most clinicians and clinical scientists do not recognize the difference between systolic and diastolic pressure in different parts of the arterial tree. Several articles have been published in which measures of distensibility were determined from central artery diameter (aorta and carotid artery) change and peripheral brachial pressure; in both, the possibility of systematic differences between central and peripheral pressure was mentioned but dismissed (Cholley *et al.*, 1995b; Salomaa *et al.*, 1995). Other publications (e.g. Pearson *et al.*, (1994)) have made the same mistake and continue to do so (Arnett *et al.*, 1999; Nagai *et al.*, 1999). This subject is of paramount importance in modern clinical practice, as recognized by the National Institutes of Health in its 1995 statement (vol. 24, no. 24, June 1995, PA-95-074) announcing the fostering of research into arterial stiffness and aortic impedance in human subjects. Various aspects of this subject are dealt with in specific chapters on aging (Chapter 19), arterial disease (Chapter 24), hypertension (Chapter 20), cardiac failure (Chapter 15), and drug therapy (Chapter 26). Just a very brief summary is given here on the effects of age and drugs on arterial properties.

Aging

Even though age-related changes in arterial properties and wave reflection characteristics are mentioned in numerous chapters of this text and one devoted entirely to arterial aging (Chapter 19), the topic will be discussed briefly here. With the passage of time, certain histologic changes that alter structure and function occur in the wall of the large central elastic arteries, but not the peripheral muscular arteries, of adult humans. Post-mortem studies show that aortic wall thickening with aging consists of an increase in intimal thickness even in populations with a low incidence of atherosclerosis (Virmani *et al.*, 1991). Several detailed studies of human arterial structural and geometric changes with age (Spina *et al.*, 1983; Hosoda *et al.*, 1994; O'Rourke *et al.*, 1987; Virmani *et al.*, 1991; Pentimone *et al.*, 1994; Alessandri *et al.*, 1995; Cattell *et al.*, 1996; Tanaka *et al.*, 1998; Schmidt-Trucksäss *et al.*, 1999; Lakatta, 2000; Dinunno *et al.*, 2001; Fleischmann *et al.*, 2001; Tanaka *et al.*, 2001; Agmon *et al.*, 2003; Ferrari *et al.*, 2003; Jacob, 2003; Lakatta, 2003; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b; Jacob, 2006; Jani and Rajkumar, 2006; Greenwald, 2007) have shown that the principal changes occur in the intima and media. Endothelial cells of the intima become more irregular in size and shape and their function is progressively depressed (see Chapter 5) (Celermajer *et al.*, 1994a; Djuric *et al.*, 1999; Smith and Hagen, 2003). The activation/upregulation of arginase appears to be an important contributor to age-related endothelial dysfunction by a mechanism that involves substrate

(L-arginine) limitation for NO synthase and therefore NO synthesis (Santhanam *et al.*, 2008) and appears to occur earlier in males than in females (Celermajer *et al.*, 1994a). In females, however, a steep decline commences at around the time of the menopause. The subendothelial layer becomes thickened with an increase in connective tissue content. In the media, the elastic fibers and laminae lose the orderly arrangement seen in earlier life and display thinning, splitting, fraying, and fragmentation. Degeneration of elastic fibers is associated with an increase in collagenous fibers and in ground substance, and often with calcium deposition in degenerate elastic material. In the human thoracic aorta, normal elastin fibers decrease from 32.14 to 20.26 percentage dry weight ($P < 0.001$) while collagen fibers increase from 21.44 to 32.41 percentage dry weight ($P < 0.001$) over the age range 20–80 years (Faber and Moller-Hou, 1952). These changes in arterial wall structure lead to increased elastic modulus (or stiffness, Figs 4.25 and 19.10) with age and a decrease in arterial distensibility (Fig. 19.13) and compliance (Gozna *et al.*, 1974; Merillon *et al.*, 1978; Langewouters, 1982; Cox, 1983b; Langewouters *et al.*, 1984; Langewouters *et al.*, 1985; Nichols *et al.*, 1985a; Hirai *et al.*, 1989; Länne *et al.*, 1992b; Sonesson *et al.*, 1993;

Pearson *et al.*, 1994; Ahlgren *et al.*, 1997; Tanaka *et al.*, 1998). This age-related increase in arterial rigidity increases PWV and causes early return of the reflected wave from the lower body to the heart causing an increase in central systolic (and pulse) blood pressure (Nichols *et al.*, 1985a; Länne *et al.*, 1992b; Sonesson *et al.*, 1993; McEniery *et al.*, 2005a; McEniery *et al.*, 2005b; Greenwald, 2007; McEniery *et al.*, 2007; Nichols *et al.*, 2008). These undesirable changes in arterial properties and wave reflection characteristics influence an increase in the incidence of stroke, renal and cardiac failure, and cardiovascular and all-cause mortality. The most striking macroscopic finding is an increase in the size of arteries with age. There is a progressive increase in elastic artery diameter (Fig. 4.26) (Learoyd and Taylor, 1966; Merillon *et al.*, 1982a; Merillon *et al.*, 1982b; Nichols *et al.*, 1985a; Länne *et al.*, 1992a; Pedersen *et al.*, 1993; Sonesson *et al.*, 1993; Sonesson *et al.*, 1994; Schmidt-Trucksäss *et al.*, 1999; Lakatta, 2000; Fleischmann *et al.*, 2001; Agmon *et al.*, 2003; Najjar *et al.*, 2005; O'Rourke and Nichols, 2005), wall thickness (Learoyd and Taylor, 1966; Pearson *et al.*, 1994), intima-medial thickness (Schmidt-Trucksäss *et al.*, 1999; Dinunno *et al.*, 2001; Tanaka *et al.*, 2001; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b), length (Sugawara *et al.*, 2008), and PWV (Fig. 19.14) (Avolio *et al.*, 1983; Avolio *et al.*, 1985a; Pannier *et al.*, 1989a;

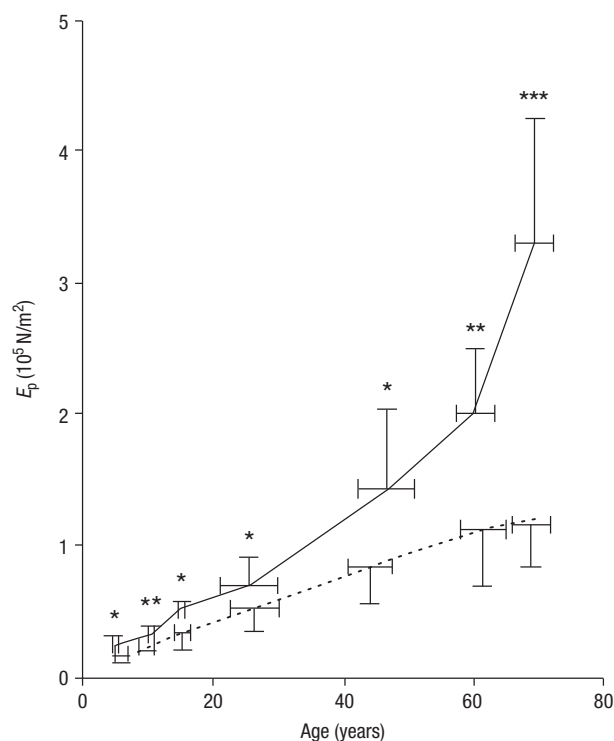


Figure 4.25 Age-related increase in the abdominal aortic pressure-strain elastic modulus (E_p) of female subjects (broken line) compared with male subjects (solid line). Note that the increase in E_p (i.e. increase in stiffness) is almost linear with age in females, whereas in males it is exponential in nature and of greater magnitude. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$. Reproduced from Sonesson *et al.* (1993)

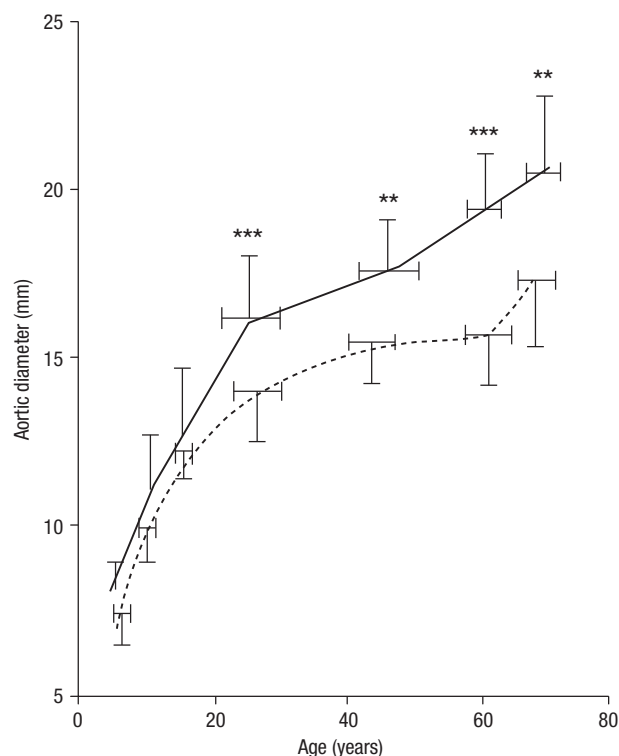


Figure 4.26 Age-related increase in mean abdominal aortic diameter (non-invasive ultrasound) of female (broken line) and male (solid line) subjects. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$. Reproduced from Sonesson *et al.* (1993)

Asmar *et al.*, 1995a; McEniery *et al.*, 2005a; McEniery *et al.*, 2005b; McEniery *et al.*, 2007; Redheuil *et al.*, 2010). In the studies by Länne *et al.* (Länne *et al.*, 1992a; Länne *et al.*, 1992b) and Sonesson *et al.* (1993), the increase in mean distal abdominal aortic stiffness (Fig. 4.24) and internal diameter (Fig. 4.26) with age was found to be gender dependent with the increase in males (5–71 years) being significantly higher than in females (4–74 years). In contrast, Hajdu *et al.* (1990) found an age-related decrease in the diameter (and cross-sectional area) and distensibility of cerebral arterioles of normotensive rats. Measurements were made at the same intra-luminal pressure in pial arterioles with deactivated SMC (Fig. 4.27).

Changes in the elastic properties of the arterial system with age were first investigated by Roy in 1880. He studied age-related changes in the pressure–volume relations of intact arterial segments and force–elongation relations of aortic strips from humans and rabbits. These studies indicate that, with advancing age, elastance (or stiffness) of arterial segments increases. Roy also observed that human arteries were stiffer than those obtained from adult experimental animals and suggested that arterial degeneration begins in childhood. Subsequent studies have shown similar results, with increased stiffness apparent as the elastic modulus of the artery increases (Butcher and Newton, 1958; Roach and Burton, 1959; Learoyd and Taylor, 1966; Gozna *et al.*, 1974; Mirsky and Janz, 1976; Cox, 1978; Langewouters, 1982; Yin *et al.*, 1983a; Langewouters *et al.*, 1984; Langewouters *et al.*, 1985; Nichols *et al.*, 1985a). Butcher and Newton (1958) found that both longitudinal and circumferential

distensibility decreased with age and that maximum distensibility occurred at age eight years. Regional differences in distensibility were also noted in these studies, with distensibility being less in the abdominal aorta than in the thoracic aorta. Physical activity and control of plasma cholesterol and body mass index may suppress the age-related increase in central arterial stiffness (Tanaka *et al.*, 1998).

Drugs

Drugs affect the arterial wall, but usually through their direct effect on SMC in the vessel wall. The most marked effect of these drugs is predominantly on the peripheral muscular arteries, especially the arterioles. As described in Chapter 26, arterial vasodilatation induced by drugs, such as nitroglycerin, that cause smooth muscle relaxation results in an active increase in arterial diameter and cross-sectional area (Fig. 4.28A) and a rightward shift of the stress–strain relation with reduced slope (and stiffness) (Fig. 4.28B) (Bank *et al.*, 1995). The dilated artery is less, rather than more, stiff. The opposite effect is seen with vasoconstrictor drugs, such as norepinephrine and phenylephrine (i.e. reduced diameter and increased stiffness). In the large central elastic arteries, dilatation causes a decrease in compliance (increase in stiffness) when this is induced either by an increase in arterial pressure (passive increase in arterial diameter) or by degeneration and fragmentation of elastin fibers in the arterial wall (Bonyhay *et al.*, 1997). Hence, different effects are seen with dilatation of central and of peripheral arteries (Nichols and Edwards, 2001; Weber *et al.*, 2003); in the former, dilatation is associated with increased stiffness, in the latter, with decreased stiffness. The paradoxical effect of vasodilator drugs in peripheral arteries has been explained on the basis of architecture of the arterial media, with SMC being in series with collagenous elements, and in parallel with elastin fibers; dilatation transfers stresses from collagenous to the more extensible elastin fibers and causes the dilated arterial wall to be less stiff and have lower elastic modulus.

An increase in stiffness (or elastance) related to arterial wall composition occurs over a long period of time in elastic arteries, e.g. with advancing age, hypertension, and arteriosclerosis, and shifts the pressure–diameter (or stress–strain) curve to the left increasing its slope (Fig. 4.29A). Acute changes can, however, occur in elastic arteries with changes in distending pressure as a result of muscular artery modification, but these changes are passive and movement is along the same pressure–diameter curve. For example, during vasodilation both pressure and diameter decrease in elastic arteries causing a passive decrease in wall stiffness and slope of the pressure–diameter curve. A change in stiffness of muscular arteries is primarily due to acute changes in smooth muscle tone and cause shifts of the pressure–diameter curve

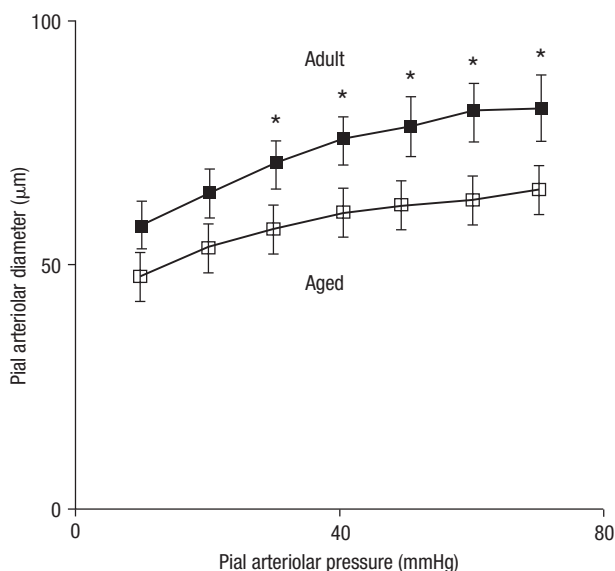


Figure 4.27 Pressure–diameter relation of pial arterioles with deactivated smooth muscle cells in young adult and aged rats. Arteriolar diameter (and cross-sectional area) is smaller in the older rats at all pressure levels. * $P < 0.01$. Reproduced from Hajdu *et al.* (1990)

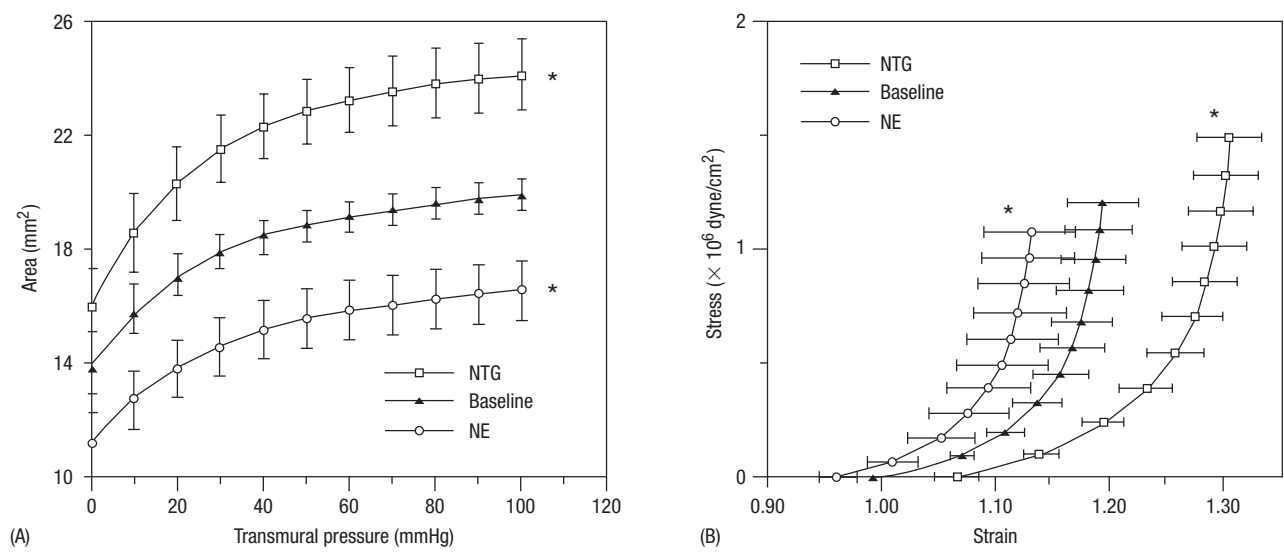


Figure 4.28 (A) Mean arterial transmural pressure versus cross-sectional area curves of the human brachial artery during baseline, smooth muscle relaxation (intra-arterial nitroglycerin, NTG) and contraction (i.e. norepinephrine, NE). Smooth muscle relaxation caused an increase and smooth muscle contraction a decrease in vessel diameter (and area) over the entire pressure range. (B) Mean stress-strain curves at baseline and after NTG and NE administration. Vasodilatation causes a decrease in arterial stiffness while vasoconstriction causes an increase ($*P < 0.001$). Reproduced from Bank *et al.* (1995)

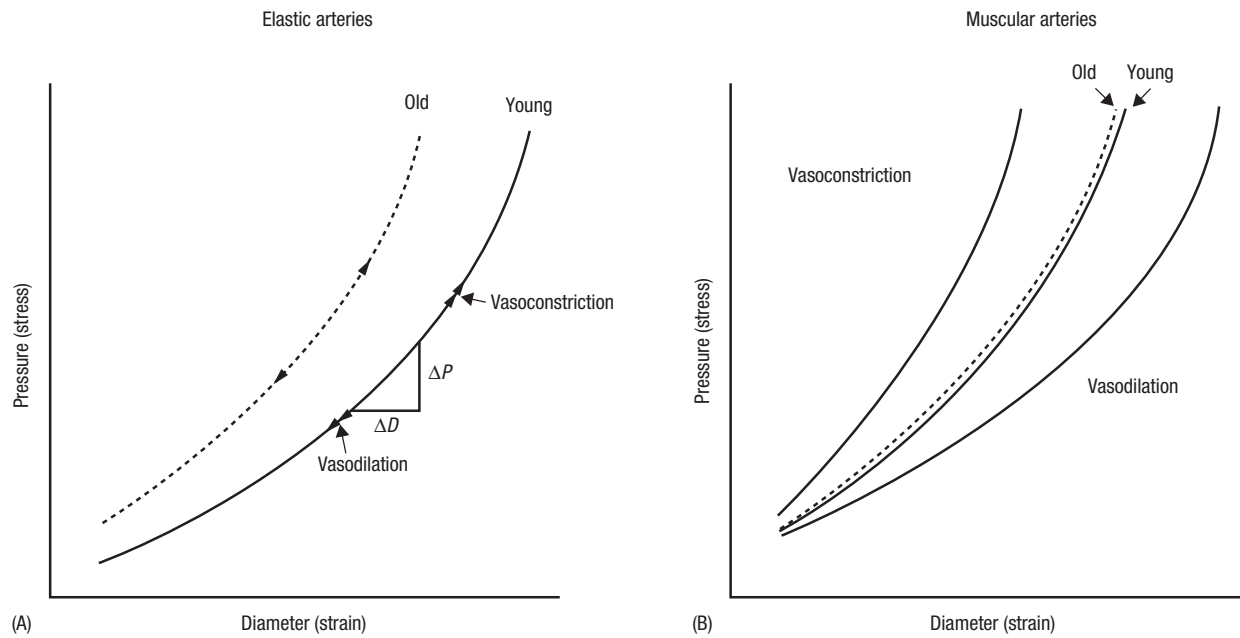


Figure 4.29 Pressure (P)-diameter (D) or stress-strain relations for elastic arteries (A) and muscular arteries (B). Elastance (or stiffness) can be estimated as the slope ($\Delta P/\Delta D$) of the pressure-diameter curve. Muscular arteries of young individuals are stiffer than elastic arteries. With advancing age, the curve for elastic arteries shifts to the left indicating increased stiffness for all diameters. Stiffness of muscular arteries changes very little with advancing age. Vasoconstrictor drugs cause a decrease in diameter and an increase in pressure and stiffness and shifts the curve to the left, while vasodilator drugs cause an increase in diameter and a decrease in pressure and stiffness and shifts the curve to the right. Drugs have little direct effect on elastic arteries, however they do have a passive effect. Vasoconstrictor drugs cause an increase in pressure, diameter, and stiffness, while vasodilator drugs cause a decrease in pressure, diameter, and stiffness. The pressure-diameter curve does not shift. Reproduced from Nichols and Edwards (2001)

(Fig. 4.29B). During vasodilation, diameter increases while pressure decreases (Boutouyrie *et al.*, 2000) causing a decrease in stiffness and a shift of the curve to the right (Bank *et al.*, 1995); vasoconstriction increases stiffness and has the opposite effect. Advancing age and hypertension have little effect on stiffness of muscular arteries in humans (Benetos *et al.*, 1993b; Bortolotto *et al.*, 1999; McEniery *et al.*, 2007; Nichols *et al.*, 2008) and drugs have little effect on stiffness of elastic arteries. However, recent evidence indicates that long-term treatment with

cholesterol-lowering drugs may reduce large elastic artery stiffness while increasing diameter (Ferrier *et al.*, 2002; Kontopoulos *et al.*, 2003).

All these issues will be taken up in the specific chapters that follow. Readers will need to appreciate that, in clinical studies, there is considerable controversy and confusion on models, principles, and results. These are a consequence of the complexity of the arterial system and the processes that affect it, and of the need to introduce some simplification.

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Endothelial function

A new role for the vascular endothelium	111	Drugs and endothelial function	123
Nitric oxide and endothelial function	113	Endothelial function and prediction of cardiovascular events	125
Measurement of endothelial function (dysfunction)	114	Endothelial function and arterial distensibility	126
Age-related changes in endothelial function	121	Wall shear stress and endothelial gene expression	126
Endothelial dysfunction and coronary atherosclerosis	122		
Exercise training and endothelial function	123		

A NEW ROLE FOR THE VASCULAR ENDOTHELIUM

Despite its very potent vasodilating action *in vivo*, acetylcholine (ACh) does not always produce relaxation of isolated preparations of blood vessels *in vitro*. We demonstrate here that relaxation of isolated preparations of rabbit thoracic aorta and other blood vessels by ACh requires the presence of endothelial cells, and that ACh acting on muscarinic receptors of these cells, stimulates the release of a substance(s) that causes relaxation of the vascular smooth muscle.

Furchgott and Zawadzki (1980)

Up until the late 1970s, the vascular endothelium was considered as a passive lining of flat cells that separated the load-bearing elements of the vascular media from the flowing blood. Over the subsequent 30 or more years, it has become apparent that the endothelium produces a host of chemical substances, notably nitric oxide (NO), endothelin, prostacyclin, and angiotensinogen, which are designed to maintain homeostasis through effects on elements of blood in the lumen, on the intimal surface which is exposed to blood, within the intima itself, and on vascular smooth muscle cells in the media (Vane *et al.*, 1990; Celermajer, 1997).

The explosion in knowledge on endothelial function began with the unexpected findings and their explanation (Fig. 5.1) (Furchgott and Zawadzki, 1980; Furchgott *et al.*, 1981; Vanhoutte, 1988; Moncada *et al.*, 1991). As research continued, all disciplines which are involved in vascular biology, including genetics and genetic engineering, new

drug discoveries and explanation of actions of older drugs, as well as arterial remodeling and pathology, pathophysiology, thrombosis, and clinical medicine were embraced. One of the first major applications was introduction of recombinant tissue plasminogen activator (rtPA) as a thrombolytic agent for treatment of coronary artery thrombosis with evolving myocardial infarction (O'Rourke *et al.*, 1988), and one of the last, use of the phosphodiesterase type 5 (PDE5) inhibitors, such as sildenafil and tadalafil, for treatment of refractory pulmonary hypertension (Wilkins *et al.*, 2001; Michelakis *et al.*, 2002; Reffelmann and Kloner, 2003; Latini *et al.*, 2008; Vlachopoulos *et al.*, 2009a; Vlachopoulos *et al.*, 2009b). In between, knowledge has been applied to better use of nitrates in disease (to potentiate the diminished supply of NO), to explain the early beneficial effect of statins (Treasure *et al.*, 1995; Celermajer, 1997; Tsiara *et al.*, 2003) and of other drugs, and to treat erectile dysfunction (impotence) with PDE5 inhibitors (Cheitlin *et al.*, 1999).

Vane won the Nobel Prize in 1982 for his discovery of prostaglandins and related biologically active substances, while Furchgott, Mourad, and Ignarro, all basic scientists, won the Prize in 1998 for their work on endothelial-derived relaxing factor as a signaling agent in the endothelium and elsewhere. Physiological and medical applications are described in many articles, with reviews by Pepine (1998a), Anderson (1999), Mombouli and Vanhoutte (1999), Vogel (1999), Bayraktutan (2002), Verma and Anderson (2002), Vita and Keaney (2002), Celermajer (2003), Reffelmann and Kloner (2003), Widlansky *et al.* (2003), Brunner *et al.* (2005), Adams (2006), Esper *et al.* (2006), Deanfield *et al.* (Deanfield *et al.*, 2005; Deanfield *et al.*, 2007), Pries *et al.* (2008),

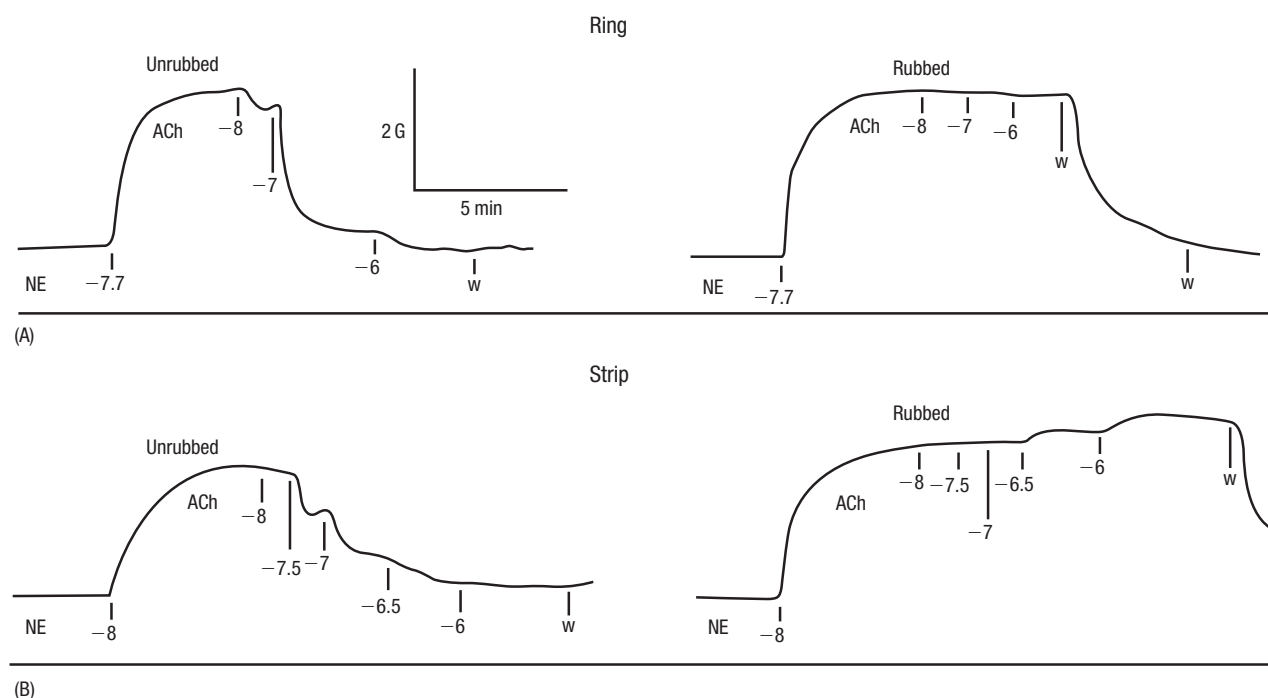


Figure 5.1 Seminal figure from Furchgott *et al.* (1981). Recordings show loss of relaxing response of rabbit aorta to acetylcholine (ACh) after complete removal of endothelial cells. In each panel, the recording on the left was made before removal and that on the right after removal of endothelial cells. The cells were removed from the ring in panel A by rubbing its intimal surface with a small wooden stick, and from the transverse strip in panel B by rubbing its intimal surface on moistened filter paper. After the procedure to remove endothelial cells, each preparation was allowed to re-equilibrate in its organ chamber for about 1 hour before being retested. Reproduced from Furchgott *et al.* (1981)

Ter Avest *et al.* (2007), Schwartz *et al.* (2009), Tomiyama and Yamashina (2010), and Vanhoutte (2009a).

Substances produced by the endothelium modify the ability of platelets to adhere to the vascular wall, and to aggregate with the formation of a clot. This mitigates against the development of a thrombus with narrowing and obstruction of the artery. Substances derived from the endothelium also inhibit entry of circulating leukocytes and lipoproteins into the vessel wall. Indeed, accumulation of monocytes and monocyte-derived phagocytes in the arterial wall causes chronic inflammation and leads to development and progression of atherosclerosis (Mestas and Ley, 2008) (Chapter 23). Endothelin-1 is a powerful vasoconstrictor (Agapitov and Haynes, 2002); its natural role appears to be in causing arterial spasm when the vessel is cut or traumatized. It plays a major role in limiting hemorrhage after trauma (Verhamme and Hoylaerts, 2006). In disease, it may be responsible for coronary artery spasm and episodes of Prinzmetal angina and it may be involved in progression of renal disease and hypertension (Dhaun *et al.*, 2006; Dhaun *et al.*, 2008). The overwhelming recent attention has been to the substance shown by Furchgott to be released from endothelium in response to acetylcholine (ACh) and other stimuli. Initially called 'endothelial-derived relaxing factor' (EDRF), it was later identified as the gas, nitric oxide (NO), by Palmer *et al.* in 1987. NO decreases smooth muscle tone

and arterial stiffness, while endothelin-1 has the opposite effect (Wilkinson *et al.*, 2002; McEniery *et al.*, 2003; Schmitt *et al.*, 2005).

Nitroglycerin and other organic nitrates have been used for over a century for treatment of angina pectoris, hypertension, and cardiac failure (Murrell, 1879; Broadbent, 1890; Mackenzie, 1902). The actions of these drugs are similar to those of NO, released from the vascular endothelium. One of the key findings of the 1998 Nobel Laureate Ignarro was to confirm that nitroglycerin and other nitrates act as NO donors following their breakdown in the vessel wall (Ignarro *et al.*, 2002). They simulate a normal mechanism which is deficient in disease. However, the therapeutically administered nitrate donors bypass the endothelium and stimulate smooth muscle cell relaxation more directly (Smulyan, 2007). This knowledge has led to increased popularity and better clinical use of nitrates, and explains the potentiation of NO and nitrate action by PDE5 inhibitors (O'Rourke and Nichols, 2002; Reffelmann and Kloner, 2003; Wessells *et al.*, 2006; Fusco *et al.*, 2009).

Attention to the endothelium had immediate application to disease, since atherosclerosis, the modern scourge, is a disease of the endothelium, begins with penetration of lipoproteins and leukocytes, continues with an inflammatory response and build up of lipids, and causes arterial obstruction and reduced or stoppage of blood flow

through adherence of platelets and thrombus formation, as well as shedding of emboli from the thrombus (Pepine, 1998b; Verhamme and Hoylaerts, 2006). Abnormalities of smooth muscle tone, including spasm, are known to accompany reduced NO production and atherosclerosis (Yasue *et al.*, 2008).

NITRIC OXIDE AND ENDOTHELIAL FUNCTION

The main role of NO is a signaling mechanism in the vessel wall. Another role is as a neurotransmitter at non-adrenergic, noncholinergic parasympathetic nerve endings, such as in the corpus cavernosum of the penis, where arterial dilation causes penile erection, and is facilitated by PDE5 inhibitors, such as sildenafil (Viagra), tadalafil (Cialis), and vardenafil (Levitra), which are widely used in the treatment of erectile dysfunction (Wessells *et al.*, 2006; Fusco *et al.*, 2009).

In the endothelium (Fig. 5.2), NO is produced from L-arginine by activity of the enzyme endothelial nitric oxide synthase (eNOS). Under basal conditions, NO is

continually released from healthy endothelial cells and serves to inhibit adhesion of platelets and leukocytes to the vessel wall, and, after diffusion to the media, to relax smooth muscle cells and maintain vascular patency and distensibility. This mechanism is most active in systemic arteries, less active in resistive arterioles, and least active in veins. Stimuli for release of NO from endothelial cells include flow shear stress exerted directly on the cell membrane *in vitro* (Fig. 5.3) or *in vivo* at the vascular interface (see below and Chapter 3) (Pohl *et al.*, 1986; Anderson and Mark, 1989; Noris *et al.*, 1995; Gnasso *et al.*, 2001; Paniagua *et al.*, 2001; Stepp *et al.*, 2001; Pyke *et al.*, 2004; Reneman *et al.*, 2006; Wessells *et al.*, 2006; Katritsis *et al.*, 2007), cytokines and chemical substances such as ACh, serotonin, and bradykinin. Arterial wall shear stress is directly related to local whole blood viscosity and pulsatile blood flow velocity and inversely related to pulsatile diameter (Samijo *et al.*, 1997; Kornet *et al.*, 1999; Malek *et al.*, 1999; Kornet *et al.*, 2000; Thury *et al.*, 2002; Dammers *et al.*, 2003; Pyke and Tschakovsky, 2005; Reneman *et al.*, 2006; Pantos *et al.*, 2007b). In general, shear stress is presented as peak systolic stress or mean

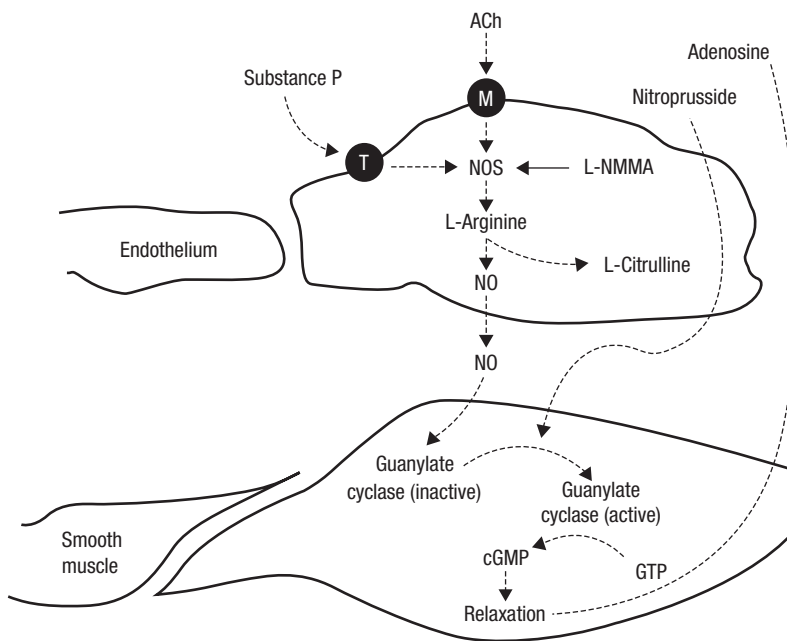


Figure 5.2 Formation of NO in the endothelium in response to acetylcholine, and its effect in causing relaxation of the smooth muscle cell after diffusion into the arterial media. Reproduced from Quyyumi (1998)

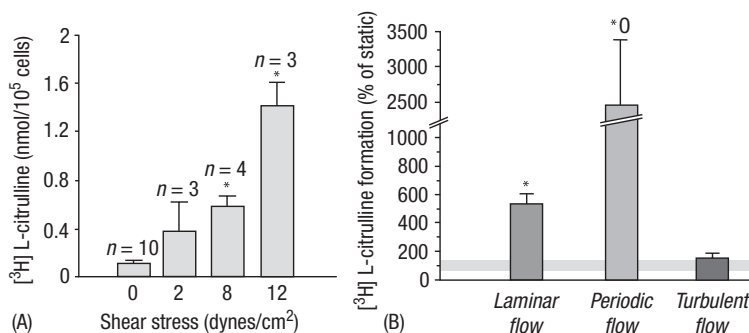


Figure 5.3 (A) Bar graph showing the dose-response relation between laminar shear stress and release of nitric oxide by cultured human umbilical vein endothelial cells. Confluent endothelial cell monolayers were incubated under laminar shear stress of 2, 8, or 12 dyne/cm²; (B) Bar graph comparing the effect of different flow conditions (steady laminar flow, 8 dyne/cm²; periodic laminar flow, from 2 to 8 dyne/cm² and turbulent flow, from 1.2 to 11.7 (average, 8) dyne/cm² on nitric oxide synthesis. Reproduced from Noris *et al.* (1995)

stress; however, a recent publication by Mitchell *et al.* (2004b) measured diastolic stress and related this measurement to flow-mediated dilation (FMD). To avoid errors in shear stress measurement, velocity and diameter must be measured at the same site and at the same instant in the cardiac cycle (or averaged over the entire cardiac cycle) (O'Rourke and Nichols, 2004b). Alternatively, shear stress can be estimated from shear rate, as derived from actually recorded velocity profiles, and blood viscosity (Dammers *et al.*, 2003; Samijo *et al.*, 1997; Reneman *et al.*, 2006). It seems logical that maximum shear stress during systole should cause the greatest release of NO. NO synthesis by eNOS from L-arginine is inhibited by the L-arginine analog, L-monomethyl arginine (L-NMMA) (Wilkinson *et al.*, 2002; Brunner *et al.*, 2005; Deanfield *et al.*, 2005; Schmitt *et al.*, 2005).

In the medial smooth muscle cells, NO activates the enzyme guanyl cyclase, which increases formation of cyclic guanosine monophosphate (cGMP), and this in turn causes smooth muscle relaxation, ultimately through sequestration of calcium ions (see Fig. 5.2). The cGMP is broken down by another enzyme, PDE5. Inhibitors of PDE5 include sildenafil, tadalafil, and vardenafil whose actions are explained on the basis of increase in cGMP, arterial dilation, reduced wave reflection amplitude, and decreased arterial blood pressure (Schofield *et al.*, 2003; Vlachopoulos *et al.*, 2003a; Vlachopoulos *et al.*, 2004b; Wessells *et al.*, 2006; Vlachopoulos *et al.*, 2009a; Vlachopoulos *et al.*, 2009b).

MEASUREMENT OF ENDOTHELIAL FUNCTION (DYSFUNCTION)

The seminal studies on endothelial function were undertaken on strips or rings of arteries, usually of rats, immersed in a water bath. Subsequent studies in living humans and experimental animals have simulated the same studies and have included:

- Use of the endothelial-independent agent nitroglycerin to assess the effects of maximal smooth muscle relaxation.
- Use of agents such as L-NMMA to inhibit production of NO by the endothelium.
- Administration of L-arginine in order to 'drive' NO production.
- Administration of oxidants and antioxidants to respectively hasten or slow the breakdown of NO.
- Administration of PDE5 inhibitors to potentiate and prolong the effects of NO.
- Use of an agent (e.g. ACh) or procedure (reactive hyperemia) to stimulate generation of NO by the endothelium.
- Measurement of an index of smooth muscle relaxation.

Several different techniques to measure endothelial function have been applied clinically in humans and in experimental animals. It is generally accepted that there is no need to determine maximal direct effect of vasodilation (and that this is difficult to ascertain), and that use of L-arginine and of L-NMMA, while useful, can have side effects and is not essential if one is only concerned with overall endothelial function and dysfunction.

At least seven types of tests that can be used to measure endothelial function in humans have been described and evaluated (Lind *et al.*, 2002; Widlansky *et al.*, 2003; Deanfield *et al.*, 2005; Pries *et al.*, 2008; Tomiyama and Yamashina, 2010; Lekakis *et al.*, 2011). Advantages and disadvantages of four of these are presented in Table 5.1.

Brachial artery catheterization with venous occlusion plethysmography

This technique was widely used in early human studies and confirmed that endothelial function is impaired with aging, hypertension, coronary atherosclerosis, heart failure, acute systemic inflammation, hypercholesterolemia, cigarette smoking, diabetes, and is associated with the risk of cardiovascular events (Creager *et al.*, 1990; Panza *et al.*, 1990; Kubo *et al.*, 1991; Cockcroft *et al.*, 1994; Taddei *et al.*, 1995; Clarkson *et al.*, 1996; Heitzer *et al.*, 1996; Taddei *et al.*, 1997; Mäkimattila *et al.*, 1999; Newby *et al.*, 1999; Hingorani *et al.*, 2000; Millgård *et al.*, 2000; Bragulat *et al.*, 2001; Heitzer *et al.*, 2001; Patel *et al.*, 2001; Perticone *et al.*, 2001; Guthikonda *et al.*, 2003; Booth *et al.*, 2004a; McCall *et al.*, 2009). Endothelial function in these studies was assessed on the basis of the capacity to increase forearm blood flow, measured by classical forearm venous occlusion mercury-in-silastic strain-gauge plethysmography after intra-arterial infusion of an endothelial-dependent agent, such as ACh, and an endothelial-independent agent, such as sodium nitroprusside (Fig. 5.4) (Panza *et al.*, 1990; Perticone *et al.*, 1998; Perticone *et al.*, 1999; Lind *et al.*, 2000; Irace *et al.*, 2001; Wilkinson and Webb, 2001). This test is reproducible (Walker *et al.*, 1999; Wassmann *et al.*, 2002; Deanfield *et al.*, 2005; Lind *et al.*, 2005; Casey *et al.*, 2010) and basically evaluates arteriolar (or microvascular) endothelial function. Also, it permits local use of eNOS inhibitors, such as L-NMMA. Its disadvantage is that it is invasive, with risk of injury to the artery or median nerve. Its advantage is that it can be performed on volunteers and permits mapping of dose-response relationships of endothelial agonists and antagonists, and following effects of treatment.

Intracoronary infusion of agonists or arterial occlusion-reperfusion

The 'gold standard' for assessing endothelial function in the coronary vascular bed is determining the changes in coronary artery diameter (macrovascular) and coronary

Table 5.1 Advantages and disadvantages of methods to quantify endothelial function in humans

Advantages	Disadvantages
1. Brachial artery catheterization with venous occlusive plethysmography More accessible circulation than coronary arteries Allows for mapping dose–response relationships of endothelial agonists and antagonists Allows for examination of basal endothelial function (with eNOS antagonist infusion) Free of operator bias	Invasive Risk of median nerve injury, infection, vascular injury
2. Intracoronary agonist infusion with quantitative coronary angiography Direct quantification of endothelial function in the vascular bed of interest Allows for mapping dose–response relationships of endothelial agonists and antagonists Allows for examination of basal endothelial function (with eNOS antagonist infusion)	Invasive Expensive Carries risks inherent with coronary artery catheterization (stroke, MI, infection, vascular injury)
3. Brachial artery ultrasound with FMD Non-invasive Safer and faster than either invasive method Reactivity correlates roughly with endothelial dysfunction in coronary circulation Flow is a physiological stimulus for vasodilation unlike agonists, such as acetylcholine	Poor resolution relative to arterial size Variability in measurements Highly operator-dependent Expensive There is no standardized method (cuff placement, time intervals)
4. Vascular tonometry and measurements of vascular stiffness and wave reflection Non-invasive Safer and faster than either invasive method Lower operator dependence than brachial artery ultrasound May reflect basal endothelial function	Importantly influenced by structural aspects of the vasculature beyond the endothelium

eNOS, endothelial nitric oxide synthase; FMD, flow-mediated dilation; MI, myocardial infarction.
 Reproduced with permission from Widlansky *et al.* (2003).

artery blood flow (microvascular). The microvascular technique for estimating endothelial function (or coronary flow reserve (CFR)) was first developed and used in the coronary circulation of animals by us (Nichols *et al.*, 1988a; Mehta *et al.*, 1989; Nichols *et al.*, 1994b; Mehta, 1995) and others (Engler *et al.*, 1983; Klein *et al.*, 1986). The technique was subsequently adapted for use in humans by Ludmer *et al.* (1986) and is most useful for studying large epicardial coronary artery atherosclerosis and small arteriolar resistance since it can be undertaken during the course of diagnostic coronary angiography and therapeutic coronary angioplasty (Reis *et al.*, 1999; Reis *et al.*, 2001; von Mering *et al.*, 2004; Fukuda *et al.*, 2006; Pepine *et al.*, 2007; Wessel *et al.*, 2007; Worthley *et al.*, 2007; Stangl *et al.*, 2008; Cassar *et al.*, 2009; Kurita *et al.*, 2009). Since the technique is invasive, it is not applicable to volunteers. Non-invasive techniques to evaluate coronary artery endothelial function in both experimental animals (Wikstrom *et al.*, 2008; Hartley *et al.*, 2009) and humans (Ikonomidis *et al.*, 2008; Kurita *et al.*, 2009; Turiel *et al.*, 2009) have been developed recently. The stimulus is usually with an agonist, such as ACh or adenosine, infused directly into the coronary artery, or with handgrip exercise, or the cold pressor maneuver,

or after arterial obstruction – undertaken as part of therapeutic balloon angioplasty. Most investigators measure coronary endothelial (or microvascular) function or CFR using a 0.014-inch Doppler flow guide wire to measure coronary artery blood flow (see Chapters 6 and 17). For an excellent recent review of methods for assessment of coronary microvascular disease in both clinical and experimental settings, see Pries *et al.* (2008). Epicardial coronary artery diameter and assessment of endothelial function is measured using quantitative coronary artery angiography (Treasure *et al.*, 1992; Treasure *et al.*, 1995; Hosokawa *et al.*, 2006; Hosokawa *et al.*, 2008; Lüscher *et al.*, 2009). In either the macrovascular or microvascular method, effect is measured as the degree of change in diameter or increase in coronary blood flow velocity above resting or baseline value in response to a stimulus (see above). The advantage is the obvious direct relevance to coronary blood flow and cardiac function. The disadvantage is the need to carry out, as part of a diagnostic or therapeutic maneuver, functional effects of contrast agents used in assessing coronary diameter, and delay of the procedure, with increased risk of stroke, myocardial infarction, etc. (Ludmer *et al.*, 1986; Cox *et al.*, 1989; Nabel *et al.*, 1990; Mills *et al.*, 1992).

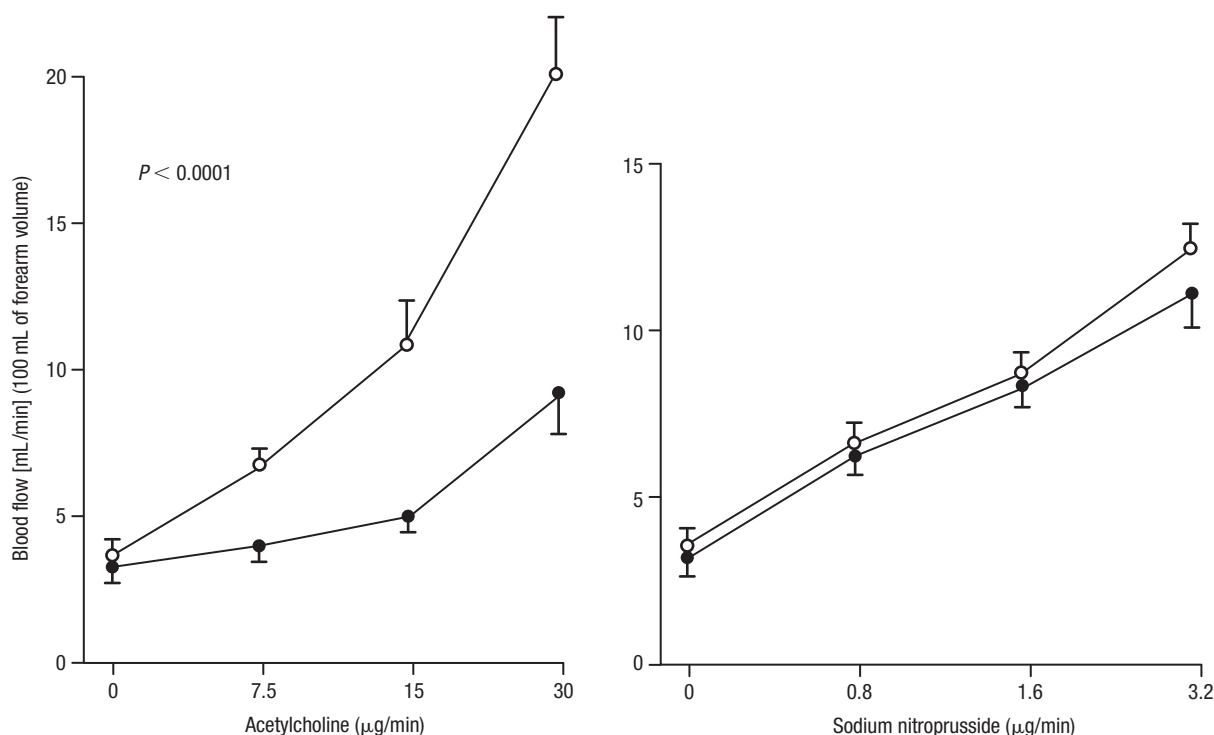


Figure 5.4 Effects of intra-arterial incremental infusion of acetylcholine and sodium nitroprusside on forearm blood flow (FBF) in hypertensive (●) and normotensive (○) individuals. As shown, the response to acetylcholine but not nitroprusside is blunted in the hypertensive individuals. From Panza *et al.* (1990)

Brachial artery ultrasound with flow-mediated vasodilation

This technique, introduced in 1992 (Celermajer *et al.*, 1992), based on earlier pioneering studies (Laurent *et al.*, 1990), has become the most widely used method to assess macrovascular endothelial function in humans. FMD (or percentage change in arterial diameter) is considered the 'gold standard' and it has confirmed that endothelial function is impaired with aging, hypertension, offspring of hypertensive parents, white coat hypertension, coronary atherosclerosis, heart failure, heart transplantation, aortic valve sclerosis, Marfan syndrome, diabetes, cigarette smoking (active and passive), obesity, sleep apnea, stroke, erectile dysfunction, and hypercholesterolemia (Celermajer *et al.*, 1994a; Celermajer *et al.*, 1994b; Clarkson *et al.*, 1996; Iiyama *et al.*, 1996; Celermajer *et al.*, 1996; Poredos *et al.*, 1999; Wilson *et al.*, 1999; Hirai *et al.*, 2000; Kuvin *et al.*, 2001a; Mullen *et al.*, 2001; Patel *et al.*, 2001; Zizek *et al.*, 2001; Gokce *et al.*, 2002; Gómez-Cerezo *et al.*, 2002; Neunteufl *et al.*, 2002; Poggianti *et al.*, 2003; Schofield *et al.*, 2003; Benjamin *et al.*, 2004; Fathi *et al.*, 2004; Mancini, 2004; Vlachopoulos *et al.*, 2004b; Vlachopoulos *et al.*, 2006b; Meyers and Gokce, 2007; Kohler *et al.*, 2008; Vlachopoulos *et al.*, 2008; Kim *et al.*, 2009; Lojanapiwat *et al.*, 2009; Yeboah *et al.*, 2009; Casey *et al.*, 2010) and is a target for prevention of cardiovascular disease (Versari *et al.*, 2009). According to Kuvin *et al.* (2001a), to enable accurate data collection and

interpretation, the assessment of endothelial function using FMD should be standardized. These guidelines are discussed in several articles (Vogel *et al.*, 2000; Vogel, 2001; Verma and Anderson, 2002), including the excellent report of the International Brachial Artery Reactivity Task Force (Corretti *et al.*, 2002). Ultrasound images of the brachial arterial and blood flow velocity through it are obtained above the antecubital fossa before (baseline) and during reactive hyperemia and nitroglycerin administration to distinguish between NO-dependent and NO-independent vasodilation.

Peripheral artery reactive hyperemia (endothelial-dependent)

To create a flow (or shear stress) stimulus in the brachial (or femoral) artery, a sphygmomanometric (blood pressure) cuff is first placed either above the antecubital fossa, and the ultrasound probe, or on the forearm below the probe (Fig. 5.5). After a rest period, a baseline rest image is acquired, and blood flow is estimated by time-averaging the pulsed Doppler velocity signal obtained from a mid-artery sample volume. Thereafter, arterial occlusion is created by cuff inflation to suprasystolic pressure. Typically, the cuff is inflated to at least 50 mmHg above systolic pressure to totally occlude arterial inflow for a standardized length of time; the desired pressure level can easily be maintained using a constant pressure regulator

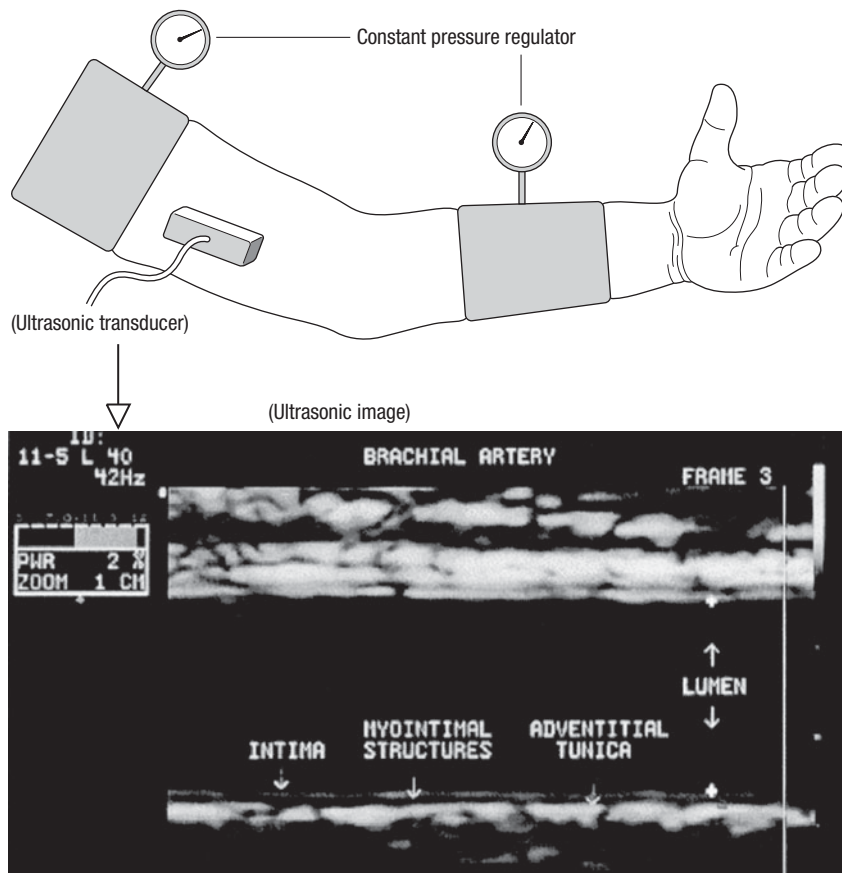


Figure 5.5 Schematic diagram to show the flow-mediated dilation (FMD) method for assessment of endothelial function. Top: placement of occlusive sphygmomanometer cuffs on either the upper limb or the forearm, and the ultrasonic sensor on the brachial artery. Bottom: measurement of arterial diameter by ultrasound. Reproduced from Corretti *et al.* (2002)

(Schofield *et al.*, 2003). Interruption of blood flow causes ischemia and consequent dilation of downstream resistance vessels via autoregulatory mechanisms. Subsequent cuff deflation induces a brief high-flow state (Pohl *et al.*, 1986) through the brachial artery (reactive hyperemia) to accommodate the dilated resistance vessels. The resulting increase in wall shear stress causes release of NO from the endothelium and brachial artery dilation (Fig. 5.6) (Anderson and Mark, 1989). The longitudinal image of the artery is recorded from 30–60 s before inflation, which is usually 4.0, 4.5, or 5.0 minutes, to 2.0 minutes after cuff deflation. A mid-artery pulsed Doppler signal is obtained upon immediate cuff release to assess reactive hyperemic blood flow velocity. A recent comprehensive tutorial summarized the most recent advancements in the ultrasonic measurement of FMD (Harris *et al.*, 2010).

Studies have variably used either upper arm or forearm cuff occlusion, and there is no consensus as to which technique provides more accurate or precise information. When the cuff is placed on the upper part of the arm, reactive hyperemia typically elicits a greater percentage change in diameter compared with that produced by the placement of the cuff on the forearm (Uehata *et al.*, 1997; Mannion *et al.*, 1998; Vogel *et al.*, 2000; Betik *et al.*, 2004). This may be due to a greater flow stimulus resulting from recruitment of more resistance vessels or possibly to direct effects of ischemia on the brachial artery. However, upper-arm occlusion is technically more challenging for accurate

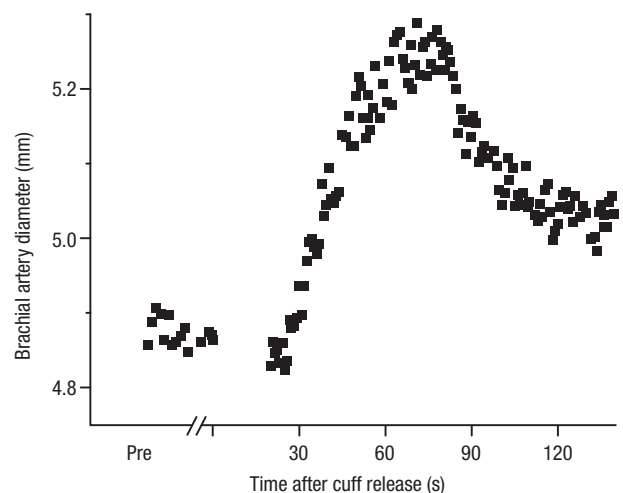


Figure 5.6 Change in brachial artery end-diastolic diameter after release of 5 minutes arterial occlusion (upper arm) in a normal subject. Peak diameter increase was approximately 8.5 percent and occurred between 60 and 70 s. Reproduced from Stadler *et al.* (1996)

data acquisition as the image is distorted by collapse of the brachial artery and shift in soft tissue. The change in brachial artery diameter after cuff release increases as the duration of cuff inflation increases from 30 s to 5.0 min. The change in diameter is similar after 5.0 to 10 minutes

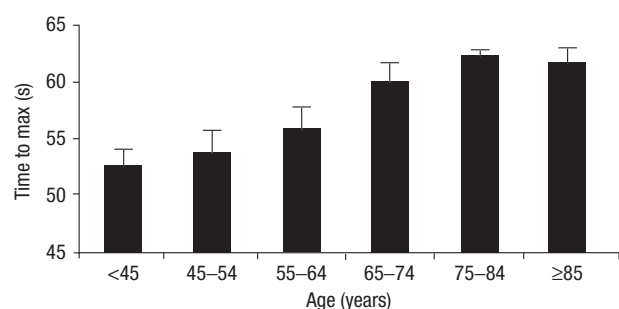


Figure 5.7 Plot of time to peak brachial artery diameter after cuff deflation stratified by deciles of age. Reproduced from Herrington *et al.* (2001)

of occlusion; therefore, the more easily tolerated 5.0 minute occlusion is typically used and diameter measurement collected at end diastole after cuff deflation. In Fig. 5.6, peak diameter occurs at 60–70 s and calculated FMD is 8.5 percent. If diameter is measured at 90 s after cuff deflation, as some studies have done, calculated FMD is only 6.5 percent, well below the accurate value (Stadler *et al.*, 1996; Donald *et al.*, 2008). The time course of FMD differs significantly between young and older subjects (Fig. 5.7) (Herrington *et al.*, 2001; Black *et al.*, 2008). Studies assuming that peak dilation occurs at an arbitrary time point, or within limited time windows, may draw misleading conclusions regarding differences between groups. Thus, automated systems, based on image-processing algorithms to measure brachial artery diameter, were developed to simplify the assessment of FMD and obtain the best FMD reproducibility (Faita *et al.*, 2011). FMD may be studied in the brachial, radial, axillary, superficial femoral, and coronary arteries. Notable caveats are that diameter changes in arteries smaller than 2.5 mm in diameter are difficult to measure precisely, even with the current high resolution ultrasound systems. Vasodilation is generally less difficult to perceive in vessels larger than 5.0 mm in diameter (Corretti *et al.*, 1995; Sorensen *et al.*, 1995).

Despite the observation that brachial artery FMD during reactive hyperemia could be used as an indicator of endothelial (dys)function, its significance has been challenged more recently (Corretti *et al.*, 2002; Alam *et al.*, 2005; Bots *et al.*, 2005; Reneman *et al.*, 2006). The brief transient response (Corretti *et al.*, 1995; Corretti *et al.*, 2002; Black *et al.*, 2008), and the absence of any steady state for both blood flow velocity and brachial artery diameter make it difficult (Pyke and Tschakovsky, 2005; Widlansky, 2009) to relate unequivocally the measured FMD to the imposed stimulus magnitude. This makes normalization of the dilatory response to the induced increase in blood flow velocity or derived shear stress, necessary for unambiguous quantitative comparison between subjects, difficult (Pyke and Tschakovsky, 2005). To achieve a well-defined stimulus–response relation, requiring a steady state for both the velocity and the

diameter change, alternative techniques have been introduced, e.g. forearm heating (Pyke and Tschakovsky, 2007) and hand-grip exercises (Pyke *et al.*, 2008). The problems encountered in brachial artery FMD assessment have been discussed in an extensive dispute by a variety of colleagues recently (Wong *et al.*, 2009) in response to an article by Parker *et al.* (2009). This discussion included such aspects as the difficulties encountered in determining wall shear stress non-invasively, among others, due to the non-parabolic shape of the velocity profile and the changing viscosity during non-steady conditions, and in normalizing FMD to blood flow velocity or shear stress. In this dispute, it was also indicated that shear rate in itself independently reflects vascular health. An additional problem encountered in testing endothelial function in the brachial artery is that this artery is practically devoid of atherosclerosis. The question can be asked as to what extent the endothelial function assessed in the brachial artery is representative of that in arteries prone to atherosclerosis. Although the brachial and coronary arteries have shown parallel FMD responses (Anderson *et al.*, 1995), the correlation is insufficient for brachial responses to predict coronary disorders. Also, discrepancies in FMD were reported between brachial and carotid arteries (Lavalée *et al.*, 2006), although they could be partially attributed to the technique used (hyperemia versus hypercapnia). Moreover, it has been disputed whether the FMD, as seen in relation to cardiovascular risk factors, represents disturbed endothelial cell function or results from a reduced stimulus (Mitchell *et al.*, 2004b).

Nitroglycerin administration (endothelium-independent)

At least 10 minutes of rest is needed after reactive hyperemia before another image is acquired to reflect the re-established baseline conditions. In most studies to date, an exogenous NO donor, such as a single high dose (0.4 mg) of nitroglycerin (NTG) spray or sublingual tablet, has been given to determine the maximum obtainable vasodilator response, and to serve as a measure of endothelium-independent vasodilation reflecting vascular smooth muscle function (Ducharme *et al.*, 1999). Peak vasodilation (or increase in diameter) occurs 3.0 to 4.0 minutes after NTG administration; images should be continuously recorded during this time, and NTG should not be administered to individuals with clinically significant bradycardia or hypotension. Determining the vasodilator responses to increasing doses of NTG, rather than a single dose, may further elucidate changes in smooth muscle function or arterial compliance that might be playing a role in any observed changes in FMD (Fig. 5.8).

The advantages of this technique are that it is non-invasive and reproducibility is good in experienced hands (Sorensen *et al.*, 1995; Corretti *et al.*, 2002). The

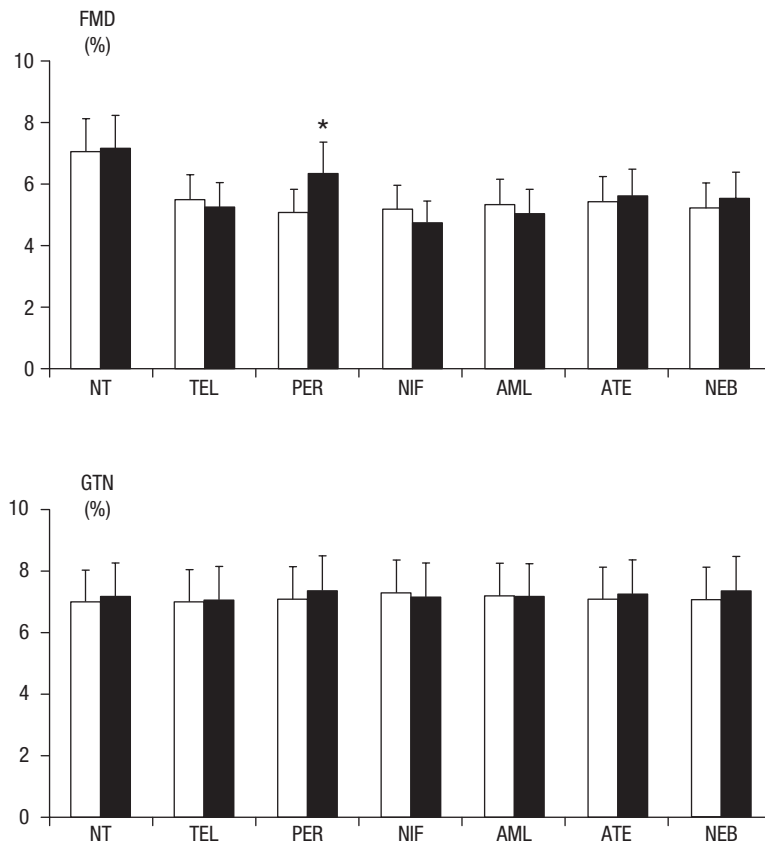


Figure 5.8 Graph showing flow-mediated and nitroglycerin-induced dilation, expressed as maximal percentage change in brachial artery diameter before (■) and after (□) six months in normotensive subjects and in patients with essential hypertension treated with the different study drugs. Data are shown as mean $\pm$ s.d. NT indicates normotensive subjects; PER, perindopril; TEL, telmisartan; NIF, nifedipine; AML, amlodipine; ATE, atenolol; and NEB, nebivolol. From Ghiadoni *et al.* (2003)

disadvantage is that it can be highly operator-dependent. Resolution of change in diameter relative to baseline diameter is difficult. There is often considerable variability in measurements. Also, there is potential for bias in diameter measurement (Boutouyrie *et al.*, 2004a).

Recent studies found a significant correlation between the strain gauge plethysmography technique (microvascular) and the brachial artery ultrasound (FMD) technique (macrovascular) regarding the evaluation of endothelial-independent vasodilation by NO donors, but no correlation was found between the two techniques regarding the evaluation of endothelial-dependent vasodilation (Lind *et al.*, 2000; Eskurza *et al.*, 2001; Irace *et al.*, 2001).

A number of new techniques have recently been proposed as potentially applicable screening tools for endothelial function testing in humans. Each novel test proposed should meet several criteria, if it is to be clinically useful for assessing cardiovascular risk in individual subjects (Celermajer, 2008). The test should:

- be simple, non-invasive, and widely applicable;
- be reproducible, with low interobserver error;
- be standardizable between laboratories and have population normal data to inform interpretation;
- be able to predict risk;
- add to the predictive value of established risk factor measurement, particularly in intermediate-risk subjects;

- be able to demonstrate that improvement with the new test predicts a reduction in subsequent cardiovascular risk.

Arterial tonometry, pulse waveform analysis, and arterial stiffness

The technique of arterial tonometry with pulse wave analysis (PWA), discussed exhaustively in Chapters 19, 20, 26, and 30 with respect to arterial stiffness and wave reflections, medial degeneration of large arteries, assessment of drug effect on heart and arteries, and on prognosis, has been further adapted by Hayward *et al.* (2002), Ravikumar *et al.* (2002), Wilkinson *et al.* (2002a), Lind *et al.* (2003), Oliver and Webb (2003), Widlansky *et al.* (2003), Suh *et al.* (2005), Lane *et al.* (2006), Weber *et al.* (2007b), Plantinga *et al.* (2008), Soga *et al.* (2008), and Merchant *et al.* (2009) for measurement of endothelial function.

With this method, tonometry enables the radial artery pressure waveform to be measured directly, and the central aortic pressure waveform to be synthesized. Rather than measuring endothelial function in a specific vascular bed, this technique permits assessment of global endothelial function of the whole systemic arterial tree. In some studies, the response is the reduction in central or radial augmented pressure (or inflection point) in

response to a β -2 agonist. This is salbutamol (albuterol) (Hayward *et al.*, 2002; Wilkinson *et al.*, 2002a; Covic *et al.*, 2004b; Suh *et al.*, 2005; McEniery *et al.*, 2006) inhaled or terbutaline administered subcutaneously (Lind *et al.*, 2003). The response can be compared to the effect of nitroglycerin, administered sublingually, on augmented pressure, augmentation index, or inflection point. Other studies have compared measurements of arterial wave reflections directly with those of endothelial function in various populations (Ravikumar *et al.*, 2002; Malik *et al.*, 2008; Plantinga *et al.*, 2008; Soga *et al.*, 2008; and Merchant *et al.*, 2009) and others compared plasma levels of asymmetric dimethylarginine, an endogenous inhibitor of eNOS, with wave reflections (Weber *et al.*, 2007). This technique is simple and non-invasive and the equipment required is relatively inexpensive and portable, and can be used for other purposes to assess effects of arterial stiffness and of other drugs. It correlates well with the venous occlusion plethysmography (Wilkinson *et al.*, 2002a) and FMD techniques (Ravikumar *et al.*, 2002; Schofield *et al.*, 2003; Malik *et al.*, 2008; Plantinga *et al.*, 2008; Soga *et al.*, 2008; Merchant *et al.*, 2009). In these studies, endothelial dysfunction was noted in patients with coronary atherosclerosis, hypertension, hypercholesterolemia, diabetes, and cigarette smoking.

Another technique for assessing endothelial function from the finger pulse amplitude tonometry (PAT) was recently evaluated by Kuvin *et al.* (2003) and Aizer *et al.* (2009). The technique is relatively operator independent making it a potentially convenient measure of endothelial

function (Nohria *et al.*, 2006; Aizer *et al.*, 2009). In contrast to measuring FMD, which requires a trained ultrasonographer to obtain accurate and reliable serial images of the brachial artery, PAT measurement requires insertion of one finger of each hand into thimble-like peripheral arterial tonometers (Fig. 5.9). Pressure change associated with change in volume pulse amplitude following the release of a 5 minute occlusion of the upper arm by a blood pressure cuff (inflated to supersystolic pressures) are detected by the tonometers and transmitted to a computer for recording and analysis. As with FMD, PAT is reduced in the presence of cardiovascular risk factors in a dose-dependent fashion, as well as with coronary atherosclerosis (Kuvin *et al.*, 2003; Hamburg *et al.*, 2008). In an ambulatory clinic setting, PAT values have been found to be lower in individuals with more than two cardiac risk factors, as well as in individuals with coronary artery disease (Kuvin *et al.*, 2007). It should be stressed, however, that in a recent analysis from the Framingham Heart study, FMD and PAT had differing relations with cardiovascular risk factors and were nearly uncorrelated with each other. Studies using PAT have demonstrated altered endothelial function in children at cardiovascular risk because of type 1 diabetes mellitus (Haller *et al.*, 2007) and have shown that impaired PAT responses are present in patients with coronary microvascular endothelial dysfunction measured *in vivo* (Bonetti *et al.*, 2004) and falciparum malaria (Yeo *et al.*, 2007; Yeo *et al.*, 2008). PAT has been shown to identify women with ischemic heart disease (Matsuzawa

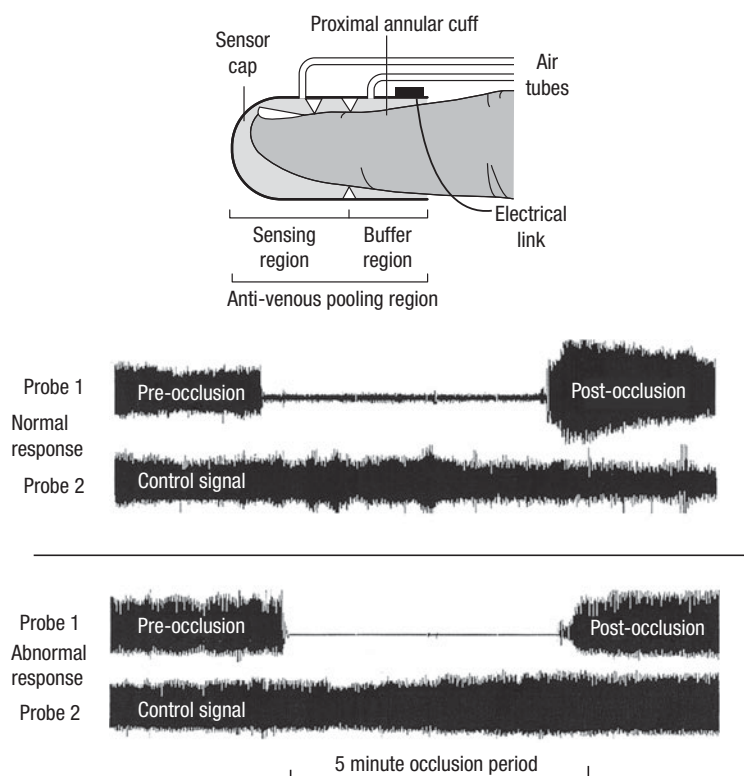


Figure 5.9 Representative reactive hyperemia peripheral arterial tonometry recordings of subjects with normal and abnormal reactive hyperemic response. Normal response is characterized by a distinct increase in the signal amplitude after cuff release compared with baseline. From Bonetti *et al.* (2004b)

et al., 2010) and predict late cardiovascular events (Rubinshtein *et al.*, 2010). Bonetti *et al.* (2003b) and Fisher *et al.* (2003) have investigated reversibility of peripheral endothelial dysfunction with interventions such as enhanced external counterpulsation (EECP) and flavonol-rich cocoa.

Both endothelial dysfunction and increased arterial stiffness commonly coexist in healthy subjects (McEniery *et al.*, 2006) and in patients at increased risk of cardiovascular disease, for example in diabetes (Wilkinson *et al.*, 2000b) and hypertension (Wallace *et al.*, 2007). Indeed, some studies in children have directly related increased stiffness with impaired endothelial function, for example in low birth weight (Martin *et al.*, 2000), hypercholesterolemia (Martino *et al.*, 2008), and severe obesity (Tounian *et al.*, 2001). These observations lead to the hypothesis that cardiovascular risk factors may exert their detrimental effects on arterial stiffness through endothelial dysfunction.

These techniques, while attractive, have not been applied as widely or for as long as other methods and need further clinical evaluation.

Laser Doppler flowmetry

Laser Doppler flowmetry (LDF) is a technique enabling the monitoring of skin microvascular blood flow (Gush and King, 1991). The assumption is that the response observed in the cutaneous circulation is a window towards the responses in other vascular beds (Khan *et al.*, 2008). During LDF, the original beam of coherent light changes in contact with moving tissues (red blood cells) and a photodiode measures the emerged beam. The fraction of shifted light depends on the concentration of moving red blood cells, whereas the magnitude of the frequency broadening depends on their average velocity (Fredriksson *et al.*, 2009). LDF has been correlated with FMD (Hansell *et al.*, 2004) and used as a research tool (Strain *et al.*, 2005), however it remains uncertain if this technique reflects the same pathobiology as that of conduit vessel endothelial function.

Microparticles

Microparticles (MPs) are vesicles (100 nm to 1 μ m diameter) shed from plasma membranes following cell activation or apoptosis (Hugel *et al.*, 2005). MPs of different cellular origins (platelets, leukocytes, red blood cells, endothelial cells, etc.) are found in the plasma of healthy subjects and their circulating levels augment in atherothrombotic diseases. Although MPs originating from endothelial cells (EMP) represent a small fraction of the overall pool of plasma MPs, increased circulating EMP levels constitute an emerging surrogate marker of endothelial dysfunction. Circulating EMP levels increase in several cardiovascular and atherothrombotic diseases

(Boulanger *et al.*, 2006) and are independent predictors of future events in patients at high risk for coronary heart disease (CHD) (Nozaki *et al.*, 2009).

Endothelial progenitor cells (EPC)

Cardiovascular risk factors damage the endothelial monolayer by inducing endothelial cell senescence and apoptosis leading to endothelial dysfunction. The restoration and reconstitution of the damaged endothelial monolayer may be a prerequisite for the prevention of endothelial dysfunction and atherosclerotic lesion formation. EPC derived from bone marrow circulate in the peripheral blood and have been implicated in neoangiogenesis after tissue ischemia has occurred (Asahara *et al.*, 1997). EPC are capable of proliferating and differentiating into endothelial cells and are therefore ideal candidates for vascular regeneration. EPC can be measured from peripheral blood either by direct enumeration using flow cytometry (CD34/KDR, CD34/KDR/CD133-positive EPC) (Duda *et al.*, 2007) or after cultivation of mononuclear cells (early and late outgrowth EPC) (Yoder *et al.*, 2007). EPC in most studies correlate with FMD and are independent predictors of future cardiovascular events (Schmidt-Lucke *et al.*, 2005; Hill *et al.*, 2003; Werner *et al.*, 2005).

These techniques (arterial tonometry, laser Doppler flowmetry microparticles and EPC) while attractive, have not been applied as widely or for as long as other methods and need further clinical evaluation.

AGE-RELATED CHANGES IN ENDOTHELIAL FUNCTION

Endothelial function is optimal in the young and in the absence of cardiovascular disease. With aging in experimental animals and in humans, there is a progressive deterioration in endothelial function as assessed by all the different methods (Celermajer *et al.*, 1994a; Gerhard *et al.*, 1996; Herrington *et al.*, 2001; Urbina *et al.*, 2009b) (Fig. 5.10). In contrast, responses to exogenous nitro-mediated vasodilators are little affected by aging (Egashira *et al.*, 1993a; Taddei *et al.*, 1995; Lind *et al.*, 1999; Sung *et al.*, 2002). Functional deterioration is associated with reduction of the bioavailability of NO and anatomical change, there being a generalized thickening of the endothelial layer, associated with hyperplasia of endothelial cells, together with hypertrophy of the basement membrane (Brandes *et al.*, 2005; Halcox *et al.*, 2009; Herrera *et al.*, 2010). The innermost layer of endothelial cells in contact with the flowing blood is the most active site of NO synthesis in response to various stimuli. Decreasing efficacy of endothelial function may be due to decreased synthesis in response to various stimuli, or

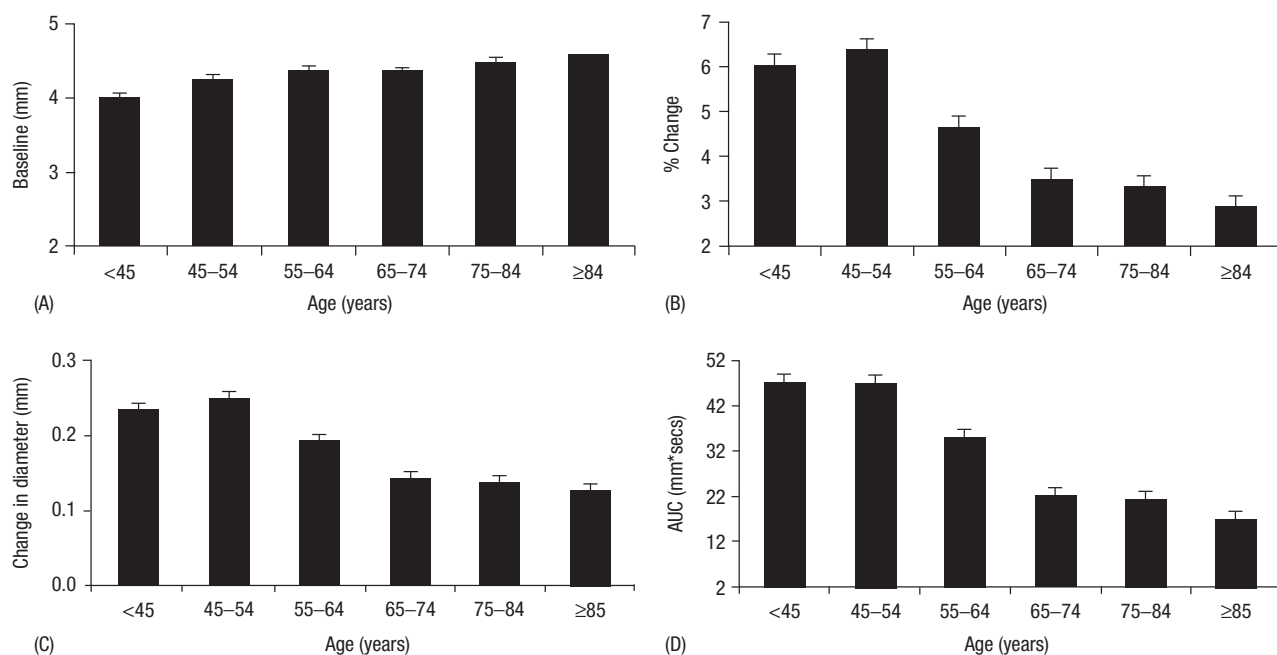


Figure 5.10 Age-related changes of various brachial artery measures stratified by decile. (A) Diameter; (B) percentage increase in diameter after 4.0 min occlusion; (C) change in diameter; and (D) area under the diameter curve (AUC). Reproduced from Herrington *et al.* (2001)

simply due to degradation of the highly volatile gaseous NO over the greater distance it needs to travel before reaching its site of action in vascular smooth muscle.

The most obvious dysfunction of the endothelium is seen in diseases which accompany aging, such as atherosclerosis, and in people with risk factors for atherosclerosis including diabetes mellitus and hypertension (Brunner *et al.*, 2005; Esper *et al.*, 2006). The active site of NO production becomes separated further from the active site of NO action by thickened atherosclerotic plaques, and hence there is less capacity of the endothelium to actively modulate the tone of medial smooth muscle.

Functional deterioration of the endothelium occurs with smoking (active and passive) and with a number of toxins (Poredos *et al.*, 1999; Rehill *et al.*, 2006; Kallio *et al.*, 2007). In the absence of endothelial thickening, an abnormality of endothelial function is attributed to decreased NO production by the endothelium or more rapid breakdown of NO by superoxide anions. Decreased sensitivity of the medial smooth muscles cells to NO is rare.

ENDOTHELIAL DYSFUNCTION AND CORONARY ATHEROSCLEROSIS

Endothelial function, as measured in the coronary and upper limb arteries, is defective not only in patients with established coronary atherosclerosis, but also in people with risk factors for atherosclerosis, including

hypertension, hypercholesterolemia-impaired glucose tolerance, obesity, and smoking (Celermajer, 1997; Kuvin *et al.*, 2001b; Bonetti *et al.*, 2003a; Widlansky *et al.*, 2003; Deanfield *et al.*, 2005; Lane *et al.*, 2006; Rehill *et al.*, 2006; Hayoz and Mazzolai, 2007; Urbina *et al.*, 2009b). Impaired endothelial function predates the appearance of coronary atherosclerosis, and may contribute to its development and progression through enabling adherence of monocytes and lipoproteins to the vessel wall, their entry into the wall, and the inflammatory response and lipid disposition that follows (Chatzizisis *et al.*, 2007; Stone *et al.*, 2007; ter Avest *et al.*, 2007; Chatzizisis *et al.*, 2008; Mestas and Ley, 2008; Lüscher *et al.*, 2009; Mehdizadeh and Norouzpour, 2009; Vanhoutte, 2009b). In the coronary arteries, endothelial dysfunction is most marked at the sites of low shear stress and atherosclerosis (Chapters 2 and 23), but is present also in apparently non-affected segments. It is often apparent as arterial constriction with further arterial narrowing in response to ACh infusion into a coronary artery, or to increased adrenergic tone and narrowing as induced by exercise or a cold pressor test (Ludmer *et al.*, 1986; Nabel *et al.*, 1990). Atherosclerosis progression is attributable to further deterioration in endothelial function with loss of the ability to prevent leukocyte or lipid entry or inflammation; thrombosis is favored by the inability to control platelet adhesion and aggregation at the arterial wall.

Impairment in endothelial function (dysfunction), measured by different methods and at different sites, is predictive of clinical events in people with no evidence of coronary atherosclerosis (Bonetti *et al.*, 2003a; Celermajer,

2003; Lippincott *et al.*, 2008; Shechter *et al.*, 2009) and in patients with established disease (Yeboah *et al.*, 2007; Yeboah *et al.*, 2009). Further, lifestyle modification and other measures, which reduce risk in preclinical or established coronary disease, have been shown to improve endothelial function (Harrison *et al.*, 1987; Heistad *et al.*, 1995; Modena *et al.*, 2002; Edwards *et al.*, 2004a; Vlachopoulos *et al.*, 2005; Alexopoulos *et al.*, 2008; Seals *et al.*, 2009; Johnson *et al.*, 2010). For a detailed discussion of arterial atherosclerosis, see Chapter 23.

EXERCISE TRAINING AND ENDOTHELIAL FUNCTION

Physicians generally accept the view that exercise enhances cardiovascular health. There is an inverse relationship between the level of physical activity and incidence of cardiovascular events, independent of pre-existing cardiovascular disease and of other risk factors. Such knowledge underlies the inclusion of exercise programs for primary and secondary prevention of cardiovascular disease (Chapter 27).

The mechanism underlying the value of exercise training has been investigated in different ways and implicates an increase in blood flow and endothelial shear stress resulting in modulation of vascular tone and improvement in endothelial function in both conduit arteries and arterioles (Niebauer and Cooke, 1996; DeSouza *et al.*, 2000; Hambrecht *et al.*, 2000b; Vita and Keaney, 2000; Maeda *et al.*, 2001; Goto *et al.*, 2003; Edwards *et al.*, 2004a; Green *et al.*, 2004; Franzoni *et al.*, 2005; Casey *et al.*, 2007; Leung *et al.*, 2008; Pierce *et al.*, 2008; Black *et al.*, 2009; Seals *et al.*, 2009; Tinken *et al.*, 2010). Most likely, this improvement is not due to self-selection (people with subtle symptoms from disease being discouraged from exercise) since it is seen when people are randomly entered into exercise programs or control programs which entail similar modification of risk factors, but not of exercise (Vaitkevicius *et al.*, 1993; Kingwell, 2000). Exercise is associated with improvement of other risk factors in general – cessation of smoking, improvement of blood pressure and cholesterol levels, and glucose tolerance and weight loss, but this cannot explain all the benefits.

Focused studies such as those of Hambrecht *et al.* (2000b) have shown that an exercise program lessened ACh-induced constriction of coronary arteries in patients with atherosclerosis, and improved coronary blood flow, with an increased flow through the microvasculature in response to adenosine.

It appears that the improved endothelial dysfunction ‘toning up the endothelium’ (Vita and Keaney, 2000) is due to increased expression of the enzyme eNOS in the endothelium in response to increased endothelial shear stress caused by higher blood flow in exercise (just preceding, during, and after). This beneficial effect appears to

persist during conduct of an exercise program, and for weeks or months thereafter. It appears to be responsible, not only for improved coronary blood flow, but for reduction in central pressure augmentation and by reduced tone in and wave reflection from peripheral conduit arteries (Vaitkevicius *et al.*, 1993; Tanaka *et al.*, 2000; Edwards *et al.*, 2004b; Otsuki *et al.*, 2007a; Tabara *et al.*, 2007; Munir *et al.*, 2008) (see also Chapter 30). A similar mechanism may underlie the benefits of EECF in patients with refractory angina pectoris (Nichols *et al.*, 2006; Casey *et al.*, 2008a). Exaggerated arterial pulsations in diastole may likewise induce expression of eNOS in the endothelium and improve vascular tone (Bonetti *et al.*, 2003b; Akhtar *et al.*, 2006).

DRUGS AND ENDOTHELIAL FUNCTION

Endothelial dysfunction has become a ‘buzz word’ in recent years, and benefits of many drugs ‘beyond blood pressure lowering’, etc., have been attributed to improvement in endothelial function. The fact that there is no clear consensus for many drugs attests to the variability in current tests of endothelial function and the absence of any one clearly superior test. Endothelial function (coronary, conduit artery, and arteriolar) improvement has been described for a multiplicity of drugs as used in hypertension, coronary artery disease, myocardial infarction, hypercholesterolemia, cardiac failure, and smoking (Dohi *et al.*, 1994; Heistad *et al.*, 1995; Frielingsdorf *et al.*, 1996; Kinlay *et al.*, 1999; Koh *et al.*, 1999; Patel *et al.*, 1999; Anderson *et al.*, 2000; Katz *et al.*, 2000; Matsuda *et al.*, 2000; Ong *et al.*, 2000; Higashi *et al.*, 2001b; Cardillo *et al.*, 2002b; de Jongh *et al.*, 2002; Frick *et al.*, 2002; Halcox *et al.*, 2002a; Mancini, 2002; Rajagopalan *et al.*, 2002; Wassmann *et al.*, 2002; Ghiadoni *et al.*, 2003; Schofield *et al.*, 2003; Heiss *et al.*, 2005; Hosokawa *et al.*, 2006).

Several lines of evidence link endothelial dysfunction, characterized by decreased bioavailability of NO, with the development of many pathological conditions, such as heart failure, hypertension, diabetes, and atherosclerosis (Desjardins and Balligand, 2006; Chatterjee *et al.*, 2008). Furthermore, the renin–angiotensin system activity may be a key factor in the pathophysiology and development of these diseases (Volpe *et al.*, 2002). It is generally believed that angiotensin-converting enzyme inhibitors (ACEIs) and angiotensin receptor blockers (ARB) improve endothelial function by improving NO bioavailability, as does exercise (Mancini *et al.*, 1996; Koh *et al.*, 1999; Vita and Keaney, 2000; Wassmann *et al.*, 2002; Green *et al.*, 2004; Leung *et al.*, 2008). Studies in rabbits have shown that ACEIs, ARBs, and direct renin inhibitors improve impaired NO bioavailability and protect against atherosclerotic changes (Ikejima *et al.*, 2008; Imanishi *et al.*, 2008a; Imanishi *et al.*, 2008b; Verma and Gupta, 2008) and other studies have shown improvement in endothelial

function in hypertensive rats (Richard *et al.*, 1996; Hayakawa *et al.*, 1997; Thuillez and Richard, 2005). Some early studies in humans failed to show any improvement in FMD with an ARB (Rajagopalan *et al.*, 2002; Ghiadoni *et al.*, 2003), however, other more recent studies have shown improvement in FMD in hypertensive patients (Benndorf *et al.*, 2007; Jung *et al.*, 2009). Also, the BANFF study (Anderson *et al.*, 2000) showed that, compared to an ARB or a calcium channel blocker (CCB), an ACEI was the only agent that improved FMD in patients with coronary artery disease. Studies that have compared the effects of antihypertensive medication (ACEIs, ARBs, CCBs, β -blockers, and diuretics) at the same reduced pressure level in essential hypertensive patients found that ACEIs were superior in improving endothelial function (Higashi *et al.*, 2001a; Ghiadoni *et al.*, 2003). The benefits of ACEIs and ARBs in clinical trials such as HOPE (Yusuf *et al.*, 2000) and LIFE (Dahlöf *et al.*, 2002) may be due to amplitude and timing of arterial pressure wave reflections (Mourad *et al.*, 2002; Nichols and Singh, 2002; Hirata *et al.*, 2005b; Williams *et al.*, 2006) rather than to any specific effect on endothelial function. Currently, pharmaceutical companies are jockeying for explanations of perceived benefits of particular drug classes or particular drugs within a class, and improvement of endothelial function is frequently claimed. The overall picture for antihypertensive medication is discussed in Chapter 20 where we point out that actual reduction in arterial pressure can explain most benefits, especially when this is considered in terms of central arterial pressure reduction. This view is enhanced by ALLHAT (2002) and other major clinical trials whose implications are discussed in the new JNC7 and ESH/ESC guidelines. A cynical view (which we largely share) has been expressed by Steve Nissen of Cleveland, that the road to hell is paved with plausible hypotheses, notably on endothelial function. One new hypothesis, however, for which there are some positive data is the 'French paradox' (low prevalence of atherosclerosis despite generally high risk factors); this is explicable on the basis of red wine increasing expression of endothelial eNOS and improving endothelial function (Vogel, 2003; Wallerath *et al.*, 2003; Whelan *et al.*, 2004; Karatzi *et al.*, 2008; Ghosh and Scheepens, 2009), arterial stiffness and wave reflections (Karatzi *et al.*, 2005; Papamichael *et al.*, 2006; Papamichael *et al.*, 2008; Karatzi *et al.*, 2009).

There are drug classes whose influence on endothelial function must be considered implicated – the statins, PDE5 inhibitors, and nitrates. Statins effectively reduce cholesterol and improve prognosis in people with established atherosclerotic disease, and in asymptomatic people at high risk (Anderson *et al.*, 1995a; Treasure *et al.*, 1995; Schwartz *et al.*, 2001; Hognestad *et al.*, 2004). Benefits are seen before any anatomical reduction in lumen narrowing occurs, and are usually associated with reduction in progression of atherosclerotic narrowing rather than reduction in narrowing itself. The benefit cannot be explained

only on the basis of leaching of cholesterol from atherosclerotic plaques, and is associated in some way with increased NO production and improvement in endothelial function (Leung *et al.*, 1993; Vogel *et al.*, 1996; Celermajer, 1997; Dupuis *et al.*, 1999; Asberg *et al.*, 2001; de Jongh *et al.*, 2002; Frick *et al.*, 2002; Celermajer, 2003; Hosokawa *et al.*, 2006; Simionescu, 2007; Vlachopoulos *et al.*, 2007b; Li and Förstermann, 2009; Ostad *et al.*, 2009). The effect may be due to reduction in inflammation within a plaque, and stabilization of the plaque so that it is less likely to rupture and initiate occlusive thrombosis or emboli shedding. It may also be associated with upregulation of eNOS activity and increased NO production (Laufs *et al.*, 1998; Kaesemeyer *et al.*, 1999; Li and Förstermann, 2009), as occurs with exercise. This possibility is supported by the finding that use of statins and PDE5 inhibitors are associated with reduction in peripheral wave reflection and decrease in central pressure augmentation (Nichols and Singh, 2002; Schofield *et al.*, 2003; Vlachopoulos *et al.*, 2003a). Such an effect has not been studied as extensively as has exercise, and the effect appears not to be as marked as with exercise, but it does indicate some improvement in endothelial function throughout the body in normal, as well as abnormal arteries. The early vascular benefits of statin therapy are discussed in excellent reviews by Tsiara *et al.* (2003) and by Liao and Laufs (2005).

PDE5 inhibitors permit amplification of NO action through reduced breakdown of cyclic GMP, the substance that causes vasodilation in consequence of NO-induced production of guanylate cyclase. Benefit is seen as apparent improvement in endothelial function (Katz *et al.*, 2000; Halcox *et al.*, 2002a; Schofield *et al.*, 2003; Sommer and Schulze, 2005; Aversa *et al.*, 2008a). PDE5 inhibitors are the most logical for treatment of endothelial dysfunction as this affects vascular smooth muscle, and these drugs enhance and prolong the effects of the smaller amounts of NO which reach the muscle (Vallance and Chan, 2001). To date, PDE5 inhibitors have not been specifically studied for this indication, but are widely and successfully used for treatment of erectile dysfunction in which there is reduced NO release from parasympathetic penile nerves, as well as from vascular endothelium. PDE5 inhibitors are now being investigated as a potent long-term treatment for chronic heart failure and primary pulmonary hypertension (Wilkens *et al.*, 2001; Patel and Katz, 2005; Kumar *et al.*, 2009; Movsesian *et al.*, 2009; Vlachopoulos *et al.*, 2009a; Vlachopoulos *et al.*, 2009b; Ryerson *et al.*, 2010) and in association with inhaled NO for more acute forms of pulmonary hypertension (Michelakis *et al.*, 2002). When initially used for erectile dysfunction, there were concerns that PDE5 inhibitors may have adverse effects in patients with cardiac or coronary disease, but these views have not been substantiated. Interaction with nitrates is predictable from mode of action and explicable as an approximate doubling of nitrate effect (O'Rourke and Nichols, 2002).

Problems in clinical practice are now identifiable as being due to the manner in which nitrates are used (relatively large doses absorbed over a short time period (Jiang *et al.*, 2002)) rather than use of a PDE5 inhibitor. This is discussed further in Chapter 26.

Attention to endothelial function and NO release and action has put a new focus on nitroglycerin – the substance that provided Alfred Nobel's fortune, and which he refused to take for his own angina and hypertension (Marsh and Marsh, 2000; Smulyan, 2007). If ill effects of endothelial function are largely due to decreased NO action, might not this be overcome through administration of NO donors, such as nitroglycerin? Might not this drug (and the PDE5 inhibitors) provide the most logical treatment of established endothelial dysfunction? Problems in metabolism of nitroglycerin to NO are responsible for its tolerance and need for intermittent administration. These are sure to be overcome in development of new nitrate analogs. Better use of nitrates is aided by better monitoring of drug action, especially on peripheral wave reflection and central pressure augmentation. This is discussed in Chapter 26 (see Figs 26.3 and 26.8).

ENDOTHELIAL FUNCTION AND PREDICTION OF CARDIOVASCULAR EVENTS

A high priority is currently given to methods which may better predict future cardiovascular events (Chapters 21, 22, and 28), and so permit better stratification of asymptomatic people into level of risk. Such characteristic

would enable better focus of preventive strategies and their application at lower cost to the community. Currently, stratification is based on the conventional risk factors as established by the Framingham and other major trials. Other factors, including circulating levels of C-reactive protein (CRP), measures of arterial stiffness, and tests of endothelial function, are being investigated. A large body of literature supports the idea that circulating CRP may help to estimate risk for initial cardiovascular events and may be used most effectively in people at intermediate risk for vascular events, offering moderate improvement in reclassification of cardiovascular risk (Fichtlscherer *et al.*, 2000; Yasmin *et al.*, 2004; Maki-Petaja *et al.*, 2006; Wilson *et al.*, 2008; Schumacher *et al.*, 2009). Endothelial dysfunction is increasingly recognized to be a biomarker of cardiovascular risk in apparently healthy subjects, as well as in patients with cardiovascular disease (Celermajer *et al.*, 1994b; Celermajer, 2003; Widlansky *et al.*, 2003; Bonetti *et al.*, 2004; Cohn *et al.*, 2004; Brunner *et al.*, 2005; Lerman and Zeiher, 2005; Lippincott *et al.*, 2008; Anderson, 2007; Shechter *et al.*, 2007; Yeboah *et al.*, 2007; Kim *et al.*, 2009; Kwagyan *et al.*, 2009; Shechter *et al.*, 2009; Yeboah *et al.*, 2009) (Fig. 5.11). The most recent publication on FMD and peripheral (digital) artery tonometry in the Framingham study showing that these techniques had different relations with cardiovascular risk factors and were not correlated with each other (Hamburg *et al.*, 2011), clouds this issue and emphasizes that the technique used may be of the essence for the prediction of events. The predictive value of endothelial function is discussed in Chapter 22. Excellent in-depth reviews of endothelial function and CPR and their

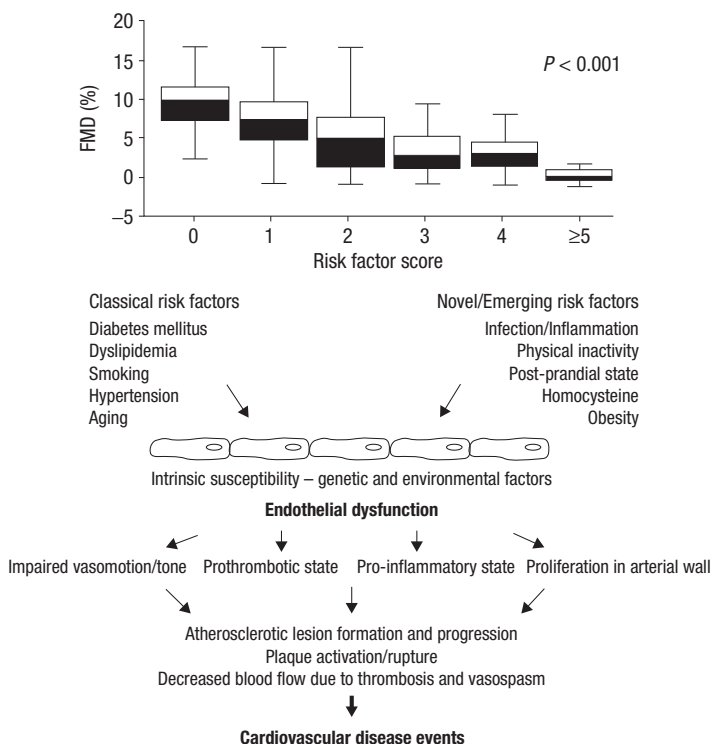


Figure 5.11 The role of endothelial dysfunction (FMD) in the pathogenesis of cardiovascular disease events. Cardiovascular disease risk factors adversely affect a diverse range of endothelial homeostatic functions and contribute mechanistically to the development, progression, and clinical expression of atherosclerosis. The response of the endothelium to the cumulative effects of risk factors may, in part, relate to intrinsic and environmental factors, such as genetic polymorphisms, dietary factors, exercise, and other factors. Thus, endothelial function may serve as a barometer for cardiovascular risk. From Celermajer *et al.* (1994b) and Widlansky *et al.* (2003)

importance in predicting cardiovascular events have been published recently (Calabrò *et al.*, 2009; Hingorani *et al.*, 2009).

Oliver and Webb (2003), Chirinos *et al.* (2005), Sutton-Tyrrell *et al.* (2005), Laurent *et al.* (2006), Mattace-Raso *et al.* (2006), Nilsson *et al.* (2009), Vlachopoulos *et al.* (2010a), and Vlachopoulos *et al.* (2010b) have pointed to the potential value of measures of arterial stiffness, such as pulse wave velocity and augmentation index, as prognostic markers of cardiovascular risk. As described in Chapter 21, these have shown independent diagnostic value in people with coronary atherosclerosis, with hypercholesterolemia, with diabetes, with hypertension, with end-stage renal disease, and in the aged. Augmentation of the pressure pulse was used by life insurance companies over 100 years ago to detect asymptomatic arterial degeneration, and to decline life insurance (Postel-Vinay, 1996). Augmentation index (see Chapter 9) has been used by Wilkinson *et al.* (2002a), Hayward *et al.* (2002), Covic *et al.* (2004b), Suh *et al.* (2005), and McEniery *et al.* (2006) (see above) as a test of endothelial function, through examination of its capacity for reduction by salbutamol (albuterol). These tests have been shown to correlate with the more complex investigations of endothelial function, and in preliminary studies (Hayward *et al.*, 2002; Agabiti-Rosei *et al.*, 2007; Weber *et al.*, 2007) have shown clear differences in groups of people with risk factors for coronary artery disease. Such measurements of stiffness parameters (Augmentation Index and Pulse Wave Velocity (Laurent *et al.*, 2006)) have the potential to show both anatomical derangement from study at rest, and function derangement from study with an agent which provokes NO release from the endothelium.

ENDOTHELIAL FUNCTION AND ARTERIAL DISTENSIBILITY

Effects of vasodilator drugs on arterial distensibility are discussed in Chapters 4 and 26. Relaxation of smooth muscle tone transfers stresses from collagenous to elastic components in the peripheral muscular conduit arteries and renders the arteries more compliant, and less stiff than they would otherwise be at their new increased diameter (Bank *et al.*, 1995; Bank *et al.*, 1999; Nichols and Edwards, 2001). Direct effects are most apparent when nitrates are used and when arterial caliber and blood pressure are measured at the same site (Bank *et al.*, 1999; O'Rourke *et al.*, 2002; Nichols and Edwards, 2001), but are readily evident, also, as reduction in wave reflection amplitude through a decrease in augmentation of the pressure waveform in peripheral and central arteries (Chapter 26).

The new phenomena have been demonstrated in studies of endothelial function. Investigations in humans (Joannides *et al.*, 1997; Kinlay *et al.*, 2001; Cheung *et al.*, 2002; Wilkinson *et al.*, 2002a) and in experimental

animals (Fitch *et al.*, 2001; Wilkinson *et al.*, 2002b) have confirmed that arterial stiffness is reduced and caliber increased by nitrates, but that inhibition of eNOS by L-NMMA has the opposite effect. Findings are consistent with continuous regulation of arterial elasticity, as well as caliber by normal endothelium, with the possibility of impaired elasticity (increased stiffness) in the presence of endothelial dysfunction (Cockcroft *et al.*, 1997; Barenbrock *et al.*, 1999; Kosch *et al.*, 2001; Kidawa *et al.*, 2003; Wilkinson *et al.*, 2004; McEniery *et al.*, 2006; Wallace *et al.*, 2007; Martino *et al.*, 2008).

WALL SHEAR STRESS AND ENDOTHELIAL GENE EXPRESSION

Besides regulating arterial diameter by modifying the production of vasoactive mediators, as discussed above, wall shear stress is also an important determinant of endothelial gene expression (Davies and Tripathi, 1993; Gimbrone and Topper, 1999; Brooks *et al.*, 2002; Chien, 2003; Dekker *et al.*, 2006; McCue *et al.*, 2006; Parmar *et al.*, 2006), wall shear stress being defined as the drag of the flowing blood along the vessel wall. It basically is a tangential stress. For its estimation the reader is referred to Chapter 8.

Endothelial genes upregulated by shear stress include transcription factors, growth factors, adhesion molecules, and enzymes; they can be upregulated by shear stress transiently or more permanently (Topper and Gimbrone, 1999). The genes expressed are dependent on the level as well as the type of shear stress applied. For example, in *in vitro* studies, upregulation of VCAM-1 is observed when the cells are exposed to steady shear stress (Ando *et al.*, 1994), but downregulation when they are exposed to oscillatory shear stress (Chappell *et al.*, 1998). Also, the redox state of endothelial cells was found to be dependent on the type of shear stress applied (De Keulenaer *et al.*, 1998). Walpola and colleagues (Walpola *et al.*, 1995) showed that *in vivo* the effect of shear stress on the expression of ICAM-1 and VCAM-1 is shear level dependent. In their experiments, low shear stress enhanced VCAM-1 expression substantially and suppressed ICAM-1 expression, while high shear stress enhanced ICAM-1 expression substantially and VCAM-1 expression only mildly. It should be kept in mind that these *in vivo* findings do not necessarily comply with *in vitro* observations. *In vitro* shear stress downregulates endothelin-1, a vasoactive molecule (Malek and Izumo, 1992), and thrombomodulin (Malek and Izumo, 1994); the latter downregulation being completely reversible (Malek and Izumo, 1994). These findings indicate that endothelial cells can discriminate between subtle variations in loading conditions which may include differences in temporal and spatial gradients of shear stress (Barbee, 2002).

Shear stress and endothelial cell shape

Biomechanical forces also affect endothelial cell structure. Endothelial cells tend to align with wall shear stress: the higher the wall shear stress, the more elongated the cells will be (Barbee *et al.*, 1994; Sato and Ohshima, 1994). These changes are associated with redistribution and rearrangement of intracellular stress fibers (Barbee *et al.*, 1994) and with their number (Walpole *et al.*, 1993). The shape change may also depend on the type of shear stress applied (Helmlinger *et al.*, 1991). By means of atomic force microscopy (Barbee *et al.*, 1994) combined with computational fluid mechanics (Barbee *et al.*, 1995), it could be demonstrated that the shear stress-induced reorganization of endothelial cells results in flattening of these cells, thereby reducing the shear stress difference along endothelial cells; this difference will be smaller, the higher the wall shear stress.

Mechanotransduction: the possible role of the glycocalyx

The transduction of biomechanical forces into biomolecular responses is called 'mechanotransduction'. This mechanism has been studied extensively over the past decades (Davies and Tripathi, 1993; Davies, 1995; Resnick and Gimbrone, 1995; Chien, 2003; Moon *et al.*, 2005). In mechanotransduction, the mechanical forces acting on the luminal side of the endothelial cells, thereby deforming these cells, are transmitted through the cytoskeleton to other areas in the cell (Helmke and Davies, 2002). The distribution of these forces throughout the cell are especially sensed at the basal adhesion points, where the endothelial cell is attached to the extracellular matrix, cell junctions, and the nuclear membrane (Davies, 1995), inducing acute and more delayed processes in these cells (Davies and Tripathi, 1993; Davies, 1995). In the cellular membrane that may respond to the induced deformation directly, shear stress activates among others stretch-sensitive ion channels, phospholipids, and integrins (Davies and Tripathi, 1993; Davies, 1995). The biomechanical forces transferred to the nuclei can be sensed by the shear stress-sensitive sequences in the promoter of a variety of genes (Resnick and Gimbrone, 1995; Gimbrone and Topper, 1999).

There is increasing evidence that the glycocalyx, a mesh of glycoproteins, sulfated proteoglycans and associated glycosaminoglycans of a few hundred nanometers to a few micrometers (Vink and Duling, 1996; Gouverneur *et al.*, 2006; Reitsma *et al.*, 2007), senses fluid dynamical forces (Florian *et al.*, 2003; Moon *et al.*, 2005; Gouverneur *et al.*, 2006; Tarbell and Pahakis, 2006). It has been proposed that the core proteins in the glycocalyx are especially suited to act as a transducer of fluid shearing stresses (Weinbaum *et al.*, 2003) and that the glycocalyx transfers the fluid dynamical forces into tensile stress (Smith *et al.*, 2003). The role of the glycocalyx in mechanotransduction is supported by the recent findings in *in vitro* studies, demonstrating attenuated shear stress-induced release of nitric oxide by (Mochizuki *et al.*, 2003) and shape change modulation of endothelial cells (Yao *et al.*, 2007) exposed to such glycocalyx degrading enzymes as heparitinase and hyaluronidase. Loss of glycocalyx apparently contributes to impaired sensing and transduction of fluid shear stress. It is as yet unclear whether the glycocalyx amplifies the shear forces that are substantially lower than the pressure forces, another important determinant of endothelial cell function (i.e. 10 dyne/cm^2 versus 10^4 dyne/cm^2). Invasive (using labeled erythrocytes) and non-invasive (imaging of superficial, particularly sublingual vasculature) techniques are being developed to measure glycocalyx in the clinical setting (van den Berg *et al.*, 2006).

In a recent study, it was shown that along the murine carotid artery bifurcation, the glycocalyx is significantly thinner opposite the flow divider, the area of disturbed flow associated with low mean wall shear stress and prone to atherosclerosis, than on the side of the flow divider and in the common carotid artery (van den Berg *et al.*, 2006). The thinner glycocalyx opposite to the flow divider was associated with a significantly thicker intima (van den Berg *et al.*, 2006), resulting from low-density lipoprotein (LDL) leakage into this layer (van den Berg *et al.*, 2009). These observations indicate that shear stress-dependent changes in glycocalyx dimensions are involved in early atherogenesis. The small glycocalyx dimensions in the carotid artery bifurcation exposed to low mean wall shear stress could be explained by the finding that shear stress is a stimulus for endothelial glycocalyx production (Gouverneur *et al.*, 2006).

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General principles for measuring arterial waves

Measurement of blood flow	129	Measurement of arterial diameter	160
Measurement of blood flow, stroke volume, and cardiac output	146		

Transducers are used to convert time-varying physiologic events (or variables) into electric signals. With the event available as an electric signal, the maximum amount of information can be obtained. In cardiology and cardiovascular physiology, the hemodynamic events most often measured are phasic pressure, flow (and velocity), volume, and length (or diameter). The transducer used to measure these variables should obey the three criteria for the faithful reproduction of an event: amplitude linearity, adequate frequency response, and phase linearity (Nichols *et al.*, 1994a; Nichols *et al.*, 1994b; Geddes and Baker, 1975).

MEASUREMENT OF BLOOD FLOW

The first to record the human pulse with a transducer was Vierordt in 1855. The recording was obtained using a lever with one end placed on the radial artery and the other attached to a pencil that had been dipped in Indian ink. The first high-fidelity recording of the arterial pulse was obtained using a counter-pressure method and reported by Marey (1860), champion of the graphic method in physiology and medicine. Such studies led to the development of the sphygmogram by Marey during the early 1860s and is discussed in detail in Chapter 26.

The study of time-varying or pulsatile pressure in the circulation preceded that of pulsatile flow by many decades because the technical development of manometers was simpler than that of flow-measuring systems (flowmeters and transducers). The theory of manometer design capable of producing adequate instruments for measuring pulsatile pressure was initiated by Otto Frank (1903) and remains essentially the same today. Frank's manometers used short, wide cannulae and rubber membranes, and were used extensively in Wiggers' exhaustive studies of pressure waveforms such as those reported in his classic monograph of 1928, *The Pressure Pulses of the Cardiovascular System*. In the 1930s, a substantial improvement

was made by Hamilton, who replaced the membrane with thin beryllium-copper and increased the sensitivity of the optical recording system (Hamilton *et al.*, 1934). This considerably improved the frequency characteristics of the system. The technical advances in electronics in the 1940s, when applied to manometry, made dramatic changes in design with respect to miniaturization and frequency response. This allowed pressure manometers to be used with long, fine catheters so that pressure measurements over a large part of the cardiovascular system could be made from a single insertion into a peripheral artery; this immediately opened the way to their extensive clinical use in humans (Cournand *et al.*, 1945). During his Nobel lecture in 1956, Cournand stated that 'the cardiac catheter was ... the key in the lock' (Liljestrand, 1957). In doing so, he summarized the powerful influence that the development of cardiac catheterization has had on the diagnosis and treatment of coronary artery disease and of valvular and congenital heart disease. The history of interventional cardiology, including cardiac catheterization, has been reviewed recently by Mueller and Sanborn (1995), Pepine *et al.* (1996), Ryan (2002), Baim (2005), Mitchell *et al.* (2008), and Braunwald (2009). It is still often not fully appreciated that optimal performance of catheter-manometer systems can only be achieved by exercising great care in the setting-up procedure and realizing that the frequency characteristics of the system deteriorates with time (Promonet *et al.*, 2000). As the quantitative analysis of arterial pressure waves requires a detailed knowledge of the accuracy of the measurements made with regard to the catheter-manometer response in terms of frequency, it is necessary first to survey the theory of manometer design.

Manometer theory

A manometer may, in the first instance, be regarded as analogous to a simple oscillating system consisting of a

mass M suspended from a spring of stiffness S ; a disk is attached to the mass and immersed in a liquid (Fig. 6.1). If the mass is displaced and released, it will oscillate in damped simple harmonic motion. The equation that describes this oscillation is given by Newton's second law of motion:

$$F = Ma, \quad 6.1$$

in which F is the sum of the restoring force $-Sx$ and the damping force $-R(dx/dt)$; R is viscous damping (or resistance). Thus

$$-Sx - R \frac{dx}{dt} = M \frac{d^2x}{dt^2},$$

and so

$$M \frac{d^2x}{dt^2} + R \frac{dx}{dt} + Sx = 0. \quad 6.2$$

The solution (Pipes, 1958) of eqn 6.2 is

$$x = Ae^{-Rt/2M} \cos\left(\left(\frac{S}{M} - \frac{R^2}{4M^2}\right)^{1/2} (t - \delta)\right). \quad 6.3$$

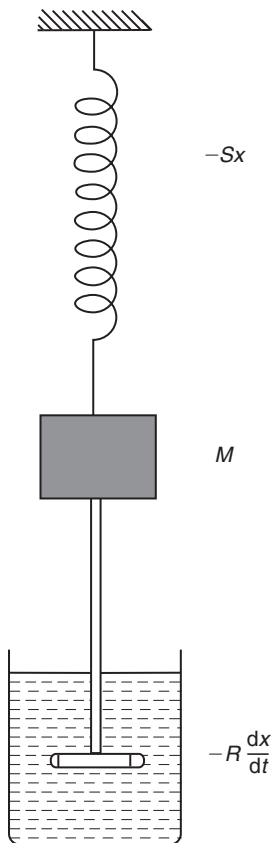


Figure 6.1 An example of a damped oscillator. A disk is attached to the mass (M) and immersed in a viscous liquid which exerts a damping force $-Rdx/dt$. The elastic restoring force is $-Sx$

The constant A is the amplitude of the motion, and

$$\omega_d = 2\pi f_d = \left[\frac{S}{M} - \left(\frac{R}{2M} \right)^2 \right]^{1/2} \quad 6.4$$

is the damped angular frequency, in radians per second, of the motion; f_d is the corresponding damped natural frequency in cycles per second (or Hz). If no friction were present, R would equal zero, and

$$\omega_0 = \left(\frac{S}{M} \right)^{1/2}, \quad 6.5$$

which is the angular frequency of undamped motion. The logarithmic decrement or damping constant of the oscillation is given by

$$\beta_0 = R/2M. \quad 6.6$$

Equation 6.3 can now be written as

$$x = Ae^{-\beta_0 t} \cos \omega_d(t - \delta), \quad 6.7$$

and eqn 6.4 as

$$\omega_d = (\omega_0^2 - \beta_0^2)^{1/2}. \quad 6.8$$

The damping coefficient is defined as

$$\beta = \beta_0/\omega_0. \quad 6.9$$

The behavior of the system on displacement and release of the mass will depend on the relative magnitudes of ω_0 and β_0 and thus β . There are four possibilities:

1. If $\beta_0 = \omega_0$, $\beta = 1$ and the mass will not oscillate after release but will return exponentially to its equilibrium position (Fig. 6.2). The equation describing this motion is

$$x = Ae^{-\beta_0 t}, \quad 6.10$$

and the system is said to be 'critically damped'.

2. If $\beta_0 > \omega_0$, $\beta > 1$ and the rate of return to equilibrium is also non-oscillatory but is slower and the system is said to be 'overdamped'.
3. If $\beta_0 \leq \omega_0$, $\beta < 1$ and the mass will oscillate but the amplitude of the oscillation will decay exponentially until it comes to rest (Fig. 6.2). A system with these characteristics is said to be 'underdamped' and its motion is

$$x = Ae^{-\beta_0 t} \cos \omega_d t. \quad 6.11$$

4. If $\beta_0 = 0$, $\beta = 0$ and the mass will oscillate sinusoidally indefinitely without exponential decay. The system is 'undamped' and its motion is

$$x = A \cos \omega_0 t. \quad 6.12$$

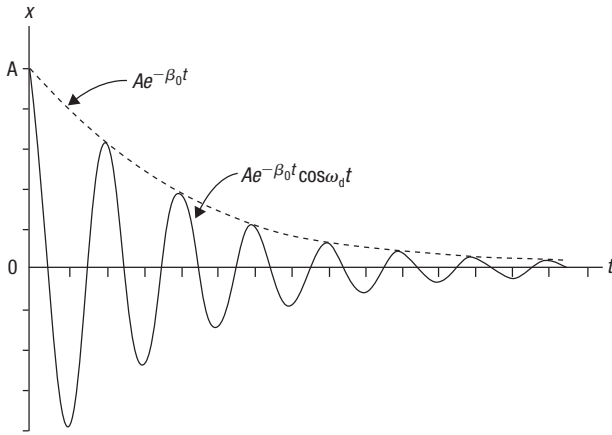


Figure 6.2 Response of an oscillator under different conditions of damping. Two situations are shown: overdamped (dashed line), when the response function falls exponentially to zero; and underdamped (solid line), when the response function oscillates, with the amplitude of oscillation decaying exponentially until it comes to rest at zero

When applying these concepts to a catheter–manometer system, the term S represents the ‘stiffness’ of the transducer; this will be, in essence, a liquid-filled chamber with a deformable membrane at the wall. The volume elasticity E of the system (manometer plus attached catheter) ($E = \Delta P / \Delta V$) replaces the spring constant S . Hence, increasing the stiffness of the membrane increases the natural frequency of the manometer. The mass M is now the mass of the liquid that moves. As the cross-sectional area of the transducer is usually very large in relation to the cross-sectional area of the catheter, it may normally be defined solely in relation to the kinetic mass of the liquid in the catheter; that is, its velocity is inversely proportional to the cross-sectional area. Hence M is given by

$$M = L\rho/\pi r^2, \quad 6.13$$

where L and r are the length and internal radius of the catheter and ρ is the density of the liquid (which can usually be taken as 1.0).

Rewriting eqn 6.5, we then obtain

$$\omega_0 = \left(\frac{\pi r^2 E}{\rho L} \right)^{1/2} = r \left(\frac{\pi E}{\rho L} \right)^{1/2}. \quad 6.14$$

Thus, the natural frequency of a catheter–manometer system varies as the radius of the catheter and inversely as the square root of its length with any given manometer – provided that the stiffness of the catheter wall is sufficiently great to maintain the elastance of the whole system. The natural frequency of the manometer itself can be increased by making the membrane stiffer, although this increase will only be proportional to the square root of the increase in the value of E . The smaller deflection of

the stiffer membrane will lead to a decrease in sensitivity and must be compensated.

The damping β_0 constant can similarly be written in a form applicable to a liquid-filled system. The value for the resistance R can be obtained using Poiseuille’s equation:

$$R = 8\mu L / \pi r^4. \quad 6.15$$

Hence, taking M again as given by eqn 6.13, we get (assuming $r = 1.0$)

$$\beta_0 = R/2M = 4\mu/r^2. \quad 6.16$$

This value is only significant in relation to the natural frequency of the manometer and is usually described as a fraction, or percentage, of critical damping, the condition where $\beta_0 = \omega_0$ or $\beta = 1.0$. The damping coefficient β can be derived from eqns 6.14 and 6.16

$$\begin{aligned} \beta &= \beta_0 / \omega_0 \\ &= \frac{4\mu}{r^2} \left(\frac{\rho L}{\pi r^2 E} \right)^{1/2} = \frac{4\mu}{r^3} \left(\frac{\rho L}{\pi E} \right)^{1/2}. \end{aligned} \quad 6.17$$

Hence, with respect to the choice of catheter, it can be seen that the damping varies not only as the square root of the length but inversely as the cube of the radius. This sensitivity of the damping to changes in the internal radius of the catheter must be borne in mind when considering the possible effects of fibrin deposition on the inside of the catheter. The fact that the term E appears in the equation also emphasizes that, if the effective elastance is decreased, as by even a minute air bubble in the system, not only will the resonant frequency be reduced, but also the damping will be increased.

The derivation of eqn 6.17 is not, however, strictly accurate, for the value of the Poiseuille resistance is only valid for steady flow. The correct solution for oscillatory flow was first obtained by Lambossy (1952a). From eqn 7.21 and Fig. 7.9, it can be seen that the resistance rises with frequency; if plotted against the parameter α , the resistance rises linearly, relative to the Poiseuille resistance, for values of α above 3.0 (McDonald, 1974).

From the expression derived for the oscillatory resistance (eqn 7.21), we may write it in a different form:

$$\begin{aligned} \beta_0 &= \frac{\mu}{2\rho r^2} \frac{\alpha^2}{M'_{10}} \sin \varepsilon_{10} \\ &= \frac{\omega \sin \varepsilon_{10}}{2M'_{10}} \end{aligned} \quad 6.18$$

Manometers that are used with catheters fall into a special class of recording instruments in that their frequency response should be checked each time they are used. The reason for this can be seen from the theoretic analysis given above because their behavior depends on the effective volume elasticity of the catheter–manometer system. Any air trapped in the system on filling will obviously change the volume elasticity and hence the

resonant frequency and damping. Furthermore, during its use in the bloodstream, the effective lumen of the catheter may change owing to the deposition of fibrin, so that the damping will increase and the resonant frequency will decrease from their initial values. To obviate this, flushing of the system at intervals is usually undertaken; this, however, creates the risk of introducing small air bubbles into the system.

A brief discussion of the magnitude of the air bubble effect will help to emphasize the vital importance of this artifact. The volume displacement of the Statham P23Db manometer, for example, is 0.04 mm^3 . It seems intuitively obvious that an air bubble of minute size will drastically reduce the effective volume elasticity of the system. Even if it is in one of the transparent parts of the manometer, it will be difficult to detect. In fact, it is much more likely to be trapped at a connection or in a stopcock. Inspection of the system will, therefore, be of no value.

Assuming the volume V of the liquid in a P23-type manometer with catheter is 1.0 mL, the effective bulk modulus B for the liquid-filled manometer will range from 0.5 to $1.1 \times 10^{10} \text{ dyne/cm}^2$. The bulk modulus or modulus of volume elasticity is defined as $B = -VdP/dV$; its reciprocal is called the 'compressibility'. The bulk modulus of water at body temperature is about $2.2 \times 10^{10} \text{ dyne/cm}^2$. That of a gas may easily be derived from the gas laws; as temperature is held constant, we may simply use Boyle's law for an ideal gas:

$$PV = \text{constant.} \quad 6.19$$

Hence $dP/dV = -C/V^2$

$$B = V dP/dV = C/V = P. \quad 6.20$$

Under the oscillatory conditions in a manometer, the actual relation will more closely approximate to an adiabatic compression, where

$$PV^\xi = C, \quad 6.21$$

and then

$$B = \xi P. \quad 6.22$$

For air, the value of ξ is 1.41. The value of P is that of atmospheric pressure, or about 10^6 dyne/cm^2 , so that we may take the bulk modulus of air as approximately $1.5 \times 10^6 \text{ dyne/cm}^2$, i.e. smaller than that of the manometer by a factor of at least 3.0×10^3 . In other words, a bubble of air which is this fraction of the volume of the catheter-manometer system will undergo the same volume change and will double the volume displacement. For the system of 1.0 mL capacity we have been considering, this would represent a bubble of 0.3 mm^3 volume. Doubling the volume displacement reduces the volume elasticity to half its previous value and reduces the resonant frequency to approximately 70 percent of its

previous level. This analysis is simplified to illustrate magnitudes; in practice, the application of the manometer equations does not apply well when air and liquid are both present. For example, the presence of an air bubble increases the damping more than would be predicted from these equations by the change in frequency response.

In practice, it would seem that air bubbles of this order of size are normally the smallest with which we have to cope. This is deduced from the fact that with successive flushing, etc., to improve a frequency response, progress is usually made by a series of jumps of at least 5.0–10 Hz and is therefore, after a little experience, obvious on quite a cursory inspection of the record.

There is no very simple or quick way to eliminate air bubbles. The most important step is to use saline from which all, or most, of the dissolved air has been removed by boiling (the use of sterilized saline will normally achieve this in any case). Desaturation of the liquid has the result that small bubbles will then tend to dissolve spontaneously. The spontaneous solution of air may be rather slow, and filling the catheter-manometer system overnight will usually save time at the start of an experimental day. Carbon dioxide is more soluble than air, and for testing procedures where catheters need to be changed at intervals, the displacement of all air with CO_2 before filling with liquid is recommended. In the completely liquid-filled pump system, it is an invaluable technique.

The use of preliminary cleansing with alcohol or detergent has been recommended by some workers. It only appears to be important where new plastic catheters are used for the first time, in order to get their inner surface thoroughly wet. Very small amounts of detergent also seem to minimize the trapping of small bubbles at connecting joints; exuberant use of wetting agents, however, may create foam, which is very difficult to remove.

The methods used to determine the dynamic frequency characteristics of a catheter-manometer system are either the free-vibration method using a step function of pressure, or the forced-oscillation method using some form of a sine-wave generator with a wide range of frequencies.

Determination of frequency response of pressure measuring systems

FREE-VIBRATION OR TRANSIENT 'POP' METHOD

A sudden step (or transient) change in pressure imposed at the end of a catheter-manometer system will cause the system to oscillate at its damped natural frequency. The oscillations will be sinusoidal and will decay exponentially in accordance with the damping coefficient (see Fig. 6.2). The pressure step can also be applied as a square wave from a multifunction generator or as a transient using an apparatus similar to the one shown in Fig. 6.3 (Nichols and O'Rourke, 2005). The design of such an apparatus can be very varied. The one used in most laboratories consists

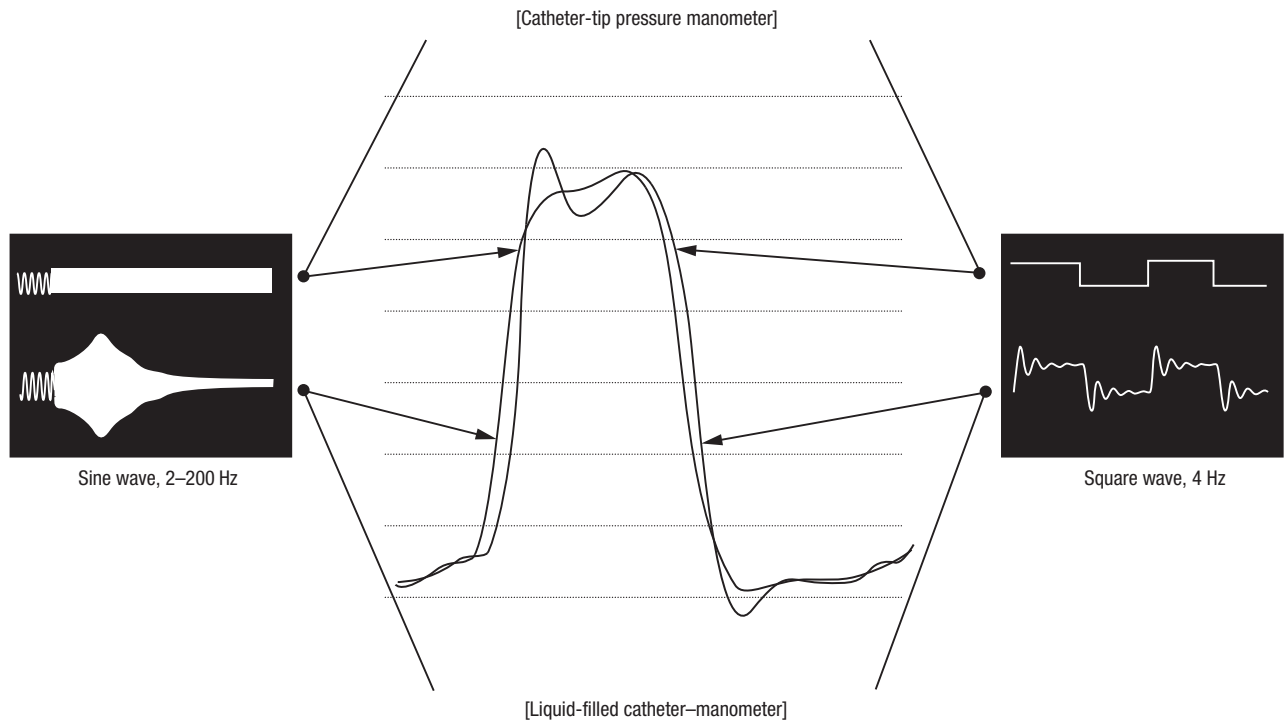


Figure 6.3 Dynamic frequency response of a liquid-filled catheter-manometer system. Left: A loudspeaker was driven by continuously increasing the frequency of a sine-wave forcing function from a multifunction generator. The natural (or resonant) frequency of the liquid-filled system was 37 Hz, but was flat (65 percent) to only 12 Hz. Right: The response of each system to a square-wave forcing function of 4 Hz. Middle: Left ventricular pressure measured in a patient using the two systems. Reproduced from Nichols *et al.* (1978)

of a section of sturdy lucite cylinder with an internal diameter of about 4.0 cm and 10–12 cm in length. The cylinder is clamped vertically to a laboratory stand. The lower end of the cylinder is equipped with an adaptor that is leakproof when a catheter is inserted through it. The cylinder is usually filled by flushing saline through the catheter. The level of the liquid should be above the tip of the catheter, but below the inlet to the sphygmomanometer bulb. The top of the cylinder is sealed with a thin rubber membrane and O-ring and then pressurized with the bulb (a large syringe can also be used). For the best results, the entire system should be filled with boiled saline and left overnight before testing. The rubber membrane used is commonly cut from discarded surgical gloves but is better purchased in flat sheets (toy balloons can also be used). This needs to be stretched lightly, without an air leak, over the end of the chamber and inflated. With an apparatus such as that described, inflation produces a pressure of 40–50 mmHg when the rubber is distended (Gabe, 1972; Yanof, 1972).

The most rapid and reliable method of rupturing the membrane according to McDonald (1974) appears to be burning (a lighted match is very effective). This breaks the membrane explosively and causes an audible ‘pop’ (hence the nickname for the technique). The membrane can also be ruptured with a sharp needle or scalpel blade. The time for the pressure to drop to atmospheric is less than 1.0 ms judged by a high-fidelity catheter-tip pressure manometer.

The pressure is displayed on an oscilloscope triggered to sweep when the pressure begins to drop and the sweep photographed, or it is recorded on a high-frequency response strip-chart recorder or computer. To save time (and rubber), an approximate idea of the resonant frequency can be obtained by giving a sharp tap to the catheter; if the same sweep speed is used (we standardize at 10 ms/cm), a little practice soon enables one to estimate the time between peaks. Thus, if an end result of 80–100 Hz is expected, it is easy to spot when the oscillations are markedly lower than this in the early stages of setting up. A recording is necessary for any accurate measurement of resonant frequency and for measuring the damping.

The main artifact in this technique is due to vibration of the cylinder support, and this needs to be made extremely rigid so that its natural frequency is far higher than that of the catheter-manometer system. Even vibration of the chamber may introduce errors into the measurement of damping, and this is discussed below under Analysis of results.

Another apparatus used to test the dynamic frequency characteristics of catheter-manometer systems consists of a multifunction generator and a loudspeaker with an attached test chamber constructed from a section of lucite cylinder. The chamber is filled, as before, by flushing saline through the catheter, which is inserted through a leakproof adaptor at the chamber’s bottom (Nichols *et al.*, 1978; Nichols *et al.*, 1985b). The diaphragm of the

loudspeaker is driven by a square-wave forcing function of 2.0–4.0 Hz from the multifunction generator. The change in pressure is detected by the manometer and recorded on an oscillograph recorder at a fast paper speed. The response, which is shown in Fig. 6.3, is similar to the one obtained using the ‘pop’ technique.

ANALYSIS OF RESULTS

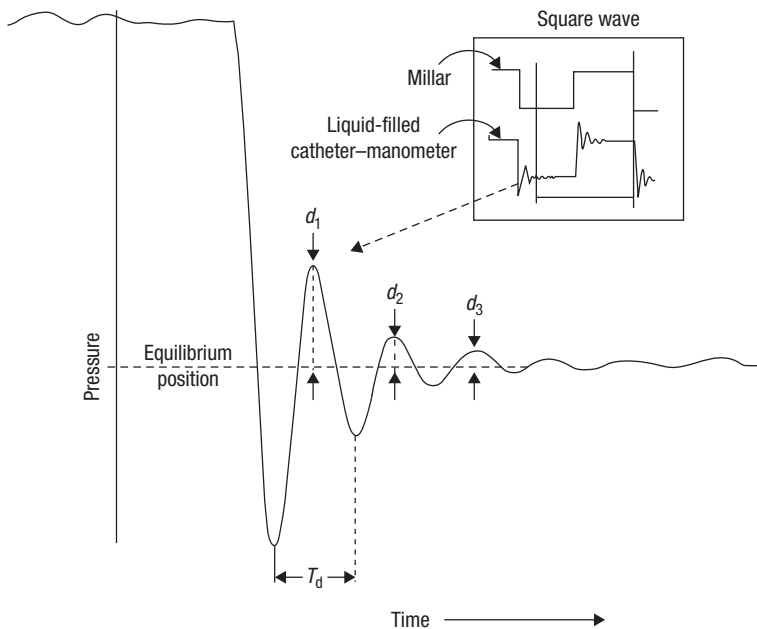
A typical step response recorded with a catheter–manometer system of low-frequency response and damping is shown in Fig. 6.4, which has the characteristic oscillations decreasing exponentially. The damped natural (or resonant) frequency f_d is derived directly from the period T_d between points of like phase in successive oscillations:

$$f_d = 1/T_d. \quad 6.23$$

In cases where the damping is low, as in the example in Fig. 6.4, the difference between the value of f_d and f_0 (the frequency if there were no damping) is not usually significant experimentally.

The damping is calculated from the logarithmic decrement Λ from cycle to cycle. Yanof (1972) and Gabe (1972) measured the deviation from the equilibrium position; McDonald (1974) and Milnor (1989) found it easier to measure the total excursion every half-cycle (from maximum deviation in one direction to maximum deviation in the other; Gabe also reviews this method). The ratio of the first overshoot d_1 to the second overshoot d_2 , is (Webster, 1978)

$$\frac{d_1}{d_2} = \exp \left[\frac{2\pi\beta}{(1 - \beta^2)^{1/2}} \right]. \quad 6.24$$



Hence

$$\log_e \left(\frac{d_1}{d_2} \right) = \Lambda = \frac{2\pi\beta}{(1 - \beta^2)^{1/2}}. \quad 6.25$$

The damping coefficient β is found by solving eqn 6.25:

$$\beta = \frac{\beta_0}{\omega_0} = \frac{\Lambda}{(4\pi^2 + \Lambda^2)^{1/2}}, \quad 6.26$$

since

$$\beta_0 = \Lambda/T_d \quad 6.27$$

and

$$\omega_0 = \frac{(4\pi^2 + \Lambda^2)^{1/2}}{T_d}. \quad 6.28$$

When Λ is small, the damped and undamped natural frequencies are similar.

FORCED-OSCILLATION (OR SINUSOIDAL) METHOD

A catheter–manometer system that is being driven by a sinusoidal forcing function will respond only to the applied frequency, ω . In an underdamped system, the amplitude of the response will increase as the applied frequency increases, until it reaches a peak; then it is said to resonate. The resonant frequency will be close to, but less than, the undamped natural frequency, ω_0 . Thereafter, the amplitude of response falls off rapidly with frequency (Fig. 6.3). In addition to the amplitude response, there is also a phase lag in the response. This increases with frequency, but the rate of increase varies with the damping

Figure 6.4 Example of catheter–manometer system response to a square-wave (or transient) change in pressure applied by a multifunction generator. An underdamped system will oscillate several times before stabilizing at an equilibrium position (broken line). The resonant frequency is calculated from the period (T_d) of each completed cycle. The measurements of d_1 and d_2 are used to determine the system's damping (see text)

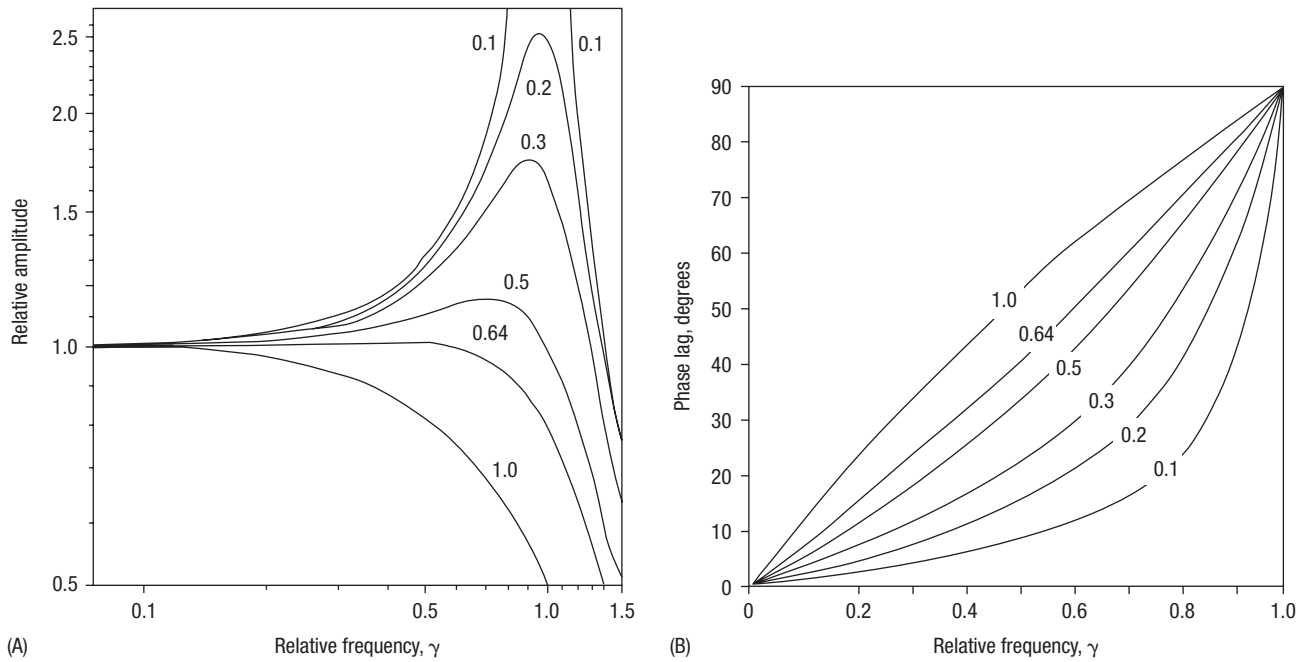


Figure 6.5 (A) The dynamic amplitude response of a catheter–manometer system to a sinusoidal pressure forcing function of unit amplitude. Relative frequency (logarithmic scale) is the ratio of the forcing frequency to the undamped natural frequency of the system. Response curves are shown for systems with different damping coefficients. (B) The phase-lag versus relative frequency of a catheter–manometer system subjected to a sinusoidal forcing function. Reproduced from Gabe (1972)

(Fig. 6.5). In all cases, however, the phase lag is 90° when the driving frequency equals the resonant frequency.

The equation of motion describing the output of a catheter–manometer system subjected to a continuous driving force follows from Newton's second law of motion (eqn 6.1). In addition to the restoring force $-Sx$ and the damping force $-R(dx/dt)$, there is an applied oscillating driving force. For simplicity, let this external force be given by $\sin \gamma t$, where γ is the ratio of the driving frequency to the undamped natural frequency. From eqn 6.1, we obtain

$$-Sx - R \frac{dx}{dt} + \sin \gamma t = M \frac{d^2x}{dt^2}$$

or

$$M \frac{d^2x}{dt^2} + R \frac{dx}{dt} + Sx = \sin \gamma t. \quad 6.29$$

The solution to this equation is

$$x = \frac{1}{[(1 - \gamma^2) + 4\beta^2\gamma^2]^{1/2}} \sin\left(\gamma t - \tan^{-1} \frac{2\beta\gamma}{1 - \gamma^2}\right), \quad 6.30$$

so that the amplitude A relative to its true value is determined by the term

$$A = \frac{1}{[(1 - \gamma^2) + 4\beta^2\gamma^2]^{1/2}}. \quad 6.31$$

It is this function that is shown graphically in Fig. 6.5 for various values of the damping coefficient β .

From eqn 6.31, it can be seen that when the catheter–manometer system is driven at its undamped natural frequency (i.e. $\gamma = 1.0$), then

$$A = \frac{1}{2}\beta. \quad 6.32$$

With small values of β (e.g. 0.2 or less), this will be very close to its true maximum at the amplitude excursion at resonance. Thus it will be, from this equation, a good measure of the damping at that frequency. This damping will be rather higher than it is at low frequencies because, as we have seen above, the resistance term in oscillatory flow increases with frequency. In other words, a liquid-filled catheter–manometer system never follows any of the plotted curves in Fig. 6.5 exactly, but will progressively move on a curve of higher damping as the frequency approaches resonance.

The true maximum amplitude occurs when

$$\gamma = (1 - 2\beta^2)^{1/2}, \quad 6.33$$

and it will be seen that, when $\beta = 0.707$, γ is zero. At this degree of damping, the recorded amplitude never rises above its true value, but remains within about 2.0 percent of this value up to about 50 percent of its undamped natural frequency. This gives a curve that resembles the behavior of a free vibrating system with critical damping and is often confused with it when manometer behavior is

discussed. In fact, with true critical damping ($\beta = 1.0$), the recorded response falls continuously from zero frequency upwards. Considering amplitude response alone, a curve that does not deviate by more than 2.0 percent can be obtained with a damping factor of 0.64, up to 67 percent of the undamped natural frequency. When β is 0.64, the amount of overshoot in the transient is 7.0 percent, as shown in Fig. 6.5A. A system with this degree of damping is 'optimally' damped (Gabe, 1972; Geddes and Baker, 1975).

The phase lag ϕ of the manometer response in relation to the driving pressure is shown in eqn 6.30 to be given by

$$\phi = \tan^{-1} \frac{2\beta\gamma}{1 - \gamma^2}, \quad 6.34$$

and is shown graphically in Fig. 6.5B for various degrees of damping. It can be seen that only with low damping is the phase lag small over any part of the frequency range, and, indeed, if accurate phase recording is required, the damping should be less than 0.1. At this value, for example, the phase lag is some 2.3° at one-tenth of the natural frequency. With modern catheter-tip manometers with resonant frequencies from 5.0 to 15 kHz, the problem does not arise.

The errors in phase introduced by a catheter-manometer system were not regarded as very important until the measurement of such complex quantities as arterial impedance and external ventricular power became of interest. To eliminate or minimize the problems associated with liquid-filled catheter-manometer systems in these measurements, careful investigators now use high-fidelity catheter-tip pressure manometers to measure arterial pressure (Murgo *et al.*, 1980a; Murgo *et al.*, 1980b; Murgo *et al.*, 1981a; Murgo *et al.*, 1981b; Laskey *et al.*, 1985; Nichols *et al.*, 1985a; Nichols *et al.*, 1986a; Nichols *et al.*, 1986b; Laskey and Kussmaul, 1987; Pepine and Nichols, 1988; Kelly *et al.*, 1992a; Kelly *et al.*, 1992b; O'Rourke *et al.*, 1992a; Sharp *et al.*, 2000; Cholley *et al.*, 2001). Even when only the form of the pressure wave is of interest, distortion of shape will result from shifts of phase of the harmonic components relative to each other. An extreme example is shown in Fig. 11.17, where there are two curves composed of a fundamental and a second harmonic component; in one curve, the second harmonic is shifted 90° in relation to the other, and it can be seen that the combined waveform is drastically altered. On the other hand, the total excursion is not much changed. Phase distortion will be progressively more marked at the higher frequencies, but the harmonics above the fifth are relatively small (see Chapter 11), so that if the larger distortions are confined to these components, neither the form nor the amplitude of the wave will be appreciably altered.

Ideally, one would wish to have no phase errors at all. The nearest approach to this is achieved with catheter-tip manometers that commonly have their resonant frequencies in the kilohertz range; the relevant frequencies in

the pulse wave are then only 1.0 or 2.0 percent of the resonant frequency, and the phase lag can hardly be discriminated from zero. When liquid-filled catheter-manometer systems must be used, a damping coefficient of 0.64 should be sought. It can be seen from Fig. 6.5 that, in this case, the phase lag varies virtually linearly with frequency from zero to resonance. In a recording from such a system, all the harmonics will be recorded with the same delay in actual time. This will be equivalent to recording the composite wave undistorted but delayed in time. If the precise time relations of the wave are not important, this is a satisfactory solution; if, however, the transmission time were being related, for example, to the ECG, it would introduce an error. It is because of this lack of relative phase distortion, coupled with the relatively flat response, that a damping coefficient of 0.64 is considered optimal. However, unless special manometers are built which allow for adjustable damping, this situation is very difficult to achieve, and once achieved is difficult to maintain with any assurance throughout an experiment.

In situations where a catheter-manometer system cannot achieve a satisfactorily high-frequency response, some workers have corrected their pressure curves by calculating the true amplitude response from the equations governing the theoretic curves shown in Fig. 6.5. This is quite easily done when a computer is available; the pressure curve is first subjected to an harmonic analysis, and the amplitude and phase correction are applied to each frequency component. It is in this situation that errors will be introduced unless the oscillatory resistance is derived from Womersley's equation and incorporated into the term β_0 as shown in eqn 6.7.

Problems associated with catheter-manometer systems

It is apparent that it is impossible to have a catheter-manometer system that does not distort the amplitude and/or phase if it is necessary to measure a wide band of frequencies. The best approach to simultaneous accuracy for both these characteristics is to have a manometer with a high-frequency response and very small damping. The highest frequency response that we normally need to record ventricular and arterial pressures accurately is 25–30 Hz in dogs and probably less than this in humans (see Chapter 11) (Nichols *et al.*, 1994c). The most striking technical advance in recent years has been in the construction of miniature manometers, which can be placed on the tip of a catheter or wire and thus eliminate the hydraulic damping consequent on having a liquid-filled catheter between the manometer and the point in the vascular system where the pressure is measured. The use of semiconductor transducers has revolutionized the development of catheter-tip manometers. These will be discussed below.

Nevertheless, there is still a large usage of manometers with a catheter attached for routine clinical work (Baim, 2005) and in animal experiments (Kurtz *et al.*, 2005) where small vessels are involved. In these cases, the recording characteristics discussed above must be considered carefully, and frequency calibration must be carried out each time they are used. As a manometer in use is actually being 'forced' by the arterial pressure, it would appear simpler and more direct to test the response over the frequency range it is desired to record. This involves using a pump, and the technical problems of building a suitable pump have, in fact, been the main reasons why this method is not in general use. Pump systems will be considered in two categories: (1) completely liquid-filled and (2) with an air chamber above the liquid.

It is preferable to use a pump that develops a pressure oscillation which is known to be of constant amplitude over its whole frequency range. In a liquid-filled system, this requires a considerable amount of power, and such constancy is most easily obtained with a mechanically driven pump. As the pressure is created by compressing the liquid in the system, the volume displaced by the pump must be very small if the pressure oscillation is kept to a reasonable size (Taylor, 1957a; Taylor, 1957b; Yanof *et al.*, 1963; McDonald, 1974). Yanof *et al.* (1963) used a sinusoidal pump with a frequency range of about 0.5 to 90 Hz. The eccentricity of the pump was of the order of $3.0\text{--}6.0 \times 10^{-4}$ cm, and the stroke volume with a small plunger was about 0.23 mm^3 . The pressure oscillation this produced was about 60 mmHg with a pump chamber of some 60 mL capacity. Figure 6.6 shows records made with this pump. The essential features of the design are given by Yanof *et al.* (1963).

A much wider frequency range can be more easily achieved with a diaphragm that is electrically driven. An apparatus of this kind was designed by Linden (1958), in

which a brass diaphragm was driven by a barium titanate crystal made to oscillate by an applied sinusoidal voltage. The behavior of the titanate unit he originally recommended was, however, below 20 Hz and needed a sinusoidal voltage of some 6200 V to produce a reasonable pressure. Vierhout and Vendrick (1961) also described an electrically driven pump with circuits designed to maintain a constant displacement in spite of variations in load, and this is reported to give good results. Hansen and Warburg (1950) described five different devices, and Hansen (1949) designed and used a magnetic pump. The oscillator-speaker system used by us (Nichols *et al.*, 1985b) produces sinusoidal signals flat from 1.0 to 200 Hz, as well as square waves (see above). It is similar to those described by Stegall (1967) and Henry *et al.* (1967) and used by McCutcheon *et al.* (1972) and Jackson and Vinegar (1979). It consists of a multifunction generator, a constant-current amplifier and a loudspeaker with an attached test chamber composed of a lucite cylinder. The cylinder is designed to test several catheter-manometer systems simultaneously; a catheter-tip pressure transducer is inserted through one of the ports and used as a reference signal.

Some of the problems relating to a completely liquid-filled system can be avoided by using an air chamber over the liquid. Such a system may be driven mechanically or electrically (e.g. Fry *et al.*, 1957a). In this case, however, the pressure generated by a pump of constant stroke varies with frequency in the lower frequency range owing to an adiabatic compression effect. Thus, the pressure oscillation increases with frequency until it reaches a steady value for an adiabatic change. This means that the pressure generated by the pump must be monitored by a second manometer, known to be accurate over the range being tested, attached to the pump chamber. Such a system was used by Fry *et al.* (1957a) in their thorough, and

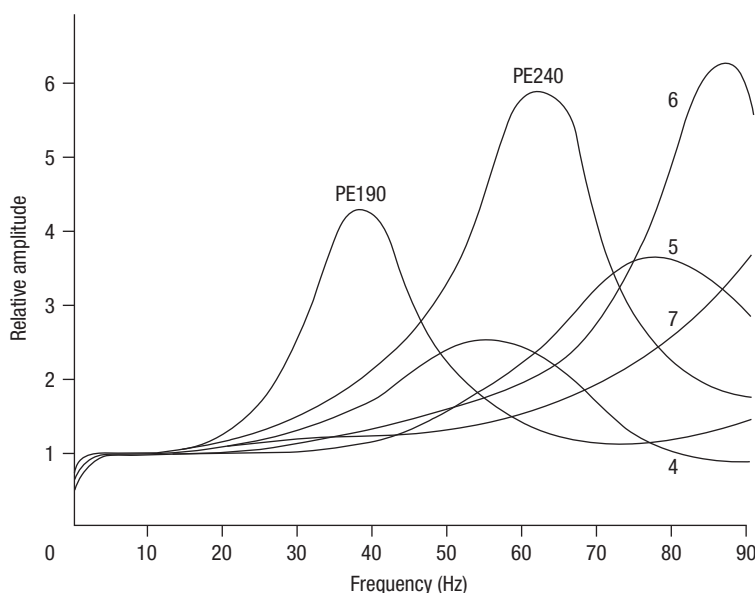


Figure 6.6 Amplitude/frequency response of several liquid-filled catheters of the same length (100 cm) attached to a Statham P23 Db manometer. The catheter-manometer system was driven by a hydraulic liquid-filled sinusoidal pump. Single numbers on the curves refer to French sizes of USCI cardiac catheters; the two PE numbers refer to polyethylene tubing (PE190, internal diameter 6.0 mm; PE240, 8.4 mm), which were larger in bore than any of the cardiac catheters. The low resonant frequency in these cases is due to the greater distensibility of the polyethylene tubing compared with the smaller, stiffer cardiac catheters. Reproduced from Yanof *et al.* (1963)

pioneering, investigation of a wide range of manometers available at that time. A monitor was also deemed necessary in this case to ensure that the pressure created by the electrically driven diaphragm was accurately known. These authors incorporated the membrane of a Lilly capacitance manometer into the wall of the compression chamber; they state that this device had a flat response to 1000 Hz.

Artifacts in pressure recording

END-PRESSURE ARTIFACT

Pressure is defined as the force exerted per unit area on the walls enclosing a fluid. A flowing liquid also has a kinetic energy component by virtue of its motion; this is discussed more fully in Chapter 2. Thus, if the tip of a catheter points upstream, it will record the sum of the lateral pressure and the converted kinetic energy. The kinetic energy pressure equivalent in these circumstances is given by

$$P_k = \frac{\rho V^2}{2} \quad 6.35$$

where V is the velocity of blood in the streamlines impacting on the catheter tip and ρ is the density of the liquid. The pressure equivalent of this energy is relatively small. If V is 100 cm/s and the density of blood is taken as 1.0, the equivalent pressure is given by

$$\begin{aligned} P_k &= \frac{(100)^2}{2} \text{ dyne/cm}^2 \\ &= \frac{10^4}{2(1333)} \text{ mmHg} = 3.76 \text{ mmHg.} \end{aligned}$$

Similarly, for a velocity of 50 cm/s, it will be one-quarter this (i.e. less than 1.0 mmHg), but for a flow velocity of 200 cm/s it would be 15 mmHg. Flow velocities of the order of 100 cm/s or higher are only found as peaks (see below) in the large vessels close to the heart, but are more likely to be seen if the artery is narrowed. In more peripheral vessels, the kinetic energy will be very small in terms of absolute magnitude. However, the peak velocity of flow (e.g. Fig. 7.1) commonly precedes the pressure peak, and hence an end-pressure artifact will show as a hump on the rising phase of the pressure limb. Undoubtedly, a great deal of the discussion on the 'anacrotic wave' of the pulse wave in the older literature was, in fact, concerned with such an artifact (McDonald, 1974).

The actual magnitude of the kinetic energy artifact will be more unpredictable than the simple exposition above suggests. Unless the velocity profile is known, the velocity of the flow impacting on the catheter end will not be known. In all flow patterns – even in the very 'flat profiles' that almost certainly occur in the large vessels close to the

heart – the velocity falls rapidly close to the wall. Therefore, if the catheter tip lies against the wall, the kinetic energy there will be small. Usually, a long, moderately stiff catheter introduced from a peripheral artery will be held against the wall by its springiness on passing around any bend, such as the junction of the iliac artery and the aorta. This is not, however, at all certain. As it is difficult to allow for an artifact of unpredictable amplitude, it is best avoided. This is achieved by closing the tip and having lateral holes.

Even so, it is possible, when recording in the ascending aorta, that a catheter passing around the arch of the aorta may lie obliquely across the lumen so that a 'lateral' hole may be subject to some flow impact. Very precise measurements in this area, as for the determination of pressure gradients, are thus subject to some uncertainty when using liquid-filled catheter–manometer systems to measure pressure.

To summarize, the end-pressure artifact is usually small, but should be guarded against as a source of distortion of the shape of a pressure curve; for accurate quantitative analysis, it clearly needs to be eliminated.

CATHETER IMPACT ARTIFACT

Whenever the liquid-filled catheter is hit, a transient pressure is created by the acceleration; any frequency component of this transient that coincides with the resonant frequency of the manometer system will create a pressure oscillation superimposed on the recorded pressure waveform. When intracardiac catheters are used, every heart beat creates such a transient. The frequencies that cause this effect rarely exceed 50 Hz; the higher the resonant frequency of the system above 50 Hz, the less intrusive this distortion is. In addition, very rapid changes of pressure, such as in the rising phase of intraventricular pressure, or the 'incisura' caused by closure of the aortic valves, can also cause free vibration of this kind. Inspection of a record by looking for such transient vibrations will give a very good estimate of the resonant frequency of the system in use.

It is not always appreciated how much the superimposition of such transients may distort pressure waveforms. An example is shown in Fig. 6.3 above, in which the pressure in a human left ventricle is recorded simultaneously with a catheter-tip (Millar) manometer and with a low-frequency response liquid-filled catheter–manometer system. From the small oscillations seen throughout the cycle, it can be deduced that the resonant frequency of this liquid-filled system was about 37 Hz. Such a resonant frequency is far better than usually achieved in most clinical cardiac catheterization laboratories, yet clearly has caused amplitude distortion and delay of the recorded pressure wave.

In spite of the general belief that high rates of pressure change are only recorded by high-frequency manometers,

it can be seen that the apparent rate of rise is higher in the liquid-filled recording. Coupled with the spike at the top of the rise, it is clear that superimposition of a transient oscillation has given a spuriously high value to the rate of pressure rise that is recorded correctly by the catheter-tip transducer. In fact, maximum LV (dp/dt) can be as much as 100 percent too high in low-frequency response liquid-filled systems (McDonald, 1974; Nichols *et al.*, 1985b).

THE USE OF INCREASED DAMPING

In so far as the above artifacts are due to resonance, an attractive way of eliminating them appears to be by increasing the damping. In the initial theoretic section above, it was pointed out that optimal damping occurs with a damping coefficient of 0.64. To this end, electric damping circuits are often incorporated in commercial amplifiers for blood pressure recording; methods of hydraulic damping are also variously used. The practical difficulties that this introduces, if precise analysis is to be made, are not, however, often appreciated. An 'optimally' damped response is only 'flat' (within 5.0 percent of its true value) to about 67 percent of the undamped resonant frequency. The phase shift is only linear with frequency (see Fig. 6.5) if the damping is very close to 0.64. The damping coefficient is very sensitive to catheter diameter; eqn 6.17 above shows that it varies inversely as the cube of the radius. Thus, if a layer of fibrin is deposited in the catheter, it may change markedly during the course of an experiment, or in a long series of clinical observations. To take an example, if a film of fibrin only 20 μm thick were deposited in a No. 4 French (4F) catheter (internal radius 0.23 mm), the damping would be increased such that an initial measured damping of 0.7 would be increased to 0.94. Reference to Fig. 6.5 above shows that the recorded amplitude would then fall continuously with increasing frequency, and the phase lag (see Fig. 6.5) would also increase markedly if the initial damping was 0.2; however, the increase would be 0.27 and would produce little amplitude change over the working frequency range (but would increase the phase lag by about 30 percent in this range). Teflon catheters of approximately this bore are commonly left *in situ* for long periods in clinical investigations. In such a catheter, a film of 50 mm thickness would more than double the damping, and if the initial situation were with 0.7 damping, the effect would be disastrous, but would be tolerable at least in terms of amplitude. With a larger catheter, the effect would naturally be smaller. In a 7F catheter (internal radius 0.56 mm), a 50 mm film would produce a result similar to that of a 20 mm film in a 4F catheter.

Thin layers of fibrin are very difficult to prevent by the use of anticoagulants, if blood is allowed to enter the catheter at any time. Use of continuous flushing with a slow flow of saline can be used; such a flow in a long, narrow catheter requires an appreciable head of pressure,

which will distort the measured mean pressure unless it is very carefully controlled.

To summarize, modern transducers used with standard cardiac catheters have a high resonant frequency with low damping when scrupulously filled with liquid. This allows accurate amplitude measurements with very small phase shifts up to rather more than 20 percent of the resonant frequency. This usually covers the physiologically significant frequency range. The resonant frequency and damping in these circumstances are easily measured.

A flat amplitude response up to 67 percent of the undamped natural frequency can be obtained by increasing the damping to 0.64; the phase lag is then linear with frequency and is thus predictable. Damping of this order is more difficult to measure yet must be known fairly accurately because the characteristic response curves change rapidly with deviations from this damping. Either condition is capable of producing accurate results if the system is well calibrated. The use of increased damping to decrease catheter-manometer artifacts is illustrated in Fig. 6.7. The danger with arbitrarily increasing the damping is that it may be used as a short-cut that avoids the tedium of obtaining the maximal response from the catheter-manometer system in the first place; a heavily damped system with an initially low resonant frequency will have very poor performance. Clinicians are often unaware of these problems in their own recording apparatus. The damping in a commercial cardiac catheterization laboratory system can be sufficient to eliminate the high frequency components which identify the systolic shoulder of an aortic or left ventricular (LV) wave recorded by a Millar micromanometer so that pressure augmentation cannot be identified (Smulyan *et al.* (2003) and personal communication). Ironically, pressure recording in the cardiac catheterization laboratory is less accurate now than it was 50 years ago, when principles were known and applied, and recordings were taken on optical or mechanical systems with frequency response high, and adequate enough to record an accurate electrocardiogram (Wood *et al.*, 1954).

Types of manometers in use

A brief survey of the principal techniques that are applied in pressure manometers in current use is given here. Additional information is given in reviews and texts by Geddes (Geddes, 1970; Geddes, 1991), Gabe (1972), Webster (1978), Milnor (1989), Nichols *et al.* (1985b), and Nichols and O'Rourke (Nichols and O'Rourke, 1998; Nichols and O'Rourke, 2005), which illustrate and discuss constructional details and the principles of operation of the different types of transducers used in the past.

The change in ohmic resistance that occurs in a wire when it is strained has been widely used in a large variety of both intravascular and extravascular manometers. The change in resistance is measured by setting the gauge as the arm of a Wheatstone bridge; straining the wire by an

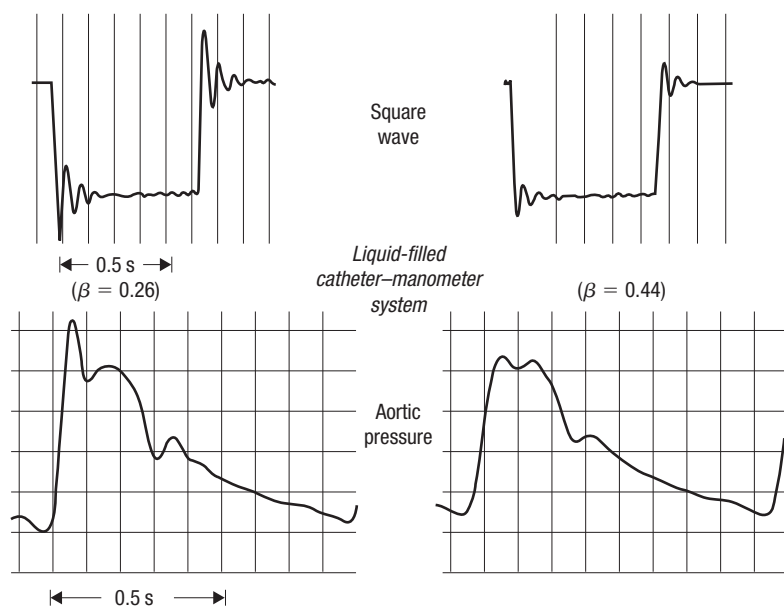


Figure 6.7 Example to illustrate the effect of increased damping on the aortic pressure waveform recorded with a liquid-filled catheter–manometer system. The response of the system to a square-wave forcing function is shown at the top. When the damping coefficient (β) is increased from 0.26 to 0.44, the amplitude of the oscillations on the square wave is decreased by more than 100 percent, and the overshoot in the pressure wave is almost totally eliminated without distorting the true shape of the wave

applied pressure unbalances the bridge, and the voltage produced is a measure of the pressure. This method has the advantages of stability and simplicity in its electric circuits and was first applied to the measurement of blood pressure by Lambert and Wood (1947). The voltage applied to the bridge has usually been obtained from an AC source to avoid problems, such as zero drift, associated with DC amplifiers. In this case, a phase-sensitive demodulation circuit needs to be included and, of course, the carrier frequency needs to be filtered out. The technical aspects of the circuitry were discussed by Geddes (Geddes, 1970; Geddes, 1991) and Webster (1978). The carrier frequency used limits the frequency range that can be recorded accurately; this is always ample for the frequencies recorded in the arterial pulse, but it often limits calibration procedures, which depend on determining the resonant frequency of the manometer system, especially with the advent of catheter-tip transducers. As filling, with its consequent problems with trapped air bubbles, is no difficulty with catheter-tip manometers and pressure wires, they do not often require frequency calibration, so that this limitation is not serious. Nevertheless, good DC amplifiers are now available, and DC excitation of the strain gauge is feasible and satisfactory. The manometers of this type that have been the most widely used in the past are all unbonded-wire strain gauges, in which the wire is wound around pegs on a sliding element and on a fixed frame. The sliding element moves when the flexible diaphragm is deformed by the applied pressure. The strain-wire resistance is balanced against a similar wire which is not strained, to compensate for temperature effects. Wire can only be subjected to tensile strains; under a compressive stress, they will buckle. Hence, the wire is pre-stretched to allow for recording below atmospheric pressure. With a volume of liquid in it of approximately 1.0 mL, the volume elasticity of a transducer is 1.3×10^{10}

dyne/cm⁵, while that of liquid at 20°C is about 2.3×10^{10} dyne/cm⁵. It is clear that very little increase in frequency response can be made by making the membrane stiffer; it was calculated by Yanof *et al.* (1963) that about 60 percent of the reported volume displacement was taken up by compression of the liquid and only 40 percent by displacement of the membrane. It was also estimated from this calculation that the membrane displacement for an applied pressure of 100 mmHg was of the order of 260×10^{-6} mm.

Semiconductor catheter-tip pressure transducers, with silicon replacing the materials usually used in strain gauge manometers, were first introduced by Angelakos (1964). He reported a natural frequency of 8.0 kHz for his manometer, and a frequency response that was flat from 0.0 to 2.0 kHz. Disk-type transducers with piezoresistive sensors of similar design performance have also been developed (Biotec, Whittaker, Micron, Konigsberg) and used for chronic implantation for long-term studies in dogs, goats, rats, and primates. An evaluation of several miniature pressure transducers that use solid-state elements was reported by McCutcheon and colleagues (McCutcheon *et al.*, 1972; McCutcheon, 1973). Of these, the Konigsberg pressure disk seems to be the most widely used for chronic implantation. The technique of implantation (Stone, 1973) and the reliability of the transducer for long-term animal studies (Stevens *et al.*, 1973; Chaer *et al.*, 2006; Regan *et al.*, 2009) have also been reported. We have used Konigsberg disks in chronic preparations to measure LV and aortic pressures in dogs and goats. These gauges are available in a variety of sizes; the ones used by us were 4.0 mm and 6.0 mm in diameter. The Konigsberg pressure disks are very reliable and can be used with confidence in chronic preparations for several months (Fuller *et al.*, 1978; Heard *et al.*, 1996). Signals from these chronically implanted transducers can be telemetered to an amplifier

several meters away, thus making it ideal for free ranging animal studies (Kurtz *et al.*, 2005; Regan *et al.* 2009). Other long-term wireless systems use miniature implantable micro-electro-mechanical (MEMS) pressure sensors for tonometric blood pressure measurements. The system is designed for use in small animals and employs a capacitive pressure sensor mounted inside a cuff and wrapped around the aorta (Cong *et al.*, 2006).

A long-term implantable hemodynamic monitor designed to record right ventricular pressure in humans is now available from Medtronic (Ohlsson *et al.*, 1998; Adamson *et al.*, 2003), CardioMems (Verdejo *et al.* 2007), and Remon Medical Technologies (Hoppe *et al.* 2009). The implantable procedure for the Medtronic system is similar to a single lead permanent pacemaker system with a programmable memory device placed in the pectoral area and a transvenous electrode carrying a pressure sensor in the right ventricular outflow tract. The device continuously measures right ventricular systolic and diastolic pressures, estimated pulmonary artery diastolic pressure and maximum rate of change of right ventricular pressure. The sensor's accuracy and precision was validated by Magalski *et al.* (2002) in the cardiac catheterization laboratory. Comparisons to simultaneous pressures obtained with a balloon-flotation catheter at 3, 6, and 12 months indicated that no significant drift occurred in pressures measured with the implanted device. Long-term performance of the implantable system has been recently established in a multicenter 12-month follow-up study in patients with chronic heart failure (Magalski *et al.*, 2002) and recently used, with success, in the COMPASS-HF clinical trial (Bourge *et al.*, 2008; Zile *et al.*, 2008) and in heart failure patients participating in a 6-minute walk test (Braunschweig *et al.*, 2009). The system has also been used in patients undergoing maintenance hemodialysis (Kjellström *et al.*, 2009). Another implantable wireless system designed for use in heart failure patients to measure pulmonary artery pressure and in patients with abdominal aortic aneurysms to measure sac pressure is made by CardioMemS. The system has been evaluated in both animals (Chaer *et al.*, 2006) and humans (Verdejo *et al.*, 2007) and used in the clinic to measure aneurysm sac pressure in patients post-endovascular aneurysm repair (Hoppe *et al.*, 2008). This system does not contain any internal energy source or battery, an external antenna activates the sensor over a radiofrequency impulse and receives the pressure-dependent change in the resonant frequency of the sensor. The Remon Medical Technologies' system comprises an implant and an external communication/analysis unit. The implant is housed in a small titanium case which contains a piezoelectric pressure transducer and a custom-built, low-power control chip (Rozenman *et al.*, 2007; Hoppe *et al.*, 2009). More details on theory, construction, and implantation of these wireless systems are presented in reviews by Potkay (2008) and Springer *et al.* (2008).

A number of excellent catheter-tip pressure manometers that operate on strain gauge and bridge circuit principles are now commercially available (Millar, ScisenseCordis, Gaeltec). However, the Millar catheter-tip micromanometer is the most widely used and is considered the 'gold standard' for high-fidelity pressure waveform recordings. Numerous other investigators compare their measured pressure waves to those measured with the Millar catheter (Rozenman *et al.*, 2007; Hoppe *et al.*, 2009). The problems associated with earlier catheter-tip pressure manometers (temperature sensitivity, fragility, etc.) have been overcome in these transducers which have been produced for more than 35 years. The pressure sensor is mounted at the tip of a variety of different cardiovascular catheters and is positioned so that it measures lateral pressure instead of impact, or end-on pressure. For a detailed description of the Millar Mikro-tip manometers, see Millar and Baker (1973), Nichols and Walker (1974), Nichols *et al.* (1978), and Zimmer and Millar (1998). These catheter-tip pressure manometers are available in a number of sizes (from 0.335 mm (1F) to 2.67 mm (8F) external diameter) and shapes, and are used extensively in both clinical and experimental laboratories around the world. The smaller manometers can be inserted through the lumen of a liquid-filled cardiac catheter (Haber *et al.*, 1993) or inserted directly into arteries of animals as small as adult rats (Liu *et al.*, 1991) and mice (Hartley *et al.*, 2008). They can also be threaded through a transeptal catheter and positioned in the left atrium or left ventricle of humans, or they can be inserted percutaneously into the femoral or brachial artery and passed into the left ventricle (Nichols *et al.*, 1978). These manometers have a resonant frequency response well into the kHz range, so that cardiovascular sounds can also be obtained along with the pressures (Nichols *et al.*, 1983; Murgo, 1998). Recordings obtained by Murgo with a Millar multisensor pressure catheter in two patients, one with pulmonary stenosis and the other with aortic stenosis, are shown in Fig. 6.8. Intravascular sounds obtained from the pressure sensors are also shown. A flotation catheter, similar to the Swan-Ganz catheter and produced by Millar Instruments, but with a high-fidelity pressure sensor mounted at the tip, was evaluated in the human cardiac catheterization laboratory, and is presented in Chapter 16. The most recent addition to the Millar family of multisensors catheters is the pressure-volume conduction catheter that can be used in small animals and humans to measure instantaneous ventricular pressure and volume simultaneously (Segers *et al.*, 2005a; Wu *et al.*, 2007). Left ventricular pressure-volume (P-V) loops recorded in two patients during cardiac catheterization using the conductance catheter are shown in Fig. 6.9; one had normal LV function while the other had severe LV dysfunction causing a right shift of the P-V loop (Wu *et al.*, 2007).

The most recent device developed to measure high-fidelity pulsatile pressure in small arteries is the RADI

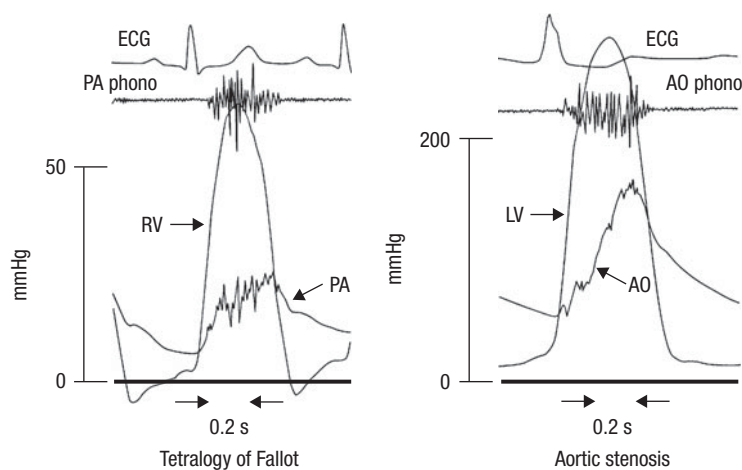


Figure 6.8 High-fidelity right and left heart pressures recorded with Millar multisensor catheters in a patient with pulmonary stenosis and another with aortic stenosis. The dynamic frequency response of the pressure sensors is high enough that cardiovascular sounds can be obtained along with the pressures. Courtesy of JP Murgo

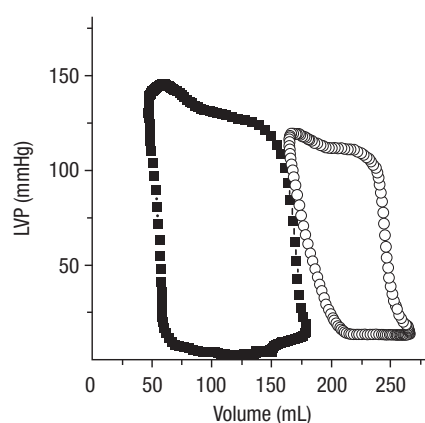


Figure 6.9 Cardiac catheterization recordings of simultaneous pressure (micromanometric) and volume (conductance sensors) for one cardiac cycle in a patient with normal left ventricular (LV) function (control (closed square)) and a patient with systolic heart failure (HF) (open circle). Reproduced from Wu *et al.* (2007)

pressure guide wire. Pressures are recorded by a fiberoptic pressure sensor located laterally and 3.0 cm from the distal tip. The basic principle is that the element modulates an optical reflection by pressure-induced elastic movement. The light source is an emitting diode in the control unit (Gorge *et al.*, 1993). The early pressure guide wires were 0.018 inch in diameter, but the most recent ones are smaller (0.014 inch diameter) (Dias *et al.*, 2004; Jensen *et al.*, 2007; Vavuranakis *et al.*, 2007). For an *in vitro* evaluation of the Millar, Scisense, and RADI pressure sensors, see Hartley *et al.* (2008). Gorge *et al.* (1994) found the RADI wires to be very reliable in measuring both absolute pressure and gradients across stenoses in an isolated pig model. Also, the pressure wire has been used successfully to measure stenotic pressure gradients (Pijls *et al.*, 2000; Marques *et al.*, 2002; Jensen *et al.*, 2007) and fractional flow reserve (Lopez-Palop *et al.*, 2002; Akasaka *et al.*, 2003; Legalery *et al.*, 2003; Trivedi *et al.*, 2005; Meuwissen *et al.*, 2008) in coronary arteries, and stenotic pressure gradients in renal arteries (Gross *et al.*, 2001) and

direct intra-aneurysm sac pressure in humans (Dias *et al.*, 2004). The practice and pitfalls of using the pressure wire to measure coronary artery pressure are discussed by Pijls *et al.* (2000). A disadvantage of the catheter tip over the fluid-filled catheter–manometer system is in measurement of mean pressure. With the fluid-filled system, zero pressure is at the external manometer, and this is usually elevated to the level of the mid right atrium. With the catheter tip manometer, pressure is measured at the tip, so that the mean pressure recording depends on its level in the chest – higher when the tip is lower, and lower when the tip is higher than the mid right atrium. Again, clinicians using these expensive devices are often unaware of the vertical level of the tip in the chest, and attribute changes in mean pressure to drift or instrument error.

Indirect (non-invasive) measurement of arterial blood pressure

Hypertension or high blood pressure, which is a major cardiovascular risk factor (see Chapter 20), is estimated to affect more than 65 million American adults and is a major health problem in the United States and many other countries of the world (Chobanian *et al.*, 2003; Pickering *et al.*, 2005a; Pickering *et al.*, 2005b); a large percentage of individuals with hypertension go undiagnosed. Therefore, the importance of measuring blood pressure non-invasively at home and in the clinic cannot be over-emphasized. A summary of the various techniques used to measure blood pressure indirectly (non-invasive) are discussed in this section.

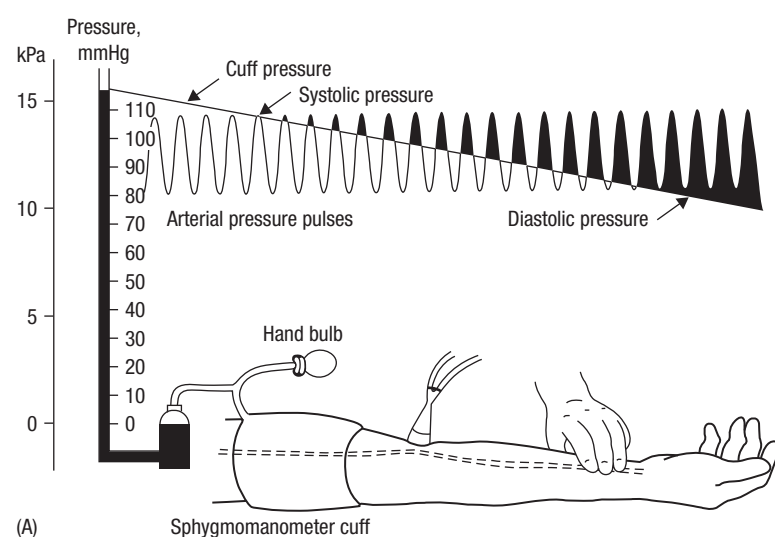
In 1896, Riva-Rocci introduced the blood pressure cuff into clinical sphygmomanometry. Arterial blood pressure is determined by either palpation (Riva-Rocci, 1896) or audible detection (Korotkoff, 1905) of the arterial pulse distal to a sphygmomanometer cuff. For a detailed history, description of the technique and use to measure blood pressure, see Geddes (Geddes, 1970; Geddes, 1991), Perloff *et al.* (1993), Beevers *et al.* (Beevers *et al.*, 2001a; Beevers *et al.*, 2001b), Jones *et al.* (2003), and Pickering *et al.* (2005a; 2005b).

In 1905, Korotkoff described the sound heard over an artery at a point immediately below a compression cuff. He divided the sounds into three phases. Later, they were divided into five phases. These sounds are known as Korotkoff (or just 'K') sounds. The point at which the sounds first appear on deflation of a sphygmomanometer cuff has been accepted universally as the systolic blood pressure (Fig. 6.10A). However, the changes in the sounds that indicate the diastolic pressure have been a subject of controversy (Rushmer, 1970). Some consider the diastolic pressure to be equal to the cuff pressure when the sound becomes muffled (i.e. at the beginning of phase 4). This has been found to be somewhat higher than the arterial pressure measured directly and simultaneously (Roberts *et al.*, 1953). The diastolic pressure recorded at the cessation of sound (i.e. at the beginning of phase 5) is somewhat lower than intra-arterial pressure and is now recommended by the American Heart Association.

In order to use the auscultatory technique with confidence, the hearing acuity of the user must be good for

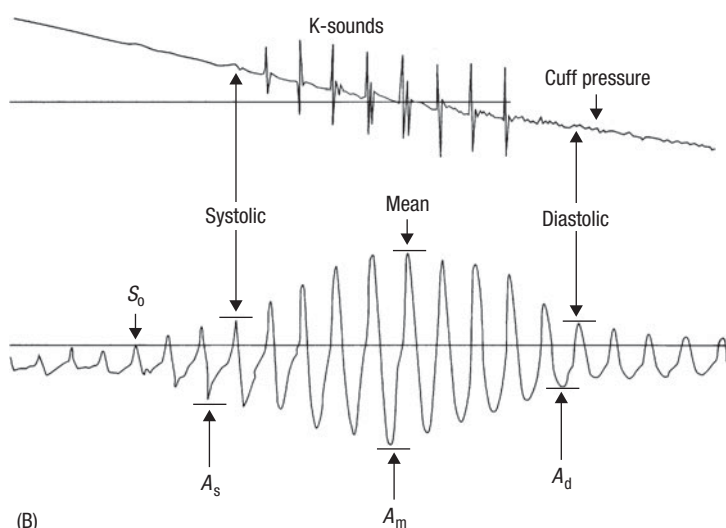
frequencies from 20 to 300 Hz, the bandwidth required for these measurements. When the arterial pressure is low (e.g. in patients with hypovolemic or cardiogenic shock), the sound spectrum shifts to lower frequencies (Geddes, 1991) and the auscultation technique becomes unreliable, possibly because of the low sensitivity of the human ear to these frequencies (Geddes, 1970; Geddes, 1991).

Several techniques have been proposed to automatically determine absolute arterial blood pressure non-invasively. Four of these are in use today. The first is an automated auscultatory device that uses a microphone or ultrasound instead of a stethoscope to detect the Korotkoff sounds and determine systolic and diastolic blood pressure (White *et al.*, 1990). The second method uses the oscillometric method (Geddes, 1991; Ohkubo *et al.*, 1998; Segers *et al.*, 2000; El Feghali *et al.*, 2007) to determine arterial blood pressure (systolic, diastolic, and mean), usually in the upper (Yarows and Amerena, 1999; Yarows and Brook, 2000; El Assaad *et al.*, 2002) or lower arm or the finger (Watson *et al.*, 1998; Yarows and Brook,



(A)

Figure 6.10 (A) Indirect blood pressure measurement system. The sphygmomanometer cuff is inflated with the hand bulb until the cuff pressure exceeds intra-arterial systolic blood pressure. The cuff pressure is then slowly released while monitoring the sound of blood flow in a downstream artery (usually the brachial) with a stethoscope. The first Korotkoff sound detected indicates systolic blood pressure, while the transition from muffling to silence (or total cessation of sound) indicates diastolic blood pressure. Reproduced from Rushmer (1970). (B) Cuff pressure with superimposed Korotkoff (K) sounds and amplified cuff-pressure oscillations. S_0 is the point at which cuff-pressure oscillations start to increase. A_s is the amplitude corresponding to auscultatory systolic blood pressure, and A_d is the amplitude corresponding to auscultatory diastolic blood pressure. A_m is the maximum oscillation amplitude that corresponds to mean blood pressure. Reproduced from Geddes (1991)



(B)

2000; Jagomagi *et al.*, 2001). The third measures the arterial blood pressure waveform continuously in the finger using the arterial volume clamp method of Peñáz (1973) or photoplethysmographic method (Philippe *et al.*, 1998) or acceleration plethysmographic method (Takazawa *et al.*, 1993a). The fourth measures the pressure waveform using a high-fidelity external tonometer (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; O'Rourke *et al.*, 1992a). A detailed description of this method is given in Chapter 30, since it relates to measurement of the arterial pressure waveform non-invasively. When measuring blood pressure it should be remembered that systolic pressure in a peripheral artery is higher than in a central artery (Chapter 9). Therefore, when evaluating non-invasive methods of measurement, the pressures (non-invasive and invasive) should be measured at the same site. In the evaluation of the photoplethysmographic method, Philippe *et al.* (1998) compared finger systolic pressure with aortic systolic pressure. They noted a 30 mmHg higher pressure in the finger compared to the aorta.

The oscillometric method of determining arterial blood pressure uses a principle first reported by Marey (1875) and later used by Erlanger (1921). Pulsatile blood flow through an artery produces oscillations of the vessel wall, which are transmitted to a blood pressure cuff encircling an extremity containing the artery. Initially, the cuff is inflated to a pressure above systolic pressure (Fig. 6.10B). At this pressure, only small-amplitude oscillations are transmitted to the cuff. As the air pressure in the cuff is decreased, the amplitude of pressure oscillations increases and reaches a maximum as the cuff pressure passes through the pressure equivalent to mean arterial pressure (Mauck *et al.*, 1980). Once the cuff pressure is reduced below diastolic pressure, the arterial wall movement decreases and the amplitude of the pressure oscillations in the cuff is greatly reduced. The cuff pressure at the time of the initial increase in arterial oscillations corresponds to maximum systolic blood pressure, and the lowest cuff pressure occurring just prior to the time that oscillations stop decreasing in amplitude corresponds to the minimum diastolic pressure (Borow and Newburger, 1982).

The oscillometric method can be used for office measurement, but only devices independently validated according to standard protocols should be used, and individual calibration is recommended (Turner *et al.*, 2006; Turner *et al.*, 2008). Lack of calibration can cause over and underdetection of hypertension. These automated pressure monitors have the advantage of being able to take multiple measurements. Proper training of observers, positioning of the patient, and selection of cuff size are all essential. It is increasingly recognized that office measurements correlate poorly with blood pressure measured in other settings, and that they can be supplemented by self-measured readings taken with validated devices at home. There is increasing evidence that home readings predict cardiovascular events and are particularly

useful for monitoring the effects of treatment. Twenty-four-hour ambulatory monitoring gives a better prediction of risk than office measurements and is useful for diagnosing white-coat hypertension. Automated blood pressure devices are commonly found in public places and represent a potential mechanism for increased screening for hypertension. In 1995, Whitcomb *et al.* reported that since the introduction of the VitaStat device in 1976, about 8000 were in use in the United States, providing more than ten million measurements per year. These devices, which were never tested by approved protocols, were found to give very inconsistent results, particularly for systolic blood pressure (Pickering *et al.*, 2005a; Pickering *et al.*, 2005b). A later model was tested in a community setting and failed to meet the Advancement of Medical Instrumentation or the British Hypertension Society criteria for accuracy. Other potential problems with these devices are that the cuff size (23 × 33 cm) is inadequate for patients with large arms, and that they are not labeled to show when or if there has been recent maintenance and revalidation of the device's performance. These and other recommendations for blood pressure measurements were recently reviewed by Pickering *et al.* (2005a; Pickering *et al.*, 2005b), and are available on the regularly updated website (www.dableducational.org).

Accuracy studies have shown that, if used properly, the Dinamap, Artsana, Seinox, Spacelabs, Microlife, and Omron systems, which use the oscillometric method, usually measure blood pressure within 5.0 mmHg or less of that measured with a centrally placed arterial catheter (Ramsey, 1991; Watson *et al.*, 1998) or mercury sphygmomanometer (Topouchian *et al.*, 2005; Stergiou *et al.*, 2006; Pini *et al.*, 2008). However, oscillometric blood pressure devices tend to overestimate systolic and underestimate diastolic blood pressure and are inaccurate in patients with atrial fibrillation (Anastas *et al.*, 2008) when compared to the auscultatory method (Goonasekera and Dillon, 1995; van Ittersum *et al.*, 1998). The overestimation in systolic blood pressure is more pronounced in older individuals (Brinton *et al.*, 1998) and patients with diabetes (van Ittersum *et al.*, 1998) and may be due to increased arterial stiffness (van Popele *et al.*, 2000a). The oscillometric signal derived using a cuff sphygmomanometer has also been used to estimate arterial compliance (Brinton *et al.*, 1996; Brinton *et al.*, 1997). The applicability of home blood pressure measurements in clinical practice has been reviewed recently by Verberk *et al.* (2007).

Multiple studies have shown that the error in blood pressure measurement is larger when the cuff is too small relative to the patient's arm circumference than when it is too large (Pickering *et al.*, 2005a; Pickering *et al.*, 2005b; Osthega *et al.*, 2006). Recognition of the increasing need for the 'large adult' cuff, or even the thigh cuff, for accurate blood pressure measurement is critical, because frequently in practice only the standard adult size cuff is available. The 'ideal' cuff should have a bladder length that

is 80 percent and a width that is at least 40 percent of arm circumference (a length-to-width ratio of 2:1). A recent study comparing intra-arterial and auscultatory blood pressure concluded that the error is minimized with a cuff width of 46 percent of the arm circumference (Marks and Groch, 2000). The recommended cuff sizes are:

- For arm circumference of 22–26 cm, the cuff should be small adult size: 12 × 22 cm
- For arm circumference of 27–34 cm, the cuff should be adult size: 16 × 30 cm
- For arm circumference of 35–44 cm, the cuff should be large adult size: 16 × 36 cm
- For arm circumference of 45–52 cm, the cuff should be adult thigh size: 16 × 42 cm.

It is well recognized that the predictive power of multiple blood pressure determinations is much greater than a single office reading (Wingfield *et al.*, 2002). When a series of readings is taken, the first is typically the highest. A minimum of two readings should be taken at intervals of at least 1 minute, and the average of those readings should be used to represent the patient's blood pressure. If there is > 5 mmHg difference between the first and second readings, additional (one or two) readings should be obtained, and then the average of these multiple readings is used.

Mean arterial blood pressure is the product of cardiac output (CO) and total peripheral resistance and is considered as the perfusion pressure through each tissue bed. The overall function of the cardiovascular system is to provide all organs with constant nutrient flow which is closely related to mean arterial pressure and is the time-averaged aortic pressure throughout cardiac cycle length. Mean arterial pressure is traditionally calculated by adding one-third (33.3 percent) of the pulse pressure to the diastolic pressure, however, recent studies have questioned the general validity of this formula (Chemla *et al.*, 2005) and recommend using 40 percent of the pulse pressure added to the diastolic pressure (Bos *et al.*, 2007a; Bos *et al.*, 2007b). O'Rourke and Adji (2010) have reviewed this issue as it relates to 'Form Factor' and calibration of non-invasive pressure waves, and note that use of any set Form Factor as described by Chemla *et al.* (2002) is not appropriate, since it depends on the shape of the pressure wave within the artery (Fig. 6.11). Several devices are now commercially available that measure beat-to-beat blood pressure non-invasively. The Mediwatch and Colin CBM-7000 use a tonometer to measure pressure (Ng *et al.*, 2004; Normand *et al.*, 2007), while the Vasotrac (Findlay *et al.*, 2006) uses a modified tonometric technique and the Finapres, which is the most widely used, uses a volume-clamp method to measure blood pressure (see below). Some investigators have used the pulse transit time measured from the R-wave of the electrocardiogram to a peripheral pulse to obtain beat-to-beat blood pressure, however, this method has proved to be unreliable (Payne *et al.*, 2006).

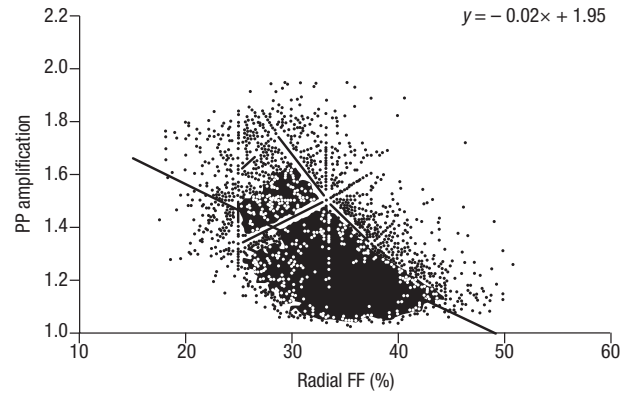


Figure 6.11 Relationship between radial artery 'Form Factor' and pulse pressure amplification from ascending aorta to radial artery in a clinical cohort of 1505 patients and volunteers. Reproduced from O'Rourke and Adji (2010)

In 1973, Jan Peñáz, a Czech physiologist, described a new approach to continuous non-invasive recording of pulsatile blood pressure in the human finger using a volume-clamp method. This device was subsequently further developed and used by Peñáz *et al.* (1976) and other investigators (Wesseling *et al.*, 1982; Takazawa *et al.*, 1993a; Imholz *et al.*, 1998; Takazawa *et al.*, 1998; Millasseau *et al.*, 2000). The Dutch physicist Karel H Wesseling improved the technical aspects of the device (finger plethysmograph of reduced dimension, feedback system for finger volume control, and automatic calibration) in an instrument called Finapres (from FINGER Arterial PRESSure) and more recently the Portapres (for ambulatory monitoring). For an historical review of the method, which has been used in the United States Space Program, see Wesseling (1995) and Imholz *et al.* (1998).

PRINCIPLE OF OPERATION AND ACCURACY OF FINAPRES, PORTAPRES, AND FINOMETER

In the volume-clamp method, the size of a finger artery is gauged with an infrared transmission plethysmograph mounted inside an inflatable cuff. When the size of the artery increases due to a blood volume (and pressure) increase, the air pressure in the cuff also increases, just enough to keep the arterial size constant. Arterial size and transmural pressure are kept constant by modulating cuff pressure in parallel with blood pressure in the finger using a pneumatic servosystem (Wesseling *et al.*, 1995).

Beat-to-beat intra-arterial and non-invasive Finapres (and the improved Finapres, known as the Finometer) cuff pressures' waveforms have been compared in a number of studies in anesthetized and awake humans (Schutte *et al.*, 2004; Yiallourou *et al.*, 2006). Arterial blood pressure variations were induced by surgical maneuvers (Van Egmond *et al.*, 1985), vasoactive drug administration (Imholz *et al.*, 1992; Millasseau *et al.*, 2000), and cardiovascular maneuvers, such as Valsalva straining (Imholz *et al.*, 1988), orthostasis (Imholz *et al.*, 1990a;

Imholz *et al.*, 1990b), and exercise (Eckert and Horstkotte, 2002). Comparisons were made in adults (Parati *et al.*, 1989), in the elderly (Rongen *et al.*, 1995), and in ambulant subjects (Imholz *et al.*, 1993). For the latter, a lightweight, battery-fed Finapres was developed (the Portapres; Langewouters *et al.*, 1990). Recent studies have used finger blood pressure, measured with the Finapres, and radial artery diameter (echotracking) to obtain vessel wall distensibility (Kool *et al.*, 1993; Kool *et al.*, 1994; Giannattasio *et al.*, 1995) in adult humans.

Pressure waveforms recorded with the Finapres compare favorably with those recorded in the radial artery with an intra-arterial catheter–manometer system during baseline and during vasodilatation with nitroprusside (Bos *et al.*, 1995). It has been demonstrated that Finapres (Portapres) can track intrabrachial pressure up to 24 hours without noticeable drift or change in accuracy (Imholz *et al.*, 1993). Also, blood pressure and pulse can be determined reliably and accurately using the Portapres, both at rest and during low-to-moderate levels of bicycle exercise (Eckert and Horstkotte, 2002). Compared to intrabrachial artery pressure over a 24-hour period, Portapres overestimates systolic pressure but not diastolic and mean pressure (Omboni *et al.*, 1998). This overestimation may be due, in part, to the amplification of the pressure pulse as it travels from the brachial artery to the finger. In addition to ambulatory blood pressure monitoring, the Portapres and Finometer have been used to study baroreflex and chemoreflex function (Timmers *et al.*, 2003; Yiallourou *et al.*, 2006; Smith *et al.*, 2008), cerebral critical closing pressure estimation (Panerai *et al.*, 2006), and deep breathing tests (Jagomagi *et al.*, 2003). The most recent development using the Finapres methodology is the Nexfin monitor. Validation against auscultatory blood pressure indicates that Nexfin provides accurate measurements of blood pressure (Eeftink-Schattenkerk *et al.*, 2009). For both types of finger monitors (oscillometric and volume clamp) it is recommended that intensive vasoconstriction in the subject be avoided during measurements (Jagomagi *et al.*, 2001).

With Huntly Millar and Dean Winter, O'Rourke and Kelly *et al.* have developed, evaluated, and used an external tonometer for indirect measurement of the arterial blood pressure waveform. This incorporates a Millar high-fidelity strain gauge micromanometer mounted in the flattened end of a pencil-type probe that can be held by hand or with a micromanipulator and applied over an artery (applanation tonometry). Other designs of the pressure sensor are under development. This instrument permits accurate registration of arterial pressure waves (Avolio *et al.*, 1988; O'Rourke *et al.*, 1992a); with it, Kelly *et al.* (Kelly *et al.*, 1988a; Kelly *et al.*, 1988b; Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; Kelly *et al.*, 1989c; Kelly *et al.*, 1989d) and O'Rourke *et al.* (O'Rourke *et al.*, 1989a; O'Rourke *et al.*, 1989b) have established pressure wave patterns at different ages in the radial, carotid and/or femoral arteries of over 1000 normal subjects (see Chapter 30). The tonometer can be used to record pressure waveforms in any superficial artery in

which a pulse can be palpated (e.g. superficial temporal, carotid, subclavin, brachial, radial, superficial branch of the radial, femoral, popliteal, dorsalis pedis), but for accurate recording of the arterial pressure wave within the artery beneath, and so, accurate recording of Form Factor and mean pressure, the principle of applanation must be achieved, and this has only been proven for the radial and carotid arteries (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; Chen *et al.*, 1996; O'Rourke and Adji, 2010).

In early studies, London *et al.* (London *et al.*, 1992; London *et al.*, 1995), Saba *et al.* (1993), and Ting *et al.* (1995a; 1995b) have used a tonometer to study wave reflections and LV hypertrophy in hypertensive patients and Schofield *et al.* (Schofield *et al.*, 2002a; Schofield *et al.*, 2002b; Schofield *et al.*, 2003) have used it to record pressure waves in heart transplant recipients (see Chapter 9 for a more detailed discussion of wave reflections). Vaitkevicius *et al.* (1993) used the instrument to study the effects of age and aerobic capacity on carotid artery stiffness, and Kelly and Fitchett (1992) used it (along with ascending aortic blood flow velocity) to obtain aortic input impedance non-invasively. More sophisticated devices using an array of tonometers have recently been described and used by Kemmotsu *et al.* (Kemmotsu *et al.*, 1989a; Kemmotsu *et al.*, 1989b; Kemmotsu *et al.*, 1989c), Sato *et al.* (1993), and Takazawa *et al.* (Takazawa *et al.*, 1995a; Takazawa *et al.*, 1995b) and sold by the Nippon-Colin and Omron health statistics companies (Ng *et al.*, 2004). Also, a tonometric arterial pulse sensor with noise cancellation has been developed recently (Ciaccio and Drzewiecki, 2008).

Tonometers are used primarily for pressure waveform analysis (Adji *et al.* 2007; Chemla *et al.*, 2008) and measuring arterial pulse wave velocity (Salvi *et al.*, 2004; Khoshdel *et al.*, 2006; Levy *et al.*, 2007; Alecu *et al.*, 2008; Rajzer *et al.*, 2008; Salvi *et al.*, 2008; Hickson *et al.*, 2009a; Hickson *et al.*, 2009b; Jatoi *et al.*, 2009; Benetos *et al.*, 2010), and calculating arterial impedance (Segers *et al.*, 2007a; Desai *et al.*, 2009). These topics and others are reviewed by O'Rourke and Hashimoto (2007), Nichols *et al.* (2008), Safar (2008), Ghiadoni *et al.* (2009), Lacolley *et al.* (2009), and Nichols and Epstein (2009) and discussed in individual chapters throughout the text.

Other non-invasive methods for obtaining central (Sharir *et al.*, 1994) and peripheral (Blank *et al.*, 1988; Blank *et al.*, 1991; Blank *et al.*, 1995; Belani *et al.*, 1999a; Belani *et al.*, 1999b; Ng *et al.*, 2004) arterial blood pressure have also been tested and used (Hickson *et al.*, 2009a; Hickson *et al.*, 2009b; Nemcsik *et al.* 2009).

MEASUREMENT OF BLOOD FLOW, STROKE VOLUME, AND CARDIAC OUTPUT

In addition to describing the circulation of blood, William Harvey (1628) also made post-mortem measurements

of LV diastolic volume; this volume was approximately 92 mL. He reasoned that about one-fourth of this amount (i.e. 23 mL) was ejected with each contraction, and by recording heart rate he estimated CO to be 3.0 L/min. Harvey's book *Exercitatio Anatomica de Motu Cordis et Sanguinis in Animalibus* initiated modern cardiovascular physiology, and it was relatively soon after this that the Reverend Stephen Hales (1733) set out to investigate the dynamics of the circulation. Hales estimated the circulation rate, measuring the velocity of the blood as it passed along the veins, arteries, and capillaries. Hales also measured CO, using techniques similar to Harvey's, and introduced the concept of peripheral resistance in arterioles (see Chapter 1).

FICK PRINCIPLE

Methods of measuring CO in live animals did not become available until the late seventeenth century. Adolph Fick published a brief paper in 1870 describing a principle that could be used to measure CO. The principle states that the flow in a given period of time is equal to the amount of substance entering the stream of flow in the same period of time divided by the difference between the concentrations of the substance before and after the point of entry. Thus the 'Fick principle' was introduced, but Fick himself never used it to measure CO. Instead, it was first used by Grehart and Quinquardt in 1886 to measure CO in six dogs. One of the first (and still the most popular) substances used to calculate the CO using the Fick principle is oxygen (Guyton, 1963). The formula for calculating CO using oxygen (O_2) is

$$CO(L/min) = \frac{O_2 \text{ consumption (mL/min)}}{\text{Arterial } O_2(\text{mL/L}) - \text{venous } O_2(\text{mL/L})}$$

O_2 consumption is measured as the oxygen content difference between the inspired and expired gas.

In order to use this formula with confidence, the samples of blood taken on both sides of the heart must be well mixed. The arterial sample may be taken from any systemic artery since the blood is well mixed in the left ventricle. However, in order to get a well-mixed sample of venous blood, the sample must be taken from the right ventricle or the pulmonary arterial system. The usual method for doing this is through a catheter. Forssmann (1929), a 25-year-old German surgical resident, first proved that this could be done by passing a ureteral catheter into his own left basilic vein and threading it into his right atrium. Cournand and his associates (Cournand and Ranges, 1941; Cournand *et al.*, 1945) established the safety of the procedure and stimulated widespread utilization of the method. It should be emphasized that the Fick principle is only valid in a steady state when both CO and O_2 consumption are constant.

PARTIAL CO_2 REBREATHING METHOD

Carbon dioxide (CO_2) can be used instead of O_2 in a 'partial rebreathing' technique using the Fick principle. The first such monitor to employ this principle is called NICO and was introduced in 1999. The monitor estimates CO using respiratory gas analysis and pulse oximetry. The partial CO_2 rebreathing technique uses the differential form of the Fick principle, applied to CO_2 produced by the body and eliminated through gas exchange in the lungs, for CO measurement. With partial rebreathing, a change in CO_2 consumption and an associated change in end-tidal CO_2 in response to a change in ventilation, are used in the Fick equation (Cholley and Payen, 2005; Ng *et al.*, 2007; Mathews and Singh, 2008; Kotake *et al.*, 2009). Several studies have compared CO measurements using the NICO system with those obtained using other techniques, including thermodilution. Initial validation studies reported a moderate agreement between values measured with the NICO system and thermodilution in patients during thoracic surgery (Ng *et al.*, 2007). A more recent comparison used an updated version of the NICO monitor software and reported better results (Kotake *et al.*, 2009).

STEWART-HAMILTON DILUTION METHOD

In the simplest of the indicator-dilution methods for determining CO, a known amount of indicator substance is injected rapidly into the venous blood, and then its concentration in the blood is measured continuously as it passes into the arterial system. The greater the CO, the more rapidly the 'dilution' curve of the substance appears and disappears in the arterial blood. This is the origin of the term 'indicator-dilution' method. Any type of substance that can be injected into the circulation without altering circulation dynamics, that does not disappear from the blood as it passes to the point of sampling, and that can be analyzed in the blood, can be used as the indicator. The indicator commonly used today is indocyanine green and the formula for calculating CO is

$$CO(L/min) = \frac{\text{Amount of indicator injected (mg)} \times 60}{\text{Average arterial concentration (mg/L)} \times \text{duration of curve (s)}}$$

The first recorded use of the single-injection indicator-dilution method for measuring CO was by Henriques (1913). He derived the formula that is used today in the method, and also recognized and discussed the importance of recirculation of the indicator as a possible error. In 1928, Hamilton and his colleagues (Hamilton, 1953; Hamilton, 1962) began a long series of studies to prove the validity of the method for accurate measurement of CO.

LITHIUM INDICATOR DILUTION (LIDCO) METHOD

Lithium chloride has also been used as an indicator in a modified form of the Fick equation (LiDCO). In this technique, a bolus of isotonic solution lithium chloride (LiCl) is injected into a central or peripheral vein and the resulting arterial lithium concentration is measured in a peripheral artery with a lithium sensor. The CO is calculated from the lithium dose and the area under the concentration curve using the formula:

$$CO = \frac{\text{LiCl dose (mmol)} \times 60}{\text{Area} \times (1 - \text{PCV}) \text{ (mmol/sec)}}$$

where PCV is packed cell volume which is calculated as hemoglobin concentration (g/dL)/34. This correction is needed because lithium is distributed in the plasma and not into the blood cells on the first pass to the arterial circulation (Mathews and Singh, 2008).

THERMODILUTION METHOD

The method that is currently used mostly in the clinical setting for use in measuring CO is that of injecting a bolus (usually 10 mL) of ice-cold or room temperature (Lyons and Dalbow, 1986) saline as an indicator. A special four-lumen catheter, usually a Swan-Ganz (Forrester *et al.*, 1972; Nichols *et al.*, 1978; Kamath *et al.*, 2009) is floated blindly through a peripheral vein (jugular, brachial, or femoral), while monitoring pressure through one lumen, and positioned in the pulmonary artery. The saline is injected through one lumen of the catheter into the right atrium, the drop in temperature of the blood is detected by a thermistor located near the tip of the catheter, and a dilution curve is recorded.

TRANSPULMONARY THERMODILUTION METHOD

A relatively new technique is arterial thermodilution or transpulmonary thermodilution. In this technique, a bolus of cold indicator is injected into a central vein and the temperature change across the cardiopulmonary system is measured from a peripheral artery (usually the femoral or radial), from which CO is calculated with the use of the indicator dilution equation above (Compton *et al.*, 2008a).

The methods described above measure the volume of blood ejected by the ventricle over a given period of time and are discussed in detail by Winniford and Lambert (Winniford and Lambert, 1994; Mathews and Singh, 2008). Six basic methods that estimate ventricular ejection on a beat-to-beat basis are the pressure pulse contour method, the echocardiographic method, the impedance cardiographic method, Doppler ultrasound method, ultrasound trans-time method, and the magnetic resonance imaging (MRI) method.

PRESSURE PULSE CONTOUR METHOD

The pressure pulse contour method can, in theory, use the pressure contour from any artery measured invasively (e.g. pulmonary artery, femoral, radial, or aorta) or non-invasively (aorta, carotid, radial, or finger) to measure beat-to-beat stroke volume (SV) and CO. SV is estimated from the pressure waveform using several methods based on knowledge obtained from two-, three-, and four-element Windkessel models representing the arterial circulatory system (Fig. 6.12) (Frank, 1899; Westerhof *et al.*, 2009). These methods are somewhat reliable until wave reflection alters the pressure wave shape. When this happens systolic pressure area increases, but SV decreases, and thus inaccuracies in the given method occur.

The first attempt to relate SV to the arterial pulse can be tracked back to the publication by Erlanger and Hooker in 1904. The early pulse contour methods used the area under the systolic pressure curve above end-diastole pressure (PSA) to calculate stroke volume SV (Kouchoukos *et al.*, 1970; Deloskey *et al.*, 1978; Wesseling *et al.*, 1982; Rodig *et al.*, 1999; Mukkamala *et al.*, 2006; Compton *et al.*, 2008b) and were somewhat unreliable during interventions that increased aortic characteristic impedance and wave reflections. These methods use the so-called Wesseling model ($SV = PSA/Z_0$), where Z_0 is characteristic impedance (Chapter 12) and is pressure- and age-dependent (Langewouters *et al.*, 1984). In order to obtain the absolute value of SV, the method must be calibrated against SV measured by a conventional method such as Fick or thermodilution or an estimated value of Z_0 inserted into the equation.

THE PICCO SYSTEM

The PiCCO system is a continuous CO monitor whose working principle is based on the Wesseling model and

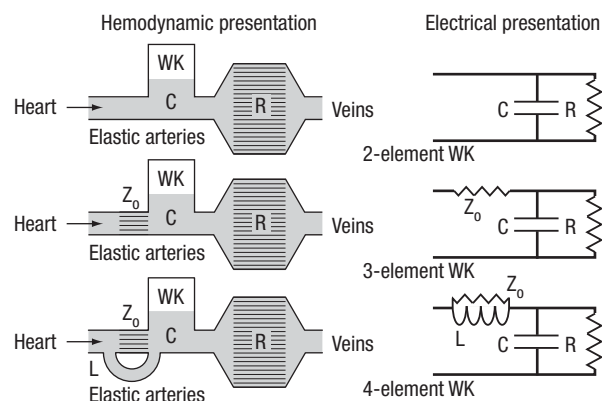


Figure 6.12 The two-element Windkessel (WK), the three-element Windkessel, and the four-element Windkessel models presented in hydraulic and electrical form. Z_0 is aortic characteristic impedance, C is total arterial compliance, L is inertance, and R is total peripheral resistance. Reproduced from Westerhof *et al.* (2009)

the software uses the original Wesseling algorithm. However, it deviates from Wesseling's modified method in that no age-related corrections for pressure-dependent non-linear changes in aortic cross-sectional area are incorporated. In the second generation PiCCO equipment, a more sophisticated algorithm that analyzes the actual shape of the waveform in addition to the PSA is used. Several clinical studies and an experimental study showed good agreement between PiCCO and thermodilution (Bein *et al.*, 2004; Mathews and Singh, 2008).

When using any pulse contour method to estimate SV, it should be kept in mind that the measured pressure wave is the sum of an incident pressure (or forward) wave (ejected from heart) and a reflected (or backward) wave (from the periphery) (see Chapter 10) (Nichols and O'Rourke, 2005). Knowledge of both waves is a necessity when calculating SV from the pressure waveform. SV is positively associated with amplitude and duration of the incident wave and negatively associated with amplitude and duration of the reflected wave (Nichols and Edwards, 2001). The algorithm used in the pulse power analysis technique supposedly takes care of this situation (Cecconi *et al.*, 2008). The algorithm used is based on the assumption that the net power change in a heart beat is SV minus blood lost to the periphery during the beat and there exists a relation between net power and net flow. Since the whole beat is taken for analysis the method is independent of the position of the sampling site. In this method, the pressure wave is transformed into a volume wave and using the mathematical technique of autocorrelation of the volume waveform the net power and beat period are derived. Net power is proportional to SV. The commercial equipment based on the pulse power analysis is the LiDCO plus (LiDCO, Cambridge, UK). Several studies have compared thermodilution and LiDCO plus and found a fairly good agreement.

THE FLOTTRAC METHOD

The recently introduced CO monitoring FloTrac/Vigileo device allows CO determination from the arterial pressure waveform of any existing peripheral arterial line without external calibration. Ease of use and minimal invasiveness render the device suitable for use inside and outside the intensive care unit. The system calculates CO, by using arterial pressure waveform characteristics in conjunction with patient demographic data. Preparation for monitoring consists only of entering the age, height, weight, and gender into the unit and zeroing the transducer. The algorithm works on the principle that the pulse pressure is proportional to SV and inversely proportional to aortic compliance. A recent study indicated that the algorithm works satisfactorily when compared with intermittent and continuous thermodilution techniques in post-operative cardiac surgical patients (Manecke and Auger, 2007; Mathews and Singh, 2008; Biancofiore *et al.*, 2009), however, other studies showed that the device systematically underestimated CO (Compton *et al.*, 2008b).

PRESSURE RECORDING ANALYTICAL METHOD

The pressure recording analytical method (PRAM) is a technique based on the mathematical analysis of peripheral (radial or femoral) pressure pulse waveform changes based on the theory of perturbations, by which any physical system under the effect of a perturbation tends to react to achieve its own state of minimum energy. The basic principle of PRAM is that, in any given vessel, the volume changes occur mainly because of radial expansion of the artery in response to variations in pressure. PRAM measures absolute SV without external calibration by determining the parameters able to characterize the elastic properties of the arteries from the objective analysis of the pressure wave contour (Giomarelli *et al.*, 2004; Romagnoli *et al.*, 2009). SV is calculated using the formula

$$SV = A/(P/t) \times K$$

where A is the whole area under the systolic portion of the pressure curve, P/t is the pressure wave contour expressed as the variation of pressure over time during the entire cardiac cycle. K is a factor inversely proportional to instantaneous acceleration of the vessel's cross-sectional area. The value of K is obtained from the expected (theoretical) and measured mean arterial blood pressure (Scolletta *et al.*, 2005).

Only a few studies have been carried out so far to prove its efficacy to determine absolute CO values. These studies were performed in animals and humans under various hemodynamic conditions and CO values were comparable to those obtained by electromagnetic flowmetry and conventional thermodilution method (Giomarelli *et al.*, 2004; Scolletta *et al.*, 2005; Romagnoli *et al.*, 2009).

Results using pulse contour methods to measure CO are improved somewhat when the pulmonary artery pressure waveform is used instead of a systemic artery pressure waveform because of less effects of wave reflection (Deloskey *et al.*, 1978; Karamanoglu and Bennett, 2006).

ELECTRICAL IMPEDANCE CARDIOGRAPHY

Impedance cardiography (ICG) is a method which calculates CO from the measurement of changes in impedance across the chest during the cardiac cycle. Lower impedance indicates greater intrathoracic fluid volume, and as the only fluid volume which changes beat-to-beat within the thorax is the blood, the change in impedance can be used to calculate SV and CO (Mathews and Singh, 2008). This technique has progressed clinically and allows non-invasive estimations of CO and total peripheral resistance using only four paired skin electrodes (Keren *et al.*, 2007; Gujjar *et al.*, 2008). While the method is desirably non-invasive and inexpensive, it has not achieved the reliability and reproducibility required of a useful clinical tool, and the evolution of algorithms to

convert impedance signals to SV across a variety of COs and in a variety of diseases continues.

ELECTRICAL CARDIOMETRY

Electrical cardiometry is a non-invasive method similar to impedance cardiography, in the fact that both methods measure thoracic electrical bioimpedance. Four standard ECG electrodes are required for measurement of CO. Electrical cardiometry is a method trademarked by Cardiotronic Inc., and shows promising results in a wide range of patients (it is currently US market-approved for use in adults, pediatrics, and neonates). Electrical cardiometry monitors have shown promise in post-operative cardiac surgical patients (both hemodynamically stable and unstable) (Funk *et al.*, 2009).

MAGNETIC RESONANCE IMAGING

Velocity-encoded phase contrast MRI is an accurate technique for measuring phasic blood flow in large vessels in mammals. MRI flow measurements have been shown to be highly accurate compared to measurements with a beaker and timer (Arheden *et al.*, 1999) and less variable than both the Fick principle (Razavi *et al.*, 2003) and thermodilution method (Kuehne *et al.*, 2004).

Velocity-encoded MRI is based on detection of changes in the phase of proton precession. These changes are proportional to the velocity of the movement of those protons through a magnetic field with a known gradient. When using velocity-encoded MRI, the result of the MRI scan is two sets of images for each time point in the cardiac cycle. One is an anatomical image and the other is an image where the signal intensity in each pixel is directly proportional to the through-plane velocity. The average velocity in a vessel, i.e. the aorta or pulmonary artery, is hence quantified by measuring the average signal intensity of the pixels in the cross-section of the vessel, and then multiplying by a known constant. The volume flow is calculated by multiplying the mean velocity by the cross-sectional area of the vessel (Fig. 6.13). These flow data can be used to graph flow versus time. The area under the flow curve for one cardiac cycle is the SV. The length of the cardiac cycle is known and determines heart rate, and thereby CO can be calculated as the product of SV and heart rate. MRI is typically used to quantify the flow over one cardiac cycle as the average of several heart beats, but it is also possible to quantify SV in real time on a beat-for-beat basis (Petzina *et al.*, 2007). MRI has been used successfully in a multitude of adverse cardiovascular alterations in both research animals and humans (Heckbert *et al.*, 2006; Petzina *et al.*, 2007; Sugawara *et al.*, 2008; Cawley and Otto, 2009).

MODELFLOW METHOD

More recent modifications of the method by Wesseling *et al.* (1993) derive an aortic flow waveform from arterial

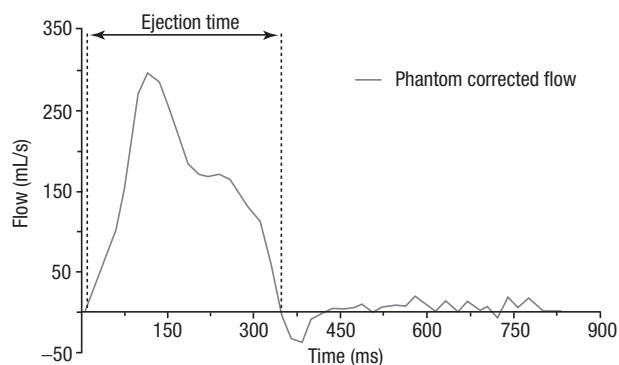


Figure 6.13 Flow curve measured in the main pulmonary artery using magnetic resonance imaging during a cardiac cycle in a patient with primary pulmonary hypertension. Stroke volume is obtained by integration of the flow curve during ejection time (323 ms) and is 58 mL. Reproduced from Mauritz *et al.* (2008)

pressure (finger or radial) by simulation of a non-linear three-element aortic input impedance model and integrates stroke volume from the flow waveform (Harms *et al.*, 1999; Jellema *et al.*, 1999; Gerhardt *et al.*, 2000; Jansen *et al.*, 2001). These methods require initial calibration against another method such as thermodilution to give meaningful values of stroke volume. Wesseling *et al.* (Wesseling *et al.*, 1993; Imholz *et al.*, 1998) computed pulsatile aortic blood flow from arterial pressure (finger or radial) by simulating a non-linear, time-varying, three-element model of aortic input impedance (Burkhardt *et al.*, 1988). Integrating the computed aortic flow waveform per beat provides left ventricular SV (Harms *et al.*, 1999; Jellema *et al.*, 1999; Gerhardt *et al.*, 2000; Jansen *et al.*, 2001). The method requires initial calibration against another method, such as thermodilution, to give meaningful values of stroke volume. In an improved version of modelflow (COMfc) the calculated flow velocity curve was calibrated using descending aortic cross-sectional area measured by transesophageal echo (de Vaal *et al.*, 2005). The modelflow method for measuring non-invasive pulsatile ascending aortic blood flow was reviewed recently by Bogert and van Lieshout (2005) and Mathews and Singh (2008). The model elements represent aortic characteristic impedance, arterial compliance, and systemic vascular resistance (see Fig. 6.12). Parameter values for the first two elements are computed from a published, age-dependent, aortic pressure–area relationship (Langewouters *et al.*, 1984). Peripheral resistance is predicted from mean pressure and model mean flow. Modelflow pulsations from aortic pressure show the visual aspects of an aortic flow curve (Fig. 6.14). For evaluation, mean modelflow from radial arterial pressure was compared with thermodilution CO estimations in open-heart surgical patients. The pooled mean difference was ± 7.0 percent, with a standard deviation of ± 22 percent. After using one comparison per patient to calibrate the model,

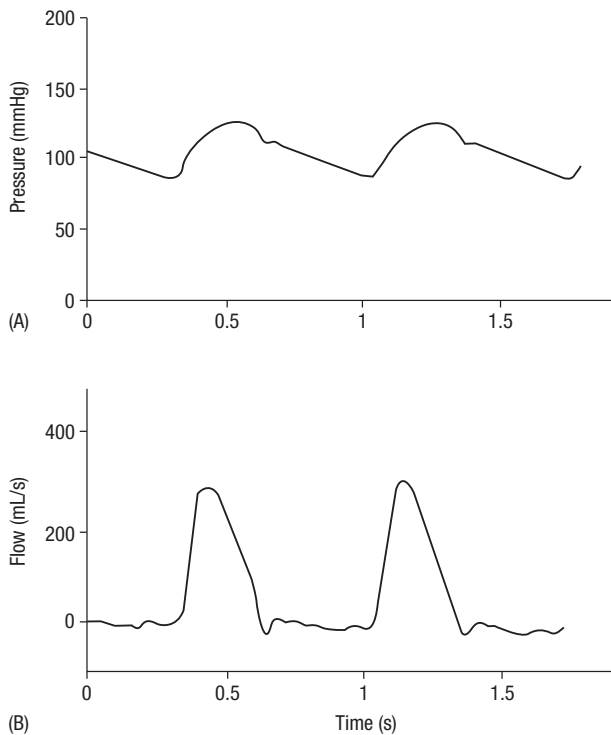


Figure 6.14 (A) Arterial pressure measured in the ascending aorta of a human. (B) Simulated model flow from aortic pressure. Model flow computed from aortic pressure shows typical characteristics of an aortic flow wave: a steep upslope at the beginning of systole, and a gradual downslope, terminating in a steep final zero flow in late diastole. Reproduced from Wesseling *et al.* (1993)

however, it followed quantitative changes in CO that occurred either during changes in the state of the patient or vasoactive drug administration. The mean deviation from thermodilution CO was ± 2.0 percent, and the standard deviation was ± 8.0 percent. Given these small errors, the method can be used to calculate a pulsatile flow curve and monitor CO continuously. Under some conditions, in situations such as exist in intensive care units where there are gross changes in ejection period and wave reflections, such a simple model may have serious shortcomings. Several studies have used the 'modelflow' of Wesseling to estimate continuous SV in humans under orthostatic stress (Harms *et al.*, 1999), in critically ill patients (Hirschi *et al.*, 1997), and in patients with septic shock (Jellema *et al.*, 1999). The method is now fully automatic, self-recording, and has a fast response, and its precision appears substantially improved compared with earlier methods (Jansen *et al.*, 2001).

Most of the blood flowmeters in use today that measure pulsatile blood flow directly are of the electromagnetic or ultrasonic types (Mills, 1972; Webster, 1978; Milnor, 1989; Sloth *et al.*, 1997; Tortoli *et al.*, 2002; Dobson *et al.*, 2006; Ioannou *et al.*, 2009), but other methods, including magnetic resonance imaging and Modelflow, are being explored (see above) (e.g. Sloth

et al., 1997; Vulliemoz *et al.*, 2002; Mauritz *et al.*, 2008). The measurement of blood flow in microvessels uses photometric and electronic methods of detecting erythrocyte velocity (Wu *et al.*, 2009; Wu *et al.*, 2011). The technique times the passage of erythrocytes between two slits in an optical image of the vessel (Wayland and Johnson, 1967). The original method devised by Wayland and Johnson (1967) has been modified to use laser (Le-Cong and Zweifach, 1979) and spectroscopic analysis (Nilsson *et al.*, 1980). Microvessel center-line measurements of erythrocyte velocity can also be made with an optical Doppler intravital velocimeter (see also Chapter 2) (Borders and Granger, 1984). Also, cutaneous microcirculatory assessment using laser Doppler flowmetry can provide immediate results for early determination of burn wound depth and is useful in selecting patients for conservative treatment of their burn wounds (Wu *et al.*, 2011).

Electromagnetic flowmeters

If a bar magnet is moved through a conductor coil like the one shown in Fig. 6.15 (left), a current is induced in the coil as indicated by the ammeter. If the magnet is held stationary with respect to the coil, the meter does not deflect. If the magnet is withdrawn from the coil, the meter again deflects, but in the opposite direction. It makes no difference whether the magnet is moved toward the coil or the coil toward the magnet. This is described as Faraday's law (Faraday, 1832), but not until 1930 was the possibility investigated of using the technique as a flowmeter (by EJ Williams, cited by Shercliff, 1962).

As blood is a conductor of electricity, its velocity of movement may be measured by aligning a blood vessel across a magnetic field and recording the induced electromagnetic flowmeter (EMF) (Fig. 6.15, right). Provided that (1) the field is uniform, (2) the conductor moves in a plane at right angles to the field, and (3) the length of the conductor extends at right angles to both field and direction of motion, the potential difference is given by the following equation (Jochim, 1948):

$$E = Hd\bar{V} \times 10^{-8} \quad 6.36$$

where E is the potential difference in volts, H the strength of the magnetic field in gauss, d the internal diameter of the vessel in centimeters, and $\bar{V}$ the mean velocity of the conductor in cm/s. If H and d are kept constant, it can be seen that the potential difference recorded varies linearly with the velocity of flow and, in fact, the velocity recorded is the average velocity across the tube (Kolin, 1945; Shercliff, 1962). The calibration curve is thus a straight line passing through the origin.

After a preliminary model made by Fabre (1932), which attracted little attention, the instrument was introduced into physiology independently by Kolin

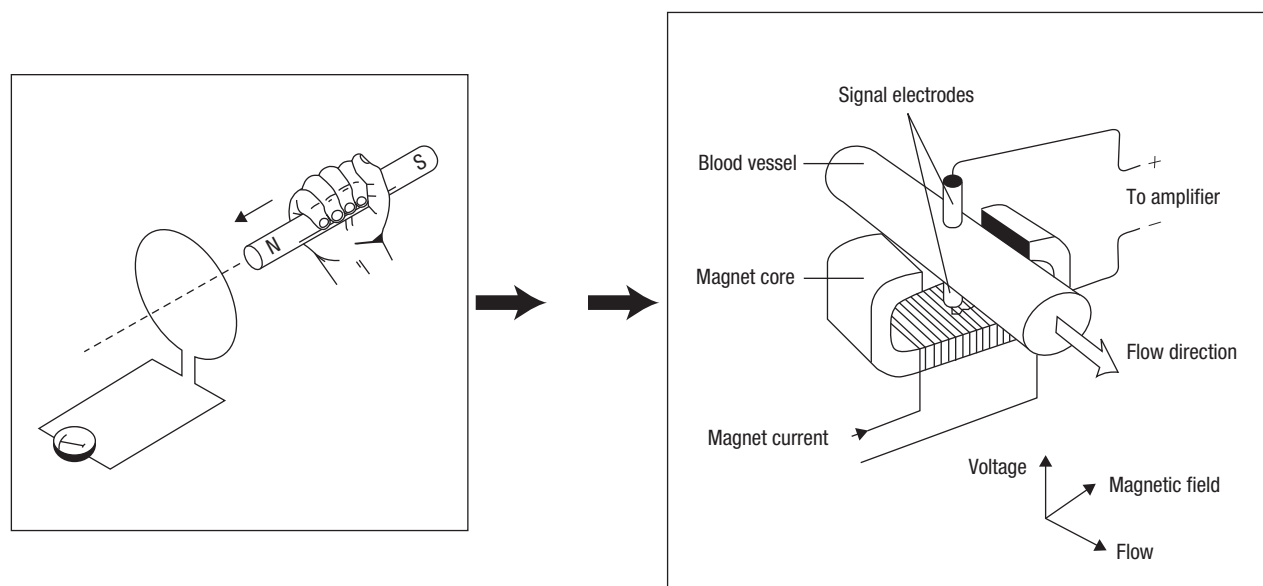


Figure 6.15 Diagrammatic illustration of the principle used to develop the electromagnetic flowmeter. Left: When a bar magnet is moved through a conductor coil, an EMF (voltage or current) is induced as detected by the meter (Faraday's law). Right: Application of Faraday's law to the measurement of instantaneous blood flow. When blood (a conductor) flows in the vessel and passes through a magnetic field, the induced EMF is detected by the electrodes and then amplified. Illustration courtesy of Skalar, Delft

(Kolin, 1936; Kolin, 1937) in the United States and Wetterer (1937) in Germany. Its advantages are the linear calibration noted above and that it can be applied to an unopened vessel. It also records back flows as well as forward flows, and the development of intravascular catheter probes makes measurements possible without exposing the vessel.

The EMF is considered the 'gold standard' for measuring pulsatile blood flow and CO. Small electromagnetic probes have been developed that can be applied to any reasonably accessible artery, such as the carotid (Hosoda *et al.*, 2001), femoral (Milnor and Nichols, 1975), coronary (Gregg and Fisher, 1963; Feldman *et al.*, 1979a; Feldman *et al.*, 1979b; Grossini *et al.*, 2007), renal (Lisik *et al.*, 2007; Mustafa *et al.*, 2007), superior mesenteric (Nakamura *et al.*, 1989), splenic (Nakamura *et al.*, 1989), and hepatic (Noguchi *et al.*, 2006; Zhu *et al.*, 2008). Interference due to the electric potentials created by the heart, which was earlier thought to limit its use on the aorta and pulmonary artery, was soon overcome, for Wetterer and Deppe (1940) used it successfully on the dog-ascending aorta, and its use there (Fig. 6.16) (Wetterer and Deppe, 1940; Milnor *et al.*, 1966; O'Rourke and Taylor, 1967; Feldman *et al.*, 1979a; Feldman *et al.*, 1979b; Murgo *et al.*, 1980a; Murgo *et al.*, 1980b; Chikada *et al.*, 2005) and on other arteries is now commonplace in both animals and humans.

The simplest form of EMF has a constant magnetic field (the DC type), and the recording system has to use a direct-coupled amplifier. Surface potentials formed at the electrodes are relatively large compared with those induced in the flowing blood, and it is difficult to discriminate between them in a DC system. The DC meter

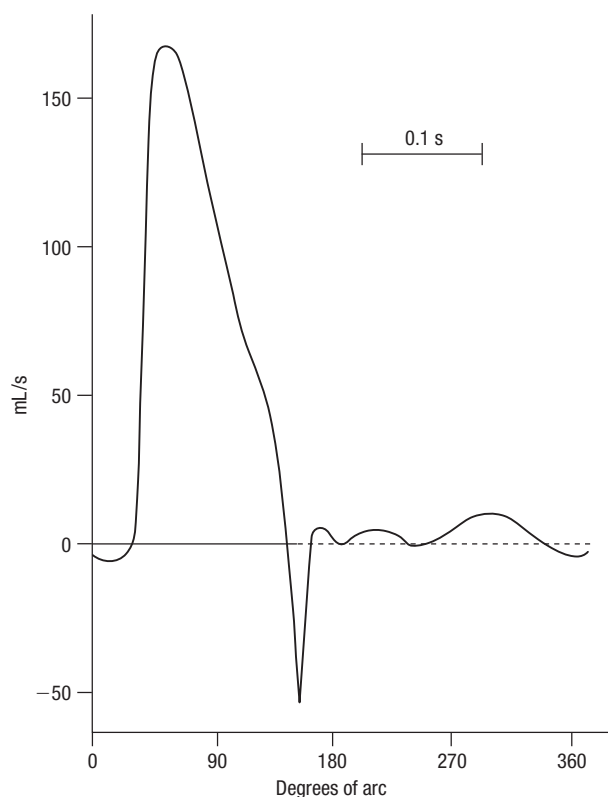


Figure 6.16 One of the first flow curves recorded in the dog ascending aorta with an electromagnetic flowmeter. Forward flow is virtually confined to the systolic ejection period, the end of which is marked by a sharp but brief back flow. Reproduced from Wetterer and Deppe (1940)

also exhibits some zero drift owing to the polarization caused by the small current generated. Jochim (1948) attempted to minimize the electrode potentials by embedding them in Perspex blocks that fitted closely around the artery. A similar apparatus was used by Richards and Williams (1953). Inouye *et al.* (1955) and Inouye and Kosaka (1959) also used this type of meter, as did Feder and Bay (1959). They found it satisfactory for intermittent short runs that do not allow significant polarization.

To avoid polarization effects, some form of alteration of the current generating the magnetic field is now virtually always used in meters for physiologic use. This alternation may be sinusoidal (the AC meter) or use a specific waveform, as in the square-wave and trapezoid-wave flowmeters. Kolin introduced an AC meter as early as 1941 and later developed and improved this type (Kolin, 1960; Kolin and Wisshaup, 1963) to the satisfactory form generally known as the gated sine-wave meter. A major difficulty has been with the so-called transformer effect. This is due to the fact that the electrode leads, electrodes, and the blood form a conductive loop that is cut by some magnetic lines of force and produces undesired potentials. Good design of the probe can minimize this, but it is eliminated by only sampling the signal through an electronic gate when the transformer potential is zero. In the AC meter, phase discrimination is necessary to distinguish forward flow from back flow. Other problems arise through potentials caused at the electrode-artery interface and coupling between the electrodes and the magnet winding. These technical aspects are discussed by Bergel and Gessner (1966), who also give an extensive bibliography of various modifications that have been introduced; Mills (1972) and Webster (1978) have also made a lengthy review of the technical problems and source of artifacts.

The probes used with the gated sine-wave flowmeters are made of plastic of the order of about 2.0 mm thick, in which coreless windings to create the magnetic field and the electrodes are incorporated; standard sizes available range from 0.5 to 26 mm internal diameter. The elimination of an iron core removes errors due to hysteresis, but greatly reduces the strength of the magnetic field created (to increase sensitivity, they are retained in the small sizes), and the signal to noise ratio, of a signal of only a few microvolts, is very small. The development of a meter and probes for use on the coronary arteries and small vessels was studied by Khouri and Gregg (1963) and Khouri *et al.* (1968).

The square-wave EMF was introduced by Denison *et al.* (1955) as a way to avoid the problems they had found with an AC meter (Richardson *et al.*, 1952). The magnet's current is pulsed in a square-wave form in which the polarity is reversed and the signal is sampled by a gate during the stable part of the field at the top of the square wave. As this model uses iron cores in the probes, there is considerable oscillation of the field each time the current is reversed.

STATIC CALIBRATION

Electromagnetic flow probes are not identical in sensitivity; therefore, each probe should be calibrated and a probe factor determined before use. The calibration system should be filled with blood (instead of saline) and the probe placed on a section of excised artery or vein. Calibration should be made over the range of flows to be measured with a particular probe. In practice, considerable change in artery size can occur without any measurable change in effective contact. Some workers (Spencer and Denison, 1959) deliberately choose a probe that will produce a marked constriction of the artery to minimize this effect. Such a constriction, however, produces distortions both of pressure and flow patterns (Nichols, 1970; McDonald, 1974; O'Rourke, 1982a). Always aim to produce as good a fit as possible without visible constriction of the artery. If experiments are planned in which large decreases in mean pressure are anticipated for any length of time, as in studying the effects of hemorrhage or vasodilatation, it is better to change to a smaller probe when the artery becomes visibly too small for the larger probe. In chronic preparations, the probe should fit loosely on the vessel when implanted, because a thin film of connective tissue anchors the probe to the vessel wall in a week or so (this also applies to ultrasonic probes). This procedure minimizes vessel rupture, lessens the possibility of pressure artifact and extends the life of the preparation (Fuller *et al.*, 1978; Heard *et al.*, 1996).

DYNAMIC CALIBRATION

The dynamic frequency characteristics of an EMF can be determined by pumping sinusoidal flows (of blood or saline) through a vessel (or dialysis tubing) with the flow probe in place. This method, however, has certain limitations; for example, it is difficult to produce flows of sufficient amplitude at frequencies greater than about 20 Hz (Gessner and Bergel, 1964; O'Rourke, 1965b). An alternative method that may be easier (Mills, 1972) is to oscillate the probe sinusoidally up and down in a saline bath. This procedure is especially suitable for catheter-tip velocity probes.

Gessner and Bergel (1964) and Bergel and Gessner (1966) tested the frequency characteristics of gated sine-wave and square-wave EMF hydraulically with a sinusoidal pump and electrically by imposing a sinusoidal voltage across the electrodes. They found no significant difference between the two methods, indicating that the dynamic characteristics were determined by the electric circuitry of the flowmeter. The same conclusion was reached by O'Rourke (1965b).

The frequency modulation or electric technique is used to determine the frequency characteristics of flowmeters. The modulation signal is fed to a multiplier from a

sine-wave generator that imitates the flow signal. The multiplier output is suitably attenuated to provide a signal to the flowmeter input. The flowmeter output is compared with the sine-wave generator output on an oscilloscope capable of Lissajous loop production, from which amplitude and phase response versus frequency can be obtained. We used this method to determine the frequency characteristics of the gated sine-wave flowmeter. The flowmeter was tested at each 'nominal' filter setting, and the results are shown in Fig. 6.17. The amplitude response at nominal 25 Hz, 50 Hz, and 100 Hz was flat ± 5 percent to 14 Hz, 23 Hz, and 49 Hz, respectively, and the phase was relatively linear with frequency at each filter setting. These results are similar to those found by Milnor and Nichols (1975) for the Biotronex BL 610 'pulsed-logic' flowmeter. The amplitude–frequency response of flowmeters is often given as '3 dB down', which means absolutely nothing to most people in the medical field. The frequency at which the amplitude response is decreased by approximately 30 percent is considered to be the 3 dB (decibel) frequency. The filter settings on a flowmeter are usually given as '3 dB' frequencies, but even

these are not always correct. Therefore, it is essential that the dynamic frequency characteristic of a flowmeter be determined before it is used.

A flat amplitude–frequency response up to about the tenth harmonic is sufficient to reproduce the pulsatile flow (and pressure) observed in most arteries of small animals and humans (Ferguson and Wells, 1959; Guz, 1968; Nichols, 1970; McDonald, 1974; Nichols *et al.*, 1985b; Milnor, 1989; Nichols *et al.*, 1994c). If blood flow acceleration (i.e. the time rate of change of velocity) is desired, the frequency response of the flowmeter should be much higher (to about the thirtieth harmonic). The flat amplitude–frequency response of a flowmeter necessary for recording these harmonics is obtained by multiplying the harmonic number by the heart rate (in Hz). Therefore, for accurate recordings of flow waves in the dog or human, the amplitude response should be flat to at least 20 Hz (see Chapter 11) and the phase should be linear over this frequency. For a more detailed account of static and dynamic calibration of EMF flowmeters, see Nichols and O'Rourke (1998).

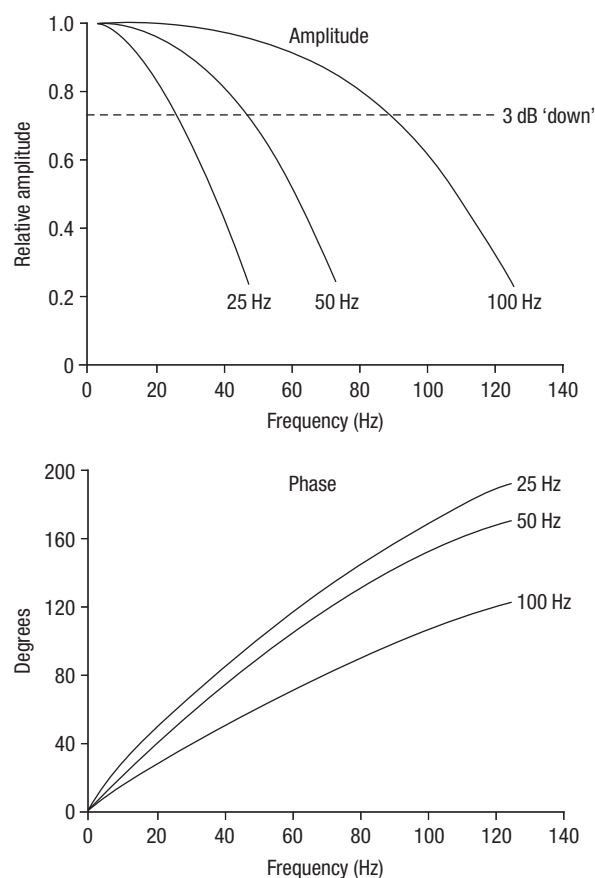


Figure 6.17 Relative amplitude and phase lag versus frequency response of a gated sine-wave electromagnetic flowmeter. Response curves are shown for nominal filter settings of 25, 50, and 100 Hz. The broken line represents the 3 dB down amplitude

Ultrasonic flowmeters

A method for measuring pulsatile blood flow velocity in both experimental animals and human subjects that has become very popular in recent years employs high-frequency sound waves (2–20 MHz) (Webster, 1978; Altobelli and Nerm 1885; Sloth *et al.*, 1997; Tortoli *et al.*, 2002). The popularity of using ultrasound to detect blood flow velocity has advanced because the technique can be used to obtain velocity waveforms (as well as vessel diameter, velocity profiles, and pressure gradients) invasively and non-invasively. An ultrasonic beam is passed through the vessel, and the instantaneous blood flow velocity within the vessel is detected from the difference in time taken for the ultrasound to pass in one direction as opposed to the other (the transit-time method), or from the frequency of the sound backscattered from the moving red blood cells (the Doppler method) (see Chapter 8). Catheter-tip velocity catheters using both methods have been developed and will be discussed below. The use of Doppler ultrasound in the study of arterial blood flow non-invasively in human subjects is covered in Chapter 8. The present chapter will cover ultrasonic methods used for invasive measurements of arterial blood flow velocity in research animals (and in human patients).

THE TRANSIT-TIME METHOD

The pulse transit time is measured in the type of meter that has come to be known as the pulsed ultrasonic flowmeter. First reported by Franklin *et al.* (1959a), later versions have been described in more detail by Franklin *et al.* (1961), Baker (1966), and Rader *et al.* (1973), and

evaluated by Gessner (1969) and Studer *et al.* (1970). The principle is very simple. A cuff is placed around an artery; this incorporates two crystals diagonally placed on either side of the vessel. Both crystals can act as transmitter and receiver. A short burst of ultrasound is emitted simultaneously from both crystals. If the burst is short compared with the transit time of sound across the vessel, each transmitter can also act as a receiver for the burst from the opposite side. Transit time is measured by starting a ramp voltage rising with each burst emitted and which is stopped by the reception of the pulse.

Baker (1966) reported that the response of the transit time ultrasonic flowmeter is linear for steady flows from 0.0 to 180 mL/s and that the electric circuitry has a uniform frequency response from 0.0 to 50 Hz.

The rate of sampling can be varied and appears to be normally done a few hundred times per second. Baker (1966) states that the cycle from pulse emission to reception and recording takes only 120 ms, so that several probes on various arteries can be used simultaneously on a 'time-sharing' basis with a single electronic unit.

A blood flowmeter that uses the ultrasonic transit-time principle is available from Transonic Systems Inc. (Ithaca, NY, USA). The flowmeter, developed at the New York State College of Veterinary Medicine at Cornell University and introduced to the Physiological Society in Oxford in 1983 (Barnes *et al.*, 1983), was designed primarily for long-term implants. The design of the probes allows for measurement of volume rate of blood flow essentially independent of vessel diameter (Latson *et al.*, 1988). Flow probes are available for use on vessels ranging in size from 1.0 to 24 mm outside diameter and are designed for use in humans (Gordon *et al.*, 1994; Hoffman *et al.*, 1995), a large variety of experimental animals (Downing *et al.*, 1993; Evans *et al.*, 1993; Comeau and Hicks, 1994; Tominaga *et al.*, 1994; Wilkins *et al.*, 1994; Choshniak *et al.*, 1995; Shen and Vatner, 1995; Dobson *et al.*, 2006; Forsberg *et al.*, 2006; de Boode *et al.*, 2009; Ioannou *et al.*, 2009), and on the external surface of tubing (Krivitski, 1995).

THE DOPPLER METHOD

Blood flow velocity measurement using the Doppler-shift principle was first reported by Franklin *et al.* (1961); later reports came from Baker *et al.* (1978), Angelsen (1980), and Hartley and Cole (Hartley and Cole, 1974a; Hartley and Cole, 1974b). Red blood cells are the scattering elements and act as moving reflectors; the transmitting crystals project an ultrasonic beam downstream, and waves reflected back upstream are detected by a receiving crystal (see Chapter 8). Calibration of the Doppler flowmeter for measuring steady flows was reported to be linear (Stegall *et al.*, 1966), and for pulsatile flow the meter compares favorably with the EMF (Spencer *et al.*, 1991).

As a meter that can be used non-invasively, it has great potential for qualitative clinical use, as discussed in

Chapter 8. Such applications as detecting fetal blood flow, monitoring cerebral blood flow velocity, detecting regions of occlusion in diseased peripheral arteries, and making records from human arteries, ranging from the pulmonary artery and aorta to digital arteries in the hand, have been discussed by Rushmer *et al.* (1967), Fronek (1972), Tunstall-Pedoe (1972), Lewis *et al.* (1984), Dubin *et al.* (1990), Sloth *et al.* (1997), and Fisher *et al.* (2008). Craig Hartley's group has developed and evaluated a high-frequency, real-time pulsed Doppler acquisition and analysis system specifically for measuring blood flow velocity in different arteries, including the main coronary artery of mice. The system consists of multiple 10/20 MHz ultrasound transducers, analog electronics for Doppler displacement and velocity measurement, signal acquisition, and processing electronics and personal computer-based software for real-time and off-line analysis. The system can also be used to measure diameter changes, augmentation index, impedance spectra, pulse wave velocity, characteristic impedance, forward and backward waves, reflection coefficients, and coronary flow reserve (Hartley *et al.*, 2007; Hartley *et al.*, 2008; Reddy *et al.*, 2009a; Reddy *et al.*, 2009b). Examples of blood flow velocities measured in the left main coronary artery, left common carotid artery, and aorta using the system are shown in Fig. 6.18.

Several groups have successfully compared invasive measurements of stroke volume using thermodilution, direct Fick and dye-dilution techniques with non-invasively obtained pulsed or continuous wave Doppler data (Huntsman *et al.*, 1983; Chandraratna *et al.*, 1984; Ihlen *et al.*, 1984; Rose *et al.*, 1984; Bouchard *et al.*, 1987). Studies have also compared beat-to-beat aortic Doppler data with data obtained using an EMF (Steingart *et al.*, 1980; Spencer *et al.*, 1991). The study by Spencer *et al.* (1991) critically compared instantaneous aortic blood flow characteristics

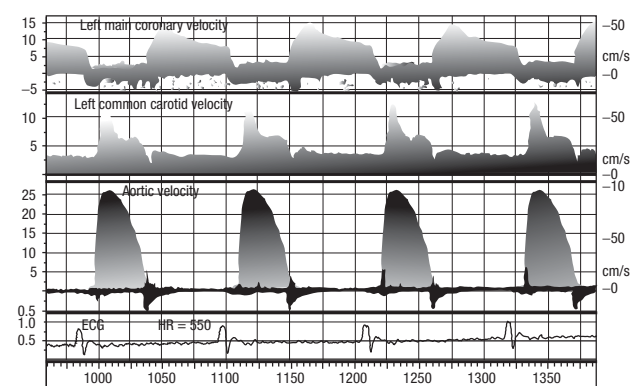


Figure 6.18 Simultaneous Doppler signals from the coronary artery, carotid artery, and aorta in a mouse anesthetized with 2 percent isoflurane showing the timing relative to the ECG and the similarity to human signals in each vessel. Blood velocities in cm/s were calculated from the Doppler shifts using the Doppler equation with a Doppler angle of 0 degrees for the coronary artery and the aorta and 45 degrees for the carotid artery. Reproduced from Hartley *et al.* (2008)

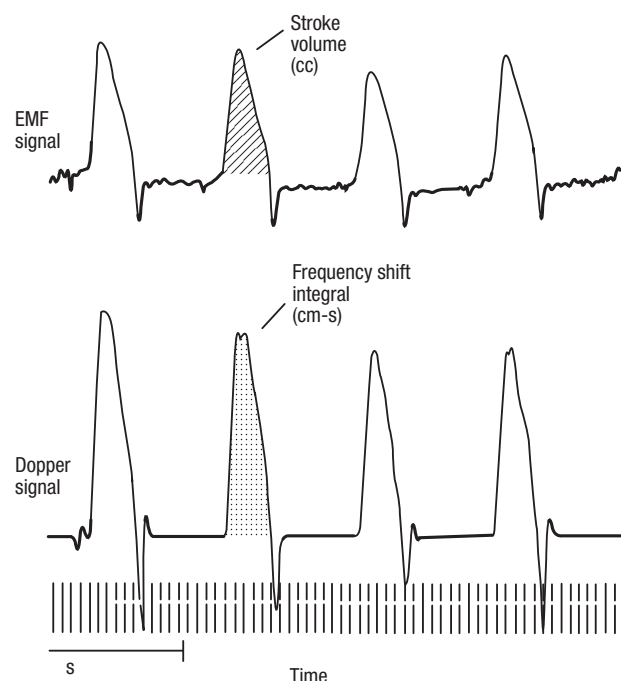


Figure 6.19 Simultaneous electromagnetic flowmeter (EMF, upper panel) and pulsed Doppler (lower panel) recordings of ascending aortic blood flow in a dog. From Spencer *et al.* (1991). The area indicated by the diagonal lines is stroke volume (upper) and the area indicated by the dotted area (lower) represents the frequency shift interval. Reproduced from Steingart *et al.* (1980)

and wave morphology acquired simultaneously in monkeys using EMF and continuous-wave Doppler techniques. In an earlier study, Steingart *et al.* (1980) compared simultaneous measurements of pulsatile blood flow and stroke volume in dogs using electromagnetic and pulsed Doppler probes implanted on the ascending aorta. An example from their publication is shown in Fig. 6.19.

Built into a rigid cuff that is implanted on an artery where the relation between the crystals and the arterial wall can be kept constant, the Doppler flowmeter can be used as a quantitative instrument. It has been used in both acute (Nichols *et al.*, 1988a; Nichols *et al.*, 1988b; Nichols *et al.*, 1993; Nichols *et al.*, 1994a; Nichols *et al.*, 1994b) and chronic experiments by several laboratories (Franklin *et al.*, 1966; Van Citters and Franklin, 1969a; Van Citters and Franklin, 1969b; Vatner *et al.*, 1971a; Vatner *et al.*, 1971b; Heard *et al.*, 1996). Because a modulated frequency signal is easier than other types of signal to telemeter, Franklin *et al.* (1966) and Van Citters and Franklin (Van Citters and Franklin, 1969a; Van Citters and Franklin, 1969b) used Doppler probes to measure blood flow in Alaska sled dogs and free-ranging wild animals.

A Doppler flowmeter in which there is good discrimination of the direction of flow was reported by McLeod in 1967 and has been further developed (McLeod, 1970). Kalmanson and his colleagues in Paris developed an instrument that enabled discrimination of the direction of flow to be made; it was used extensively in humans

(Kalmanson *et al.*, 1968; Kalmanson *et al.*, 1970; Kalmanson *et al.*, 1972) to record arterial and venous flows outside the thorax. They also made some records of the flow in the output tract of the right ventricle. While calibration in terms of volume flow was difficult, they showed what a valuable addition to cardiological diagnosis this instrument could be in studying valvular incompetence.

A pulsed Doppler flowmeter was developed by Peronneau and Leger (1969) and used by Safar *et al.* (1981a, 1981b) and Simon *et al.* (1985) to measure bidirectional blood flow velocity in humans (see also Chapter 8). Whereas the Doppler device uses a continuous emission of ultrasonic frequency, in this modification the tone emission is divided in short (microsecond) bursts at rapidly repeated intervals. The emitting and receiving crystals are applied to the vessel and angled into the bloodstream. The pulses of reflected waves will vary in their time of arrival back at the wall according to the distance the wave travels into the lumen before being reflected back to the source. The distance from the wall at which flow is being recorded can be determined from the velocity of sound in blood; by varying the timing of the receiving gate, a complete scan of the interior of the blood vessel can be obtained. The velocity of flow at each point is determined by the Doppler frequency shift of each pulse; in spite of the short pulse duration when emitting frequencies of 3.0–8.0 MHz are used, it will be seen that each pulse contains a long series of waves and may be treated as being in steady-state oscillation. The ability to scan the flow velocities within the blood vessel has led to its use for recording the velocity profiles that are discussed in Chapter 2.

Esophageal Doppler ultrasound monitoring is a widely used technique of minimally invasive CO monitoring performed currently in critically ill patients (Bein *et al.*, 2004; Dark and Singh, 2004; Koliopanos *et al.*, 2005; Chen *et al.*, 2009). It is based on measurement of blood flow velocity in the descending thoracic aorta by means of a Doppler transducer and either an estimated aortic cross-sectional area close to the mean value during systole (derived from a nomogram stored in the computer based on age, sex, height, and weight) or a measured cross-sectional area using an M-mode echo-transducer incorporated into the probe. This provides immediate, comprehensive, beat-to-beat SV, and continuous. The Doppler probe is inserted into the esophagus to an approximate depth of 35 cm from the incisors in an intubated patient. The probe is then rotated so that the transducer faces the aorta and characteristic aortic blood flow velocity signal is obtained (Fig. 6.20).

The major assumption made using the esophageal Doppler monitoring is that a constant percentage of CO enters the descending thoracic aorta as measurements made in the descending thoracic aorta exclude blood flow going to the heart, head, and arms. However, the percentage of the CO that passes into the descending thoracic aorta varies with some disease states. For example in

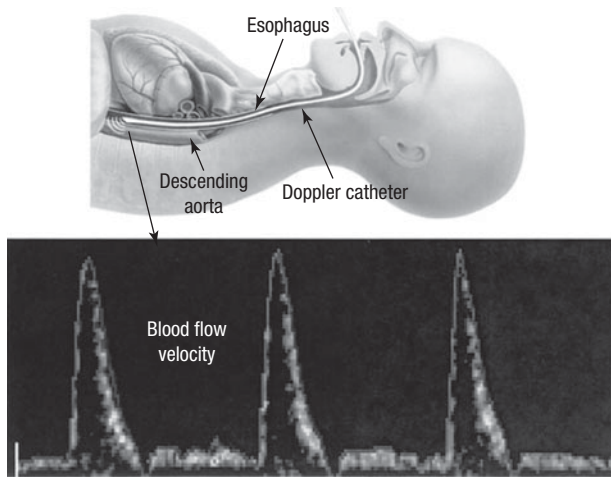


Figure 6.20 (Top panel) Schematic representation of esophageal Doppler catheter in a patient demonstrating the close relation between esophagus and descending aorta. (Lower panel) The spectral representation of descending thoracic aorta blood flow velocity. Reproduced from Cholley and Singer (2003).

hypovolemic state, there will be more flow to coronary and cerebral circulations so a greater portion of the blood will leave the aorta proximal to the descending thoracic aorta. Conversely, with lower limb regional block, vasodilatation due to loss of sympathetic tone will cause a greater portion of the blood to be delivered beyond the descending thoracic aorta. The presence of an intra-aortic balloon pump or severe aortic coarctation will produce turbulent flow in the descending aorta and introduces an error in the measurement (Mathews and Singh, 2008). A thorough description of the technique, its advantages, and limitations has been published (Cholley and Singer, 2003; Mathews and Singh, 2008; Funk *et al.*, 2009). Various studies comparing CO values obtained with the esophageal Doppler ultrasound monitoring system and other techniques of measuring CO report acceptable correlation and recommend its clinical usefulness.

Intravascular velocity transducers

The accurate measurement of blood flow with an intravascular transducer involves knowledge of the distribution of the velocity profile across the artery and, in particular, the relation of the velocity measured by the meter to mean velocity across the vessel. It has been established by several investigators (e.g. Schultz *et al.*, 1969; Seed and Wood, 1971; McDonald, 1974) that the velocity profile is fairly 'flat' in vessels as large as the aorta and main pulmonary artery (i.e. uniform across most of the artery), with the region of shear limited to a narrow region at the wall (see Chapter 2). If the velocity sensor is mounted at the tip of the catheter and the catheter is unrestrained, the sensor may usually rest against or near the inner wall of the systemic artery, and the measured velocity will be very

small. This is especially true for the ascending aorta since the catheter, inserted from a peripheral artery, must make a bend on its passage toward the heart. In the pulmonary artery, an intravascular velocity catheter will pass through the pulmonary valve and thus tend to remain in the more central part of the bloodstream. Most of the recent catheter transducers have a 5.0 cm leader at the tip that serves the same purpose in the ascending aorta (see below). The catheter is advanced until the leader is through the aortic valve cusps, and this stabilizes the velocity sensor within the central portion of the bloodstream (at the position of the profile where the velocity is maximum). An unsupported central position will tend to show catheter whip, which may cause artifacts of itself.

ULTRASONIC TRANSDUCERS

Intravascular ultrasonic (IVUS) Doppler transducers have been reviewed by Hartley (1989), Kern *et al.* (1994), Winniford and Lambert (1994), and Kern (1998). One of the earliest catheter-mounted ultrasonic velocity probes was based on the differential transit-time principle (Plass, 1964). It used two 5.0 MHz crystals mounted facing each other across a flow-through section 3.0 cm long near the tip of a 4.0 mm diameter catheter (Studer *et al.*, 1970). The large physical size of the transducer limited its use to the larger vessels such as the aorta, and it was not practical for use in humans.

The Doppler-shift principle described by Franklin (1961) has also been used in the development of IVUS catheters. Briefly, continuous ultra-high-frequency sound waves are emitted from one of the piezo crystals (sensor). Part of the emitted sound is back-scattered from the blood cells passing through the vessel and around the catheter. These back-scattered waves are received by the other crystal. The frequency shift of the back-scattered sound waves is proportional to the velocity of the blood. This instrument was slightly less than 3.0 mm in diameter, but was reduced in later models. The velocity waveform recorded with the transducer was found to agree to within 10 percent of an externally applied Doppler probe, and its frequency response was good up to 50 Hz. One advantage of this type of probe is that zero flow velocity can be obtained by disconnecting the input signal to the frequency meter, since zero frequency shift corresponds to zero velocity.

Benchimol and associates used a catheter probe (external diameter 1.5 mm) of similar design, but without a pressure sensor, to record pulsatile blood flow velocities at several different sites in the cardiovascular system of patients. The majority of the patients had some type of heart disease, but some had normal cardiovascular function. Velocities were recorded in the inflow and outflow tracts of the left ventricle (Benchimol *et al.*, 1973; Benchimol *et al.*, 1974), the carotid (Benchimol *et al.*, 1970), renal (Benchimol and Desser, 1974), coronary

arteries (Benchimol *et al.*, 1971; Benchimol *et al.*, 1972), and the aorta (Benchimol *et al.*, 1969). Two disadvantages of this probe were its large size and its inability to discriminate between forward and backward flow velocity. These problems were solved by Kalmanson *et al.* (1972) and Reid *et al.* (1974) in an instrument that uses only one ultrasonic crystal. The catheter developed by Reid was small enough (3F or 1.0 mm external diameter) to be used in the coronary arteries of dogs.

Single crystal transducers can be used with either continuous wave (Kalmanson *et al.*, 1972; Reid *et al.*, 1974) or pulsed Doppler. Pulsed Doppler instrumentation was introduced by Peronneau and Leger (1969). During the last 20 years, several types of catheters based on pulsed Doppler ultrasound have been designed, evaluated, and used to measure coronary blood flow velocity in humans.

The catheter-tip transducer developed by Hartley and Cole (1974b) is capable of on-line continuous measurements of coronary blood flow velocity and coronary blood flow reserve in humans (Cole and Hartley, 1977; Dole *et al.*, 1984b). In this early version, a 20 MHz crystal was placed at the tip of a 5F (1.67 mm) Sones catheter. The catheter, however, was too large to be placed subselectively in the coronary circulation. This is also true for the Judkins-style Doppler catheter designed by Kern (1990). More recently, Wilson and colleagues (Wilson *et al.*, 1985; Wilson and White, 1986) developed a 3F (1.0 mm) catheter-tip velocity transducer with a side-mounted crystal for measuring coronary blood flow velocity in humans. A major limitation of this catheter was the inability to steer it away from small branches and the arterial wall. This problem has been solved in catheter-tip transducers recently developed by Sibley *et al.* (1986) and Serruys *et al.* (1989). The crystal is mounted at the tip of a cardiac catheter with an internal lumen that accommodates a standard angioplasty steerable guidewire (0.014 in). The guidewire allows precise placement and easy adjustment of the catheter in the coronary circulation. All of these catheters use 20 MHz Doppler crystals, are 1.0 mm (3F) in external diameter at the distal tip, and have a central lumen for a guide wire. There are, however, major differences in their design characteristics. The transducer produced by Millar Instruments, Inc. is mounted on the 1.0 mm tip of a USCI Rentrop reperfusion catheter. The crystal has an annular shape with a central hole for the guidewire to pass through, and the sound beam extends from the tip in the direction of the catheter. In contrast, the transducer produced by Numed (the Nuvel) has a side-mounted rectangular crystal 5.0 mm from the tip, which emits sound at an angle to the catheter. The device used by Serruys *et al.* (1989) was a balloon angioplasty catheter with a crystal mounted at its tip. All of the catheters allow intracoronary administration of agents such as vasodilators, either through the guiding catheter in the coronary ostia or directly through the lumen of the Doppler catheter. Numerous studies, including stenoses severity (Johnson *et al.*, 1989),

coronary blood flow reserve (Porter *et al.*, 1993; Rossen and Winniford, 1993), endothelial function (Mills *et al.*, 1992; Egashira *et al.*, 1993a; Egashira *et al.*, 1993b; Egashira *et al.*, 1993c), success of coronary artery angioplasty (Geschwind *et al.*, 1994), effects of intra-aortic balloon pumping (Ishihara *et al.*, 1992a; Ishihara *et al.*, 1992b), and success of thrombolysis (Crea *et al.*, 1993), have been conducted in human coronary arteries. The side-mounted ultrasonic velocity catheter has also been used to study endothelial function in the pulmonary artery of children with abnormal pulmonary hemodynamics (Celermajer *et al.*, 1993b). In 1989, Segal *et al.* developed a flow-directed Doppler pulmonary artery catheter that could measure instantaneous and continuous CO. The catheter was evaluated in both an *in vitro* pump model and an animal model and subsequently used in humans (Segal *et al.*, 1990; Segal *et al.*, 1991). The Doppler catheter CO correlated well with CO obtained with the EME, thermol dilution, and Fick methods. In addition to instantaneous blood flow velocity, the catheter system also provides instantaneous diameter and velocity profile measurements.

The most recent addition to the family of intravascular ultrasonic (IVUS) velocity devices, and now the most widely used, is the Doppler-tipped guidewire produced by Cardiometrics, Inc. (Mountain View, CA, USA, gated sine-wave). The 0.014 or 0.018 inch diameter angioplasty guidewire has an end-mounted 12 MHz crystal (Fig. 6.21). The Doppler-tipped guidewire has been validated in the coronary artery of dogs (Doucette *et al.*, 1992; Sudhir *et al.*, 1993) and used extensively to study coronary artery physiology in humans (see Chapter 17) (Ofili *et al.*, 1991; Segal *et al.*, 1992; Kern *et al.*, 1993a; Kern *et al.*, 1993b; Ofili *et al.*, 1993; Miller *et al.*, 1994; Kume *et al.*, 2008; Torii *et al.*, 2009), and was the topic of a book by Kern (1998). One of the most recent uses of the IVUS velocity guidewire is the determination of coronary blood flow reserve (von Mering *et al.*, 2004; Pepine *et al.*, 2007) in patients undergoing elective cardiac catheterization. Coronary blood flow and the reactivity of the microcirculation can also be assessed non-invasively with transthoracic echocardiology (Kiviniemi, 2008).

ELECTROMAGNETIC TRANSDUCERS

Although the catheter-tip electromagnetic flow probe is seldom used now, we believe it is important to review its development and use. There was, however, a recent publication by Noda *et al.* (2006) which studied the effects of nifedipine on systemic and pulmonary vascular impedance using the Millar multisensory (pressure and velocity) catheter. An electromagnetic flow probe mounted at the tip of a cardiac catheter was designed by Mills (1966). The physical analysis of such a system was made by Kolin (1944). This intravascular probe uses the same principle as the external electromagnetic cuff probe (Mills, 1972). The magnetic field is external to the probe, so the signal is

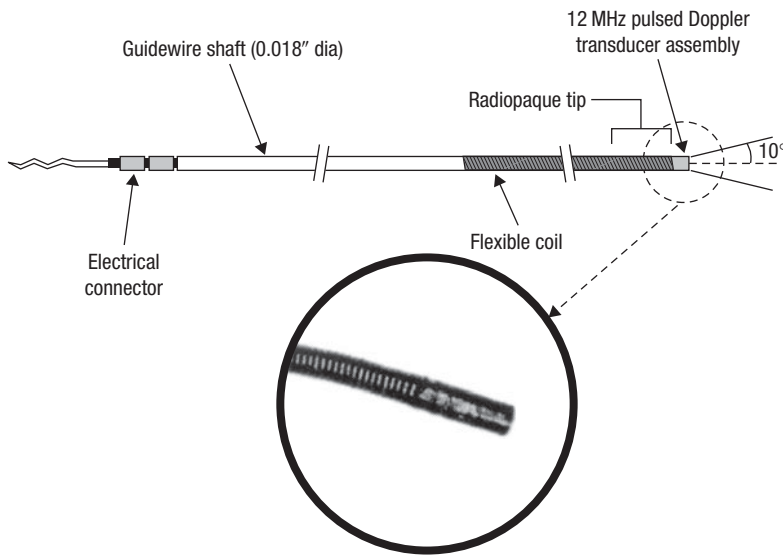


Figure 6.21 Drawing of the Cardiometrics velocity guidewire (0.014 or 0.018 in). The Doppler crystal is mounted at the tip of a flexible coil, which may be shaped into a curve to provide steerability within the arterial tree. Reproduced from Doucette *et al.* (1992)

generated by the velocity of blood flowing around and not through it.

The Mills catheter-tip probe was used to measure blood velocity in the venous (Wexler *et al.*, 1968) and arterial (Gabe *et al.*, 1969) systems of humans. The measurements of arterial pressures and velocity made by Gabe and associates led to studies of arterial pressure–flow relationships and vascular impedance by Mills *et al.* (1970) and later by Nichols *et al.* (1977b). The applications of the probe in the study of the central circulation in humans under control conditions and during the introduction of several different interventions have also been investigated (Mason *et al.*, 1970).

Nichols *et al.* (1977a) used a velocity catheter first introduced by Bond and Barfoot (1967) to assess LV performance in humans. In the study, LV volume was determined by subtracting the integral of the flow trace at each sampling interval from the end-diastolic volume measured from a biplane left ventriculogram. By utilizing a thin-walled spherical model for the ventricle, instantaneous velocity of circumferential fiber shortening, wall tension, and circumferential length were calculated. It was concluded that a three-dimensional construction of these variables during ejection was readily obtainable during diagnostic cardiac catheterization to provide a sensitive and comprehensive description of myocardial function in humans. We (Nichols *et al.*, 1977a; Nichols *et al.*, 1977b; Christie *et al.*, 1978; Deloskey *et al.*, 1978; Pepine *et al.*, 1978; Pepine *et al.*, 1979) also used a modified version of the probe to measure blood flow velocity at various sites in the vascular system of patients undergoing diagnostic cardiac catheterization. The probe had a 5.0 cm leader similar to the one used by Uther and co-workers (Uther *et al.*, 1973), as well as a lumen for measuring pressure simultaneous with the velocity. Millar pressure sensors have been mounted at the tip of the same catheter

that carries the velocity probe (Jacobs *et al.*, 1972; Murgo *et al.*, 1975). Recordings obtained with this instrument in the aorta and left ventricle of patients were reported by Murgo *et al.* (1975). Recent advances by Millar have yielded multisensor catheters that include not only multiple pressure sensors, but also catheter-mounted electromagnetic flow velocity probes. The Millar velocity probes were evaluated in dogs (Nichols *et al.*, 1985b) and humans (Nichols *et al.*, 1980b), and can be used in conjunction with any commercially available EMF. The multisensor pressure–velocity catheters are designed for use in the pulmonary artery or aorta (Fig. 6.22). Since the introduction of the multisensor pressure–velocity catheter, it has been used to measure pressures and flows in normal humans (Murgo *et al.*, 1980a; Murgo *et al.*, 1980b; Murgo *et al.*, 1981a; Murgo *et al.*, 1981b; Merillon *et al.*, 1984; Murgo and Westerhof, 1984; Nichols *et al.*, 1985a; Condos *et al.*, 1987; Murgo and Westerhof, 1987), kangaroos (Avolio *et al.*, 1984; Avolio *et al.*, 1985b; Nichols *et al.*, 1986a; O'Rourke *et al.*, 1986b), dogs (Nichols *et al.*, 1985b), and primates (Latham *et al.*, 1987; Latham *et al.*, 1988a). Other studies with the multisensor catheter in humans include various cardiac diseases (Murgo *et al.*, 1980b; Nichols *et al.*, 1981; Brin and Yin, 1984; Merillon *et al.*, 1984; Laskey *et al.*, 1985; Yaginuma *et al.*, 1986; Laskey and Kussmaul, 1987; Kussmaul *et al.*, 1988; Kelly and Fitchett, 1992; Nichols and Pepine, 1992), hypertension (Merillon *et al.*, 1982a; Merillon *et al.*, 1982b; Merillon *et al.*, 1982c; Nichols *et al.*, 1986b; Ting *et al.*, 1986; Ting *et al.*, 1995a; Ting *et al.*, 1995b), and the cardiovascular effects of various drugs (Klinke *et al.*, 1980; Gundel *et al.*, 1981; Yin *et al.*, 1983a; Yin *et al.*, 1983b; Merillon *et al.*, 1984; Yaginuma *et al.*, 1986). Examples of pressure and flow waves recorded in two patients using the multisensor catheter are shown in Figs 6.23 and 6.24. One of the patients had aortic

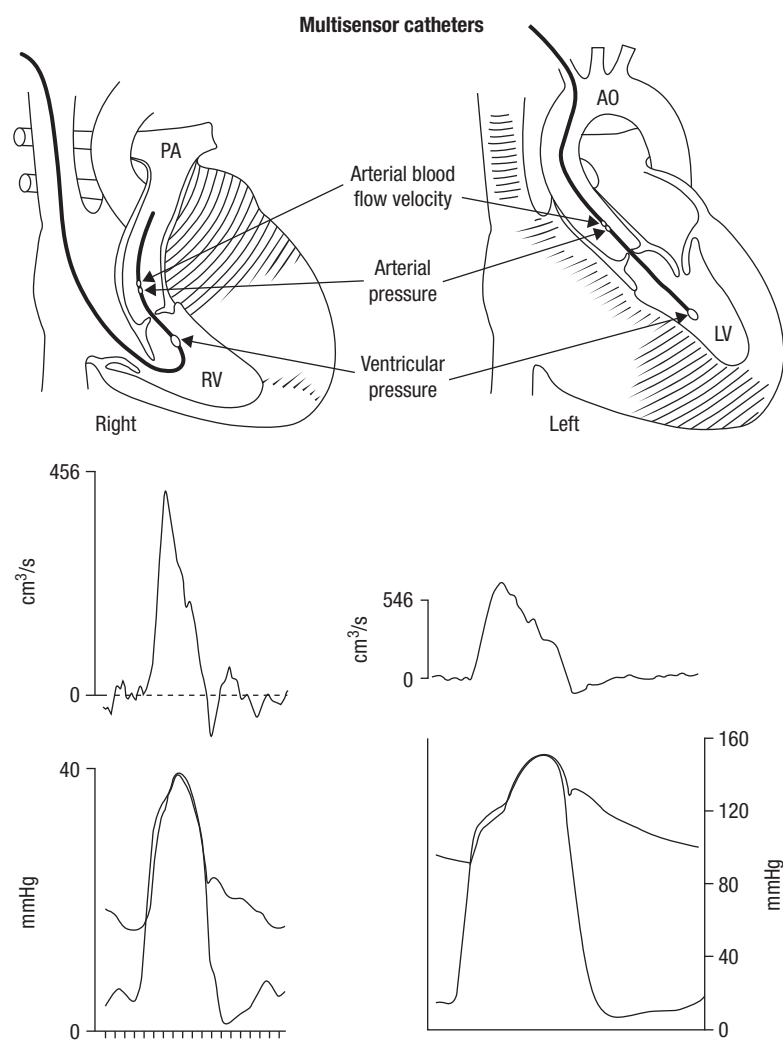


Figure 6.22 Pressure and flow waves measured with Millar multisensor pressure–velocity catheters in the right ventricle and pulmonary artery (left) and the left ventricle and ascending aorta (right). The arterial pressure and velocity sensors are mounted at the same site in each catheter

insufficiency (Fig. 6.23) and the other had hypertrophic cardiomyopathy (Fig. 6.24).

MEASUREMENT OF ARTERIAL DIAMETER

It is clear from Chapters 3 and 4 that the measurement of instantaneous diameter change is essential in the study of dynamic elastic properties of arteries (McDonald, 1974; Dobrin, 1978; O'Rourke, 1982a; O'Rourke, 1982b; Milnor, 1989; Pearson *et al.*, 1994; Stefanadis *et al.*, 1995a; O'Rourke and Nichols, 2005; Reneman *et al.*, 2005). The arterial diameter waveform is almost identical to the pressure waveform (Fig. 6.25, page 162), so the instrumentation used to measure this variable must have dynamic frequency characteristics similar to those used to measure other pulsatile hemodynamic variables (see Chapter 11). The mass and/or stiffness of strain gauge-type diameter transducers that attach directly to the artery must be kept to a minimum to prevent distortion of the

natural movement of the vessel wall. This presents problems since arterial walls change their radius by less than about 10 percent, while the pressure within the arteries may change by 50 percent with each cardiac cycle (Peterson, 1966).

The instruments used to measure phasic arterial diameter are extremely varied. The major categories are electrical (resistance and inductance), optical, ultrasonic, and magnetic resonance (MRI), and computed tomography (CT). Some of the earlier techniques (electrical and optical) used in animal models are almost or totally obsolete, but will still be briefly discussed here. Most of the techniques have been used both *in vivo* and *in vitro*, and some have been modified for intra-arterial diameter measurements. The methods designed and used predominately in humans are ultrasonic (B-mode and transesophageal echocardiography (TEE)) (Rydén *et al.*, 2005; Giannattasio *et al.*, 2008; Vermeersch *et al.*, 2008; Leye *et al.*, 2009; Shiran *et al.*, 2009), MRI (das Chagas de Azevedo *et al.*, 2005; El-Barhoun *et al.*, 2009), and

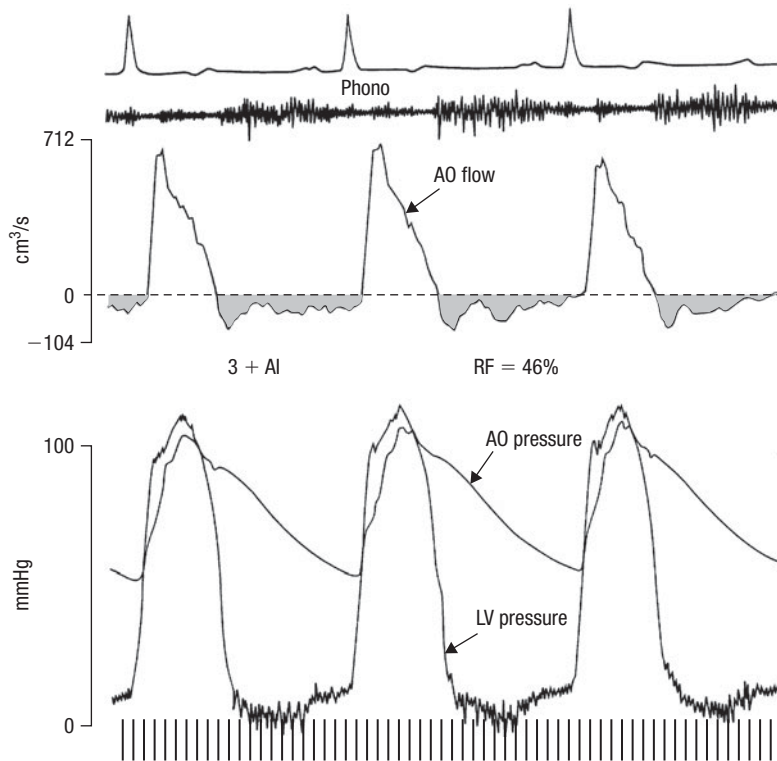


Figure 6.23 Left ventricular and ascending aortic pressures, aortic blood flow and heart sounds recorded with a Millar multisensor catheter in a patient with aortic insufficiency (AI, 3+). Total forward stroke volume is the area under the flow curve above the zero baseline, and regurgitant flow the area below the baseline (shaded area). Effective stroke volume is the difference between total and regurgitant volumes. Regurgitant fraction (RF) was 46 percent

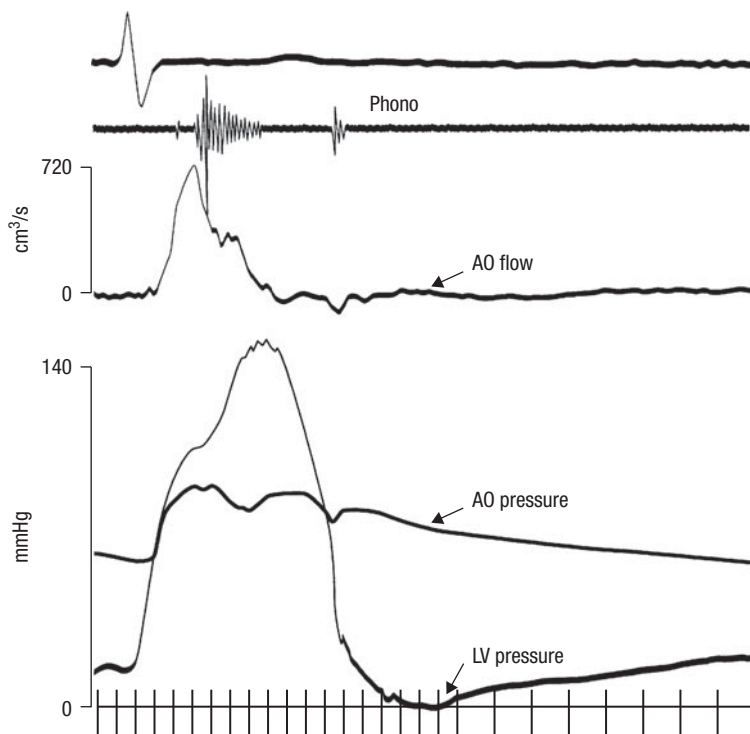


Figure 6.24 Pressures, flow, and heart sounds recorded with a Millar multisensor catheter in a patient with hypertrophic cardiomyopathy. More than 90 percent of the stroke volume is ejected during the first half of ventricular ejection

computed tomography (Mao *et al.*, 2008; Blount and Hagspiel, 2009; Wolak *et al.*, 2009). For detailed reviews of the various methods used to measure organ dimensions, see Peterson (1966), Bergel (1972), Webster (1978), Milnor (1989), and Cassottana *et al.* (1997).

Resistance strain gauges

As stated above, when a wire (or foil) is stretched or strained, a change in cross-sectional area (and thus resistance) is produced. If the wire is set as one arm of a

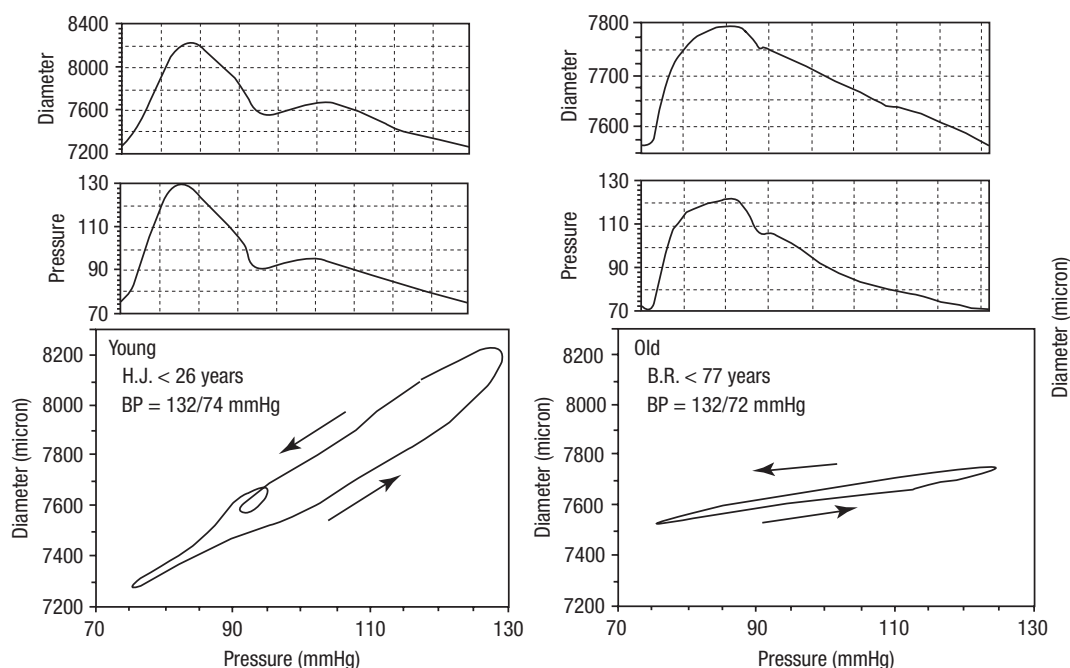


Figure 6.25 (Upper) Common carotid artery diameter and pressure waveforms recorded in a 26-year-old subject (left) and compared to measurements recorded in an elderly subject (right) during one cardiac cycle. (Lower) Representative pressure–diameter relations for each subject. Owing to viscoelasticity, the pressure–diameter relations (loops) recordings exhibit hysteresis. Reproduced from Giannattasio *et al.* (2008)

Wheatstone bridge, when the wire is strained by an applied force the bridge is unbalanced, and the resultant change in voltage is directly related to the applied force. Patel and his associates (Patel *et al.*, 1961; Patel *et al.*, 1973) used a resistance strain gauge designed by Mallos (1962) to study arterial pressure–diameter relations in dogs. The instrument utilized a metallic foil strain gauge cemented to a thin metal bridge. The bridge was arched between the legs of a caliper, which could be sutured to the outer wall of an artery. The static calibration of the instrument was essentially linear from 3.0 to 16 mm, and the dynamic frequency response was flat (65 percent) to 25 Hz. Greenfield and Patel (1962) and Greenfield and Griggs (1963) used a similar gauge to record the diameter changes of the ascending aorta and main pulmonary artery of humans.

Miniature, silicon semiconductor strain gauges have also been used in the development of instruments to measure vessel displacement. Peterson (1966) used tiny spring-temper, stainless steel cantilever beams to obtain vessel radius; Murgo *et al.* (1971), using a similar approach, developed a strain gauge device that could be inserted through a catheter and used to measure phasic internal arterial diameter changes.

Inductance strain gauges

A differential transformer is composed of a primary and a secondary coil within which a high-permeability alloy core

(or slug) can move freely. Movement of the core induces a voltage that is directly related to its motion. Gauges using differential transformers as displacement transducer elements are possibly the most successfully used variable-inductance devices (Neubert, 1963; Bergel, 1972; Milnor, 1989). The transformer can be made extremely small and fitted into the legs of a small, lightweight caliper (Gow, 1966; Dobrin and Rovick, 1969; Dobrin, 1978). Gow's (1966) electric caliper diameter gauge used jewel bearings as pivots to ensure that frictional forces were negligible. The static calibration of the gauge was linear, and the dynamic frequency response was flat (65 percent) to 20 Hz with minimal phase distortion. Differential transformers can be made small enough to be mounted on the tip of a catheter for intravascular diameter measurements (Pieper and Paul, 1969). The coupling of the gauge transformer is determined by the position of a ferromagnetic sleeve that slides over the transformer coils. This sleeve is mechanically linked to three braces that are spaced 120° apart around the instrument tip.

These instruments can only be used in large (>10 mm diameter) vessels and are probably not safe for use in humans (Bergel, 1972; McDonald, 1974).

Van Citters (1966) developed a mutual-inductance gauge to measure vessel diameter. The major advantage of this technique is that diameter measurement is made with small coils attached to either side of the vessel, thus eliminating the iron core. Alternating current is passed through one coil, and the magnitude of the current induced in the other is a function of the distance between

the two coils, which is a measure of external arterial diameter.

Optical methods

One optical method that has been used employs a television camera and video tracking circuits (Ngai and Winn, 1995). The television camera records a magnified image of the vessel, which is displayed on a television monitor, with the tracking band superimposed. The band is positioned across the diameter of the vessel. The output is a voltage analog proportional to the distance between the boundaries of the vessel's image. Measurements can be made of vessels of almost any size using this technique. Diameter of subendocardial arterioles and venules of a beating heart can be measured with a newly developed needle-probe videomicroscope with a charged-coupled device camera (Hiramatsu *et al.*, 1994).

A photoelectric device for contact-free recording of phasic diameter waveforms of exposed arteries *in situ* was developed by Wetterer *et al.* (1977b). The device has been used to study the dynamic elastic properties of various arteries in both dogs and humans. The light emitted by a 6W light bulb is collimated by a projection lens and directed onto the surface of a silicon photocell. The voltage drop caused by the photocell current flowing through an adjustable load resistor represents the signal voltage. This voltage decreases linearly with increasing area of the

shadow cast by the artery on the photocell, and is thus proportional to the diameter of the artery.

Ultrasonic methods

Perhaps the most widely used method of measuring phasic arterial diameter waveforms at present, especially in conscious animals and humans, uses high-frequency sound waves. An extensive discussion of the cardiovascular applications of ultrasound and the measurement of arterial diameter are given by Wells (1969), Webster (1978), and Reneman *et al.* (2005). The application of ultrasound to the measurement of blood flow velocity and diameter are also discussed in Chapter 8.

Two different techniques employing ultrasonic beams to measure arterial diameter have been reported. One depends on the time that elapses between emission of a short burst of ultrasound, and reception of the 'echo' from the arterial walls (Eriksen, 1987). This technique has been used to measure (non-invasively) diameter changes in both central (Stefanadis *et al.*, 1990; Lacombe *et al.*, 1992; Pedersen *et al.*, 1993; Astrand *et al.*, 2005; das Chagas de Azevedo *et al.*, 2005; Leye *et al.*, 2009; Shiran *et al.*, 2009) and peripheral arteries (Safar *et al.*, 1981a; Safar *et al.*, 1981b; Kelly and Fitchett, 1992; Pasierski *et al.*, 1992; Kool *et al.*, 1994; Pearson *et al.*, 1994; Reneman *et al.* (2005); Giannattasio *et al.*, 2008; Vermeersch *et al.*, 2008) of humans (Fig. 6.26). The reports by Pasierski *et al.* (1992) and Pearson *et al.* (1994) used a gastroscope

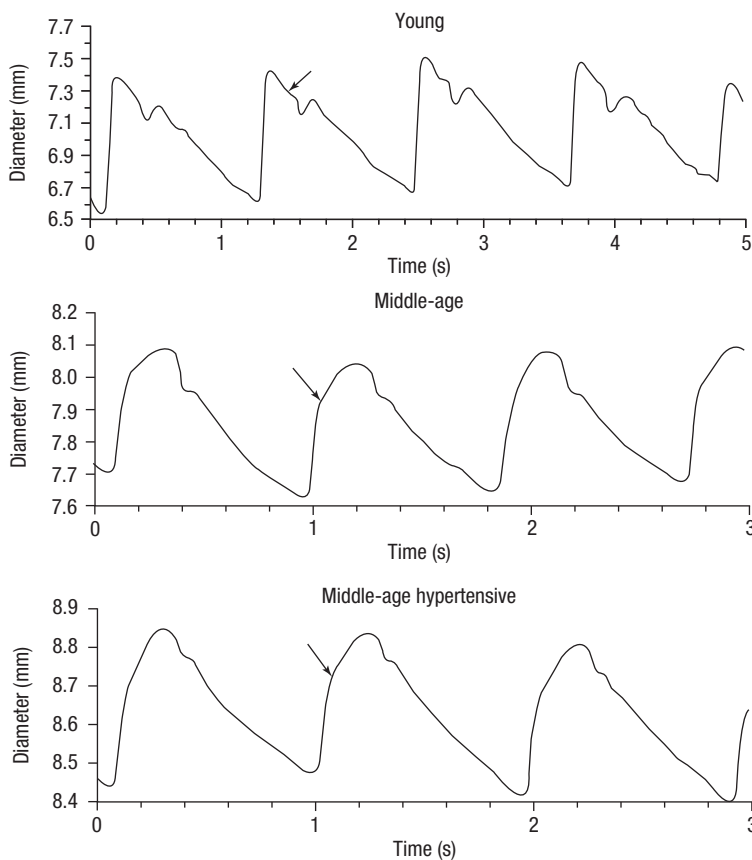


Figure 6.26 Diameter waveforms as recorded in the common carotid artery of a young (top) and a middle-aged normotensive volunteer (middle), and a middle-aged hypertensive patient (bottom). (Arrows indicate beginning of reflected wave). During the recordings the volunteers and the patient were fully relaxed. Distension (Δd) and circumferential strain ($\Delta d/d$; d , diameter) were reduced at even a higher blood pressure, in normotensive volunteer at older age, reflecting increased arterial stiffness. In hypertension, arterial stiffness is further increased as indicated by the slightly reduced distension and the substantially reduced circumferential strain at a significantly higher blood pressure. Reproduced from Reneman *et al.* (2005)

with a 5.0 MHz crystal mounted at the tip to measure descending aortic diameter and Leye *et al.* (2009) and Shiran *et al.* (2009) used (transesophageal echocardiography) to measure LV outflow tract diameter. The phase-locked echo-tracking systems developed and utilized by Sonesson *et al.* (Sonesson *et al.*, 1993; Sonesson *et al.*, 1994), Länne *et al.* (Länne *et al.*, 1992a; Länne *et al.*, 1992b) and Hansen *et al.* (1995) hold particular promise for accurate measurement of pulsatile arterial diameter change non-invasively. Briefly, an echo-tracking phase-locked loop restores the position of an electronic gate relative to the moving echo, and the compensatory movement of the gate yields the movement of the echo. Two electronic markers, representing the electronic gates, automatically lock to the luminal interfaces of the echoes from the anterior and posterior vessel wall. The markers are shown in the real-time picture, to indicate the registration level. The output signal from the echo-tracking circuits represents the distance between the vessel walls. The smallest detectable movement is 8.0 μm (Benthin *et al.*, 1991).

The other technique measures the transmission time between two crystals positioned on each side of the arterial wall and is best suited for studies in animals (Hartley *et al.*, 1978; Vatner, 1985; Barra *et al.*, 1993; White *et al.*, 1994; Armentano *et al.*, 1995). In both cases, the ultrasound is emitted from crystals made of a piezoelectric material. Phasic-diameter waveforms can be recorded accurately in arteries as small as 2.0 mm outside diameter using the transit-time method. This makes the method suitable for recording coronary artery diameter changes in dogs (Vatner, 1985). The ultrasonic transit-time method has also been employed to obtain intra-arterial diameter using a catheter-tip device (Stegall, 1974). Devices have also been developed for intravascular imaging (Tobis *et al.*, 1989; Bank *et al.*, 1995) and are being produced commercially by Inter-therapy (Costa Mesa, CA, USA).

Two of the most recent non-invasive methods to measure aortic diameter use MRI (Fig. 6.27) (Vulliemoz *et al.*, 2002; das Chagas de Azevedo *et al.*, 2005; El-barhoun

et al., 2009) and computed tomography (Mao *et al.*, 2008; Blount and Hagspiel, 2009; Wolak *et al.*, 2009). In addition to time-varying diameter, the method also provides cross-sectional area, volume flow, pulse wave velocity, and local vessel wall elastic properties (Fig. 6.28). These methods which are gaining in popularity are, however, very expensive and data collection is time consuming.

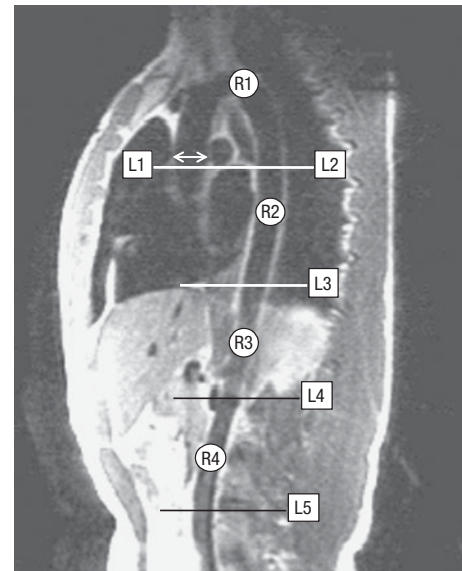


Figure 6.27 Sagittal image of the aorta from the aortic root to the level of bifurcation. L1–L5 indicate the level of cine phase contrast magnetic resonance image (MRI) acquisition. L1, the ascending aorta, located 1 cm distal to the aortic valve; L2, the descending aorta, in line with L1; L3, at the level of the diaphragm; L5, 3 cm above the level of the aortic bifurcation; and L4, midway between L3 and L5. R1–R4 indicate the region of pulse wave velocity (PWV) measurement. R1, the aortic arch; R2, the thoracic-descending aorta; R3, the mid-descending aorta; and R4, the abdominal aorta. The image was acquired by a blood suppressed fast spin echo sequence, averaged over 16 heart beats. Courtesy of Dr Stacey Hickson

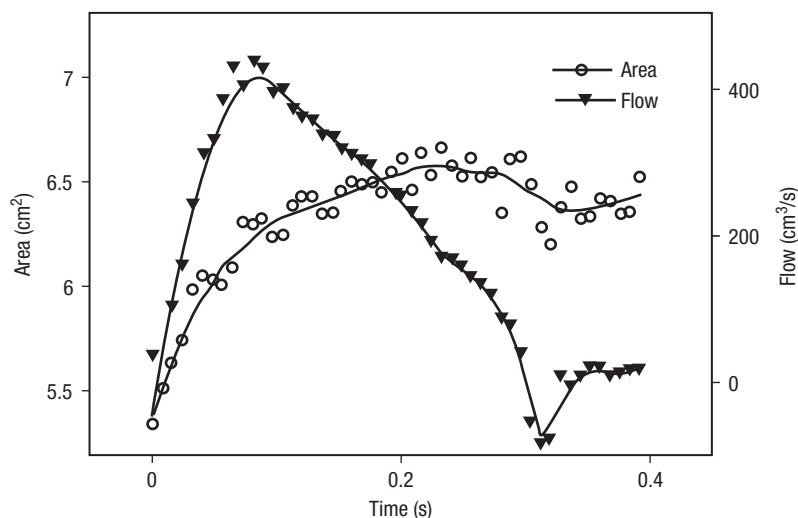


Figure 6.28 Cross-sectional area and flow measured in the ascending aorta during systole and early diastole of a middle-aged healthy man using magnetic resonance imaging. Reproduced from Vulliemoz *et al.* (2002)

Pulsatile pressure–flow relations

The relation of flow to an oscillatory pressure gradient	168	Development of the pressure gradient time-derivative method	171
The calculation of flow from the pressure gradient	169		
The effect of the size of the artery on the flow pattern	171	The validity of the Womersley equations when applied to arterial blood flow	174
The values of α found in the circulation	171	The concept of vascular impedance	175

One of the conditions for the application of Poiseuille's equation (eqn 2.7) is that the flow is 'steady', i.e. that it does not vary with time (a condition also known as 'stationary flow'). It has long been realized that the equation could not be applied to arterial flow, which is highly pulsatile. While flow measurements were confined to the determination of mean flow only with the use of such instruments as Ludwig's stromuhr, the problem aroused little interest. This probably accounts for the complete neglect of pioneering mathematical analyses such as that of Witzig (1914). A somewhat empirical approach to the problem was made by Shipley *et al.* (1943) as a result of their detailed studies of pulsatile flow patterns in a wide range of arteries using an orifice meter. They could find no clear relation between the pressure and flow waveforms and, with considerable insight, suggested that flow was more closely related to the derivative of the pressure rather than to the pressure itself. They did not pursue the quantitative relationship and their comment seems to have escaped notice as it was generally assumed that an increase in pulsatile flow would cause an increase in pulsatile pressure. This is indeed true when there are no reflections in the system.

In the early 1950s, when McDonald became interested in pulsatile pressure–flow relations, most hemodynamics were based on simple applications of Windkessel theory, which treated the arterial tree as a chamber in which pressure rose as blood was pumped into it. As an approximation for calculating the stroke volume, this analysis was quite useful, but it was quite inadequate to predict the instantaneous flow rate, as Peterson (1954) showed. Later, however, several studies have shown the suitability of three- and four-element models of the arterial input impedance to describe the relationship between aortic pressure and flow (Burkhoff *et al.*, 1988; Cholley *et al.*, 1995b; Westerhof *et al.*, 2006; Lankhaar

et al., 2006; Burattini and Di Salvia, 2007; Mohiuddin *et al.*, 2007; Westerhof *et al.*, 2009). Once model parameters are found, flow can be computed from the measured pressure by simulating the model (Wesseling *et al.*, 1993; Harms *et al.*, 1999; Jellema *et al.*, 1999; Gerhardt *et al.*, 2000; Jansen *et al.*, 2001). When pulsatile blood flow velocity and pressure are measured simultaneously and continuously at the same site in the femoral artery, we find a situation like that illustrated in Fig. 7.1. Although the increase in flow starts more or less synchronously with the pressure rise, the peak flow velocity precedes the peak pressure. A similar relationship also exists in the ascending aorta (Perlini *et al.*, 1991). However, the gap between peak flow and peak pressure is more pronounced, especially in older individuals and those with hypertension. On the basis of previous reasoning, this would appear to violate the laws of inertia. The apparent anomaly was resolved when it was realized that it is the pressure gradient along the artery that determines pulsatile flow (McDonald, 1955) just as it determines steady flow in Poiseuille's equation. This elementary observation can only be described adequately, in the late JBS Haldane's phrase, as a 'blinding glimpse of the obvious'.

The pressure gradient may be determined by recording the pressure at two sites a short distance apart along an artery (Fig. 7.2A) and subtracting the pressure at the downstream site from that at the upstream site at each moment (the derivation of the gradient over a finite distance from the gradient at a point, dP/dz , is discussed below). Owing to the fact that the pressure wave generated in the aorta by the expulsion of blood by the ventricle travels along the arteries, the crest of the wave reaches the upstream recording site a short time before it reaches the downstream site. At this time, the pressure is lower at the first than at the second site and the pressure gradient slopes in this direction. The situation rapidly reverses, and

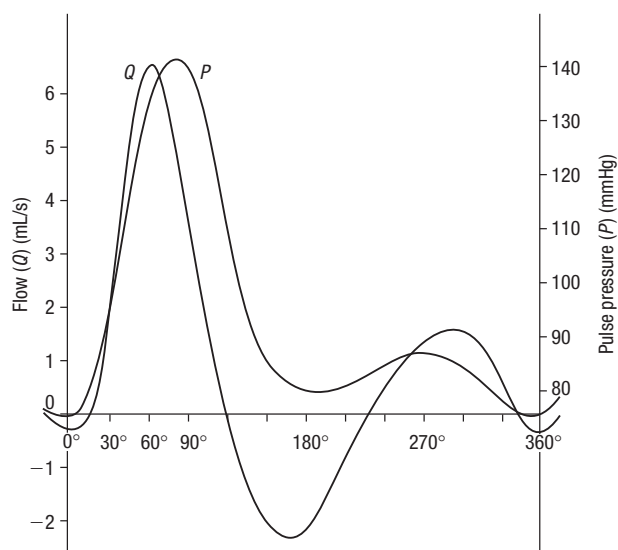


Figure 7.1 A flow velocity pulse (Q) and the arterial pressure pulse (P) recorded simultaneously in the femoral artery of a dog. Flow velocity was recorded by high-speed cinematography, but the corresponding volume flow (mL/s) is plotted as the ordinate (left). Although superficially similar in shape when plotted on a comparable scale, the fact that the peak flow occurs before the pressure peak shows that there is no simple relation between these two curves. The flow is, in fact, determined by the pressure gradient (see text). The pulse frequency was 2.75 Hz in this experiment. The abscissa representing time is plotted as fractions of the cycle length, i.e. as degrees of arc. With pulse frequencies of this order one degree of arc is about 1.0 ms

when the crest has reached the second site the pressure gradient is in the opposite direction. Any secondary ‘bumps’ in the waveform will cause similar oscillations. The resultant pressure gradient, therefore, is one that oscillates about a mean shown in Fig. 7.2B. As we have a traveling wave in all arteries, all arterial pressure gradients will be of this form. This pressure gradient is directly related to the flow that is occurring just as the gradient and flow always occur together in Poiseuille flow.

The factors involved in moving a fluid by a pressure that oscillates in this way are clearly much more complex than those required for steady flow. The mass of fluid may be thought of as being at rest at the start of the cycle – as indeed it is, virtually, in the ascending aorta at the end of diastole. When a force in the form of a pressure is applied, it will first appear to resist movement because of its inertia. As it accelerates, it acquires ever-increasing momentum; it will continue to accelerate as the pressure gradient increases. With increasing velocity, the viscous drag also increases. If the pressure gradient suddenly fell to zero, the momentum of the blood would keep it moving until the opposing viscous forces brought it slowly to rest. In fact, because we are dealing with a traveling wave, the gradient reverses when the pressure

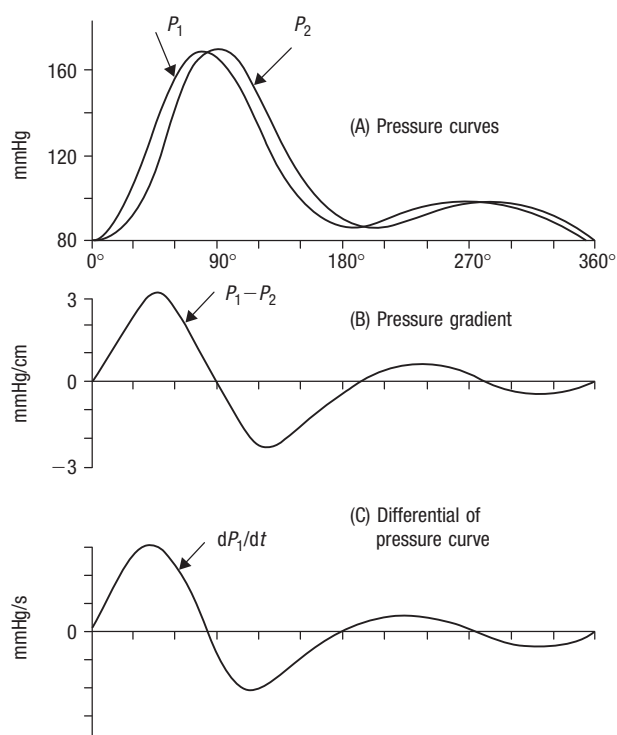


Figure 7.2 A diagram that shows how a traveling pressure wave creates an oscillatory pressure gradient. (A) Two pressure waves recorded a short distance apart in the femoral artery of a dog. The downstream wave is identical to the upstream one. (B) The pressure gradient (mmHg/cm) derived by subtracting the pressure at the downstream site from that at the upstream one at 15° intervals and dividing by the distance between recording sites. (It should be noted that this gives a gradient opposite in sign to the usual mathematical convention for slopes.) (C) The derivative with respect to time of the upstream pressure wave (dP_1/dt). The form of this curve and that of the pressure gradient (which in the limiting case of a very small interval is the derivative with respect of distance, dP/dz) is very similar. In this case, the only difference is due to the transmission time over the interval necessary to determine the gradient. The vertical lines demonstrate the small phase differences that this creates. In the presence of the usual distortion of the wave as it travels, this similarity between time derivative and space derivative no longer holds with any precision (see text)

has passed its peak, as illustrated in Fig. 7.2B. This reversal causes a rapid deceleration of the flow; only if the reversal of the gradient continues after the forward flow has been brought to a halt does it lead to a reversal of flow. This is commonly seen in the femoral artery and is illustrated in Fig. 7.7 (p. 170). In the ascending aorta, however, this reversal does not occur to any significant extent owing to the presence of valves. Noble (1968), Murgo *et al.* (Murgo *et al.*, 1975; Murgo *et al.*, 1980a; Murgo *et al.*, 1980b), Nichols *et al.* (Nichols *et al.*, 1980a; Nichols *et al.*, 1980b), Pasipoularides *et al.* (1984), Takazawa (1987), and Murgo (1998) made high-fidelity

recordings of simultaneous pressures in the outflow tract of the left ventricle and the root of the aorta of dogs and humans. Initially, the ventricular pressure is higher than that in the aorta immediately after the aortic valve opens (Fig. 7.3). During the latter part of systole, as the ventricular pressure begins to fall, the aortic pressure is higher, i.e. the pressure gradient has reversed. The point of reversal occurs close to the first systolic shoulder of the aortic and left ventricular pressure waves; the inflection point occurs later. Murgo *et al.* referred to the left ventricle–aorta gradient as the ‘impulse’ gradient. As the forward pressure gradient is greatly in excess of that required to maintain a steady flow, there is initially a rapid acceleration and the reversal of the gradient is related to the deceleration of flow. Noble (1968) has attempted to interpret this entirely in terms of relation of the duration of active ventricular contraction and the momentum imparted to the blood, but this biphasic pattern of the pressure gradient will occur wherever a pressure wave is propagated along a vessel. Confusion on this matter has been widespread in the physiologic literature. The pressure difference between ventricle and aorta is very small compared with the large pressure swing inside the ventricle, and Wiggers (1928) had not found it possible to define the difference clearly. In the diagram based on his records, which became standard in

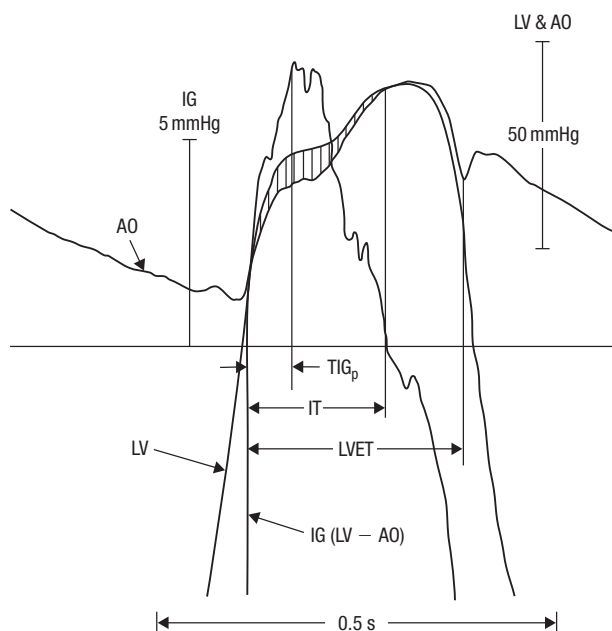


Figure 7.3 Simultaneous high-fidelity measurement of left ventricular (LV) and aortic (AO) pressures in a human during cardiac catheterization. The hatched area represents the normal impulse gradient (IG), which is further displayed by an electronically derived signal representing the difference between LV and AO pressure. TIG_p, time to peak impulse; IT, impulse time; LVET, left ventricular ejection time. Reproduced from Murgo *et al.* (1975)

physiologic textbooks, the pressure difference was assumed to be analogous to that of steady flow. The ventricular pressure was thus always shown as exceeding aortic pressure throughout the whole period of systolic ejection. Using strain gauge manometers attached to fluid-filled catheters, Spencer and Greiss (1962) reported that the pressure in the ventricle exceeded the aortic pressure during a brief period after the opening of the aortic valves, and thereafter the ventricular pressure was less than the aortic pressure. The brief period was a spike in the pressure recording and due to manometer artifact in a system that was underdamped and of a relatively low resonant frequency. This is exactly analogous to over-swing of such a system in response to a step function of pressure (and the very rapid rise in ventricular pressure is, in effect, such a function) succeeded by a train of oscillations that are used to determine the frequency response of manometers (see Chapter 6). In fact, such after-oscillations can be seen in the Spencer and Greiss recordings. With the introduction of the high-fidelity catheter-tip and disk manometers used by Noble (1968), Murgo *et al.* (1975), Nichols *et al.* (1978), and Wang *et al.* (2003), the true form of aortic and ventricular pressure curves has been established. Using high-fidelity pressure catheters, Pasipoularides *et al.* (1984) and Murgo (1998) found similar pressure gradients within the ventricle itself between the left ventricular cavity pressure and outflow tract pressure (Fig. 7.4). During contraction, there is an early systolic pressure gradient that is rapidly generated between the left ventricular cavity and the outflow tract. The peak gradient occurs just before the peak of aortic outflow and probably coincides with maximum blood flow acceleration (Fig. 7.4). O'Rourke *et al.* (1968) showed that the differential pressure can be distorted if pressure is measured end-on, rather than side-on to the direction of flow, and when measured side-on, may be distorted by a Venturi effect at high flow velocity (> 80 cm/s) (Fig. 2.10). O'Rourke and Milnor (1971) showed that differential pressure in the pulmonary artery could be distorted by geometric taper of this vessel. Takazawa (1987) showed that high flow velocity in the human aorta could contribute to the early systolic shoulder on the aortic pressure wave from which augmentation index and return of wave reflection are calculated.

We saw in Chapter 2 that the solution of the factors involved in the laws for steady flow could not be determined without an adequate mathematical theory. In the case of oscillating flow, it was essential to have a mathematic solution of the problem from the beginning. Such a solution was made by Womersley (Womersley, 1955a; Womersley, 1955b) and has been discussed in relation to the velocity profile in Chapter 2. The flow in a tube, in which any dilatation is neglected, was considered when the pressure gradient varied periodically in the form of a sine wave. With a Fourier series, the flow related to any periodic pressure gradient could be described.

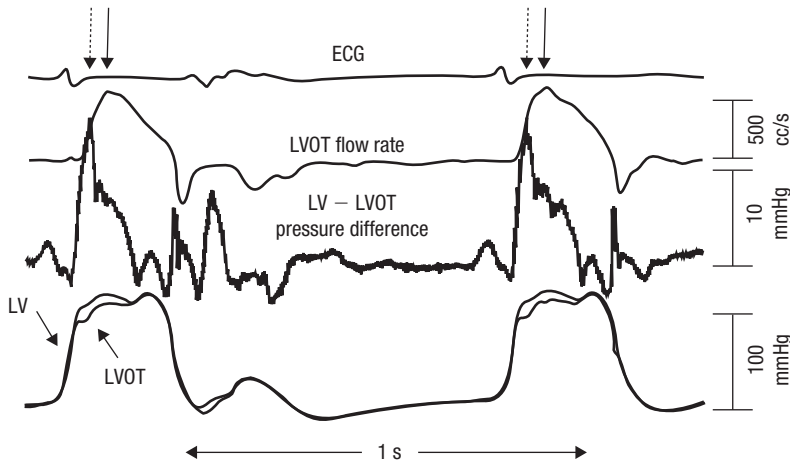


Figure 7.4 Simultaneous high-fidelity pressure signals measured in the left ventricular (LV) cavity and the outflow tract (LVOT) in a human with normal LV and valvular function during a sinus beat, a premature ventricular contraction and a subsequent sinus beat. Also shown is LVOT blood flow velocity and the LV – LVOT pressure difference (or gradient). The broken arrows point to the peak of the pressure difference and solid arrows point to the peak LVOT blood flow velocity. Reproduced from Murgu (1998)

THE RELATION OF FLOW TO AN OSCILLATORY PRESSURE GRADIENT

The equation of motion of a liquid, if the pressure gradient varies with time, is

$$\frac{\partial^2 w}{\partial r^2} + \frac{1}{r} \frac{\partial w}{\partial r} - \frac{1}{v} \frac{\partial w}{\partial t} = -\frac{1}{\mu} \frac{\partial P}{\partial z}. \quad 7.1$$

If we take the pressure difference over a distance L as the pressure gradient and represent it by a simple harmonic motion, then

$$\frac{P_1 - P_2}{L} = \frac{\partial P}{\partial z} = A^* e^{i\omega t},$$

and the solution for the velocity, w , of the lamina at a distance ($y = r/R$) from the axis is

$$w = \frac{A^*}{i\omega\rho} \left(1 - \frac{J_0(\alpha y i^{3/2})}{J_0(\alpha i^{3/2})} \right) e^{i\omega t},$$

since

$$\alpha^2 = \frac{R^2 \omega \rho}{\mu} = \frac{R^2 \omega}{v}.$$

To obtain the volume flow Q (in cm^3/s) it is necessary to integrate the velocity across the lumen of the tube:

$$Q = \frac{\pi R^2 A^*}{i\omega\rho} \left(1 - \frac{2J_1(\alpha i^{3/2})}{\alpha i^{3/2} J_0(\alpha i^{3/2})} \right) e^{i\omega t}, \quad 7.2$$

where J_0 and J_1 are Bessel functions of order zero and one, respectively. The expression in the braces was termed $[1 - F_{10}]$ by Womersley (Womersley, 1955a; Womersley, 1955b) and was tabulated by him in his monograph of 1957a.

When the real part of the pressure gradient is written as $M \cos(\omega t - \phi)$, eqn 7.2 can be written as

$$Q = \frac{\pi R^2}{\omega\rho} M [1 - F_{10}] \sin(\omega t - \phi) \quad 7.3$$

and may be calculated in this form using the real and imaginary parts of $[1 - F_{10}]$. The physical interpretation of the equation is, however, easier if the function $[1 - F_{10}]$ is expressed in modulus (M'_{10}) and phase (ε_{10}) form, so that

$$[1 - F_{10}] = M'_{10} e^{i\varepsilon_{10}}.$$

Thus

$$Q = \frac{\pi R^2}{\omega\rho} M M'_{10} \sin(\omega t - \phi + \varepsilon_{10}). \quad 7.4$$

The analogy with Poiseuille's equation is more easily seen if the term α^2 is substituted:

$$Q = \frac{\pi R^4 M M'_{10}}{\mu \alpha^2} \sin(\omega t - \phi + \varepsilon_{10}), \quad 7.5$$

because $M'_{10}/\alpha^2 \rightarrow 1/8$ as $\alpha \rightarrow 0.0$ and $\varepsilon_{10} \rightarrow 90^\circ$ so that $M \sin(\omega t - \phi + 90^\circ)$ becomes $M \cos(\omega t - \phi)$ which is $\Delta P/L$ or dP/dz .

The kinematic similarity of oscillatory flows in tubes is characterized by the non-dimensional parameter α which, as seen above, is a function of the radius, frequency, and kinematic viscosity. Assuming the viscosity to be constant, it is clear that the same α may arise from a lower frequency in a wide tube as from a higher frequency in a narrow tube. The velocity profiles will then be identical; the volume flow from a given gradient, however, is also a function of the cross-sectional area, so will only be similar in its phase relations. Although the dimensionless parameter α does not appear explicitly in eqn 7.4, it must be remembered that both M'_{10} and ε_{10} are determined by it; they should properly be written $M'_{10}(\alpha)$ and $\varepsilon_{10}(\alpha)$, but the form has not been used in order to avoid making the equations more cumbersome than they need to be. The variations of M'_{10} , M'_{10}/α^2 and ε_{10} with α are shown in Fig. 7.5. The ratio M'_{10}/α^2 is plotted as a ratio of the Poiseuille constant $1/8$.

To consider the behavior of the oscillating components of flow in an artery, the best physical analog is that of a tube connecting two water reservoirs. If these reservoirs

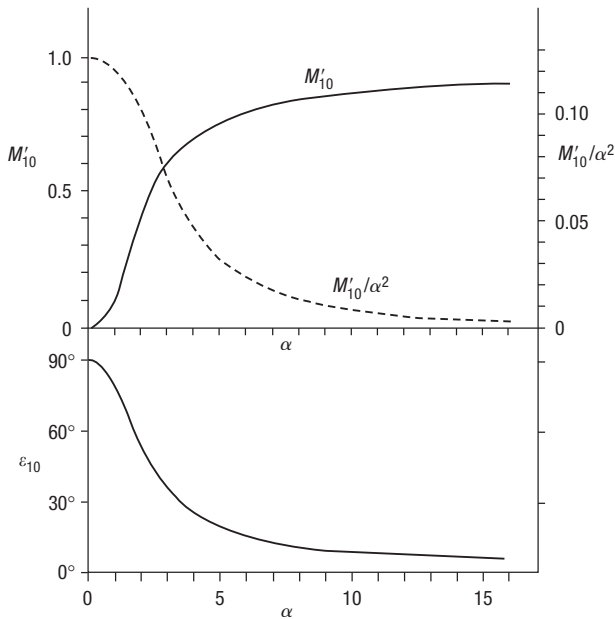


Figure 7.5 The parameters M'_{10} , M'_{10}/α^2 , and ε_{10} as functions of the non-dimensional constant α . As α approaches zero, M'_{10}/α^2 approaches $1/8$, the numerical constant in the Poiseuille equation. The parameter ε_{10} approaches 90° as α approaches zero, whereas at high values of α , ε_{10} approaches 0° . M'_{10} approaches 1.0 at high values of α . Reproduced from McDonald (1974) as modified by Minor (1989)

are supported on either end of a beam that is rocked about a fulcrum in the center, the head of water, and thus the pressure gradient, will vary rhythmically. It is clear that as one reservoir rises, water will begin to flow out of it along the tube into the other reservoir. The flow, however, will build up more slowly than the upward movement of the reservoir, i.e. more slowly than the applied pressure, because of its inertia. Hence there is a phase lag, $(90 - \varepsilon_{10})^\circ$, of flow behind pressure. Equally, when the movement of the reservoir is reversed, the flow will continue in its original direction due to its momentum. As the rate of oscillation increases, it is easy to see that the amount of flow generated in the tube will also become smaller at the same head of water due to the inertial effect, and this is expressed quantitatively by the upper broken curve in Fig. 7.5. It will lag more and more behind the changes in the pressure head (lower curve, Fig. 7.5). This is a picture of the change in the type of flow when α increases due to the increase in frequency, and will be found in the higher harmonics of the arterial pressure gradient. The reason that an increase in α , due to an increase in the size of the tube, produces effects similar to those of increased frequency, has already been discussed (see Chapter 2) when we considered the velocity distribution across the tube – the velocity profile. Reference to Figs 2.19 and 2.20 will show that, with a large α , the velocity profile becomes flat across the central region of the tube with a region of disproportionately increased

shear at the wall. We may describe this simply as showing that in a large tube, the effect of viscosity is most evident near the wall while the central region moves almost like a solid mass. Hence, the inertia of this mass dominates the flow pattern in a large tube in much the same way as it does in a similar tube at a higher frequency. The reduction in oscillating flow with a given pressure gradient compared with that predicted by Poiseuille's law had been noted occasionally by previous workers and was explained as being due to an increase in 'apparent viscosity'. The analogy drawn here makes it clear that this 'damping' of flow is an inherent property of oscillating flow in any liquid and does not require the assumption of strange viscous properties. It should be remembered that the concept of 'damping' in this sense is only relative to the corresponding Poiseuille flow. The actual volume flow, Q , does in fact increase in a larger tube with oscillating flow because R^4 increases in magnitude much faster than M'_{10}/α^2 decreases.

THE CALCULATION OF FLOW FROM THE PRESSURE GRADIENT

With a characteristic arterial pressure gradient such as that shown in Fig. 7.2B measured in the femoral artery of a dog, the first step necessary is to resolve it into its harmonic components by a Fourier analysis (see Chapter 11). The first four harmonic terms of such a curve obtained using eqn 7.5 are shown graphically in Fig. 7.6A together with their resynthesis to form the original curve. Each pressure oscillation has a corresponding flow term that is also sinusoidal in form. To derive the total flow curve due to the pressure gradient as recorded, the harmonic components of the curve are added together and the result is shown in Fig. 7.6B. The final curve requires some further explanation because a mean flow has been added to it, and this is shown by lowering the line representing zero flow (Nichols, 1970; McDonald and Nichols, 1973; McDonald, 1974; Nichols and O'Rourke, 1990). This is necessary from the nature of the analysis. The harmonic terms are all sine waves, i.e. they have a mean value of zero, so the summation of a series of sine waves also can only sum to zero over a complete cycle. Where the mean value is not zero, it is represented in the Fourier series (eqn 11.2) by the constant term $A_0/2$. The flow through an artery must have a positive mean value because its function is to transport blood from the heart to the tissues of the body. The simplest way of representing this has been to take the mean value of the pressure gradient over the whole cycle and assume that this generates a steady flow of Poiseuille type that is added to the oscillating term. This method has two defects, one practical and one theoretic. The practical difficulty is that the mean positive pressure gradient is of very small dimensions – in the case illustrated, the mean flow is only 1.0 mL/s , which corresponds to a mean gradient of only

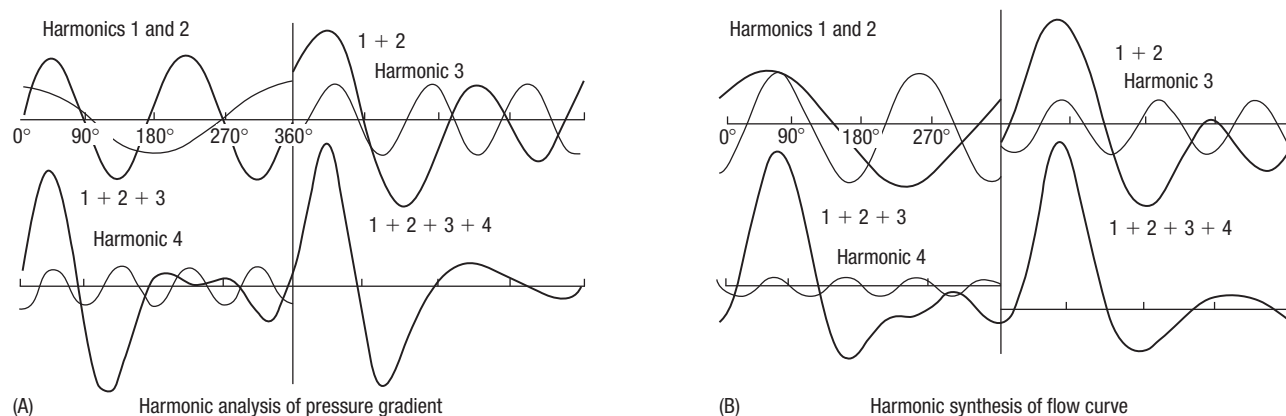


Figure 7.6 (A) The first four Fourier components of the pressure gradient of one experiment in the form $M \cos(\omega t - \phi)$. Upper left, 1 and 2 are the sine waves representing the first, or fundamental, harmonic ($n=1$) and the second harmonic ($n=2$). The modulus (M) of the second harmonic is 1.32 and that of the fundamental is 0.78. ϕ is $10^\circ 39'$ for the first harmonic and $82^\circ 45'$ for the second. Upper right, 1 + 2 represents the sum of the first two harmonics, and the third harmonic is superimposed ($M = -0.74$ and $\phi = +26^\circ 30'$). Lower left, 1 + 2 + 3 is the sum of the first three harmonics, and the fourth is superimposed ($M = -0.41$, $\phi = -16^\circ 30'$). Lower right, 1 + 2 + 3 + 4 is the sum of four harmonics. (B) The flow components shown singly and then summed successively. In the final summation 1 + 2 + 3 + 4, the zero line has been moved down to represent the addition of the mean flow as the sine waves only represent the oscillating part of the curve. Because of the damping of the flow in the higher harmonics, the flow due to the fifth and sixth harmonics is too small to show on the graph. The ordinates are on an arbitrary scale

0.13 mmHg/cm of artery. It is, therefore, at the mercy of manometric errors for, even measuring over an 8.0 cm length of femoral artery, this only amounts to a pressure drop of about 1.0 mmHg. This accuracy can only be attained by using a reliable differential manometer. In the illustrations, the mean flow has been checked by direct observation. The theoretic objection is that the system is non-linear, so that there is an interaction between the oscillating flow and the steady flow; thus they cannot truly be treated as separate terms. At the peak of the systolic flow in one case, Womersley (1955b) calculated that the alteration in steady flow would be of the order of 15 percent. In terms of the shape of the pulsatile flow curve, this is negligible because the steady flow is itself small compared with the oscillating components of the flow. As regards the steady flow itself, the correction cannot be disregarded. As the complexity of the calculation of the correction is coupled with the inherent inaccuracies of the measurement of the mean pressure gradient, we can see that this method is of no practical application in the prediction of mean flow in an artery. The work surveyed later, however, shows that important physiologic data may be derived from the phasic variations in the pattern of flow, and these are well predicted by the calculated curves.

Figure 7.7 shows the comparison between an observed flow curve and one calculated from the 'pressure gradient' measured at the same time, and it can be seen that the correlation, allowing for experimental error, is fairly good. Mathematically, the convincing aspect of these two curves is that the phase relations are accurately predicted. Discrepancies in amplitude may easily arise from experimental error in view of the very precise manometry required. The accuracy of comparison between measured

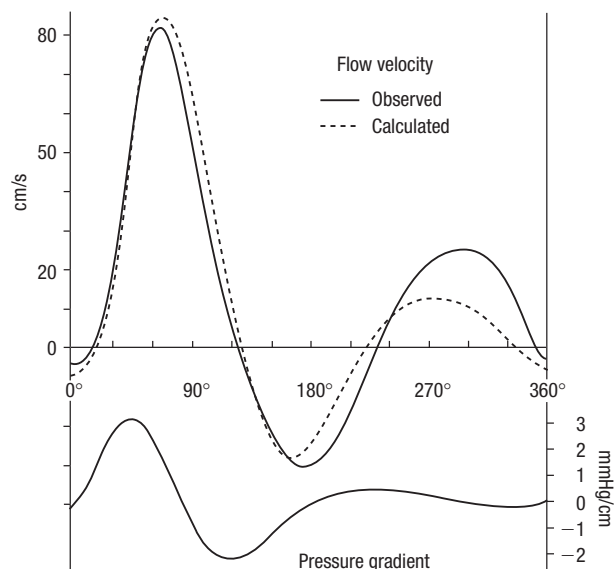


Figure 7.7 The flow-velocity curve in the dog femoral artery (upper curve) recorded by high-speed cinematography and compared with that calculated from a derived 'pressure gradient' (lower curve). The pressure gradient was calculated from the time-derivative of the pressure and the value for the 'peak-to-peak' wave velocity recorded in a series of animals by Hale *et al.* (1955). The correspondence between the two curves, which is reasonably good, suggests that the equations on which the calculation is based are satisfactory. The pulse frequency in this experiment was 2.85 Hz. The abscissa is expressed as degrees of arc of a cycle (360°) because of the method of calculation (see Fig. 7.1)

and calculated flow curves is maintained during peripheral vasodilatation (McDonald, 1974; Nichols and O'Rourke, 1990). The first tests of the Womersley

equation were made by Helps and McDonald (1954), Hale *et al.* (1955), McDonald (1955), and McDonald and Taylor (1959) on measurements of the pressure gradient in the femoral artery of dogs. Since then, it has been applied to the canine ascending aorta (Nichols, 1970; McDonald and Nichols, 1973), and pulmonary artery (O'Rourke and Milnor, 1971), and the human aorta (Gabe *et al.*, 1964), the common carotid artery (Sugahara, 1968), the external iliac artery (Gabe, 1965b), and the pulmonary artery (Milnor *et al.*, 1969). Excellent reviews of this method have been written by Oka (1967), Nichols (1970), and Milnor (1989).

THE EFFECT OF THE SIZE OF THE ARTERY ON THE FLOW PATTERN

A return to the equation (eqn 7.5) for the oscillating flow shows that it differs from the formula for Poiseuille's law. For oscillating flow

$$Q = \frac{\pi R^4}{\mu} \frac{M'_{10}}{\alpha^2} M \sin(\omega t - \phi + \varepsilon_{10}),$$

whereas for steady flow (eqn 2.7)

$$Q = \frac{\pi R^4}{\mu} \frac{1}{8} \frac{(P_1 - P_2)}{L}.$$

With an oscillating pressure, the amplitude of the flow no longer varies linearly with the pressure gradient but is modified by the term M'_{10}/α^2 . Furthermore, the flow lags behind the pressure by an angle $(90 - \varepsilon_{10})^\circ$. When α is small, M'_{10}/α^2 approximates to the value $1/8$, which is the constant in Poiseuille's law (Fig. 7.5). At the same time, the value of ε_{10} tends to 90° so that the term $M \sin(\omega t - \phi + \varepsilon_{10})$ becomes $M \sin(\omega t - \phi + 90^\circ)$ or $M \cos(\omega t - \phi)$, i.e. $(P_1 - P_2)/L$. The formula for the flow thus approximates to Poiseuille's law in small vessels. When α is 0.5 or less for the fundamental harmonic, the difference between the two equations is negligible. The approximation is quite good in a vessel such as the saphenous artery ($\alpha = 0.8$), so that in small vessels even if the pressure gradient varies the flow will vary, approximately linearly, with it.

With large arteries, the oscillating flow deviates very greatly from the Poiseuille formula. When α is 10, it can be seen from Fig. 7.5 that the amplitude of the flow is about $1/15$ of that predicted by Poiseuille's formula and furthermore is more than 80° out of phase. The limiting condition as α gets very large is that the flow excursion decreases to a very small value and is 90° out of phase. The phase relation is then analogous to the relation of an alternating voltage to the current it produces in an inductance. Such a situation will be reached in the aorta where the α of the fundamental will be between 10 and 15 so that the fourth harmonic can reach 30.

THE VALUES OF α FOUND IN THE CIRCULATION

From values calculated for dogs, we have seen above that α at the pulse frequency was less than 1.0 in the saphenous artery, about 3.0 in the femoral artery, and about 10 or higher in the aorta. These values were for particular individuals and they will show variation in the same individual with changes of pulse frequency and between individuals because of differences in caliber. It is interesting to compare the range of values found in different species. Because small animals, which have small arteries, tend to have fast heart rates, and vice versa, the value of α tends to vary over a surprisingly small range at comparable anatomic sites. Some estimated values of α are given in Table 7.1. Although the larger species tend to have somewhat larger values of α , there is quite a remarkable correspondence of values. This suggests that the circulation in various mammals has a dynamic or kinematical similarity and lends comforting theoretic support to the application of results derived from lower animals to the interpretation of observations made in humans.

DEVELOPMENT OF THE PRESSURE GRADIENT TIME-DERIVATIVE METHOD

The assumption that the flow in the proximal aorta may be regarded as largely unsheared was used implicitly in dogs by Fry *et al.* (Fry *et al.*, 1956; Fry *et al.*, 1957b), and Greenfield and Fry (Greenfield and Fry, 1965a; Greenfield and Fry, 1965b) in deriving a pressure-flow relation that is simpler than the more general form given by Womersley (Womersley, 1955a; Womersley, 1955b). Fry's derivation is based directly on the Navier-Stokes equations. The pressure gradient along the aorta is treated as the sum of two components – a large inertial term resulting from the acceleration and deceleration of the blood, and a smaller frictional term that is taken as proportional to the velocity of the blood. This may be written as

$$-\frac{\partial P}{\partial z} = I \frac{\partial Q}{\partial t} + RQ \quad 7.6$$

where P is the lateral pressure, z the axial coordinate of the blood vessel, Q the instantaneous flow, and t is time. I and R are the fluid inductance and resistance, respectively, and are given by

$$I = 1.1 \frac{\rho}{g\pi r^2} \quad 7.7$$

and

$$R = 1.6 \frac{8\mu}{gr^4}, \quad 7.8$$

where g is the gravitational constant, r the radius of the vessel, ρ the density, and μ the viscosity of the blood. The constants 1.1 and 1.6 are experimentally derived. Since the

Table 7.1 Values of α for the fundamental in various mammals

Vessel	Species	Weight (kg)	Radius (cm)	Heart rate (per min)	α
Proximal aorta	Mouse	0.017	0.035	650	1.5
	Rat	0.6	0.10	400	3.3
	Ferret	1.5	0.15	225	3.7
	Rabbit	3	0.17	200	4.0
	Cat	4	0.20	180	4.4
	Dog	20	0.78	90	13.1
	Man	75	1.5	70	22.2
	Ox	500	2.0	52	25.6
	Horse	850	3.8	44	41.8
	Elephant	2000	4.5	38	49.2
Femoral artery	Rat	0.6	0.04	350	1.5
	Rabbit	4	0.08	280	2.4
	Dog	20	0.23	90	3.9
	Man	75	0.27	70	4.0

pressure gradient cannot be measured at a point, $\Delta P/\Delta z$ is substituted for $-\partial P/\partial z$ in the equation, which is then solved continuously for the flow Q by a simple analog computer (Fry *et al.*, 1957b). This method was used for flow measurements in the aorta of humans by Greenfield *et al.* (Greenfield *et al.*, 1968; Greenfield *et al.*, 1971), and Kendall *et al.* (Kendall *et al.*, 1971; Kendall *et al.*, 1973). In the animal studies, Greenfield and Fry (Greenfield and Fry, 1965a; Greenfield and Fry, 1965b) found that there were insignificant differences in the flow data obtained by the more elaborate computations using Womersley's equation and those obtained using this simpler formulation. They also found that the instantaneous flows obtained by either method compared reasonably well with the flows measured with an electromagnetic flowmeter.

Porjé and Rudewald (Porjé and Rudewald, 1957; Porjé and Rudewald, 1961) and Rudewald (1962) used the following equation to compute the volume flow curve (Q) in humans (Fig. 7.8).

$$Q = \frac{\pi r^2}{\rho L} \int (P_1 - P_2) dt + C, \quad 7.9$$

in which $(P_1 - P_2)$ is the pressure drop along the length L of the vessel of radius r , and C is the constant of integration. Thus, simple numerical or electrical integration of the measured pressure gradient will produce an estimate of volume blood flow.

In 1959, Jones *et al.* introduced a method for computing blood flow velocity from the pressure which is more attractive in that it uses only one pressure. These investigators took the original Fry equation and made the substitution.

$$\frac{dP}{dz} = \frac{dt}{dz} \frac{dP}{dt} = -\frac{1}{c} \frac{dP}{dt}, \quad 7.10$$

so that it became

$$\frac{1}{c} \frac{dP}{dt} = \rho \frac{dV}{dt} + aV, \quad 7.11$$

where V is the instantaneous blood flow velocity and c is the pulse wave velocity (considered to be constant). This equation was solved by an analog computer similar to the one used by Fry *et al.* (1957b). The 'resistance' term a was found by adjustment until the flow velocity was zero during diastole. The integral of the flow wave was then matched against the stroke volume measured by the indicator dilution or electromagnetic flowmeter method, thus producing a calibration constant K which converted the analog results to milliliters. The stroke volumes obtained in this manner in dogs were compared with those from an electromagnetic flowmeter (Jones and Griffin, 1962) and the dye-dilution method (Boyett *et al.*, 1966). Jones *et al.* (1966) and Jones and Reeves (1968) have also used this method to compute the stroke volume in humans.

There are three major disadvantages in using the methods mentioned above for deriving the instantaneous ascending aortic blood flow from the pressure gradient. First, an initial stroke volume must be determined by another method, for example electromagnetic flowmeter, indicator dilution, etc., in order to obtain a calibration constant. Second, the pressure gradient must be measured over a very small interval and this requires the precise calibration matching of the two manometers. This pressure gradient is very small. For example, calculation from the Poiseuille equation shows that, in the human ascending aorta with a mean flow of 5.0 L/min and a diameter of 2.5 cm, the mean pressure drop over an interval of 5.0 cm will be only 0.014 mmHg (McDonald, 1974), and, third, the pulse wave velocity is considered to be constant. The conditions for the use of this substitution are specifically for a wave that is undistorted in travel. In

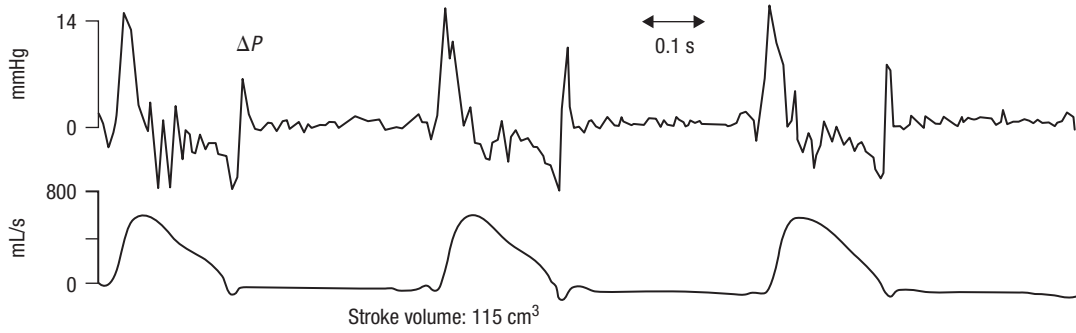


Figure 7.8 Blood flow waveform (bottom) computed from the pressure gradient (ΔP) in the human thoracic aorta. Reproduced from Rudewald (1962)

particular, the presence of reflected waves was shown by Taylor (Taylor, 1957a; Taylor, 1957b) to cause the measured transmission velocity of a harmonic wave to vary greatly with frequency. This was thus named the ‘apparent phase velocity’, c' , and both Taylor (1959c) and McDonald (1974) concluded that this was the appropriate value of c to use in eqn 7.11. Nichols and McDonald (1972) made an extensive study of the wave velocity in the ascending aorta of dogs and found that the wavefront velocity c is very close to the mean apparent phase velocity c' above 2.0 Hz; however, below 2.0 Hz, c' increases markedly due to reflected waves (see Fig. 3.10).

In order to eliminate these disadvantages, Nichols (1970) and McDonald and Nichols (1973) investigated the substitution of the apparent phase velocity in the derivation of pulsatile flow from the pressure gradient in the ascending aorta of dogs. The harmonic components of the pressure gradient are substituted by a modification of eqn 7.10, where the velocity c is represented by the apparent phase velocity c' , so that we have

$$-\frac{dP_1}{dz} = \frac{dt}{dz} \frac{dP_1}{dt} = \frac{1}{c'} \frac{dP_1}{dt}, \quad 7.12$$

where P_1 is the proximal pressure given by $|P_1| \cos(\omega t - \phi_1)$ (likewise the distal pressure is $P_2 = |P_2| \cos(\omega t - \phi_2)$), and

$$c' = \omega \Delta z / \Delta \phi. \quad 7.13$$

Δz is the distance between the two pressure measuring points and $\Delta \phi = \phi_2 - \phi_1$ is the measured phase difference between the two pressures for a particular harmonic component. Differentiating the expression for P_1 gives

$$\begin{aligned} \frac{dP_1}{dt} &= -\omega |P_1| \sin(\omega t - \phi_1) \\ &= \omega |P_1| \cos(\omega t - \phi_1 + \pi/2). \end{aligned}$$

Therefore, from eqns 7.12 and 7.13 (see Fig. 7.2C), we obtain

$$-\frac{dP_1}{dz} = \frac{\Delta \phi}{\omega \Delta z} \omega |P_1| \cos(\omega t - \phi_1 + \pi/2) \quad 7.14$$

so that, by substitution, eqn 7.4 becomes

$$Q = \frac{\pi R^2 \Delta \phi}{\omega \rho \Delta z} |P_1| M'_{10} \sin(\omega t - \phi_1 + \varepsilon_{10} + \pi/2), \quad 7.15$$

and this is calculated for each of the ten harmonic terms with appropriate values of ω , $|P_1|$, ϕ and the Bessel function terms M'_{10} and phase ε_{10} . It can be seen from Fig. 3.6 and eqn 7.13 that the phase velocity c' is frequency-dependent, especially at frequencies below 2.0 Hz. If a constant value is used in deriving the flow curve, the results will be in error to some degree.

The parameter α is calculated for each harmonic component, and the values of M'_{10} and ε_{10} are given by the series expansions

$$M'_{10} = 1 - \frac{\sqrt{2}}{\alpha} + \frac{1}{\alpha^2} \quad 7.16$$

and

$$\varepsilon_{10} = \frac{\sqrt{2}}{\alpha} + \frac{1}{\alpha^2} + \frac{19}{24} \frac{1}{(\sqrt{2})\alpha^3}. \quad 7.17$$

Equation 7.17 is in radians, which are used as the angular measure throughout.

Q is calculated for each of the ten harmonic terms, and the pulsatile flow curve is synthesized by summing these terms. As pointed out above, the mean of the synthesized flow curve is zero, since the mean term in the Fourier series expansion is zero (inspection of eqn 7.15 shows that only the oscillatory terms of the curve are calculated). The mean flow is, therefore, obtained by assuming that flow in diastole is zero; i.e. mean flow is calculated as the average absolute value of the negative ordinates in diastole sampled at 10 ms intervals, and the flow curve is then adjusted by adding this average value to each of the flow ordinates.

Ten different interventions were used in the test of eqn 7.15 by Nichols (1970) and McDonald and Nichols (1973). Peak flows, as well as mean flows, were compared with simultaneous results from an electromagnetic flowmeter. Among these interventions were vagal stimulation

and various vasoactive drug infusions that cause most pressure-based methods to fail, especially the pulse contour methods. The results for mean flow comparisons using the modified Womersley equation gave a correlation coefficient of 0.97; for peak flow the correlation coefficient was 0.91. This shows that the accuracy of prediction of the peak flow is far less than that for mean flow. The results for the control values alone were considerably better, but with vagal stimulation they were worse.

THE VALIDITY OF THE WOMERSLEY EQUATIONS WHEN APPLIED TO ARTERIAL BLOOD FLOW

As with the derivation of the Poiseuille equation for steady flow, it is necessary to review the basic assumptions upon which the mathematical analysis of Womersley is based. The two derivations share many of these assumptions, although they need to be stated in a somewhat different form.

1. **That the flow is laminar.** The evidence of the distribution of laminar flow in arteries has been discussed in Chapter 2, where it was seen that in peripheral vessels, such as the femoral artery, this is almost certainly the case. McDonald (1974) suggested that, because of the high Reynolds numbers attained during systole, the laminar pattern probably breaks down in large arteries near the heart during part of the cycle. The evidence of Schultz *et al.* (1969) indicates that even in the human ascending aorta, any appearance of turbulence, if detected, is very brief in normal circumstances. Only in diseased conditions, notably with stenosis of the aortic valve, is any large-scale disturbance seen. As with the high values of α found in these large vessels, the resistive component is small compared with the inertial components, one would expect that deviations from laminar flow would only make a small difference in the predicted flow.
2. **That the length of the tube is long compared with the region being studied.** This is the ‘inlet length’ problem that was considered in Chapter 2. While the length of the ‘inlet’ for the oscillatory flow in the systemic and pulmonary arterial trees has not been determined, clearly the ascending aorta and pulmonary artery (the regions of special interest) will be fully within this length. The theoretic and experimental work of Chang and Atabek (1961) shows that this will introduce an error, but it is small compared with the probable error in pressure measurement and may be neglected.
3. **That the fluid is homogeneous and does not have anomalous viscous properties.** While this assumption as stated is untrue, it was seen in Chapter 2 that the effects of anomalous (non-Newtonian) viscosity of blood at high shear rates only became important in very small vessels of arteriolar dimensions. In oscillatory flow, however, there are periods in every cardiac cycle when the shear rate becomes very low. While no experimental work has been done to test the effects of this, a detailed theoretic investigation by Taylor (1959c) showed that the error would not be more than 1.0–2.0 percent.
4. **That the flow is through a cylindrical tube whose diameter remains constant.** While superficially this assumption would appear to involve a large approximation, the experimental evidence shows that the actual radial dilatation of arteries during the cardiac cycle is in fact small (Hirooka *et al.*, 1988). The error due to this approximation is small compared with the probable error of measuring flow.
5. **That the flow may be expressed as a sum of harmonic components calculated from the individual harmonic terms of the pressure gradient.** This is discussed in terms of non-linearity in relation to hydrodynamic equations in Chapter 11 and to the non-linear elastic behavior of the arterial wall in Chapter 4. In terms of pressure and flow, it is generally agreed that non-linear terms are very small.
6. **That no ‘slip’ occurs at the wall.** The justification for this was discussed with the assumptions underlying Poiseuille’s equation in Chapter 2, and it is clear that this assumption is always true for liquid flow.

The Womersley equations have been tested experimentally in rigid tubes by Linford and Ryan (1965) over a wide range of values for α . They found a very close agreement between the theoretic and measured flows; they further showed that, as anticipated, only very small differences occurred when turbulent flow was present at values of $\alpha > 10$. The comparison made by Kunz and Coulter (1967), using blood in tubes, has been discussed in Chapter 2. Again, over the great part of the range, the agreement between theory and experiment was very close but, at very low frequencies, discrepancies appeared that were attributable to the anomalous viscous properties of the blood. Greenfield and Fry (1965a) made a detailed comparison between flow measured in the thoracic aorta with an electromagnetic flowmeter and the flow predicted by both the full Womersley equation and the simpler approach introduced by Fry *et al.* (1956). The predicted flow showed close agreement between the two methods. While the comparison of predicted and measured values was good, there was found to be a small, random difference of between 2.0 and 10 percent. This, however, is most likely to be attributable to the experimental error in measuring the pressure gradient. The experimental

situation in a living animal will always make a precise testing of the basic equations used extremely difficult.

In summary, the Womersley equation cannot be rigorously applied to arteries; nevertheless, it is almost certainly as close, or closer, a practical approximation as Poiseuille's equation is for steady flow in arteries.

THE CONCEPT OF VASCULAR IMPEDANCE

Because the behavior of oscillating liquid flow is not generally familiar, we do not have a set of technical terms peculiar to this branch of hydrodynamics. Inevitably, the terms have been borrowed from other branches of physics, usually from the language of electricity. In this way, we treat an alternating pressure gradient as analogous with an alternating voltage, and the oscillatory pressures and flows outlined in this book may be called the 'alternating current (AC) theory of flow', in contrast with the 'direct current (DC) theory' represented by steady flow (McDonald, 1974). Treated purely as an analogy, this makes a useful distinction that emphasizes the fact that phase differences between pressure and flow are important, as they are between alternating voltage and current in electrical circuits (Halliday and Resnick, 1988). It was suggested by McDonald (1955) that the term 'impedance' should be used when considering 'dynamic' pulsatile flow and pressure in arteries, and that the term 'resistance' should be confined to the 'static', or mean flow, terms (Chapter 12). This use of impedance is also standard in acoustics (which, in dealing with oscillating flows of a compressible fluid, is the best analogy for liquid flows) and in mechanics. Womersley (Womersley, 1957a; Womersley, 1957b) has shown that the equations considered in this chapter (and others) can be given quantitative expression in terms analogous to those of electrical theory.

In theoretic treatments of arterial flows by analogy with electrical networks (e.g. Landes, 1949; Ronniger, 1954), the analogy of current was always treated as velocity of flow and not volume flow. Impedance, like resistance, is pressure/flow; as the units of pressure are dyne/cm² and those of velocity are cm/s, this leads to the unit of impedance as dyne-s/cm³. This is the same unit as that used for acoustical and mechanical impedance, both of which are closer analogies of the impedance to liquid flows than is electrical impedance. However, it has become common practice in clinical hemodynamics to record the blood flow as volume flow (cm³/s). Hence, impedances are normally recorded in terms of volume flow using the unit of dyne-s/cm⁵. Working with volume flow also has the advantage that the product of pressure × flow gives us the power directly, dyne-cm/s. This is analogous to the derivation of electrical power as voltage × current. The conversion from one to the other is simple in that the conversion factor only involves the cross-sectional area of the vessel. In later chapters, the impedance is usually expressed in terms of volume flow. Where generalizations about the whole vascular bed are being made, however, there is an advantage in using the

linear velocity of flow in calculating the impedance because, as was seen in Chapter 2, the changes in velocity of flow are determined by changes in the total cross-sectional area of the branches at any level; the impedance calculated from volume flow only expresses the impedance of a single vessel out of an array of vessels in parallel.

Writing the pressure gradient in exponential form, we have

$$\frac{P_1 - P_2}{L} = A^* e^{i\omega t},$$

and the corresponding term for the average velocity across the tube is

$$\bar{V} = \frac{Q}{\pi R^2} = \frac{A^* R^2 M'_{10}}{i\mu \alpha^2} e^{i\varepsilon_{10}} e^{i\omega t}. \quad 7.18$$

The longitudinal impedance, Z_L , is defined as the impedance per unit length L of the tube and is thus given by

$$\begin{aligned} Z_L &= \frac{A^* e^{i\omega t}}{\bar{V}} = \frac{i\mu \alpha^2}{R^2 M'_{10}} e^{-i\varepsilon_{10}} \\ &= \frac{\mu \alpha^2}{R^2 M'_{10}} \sin \varepsilon_{10} + \frac{i\mu \alpha^2}{R^2 M'_{10}} \cos \varepsilon_{10} \end{aligned} \quad 7.19$$

(or, when volume flow is used, the R^2 is replaced by πR^4).

The electrical complex impedance analogous to flow in a rigid tube is written as

$$Z = R + i\omega L, \quad 7.20$$

where R is resistance and L is the inductance; hence the real part of eqn 7.19 may be called the resistance and the imaginary part may be called the inductive term of reactance. Thus

$$\text{Resistance} = \frac{\mu \alpha^2}{R^2 M'_{10}} \sin \varepsilon_{10} \quad 7.21$$

and

$$\text{Reactance} = \frac{\mu \alpha^2}{R^2 M'_{10}} \cos \varepsilon_{10}. \quad 7.22$$

Expanding the term α^2 ($= R^2 \omega \rho / \mu$) we can also write

$$\text{Resistance} = \frac{\omega \rho}{M'_{10}} \sin \varepsilon_{10} \quad 7.23$$

and

$$\text{Reactance} = \frac{\omega \rho}{M'_{10}} \cos \varepsilon_{10} \quad 7.24$$

so that

$$\text{Inductance} = \frac{\rho}{M'_{10}} \cos \varepsilon_{10} \quad 7.25$$

Normalized values for the inductance and the resistance with changing α are shown in Fig. 7.9. It can be seen

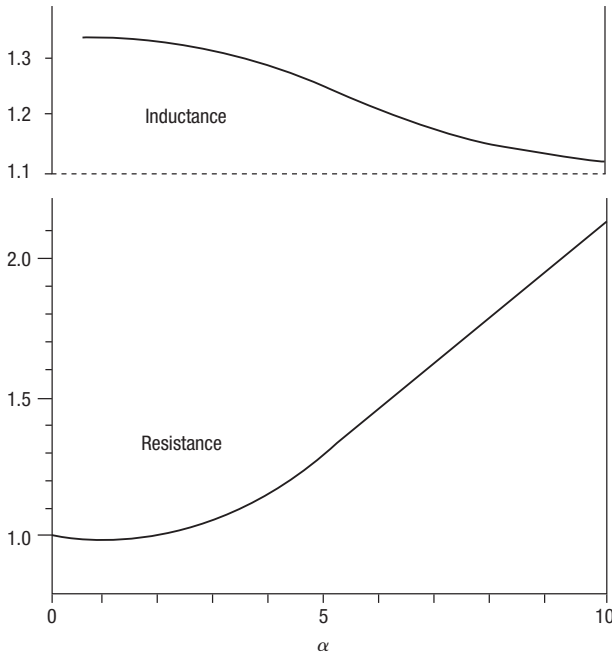


Figure 7.9 Inductance (eqn 7.25) and resistance for oscillatory flow in a rigid tube plotted as a function of α . The values for resistance are given relative to the steady-flow, or Poiseuille, resistance and are $1/8$ times that given by eqn 7.21. As α is a function of the square root of the frequency, it can be seen that both these components of the impedance are frequency-dependent, in contrast to the analogous electrical quantities. The application of the resistance in oscillatory flow to the manometer equations is discussed in Chapter 6

that the inductance falls slowly with increasing α and tends, in the limit, to 1.0. The limiting value of the reactance is, therefore, ω . The resistance on the other hand remains close to the Poiseuille resistance up to $\alpha = 3.0$ and then rises steadily with α (i.e. with $\omega^{1/2}$); with $\alpha > 4.0$, the relationship is almost exactly linear. Fry *et al.* (1964) found that, in the proximal aorta, they could substitute a weighted mean value of 1.1 for the inductance and 1.6 times the Poiseuille resistance for the resistance term over the frequency range represented in the pressure gradient. These are approximately the values at $\alpha = 7.0$.

When considering a conducting system along which the wave of oscillatory flow is propagated, we define the

characteristic impedance Z_0 and the velocity of propagation c so that

$$Z_0 = \frac{cZ_L}{i\omega}; \quad 7.26$$

so that from eqn 7.19, after expanding α^2 , we obtain

$$Z_0 = \frac{\rho c}{M'_{10}} e^{-i\epsilon_{10}}. \quad 7.27$$

Applying the relation between the complex wave velocity c and the real part of the velocity in a longitudinally restrained elastic tube derived by Womersley (Womersley, 1957a; Womersley, 1957b), we have

$$Z_0 = \frac{\rho c_0}{(1 - \sigma^2)^{1/2}} \frac{1}{(M'_{10})^{1/2}} e^{-i\epsilon_{10}^2}, \quad 7.28$$

where c_0 is the velocity derived from the Moens–Korteweg equation for an elastic tube filled with an inviscid liquid.

The characteristic impedance is defined as the ratio of oscillatory pressure to flow at the input of a tube in which no reflected waves return to the origin, i.e. that the tube ends in a matched impedance, or is so long that any reflected waves arising from the termination are completely attenuated before they reach the origin (see Chapter 9).

The term ‘input impedance’ is used to express the ratio of oscillatory pressure to flow at the input to any region of the circulation and is modified by reflected waves. The frequency dependence of the input impedance due to the interaction with reflected waves is discussed in Chapter 10 and the experimental and clinical work on the subject is reviewed in Chapter 12. The input impedance in rubber tubes was analyzed by Hardung (1958) and the components of the characteristic impedance were investigated in detail by Taylor (1959c). Excellent reviews of the topic in viscoelastic tubes (Hardung, 1962; Papageorgiou and Jones, 1989), Windkessel models (Westerhof *et al.*, 2009), and distributed models (Avolio, 2009) go into greater detail than is considered here.

Ultrasound

Physical principles	177	Doppler principle	181
Echo resolution	178	Continuous wave Doppler systems	182
Transducers	178	Pulsed Doppler systems	185
Beam forming	179	Applications	189
Image generation	180		

PHYSICAL PRINCIPLES

Sound is a mechanical interaction between local pressure perturbation p and particle velocity v :

$$p = Zv \quad 8.1$$

The proportionality constant Z in the above expression is called the acoustic impedance and is directly related to the elastic characteristics of the medium. Because of the mechanical interaction, particle motion will induce a pressure variation and vice versa, creating a wave propagating in the direction of the motion and pressure variation, i.e. a longitudinal wave. For a homogeneous medium the acoustic impedance can be expressed as the product of mass density ρ (kg/m^3) of the medium and sound propagation speed c (m/s):

$$Z = \rho c \quad 8.2$$

Consequently, the unit of Z is $\text{kg}/(\text{m}^2\text{s})$, but in most articles it is referred to as MRayl. Table 8.1 lists for some (biological) materials their sound-related characteristics. The above expressions are rather independent of the sound frequency f ($\text{Hz} = 1/\text{s}$). Defining the wavelength λ (m) as the spatial distance along the propagation direction between similar points in subsequential cycles, then it follows that:

$$\lambda = c/f \quad 8.3$$

The wavelength is inversely proportional with the frequency; a higher sound frequency automatically implies a shorter wavelength if the sound propagation speed is the same.

At boundaries of media with different acoustic impedances, the incident wave is partially reflected and partially

transmitted (Fig. 8.1). Particles at both sides of a boundary between media with different acoustic impedances Z_1 and Z_2 should experience the same pressure and particle velocity (Wells, 1969). Moreover, the propagating wave does not lose energy when it meets a boundary. Combining those prerequisites results in the following expressions for the reflection (γ_r) and transmission (γ_t) coefficients for perpendicular incidence:

$$\gamma_r^2 = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}, \quad \gamma_t^2 = \frac{(Z_1 - Z_2)^2}{\sqrt{2Z_1Z_2}} \quad 8.4$$

and $\gamma_r^2 + \gamma_t^2 = 1$

In the above expressions, the reflection and transmission coefficients are defined as the reflected or transmitted pressure with respect to the incident pressure. It follows that a homogeneous medium does not exhibit reflection and that the reflection coefficient increases for a higher relative impedance mismatch.

Table 8.1 The acoustic impedance Z , sound speed c , and attenuation coefficient α (approximate values) of some human tissues at 37°C. Water and air are included to illustrate differences in acoustic impedance and attenuation coefficient (Goss *et al.*, 1978)

Tissue	$Z \times 10^6$ ($\text{kg}/(\text{m}^2 \cdot \text{s})$)	c (m/s)	α (dB/(cm · MHz))
Blood	1.5	1580	0.2
Bone	7.8	3500	10
Brain	1.58	1540	0.8
Liver	1.65	1550	1.2
Muscle	1.6	1560	1.1
Fat	1.38	1480	0.6
Air	0.0004	330	10
Water (20°C)	1.48	1480	0.002

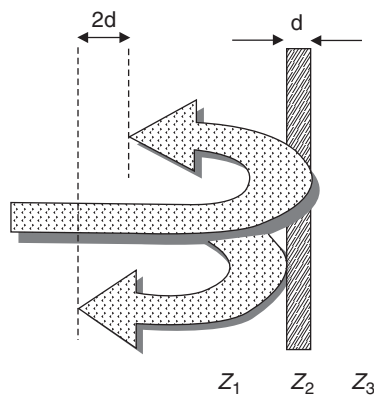


Figure 8.1 At acoustic interfaces, sound is partially reflected and transmitted. The round trip distance between layers is twice the layer thickness d

Reflection only occurs at interfaces with a dimension that is a few times larger than the wavelength. If the dimension is much smaller than the wavelength, then the wave will be dispersed in all directions, termed ‘scattering’. In the intermediate region, the radiation pattern gradually changes from a single peak to a circular pattern as the sound frequency or the interface dimension decreases. Most tissues exhibit a combination of reflection and scattering (Evans *et al.*, 2000; Cobbold, 2007).

While sound waves travel through a medium they will be attenuated. Sound energy is lost because of non-elastic particle interactions, but also because of scattering and reflection. Table 8.1 lists the attenuation coefficients expressed as dB/(cm · MHz) for several (biological) media and illustrates that acoustic impedance and attenuation coefficients are quite different quantities. Both bone and air have a high attenuation, but air has an extremely low acoustic impedance. Higher frequencies will be more attenuated, so greater penetration depths require a lower frequency than superficial applications. In general, the sound frequency for biological applications will be in the range from 2 to 40 MHz, which is a frequency far beyond the upper frequency that is audible for the human ear, and, henceforth, termed ‘ultrasound frequency’.

Considering the acoustic mismatch between bone and water (Table 8.1), it is obvious that a water–skull interface exhibits a high reflection and a poor transmission as in transcranial applications. Apart from the high (frequency-dependent) attenuation within the bone, the transmitted wave is further attenuated at the next bone–brain interface. Any reflected wave from inside the brain has again to pass twice a water–bone interface. Similarly, air layers and air bubbles are serious obstacles for ultrasound transmission (Evans *et al.*, 2000; Cobbold, 2007).

ECHO RESOLUTION

In theory, it would be possible to perform ultrasound imaging in transmission mode, similar to röntgen

imaging, but the diffraction effects and high attenuation make this approach unattractive. However, the sound speed is almost the same for most tissues. Henceforth, the distance Δd traveled for a given time Δt may be assumed to be independent of tissue type:

$$\Delta d = \Delta t \cdot c \quad 8.5$$

For a constant propagation speed, e.g. $c = 1540$ m/s, imaging can be performed in reflection mode: a sound signal is emitted, reflected by acoustic inhomogeneities and sensed, where the time delay between emission and reception is directly related to the depth of the interface. Because the round trip distance is twice the depth (Fig. 8.1), expression 8.5 transforms into:

$$\Delta d = \Delta t \cdot c/2 \quad 8.6$$

The concept of reflection mode only works satisfactorily, if closely spaced objects (with different acoustic impedances) can be discerned. For the depth direction, the shorter the sound signal used, the shorter the distance between subsequent interfaces can be without ambiguous overlap of the reflected signals of both interfaces. High resolution ultrasound systems utilize pulses of about two periods, which converts to a depth resolution of one wavelength: a higher frequency translates into a better resolution for the same system quality (Evans *et al.*, 2000; Cobbold, 2007). In the transversal direction (perpendicular to the propagation direction), the wave should be confined to a narrow region, termed the ‘sound beam’, i.e. the path followed by the generated sound pulse. As will be discussed below, this can be trimmed down by focusing on about five wavelengths (independent of the actual sound frequency used) over the depth range of interest.

The bandwidth B of a signal is defined as the region between the frequencies where the spectral amplitude is half the peak amplitude. The bandwidth of an ultrasound pulse is inversely related to its effective duration T (Gabor, 1946):

$$BT = 1 \quad 8.7$$

So a short pulse requires a large bandwidth. The quality factor Q stands for the emission center frequency divided by bandwidth and equals the effective burst length expressed in cycles. The frequency dependent attenuation mainly affects the upper frequencies, with the consequence that the mean frequency of a pulse gradually shifts to a lower frequency for increasing path lengths (Fig. 8.2).

TRANSDUCERS

Ultrasound is generated by piezoelectric transducers, converting electrical energy into acoustic energy and

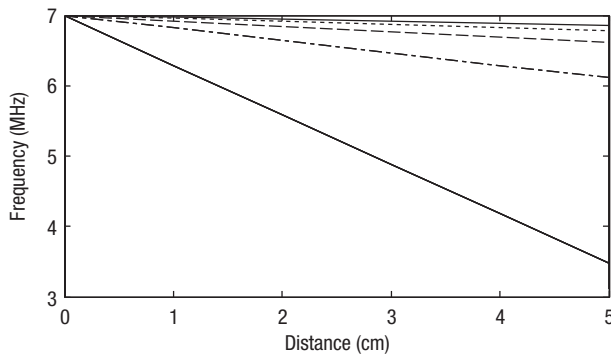


Figure 8.2 Effect of frequency and depth dependent attenuation of 1 dB/(cm · MHz) on the received center frequency for a Gaussian-shaped ultrasound burst with an initial center frequency of 7 MHz and a duration of 10, 8, 6, 4, and 2 periods (from top to bottom)

vice versa. The technology to generate efficiently a short acoustic pulse with an effective duration of one to two periods is still under development, considering selection of material to extend bandwidth and efficiency, matching layers at the transducer face to improve acoustic coupling and to extend bandwidth, and backing to reduce ringing. The simplest form is the flat disk-shaped transducer as currently still in use for transcranial Doppler applications where changing beam direction and positioning is less relevant. However, to compose an image by scanning a plane requires fast manipulation of transducer position or orientation. That is why phased (Somer, 1968) and linear (Bom *et al.*, 1971) arrays emerged. A phased array has a height and width of about 30 wavelengths, but in the width direction (across the scan plane) the transducer is subdivided into small elements (typically 64 elements) with a width less than the wavelength. Because of these small dimensions, each element will act as a scatterer, but altogether they will generate beams that can be steered in a preset direction by activating the elements with respective time delays. A phased array produces inherently a sector image. The small footprint allows beam steering between ribs which makes the phased array transducer attractive for cardiac applications.

A linear array is much longer than a phased array and contains from 128 to 256 elements with a width of about a wavelength. To generate an echo-beam, a subgroup of 16–32 neighboring elements is activated, after which the same group is used for reception. After reception is completed, the active group is shifted in position over one element and the procedure of emission and reception is repeated, eventually resulting in a rectangular image (Evans *et al.*, 2000; Cobbold, 2007). Because of its elongated shape, linear array probes are suited for superficial locations and soft tissue regions (abdomen), where the scan field is extended by curving the linear array, resulting in a trapezoid scan image.

BEAM FORMING

The shape of the generated beam pattern highly depends on the dimensioning of the transducer and the way transducer elements are activated. To direct the ultrasound into a particular direction, the active area of the transducer generally has dimensions d_t in the order of 30 wavelengths. A smaller height or width results in a more omnidirectional radiation pattern and generates a smaller amount of energy, while a larger diameter produces a wider beam. Close to the transducer, many field spots can be found from where the path lengths to points at the transducer elements differ a multiple of half a wavelength, resulting in spatial cancellation or enhancement of the ultrasound signal. Consequently, the evolving ultrasound distribution close to the transducer exhibits many minima and maxima (speckle pattern), but this irregularity will gradually disappear with increasing distance. The last maximum where the entire transducer surface best contributes to the local sound intensity is reached at a distance

$$d = \frac{d_t^2}{4\lambda} \quad 8.8$$

This distance marks the transition from the near to the far field and at this point the width of the ultrasound beam, at 50 percent (−6 dB) of the axis level, reaches a minimum of 60 percent of the active transducer dimension, being a natural focus.

Within the near field, the beam can be narrowed by focusing. For both linear and phased arrays, focusing in the plane perpendicularly to the scan plane is achieved mechanically by a curved cover with a sound speed deviating from that of tissue. If the cover sound speed is lower, the sound at the center will be more delayed than the sound closer to the edge, creating a curved wavefront. Only at a specific distance all signals will arrive in phase, setting the focal distance. The curvature is designed such that the actual position of the focal point will be about halfway the natural focus, acting for both emission and reception. For focusing within the scan plane, the elements effectively contributing to the beam are activated with respective time delays, again resulting in a curved wavefront (Fig. 8.3). The imposed time delays can be manipulated electronically, allowing a free choice of the focal point. Unlike at emission where the time delays set the focal distance, at reception the time delays between elements are dynamically changed with the time elapsed after emission, creating a focal point that dynamically shifts across the depth range (Evans *et al.*, 2000; Cobbold, 2007). The overall sound profile is the product of the emission and reception beam profiles. An effective beam width at −6 dB of five wavelengths is possible over a considerable depth range (but within the near field) if a large number of elements is utilized and tuning is performed carefully. The overall performance will degrade if

the sound passes obliquely through tissue layers with sound speeds slightly different from the assumed sound speed. For the same reason, the assumed sound speed should be adjusted, if the system is used in a phantom set up.

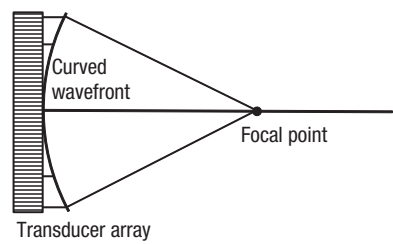
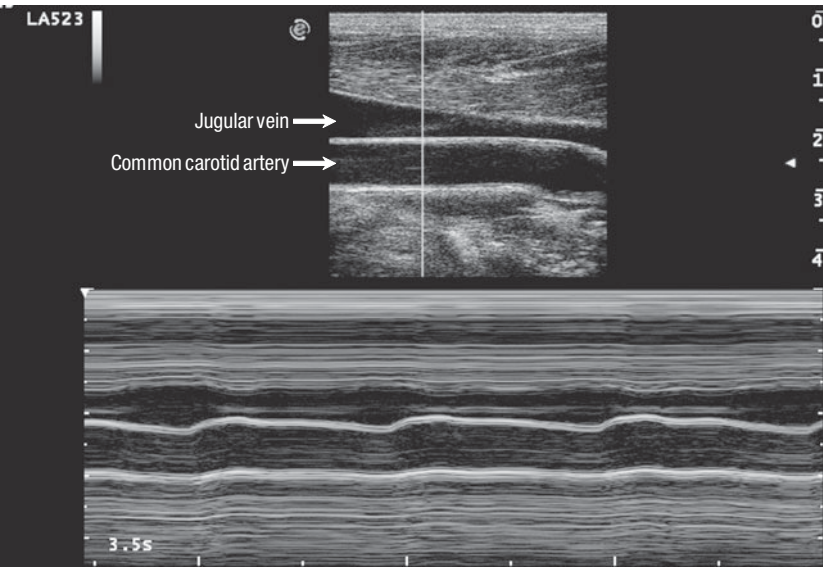


Figure 8.3 Imposing time delays between activation of transducer elements creates a curved wavefront focusing on a preset depth

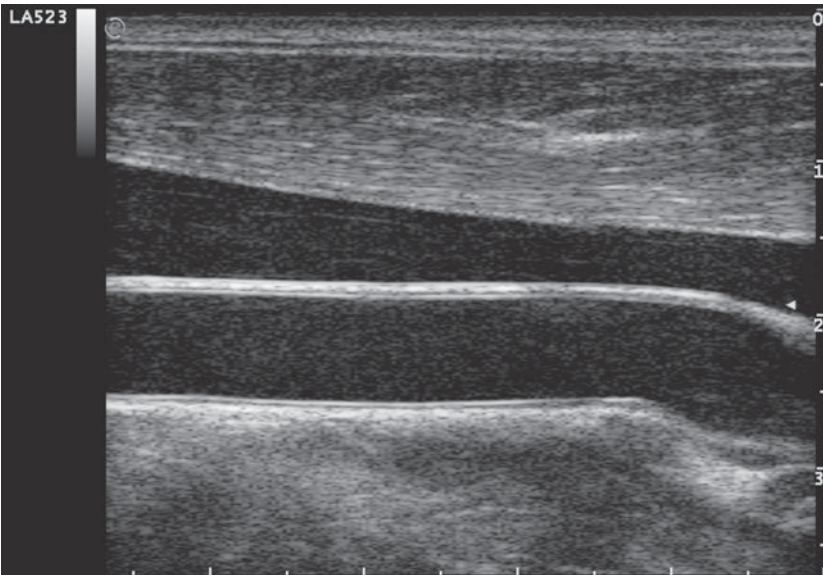
IMAGE GENERATION

The eventual radiofrequency (RF) signal is the sum of the signals picked up by the elements and is referred to as an echo line. The display of the amplitude of the RF signal (the envelope) versus depth (eqn 8.6) is called A-mode (A, amplitude). Presenting echo-amplitudes of subsequent echo lines along the same path as shades of gray reveals the dynamic behavior of structural interfaces (Fig. 8.4) and is indicated as M-mode (M, motion). A two-dimensional (2D) image is obtained by rotating (phased array) and/or displacing the transducer group (linear array; Fig. 8.4) and displaying for each angle/position the amplitude of the RF-signal (the envelope) in shades of gray (B-mode, brightness).

The lower signal amplitude at greater depth is dynamically corrected by an assumed depth-dependent gain



(A)



(B)

Figure 8.4 Linear array M-mode registration (A) along highlighted line in two-dimensional B-mode image (B) crossing jugular vein and common carotid artery. The M-mode image clearly provides the wall motion, specifically that of the anterior wall, during the cardiac cycle. The blood vessels appear as anechoic regions

which can be manually adjusted by slider controls (depth gain). A smooth image is achieved by converting and interpolating the signal samples to the image grid of the presentation monitor. The display frame rate may easily exceed 25/30 Hz for a frame composed of 128 lines, matching the acquisition frame rate of a standard video recorder. To avoid the image only revealing white blots from hard reflectors (a large mismatch in acoustic impedance), the signal is (logarithmically) compressed, enhancing weak signals relatively to strong signals. Too weak signals, probably noise, are set to zero (thresholding). Also blood has a low echogenicity and will appear as black. Signals from reflectors and scatterers with an interspacing smaller than the local resolution will interfere, creating also the grainy speckle (Burckhardt, 1978). Within the focal zone, the speckle dimensions betray the resolutions in both directions. Speckle can be effectively reduced by considering targets from slightly different directions and summing the images (compounding) (Berson *et al.*, 1981; Lightvoet and Eversdijk, 1981; Adams, 2006). This concept works quite neatly for linear arrays where the element size allows some steering and multiple non-overlapping element groups can be activated simultaneously such that compounding would not sacrifice frame rate (Fig. 8.4). Because of differences in tissue composition along slightly displaced echo beams, and, hence, different propagation times, cancellation might be suboptimal while the distances observed between structure interfaces slightly change compared to perpendicular insonation. The image smoothness is further improved by image processing algorithms, involving local smoothing based on the statistics of the pattern and averaging over subsequent images (persistence).

Perpendicular insonation at parallel tissue layers causes reverberations because echoes are reflected again and again. Despite the fact that subsequent reflections have considerably lower amplitudes than the incident pulse, the reverberations cause a trail of echoes that will extend into echo-poor regions, e.g. the cardiac ventricles or the vessel

lumen, degrading, for example, the contrast between anterior wall and lumen. For deep lying structures reverberations can be effectively suppressed by harmonic imaging. In this approach, use is made of the property that high amplitude ultrasound pressures (> 0.3 MPa) gradually exhibit nonlinear distortions due to the fact that the wave at high pressure travels faster than during rarefaction. This causes a gradual development of higher harmonics (Kono *et al.*, 1997; Tranquart *et al.*, 1999). At reception, the higher harmonics are separated from the fundamental ones, for instance, by alternatively switching the phase of the emitted burst and adding subsequent radiofrequency signals. The native fundamentals will be canceled leaving the tissue harmonics (Averkiou and Hamilton, 1995; Averkiou and Powers, 2001). The process of tissue harmonic generation only holds for the incident wave, but not for the reverberation because of its inherently low amplitude. Moreover, harmonic generation will only occur close to the beam axis (high amplitude), effectively improving lateral resolution (Cherin *et al.*, 2002; Tact and Jirik, 2004).

In the above summary of ultrasound imaging, it is assumed that the sound propagation is tissue independent and that the relevant dimensions can be expressed as a multiple of wavelengths. The general rules for transducer dimensions and the consequences for penetration depth, focusing, and resolution are summarized in Table 8.2.

DOPPLER PRINCIPLE

Let us consider a sound wave of one period with an incident frequency f_i impinging on a single reflector moving in the opposite direction with velocity v . At the start of the reflection, the end of the wave and the reflector are separated by one wavelength. They will meet somewhere in between after some time T_r and the particle has

Table 8.2 Echo and Doppler transducer, and signal characteristics for an assumed sound speed of $c = 1540$ m/s. The dominating parameter is the center frequency (top line)

Center frequency	MHz	2	3.5	7	10	15	20	30	40
Wavelength λ	μm	770	440	220	154	103	77	51	39
Penetration (300λ)	mm	231	132	66	46	31	23	15	12
Attenuation at 1 dB/(cm · MHz)	dB	92	92	92	92	92	92	92	92
PRF	kHz	3	6	12	17	25	33	50	67
Transducer diameter D (30λ)	mm	23.1	13.2	6.6	4.6	3.1	2.3	1.5	1.2
Natural focus (in plane), eqn 8.8	mm	173	99.0	49.5	34.7	23.1	17.3	11.6	8.7
Natural focus/max depth	%	75	75	75	75	75	75	75	75
Beamwidth at natural focus	mm	13.9	7.92	3.96	2.77	1.85	1.39	0.92	0.69
RF echo bandwidth (at 2λ)	%	50	50	50	50	50	50	50	50
Axial echo resolution (at 2λ)	μm	770	440	220	154	103	77	51	39
Max unambiguous Doppler velocity	m/s	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64
Segment length 64 points fast Fourier transform (FFT)	ms	19.2	11.00	5.49	3.84	2.56	1.92	1.28	0.96

covered a distance d (and the wave end $\lambda - d$):

$$T_r = \frac{\lambda - d}{c} = \frac{d}{v} \rightarrow d = \frac{\lambda v}{c + v} \quad 8.9$$

The end of the reflected wave has made a shortcut of $2d$, reducing the wavelength to $\lambda - 2d$, which converts to a frequency shift f_d :

$$f_i + f_d = \frac{c}{\lambda - 2d} = \frac{c(c + v)}{\lambda(c - v)} \rightarrow f_d = \frac{2f_i v}{c} \quad 8.10$$

For the derivation of eqn 8.10, use is made of $c = \lambda f$ (eqn 8.3) and the assumption that the particle velocity (up to 2 m/s) is much smaller than the sound speed (around 1540 m/s). The right hand part of eqn 8.10 is the famous Doppler equation, which was originally derived by Doppler for light (Doppler, 1842). In 1845, the concept of the Doppler (frequency) shift for a reflector moving with respect to a receiver was explored by Buijs Ballot for sound (Buijs Ballot, 1845a; Buijs Ballot, 1845b). For that purpose, a musician on a moving locomotive blew on a horn while another musician along the railway track had to indicate the tone he perceived.

If the pathway of the reflector is inclined with respect to the ultrasound beam, then v is replaced by its projection $v \cos \alpha$ with α the inclination angle (Fig. 8.5). If the reflector is moving away from the receiver, then the Doppler frequency shift f_d will be negative.

Assuming a relative velocity of blood of $v = 1$ m/s and an incident frequency of 5 MHz the Doppler shift frequency f_d will be 6.1 kHz ($c = 1500$ m/s). Generally, blood flow velocities are below 1 m/s; only in and immediately distal to a stenosis, velocities up to 5 m/s may be reached. Hence, the Doppler shift frequencies are in the audio range and can be evaluated aurally. Observation angles unequal to zero will only lead to a received signal, if scattering is involved (small dimension of object), as is the case for red blood cells with an ellipsoid dimension of 5–8 μm . If they travel with different velocities, a spectrum of Doppler frequencies, rather than a single frequency component will be received (Reneman and Spencer, 1979; Evans *et al.*, 2000; Cobbold, 2007).

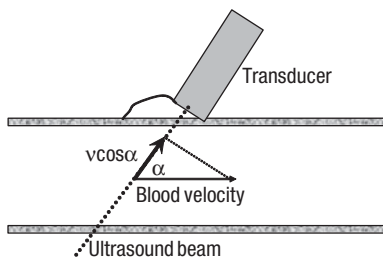


Figure 8.5 Only the velocity component of a moving particle relative to the transducer ($v \cos \alpha$) contributes to the Doppler effect

In vascular studies and in vascular medicine, the blood flow velocity is measured by placing a probe, containing the transducers for emission and reception of ultrasound, on the skin. The direction of the probe is interactively varied to locate an artery until a balance is found between the amplitude and the pitch of the Doppler ultrasound. The Doppler technique is applied in systems based upon continuous wave (CW) and pulsed approaches.

CONTINUOUS WAVE DOPPLER SYSTEMS

Transducer

The simplest configuration to extract the Doppler shift information from the blood velocity distribution is a CW system. It employs separate transducers (Fig. 8.6) for emission and reception of ultrasound. The sensitivity range of the probe is confined to the intersection of the emitted beam and the sensitivity range of the receiver crystal (Fig. 8.6). Hence, the sample volume of a CW Doppler system is quite large and does not convey information about the depth from which the Doppler signals are originating (Evans *et al.*, 2000; Cobbold, 2007). On the other hand, the system exhibits a high sensitivity, making it easy to locate a vessel of interest by listening to the Doppler sound. Nowadays, the Doppler unit to emit and process the received signal is the size of an average-sized pocket calculator and is therefore popular with medical doctors to check rapidly whether an artery is still patent.

Receiver, demodulator

The basic configuration of a CW Doppler system consists of an emitter, receiver, demodulator, bandpass filter, and Doppler frequency detector (Evans *et al.*, 2000). An oscillator provides the emitter with a sine wave with a constant amplitude and frequency, the emission frequency. The receiver amplifier has a large dynamic range to accommodate the large echoes from structures, e.g. vessel walls, as well as signals backscattered by red blood cells with a factor of 100 (40 dB) lower amplitude. The demodulator is a crucial element in Doppler systems, because it shifts the frequency spectrum centered at the emission frequency down to the audio range by

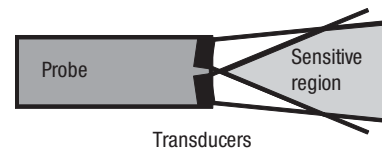


Figure 8.6 A continuous wave probe has separate transducers for continuous emission and reception of ultrasound waves. The sensitive region of the probe is where both beams overlap

multiplying its input signal by the oscillator signal (Evans *et al.*, 2000; Cobbold, 2007).

Before we further discuss the details of signal processing in Doppler systems, we will first address the concept of complex signals (see Chapter 11). Any cosine (or sine) wave is described by its amplitude A and frequency f . The argument α of $\cos(\alpha)$ is called the (instantaneous) phase of the signal. Since the signal has two degrees of freedom, it is impossible to extract unambiguously the current phase or amplitude from a sample (or a few samples) of an unknown signal. This problem is solved by considering the sample as a projection of a complex signal on the (real) x -axis (Fig. 8.7), where the sample point rotates counterclockwise for positive frequency and clockwise otherwise. The sequence of x -axis samples is called the in-phase signal as opposed to the in-quadrature signal sequence of corresponding projections on the y -axis (imaginary axis). The amplitude of the signal is the radius, while the instantaneous phase follows from the arctangent of the imaginary and real component. The combination of a real and imaginary signal is called a 'complex signal'. The in-quadrature signal is derived from the in-phase signal by shifting the phase over 90° independent of the frequency, i.e. the (sum of) cosine-components is converted into the (sum of) sine-components. Demodulation of the received signal by a complex reference signal results in a complex audio signal (Evans *et al.*, 2000; Cobbold, 2007).

Wall Doppler filter

Spurious noise and the signal components at a double emission frequency at the output of the demodulators are effectively suppressed by low-pass audio filters with a cut-off frequency of about 15 kHz, depending on the emission frequency employed and the anticipated maximum velocity (see Doppler equation). The demodulation process applies for all frequency components of the received signal, so also for signal components with zero or a very low Doppler frequency and with relatively high amplitudes, originating from stationary or slowly moving structures as vessel walls. To suppress these low frequencies, both audio low-pass filters are replaced by bandpass filters with a lower cut-off frequency in the range of 100–300 Hz,

setting also the lower bound for the (blood flow) velocities that can be detected (Doppler equation).

Audio output

The demodulated bandpass filtered quadrature signals contain Doppler shift frequencies in the audio range. Both quadrature channels sound exactly the same. Aural evaluation may be impaired if the Doppler information of both a vein and an artery is conveyed. Generally, however, the vein and artery will have an opposite flow direction with respect to the probe. Under this condition, the positive and negative Doppler signals can be separated by proper manipulation of the quadrature signals (McLeod, 1967; Evans *et al.*, 2000) and presented via stereo speakers.

Velocity estimation

An important characteristic of a Doppler system is its capability to estimate the instantaneous Doppler frequency averaged over a short interval (e.g. 20 ms). The Doppler equation relates the average Doppler frequency to an average (blood) velocity. Usually in CW Doppler systems, the principle of a zero-crossing counter is used to estimate the average frequency (Evans *et al.*, 2000; Cobbold, 2007). For this purpose, only the sign of the quadrature signals is retained (positive/negative), so at any instant it is only known in which quadrant the phase vector is located, but its precise position (phase) within the quadrant remains unknown (Fig. 8.7B). If the phase vector changes a quadrant, either a positive or negative short pulse (duration about 0.1 ms) is generated, depending on the rotation direction. The positive and negative pulses are low pass filtered with a cut-off frequency of about 20–30 Hz (response time, 50–30 ms), yielding a velocity output (after proper calibration) as a continuous function of time. The filter process may be applied to the sum of positive and negative pulses or for both outputs separately. One should realize that the sum of the velocity outputs is not equal to the velocity of the summed outputs, because in the first case the relative amplitude information is not accounted for.

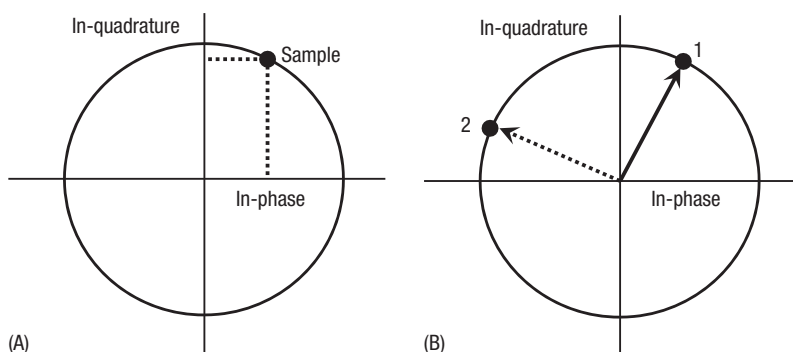


Figure 8.7 A complex signal is composed of a real (x -axis, in-phase) and an imaginary (y -axis, in-quadrature) component (A). The phase vector rotates counterclockwise for positive frequencies (B). Unambiguous phase change between observations is limited to 180°

The output of a zero-crossing detector is only accurate for signals with a narrow bandwidth with respect to the center frequency and the error will become substantial (10 percent) for wide band signals (Rice, 1944; Peronneau *et al.*, 1974; Gill, 1979), which is generally the case when observing a velocity distribution within an artery with a CW system. The situation becomes unreliable if simultaneously positive and negative flow velocities are observed (extremely wide bandwidth relative to the mean Doppler frequency). To avoid that under poor signal-to-noise conditions large errors in the velocity estimates occur, the zero-crossing detector is disabled when the Doppler signal amplitude falls below a preset threshold (Evans *et al.*, 2000; Cobbold, 2007).

Spectral analysis

Any signal in the time domain can be decomposed into its constituent components in the frequency domain (Fourier analysis). To allow proper interpretation, the signal should be stationary within a given time window, i.e. its statistical characteristics should be constant. Due to the pulsatile nature of blood flow, Doppler signals only comply with this requirement for a short period of time. Therefore, the quadrature Doppler signal is broken down into short (overlapping) segments of 10 ms that are subjected to Fourier analysis (Chapter 11). For that purpose, the quadrature signals are sampled and converted to digital numbers at a sample frequency that should be at least twice the maximally anticipated frequency to avoid aliasing. In the latter case, the phase vector rotates over more than 180° (Fig. 8.7B) between observations because of a displacement larger than a quarter of the wavelength. This is falsely interpreted as a phase shift of less than 180° in the opposite direction and the observed Doppler frequency deviates in magnitude

and in sign from the true one (aliasing) (Newhouse *et al.*, 1980; Hartley, 1981).

The spectral amplitude distribution of the subsequent time segments constitutes the spectrogram with time on the horizontal axis and the (Doppler) frequency scaled to velocity on the vertical axis. The amplitude of each spectral component is displayed as shades of gray (Fig. 8.8). The selected length of the time window dictates the width of each spectral bin (spectral resolution); for 10 ms segments this will be 100 Hz ($BT=1$). Increasing the length of the time segments will improve frequency resolution at the expense of blurring the time-dependent features of the signal, e.g. acceleration time. To avoid this problem, the time segments are normally half-overlapping, resulting in twice as many spectra for the same total length of the signal. Spectral analysis is rather insensitive to the signal-to-noise ratio, because noise is evenly distributed over the entire spectral range, while the Doppler signal of blood and specifically of tissue is concentrated within a narrow frequency band. Consequently, Doppler signals from vibrating vessel walls may incidentally show up in spectrograms, especially in the early systolic phase when acceleration is high.

As blood flow velocity accelerates from end-diastole to peak-systole, the number of observed blood cells will remain more or less the same and one would expect that the Doppler signal amplitude will hardly change. However, the increased friction on blood cells due to shear stresses causes spatially a more inhomogeneous density distribution and a concomitant increase in amplitude (Paeng *et al.*, 2001; Paeng and Shung, 2003; Paeng *et al.*, 2004). On the other hand, in the frequency domain at peak-systole the signal power is distributed over a wider range, causing substantial modulation of the spectral amplitude over the cardiac cycle. This modulation is attenuated by displaying the logarithm of the spectral amplitude (compression) with the additional advantage

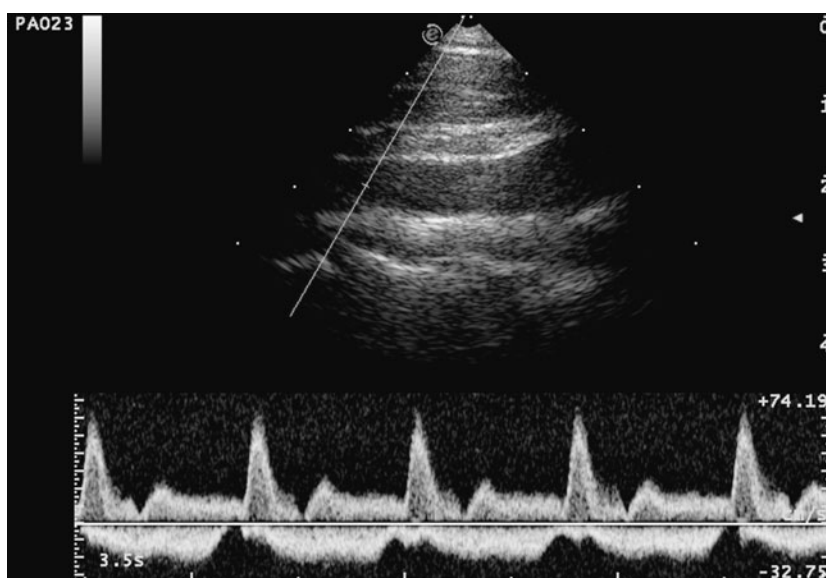


Figure 8.8 Continuous wave (CW) spectrogram of the common carotid artery (CCA; positive velocity) and jugular vein (negative velocity) as recorded with a phased array system. Because a large part of the artery cross-section contributes to the Doppler signal, the sonogram contains all velocities up to the maximum velocity (maximum Doppler frequency), although velocities close to the maximum dominate for the arterial blood velocity signal

that the appearance of the spectrogram is rather insensitive to the gain settings.

The angle of observation required to eventually convert Doppler frequency to velocity (eqn 8.10), follows from the superimposed presentation of an echo-image and the angled Doppler line (Duplex system). Prior to activation of the CW, a cursor is manually aligned with the assumed flow direction of the artery, providing direct system feedback for automatic conversion from Doppler frequency to velocity.

PULSED DOPPLER SYSTEMS

Continuous wave Doppler systems have the advantages of an unlimited velocity range and their easy use. On the other hand, they have a poor range selectivity which makes them less suited to identify regions with different flow patterns, as in cardiac cavities or extremities where arterial blood flow may reverse in end-diastole. Pulsed Doppler systems, like echo systems, allow interrogation of blood flow at a selected depth. Moreover, they can easily be combined with echo systems to provide visual feedback about the site of measurement and the angle of observation (Evans *et al.*, 2000; Cobbold, 2007).

Single gate pulsed Doppler system

The concept of pulsed Doppler systems is based on the notion that signals are fully determined by samples, provided that, according to the Nyquist theorem, the sample frequency exceeds twice the maximally anticipated signal frequency. This observation implies that the received signal can be applied intermittently rather than continuously to the input of the demodulator, i.e. the ultrasound can be emitted in short bursts with a corresponding increase in receiver and demodulator bandwidths ($BT=1$). Theoretically, the best depth selectivity will be obtained with echo-mode pulses (one to two periods). However, in practice, long bursts up to ten periods will be used in pulsed Doppler mode to avoid the effect of depth- and frequency-dependent attenuation on the center frequency (Fig. 8.2). Because the signal from the selected depth range is only intermittently present at the output of the demodulator, it has to be captured with a sample-and-hold to retain the value until it is updated in the next emission-reception cycle. The sample-and-hold is activated at a preset delay with respect to the time of emission, corresponding to the desired depth of interest, and acts as a gate for the demodulated and filtered signal. The receiver and demodulator bandwidths in pulsed Doppler mode set the axial length of the sample volume while its lateral dimensions are determined by the local shape of the ultrasound beam (Hoeks *et al.*, 1984).

It should be stressed that the demodulator of a pulsed Doppler system cannot provide the Doppler shift frequency, because of the short observation time. Instead

it provides the instantaneous phase of the RF signal with respect to the oscillator reference signals as function of depth. Only the output of the sample-hold reveals the Doppler signal over time. It has a staircase appearance and is directly suited for conversion to digital format to perform wall filtering or Fourier analysis to obtain the sonogram. In a digital filter, the cut-off frequency is directly related to its update frequency, i.e. the pulse repetition frequency (PRF) of the pulsed Doppler system.

A common parameter to extract from the spectrogram is the frequency envelope (Fig. 8.9), because it is rather insensitive to the observed velocity distribution and noise (Hoskins *et al.*, 1998; Hoskins, 1999) and can be directly associated with the center stream velocity for a straight artery. However, there is never a sharp cut-off for the maximum frequency. The transit time of a scatterer through the sensitive region of a Doppler system depends on the path of the scatterers and the velocity (Newhouse *et al.*, 1977; Newhouse *et al.*, 1980). Because of the time bandwidth product, the transit time (Willink and Evans, 1996) will cause blurring of the frequency spectrum (geometric broadening) which will vary throughout the cardiac cycle. In the past, several maximum frequency estimators have been proposed (D'Alessio, 1985; Nowicki *et al.*, 1985; Mo *et al.*, 1988; Marasek and Nowicki, 1994; Moraes *et al.*, 1995) and each of them uses some arbitrary criterion for the maximum frequency with respect to the spectral Doppler amplitude and the noise background.

One might argue that for a pulsed Doppler system with the sample volume placed properly within the lumen, the problem of 'wall filtering' is less compelling, because the simultaneous presence of structures and blood cells is quite unlikely. The real and much overlooked problem, however, is caused by the reverberations causing virtual echoes within the lumen. As discussed, reverberations originate from parallel oriented structures perpendicularly to the ultrasound beam direction. As the artery expands, these structures will be both displaced and compressed. The combined result is that reverberation echoes move approximately twice as fast, which demands for a higher cut-off frequency of the wall filter. Fortunately, Doppler investigations are angled with respect to the artery orientation, alleviating the problem.

A pulsed Doppler system samples the Doppler signal from a specific depth at a rate equal to the PRF of the system. This is sufficient to accommodate Doppler signals from physiological flows without velocity aliasing (peak velocities up to 1.5 m/s), but inappropriate for pathophysiological velocities as in arterial stenoses or stenosed cardiac valves. Aliasing will show up on the spectrogram as a roll-over: frequencies exceeding the upper limit of $PRF/2$ will appear at the bottom near $-PRF/2$ and vice versa (Fig. 8.9). This can easily be solved by shifting the baseline upward or downward, depending on the direction requiring the largest range. Baseline shifting will extend the frequency range to maximally PRF for unidirectional flow. Some systems provide a further

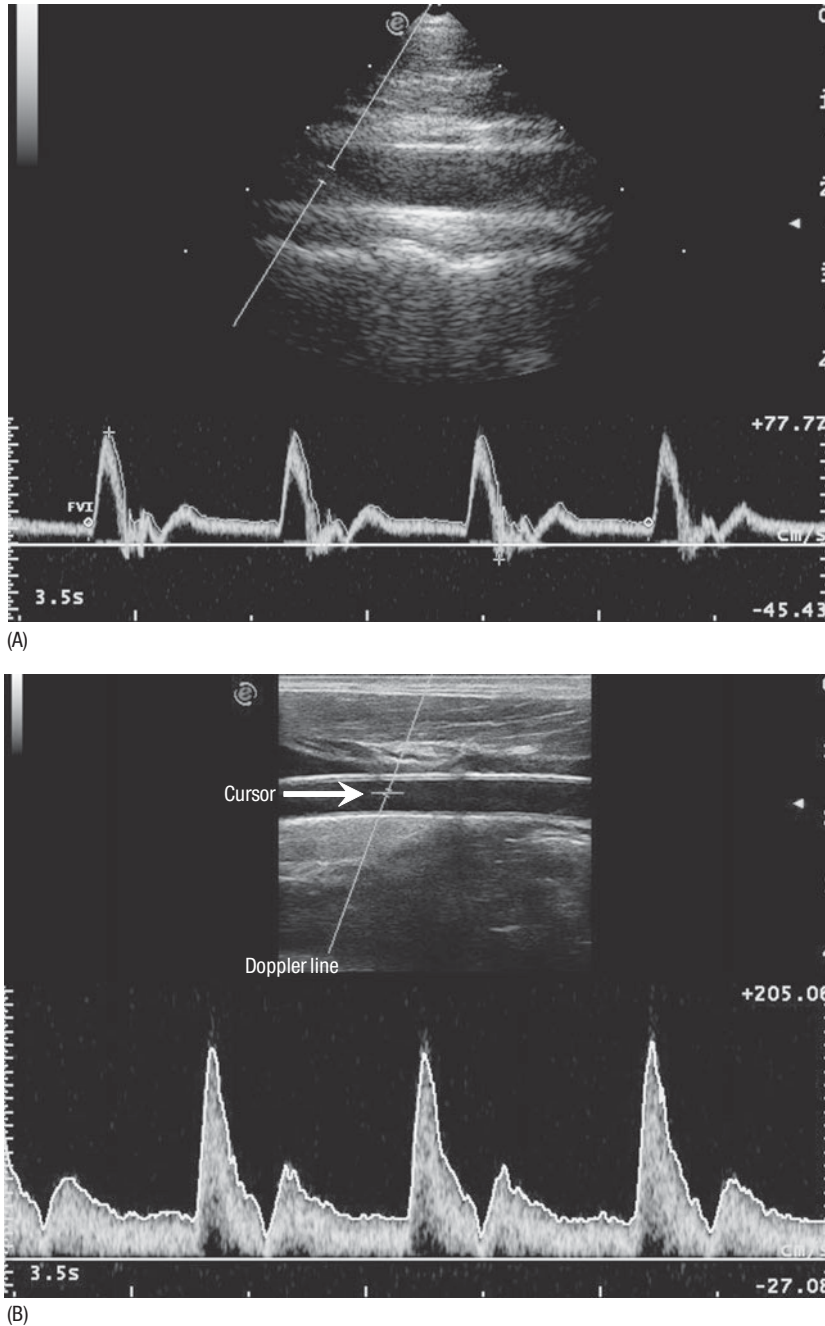


Figure 8.9 Pulsed Doppler recordings of the common carotid artery velocity with a phased (A) and linear (B) array. The cursor within the sample volume is manually aligned with the assumed flow direction. The envelope of the Doppler spectrogram is used to quantify the shape of the velocity waveform

extension by a high PRF (Hi-PRF) option where the PRF is set intentionally to a too high value. Then the sampled Doppler signals will originate from two depth locations (depth aliasing). If only one location contains flow, then the Doppler signal is appropriately obtained at a double observation rate, thereby extending the velocity range (Evans *et al.*, 2000; Cobbold, 2007). The drawback is that the depth-depending gain is restricted, putting a high demand on the dynamic range of the system, especially if the PRF is tripled or quadrupled.

After wall filtering, to eliminate the high amplitude low frequency signals generated by the artery walls and reverberations, the phase change of the complex sampled Doppler signal is a direct measure of the displacement of

scatterers within the sample volume per sample interval ($1/\text{PRF}$), i.e. velocity. If the Doppler signals are retained in their sampled (digital) form, then a zero-crossing counter is unsuited to estimate the average Doppler frequency, because transitions to opposite quadrants will occur for instantaneous Doppler frequencies exceeding $\text{PRF}/4$. In 1985, Kasai (Kasai *et al.*, 1985) proposed an alternative to determine average Doppler frequency, based on the argument of the complex autocorrelation $C(1)$ at time lag 1 of the sampled quadrature Doppler signals $d(t)$:

$$C(1) = \frac{1}{PL-1} \sum_{j=1}^{PL-1} d^*(j)d(j+1) \quad 8.11$$

In eqn 8.11, PL is the package length (the number of Doppler signals considered) and $d^*(j)$ is the conjugate of $d(j)$, i.e. the sign of the imaginary part is reversed. The precision of the velocity estimate improves with the square root of the package length (Hoeks *et al.*, 1992; Torp *et al.*, 1994; Loupas *et al.*, 1995) and will generally be better than 5 percent of the PRF. To maintain the dynamic characteristics of the velocity waveform, the observation window should be about 10 ms, which converts to a package length of 100 for a PRF of 10 kHz (maximum unambiguous depth of 7.5 cm). This quite large number ensures precise detection of the average velocity, even if sophisticated wall suppression filters, like singular value decomposition, are applied (Ledoux *et al.*, 1997; Torp, 1997).

Multigate Doppler system

The digital and computer techniques, emerging in the 1980s, facilitated reliable data processing of the sampled Doppler signals, regarding vessel wall suppression, mean velocity detection, and spectral analysis. The high processing power allowed integration of parallel processing channels for each sample volume into single circuitry with serial processing, accommodating many sample volumes (multigate pulsed Doppler system). Given this technology, it was a small step to drastically extend the number of consecutive sample volumes to 64 and even 128 samples along a line of insonation (Brandestini, 1978; Hoeks *et al.*, 1981; Nowicki and Reid 1981; Tortoli, 1996). A prerequisite is a large dynamic range for the analog-to-digital conversion of the outputs of the demodulator, because at that stage the weak Doppler signals from blood particles are still mixed with the large amplitude echo signals from stationary and slowly moving structures like vessel walls.

Multigate Doppler systems allow measurement of the time-dependent velocity distribution over the cross-section of an artery lumen as a function of time (Reneman *et al.*, 1985; Reneman *et al.*, 1986a; Reneman and Hoeks, 2000; Reneman *et al.*, 2006). The detail of the eventually obtained instantaneous velocity distribution largely depends on the spatial resolution that can be attained. As stated before, a short duration of the emitted acoustic pulse requires a large bandwidth, introducing the effects of depth- and frequency-dependent attenuation. Hence, the central frequency of the received signals will be lower than the emission frequency and, consequently, the velocity will be overestimated. To solve this problem, the local central frequency of received radiofrequency signals is estimated simultaneously from the demodulated RF signals (Loupas *et al.*, 1995; Rabben *et al.*, 2002), using autocorrelation at depth lag 1 over short depth segments. However, the required roll-off of the phase detection filters following the demodulators sets a lower bound for the effective pulse duration. For short emission, as in echo

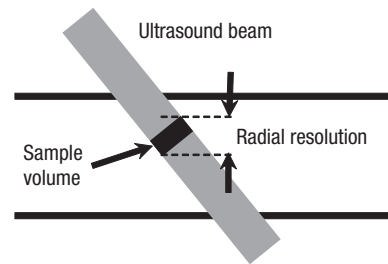


Figure 8.10 The spatial resolution in the radial range is affected by both the axial (length sample volume) and lateral (local beamwidth) resolution of a pulsed ultrasound system

mode, preference is given to direct processing of the RF signals (Brands *et al.*, 1997), with the advantage that the phase characteristics within the sample gate are retained, contributing to noise suppression.

One should realize that both pulse duration and local beamwidth in combination with the angle of observation affect the radial resolution of the velocity estimate (Fig. 8.9). For an observation angle of 60–70° between flow direction and incident beam, it does not have sense to make the pulse much shorter than 1/4 of the local beamwidth, since further shortening will hardly affect the observed range in the radial direction of the blood vessel (Fig. 8.10).

Color Doppler systems

Blood cells exhibit a low echogenicity compared to structures like vessel walls. Consequently, blood appears black on an echo-image, which makes it rather easy to locate normal arteries. The situation changes for arteries with calcified walls and a suspected low or absent blood flow. Moreover, thrombotic formations as a result of vulnerable plaques have a higher echo level than blood, blurring the contrast with vessel walls. That is why some decades ago it was proposed to extend the potentials of multigate Doppler systems to two-dimensional flow maps where the velocity information is superimposed in color (color Doppler) on the echo-image in B-mode (Kasai *et al.*, 1985). Direct visualization of the anatomical relationship between structures and blood flow makes it directly clear where anomalies are to be expected.

To obtain flow velocity maps, a large number of sample volumes distributed along the ultrasound beam are repetitively interrogated in pulsed Doppler mode. An observation time of 1 ms (package length of 6–12 subsequent emission and processing cycles) will do to obtain a crude estimate of the velocity distribution (Switzer and Nanda, 1985; Kurjak *et al.*, 1988; Holen *et al.*, 1990; Kitabatake *et al.*, 1990; Steinke *et al.*, 1990; Ranke *et al.*, 1992; Torp *et al.*, 1994; Wells, 1995; Herment *et al.*, 1996; Ferrara and DeAngelis, 1997; Torp, 1997; Hoskins, 1999), after which the beam is switched to another direction and/or position and the process is repeated. In this way,

16 velocity maps composed of 64 lines can be produced per second. A higher frame rate is possible by selecting a narrower region of interest for color Doppler. The time skew across the image is reciprocal to the frame rate. For linear array systems, the frame rate can be further boosted by activating simultaneously non-overlapping transducer groups. Although pulsed Doppler processing is subject to aliasing, it is not really a problem for color mode systems, because aliasing artifacts will show up as a mosaic pattern: the random distribution of high velocities with opposite directions are easily identified as jets with a high velocity as may occur in and distal to stenotic heart valves and artery stenoses.

Because of the short observation time, the estimated color Doppler velocities are rather noisy, requiring extensive smoothing in both temporal and spatial direction. Moreover, the short observation time sets a lower limit to the cut-off frequency of the vessel wall filter, with the risk of a void between the color map and the wall. Especially in the absence of a blood flow Doppler signal, the velocity estimates will be very noisy. That is why the color Doppler only is displayed if the local echo level is low and the Doppler amplitude exceeds a preset threshold. For accurate assessment of flow velocity anomalies, the system is switched back to single M-line color mode or to single gate Doppler (Duplex mode), permitting detailed frequency analysis for a sample volume positioned according to the acquired echo and flow map information.

It depends on the application, but in most situations one is interested in the presence and direction of blood flow rather than its magnitude. For those situations, it is sufficient to detect the amplitude of the Doppler signals and to display its two-dimensional distribution in color mode (power Doppler, angio Doppler). The information about flow direction (positive/negative) is retained in the color. Because Doppler amplitudes vary slowly, one may average over a longer time period, reducing estimation artifacts in the image (Moehring and Spencer, 2002; Saqqur *et al.*, 2004). The different scanning modes and how they have been interlaced are summarized in textbooks (Evans *et al.*, 2000; Cobbold, 2007).

RF processing

Estimation of the average velocity (Doppler frequency) after demodulation in a system with a high spatial resolution requires correction for the effects of depth- and frequency-dependent attenuation and will also make the velocity estimate noisier due to a decreased signal-to-noise ratio (SNR). These problems are circumvented by processing of the received radio frequency signals in echo mode (high axial resolution), captured digitally with a relatively high conversion frequency. The average displacement of reflectors within a depth window over repeated observations then follows from the location of

the peak of the cross-correlation at time lag 1 (Bonnefous and Pesque, 1986) of the two-dimensional RF matrix as function of depth and time. As opposed to conventional Doppler processing, displacement detection is performed by cross-correlating subsequent RF signals over a depth window rather than by correlating RF signal and reference (oscillator) signal. The position of the peak of the cross-correlation is a direct measure of the displacement. Its estimation improves with a higher sampling frequency and interpolation (Jensen, 1993a; Lai and Torp, 1999). This will put a high demand on processing power, also because for each sample point in depth the reflections of stationary and slowly moving reflectors should be removed prior to cross-correlation. A simple way to perform the latter operation is the removal of the baseline offset, as observed over the time window, but also higher order approaches have been suggested (Hoeks *et al.*, 1992; Jensen, 1993b; Schlaikjer *et al.*, 2003). The phase detector in traditional pulsed Doppler systems smoothes phase transitions prior to velocity estimation. The RF processing technique, however, retains in the displacement detection algorithm the information on phase transitions, originating from the randomly distributed scatterers (Hoeks *et al.*, 1992) by reversing the sequence of operations and selective averaging in the depth direction.

Weak points of the displacement estimation as outlined above are the high RF sampling frequency and the required range of correlation lags, spanning a displacement range of one to two wavelengths. However, the number of lags required can be reduced to only three if the RF signals are converted to a complex signal and the shape of the RF spectrum is known (Brands *et al.*, 1997). The characteristic parameters of the RF spectrum, i.e. RF signal power, noise power, RF center frequency, RF bandwidth, and Doppler frequency, follow directly from the estimated cross-correlation coefficients. The use of a complex correlation rather than a real correlation function (De Jong *et al.*, 1990) considerably attenuates the velocity-dependent bias (Brands *et al.*, 1997), which otherwise would cause drift of the estimated displacement waveform over a cardiac cycle (Hoeks *et al.*, 1992; Stadler *et al.*, 1997).

As stated above, the axial resolution of the system is set by the RF bandwidth, which directly follows from the analysis together with the local RF center frequency. To achieve the best performance, the length of the depth window for the cross-correlation in depth is set to the axial resolution of the system in echo mode (one wavelength). To improve robustness and detail in the velocity distribution, data windows are chosen half-overlapping in depth and in time, rendering a large number of velocity estimates per second for a depth range of 1–2 cm.

The common wall filter is a highpass filter and, hence, suppresses Doppler frequencies within a range around zero frequency. Considering a slowly moving structure like a vessel wall (or more importantly its reverberations within the lumen), the cut-off frequency and roll-off

should be set high to avoid that in the early systolic phase the filter will leak signals from the structure. Clearly a zero order filter (removing mainly the mean level) will not do. An alternative is to use the complex RF velocity estimator to provide the Doppler shift frequency of structures as function of time (Brands *et al.*, 1999). The wall Doppler shift is subsequently used to shift adaptively the complex RF signals in frequency, maintaining the spectral peak of the Doppler signals at zero frequency and allowing the highpass filter to function optimally. Since the spectral dispersion of the structures is limited, the selection of the filter cut-off frequency is only limited by the length of the time window (Ledoux *et al.*, 1997; Torp, 1997; Bjaerum *et al.*, 2002; Yoo *et al.*, 2003). After blood velocity detection, the estimated velocities have to be corrected for the imposed shift in signal frequency.

Unlike conventional Doppler techniques, employing long ultrasound bursts, echo RF correlation techniques combine the best spatial position and displacement resolution. For large structures, the color Doppler techniques supply a viable alternative to evaluate tissue motion. The only difference with blood Doppler techniques is that the wall Doppler filter is discarded, requiring a minor modification of color Doppler systems (McDicken *et al.*, 1992). Tissue Doppler techniques do not have problems with the signal-to-noise ratio. The observed tissue velocities are acquired relative to the transducer and eventually velocities should be expressed relative to a suitable reference point, e.g. for left ventricular wall contraction the center of the left ventricle.

APPLICATIONS

Diameter estimation

To identify arterial wall positions, necessary for the estimation of arterial diameter, one may start from somewhere within the lumen and allow the system to search for the wall interfaces, characterized by an edge with a relatively high amplitude, originating from the media-adventitia interface (Williamson *et al.*, 2008). Because its echo level varies with the system gain settings, the position estimate will be amplitude dependent when using an absolute threshold for the lumen-wall interface. Therefore, a two-step approach is followed. First, the maximum echo amplitude in a local region is determined dynamically and then the threshold level is set at 50 percent of the local maximum and the position from where the echo levels pass the threshold is searched (Meinders *et al.*, 2001a). This technique is applied to both the anterior and the posterior wall, yielding the diameter. In this approach, in the common carotid the error is about 0.1 mm for a 7.5 MHz system and considerably smaller than for edge detection with echo gradients (Rossi *et al.*, 2010).

The diameter of the common carotid artery gradually increases with age (Reneman *et al.*, 1985; Reneman *et al.*, 1986b; Samijo *et al.*, 1997). The diameter of the femoral artery varies strongly with sustained resistance variations of the peripheral vascular bed (Schmidt-Trucksäss *et al.*, 2000; De Groot *et al.*, 2006). A reliable wall detection algorithm is also of interest for studies on flow-mediated dilation in the brachial artery, as discussed in Chapter 5 (Celermajer *et al.*, 1992; Sorensen *et al.*, 1995; Stadler *et al.*, 1996; Black *et al.*, 2007; Thijssen *et al.*, 2008). However, edge detection algorithms suitable for the carotid artery cannot directly be transposed to the brachial artery. In muscular arteries, the lumen-intima transition generates a higher echo; the media is more echogenic, while the media-adventitia transition is less distinct. For a brachial artery with a diameter of 4 mm an absolute estimation error of 50–100 μm converts to a relative error of 1–2 percent (Sorensen *et al.*, 1995), which is substantial for an anticipated increase in end-diastolic diameter of 6 percent in response to flow stimulation.

Intima-media thickness

The major layers of the arterial wall appear on a B-mode image as a single reflection from the lumen-intima interface, a hypoechoic region associated with the media and hyperechoic reflections from the media-adventitia interface and the adventitia (Fig. 8.4). The outer bound of the arterial wall is not distinguishable on ultrasound scans, prohibiting proper evaluation of wall thickness, but as shown in Fig. 8.11 the intima-media thickness (IMT) is clearly visible as a double-line pattern (Touboul *et al.*, 2006). The echo of the lumen-intima transition (posterior wall) covers the intima and part of the media. Conversely, at the anterior wall, the trailing edge of the large adventitial echoes may obscure the relatively weak media signals, while the intima-lumen transition is blurred by reverberations. Therefore, IMT measurements are

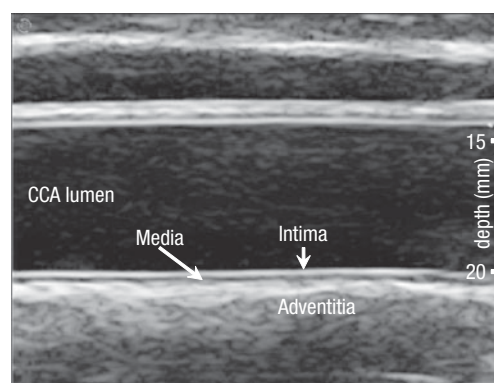


Figure 8.11 The lumen-wall echo has a typical double line pattern, composed of the lumen-intima echo and the media with a relatively low echo signal

generally executed for the posterior wall, preferably at perpendicular incidence (Touboul *et al.*, 2006). Starting with the basic study of Pignoli (Pignoli *et al.*, 1986), IMT has been derived from B-mode images (Selzer *et al.*, 1994; Wikstrand and Wendelhag, 1994; Gariepy *et al.*, 1996; Bae *et al.*, 2006), or from scans based on RF signals without compression (Willekes *et al.*, 1999). IMT estimation error is generally lower than the error in artery wall location determination, because any error made in media–adventitia interface identification is correlated with that of the lumen–intima interface. The IMT intrasubject inter-registration variation (reproducibility) can be as low as 40 μm for end-diastolic IMT averaged over the image (Stensland-Bugge *et al.*, 1997; Kanters *et al.*, 1998; Liang *et al.*, 1998; Baldassarre *et al.*, 2000b; Tang *et al.*, 2000; Schmidt-Trucksäss *et al.*, 2001; Selzer *et al.*, 2001; Hasegawa *et al.*, 2004).

It has been shown that the IMT of the common carotid artery is associated with age, and metabolic and vascular disease (Riley *et al.*, 1992; Riley *et al.*, 1996; Bots *et al.*, 1999; Baldassarre *et al.*, 2000a; Sun *et al.*, 2000; Tanaka *et al.*, 2000; Davis *et al.*, 2001; De Groot *et al.*, 2004; Tropeano *et al.*, 2004; Bots *et al.*, 2005; Rosvall *et al.*, 2005; Mattace-Raso *et al.*, 2006; Wang *et al.*, 2006; Cao *et al.*, 2007; Lorenz *et al.*, 2007; Kastelein *et al.*, 2008) and may have predictive power for future cardiovascular events (Zureik *et al.*, 2000). At younger age (24–39 years), the average increase in IMT per year in the common carotid artery is close to 6 $\mu\text{m}/\text{year}$ (Juonala *et al.*, 2008). For a wider age range, the average increase in IMT in this artery appeared to be 4 $\mu\text{m}/\text{year}$ for healthy controls and 9 $\mu\text{m}/\text{year}$ for subjects with familial hypercholesterolemia (De Groot *et al.*, 2004). An earlier study showed an increase of 9 $\mu\text{m}/\text{year}$ for a healthy population in the age range of 21–105 years (Homma *et al.*, 2001).

Despite the wide interest in IMT, it remains questionable whether the IMT measured in the common carotid artery indeed has predictive power in addition to standard risk factors (Simon *et al.*, 2010). Recent studies tend to suggest that for the common population IMT highly depends on subject age and that higher values can be attributed to the presence of plaques or fatty streaks, causing an irregular appearance which is coincident with and confirmative for vascular disease elsewhere (Zureik *et al.*, 2002; Schmidt-Trucksäss *et al.*, 2003; Hermans *et al.*, 2007; Junyent *et al.*, 2008; Graf *et al.*, 2009). It should be realized that despite the relation between an increase in IMT in the common carotid artery and vascular disease elsewhere, this increase cannot be considered as a causal step in the development of atherosclerosis.

Distension measurements

The assessment of vessel wall motion with pulsed Doppler systems requires that the wall remains within the sample volume. This should be irrespective of the dynamic

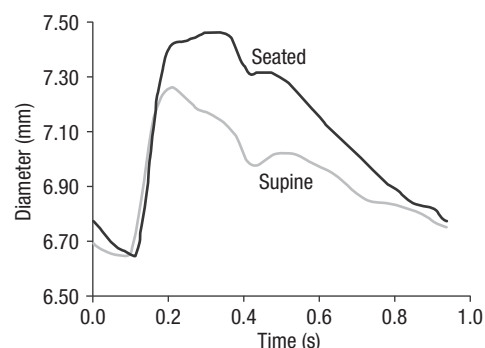


Figure 8.12 The distension waveform as recorded in the common carotid artery in supine and seated positions reflects the effect of changes in local pulse pressure

characteristics of the wall, implying a large sample volume relative to the artery caliber (Hartley *et al.*, 2005). Because the arterial wall will be compressed with the same percentage as the diameter expands (Hoeks *et al.*, 2008), a large sample volume will result in underestimation of the displacement. Knowing the observed displacement in the previous observation interval, the position of the sample volume can be adjusted accordingly (Groves *et al.*, 1982; Hoeks *et al.*, 1990; Okamoto *et al.*, 2008), i.e. the wall interface is tracked over time (phased locked echo-tracking). When applied simultaneously to the posterior and anterior walls, direct information over the change in diameter over time (distension waveform) is provided. For elastic arteries, the artery distension waveform during the cardiac cycle (Fig. 8.12) is linearly related to the local transmural pressure waveform.

The volume of blood temporally stored in the early phase of the cardiac cycle per unit of length, i.e. the compliance coefficient (CC), equals:

$$CC = \frac{\Delta A}{\Delta p} [\text{mm}^2/\text{MPa}] \quad 8.12$$

In this expression, Δp is the increase in pressure and ΔA is the associated increase in cross-sectional area, which follows directly from the end-diastolic diameter and the increase in diameter, assuming a circularly shaped cross-section. Generally, pulse pressure is used in this equation, which is not without difficulties for the common carotid artery (see below). A higher arterial compliance buffers the blood volume ejected by the heart with a lower increase in pulse pressure and, hence, a lower cardiac load (Safar *et al.*, 2003).

The extending arterial wall strains the wall, which is quantified by the distensibility coefficient (DC):

$$DC = \frac{\Delta A/A}{\Delta p} [1/\text{MPa}] \quad 8.13$$

The DC is a material property. Unlike the compliance coefficient, the distensibility coefficient is quite similar

across species. The DC of the common carotid artery degrades as a function of age due to a lower relative diameter change for an increasing pulse pressure (Reneman *et al.*, 1985; Reneman *et al.*, 1986b), indicating that the artery becomes stiffer. Because the diameter increases simultaneously, the effect of reduced distension is attenuated for the compliance (Reneman *et al.*, 1986b; Reneman *et al.*, 2006).

The carotid artery pulse pressure cannot be estimated non-invasively, but can be derived from the pulse pressure elsewhere, e.g. the brachial artery, assuming that the mean pulse pressure at both locations is the same and that the mean pulse pressure and mean distension only differ by a scale factor (Van Bortel *et al.*, 2001). Under those conditions, the estimated pulse pressure in the common carotid artery deviates about 1 mmHg from the actually measured one.

The estimation of CC and DC becomes more complex if the pressure–diameter relationship assumes an exponential relationship as likely occurs for stiffened arteries, but it remains possible to derive the local pressure waveform under those circumstances (Meinders and Hoeks, 2004; Vermeersch *et al.*, 2008). However, the differentials in the above expressions for CC and DC have to be replaced by derivatives, implying that CC and DC will become pressure-dependent. Evaluation of the pressure-dependent DC has shown that stiffening of arteries starts at higher pressures and gradually extends to lower pressures (Hermeling *et al.*, 2010). The findings in this study also indicate that the severity of artery stiffening is better evaluated by local pulse wave velocity as determined at systolic pressure than at diastolic pressure (Hermeling *et al.*, 2010). Patients with a similar pulse wave velocity in diastole appeared to have a significantly higher one at systolic pressure. These observations imply that the degree of systolic stiffening has to be considered when studying the consequences of arterial stiffening for left ventricular function. The study by Hermeling *et al.* also shows that after correction for time delay carotid artery distension versus aortic pressure does not show viscoelastic effects (closed loop), corroborating earlier observations that an open pressure–diameter loop stems from a mismatch in frequency characteristics of the recording systems (Hoeks *et al.*, 2000).

Local pulse wave velocity

Since the onset of the distension wave is coincident with that of the pressure wave, local pulse wave velocity can be directly extracted from the time delay between the feet of simultaneously recorded distension waves. The latter is possible with two-dimensional ultrasound imaging where the frame rate is boosted by a limited number of echo lines spaced at a larger interline spacing (Stale *et al.*, 1990; Meinders *et al.*, 2001b; Vappou *et al.*, 2010). In the approach followed by Hermeling *et al.* (Hermeling *et al.*,

2009) for carotid application, in total 14 echo lines were spaced at 1.3 mm (covering 18 mm) for a frame rate of 800 Hz. The slope of the regression line through the line position versus the onset of the distension waves is a direct estimate of the local pulse wave velocity, being 5.5 m/s for the common carotid artery independent of age (Hermeling *et al.*, 2009).

Volume flow and perivascular Doppler

Pulsed Doppler systems in combination with B-mode imaging (duplex systems), as described above, are specifically suited for transcutaneous examinations. With spectral analysis, it is quite easy to obtain relatively accurately the maximum frequency envelope as a function of time. Because of the large size of the sample volume with respect to the vessel diameter, the center stream blood velocities are quite likely captured. It is far more complicated to obtain transcutaneously realistic information about instantaneous volume flow, being the product of cross-sectional lumen area and average velocity across the same artery cross-section. Because beam diameter is generally smaller than lumen diameter, the average flow velocity along a central chord has to be extrapolated to average cross-sectional flow, assuming rotation symmetry which highly depends on the ratio of beam and artery diameter (Gill, 1985). Moreover, during the registration, the angle of observation should remain constant.

For invasive (animal) studies, the above problems are circumvented by a probe clipped on an artery (Transonic Systems, Ithaca, NY, USA). Two transducers are positioned on one side at some interspacing, while a flat hard reflector is mounted on the opposite side, such that the sound produced by one transducer after reflection reaches the other one. The transducers have a large dimension and produce a wide beam to achieve a rather homogeneous insonation over the artery cross-section. The ultrasound transit time from one transducer to the other varies with the direction and magnitude of the velocity of the intermediate medium. The transit time will be longer in the upstream than in the downstream direction. Their difference excludes the contribution of stationary vessel walls and is a direct measure of the average velocity of the moving medium (Drost, 1978). Because the cuff size determines both the artery caliber and the ultrasound sensitivity distribution, the technique can be calibrated such that it inherently accounts for possible sources of bias in the estimated instantaneous volume flow.

Wall shear stress *in vivo*

The endothelial cells, lining the artery wall on the luminal side, sense a pressure pulse, i.e. the difference between diastolic and systolic blood pressure, and a tangential

stress exerted by the flowing blood. The tangential stress is known as ‘wall shear stress’ (WSS). It can be estimated as the product of wall shear rate (WSR) and blood viscosity, wall shear rate being defined as the radial derivative of blood flow velocity at the wall. As shown in Chapter 5, wall shear stress is an important determinant of endothelial function and gene expression and plays a role in atherogenesis, possibly by modifying glycocalyx dimensions (see Chapter 5).

In *in vitro* and animal experiments, a steady-state flow condition is frequently assumed (Poiseuille flow) where shear stress τ is derived from measured flow q , the lumen radius r , and the medium viscosity η according to

$$\tau = \frac{4\eta q}{\pi r^3} \quad 8.14$$

In clinical studies, shear stress is often calculated from whole blood viscosity η and shear rate γ , as estimated from the measured average blood flow velocity, and internal diameter of the artery, assuming a parabolic velocity profile according to (Hoeks and Reneman, 2001)

$$\tau = \eta\gamma = \frac{4\eta v_m}{d} \quad 8.15$$

where v_m is the center stream blood flow velocity averaged over a cardiac cycle and d the end-diastolic internal arterial diameter.

In the *in vivo* situation, we are dealing with non-Newtonian fluid under pulsatile conditions, distensible vessels and, due to the branching arterial tree, the effect of entrance lengths. Consequently velocity profiles, i.e. the velocity distributions over the cross-sectional area of the artery will not develop to a full parabola. Therefore, the equations (8.14) and (8.15) can only be considered as a first order approach to calculate shear stress averaged over the cardiac cycle (mean shear stress). A more precise estimation of pulsatile shear stress *in vivo* is derived with a Womersley expansion (Leguy *et al.*, 2009a; Leguy *et al.*, 2009b) or from actually recorded velocity profiles.

In vivo wall shear stress can be estimated from WSR and blood viscosity, WSR being derived from velocity profiles recorded with either ultrasound or magnetic resonance imaging (MRI). Ultrasound is an attractive option, especially because of its better temporal resolution compared to MRI. To achieve a good spatial resolution across the artery lumen for a multigate pulsed Doppler system, a large number of sample volumes with small dimensions is required. This is possible by operating a Doppler system with a large bandwidth, accommodating short ultrasound bursts (Tortoli *et al.*, 1985; Tortoli *et al.*, 1989; Loupas *et al.*, 1995; Tortoli *et al.*, 2006), or processing the RF signals acquired in echo mode (Brands *et al.*, 1997). For example, RF signals obtained from the human common carotid artery with a 7.5 MHz two-dimensional imaging device in echo mode permit a sample volume resolution of 300 μm in depth

(Brands *et al.*, 1999). Considering half-overlapping sample volumes with a spacing of 150 μm provides a detailed time-dependent velocity distribution (Fig. 8.13). The shear rate distribution is obtained by determining the radial derivative of blood flow velocity at each site and each time instant (Fig. 8.13). WSR is high near the wall and drops non-linearly towards the center of the vessel, which is consistent with a flattened parabolic velocity profile, a shape found in most large arteries (Reneman *et al.*, 2006). Because blood flow velocities cannot be determined exactly at the wall, the maximum value of the radial derivative of the velocity profile is considered as the estimate of instantaneous WSR. From the shear distribution mean WSR, the time-averaged shear rate over one cardiac cycle, and peak WSR, the value at peak systole, can be determined.

In large arteries, whole blood viscosity is used to calculate WSS, because the plasma layer of a few micrometers in thickness (Wazer, 1963) can be ignored relative to the size of the sample volume of the ultrasound system (250–300 μm). Whole blood viscosity (WBV) can be estimated, using the equation as proposed by Weaver *et al.* (1969):

$$\begin{aligned} \log(\text{WBV}) = & \log(\eta_0) \\ & + (0.03 - 0.0076 \log(\gamma))Ht \end{aligned} \quad 8.16$$

where η_0 is plasma viscosity and Ht is hematocrit. Under the shear rate conditions found in large arteries *in vivo* (Reneman *et al.*, 2006), the effect of incidental changes in plasma viscosity on this estimation of WBV is negligible (Dammers *et al.*, 2003), leaving shear rate and hematocrit as the most relevant parameters. Mean WSR varies along the arterial tree, being substantially higher in the common carotid artery (310–414 s^{-1} in the various studies) than in the femoral (130 s^{-1}) and brachial arteries (95 s^{-1}) (Reneman *et al.*, 2006). This implies, that, for the same subject whole blood viscosity is higher in the brachial (5 $\text{mPa} \cdot \text{s}$) than in the common carotid artery (3 $\text{mPa} \cdot \text{s}$). Consequently, no average whole blood viscosity value can be taken to calculate WSS. If determined *in vitro* with a cone plate viscometer, it should be measured at appropriate and sufficiently high shear rates, which is not always possible (Gnasso *et al.*, 1996).

WSS values across arteries and across species

In presumed healthy volunteers, mean WSS is far from constant along the arterial tree as predicted by theory (Reneman *et al.*, 2006). It varies on average from 1.2–1.3 Pa in the common carotid artery, to 0.3–0.5 in the femoral artery bifurcation and around 0.5 in the brachial artery (Table 8.3). In normal coronary arteries, mean shear stress, derived from intra-coronary velocity measurements and coronary angiography, was found to be on

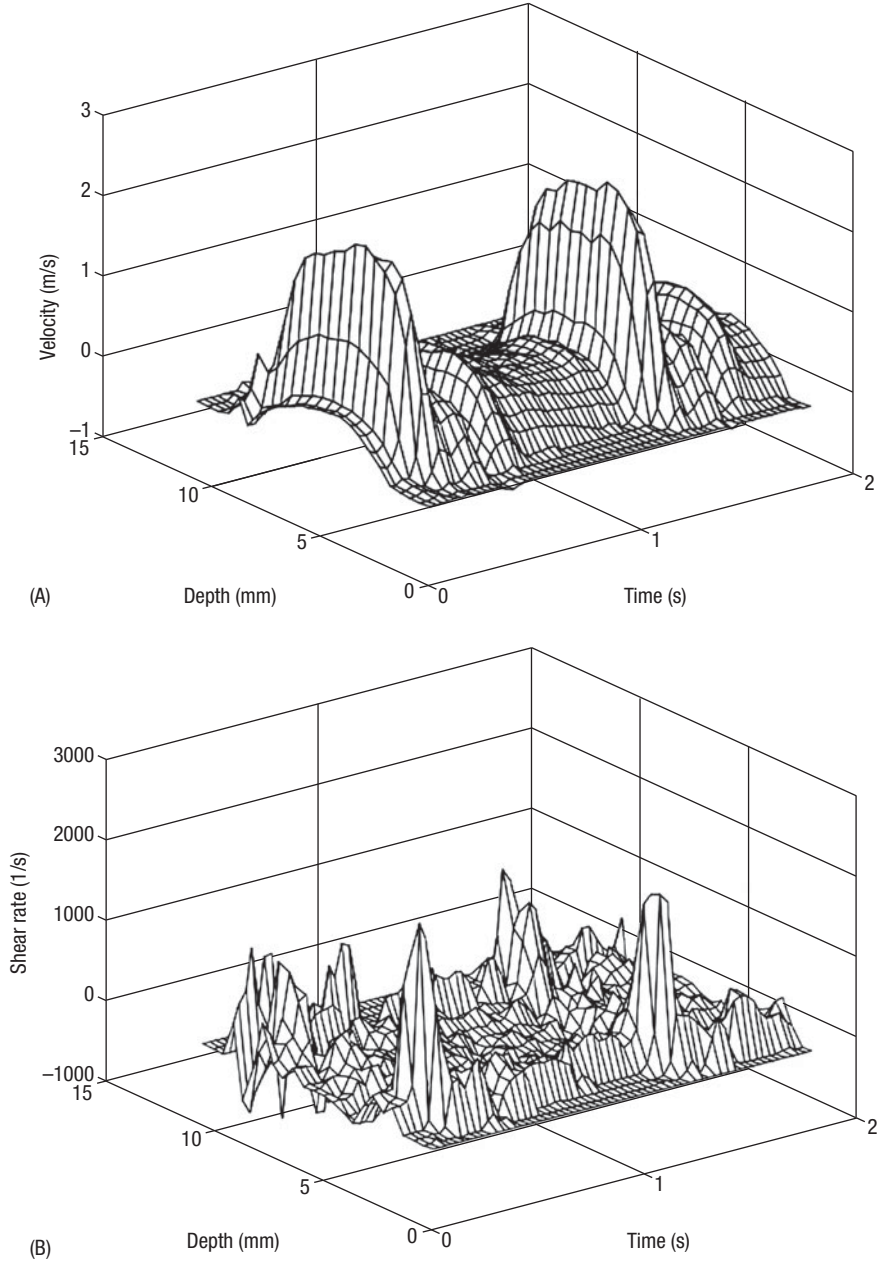


Figure 8.13 Velocity (A) and shear rate (B) distribution in the common carotid artery of a young healthy volunteer

Table 8.3 The average values of mean and peak wall shear stress (WSS; Pa) in the common carotid artery (CCA), the common (CFA) and superficial femoral (SFA) arteries and the brachial artery (BA) at rest, as determined in the various studies on presumed healthy volunteers (Reneman *et al.*, 2006)

	Mean WSS		Peak WSS	
	Young	Old	Young	Old
CCA	1.3	1.2	3.8	2.6
CFA	0.4	0.3	4.0	3.8
SFA	0.5	0.5	3.4	4.0
BA	0.5	0.5	3.6	3.3

the average 0.68 Pa (Doriot *et al.*, 2000). Only in the common carotid artery is the value close to the theoretically predicted value of 1.5 Pa (Kamiya *et al.*, 1984). The lower mean WSS in these conduit arteries at rest can be explained by the high peripheral resistance in these vessels, reducing mean volume flow and inducing reflections. Dilation of the femoral artery vascular bed results in mean WSS values not significantly different from those in the common carotid artery (Kornet *et al.*, 2000). This observation indicates that, at rest, mean WSS is largely determined locally. The low mean WSS in these conduit arteries allows for an increase in mean WSS during exercise without reaching levels that could be damaging to the

endothelial cells. The pronounced increases in flow under these circumstances, after all, are not compensated for by an adequate enlargement of the arterial diameter. Mean WSS is lower in the common than in the superficial femoral artery. Although small, this difference in mean WSS is significant (Kornet *et al.*, 2000), the common femoral artery seeing reflections from both the deep and the superficial femoral artery, while the latter one only sees reflections from its own vascular bed. Also, in the carotid artery, bifurcation differences in mean WSS have to be appreciated (Kornet *et al.*, 1998). Mean WSS in the elastic common carotid artery decreases slightly, but significantly, with age (Reneman *et al.*, 2006). This can be explained by the increase in diameter of this artery at older age (Reneman *et al.*, 1985; Reneman *et al.*, 1986b) to limit the loss of arterial compliance as a consequence of the stiffening of this artery with increasing age (Reneman *et al.*, 2006). This is an example of competition between two regulatory mechanisms. Peak WSS (Table 8.3) does not vary significantly along the arterial tree, albeit that peak WSS in the carotid artery tends to decrease with age (Reneman *et al.*, 2006).

Recent studies have shown that in a particular artery, mean WSS differs significantly across species. For example, in the infrarenal aorta mean WSS increases linearly with decreasing body mass on a log/log scale from 0.5 Pa in humans to about 7.0 Pa in rats and about 8.8 Pa in mice (Greve *et al.*, 2006). In their study, WSS was estimated similarly to the approach described above, except that the velocity profiles were recorded with MRI. In their review, Cheng *et al.* (2007) collected a similar pattern for the common carotid artery, mean WSS on the average varying from 1.2 Pa in humans to 4.7 Pa in rats and 6.5 Pa in mice. The substantially higher WSS values in small animals indicate that their endothelial cells are less sensitive to shear forces or that they function at a different set-point.

CONSEQUENCES OF THE MEAN WSS FINDINGS *IN VIVO*

The differences in mean WSS as observed *in vivo* along the arterial tree and in a particular artery between species imply that in studies on the effect of shear stress on endothelial cell function and structure, these cells have to be studied at the shear stress levels they are exposed to in real life. No general shear stress value can be taken for endothelial cells derived from different vascular areas or from different species. What is normal for one type of endothelial cell can be too high or too low for another. Application of the right level of shear stress to specific endothelial cells is of utmost importance, because the pattern of gene expression has been shown to be dependent on the level of shear stress applied (see Chapter 5).

The *in vivo* observation that WSS is far from constant along the arterial tree is in disagreement with the theoretical assumption of constancy independent of the location in the arterial system (LaBarbera, 1990; Kassab and Fung, 1995). The concept of constant mean WSS is based upon the principle of minimal work according to Murray's law (Murray, 1926), which implies that the cube diameter of a vessel should be proportional to the flow through it. Consequently, the cube of the diameter of a parent vessel equals the sum of the cubes of the diameters of the daughter vessels. In Murray's law, however, it is assumed that Poiseuille's law can be applied to the arterial system. The validity of Murray's 'cube law' for the whole arterial tree was first challenged by Zamir and colleagues (Zamir *et al.*, 1992). At the present state of the art, it may be concluded that Murray's law does not hold for the whole arterial system and that at rest the exponent of the power law varies along the arterial tree, possibly from 2.0 in larger arteries to 2.55 in coronary arteries and about 3.0 in arterioles (Reneman and Hoeks, 2008).

Wave reflections

Introduction	195	Reflections of pressure and flow waves in the arterial system	209
The reflection of a transient wave	198	Classification of arterial pressure waves	210
Reflected waves in a steady-state oscillation	201	Wave reflection and myocardial oxygen supply and demand	215
Reflection in a transmission line	202	Sites of wave reflection	216
The effect of reflections on apparent phase velocity	204	Secondary reflections of reflected waves	218
Reflection coefficient	204	Wave reflection in clinical trials	220
Reflection at a single arterial junction	206	Modification of wave reflection with drugs	222
Reflections due to a continuous change in elastic properties	208		

The great authority of Galen is so all-powerful that I see several hesitate over his experiment about the tying of an artery above a reed introduced into its cavity. The object of the experiment is to show that the pulsation of arteries is brought about by a property transmitted from the heart through their coats and not by the impact of blood within their cavities ... For, so soon as you have tied the artery above the reed or tube with a string, the vessel in question promptly dilates heartwards from the end of the reed under the impact of the blood from above. In consequence, the (forward) flow of blood is impeded and its impact is reflected backwards.

Harvey, 1649

INTRODUCTION

This chapter in the jubilee edition introduces wave reflection and largely explains differences in pressure and flow waves in different arteries under different conditions, as set out in the next chapter. It also refers to other chapters where change in wave reflection explains apparent paradoxes in management of heart failure (Mullens *et al.*, 2008; O'Rourke and Nichols, 2009a) and differences between achieved blood pressure and outcome in clinical trials (Yusuf *et al.*, 2000; ALLHAT, 2002; Dahlöf *et al.*, 2002; Sever *et al.*, 2005; Williams *et al.*, 2006; Jamerson *et al.*, 2008) and smaller trials (Asmar *et al.*, 2001b; Matsui

et al., 2009). It provides a mechanism to explain effects of aging and of diseases such as aortic coarctation (Chapters 19 and 24). It can also explain differences in severity of hypertensive disease in black and white people (Heffernan *et al.*, 2008), and the greater susceptibility of black people to heart failure (Bibbins-Domingo *et al.*, 2009), and differences between males and females (Cecelja *et al.*, 2009). It explains why the same classes of drug are so effective in improving outcomes of patients with hypertension and with systolic heart failure, and probably diastolic heart failure as well – on the basis of reduction in early wave reflection (Chapter 26). Yet wave reflection is barely mentioned in textbooks of medicine or cardiology, and rarely discussed on the main stage of scientific cardiac meetings. And, there is still confusion in the existing literature on the intensity and importance of wave reflection (Table 9.1). Understanding of wave reflection played a key part in the development of scientific surgery.

The father of modern scientific surgery, John Hunter, took note of Galen's and Harvey's observations in his famed descriptions of a new operation for cure of popliteal aneurysm (Moore, 2005). Previous attempts to tie the femoral artery above the aneurysm usually resulted in failure, either from gangrene of the leg below the knee or from secondary hemorrhage at the site of the ligature. Hunter overcame this by applying a series of ligatures to the femoral artery causing this to taper, so that reflection occurred at the ligatures above, and not just from the distal site of occlusion that occluded the artery.

Table 9.1 Views on magnitude of wave reflection in systemic arteries

Era	Method	Authors	Study	Reflection intensity
180	PWC	Galen (quoted by Harvey (1649))	Animal	Strong reflection when artery occluded
1648	PWC	Harvey (Harvey (1649))	Animal	Strong reflection when artery occluded
1785	PWC	Hunter (Moore, 2005)	Human	Wave reflection decrease with multiple occlusion
1908	T, PWC	Frank (1905)	Animal	Strong wave reflection 'Grundswingung'
1939	PWC	Hamilton and Dow (1939), Remington (1963)	Animal	Reflection 100%, 'standing wave'
1955	T, FA	Womersley (Womersley, 1957a; Womersley, 1957b; Womersley, 1958), McDonald and Taylor (1959)	Animal	Wave reflection definite but low
1965	T, FA	Taylor (Taylor, 1964; Taylor, 1966a; Taylor, 1966b)	Animal	High for low frequency component, low for high frequency components
1966	T, FA, PWC	O'Rourke and Taylor (1966), O'Rourke (1967a)	Animal	High wave reflection
1972	FA, PWC	Murgo <i>et al.</i> (1980a)	Human	Wave reflection ↑ with age
1995	PWA, T	Westerhof and O'Rourke (1995)	Animal	Wave reflection high, adds to pressure or subtracts from flow
2004	T, FA, PWA	Mitchell <i>et al.</i> (2004a)	Human	Wave reflection ↓ with age
2005	T, FA, PWA	O'Rourke and Nichols (2005)	Human	Wave reflection ↑ with age
2008	PWA, WIA	Davies <i>et al.</i> (2007)	Human	Low for low frequency component, high for high frequency component
2009	PWA	Namasivayam <i>et al.</i> (2009a)	Human	Wave reflection high, at all ages

FA, Fourier analysis; PWA, pulse wave analysis; PWC, pulse wave contour; T, theory; WIA, wave intensity analysis.

An observer, Paolo Assalini, described the procedure as follows:

John Hunter, having placed the patients in a convenient position, made an incision about five inches long in the skin of the inner, lower part of the thigh with a small knife; he exposed the artery, then grasped it in his fingertips and separated it from the cellular membrane, the vein and the nerve for a distance of eight inches, making use only the slender handle of his knife. This done, he passed under the artery a grooved probe, and ran along this a silver needle threaded with a thick ligature which he tied on the lower part of the artery to block it off. The pulsation in the tumour ceased immediately but the blood thumped with such force against the ligature that to diminish the knocking and the danger of the thing bursting he made another ligature two inches away from the first one and tied it less tightly. Above this second one he made a third, tied even more loosely, and then a fourth ... Then, after pulling the threads to the outside, and separating them, he closed the lips of the wound.

Mariners and yachtsmen taking to the seas need know about wave travel and reflection. They seek refuge on stormy days inside the harbour wall and know the hazards that abound outside the wall whence waves are reflected. To cardiologists as to mariners, wave reflections are a fact of life, yet many do not see the waves or seek to interpret them. Further editions of this book undoubtedly will contain more information on wave reflection.

So far, we have considered pulsatile blood flow in arteries as though they were long uniform tubes with a remote end. The actual situation, with the repeated occurrence of branches ending in the capillary bed and arterial non-uniformity, makes it necessary to consider what effects, if any, changes in the physical properties of the arteries beyond the site of measurement will have on the pressure and flow waveforms. Such effects do indeed exist and are primarily attributable to the presence of reflected waves, and it is generally agreed that such waves do occur (Milnor, 1989; O'Rourke *et al.*, 1992a; O'Rourke *et al.*, 1992b; O'Rourke *et al.*, 1992c). Even though the concept of pulse wave reflections in hemodynamics was first referred to by William Harvey in his description of the circulation in the seventeenth century, it is still omitted from most modern textbooks dealing with the heart and arterial system. For this reason, most cardiologists and cardiovascular physiologists are not familiar with the concept of wave reflections. Numerous articles have been published on the topic since its introduction, including several reviews (Westerhof *et al.*, 1978; O'Rourke and Yaginuma, 1984; O'Rourke and Kelly, 1993; O'Rourke and Gallagher, 1996; Nichols, 2005; Agabiti-Rosei *et al.*, 2007; Laurent and Boutouyrie, 2007; O'Rourke and Hashimoto, 2007; Nichols *et al.*, 2008; Swillens and Segers, 2008), books (Milnor, 1989; Nichols *et al.*, 1993; Safar and O'Rourke, 2006; Namasivayam *et al.*, 2009a), and journal supplements (see, for example, *Medical, Engineering, Biological and Computing* 2009: 47, 107–239). Its importance in cardiovascular medicine cannot be overemphasized. Numerous observational epidemiology studies and clinical trials have shown a strong

association between indices of arterial stiffness and pulse wave reflections (pulse pressure) and adverse cardiovascular events including mortality (Amar *et al.*, 2001; Franklin *et al.*, 2001; Laurent *et al.*, 2001a; van Bortel *et al.*, 2001a; Boutouyrie *et al.*, 2002; Klassen *et al.*, 2002; Mackey *et al.*, 2002; Meeks, 2002; Tozawa *et al.*, 2002; Roman *et al.*, 2009; Wang *et al.*, 2009; Benetos *et al.*, 2010; Manisty *et al.*, 2010; McEniery *et al.*, 2010a; McEniery *et al.*, 2010b; Vlachopoulos *et al.*, 2010a; Vlachopoulos *et al.*, 2010b; Wang *et al.*, 2010; Weber *et al.*, 2010). Independent studies have shown that central elastic artery stiffness and wave reflections are increased in older individuals (Nichols *et al.*, 1985a; Franklin *et al.*, 1997; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b; McEniery *et al.*, 2005a) and in patients with coronary artery disease (Gatzka *et al.*, 1998), myocardial infarction (Hirai *et al.*, 1989), hypertension (Benetos *et al.*, 2002), stroke (Lehmann *et al.*, 1995), diabetes mellitus (Wilkinson *et al.*, 2000a; Wilkinson *et al.*, 2000b), end-stage renal disease (Blacher *et al.*, 1999b), and hypercholesterolemia (Wilkinson *et al.*, 2002). These studies are discussed in other chapters.

This chapter will review the theoretic aspects of wave reflections and their occurrence and importance in the mammalian arterial system in health and disease and their modification with vasoactive drugs.

A traveling wave, such as the pressure or flow wave, will be reflected to some extent wherever there is a discontinuity (or impedance change) in the system. Possible reflecting sites include branching points, areas of alteration in arterial elastance (or stiffness), and the high-resistance arterioles (Hamilton, 1944; Westerhof *et al.*, 1972; McDonald, 1974; O'Rourke, 1982a; O'Rourke, 1982b; Milnor, 1989; Khir and Parker, 2002; Nichols, 2005). The extreme cases are where a tube with pulsatile liquid flow is either completely blocked or opens into a large reservoir, either of which causes total wave reflection, although in the latter case it is in the opposite sense (180° out of phase with the former). Grashey (1881) and Von Kries (1892) defined pressure reflections at a closed end as 'positive reflection' and at an open end as 'negative reflection'. In the arterial system, we are obviously dealing with intermediate conditions causing only partial reflection, but the terms 'closed' and 'open' end are convenient to describe their character. A discontinuity may be a change in caliber, as at a point of branching or taper, or merely a change in the elastic properties of the arterial wall. It might, therefore, seem obvious that the arterial tree, which has many discontinuities of this sort, would create large reflections and that there are no grounds for argument. The experiments of Peterson and Shepard (1955), however, showed that a large pressure wave created by a rapid injection pulse of blood retrogradely into a femoral artery had no detectable effect on the pressure curve recorded at the root of the aorta. Their interpretation was that any reflected waves created in the periphery (which would undoubtedly be very much smaller than their experimental ones) would be very rapidly damped out. This was in

agreement with earlier findings of Peterson (1954), that the pressure wave in the arch of the aorta due to an injection pulse contained no components that could be attributed to discrete reflected waves. This is not, however, a good argument against wave reflection. Reflected waves return from multiple sites in many arteries, the femoral artery being just one of these. A more realistic test would have been simultaneous generation of pressure waves in several peripheral arteries. Starr (1955) also performed experiments that questioned the existence of reflections. His experiments were performed on human cadavers. After artificially generating a pressure wave in the aorta, he measured only a small retrograde reflected wave and found no evidence of secondary reflections at the aortic valve or in the upper part of the body. These results, however, were obtained under conditions of severe peripheral vasodilatation (death) and are similar to those obtained in living patients during administration of vasodilator drugs (Kelly *et al.*, 1990a).

The existence of wave reflections within the arterial tree is supported by several observations. The shapes of the pressure and flow waves measured, for example, in the ascending aorta are dramatically different (Fig. 9.1). During ventricular ejection, the aortic pressure often continues to increase while the flow is decreasing (Fig. 9.1A). This secondary rise in pressure must be due to wave reflection (Wiggers, 1928; Hamilton and Dow, 1939; Hamilton, 1944; Westerhof *et al.*, 1972; Sipkema *et al.*, 1980; O'Rourke, 1999; Nichols and Edwards, 2001; Davies and Struthers, 2003). Large secondary pressure waves are also seen in some animals during diastole when the aortic valve is closed and flow is zero (Fig. 9.1B) (O'Rourke, 1967a; Nichols *et al.*, 1986a), and in patients with cardiogenic shock in whom hypotension and vasoconstriction are combined (O'Rourke, 1971). Also, the amplitude of the pressure pulse increases and the flow pulse decreases as they travel towards the periphery (see Figs 10.1, 10.6, and 10.8A) (Wiggers, 1928; Hamilton and Dow, 1939; Hamilton, 1944; O'Rourke, 1982a; Avolio *et al.*, 1984). These changes in peak pressure and flow can only result from wave reflections.

Wave reflection is the major cause of increasing systolic blood pressure and decreasing ventricular ejection with age (O'Rourke, 1982a; O'Rourke, 1982b; Nichols *et al.*, 1985a; Nichols *et al.*, 1987; O'Rourke *et al.*, 1991; Nichols *et al.*, 1993; Nichols, 1994; Westerhof and O'Rourke, 1995) and is responsible in youth for substantial differences in systolic blood pressure between central and peripheral arteries (Figs 9.2 and 10.1). Such differences in systolic blood pressure and ventricular ejection are due to the amplitude and timing of wave reflection (O'Rourke *et al.*, 1991; Nichols *et al.*, 1993; Nichols and Edwards, 2001; Segers *et al.*, 2007a). Wave reflection is also responsible for the characteristic pressure waveforms of patients with elevated arterial blood pressure (O'Rourke, 1970a; Nichols *et al.*, 1986b; Ting *et al.*, 1990; O'Rourke *et al.*, 1991; Nichols *et al.*, 1992; Nichols *et al.*, 1993; Nichols *et al.*, 2008) as first described by Frederick Mahomed

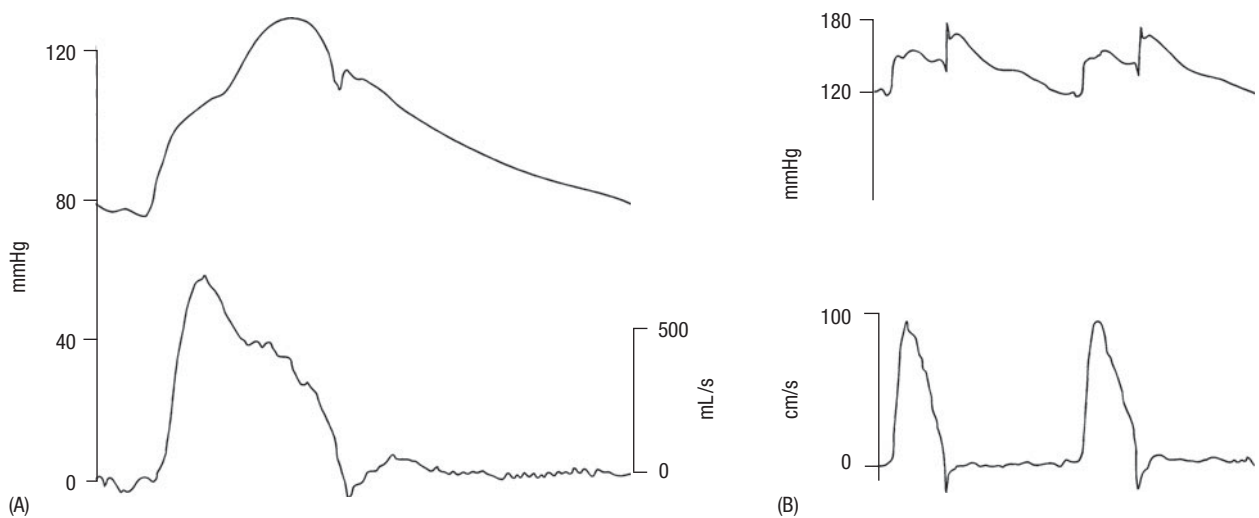


Figure 9.1 High-fidelity ascending aortic pressure (top) and blood flow velocity (bottom) waves recorded in (A) a human and (B) a kangaroo. In the pressure waveform of the human, there is a large secondary wave that begins just after peak blood flow velocity (note the inflection on the pressure wave) and continues throughout the remainder of systole. This increase in pressure occurs even though the flow is decreasing. In the kangaroo, the secondary pressure wave occurs in early diastole, while blood flow is zero. The peak of this diastolic pressure wave is greater than peak systolic pressure

(Mahomed, 1872; Mahomed, 1874). Although wave transmission and reflections are known to exist, they are ignored, and simplistic inaccurate models of the arterial system are relied upon (the Windkessel) (Frank, 1899; Wezler and Boger, 1939; Westerhof *et al.*, 1969; Noordergraaf, 1978; Burkhoff *et al.*, 1988). In this model, the pressures generated by the heart are assumed to be transmitted instantaneously throughout the Windkessel, whereas a finite, though relatively short, time is actually required for transmission in the major arteries. Moreover, it is assumed that the pressure pulse generated by cardiac contraction dies away before the next beat, and that there is a single ‘systolic’ and a single ‘diastolic’ blood pressure that is the same in all major arteries.

THE REFLECTION OF A TRANSIENT WAVE

In a string

Suppose that a transverse wave (or pulse) travels down a stretched string that is ‘fixed’ at one end to a stationary support (Fig. 9.3, left). This centrifugal wave will travel toward the fixed end at a velocity c_0 which is dependent on the elasticity (E) and density (ρ) (or mass) of the string:

$$c_0 = \sqrt{E/\rho} \quad 9.1$$

When the wave arrives at the fixed end, it exerts an upward force on the support, and by Newton’s third law of motion an equal but opposite force is exerted on the string by the support. This reaction force generates a wave

that travels at the same velocity as the centrifugal wave, but backward along the string in a direction opposite to that of the forward (or centrifugal) wave. This centripetal (or backward) wave is referred to as a ‘reflected wave’. The transverse displacement of the centripetal wave is reversed or 180° out of phase with respect to the centrifugal wave. If a series of periodic centrifugal waves (a wave train) is generated in the string, each wave will be reflected at the fixed end. The displacement of any point along the string is the algebraic sum of the displacements caused by the centrifugal and centripetal waves. Since the end point of the string is fixed, centripetal waves subtract from the centrifugal waves.

Let us now consider reflection of a wave at a ‘free’ end of a stretched string (i.e. at an end that is free to move transversely). This can be achieved by attaching the end of the string to a ring that slides without friction along a vertical rod (Fig. 9.3, right). When the centrifugal wave arrives at the free end, an upward force is exerted on the ring, and again a centripetal wave is generated, this time in phase with the centrifugal wave. Since the string terminates in a free end, the centripetal wave adds to the centrifugal wave.

In the treatment given above, it is assumed that the string’s elasticity is uniform throughout its length and that there is total reflection of the centrifugal wave at the end (or boundary); the amplitude of both centrifugal wave and centripetal wave decrease (i.e. they are attenuated) somewhat during travel due to damping. Suppose now that instead of being attached to a rigid wall, the string is attached to another string of a different elasticity. At the boundary joining the two strings, the centrifugal (i.e. measured, S_m) wave will be partially transmitted

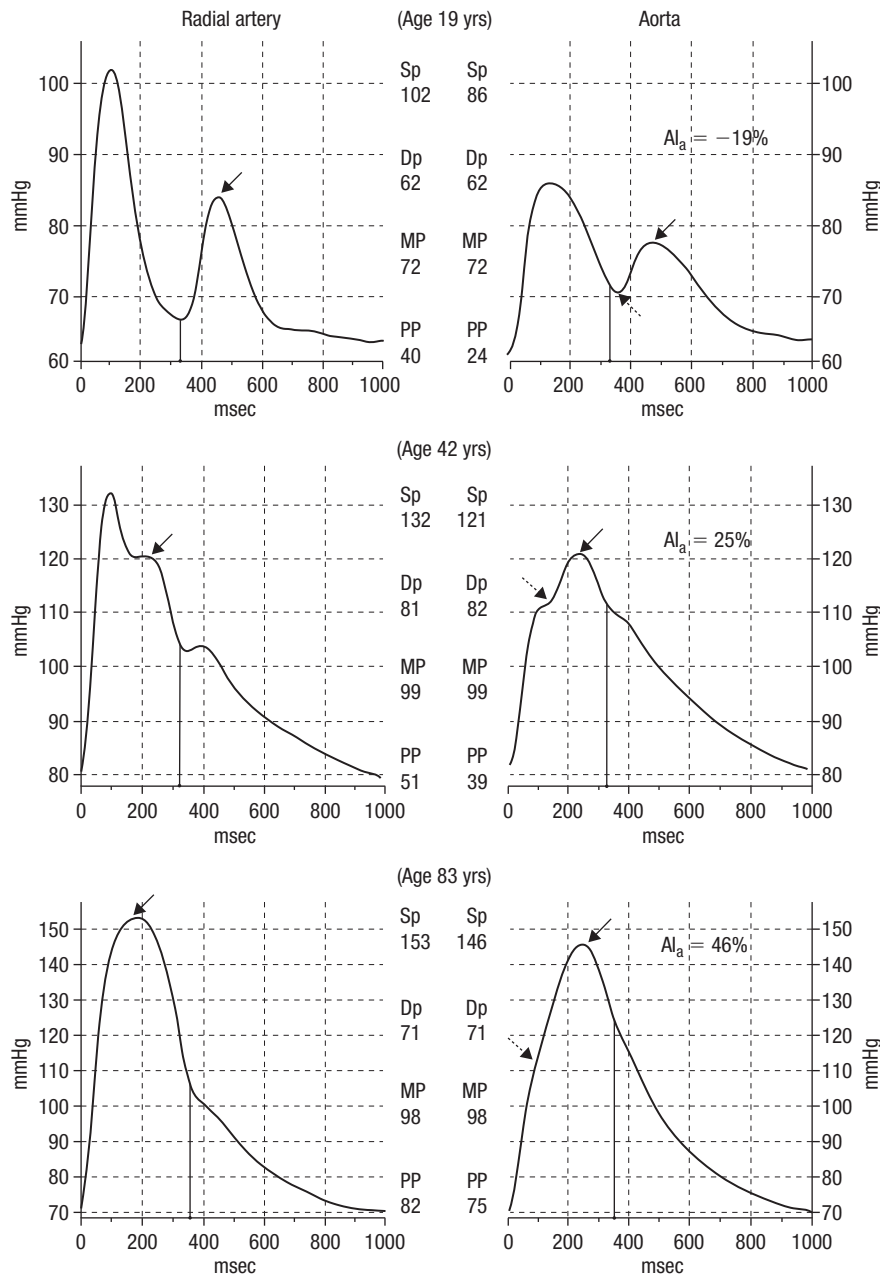


Figure 9.2 Recordings of radial artery pressure waves (applanation tonometry) and synthesized aortic pressure waves (mathematical transfer function) (see Chapter 26) in three healthy individuals to illustrate the age-related changes in amplitude and timing of the reflected pressure wave. Solid arrows identify the peak of the reflected wave and broken arrows indicate the beginning upstroke of the reflected wave (inflection point). In youth, the reflected wave occurs in diastole and aids coronary artery perfusion, but with increasing age the elastic arteries stiffen, increasing pulse wave velocity and causing the reflected wave to occur in systole with greater amplitude. This modification in wave reflection properties causes a decrease stroke output and a decline in coronary artery perfusion. In these three individuals, pulse pressure in the radial artery doubled while pulse pressure in the ascending aorta tripled causing amplification (brachial/aortic pulse pressure) to decrease from 1.7 to 1.1 while aortic augmentation increased from -19 to +46 percent. Reproduced from Nichols and Singh (2002)

(incident, S_i) and partially reflected (S_r). The amplitude of the reflected wave will be less than that of the incident wave because of energy dissipation due to damping. The incident wave and reflected wave travel with the same velocity, however the velocities are different in the two strings because of the difference in elasticity. The

relationship among measured, incident, and reflected waves is given by:

$$S_m = S_i + S_r \quad 9.2$$

and the 'reflection factor' (R_f) is the ratio of reflected to incident wave (i.e. S_r/S_i). Similarly, the ratio S_r/S_m is

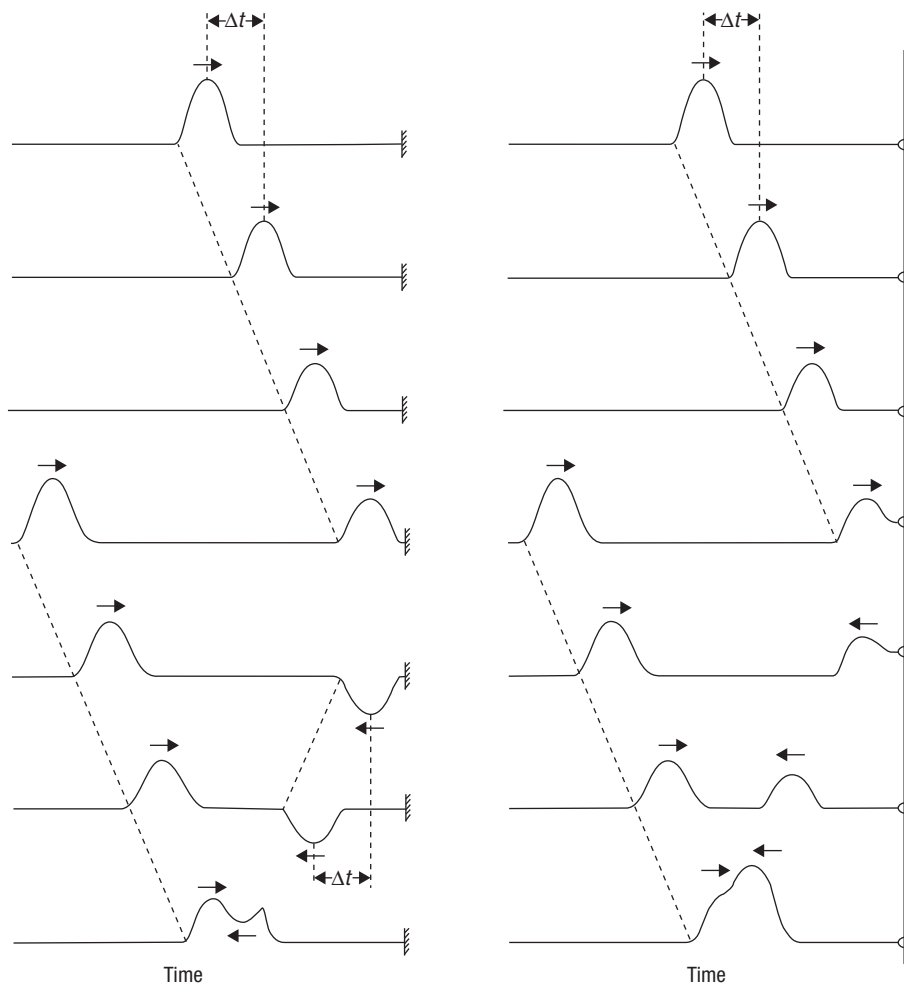


Figure 9.3 Wave travel in a string with a 'fixed' end (left) and a 'free' end (right). When the incident wave ($\rightarrow$) arrives at the end, it is reflected ($\leftarrow$). If the end of the string is fixed, the reflected wave will be inverted (or negative) and subtract from a second incident wave traveling down the string (reflection coefficient negative). If the end of the string is free to move in a transverse direction, the reflected wave will be positive and add to a second incident traveling down the string (reflection coefficient positive). The velocity of the incident waves and reflected waves are the same. The decrease in amplitude of the waves during travel is due to damping

termed the 'transmission coefficient'. By analogy with the theory of light rays, the 'reflection coefficient' (R_c) may be defined by the characteristic impedance on either side of the boundary and is directly related to the reflection factor. Thus,

$$R_f = (S_r/S_i) = (Z_T - Z_0)/(Z_T + Z_0) = R_c \quad 9.3$$

where Z_0 is the characteristic impedance (index of elastance) of the proximal part of the string and Z_T that of the terminal (or distal) part.

In a tube

The simplest case to consider is that where a single pressure wave is generated by an injection of liquid into one end of a long tube filled with liquid. The wave will

travel down the tube with a velocity that depends on the physical properties of the tube and will diminish in amplitude, due to the viscous dissipation of energy, as it travels. When the incident wave reaches the end of the tube, it will be reflected and will not be recorded again at the origin until it has completed a second transit. If the termination is closed, it will be returned as a positive wave. In comparison with the string, the 'closed' end corresponds to a 'free' end and the 'open' end to a 'fixed' end.

Figure 9.4 shows pressure waves generated at the origin of a tube with total occlusion at shorter and shorter distances from its origin (Fig. 9.4, left, A–D). Wave reflections from the occlusion are seen as echoes after the initial or incident wave. These reflected waves occur at shorter intervals when the occlusion is closer to the tube's origin. After the pressure wave is generated, both the origin and the occlusion behave as closed-end reflecting sites. The

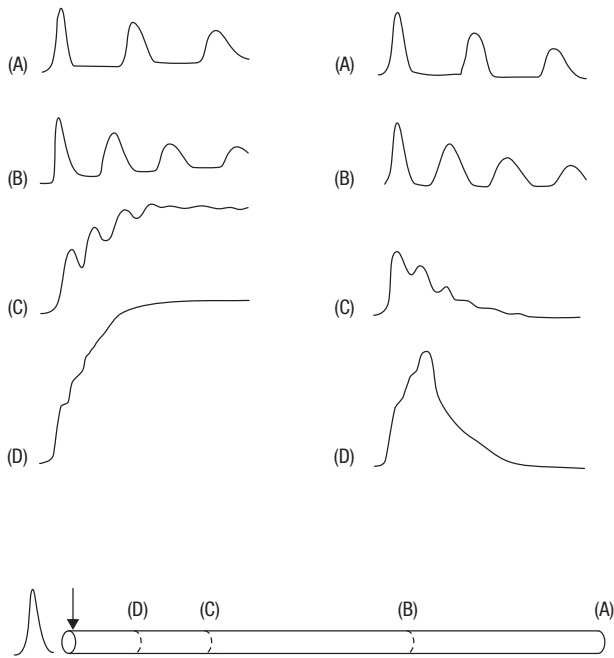


Figure 9.4 (A) to (D) show pressure waves measured at the origin of a distensible tube after a single injection of liquid (arrow), when the tube is occluded closer and closer to its origin. The first wave results from injection, and the other waves occur as a result of wave reflections. Tracings at the left result from complete occlusion and those at the right from incomplete occlusion. Reproduced from O'Rourke and Yaginuma (1984)

wave bounces back and forth between the origin and occlusion of the tube until it is damped to extinction as a result of incomplete reflection and attenuation caused by liquid and wall viscosity. With successive reflections, the mean pressure rises as a result of the tube's distensions by the volume of liquid injected.

A steady blood flow leaves the arterial system through the high-resistance arterioles, so that a high pressure is maintained throughout the cardiac cycle. In the tubular model discussed, allowance can be made for the 'run-off' from the arterial system by partially occluding the tube at points A–D (Fig. 9.4, right). Pressure waves in this model now resemble those recorded in the ascending aorta in different circumstances. In cases A–C, there are obvious wave reflections. In case D, reflections are not apparent because, as we saw above for the string, the reflected wave summates with the incident wave.

A second feature of a transient wave traveling in a viscous medium is also illustrated in Fig. 9.4. It is seen that the reflected wave is somewhat altered in shape as well as in amplitude when it returns. It is less sharply peaked and more spread out in time. From consideration of the frequency components of a transient pulse in Chapter 11, it is noted that such a pulse is made up of a wide spectrum of frequencies. In a viscous medium, it is seen in Chapter 4 that the wave velocity will vary somewhat with frequency. In travel over a considerable distance, as in the upper trace

in Fig. 9.4, this leads to the spreading of the pulse that is seen. This phenomenon is known as 'dispersion'. In a short tube, as in the lower trace, it does not normally have much effect and does not appear to play a significant role in determining the form of the traveling pressure wave, because at the frequencies normally involved the significant components of the arterial pulse wave all travel at approximately the same velocity.

REFLECTED WAVES IN A STEADY-STATE OSCILLATION

When considering wave reflections in the arterial system where the heart beats regularly, a steady-state oscillation is considered rather than the behavior of a single transit. For example, consider a pressure wave in the form of a simple harmonic oscillation (or sine wave) that is not being damped (i.e. no attenuation) and which is completely reflected at a 'closed' end. There are two components of this pressure wave: an incident wave traveling away from the origin, and a reflected wave traveling backwards toward the origin. Two such waves may be represented by the equations

$$Y_1 = Y_m \sin(kx - \omega t) \quad 9.4$$

and

$$Y_2 = Y_m \sin(kx + \omega t), \quad 9.5$$

where $k = 2\pi/\lambda$ is the wave number, $\lambda = c/f$ is the wavelength, and $\omega = 2\pi f$ is the angular frequency, c is wave velocity, and f is frequency. From the trigonometric identity for the sum of the sines of two angles, we obtain

$$Y = Y_1 + Y_2 = 2Y_m \sin kx \cos \omega t. \quad 9.6$$

Equation 9.6 is the equation of a 'standing wave'. The amplitude, $2Y_m \sin kx$, has a maximum value of $2Y_m$ at positions where

$$kx = \pi/2, 3\pi/2, 5\pi/2, \text{ etc.},$$

or

$$x = \lambda/4, 3\lambda/4, 5\lambda/4, \text{ etc.}$$

These points are called 'antinodes' and are spaced one half wavelength apart (Fig. 9.5). The amplitude has a minimum value of zero at positions where

$$kx = \pi, 2\pi, 3\pi, \text{ etc.},$$

or

$$x = \lambda/2, \lambda, 3\lambda/2, \text{ etc.}$$

These points are called 'nodes' and are also spaced one half-wavelength apart. The separation between a node and an adjacent antinode is therefore one quarter of a wavelength.

Since these hypothetical (sine) waves are obtained in a 'closed' end system (or tube), the resultant pressure wave is the sum of the incident and reflected waves (Fig. 9.5). This resultant wave is also sinusoidal with the same frequency as the incident and reflected waves, but the amplitude varies as a function of the phase difference between the two waves. In this example, the maximum amplitude, with incident and reflected waves in phase, is 1.8 (1.0+0.8) arbitrary units. As the phase difference between the two waves increases from zero, the amplitude of the resultant wave decreases (Fig. 9.6) and becomes nearly zero at almost 180° . At this point, the tube begins to behave as an open-end system, and the reflected wave is inverted with respect to the incident wave and therefore subtracts from it (see Fig. 9.3).

When dealing with tubes filled with liquid we must, however, take into account the effect of damping. In this situation, the amplitude of a propagated wave diminishes exponentially as it travels; this applies equally to the reflected wave. Therefore, even with complete reflection, the reflected wave will always be smaller than the incident wave in amplitude (with the exception of the point of reflection

where they will be the same). This discrepancy will increase as the distance between measuring site and reflection site increases. Therefore, there can never be complete cancellation even at the first node from the end, and each successive node will have an increasing, although relatively small, amplitude. The nodes and antinodes in this situation are more properly called 'relative nodes' and 'antinodes'. The condition of a standing wave also cannot be achieved, because adjacent portions of the wave will never be completely synchronous and there will be an apparent movement of the wave. At a relative antinode, the amplitude is large but the phase shift is small; expressed as a velocity, this gives a high apparent phase velocity c' . At the nodes, the phase shift is large and c' , like the amplitude, is very low.

This attempt to describe the interaction of incident and reflected waves in simple conceptual terms is given as an introduction to a problem that is unfamiliar in biology, but it is inadequate for more precise analysis. It is thus necessary to formulate mathematically.

REFLECTION IN A TRANSMISSION LINE

The mathematical analysis has been set out by Taylor (1957a) by a treatment of the analogy of an electric

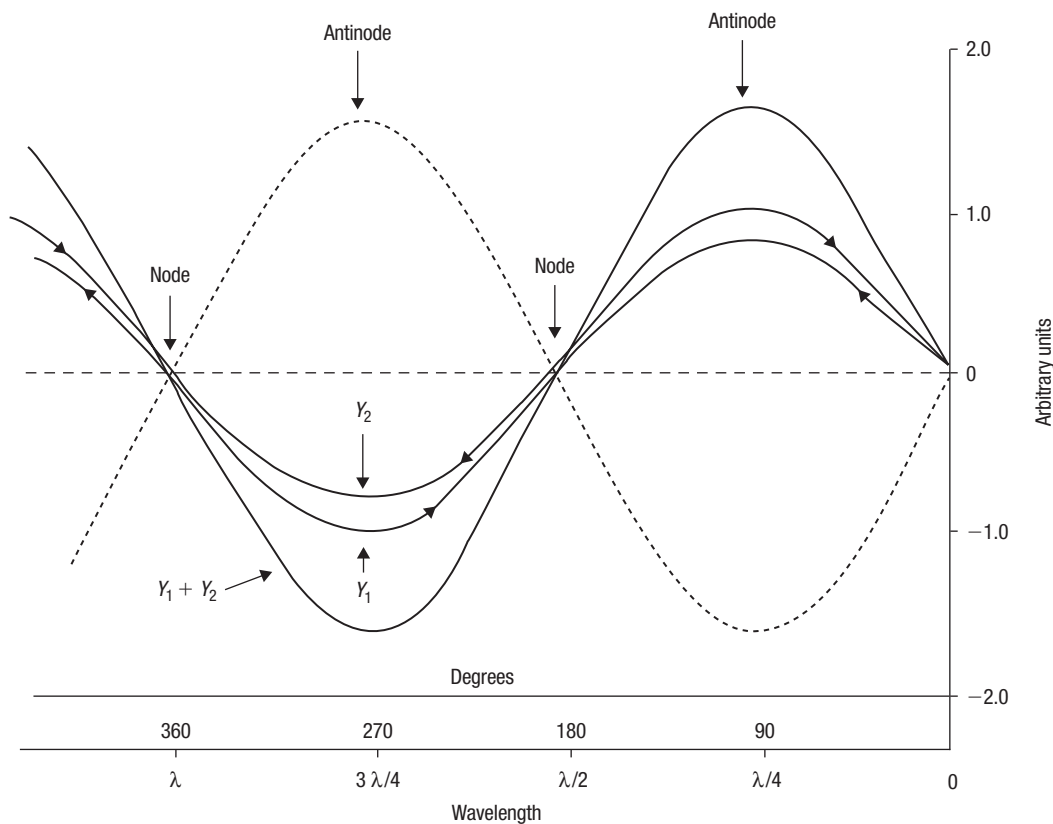


Figure 9.5 The superposition of an incident (Y_1) and a reflected (Y_2) wave at a 'closed' end. For clarity, it is assumed that only 80 percent of the wave is reflected (i.e. reflection coefficient = 0.8). Since the reflected wave occurs at a closed end, the two waves sum together to give the resultant wave $Y_1 + Y_2$. A time exposure of the motion of these two waves would result in an envelope (solid line and broken line) of a standing wave with both nodes and antinodes

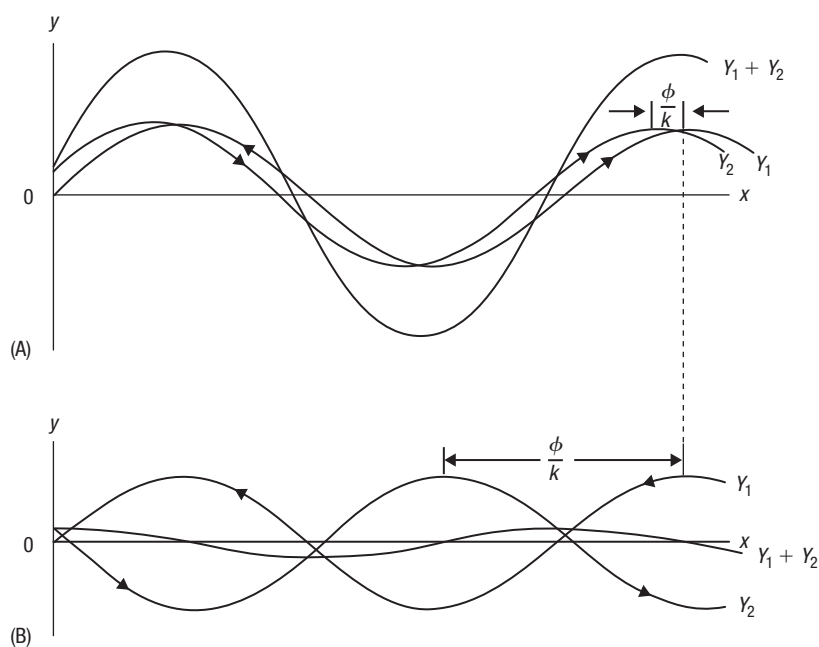


Figure 9.6 (A) The superposition of incident (Y_1) and reflected (Y_2) waves of equal frequency and amplitude that are almost in phase results in a wave ($Y_1 + Y_2$) of almost twice the amplitude of either component. (B) The superposition of incident and reflected waves of equal frequency and amplitude and almost 180° out of phase results in a wave whose amplitude is nearly zero

transmission line. The analogies taken are that voltage is equivalent to pressure (P) and current to flow (Q). The solutions were verified by testing in a simple hydraulic model (Taylor, 1957b). From the treatment discussed in Chapter 4, we can define a propagated wave P using the propagation constant γ as

$$P_{(z)} = P_0 e^{-\gamma z}, \quad 9.7$$

where P_0 is the wave at the origin and $P_{(z)}$ is the wave at any distance z along the tube. If the length of the tube is l and the reflection coefficient is R_c (see below), the reflected wave at a distance z will be

$$P_{(z)} = R_c P_0 e^{-\gamma(2l-z)} = P_0 R_c e^{\gamma(z-2l)}, \quad 9.8$$

and the composite term will be

$$P_{(z)} = Q_0 Z_0 \left(\frac{e^{-\gamma z} - R_c e^{\gamma(z-2l)}}{1 + R_c e^{-2\gamma l}} \right), \quad 9.9$$

where Q_0 is the flow at the origin and Z_0 is the characteristic impedance of the tube (see Chapter 12).

If the end is completely open, the reflection coefficient R_c will be 1.0. Equation 9.9 then becomes

$$P_{(z)} = \frac{Q_0 Z_0 \cosh \gamma(l-z)}{\sinh \gamma l} \quad 9.10$$

and

$$P_0 = Q_0 Z_0 \coth \gamma l. \quad 9.11$$

Since $\gamma = a + ib$, eqn 9.10 can be expanded and the amplitude $|P_{(z)}|$ of the pressure is then given by

$$|P_{(z)}| = |Q_0 Z_0| \left[\frac{\cosh 2(l-z)a + \cos 2(l-z)b}{\cosh 2la - \cos 2lb} \right]^{1/2}. \quad 9.12$$

The input impedance Z_i of a line of length l , closed at the end, is then expressed in relation to the characteristic impedance Z_0 where there are no reflections as

$$\frac{|Z_i|}{|Z_0|} = \left(\frac{\cosh 2la + \cos 2lb}{\cosh 2la - \cos 2lb} \right)^{1/2}. \quad 9.13$$

The input impedance as a function of characteristic impedance is plotted versus wavelength in Fig. 12.5; the theoretic curve (i.e. the one obtained in a transmission line) is shown at the top and the experimental verification (i.e. the curve obtained in a rubber tube model) at the bottom.

The change in amplitude of the input impedance with change in the physical length of the tube will also be duplicated by changing the pump frequency while the length of the tube is kept constant. This is due to the fact that the length of the tube changes in terms of wavelength because the wavelength varies inversely with the frequency. This is inherent in eqn 9.13 because the cosine terms involve b . In Chapter 4, this was defined as $b = \omega/c_1$. As the wavelength $\lambda = c/f$, we may rewrite $b = 2\omega/c_1 = 2\pi/\lambda$. Thus, the term $lb = 2\pi l/\lambda$. In the arterial system, where the total length is constant, the investigation of the reflection effects needs to be made by the changes in the input impedance spectra at the frequencies of each harmonic component. This is discussed in detail in Chapter 12.

THE EFFECT OF REFLECTIONS ON APPARENT PHASE VELOCITY

When considering the propagation velocity of a single sinusoidal oscillation or wave, it must be expressed as the velocity of travel of points of identical phase. It is thus dependent on the rate of change of phase in respect to distance (i.e. $d\phi/dz$), and since the angular frequency is T , then the wave velocity is given by

$$c' = \omega dz/d\phi. \quad 9.14$$

As the phase shift is measured over a finite interval, Δz , the equation becomes

$$c' = 2\pi f \Delta z / \Delta \phi. \quad 9.15$$

If there is no reflection of the wave, the wave velocity is determined by the elastic properties of the tube; but, as was seen in Chapter 4, it will vary with frequency owing to the viscous properties of the wall and the viscosity of the liquid within it.

If there is a reflected wave component, however, we have already seen that this interacts with the incident wave, and the resultant wave shows waxing and waning of amplitude at different regions of the tube. Similarly, the rate of change of phase also varies, and hence the apparent phase velocity. This is termed the 'apparent phase velocity' to distinguish it from the true wave velocity, which is determined by the physical properties of the wall and the contained liquid. An appreciation of the modification of phase shift due to reflections is not easy to accept intuitively, but was determined mathematically by Taylor (1957a). He found that the apparent phase velocity in the presence of reflections varied in the same manner as the amplitude of the pressure oscillations, with the exception that close to an open end it continued to rise when the pressure oscillation would be falling.

The equation for the ratio of the apparent phase velocity c' to the characteristic (or true) wave velocity c_0 , which is determined by elastic properties, is given here in the generalized form for a complex reflection coefficient

$R_c = \exp 2(a + ib)$, so that

$$\frac{c'}{c_0} = \frac{\cosh 2[(l-z)a - a] - \cos 2[(l-z)b - b]}{\sinh 2[(l-z)a - a] - (a/b) \sin 2[(l-z)b - b]}. \quad 9.16$$

With an open end, the terminal impedance $Z_T = 0$, $R_c = 21.0$ and $a = b = 0.0$; for a closed end $Z_T = \infty$, $R_c = +1.0$, $a = 0.0$, and $b = \pi/2$.

The variation of apparent phase velocity at different points along a tube was measured by Taylor (1957b), which verified the theoretic curves in Taylor (1957a). This is shown in Fig. 9.7.

Apparent phase velocity can be calculated *in situ* from the measurement of two pulsatile pressures a known distance, Δz , apart using eqn 9.15. Each pressure wave is subjected to Fourier series analysis and $\Delta \phi$ calculated for each harmonic. This method of measurement gives a frequency spectrum similar to the impedance spectrum (Fig. 9.8). c' for the first harmonic is related to wave reflection amplitude and in this example is almost 750 cm/s. The average of the apparent phase velocities above 2.0 Hz is 475 cm/s and is similar to the true wave velocity (Nichols and McDonald, 1972; Latham, 1989), which is dependent upon the physical properties of the artery.

REFLECTION COEFFICIENT

Reflection of a traveling wave in a viscous liquid will occur to some degree whenever there is a change in the impedance or elastic properties of the tube down which it is propagated. We have already defined the characteristic impedance (Z_0) of an elastic tube in eqn 7.28:

$$Z_0 = \frac{\rho c_0 e^{-2i\epsilon/2}}{(12 - \sigma^2)^{1/2} (M'_{10})^{1/2}}$$

Thus it can be seen that a change in the wave velocity c_0 (which depends on the elastic properties and relative thickness of the wall) and/or a change in the value of the parameter α on which the values of the Bessel function terms M'_{10} and ϵ depend, will change with the impedance. As $\alpha = R(\omega/\nu)^{1/2}$, this change in the value of α will be

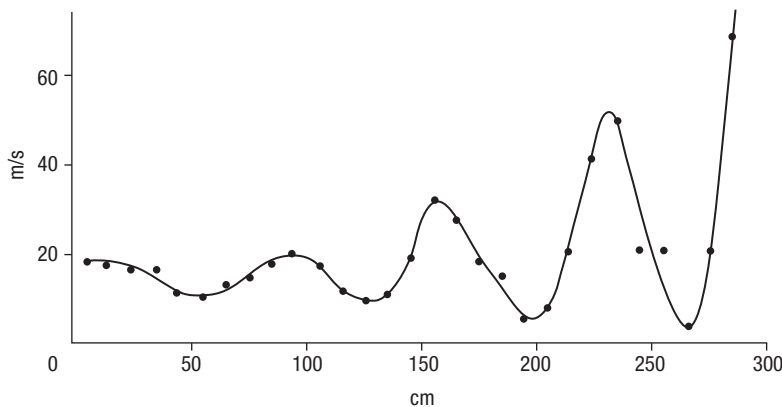


Figure 9.7 Apparent phase velocity measured in a rubber tube occluded at 300 cm from the end. The frequency was 10 Hz and the true phase velocity in the tube was 14.1 m/s. The apparent phase velocity, like the input impedance, oscillates around the value found in the absence of reflections with the first minimum approximately one-quarter wavelength from the distal end of the tube. The diminution in the oscillation in velocity, as measurements are made further from the end, is due to damping. Reproduced from Taylor (1957b)

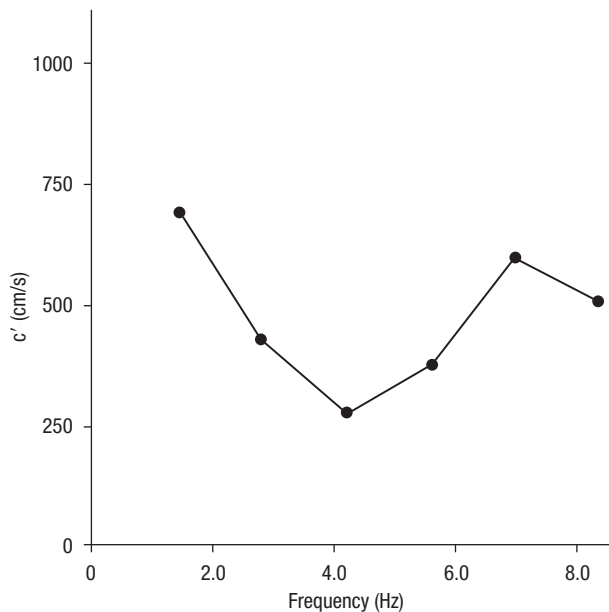


Figure 9.8 Apparent phase velocity (c') spectrum obtained in the ascending aorta of a middle-aged normotensive male patient with coronary artery disease. The two pressures from which c' was calculated were measured with a cardiac catheter with pressure sensors 5.0 cm apart. The oscillation in the spectrum is due to the effects of wave reflection

brought about by a change in the radius R , as the angular frequency ω and the kinematic viscosity ν are the same throughout. If we treat a discontinuity at which a change in wave velocity or a change in radius occurs as a 'termination' and symbolize the impedance from that point onwards as Z_T , then the usual convention for defining the reflection coefficient R_c is given in eqn 9.3.

When the end is completely closed, the terminal impedance will be infinitely large ($Z_T = \infty$) and R_c will be $+1.0$, i.e. the incident wave will be 100 percent reflected. When the end is open, $Z_T = 0.0$ and R_c will be -1.0 , i.e. the reflected wave will be 180° out of phase with the incident wave (see Fig. 9.3).

Since the arterioles are the major resistance vessels, Hamilton (1944) identified them as the major sites of wave reflection in the circulation. This led Taylor (Taylor, 1957a; Taylor, 1957b) and colleagues McDonald (McDonald and Taylor, 1959) and O'Rourke (O'Rourke and Taylor, 1966) to calculate the reflection coefficient (R_c) in the femoral vascular bed from measurements of femoral vascular resistance (Z_T) and femoral characteristic impedance (Z_0). R_c was found to be approximately 0.8; similar values were found for other vascular beds, including the whole systemic circulation but not the renal and other vasodilated beds (see Chapter 10). In further investigations, Z_T was considered to be the complex input impedance and was used instead of the resistance (Westerhof *et al.*, 1972; Sperling *et al.*, 1975; Van den Bos *et al.*, 1976; Laskey and Kussmaul, 1987; Fitchett, 1991).

For any given discontinuity (or impedance change), as characterized above, where partial reflection occurs, the remaining fraction of the wave will be transmitted past the reflection point (see above). From this physical concept, Wetterer (1956) proposed that pressure and flow waves can be regarded as a summation of incident (or forward) and reflected (or backward) waves. Thus if we call the amplitude (or modulus) of the incident pressure wave P_i and the modulus of the reflected wave P_r , the modulus of the measured pressure wave, P_m , for a given harmonic will be

$$P_m = P_i + P_r, \quad 9.17$$

and the reflection factor (as defined above for a string) is the ratio P_r/P_i (Li, 1986; Berger *et al.*, 1993; Berger *et al.*, 1994), then the ratio of the incident wave to the measured wave is P_i/P_m and is termed the 'transmission coefficient' (see above). Figure 9.9 is a plot of the reflection factor versus frequency measured in the ascending aorta of a dog during various interventions that cause a change in aortic impedance and wave reflections (Westerhof *et al.*, 1972). It is seen from the figure that, at low frequencies, the reflection factor is much larger than for high frequencies. For low frequencies, the reflection factor is different for each intervention, but for high frequencies the reflection factors are similar regardless of the intervention. High resistance and aortic occlusion increase the reflection factor, while low resistance decreases it (Westerhof *et al.*, 1972).

The ratio P_r/P_i is sometimes called the 'reflection coefficient', but the term invites confusion with eqn 9.3 (RHS), which is the general expression used for the reflection coefficient; therefore, the ratio should be called the 'reflection factor' (or reflectance). This quantity is a complex number with both modulus and phase; for pressure, the modulus ranges from 0.0 to $+1.0$, and for flow from 0.0 to -1.0 . The phase angle ranges from 190° to 290° , with the reflected pressure and flow waves 180° out of phase.

In a system like the arterial tree where there are a succession of branching points as the pressure waves travel away from the heart, the possibility of repeated reflections makes a precise analysis extremely difficult. Remington (1965) attempted to carry out such an analysis for the individual branching points of the arteries of the fore part of the dog by following discrete portions of the pulse pressure wave recorded at several different points (based on the measurements of Meisner and Remington (1962)). The labor involved was very great and, even so, the values obtained are doubtful in the absence of good evidence of such factors as wave attenuation. Taylor (Taylor, 1966a; Taylor, 1966b) approached the same problem with computer simulation by setting up a 'model' of branching transmission lines. For the present, we will confine ourselves to assessing the reflection coefficient at points of simple arterial branching.

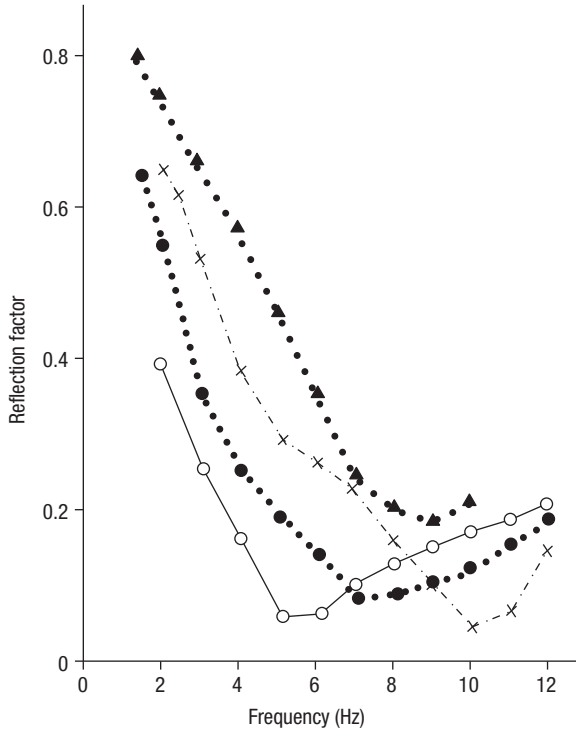


Figure 9.9 Moduli of the reflection factor (ordinate) plotted as a function of frequency in the dog ascending aorta during several interventions that alter arterial impedance and wave reflections. Control (closed circles), low resistance (open circles), high resistance (crosses), and aortic occlusion (closed triangles) are shown. Note that the spectrum minimum shifts to lower frequencies with vasodilatation and to higher frequencies with vasoconstriction and aortic occlusion. Frequency shifts are attributable to concurrent change in arterial pressure and pulse wave velocity. Reproduced from Westerhof *et al.* (1972)

REFLECTION AT A SINGLE ARTERIAL JUNCTION

For an artery of radius r_1 and wave velocity c_1 making a junction with an artery of radius r_2 and wave velocity c_2 , from the 'water hammer' equation $Z_0 = \rho c_0 / \pi r^2$, we obtain $Z_1 = \rho c_1 / \pi r_1^2$, and $Z_2 = \rho c_2 / \pi r_2^2$, so that

$$Z_1/Z_2 = (r_2^2/r_1^2)(c_1/c_2). \quad 9.18$$

Now, if $c_1 = c_2$ (i.e. the elastic properties of the arterial wall do not change), it can be seen that Z_1/Z_2 is the ratio of the cross-sectional areas of the vessels on either side of the junction. As when $Z_1/Z_2 = 1.0$ the reflection coefficient $((Z_1 - Z_2)/(Z_1 + Z_2))$ is zero and no reflection occurs, then, in the simplest case, partial reflection of a closed type will occur when the cross-sectional area increases. In practice, however, the wave velocity in arteries increases as they pass peripherally (i.e. c_1, c_2), so that the condition of zero reflection is at a bifurcation with an area ratio > 1.0 . Karreman (1952) calculated this assuming that the wall thickness remained unchanged, so that from the

Moens–Korteweg equation (3.51)

$$c_1/c_2 = (r_2/r_1)^{1/2}, \quad 9.19$$

and then the point of zero reflection is at an area ratio, for a bifurcation, of approximately 1.15.

Womersley (1958) extended this analysis by incorporating the full equation of the flow for a viscous liquid when, for the tethered elastic tube, he showed that from the characteristic impedances calculated from eqn 7.28

$$\frac{Z_1}{Z_2} = 2 \left(\frac{r_2}{r_1} \right) \left(\frac{c_1}{c_2} \right) \left[\frac{M'_{10}(\alpha_2)}{M'_{10}(\alpha_1)} \right]^{1/2} e^{i(\epsilon\alpha_2 - \epsilon\alpha_1)/2}, \quad 9.20$$

where α_1 and α_2 are the values for the main artery and for a branch, respectively. The general impression is that for relatively large values of α (e.g. $\alpha = 10$), the reflection conditions for blood are not very different from those for a non-viscous liquid. Points of considerable interest are:

- With a viscous liquid, modulus and phase of the incident and reflected waves change in such a way that there is never exact matching of the systems, although the minimum reflection at the point of 'impedance matching' is only a few percent (Li, 1985).
- The area ratios of major branches that are found in the body, which are about 0.85–1.31 (Papageorgiou *et al.*, 1990), represent very small reflection coefficients in arteries of the caliber of those at the aorto-iliac bifurcation (Greenwald *et al.*, 1990; Papageorgiou *et al.*, 1990).
- The phase change from a lag of 0° to a lag of 180° occurs more gradually in vessels containing a viscous liquid, so that in the vicinity of the minimal reflection, conditions of intermediate phase shift can occur.
- If we regard α as a function of frequency, the amount of reflection is not markedly frequency dependent.

It should be emphasized that the calculation above is an abstraction concerned only with a single junction, with the assumption that the branches are themselves infinitely long tubes so that no subsequent reflection occurs. Figure 9.10 shows corresponding curves for bifurcations in which the parent branch has $\alpha = 1.0$ (in the dog, this represents the fundamental component in an artery of the order of the saphenous artery) and $\alpha = 5.0$; α for a non-viscous system ($\alpha = \infty$) is also shown in the figure. The same ratio for c_1/c_2 has been taken arbitrarily, in the absence of an exact knowledge, because this approximately offsets the retarding effect on wave velocity that viscosity has in small tubes. As α decreases, the point of minimal reflection is shifted toward larger area ratios; for $\alpha = 1.0$, this is approximately 1.35 and any change in area at a junction smaller than this will cause a closed-end type of reflection.

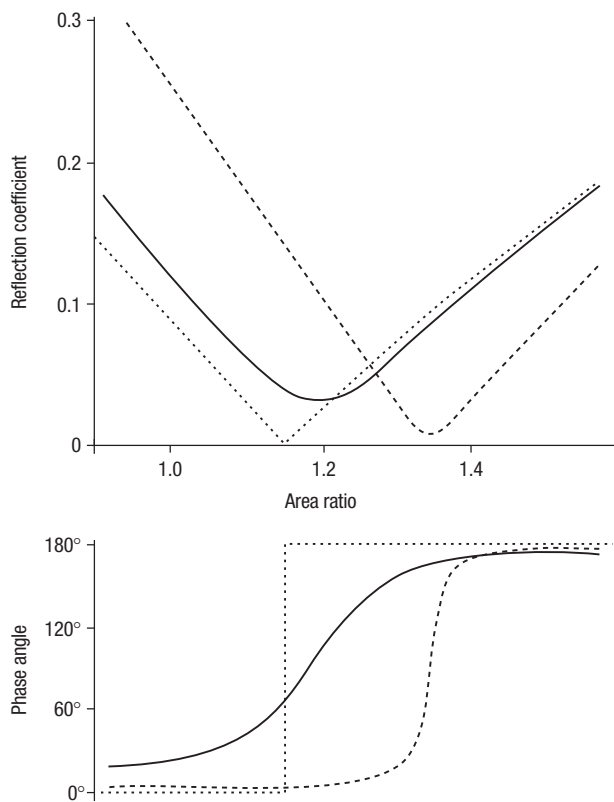


Figure 9.10 Theoretical relationship between local reflection coefficient and area ratio (ratio of branches to parent vessel cross-sectional areas). Curves are shown for $\alpha = 1.0$ (dashed line), $\alpha = 5.0$ (solid line) and $\alpha = \infty$ (dotted line). As α decreases, the minimum of the reflection coefficient shifts to higher values of the area ratio. The phase angle lies near 0° for low area ratios and near 180° for high area ratios. Reproduced from Womersley (1958)

This value, it will be seen, is approaching that of the area ratio for identical liquid resistance of steady flow across a bifurcation, which is 1.414 (see Chapter 2). Just as it was shown that the resistance changes increase with the number of branches, so will the impedance change increase. This effect is much less with larger branches, and if viscosity is neglected the reflection is determined by the area ratio irrespective of the number of branches. This is to be expected as the oscillatory flow in small vessels has a pressure–flow relation very close to that of Poiseuille’s equation (i.e. it is largely determined by the viscosity, whereas in large vessels it is mostly determined by the inertia of the liquid). This is also illustrated by the fact that the phase lag changes from zero to 180° very rapidly at both small and large values of α , but is much more gradual at intermediate values (e.g. $\alpha = 5.0$).

Although it must be recognized that in still smaller vessels, prediction of the amount of reflection at individual junctions must be highly speculative, nevertheless, unless dramatic and unforeseen changes in the elastic behavior of the wall occur, it may be expected that reflection

of pressure waves will always be of the closed-end type. If the average area ratio change at each branching is 1.15, the reflection coefficient in vessels of 1.0 mm diameter or less that bifurcate will probably be about 13.0 percent. The total reflection from an arterial termination will be the summation of a succession of branches very close together (close, that is, in terms of wavelength) but with divisions at each site into many more than two subsidiaries and a reflection coefficient greater than for a simple bifurcation. For an area ratio of 1.15, the reflection coefficient for $\alpha = 5.0$ is about 4.0 percent and for $\alpha = 10$ only about 2.0 percent. However, calculations based on experimental estimations of branch size (Womersley, 1958) have suggested a reflection coefficient of 14 percent (phase shift 40°) at the terminal division of the aorta and about 7.0 percent amplitude (phase approximately 90°) at the celiac axis. From direct measurements of the aorto-iliac bifurcation, Li (1985) found an area ratio of 1.0 and a local reflection coefficient of 6.0 percent. When one iliac artery was occluded, the reflection coefficient increased to 22 percent. Local reflection coefficients determined from area ratios alone, assuming equal wave velocities for the iliac branches, were very close to those calculated from characteristic impedances.

Experimental area ratio measurements of the aorto-iliac bifurcations were made from radiographs by Gosling *et al.* (1971). They found the area ratio to be 1.16 in the chicken and 1.23 in the dog, values very close to those predicted by theory. In the same study, Gosling *et al.* found that the area ratio of the aorto-iliac bifurcation in living normal humans was age dependent, decreasing from a value of 1.06 in the first decade of life to 0.76 in the fifth (Fig. 9.11). These results are similar to those obtained post-mortem by Greenwald *et al.* (1990). Laogun *et al.* (1979) observed a similar decline with age in Nigerians. The ratio remained close to 0.6 in atherosclerotic patients who were 40–80 years of age (Newman *et al.*, 1971; Lallemand *et al.*, 1972). Greenwald *et al.* (1990) also observed little change in older subjects, although the mean for the age range of 50–88 years was 0.78. A reduction in the area ratio of atherosclerotic specimens has also been observed by Sauvage *et al.* (1973), but age was not reported. Recently, Cecelja *et al.* (2009) have presented data which suggest that greater wave reflection in women is due to greater physical taper of major arteries, on a genetic basis. These age-related changes in proximal to distal diameter ratio may be due to greater degeneration (and increased diameter) of the proximal aorta with age (Lam *et al.*, 2010), and can account for increase in wave reflection from its low value in youth to a higher value in older people. This may contribute to the apparent change in position of the summed reflection site in older people with isolated systolic hypertension (ISH) (O’Rourke and Nichols, 2005) while greater taper in women (Cecelja *et al.*, 2009) may, with shorter body length, account for greater AIx than in males, despite similar arterial pulse wave velocity (PWV) (Hayward and Kelly, 1997; Smulyan

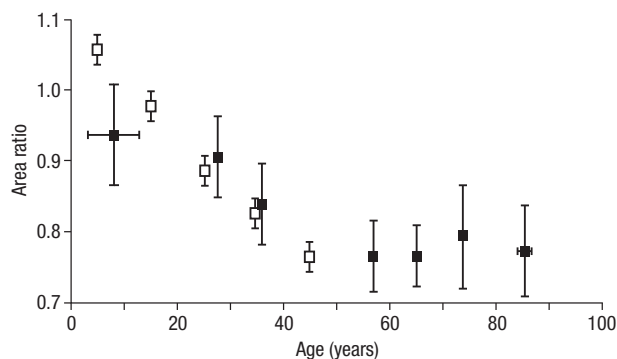


Figure 9.11 Age-related changes in area ratio of the aorto-iliac bifurcation in humans. Open squares, data from Gosling *et al.* (1971); closed squares, data from Greenwald *et al.* (1990)

et al., 1998). However, other studies found higher augmentation index in age-matched young girls compared to boys of similar height and aortic PWV (Ayer *et al.*, 2010). The time to the onset of the reflected wave was shorter in girls. The changes in wave reflection characteristics may be associated with reduced arterial diameter and increased aortic tapering in women compared to men (London *et al.*, 1995).

REFLECTIONS DUE TO A CONTINUOUS CHANGE IN ELASTIC PROPERTIES

As discussed in Chapter 4, the modulus of elasticity of arteries increases with the distance from the heart (Fig. 4.5). This causes a progressive increase in PWV, rising, in the dog, from a value of about 4.0 m/s in the ascending aorta (Nichols and McDonald, 1972) to some 14–15 m/s in the tibial artery of the distal part of the hind leg (Fig. 4.22) (McDonald, 1968a). Increases in regional PWV along the aorta are also seen in humans (Latham *et al.*, 1985) and primates (Latham *et al.*, 1987), but not in kangaroos (Avolio *et al.*, 1985a). As this increase in elastic modulus will clearly cause a change of impedance, it can be expected to give rise to wave reflection (McEniery *et al.*, 2005a). Initially, it may be thought of as a large series of small discontinuities spaced closely along a tube, but this is cumbersome to analyze. Taylor (Taylor, 1964; Taylor, 1965) instead treated it as analogous to a non-uniform electric transmission line and, in his initial papers, neglected viscous effects. In his later computer model of an array of branching tubes, the viscosity terms were incorporated (Taylor, 1966a; Taylor, 1966b). Fich *et al.* (1966) also analyzed an electric analog model of the aorta, which treated it as a single tube that tapered with a diminishing diameter, and the net effect was similar.

The mathematical treatment was essentially the same as that used by Taylor (1965), which has been discussed above. Taylor illustrated the changes by computing the

effects for a transmission line in which the impedance (and the wave velocity) increased by a factor of three from the origin to the end. When the line (or tube) then terminates in a matched impedance, so that no reflections will arise at this termination alone, it is interesting to note that the amplitude of the pressure wave will, nevertheless, rise and fall as if through the creation of nodes and antinodes (Fig. 9.12). These are, however, superimposed on a general rising trend of amplitude along the tube. The final level of the mean pressure oscillation in this example (where the impedance increases by a factor of 3.0) rises to $\sqrt{3.0}$ times that at the origin. The rate of rise in wave velocity along the tube was represented by a cosine curve, so that the velocity c at any distance z from the origin was

$$c(z) = c(0)(2 - \cos \pi z). \quad 9.21$$

As the value of c rose smoothly from $c(0) = 1.0$ to $c(1) = 3.0$, the elastic modulus correspondingly would rise by a factor of 9.0 from the Moens–Korteweg equation (eqn 3.51) if it was assumed that the wall thickness/diameter ratio remained constant.

When, in addition, there is superimposed a terminal reflection, the formation of clear maxima and minima of pressure oscillations are seen (Fig. 9.12) at the origin with changes of frequency, and similarly at varying distances along the tube. In a uniform transmission line (or tube), we have seen that these are spaced at regular intervals of one-quarter wavelength from the termination. When the wave velocity is increasing with distance from the origin, this changes the wavelength, and thus the spacing between a maximum and a minimum, and they are separated by longer intervals at the end of the tube than they are nearer to the origin. These maxima and minima are, furthermore, superimposed on the same rising trend of pressure oscillation as occurs when there is no terminal reflection. In the latter case, the pressure amplitude is determined by the relation at any distance z from the origin:

$$P(z) = P_{(0)} |\tilde{Z}_{0(x)} / \tilde{Z}_{0(0)}|^{1/2}, \quad 9.22$$

where $\tilde{Z}_0$ is called the ‘nominal’ characteristic impedance at any point.

This concept has been of great value in giving a deeper insight into the causes for the well-known phenomenon of the rise in the amplitude of the pressure wave (see Figs 10.1, 10.6, and 10.8A) and its harmonic components (see Figs 10.7 and 10.8B) as it travels along the arterial tree. The experimental analysis of this ‘peaking’ of the pressure is discussed in Chapter 10. It is also important in interpreting the input impedance of the whole arterial tree, which is, as a consequence of the non-uniformity of the system, maintained at a low level at the origin in spite of the fact that the blood has to be pumped through the very high terminal impedance of the arterioles and capillaries (see Chapter 12).

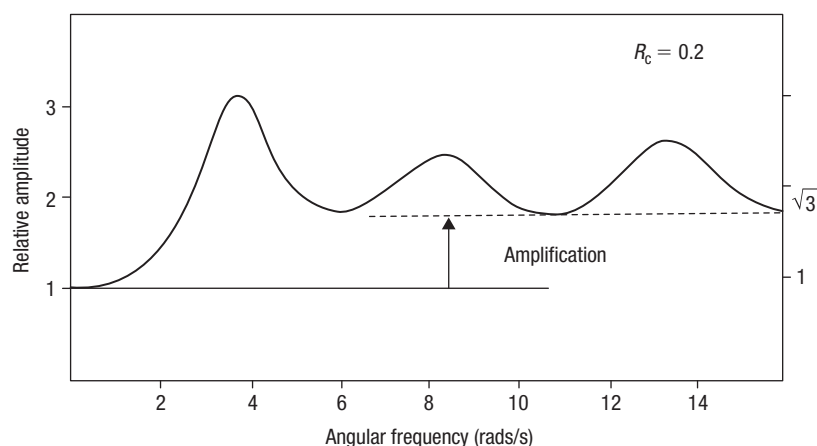


Figure 9.12 Behavior of the relative amplitude (ratio of peripheral to central amplitude of pressure oscillations) of pressure wave reflection in a non-uniform transmission-line model of the arterial tree. The elastic modulus increases distally causing the pulse wave velocity to increase by a factor of 3.0 from the origin to the termination where the reflection coefficient (R_c) is 0.2. Reproduced from Taylor (1964)

REFLECTIONS OF PRESSURE AND FLOW WAVES IN THE ARTERIAL SYSTEM

The arterial tree is made up of a system of distensible tubes along which pressure and flow waves, generated by the heart, are transmitted and reflected. Since the PWV of these waves is in meters/second, it is obvious that in one cardiac cycle the generated wave has sufficient time to travel to the periphery and back. If the reflected waves were damped out completely, the pressure and flow waves generated by the heart would be identical in contour. Reflection of pressure waves has been discussed in considerable detail in the literature (McDonald and Taylor, 1956; McDonald, 1974; O'Rourke, 1982a; O'Rourke, 1982b; O'Rourke and Yaginuma, 1984; Latham *et al.*, 1985; Nichols *et al.*, 1986a; Latham *et al.*, 1987; Fitchett *et al.*, 1988a; Fitchett *et al.*, 1988b; Campbell *et al.*, 1989; Milnor, 1989; Nichols *et al.*, 1992; O'Rourke *et al.*, 1992a; O'Rourke *et al.*, 1992b; O'Rourke *et al.*, 1992c; Nichols *et al.*, 1993; O'Rourke, 1993; Berger *et al.*, 1994; O'Rourke and Hashimoto, 2007; Nichols *et al.*, 2008; Safar, 2008; Ghiadoni *et al.*, 2009; Nichols and Epstein, 2009; Wang *et al.*, 2010), but the importance of reflected flow waves was not emphasized until more recently (Westerhof *et al.*, 1972; Van Den Bos *et al.*, 1976; Murgo *et al.*, 1981a; Murgo *et al.*, 1981b; Brin and Yin, 1984; Yaginuma *et al.*, 1985; Laskey and Kussmaul, 1987; Murgo and Westerhof, 1987; Nichols *et al.*, 1987; Takazawa, 1987; Nichols *et al.*, 1993; Nichols, 1994; Westerhof and O'Rourke, 1995; Nichols and Edwards, 2001). The analysis of wave reflections and their influence on pressure and flow waveforms are outlined below.

Since pressure and flow signals are periodic functions, they can be represented as a Fourier series (see Chapter 11) (Attinger *et al.*, 1966a). In the following analysis, each harmonic component will be considered separately, and the mean pressure and flow terms will be disregarded.

As was pointed out earlier, pressure and flow waves measured in the arterial system are markedly different because reflected waves influence their contour. Reflected

pressure waves add to the incident or forward traveling wave, while reflected flow waves subtract from the incident wave (Wetterer, 1956; Wetterer and Kenner, 1968; Westerhof *et al.*, 1972; McDonald, 1974; Van den Bos *et al.*, 1976; O'Rourke, 1982a; O'Rourke, 1982b; Burattini and Carlos, 1988; Milnor, 1989; Nichols and Edwards, 2001). The measured pressure and flow waves can be dissected into their incident (or forward) and reflected (or backward) components with use of mathematical techniques (Westerhof *et al.*, 1972; Van den Bos *et al.*, 1976; Laxminarayan, 1979; Murgo *et al.*, 1981a; Murgo *et al.*, 1981b; Brin and Yin, 1984; Li, 1986; Ting *et al.*, 1986; Yaginuma *et al.*, 1986; Laskey and Kussmaul, 1987; Murgo and Westerhof, 1987; Nichols *et al.*, 1987; Parker and Jones, 1990; Berger *et al.*, 1993; Berger *et al.*, 1994; Westerhof *et al.*, 2006; Wang *et al.*, 2010):

$$P_m = P_i + P_r \quad 9.23$$

$$Q_m = Q_i + Q_r \quad 9.24$$

$$P_i = Z_0 Q_i \quad 9.25$$

$$P_r = -Z_0 Q_r, \quad 9.26$$

where P is pressure and Q is flow, and the subscript 'm' represents measured, 'i' incident (or forward), and 'r' reflected (or backward) (see above). Thus, the measured pressure ($P(t)$) and flow ($Q(t)$) waves sensed by transducers (see Fig. 9.1) are functions of time and are equal to the sum of an incident wave and a reflected wave. The pressure and flow waves are related by the characteristic impedance (Z_0) (eqns 9.17 and 9.18). In a system long enough that reflected waves are completely damped out before they reach the origin, as in Fig. 9.4A and B, input impedance is constant and measured pressure and flow waves are identical in shape (McDonald, 1974; O'Rourke, 1982a; O'Rourke, 1982b; O'Rourke and Yaginuma, 1984; Milnor, 1989). Since measured pressure and flow waves

are both altered by reflected waves (see Fig. 9.1), resulting in a frequency-dependent impedance spectrum (see Chapter 12), characteristic impedance may be derived and eqns 9.23, 9.24, 9.25, and 9.26 solved to yield the incident and reflected pressure and flow waves:

$$P_i = (P_m + Z_0 Q_m)/2 \quad 9.27$$

$$P_r = (P_m - Z_0 Q_m)/2 \quad 9.28$$

$$Q_i = (Q_m - P_m/Z_0)/2 \quad 9.29$$

$$Q_r = (Q_m - P_m/Z_0)/2. \quad 9.30$$

An example of incident and reflected pressure and flow waves recorded in a human (age 52 years) is shown in Fig. 9.13. In this illustration, the reflected wave occurs during systole and thus causes a late systole peak in the measured pressure wave which places an extra workload on the ventricle. This extra effort, which is wasted, increases myocardial oxygen demand and decreases ventricular efficiency (Hashimoto *et al.*, 2008a; Hashimoto *et al.*, 2008b). Effects of reflection on the flow wave are usually less dramatic (see Chapter 14) (Westerhof and O'Rourke, 1995) and in this example occurs during deceleration, causing a convex shape to the curve which usually reduces the area under the curve (see Fig. 9.1).

CLASSIFICATION OF ARTERIAL PRESSURE WAVES

Changes in arterial properties (including geometry) and PWV cause profound changes in arterial pressure and flow wave contours (Chapter 10) and in impedance spectral patterns (Chapter 12). These changes are attributed to the timing (or phase relation with respect to the incident wave) and amplitude (or energy content) of pulse wave reflections from peripheral reflecting sites (Van den Bos *et al.*, 1976; Murgo *et al.*, 1980a; Murgo *et al.*, 1981a; Murgo *et al.*, 1981b; Nichols *et al.*, 1987; O'Rourke, 1990a; O'Rourke, 1990b; O'Rourke, 1990c; Nichols *et al.*, 1993; O'Rourke, 1993b; Nichols and Edwards, 2001). In most (but not all) experimental animals and in young humans, wave reflection is so timed that the wave returns to the ascending aorta after ventricular ejection has ceased (see Fig. 9.2) (Nichols *et al.*, 1986a; O'Rourke *et al.*, 1991). Such timing is desirable, since the augmentation in ascending aortic pressure caused by the reflected wave occurs in diastole rather than systole (O'Rourke, 1970b; Murgo *et al.*, 1980a; Nichols *et al.*, 1986a; O'Rourke *et al.*, 1992a; O'Rourke *et al.*, 1992b; O'Rourke *et al.*, 1992c; Nichols and Edwards, 2001). Such a prolonged pressure rise augments diastolic pressure and increases coronary blood flow to the myocardium without increasing left ventricular afterload. Wave reflection during diastole,

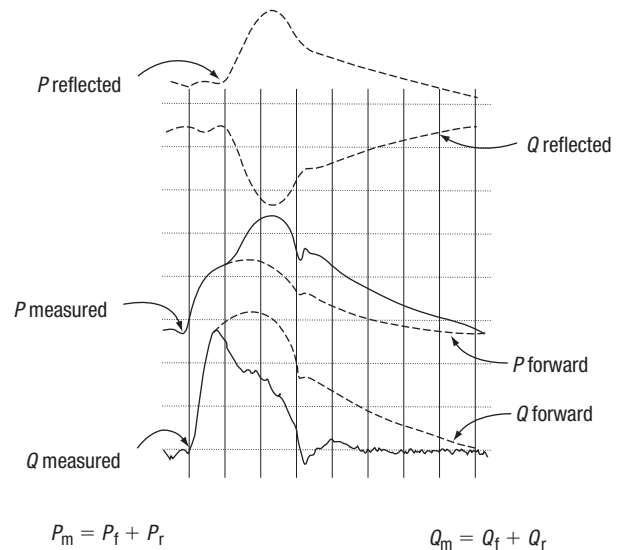


Figure 9.13 The influence of pulse wave reflections on ascending aortic pressure (P) and flow (Q) waveforms. Incident or forward (i) and backward or reflected (r) pressure and flow waves are summed to yield measured (m) pressure and flow waveforms. The forward pressure and flow waves are identical, as are the reflected waves, except that the reflected flow wave is inverted with respect to the reflected pressure wave

therefore, appears to be highly advantageous. In later years and in patients with systemic hypertension, diabetes, end-stage renal disease, and a variety of other vascular diseases (see Chapters 19, 20, 24, and 25), the elastic arteries become stiffer (i.e. less compliant) (Nichols *et al.*, 1985a; Nichols *et al.*, 1986b; O'Rourke *et al.*, 1993a; O'Rourke, 1994; McEniery *et al.*, 2005a). An increase in arterial stiffness causes an increase in PWV and early return of the reflected wave to the ascending aorta during ventricular ejection. Such timing is detrimental, since the augmentation caused by the reflected wave increases systolic pressure and ventricular afterload (see Fig. 9.2). It is also detrimental in the microcirculation of the brain and kidney by causing flow pulsations to pass more distally into the delicate small arteries and arterioles (O'Rourke and Safar, 2005; O'Rourke, 2009b). This issue is discussed further in Chapter 18.

Augmentation index

Aortic augmentation index (AIx) is defined as the ratio of the late systolic boost in the aortic pressure wave and pulse pressure $[(P_s - P_i)/(P_s - P_d)]$ and is usually expressed as a percentage (Fig. 9.14) (Yaginuma *et al.*, 1985; Takazawa, 1987; Kelly *et al.*, 1989c; Kelly *et al.*, 1989d). The configuration or contour of the ascending aortic pressure wave has been classified by Murgo *et al.* (1980a) and Nichols *et al.* (Nichols *et al.*, 1992; Nichols *et al.*, 1993) according to the following criteria.

1. **Type A.** Peak systolic blood pressure (P_s) occurs in late systole after a well-defined inflection point (P_i), and AIx is greater than 12 percent; P_d is aortic minimum diastolic pressure and ($P_s - P_d$) is pulse pressure.
2. **Type B.** Peak systolic pressure also occurs in late systole following an inflection point, but AIx is greater than 0.0 percent and less than or equal to 12 percent.
3. **Type C.** The inflection point occurs after peak systolic pressure and AIx is less than or equal to 0.0 percent. Type C waveforms can be subdivided into type C_s if the peak of the reflected wave occurs in systole and type C_d if the peak occurs in diastole (see Fig. 9.2).
4. **Type D.** There is no inflection point on the measured pressure wave because the reflected wave arrives early in systole and merges (or blends) with the incident wave.

Two visible demarcations usually occur on the initial upstroke of the central aortic pressure wave: the shoulder (P_{sho}) and the inflection point (P_{inf}) (Figs 9.2 and 9.15). Studies have shown that the shoulder occurs at peak blood flow velocity and the inflection point denotes the initial upstroke of the reflected pressure wave (Wetterer, 1956; Takazawa, 1987; Fujii, 1989; Nichols *et al.*, 1993; Nichols and Edwards, 2001). The shoulder is generated by the left ventricular ejection (or flow) wave and is an estimate of incident (or forward) wave amplitude. When the shoulder and inflection point occur simultaneously no demarcation is visible (i.e. type D wave). Therefore, if pressure and flow velocity are measured simultaneously, the initial upstroke of the reflected wave in a type D waveform can be determined and AIx calculated. Kelly *et al.* (1987) and Takazawa *et al.* (1995a) described a method using the fourth derivative of the measured pressure wave to identify the inflection point when blood flow velocity is not measured. In this method, the inflection point supposedly occurs at the first zero crossing, after minimum diastolic pressure, from positive to negative of the fourth derivative (see Fig. 9.15). Although this method has been used by several investigators, it may give erroneous results, especially if there is no identifiable inflection point (i.e. type D waveforms). Indeed, Gatzka *et al.* (2001a; 2001b) and Soderstrom *et al.* (2002) reported several values of augmentation index (both aortic and carotid) that were erroneously high (>50 percent) using this method. The fourth derivative method probably identifies the shoulder of the pressure wave (Segers *et al.*, 2008) and not the inflection point except in type D waves. Another method which uses the second derivative to identify the inflection point is used in the SphygmoCor system. In this method, the inflection point occurs at the second peak of the derivative and, in this example, occurs at a later time than the shoulder (Fig. 9.15).

Type D pressure waveforms are only seen in older individuals (over 65 years of age) or in patients with very

stiff elastic arteries (and high PWV), especially those with ISH (see Fig. 9.2) (Nichols *et al.*, 1992; Nichols *et al.*, 1993), and were not classified by Murgo *et al.* (1980a) because their subjects were all normotensive and less than 55 years of age. Type A aortic pressure waveforms are almost invariably seen in subjects between 40 and 65 years of age and are attributable to reflected waves from the lower body returning during early systole and extending throughout the remainder of ventricular ejection so that pressure in late systole is augmented. Younger adults (under age 30 years) almost invariably exhibit a type C contour with a low-amplitude reflected wave beginning in late systole (type C_s) and extending into diastole; in adolescents and young adults below about 20 years of age who have reached full height, the entire reflected wave usually occurs in diastole (type C_d) (see Fig. 9.2). Subjects between 30 and 45 years of age usually show a type B contour as a result of intermediate amplitude and timing of wave reflection. From these observations, it is clear that amplitude and timing of wave reflections are directly related to the elastic properties of the arterial tree. Figure 9.16 shows the relationship of aortic augmentation index (left) and Δt_p (right) versus carotid-femoral PWV. In young individuals from <12 years of age to infancy, the waveform types are reversed because of body length (see below), i.e. infants have a type A waveform that progresses to a type B waveform and then to a type C waveform at about age 12–15 years (see Chapter 19) (Gevers, 1994; Murakami *et al.*, 2010). The relation between the waveform type and augmentation index is shown in Fig. 9.17.

In addition to being dependent upon the number and location of reflection sites, the amplitude and timing of the reflected wave also depends upon PWV and arterial damping, i.e. the faster the wave travels or the shorter the distance traveled, the less the damping and the greater the amplitude of the reflected wave. The augmentation index has been determined (and related to age) from aortic (Murgo *et al.*, 1980a; O'Rourke *et al.*, 1992a; O'Rourke *et al.*, 1992b; O'Rourke *et al.*, 1992c; Nichols *et al.*, 1993; Nichols and Singh, 2002; McEniery *et al.*, 2005a; Roman *et al.*, 2007; McEniery *et al.*, 2008; Roman *et al.*, 2009; McEniery *et al.*, 2010a; McEniery *et al.*, 2010b), carotid (Kelly *et al.*, 1989a; van Bortel *et al.*, 2001a; Mitchell *et al.*, 2004a; Segers *et al.*, 2009), and radial (Kelly *et al.*, 1990a) pressure waves (see Figs 9.14 and 9.18). Any alteration in wall properties of elastic and/or muscular arteries with age, disease, or intervention will modify the morphology of the pressure waveform to some degree (Nichols and Edwards, 2001). The central aortic augmentation index has been used by several investigators as an index of arterial stiffness or elastance (Wilkinson *et al.*, 1998); however, when making such comparisons it must be kept in mind that augmentation index is influenced by heart rate (Wilkinson *et al.*, 2000a; Wilkinson *et al.*, 2000b), aortic taper (Cecelja *et al.*, 2009), the pattern of ventricular ejection (Westerhof and O'Rourke, 1995), and body length (Langenberg *et al.*, 2003) (see below) and is not an accurate

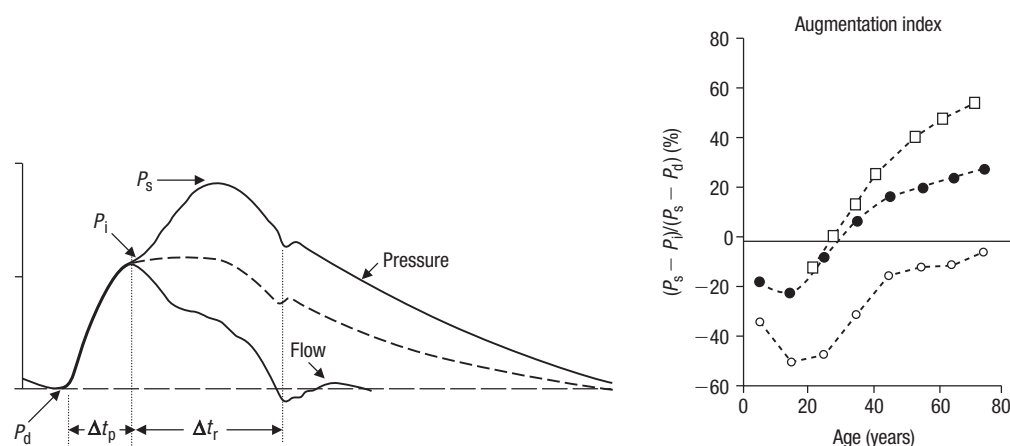


Figure 9.14 Left: High fidelity pressure and flow velocity waveforms measured in the ascending aorta of a middle-aged patient. P_s is peak systolic pressure, P_i is an inflection point that indicates the beginning upstroke of the reflected pressure wave, and P_d is minimum diastolic pressure. Δt_p is the round trip travel time of the forward (or incident) wave from the ascending aorta to the major reflecting site and back, and Δt_r is the systolic duration of the reflected pressure wave. The broken curve is the wave (pressure and flow) that would be measured in the absence of reflections. Right: Effects of age on augmentation index calculated from pressure waves recorded in the ascending aorta ($\square$), carotid artery ($\bullet$), and radial artery ($\circ$)

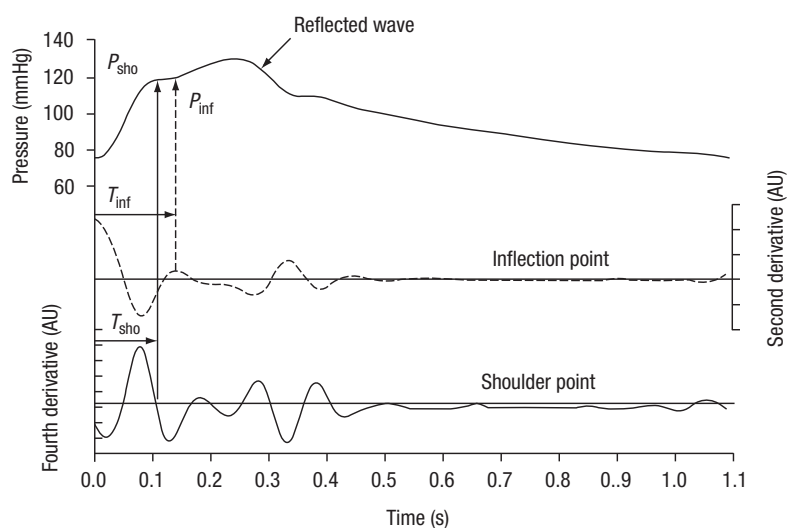


Figure 9.15 Determination of the shoulder and inflection point on a central aortic pressure wave using pressure waveform analysis. The second zero crossing of the fourth derivative identifies the shoulder (P_{sho}) of the pressure wave and the second peak of the second derivative identifies the inflection point (P_{inf}). Reproduced from Segers *et al.* (2007c)

surrogate of arterial stiffness (see Fig. 9.16) (Gurovich *et al.*, 2009). With advancing age, PWV and aortic stiffness increase and the reflected wave returns to the heart earlier with greater amplitude and systolic duration (see Chapter 19) (Fig. 9.18). These modifications in wave reflection characteristics cause an increase in augmentation index and wasted ventricular pressure energy (or effort) (Nichols and Singh, 2002; Hashimoto *et al.*, 2008a). However, after about age 60–65 years, the increase in augmentation index ceases and levels off (but does not decrease), while PMV continues to rise (McEniery *et al.*, 2005a); a decrease in augmentation index occurs when the ventricular is too weak to generate the late systolic pressure boost (Denardo *et al.*, 2010). Also, other cardiovascular diseases that increase arterial stiffness or smooth muscle tone, such as hypertension, end-stage renal disease, diabetes (type 1 and 2), and hypercholesterolemia cause an increase in

augmentation index and wasted left ventricular (LV) energy (Wilkinson *et al.*, 1998; Aggoun *et al.*, 2000; London *et al.*, 2001; Scuteri *et al.*, 2001; Wilkinson *et al.*, 2002; Schram *et al.*, 2004; Weber *et al.*, 2004; Nichols, 2005; Casey *et al.*, 2008b; Nichols *et al.*, 2008; Nichols and Epstein, 2009; Martin *et al.*, 2010).

Left ventricular load and wave reflection

Left ventricular external ‘after’ load may be studied in the time domain as the pressure generated in the left ventricle and ascending aorta during ventricular ejection, or in the frequency domain as the ascending aortic input impedance (see Chapter 12). These two approaches are complementary and represent different ways of looking at the same information (O’Rourke *et al.*, 1993a; Nichols and

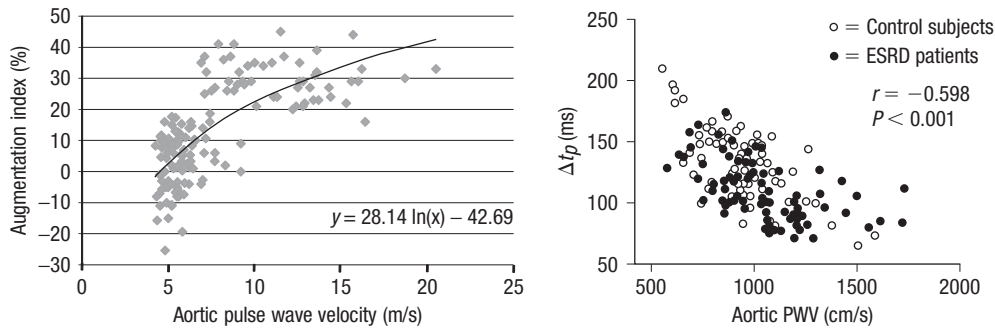


Figure 9.16 Relationship of aortic augmentation index (left) and Δt_p (right, reproduced from London *et al.* (1992)) versus carotid-femoral pulse wave velocity

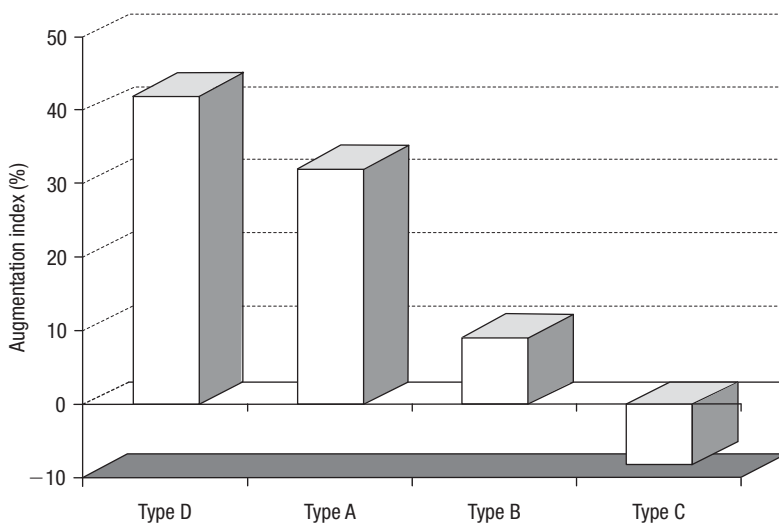


Figure 9.17 Relationship between average aortic augmentation index and waveform type

Edwards, 2001). Left ventricular load has two components, one steady and the other pulsatile. The first is represented by mean arterial pressure (or peripheral resistance), and the second by the additional pressure that the ventricle generates during systole on account of the pulsatile nature of its ejection into the ascending aorta (O'Rourke, 1967b). In addition to ejection dynamics, the pulsatile component is dependent upon arterial properties and wave reflections. However, wave reflections do not contribute to ventricular load in individuals with type C_d waveforms. Ventricular load may be represented by pressure alone, as myocardial oxygen demands and ventricular ejection are, for a given ventricular size, shape and contractility, exclusively dependent on the pressure against which the ventricle pumps. The higher the pressure, the higher the oxygen consumption and energy cost of contraction (Kelly *et al.*, 1992a; Kelly *et al.*, 1992b), but the higher the pressure, the lower the volume of blood the ventricular can expel (Ross *et al.*, 1966; Elzinga and Westerhof, 1979; Nichols *et al.*, 1980b; Nichols and Pepine, 1982; Westerhof and O'Rourke, 1995; Nichols and Edwards, 2001). The pressure generated by the ventricle ($P_s - P_i$) in response to the reflected wave is wasted energy

($E_w = 0.25\Delta t_r(P_s - P_i)\pi$) since it reduces ejection instead of enhancing it (see Fig. 9.14, left) (Nichols and Singh, 2002). This energy expended by the ventricle takes into account both amplitude and systolic duration of the reflected wave. During systole, aortic pressure approximates left ventricular pressure, differing only by a small amount caused by the Venturi effect, and by inertial and viscous factors involved in transfer of blood from one chamber to the other (Fig. 7.3).

In normal circumstances in adolescent humans, peak left ventricular and aortic pressure are seen at the time of peak flow into the aorta, whereafter pressure falls slowly, then more quickly to the incisura when the aortic valve closes. As described in Chapter 19, increasing age is associated with progressive rise in aortic and left ventricular systolic pressure as a consequence of aortic stiffening and increased wave reflection. Aortic stiffening alters left ventricular systolic pressure in two ways – first, by causing a greater rise of pressure at the time of peak aortic flow (unaugmented pressure), and second, by increasing PWV in the aorta and so causing the reflected wave from peripheral sites to return early and boost pressure in late systole (Fig. 9.19). These effects on the

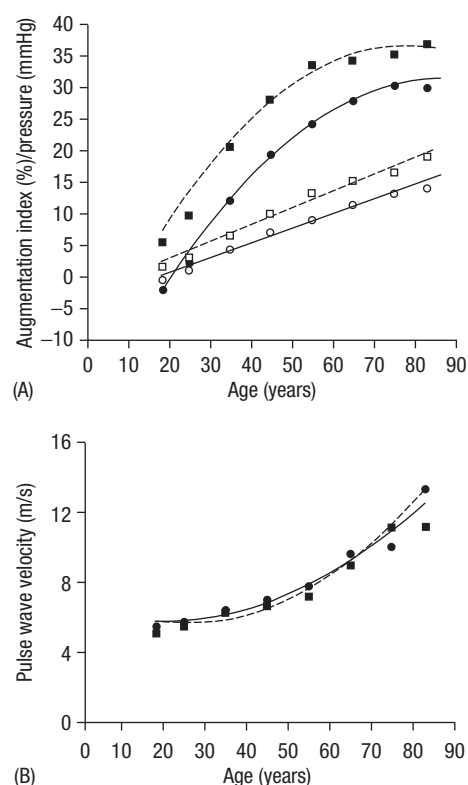


Figure 9.18 Regression curves representing the effect of age on parameters of arterial stiffness and wave reflection for males (circles, solid lines) and females (squares, dashed lines). Panel A represents augmentation pressure (open circles/open squares) and augmentation index (closed circles/closed squares). Panel B represents aortic pulse wave velocity. Reproduced from McEniery *et al.* (2005a)

ascending aortic and left ventricular pressure wave are usually readily apparent by the fourth decade of life and are very obvious by the seventh decade, where in addition, the prevalence of hypertension, coronary artery disease, and cardiac failure are very high. The effects of arterial stiffening are further increased in hypertension by increased aortic distending pressure (O'Rourke, 1970a; O'Rourke, 1976; Nichols *et al.*, 1986b), in atherosclerosis by involvement of the descending aorta and other systemic arteries (Hirai *et al.*, 1989) and in cardiac failure through increased vascular tone of muscular arteries and arterioles (Pepine *et al.*, 1978; Brin and Yin, 1984; Laskey and Kussmaul, 1987; Binkley *et al.*, 1990; Nichols and Pepine, 1992; Eaton *et al.*, 1993; Eaton *et al.*, 1994; O'Rourke and Hashimoto, 2007; Denardo *et al.*, 2010).

Wave reflection increases left ventricular load by increasing aortic pressure in late systole. The boost is substantial and accounts for an increase in aortic and left ventricular systolic pressure of some 25 mmHg in normal subjects between ages 30 and 60 (Nichols *et al.*, 1987; Kelly *et al.*, 1989a) and is correlated with left ventricular mass

(Fig. 9.20) (Marchais *et al.*, 1993; London and Guerin, 1999; Ioannou *et al.*, 2009). In older subjects, and in disease states such as arteriosclerosis and hypertension (especially 'systolic' hypertension), the boost to left ventricular systolic pressure may exceed 40 mmHg (Yaginuma *et al.*, 1985; Takazawa, 1987; Nichols *et al.*, 1992; Nichols, 1994; Nichols, 2005). In such circumstances, the mean pressure during systole may approach 50 percent above mean pressure during the whole cardiac cycle, denoting inefficiency of the arterial system in accepting pulsatile blood flow from the heart. Here, the heart requires some 50 percent more energy for ejection than it would require if its output were continuous. The extra pulsatile energy expenditure is normally around 10 percent, but can increase to more than 20–30 percent in old age and hypertension (O'Rourke *et al.*, 1968; Nichols *et al.*, 1977a) and diseases that increase arterial stiffness and wave reflection.

In the frequency domain, left ventricular load is expressed as ascending aortic impedance (Chapter 12) (O'Rourke and Taylor, 1967; Milnor, 1975; Nichols *et al.*, 1980a; Nichols and Pepine, 1982). Effects of early wave reflection are apparent as a shift of impedance curves (of modulus and phase) to the right, so that minimal values and fluctuations are seen at higher frequencies (see Chapter 12). The effects of early wave reflection are very apparent in humans by the age of 40. In a series of 18 normal human subjects, Murgu *et al.* (1980a) showed that impedance fluctuations occurred at the same frequencies as had been described for rabbits (Avolio *et al.*, 1976). The shorter distance to reflecting sites in rabbits was completely offset by the higher aortic PWV in humans. The significance of these findings is not appreciated until impedance curves are compared with the ascending aortic flow wave, when gross mismatch is apparent. While in the rabbit, the peak energy of the left ventricular ejection wave occurs at a frequency at which ascending aortic impedance is least, denoting optimal ventricular–vascular interaction, the peak energy of the left ventricular ejection wave in humans occurs at a frequency where the impedance modulus is quite high, denoting far from optimal interaction (see Fig. 19.19).

Looked at in the time or frequency domain, early wave reflection during systole is seen in adult humans even at 40 years of age and creates a substantial increase in left ventricular load and wasted pressure energy (see Fig. 9.19). The diseases of the cardiovascular system regularly seen in clinical practice occur in people of this age or greater. Furthermore, the diseases themselves may hasten wave reflection and provide a further increase in left ventricular load. Hence, modification of wave reflection must constitute a rational approach to therapy (see Chapter 26). It is fascinating to note the long appreciation of arterial stiffness and left ventricular function. In his influential text, Mackenzie (1902) related decreasing exercise capacity from the fourth decade of life to cardiac dysfunction caused by arterial stiffening.

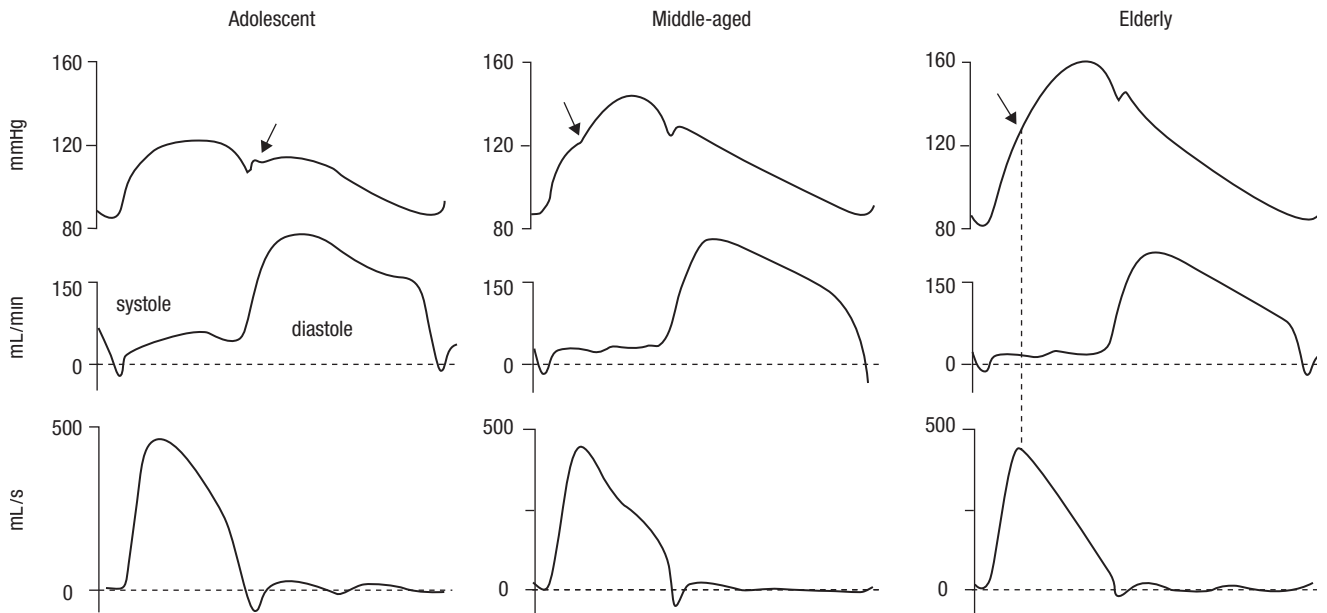


Figure 9.19 Schematic illustration showing pressure (top) and flow waves recorded in the ascending aorta (bottom) and coronary artery (middle) in normotensive subjects (adolescent, middle-aged, and elderly). In adolescents, the reflected pressure wave (beginning at the arrow) occurs in diastole and enhances coronary blood flow. With advancing age, the reflected wave arrives earlier, occurs in systole during ventricular ejection and causes a decrease in coronary blood flow. The deceleration phase of the flow wave is convex in adolescents, concave in middle age and linear in the elderly

WAVE REFLECTION AND MYOCARDIAL OXYGEN SUPPLY AND DEMAND

Myocardial oxygen requirements can be estimated from the area beneath the systolic part of the left ventricular pressure wave, as tension–time index (Sarnoff *et al.*, 1958). The value may be approximated (ignoring isovolumic contraction and relaxation) by the area beneath the systolic part of the ascending aortic pressure wave (systolic pressure–time index) (O’Rourke, 1982a; O’Rourke, 1982b). Myocardial oxygen supply is determined by the patency of coronary arteries and by the pressure gradient across the coronary bed. As coronary vessels in the left ventricle are squeezed shut by contraction during systole, the walls of this chamber can only be perfused during diastole (Buckberg *et al.*, 1972a; Buckberg *et al.*, 1972b; Berne and Rubio, 1979). Left ventricular coronary perfusion pressure is that pressure maintained in the aorta throughout diastole and can be estimated as the area beneath the diastolic part of the aortic pressure wave (diastolic pressure–time index) (see Fig. 9.19) (when venous pressure is negligible). These areas are markedly influenced by systolic and diastolic time duration.

In addition to serving as a conduit that delivers blood to peripheral vascular beds, the function of the elastic arteries (aorta, carotid, iliac, etc.) is to cushion flow pulsation from the ventricle so that the heart may function most efficiently. Efficient function is apparent as a small pulse pressure in the ascending aorta, with average pressure during systole being only slightly greater (about

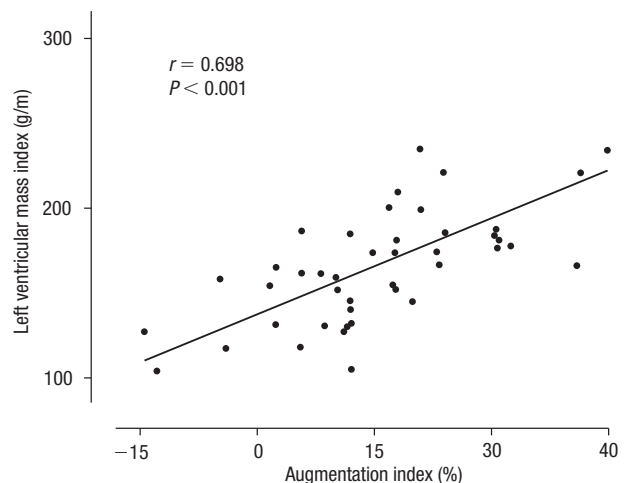


Figure 9.20 Illustration showing the effects of wave reflection (carotid augmentation index) on echocardiographically determined left ventricular mass index in a group of humans ranging in age from 20 to 70 years. Reproduced from Marchais *et al.* (1993)

5.0 mmHg) than mean cycle pressure, and average pressure during diastole only slightly less (by about 5.0 mmHg). Design of the normal arterial system is such that pressure rise during systole is minimized (so that myocardial oxygen demands are minimized) and pressure is maintained as high as possible during diastole (so that coronary blood flow may be maintained). In adolescent humans, and in most experimental animals (but not all),

wave reflection is seen to be so timed as to enhance pressure during diastole without boosting systolic pressure. Appropriately timed wave reflection is thus an important factor in minimizing myocardial oxygen requirements and maximizing the capacity for myocardial blood flow (Fig. 9.19).

Early wave reflection distorts the favorable shapes of the ascending aortic pressure and flow waves, causing a relative increase in pressure and a decrease in flow during systole and a relative decrease in pressure during diastole (Fig. 9.19). Early wave reflection thus distorts arterial and ventricular function, not only by increasing aortic and left ventricular pressure, and thus myocardial oxygen requirements (Kelly *et al.*, 1992a; Kelly *et al.*, 1992b), but also by decreasing aortic pressure during diastole, thus reducing the capacity for left ventricular myocardial perfusion and ejection.

Early wave reflection predisposes to myocardial ischemia by increasing the requirement of the myocardium for blood while decreasing capacity for supply of blood through the coronary arteries. The deleterious effects of early wave reflection are the opposite to the beneficial effects of arterial counterpulsation. With counterpulsation achieved by intra-aortic balloon pumping or external cuff inflation around the legs, the timing of balloon inflation and deflation is so adjusted as to artificially decrease ascending aortic and left ventricular pressure during systole, while increasing aortic pressure during diastole (see Chapter 17).

SITES OF WAVE REFLECTION

As mentioned earlier, wave reflections arise from any discontinuity in caliber or elastic properties along the arterial tree where there is a change (or mismatch) in impedance. This mismatch can occur acutely or chronically. Major changes in the central elastic arteries (aorta, carotids, iliacs, etc.) occur over time and acute alterations in wall properties are passive, while changes in the muscular arteries (femorals, popiteals, posterior tibials, brachials, etc.) and arterioles most often occur acutely and alterations in wall properties are active (Boutouyrie *et al.*, 2000; Nichols and Edwards, 2001). For example, the aging process and hypertension predominantly modify elastic arteries not muscular arteries (Bortolotto *et al.*, 1999) and vasoactive drugs predominantly modify muscular arteries and arterioles, not elastic arteries (Latson *et al.*, 1988; Safar *et al.*, 2003; Stewart *et al.*, 2003). Possible reflecting sites are branching points, areas of alteration in arterial elastance, and high-resistance arterioles (Campbell *et al.*, 1989; Safar and O'Rourke, 2006; Cecelja *et al.*, 2009).

The high-resistance arterioles are considered to be the major sites of wave reflection in the circulation (McDonald, 1974; O'Rourke, 1982a; O'Rourke, 1982b; Milnor, 1989). As blood flows through the circulation, arterial blood pressure falls very little over long distances in the

arteries, then falls precipitously over a short distance in the arterioles (see Fig. 16.2). This fact, together with the observed vasodilator effects on arterial pressure waves, led Hamilton (1944), McDonald (1974), Milnor (1989), and O'Rourke (O'Rourke, 1982a; O'Rourke, 1982b) to this conclusion. Later evidence has reinforced these views, especially the calculation of peripheral reflection coefficient from impedance determinations, and the finding that vasodilator agents markedly attenuate the influence of wave reflection on pressure and flow waves and input impedance spectra (O'Rourke and Taylor, 1966; Pepine *et al.*, 1979; Merillon *et al.*, 1982a; Van den Bos *et al.*, 1982; Brin and Yin, 1984; Ting *et al.*, 1986; Yaginuma *et al.*, 1986; Fitchett *et al.*, 1988b; Latson *et al.*, 1988). Vasodilator agents, in general, reduce peripheral wave reflections through their effects on muscular arteries, as well as arterioles (Fitchett *et al.*, 1986; Yaginuma *et al.*, 1986; Fitchett *et al.*, 1988a; Fitchett *et al.*, 1988b; Latson *et al.*, 1988; Asmar *et al.*, 2001b; Dart *et al.*, 2001; Kelly *et al.*, 2001; Vlachopoulos *et al.*, 2001; Wilkinson *et al.*, 2001; Klingbeil *et al.*, 2002; Pannier *et al.*, 2002; Nichols *et al.*, 2008; Nichols and Epstein, 2009; Miyashita *et al.*, 2010), and this decreases ventricular load in humans with inappropriately early wave reflection (see Chapters 26 and 27).

Several studies, past and present, emphasize branching points and taper of major arteries as important sites of wave reflection (Remington, 1963; Wetterer *et al.*, 1977a; Westerhof *et al.*, 1978; Westerhof *et al.*, 1979; Murgo *et al.*, 1980a; Latham *et al.*, 1987; London *et al.*, 1995; Khir and Parker, 2002; Myers and Capper, 2004). Early evidence of pressure wave transmission suggested that wave reflection arose from the region of the aortic bifurcation or pelvic branches of the iliac artery (Frank, 1905; Hamilton and Dow, 1939; Alexander, 1953; Remington, 1963). Latham *et al.* (1985), using a catheter with six micromanometers to measure pressure, found evidence to suggest a significant reflecting site at the level of the renal arteries in humans. Studies such as these draw attention to the distal aorta and suggest some anatomic, histologic, or physiologic change in the aorta, renal, or iliac arteries in this region. As seen from the ascending aorta, this site appears to behave as a functionally discrete reflecting site, but so does the upper thigh when impedance measurements are taken in the descending abdominal aorta (O'Rourke and Taylor, 1966; O'Rourke, 1982a; O'Rourke, 1982b; O'Rourke and Yaginuma, 1984), and the knee area when measurements are made in a femoral artery. In all cases, the functionally discrete (or effective) reflection site appears to represent nothing more than the resultant or average of all individual reflecting sites in the vascular bed downstream (Frank, 1930; O'Rourke, 1982a; O'Rourke, 1982b). Modeling studies (Taylor, 1966a; Taylor, 1966b; Campbell *et al.*, 1989) have shown that, when individual reflecting sites are closely grouped in a vascular bed, the reflection coefficient at low frequencies appears to be almost as high as at each individual termination. However, when there is wide spatial dispersion of reflecting

sites, discrete wave reflection is less apparent. Comparative physiologic studies support this concept, as discrete wave reflection is most pronounced in the Australian wombat (O'Rourke, 1967a; O'Rourke, 1967b) and kangaroo (Avolio *et al.*, 1984; Avolio *et al.*, 1985b; Nichols *et al.*, 1986a), animals with large muscular groups in the lower body, and least pronounced in snakes (Avolio *et al.*, 1982) and alligators (Syme *et al.*, 2006), animals with distributed muscular groups over a long distance. Other information suggests that wave reflection is minimal at bifurcations and branches of major arteries and maximal at arterial/arteriolar junctions where resistance to flow increases abruptly. When pressure waves are recorded between aorta and peripheral arteries such as the femoral or aphenous, there is a gradual increase in pulse pressure and in lower harmonic components, indicative of a major reflecting site downstream beyond the femoral or saphenous artery with no indication of such a site upstream (Figs 10.1, 10.6, and 10.7) (McDonald and Taylor, 1959). Furthermore, vascular impedance in the femoral artery signifies presence of a reflection site downstream, in the region of the knee (Figs 12.11 and 12.12), and when a vasodilator (acetylcholine) is injected intra-arterially (Fig. 12.14), there appears to be general reduction in wave reflection (Figs 12.12 and 12.13) which corresponds with marked increase in femoral flow (Fig. 12.14), but no change in the timing of wave reflection (Figs 12.12 and 12.13).

The concept of 'effective' length of the arterial system implies that it may be represented by a single uniform viscoelastic tube (or transmission line) terminated by an impedance, creating a single reflection site (Campbell *et al.*, 1989). The distance L_z to this 'effective' reflecting site from the point of pressure and flow measurement (usually the ascending aorta) can be calculated by the quarter-wavelength relationship (Mills *et al.*, 1970; McDonald, 1974; Nichols *et al.*, 1985a; Milnor, 1989):

$$L_z = \frac{\lambda}{4} = \frac{c_0}{4f_{\min}}, \quad 9.31$$

where c_0 is the pulse wave velocity (or phase velocity) and $f_{\min}$ is the frequency of the first aortic input impedance modulus minimum (or node) (see Chapter 12). Although this simple equation has been used by numerous investigators, its accuracy has been questioned by others (Sipkema and Westerhof, 1975; Burattini and Carlos, 1988). The distance from the heart to the effective reflecting site calculated using the above relation is 30–45 cm in a 20 kg dog (Nichols and McDonald, 1972; McDonald, 1974), 35–50 cm in a 20 kg kangaroo (Nichols *et al.*, 1986a) and 40–55 cm in an adult human of < 55 years of age (Murgo *et al.*, 1980a; Nichols *et al.*, 1987). It should be noted that the reflection site distance L_z is age-related and depends upon arterial elasticity (i.e. pulse wave velocity) and size (i.e. length) of the arterial tree (which affects $f_{\min}$), which is discussed in detail in Chapters 3 and 4. Since wave reflections alter the shape of the pressure wave, Van den

Bos *et al.* (1976), Murgo *et al.* (1980a), and Latham *et al.* (Latham *et al.*, 1985; Latham, 1989) derived 'effective' reflection site distance L_p from the aortic pressure waveform using a 'lumped' element uniform tube model and the relationship

$$L_p = c_0 \Delta t_p / 2, \quad 9.32$$

or

$$c_0 = 2L_p / \Delta t_p,$$

where Δt_p is the time from the initial upstroke of the pressure wave to the inflection point (or the upstroke of the reflected wave) (see Figs 9.14 and 9.15) and represents the time of travel of the pressure wave from the measuring point to the 'effective' reflection site and back. Since Δt_p is inversely related to PWV and directly related to effective reflection site distance, it follows that changes in the physical properties (i.e. stiffness, length, and taper) of the arterial system will influence Δt_p . London *et al.* (1992) confirmed this in a group of patients with systolic hypertension resulting from end-stage renal disease (Fig. 9.16, right). The calculation of effective reflection site distance from the pressure waveform correlates well with that calculated from the impedance spectra (Murgo *et al.*, 1980a; Nichols *et al.*, 1986a) and that measured directly (Van den Bos *et al.*, 1976). Murgo's study included mostly young subjects < 40 years of age (only three of 18 subjects were > 40 years). Effective reflection site distance was similar in all three study groups (types A, B, and C waveforms) using both of the above equations (9.31 and 9.32). Data from McEniery *et al.* (2005a) and Sugawara *et al.* (2010a; 2010b) confirm and extend this conclusion (see Fig. 9.21). During the aging process in individuals of 55 years of age, aortic PWV increases exponentially while Δt_p decreases very little, therefore, effective reflection site distance calculated using the 'lumped' model (i.e. eqn 9.32) increases instead of decreases as one might expect (Fig. 9.21). Although reflection site distance increases, augmented pressure and augmentation index paradoxically increase instead of decrease (see Fig. 9.18) which raises the possibility of inaccuracy of the equation (eqn 9.32) as suggested at the recent Artery 10 meeting in Verona (Westerhof and Westerhof, 2010).

Short height has been associated with an increased risk of cardiovascular disease, specifically coronary artery disease, in both males and females in a large number of studies and clinical trials (Hebert *et al.*, 1993; Rich-Edwards *et al.*, 1995; Forsen *et al.*, 2000; Jousilahti *et al.*, 2000; Smith *et al.*, 2000; McCarron *et al.*, 2002; Paajanen *et al.*, 2010). Also, cardiovascular disease mortality is increased in lower limb amputees compared to age-matched controls (Hrubec and Ryder, 1980; Modan *et al.*, 1998). Several theories have been suggested as possible causes for the adverse events, but the reasons why body shortness influences cardiovascular disease risk remain

largely unknown. There are published data to suggest a physiologic explanation for this phenomenon. A short arterial tree places the major reflection site closer to the heart and, therefore, causes a decrease in Δt_p (Smulyan *et al.*, 1998; McEniery *et al.*, 2005a; Ayer *et al.*, 2010) (Fig. 9.22, left) and an increase in augmentation index (Figs 9.18 and 9.22, right) (London *et al.*, 1992; Marchais *et al.*, 1993; Hayward *et al.*, 1997; McEniery *et al.*, 2005a; Lieber *et al.*, 2010), pulse pressure (Langenberg *et al.*, 2003), and ventricular afterload. The observed changes in amplitude and timing of wave reflection in shorter individuals are independent of changes in aortic stiffness and PWV (Vermeersch *et al.*, 2008; Cecelja *et al.*, 2009; Ayer *et al.*, 2010). These observations were recently confirmed in an isolated rabbit heart model pumping into rubber tubes of different lengths (Zannoli *et al.*, 1999). As tube length was shortened, Δt_p decreased, on a pure mechanical basis, and was associated with a reduction in stroke volume and coronary artery driving pressure. Similar changes were observed in a dog model that was designed to determine the role of wave reflection on the patterns of pressure and

flow waves in the ascending aorta (Van den Bos *et al.*, 1976). Aortic occlusions with balloon inflation at three different sites caused pressure pulses with secondary systolic rises and flow pulses with biphasic deceleration. As the occlusion occurred closer to the heart Δt_p decreased and approximated the time in which the pulse traveled from the manometer sensor in the ascending aortic to the inflated balloon and back. Replacing the aorta with a stiff tube increases PWV and shifts the reflection site distally, but still causes the reflection wave to arrive earlier at the heart and increase pressure in late systole and flow pulses with biphasic deceleration (Randall *et al.*, 1984; Nichols and Edwards, 2001).

SECONDARY REFLECTIONS OF REFLECTED WAVES

Work performed by O'Rourke and Taylor (1967) and O'Rourke (O'Rourke, 1982a; O'Rourke, 1982b) suggested that the systemic circulation could be represented by an

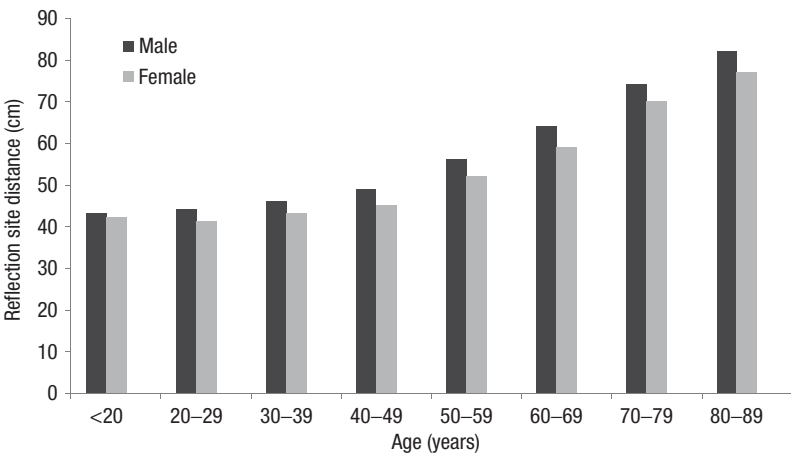


Figure 9.21 Illustration showing the age-related change in 'effective' reflection site distance in males and females included in the Anglo-Cardiff Collaborative Trial (ACCT). Reproduced from data published by McEniery *et al.* (2005a)

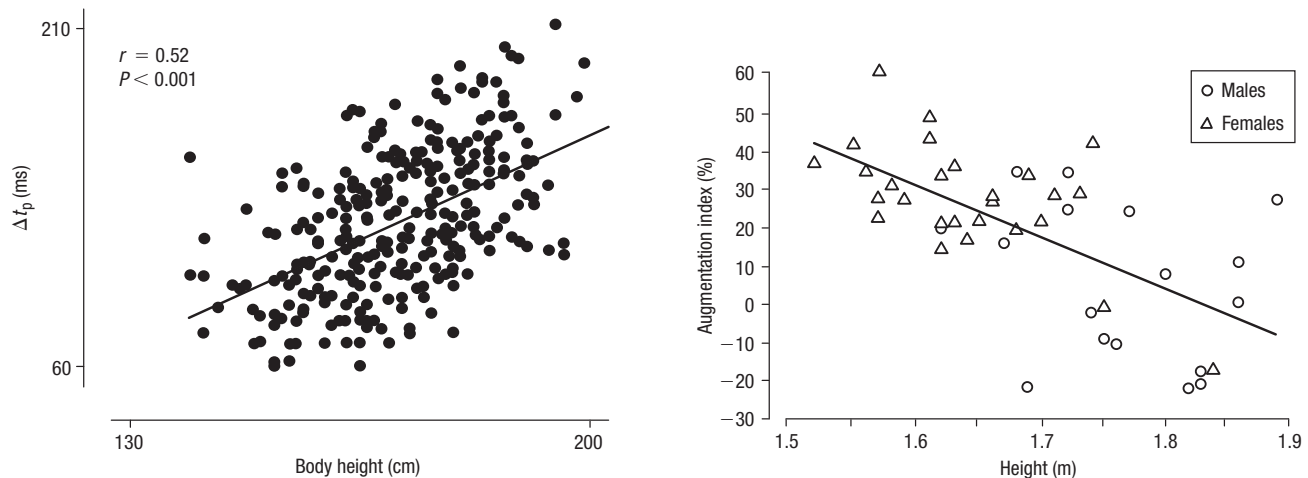


Figure 9.22 Relationship between travel time of the pressure wave and body height (left) and carotid augmentation index and body height for a single decade (right). Reproduced from Smulyan *et al.* (1998) and Hayward *et al.* (1997)

asymmetric T-tube (see Fig. 12.26), whose short and long arms behave, respectively, like the apse and nave of a cathedral, with reflection and re-reflection of the pulse at both ends. Other work (McDonald, 1968b; McDonald, 1974; Avolio *et al.*, 1976; O'Rourke and Avolio, 1980; Nichols *et al.*, 1986a; Burattini and Campbell, 1989; Burattini *et al.*, 1991; Burattini and Campbell, 1993) has provided further support for this general concept of the arterial system.

The presence of prominent diastolic fluctuations in the earliest recorded accurate pressure waveforms led Otto Frank (Frank, 1899; Frank, 1905) and his followers to refer to arterial resonance, and even to calculate the length of the arterial resonating system. In dogs, when the PWV averages 6.0 m/s between proximal aorta and femoral artery and the period of the 'resonant' wave is 0.2 s, this is calculated as 60 cm. Frank believed that this resonance was superimposed on the basic shape of the pressure waveform, which he considered to decay exponentially from the time of aortic valve closure, as in a Windkessel. This concept was carried further by Hamilton and Dow (1939) who introduced the concept of aortic 'standing waves' to explain apparently simultaneous peaks of pressure waves in lower body arteries, the diastolic pressure wave and the reciprocal fluctuations of pressure in upper and lower body about a 'node' in the descending thoracic aorta.

These concepts of Frank and followers, and of Hamilton and colleagues, assume not only intense reflection of pressure waves from peripheral sites, but also re-reflection from the aortic valve or from peripheral sites at the opposite end of the body (Berger *et al.*, 1993).

McDonald (1974) criticized the concept of resonance and of standing waves as being physically impossible since the reflected wave is being damped considerably during its travel between reflecting sites. Resonance certainly cannot occur in a biological system with incomplete reflection and strong attenuation. The concepts of resonance, standing waves, and nodes are, however, of some value provided one does not take these terms literally as meaning what they would in a physical system (O'Rourke, 1982a; O'Rourke, 1982b). O'Rourke (O'Rourke, 1982a; O'Rourke, 1982b) suggested a compromise between the views of these distinguished physiologists. The compromise (O'Rourke, 1967a; O'Rourke, 1967b; O'Rourke, 1971) is similar to that advanced by Wetterer and Kenner (1968), that re-reflection of reflected waves does occur, and that a form of damped resonance does develop between reflecting sites in the upper and lower parts of the body. This concept, with wave reflection occurring between lumped arteriolar terminations in the upper and lower parts of the body (Fig. 9.23), was advanced to explain the major features of pressure waves, flow waves, and impedance patterns in different arteries of different animals under different conditions (see Chapters 10 and 12). This compromise was considered, and rejected, by McDonald (1974) on the basis of significant attenuation of the pulse over long lengths of artery. He considered the

pulse as traveling to the feet and back, rather than to a lumped (or discrete) reflecting site in the region of the pelvis, some 40–50 cm (not 100 cm) distant. When the pulse is considered to travel over some 60 cm in the asymmetric T-model of the arterial system suggested by impedance studies (see Chapter 12), and when experimentally determined values of reflection coefficient and attenuation are used, the compromise appears to be quite reasonable.

The amplitude (or intensity) and timing of reflected waves alter pressure and flow waveforms and input impedance spectra (see above). These concepts are discussed more fully in Chapters 10, 12, 19, and 20. In most animals (but not all) and adolescent humans, the reflected wave occurs in early diastole (in the ascending aorta), while in older humans the reflected wave occurs in systole (see Fig. 9.2). Vasodilatation (Pepine *et al.*, 1979; Merillon *et al.*, 1982a; Brin and Yin, 1984; Nichols *et al.*, 1986b; Ting *et al.*, 1986; Yaginuma *et al.*, 1986; Laskey and Kussmaul, 1987; Fitchett *et al.*, 1988a; Fitchett *et al.*, 1988b; Latson *et al.*, 1988; Chang *et al.*, 1990; Kelly *et al.*, 1990a; Laskey *et al.*, 1993; O'Rourke *et al.*, 1993a) increases travel time and reduces the amplitude of the reflected wave, and decreases the fluctuations of modulus and phase in the input impedance spectra. Any accompanying reduction in mean arterial pressure decreases PWV and shifts the impedance phase zero-crossing (and minimum of modulus) to a lower frequency (see Figs 12.16 and 20.5). Hypertension, the arterial degeneration that accompanies aging in humans, and aortic coarctation all decrease arterial distensibility and increase pulse wave velocity and have the opposite effect of reduction in mean arterial pressure, shifting impedance curves to the right (Alexander *et al.*, 1989). In these conditions, the impedance modulus is higher at low frequencies as a consequence of early wave reflection (see Figs 19.19 and 19.22) (O'Rourke and Cartmill, 1971; Merillon *et al.*, 1982a; Merillon *et al.*, 1982b; Merillon *et al.*, 1982c; Nichols *et al.*, 1985a; Nichols *et al.*, 1986b; Ting *et al.*, 1986; Nichols *et al.*, 1987). In aortic coarctation, reflection occurs early as a consequence of the new reflecting site provided by the coarctation, as well as the accompanying hypertension (O'Rourke and Cartmill, 1971). When, as with aging in man, wave reflection occurs almost simultaneously from the lower and upper body (as a consequence of relatively greater increase in PWV to the lower body), the two effective reflecting sites appear as one (Burattini *et al.*, 1991). Physiologic maneuvers in humans, such as the Mueller and Valsalva maneuvers, alter the timing and amplitude of wave reflection and cause marked changes in the ascending aortic impedance spectra and proximal aortic pressure and flow waveforms (Murgo *et al.*, 1981a; Murgo *et al.*, 1981b; O'Rourke, 1981b; Condos *et al.*, 1987). Pressure wave reflections occurring during systole boost systolic pressure in the ascending aorta and increase ventricular load, while those occurring during diastole favor myocardial perfusion (see above).

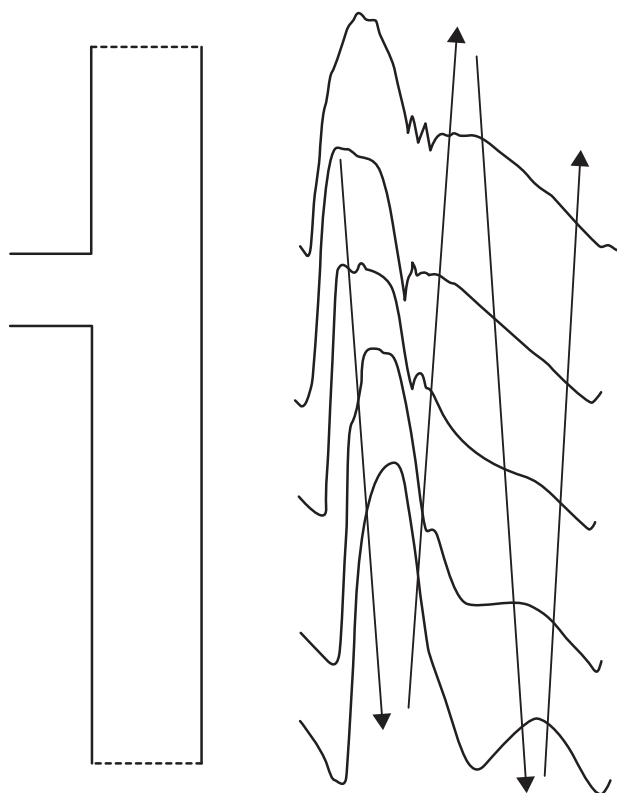


Figure 9.23 The asymmetric T-tube model of the arterial system, and pressure waves recorded between the aortic arch (top trace), iliac artery (bottom trace) and intermediate sites in a dog. A line is drawn parallel to the foot of the waves to indicate travel of the initial impulse generated by left ventricular ejection. Reflection and re-reflection of this impulse are indicated by the other lines and explain wave contour at different sites. Reproduced from O'Rourke (1970a)

Wave reflections also alter the flow waveform (see Fig. 9.19). In late systole, the deceleration portion of the flow wave is convex to the right in adolescents. By middle age, this convexity has become almost linear, and in the elderly the deceleration portion of the curve is usually concave to the right from peak flow to zero (Sakai *et al.*, 1992; Nichols *et al.*, 1993; Miyashita *et al.*, 1994).

WAVE REFLECTION IN CLINICAL TRIALS

The basic principles of wave reflection, previously discussed, were established largely by Michael Taylor, the principal protégé of Donald McDonald, initially in London, then independently in Sydney from 1955 to 1975. Taylor reconciled the views of Hamilton, Dow, Remington, Alexander, and Peterson on the pulse in the time domain to McDonald's view of the pulse in the frequency domain, then advanced well beyond all (Taylor, 1959a; Taylor, 1964; Taylor, 1965; Taylor, 1966a; Taylor, 1966b; Taylor, 1966c; Taylor, 1966d; Taylor, 1969; Taylor, 1973; Taylor, 1994).

Another major (and continuing) contributor is Nico Westerhof whose hemodynamic work was initiated in Philadelphia, continued in Amsterdam, and now involves his son, Berend. Nico tackled specifically the task of separating forward and backward traveling waves, together with the effects of wave reflection on flow, as well as pressure (a negative flow effect), which is potentially relevant when the left ventricle is weakened and behaves as a 'pressure source' with output dependent on pressure (Westerhof *et al.*, 1972; Westerhof and O'Rourke, 1995; Denardo *et al.*, 2010).

Since 1970, this work increasingly was taken up by physicians such as Murgo and colleagues (Murgo *et al.*, 1980a; Murgo *et al.*, 1980b), whose major work relating pressure waveforms to impedance patterns will be discussed in the next chapter. The Framingham study drew attention from 1971 to the effects of arterial stiffening in older people (Kannel *et al.*, 1971; Kannel *et al.*, 1981), and led to the SHEP study in 1991 which established the value of treating older people with ISH (SHEP Investigators, 1991). With reintroduction of pulse waveform analysis through non-invasive arterial applanation tonometry (Chapter 30), many new studies were directed to factors which influence the arterial waveform through alteration of arterial stiffness or wave reflection (Laurent *et al.*, 2006; Safar and O'Rourke, 2006; Agabiti-Rosei *et al.*, 2007). From the early 1990s, Safar and colleagues in Paris began their outcome studies on aortic PWV and carotid waveforms in patients with hypertension and end-stage renal disease. Kelly and Fitchett (1992) introduced non-invasive methods for measuring ascending aortic impedance (as now applied by Mitchell *et al.* (2001) and by Segers *et al.* (2007b) in the Framingham and Asklepios trials, respectively), together with applanation tonometry for studying aging change and effects of drugs on wave reflection (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; Kelly *et al.*, 1989c; Kelly *et al.*, 1990a; Kelly *et al.*, 1990b; Kelly *et al.*, 1992a; Kelly *et al.*, 1992b). The SphygmoCor system was introduced in 1993 for generating the aortic pressure waveform from the radial artery pressure waveform (O'Rourke *et al.*, 1993a) following acknowledgment that a generalized transfer function can be used to synthesize the aortic from the radial or brachial pressure wave (Karamanoglu *et al.*, 1993; Chen *et al.*, 1997; Sharman *et al.*, 2006; Westerhof *et al.*, 2008a; Westerhof *et al.*, 2008b; Segers *et al.*, 2009). Studies of arterial stiffness and wave reflection have burgeoned since, as have instruments used to measure arterial stiffness, wave reflection, and the effects of aging, disease, and therapy (see Chapter 4) (Laurent *et al.*, 2006; Safar and O'Rourke, 2006; Agabiti-Rosei *et al.*, 2007).

One of the first such devices was based on a Windkessel model of the arterial system and sought to measure distal and proximal arterial compliance from the diastolic part of the radial artery waveform (Cohn and Finkelstein, 1992). As with other Windkessel models, this device ignored effects of wave reflection. Multiple problems were found with it (Manning *et al.*, 2002) and it has not found

favor with consensus groups (Laurent *et al.*, 2006; Mancia *et al.*, 2007a) as being based on an unrealistic model. A recent example is a hybrid model of Windkessel and transmission line (Davies *et al.*, 2007). This, using wave-intensity analysis, requires separation of a Windkessel component from the pressure wave and treating the remnant as traveling in a transmission line. This hybrid can be included as among the ‘... unwieldy mathematical edifices’ dismissed by McDonald in the early editions of this book. Unsurprisingly, this method concludes that wave reflection is minute in systole and does not contribute to the pressure waveform during diastole.

Clinical consensus groups (Laurent *et al.*, 2006; Agabiti-Rosei *et al.*, 2007; Mancia *et al.*, 2007a) agree that methods directed at the study of arterial stiffness and wave reflection must include measurement of arterial stiffness (especially in the proximal aorta) and of wave reflection, and must allow for effects of wave reflection on arterial pressure when measured in the brachial artery. Beneficial effects of drugs in heart failure and hypertension can only be applied fully when effects on wave reflection are taken into account. Clinical interest in wave reflection has re-awakened some of the old issues on the site or sites of wave reflection, their effects on the macrovasculature and microvasculature, their change with age, and the effects of drugs and of disease. These issues will be addressed in later chapters. To conclude this chapter, we will address some of the specific and contentious issues that clinicians are dealing with at the moment, and their possible explanation to help explain confusion on views, published over the years (Table 9.1), together with Harvey’s original description at the beginning of this chapter.

Most of the information available to date on wave reflection has come from detailed, directed, invasive studies by physiologists on animals, and from computer modeling of vascular beds. Clinical information on humans has come from use of less accurate, non-invasive techniques in normal and diseased subjects, and from invasive studies undertaken during the course of diagnostic cardiac catheterization. In the latter, operators were not always able to separate true pressure features from artifact caused by low manometric frequency response or from position of the catheter and movement of the catheter within the heart or arteries as a consequence of the heart’s movement. Clinicians, in general, have not been as aware as physiologists of methodological limitations or artifact.

A specific issue has arisen in the last ten years over the contribution of aortic stiffness and wave reflection to changes in the pulse waveform with aging, as seen in the Framingham cohort. In the Framingham study, carotid-femoral PWV and carotid augmentation index have been measured, together with non-invasive estimations of central aortic impedance from carotid pressure and LV outflow tract flow. Over 150 events occurred in subjects with such measures and outcome associations have been reported (Mitchell *et al.*, 2010). There was no relationship

found between outcomes and measures of wave reflection. Mitchell’s study ran counter to those of Wang *et al.* (2010), Weber *et al.* (2010), Vlachopoulos *et al.* (Vlachopoulos *et al.*, 2010a; Vlachopoulos *et al.*, 2010b), and Benetos *et al.* (2010), all of whom showed ill effects of early wave reflection. Mitchell’s methods again came under criticism on the basis of inappropriate use of brachial artery tonometry (O’Rourke and Adji, 2010a) and detection of the inflection point. Prior to this, the Framingham investigators had reported on change in carotid-femoral PWV, carotid augmentation index, and aortic diameter with age (Mitchell *et al.*, 2004a; Ingelsson *et al.*, 2008), with contradictory results on aortic diameter and augmentation index. Preliminary results on aortic non-invasive impedance have also been reported, and from these, contributions of incident and reflected waves have been estimated (Mitchell *et al.*, 2007a; Mitchell *et al.*, 2008; Wang *et al.*, 2010). Mitchell’s most recent publication found no relationship of these indices with outcomes (Mitchell *et al.*, 2010).

Mitchell *et al.* (Mitchell *et al.*, 2004a; Mitchell *et al.*, 2008) have argued on the basis of their data, that stiffening of the proximal aorta (estimated as characteristic impedance in volume terms) is the principal cause of increase in aortic pulse pressure with age, especially over age 60, and that wave reflection, estimated as carotid augmentation index, remains constant or even decreases with age. Such a view contrasts with the traditional view that both aortic impedance and the amplitude of the primary wave generated by ventricular contraction increase with age, but that wave reflection is the dominant factor, and that which is most amenable to treatment (O’Rourke, 1976; O’Rourke and Nichols, 2004; Safar and O’Rourke, 2006; Cecelja *et al.*, 2009; Namasivayam *et al.*, 2009a; Namasivayam *et al.*, 2009b; Namasivayam *et al.*, 2010).

Mitchell’s view is based on relative flattening of carotid augmentation index with age > 60 years as being a specific measure of wave reflection intensity. Furthermore, he states that a marked age-related increase in aortic PWV in subjects > 60 years with little change in brachial PWV leads to impedance matching between central aorta and proximal muscular arteries, which reduces proximal wave reflection. These views are simplistic and inaccurate. As measured from normal population data (McEniery *et al.*, 2005a), both aortic pulse pressure and amplitude of this reflected wave rise linearly with age (see Fig. 9.18). Each, however, has a different intercept on the pressure axis. On this account, this ratio must be curvilinear (Namasivayam *et al.*, 2010). This ratio is expected to reach an asymptote below 50 percent; it does in the population data of McEniery *et al.* (2005a) and Mitchell *et al.* (2004a). If the ratio of reflected wave to pulse pressure (which is augmentation index) continued to rise in a linear fashion as it does below age 50, it would exceed 50 percent around 70 years of age. This is impossible (because the reflected wave cannot be greater than the incident wave). Namasivayam

et al. (2009a) have analyzed the McEniery data (which is similar to that of Framingham) using product of coefficient mediation analysis, and have shown that increase in the reflected wave dominates up to age 60, and its contribution is of similar magnitude to the incident (or forward) wave over age 60. Similar results were seen in a clinical cohort (Namasivayam *et al.*, 2009b). Mitchell's views are also based on the view that aortic diameter remains constant or decreases with age such that characteristic impedance (expressed by them in volumetric terms as $\text{dyne} \cdot \text{s}/\text{cm}^5$) increases to a greater degree than if the aorta dilated with age and impedance were expressed in velocity terms, which can be compared directly to PWV as $\text{dyne} \cdot \text{s}/\text{cm}^3$ through the Waterhammer formula. Subsequent Framingham data confirm other studies and show that the aortic diameter does indeed increase, not decrease, with age (Ingelsson *et al.*, 2008; Lam *et al.*, 2010).

Murgo *et al.* (1980a) estimated distance to the 'functionally discrete' reflecting site from the studies of ascending aortic pressure, flow, and impedance and placed this some 44 ± 2.0 cm from the aortic root. They obtained similar values from the inflections of the rising limb of the aortic pressure wave (48 ± 2.0 cm), and from the frequency modulus and phase of impedance. As stated above, the work of Murgo *et al.* (1980a) found that major reflection site distance was similar in all three of their study groups using both methods. Their study included only a limited number of patients ($n=18$) and only two of these were over 50 years of age. Results from Sugawara *et al.* (2010a; 2010b) and the data presented in Fig. 9.21 which was taken from the Anglo-Cardiff Collaborative Trial (ACCT) show little change in reflection site distance below 60 years of age in both males and females, but increases in parallel with PWV after age 60 years. The aging change conforms with the evidence of greater physical taper of the aorta with age (Gosling *et al.*, 1971; Greenwald *et al.*, 1990). It is difficult to explain Namasivayam *et al.*'s (2009a) results (increasing effects of wave reflection with age despite minor changes of PWV up to the age of 60) unless the reflected wave returns earlier to the heart and increases systolic pressure. But Mitchell *et al.* argue that a better 'match' of aortic with peripheral wave velocity and impedance will result in lower wave reflection. Because of the dominance of reflection at peripheral arterial/arteriolar sites, we believe that any such effect will be modest or offset by physical taper of the aorta. In equating carotid or aortic augmentation indices with wave reflection, Mitchell *et al.* assume that contour of the aortic flow wave remains constant with age. This has been questioned on the basis of Japanese studies that have shown reduction in late systolic flow in older people (Chapter 19) (Sakai *et al.*, 1993; Miyashita *et al.*, 1994). Adji *et al.* (2009) and Namasivayam *et al.* (2009a) in modeling studies have identified this as a contributing cause of decreasing aortic augmentation index with age.

Westerhof *et al.* (2008a; 2008b) have raised other qualifications in the estimation of reflecting site from an

inflection of the aortic pressure wave, questioning whether the phase of wave reflection in the lower thoracic aorta may not induce an apparent delay such that the reflected wave may appear to travel at a lower speed than the incident wave. Schillaci *et al.* (2009) questioned this as did we (see Fig. 10.9) (Nichols and O'Rourke, 2009a). All, however, have misgivings on the identification of a point on the aortic pressure wave that corresponds to the return of a reflected wave from the lower aorta, especially in older subjects. This is apparent as a change in direction of the pressure wave, such that it moves in the opposite direction to flow during systole (i.e. concavely to the left, Fig. 9.19) or onset of a positive wave during diastole when flow is zero. In older people, this may be hidden in the initial wave shoulder caused by peak flow velocity. It may be hidden in the peak of the aortic pressure wave, or in very young subjects (or in hypotension or with tachycardia) may be hidden in the incisura caused by aortic valve closure. In some experimental animals, the incisura may clearly be seen before onset of the reflected wave under control conditions (Fig. 10.6). In humans under basal normal conditions, the onset of the reflected wave usually precedes aortic wave closure (Fig. 10.4).

Westerhof's principal concern (Westerhof *et al.*, 2008a; Westerhof *et al.*, 2008b; Kips *et al.*, 2009) is on using the inflection point on the pressure wave to generate a reliable measure of PWV under different conditions and at different ages, and that which would eliminate need for the more accurate but more time-consuming measurement of PWV. Most would readily agree with this concern.

In human disease, wave reflections may have a marked effect on the ascending aortic pressure wave. This is most apparent in aortic coarctation (Chapter 24) or severe arteriosclerosis (Chapter 23). In aortic coarctation (see Chapter 24) (O'Rourke and Cartmill, 1971; O'Rourke, 2008), the reflecting site must be <20 cm from the aortic valve, so that with wave velocity in the aorta ~ 10 m/s, wave reflection must return to the heart <50 ms after aortic valve opening, and so prior to the peak of aortic flow, and to the usual initial pressure peak. Bilateral occlusion of the femoral arteries increases the amplitude of the late systolic peak (or reflected wave) of the ascending aortic pressure wave, but does not alter the early part (unaugmented pressure or shoulder) of the wave (Murgo *et al.*, 1980a).

MODIFICATION OF WAVE REFLECTION WITH DRUGS

In the early work conducted by Taylor and O'Rourke, reduction in wave reflection in a vascular bed through intra-arterial dilator or reactive hyperemia was always accompanied by marked reduction in peripheral resistance, with increase in flow (Figs 12.12, 12.13, and 12.14) (O'Rourke and Taylor, 1966). Reduction in wave reflection was easily explained by arteriolar vasodilation. It was

assumed that all reduction in wave reflection came from arteriolar vasodilatation, and effects of drugs were explained in this way. However, this view had to be changed when Yaginuma *et al.* (1986) showed that nitroglycerine could have a marked effect on wave reflection (measured from ascending aortic input impedance with ascending aortic pressure and flow waves at cardiac catheterization), at doses which had no effect on peripheral resistance. The change in wave reflection, however, was explicable on the known effects of the drug on muscular arteries – where dilation occurs with small doses of nitroglycerine (as in the coronary system) and to a greater degree in smaller than larger muscular arteries (Fig. 26.2) (Feldman *et al.*, 1981). This explanation is now generally accepted and has been explained as ‘trapping’ of wave reflection in the peripheral vasculature. The explanation brings to light the differential effects of vasodilator drugs on muscular arteries and of arterioles (discussed again in Chapter 26). However, it includes another paradox – that this effect appears to have little or no effect on the upper limb, where the primary wave of pressure retains its amplitude (Kelly *et al.*, 1989a; Simkus and Fitchett, 1990; Takazawa *et al.*, 1995b), and the transfer function for pressure between aorta and brachial/radial arteries does not change appreciably (Karamanoglu *et al.*, 1993). These issues, discussed further in Chapter 26, make it clear that description of a drug as a vasodilator must be clarified as predominantly venodilator, predominantly arteriolar dilator, or predominantly arterial dilator (O’Rourke *et al.*, 1993a).

The effect of nitroglycerine on wave reflection is explained on the basis of change in arterial caliber, reduced PWV, and later return of the reflected wave to the heart (Yaginuma *et al.*, 1986). Gosling *et al.* (1971) and Greenwald *et al.* (1990) have shown change in area ratio at the aortic bifurcation which is the opposite of that seen with nitroglycerine – i.e. an increase in luminal taper. If this were to occur progressively in the proximal aorta, and in other arteries, it could explain greater reflection at the proximal points, and movement of the summated reflecting site closer to the heart with increasing age. Such an explanation has been advanced to explain apparently greater reflection in women than men at the same age (Cecelja *et al.*, 2009), independent of smaller body size (Smulyan *et al.*, 1998). An opposite effect has been postulated by Mitchell *et al.* (2007b) and Sugawara *et al.* (2010a; 2010b) with functionally discrete reflecting site moving distally with increasing age (see above). Effects of

taper can explain differences in apparent reflecting site and of pressure wave augmentation between males and females (Cecelja *et al.*, 2009; O’Rourke and Hayward, 2009). This is discussed further in Chapter 19. A multi-center Japanese study has reported effects of anti-hypertensive agents on central and brachial pressure (Miyashita *et al.*, 2010) and showed similar beneficial effects of angiotensin-converting enzyme inhibitors (ACEI), angiotensin receptor blockers (ARB), and calcium channel blockers (CCB) on central pressure; this supports previous studies which also failed to show benefit of atenolol or diuretics on central pressure.

Mitchell *et al.* (Mitchell *et al.*, 2004a; Mitchell *et al.*, 2008) refer to arterial ‘remodeling’ with age, noting that progressive increase in aortic PWV with age and maintenance of high PWV in muscular arteries, results in an arterial stiffness gradient and favorable ‘matching’ in the elderly. They infer that such ‘matching’ reduces central wave reflection and is good. Taylor (Taylor, 1964; Taylor, 1966c; Taylor, 1967), however, had shown that mismatch of PWV along the aorta into muscular arteries was a feature of younger animals, and even seen in the sheep fetus. Learoyd and Taylor (1966) showed that this gradient of elasticity (or stiffness), measured as PWV, disappeared or reversed in older humans; this was confirmed by Avolio *et al.* (Avolio *et al.*, 1983; Avolio *et al.*, 1985a), and in all human aging studies since (McEniery *et al.*, 2007). Taylor (Taylor, 1964; Taylor, 1965; Taylor, 1966a) (see Chapter 14) saw this ‘non-uniform elasticity’ from central to peripheral arteries of younger animals and humans to be a favorable factor, since it provided the LV with a low proximal aortic characteristic impedance, while restricting volume changes of peripheral arteries during exercise, changes in arterial pressures, etc. He named ‘non-uniform arterial elasticity’ as one of the factors responsible for arterial efficiency, along with low arterial resistance, and with dispersed wave reflections returning to the heart after LV ejection had ceased, but at a time when they could aid perfusion of coronary arteries (Taylor, 1967). Mitchell did not reference Taylor, nor deal with his data when presenting an opposing viewpoint. This is still promoted by Mitchell *et al.* (Mitchell, 2009a; Mitchell, 2009b; Mitchell *et al.*, 2010). In line with Taylor, we consider that aging changes represent degeneration, not ‘remodeling’ – a euphemistic term which has little place in science, although encouraged by advertising in contemporary journals (O’Rourke and Hashimoto, 2007; Hashimoto and O’Rourke, 2008; O’Rourke and Hashimoto, 2008).

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Contours of pressure and flow waves in arteries

The ascending aorta	226	Pressure and flow waves in human upper body arteries	239
The aortic arch and its branches to head, neck, and upper limbs	227	Relationship between pressures in the brachial artery and ascending aorta of humans	243
The thoracic aorta, abdominal aorta, and lower limb vessels	229	The pulse in disease	246
Interventions	233	Arterial stenosis	248
Explanation of pressure and flow contours in major systemic arteries	239	Non-invasive pressure recordings: clinical implications	249

The first edition of *Blood Flow in Arteries* contained separate chapters on arterial flow waves and arterial pressure waves. These were excluded from the second edition on the basis that there was general agreement on the contour of, and mechanisms responsible for, pressure and flow waves in different arteries of experimental animals. This is basically true. However, there has been, over the past 40 years, increasing interest in the contour of pressure and flow waves in human arteries under different conditions, and thus more need to know what is responsible for determining the contour of the pulse in central arteries and its modification in transit to peripheral vessels (Fig. 10.1) (Oparil and Izzo, 2006). Over this period, it has become routine practice to monitor the arterial pressure pulse contour in operating rooms and in the intensive care units, as well as to record it in the cardiovascular diagnostic laboratory. New sphygmographs, utilizing high-fidelity pressure sensors and the principle of applanation tonometry, are now available commercially (see Chapters 26 and 30). These permit accurate registration of arterial pressure waves in central (carotid) and peripheral arteries of humans (see Chapter 9) (Avolio *et al.*, 1988; Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; Kelly *et al.*, 1990a; Kemmotsu *et al.*, 1991a; Kemmotsu *et al.*, 1991b; O'Rourke *et al.*, 1992a; O'Rourke *et al.*, 1992b; O'Rourke *et al.*, 1992c; O'Rourke *et al.*, 1993a; Sato *et al.*, 1993; Chen *et al.*, 1996; Chen *et al.*, 1997; Wilkinson *et al.*, 1998; Fetis *et al.*, 1999; Siebenhofer *et al.*, 1999; Filipovsky *et al.*, 2000; Avolio *et al.*, 2001; Brooks *et al.*, 2001; Pauca *et al.*, 2001; Nichols and Singh, 2002; Jankowski *et al.*, 2004; Mitchell, 2009a; Mitchell, 2009b). Their use has led to re-exploration of the

relationship between central and peripheral pressure in humans and questioning of sphygmomanometric values of arterial pressure (Smulyan *et al.*, 2003; Smulyan *et al.*, 2008; Zuo *et al.*, 2010). With the application of ultrasound to diagnostic studies, flow pulses can be recorded non-invasively from the great vessels of the chest and abdomen and from peripheral arteries in humans, and even from the cerebral blood vessels of humans and the umbilical arteries of the fetus (Perlman *et al.*, 1983; Fleischer *et al.*, 1985; Trudinger *et al.*, 1985; Shimamoto and Shimamoto, 1995) (see Chapter 6). Disease is being diagnosed and the effects of therapy monitored and altered on the basis of interpretation of pressure and flow wave characteristics. This sixth edition of *Blood Flow in Arteries* cannot be a text for such interpretations, but it can provide a background for these. Accordingly, this chapter will describe normal features and basic mechanisms relevant to the pulse in experimental animals and humans, while Chapters 19, 20, 21, 22, 23, 24, 25, and 26 deal with effects of aging in humans, specific disease, and the influence of therapies. Chapter 30 deals specifically with non-invasive pressure recording and its value and potential in humans.

The arterial pulse has been recognized from antiquity as the most fundamental sign of life. Galen devoted a large part of his writing (17 books) to the pulse and to a complicated system of classification and nomenclature that still haunts clinical textbooks and plagues medical practitioners (Broadbent, 1890; Bedford, 1951; O'Rourke, 1971; Harris, 1973; O'Rourke *et al.*, 1992a; O'Rourke *et al.*, 1993a). Marey (Marey, 1860; Marey, 1863; see Snellen, 1980) was the first to obtain accurate non-invasive records of the arterial pressure pulse; his work

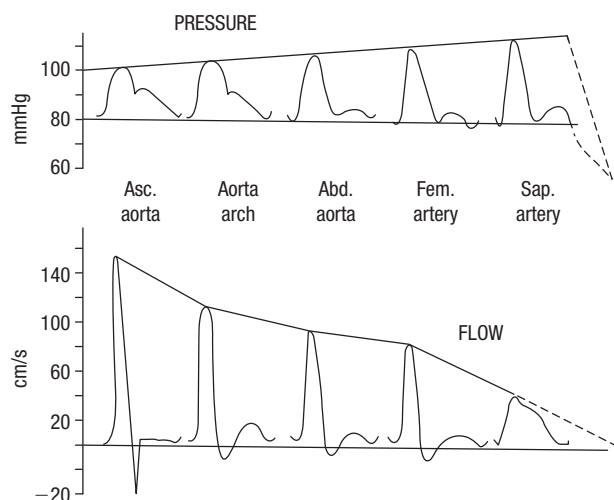


Figure 10.1 A diagrammatic comparison of the behavior of the arterial pressure and flow pulses in arteries as they travel away from the heart. Mean pressure falls slowly, but the pulsatile pressure variation increases (due to increase in systolic pressure) until, in the saphenous artery, it may double that at the root of the aorta. The flow oscillation, on the contrary, diminishes markedly. Such behavior can only be accounted for by the presence of a closed-end type of reflection in the small peripheral vessels. In the absence of reflections, damping would cause a parallel fall in pressure and flow oscillations. Ultimately, the pressure oscillations must also be damped out; the probable region of this fall is in the smallest arteries and proximal arterioles, as indicated by the broken lines. The increase in the ratio of the pulsatile pressure amplitude to that of the flow amplitude is largely determined by the increase in impedance of the low-frequency components; the change in shape of the pressure wave to that of the flow wave depends on the changes in impedance of the various frequency components in terms of their distance from the main reflection sites, since the impedance is at a minimum at one-quarter wavelength distance from these peripheral sites

and that of Mahomed (Mahomed, 1872; Mahomed, 1874), Broadbent (1890), Mackenzie (1902), and others (see O'Rourke, 1982b; O'Rourke *et al.*, 1992a) introduced a new rigor to clinical studies, a rigor that was blunted by uncritical acceptance of the sphygmomanometer in subsequent years. Otto Frank (Frank, 1899; Frank, 1905) introduced the first accurate manometer for registering intra-arterial blood pressure. This was subsequently refined by Hamilton *et al.* (1934) and Wiggers (1928) and then applied to cardiovascular diagnosis in humans when successful cardiac surgery provided the stimulus and need for accurate data in the 1940s (Salans *et al.*, 1951; Kroeker and Wood, 1955; Kroeker and Wood, 1956; Thomas, 1985). The current state-of-the-art instrument is the catheter or guidewire-mounted micromanometers (Chapter 6) (Murgo and Millar, 1972; Nichols and Walker, 1974; Jensen *et al.*, 2007), which can accurately record pressure (and sound) fluctuations up to 2.0 kHz. An early goal with manometers was to find ways of calculating flow

from the contour of pressure (Frank, 1930; Remington *et al.*, 1948; McDonald and Nichols, 1973), but this proved unreliable (Remington *et al.*, 1948; Brotmacher, 1957), and unnecessary too with the introduction of electromagnetic and ultrasonic flowmeters. Flow waves from humans, as well as from experimental animals, are now readily obtained using catheter-mounted electromagnetic (Murgo and Millar, 1972; Nichols *et al.*, 1977a; Nichols *et al.*, 1977b; Nichols *et al.*, 1980a; Nichols *et al.*, 1980b; Nichols *et al.*, 1981; Nichols *et al.*, 1983; Nichols *et al.*, 1985a; Nichols *et al.*, 1986a; Nichols *et al.*, 1986b; Kelly and Fitchett, 1992; Nichols *et al.*, 1992; Markus *et al.*, 1994) and ultrasonic transducers (Sibley *et al.*, 1986; Mills *et al.*, 1992; Kern *et al.*, 1993a; De Bruyne *et al.*, 1996) and from probes (electromagnetic and ultrasonic) placed externally on the vessel (Nichols *et al.*, 1985b; Nichols *et al.*, 1988a; Nichols *et al.*, 1988b; Nichols *et al.*, 1994a; Nichols *et al.*, 1994b; Heard *et al.*, 1996). These instruments, their uses and shortcomings are described in Chapters 6 and 8. Flow waveforms can also be measured non-invasively by magnetic resonance imaging (see Fig. 19.17) (Redheuil *et al.*, 2010).

THE ASCENDING AORTA

The pattern of left ventricular ejection determines the contour of the flow wave in the ascending aorta. The pattern of ejection depends on ventricular filling (pre-load), ventricular properties, and their modification by the physical properties and wave reflection characteristics of the arterial tree (afterload). Under normal conditions, the ascending aortic flow waves are surprisingly similar in a wide range of animals, with peak ejection velocity around 70–100 cm/s achieved in the first third of systole, and with ejection comprising some 30–40 percent of the cycle length irrespective of body size (Fig. 10.2). Changes in heart rate affect diastole more than systole, the duration of ejection increasing relative to the diastolic period when the heart rate quickens and decreasing when the heart rate slows. Changes in flow contour with changes in heart rate and wave reflection amplitude are accompanied by relative changes in amplitude of harmonic components, with the interesting consequence that flow components are maintained within the same frequency band irrespective of heart rate (see Fig. 13.3).

While (with allowance for 'scaling' and differences in body size and resting heart rate) ascending aortic flow waves are similar in different animals, two different patterns are seen in ascending aortic pressure (Fig. 10.2). In the first, typical of dogs, sheep, rabbits, kangaroos, ferrets, and human adolescents, the pressure wave has a rounded systolic peak while diastole shows a secondary positive wave after the incisura caused by aortic valve closure. In the second, typical of guinea pigs and older adult horses and humans, aortic pressure rises to a late systolic peak and pressure falls after the incisura in an almost

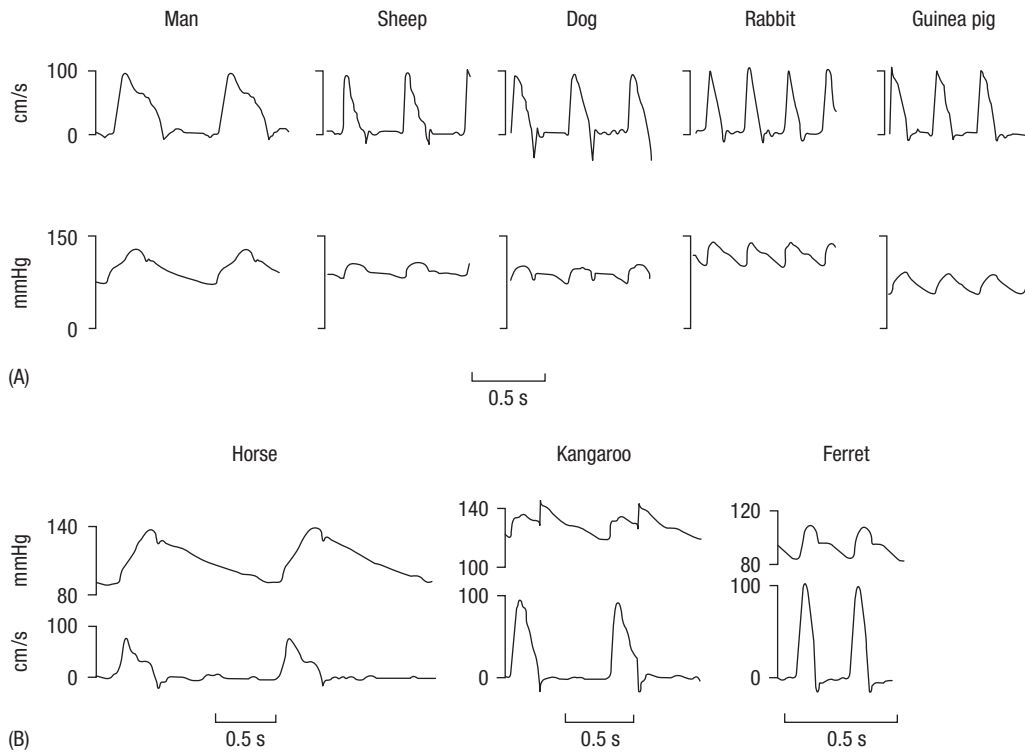


Figure 10.2 (A) Ascending aortic flow (above) and pressure (below) waves in a human and in a variety of animals. Reproduced from O'Rourke (1982b). (B) Ascending aortic pressure (above) and flow (below) waves in a horse, kangaroo, and ferret

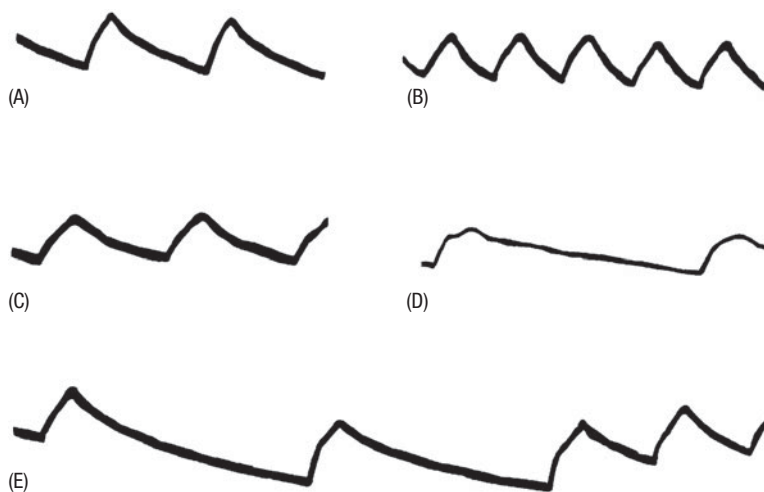


Figure 10.3 Proximal arterial pressure waves in birds and small mammals: (A) starling (blood pressure 210/160 mm Hg, heart rate 375/min); (B) mouse (160/120, 590); (C) rat (180/140, 430); (D) pigeon (140/100, 200); (E) mouse with trachea occluded (150/100, heart irregular). Reproduced from Woodbury and Hamilton (1937)

exponential fashion without any secondary diastolic wave. This pattern is seen in cold-blooded animals (turtles, frogs, and snakes), in birds such as turkeys (McLachlan, 1962; Taylor, 1965; Avolio *et al.*, 1982) and emus (Avolio *et al.*, unpublished data), and in some small animals such as rats, mice, and birds (Fig. 10.3) (Woodbury and Hamilton, 1937).

The shape of the arterial pressure wave is dependent on a number of factors, including duration of systole, mean arterial pressure, vasomotor tone, pulse wave velocity, and wave reflections. In humans, during the Valsalva maneuver, the second pattern is observed in the beats with longer duration of ejection, and the first when ejection is

short and arterial pressure is low (Fig. 10.4) (Mills *et al.*, 1970; O'Rourke, 1971; Murgu *et al.*, 1981a). An explanation of these patterns, and of their changes, is available on the basis of the timing of echoes (reflected waves) from peripheral arterial terminations (see below and Chapter 9).

THE AORTIC ARCH AND ITS BRANCHES TO HEAD, NECK, AND UPPER LIMBS

The pressure wave shows little or no perceptible change in transmission to the upper descending thoracic aorta and

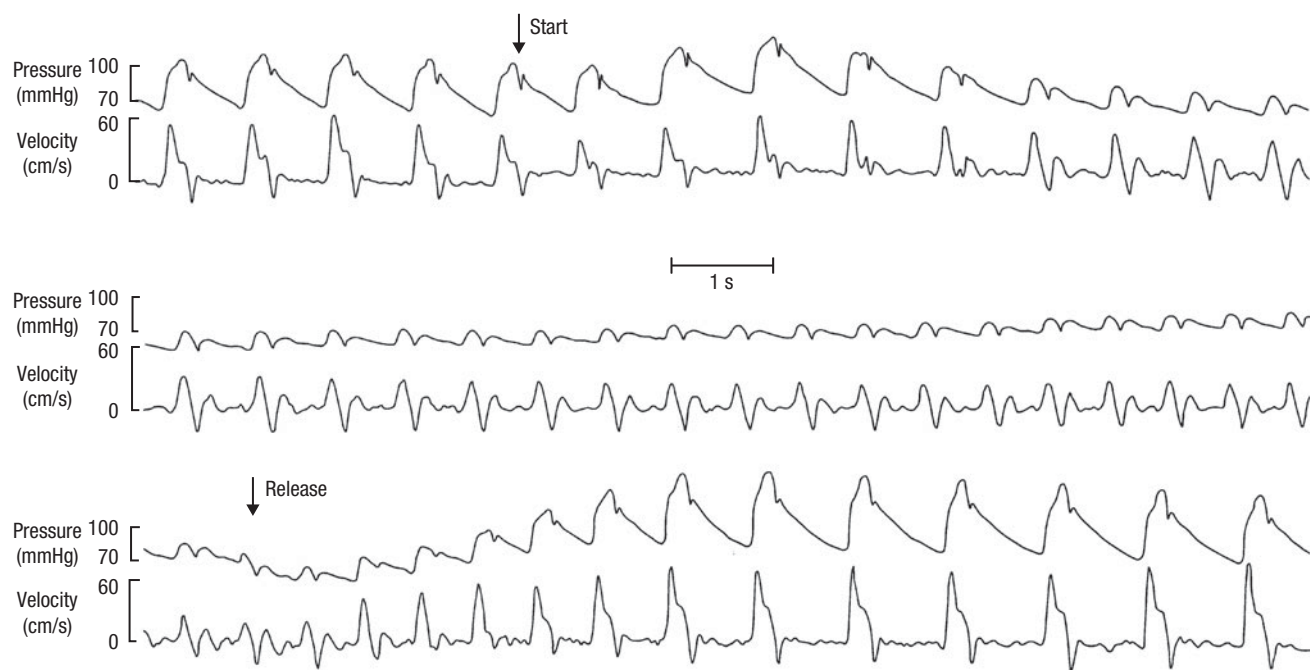


Figure 10.4 Pressure (above) and flow (below) waves in the brachiocephalic artery of a human during the Valsalva maneuver. Changes in pressure and flow waves can be explained on the basis of amplitude and timing of wave reflection in relation to ventricular ejection. Reproduced from Mills *et al.* (1970)

to the orifices of the brachiocephalic and subclavian arteries. Patterns of pressure are virtually identical in the proximal part of all these vessels, but the patterns of flow are quite different, and different from that in the ascending aorta (Fig. 10.5) (Attinger *et al.*, 1966; O'Rourke, 1967a; Avolio *et al.*, 1976; Hamilton, 1944). In the ascending aorta, there is forward flow only in systole, with peak flow in the first third of systole and no flow in diastole. In the upper descending thoracic aorta, the peak of the flow wave is more rounded, often with a second late systolic rise, the incisura is replaced by a wider period of back flow and there is a secondary positive flow wave in diastole (Fig. 10.5). This pattern is most apparent in dogs, sheep, and rabbits, that regularly show a secondary diastolic pressure wave. In the brachiocephalic and subclavian arteries, the period of forward flow is abbreviated, with back flow often preceding the incisura and with a prominent early diastolic secondary flow wave that corresponds to back flow in the descending aorta (Fig. 10.5). This pattern is typical of dogs, sheep, and rabbits, but it is seen in some humans as well, particularly in young adults (Murgo *et al.*, 1980a).

When flow waves are recorded simultaneously, or lined up as though recorded simultaneously (Fig. 10.5), it is apparent that there are reciprocal fluctuations of flow in late systole and in diastole between the upper and lower parts of the body. Quite extensive studies of this phenomenon have been conducted in humans by Mills *et al.* (1970) and Murgo *et al.* (Murgo *et al.*, 1980a; Murgo *et al.*, 1981a; Murgo *et al.*, 1981b). It has been shown that the different patterns of brachiocephalic flow can, like aortic

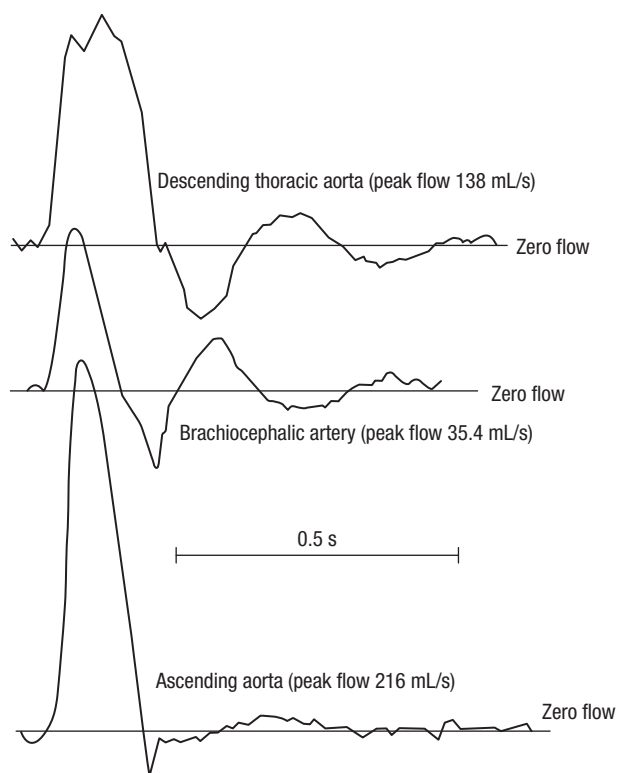


Figure 10.5 Flow waves in the descending thoracic aorta, brachiocephalic artery and ascending aorta of a dog. Records were taken with a single-channel flowmeter at intervals of approximately 15 s, then drawn from the digital voltmeter output and displayed as though recorded simultaneously. Reproduced from O'Rourke (1967a)

pressure, be explained on the basis of duration of ejection and the timing of wave reflection primarily from peripheral sites in the lower parts of the body (O'Rourke and Avolio, 1980; O'Rourke, 1982a; O'Rourke, 1982b; Burattini *et al.*, 1991).

Secondary flow fluctuations are more apparent in the brachiocephalic artery than in the descending thoracic aorta, partly because the former has a narrower caliber. The relationship between the brachiocephalic flow contour and the contour of the arterial pressure wave in humans has been examined by O'Rourke and Avolio (1980) and by Hirata *et al.* (Hirata *et al.*, 2006; Hirata *et al.*, 2008a). When the diastolic pressure wave was prominent, there was a prominent diastolic flow wave as well. When the pressure wave showed no diastolic wave, the flow wave showed a late systolic shoulder. Differences in the flow waves and in the pressure waves were studied in humans and in a model of the systemic arterial tree, and were explained by O'Rourke and Avolio (1980) on the basis of differences in the timing of wave reflection from the lower part of the body relative to the upper. These studies on arterial models have been verified and extended by Karamanoglu *et al.* (Karamanoglu *et al.*, 1994; Karamanoglu *et al.*, 1995). Implications to cerebrovascular disease are discussed in Chapter 18.

THE THORACIC AORTA, ABDOMINAL AORTA, AND LOWER LIMB VESSELS

Pressure

Most of the studies on pressure wave transmission have concentrated on changes along the aorta and into the lower limb arteries. Frank (Frank, 1899; Frank, 1905) had stressed the amplification of the systolic wave and exaggeration of the diastolic wave, and likened the arterial system to an underdamped manometer (Wiggers, 1928). Hamilton and Dow (1939) noted that, in the abdominal aorta and lower limb arteries, the foot of the wave is progressively delayed, whereas the peak of the pressure wave is almost synchronous in time, while diastolic pressure fluctuations are reciprocal (i.e. occur 180° out of phase) with those in the proximal aorta. Hamilton and Dow introduced the concept of 'standing waves' in the lower aorta, with reflection between lower-limb arteries (Hamilton, 1944; Remington and Wood, 1956) and the closed aortic valve, creating a node of pressure in the lower thoracic aorta and resonance of the whole arterial system. This concept was challenged by McDonald in earlier editions of this book and by Taylor (McDonald and Taylor, 1959) on the basis that wave reflection needs to be total and attenuation absent for resonance to occur. While this argument is, strictly speaking, true, most now agree that wave reflection is stronger and attenuation weaker than believed at the time. This old objection must be revised. It is impossible to explain data on pressure wave

transmission along the aorta without admitting wave reflection and conceding that a type of damped resonance does indeed occur (see Chapter 9).

Figure 10.6 shows pressure waves recorded at 5.0 cm intervals between the ascending aorta and iliac artery of a wombat – an Australian marsupial similar in size to a dog, but having a relatively long body and short, stubby limbs. In these animals, changes in amplitude and contour of pressure waves were more pronounced than those usually seen in dogs (O'Rourke, 1967a). Such differences were attributed to body shape, wombats having less spatial dispersion of peripheral reflecting sites and thus more obvious reflection phenomena. The influence of wave reflection on wave contour was inferred from the study of individual harmonic components of the pressure waves recorded at the different sites (Fig. 10.7). There is a progressive increase in amplitude of all harmonics between the central and peripheral arteries, as would be expected in the presence of a 'closed-end' reflecting site (see Chapter 10) and fluctuations of modulus of a type that would be expected in the presence of such a site. These data, while supporting the concept of reflection and damped resonance, confirm one of McDonald's objections – that there is no anatomic location of a pressure 'node', since the nodes of different harmonics of the wave contour occur at different positions along the aorta.

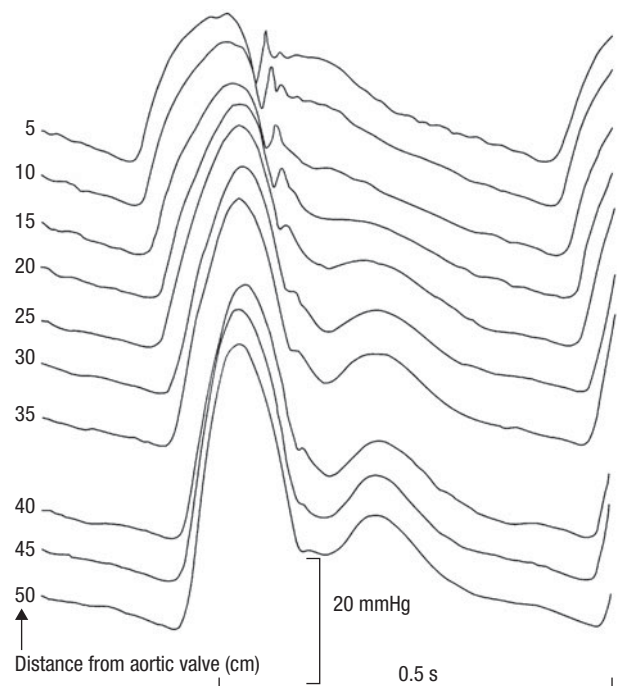


Figure 10.6 Pressure waves recorded sequentially at 5.0 cm intervals between the aortic arch (5.0 cm from the aortic valve) and the internal iliac artery (50 cm from the aortic valve) in a 16.5 kg wombat through a catheter inserted in the femoral artery. Note that the incisura caused by aortic valve closure can be seen in all tracings, separate from the foot of the diastolic reflected wave, with the two coinciding only in the ascending aorta. Reproduced from O'Rourke (1967a)

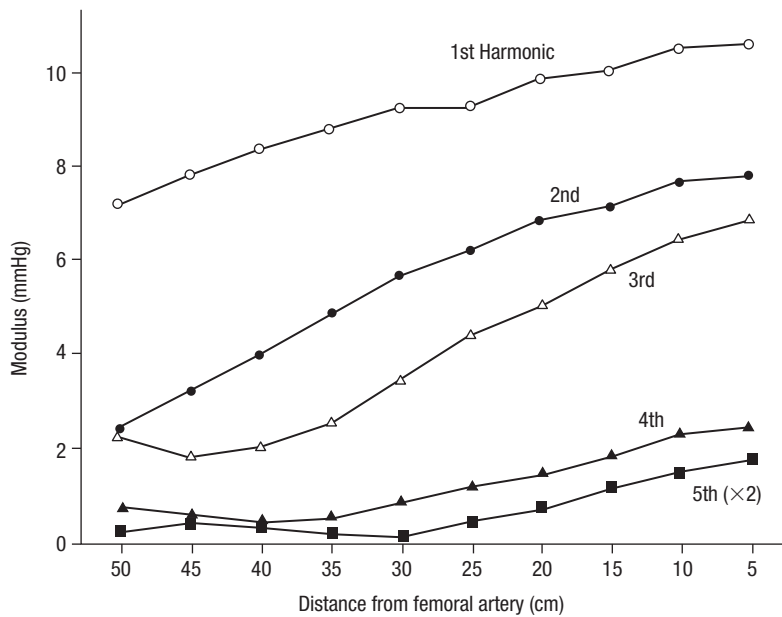


Figure 10.7 Amplitude of the first five harmonics of pressure waves (from Fig. 9.6) recorded between the aortic arch and iliac artery of a wombat. Distance recorded is from the catheter entry site into the femoral artery. Reproduced from O'Rourke (1967a)

Changes in pressure waves similar to those in Fig. 10.6 above have been noted in rabbits, sheep, and young humans, as well as in dogs, wombats, and kangaroos (Rollo, 1963; Taylor, 1965; O'Rourke *et al.*, 1968; Avolio *et al.*, 1984, 1985b; Nichols *et al.*, 1986a). They are particularly prominent in kangaroos, whose proximal aortic diastolic pressure wave is so large that peak pressure in diastole may exceed peak systolic pressure (Figs 10.2 and 10.8). These changes, however, have not been noted in the animals and adult humans who show the second type of pressure pattern noted in the ascending aorta: a late systolic peak and near exponential decay of pressure during diastole. When the ascending aorta evidences this type of pattern, pressure waves in other arteries showed the same or similar amplitude and contour, with little or no amplification of the wave, no diastolic wave and near exponential decay of pressure throughout diastole (see Fig. 4.21). Intermediate patterns are of course seen and are the rule in normal human adults and in baboons (Latham *et al.*, 1987; Latham *et al.*, 1988a). Changes in pressure wave propagation in humans have attracted special interest, as by Latham *et al.* (1985), who measured simultaneous pressure waves at six sites along the aorta with micromanometers (see Fig. 4.23). Older children and young adults (O'Rourke *et al.*, 1968; O'Rourke *et al.*, 1992a) show the most prominent reflection phenomena (i.e. change in pressure contour with transmission), while elderly hypertensive subjects show the least (see Fig. 4.21) (Nichols *et al.*, 1993). These findings were initially explained on the basis of differences in peripheral reflection coefficient (O'Rourke *et al.*, 1968). It now seems clear that this is not always the case (O'Rourke, 1971; O'Rourke and Yaginuma, 1984; O'Rourke *et al.*, 1992a; Nichols *et al.*, 1993; Wilkinson *et al.*, 2001). It is possible to explain the observed findings on the basis of differences in pulse wave velocity and wave reflection (Fig. 10.9). In the case of

rabbits, dogs, and children, the initial impulse generated by the heart travels back and forth over the arterial system with retention of a distinct identity. This is made possible by a combination of the relatively short duration of ejection, relatively long body length, and relatively slow pulse wave velocity. With aging in humans, body length remains relatively constant (from adolescence) and duration of left ventricular ejection likewise, while pulse wave velocity increases markedly in elastic arteries (Avolio *et al.*, 1983). This single factor accounts for the change in arterial pressure wave contour and pressure wave transmission with aging in humans (O'Rourke and Yaginuma, 1984; Kelly *et al.*, 1989a; Nichols *et al.*, 1993). When, in adult humans, pulse wave velocity is reduced either by hypotension or the Valsalva maneuver, or when duration of systolic ejection is reduced as with an extra systole or with tachycardia, a diastolic wave reappears in the arterial pulse (Figs 10.4 and 10.10) (Lewis, 1906; Frank, 1927; Kroeker and Wood, 1956; Mills *et al.*, 1970; Murgo *et al.*, 1981a; Denardo *et al.*, 2010). The most prominent diastolic wave and the most marked peripheral amplification of the pulse in man is seen when these two features are combined, as in low output heart failure and shock (Fig. 10.10) (Ewy *et al.*, 1969; Meadows *et al.*, 1971; O'Rourke, 1971; Orchard and Craigie, 1980; O'Rourke, 1982a; O'Rourke, 1982b; O'Rourke *et al.*, 1992a; Denardo *et al.*, 2010).

In other animals where discrete wave reflection phenomena are not apparent – turkeys, emus, guinea pigs, rats, and other small mammals and birds, and in frogs and turtles – a similar explanation can be offered. This is that a mix of relatively short body length, and/or relatively long duration of ejection, and/or relatively fast wave velocity results in reflected waves being merged with the incident wave so that they lose their discrete identity (type D beat, see Chapter 9).

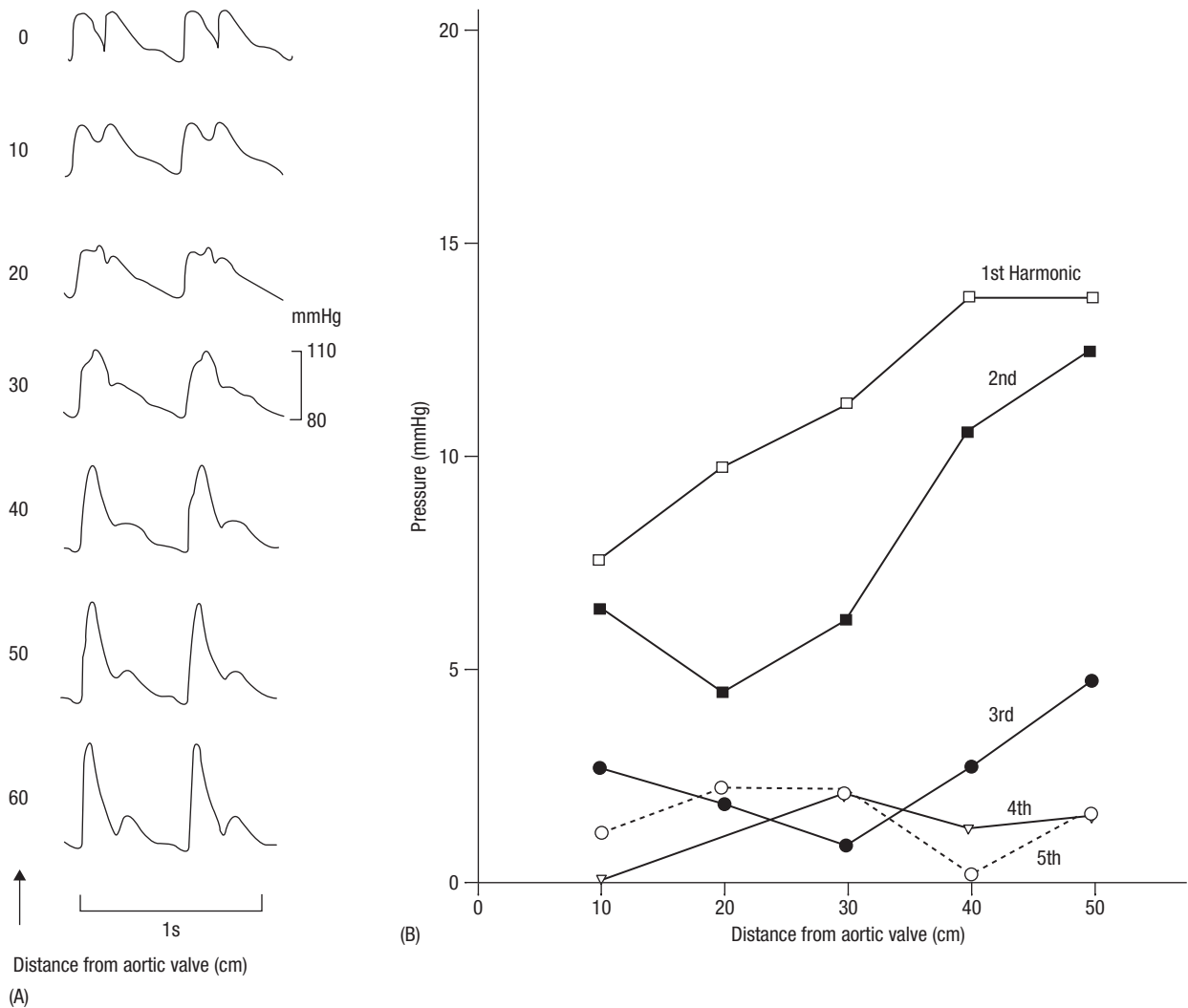


Figure 10.8 (A) Pressure waves recorded between the aortic root and iliac artery of a 22 kg kangaroo. (B) Change in amplitude of the first five harmonics of the pressure waves in the kangaroo between aortic arch and the iliac artery. Reproduced from Avolio *et al.* (1985b)

It is easy to underestimate the value of studies in comparative physiology as pointless and meddlesome with nature. Harvey's studies on the circulation of blood depended on such experiments and were vigorously defended by him (see Fig. 1.1). In the present study, such investigations have helped to provide general explanations of complex phenomena on the basis of body size and shape, as these affect vascular lengths and branching patterns, even though they do raise other queries that have not yet been settled. They provide a sound basis for understanding the effects of disease in humans, as this affects arterial caliber, length, and stiffness.

One example of such benefit is the information that has come from the study of pressure wave transmission in snakes. The long tapering body of these animals leads to wide physical distribution of individual peripheral reflecting sites, so that there is no single functionally discrete site as in other animals; they represent the opposite extreme to wombats and kangaroos (Avolio

et al., 1982; Avolio *et al.*, 1984; Avolio *et al.*, 1985b; Nichols *et al.*, 1986a). However, studies of snakes have also helped to explain the lack of appreciable wave attenuation over long lengths of artery. The diamond python has a long, narrow aorta of approximately 2.0 mm diameter; here, there is virtually no change in pressure wave amplitude over 50 cm or more (Avolio *et al.*, 1982). This may be due in part to non-uniform elasticity of the aorta, with distal stiffening tending to increase peripheral pressure pulsation. These findings help to illustrate how attenuation has been overemphasized in the past. Recent calculations (Nichols, unpublished data) suggest that the earlier estimates of McDonald (1974) were far too high. This subject is discussed further in Chapter 9.

Explanations of pressure wave contour, given above, imply not only that the pressure wave is reflected at peripheral sites and can retain a distinct identity but also that this wave can retrace the arterial system and be

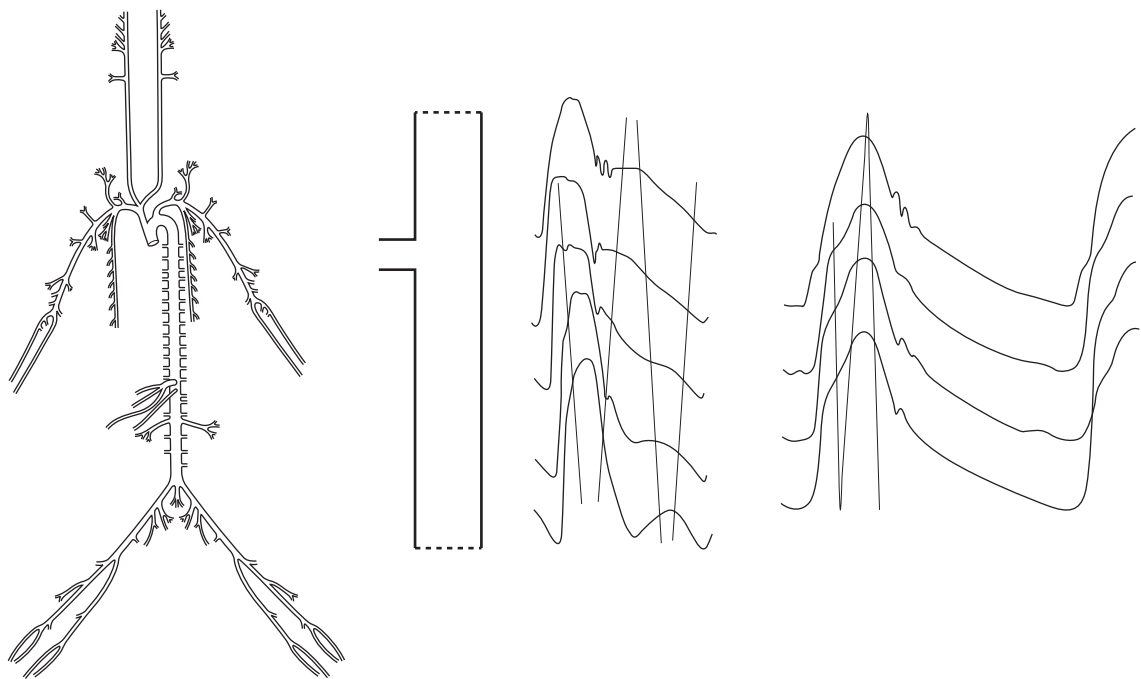


Figure 10.9 Explanation of pressure wave amplitude and contour between the central aorta and peripheral arteries on the basis of wave travel and reflection. The systemic arterial system on the extreme left is likened to an asymmetric T-tube, with the short limb representing all arteries in the upper part of the body and the long limb representing the descending aorta and all arteries in the lower part of the body. The ends of the T-tube represent the resultant of all individual arterial terminations and reflecting sites in the upper (top) and lower (bottom) parts of the body, respectively. Pressure waves in different arteries of a dog (second from right) and in a patient with arteriosclerosis (far right, half amplitude) can be explained on the basis of reflection and re-reflection of the pulse and on the timing of wave reflection with respect to ventricular ejection. Reproduced from O'Rourke and Yaginuma (1984)

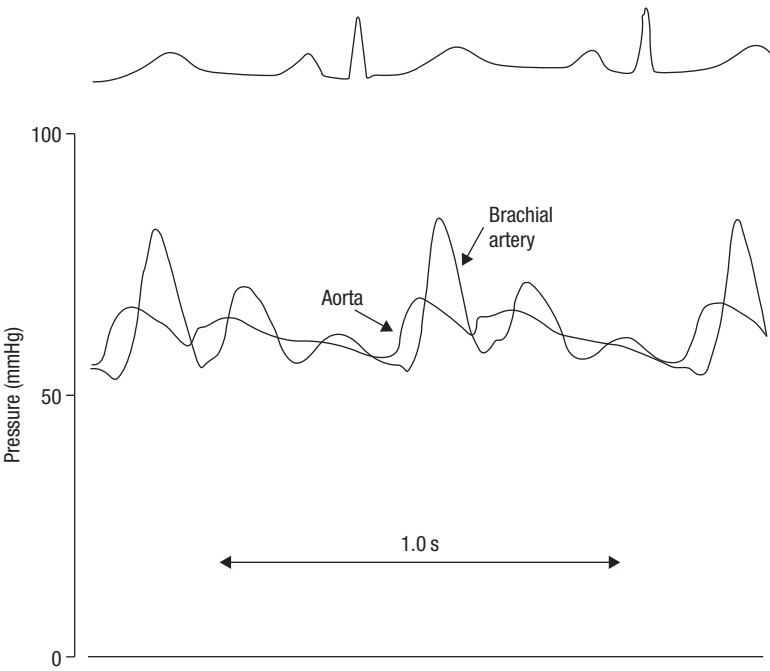


Figure 10.10 Pressure waves recorded simultaneously in the aortic arch (lower amplitude wave) and brachial artery (higher amplitude wave) of a human with hypotension and clinical features of peripheral vasoconstriction. Reproduced from O'Rourke (1970b)

reflected again and again. Earlier (McDonald and Taylor, 1959), this possibility was rejected on the basis that it requires a higher degree of reflection and lower degree of attenuation than appeared credible. Other studies (see Chapter 9) have provided such evidence and now permit

the explanation to be promoted. It seems, however, that this was inevitable; how else can one explain a second and sometimes a third diastolic wave when the heart contracts but once and the arteries do not contract rhythmically at all (Fig. 10.10)?

Flow

There is now general agreement on the contour of flow waves in systemic arteries (Wetterer, 1954; O'Rourke, 1965a; Attinger *et al.*, 1967; O'Rourke, 1967a; Mills *et al.*, 1970; Kenner, 1972; Busse *et al.*, 1975; O'Rourke, 1982b). In dogs, and in mammals of similar size, diastolic flow waves are usually apparent in the thoracic and abdominal aorta, and in the iliac and femoral arteries (Figs 10.1 and 10.5). The diastolic flow wave is associated with back flow in these vessels and is related to the diastolic pressure wave caused by the same mechanism. Back flow is usually more apparent below rather than above major branches of the abdominal aorta, suggesting that diastolic blood flow into these vessels is a contributing factor. Such back flow is reduced when these vessels are clamped and reappears when they are unclamped. This point may be of relevance to the predisposition of the infrarenal aorta to atherosclerosis and aneurysm formation when diastolic flow reversal is highest (see Chapters 23 and 24). In small peripheral arteries, such as the saphenous (Fig. 10.1), there is no back flow and the flow fluctuations are very small; this can be explained on the basis of proximity to peripheral reflecting sites and high impedance at the frequency of the lower harmonics of the wave. In the splanchnic branches of the abdominal aorta – renal, superior, and inferior mesenteric – there is no back flow and the flow contour is similar to that of pressure; this has been attributed to low peripheral resistance and hence to a lowish reflection coefficient in vascular beds supplied by these arteries. When vasodilatation is produced pharmacologically in a peripheral vascular bed, the flow wave comes to resemble the wave of pressure (see Fig. 12.14). This is explicable on the basis of a decrease in peripheral resistance and in peripheral wave reflection (see Chapter 9).

When, as in adult humans, there is no diastolic pressure wave of consequence in the peripheral pressure pulse, there is no diastolic wave either in aortic or femoral flow and the lower body flow waves resemble those in the ascending aorta, with forward flow in systole and little or no flow in diastole (Fig. 10.11). Diastolic flow persists, however, in such vessels as the renal artery, while late systolic fluctuations in the brachiocephalic artery pay testimony to the fact that wave reflection is still present but occurring unusually early (Fig. 10.11) from blood vessels in the arms, neck, and head, including the brain.

INTERVENTIONS

Pressure and flow waves have been recorded during a variety of interventions in humans and in experimental animals. Changes in wave contour can be explained on the basis of concepts already presented, aided by a consideration of vascular impedance (see Chapter 12), which

provides quantitative information on the properties of the vascular bed downstream. The major factors contributing to a change in wave contour are cardiac (pattern and duration of ventricular ejection) and vascular (velocity of wave travel in large arteries, and degree of vasomotor tone and of wave reflection in peripheral beds) (O'Rourke, 2000b; London *et al.*, 2001; Nichols and Edwards, 2001) (see Chapter 9). In the interventions described below, more than one of these changes was occurring at the time of study, so that interpretations had to be based on efforts to control the major variables while altering one at a time. This subject arises again in Chapters 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, and 30 where the effects of disease and of therapies are discussed. With respect to vascular factors and measurements taken from the upper limb, it is now clear that one must consider separately the different effects of wave travel and reflection in upper and lower body vessels.

Vasodilatation

The effects of arteriolar vasodilatation in a vascular bed on pressure and flow wave contours in arteries of supply have been studied by intra-arterial injection of nitroglycerin or acetylcholine (Pritchard *et al.*, 1943; Hamilton, 1944; O'Rourke and Taylor, 1966), or during reactive hyperemia (O'Rourke, 1965a; Rowell *et al.*, 1968). When dilatation is confined to a single vascular bed (e.g. the femoral), changes in mean pressure and arterial pulse wave velocity are minimal. After intra-arterial injections, records need to be taken before recirculation and systemic effects have become apparent.

The effects of arteriolar vasodilatation are most obvious in vascular beds with a high resting vasoconstrictor tone, and comprise a rise in mean flow, a disappearance of back flow and a change of inflow wave (Fig. 12.14). The pressure wave shows less diastolic fluctuation.

These changes are readily explained from impedance spectral analysis (see Figs 12.12 and 12.13) as resulting from decreased peripheral wave reflection. Rowell *et al.* (1968) noted that amplification of the arterial pulse between the aorta and brachial artery during leg exercise could be eliminated by reactive hyperemia in the arm. This finding supports the concept that peripheral amplification of the pulse is caused by peripheral vasoconstriction and high wave reflection. Kroeker and Wood (1955) found that amplification of the pulse between the aorta and femoral artery was markedly reduced during leg exercise; this finding is easily explicable on the basis of decreased arteriolar tone and decreased reflection in the femoral bed. Similar findings have been reported for the upper limb (Karamanoglu, 1992; Gallagher, 1993; Gallagher *et al.*, 2004), during reactive hyperemia, and explained on the basis of decrease in wave reflection (Karamanoglu *et al.*, 1995).

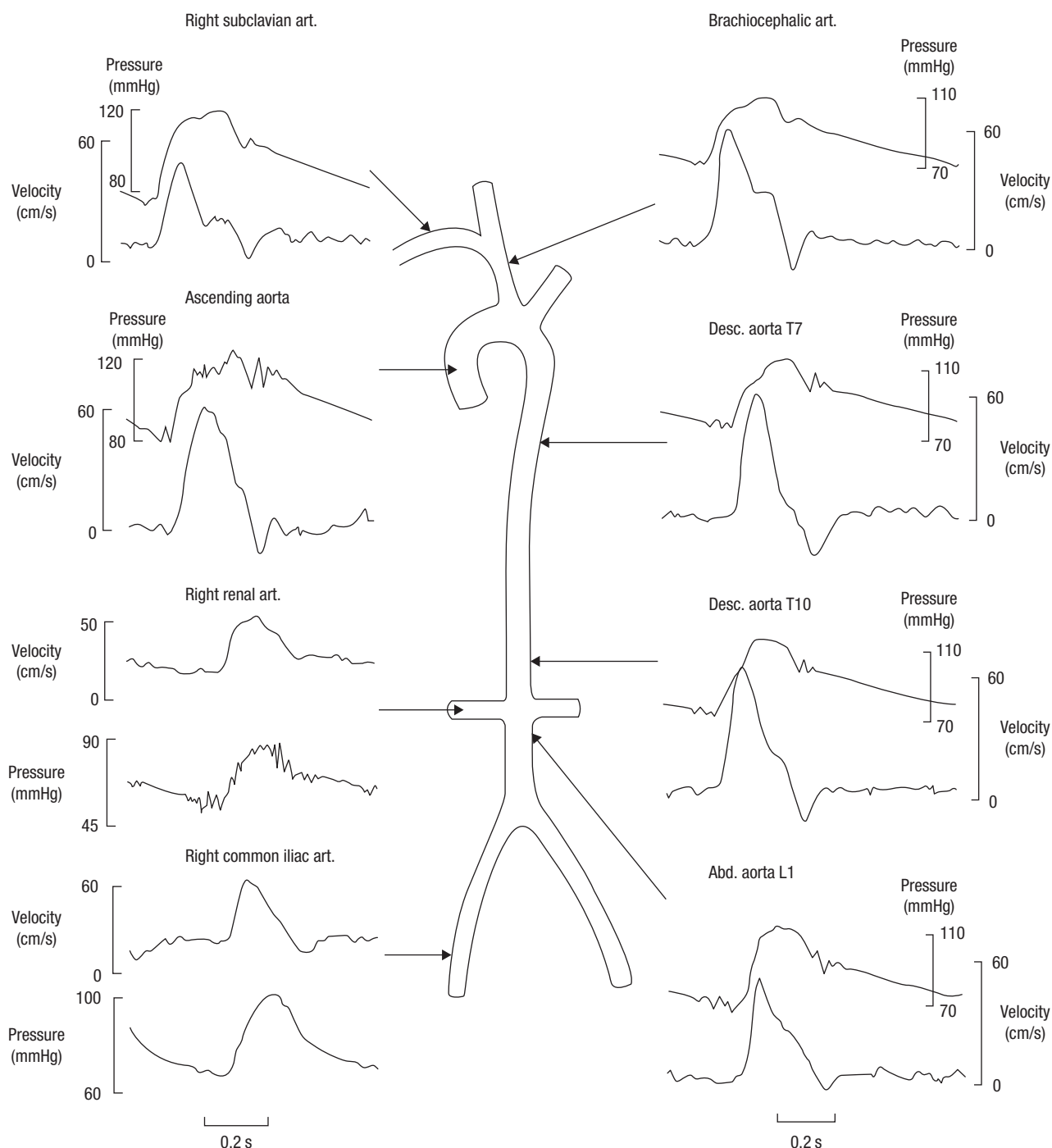


Figure 10.11 Pressure and velocity waveforms in different arteries recorded in a human undergoing diagnostic cardiac catheterization with an electromagnetic catheter transducer with a fluid-filled lumen for pressure measurement. Reproduced from Mills *et al.* (1970)

Vasoconstriction

Localized vasoconstriction in a vascular bed has been studied by intra-arterial infusion of epinephrine or norepinephrine (Fig. 10.12) (Pritchard *et al.*, 1943; O'Rourke and Taylor, 1966).

Changes are most marked in vascular beds with low resting vasomotor tone (such as the renal), where pressure and flow waves (which normally resemble those recorded

in the femoral artery after intra-arterial acetylcholine) come to resemble those seen in normal circumstances in the femoral artery (Fig. 12.14). In the femoral artery, intra-arterial norepinephrine has little effect on pressure or flow wave contours, but markedly reduces mean flow so that the back flow is more pronounced than under control conditions. Changes are explicable from studies of vascular impedance (see Chapter 12). The relatively minor effects on the femoral vascular bed are explicable on the basis that

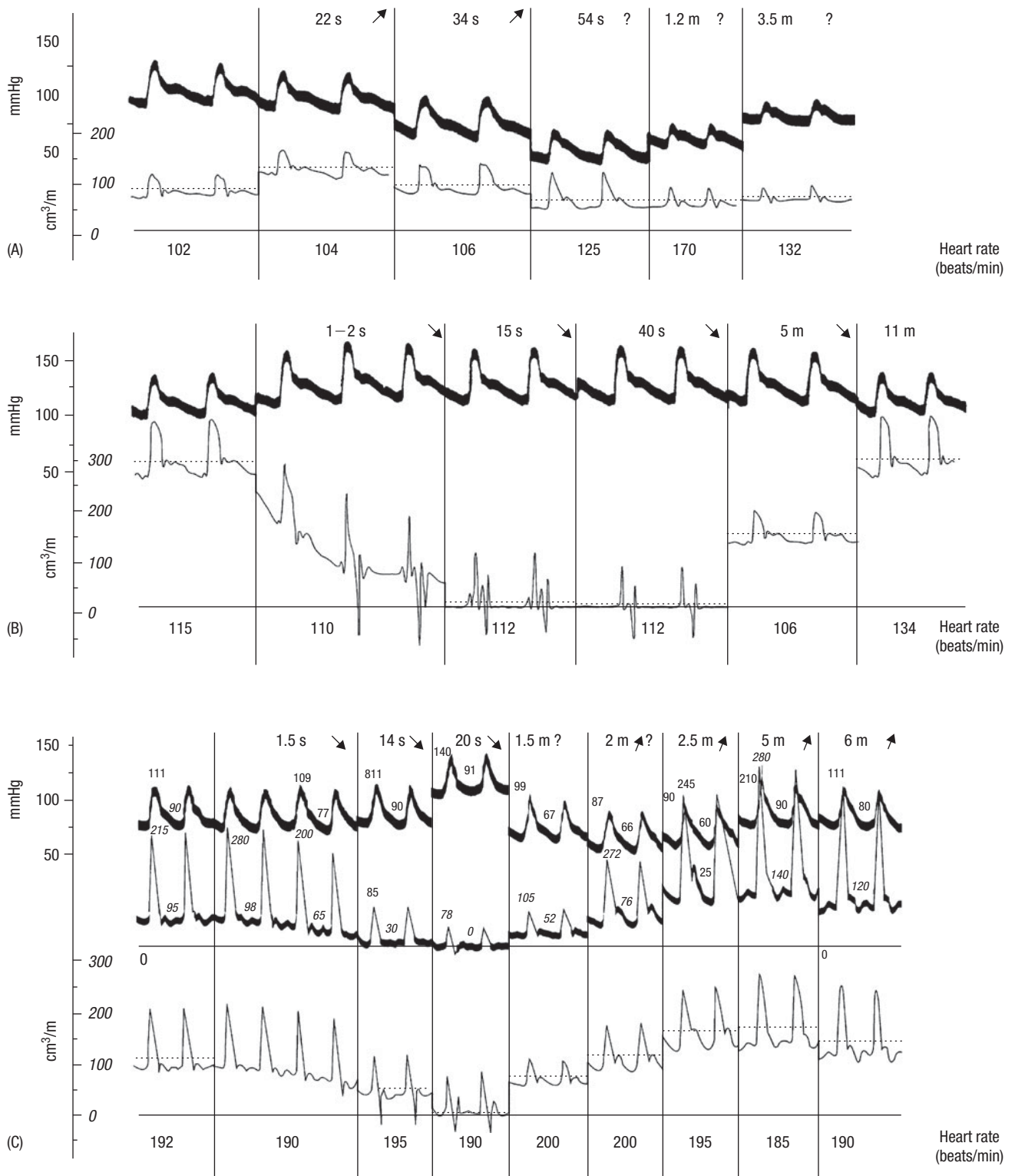


Figure 10.12 Some of the earliest studies of drug effects on vascular beds were conducted with an orifice meter that utilized the Bernoulli principle for flow recording. Results here from Pritchard *et al.* (1943) show the effects on pressure (heavy lines) and flow (thin lines): (A) during intra-arterial injection of nitroglycerin into the renal artery of a dog; (B) following injection of suprenen into the renal artery; and (C) following injection of suprenen into the hepatic artery. In (C), the original (center) and rectified (lower) flow curves are shown. In the normally vasodilated beds of the renal and hepatic artery, intra-arterial injection of a vasodilator causes little change in flow contour apart from what might be predicted from hypotension and shortening of ventricular systole. Vasoconstriction therapy, on the contrary, causes a marked reduction in mean and pulsatile flow in these vessels. (↗) Denotes constriction; (↘) denotes dilatation and (?) not known

arteriolar tone is already high, resulting in a reflection coefficient of approximately 0.8, and this is increased only marginally to around 0.95 by intense vasoconstriction.

Arterial obstruction

Arterial obstruction causes little perceptible change to the proximal pressure wave contour – as might be expected for the reason given above. The flow contour is, however, markedly altered. When the obstruction is close, pulsatile as well as mean flow may be eliminated. This is discussed below under Arterial stenosis. When the obstruction is at a distance, changes to the flow wave may resemble those seen during peripheral vasoconstriction, with a fall in the entire flow wave so that mean flow is seen to be markedly reduced and back flow created, but with little perceptible change in the flow wave contour (Fig. 10.13). A similar explanation can be offered for this as for peripheral vasoconstriction – that wave reflection has been increased from around 0.8 to near unity at a new site somewhat closer to the recording site than that which normally represents the summation of all individual reflecting sites in the vascular bed.

Tilting

The effects of change in posture on arterial pressure waves have been studied in humans by Kroeker and Wood (1955) and by Rowell *et al.* (1968). Tilting from the recumbent position causes an increase in amplification of the brachial artery pressure pulse, compared with the aortic, from 46 to 65 percent in the Mayo series. This change was

associated with and is probably due to the decrease in ejection period (from 311 to 231 ms). Kroeker and Wood (1955), O'Rourke (1971), and Wilkinson *et al.* (2001) had found that brachial pressure pulse amplification was very sensitive to changes in the duration of ventricular ejection, as is amplification between the brachial and finger arteries (Bos *et al.*, 1995).

Exercise

Exercise (see Chapter 27) is associated with a decrease in amplification of the pressure wave between the aorta and active limb (Kroeker and Wood, 1955), as a result of vasodilatation in that limb, but may lead to greater amplification of the pulse in the inactive limb.

This subject was studied specifically by Rowell *et al.* (1968), who showed that marked changes in pressure wave amplification from the aorta to the brachial and radial artery occur during leg exercise (see Fig. 10.14). They found that the ratio of radial to aortic pulse pressure increased from 52 to 124 percent during leg exercise, so that the systolic pressure in the radial artery exceeded the aortic pressure by 82 mmHg. As noted, these effects are abolished by reactive hyperemia in the arm. The authors' warning, 'that estimates of stress on aortic and cerebral vessel walls and central baroreceptors would be grossly overestimated by use of peripheral pulse pressures', also applies to the left ventricle but has largely fallen on deaf ears. Measurement of arterial pressure waves non-invasively in the carotid artery (or radial artery – see Chapter 30) assists in the interpretation of arterial pressure in central and peripheral arteries (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; O'Rourke *et al.*, 1989a;

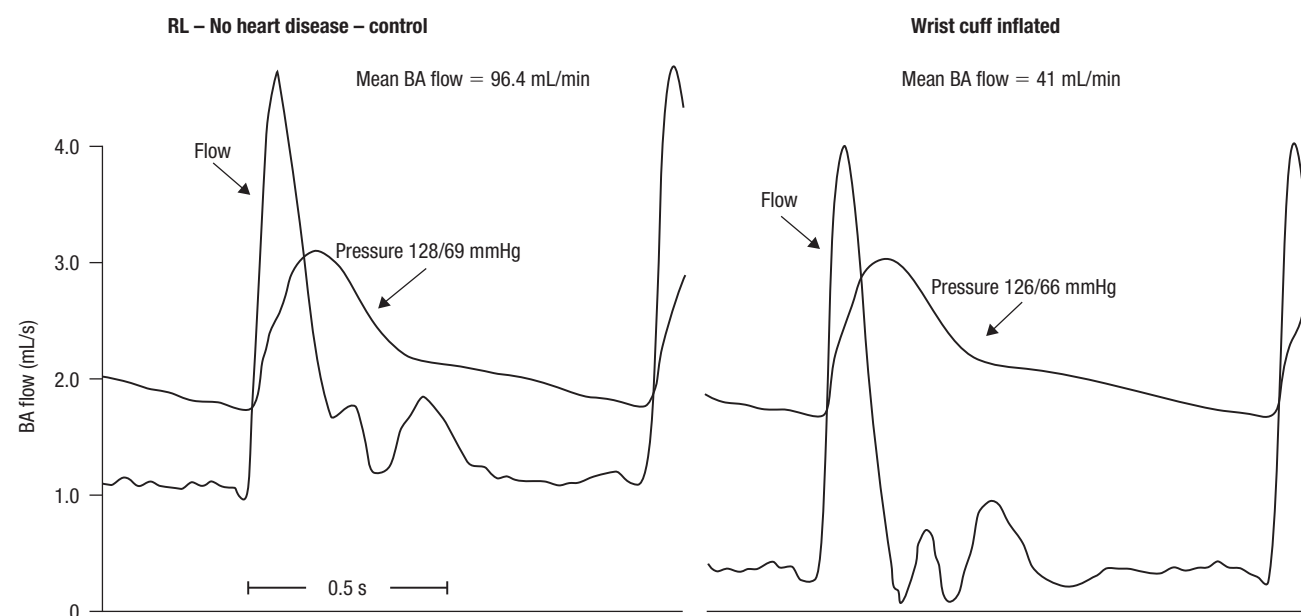


Figure 10.13 Phasic brachial artery flow patterns in a patient without left heart or systemic vascular disease, in normal circumstances and during inflation of a wrist cuff to above arterial pressure. Reproduced from Gault *et al.* (1966)

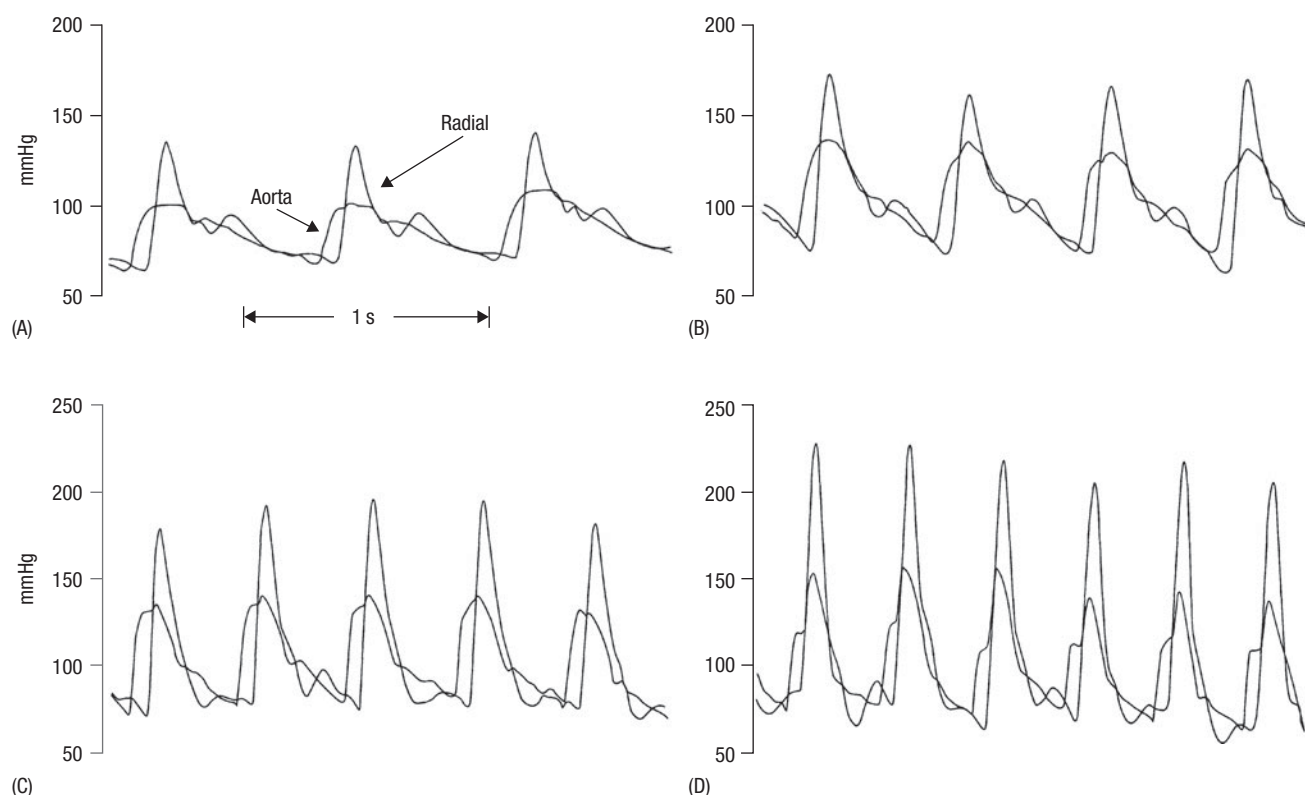


Figure 10.14 Simultaneous recordings of aortic and radial arterial pressure waves: (A) during rest, and (B) at 28 per cent, (C) 47 per cent, and (D) 70 per cent of maximum oxygen uptake in a normal human subject during leg exercise. Reproduced from Rowell *et al.* (1968)

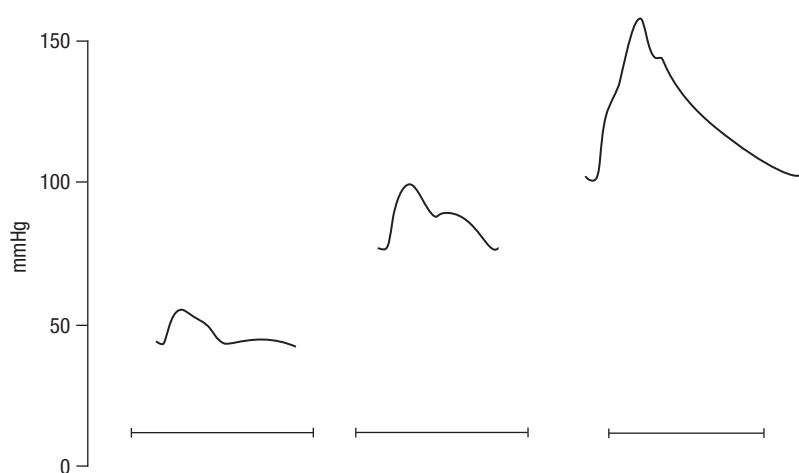


Figure 10.15 Effects of decreasing (left panel) and increasing (right panel) mean arterial pressure on the contour of the pressure wave in the ascending aorta of a rabbit. During hypotension, the diastolic wave is displaced later into diastole. During hypertension, this wave moves into systole to create a late systolic peak while pressure during diastole falls almost exponentially. Reproduced from O'Rourke (1970a) after data rearranged from Wetterer (1954)

O'Rourke *et al.*, 1992a; O'Rourke *et al.*, 1992b; O'Rourke *et al.*, 1993a; O'Rourke *et al.*, 1993b; Chen *et al.*, 1996; Chen *et al.*, 1997; Avolio *et al.*, 2001; Kass *et al.*, 2001; Van Bortel *et al.*, 2001b; Sharman *et al.*, 2006).

Mean arterial pressure change

Changes in mean arterial pressure can be induced by graded intravenous infusion of norepinephrine or by reduction in venous return, without marked change in

calculated peripheral resistance or in calculated peripheral reflection (O'Rourke, 1965a). Such interventions are, however, usually accompanied by changes in heart rate and in ejection duration that may complicate the interpretations (Fig. 10.15). The main changes are usually very obvious and are attributable to greater changes in tension on collagenous elements of the arterial wall and increased pulse wave velocity with elevation in arterial pressure (see Chapter 30). Increased wave velocity causes wave reflection to return earlier and often leads to fusion of incident and reflected waves with generation of a pressure wave

with a later systolic peak in all major arteries (Wetterer, 1954). Increase in arterial pressure has the same effects as arterial stiffening with age and is discussed in Chapter 20. Changes in arterial pressure are largely responsible for the changes in pressure and flow contours seen during the Valsalva maneuver (Mills *et al.*, 1970; Chen *et al.*, 1997).

Valsalva maneuver

Changes in pressure and flow contours during the Valsalva maneuver have long intrigued physiologists and clinicians (Frank, 1926a; Frank, 1926b; Frank, 1926c; Kroeker and Wood, 1956; Greenfield *et al.*, 1967; Mills *et al.*, 1970; Murgo *et al.*, 1981a; Latham *et al.*, 1985; Pepine and Nichols, 1988; Chen *et al.*, 1997). The Valsalva maneuver entails forced expiration against a fixed pressure, with an increase in intrathoracic and intra-abdominal pressure. Such an increase in pressure decreases the transmural aortic pressure and causes a decrease in pulse wave velocity along the length of the aorta and in all intrathoracic and intra-abdominal arteries (Hamilton *et al.*, 1936; Wezler and Knebel, 1937). An increase in thoracic

pressure also decreases venous return and causes a fall in cardiac output with consequent cardio-acceleration and peripheral vasoconstriction.

The Valsalva maneuver conveniently demonstrates the effects of peripheral wave reflection on the arterial pulse in humans, by generating the conditions that predispose to the separation of incident from reflected waves – short ventricular ejection and slow wave velocity over a long arterial segment.

The most obvious effects of the Valsalva maneuver are seen in the upper limb vessels and comprise a marked increase in diastolic fluctuations of pressure and flow and a marked increase in the relative amplitude of the brachial and radial pulses (see Figs 10.4 and 10.16). The radial pulse pressure is increased by 150 percent compared with the central aortic (46 percent under control conditions). The timing between successive pressure peaks in the brachial artery does not alter appreciably during the Valsalva maneuver, whereas the time interval between peaks in the femoral artery increases substantially (Fig. 10.16B), and amplification of the pressure wave between the central aorta and femoral artery tends to decrease (Kroeker and Wood, 1956).

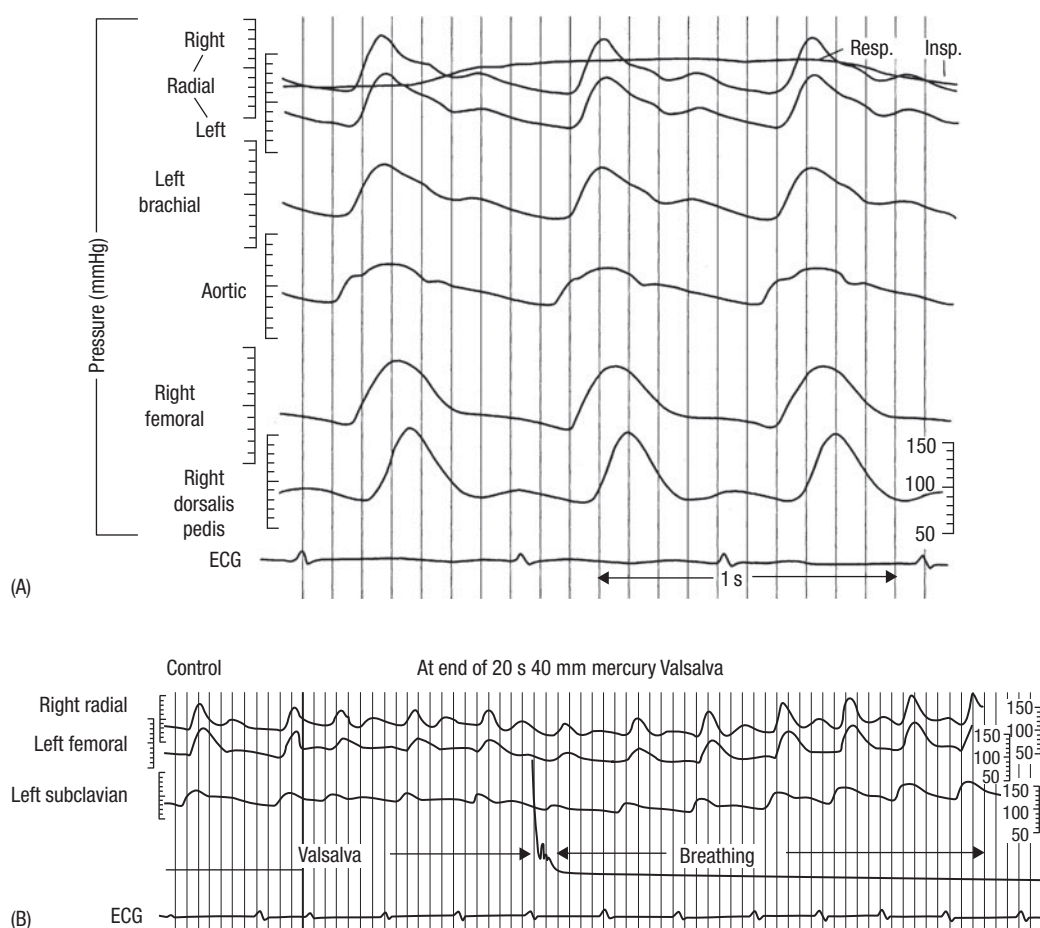


Figure 10.16 (A) Pressure waves recorded simultaneously from different arteries in a normal young human adult. (B) Pressure waves recorded simultaneously in a normal young adult human under control conditions (left) and at the end of a Valsalva maneuver. Reproduced from Remington and Wood (1956)

The main features of the Valsalva maneuver can be explained on the basis of the principles of wave travel and reflection, as discussed earlier. They help in turn to understand the changes of pressure and flow wave contours as seen in human disease, particularly shock and heart failure (Figs 10.10 and 14.13).

Latham *et al.* (Latham *et al.*, 1985; Latham *et al.*, 1988b) and Condos *et al.* (1987) have studied the effects of the Mueller maneuver, which requires forced inhalation against a fixed negative pressure and which increases transmural pressure across the thoracic aorta. As might be predicted, this simulates the effects of arterial stiffening from age or hypertension. The Muller maneuver reproduces the effects of obstructive sleep apnea (Phillips *et al.*, 2005) and is discussed in Chapter 16.

EXPLANATION OF PRESSURE AND FLOW CONTOURS IN MAJOR SYSTEMIC ARTERIES

Clearly, the contour of pressure and flow waves in major systemic arteries can be explained on the basis of wave reflection, but there are serious objections to the concepts of Frank (1905) and of Hamilton and Dow (1939) that such reflection occurs between arterioles in the periphery of the body and the closed aortic valve. This concept cannot explain why the 'node' of pressure (above and below which pressure fluctuations appear to be 180° out of phase) is located in the descending thoracic aorta, closer to the heart than the peripheral termination (see Fig. 10.6), nor the reciprocal flow fluctuations between the brachiocephalic and subclavian arteries on the one hand and the descending thoracic aorta on the other (see Fig. 10.5). Neither can this concept explain the patterns of impedance in major arteries (see Chapter 12), which give evidence of two functionally discrete reflecting sites in the peripheral circulation – one relatively close, representing all arterial terminations in the upper part of the body, and the other further away, representing all arterial terminations in the lower body. Findings were reconciled when the original concept was modified to downplay reflection at the aortic valve and to consider 'resonance' as arising from reflection and re-reflection of the pulse between sites in the upper and lower parts of the body. The arterial tree of the whole body was represented by an asymmetric T-tube (see Fig. 10.9) whose upper limb represented all arteries in the upper part of the body, whose end represented all arterial–arteriolar junctions in the upper body, and whose lower limb represented the descending aorta and all lower limb arteries, with the lower end representing all arterial–arteriolar junctions in the lower body. This concept is generally attributed to O'Rourke and Taylor (1967), but the embryonic idea was stated in the first edition of this book (McDonald, 1960). It has gained independent support from the impulse function studies of Sipkema *et al.* (1980) (Fig. 10.17) and modeling studies of Avolio (O'Rourke and Avolio, 1980), Burattini *et al.* (1991),

and Karamanoglu *et al.* (Karamanoglu *et al.*, 1994; Karamanoglu *et al.*, 1995). It is now clear that differences in body size and shape can be explained on the basis of absolute and relative differences in the two limbs of the T (O'Rourke, 1982a; O'Rourke, 1982b; O'Rourke *et al.*, 1992a). Wave reflection in the arterial system has been likened to sound reflection in a cathedral whose apse corresponds to the shorter limb of the T and whose nave corresponds to the longer limb.

This concept is an oversimplification of wave reflection in the systemic arterial system. It does explain the main features of pressure and flow waves in major arteries, but it downplays the role of the closed aortic valve in wave reflection and does not satisfactorily explain the change in contour of the pressure pulse in upper limb arteries (Karamanoglu *et al.*, 1994; Karamanoglu *et al.*, 1995). Neither does it explain the work of Latham *et al.* (1985), which suggests that the abdominal aorta shows relative constriction in older adult humans and provides an additional reflecting site in the lower body (see Chapter 9). However, the concept is useful to a first approximation, although it will need to be modified if detail is to be explained satisfactorily.

PRESSURE AND FLOW WAVES IN HUMAN UPPER BODY ARTERIES

There has been less interest in pulse wave transmission through upper body arteries than through the aorta and lower body vessels. Up until 1950, there were virtually no data on the subject, and even now there has been little attempt to model the head and fore limb circulation (Remington and O'Brien, 1970; Karamanoglu *et al.*, 1994; Karamanoglu *et al.*, 1995) – in contrast to the numerous attempts on the aorta. There are acknowledged difficulties: head and upper limb arteries are smaller and more difficult to catheterize in experimental animals than those in the lower limbs, and it is often not possible to pass catheters without occluding the major arteries downstream or obstructing the vessel upstream. Hence, data are more likely to be suspect than those recorded in the aorta and its major branches. The subject is clearly important, especially in humans, where the peak and nadir of the brachial artery pressure wave are usually considered by clinicians to represent systolic and diastolic pressure throughout the whole arterial system, and where palpation of the radial arterial pulse is an important part of the traditional clinical examination. The radial artery, in contrast to most other major arteries, can be easily compressed against bone, allowing the artery to be flattened in such a way that conventional tonometric apposition theory can apply for interpretation of pulsations transmitted to the finger (Kelly *et al.* 1987; Avolio *et al.*, 1988; Kelly *et al.*, 1988a; Kelly *et al.*, 1989a; O'Rourke *et al.*, 1989a; O'Rourke *et al.*, 1989b; Kelly *et al.*, 1990a; O'Rourke *et al.*, 1992a; O'Rourke *et al.*, 1992b;

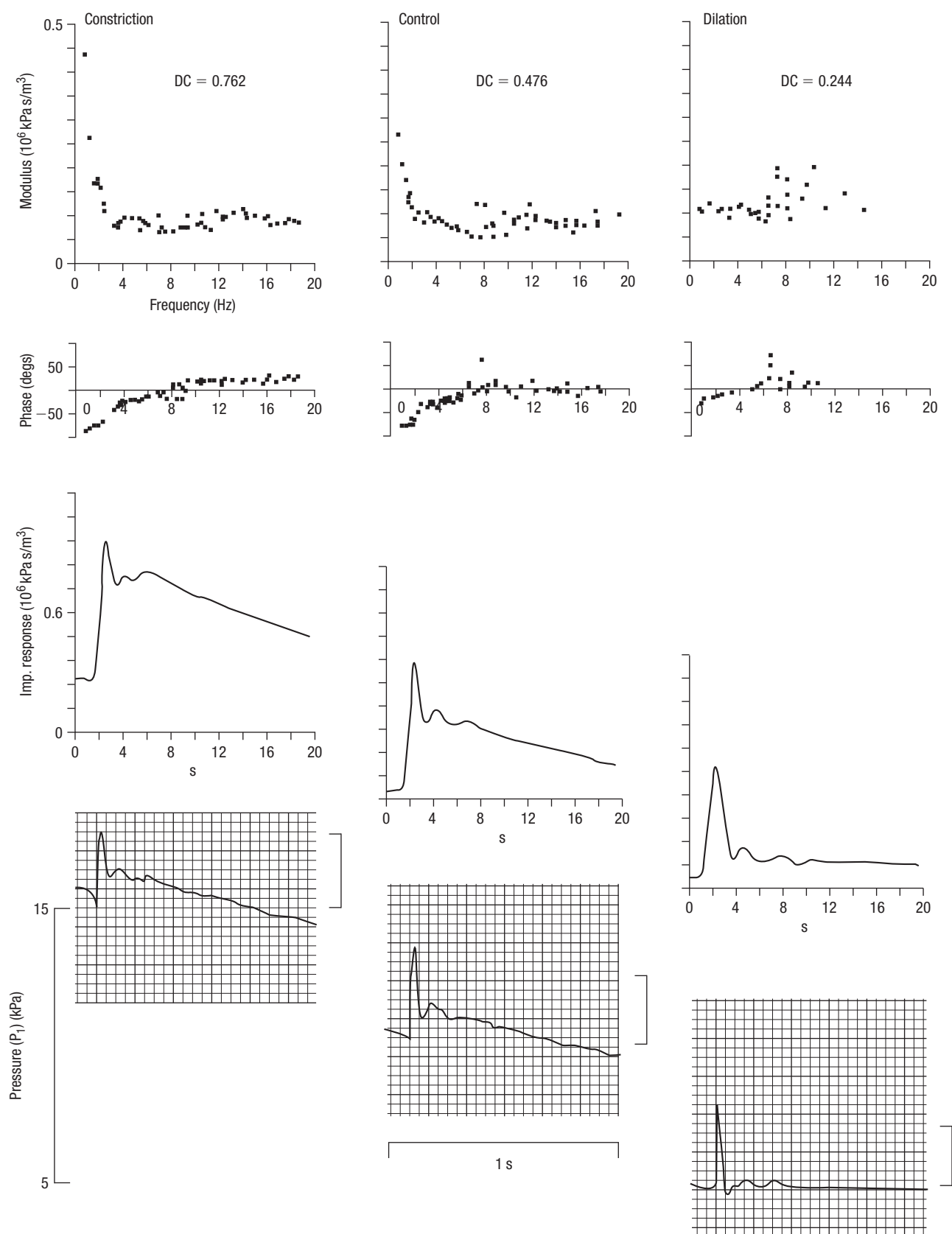


Figure 10.17 Ascending aortic input impedance in a dog (top) under three different conditions, with (middle tracings) the impulse response calculated from respective impedance determinations and (lower tracings) measured impulse response under the three conditions. Both impedance and impulse response, determined experimentally, provide evidence of early and late wave reflection from peripheral arterial terminations. Reproduced from Sipkema *et al.* (1980)

O'Rourke *et al.*, 1992c; Chen *et al.*, 1997). These studies have provided fresh information for modeling the upper limb vasculature in humans (Gallagher, 1993; Karamanoglu *et al.*, 1993; Karamanoglu *et al.*, 1994; Karamanoglu *et al.*, 1995; Gallagher *et al.*, 2004; Segers *et al.*, 2007b; O'Rourke and Adji, 2010b; Adji *et al.*, 2011). This topic is also addressed in Chapter 30.

Pressure waves in humans

Remington (Remington, 1963; Remington, 1965) reported studies of pressure wave transmission in upper body arteries. He showed that while the pressure wave in the proximal subclavian and brachiocephalic arteries showed oscillations reciprocal to those recorded in the lower body, further changes occurred in the amplitude and contour of the wave as transmitted centrifugally; these could not be explained on the basis of a single functionally discrete reflecting site in the upper body.

Since the advent of diagnostic cardiac catheterization, many studies have been reported in humans (Salans *et al.*, 1951; Kroeker and Wood, 1955; Kroeker and Wood, 1956; Remington and Wood, 1956; Rowell *et al.*, 1968; O'Rourke, 1970b; Gallagher, 1993; Karamanoglu *et al.*, 1993; Bos *et al.*, 1995; Bos *et al.*, 1996; Chen *et al.*, 1996; Chen *et al.*, 1997; Fetcs *et al.*, 1999; Segers *et al.*, 2007b; O'Rourke and Adji, 2008). The classic studies were conducted in Earl Wood's laboratories at the Mayo Clinic on the investigators themselves, as well as on their patients. These studies confirmed that the contour of the pressure wave in the upper limb is quite different from that in the femoral artery and its peripheral branches. There are usually two systolic peaks in the brachial artery, because of wave reflection from the lower body, but only one in the

femoral. The time period between the initial systolic peak and diastolic wave is shorter in the upper limb arteries than in the femoral, suggesting 'resonance' in a shorter system (Fig. 10.13). Salans *et al.* (1951) noted differences between pressure waves in the aortic arch, axillary artery, and brachial artery of man, and suggested that in the upper limb, resonance was occurring in an even shorter system. They showed, too, that such evidence of amplification and of wave reflection was not always apparent, particularly when arterial pressure and pulse wave velocity were high (Fig. 10.18). Remington and Wood (1956) conducted detailed studies of pulse wave contour in normal subjects. They showed how the normal rounded systolic pressure wave is progressively altered during transmission, through development of an early systolic peak and second systolic shoulder in the proximal subclavian artery, and how, with transmission through the limb itself to the radial artery, the secondary systolic peak is blunted, but the diastolic wave is accentuated (Fig. 10.19). They attributed their findings to resonance in the limb, with a 'node' in the axillary region and 'standing waves' above and below; their findings and explanation were in keeping with those of Salans *et al.* (1951). These interpretations are subject to the same criticisms as voiced by McDonald and others (McDonald and Taylor, 1959) in relation to aortic 'standing waves'. It would be much more satisfactory to interpret reflection phenomena from a description of pressure-flow relationships described as impedance (see Chapter 12) or pressure/pressure transfer functions (O'Rourke and Avolio, 2008; Segers *et al.*, 2005a; Segers *et al.*, 2007b).

In normal circumstances, the arterial pressure wave is markedly amplified in transit from the ascending aorta to the radial artery (Rowell *et al.*, 1968; O'Rourke, 1970b; Karamanoglu *et al.*, 1993; O'Rourke *et al.*, 1993a;

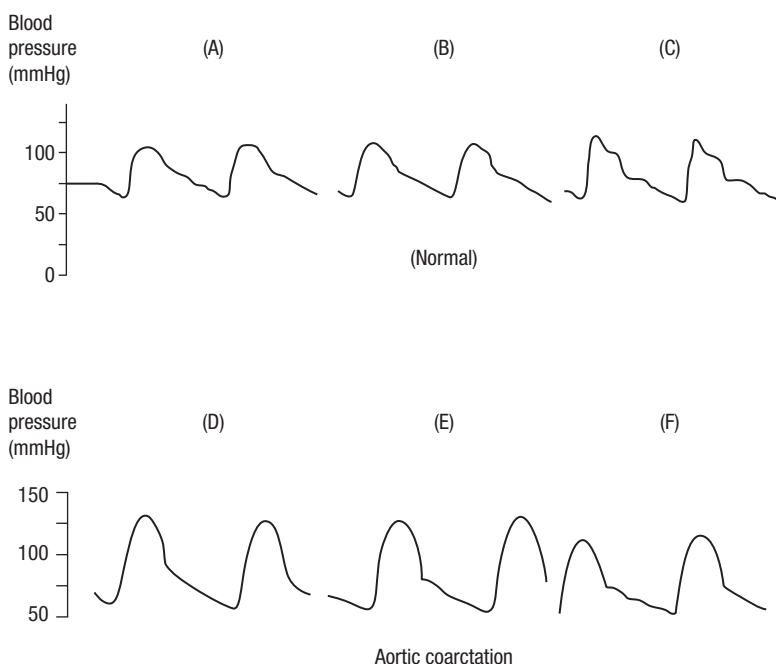


Figure 10.18 Pressure waves recorded at short time intervals apart in (A) the central aorta, (B) the axillary artery, and (C) the brachial artery, in a normal subject and (D, E, and F) at corresponding sites in a patient with aortic coarctation and elevated arterial pressure. Reproduced from Salans *et al.* (1951)

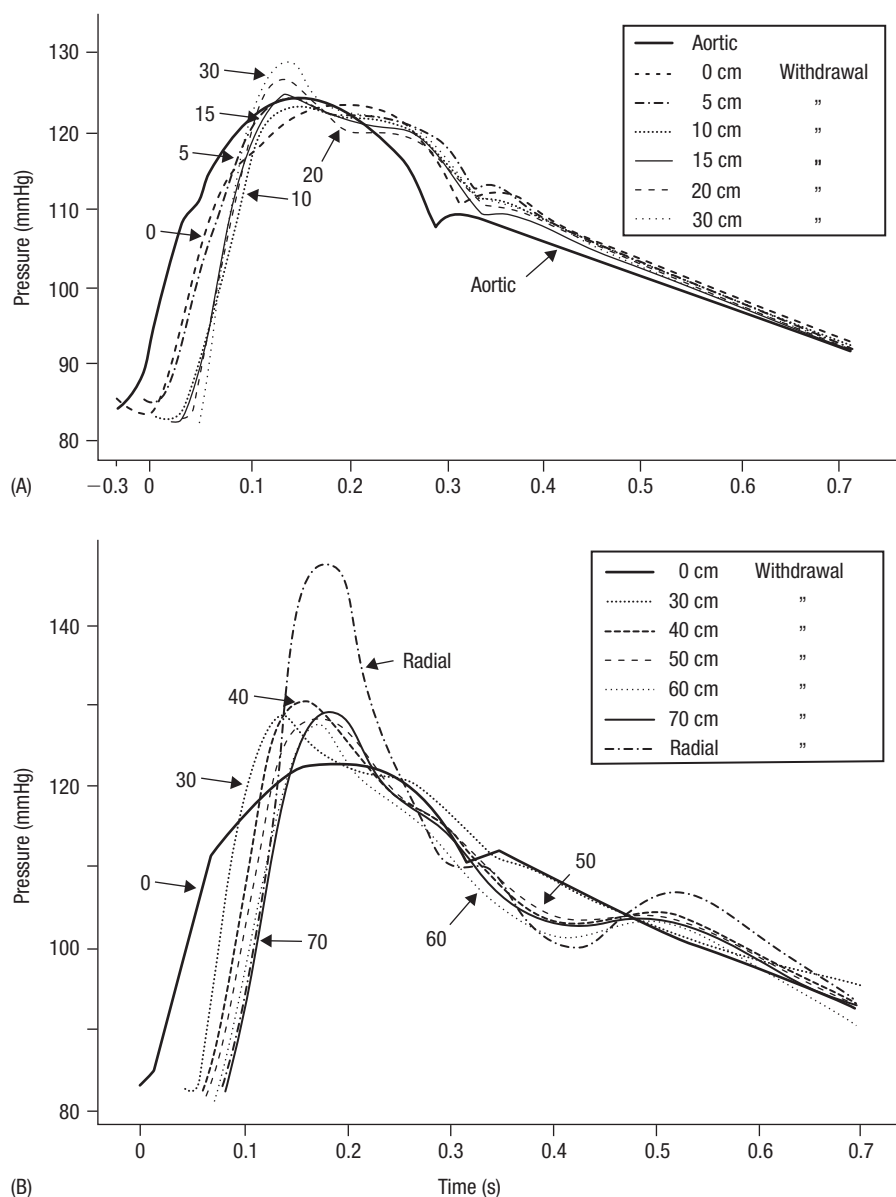


Figure 10.19 Transformation of the pressure wave in the upper limb of a human. (A) Arterial pressure waves recorded at 5.0 cm intervals of withdrawal. The final curve is that in the brachial artery, 30 cm downstream. (B) Arterial pressure waves with continued withdrawal into the lower arm. The heavy solid line represents the pulse in the most proximal position within the subclavian artery. The 30 cm withdrawal pulse is the final curve in (A), and subsequent waves are those recorded during withdrawal of the catheter into the radial artery system. Reproduced from Remington and Wood (1956)

Chen *et al.*, 1997), while mean pressure falls by 2.0 mmHg at most (Pauca *et al.*, 1992; Pauca *et al.*, 2001). The factors responsible for amplification are similar to those described for the femoral artery. Age, however, appears to be less important in the upper limb, presumably because of the different flow wave contour in the brachiocephalic and subclavian arteries compared with the descending thoracic aorta, and the smaller change in upper limb pulse wave velocity than in aortic pulse wave velocity that occurs with age (see Figs 10.5, 10.11, 19.13, and 19.14) (Avolio *et al.*, 1983; Avolio *et al.*, 1985a; Gallagher, 1993; Karamanoglu *et al.*, 1995; Gallagher *et al.*, 2004).

The new method of arterial tonometry (Avolio *et al.*, 1988; Kelly *et al.*, 1988a; Kelly *et al.*, 1988b; Saba *et al.*, 1993; Vaitkevicius *et al.*, 1993; London *et al.*, 1995; Chen *et al.*, 1996; Chen *et al.*, 1997) has provided fresh information on pressure wave transmission to the upper limbs in humans and indeed fresh insight into the effects of aging and of vasoactive drugs on the arterial pressure wave (O'Rourke, 1988; Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; O'Rourke *et al.*, 1989a; O'Rourke *et al.*, 1989b; O'Rourke *et al.*, 1992a; O'Rourke *et al.*, 1992b; O'Rourke and Kelly, 1993; O'Rourke *et al.*, 1993a). This is discussed later (see Chapter 30) and has important implications

for interpretation of sphygmomanometric pressures with different therapies in humans (O'Rourke, 1990a; O'Rourke *et al.*, 1993a; Chen *et al.*, 1996; Chen *et al.*, 1997; Williams *et al.*, 2006; Miyashita *et al.*, 2010).

Flow waves in humans

In their own classic human studies, Mills *et al.* (1970) stressed the curious pattern of flow waves in the arteries supplying the head, neck, and upper limbs. These have been explained on the basis of early wave reflection from the upper body causing deceleration of systolic flow in these vessels, and thus systolic notching (see Figs 10.4 and 10.11) or a pronounced dicotic notch and prominent diastolic wave (O'Rourke and Avolio, 1980). The changes with extrasystoles or atrial fibrillation or with alteration of arterial pressure, or with the Valsalva maneuver or prolonged expiration, can be explained on the same basis (Fig. 10.20). Avolio's model (O'Rourke and Avolio, 1980) has permitted simulation of normal circumstances and the possible result of doubling distances to reflecting sites in the upper part of the body; the latter simulation creates flow waves of similar contour in the brachiocephalic artery and descending thoracic aorta. Fig. 10.21 shows the value of harmonic analysis in the interpretation of pressure and flow wave contour in the brachiocephalic artery of two human subjects. These patients showed almost identical brachiocephalic impedance patterns, indicating that the properties of the vascular beds were virtually

identical. The differences in contour of the pressure and flow waves at the input of the bed can only be explained on the basis of different excitation, with ventricular ejection in patient (A) being more prolonged and generating a higher mean pressure with higher aortic pulse wave velocity than in patient (B).

Flow waves in the brachial artery of humans have been recorded by Gault *et al.* (1966) using an electromagnetic flow transducer applied to the exposed arteries, and by Kalmanson *et al.* (1968), Safar 1981a; Safar 1981b, and others using transcutaneous Doppler techniques. These show that the unusual flow pattern of the brachiocephalic artery is maintained in the arm (Fig. 10.22). The flow contour in the common carotid artery is similar but with a bigger mean flow offset. The correspondence between features of the carotid flow and pressure waves was pointed out first by Hirata *et al.* (2006) (Fig. 10.23), and used by them to associate pulsatile cerebral flow fluctuations to lacunar infarcts, and to suggest a new strategy for preventing their development (Hirata *et al.*, 2008a; Hirata *et al.*, 2008b) (see also Chapter 18).

RELATIONSHIP BETWEEN PRESSURES IN THE BRACHIAL ARTERY AND ASCENDING AORTA OF HUMANS

Use of the sphygmomanometer is firmly rooted in clinical practice. Clinicians refer to 'the' systolic and 'the' diastolic

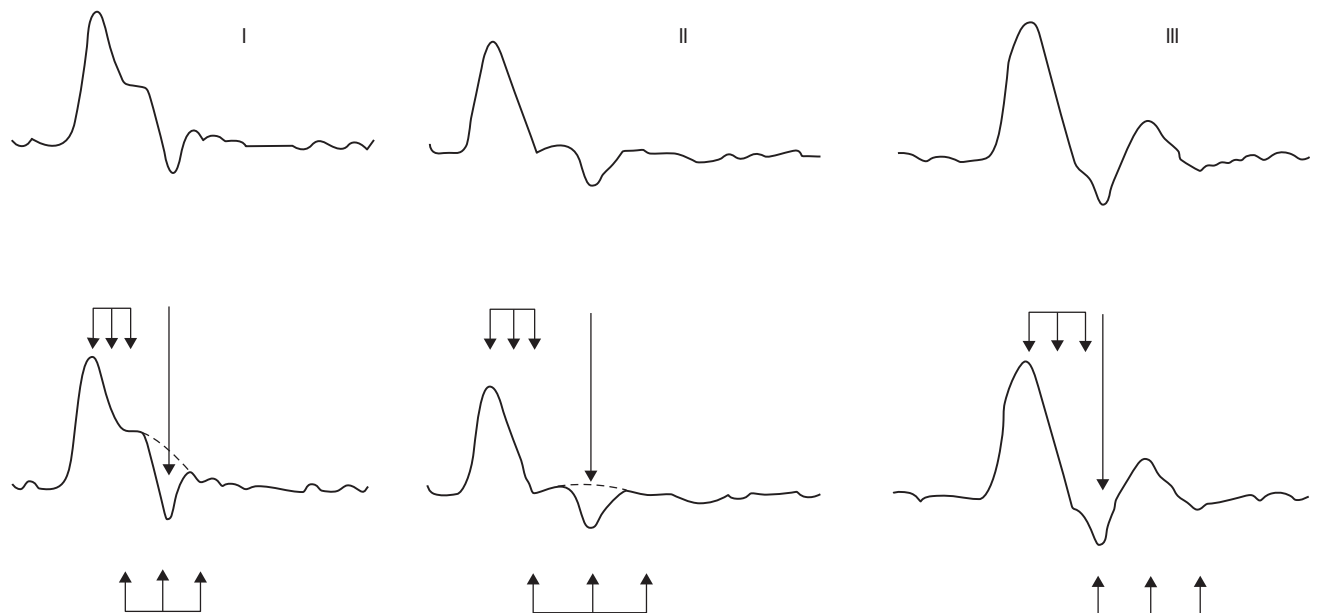


Figure 10.20 Types I, II, and III brachiocephalic flow waves redrawn from Mills *et al.* (1970) with the suggested mechanism outlined below. The single arrow represents aortic valve closure. The early triple arrow represents wave reflection from the upper body, and the later triple arrow wave reflection from the lower part of the body. The dotted line indicates the expected wave contour in the absence of an incisura caused by back flow through the aortic valve. Wave reflection from the upper body decelerates flow in the brachiocephalic artery, while reflection from the lower body accelerates forward flow in the artery. The three different flow patterns are readily explicable on the basis of timing of the wave reflection from upper and lower body in relation to duration of ventricular ejection. Reproduced from O'Rourke and Avolio (1980)

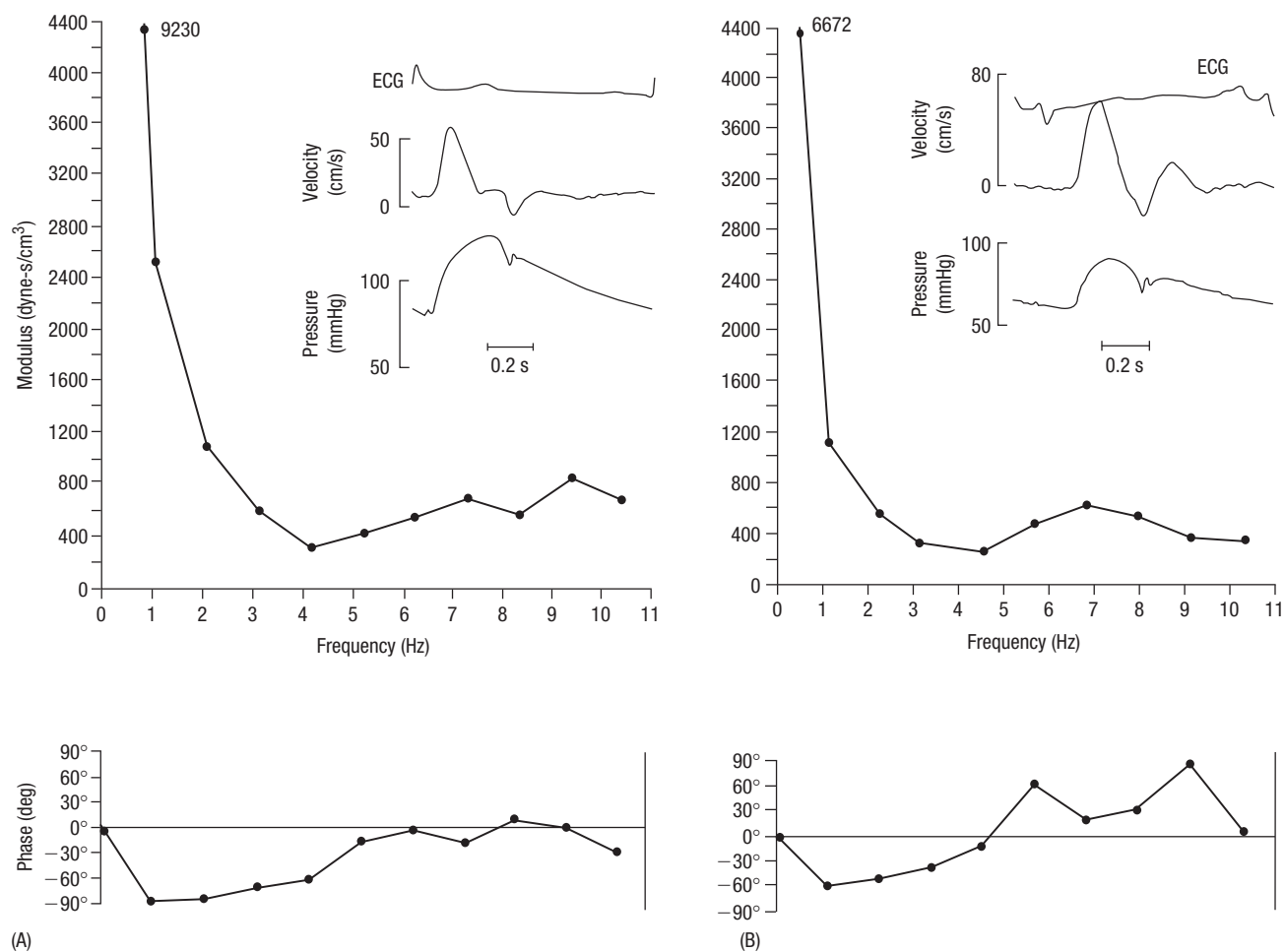


Figure 10.21 Pressure and velocity waves (inset) in the brachiocephalic artery of two patients, together with the modulus (above) and phase (below) of impedance spectra determined from the waves. Similarity in the patterns of impedance indicates similarity of the physical properties of the vascular beds in the two patients, indicating that the different contours of flow and pressure waves are due to differences in duration of ventricular ejection, this being longer in subject (A) with a type II wave and shorter in patient (B) with a type III brachiocephalic flow. Reproduced from Mills *et al.* (1970)

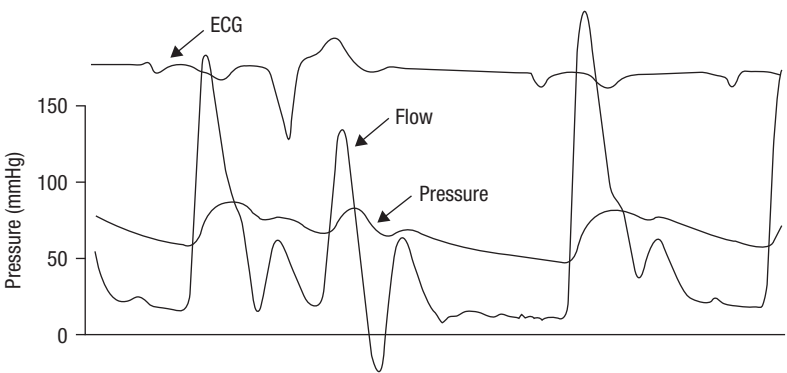


Figure 10.22 Phasic brachial artery flow recorded with a cuff-type electromagnetic flowmeter, together with brachial artery pressure in an apparently normal human, showing a similar pattern to the brachiocephalic artery, and variation in wave contour brought about by differences in ventricular ejection duration with extrasystoles. Reproduced from Gault *et al.* (1966)

pressure as though these are the same in all arteries under all conditions. Attention has been paid to the inaccuracy of the sphygmomanometer in recording true brachial intra-arterial systolic and diastolic pressure as a result of inadequate cuff size, a large arm, a sclerotic artery (Messerli *et al.*, 1985b) (see Chapter 6), or other reasons (Watson *et al.*, 1998; Lane *et al.*, 2002), but few clinicians

realize that brachial systolic and diastolic pressure do not always reflect the corresponding pressures in the ascending aorta. This was studied by Rowell *et al.* (1968), O'Rourke (1971), Lasance *et al.* (1976), Karamanoglu *et al.* (1993), Gallagher (1993), Chen *et al.* (1997), Fetis *et al.* (1999), Sharman *et al.* (2006), and Westerhof *et al.* (2008a). Remington and Wood (1956) showed that the

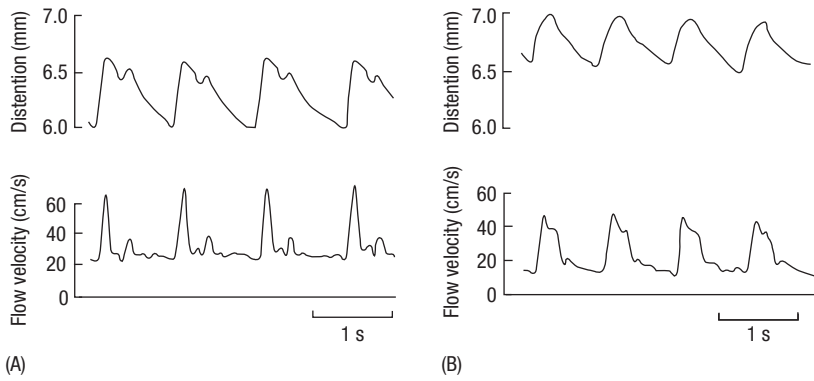


Figure 10.23 Internal carotid artery pressure (top) and ultrasonic flow velocity (bottom) waves in a healthy 23-year-old subject (A) and in a healthy 65-year-old subject (B). Reproduced from Hirata *et al.* (2006)

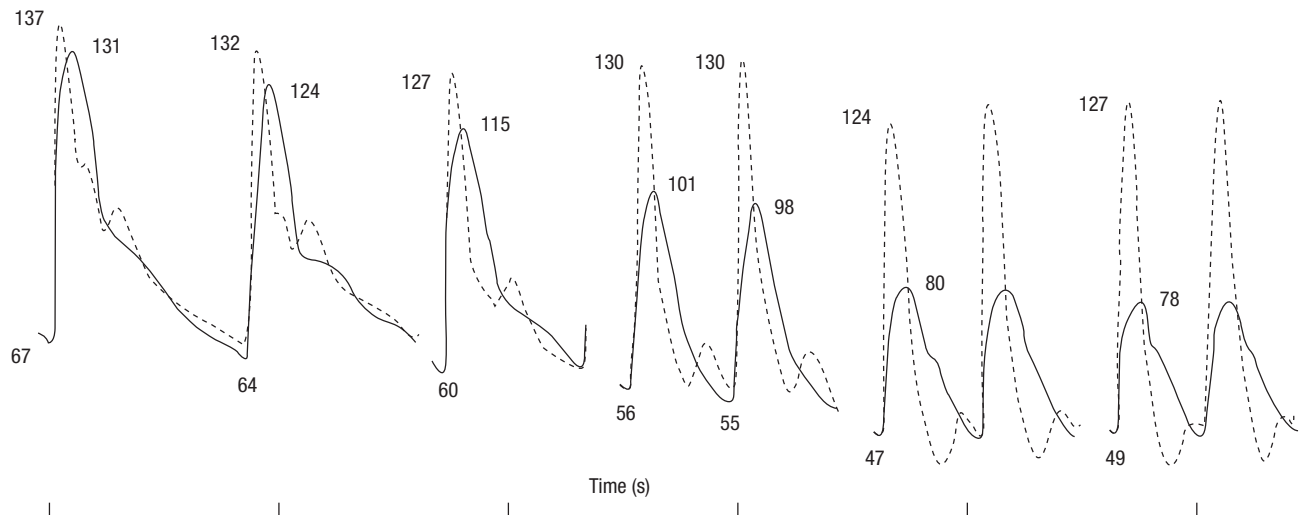


Figure 10.24 Simultaneously recorded pressure waves from the radial artery through a needle (dashed line) and from the abdominal aorta through a catheter (solid line) during the course of an hypotensive reaction to rapid intra-aortic injection of isotonic electrolyte solution (numbers represent mmHg). Reproduced from Remington and Wood (1956)

brachial artery systolic pressure may overestimate the central aortic systolic pressure by up to 50 mmHg under conditions that simulate human disease (Fig. 10.24), while Rowell *et al.* (1968) showed differences up to 80 mmHg during exercise (Fig. 10.14). These studies emphasize the importance of monitoring central arterial pressure when cardiac afterload is to be determined precisely – as with shock (Fig. 10.10), with the use of vasodilator therapy, during use of intra-aortic balloon or enhanced external counterpulsation.

A number of recent studies (Mitchell *et al.*, 2002; Mitchell *et al.*, 2003; Mitchell *et al.*, 2004a; Mitchell *et al.*, 2005a; Mitchell *et al.*, 2005b; Mitchell *et al.*, 2006; Dart *et al.*, 2007; Mitchell *et al.*, 2007a; Mitchell *et al.*, 2007b; Segers *et al.*, 2007b; Mitchell *et al.*, 2008; Mitchell, 2009a; Mitchell 2009b; Segers *et al.*, 2009; Mitchell *et al.*, 2010) report little or no difference between systolic pressure in the aorta and brachial artery. All these used brachial applanation tonometry, and results are attributable to failure of applanation (O'Rourke and Adji, 2010a). All three authors showed similar form factors (O'Rourke and Adji, 2010a) in carotid and brachial arteries, with diastolic and mean pressure assumed equal and systolic pressure within 1–2 mmHg at the two sites. When systolic carotid pressure

is generated from the radial artery tonometry, either from use of generalized transfer function, by considering mean and diastolic pressure equal, or from the second systolic shoulder of the radial pressure pulse, differences of 10 mmHg in systolic pressures are regularly seen, as in invasive studies (Fig. 10.25) (O'Rourke and Adji, 2010b). Anomalies are attributable to failure of applanation at the brachial site (O'Rourke and Adji, 2010a).

Following earlier work by Yaginuma *et al.* (1986) on the effects of nitroglycerin on ascending aortic impedance and pressure wave contour, Kelly *et al.* (1990a) studied the effects of this drug on wave contour in the ascending aorta and brachial artery (Fig. 10.26). They showed that reduction in brachial systolic pressure underestimated reduction in central aortic systolic pressure by an average of 11 mmHg, and up to 20 mmHg. These findings were explained on the basis of a difference in pressure wave contour at the two sites, with the peak of the aortic pulse normally being created by wave reflection from the lower body, while that in the brachial artery was not (Kelly *et al.*, 1990a). The important implication was that a drug such as nitroglycerin, when assessed by conventional sphygmomanometry, has a far more beneficial effect on

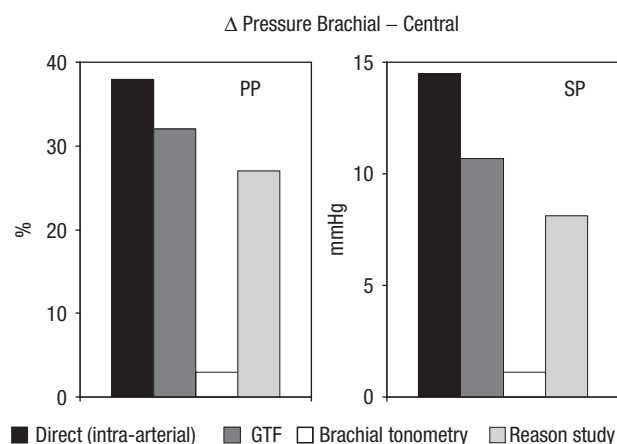


Figure 10.25 Amplification of the arterial pulse in the upper limb from different publications, using different methods and in different cohorts. (1) Black: Pressure recorded directly (intra-arterial) by matched catheter–manometer systems in the ascending aorta and radial artery of young human volunteers in physiological studies and in patients undergoing cardiac surgery prior to initiation of cardiopulmonary bypass. (2) Dark gray: Pressure recorded in the radial artery and calculated for the ascending aorta in clinical trials, using the SphygmoCor transfer function system (GTF). (3) White: Form Factor Method. Pressures measured in the carotid artery and brachial artery by applanation tonometry, with the brachial systolic and diastolic cuff values applied to the brachial wave, with brachial mean pressure calculated by integration of the brachial waveform, and with the carotid wave calibrated from the brachial, assuming that the mean pressure and diastolic pressure wave is the same at the two sites and with carotid systolic pressure extrapolated. (4) Light gray: Form Factor method as used by Asmar *et al.* for calculation of carotid waveform using the Form Factor (FF) method, but applying the brachial cuff calibration to the radial pressure wave recorded by applanation tonometry. Left panel: Amplification expressed as percentage of peripheral pulse pressure divided by central pulse pressure. Right panel: Amplification expressed as mmHg difference between peripheral and central systolic pressure. The method used by Mitchell, Segers, Cameron/Dart provides values of amplification which are less than 20 percent of those published by others. From Asmar *et al.* (2001b) and O'Rourke and Adji (2010)

left ventricular afterload than is apparent clinically. This work has been verified by Simkus and Fitchett (1990) and Takazawa *et al.* (1995b). It was originally suggested by non-invasive tonometric studies that had shown similar differential effects on the carotid and radial artery pulse (Fig. 10.27). It provided the impetus for clarifying the relationship between ascending aortic pressure and the pressure waveform in the brachial and radial artery (Gallagher, 1993; Karamanoglu *et al.*, 1993; Karamanoglu *et al.*, 1994; Karamanoglu *et al.*, 1995) and for development of a method for generating the ascending aortic pressure waveform from the peripheral pressure pulse (O'Rourke *et al.*, 1992c; O'Rourke *et al.*, 1993a; O'Rourke and Kelly, 1993; Chen *et al.*, 1997). This method is best utilized with non-invasive measurement of the radial

artery pulse by applanation tonometry. It is discussed in detail in Chapter 30, where near-constancy of transfer function in the upper limb under different conditions is highlighted and validation data are presented.

These issues are relevant to claims on or implications of 'benefits beyond systolic blood pressure lowering' in the HOPE, LIFE, ACCOMPLISH and other recent clinical trials (Yusuf *et al.*, 2000; Brunner and Gavras, 2002; Dahlöf *et al.*, 2002; Hirata *et al.*, 2005b; Jamerson *et al.*, 2008; Matsui *et al.*, 2009) and have generated brisk correspondence (O'Rourke, 2000b; Mourad *et al.*, 2002; Nichols *et al.*, 2002; Sleight and Yusuf, 2002; Griffin *et al.*, 2003; Kurtz, 2003) (see also Chapter 26).

While the pressure pulse is normally amplified in the upper limb, it may under some conditions be decreased in peripheral arteries. Effects diametrically opposite to those described above are sometimes seen after hypothermic cardiac surgery, when as a consequence of conduit arterial constriction and/or arteriolar dilatation in the hand, radial and brachial artery pressures may seriously underestimate systolic pressure in the ascending aorta and lead to inappropriate therapy (Wesseling *et al.*, 1985; Gravlee *et al.*, 1989a; Gravlee *et al.*, 1989b; Imholz *et al.*, 1992).

Issues may also be relevant to subtle ill effects of drugs which are only apparent when central pressure is measured in addition to brachial pressure (O'Rourke, 2005). They may also explain greater susceptibility of African-Americans to cardiovascular disease despite similar brachial pressures (Heffernan *et al.*, 2008).

THE PULSE IN DISEASE

Palpation of the arterial pulse is a routine part of the clinical examination, rooted in antiquity. This book cannot hope to give a comprehensive account of the pulse if it is to retain its original character. Some comments based on previous discussions are, however, in order.

Changes in contour of the arterial pulse are usually due to changes in vascular properties, with relatively less or more, or relatively early or later, wave reflection. Three exceptions need to be mentioned. The first is hypertrophic cardiomyopathy, in which flow ejection from the heart often has a biphasic pattern, almost like that seen in the brachiocephalic artery of older human subjects (Hernandez *et al.*, 1964; Murgu *et al.*, 1980b). Such an unusual pattern may, in subjects with normal vascular properties and ascending aortic impedance, generate a biphasic or 'bisferiens' pressure pulse (Fig. 6.24). The second is aortic valve stenosis or the combination of valvular stenosis and incompetence. Here, a ventricular ejection wave of normal contour can generate such high velocities in the ascending aorta that a prominent Venturi effect is created, with a biphasic pressure wave generated in and propagated from the ascending aorta (Dow, 1941; O'Rourke, 1982b). This is the basis of the 'anacrotic' pressure pulse in valvular aortic stenosis and of the

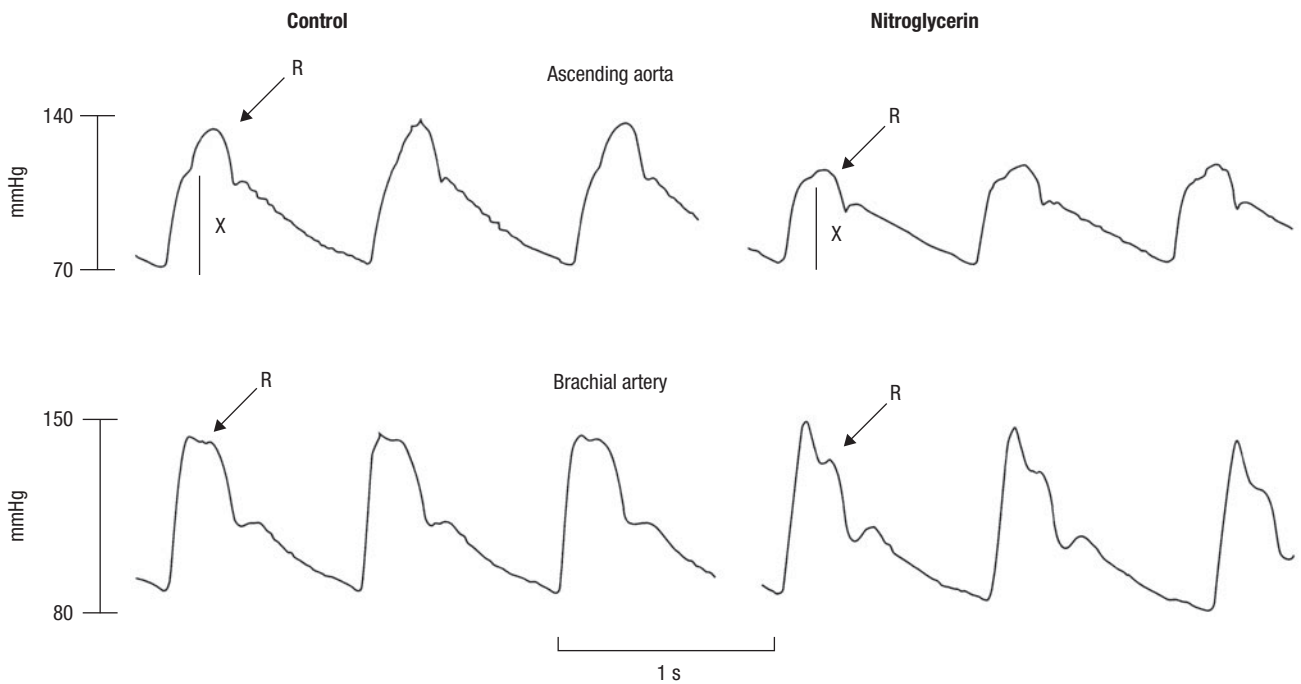


Figure 10.26 Pressure waves recorded directly in the ascending aorta (top) and brachial artery (bottom) under control conditions (left) and after 0.3 mg sublingual nitroglycerin (right) in a human adult. X, height the pressure pulse would be without reflection (R). Reproduced from Kelly *et al.* (1990a)

'bisferiens' pulse in combined stenosis and incompetence. In aortic valve disease, the duration of ventricular ejection is relatively long, and the lower harmonics of the pressure pulse are far greater than the higher harmonics. These are the reasons given for low amplification of the pulse between the ascending aorta and brachial artery in aortic valve disease (O'Rourke, 1971; O'Rourke *et al.*, 1992c). The third is myocardial weakness, where wave reflection has a negative effect on flow, and accounts for reduction in late systolic flow or abbreviation of systole (see Chapter 14) (Westerhof and O'Rourke, 1995).

Note has been made of the change in arterial pressure and flow contour with aging. Such changes are explicable on the basis of increased pulse wave velocity, with merging of reflected waves into the incident wave (see Fig. 10.9). A similar explanation can be given for changes induced by high arterial pressure (see Fig. 10.15). Changes in the arterial pressure pulse with hypertension and aging were first noted by Mahomed (Mahomed, 1872; Mahomed, 1874) over 100 years ago (O'Rourke, 1992; Swales, 1996a). In the absence of a reliable sphygmomanometer, he diagnosed high blood pressure on the basis of increased tension of the pulse and presence of a late systolic peak, but he noted that such a pulse may be felt without high tension in the elderly. More recent studies (Freis *et al.*, 1966; Puls and Heizer, 1967; O'Rourke *et al.*, 1968; Kelly *et al.*, 1989a; O'Rourke and Gallagher, 1996; O'Rourke and Hashimoto, 2007) have confirmed these interpretations.

In shock, the three conditions are present that predispose to prominent evidence of wave reflection (Frank,

1926a; Frank, 1926b; Frank, 1926c; Kroeker and Wood, 1956; O'Rourke, 1971), as in the hypotensive phase of the Valsalva maneuver – shortened ventricular systole, low arterial pressure with slow pulse wave velocity, and intense peripheral vasoconstriction. In shock, diastolic waves are often very prominent (Fig. 10.10). On the basis of his studies of hemorrhagic shock, Hamilton (1944) suggested that the arterial pressure pulse contour may have diagnostic utility in confirming whether hypotension is compensated by peripheral vasoconstriction or is caused by peripheral vasodilatation – the former state being associated with prominent diastolic waves and the second with none.

The obvious clinical implications of pulse wave analysis have not yet been tackled in a realistic way by the major professional bodies. Important blocks to progress are the nomenclature and classification of pulses, and the imperfect knowledge of underlying mechanisms (Vasan, 2008). Leading articles on the pulse in authoritative clinical journals (Stead, 1965; Perloff, 1983) contained a host of errors, as do the relevant sections of most contemporary cardiologic texts. One hopes that either the American Heart Association, the American College of Cardiology, or the European Society of Cardiology will establish some new standards, at least for nomenclature, that will create a clean break with Galenic mystery.

As noted, Doppler flow records are being used increasingly in studies of arterial disease. Localized arterial stenoses upstream are suggested by flow pulse contours of low amplitude with little or no back flow and low calculated pulsatility index. A different type of interpretation

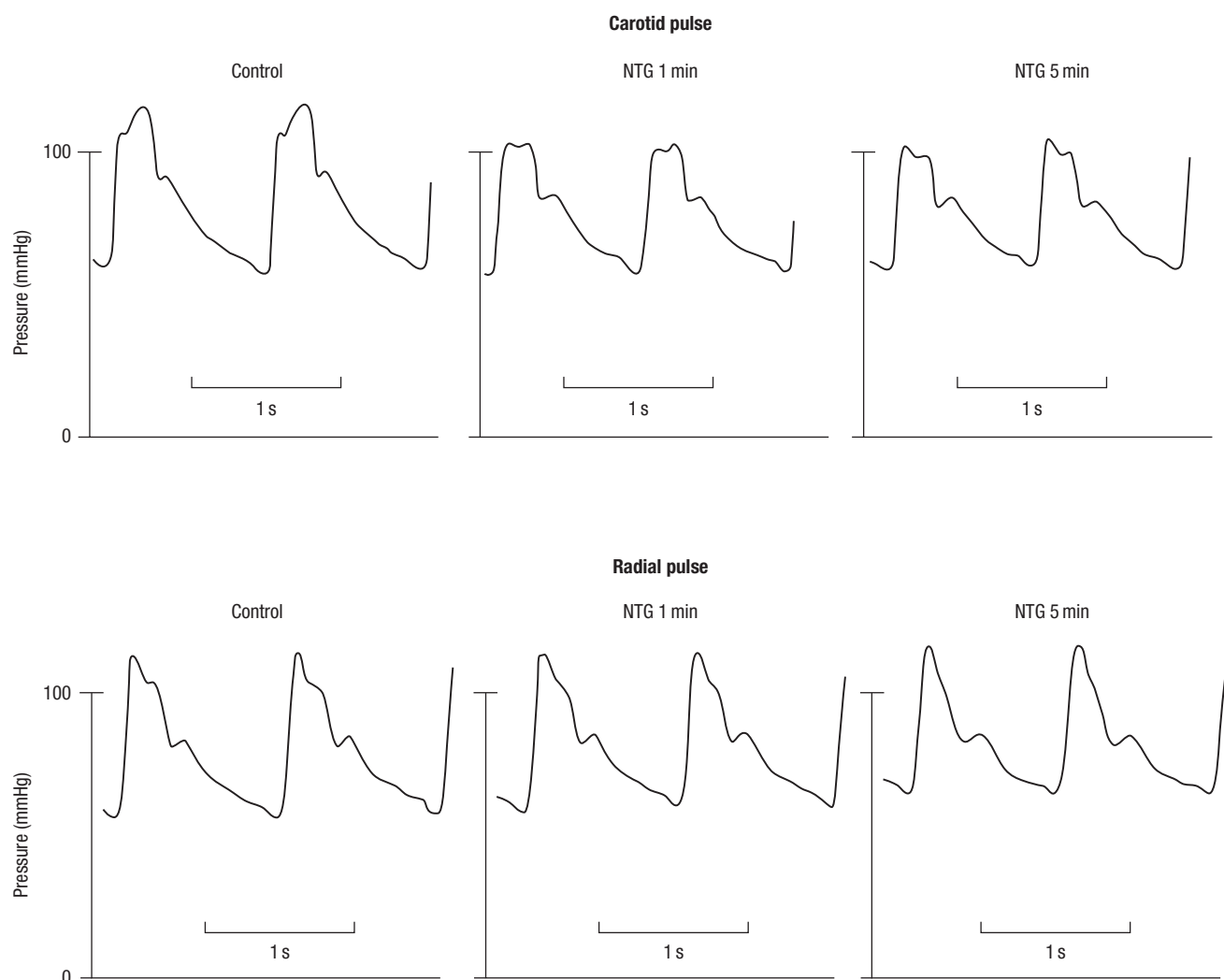


Figure 10.27 Pressure waves recorded indirectly from the carotid artery (top) and radial artery (bottom) of a 50-year-old normotensive man under control conditions (left), and 1.0 minute (center) and 5.0 minutes (right) after sublingual administration of 0.3 mg nitroglycerin (NTG). Reproduced from O'Rourke *et al.* (1989a)

has been applied to the flow patterns from the umbilical artery of the fetus (Trudinger *et al.*, 1985). The artery normally supplies a dilated vascular bed, the placenta, at a considerable distance from the fetus itself. The flow pattern in this artery is normally of low amplitude about a relatively high mean flow (Fig. 10.28). Greater relative pulsatility and appearance of back flow is usually due to placental degeneration with increased placental resistance (Fig. 10.29) (Trudinger *et al.*, 1985). Changes in the flow pattern are thus similar to those seen in a low-resistance vascular bed, such as the renal during infusion of a vasoconstrictor agent (Fig. 10.12), or in diffuse small vessel disease, as in a granular contracted kidney.

ARTERIAL STENOSIS

There has been great interest in the effects of arterial stenosis on the patterns of pressure and flow waves above

and below the lesion. The hope has been to gain information on the presence, location, and severity of 'sub-clinical' lesions, as well as those which are responsible for symptoms. This subject has been discussed in textbooks (Strandness and Sumner, 1975; Lord, 1986; Veith *et al.*, 1994; Loscalzo *et al.*, 1996), reviews (Yao *et al.*, 1984), a host of original articles and in Chapter 24. The field is vast and only general principles can be addressed here (see also Chapter 8 regarding the use of ultrasonic techniques in clinical diagnosis).

Narrowing of an artery causes a reduction in the amplitude of flow waves above and below the lesion, and of the pressure wave below; there is little change in the amplitude or contour of the pressure wave above (Fig. 10.30) (Farrar *et al.*, 1979). These changes are attributable to alterations of impedance caused by the stenosis (Fig. 10.31). The pressure and flow pulsations are most sensitive to changes in the caliber of the artery; changes in the waves are noted with lesser degrees of

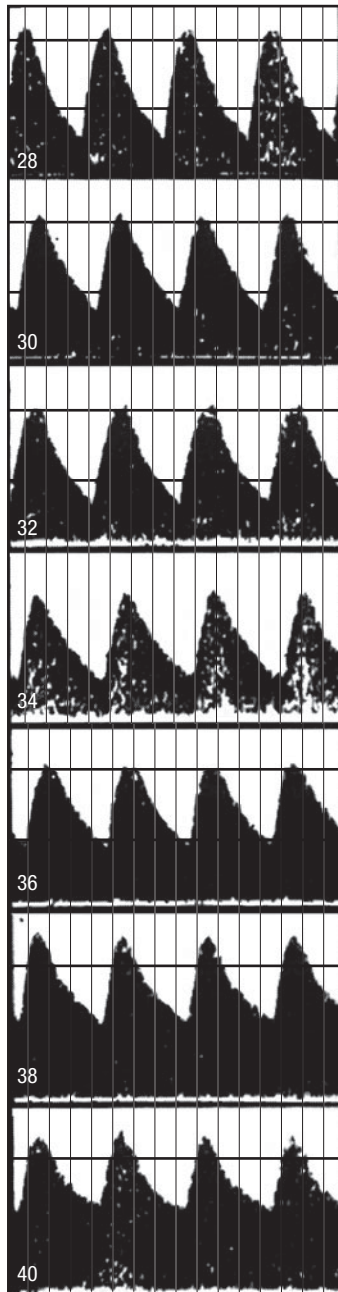


Figure 10.28 Doppler flow waveforms from the umbilical artery of a normal human fetus at between 28 and 40 weeks after conception. Reproduced from Trudinger *et al.* (1985)

stenosis than cause change in mean flow or pressure gradient across the stenosis. Changes in Doppler flow velocity waveforms (Fig. 10.32, page 252) are utilized to characterize the severity of the stenosis. The most commonly used method is the pulsatility index, defined as the peak-to-peak reading of the flow wave divided by the mean of the wave (Yao *et al.*, 1984; Downing *et al.*, 1991; Downing *et al.*, 1993); a value of 5.0 or more in a limb artery is considered normal. Stenoses create disturbance in the flow waveform downstream. This becomes apparent as spectral broadening of the recorded Doppler flow wave.

Indices of such perturbation of flow have been explored to quantify the degree of arterial stenosis (Moutet *et al.*, 1983; Campbell *et al.*, 1985).

NON-INVASIVE PRESSURE RECORDINGS: CLINICAL IMPLICATIONS

Prior to the introduction of the sphygmomanometer as used today, clinical studies of arterial pressure and of circulatory function were conducted with the aid of sphygmographs introduced by Marey (1863) (see also Snellen (1980)) and improved by Mahomed (1872) and Mackenzie (1902). Using these, Mahomed established the concept of essential hypertension and described its natural history, and Mackenzie wrote his influential text. Sphygmographs subsequently lost favor and were replaced by the cuff sphygmomanometer (for recording brachial, systolic, and diastolic pressure) and the electrocardiogram (for recording the heart beat).

Sphygmography attained a new lease of life in the 1970s, in association with phonocardiography for the measurement of systolic and diastolic time intervals. These were used to assess left ventricular function and coronary perfusion (Weissler *et al.*, 1963; Weissler *et al.*, 1968; Boudoulas *et al.*, 1979; Boudoulas *et al.*, 1981; Lewis *et al.*, 1981; Boudoulas *et al.*, 1982; Boudoulas, 1990; Boudoulas and Stefanadis, 2008). The work of Ferro *et al.* (1995) (see Chapter 17) draws further attention to the value of measuring diastolic time for assessment of patients with angina. In earlier studies, carotid pressure waves were measured with air-filled tambours or with microphones; the waves could not be calibrated, and attention was directed to the wave foot and incisura. This field lapsed with the introduction of echocardiography, which could provide the same information in a more comprehensive way (see Ferro *et al.*, 1995). The introduction of tonometry has provided a fresh impetus to this field.

The use of tonometry for measuring the peripheral arterial pressure wave, and for synthesizing the aortic pulse, is described in Chapters 6 and 30. Kelly, O'Rourke and colleagues in Sydney worked with Huntly Millar in Houston to develop, validate, and apply hand-held tonometry as an extension of the Millar micromanometer, which had long been the standard for the accurate measurement of the intra-arterial pressure wave contour (Chapter 6). Kelly *et al.* (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b) described, validated, and used the instrument to show aging change in a group of 1005 normal human subjects (Fig. 10.33). This study showed the characteristic differences between carotid and radial artery pressure waves originally described by Mahomed (1872) (Fig. 10.34), with the secondary systolic wave lower in the radial than in the carotid artery, and the gradual rise of this secondary systolic wave with age, such that it constituted the systolic peak of the carotid pressure wave by the fourth decade of

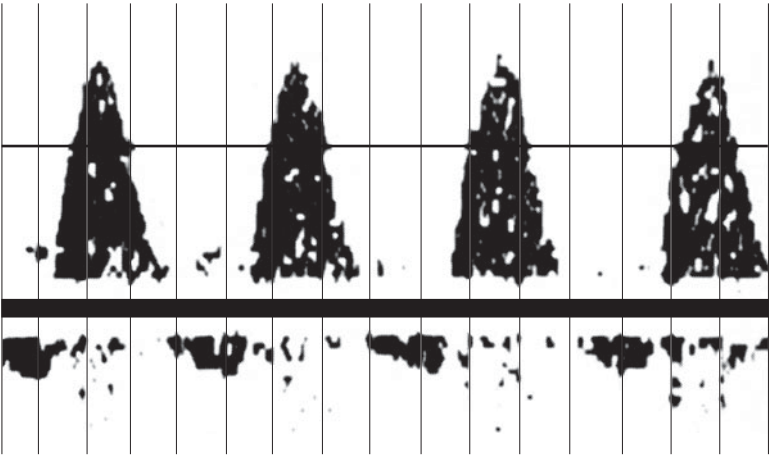


Figure 10.29 Doppler flow waveforms from the umbilical artery of a 33-week fetus with evidence of placental degeneration. Reproduced from Trudinger *et al.* (1985)

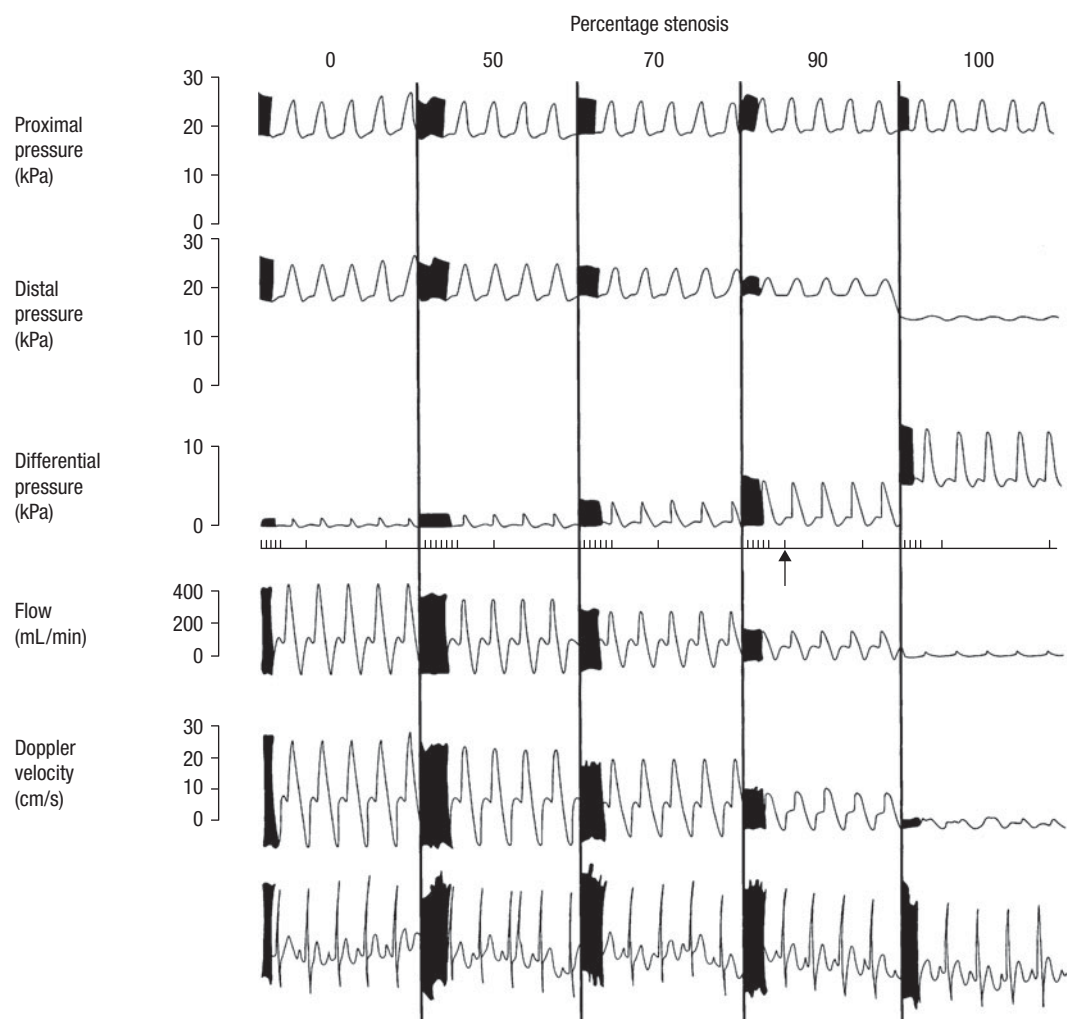


Figure 10.30 Recordings taken in a dog with graded occlusion of the femoral artery (left to right, with stenosis expressed in terms of decrease in cross-sectional area). Pressure recorded proximal and distal to the stenosis is displayed (in the first two tracings), together with differential pressure (third tracing). Also shown is volume flow (fourth tracing) measured with an electromagnetic flowmeter proximal to the stenosis, together with velocity flow measured with a Doppler instrument distal to the stenosis. Reproduced from Farrar *et al.* (1979)

life, but in the radial artery remained lower than the systolic peak, even in the eighth decade. This work on aging affirmed the previous findings of Mahomed (Mahomed, 1872; Mahomed, 1874), who also pointed out

the gradual increase of the secondary systolic wave in the radial pulse with aging and in hypertension. These differential findings for the radial and carotid pulse are attributable to the later arrival of reflected waves from the

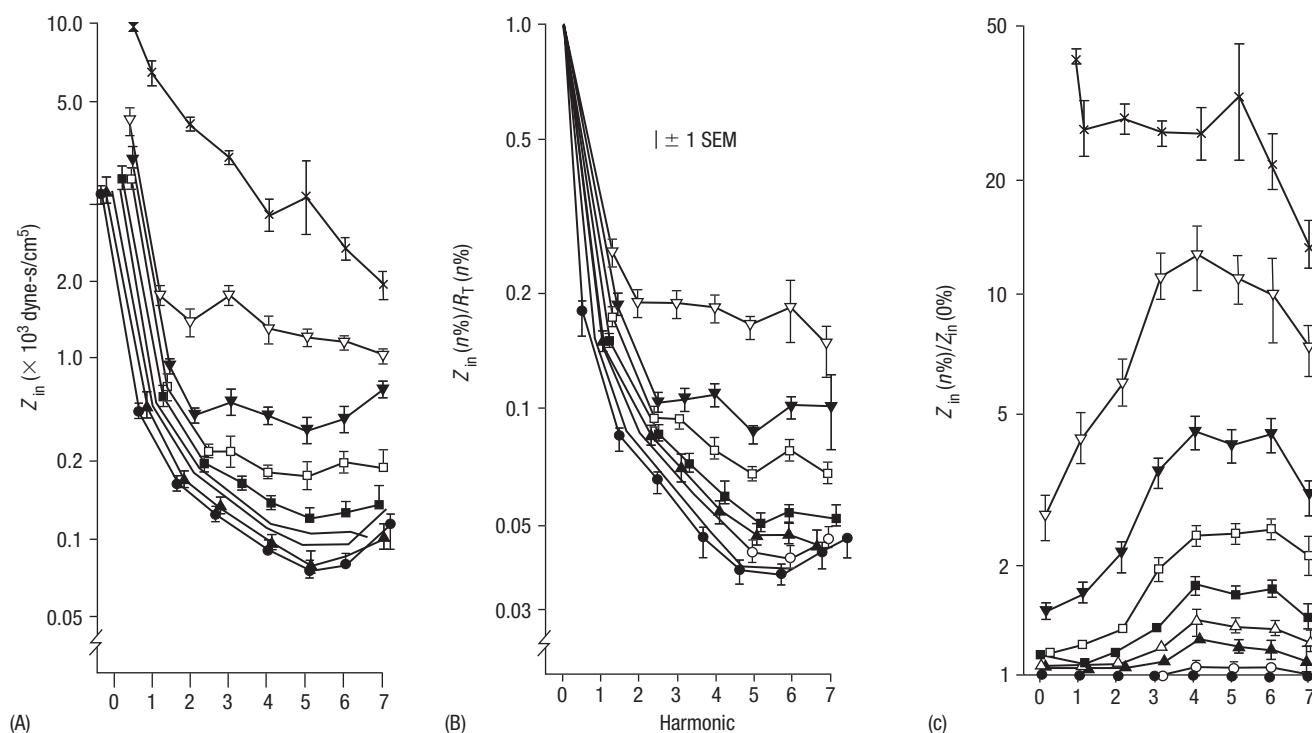


Figure 10.31 Femoral arterial impedance moduli spectra during graded occlusion of the femoral artery of a dog immediately downstream from the site of measurement. The modulus (Z_{in}) is expressed in three different ways: (A) in actual impedance units (pressure/flow), (B) as a ratio of impedance modulus to total resistance (R_T), and (C) as a ratio of the impedance modulus at n percent stenosis to that at zero stenosis. The stenosis percentage is expressed as an area change: (●) 0; (○) 20; (▲) 40; (△) 50; (■) 60; (□) 70; (▼) 80; (▽) 90; (×) 100. Reproduced from Farrar *et al.* (1979)

lower body at the radial than at the carotid site. The differences account for the differential effects of vasodilator agents, and especially nitroglycerin on systolic pressure in the central arteries (ascending aorta and carotid) compared with the radial and brachial (see Figs 10.26 and 10.27), as previously discussed.

Figure 10.33 shows greater prominence of the second systolic wave of both the radial and carotid pulse in the first compared with the second decade of life. This is attributable to the earlier return of wave reflection on account of the small body dimensions in children. Short body length (with a relatively long ejection duration) is the reason for the pressure wave in infants looking like that recorded in older human subjects (Hsieh *et al.*, 1989; Gevers, 1994) (see Chapter 19); early wave reflection is in one case due to short body length and in the other to arterial stiffening with greater wave velocity. London and colleagues have pointed out that early wave reflection is also a feature of people with short stature (Marchais *et al.*, 1993) and is a factor in the difference of pulse wave contour between men and women (see Chapter 9) (London *et al.*, 1995; Saba *et al.*, 1995a; Hayward and Kelly, 1997; Smulyan *et al.*, 2001).

Tonometry is now widely used for the measurement of the carotid pressure wave and for the determination of pressure wave augmentation at this site (Gatzka *et al.*, 2001a; Gatzka *et al.*, 2001b; Van Bortel *et al.*, 2001b; Safar *et al.*, 2002a; Mitchell *et al.*, 2004a). Augmentation has

been associated with increasing age and with hypertension, diabetes mellitus, and renal failure and also independently with carotid intima-medial thickness and left ventricular hypertrophy (London *et al.*, 1992; Saba *et al.*, 1993; Vaitkevicius *et al.*, 1993; Roman *et al.*, 1995b; Gottdiener *et al.*, 1996; London *et al.*, 2001; Safar *et al.*, 2002a; Brooks *et al.*, 2001). This subject is discussed further in Chapters 9, 28, 29, and 30.

A new instrument has been devised, using an array of tonometers that are applied over the wrist with a force which depends on the blood pressure measured by an oscillometric technique in the brachial artery (Kemmons *et al.*, 1991a; Kemmons *et al.*, 1991b; Sato *et al.*, 1993; Takazawa *et al.*, 1995a; Chen *et al.*, 1997) (see also Chapter 30). This provides a continuous record of phasic arterial pressure and is well suited to clinical situations where this information is desirable, but radial artery puncture is not warranted. This technique carries great promise for continuous non-invasive blood pressure recording.

The Finapres system discussed in Chapters 6 and 30 also provides continuous non-invasive recordings of the arterial pressure wave from the finger and is widely used for this purpose. The system uses a servocontrolled method to maintain finger volume constant, with the pressure recorded as the pressure required to achieve this. In this volume-clamp method, the size of the finger artery is gauged with a photoplethysmograph mounted inside a

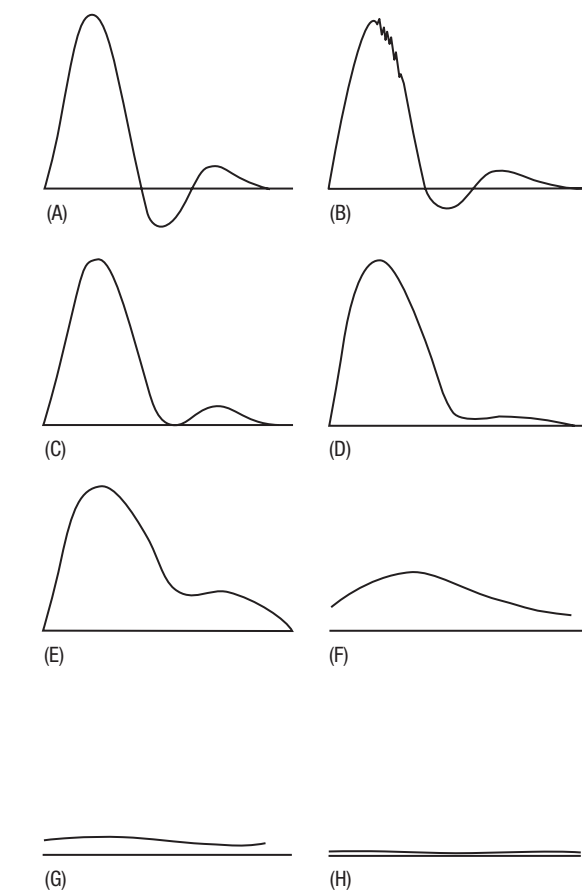


Figure 10.32 Change in typical flow velocity waveforms in a peripheral artery with progressive decrease in caliber to total occlusion (H) in a segment of artery (left, A, C, E, and G) above and below (right, B, D, F, and H) the site of occlusion. Reproduced from Johnston *et al.* (1981)

cuff. Mean pressure measured with this system tends to be lower and pulse pressure higher, than in the brachial artery, as expected from wave travel and reflection, but differences can be substantial in the presence of arterial

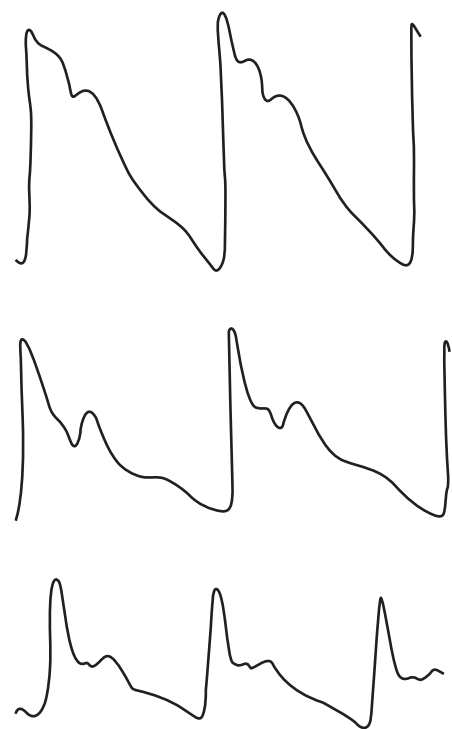


Figure 10.34 Pressure waves recorded non-invasively by sphygmography in the carotid artery (top), brachial artery (center), and radial artery (bottom) of a normal 22-year-old man. Reproduced from Mahomed (1872)

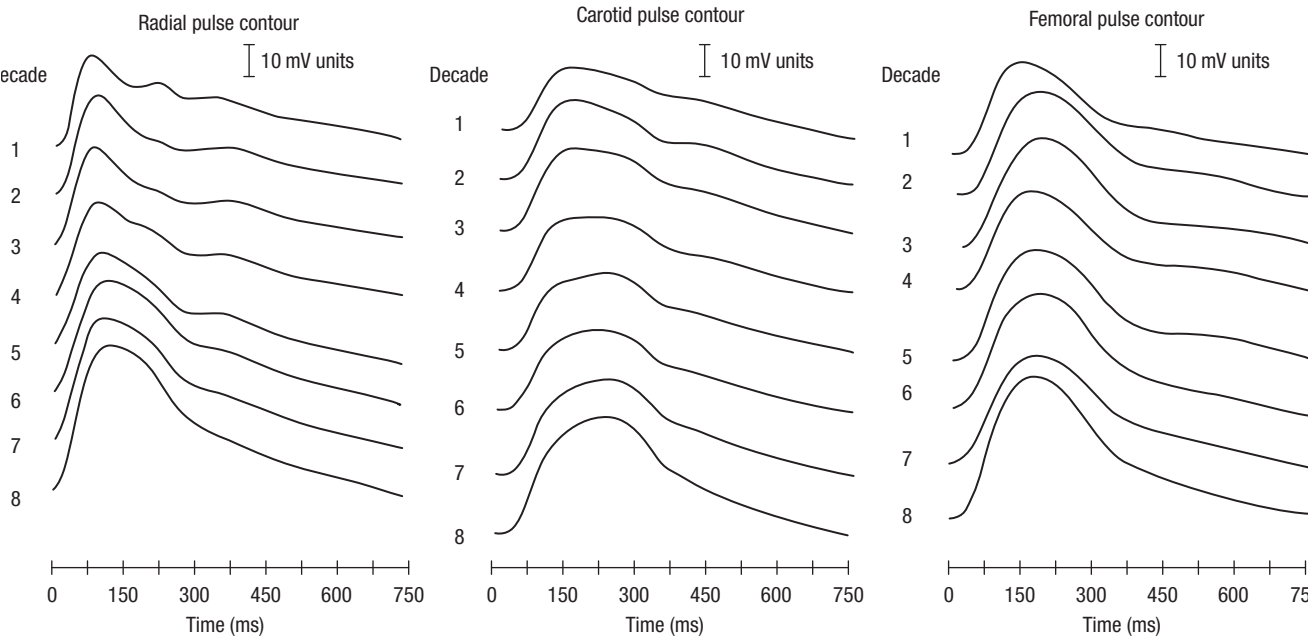


Figure 10.33 Pressure wave contour changes with age in the radial artery (left), carotid artery (center), and femoral artery (right). Data have been ensemble-averaged for eight different decades from 1005 normal human subjects. Reproduced from Kelly *et al.* (1989a)

disease or with altered arterial tone, or with change in heart rate (Imholz *et al.*, 1990a; Imholz *et al.*, 1990b; Bos *et al.*, 1992; Imholz *et al.*, 1992; Bos *et al.*, 1995; Rongen *et al.*, 1995). Criticism of the Finapres has been more severe than warranted; the same problems that pertain to pressure distortion between the brachial and finger arteries also pertain between the ascending aorta and brachial artery (O'Rourke and Gallagher, 1992), but the latter are generally overlooked.

Photoplethysmography is widely used for recording the arterial pulse by anesthetists and intensivists. It is not a recording of pressure, but a recording of volume change between sensors that are usually applied to the finger. Japanese investigators have studied the contour of the finger plethysmograph quite extensively (Chapter 30). Takazawa (1987) has shown that the finger photoplethysmograph shows systolic augmentation very similar to that which is recorded in the carotid artery, and so is

presumably similarly related to aortic augmentation and to vascular and ventricular hypertrophy. This subject is certainly worth pursuing, since finger plethysmography is far easier to apply than carotid tonometry.

The finger photoplethysmograph shows changes with age similar to those seen in the carotid arterial pulse, as recorded by tonometry (Takazawa, 1987; Millaseau *et al.*, 2002; Pannier *et al.*, 2002). A Japanese custom is to describe these changes in terms of the second derivative wave, or 'acceleration plethysmograph' (Takazawa *et al.*, 1988; Takazawa *et al.*, 1993a). Instruments are available in Japan for interpreting the finger photoplethysmograph according to this method. This field links modern medical practice with old Oriental customs and must provide fresh insights into the secrets carried by the arterial pulse (Dal *et al.*, 1985). This field has been extended by Chowienczyk *et al.* (1999) and Millaseau *et al.* (2002) in London (see also Chapter 30).

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Principles of recording and analysis of arterial waveforms

Recording arterial waveforms

255

Mathematical analysis of arterial waveforms

260

Physiological waveforms (or cardiovascular variables) recorded in the experimental or clinical cardiovascular laboratory are usually time-varying periodic functions, such as the electrocardiogram, ventricular volume, arterial diameter, blood pressure, and blood flow (mean ascending aortic blood flow is taken as cardiac output). Each variable has its own characteristic waveform, which changes relatively little from one beat to the next in a constant physiologic state and cardiac sinus rhythm. Various transducers are used to convert these physiologic events of force, motion, etc., to electrical signals that are amplified, monitored, recorded, processed, stored, and analyzed. Pressure waves measured in the periphery can be transferred to a central site where the morphology is different. A large amount of information can be obtained from simple inspection and analysis of the wave contour (Wiggers, 1928; Murgo *et al.*, 1980a; O'Rourke, 1982a; O'Rourke, 1982b; Khir *et al.*, 2001; Nichols and Edwards, 2001; Nichols and Singh, 2002; Soderstrom *et al.*, 2002; Denardo *et al.*, 2010) (see Chapters 10 and 30), but more specific quantified variables can be measured and computed. For example, the reduced rate of rise of aortic and left ventricular pressures in aortic stenosis and myocardial dysfunction can usually be recognized at a glance, but measurement of the actual derivative (dp/dt) is more informative than a qualitative description of the wave. Also, the change in shape of the central aortic pressure wave with changes in arterial stiffness and wave reflections is obvious from visual inspection, but the augmentation index is more informative (Nichols *et al.*, 2008). Recording of cardiovascular variables at the time of experimentation or cardiac catheterization is, therefore, only a first step to be followed by quantitative analysis. The goal of the analysis is to put the observations in a numerical form for easy tabulation and understanding.

RECORDING ARTERIAL WAVEFORMS

The objective of recording physiologic events is to obtain a faithful reproduction of the cardiovascular variable of interest. If this is not totally achieved, a measuring error must exist (Geddes and Baker, 1975; Nichols *et al.*, 1994c). Measuring errors may arise in any part of the recording system (e.g. the transducer, catheter, amplifier, recorder, storage device, or any other component). Therefore, when testing for accuracy, the entire recording system should be evaluated.

Transducers, the instruments used to convert force, motion, or other cardiovascular variables into an electrical signal, have already been discussed in Chapter 6. The present chapter, therefore, is concerned with amplifying, recording, displaying, processing, storing, and analyzing these signals. Any system designed for faithful reproduction of an event or variable must meet the following criteria: (1) amplitude linearity; (2) adequate frequency response; and (3) phase linearity (Geddes and Baker, 1975).

Amplitude linearity

Amplitude linearity refers to the ability of the transducer–processor–reproducer system to produce an output signal that is directly proportional in magnitude to the input signal amplitude. This condition must be satisfied for measurements both above and below the zero baseline and should include the entire range of measurement (Fig. 11.1). Thus, before static calibration and linearity testing, the approximate range of the variable to be measured should be known. This can usually be obtained from the literature or a few preliminary measurements. In Fig. 11.1, linearity of the electrical output of an

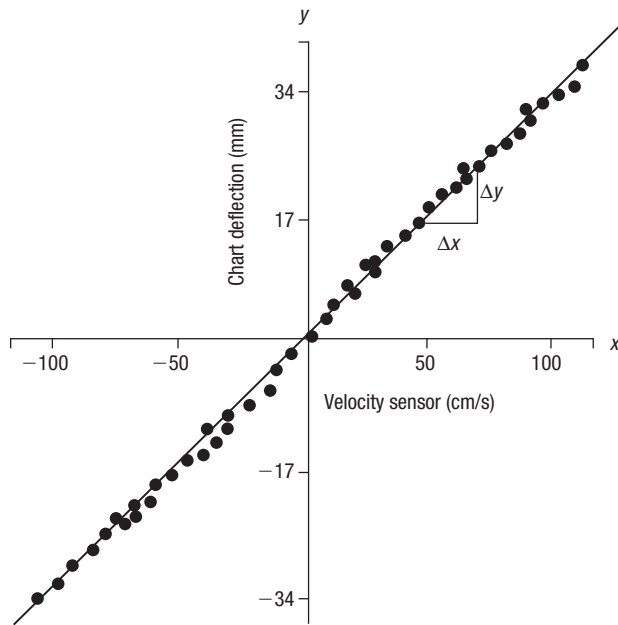


Figure 11.1 Linearity of the electrical output of a catheter-mounted electromagnetic velocity transducer/flowmeter system over the range of velocities recorded in normal humans. Linearity was determined in a bidirectional hydraulic pump model filled with physiologic saline. Reverse flow is indicated by a minus sign. The slope ($\Delta y/\Delta x$) of the regression line is the sensitivity. Reproduced from Nichols *et al.* (1985b)

electromagnetic flowmeter/catheter-tip velocity transducer system was tested in a bidirectional hydraulic pump model (O'Rourke, 1965b; Nichols *et al.*, 1985b). The velocity of the liquid (physiologic saline) detected by the electromagnetic velocity transducer, amplified by the flowmeter, and transmitted to the chart recorder was linear ($r=0.996$) over the range -103 to $+110$ cm/s, which covers the range of blood flow velocity measurements reported in the literature for normal humans (Nichols *et al.*, 1977a; Nichols *et al.*, 1977b; Murgu *et al.*, 1980a; Murgu *et al.*, 1980b; O'Rourke, 1982a; O'Rourke, 1982b; Laskey *et al.*, 1985; Nichols *et al.*, 1985a). Although the input-output characteristics of a system may be represented as a straight line, careful testing with accurate inputs usually reveals a small deviation from linearity. Under these conditions, regression analysis should be performed to obtain the line of best fit and this relation used in calibration. The slope ($\Delta y/\Delta x$) of this straight line at any point is the sensitivity.

HYSTERESIS

Another quality related to system linearity is hysteresis (the Greek word for lagging behind), which is a measure of the ability of the system to produce an output that follows the input independent of the direction or

change in magnitude of the input (Cobbold, 1974). In mechanical systems, hysteresis results from energy being absorbed by the system during loading and not being recovered during unloading. With most modern recording systems, the error due to hysteresis is less than 1.0 percent. Hysteresis characteristics also result from backlash in mechanical coupling, for example in hydraulic tube systems. If the tube is rigid and the flow sinusoidal, pressure and diameter are in phase, and the display on a dual beam oscilloscope is a straight line (Fig. 11.2A). If the tube is viscoelastic, diameter lags behind pressure and the display on the oscilloscope forms a 'hysteresis (or Lissajou) loop' (Fig. 11.2B). A greater phase lag (or difference) would increase the width of the loop (Fig. 11.2C). Applied to an artery *in vivo*, this method creates a hysteresis loop with each heartbeat (see Fig. 6.25).

NOISE

Another potential problem in recording systems is the addition of unwanted noise superimposed on the measured cardiovascular variable. The frequency of this noise is usually much higher than the variable being measured. It is part of the output of an instrument not related to the variable itself. Sometimes it is generated within the instrument; at other times, it is caused by external interference. In any recording system, it is necessary to distinguish the signal from the unwanted noise. This can be done by measuring the signal-to-noise ratio and calculating the noise figure (Yanof, 1972). The amount of noise in the output signal depends on the amount of noise amplified by the amplifier and presented to the output. The signal-to-noise ratio at the input divided by the signal-to-noise ratio at the output is the noise figure and should ideally equal unity.

Since the patient or experimental animal tends to act as an antenna, a spectral analysis of the noise (the amount of noise concentrated at each frequency) would show a rather large 60 Hz component. This is because most electrical instrumentation in the United States uses 60 Hz current, and this 60 Hz noise is radiated from a variety of wires and devices.

The best approach to handling unwanted noise is to identify its source and then eliminate it. In most cases, this can be accomplished by proper grounding. Every laboratory should be provided with a good reference ground. This means that electrostatic shields must be provided with an extremely low resistance path to the earth. In this way, excess electrons can travel to and from the earth, which acts like an infinitely large source of electrons and also an infinitely large sink into which electrons can flow (Yanof, 1972). All electronic instruments in the laboratory should be grounded to this reference ground point, care being taken to avoid an excessive number of ground leads. In some cases, it may be necessary to ground the patient

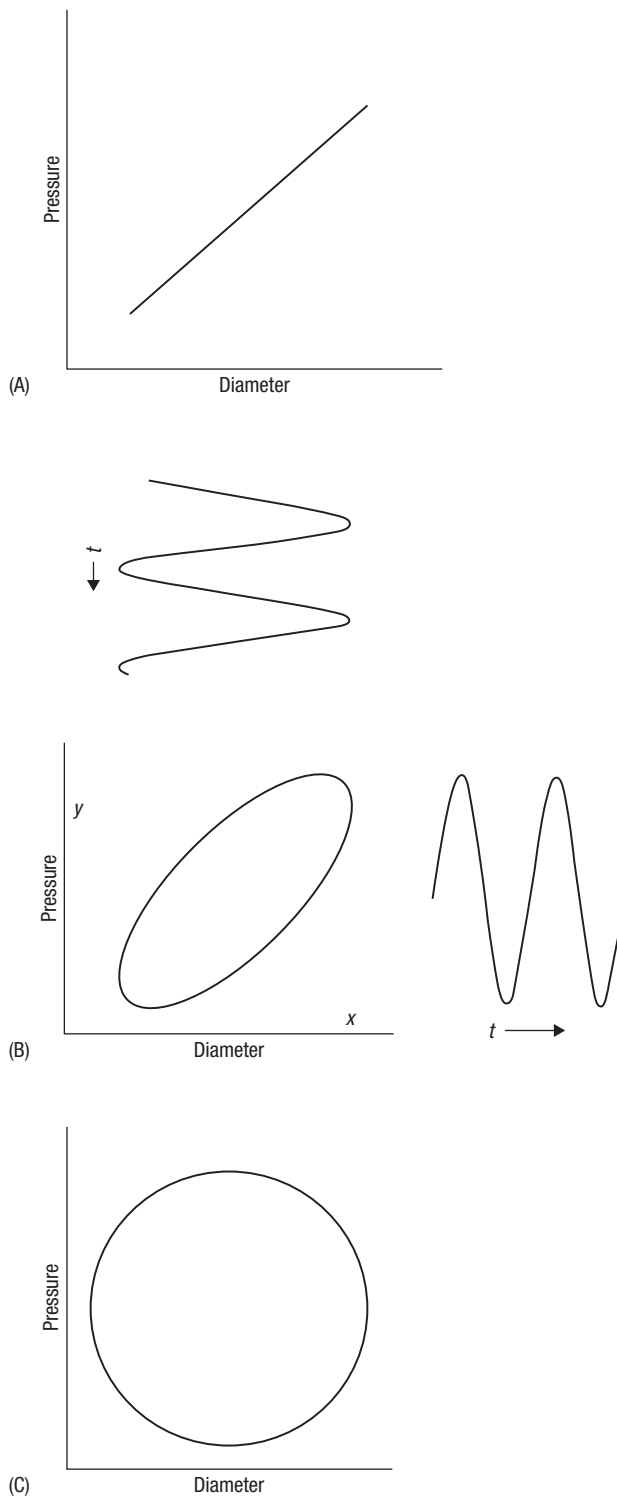


Figure 11.2 Hysteresis or Lissajou figures (or loops) obtained in a hydraulic model by displaying pressure and diameter (sinusoidal waves) simultaneously on the two axes of an oscilloscope. The sine waves are of the same frequency and amplitude. If the tube of the model is rigid, pressure and diameter are in phase, and the Lissajou figure is a straight line (A). However, if the tube is of viscoelastic material, pressure leads diameter, and the figure produced is an ellipse (B). A greater phase difference increases the width of the loop (C)

or experimental animal. However, the operating or catheterization table and the particular instrument must not be grounded to the reference point and the table then grounded to the instrument. If this is done, a ground loop will be established and the noise would increase considerably (Yanof, 1972). All ground wires should be arranged as symmetrically as possible and should be at least 12 gauge insulated wire.

High-frequency noise can also be eliminated or minimized by introducing a low-frequency bandpass electronic filter into the circuit. However, when adding electronic filters, care must be taken not to filter the cardiovascular variable itself (see below). Another approach to removal of noise from periodic signals is digital signal averaging. Signal averaging is commonly used to derive the signal-averaged electrocardiogram. In this application, efficient noise removal is crucial to record very low voltage afterpotentials. This degree of noise removal is not needed for routine hemodynamic recording; however, it is useful in research applications.

DRIFT

The tendency for recording systems to drift or gradually change their output can complicate a catheterization or experimental procedure. In most cases, instrumentation drift affects only the baseline calibration. If baseline drift does occur, the static calibration (or linearity) curve (see Fig. 11.1) moves vertically up or down without a change in slope (or sensitivity). If the calibration factor (or step) is affected, the sensitivity of the calibration curve changes. Baseline drift in a liquid-filled pressure-recording system can be detected and corrected by frequent checks of the 'zero baseline' by opening a stopcock and exposing the pressure transducer to atmospheric pressure. When measuring blood flow velocity with an electromagnetic flowmeter in the main pulmonary artery or ascending aorta, one assumes that the flat portion of the velocity curve in late diastole represents zero baseline since electronic zero cannot always be trusted (Milnor, 1989; Nichols and O'Rourke, 1990). Changes in the calibration factor can only be detected by recalibration. No matter how carefully a system was calibrated initially, erroneous measurements will be obtained if changes occur in either the zero baseline or the calibration factor. Fortunately, the recording systems used in modern laboratories are designed to minimize instrumentation drift and maintain constant calibration, but it is still unwise to assume system stability, and hemodynamic procedures should allow for periodic checking and readjustment if necessary. Some transducers, especially those used to measure blood pressure, are particularly sensitive to environmental temperature; therefore, the recording system should be turned on well in advance of any measurement. In a busy laboratory, it may be desirable to leave the recorder on during weekdays.

CALIBRATION

Calibration must be carried out over the range of input amplitudes to be measured (see above). Calibration involves determination of the response of the entire system including the recording apparatus and any other devices, such as strip chart recorders, oscilloscope monitors, magnetic tape recorders, and digital recording devices. Liquid-filled catheter–manometer systems, those most commonly used (especially in the human cardiac catheterization laboratory) for measuring blood pressure, are usually calibrated against a column of mercury. Calibration for every catheterization procedure is essential. Incremental increases in mmHg (input) are applied to the pressure transducer, and the resulting output signals are measured (Fig. 11.3). A plot of the input–output values results in a linear calibration relationship. Using the relationships of Figs 11.1 and 11.3, one obtains calibrated scales for the chart recording of arterial blood pressure and blood flow velocity signals. When the velocity profile is flat across the artery, the velocity signal can be calibrated in volumetric flow units by multiplying by the cross-sectional area of the vessel. Also, ascending aortic or main pulmonary artery velocity can be calibrated in volumetric flow units by comparison of the mean velocity signal to a simultaneous measurement of cardiac output (Nichols *et al.*, 1977a; Nichols *et al.*, 1977b).

Basic scientists and clinicians often need to know whether a new method of measurement is equivalent to an established (or ‘gold standard’) one already in clinical use. A short review of the methodology used in method-comparison studies is presented here. For a more in-depth evaluation of the methodology see reviews by Bland and Altman (Bland and Altman, 1999; Bland and Altman, 2003), Dewitte *et al.* (2002), Barnhart *et al.* (2007), and Smellie (2008). Currently available software eliminates the need for tedious statistical computation but does not reduce the burden of understanding the concepts underlying a method-comparison study and accurate

interpretation of the final results. Current guidelines for the combined graphical/statistical interpretation of method-comparison studies (Brazdzionyte and Macas, 2007; Haber and Barnhart, 2008) include a scatter plot combined with correlation and regression analysis and/or a difference plot combined with calculation of the limits of the differences between the methods (the so-called 95 percent confidence limits of the bias or ± 2 standard deviations) (Bland and Altman, 1986). The Bland–Altman plots (or Tukey mean-difference plots) are simple to construct and easy to interpret and are often used as a substitute for more rigorous statistical analysis. The general features of the Bland–Altman plot have been well described in the literature (Bland and Altman, 1986; Bland and Altman, 2003). The abscissa (x axis) shows the mean values of the two methods ($(A + B)/2$), whereas the ordinate (y axis) represents the absolute difference between the values of the two methods ($B - A$). Some investigators report their results only in Bland–Altman plots (Scolletta *et al.*, 2005; Compton *et al.*, 2008b), while others report their results in linear regression and Bland–Altman plots (Giromarelli *et al.*, 2004). A big mistake made by numerous investigators is comparing results to methods not considered as gold standard (Compton *et al.*, 2008b).

Adequate frequency response and phase linearity

The dynamic frequency response and phase linearity refer to the ability of the recording system to provide a signal that is identical (except in amplitude) to the event presented to it. The dynamic frequency response of a recording system must be great enough to detect the highest harmonic of the cardiovascular variable being recorded. Before the frequency response of such a system can be adequately discussed, it is necessary first to establish the relationship between sine waves and waves of

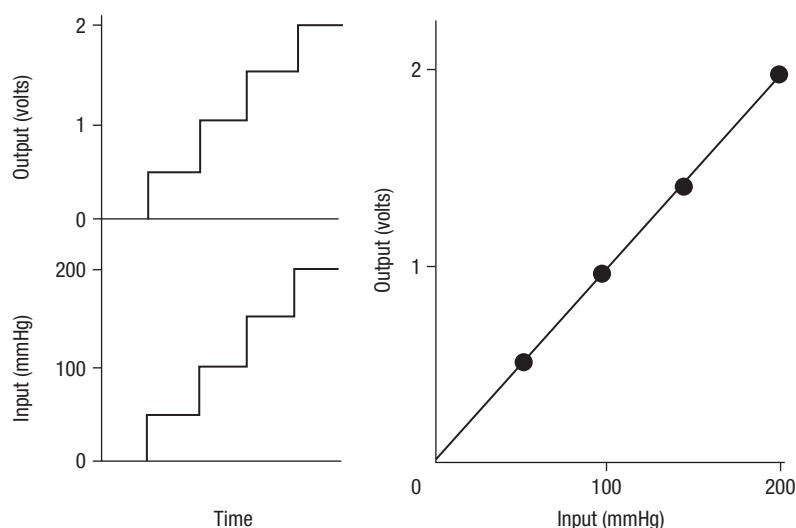


Figure 11.3 Calibration of a liquid-filled catheter–manometer system. Incremental increases in pressure are applied to the manometer and the resulting output voltage signal recorded (left). These input–output values are plotted as a linear relationship (right). Reproduced from Milnor (1989)

non-sinusoidal form. All periodic waves of non-sinusoidal form are designated complex waves (O'Rourke, 1982a; O'Rourke, 1982b; Milnor, 1989; Nichols and O'Rourke, 1998; Geddes, 1991). By use of Fourier series analysis, it is possible to show that any periodic complex wave, such as pressure, velocity, or diameter, can be dissected into a series of sine and cosine waves, which, when added, will produce the original complex wave. The technique of analyzing and synthesizing periodic waves using Fourier methods is discussed below.

The dynamic frequency characteristics of a recording system can be tested using a sine wave forcing function as input and observing the output (see Chapter 6). The output of the system should also be a sine wave of similar frequency and phase angle, but not amplitude. If the phase angle of the output differs from that of the input, it must be linear with frequency (Fig. 11.4A) so that a

correction factor can be introduced in analysis (Nichols and O'Rourke, 1990).

Phase distortion occurs when the phase angle between input and output signals is non-linear with frequency. A linear phase angle versus frequency characteristic by itself introduces no distortion of a signal; the signal is only displaced in time. As illustrated in Fig. 11.4B, the output left ventricular pressure signal is identical to that of the input signal, except that it is delayed in time by an amount that depends on the slope of the phase characteristics. A non-linear phase characteristic results in the output signal spectral components having the wrong time relationship, thereby distorting the output signal, as shown in the lower waveform in Fig. 11.4B. In multichannel recordings in which the time relation between the signals recorded on separate channels is significant, it is important that each channel has identical amplitude and phase characteristics.

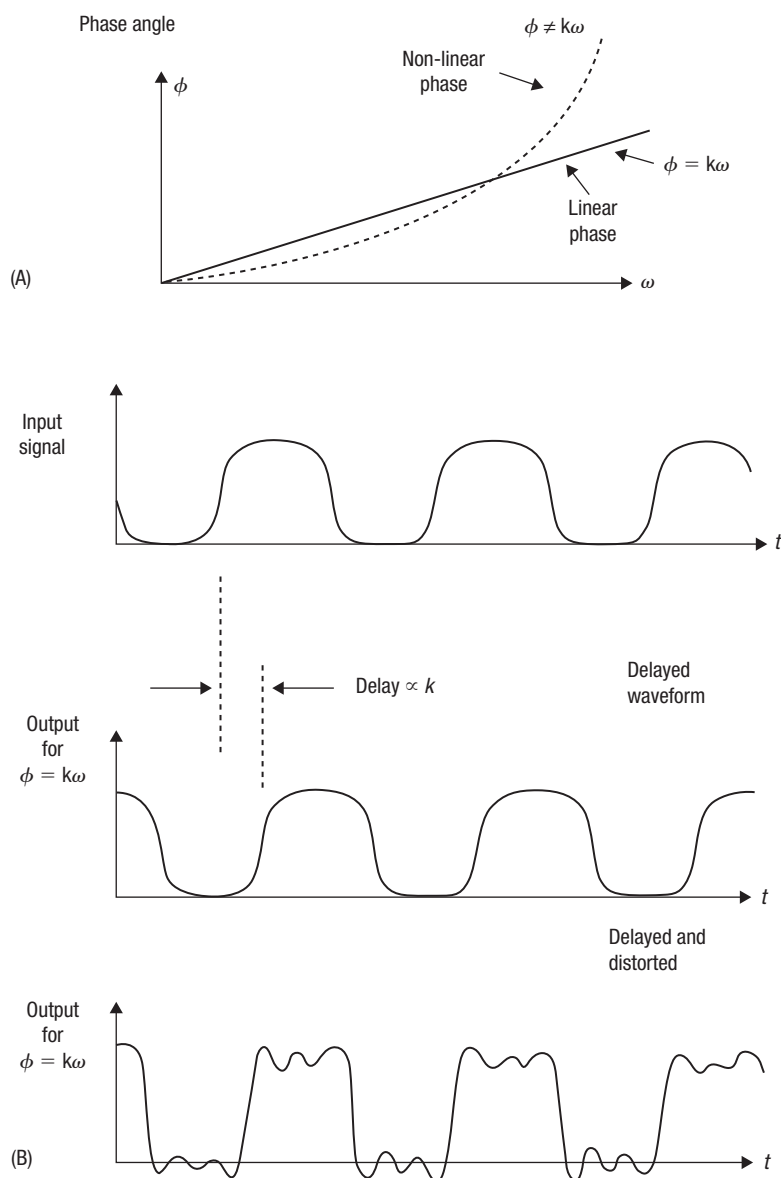


Figure 11.4 Diagram illustrating the effect of phase distortion in the absence of amplitude distortion: (A) System characteristics; (B) the effects of a linear and non-linear phase characteristic on the shape of a simulated left ventricular pressure wave

MATHEMATICAL ANALYSIS OF ARTERIAL WAVEFORMS

The computations carried out on the recorded physiologic signals are, in general, of three kinds: (1) solution of algebraic equations; (2) frequency analysis of the recorded signal; and (3) application of differential equations. Frequency analysis, which reduces periodic waveforms into the sum of certain sinusoidal waves, is a powerful technique for the investigation of linear systems and is discussed in detail below.

In Chapters 2 and 7, the problems of pulsatile liquid flow were all dealt with in the case of an oscillation in simple harmonic motion (i.e. of sinusoidal form). The conversion of repeating, or periodic, waveforms, such as pressure, flow, and diameter pulsations, to numerical form is performed by Fourier, or harmonic, analysis. This technique was named after Joseph Fourier (1768–1830), a French mathematician and confidant of Napoleon, who first undertook the systemic study of such series expansions in a memorable monograph *Theorie Analytique de la Chaleur* published in 1822. The use of such series in particular problems, however, dates from the time of Daniel Bernoulli (1700–82), who used them to solve certain problems connected with vibrating strings. Although this type of analysis was little used in cardiovascular physiology until introduced by Womersley (Womersley, 1955a; Womersley, 1955b; Womersley, 1955c; Womersley, 1955d) and McDonald (1955), it had been used theoretically by Frank (1927) in addition to the emphasis on transient phenomena in the Windkessel analysis. Fourier analysis was subsequently discussed by Wiggers (1928), Apéria (1940), and Porjé (1946), and has been used to study ventricular performance and arterial load in both experimental animals (Attinger *et al.*, 1966a; Attinger *et al.*, 1966b; Milnor *et al.*, 1966; Taylor, 1966c; Taylor, 1966d; O'Rourke and Taylor, 1967; Abel, 1971; Dobson *et al.*, 2006; Katsuda *et al.*, 2006; Syme *et al.*, 2006; Westerhof *et al.*, 2006; Reddy *et al.*, 2008a; Reddy *et al.*, 2008b; Ioannou *et al.*, 2009; Reddy *et al.*, 2009), humans (Mills *et al.*, 1970; Nichols *et al.*, 1977a; Nichols *et al.*, 1977b; Murgo *et al.*, 1980a; Murgo *et al.*, 1980b; Mazzaro *et al.*, 2005; Noda *et al.*, 2006; Segers *et al.*, 2006; Westerhof *et al.*, 2006; Morgan and Hosking, 2007; Segers *et al.*, 2007b; Senzaki *et al.*, 2007), and Windkessel (Quick *et al.*, 2006; Westerhof *et al.*, 2006; Burattini and Di Salvia, 2007) and multibranched (Westerhof *et al.*, 2006; Avolio, 2009) models (see Chapters 12 and 13). As this type of analysis is somewhat unfamiliar to many physiologists and cardiologists and others using Fourier methods, a general account of the nature and application of the Fourier series is given here, without any attempt at mathematical rigor, especially in relation to the type of curves we are concerned with, which cannot be defined as a function of an algebraic variable on which the processes of integration can be performed, but have to be dealt with as a finite sum of measured values.

A time-varying periodic function $F(t)$ is defined as one which repeats itself after a given time T , the period. This may be expressed as

$$F(t) = F(t + T).$$

This is familiar in the definition of a sinusoidal wave which repeats after 2π radians, so that $\sin x = \sin(x + n2\pi)$. As the frequency f is $1/T$, the angular frequency ω , in radians per second, is given by

$$\omega = 2\pi/T = 2\pi f. \quad 11.1$$

The Fourier theorem may be stated (Sokolnikoff, 1939; Hobson, 1957; Pipes, 1958; Wylie, 1960): if $y = F(t)$ is any continuous function of the variable t that repeats with a period T , then

$$\begin{aligned} F(t) = & \frac{1}{2}A_0 + A_1 \cos \omega t + A_2 \cos 2\omega t \\ & + A_3 \cos 3\omega t + \cdots + A_n \cos n\omega t \\ & + B_1 \sin \omega t + B_2 \sin 2\omega t \\ & + B_3 \sin 3\omega t + \cdots + B_n \sin n\omega t, \end{aligned} \quad 11.2$$

where n is an integer.

To evaluate the coefficients A_n and B_n , both sides of eqn 11.2 are first 'multiplied' by dt and then integrated between the limits 0 and 2π . Thus,

$$\begin{aligned} \int_0^{2\pi} F(t) dt = & \frac{1}{2}A_0 \int_0^{2\pi} dt + A_1 \int_0^{2\pi} \cos \omega t dt \\ & + \cdots + A_n \int_0^{2\pi} \cos n\omega t dt \\ & + B_1 \int_0^{2\pi} \sin \omega t dt + \cdots \\ & + B_n \int_0^{2\pi} \sin n\omega t dt, \end{aligned} \quad 11.3$$

and every right hand side (RHS) term except the first is zero, so that

$$\int_0^{2\pi} F(t) dt = 2\pi A_0 \quad 11.3a$$

or

$$A_0 = \frac{1}{2\pi} \int_0^{2\pi} F(t) dt. \quad 11.3b$$

To evaluate A_n , both sides of eqn 11.2 are multiplied by $\cos n\omega t \, dt$ and integrated, so that we obtain

$$\begin{aligned}
 & \int_0^{2\pi} F(t) \cos n\omega t \, dt \\
 &= \frac{1}{2} A_0 \int_0^{2\pi} \cos n\omega t \, dt \\
 &+ A_1 \int_0^{2\pi} \cos \omega t \cos n\omega t \, dt \\
 &+ \cdots + A_n \int_0^{2\pi} \cos^2 n\omega t \, dt \\
 &+ B_1 \int_0^{2\pi} \sin \omega t \cos n\omega t \, dt \\
 &+ \cdots + B_n \int_0^{2\pi} \sin n\omega t \cos n\omega t \, dt,
 \end{aligned} \tag{11.4}$$

and every RHS term except the one containing $\cos^2$ is zero, giving

$$\begin{aligned}
 & \int_0^{2\pi} F(t) \cos n\omega t \, dt \\
 &= A_n \int_0^{2\pi} \cos^2 n\omega t \, dt \\
 &= \frac{A_n}{2} \int_0^{2\pi} (1 + \cos 2n\omega t) \, dt = \pi A_n
 \end{aligned}$$

so that

$$A_n = \frac{1}{\pi} \int_0^{2\pi} F(t) \cos n\omega t \, dt \tag{11.5}$$

and, similarly,

$$B_n = \frac{1}{\pi} \int_0^{2\pi} F(t) \sin n\omega t \, dt. \tag{11.6}$$

We can also see that $A_0/2$ is the mean value of the function over the period T , or from 0 to 2π . Equations 11.5 and 11.6 are known as the Euler (1707–83) or Euler–Fourier formulae, and the series 11.2 is known as the Fourier series of $F(t)$ (Wylie, 1960).

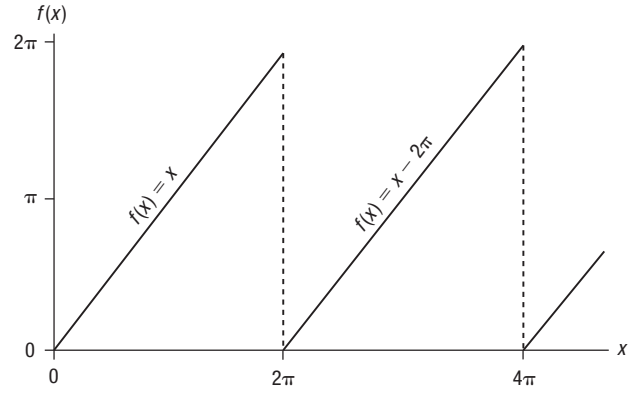


Figure 11.5 A simple repetitive 'sawtooth' function

As an example, let us take the sawtooth function (Fig. 11.5) that repeats with a period of 2π and has the form

$$f(x) = \begin{cases} x & \text{for } 0 \leq x \leq 2\pi \\ x - 2\pi & \text{for } 2\pi \leq x \leq 4\pi; \text{ etc.} \end{cases}$$

Thus, for the first linear portion of the function (for $f(x) = x$):

$$A_0 = \frac{1}{2\pi} \int_0^{2\pi} x \, dx = \pi$$

and

$$A_n = \frac{1}{\pi} \int_0^{2\pi} x \cos nx \, dx = \frac{1}{n\pi} \int_0^{2\pi} x \, d(\sin x).$$

Integrating by parts we obtain

$$A_n = \frac{1}{n\pi} \left[(x \sin x) \Big|_0^{2\pi} - \int_0^{2\pi} \sin nx \, dx \right],$$

and both these terms are zero so that there are no cosine terms in the expansion. Analogously, we write

$$\begin{aligned}
 B_n &= \frac{1}{\pi} \int_0^{2\pi} x \sin nx \, dx \\
 &= -\frac{1}{n\pi} \left[(x \cos x) \Big|_0^{2\pi} - \int_0^{2\pi} \cos nx \, dx \right] \\
 &= -\frac{2}{n} \cos 2n\pi,
 \end{aligned}$$

and so we have

$$\begin{aligned}
 f(x) = x &= \pi - 2 \left[\sin x + \frac{1}{2} \sin 2x \right. \\
 &\quad \left. + \frac{1}{3} \sin 3x + \cdots + \frac{1}{n} \sin nx \right].
 \end{aligned}$$

The progressive summation of the first three terms of this function is shown in Fig. 11.6, which demonstrates how a straight line may be approximated by a small number of terms of a Fourier series; accuracy of fit increases as the

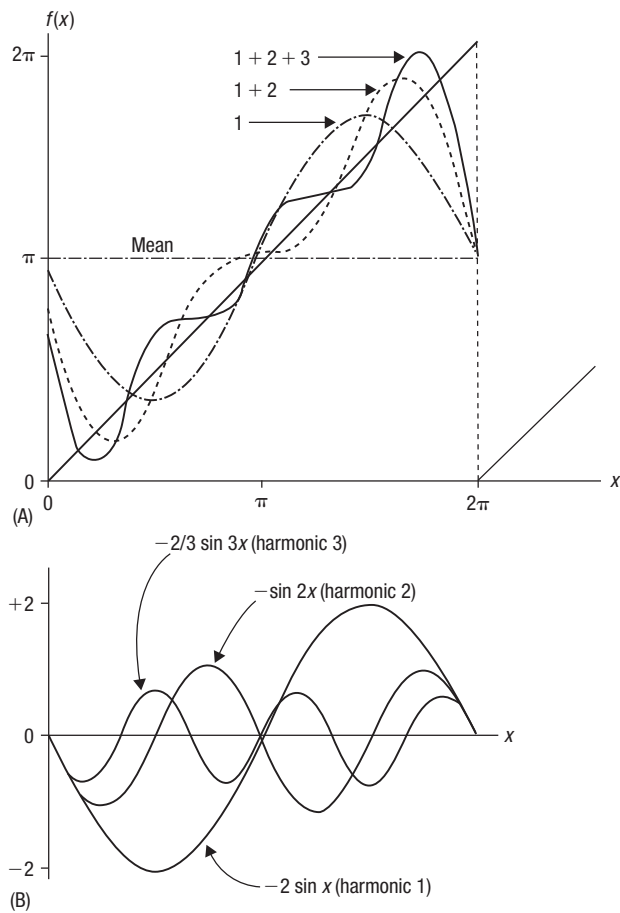


Figure 11.6 (A) The partial Fourier resynthesis of the function shown in Fig. 11.5. The curve labeled (1) represents the mean plus the first harmonic, i.e. $\pi - 2\sin x$; that labeled (1+2) has the second harmonic added, i.e. $\pi - 2\sin x - \sin 2x$; and that labeled (1+2+3) has the third harmonic added, i.e. $\pi - 2\sin x - \sin 2x - 2/3(\sin 3x)$. (B) Individual harmonic components, with the mean term omitted

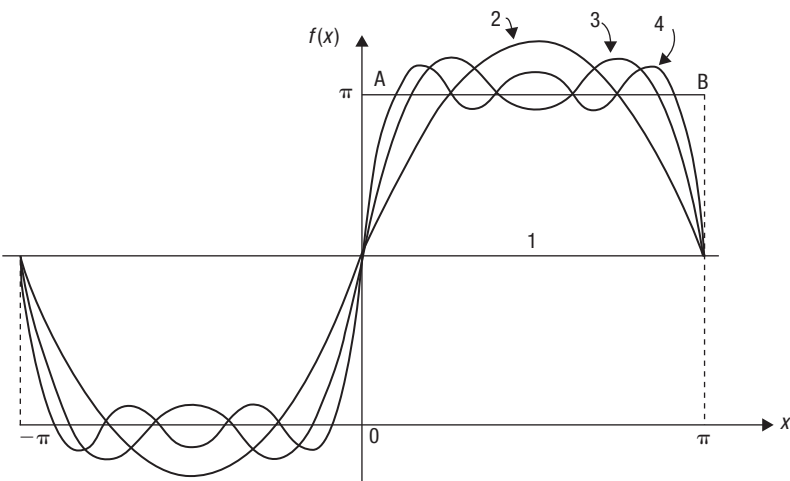


Figure 11.7 A square wave occupying half a cycle length. Note the overshoot at the discontinuity. The Fourier equation for this curve is shown in the text, and it can be seen that only odd-numbered harmonics are present; nevertheless, the sum of the first three harmonics represents 93.4 percent of the variance of the whole curve. The first term ($\pi/2$) of the equation is the mean value of the function and is represented by the number 1; the second term is $2\sin x$, etc

number of harmonic terms increase. This is true for any periodic function and is discussed below.

In a similar way, a square wave that, for example, has a value of π in the interval 0 to π and of 2π in the interval 2π to 0 can be analyzed as the sum of the respective intervals. This expansion gives

$$f(x) = \frac{\pi}{2} + 2 \left[\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right], \quad 11.7$$

and the synthesis of these terms is shown in Fig. 11.7. In both this case and that illustrated in Fig. 11.6, it can be observed that there are discontinuities in the function. Thus, the square wave has both the values of zero and $\pi/2$ when $x=0$ and $x=\pi$. The Fourier synthesis will then always have a value that is a mean of these two values at these points; this is a general property of the series. The rule concerning discontinuities is generalized as the 'Dirichlet (1805–59) condition', which states that 'a function with a finite number of maxima and minima, and a finite number of points of discontinuity may be represented by a Fourier series' (Pipes, 1958; Wylie, 1960). The illustration in Fig. 11.7 also shows that, at the jump of the discontinuity, the graph of the series overshoots at both extremes; this is known as the Gibbs phenomenon (Hobson, 1957).

The use of Fourier series to analyze a periodic function such as the pressure wave shown in Fig. 11.8 can best be demonstrated by a simple example. One pulse (or beat) is selected for analysis, and its period (or cycle length, T) is divided into N equally spaced ordinates or intervals. An easily recognizable feature of the pressure wave, such as the very beginning of the upstroke, is usually chosen as the first point to be measured, and in a perfectly repetitive (or periodic) wave, the first point of the next cycle will have the same value. Usually, the ECG is recorded simultaneously with the hemodynamic variables, and the peak of the QRS complex is used as a trigger to start and stop the digital sampling.

In this example, $T=0.94$ s, $N=12$, and $\Delta t=0.078$ s (in general, the sampling rate is 100 samples per second).

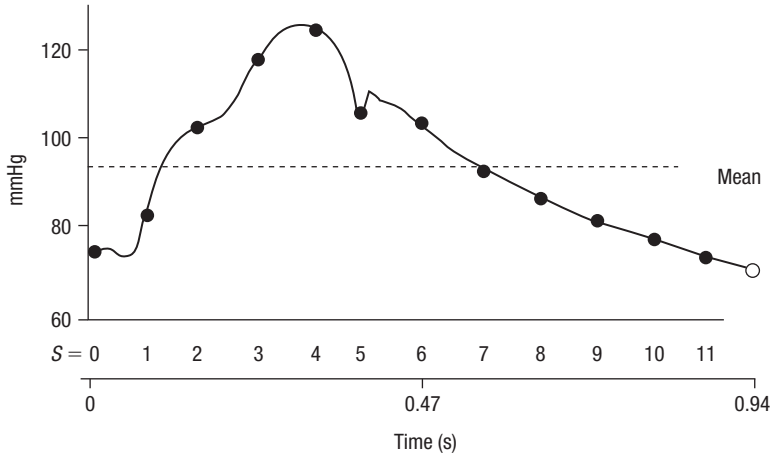


Figure 11.8 Ascending aortic pressure wave illustrating translation of a continuous curve into a series of digitized values. The first step in calculation of a Fourier series for this wave is measurement of pressures at regular intervals of time, Δt . The total duration (or period, T) of the wave is 0.94 s; a time interval (Δt) of 0.078 s has been selected for purposes of illustration, so that 12 points (closed circles) are measured. The values of pressure at these 12 points are used to calculate the Fourier coefficients and moduli and phase angles, with the results shown in Fig. 11.1

The procedure identifies 12 ordinates, designated in the figure by the letter S (0 through 11) so that the time marked by each ordinate is $t_s = S\Delta t$. The calibrated value of each ordinate (y_s) is measured, usually by electronic (or graphics tablet) analog-to-digital conversion, and the Fourier coefficients, A_n and B_n , can then be calculated from

$$A_n = \frac{2}{k} \sum_{s=0}^k y_s \cos n\omega t_s \quad \left(\text{for } n = 0, 1, 2, \dots, \frac{k}{2}\right) \quad 11.8$$

$$B_n = \frac{2}{k} \sum_{s=0}^k y_s \sin n\omega t_s \quad \left(\text{for } n = 0, 1, 2, \dots, \frac{k}{2}\right).$$

The values of y_s are now given by the finite series

$$y_s = \frac{A_0}{2} + \left\{ \sum_{n=1}^{k/2} A_n \cos n\omega t_s + \sum_{n=1}^{k/2} B_n \sin n\omega t_s \right\}. \quad 11.9$$

It will be recalled that a function of the form of a sine or cosine constitutes a simple harmonic motion (SHM), which is also defined as the function of the movement, with respect to time, of the foot of a perpendicular drawn from the radius vector of a circle, on to the x -axis, as the vector rotates counterclockwise around the circle. Each pair of terms $A_n \cos n\omega t$ and $B_n \sin n\omega t$ in eqn 11.2 may thus be represented graphically (Fig. 11.9A) as two such rotations represented simultaneously. The sum of these two functions represents the SHM of a single vector of length M starting its rotation at an angle ϕ (Fig. 11.10B). From Pythagoras' theorem, we can see that

$$M = (A^2 + B^2)^{1/2} \quad 11.10$$

and

$$\tan \phi = B/A \quad \text{or} \quad \phi = \tan^{-1}(B/A). \quad 11.11$$

Thus, we may write

$$A \cos \omega t + B \sin \omega t = M \cos(\omega t - \phi), \quad 11.12$$

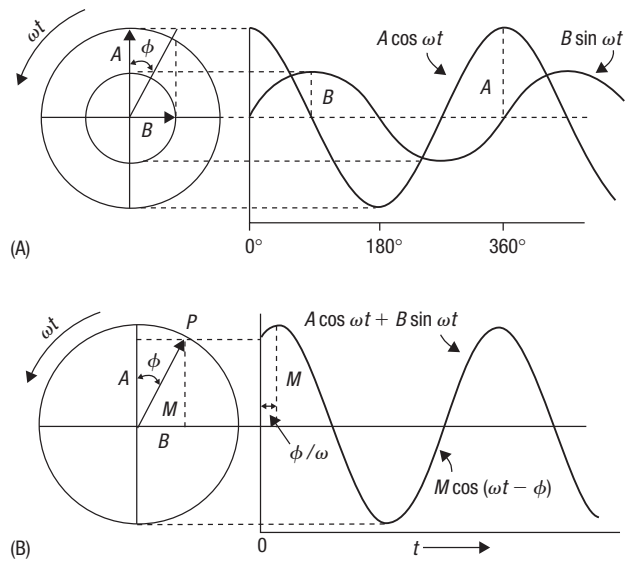


Figure 11.9 Diagrammatic representation of a simple harmonic motion in terms of its Fourier coefficients. (A) A pair of Fourier terms $A \cos \omega t$ and $B \sin \omega t$; for convenience the sine term is drawn as if it were a cosine projection 90° behind the true cosine component. Thus, at $\omega t = 0^\circ$ or 360° , $\cos \omega t = A$ and $\sin \omega t = 0$; when $\omega t = 90^\circ$ the two vectors A and B have rotated counterclockwise through that angle and $\cos \omega t = 0$ and $\sin \omega t = B$; with $\omega t = 180^\circ$, the cosine is $-A$ and the sine zero. The waves traced out by the points moving around the circles are shown at the right. The sum of these two curves is shown in the oscillation at the right of the bottom portion of the figure. (B) The circular diagram shows how the resultant single oscillation described in (A) can be described as a single wave; its initial point is at $-\phi$ and the amplitude is M . It is thus the curve of $M \cos(\omega t - \phi)$. It can be seen that $M = (A^2 + B^2)^{1/2}$ and that $\tan \phi = B/A$

where M is called the modulus, or amplitude, of the oscillation and ϕ its phase, or argument.

From Fig. 11.10, we also see that

$$A = M \cos \phi \quad \text{and} \quad B = M \sin \phi. \quad 11.13$$

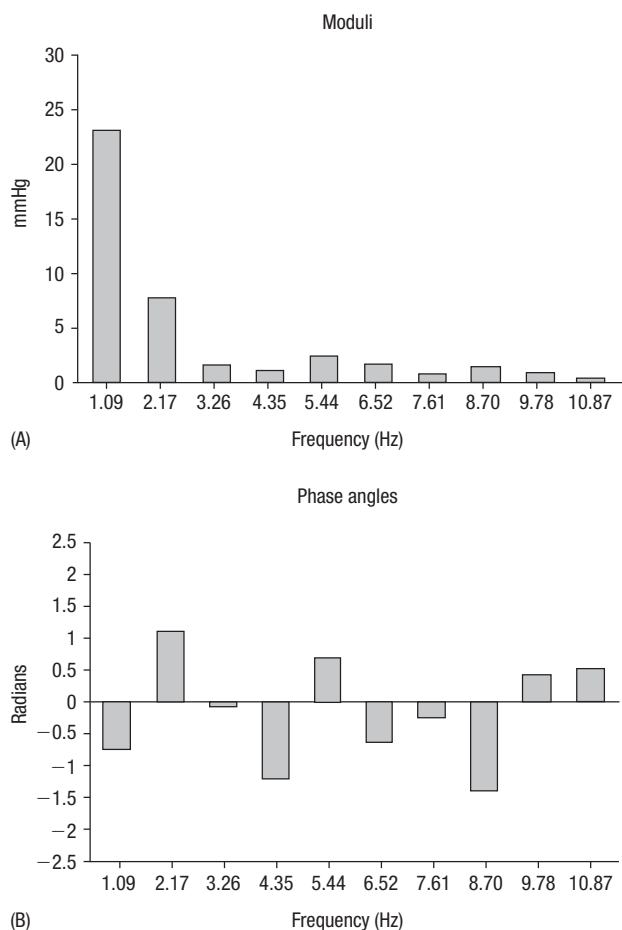


Figure 11.10 Moduli (A) and phase angles (B) calculated from the pressure wave shown in Fig. 11.8 for six harmonics. Abscissa: frequency (i.e. heart rate in Hz $\times$ harmonic number). Moduli and phase angles were calculated using eqns 11.9, 11.10, 11.11, and 11.12. For the first harmonic (or fundamental), $M = 23.1$ mmHg, and $\phi = -0.8$ radian. The mean term, calculated using eqn 11.8, indicates a mean blood pressure of 96.5 mmHg

To demonstrate the equivalence, we expand the RHS of eqn 11.12:

$$\begin{aligned}
 M \cos(\omega t - \phi) &= \cos \phi \cos \omega t + M \sin \phi \sin \omega t \\
 &= A \cos \omega t + B \sin \omega t,
 \end{aligned}
 \quad 11.14$$

which is the simplest proof that the angle ϕ in this equivalence must be written as $-\phi$. If, as is often done, eqn 11.14 is written as $\cos(\omega t - \phi)$, then $\tan \phi = -B/A$.

The results of Fourier series analysis are usually displayed as a graphic spectrum, plotting the moduli (M) and phase angles (ϕ) as functions of frequency (or harmonic number) (Fig. 11.10). Data (moduli and phase angles) are obtained and plotted only at the harmonic frequencies and are called 'line spectra'. Values of moduli and phase angles are most often connected (as in

impedance spectra), forming a continuous line (see Chapter 12) when, in fact, the spectrum of a Fourier series is not continuous.

In Chapters 2 and 7, the harmonic motion is often written in complex number notation. By use of the operator $i = (-1)^{1/2}$, the point P can be defined as $(A + iB)$. Here A is a length measured along the x , or real, axis and is the real part of the complex number; B is measured along the y , or imaginary, axis and is called the imaginary part of the complex number. The manipulation of complex numbers may be summarized by saying that this is identical to that of the more familiar algebraic symbols, but the real and imaginary parts must always be kept separate. All complex numbers have, as it were, a 'mirror image' in the x -axis, and this is called the 'complex conjugate'; for example, the complex conjugate of $(A + iB)$ is $(A - iB)$. The algebraic manipulation of complex numbers necessitates using the properties of i (i.e. $i^2 = -1$, $i^3 = -i$, $i^4 = +1$, etc.); thus, the product of $(A + iB)$ with its complex conjugate $(A - iB)$ is $A^2 + B^2$, a real number. This equivalence is used when dividing by a complex number: both numerator and denominator are multiplied by the conjugate of the latter so that the denominator becomes real.

The use of i in exponents may be explained by referring to the series expansion of e^x :

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots, \quad 11.15$$

so that

$$e^{ix} = 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \cdots + \frac{(ix)^n}{n!} + \cdots \quad 11.16$$

and

$$e^{-ix} = 1 + ix - \frac{x^2}{2!} + \frac{ix^3}{3!} + \cdots + \frac{(-ix)^n}{n!} \quad 11.17$$

so that

$$\begin{aligned}
 e^{ix} + e^{-ix} &= 2 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \right) \\
 &= 2 \cos x
 \end{aligned}
 \quad 11.18$$

and

$$\begin{aligned}
 e^{ix} - e^{-ix} &= 2i \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \right) \\
 &= 2i \sin x.
 \end{aligned}
 \quad 11.19$$

Hence

$$\begin{aligned}\cos x &= \frac{(e^{ix} + e^{-ix})}{2} \\ \sin x &= \frac{(e^{ix} - e^{-ix})}{2i}.\end{aligned}\quad 11.20$$

Thus

$$e^{ix} = \cos x + i \sin x, \quad 11.21$$

and a Fourier term such as in eqn 11.12 may be written as the real part of

$$Me^{i(\omega t - \phi)}, \text{ i.e. } A \cos \omega t + B \sin \omega t, \quad 11.22$$

while its complex number equivalent will be $A + iB$.

This relationship leads to a condensed way of writing a Fourier series in exponential form. Equation 11.2 above is the equivalent of

$$F(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} (A_n \cos n\omega t + B \sin n\omega t). \quad 11.23$$

From eqns 11.20 and 11.21, we may rewrite eqn 11.23 as

$$\begin{aligned}F(t) &= \frac{A_0}{2} + \frac{1}{2} \sum_{n=1}^{\infty} (A_n - iB_n)e^{in\omega t} \\ &\quad + \frac{1}{2} \sum_{n=1}^{\infty} (A_n + iB_n)e^{-in\omega t}.\end{aligned}\quad 11.24$$

This can be simplified by introducing negative values of n ; since $\cos(-\theta) = \cos \theta$, and $\sin(-\theta) = -\sin \theta$, then also $A_{-n} = A_n$, and $B_{-n} = -B_n$. The last term of eqn 11.24 can then be changed, thus

$$\begin{aligned}&\frac{1}{2} \sum_{n=1}^{\infty} (A_n + iB_n)e^{-in\omega t} \\ &= \frac{1}{2} \sum_{n=1}^{-\infty} (A_n - iB_n)e^{in\omega t},\end{aligned}\quad 11.25$$

and eqn 11.24 simplifies to

$$\begin{aligned}F(t) &= \frac{A_0}{2} + \frac{1}{2} \sum_{n=-\infty}^{+\infty} (A_n - iB_n)e^{in\omega t} \\ &= \frac{1}{2} \sum_{n=-\infty}^{+\infty} F(n)e^{in\omega t}.\end{aligned}\quad 11.26$$

By a similar process, writing the function in terms of the period T , we find that

$$F(n) = \frac{1}{T} \int_{-T/2}^{+T/2} F(t)e^{-in\omega t} dt. \quad 11.27$$

Equations 11.26 and 11.27 are defined as the Fourier transforms of each other. In amplitude and phase form,

$$F(n) = (A_n^2 + B_n^2)^{1/2} \exp \left[i \tan^{-1} \left(\frac{-B}{A} \right) \right]. \quad 11.28$$

TRAVELING WAVES

The exponential form of a Fourier series is also a useful way of compactly designating a traveling wave. Thus, an undamped wave traveling at velocity c in the direction of the z -axis (conventionally denoting the axis of a tube) at any distance z_1 from the origin is defined (for $n = 1$) as

$$M e^{i\omega(t - z_1/c)},$$

and later at z_2 as $M e^{i\omega(t - z_2/c)}$ or, in terms of eqn 11.22, as $M e^{i(\omega t - \phi_1)}$ and $M e^{i(\omega t - \phi_2)}$. The interval $(\phi_2 - \phi_1)$ is called the phase shift, so that

$$(\phi_2 - \phi_1) = (\omega/c)(z_2 - z_1),$$

or

$$c = \frac{\omega(z_2 - z_1)}{(\phi_2 - \phi_1)} = \omega \left(\frac{\Delta z}{\Delta \phi} \right). \quad 11.29$$

The velocity c is termed the 'phase velocity'.

Requirements and limitations of a Fourier analysis

The number of samples that need to be taken depends on the frequencies present in the waveform and on the number of harmonics that it is desired to determine. The limiting rule for the latter is simple: if N intervals are taken in one cycle, the number of harmonics that can be determined is $N/2$. All higher harmonics are indeterminate. Thus, if 12 points (see above example) are taken (a manual method given in Franklin (1958)), six harmonics would be valid. When the sampling rate is set per second, the same rule applies to the maximum frequencies that are analyzable; that is, 100 samples per second allows up to 50 Hz to be analyzed. The number of harmonics this represents depends on the heart rate; if this is 2.0 Hz (2.0 beats per second or 120 beats per minute), it is 24 harmonics, whereas with a rate of 3.0 Hz it is only 16 harmonics. If there are frequencies present beyond the limit set by the sampling rate (the 'folding' or Nyquist frequency), they will introduce errors into the components analyzed. This phenomenon is known as 'aliasing' and is discussed by Blackman and Tukey (1959) and by Attinger

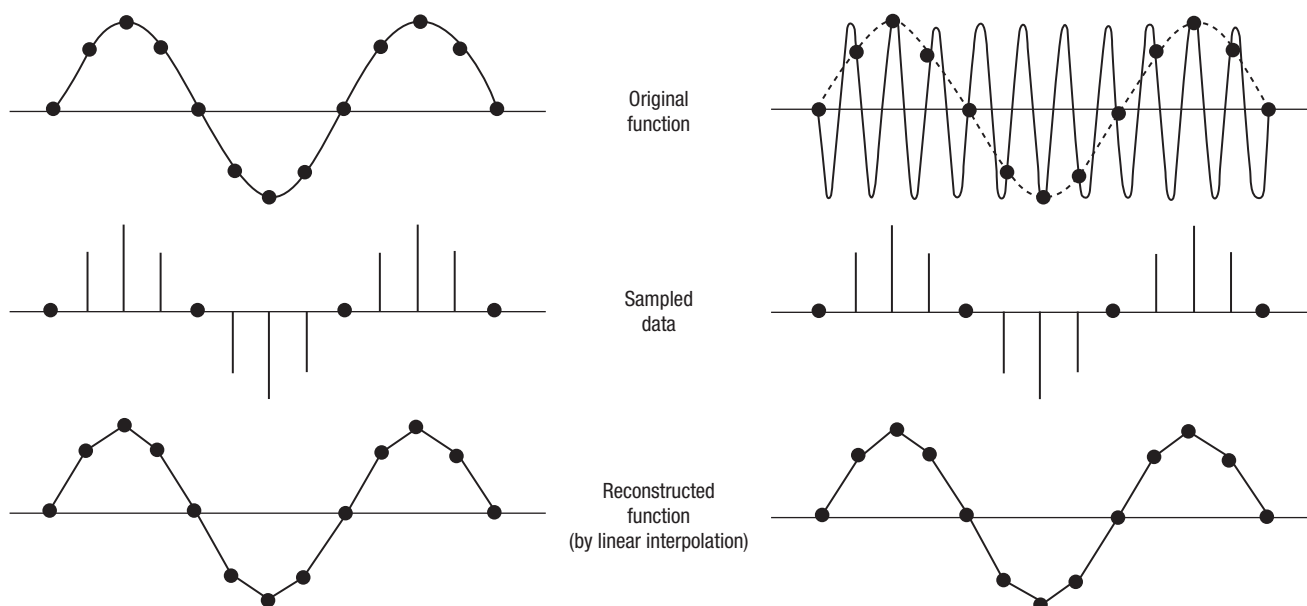


Figure 11.11 Diagram to illustrate aliasing error. The sine wave signal at the left is sampled at a rate of greater than two per cycle, while the one at the right is sampled at $7/8$ of the sampling frequency. Both give rise to identical reconstructed waveforms. Reproduced from Gardenhire (1964)

et al. (Attinger *et al.*, 1966a; Attinger *et al.*, 1966b). Aliasing errors can best be understood through the use of a graphical example of the sampling and reconstruction of two sinusoidal waveforms. Figure 11.11 shows two sine wave signals, one having a frequency of $1/8$ of the sampling frequency, f_s , the other being $7f_s/8$. It can be seen that both waveforms give rise to identical sampled data, which reconstructs to a frequency of $f_s/8$. The frequency at which the erroneous signal is produced can be found by 'folding over' all frequencies of greater than $f_s/2$ on the frequency spectrum of the signal. Thus, $7f_s/8$ gives rise to a frequency of $f_s - 7f_s/8 = f_s/8$, as shown in the figure. Such errors can be avoided by using a higher sampling frequency. Increasing the number of samples, however, carries a penalty in the time taken for computation of the series, so that a compromise has to be reached. It is preferable to insert a filter in each of the lines to be computed to cut off frequencies above those required for analysis, but this requires care in matching the filters of each channel and in ensuring that no phase distortion occurs in channels that are analyzed. Active, rather than passive, filters are to be preferred.

The accuracy of measurement of the ordinates of the samples taken is a limiting factor for the possible accuracy of the analysis. With an analog-to-digital converter, the limit would appear to be set by the number of 'bits' available in the conversion. Thus, a 6-bit converter samples to an accuracy of one part in 2.0^6 (i.e. 64), approximately 1.5 percent of the width of the channel, and accuracy would apparently be increased by changing to 8- or 10-bit conversion. This may be illusory, for with FM recording on magnetic tape, the accuracy of retrieval is often no better than 1.0 percent, although some

manufacturers claim that their recorders are accurate to 0.5 percent. While this appears to be a trivial error in biological recording, it becomes of significance in measuring the phase angle of the higher harmonics, because to derive $\tan \phi$ the ratio of two small numbers is required. Even at 1.0 Hz, a 1.0 percent error in 360° is 3.6° , so that discrimination of very small phase shifts is impossible and leads to a considerable standard deviation in the measurement of the phase velocities of those components.

The significant harmonic terms in arterial waveforms

The reproduction of the original wave from a given number of harmonic terms can, by inspection, give a qualitative check on the adequacy of the analysis.

For a more quantitative measure, the variance of the wave and of the Fourier series is compared. The variance of a curve is familiar as a statistical procedure as it is the square of the standard deviation. Thus, for a curve delineated by m ordinates with a mean value of y we have

$$\text{Variance} = \frac{1}{m} \sum_{i=1}^m (y_i - \bar{y})^2. \quad 11.30$$

This is more usually calculated as

$$\frac{1}{m} \sum_{i=1}^m y_i^2 - \left(\frac{1}{m} \sum_{i=1}^m y_i \right)^2. \quad 11.31$$

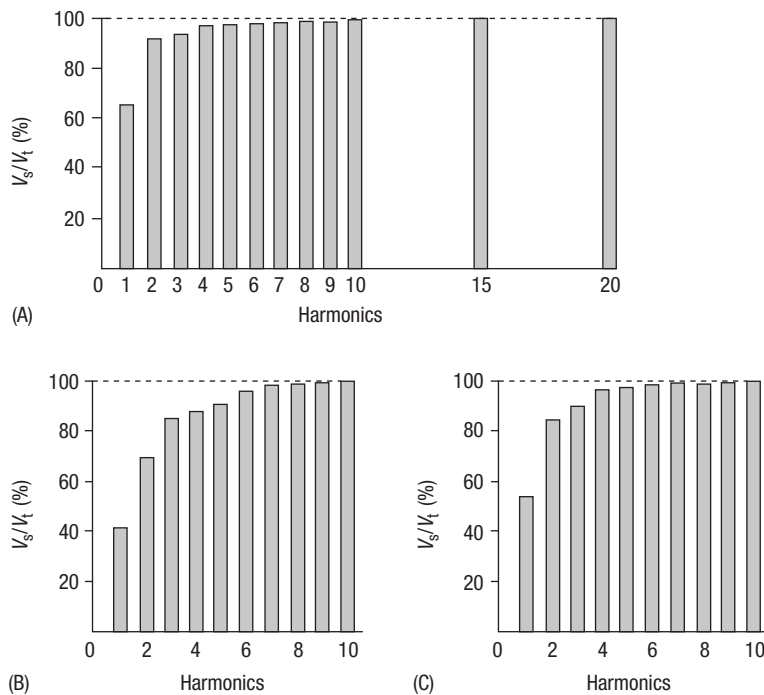


Figure 11.12 The columns represent the cumulative variance of the Fourier series (V_s) to the total variance (V_t) of a dog's aortic pressure and flow waves. (A) This example of a pressure wave was specifically taken to 20 harmonics for demonstration purposes. (B) The variance of a flow wave at a lower heart rate is shown. Ten harmonics are still more than adequate, even though the tenth harmonic is only at 12.8 Hz. (C) The variance represented by ten harmonics of the flow curve measured simultaneously in the ascending aorta with the pressure wave analyzed in (A). Although the first and second harmonics contribute a smaller proportion of the total variance than in the pressure curve, a 99 percent total is actually achieved in the flow at a lower frequency than in the pressure

The variance of a Fourier series is given from the following equation resulting from Parseval's theorem:

$$\text{Variance (series)} = \frac{1}{m} \sum_{n=1}^k M_n^2; \quad 11.32$$

that is, half the sum of the squares of the moduli (M_n) of the harmonic terms used. The graphs shown in Fig. 11.12 give the progressive sum of the variance of series representing typical pressure and flow waveforms as a percentage of the variance of the original curves. It can be seen that better than 99.5 percent of the variance is included, a point often reached by the seventh or eighth harmonic. No appreciable improvement is achieved by going to 20 harmonics, and the residual loss appears to consist entirely of noise in the recording systems and the low-energy content of cardiovascular sounds (Nichols, 1970; McDonald, 1974).

It is on such evidence that we state that no significant propagated waves in relation to blood flow exist above 30 Hz, and those above 20 Hz are small. The original ascending aortic pressure waveform and its resynthesis from six harmonics is shown in Fig. 11.13, which demonstrates the degree of accuracy that this number of harmonics gives.

The validity of Fourier analysis in the circulation

Any series of which all the terms are sine and cosine functions must be periodic. The heart beat is normally periodic, but doubts have been raised as to whether the variations in cycle length that do occur make the analysis

invalid. Although, mathematically, the phrase can only be used for a function that has repeated itself from minus infinity and will continue for an infinitely long time in the future, in physical analysis, it is applied to systems said to be in steady-state oscillation. When a train of pulses is first started in a quiescent system, the initial excitation will produce changes that will decay exponentially with time owing to the damping of the system, and which will be superimposed on the periodic excitation. Once the initial excitation has died away and has become undetectably small, the system is in steady-state oscillation. In other words, there is no way of detecting how long the oscillation has been in progress. The length of time required to achieve this is not known exactly, but we do know that the viscous damping in the arterial system is high. From studies of the after-effect of a transient imposed on the arterial system, there is strong evidence that, in the dog, the steady state is reached in a period of about 0.5 s or within the duration of one cardiac cycle (McDonald, 1974). Therefore, following a sharp discontinuity in periodicity, as with an ectopic heart beat followed by a pause, a steady state will have been reached again by the end of the normal beat.

There are also occasional fluctuations of heart rate (sinus arrhythmia). The inaccuracies due to such causes may be checked by analyzing a long train of pulses and determining its periodogram or its power spectrum.

The power spectrum

The technique of performing statistical analyses such as this was introduced into circulatory physiology by Randall (1958), and a more detailed account is given in standard texts such as Blackman and Tukey (1959) or Lee (1960).

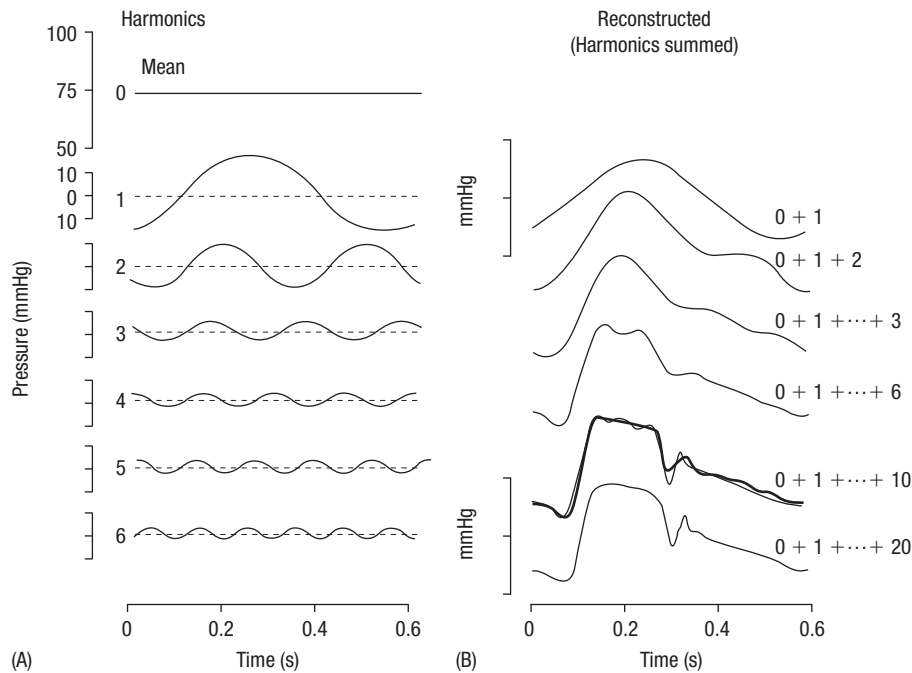


Figure 11.13 Example of Fourier series representation of a periodic wave. (A) The mean term (or 0 harmonic) and harmonics 1 through 6 of a pressure wave, recorded in the ascending aorta of a dog. (B) Individual harmonics are added to reconstruct (or resynthesize) the original (measured) wave. Agreement is relatively close with the first six harmonics ($0+1+\dots+6$) summed, and better with the first ten harmonics ($0+1+\dots+10$), while the resynthesized wave is almost identical to the original wave (thin line) with the summation of the first 20 harmonics ($0+1+\dots+20$). Reproduced from Westerhof *et al.* (1979)

The first step in analyzing a time series such as a train of pulse waves is to perform an autocorrelation. This is, in essence, a correlation of points in the curve with points occurring later delayed by a variable time lag, τ . When $\tau=0$, the correlation of each point is with itself, and the correlation is 1.0. If the function is, for example, a sine wave, the correlation will become 1.0 again when $\tau=T$, the period of the sine wave, and the autocorrelation function is itself a sine wave. If the period varies slightly, the autocorrelation function will approximate to a sine wave, but with $\tau>0$, the correlation will never be 1.0. When the function is compounded of harmonic components, as is the pulse wave and the electrocardiogram, then, equally, we get a corresponding oscillation in the autocorrelation function. An example is shown in Fig. 11.14 of an autocorrelation function of a train of pulse waves in which there was a marked respiratory variation in pulse frequency.

In mathematical terms, the performance of the autocorrelation is in terms of the product $f_1(t) \times f_1(t+\tau)$, and we write, from eqns 11.25, 11.26, and 11.27,

$$\frac{1}{T} \int_{-T/2}^{+T/2} f_1(t)f_1(t+\tau)dt = \sum_{n=-\infty}^{+\infty} |F_1(n)|^2 e^{in\omega\tau}. \quad 11.33$$

The left-hand member of eqn 11.33 is called the ‘autocorrelation function’ of $f_1(t)$ and is represented by the symbol $\phi_{11}(\tau)$, where the subscript 11 indicates that the function is correlated with itself.

The ‘power spectrum’ is represented by the symbol $\Phi_{11}(n)$, so that

$$\Phi_{11}(n) = |F_1(n)|^2. \quad 11.34$$

In terms of these symbols,

$$\phi_{11}(\tau) = \sum_{n=-\infty}^{+\infty} \Phi_{11}(n)e^{in\omega\tau}, \quad 11.35$$

and, inversely,

$$\Phi_{11}(n) = \frac{1}{T} \int_{-T/2}^{+T/2} \phi_{11}(\tau)e^{-in\omega\tau} d\tau. \quad 11.36$$

From eqns 11.35 and 11.36, we can say that the autocorrelation function and the power spectrum are Fourier transforms of each other.

The power spectra derived by autocorrelation and cross-correlation methods are shown in the left-hand side of Fig. 11.14 (co-spectral power in A, quadrature-spectral power in B). It can be seen that there is a peak at a little over 2.0 Hz and smaller ones at twice and three times this frequency. These represent the first three harmonics of the dominant pulse frequency, while the spread on either side is due to the variations in this frequency with respiration; the fluctuation at the respiratory rhythm gives rise to the terms at low frequency. Similar variations are also seen in the spectral distribution of statistical variance of the

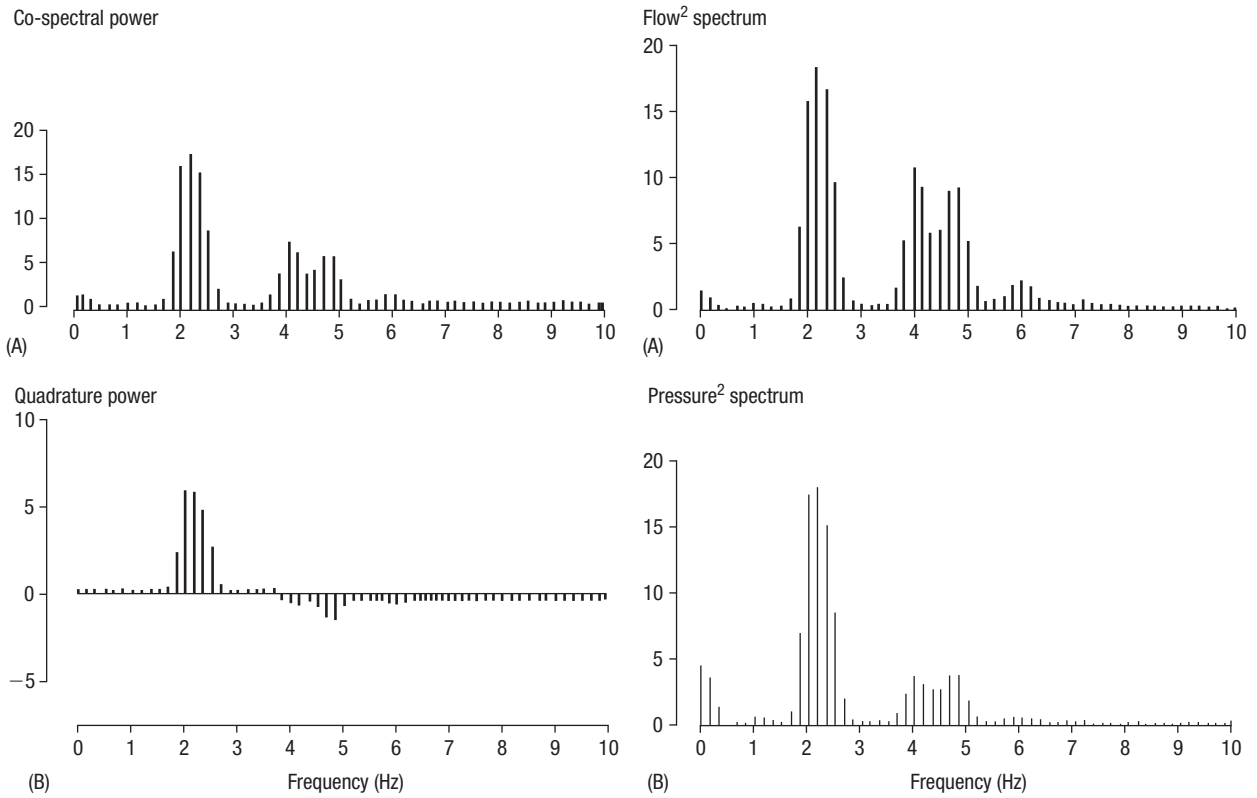


Figure 11.14 Left: Each ordinate of these curves represents the amount of pulsatile power in a frequency of 0.167 Hz. The units are mmHg $\times$ mL/min. (A) Spectral distribution of pulsatile power resulting from pressure and flow (canine femoral artery) components in phase with one another (co-spectral power). (B) Spectral distribution of pulsatile power resulting from pressure and flow components 90° out of phase with one another (quadrature power). Right: Each ordinate represents the amount of activity in a frequency band of 0.167 Hz. (A) Spectral distribution of statistical variance of flow about its mean. The units are (mL/min)². (B) Spectral distribution of statistical variance of pressure about its mean. The units are (mmHg)². Reproduced from Randall (1958)

flow (Fig. 11.14A, right) and pressure (Fig. 11.14B, right) waves.

The most recent use of power spectral analysis is in studies of heart rate variability (Ranpuria *et al.*, 2008; Almozni-Sarafina *et al.*, 2009; Buccelletti *et al.*, 2009; Montano *et al.*, 2009a; Montano *et al.*, 2009b), sympathetic nerve activity (Chapuis *et al.*, 2004; Oréa *et al.*, 2007; Furlan *et al.*, 2009), baroreceptor function (Bertram *et al.*, 2005), and low-frequency cardiovascular oscillations (or Mayer waves) (Cohen and Taylor, 2002; Bertram *et al.*, 2005; Julien, 2006).

The power spectrum derived from the autocorrelation function can be seen from eqn 11.34 to be a real function of frequency and so involves the co-spectral, cosine terms (C_{11}) only.

When it is desired to compare two separate time series such as the pressure and the corresponding flow, the similar process of cross-correlation is employed. In this case, the points on one function are correlated with those on the other over a series of delay lags, τ . The cross-correlation function is then defined as

$$\phi_{12}(\tau) = \frac{1}{T} \int_{-T/2}^{+T/2} f_1(t)f_2(t + \tau) dt \quad 11.37$$

and the cross-power spectrum of the functions is $\Phi_{12}(n)$, so that we get similar relations

$$\phi_{12}(\tau) = \sum_{n=-\infty}^{+\infty} \Phi_{12}(n)e^{in\omega\tau} \quad 11.38$$

and

$$\Phi_{12}(n) = \frac{1}{T} \int_{-T/2}^{+T/2} \phi_{12}(\tau)e^{-in\omega\tau} d\tau. \quad 11.39$$

The cross-power spectrum, Φ_{12} , derived from the cross-correlation function, ϕ_{12} , can be complex, with co-spectral and quadrature terms C_{12} and Q_{12} .

If $f_1(t)$ is the series of pressure waves and $f_2(t)$ the series of flow waves, the ratio between them is the impedance $Z = |Z|e^{-i\phi}$, where

$$|Z| = (\Phi_{11}/\Phi_{22})^{1/2}, \quad 11.40$$

and the phase is

$$\Phi = \tan^{-1}(Q_{12}/C_{12}). \quad 11.41$$

The 'coherence' is a quantity used for testing the significance of the results and is given by

$$\text{Coherence} = \frac{(C_{12}^2 + Q_{12}^2)}{(\Phi_{11} \cdot \Phi_{22})^{1/2}}. \quad 11.42$$

These methods have been used very effectively by Taylor (Taylor, 1966a; Taylor, 1966b). The spectrum of frequencies in a long train of pressure waves in the femoral artery when the heart was of a normal regularity is shown in Fig. 11.15A. A pacemaker was then used to stimulate the atrium in a random fashion so that the heart rate was very irregular, as it may be with atrial fibrillation. The spectrum then appeared as illustrated in Fig. 11.15B and now represents more nearly a continuous series of frequencies. In particular, components can be seen down to very low frequencies that are much below those that can be obtained by pacing the heart regularly or by imposing heart block. As a result, Taylor was able to explore the impedance of various vascular beds (see Chapter 12) into this interesting range of frequencies that cannot be measured by single-pulse analysis. Equally, by comparing these results with those of the harmonic analysis of a single pulse, it can be seen that, under normal conditions, slight variations of pulse frequency introduce little error into the results.

Linearity

The use of simple harmonic analysis in defining the physical relations also assumes linearity of the system. Some confusion regarding the limits of its applicability in a non-linear system tends to arise in discussion of this point, so that it is as well to redefine them. For numerical description of a periodic phenomenon such as the pulse wave, the Fourier series is perfectly valid whether the system is linear or not. A resynthesis of the harmonic components will recreate the original wave to whatever accuracy is defined by the number of harmonics taken. In this sense, it is valid as a means of describing a single wave, even if it is solitary. If, however, the Fourier series is taken term by term and related causally to the corresponding terms of the flow wave, this presupposes a linear system. In a non-linear system, there will be interaction between the harmonic components, and additional frequency components will be created.

The arterial system is clearly non-linear in the precise sense of the term. The equation of motion between the pressure gradient and the flow contains the second-order terms of the Navier–Stokes equations (see Chapter 7). The relation of pressure to flow in an elastic tube contains further non-linear terms resulting from the changes in radius with pressure (see Chapter 4). Finally, the elastic properties of the arterial wall (see Chapter 4) are markedly non-linear in that the wall gets stiffer with increasing strain. The assumption of linearity in the system is, therefore, of necessity an approximation. If the predicted

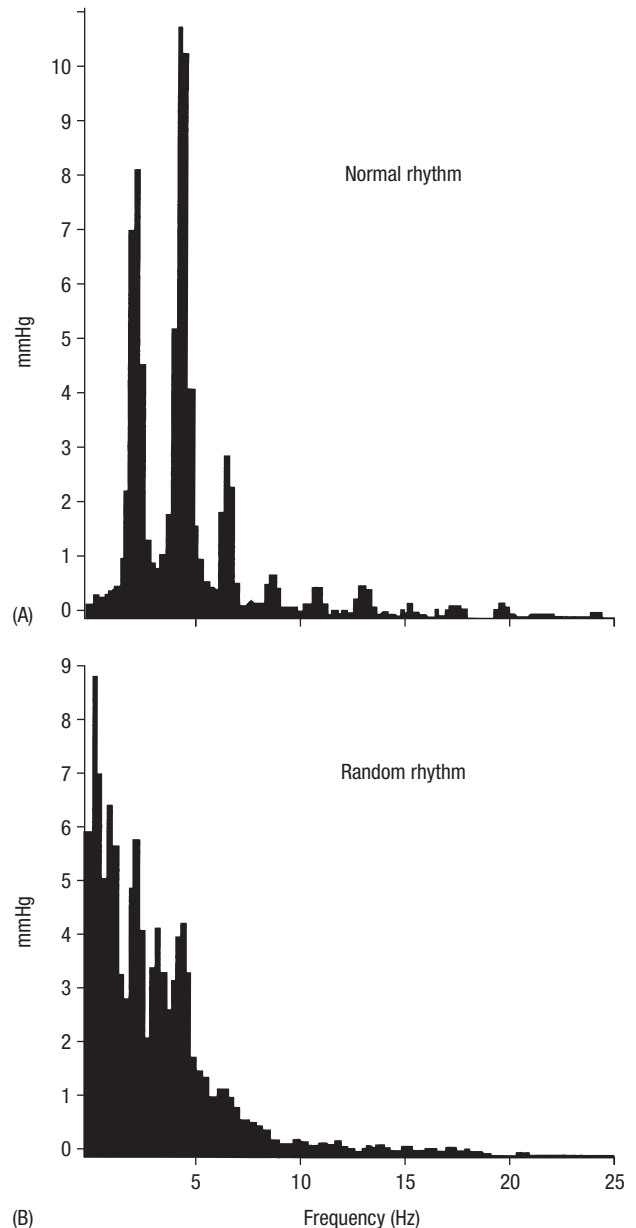


Figure 11.15 (A) Frequency spectrum of a long sequence of pressure pulses with a regular heart rhythm or rate. The peaks are seen to fall at integral multiples of the lowest frequency (about 2.0 Hz) and represent the harmonic components. The fact that a heart rate is never exactly regular is shown by the low level between the main peaks. (B) Frequency spectrum of a train of pulses while the heart is being excited at a randomly varying rate. A much more continuous band of frequency components has been created. In particular, a large number of these fall below the normal pulse frequency. This frequency region is especially useful in studying the arterial impedance. Reproduced from Taylor (1966a)

non-linearities are not greater than the order of accuracy of our measuring devices, they are virtually impossible to detect and therefore, in an experimental situation, may be deemed not to exist. This assumption has, however, to be re-examined from time to time as improvements in instrumentation occur.

The experimental evidence relating to the detection of non-linear behavior is discussed at various places throughout this book where it is relevant to the various aspects of the system being considered. The methods used may be put into three general categories.

First, the system may be excited by a single harmonic frequency and the creation of higher harmonic frequencies studied. In a simple system, this is an elegant method and was used effectively by Taylor (1959c) in a long, single rubber tube in relation to the creation of a second harmonic. He found that the second harmonic term grew in amplitude with distance from the origin and reached a maximum at 1.0–1.5 wavelengths along the tube and diminished thereafter. In the arterial system, a similar study is difficult because the whole system is short and only attains part of a wavelength at the frequency of the fourth, or higher, harmonics, which are small in amplitude. Attinger *et al.* (1966a) attempted this type of analysis in a perfused isolated aorta attached to a sinusoidal pump and could not regularly detect any higher frequencies. They concluded, therefore, that the non-linear terms did not amount to more than 5.0 percent, which they considered to be the limit of accuracy of their studies.

Second, the system *in situ* may be studied when driven by the heart at different frequencies without changing the properties of the peripheral bed (Noble *et al.*, 1967; McDonald, 1974; Nichols *et al.*, 1977a; Nichols *et al.*, 1977b). The argument underlying this approach is that the behavior of the bed (e.g. the input impedance at 3.0 Hz) should be constant and independent of the heart rate. If the heart is driven successively at 1.0, 1.5, and 3.0 Hz, the components at 3.0 Hz will be successively the third, the second, and the fundamental harmonics. If there is no consistent change in the impedance at 3.0 Hz in these cases, the system may be considered as functioning linearly. Data summarized in Chapter 12 show that this is the case.

Third, a more precise and explicit method as an extension of the first category above, and aimed solely at detecting non-linearity, is to excite the system simultaneously with two frequencies that are not integral multiples of each other, and look for the appearance of addition and subtraction frequencies. This was done by Dick *et al.* (1968), who replaced the flow input of the heart with two sinusoidal pumps; these were run at 2.0 and 3.0 Hz, and the power spectrum of the pressure was analyzed. The only 'stray' frequency of appreciable magnitude that they could detect was the addition frequency of 5.0 Hz, but that was only 1.0–2.0 percent of the input amplitude in the abdominal aorta. They therefore concluded that the arterial system, for all practical purposes, behaved as a linear system, at least in regard to pressure and flow relations.

While to biologists bred with the legend of the 'exactness' of the physical sciences this may appear as a crude approximation, it should be emphasized that such approximation is universal in physics as, strictly speaking, no precisely linear system exists in the real world (McDonald, 1974). The transmission of sound is, for

example, non-linear, but this has not hampered the widespread application of harmonic analysis to it; where sound transmission becomes markedly non-linear, as in loudspeaker output (McLachlan, 1941), the cross-terms in the Fourier series can be defined and analyzed. With digital computers available, a similar analysis could be made, but the pursuance of such a goal would have to be tempered by the decision as to whether, in terms of the available accuracy of our raw data, the extra precision would have any significant meaning or be worth the far greater computation time necessary.

Errors in resynthesis

There is a tendency in a number of abbreviated methods for obtaining the Fourier components of a waveform to concentrate on the amplitude of the components and neglect their phase relations. For many purposes, such as in electric transmission line studies where one is often concerned with detecting a small signal in the presence of noise, this is sufficient. Errors in determining the phase relation can, however, cause large changes in the form of the resynthesized curve even though the representation in terms of its variance (eqns 11.30, 11.31, and 11.32) will be unchanged. A simple example is given in Fig. 11.16, in which a first and second harmonic are added together in two ways; in curve B the phase of the second harmonic is shifted by 45° . It can be seen that the morphology of curve B is markedly different than that of curve A, with both peaks being less and delayed. This illustrates the errors that can arise in methods of computing flow curves from pressure measurements (see Chapter 7) without measuring phase shifts (which constitute the phase of the complex curves discussed in Chapter 12) (McDonald and Nichols, 1973; Nichols, 1983). A similar error can occur in phase angle calculation when using a mathematical transfer function (TF) to synthesize the central blood pressure waveform from a peripheral waveform. This error was pointed out and discussed by Segers *et al.* (2007b) in response to the article published by Hope *et al.* (2007). The TF, expressed in the frequency domain using Fourier analysis, is obtained from simultaneous measurements of pressure waves in a peripheral artery ($P_p = |P_p| \cos(\omega t - \alpha)$) and a central artery ($P_c = |P_c| \cos(\omega t - \beta)$) (see Chapters 11 and 30). The modulus (or amplitude) of the TF is obtained as $|P_p|/|P_c|$ and the phase (angle) as $(\beta - \alpha)$. The modulus of the TF expresses the amplification of the harmonic and the phase expresses the delay between the two harmonics. Only one harmonic of modulus and phase are indicated here but, in general, ten harmonics are used to obtain the entire spectra. The calculation of the moduli is straightforward, however, there is a potential danger in the treatment of the phase angles, especially for the higher harmonics (Segers *et al.*, 2007b). Phase angles of sinusoidal waves repeat ever 360° (see Fig. 11.9) and when calculating the TF, most software will calculate the phase

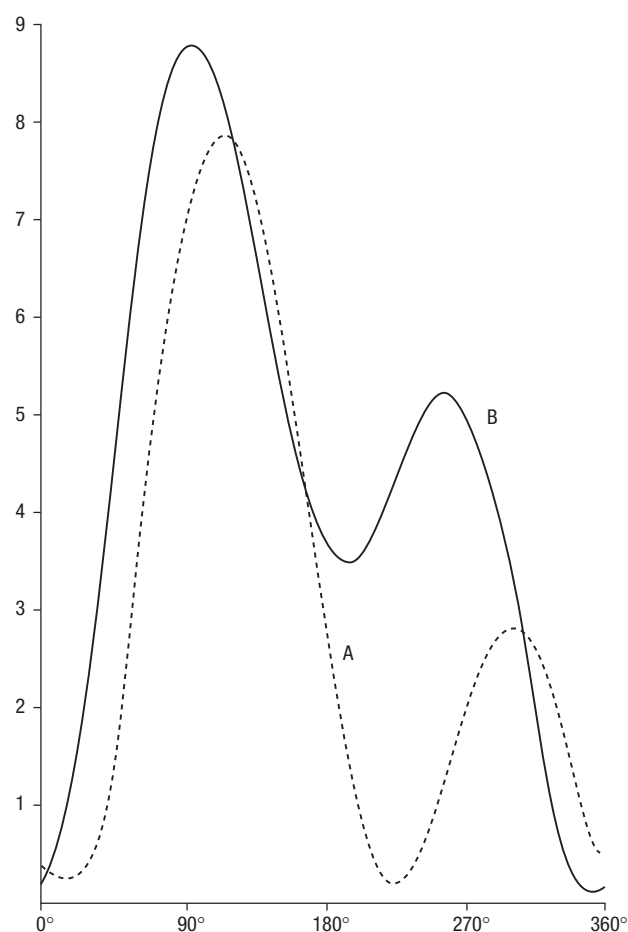


Figure 11.16 A simple example to show that errors in measuring the phase of harmonic terms can produce a great distortion in resynthesizing a curve. (A) and (B) are a combination of a first and second harmonic of equal magnitude; however, in (B) the second harmonic has been shifted by 45° compared with (A)

angle between -180° and $+180^\circ$ or -3.14 and $+3.14$ radians, which will lead to phase jumps of 360° in the TF phase angle. This process is referred to by Segers *et al.* (2007a) as 'wrapping'. To obtain the monotonically decreasing phase needed for the TF, these phase jumps must be corrected by 'unwrapping' the phase angles. The effects of this phase angle 'unwrapping' on the morphology of the synthesized carotid pressure waveform from the radial waveform is shown in Fig. 11.17. In the calculations of Hope *et al.* (2007) (Fig. 11.17, 'wrapped'), phase deviated from that calculated by Segers *et al.* (2007a) from the same data above 4 Hz, crossing zero, then becoming positive, i.e. radial pressure wave preceded the aortic wave components! This is impossible, and can explain the inability of Hope *et al.* to generate an accurate aortic waveform from radial pressure waveform. This issue together with inappropriate use of brachial tonometry (Chapters 28 and 30) explains failure of the method used by Hope *et al.* to show superiority of central over brachial pressure in previous studies (Dart *et al.*, 2006; Dart *et al.*, 2007; O'Rourke and Avolio, 2008; O'Rourke and Adji, 2010).

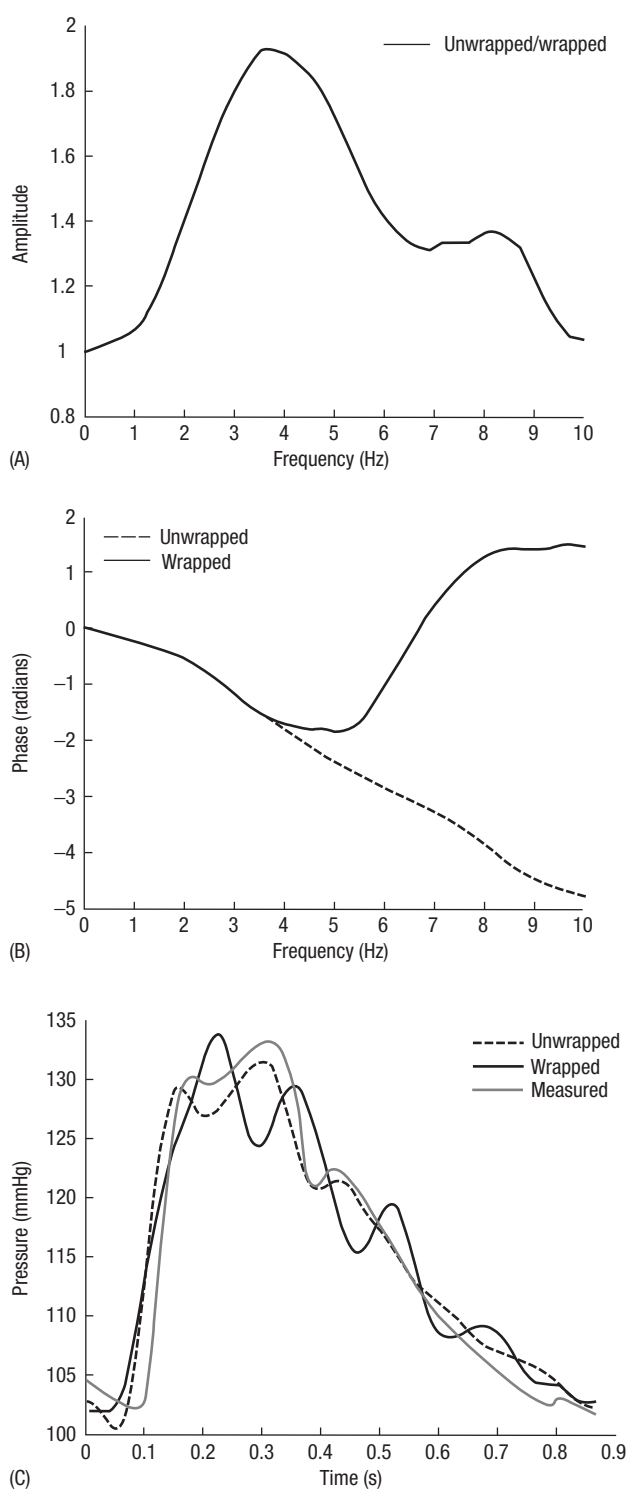


Figure 11.17 (A) Amplitude (or modulus) and (B) phase of an averaged radial/carotid transfer function (TF). It can be seen that the pattern of the phase is totally different when accounting for the phase jumps (unwrapped). In (C) these averaged TFs are used to synthesize the carotid waveform from the measured radial pressure waveform using 'wrapped' and 'unwrapped' phases. Unwrapping the phases has no effect on the amplitude of the TF. Reproduced from Segers *et al.* (2007a)

Vascular impedance

The concept of impedance	273	Ascending aortic impedance: the input impedance to the	
The theory of vascular impedance	276	systemic circulation	289
Input impedance and models of the arterial tree	278	Pulmonary vascular impedance	295
Assumptions in the application of impedance theory to		Effects of body size and shape on input impedance in	
the arterial system	282	systemic arteries	296
Impedance in a typical peripheral artery	283	Vascular impedance in humans	300
Changes in input impedance between peripheral arteries		Theoretical and technical problems in the determination	
and the central aorta	288	and interpretation of vascular impedance	306

THE CONCEPT OF IMPEDANCE

As its name implies, the term 'impedance' is the measure of the opposition to flow presented by a system. By conventional use in physics, and in its most familiar form in the terminology of electric current theory, the term is confined to oscillatory motions or alternating current. Thus, usage is not confined to electric current, but is also used as 'mechanical impedance' in vibrating solid systems and as 'acoustic impedance' in gas-filled systems. Etymologically, the term 'resistance' conveys the same sense as impedance but is, by conventional definition, confined to non-oscillatory or steady motions. Resistance may thus be considered as impedance at zero frequency. All the above terms have a precise physical meaning and can be expressed in mathematical terms. The term 'impedance' is frequently used by clinicians in another sense: in a general way, as referring to the impediment or hindrance to blood flow (Wilcken *et al.*, 1964; Miller *et al.*, 1976; Mason, 1978; Finkelstein *et al.*, 1985). Some have described impedance as instantaneous pressure divided by instantaneous flow throughout the cardiac cycle (Urshel *et al.*, 1968a; Urshel *et al.*, 1968b). This concept has no counterpart in the physical sciences and is inappropriate for describing pressure-flow relationships in a system where at any one time pressure depends not only on flow at the same instant, but also on flow at previous instants (this being a consequence of blood inertia, arterial distensibility, and pulse wave travel and reflection). So-called 'instantaneous impedance' is a contradiction in terms because it is a frequency-dependent quantity rather than a time-dependent quantity.

When the general term 'impedance' is applied to the vascular system, it is usually in reference to input impedance, this being the relationship between pulsatile pressure and pulsatile flow recorded in an artery feeding a particular vascular bed. However, four different types of impedance have been defined. These are:

1. Longitudinal impedance $Z_L = (P_1 - P_2)/Q$
2. Input impedance $Z_i = P/Q$, (where P and Q are influenced by wave reflection)
3. Characteristic impedance $Z_0 = P/Q$, (where P and Q are not influenced by wave reflection)
4. Terminal impedance $Z_T = \bar{P}/\bar{Q}$, (where P and Q are measured at the termination of the system immediately upstream from the reflecting site).

Figure 12.1 illustrates these different types of impedance as they apply to a simple tubular model of a vascular bed, together with the form taken by impedance values when expressed (like electric and all other types of impedance) in terms of modulus (modulus of pressure divided by modulus of flow) and phase (delay between pressure and flow waves).

Longitudinal impedance

This is the ratio of the pressure gradient to flow and is thus the pulsatile analog of longitudinal arterial resistance. It is the opposition to flow along a segment of artery; longitudinal resistance of this arterial segment is the relationship of mean pressure to flow along its length, so it is longitudinal impedance at zero frequency. Longitudinal

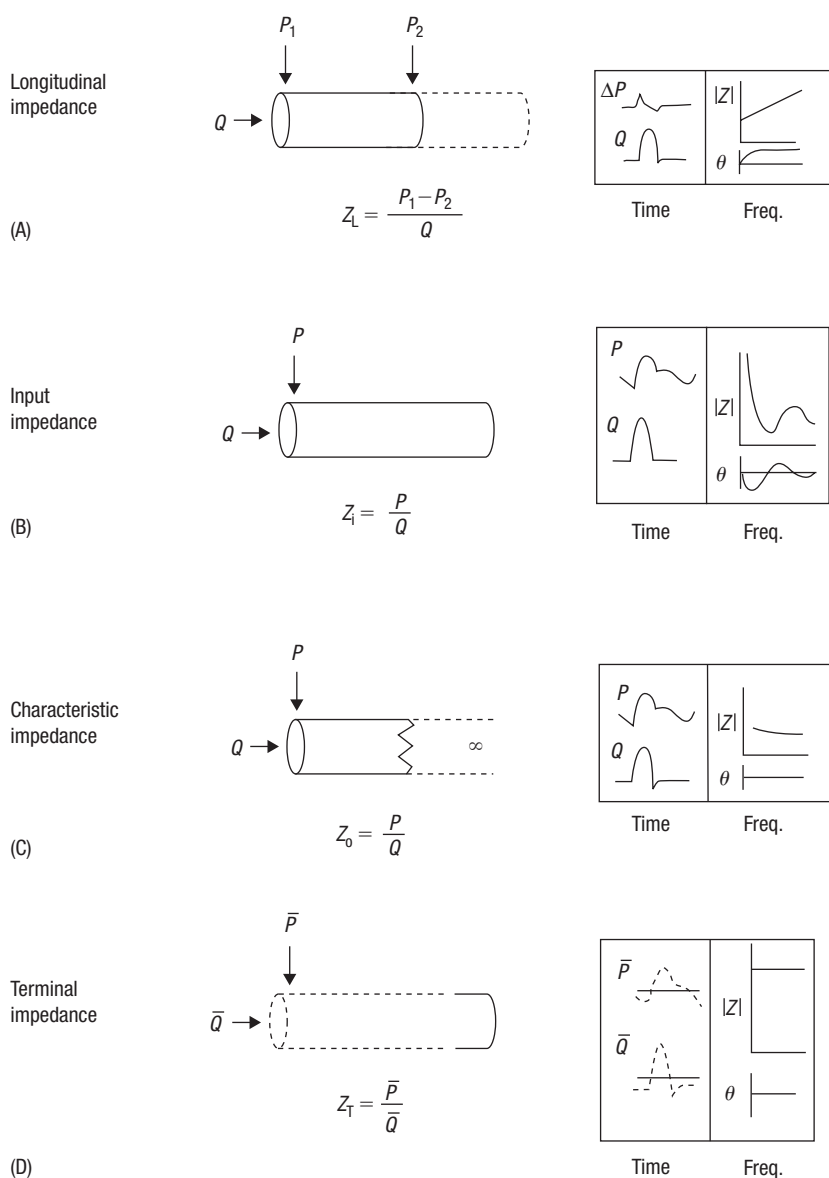


Figure 12.1 Diagrammatic representation of impedance concepts (left) and their graphic representation as modulus and phase (right)

impedance depends only on the local properties of the vessel and the blood within it. It is not influenced at all by the properties of the vascular bed downstream (Curi *et al.*, 2002). Siewiorek *et al.* (2009) used longitudinal impedance measurements to evaluate the performance assessment of distal protection filters in a physiologically realistic *in vitro* model of the human common carotid artery circulation. Longitudinal impedance is discussed in Chapter 7 and was the focus of interest of McDonald and Womersley in the 1950s. Taylor broadened this by introducing the other terms.

Input impedance

This is the ratio of pulsatile pressure (as distinct from the pressure gradient) and pulsatile flow at a particular arterial site, which may be regarded as the input to all

the vascular tree beyond this site; it depends not only on the local arterial properties, but also on the properties of all vessels in the vascular bed beyond, down to the points where all pulsations generated by the heart have been attenuated. These points probably correspond to the arterial end of the capillaries, where flow and pressure are virtually continuous, without significant pulsation transmitted from the heart. Both longitudinal impedance and input impedance can be determined readily from experimental data and expressed as modulus and phase (Fig. 12.1). Such measurements in an artery would relate the same flow wave to the pressure gradient over, say, 1.0 cm of artery (for longitudinal impedance) and to the complete pressure wave (for input impedance). Since the amplitude of the pressure gradient is far smaller than that of the pressure wave, it is clear that the modulus of longitudinal impedance is far smaller than the modulus of input impedance.

While input and longitudinal impedance can be displayed in graphical form from experimental data, terminal impedance and characteristic impedance usually cannot, so are usually estimated. Nonetheless, they have definite theoretic and practical meanings.

Characteristic impedance

This is the relationship between pulsatile pressure and pulsatile flow in an artery when these waves are not influenced by wave reflection. There is, in fact, no circumstance in life where pressure and flow are not altered by wave reflection; this is a consequence of the varying properties of the vascular bed along its length, its branching pattern and particularly the steep rise in resistance within the muscular arterioles (see Chapter 9). Input impedance may approximate the characteristic impedance when the vascular bed is maximally dilated and wave reflection effects are minimal (O'Rourke and Taylor, 1966). The usual way of estimating characteristic impedance is by averaging impedance moduli over a frequency band where one might expect fluctuations due to wave reflection above the characteristic impedance modulus to cancel out oscillations below. This commonly used method is anything but precise (O'Rourke, 1982a; O'Rourke, 1982b; O'Rourke, 1982c). Another method of estimating characteristic impedance is to average moduli at high-frequency values of impedance where fluctuations caused by reflections have settled. This method has practical shortcomings since values of moduli used are calculated from values of pressure and flow close to the noise level of recording instruments (Taylor, 1966a; Taylor, 1966b; Taylor, 1966c; O'Rourke, 1982a; O'Rourke, 1982b; O'Rourke, 1982c).

A different method of calculating characteristic impedance was developed by Dujardin and Stone (1981) and used by them to determine the effects of interventions on properties of the aorta and pulmonary artery (Dujardin *et al.*, 1982; Stone and Dujardin, 1984; Lucas *et al.*, 1988) (see Chapter 4). This method relates the initial upstroke of the pressure wave to the upstroke of the simultaneously recorded flow wave; the method is based on the reasonable assumption that the effects of wave reflection are minimal or absent in the first 40 or so ms of the waves (see Fig. 4.1). Mitchell *et al.* (Mitchell *et al.*, 2001; Mitchell *et al.*, 2002; Mitchell *et al.*, 2007b; Mitchell *et al.*, 2008; Mitchell, 2009a; Mitchell, 2009b; Mitchell *et al.*, 2010) used a variation of this approach by relating peak ascending aortic flow to the simultaneously recorded carotid pressure wave from its foot to its early systolic shoulder. This method of calculating characteristic impedance is fairly accurate when both pressure and flow are measured at the same site.

Characteristic impedance, like resistance and terminal impedance, is usually considered in terms of modulus only, since phase is close to zero. Characteristic

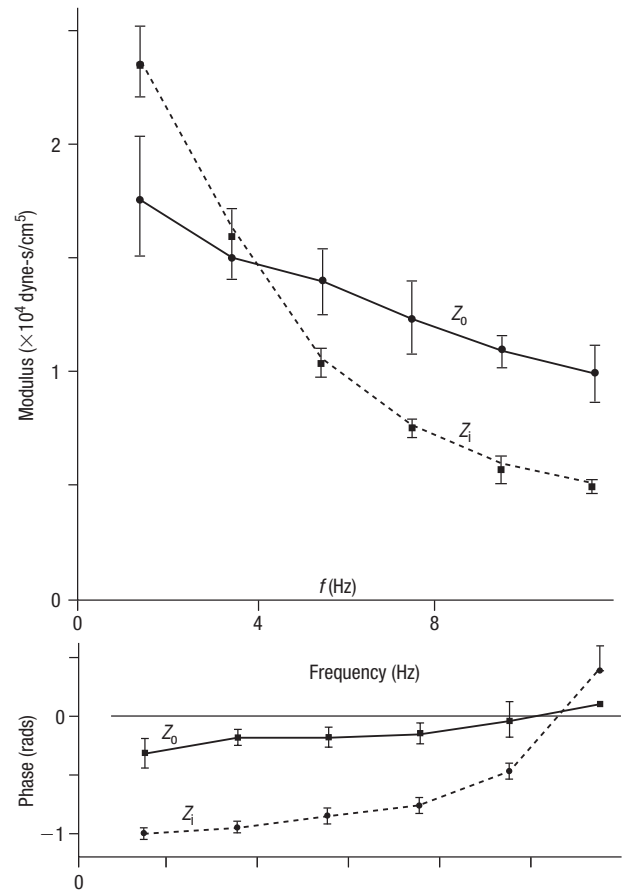


Figure 12.2 Input impedance (Z_i , broken lines) and characteristic impedance (Z_0 , solid line) in the femoral artery of a dog. Z_i was determined from measurements of pressure and flow at the proximal end of segment of the artery. Z_0 was computed by the method of Milnor and Nichols (1975)

impedance can actually be measured experimentally by methods that separate the reflected from the measured wave of pressure and flow, permitting just the antegrade components of pressure and flow to be related (see Chapter 9). The technique demands measurement of flow at one point in an artery and pressure at three points (Gessner and Bergel, 1966; Westerhof and Noordergraaf, 1970) or pressure with flow at two points (Milnor and Nichols, 1975; Milnor, 1989). The method demands impeccable accuracy in pressure and flow determinations and is only feasible for measurements in relatively long arteries of constant caliber. Experimental determinations have been published for the femoral and common carotid arteries of dogs. These show agreement with the theoretically predicted values: modulus being frequency-dependent but showing no distinct fluctuation, and phase being close to zero at all frequencies (Fig. 12.2) (Cox, 1970; Milnor and Nichols, 1975; Busse *et al.*, 1979b). Errors in phase are easily made by failure to consider the appropriate quadrant (Segers *et al.*, 2007a)

which lead to distortion of the resynthesized curve (see Fig. 11.17).

Terminal impedance

This is the opposition to flow at the termination of a vascular bed immediately upstream from its termination. This termination is taken to represent the high-resistance arterioles. Properties of the arterioles are almost purely resistive (with little effect of blood inertia or vessel distensibility), that is, resistance completely dominates reactance. Terminal impedance is thus considered in terms of modulus only and takes the form, like peripheral resistance, of mean pressure divided by mean flow. It is not, however, strictly identical to peripheral resistance, since there is some (albeit small) drop of pressure along the arterial tree from the point of pressure measurement in a large artery to arteriolar origin (Fig. 12.1), and also because mean pressure is not zero in the capillaries beyond, even though the transmission of pulsatile pressure through to these vessels may be extremely small. Because of the finite pressure drop along the arterial system and finite pressure in the capillaries, terminal impedance is somewhat less than peripheral resistance as measured from the mean arteriovenous pressure difference. This point is of considerable relevance when venous pressure is high, as is normally the case for the pulmonary circulation and for the systemic circulation in heart failure (see Chapters 15 and 16).

Impedance modulus (units)

This is the modulus of pressure (dyne/cm²) (or pressure gradient for longitudinal impedance) divided by the modulus of flow, so is expressed in the units of dyne-s/cm⁵ for volumetric flow (when flow is measured in cm³/s) or as dyne-s/cm³ for linear flow (when flow is measured in terms of linear velocity, as cm/s). There is no agreement or convention as to which should be used. In the second edition of this book, McDonald pointed out that in the early theoretic treatment of arterial flows by analogy with electric networks (Landes, 1949; Ronniger, 1954), the analogy of current was always treated as velocity of flow and not as volume flow. Impedance in terms of velocity flow (dyne-s/cm³) is analogous to acoustic and mechanical impedance where the same units are utilized. One set of units can simply be converted to the other by multiplication (or division) by the cross-sectional area (provided of course that the cross-sectional area remains constant). However, other conceptual considerations are involved; these are discussed below. In this book, we favor use of impedance modulus expressed in dyne-s/cm³, as did McDonald in 1974 (McDonald, 1974) and O'Rourke in 1982 (O'Rourke, 1982a; O'Rourke, 1982b). On the other hand, Milnor (1989) and many others have preferred volume flow (dyne-s/cm⁵).

THE THEORY OF VASCULAR IMPEDANCE

Longitudinal impedance

The concept of vascular impedance originally arose in 1955 from Womersley's linearized solutions for the equations of motion in a thin-walled, liquid-filled tube (see Chapter 5) (Womersley, 1955b; Womersley, 1957b). This was longitudinal impedance Z_L , since the flow Q was related to the pressure gradient ($P_1 - P_2$) and represents the impedance per unit length L of an artery.

Longitudinal impedance is related to the physical properties of the artery in the following manner. First, we have

$$-\frac{dP}{dL} = \frac{P_1 - P_2}{L} = A^* e^{i\omega t} \quad 12.1$$

where $A^* = M(\cos \phi - i \sin \phi)$, M is the modulus of the complex conjugate A^* , and ϕ is its phase angle. Also,

$$Q = \frac{A^* \pi R^4 M'_{10}}{i\mu\alpha} e^{\varepsilon_{10}} e^{i\omega t}, \quad 12.2$$

where R is the radius of the artery, μ is the blood viscosity, M'_{10} is the modulus of the Bessel function of the first kind, and of order zero, and ε_{10} is the phase angle of the Bessel function.

Hence,

$$Z_L = \frac{A^* e^{i\omega t}}{Q} = \frac{i\mu\alpha^2 e^{-i\varepsilon_{10}}}{\pi R^4 M'_{10}} \quad 12.3$$

or

$$Z_L = \frac{\mu\alpha^2}{\pi R^4 M'_{10}} \sin \varepsilon_{10} + \frac{i\mu\alpha^2}{\pi R^4 M'_{10}} \cos \varepsilon_{10}. \quad 12.4$$

From electric terminology, the first term on the right hand side (RHS) of eqn 12.4 is the resistive component of the longitudinal impedance and the second term is the reactive component (see Chapter 6); that is,

$$\text{Resistance} = \frac{\mu\alpha^2}{\pi R^4 M'_{10}} \sin \varepsilon_{10} \quad 12.5$$

and

$$\text{Reactance} = \frac{\mu\alpha^2}{\pi R^4 M'_{10}} \cos \varepsilon_{10}. \quad 12.6$$

Recalling that $a^2 = R^2 \omega \rho / \mu$, we can write

$$\text{Resistance} = \frac{\omega \rho}{\pi R^2 M'_{10}} \sin \varepsilon_{10} \quad 12.7$$

and

$$\text{Reactance} = \frac{\omega \rho}{\pi R^2 M'_{10}} \cos \varepsilon_{10}, \quad 12.8$$

so that

$$\text{Inductance} = \frac{\rho}{\pi R^2 M'_{10}} \cos \varepsilon_{10}. \quad 12.9$$

Inductance (or inertance per unit length) is the constant that dQ/dt must be multiplied by to give the force per area associated with the acceleration of the fluid.

The concept of longitudinal impedance was further developed by McDonald and used by him to calculate pulsatile flow from the pulsatile pressure gradient recorded in an artery (Bergel *et al.*, 1958).

Characteristic impedance

The step from longitudinal to characteristic and input impedance was first made by Michael Taylor (Taylor, 1957a; Taylor, 1957b; Taylor, 1959b; Taylor, 1959c) in his theoretic treatises on pulsatile pressure–flow relationships in arteries. These were conducted in McDonald's department at St Bartholomew's Hospital in London. They preceded his own extensive experimental and modeling studies (Taylor, 1959a; Taylor, 1959b; Taylor, 1959c; Taylor, 1964; Taylor, 1965; Taylor, 1966a; Taylor, 1966b; Taylor, 1966c; Taylor, 1966d; Taylor, 1969; Taylor, 1994). Taylor's move was from the theoretic to the practical, whereas most others in this field have subsequently worked in the opposite direction. Taylor pointed out that longitudinal impedance is related to input impedance through the complex wave velocity ($c/i\omega$) just as pressure gradient is related to absolute pressure. The simplest first step was to consider wave velocity in the absence of wave reflections (c). This, of course, gave the relationship between longitudinal impedance and characteristic impedance (Z_0).

$$Z_0 = \frac{P}{Q} = \frac{P_1 - P_2}{Q} \frac{c}{i\omega} \quad 12.10$$

or

$$Z_0 = Z_L \frac{c}{i\omega}. \quad 12.11$$

By substitution of eqn 12.3, we obtain the relationship of Z_0 to the properties of the artery.

Thus

$$Z_0 = \frac{c}{\omega} \frac{\mu \alpha^2 e^{-i\varepsilon_{10}}}{\pi R^4 M'_{10}} \quad 12.12$$

or

$$Z_0 = \frac{c \rho e^{-i\varepsilon_{10}}}{\pi R^2 M'_{10}}. \quad 12.13$$

Assuming that for an artery we are considering a longitudinally constrained tube, we substitute for c a relation containing the wave velocity c_0 given by the Moens–Korteweg equation (eqn 3.51):

$$c = c_0 / (X - iY), \quad 12.14$$

and so

$$c = \frac{c_0 (M'_{10})^{1/2}}{(1 - \sigma^2)^{1/2} e^{-i\varepsilon_{10}/2}}, \quad 12.15$$

(recall from Chapter 2 that σ is Poisson's ratio). Equation 12.13 now becomes

$$Z_0 = \frac{c_0 \rho e^{-i\varepsilon_{10}/2}}{\pi R^2 (1 - \sigma^2)^{1/2} (M'_{10})^{1/2}}. \quad 12.16$$

The characteristic impedance thus varies directly with the elastic modulus of the artery and inversely with its cross-sectional area. Therefore, the stiffer the artery and/or the smaller its radius, the higher the characteristic impedance (Cholley *et al.*, 2001).

When there are reflected waves present, the situation becomes more complicated because of the formation of nodes and antinodes (see Chapter 9). The impedance at the origin of a system in which reflected waves are present is termed the 'input impedance' (Attinger *et al.*, 1966c; Gessner, 1972; McDonald, 1974; Nichols *et al.*, 1977a; Gow, 1980; Finkelstein and Collins, 1982; O'Rourke, 1982a; O'Rourke, 1982b; O'Rourke, 1982c; Milnor, 1989; Westerhof and Elzinga, 1991; Westerhof and O'Rourke, 1995).

Input impedance

The input impedance moduli of any region of the arterial system is the ratio of harmonic terms of pressure at the input to the corresponding harmonic terms of flow; the difference in phase angles of pressure and flow determines the input impedance phase. Thus, the ratio of pulsatile pressure and pulsatile flow in the ascending aorta determines the input impedance of the systemic arterial system. Similarly, the ratio measured in the femoral artery expresses the input impedance of the femoral bed supplied by that artery.

Expressing each complex harmonic term in modulus and phase form gives the input impedance as a set of terms (modulus and phase) versus frequency (spectra) obtained by Fourier analysis (Chapter 11).

In complex notation, an harmonic component of pressure is

$$P = |P|e^{i(\omega t - \phi)} \quad 12.17$$

and flow is

$$Q = |Q|e^{i(\omega t - \beta)}. \quad 12.18$$

The input impedance for this particular harmonic term is

$$Z_i = \frac{P}{Q} = \frac{|P|e^{i(\omega t - \phi)}}{|Q|e^{i(\omega t - \beta)}} = \frac{|P|e^{i(\beta - \phi)}}{|Q|}. \quad 12.19$$

The real part of eqn 12.19 is

$$Z_i = \frac{|P|}{|Q|} \cos(\beta - \phi), \quad 12.20$$

where $|Z| = |P|/|Q|$ is the modulus and $\theta = (\beta - \phi)$ is the phase of the impedance.

In practical application to the study of the circulation, and specifically of the arteries, input impedance has been of immense value. While the concepts of longitudinal impedance, characteristic impedance, and terminal impedance are important, they are very much of subsidiary importance to input impedance. Where the term ‘impedance’ or ‘vascular impedance’ is used alone without further qualification, it is taken to mean ‘input impedance’.

INPUT IMPEDANCE AND MODELS OF THE ARTERIAL TREE

The simplest model of the arterial system is of course the Windkessel with lumped capacitance and resistance elements (see Chapter 1). This model with an added inductance was used by Patel *et al.* (1963a) to explain features of averaged impedance data in the ascending aorta of dogs of various sizes. The averaged moduli and phases showed little fluctuation at high frequency and the model appeared to fit the data reasonably well, but the authors cautioned against too literal an extrapolation. Absence of fluctuations, which might be indicative of reflection, was in this historic paper (the first on ascending aortic and pulmonary impedance) subsequently attributed to averaging of data obtained from animals of different sizes (O’Rourke and Taylor, 1967); such averaging could add maximum values of modulus in one animal to minimal values of modulus in another.

At the input of a Windkessel, the impedance modulus falls from its high value at zero frequency (which is determined by its resistive value) to low values at high frequency (which are determined by the magnitude of capacitance in relation to resistance), while phase is zero at zero frequency, becoming progressively more negative (flow leading pressure) at higher frequencies (Fig. 12.3A,

left). Impedance modulus and phase show no fluctuations in the Windkessel because there can be no wave reflection in such a system.

A modification of the Windkessel takes account of arterial characteristic impedance by having a separate inductance and capacitance in the line between the input and mean capacitive and resistive components (Fig. 12.3A). This model was extensively used by Westerhof and co-workers (Westerhof *et al.*, 1969; Westerhof *et al.*, 1973; Noordergraaf, 1978; Westerhof *et al.*, 1978; Westerhof *et al.*, 1979; Sipkema *et al.*, 1980) and is sometimes colloquially referred to as the Westkessel. This type of model has been utilized for studies of ventricular function by many, including Sunagawa and co-workers (Sunagawa *et al.*, 1982; Sunagawa *et al.*, 1985a; Latham *et al.*, 1987; Latham *et al.*, 1988a; Latham *et al.*, 1988b; Westerhof and Elzinga, 1991; Wesseling *et al.*, 1993), as a simple approximation of the systemic circulation capable of achieving some separation of the properties of proximal aorta, from more distal aorta and peripheral arteries, and from the resistive vessels. In this model, the input impedance modulus falls from a value at zero frequency which is dependent on peripheral resistance to a constant value at high frequencies which is dependent on capacitive and inductive properties of the proximal aorta, with steepness of fall dependent on capacitive properties (distensibility) of the aorta and large arteries (Fig. 12.3A).

The impedance phase is zero at zero frequency, becomes negative at low frequencies and then returns towards zero at high frequencies. As for the Windkessel, this model (Fig. 12.3A, right) shows no fluctuation of modulus, nor of phase (except for the initial early negativity returning towards zero), because there can be no wave travel or reflection in the model.

It is clear that wave reflection markedly alters pulse wave patterns (see Chapters 9 and 10) and experimentally determined impedance values, so must be considered in any realistic model of the arterial system. This was recognized by Taylor, who conducted the first definitive studies on impedance patterns in simple tubular models of the arterial system during the 1950s (Taylor, 1957a; Taylor, 1957b; McDonald and Taylor, 1959; Taylor, 1959b), then refined these with models (Fig. 12.3B) of increasing sophistication during the 1960s (Taylor, 1964; Taylor, 1965; Taylor, 1966a; Taylor, 1966b; Taylor, 1966c; Taylor, 1969; Taylor, 1973), conducting the later work with detailed experimental studies that will be presented in a later part of this chapter. More recent contributions on modeling of the arterial system and vascular impedance have been numerous, and include those by Westerhof *et al.* (Westerhof *et al.*, 1969; Westerhof *et al.*, 1973; Westerhof *et al.*, 1978), Cox (1975a), Avolio (1980), O’Rourke and Avolio (1980), and others (Noble, 1979a; Burattini and Carlos, 1988; Burattini *et al.*, 1991; Berger *et al.*, 1994; Karamanoglu *et al.*, 1994; Karamanoglu *et al.*, 1995; Burattini and Natalucci, 1998; Quick *et al.*, 1998; Sharp and Dharmalingham, 1999; Stergiopoulos *et al.*,

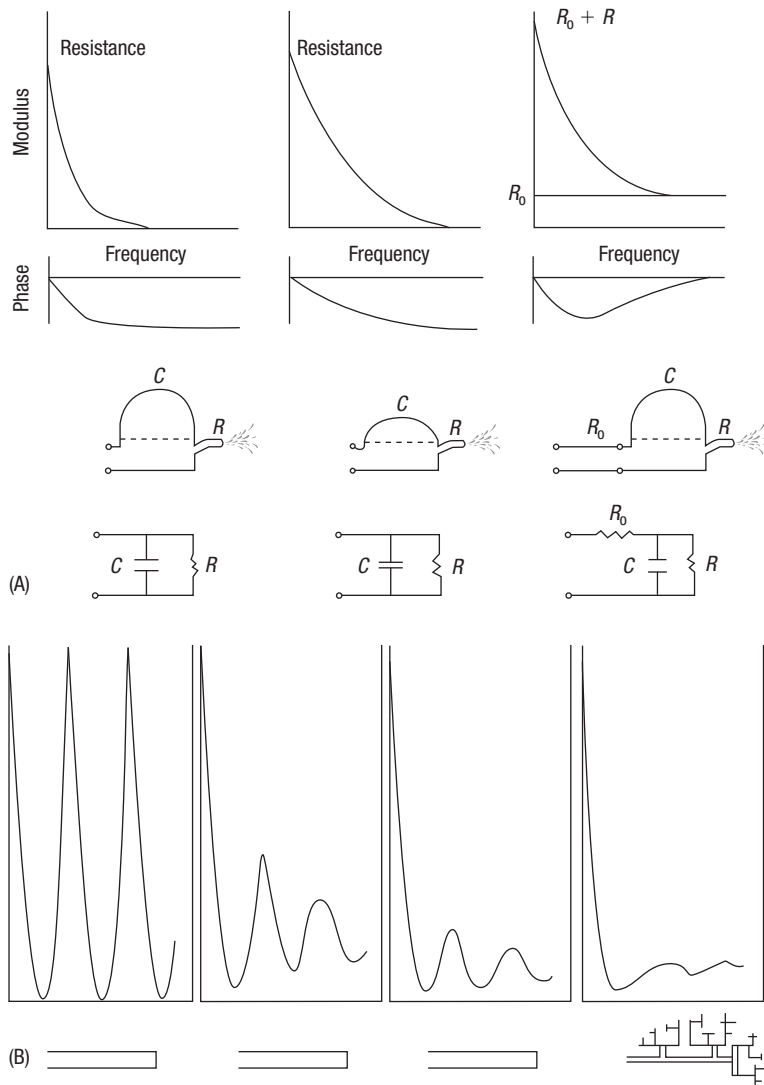


Figure 12.3 (A) Modulus and phase of impedance at the input of three simple models of the arterial system, together with the corresponding mechanical and electric models. Left: simple Windkessel with high arterial compliance. Center: simple Windkessel with lower arterial compliance. Right: modified Windkessel with proximal as well as distal resistance. Reproduced from O'Rourke *et al.* (1984). (B) Modulus of impedance at the input of four different tubular models of the arterial system, all of which permit wave travel and reflection. The first is a simple tube terminating at its closed distal end and with inviscid liquid and tubular wall. The second is a simple tube as the first, but with viscid liquid and viscoelastic wall. The third is a simple tube as in the second, but with progressively increasing wall stiffness towards the distal end. The fourth is a network of tubes, incorporating all features of the third model. Reproduced from O'Rourke *et al.* (1984)

1999; Miyashita *et al.*, 2000; Sharp *et al.*, 2000; Westerhof and Stergiopoulos, 2000; Mitchell *et al.*, 2004a; Chirinos *et al.*, 2009b; Chirinos *et al.*, 2009c). Most recent studies of vascular impedance have used non-invasive measures of pressure and flow in humans, usually in the ascending aorta or LV outflow tract (for flow) and carotid artery (for pressure) (Kelly and Fitchett, 1992; Segers *et al.*, 2007b; Mitchell, 2009a; Mitchell, 2009b). Accuracy is compromised by the need to measure flow and pressure at different sites and at different times. Various assumptions (Table 12.1) are used which make interpretations more difficult than in previous invasive studies. The electromagnetic flow/pressure catheter system is rarely used now (Murgo *et al.*, 1980a; Nichols *et al.*, 1985a; Noda *et al.*, 2006) on account of flow turbulence in the ascending aorta and artifact related to this. Cost and restrictions on reuse also pose impediments. In the Asklepios study (Segers *et al.*, 2007b), there was far more variability in calculated ascending aortic impedance at any age than in simpler measures, such as augmentation index. In the Mitchell study for Framingham (Mitchell *et al.*, 2010), there was no

apparent benefit for indices of wave reflection, such as calculated impedance in prediction of outcomes, but methodological difficulties have been noted, and are discussed elsewhere. In Murgo's classic studies of invasively determined aortic impedance (Murgo *et al.*, 1980a), effects of wave reflection could be interpreted directly from the pressure and flow waves in the time domain, as well as from impedance in the frequency domain. A variety of modeling methods has been used, including mathematical models, electric networks, and hydraulic systems. Results are directly comparable, as they should be, despite differences in approach. The benefit of mathematical models is that variables can be changed at will by altering parameter values in a digital computer. Analog and electric models are more difficult to change, while hydraulic models are hard to handle, with results too dependent on the accuracy of flow and pressure-measuring systems. This subject is addressed further in an edition of the *Journal of Biomechanics* edited by Westerhof (2003; 36(5): 621–748), and in a later journal supplement *Medical, Biological, Engineering and Computing* (2009; 47(2): 107–239).

Table 12.1 Assumptions inherent in recent non-invasive studies of ascending aortic impedance in humans (Ghent and Framingham group)

1. Brachial cuff systolic blood pressure (SBP), diastolic blood pressure (DBP) = intra-arterial SBP, DBP
2. Applanation tonometry applicable to brachial artery
3. Brachial tonometric waveform form factor (FF) $\approx$ brachial intra-arterial waveform FF
4. Brachial mean pressure = carotid mean pressure
5. Brachial diastolic pressure = carotid diastolic pressure
6. Carotid systolic pressure accurately calculated from carotid waveform FF and brachial diastolic and mean pressure
7. Carotid pressure waveform recorded accurately by tonometry
8. Carotid pressure waveform $\approx$ aortic pressure waveform
9. Aortic root does not move in relation to ultrasonic sensor during cardiac cycle
10. Left ventricular outflow tract waveform envelope = flow velocity
11. Velocity profile flat in left ventricular outflow tract and proximal aorta
12. Left ventricular outflow tract is circular and diameter is measurable accurately
13. Ascending aortic flow waveform = left ventricular outflow tract waveform
14. Ascending aortic diameter is measurable accurately
15. Carotid pressure and left ventricular flow, recorded at different times, can be aligned
16. Carotid augmentation = aortic augmentation
17. Characteristic impedance measurable in time domain from initial flow and pressure peak (Dujardin/Stone method) (Dujardin and Stone, 1981)

Taylor's first model of the arterial system was an attenuating transmission line, terminating in a high resistance. In this, modulus (and phase) showed fluctuations in input impedance that were attributed to peripheral wave reflection. Data were expressed in terms of wavelength along the abscissa, but with distance to termination and wave velocity constant, this was equivalent to frequency (Fig. 12.4). Similar results were obtained for Taylor's hydraulic model, which comprised simply a long distensible tube, completely closed at one end and activated at the other (the input) with a hydraulic sinusoidal pump.

Figure 12.5 shows the input impedance of a tube that is taken to represent the arterial system. If in such a system there is no attenuation of the traveling wave, and positive, but incomplete wave reflection at its termination (solid line), the modulus and phase of the input impedance will show repeated fluctuations with increasing frequency, as illustrated in Fig. 12.5A. Such fluctuations will likewise be seen along the tube, as in Fig. 12.5B. The positions of minima and maxima are determined by the length of the tube and its wave velocity (see Chapter 9). With input impedance plotted against frequency (Fig. 12.5A), impedance is maximum at zero frequency and at successive frequencies that correspond to one half-wavelength, with minima at values that correspond to 1/4, 3/4, and 5/4 of one wavelength, and with phase-crossing zero at multiples of one-quarter wavelength. With impedance plotted against distance from the tubular end (Fig. 12.5B), there are similar fluctuations in modulus and phase, with modulus maximal and phase at zero at the terminus itself, and with minima and maxima alternating back up the tube at multiples of one-quarter wavelength, and with phase-crossing zero also at multiples of one-quarter wavelength.

Figure 12.5 differs from Fig. 12.4 in taking no account of pulse wave attenuation. When included (dotted line in

Fig. 12.5), this contributes to decreasing fluctuation of modulus and phase at higher frequencies and at increasing distance from the tubular termination.

The model results illustrated in Fig. 12.4 are a simplification of how such a model may represent the arterial system. No account is taken of variation in phase of wave reflection, a phenomenon which is a consequence of the complex reflection factor at the tubular end (see Chapter 9). This point is discussed in detail by Milnor (1989), but in practice it appears to be of little significance. Reflection at the terminus appears to be as from a completely closed end, with pressure and flow 180° out of phase (see Chapter 9).

Taylor's subsequent analysis considered the effects of variable wave reflection at the tubular end, and of liquid and wall viscosity on input impedance. Viscosity increased wave attenuation and decreased fluctuations in the modulus and phase of impedance at increasing frequencies. His next consideration was the effects of increasing wall stiffness between the central aorta and peripheral arteries. It is a known property of all mammals that the proximal aorta is far more distensible than the distal abdominal aorta and muscular distributing arteries. This was shown to be responsible for considerable reduction in impedance values at higher frequencies in relation to impedance at zero frequency (Fig. 12.6). The significance of this as a factor that isolates the heart from the peripheral resistance is discussed below.

The effects of variable distance to individual reflecting sites was the next factor considered. It was shown (Fig. 12.7) that this reduces fluctuations of modulus and phase with increasing frequency, and that the magnitude of this dampening effect was dependent on the degree of spatial dispersion of peripheral reflecting sites. Taylor's final contribution in this area was to develop a randomly branching model of the arterial system that might be

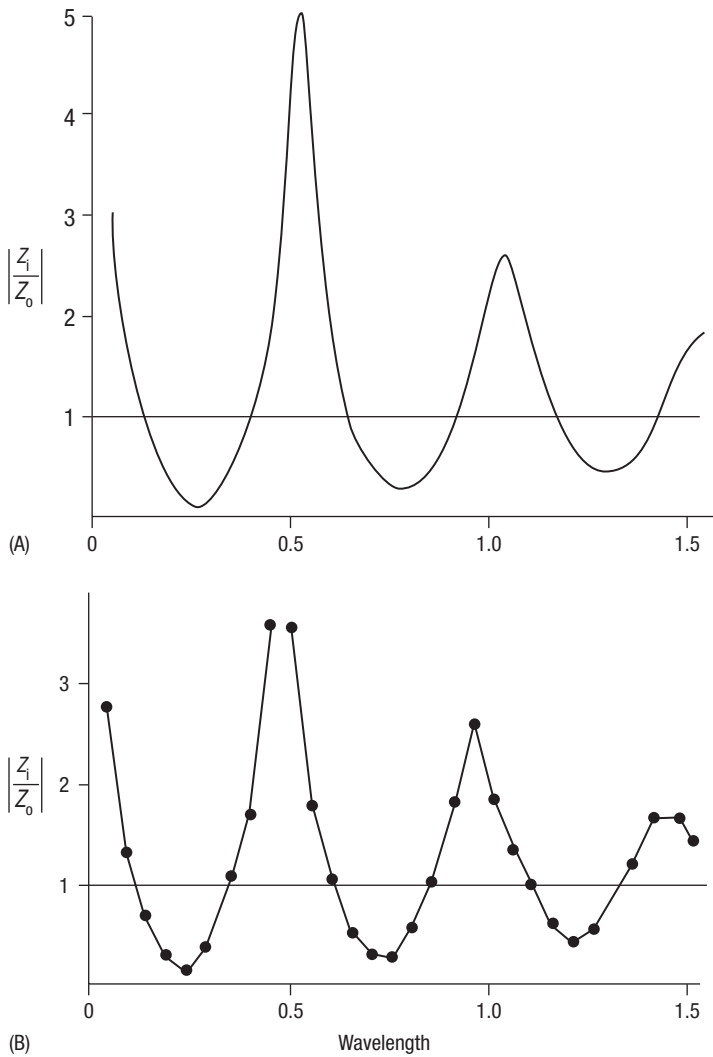


Figure 12.4 Modulus of input impedance (Z_i) as a function of characteristic impedance (Z_0), plotted against wavelength. (A) As calculated for a uniform transmission line. (B) As measured in a rubber tube. Reproduced from Taylor (Taylor, 1957a; Taylor, 1957b)

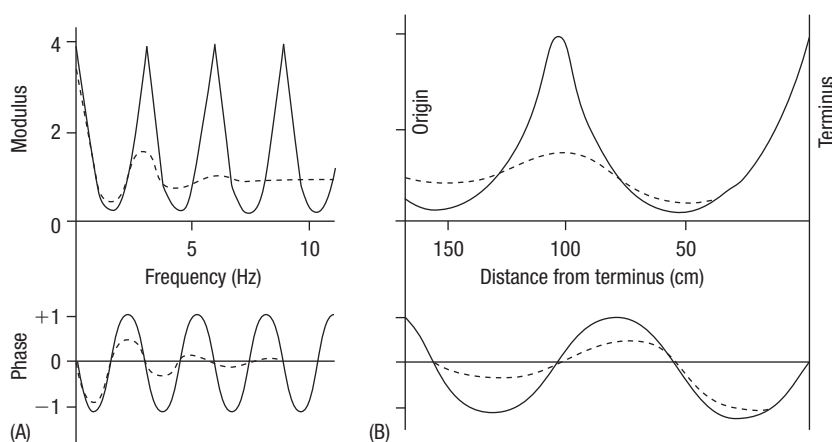


Figure 12.5 Modulus (above) and phase (below) of input impedance in a uniform elastic tube terminating in high terminal impedance. (A) As determined at the input and expressed as a function of frequency. (B) As determined at the same frequency and expressed as a function of distance from the tubular end. Unbroken lines: no liquid or wall viscosity; dotted lines: with allowance made for liquid and wall viscosity. For explanation, see text. Reproduced from Milnor (1989)

expected to mimic more closely the properties of a peripheral vascular bed (Fig. 12.8) (Taylor, 1966a). Eight consecutive generations of vessels were considered, each giving off two branches at its distal end, and the last-generation vessel terminating at a pure resistance. The equations of Womersley's constrained tube were used (see

Chapter 6) to calculate impedance and propagation coefficients. No account was taken of the anatomic and hydrodynamic peculiarities of bifurcations. Input impedance of the last generation of branches was first determined, and these were taken as the terminal impedance of the next suborder of branches, and so on up to the input

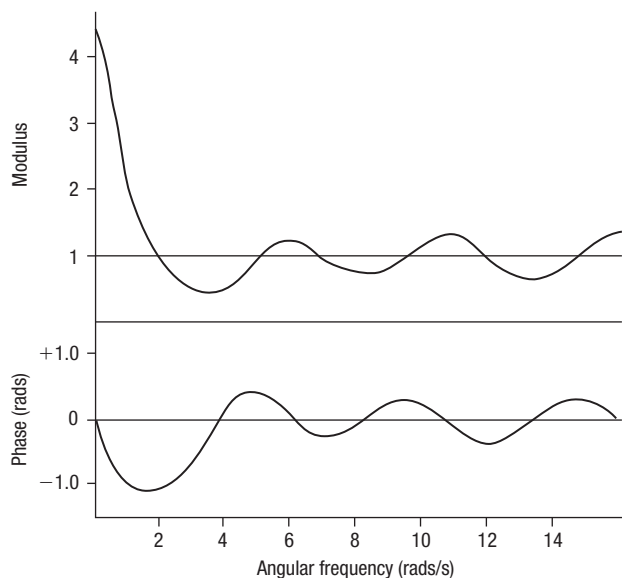


Figure 12.6 Modulus and phase of impedance at the input of a non-uniform elastic tube showing 'isolation' of input from periphery (see text). Reproduced from Taylor (1965)

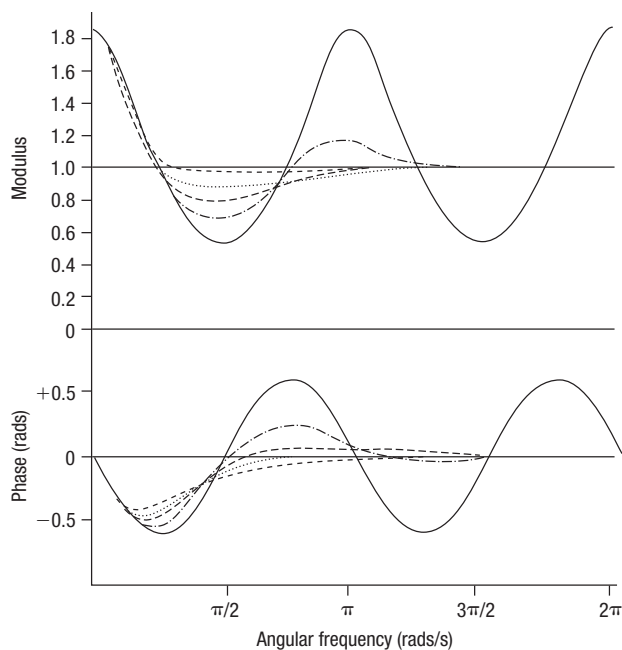


Figure 12.7 The influence of spatial dispersion on peripheral reflecting sites. The input impedance of flow assemblies of 100 tubes of random length, connected in parallel at their origins. The average length of all tubes in the one assembly is the same. The dashed line shows results in the assembly with the greatest degree of spatial dispersion; the dotted and dashed lines show results in the assembly with the least degree of spatial dispersion; the dotted line and line with short and long dashes show results in assemblies with intermediate degrees of spatial dispersion. The solid line shows results in a single equivalent tube. Reproduced from Taylor (1966a)

of the whole assembly. The whole complex assembly showed behavior that was remarkably similar to impedance calculated from pressure and flow at the input of a peripheral vascular bed. The model was next modified to represent separately the vascular tree in the upper and in the lower part of the body; this (see below) was extremely useful in explaining impedance patterns in the ascending aorta, and the differences of impedance in this vessel compared with the impedance in the descending thoracic aorta (supplying the lower body) and in the brachiocephalic and subclavian arteries (supplying the upper body).

Further developments in computer modeling have included efforts to represent realistically the dimensions and branching patterns of arteries in the systemic circulation (Westerhof and Noordergraf, 1970; Avolio, 1980; O'Rourke and Avolio, 1980; O'Rourke, 1982c; Karamanoglu *et al.*, 1994; Karamanoglu *et al.*, 1995) and lungs (Milnor, 1989; Fourie *et al.*, 1992). Results are remarkably similar to experimentally derived impedance values.

All these models have been very useful in interpreting values of input impedance determined in living animals. They have been helpful, too, in explaining differences between different vascular beds, between different-sized and different-shaped animals, with changes in mean arterial pressure, with alterations in arteriolar tone, and in disease. Modeling results are discussed later in relation to these experimental findings.

ASSUMPTIONS IN THE APPLICATION OF IMPEDANCE THEORY TO THE ARTERIAL SYSTEM

The theoretic concept of vascular impedance is based on the theoretic principles established by Womersley on the relationship between pulsatile pressure and flow in arteries. This is a more general application of Poiseuille's law as it applies to the relationship between mean pressure and flow. Just as Poiseuille's law (see Chapter 2) cannot strictly be applied to any part of the circulation (see Chapter 2), so too Womersley's concepts cannot strictly be applied either. Yet both have been applied and both have been extremely valuable in interpreting vascular properties – Poiseuille's law in interpreting change in arterial caliber and vasomotor tone from changes in peripheral resistance, Womersley's in interpreting properties of vessels that modify pulsatile phenomena. In interpreting data, however, one must remain aware of the restrictions and reservations that apply to both. All those that apply to Poiseuille's law for steady flow also apply to Womersley's concepts for pulsatile flow. In addition, Womersley's theory assesses linearity of pressure–flow relationships – or more precisely that non-linearities are sufficiently small to be neglected as a first approximation. This is discussed elsewhere (see Chapter 6); experimentally determined impedance results support the concept of linearity, and

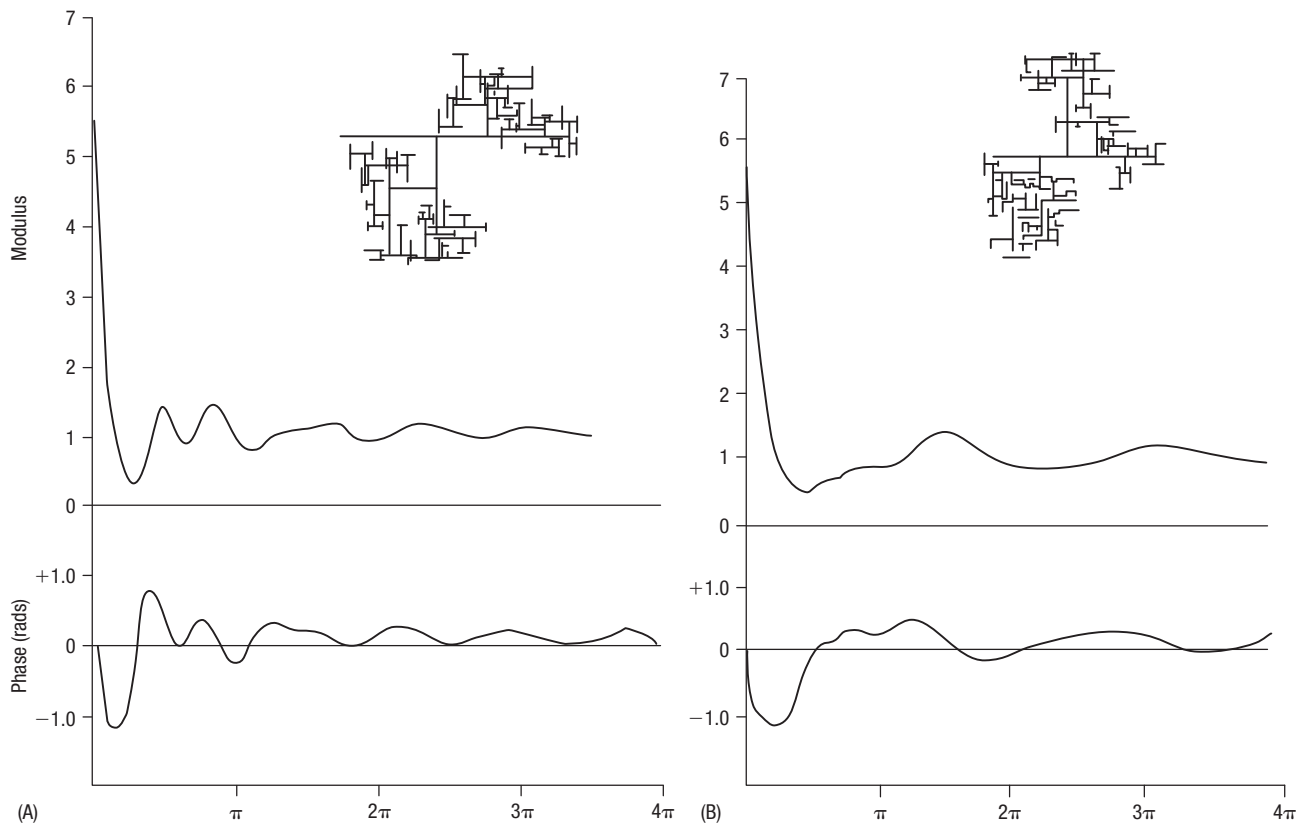


Figure 12.8 (A) Input impedance of a randomly branching network of elastic tubes. (B) Input impedance of a network comprising two separate randomly branching networks at different distances from the origin. Reproduced from Taylor (1966a)

there is far less controversy on this issue than there was in the past. However, the potential for non-linearities, even large non-linearities, is still present and must be kept in mind if the concepts are to be applied sensibly. Other practical, conceptual, and theoretic problems in the use of vascular impedance are discussed later in this chapter.

IMPEDANCE IN A TYPICAL PERIPHERAL ARTERY

Input impedance of the femoral vascular bed has been measured more frequently than that of any other, except for the whole systemic circulation (ascending aortic impedance) and pulmonary circulation (main pulmonary artery impedance). The femoral artery can be considered to supply a typical peripheral vascular bed and is, technically, an ideal site for determining impedance under a variety of conditions, since the femoral artery can readily be exposed surgically so that flow and pressure can be measured with minimal surgical trauma and under very light anesthesia. In addition, the effects of vasodilatation and vasoconstriction can be maximized by intra-arterial infusion of dilator or constrictor drugs, respectively, without large changes in total systemic resistance and thus of mean arterial pressure. Impedance patterns in other peripheral arteries show the same general features and the

same changes under different conditions as those in the femoral artery.

The first published data of vascular input impedance were those of Randall and Stacy (1956) for the femoral vascular bed. While this paper was of historic importance in appreciating the value of the impedance concept, the actual results are dubious because of the method used for recording flow. The flowmeter was a cannulating electromagnetic flowmeter connected to the artery by 15 cm of polyethylene tubing. Pressure was measured proximal to the tubing. This tubing produced great damping of pulsatile flow, and the impedance measured was in effect that of the bed plus the length of plastic tubing. McDonald and Taylor (1959) presented data on the modulus of the femoral impedance calculated from flow measured by the high-speed cinematography technique in the femoral artery of a small dog (Fig. 12.9). This showed a well-marked minimum at 13.5 Hz, which was attributed to a 'mean' position for the termination of the bed somewhere between the ankle and knee. McDonald (1960) subsequently published data on change in impedance with vasoconstriction and vasodilatation, using the same high-speed cinematographic flow technique, and, later, femoral vascular impedance from flow measured with the differential pressure technique, using the Womersley equation. These early results are very similar to those obtained subsequently with the more accurate electromagnetic

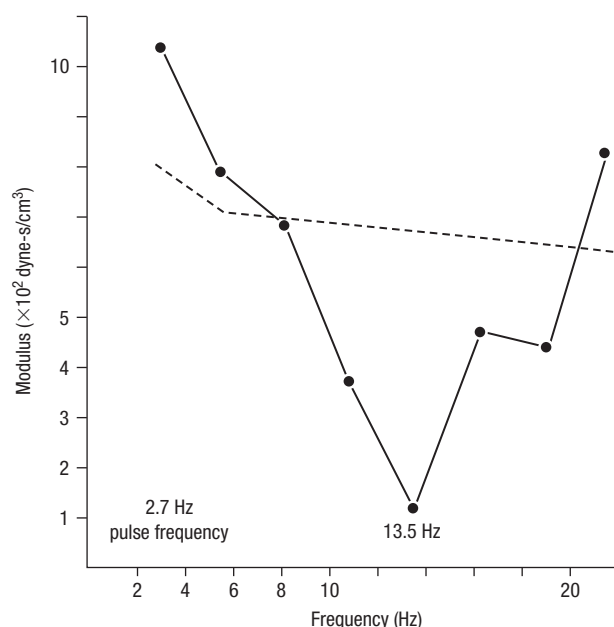


Figure 12.9 Modulus of femoral vascular impedance determined in a dog, with flow recorded by high-speed cinematography; dotted line is calculated characteristic impedance. Data originally published in McDonald (1960)

method for measuring flow. McDonald's results for femoral vascular impedance in a large dog using the differential-pressure technique were similar to those obtained by Gabe (1965b) in a human, even to the frequency of the minimal value of modulus and phase crossover. While this similarity between dog and human was reassuring to the investigators, it is really quite surprising given the fact that the human adult leg is much longer than that of the dog. This was the first of many reports on impedance in humans that showed an impedance minimum occurring at similar frequencies in corresponding arteries of dogs (Patel *et al.*, 1963a; Patel *et al.*, 1965a; Patel *et al.*, 1965b). With the body length so much longer in an adult human than in a dog, this suggested that the pulse wave velocity was considerably higher than in the dog. We are only now beginning to understand the implications of this finding – that the adult human shows signs of subclinical arterial degeneration that are not seen to nearly the same degree in adult mature dogs.

The most detailed study of input impedance of the femoral bed is that of O'Rourke and Taylor (1966). They used an electromagnetic flowmeter and were the first to use a digital computer for analysis of large data batches. A composite set of values, measured in a dog, is shown in Fig. 12.10; in this animal, heart block was induced and a variety of heart rates imposed by electrical pacing. This shows again a high value for the impedance modulus at the lowest frequencies, with a broad minimum from 8.0 to 14 Hz and a rise to a subsequent maximum at 20 Hz. The phase is initially negative for the first harmonic at the lowest heart rate, passes through zero at approximately

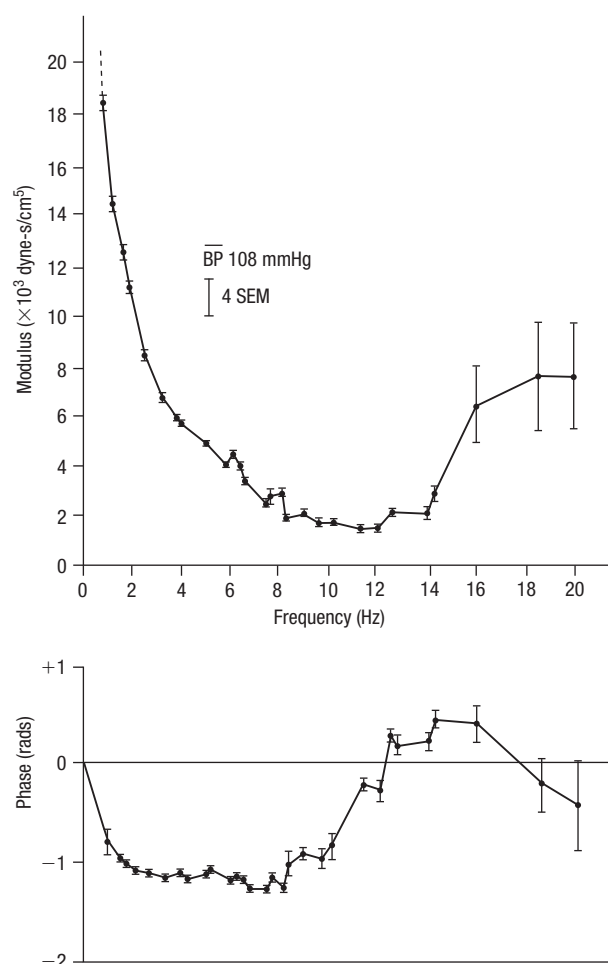


Figure 12.10 Vascular impedance in the femoral artery of a dog. Impedance was determined from Fourier analysis of a series of 57 waves recorded in this dog with heart block, whose heart was paced at different frequencies. Reproduced from O'Rourke and Taylor (1966)

12 Hz to become positive, but again passes through zero to become negative at around 18 Hz.

The fact that the impedance plots for a variety of heart frequencies all fell on the same curve with such little variation as shown in Fig. 12.10 is strong justification for the assumption that, in terms of pressure/flow relations, the arterial tree may be treated as a linear system. Similar results were repeated by the same authors for data obtained when the heart was beating quite irregularly during sinus arrhythmia (Fig. 12.11). In both cases, data points were virtually superimposed for modulus and for phase, even with data from the third harmonic of pressure and flow waves recorded for a long period corresponding with the first harmonic of waves recorded for a period one-third as long. Similar results have been repeatedly reported for the femoral artery and for other vascular beds by other authors (Bergel and Milnor, 1965; Patel *et al.*, 1965a; Patel *et al.*, 1965b; Noble *et al.*, 1967; Nichols *et al.*, 1977a; Noble, 1979b; Milnor, 1989; see also

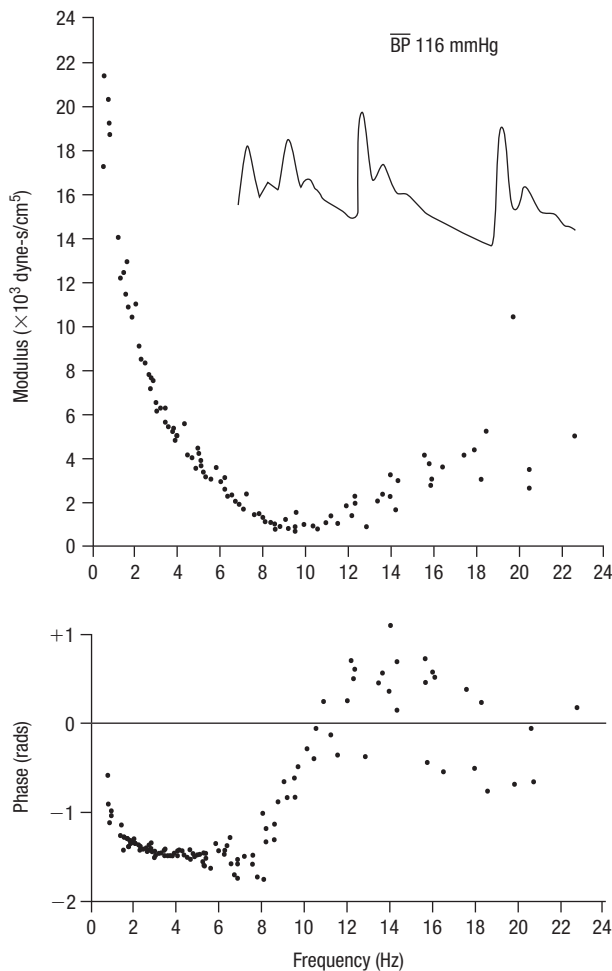


Figure 12.11 Vascular impedance in the femoral artery of a dog; impedance was determined from a series of 11 waves recorded during marked sinus arrhythmia. Reproduced from O'Rourke and Taylor (1966)

Dick *et al.*, 1968; Zuckerman and Yin, 1989a; Mitchell *et al.*, 1994).

O'Rourke and Taylor (1966) reported on the changes that occur in the femoral vascular impedance patterns with arteriolar constriction and dilatation. Figure 12.12 records the changes in modulus of input impedance to the femoral vascular bed following the intra-arterial injection of 100 mg of acetylcholine. The first control shows a similar pattern of impedance to that seen in Fig. 12.10: a high value at zero frequency falling to a minimum value between 10 and 12 Hz and thereafter increasing again. Two seconds after injection, the impedance at zero frequency has fallen dramatically, and this fall becomes maximal at between 6 and 16 s. At the same time, the impedance modulus at 10–12 Hz has increased so that the previous maximum is not seen; the whole curve is much flatter at 6.0–16 s than is the control curve. This change is typical of a situation where there is very little wave reflection. The effect of acetylcholine passes off very rapidly; the first indication is an increase in modulus at

zero frequency and in the low-frequency range, but by 30 s the maximum at 10 Hz with subsequent rise to another maximum at a higher frequency can be seen again. The full control pattern is re-established at 44 s.

Changes in impedance phase with vasodilatation caused by acetylcholine are shown in Fig. 12.13. The flattening in impedance modulus is associated with flattening in impedance phase, also indicating a decrease in peripheral wave reflection. The pressure and flow waves from which these data were derived are shown in Fig. 12.14, with waves in Fig. 12.14A recorded under control circumstances and those in Fig. 12.14B during the peak effect of intra-arterial acetylcholine. A reduction in the impedance phase is apparent, as is similarity in the contours of pressure and flow waves.

The effects of vasoconstriction caused by intra-arterial norepinephrine infusion were less dramatic than those induced by acetylcholine, and comprised a slightly greater (more negative) impedance phase at low frequencies (Fig. 12.13), a greater impedance modulus at zero frequency, but little change in the minimal value of impedance modulus at 10–12 Hz.

It is interesting that the changes in impedance modulus and phase associated with arteriolar dilatation or constriction were not associated with any shift in the frequency of the minimum of impedance modulus or phase crossover. However, these changes were accomplished without any appreciable change in mean arterial pressure (since dilatation and constriction, respectively, only involved the regional vascular bed). This was not the case, however, when the mean arterial pressure was raised or lowered by intravenous infusion of vasoactive agents or by other maneuvers. Figure 12.15 (page 288) shows the effects of altering the mean arterial pressure on femoral vascular impedance. An increase in the mean pressure shifted the impedance curves to the right, so that minimal values of modulus and phase crossover occurred at a higher frequency; lowering the mean pressure shifted the impedance curves to the left. In this case, a three-fold change in vascular resistance was not associated with any perceptible change in the minimal value of impedance modulus, and with only a small change in impedance phase. By way of contrast, the difference in vascular resistance between maximal dilatation and maximal constriction in Fig. 12.13 was more than 50-fold (a 7.5-fold reduction in resistance from control to full vasodilatation and a greater than eight-fold increase in resistance from control to full vasoconstriction).

These data for femoral vascular impedance were interpreted by O'Rourke and Taylor (1966) on the basis of Taylor's models of the vascular system (Taylor, 1957a; Taylor 1957b; Taylor, 1959a; Taylor, 1959b; Taylor, 1959c; Taylor, 1966a; Taylor, 1966d). The patterns of impedance modulus and phase under control conditions were explained on the basis of a single, functionally discrete reflecting site in the vascular bed downstream. The distance to this reflecting site was estimated from the

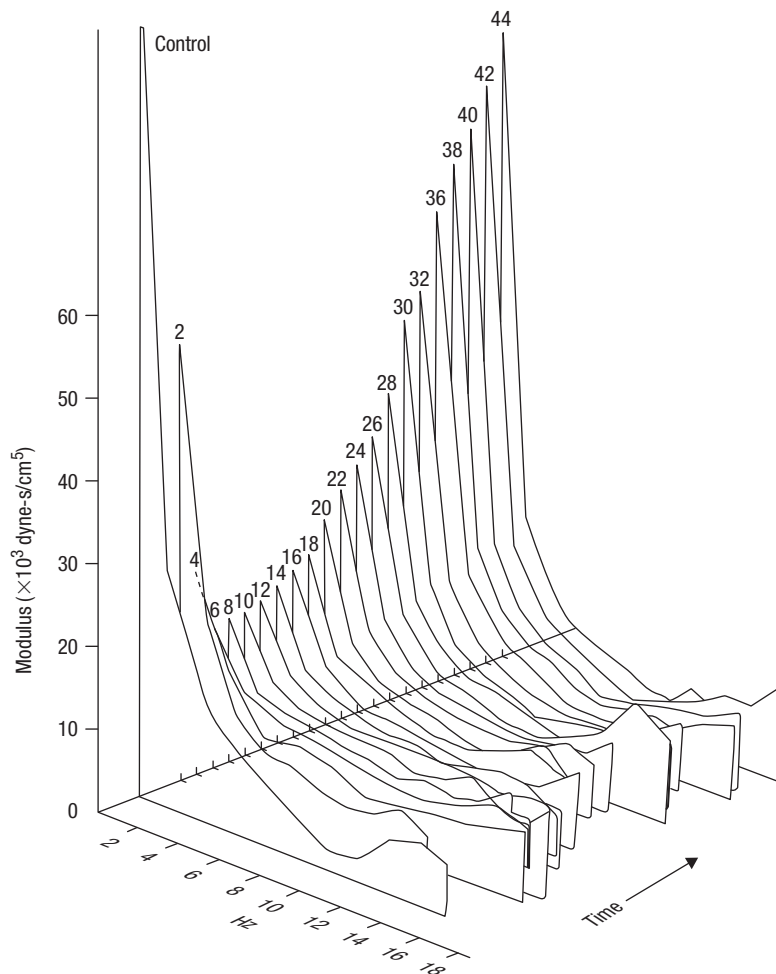


Figure 12.12 Effects of vasodilatation on the modulus of vascular impedance in the femoral artery of a dog. The time course of the impedance modulus is shown following intra-arterial injection of acetylcholine (100 mg in 1.0 mL). Each curve represents the modulus determined from one pair of pressure and flow waves. Time in seconds following injection is shown above each curve. Reproduced from O'Rourke and Taylor (1966)

minimal value of impedance modulus and phase crossover (which occurred at 9.0 Hz; Fig. 12.15) (see Chapter 9); taking this to represent one-quarter wavelength ($\lambda/4$) from the reflecting site, and using a value of 10 m/s for wave velocity, wavelength (c/f) = 1000/9 or 111 cm, and $\lambda/4 = 28$ cm. This distance in the dog concerned identified a point just below the knee. This was considered to represent a 'functionally discrete' reflecting site in the femoral vascular bed, being the resultant of all individual reflecting sites in this bed. The intensity of wave reflection in this vascular bed was calculated from eqn 10.3. With the femoral vascular resistance taken as 57×10^3 dyne-s/cm⁵ and the characteristic impedance as 10×10^3 dyne-s/cm⁵, the reflection coefficient R_c was calculated as

$$R_c = (Z_T - Z_0)/(Z_T + Z_0) \\ = 47/67 \text{ or } 0.70$$

For Figs 12.10 and 12.11, reflection coefficients were calculated as 0.89 and 0.92. A value of 0.8 for R_c seems a reasonable approximation. This value is considerably higher than previously estimated by McDonald (0.3–0.4) for the femoral bed. McDonald, however, had taken Z_T as close to the value of impedance modulus at the lowest harmonic, neglecting the initial steep rise at lower

frequencies to the value at zero frequency. This mistake was pointed out by Taylor (Taylor, 1964; Taylor, 1966a) and acknowledged by McDonald (1974) in the second edition of this book.

Intense vasodilatation induced by intra-arterial acetylcholine in the femoral bed virtually eliminated all evidence of wave reflection in the terminal vascular bed. This finding indicates two important points: first that wave reflection is almost entirely due to arteriolar tone in the peripheral vascular bed, and second that other characteristics of the vascular bed that might cause wave reflection but are not altered by acetylcholine (the myriad of branching points in the vascular bed) are unimportant as reflecting sites. This finding for intra-arterial vasodilatation provides probably the most convincing support for Hamilton's contention (Hamilton, 1944) that arterioles are the major site of peripheral wave reflection. It is probable that differences in caliber and distensibility of the aorta and other major arteries may provide additional sites of wave reflection (Taylor, 1964; Latham *et al.*, 1985; Burattini *et al.*, 1991; Latham *et al.*, 1992), as previously discussed, and as emphasized by many on a theoretic or intuitive basis (Mitchell, 2009a; Mitchell, 2009b). At least in a peripheral vascular bed such as the femoral, this does not appear to be the case. The normally high reflection coefficient in the

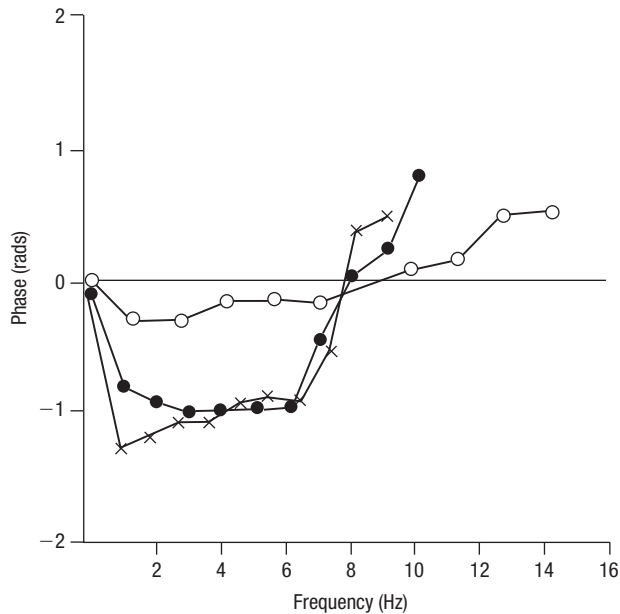


Figure 12.13 Effects of vasodilatation (following intra-arterial acetylcholine 100 mg) and of peak vasoconstriction (following intra-arterial injection of norepinephrine 6 mg) on the phase of impedance in the femoral artery of a dog. (○) Vasodilatation; (●) control; (X) vasoconstriction. Reproduced from O'Rourke and Taylor (1966)

femoral vascular bed can be attributed to normally high resting arteriolar tone. A high reflection coefficient under control conditions explains, too, the relatively small change in impedance patterns with vasoconstriction, which cannot increase the reflection coefficient beyond unity (i.e. from 0.8 towards 1.0), whereas vasodilatation can reduce this to a far greater degree (from 0.8 towards zero).

Taylor's model of the peripheral vascular bed (see Fig. 12.8) was applied to impedance results from the femoral vascular bed under control conditions and during vasodilatation and vasoconstriction. Using realistic values for vascular parameters (femoral artery diameter 0.4 cm, kinematic viscosity of blood 0.04 poise, average distance to reflecting sites 20 cm, pulse wave velocity 10 m/s in a large artery, but modified by the effects of α in the smaller vessels, reflection coefficient $R_c = 0.8$ under control conditions, zero at peak vasodilatation and 0.95 at peak vasoconstriction), impedance results were obtained which were virtually identical, in pattern and in magnitude, to those obtained under these different conditions (Fig. 12.16). Such findings increase the confidence with which explanations can be given for vascular impedance.

Impedance patterns in the subclavian and brachiocephalic artery of dogs are similar to those in the femoral artery (Fig. 12.17, page 290) (Attinger *et al.*, 1966c;

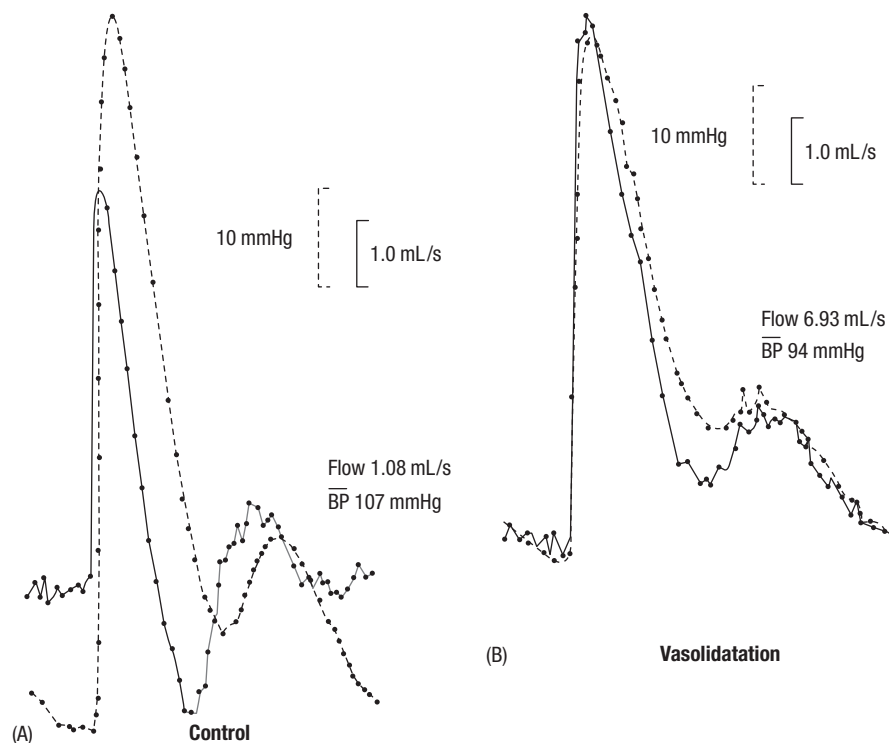


Figure 12.14 Pressure (broken line) and flow (solid line) waves recorded from the femoral artery of a dog under control conditions (left) and during vasodilatation induced by the intra-arterial injection of 100 mg acetylcholine (right). Reproduced from O'Rourke and Taylor (1966)

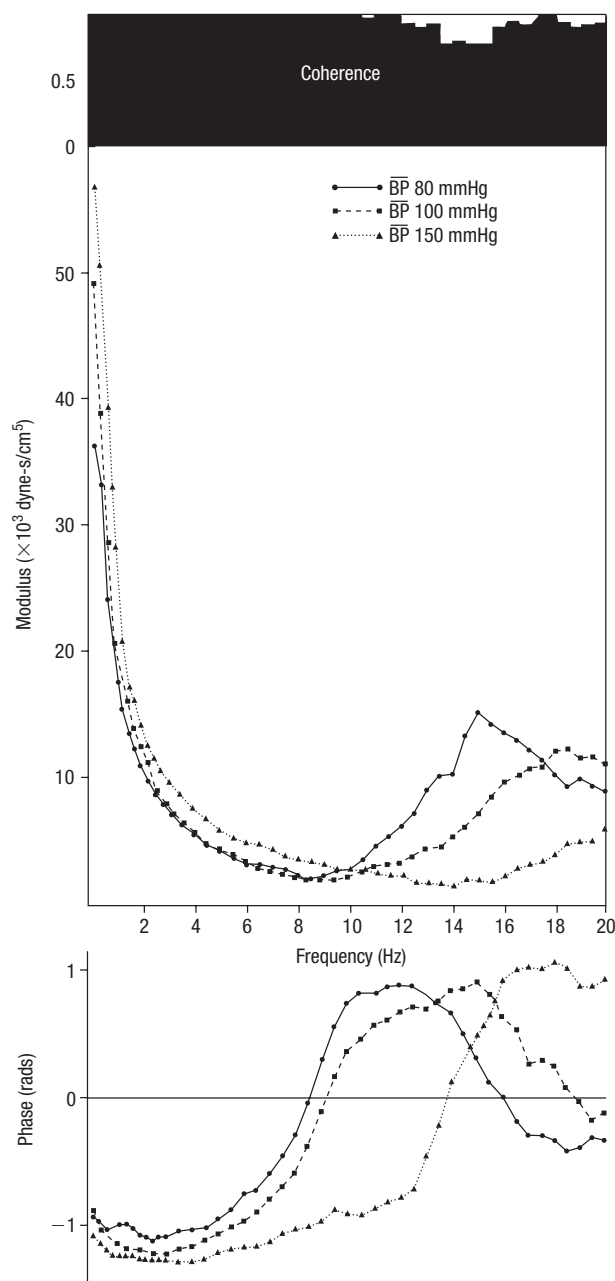


Figure 12.15 Femoral vascular impedance in a dog under control conditions with mean arterial pressure 80 mmHg, and at mean pressure 100 and 150 mmHg generated by intravenous infusion of norepinephrine. Peripheral resistances: 36.4, 57.4, and 120.8×10^3 dyne-s/cm⁵, respectively. Reproduced from O'Rourke (1970a)

O'Rourke, 1967a; Cox and Pace, 1975). The input impedance of the carotid vascular bed in dogs shows less evidence of wave reflection than does the femoral, subclavian, or brachiocephalic artery, with less-marked minima of impedance modulus and smaller variation in impedance phase (Fig. 12.17). This is attributable to the lower peripheral resistance of this bed, doubtless due to relatively low resting vasoconstrictor tone in cerebral arterioles. In the case of all the vascular beds, the distance

to the reflection site can be estimated from the frequency of the impedance modulus (and crossover in phase), and the peripheral reflection coefficient can be gauged, too, from the relationship between terminal and (estimated) characteristic impedance, just as for the femoral vascular bed. As for this bed, the functionally discrete reflecting site appears to be some half to two-thirds of the distance to the periphery of the bed and to represent the resultant of all individual arterial terminations.

In the renal artery, arteriolar tone is low and blood flow is high, while the distance to arteriolar terminations is short. Patterns of impedance modulus and phase in this vessel under control conditions (Fig. 12.18, page 291) resemble those in the femoral artery of a dog during extreme vasodilatation, while pressure and flow waves are similar in shape with little phase delay. As for the femoral artery with vasodilatation, findings are attributable to a low peripheral reflection coefficient.

CHANGES IN INPUT IMPEDANCE BETWEEN PERIPHERAL ARTERIES AND THE CENTRAL AORTA

Attinger *et al.* (1966c), O'Rourke (O'Rourke, 1967a; O'Rourke, 1982a; O'Rourke, 1982b) and Cox and Pace (1975) determined input impedance in different arteries in the same dog between peripheral arteries and the central aorta. Similar data were obtained for humans by Mills and colleagues (Mills *et al.*, 1970) and by Murgo *et al.* (Murgo *et al.*, 1980a; Murgo *et al.*, 1980b), using a catheter-tipped flow transducer with pressure sensor (see Chapter 6), and for other animals (Avolio *et al.*, 1976; Avolio *et al.*, 1982; Avolio *et al.*, 1984; Nichols *et al.*, 1986a). In Fig. 12.19, these data are compared most readily by expressing the impedance modulus in terms of linear flow velocity (i.e. as dyne-s/cm³). Since pressure and flow velocity wave amplitudes are similar in different major arteries (albeit with the former increasing and the latter decreasing somewhat from central to peripheral arteries), the impedance modulus must be similar. Volume flow, however, decreases markedly in the smaller peripheral vessels, so that for visual comparisons the modulus must be plotted on a logarithmic scale; such a scale was utilized by Cox and Pace (1975) (Fig. 12.19, page 291).

Impedance patterns between the central and peripheral arteries can be explained on the same basis as those in the femoral and other peripheral arteries. As expected, the frequency of the impedance minima and phase crossover becomes progressively lower in arteries further from the periphery and closer to the heart. This is attributable in part to the longer distance to peripheral reflecting sites from central arteries, and in part to the lower wave velocity in the thoracic aorta (Taylor, 1965; O'Rourke, 1982a; O'Rourke, 1982b). The position of functionally discrete peripheral reflecting sites in the vascular bed

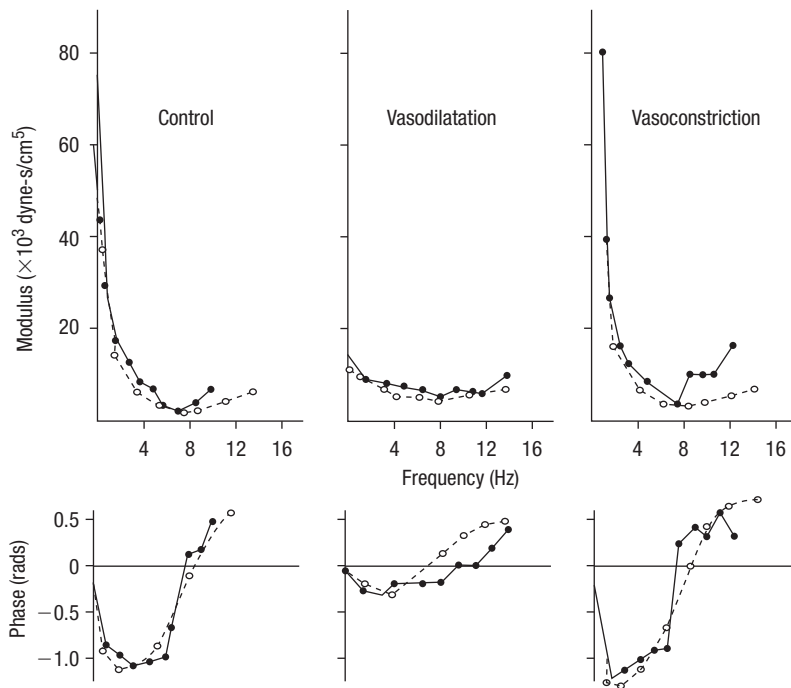


Figure 12.16 Experimentally determined values of impedance modulus (above) and phase (below) in the dog femoral artery (solid line) under control conditions, following intra-arterial injection of acetylcholine 100 mg, and after norepinephrine 6.0 mg, together with calculated values of impedance at the input of Taylor's equivalent model of the femoral vascular bed with reflection coefficient 0.8, zero, and 0.95, respectively (broken line). Reproduced from O'Rourke (1981a)

downstream can be calculated for all data sets. 'Seen' from the femoral artery, the peripheral reflecting site appears to be just below the knee; 'seen' from the abdominal aorta, this site appears to be in the upper thigh; 'seen' from the upper descending thoracic aorta, it appears to be in the region of the aortic bifurcation; and 'seen' from the origin of the brachiocephalic artery or subclavian artery from the aortic arch, the peripheral reflecting site appears to be just above the elbow (see Chapter 9). This mysterious functionally discrete reflecting site is like the end of a rainbow – clearly apparent from a distance, but receding as one approaches it and invisible up close. Reflection site distance is dependent upon arterial stiffness (and pulse wave velocity) and, therefore, changes with age and hypertension and other conditions and interventions that alter arterial properties and wave reflection characteristics (see Chapter 9).

Figure 12.17 includes the two patterns of impedance seen in the major arteries which supply the upper and lower parts of the body – the brachiocephalic artery on the one hand and the descending thoracic aorta on the other. Impedance patterns in the brachiocephalic, carotid and subclavian arteries are similar, with impedance modulus minima and phase crossover in a largish dog around 7.0 Hz, while in the descending thoracic aorta the minimum of impedance modulus and phase crossover is seen around 3.0 Hz. The first minimum in carotid and brachiocephalic arteries suggests a 'closed-end' reflecting site in the upper body that corresponds to a point around the elbow. The first minimum in the descending thoracic aorta, on the other hand, suggests a 'closed-end' reflecting site somewhat over twice as far distant – in the region of the aortic bifurcation or pelvis. The ascending aorta supplies both of these beds in parallel.

ASCENDING AORTIC IMPEDANCE: THE INPUT IMPEDANCE TO THE SYSTEMIC CIRCULATION

The vascular input impedance in the ascending aorta is of particular significance. Not only does this describe the relationship between pressure and flow at the root of the aorta, and so characterize the properties of the whole systemic circulation, but it can also be taken to represent the hydraulic load presented by the systemic circulation to the left ventricle of the heart. This latter aspect of ascending aortic impedance, together with vascular-ventricular coupling of left and right ventricles, is discussed in Chapters 13 and 14. Here we deal just with impedance in relation to vascular (not cardiac) properties.

The original, and most subsequent, experimental studies of ascending aortic impedance have been performed on dogs (Patel *et al.*, 1963a; Attinger *et al.*, 1966c; Noble *et al.*, 1967; O'Rourke and Taylor, 1967; Cox and Pace, 1975; Burattini and Campbell 2002; Dobson *et al.*, 2006); there is now general agreement on the patterns observed and the mechanisms responsible for these. The following data are from other experimental animals (Avolio *et al.*, 1976; Avolio *et al.*, 1982; Avolio *et al.*, 1984; Nichols *et al.*, 1986a; Zahka *et al.*, 1989; Zuckerman and Yin, 1989a; Mitchell *et al.*, 1994; Kobayashi *et al.*, 1996; Burattini and Campbell, 2002; Reddy *et al.*, 2003; Hu *et al.*, 2005; Segers *et al.*, 2005b; Syme *et al.*, 2006; Chen *et al.*, 2008; Ioannou *et al.*, 2009) and from humans (Mills *et al.*, 1970; Nichols *et al.*, 1977a; Merillon *et al.*, 1978; Merillon *et al.*, 1979; Murgo *et al.*, 1980a; Murgo *et al.*, 1980b; Gundel *et al.*, 1981; Murgo *et al.*, 1981a; Murgo *et al.*, 1981b; Merillon *et al.*, 1982a; Merillon *et al.*, 1982b; Merillon *et al.*, 1982c; Yin *et al.*, 1983b; Merillon *et al.*, 1984; Laskey *et al.*, 1985; Yaginuma *et al.*, 1985; Nichols *et al.*, 1985a;

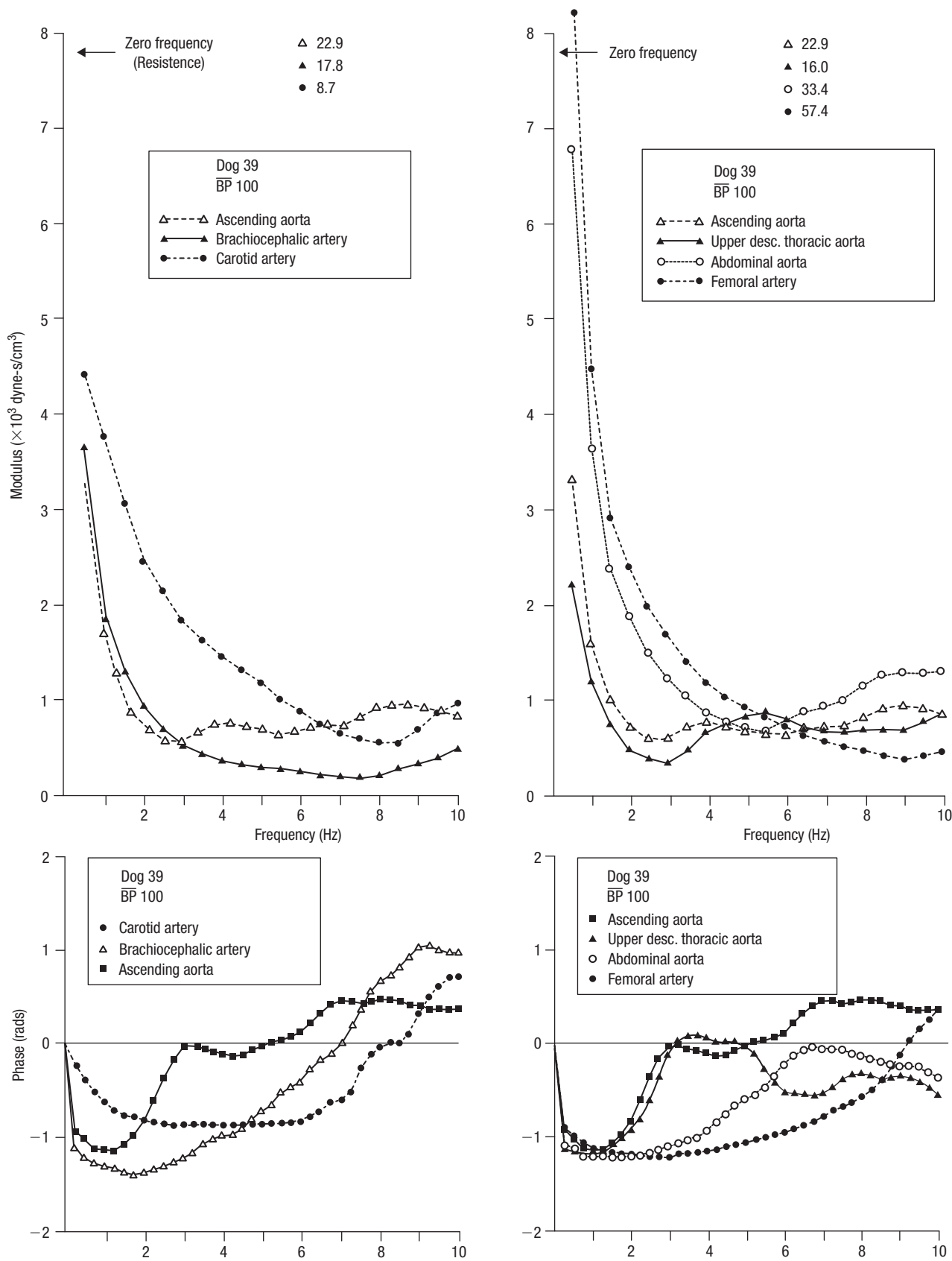


Figure 12.17 Impedance modulus (above) and phase (below) in the ascending aorta and upper body arteries (left) and lower body arteries (right) of a dog. With modulus expressed in terms of linear flow velocity, data can be displayed and compared on the same axes. Reproduced from O'Rourke (1967a)

Yaginuma *et al.*, 1986; Laskey and Kussmaul, 1987; Binkley *et al.*, 1990; Kelly and Fitchett, 1992; Kussmaul *et al.*, 1992; Nichols and Pepine, 1992; Endo *et al.*, 1994; Ting *et al.*, 1995a; Ting *et al.*, 1995b; Sharp *et al.*, 2000; Mitchell *et al.*,

2004a; Mazzaro *et al.*, 2005; Noda *et al.*, 2006; Segers *et al.*, 2007b; Mitchell *et al.*, 2008; Chirinos *et al.*, 2009a; Chirinos *et al.*, 2009b; Chirinos *et al.*, 2009c); these are, *mutatis mutandis*, comparable to the data from dogs described here.

In dogs, the patterns of impedance in the ascending aorta differ from the patterns seen in other arteries (O'Rourke and Taylor, 1967). The first minimum of impedance modulus occurs at a lower frequency than in other arteries (and at a frequency just below that in the proximal descending thoracic aorta some 5.0 cm away) (Fig. 12.17). This is as expected since the ascending aorta is further from peripheral arterial terminations than any other site of measurement could be, and also because the pulse wave velocity in the thoracic aorta is lower than in other arteries (this scenario can be reversed in older humans). The impedance modulus, however, does not show successive maxima and minima at multiples of the first minima, as seen in other arteries, most obviously the descending thoracic aorta (Fig. 12.17). After the initial first minimum, the impedance modulus either remains low over a broad band of frequencies or rises slightly before falling to a second minimum at close to twice (rather than three times) the frequency of the first minimum (Fig. 12.20). The impedance phase in the ascending aorta also differs from that in other arteries, usually remaining below zero at the frequency of the first minimum and crossing zero well above this frequency (Figs 12.20, 12.21 and 12.22). These points were stressed by O'Rourke and Taylor (1967), but they were apparent in

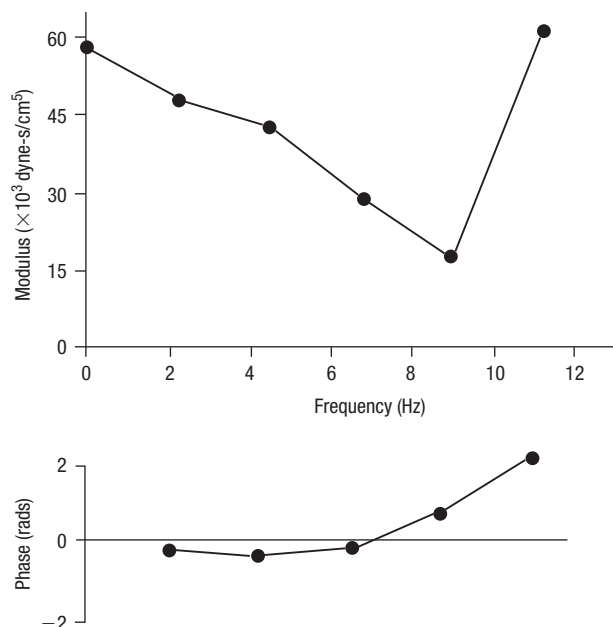


Figure 12.18 Vascular impedance in the renal artery of a dog under control conditions

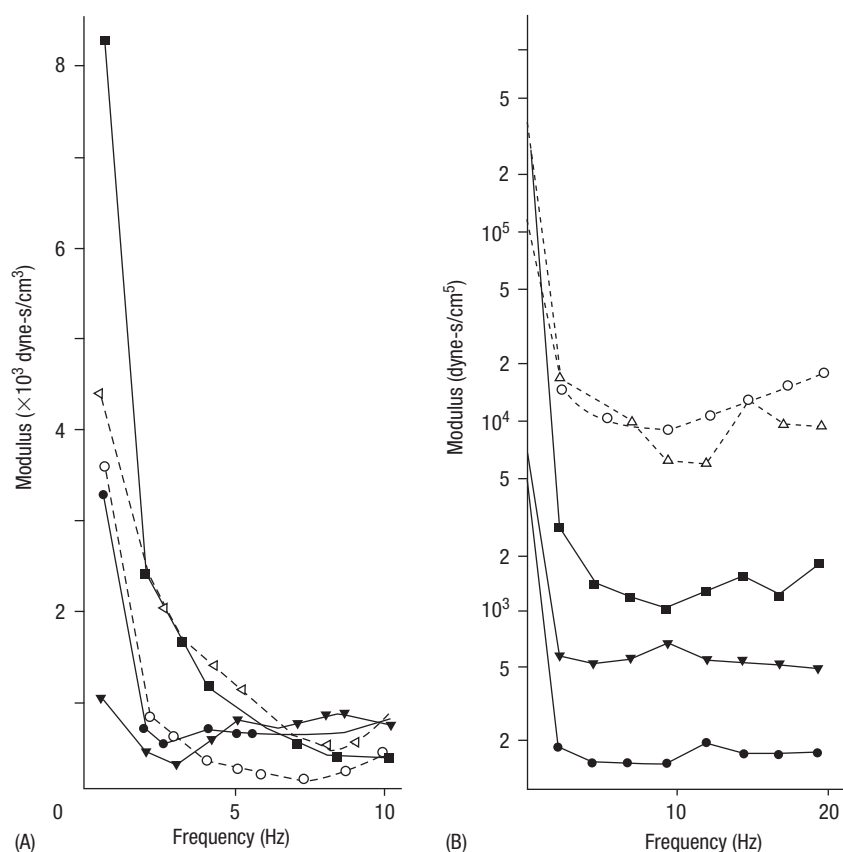


Figure 12.19 (A) Impedance modulus in different arteries of the same dog expressed in terms of linear flow velocity (dyne-s/cm^5) and displayed on a linear scale. Data from O'Rourke (1967a). (B) Impedance modulus in different arteries of the same dog expressed in terms of volume flow (dyne-s/cm^5) and displayed on a semi-logarithmic scale. Data from Cox and Pace (1975). (●-●) Ascending aorta; (▼-▼) upper descending thoracic aorta; (○-○) brachiocephalic artery; (△-△) carotid artery; (■-■) femoral artery in (A), subclavian artery in (B)

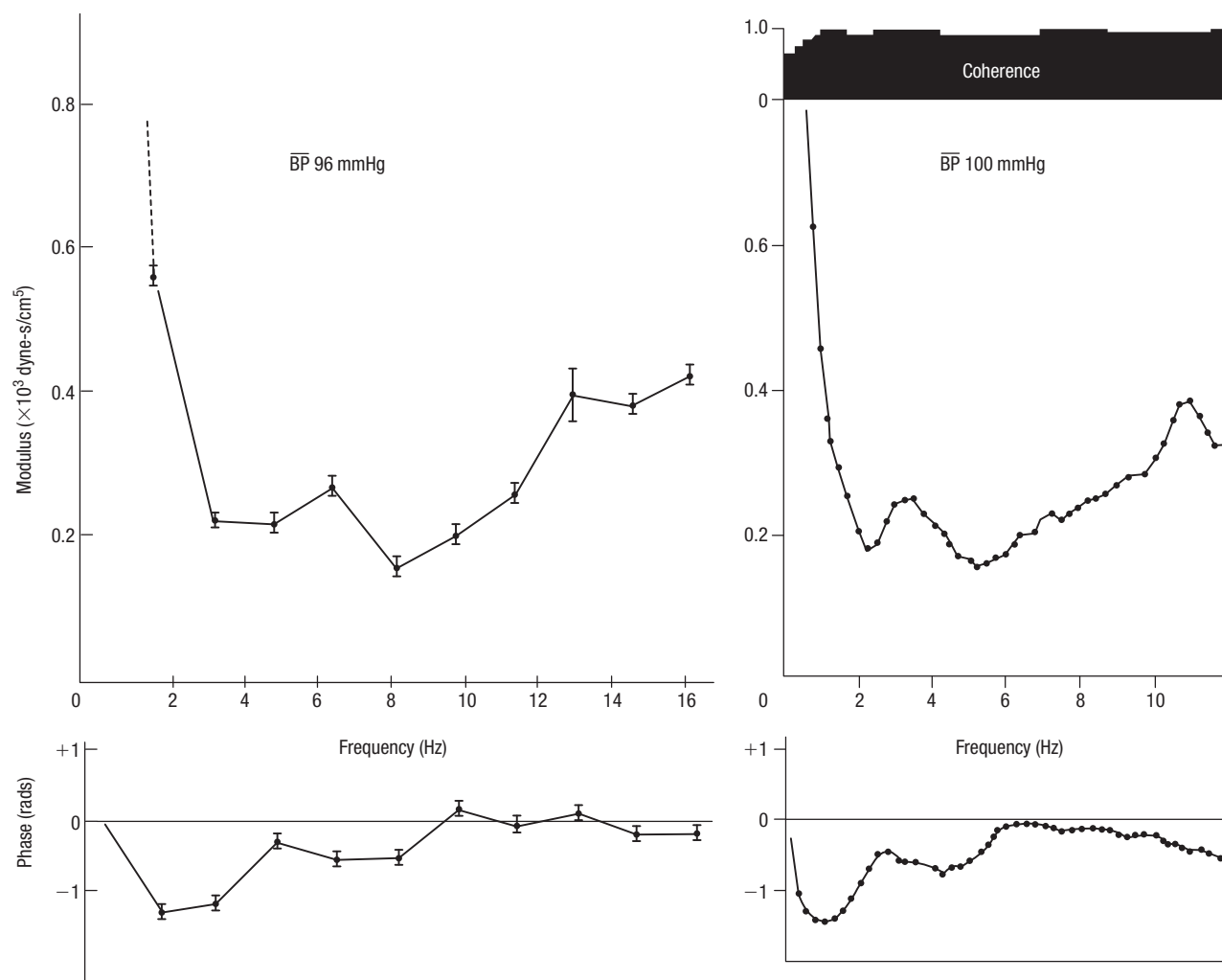


Figure 12.20 Modulus (above) and phase (below) of impedance in the ascending aorta of a small 14.3 kg dog (left) and of a large 23 kg dog (right) data obtained from Fourier analysis of waves recorded during regular heart action (left) and from frequency spectrum analysis of waves recorded during induced cardiac irregularity (right). Reproduced from O'Rourke and Taylor (1967)

data reported previously by Patel *et al.* (1963a) and subsequently by Attinger *et al.* (1967) (Fig. 12.21), McDonald (1974) (Fig. 12.22), and others.

O'Rourke and Taylor (1967) explained these impedance patterns on the basis of the two separate vascular beds perfused by the ascending aorta – that in the upper part of the body with the summation of arterial terminations relatively close, and that in the lower part of the body with summed arterial terminations much further away. They saw the ascending aorta as supplying the two vascular beds (one much shorter than the other) in parallel. Experimentally, they sought to test this hypothesis by measuring ascending aortic impedance under control conditions, then when exposed only to wave reflection in the lower body (by occluding the brachiocephalic and subclavian arteries flush with the aorta), and then with those vessels open by occluding the descending thoracic aorta flush with the subclavian artery orifice. Results were similar to those recorded in the descending aorta and

upper body vessels, respectively. When the ascending aortic impedance was compared with impedance patterns in the descending thoracic aorta and brachiocephalic arteries of the same dog, the former was readily seen as a composite of the other two (Fig. 12.23).

The ascending aortic impedance of the dog was examined in a multibranched model of the systemic circulation by Taylor (Taylor, 1966a; Taylor, 1966b). It was found here that reflection from the end of long and short arterial networks led to attenuation of impedance fluctuations, and just the sort of patterns observed experimentally (see Fig. 12.8b). Gessner (unpublished data) found similar results in a simple asymmetric T-shaped model of the systemic arterial system (Fig. 12.24), as did Sipkema *et al.* (1980) and Burattini *et al.* (1991). This model of the systemic circulation (Fig. 12.25, page 295) appears to be a realistic and simple conceptual model and is used elsewhere in this book to explain contours of pressure and flow waves in the major arteries.

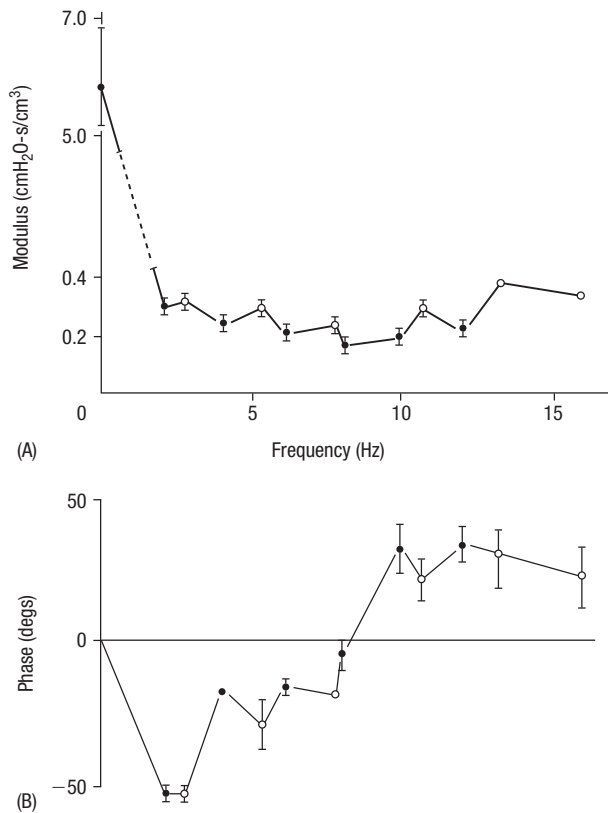


Figure 12.21 Modulus (A) and phase (B) of impedance in the ascending aorta of a dog. Reproduced from Attinger *et al.* (1967)

In the asymmetric T, the short limb represents arteries supplying the head and arms, and its termination represents all individual reflecting sites (principally arterial terminations and high-resistance arterioles in the upper part of the body), while the lower limb represents the descending aorta and arteries supplying the trunk and lower limbs, and its termination represents all individual arterial terminations in the lower part of the body.

O'Rourke and Taylor (1967) examined the effects of vasoconstrictor and vasodilator agents on the ascending aortic impedance. Results were similar to, but less clear-cut than, those seen in the femoral artery. In the ascending aorta, relative changes in peripheral resistance inducible with drugs were far lower than possible in the femoral artery and were always associated with substantial changes in the mean arterial pressure. These points probably explain why many other authors have not been able to show any effect of such agents on the frequency-dependent components of ascending aortic impedance.

Latson *et al.* (1988) overcame the problem of mean arterial pressure variation during nitroglycerin infusion by controlling venous return to the left ventricle. They were able to demonstrate that nitroglycerin has a relatively minor effect on the pulse wave velocity (decreasing this by approximately 10 percent), but a much more

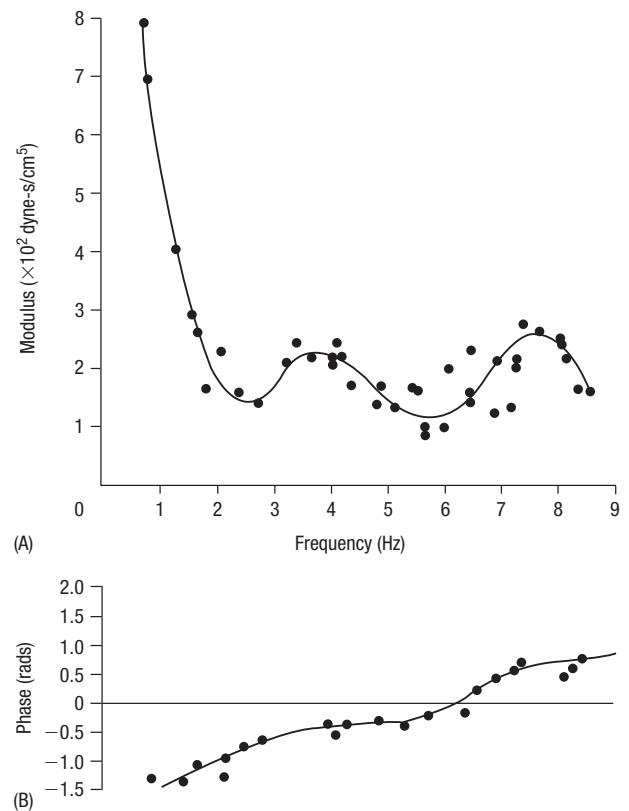


Figure 12.22 Modulus (A) and phase (B) of impedance in the ascending aorta of a dog

pronounced effect on the magnitude of wave reflection. Wave reflection was substantially reduced while peripheral resistance did not change. Latson attributed the findings to differential effects of nitroglycerin on conduit arteries of different size, with the smaller dilated to a greater degree than the larger. This work confirmed the findings and the interpretation given by Yaginuma *et al.* (1986) for humans, and the modeling experiments of Avolio described in Yaginuma's paper. The important implications of these findings in humans, subsequently confirmed by Fitchett *et al.* (Fitchett *et al.*, 1988a; Fitchett *et al.*, 1988b), are given in Chapter 26.

Changes in the mean arterial pressure caused changes similar to those seen in peripheral arteries, with a shift of the curves to the left with reduction in pressure and to the right with pressure increase. As for the femoral artery, these changes are attributable to an alteration in the pulse wave velocity between recording and peripheral reflecting sites, with a consequent change in the timing of wave reflection (see also Zuckerman and Yin, 1989a; Ting *et al.*, 1990; Ting *et al.*, 1991; Ting *et al.*, 1993; Ting *et al.*, 1995a; Ting *et al.*, 1995b). One unexpected feature of these experiments (O'Rourke and Taylor, 1967) was the apparent reduction (rather than increase) in characteristic impedance with increase in arterial pressure. This was explained on the basis that higher distending pressure

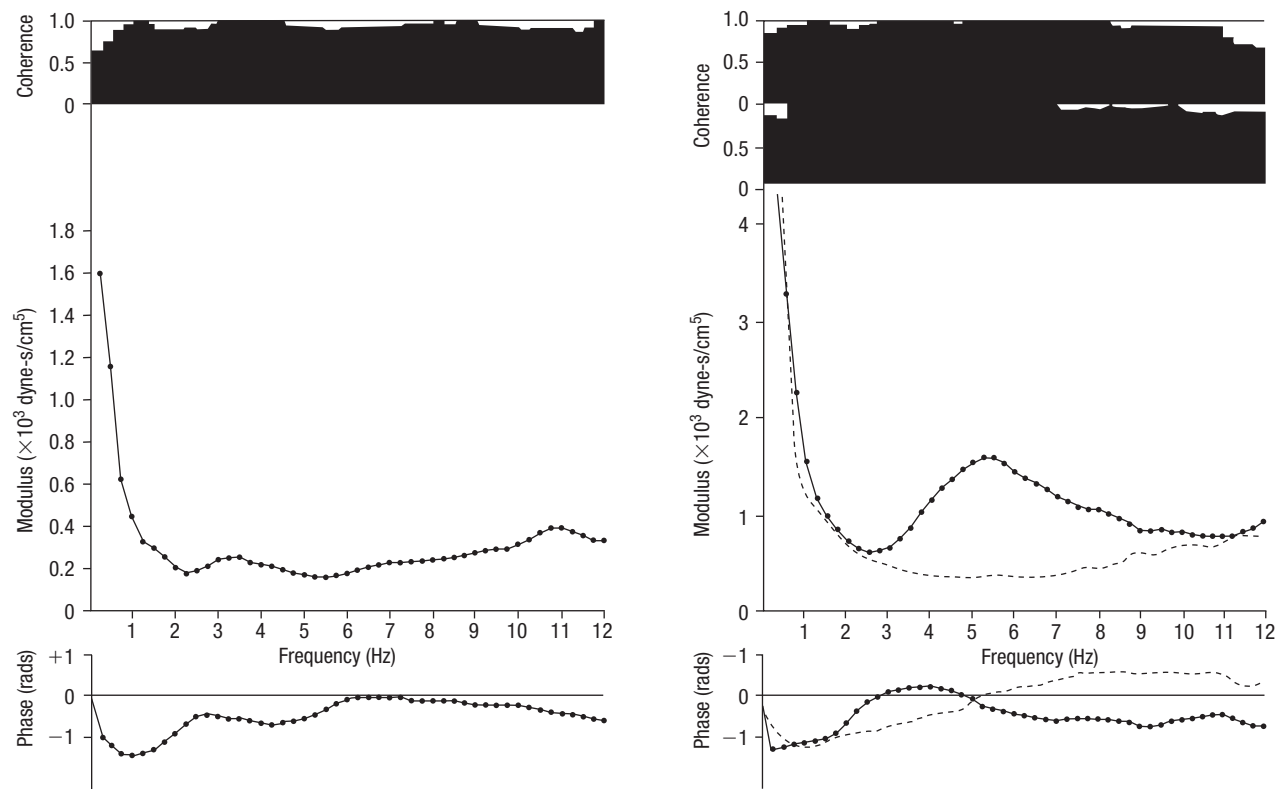


Figure 12.23 Impedance modulus (above) and phase (below) in the ascending aorta of a 23 kg dog (left) and in the descending aorta (solid line) and brachiocephalic artery (dashed line) of the same dog at the same mean arterial pressure (right). Ascending aortic impedance appears to represent the sum of these two major vascular beds in parallel. Reproduced from O'Rourke and Taylor (1967)

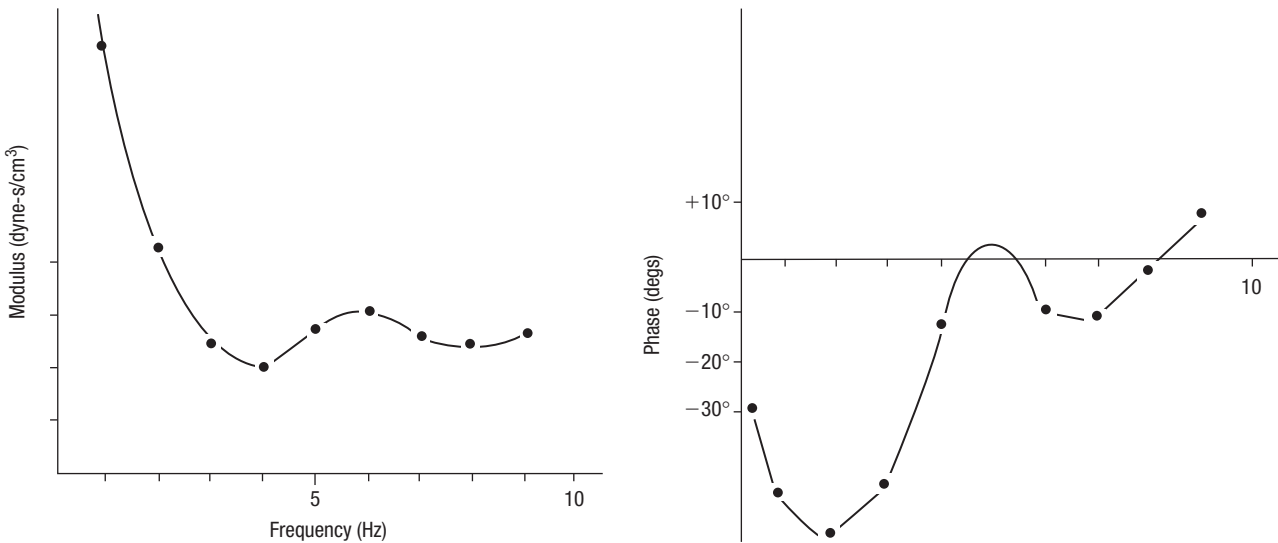


Figure 12.24 Input impedance of a simple, asymmetric T-tube model taken to represent the systemic arterial system, with the 'lower limb' system twice the length of the 'upper limb' system (U Gessner, unpublished data)

at the point of pressure measurement caused arterial dilatation, so that impedance calculated using volumetric flow (i.e. dyne-s/cm⁵) was underestimated (O'Rourke and Taylor, 1967); similar but opposite changes with a fall in

mean pressure can explain the apparent increase in characteristic impedance when blood pressure is low (Eaton *et al.*, 1993; Avolio, 1994; Eaton *et al.*, 1994). This subject is discussed further below.

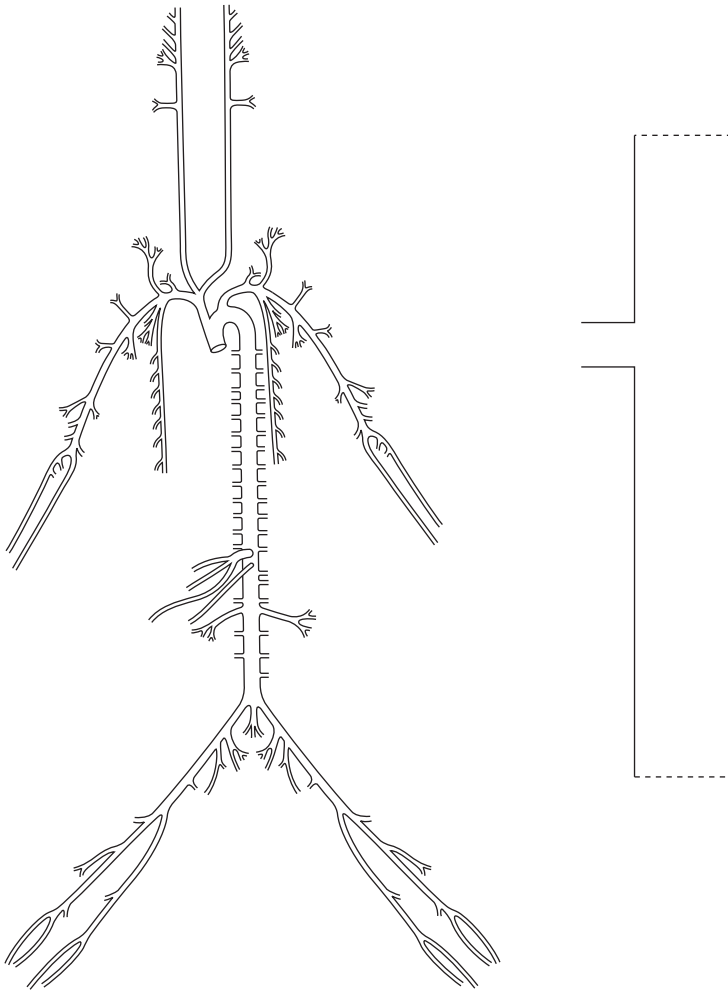


Figure 12.25 The systemic arterial system of a dog with lengths drawn to scale and diameter estimated (left), represented by a single tube (right), incompletely occluded at both ends, which receives blood from the heart one-third of the way along its length. For an explanation, see the text. Reproduced from O'Rourke (1967a)

The main features of ascending aortic impedance in dogs may be summarized as follows:

1. The impedance modulus falls more steeply than in any other artery to a minimal value that occurs at a lower frequency than in any other artery.
2. The impedance modulus shows less fluctuation than in any other artery. This is attributable to cancellation of wave reflection effects from sites close at hand in the upper body with reflection from sites further away in the lower body.
3. The impedance modulus settles at a lower value (the characteristic impedance) than in any other artery. This is attributable to the greater distensibility and lower pulse wave velocity of this vessel compared with any other.
4. The impedance phase shows early fluctuations before crossing zero and close to the frequency of the second (rather than the first) impedance minimum. As for modulus, this is attributable to the interaction of wave reflection from two functionally discrete reflecting sites in the peripheral circulation.

Implications of these impedance features of the ascending aorta with respect to left ventricular load are discussed in Chapter 13. It appears that ascending aortic impedance characteristics are extremely favorable for intermittent ejection from the left ventricle over the range of heart rates seen in life.

PULMONARY VASCULAR IMPEDANCE

Pulmonary vascular impedance (see also Chapter 16) was first determined by Caro and McDonald (1961) in rabbits, using a sinusoidal pump of variable frequency superimposed on steady flow and measuring the resulting sinusoidal pressure fluctuation with Taylor's harmonic resolver. The results obtained with this artificial perfusion technique are similar to those subsequently obtained from Fourier or frequency spectrum analysis of experimentally recorded pulmonary artery pressure and flow waves. Further studies of pulmonary vascular impedance have been performed in dogs (Patel *et al.*, 1963a; Bergel and Milnor, 1965; Milnor *et al.*, 1966; van den Bos *et al.*, 1982; Piene, 1986; Maggiorini *et al.*, 1998), in sheep (Santana

et al., 2005; Grignola *et al.*, 2006), in goats (Wauthy *et al.*, 2004), in pigs (Greenwald *et al.*, 1985) and in humans (Milnor *et al.*, 1969; Wilcox and Lucas, 1980; Murgo and Westerhof, 1984; Kussmaul *et al.*, 1988; Chen *et al.*, 1990; Kussmaul *et al.*, 1992; Leather *et al.*, 2002; Huez *et al.*, 2004; Hunter *et al.*, 2008). Noda *et al.* (2006) compared ascending aortic and pulmonary artery impedance in humans before and after administration of nifedipine. There is now general agreement as to patterns observed and underlying mechanisms.

Pulmonary vascular impedance is similar to ascending aortic impedance, but with five principal differences. The first difference is that pulmonary resistance is much lower than systemic (about one-tenth as great) so that the impedance modulus at zero frequency is far less than in the ascending aorta. The second main difference is that the pulmonary artery is more distensible at physiologic pressures than the ascending aorta. As a consequence, the pulse wave velocity is lower (approximately one-half) than in the ascending aorta, and the characteristic impedance is likewise approximately half as great. The third difference arises from the lesser difference between terminal and characteristic impedance in the pulmonary artery compared with the ascending aorta. As a consequence (and because the first minimum modulus occurs at similar frequencies), the impedance modulus falls less steeply than in the ascending aorta. The fourth difference is also a consequence of the first two. With resistance or terminal impedance some 10 percent of that in the systemic circulation, and the characteristic impedance some 50 percent of aortic, the ratio of $(Z_T - Z_0)/(Z_T + Z_0)$ will determine peripheral reflection as much lower – approximately 0.4 compared with 0.8 in the systemic circulation, when Z_T is calculated from the difference between mean pulmonary artery and mean left atrial pressure. As a consequence of this lower reflection coefficient, the impedance phase is considerably lower than in the ascending aorta, and the fluctuations in modulus and phase are less marked. The fifth point is that there is no apparent difference in timing of wave reflection from the two lungs, so that there appears to be a single functionally discrete reflecting site in the pulmonary vascular bed. The impedance modulus shows a single broad minimum at a frequency where phase crosses zero, and rises to a subsequent maximal value at approximately twice this frequency. In a medium-sized dog, this minimal value of modulus is seen at about 3.0 Hz, with the subsequent maximum at about 6.0 Hz. From these figures, and with wave velocity assumed at 200 cm/s (Bargainer, 1967), one can calculate the functionally discrete reflecting site to be some 17 cm from the main pulmonary artery. This corresponds to a site well out towards the periphery of the lung.

A point of similarity between pulmonary and ascending aortic input impedance is that the minimal value of impedance in the pulmonary artery corresponds in frequency to the broad minimum of impedance modulus in

the ascending aorta. Greater distances in the systemic circulation are compensated for by higher wave velocity, and lesser distances in the pulmonary circulation by lower wave velocity, so that wave reflection returns with the same timing from the periphery to the origin of both arterial networks. This is clearly of importance in relation to left and right ventricular performance, and is discussed in Chapter 13.

Experimental pulmonary hypertension has been produced temporarily by the infusion of serotonin (Bergel and Milnor, 1965). In their acute experiments, Bergel and Milnor reported doubling of mean pulmonary arterial pressure with pulmonary vasoconstriction, leading to a shift of the minima of impedance modulus and phase crossover from approximately 3.0 to nearly 6.0 Hz; there was no apparent change in characteristic impedance expressed in terms of volumetric flow (i.e. as dyne-s/cm⁵). The authors attributed the change in pulmonary vascular impedance to constriction of major arteries, so that reflection sites moved closer to the heart. They considered an increase in pulmonary arterial pulse wave velocity to be an unlikely cause of impedance pattern alteration. In the same laboratory, however, Bargainer (1967) subsequently reported a marked increase in pulmonary artery pulse wave velocity with serotonin infusion, and attributed this to gross dilatation of the vessel and decreased distensibility at high mean pressure. Such a change in wave velocity readily explains the effects of serotonin on impedance spectra. In proposing an altered site of wave reflection, Bergel and Milnor were apparently misled by the constant characteristic impedance when expressed in terms of volumetric flow (dyne-s/cm⁵). With arterial dilatation downstream from a cuff-type flow probe, at the point of pressure measurement, linear flow velocity is considerably decreased compared with that within the transducer, so that impedance values expressed in terms of flow velocity (dyne-s/cm³) will actually increase, while values expressed in terms of volumetric flow (dyne-s/cm⁵) remain constant. This situation parallels that already discussed with respect to impedance changes in the ascending aorta during norepinephrine infusion (see above). These matters give further weight to the argument that impedance modulus is best expressed in terms of linear flow velocity.

EFFECTS OF BODY SIZE AND SHAPE ON INPUT IMPEDANCE IN SYSTEMIC ARTERIES

This subject has been pursued in experiments on dogs, sheep (Avolio, 1976), wombats (O'Rourke, 1965a; O'Rourke, 1967a), rabbits and guinea pigs (Avolio *et al.*, 1976), snakes (Avolio *et al.*, 1982), kangaroos (Avolio *et al.*, 1984; Nichols *et al.*, 1986a), alligators (Syme *et al.*, 2006), pigs (Ioannou *et al.*, 2009), mice (Reddy *et al.*, 2003; Segers *et al.*, 2005b), ferrets (Burattini and Campbell, 2002), and rats (O'Rourke, 1981a; Zuckerman and

Yin, 1989a; Mitchell *et al.*, 1994; Hu *et al.*, 2005; Chen *et al.*, 2008) (see Chapter 9). Results show almost complete agreement with theoretic predictions.

In peripheral vascular beds, impedance patterns in different species are identical to those of dogs. The main difference is, of course, in size of the vascular bed and the distance to arterial terminations and reflection sites. Where examined, the apparent functionally discrete reflecting site in the femoral vascular beds of different sizes and animals of different species appears to be somewhere below the knee. In small animals, the impedance minimum and phase crossover is seen at a higher frequency than in large animals. Irrespective of species, the impedance patterns of a peripheral vascular bed depend on the size of the bed, and thus on the size of the animal.

The same principles apply to impedance patterns in more proximal arteries up to the aortic arch. Impedance patterns in the descending thoracic aorta of sheep, dogs, rabbits, and guinea pigs are similar (O'Rourke, 1982b), but with the frequency of impedance modulus and phase crossover higher in the smaller animals and lower in larger animals. The pattern of impedance in the ascending thoracic aorta of a guinea pig is similar to that in the femoral artery of a dog (O'Rourke, 1982b). In mammals, body size determines impedance patterns of the peripheral and central arteries up to the aortic arch.

It is in the ascending aorta that important differences of input impedance are seen in different mammals. As discussed earlier, the impedance modulus and phase in the ascending aorta of dogs are explicable on the basis of two separate vascular beds supplying the upper and lower parts of the body, but with one half the length of the other. This results in the maximal value of modulus being attenuated and occurring at less than twice the frequency of the first minimum. In sheep and rabbits (Avolio *et al.*, 1976; O'Rourke, 1982b) and in older humans (Mills *et al.*, 1970; Murgu *et al.*, 1980a; Nichols *et al.*, 1985a; Nichols *et al.*, 1993), impedance patterns of modulus and phase are similar to those seen in peripheral vascular beds, with a subsequent maximum of modulus occurring at twice the frequency of the first minimum and with the bed capable of representation by a tube with a single distal reflecting site. There appears to be a simple explanation for all these findings, an explanation which in different mammals is based on differences in body shape, and in humans on change in wave velocity with age (see Chapter 19).

In sheep and rabbits as compared with dogs, the lower body is relatively long in relation to the upper body (Avolio *et al.*, 1976; O'Rourke, 1982b). The distance to reflecting sites in the upper body appears to be closer to one-third that to reflecting sites in the lower body (in contrast to dogs, where the ratio is closer to 1.0:2.0). In contrast to dogs, the minimum of the impedance modulus in the brachiocephalic and subclavian arteries of sheep, rabbits, and guinea pigs corresponds to the second minimum of modulus in the descending thoracic aorta,

not (as in dogs) to the intervening maximum of impedance modulus. Thus, in sheep, rabbits, and guinea pigs, $\lambda/4$ in the lower body system corresponds to $3\lambda/4$ in the upper body system, whereas in dogs $\lambda/4$ in the lower body system corresponds to $\lambda/2$ in the upper body system. (In humans, a variation of this explanation is offered – that relative bodily dimensions are similar to those of dogs, but with aortic degeneration caused by age increasing wave velocity in the lower body, so that a relationship of $\lambda/4$ in the lower body to $\lambda/2$ in the upper body changes towards $\lambda/4$ in the lower body to $\lambda/4$ in the upper body (see Chapter 19 and Fig. 12.26).) The differences in ascending aortic impedance between different animals can readily be explained on the basis of differences in body shape (as well as absolute size), specifically to differences in length of the head, neck, and upper limbs in relation to the length of the trunk and lower limbs. The test of such a hypothesis will be how it explains impedance patterns in other animals of different sizes and shapes, whose ascending aortic pressure and flow waves and impedance patterns have yet to be determined. Data from kangaroos certainly support the hypothesis (Nichols *et al.*, 1986a).

This explanation of impedance patterns gains support from Avolio's modeling studies of the systemic arterial system of sheep, rabbits, and guinea pigs. When allowance is made for body size and shape in the dimensions of arterial segments, results are almost identical to those determined in life (Fig. 12.27). The explanation is also consistent with the interpretation of pressure and flow contours in different systemic arteries (see Chapter 10) and with the findings of Sipkema *et al.* (1980) and Burattini *et al.* (1991).

The kangaroo (Avolio *et al.*, 1984; Nichols *et al.*, 1986a) appears to be an extreme example of the influence of body shape on ascending aortic impedance patterns and on contours of arterial pressure and flow waves. The kangaroo has a small head and very short arms, but has a relatively large trunk and powerful, well-developed lower limbs (Fig. 12.28). In this animal, the upper body is insignificant in size and length compared with the lower body. In kangaroos, the ascending aortic impedance shows large fluctuations in modulus and phase of impedance, just as is seen in the descending aorta of many other mammals (Fig. 12.29). In the ascending aorta of kangaroos, there is evidence of just one reflecting site in the peripheral circulation, and this is, of course, in the lower body.

Snakes (Avolio *et al.*, 1982) and alligators (Syme *et al.*, 2006) provide another test of the theory developed on influence of body shape on arterial impedance. In these reptiles, the descending aorta is formed from the confluence of two aortic arches; it then runs the length of the lower body to the tail, with muscular arteries distributed almost uniformly along its length. There are no discrete muscular groups as in other animals. In contrast to mammals, snakes and alligators show no evidence of discrete wave reflection in either impedance or in wave

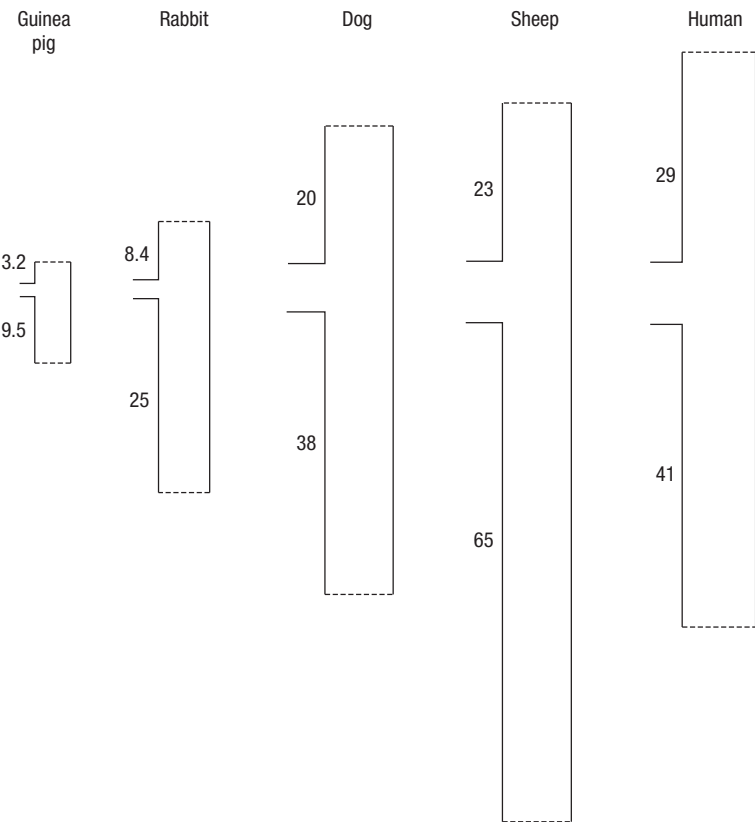


Figure 12.26 Asymmetric T-tube models of the systemic arterial tree of a variety of mammals, as calculated from determination of impedance in the ascending aorta, descending thoracic aorta, and brachiocephalic artery. Reproduced from O'Rourke (1982a)

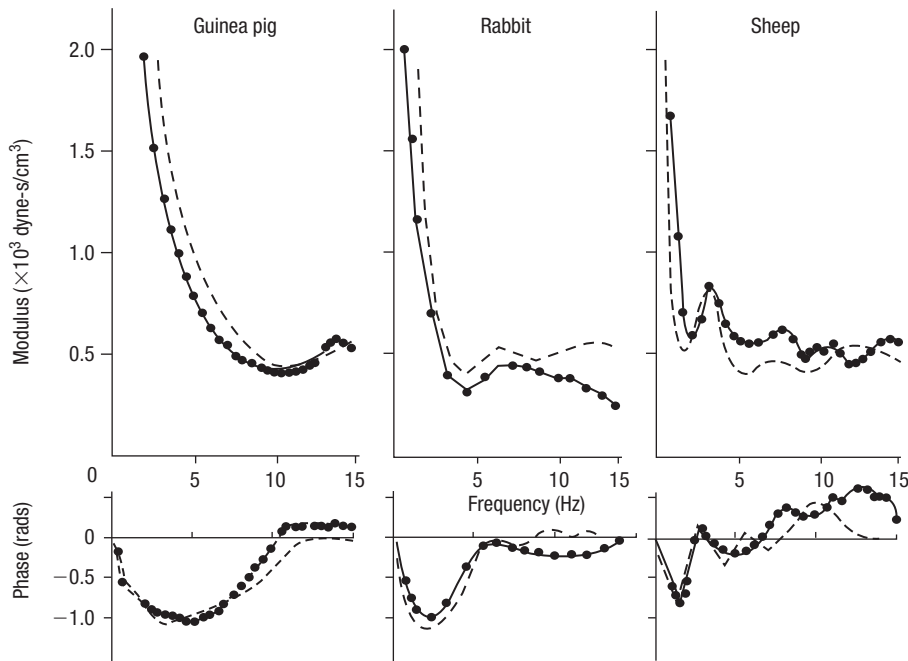


Figure 12.27 Vascular impedance (•—•) in the ascending aorta of guinea pig, rabbit and sheep, together with theoretically determined impedance (—) in the ascending aorta of these three animals using Avolio's multibranched model of the systemic arterial tree with measured vascular dimensions and realistic arterial properties. Reproduced from Avolio (1976)

contour. The impedance modulus falls from a high value at zero frequency to a relatively constant value around 5.0 Hz, while the phase, initially negative at low frequency, drifts back towards zero at about this same frequency

(Fig. 12.30, page 300). The behavior is virtually identical to that in a system of tubes with a wide distribution of lengths (see Fig. 12.7). Impedance patterns are explicable on the basis of complete cancellation of wave reflection

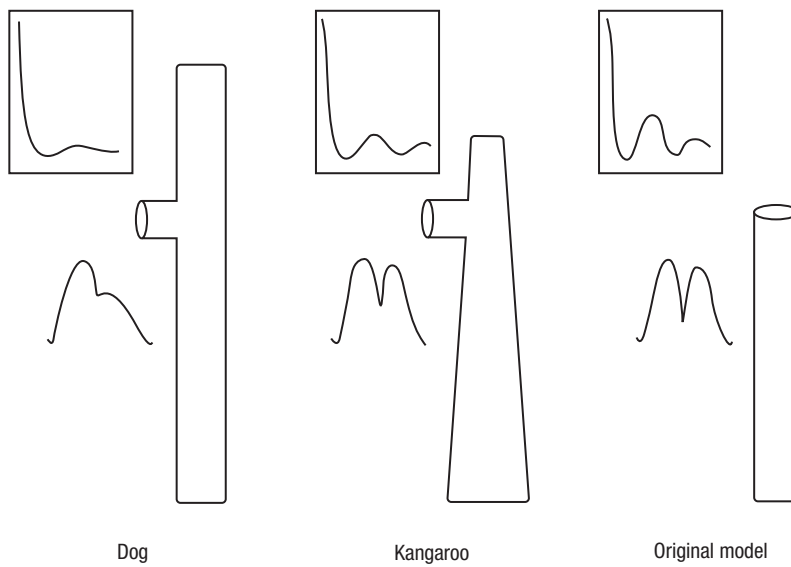


Figure 12.28 Asymmetric T-tube models of the systemic arterial trees of the dog, kangaroo, and original model. Corresponding input impedance moduli spectra and pressure waveforms are also shown

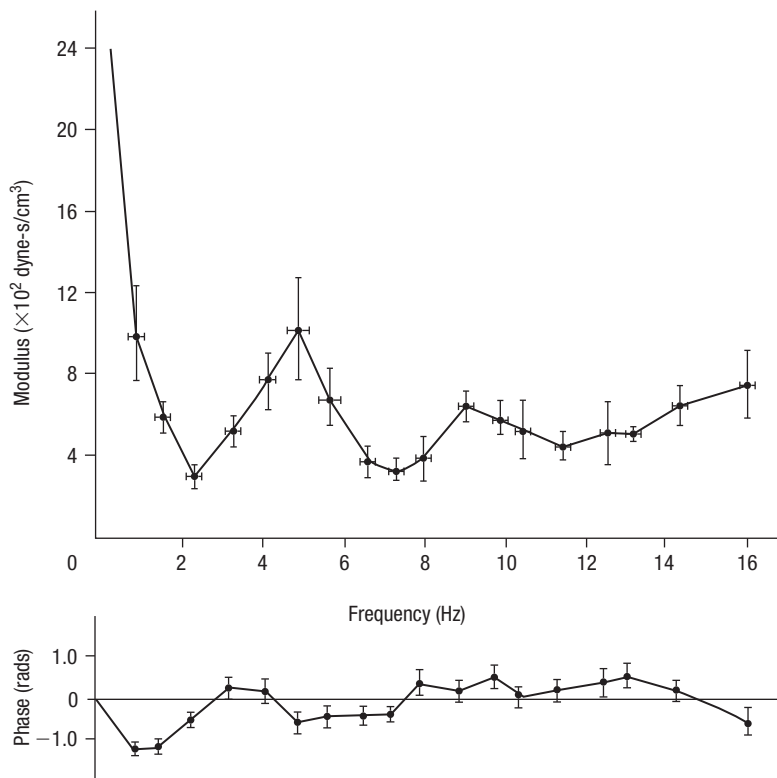


Figure 12.29 Modulus and phase of impedance in the ascending aorta of a series of 15 kangaroos. Reproduced from Nichols *et al.* (1986a)

from different sites at different distances from the aortic origin. The data from snakes provide further confirmation of the theory proposed to explain impedance patterns and pressure–flow relationships.

Impedance patterns in different animals have been explained on the basis of body size and shape, as these influence wave reflection. The timing of wave reflection can, of course, be altered by pulse wave velocity, as well as arterial length. The comparisons on the basis of length are possible because pulse wave velocity is roughly similar in

animals of different size and shape (Westerhof and Elzinga, 1991), but not with aging in humans (see above).

One consequence of wave reflection on ascending aortic impedance is that the minimum values of impedance modulus are seen at relatively low frequencies in large animals and at relatively high frequencies in small animals (O'Rourke, 1981a). Chapter 13 discusses the implications of this phenomenon in relation to ideal ventricular–vascular interaction and to the ideal range of heart rates.

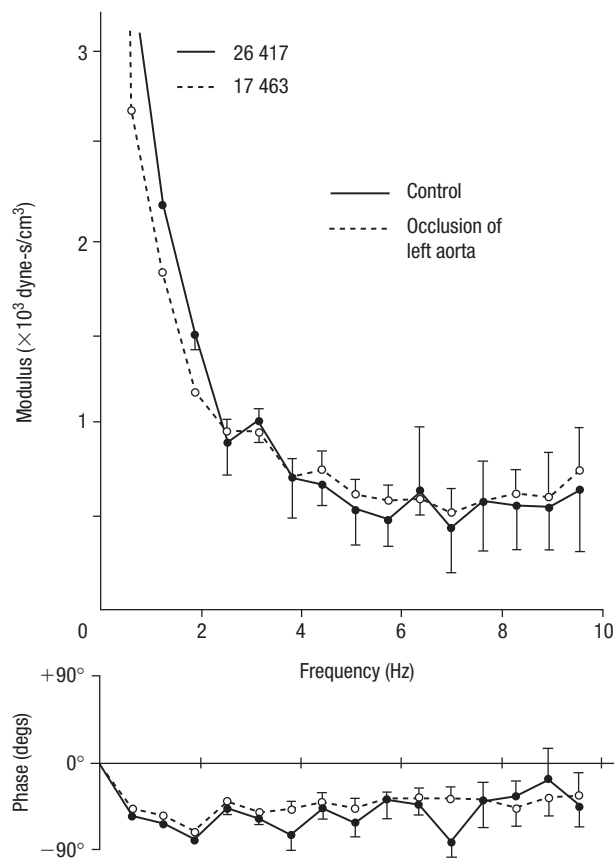


Figure 12.30 Modulus and phase of impedance in the descending aorta of diamond python snakes. In comparison with mammals, there is little fluctuation in modulus or phase, suggesting absence of discrete reflection in the vascular bed downstream. From Avolio *et al.* (1982)

VASCULAR IMPEDANCE IN HUMANS

In humans, most emphasis has been placed on data from the ascending aorta and pulmonary artery because of the obvious relevance to left and right ventricular hydraulic load (see Chapter 13). The earliest data were obtained by Gabe *et al.* (1964) using the Womersley equations to determine flow from differential pressure in the ascending aorta. A similar method was later used by Milnor *et al.* (1969) to determine flow and input impedance in the pulmonary artery. Two groups (Patel *et al.*, 1965b; O'Rourke and Avolio, 1980) determined flow with a cuff-type electromagnetic flow probe applied to the ascending aorta during open-heart surgery. Most workers, however, have used the electromagnetic velocity catheter originally developed by Mills (Mills and Shillingford, 1967), improved by Murgo and Millar (1972) with incorporation of a micromanometer into the catheter, and modified further by Uther *et al.* (1973) with addition of a tail to aid stabilization in the central stream of large vessels (see Chapter 6) (Mills *et al.*, 1970; Nichols *et al.*, 1977a; Merillon *et al.*, 1978; Pepine *et al.*, 1978; Merillon *et al.*,

1979; Pepine *et al.*, 1979; Murgo *et al.*, 1980a; Murgo *et al.*, 1980b; Gundel *et al.*, 1981; Murgo *et al.*, 1981a; Murgo *et al.*, 1981b; Merillon *et al.*, 1982a; Merillon *et al.*, 1982b; Merillon *et al.*, 1982c; Yin *et al.*, 1983b; Merillon *et al.*, 1984; Laskey *et al.*, 1985; Nichols *et al.*, 1985a; Yaginuma *et al.*, 1985; Yaginuma *et al.*, 1986; Laskey and Kussmaul, 1987; Binkley *et al.*, 1990; Nichols and Pepine, 1992; Endo *et al.*, 1994; Ting *et al.*, 1995a; Ting *et al.*, 1995b; Noda *et al.*, 2006). More recently, impedance has been determined non-invasively by transcutaneous Doppler flow recordings in association with carotid or subclavian pressure waveforms (Kelly and Fitchett, 1992; Downing *et al.*, 1993; Markus *et al.*, 1994; Mitchell *et al.*, 2002; Mitchell *et al.*, 2003; Mitchell *et al.*, 2004a; Mazzaro *et al.*, 2005; Segers *et al.*, 2007b; Mitchell *et al.*, 2008) or from the synthesized aortic pressure waveform (Martin, 1994).

There is general agreement on the impedance patterns recorded with these different methods, at least in their most obvious features. A number of points should, however, be made in interpretation of published data. First, the most frequently used method of flow recording with an electromagnetic catheter is very prone to artifacts, due to catheter movement, or to a change in position of the catheter in relation to the vessel wall when aortic velocity profile is not completely flat (see Chapter 2) or to other unidentified factors. Operators need to be careful in placement of catheters and in selection of data for analysis. Second, data have been obtained from patients undergoing cardiac surgery or undergoing diagnostic cardiac catheterization for suspected cardiac or arterial disease. While data for normal subjects have been selected from this group by exclusion of those with proven abnormalities, it is possible that patients selected from the group undergoing diagnostic catheterization were not completely normal. Third, virtually all published data have been from adults, normally in the 40–60 age group (i.e. in humans who were 38–58 years older than the other animals against whom results are compared). Such comparisons between human data and data from experimental animals should be made with attention to the caveat raised by CS Roy in 1880: 'Only in the case of young children do we find that the elasticity of arteries is so perfectly adapted to the requirements of the organism as it is in the case of the lower animals'. At present, there are only limited data available for children (Figs 12.31 and 12.32) (Endo *et al.*, 1994; Sharp *et al.*, 2000).

Ascending aortic impedances in apparently normal, conscious, adult human subjects are shown in Figs 12.33 and 12.34 (pages 302 and 303). These are similar to those seen in dogs – indeed in one sense too similar – with impedance modulus reaching a minimal value at 2.0–6.0 Hz and phase crossing zero at the same frequency, with subsequent rise in modulus, then stabilization around a nearly constant value that may be taken to represent the characteristic impedance. Impedance minimum is seen at a frequency a little lower than that in children. The similarity to dogs is that the first minimum

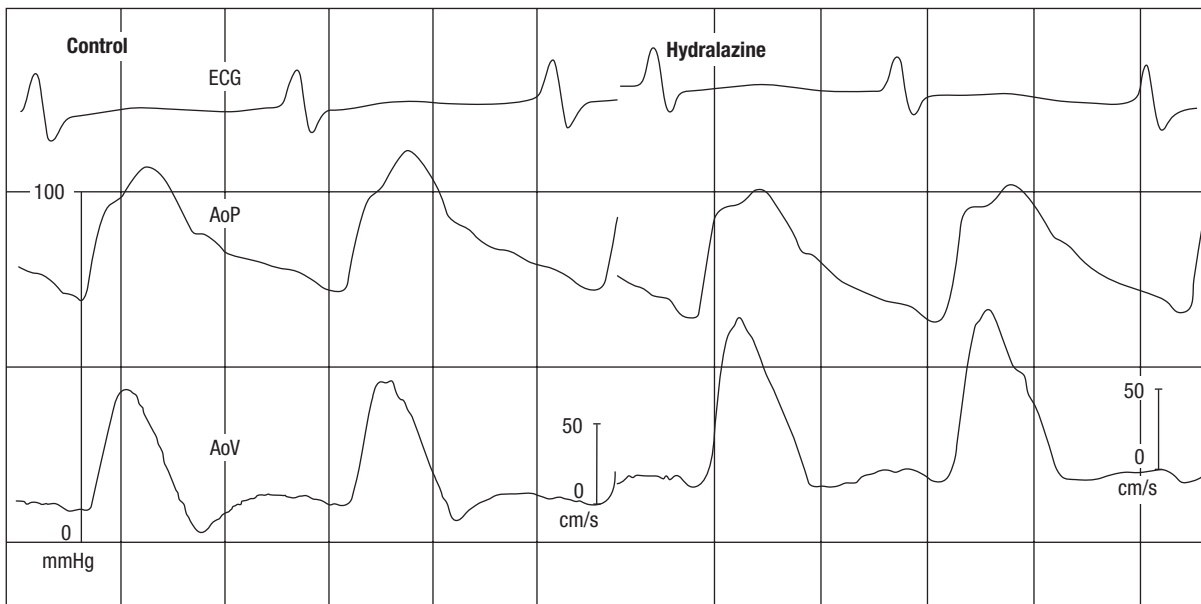


Figure 12.31 Typical recordings of aortic pressure and blood flow velocity (cm/s) in the control state and after hydralazine administration in a child with ventricular septal defect. AoP, aortic pressure; AoV, aortic flow velocity. Reproduced from Endo *et al.* (1994)

of modulus is reached at a similar frequency despite the enormous difference in body size (compare Fig. 11.20 with Figs 12.33 and 12.34). The difference is that the first impedance minimum is frequently followed by a clearly defined maximum at approximately twice the frequency of the first minimum.

The most obvious explanation of impedance fluctuations is that the human systemic circulation presents just one functionally discrete reflecting site, and some workers (Mills *et al.*, 1970; Westerhof *et al.*, 1979; Murgo *et al.*, 1980a; Latham *et al.*, 1985) considered such a site to be in the lower part of the body, neglecting wave reflection in the upper body. While such an explanation has been offered by us for the kangaroo (see, for example, Avolio *et al.*, 1984; Nichols *et al.*, 1986a), we are reluctant to apply this in humans whose head and upper limbs are relatively far better developed than in the kangaroo. There does, however, appear to be a comprehensive explanation that accounts for the relatively high frequency of the impedance minimum, the different patterns of impedance in some humans (Fig. 12.34) and the changes in ascending aortic impedance in humans during the Valsalva maneuver. The explanation is based on Roy's caveat and our studies of arterial pulse wave velocity with age (Avolio *et al.*, 1983; Avolio *et al.*, 1985a; Nichols *et al.*, 1985a).

Studies of pulse wave velocity in humans (Bramwell and Hill, 1922a; Bramwell and Hill, 1922b; Ho, 1982; Avolio *et al.*, 1983; Avolio *et al.*, 1985a) (see also Chapters 4 and 19) have shown that the aortic pulse wave velocity in humans increases markedly with age, whereas the pulse wave velocity in the upper limbs increases to a far lesser degree. The relatively high pulse wave velocity in the

descending aorta in adult humans readily explains the relatively high frequency of the impedance minimum in the ascending aorta. The explanation offered for the similarity with dogs is that the position of the functionally discrete reflecting site in the long lower body is identical, but that the speed of travel of the pulse in humans is far higher. The presence of a discrete impedance maximum at twice this frequency is based on the assumption that the wave velocity in the upper body arteries is similar to or only slightly higher than that in dogs. With more nearly even timing of wave reflection from sites in the upper and lower body, the impedance patterns will more nearly resemble that in a tube with a single reflecting site consistent with the interpretation given by others or, as we see it, an asymmetric tube with length and wave velocity such that $\lambda/4$ in the upper body corresponds to $\lambda/4$ in the lower body (see Fig. 12.35).

Murgo's data are consistent with this explanation. Murgo *et al.* (1980a) showed (Fig. 12.34) that ascending aortic impedance patterns resembling those published for dogs were seen when the ascending aortic pressure wave showed (as in dogs) no late systolic peak, whereas the fluctuating impedance patterns were seen in subjects who showed a definite late systolic peak on the ascending aortic pressure wave (for pressure wave classification (see Chapter 9). Mahomed (1874), Freis *et al.* (1966), and O'Rourke *et al.* (O'Rourke *et al.*, 1968; O'Rourke *et al.*, 1992a; O'Rourke *et al.*, 1992b; O'Rourke *et al.* 1992c) have shown that such a late systolic peak is a characteristic of human aging, and O'Rourke (1982b) has attributed this to early wave reflection caused by increased pulse wave velocity in older subjects. Explanations thus become easily

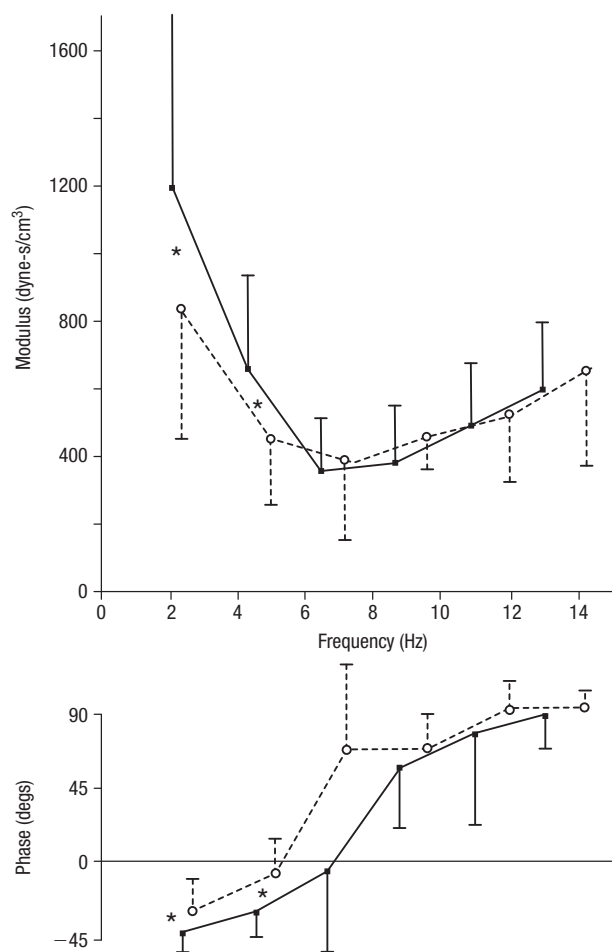


Figure 12.32 Aortic input impedance spectra obtained in eight children aged 2.2 to 8.8 (average 3.7) years with ventricular septal defect, in the control state (■—■) and after hydralazine administration (○—○), depicted as the mean and standard error of the mean. Asterisks indicate level of significant difference ($P < 0.05$) between values of the same harmonic. Reproduced from Endo *et al.* (1994)

reconcilable for the two phenomena: patterns similar to those of dogs, with lack of fluctuation in impedance modulus, are characteristic of the young, while the strongly fluctuating pattern is more characteristic of the aged. In a more detailed study combining data on impedance from different groups (including those of Murgó), Nichols *et al.* (1985a) have studied the effects of aging on ascending aortic impedance and have clearly documented aging change as an increase in the aortic characteristic impedance, an increase in frequency of the first minimum of the impedance modulus and an increase in fluctuations of the impedance modulus. This study provides confirmation for explanations previously advanced on the basis of asymptomatic arterial degeneration and increase in wave velocity, particularly in the lower part of the body. It is supported by the earlier study of Merillon *et al.* (Merillon *et al.*, 1982a; Merillon *et al.*, 1982b) and the more recent study of Mazzaro *et al.* (2005). It also

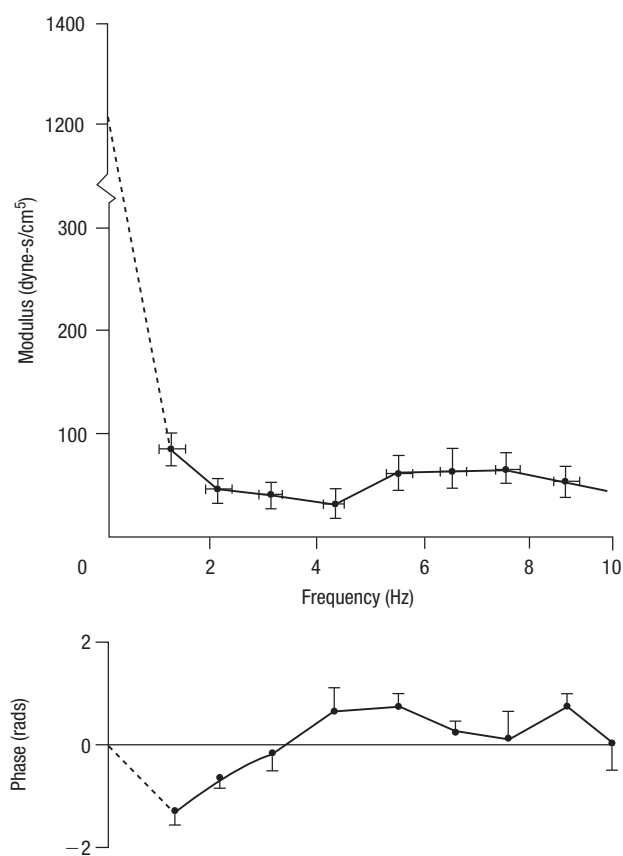


Figure 12.33 Modulus and phase of ascending aortic impedance obtained in five apparently normal adult humans ($n = 5$). Reproduced from Nichols *et al.* (1977a)

supports Roy's point that, when comparisons are made between arterial elasticity of humans and other animals, the human data should be from children. O'Rourke and Avolio (1980) sought to explain ascending aortic impedance patterns in humans on the basis of Avolio's multi-branched model (Avolio, 1980), using realistic values for arterial dimensions and published values of arterial properties. Results were in fair agreement (Fig. 12.36), but would have been better had the authors used the higher values of aortic pulse wave velocity that they subsequently measured in adult humans. Modeling studies on this subject have been published by Burattini *et al.* (1991) and by Karamanoglu *et al.* (1994). Karamanoglu *et al.* stressed that, with respect to the ascending aorta, effects of wave reflection from the lower body dominate reflection from the upper body.

Mills *et al.* (1970) and Murgó *et al.* (1981a) studied the effects of the Valsalva maneuver (forced expiration against a closed glottis) on ascending aortic impedance (Figs 10.4 and 12.37, pages 228 and 305). This work complements the earlier pressure pulse studies performed by Kroeker and Wood (1956) (see Fig. 10.16). The latter workers showed that this maneuver was associated with a reduction in amplitude of the late systolic pressure peak, with reappearance of a prominent diastolic wave during peak

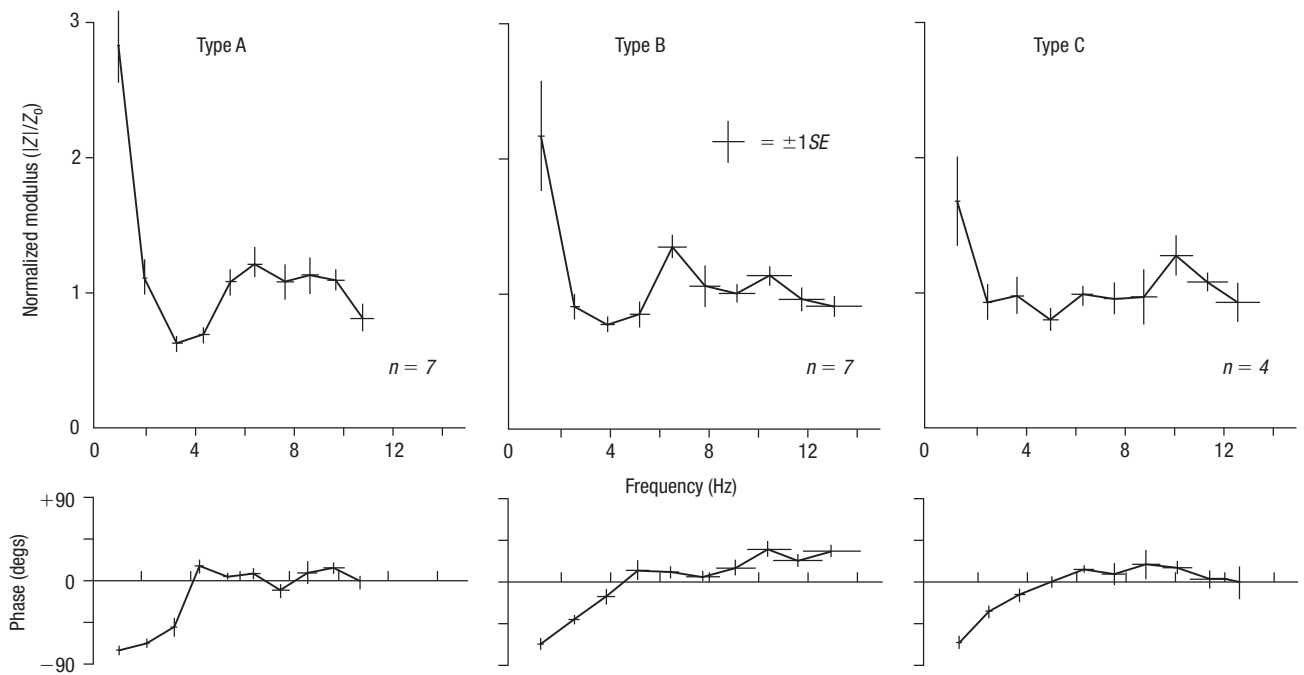


Figure 12.34 Ascending aortic impedance in 18 apparently normal humans. Subjects were divided into three different groups according to the contour of the ascending aortic pressure wave. Subjects in the group at left (type A) showed a pressure wave with late systolic peak and no diastolic wave. Subjects in the group at right (type C) were usually younger and showed no late systolic peak, but a definite diastolic pressure wave. Subjects in the group in the center (type B) showed waves of intermediate contour. Reproduced from Murgó *et al.* (1980a)

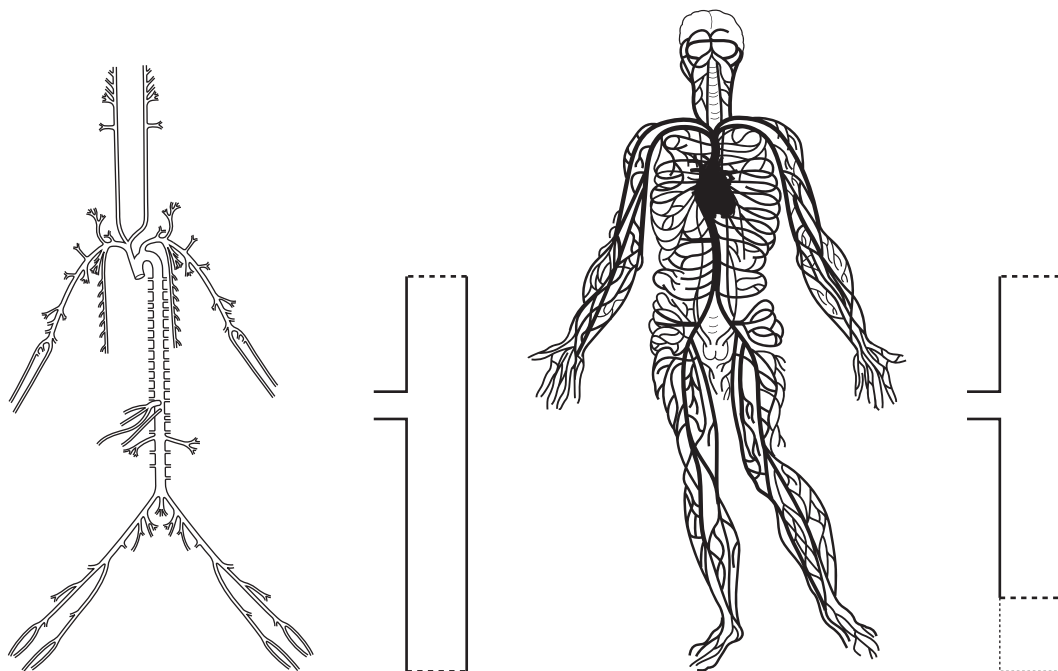


Figure 12.35 Systemic arterial tree of a dog (top left), drawn to scale from dissection, and human systemic arterial tree (top right) as drawn by Vesalius, together with equivalent T-tube models. In adult humans, the lower limb of the T appears foreshortened because of the increased pulse wave velocity caused by asymptomatic arterial degeneration in the descending aorta. Reproduced from O'Rourke (1982b)

expiratory effort; they explained the results as due to a decrease in duration of systole and delayed return of wave reflection. The latter readily explains the change in ascending aortic impedance patterns from those attributable to

early wave reflection from the lower body (due to arterial degeneration and high aortic pulse wave velocity) to those due to late wave reflection from the lower body (caused by markedly reduced transmural pressure in the thoracic and

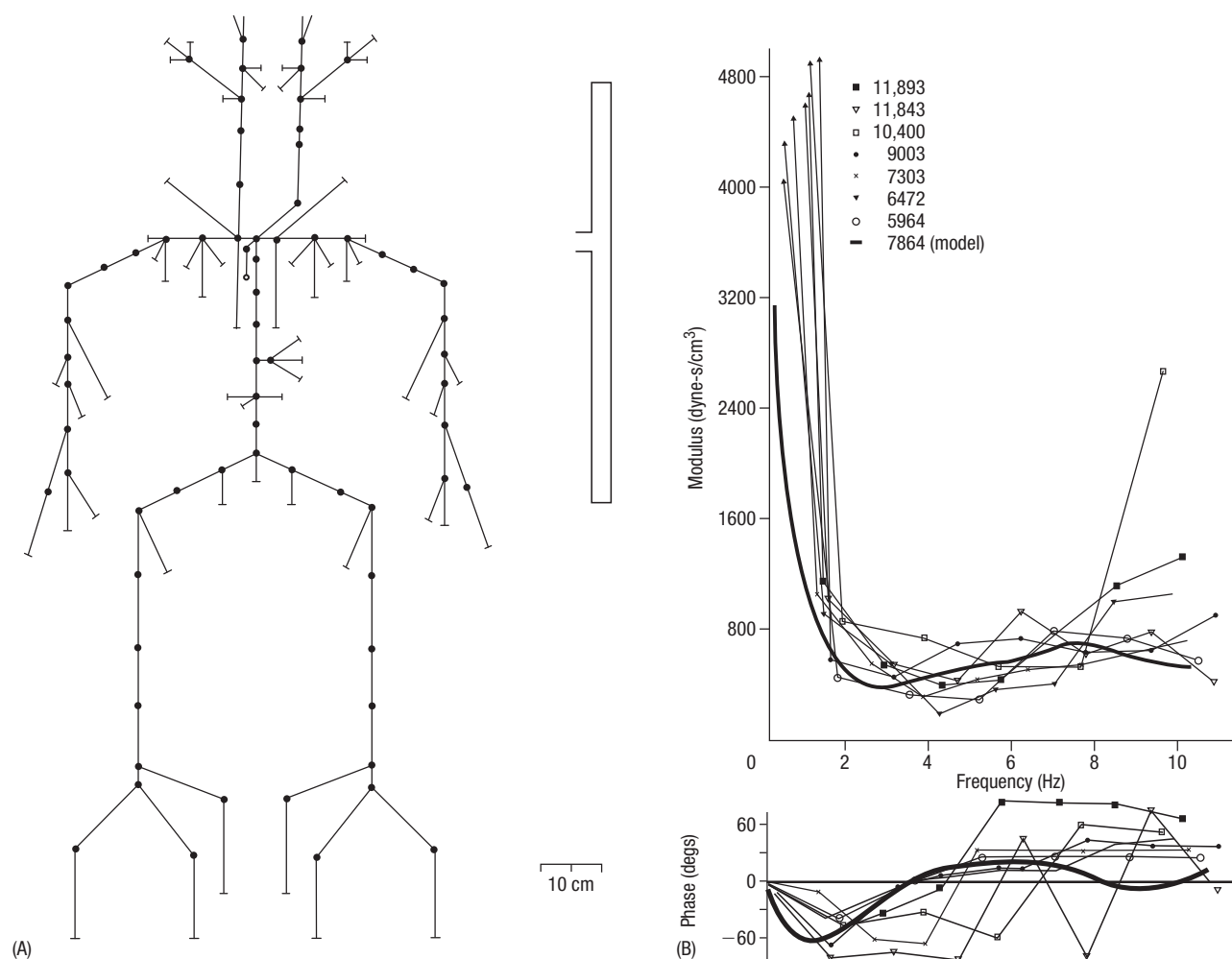


Figure 12.36 (A) Multibranched model of the systemic arterial system based on anatomic data, together with corresponding simplified asymmetric T-tube model. (B) Ascending aortic impedance in seven human subjects, and in the multibranched model. Values of peripheral resistance are given at the upper left. Reproduced from O'Rourke and Avolio (1980)

abdominal aorta during Valsalva strain). This explanation is entirely consistent with that previously offered for changing impedance patterns with age in man; the Valsalva maneuver temporarily 'undoes' the aging process. Murgo and colleagues (Murgo *et al.*, 1981a; Latham *et al.*, 1985) have been hesitant to accept such an explanation. Restricting their considerations to wave velocity in the lower body, these authors preferred to explain the Valsalva effect on the basis of decreased wave reflection in the lower body rather than on the basis of different timing of reflected waves from the upper and lower body and greater cancellation of these. We see our explanation as simpler, more logical, more comprehensive and more in accord with other data. Latham *et al.* (Latham *et al.*, 1985; Latham *et al.*, 1988b) have recently reported further data during Valsalva strain and during forced inspiration against a closed glottis (the Mueller maneuver), with emphasis on wave reflection in the region of the terminal aorta.

As part of their extensive studies of ascending aortic impedance in humans, Murgo *et al.* (1981b) studied

ascending aortic impedance during supine leg exercises. Exercise caused a fall in peripheral resistance, and an increase in characteristic impedance (expressed in terms of linear flow velocity, i.e. as dyne-s/cm³) that was attributable to an increase in mean arterial pressure, but no obvious change in fluctuation of modulus or phase of impedance to indicate reduction in wave reflection. These findings contrast with the effects of vasodilating agents, which reduce such fluctuation, although, as discussed above, vasodilatation has to be quite marked before such changes may become apparent. The findings in this paper may have been due to the relatively mild and localized nature of vasodilatation during supine leg exercise. Similar results have been reported by Laskey *et al.* (1985); exercise is considered further in Chapter 27.

Impedance was determined in a variety of systemic arteries in humans by Mills *et al.* (1970) in their classic studies that followed the introduction of the catheter-tip velocity transducer (see Figs 10.11 and 12.38). Impedance patterns were similar to those seen in experimental

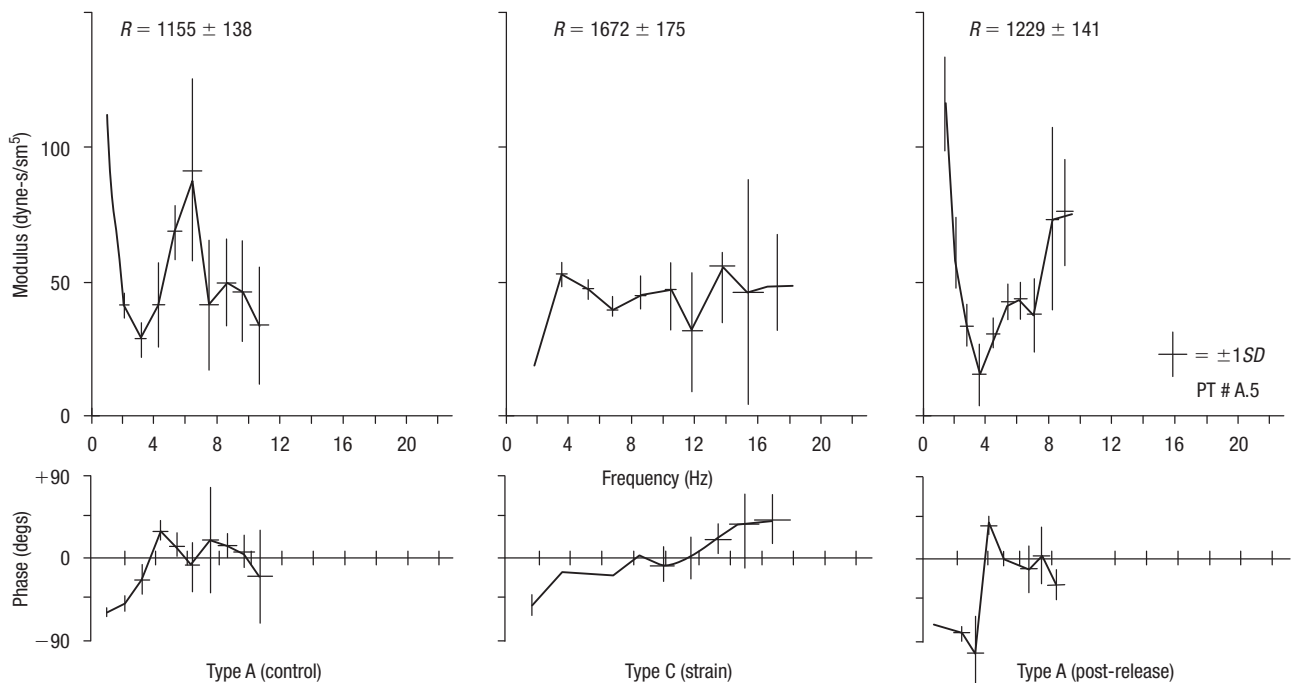


Figure 12.37 Ascending aortic impedance in eight adult human subjects before (left), during (center) and after (right) a Valsalva maneuver. The maneuver was associated with the characteristic change of aortic pressure wave contour from type A (late systolic peak, no diastolic wave) to type C (no late systolic peak, prominent diastolic wave) then back to type A (see Fig. 12.35). Reproduced from Murgo *et al.* (1981a)

animals, with the exceptions previously noted for the ascending aorta as characteristic of adult humans. Mills and co-workers confirmed a high frequency of impedance minima in the descending thoracic aorta, suggesting early wave reflection from the lower body, but reported a relatively lower frequency of impedance minimum in the iliac artery even further downstream, suggesting later wave reflection from the femoral vascular bed. This point has been further explored by Latham *et al.* (1985), who have suggested that physical tapering of the lower abdominal aorta (as shown by Langewouters *et al.* (1984); Gosling *et al.* (1971); Greenwald *et al.* (1990)) (Fig. 10.10), together with elastic non-uniformity (Taylor, 1964), present an important anatomically discrete reflecting site in the abdominal aorta of humans. This point warrants confirmation, and the whole subject further study. There is something peculiar about the lower aorta in humans which makes this particularly prone to degeneration and aneurysm formation. Some local characteristic in pulsatile pressure or flow appears to play a crucial role; this is discussed further in Chapters 23 and 24.

The largest study of human systemic arterial impedance is included in the Asklepios investigation which comprised over 2000 healthy men and women (Segers *et al.*, 2007b; Chirinos *et al.*, 2009b), aged 35–55 years. The greatest value will come from repeated non-invasive studies. Results to date show no consistent pattern for the derived indices, and the most consistent from simple indices such as carotid AIx. Mitchell *et al.* (Mitchell *et al.*,

2002; Mitchell *et al.*, 2004a; Mitchell *et al.*, 2008; Mitchell *et al.*, 2010) have conducted a battery of hemodynamic investigations with no benefit apparent in prediction of outcome for aortic impedance or other measures except aortic pulse wave velocity. Subsequent correspondence following the 2010 outcome paper points to multiple sources of error including issues in Table 12.1. Other studies, summarized in Chapter 28, show benefits of simpler parameters of central pressure for prediction of outcome (see also Benetos *et al.*, 2010; O'Rourke and Adji, 2010a; Vlachopoulos *et al.*, 2010a; Vlachopoulos *et al.*, 2010b; Wang *et al.*, 2010; Weber *et al.*, 2010). Human pulmonary vascular impedance has been determined from flow data obtained with the differential pressure technique (Milnor *et al.*, 1969) and the electromagnetic velocity catheter (Mills *et al.*, 1970; Yin *et al.*, 1983b; Murgo and Westerhof, 1984; Kussmaul *et al.*, 1988; Chen *et al.*, 1990; Kussmaul *et al.*, 1992; Noda *et al.*, 2006) (see Chapter 16). Results are entirely similar to those in dogs, with relatively low values of pulmonary vascular resistance and of characteristic impedance, a less steep fall in modulus with frequency than in the ascending aorta, a minimal value of modulus and phase crossover at 2.0–5.0 Hz and a maximal value of modulus at approximately twice this frequency, and with impedance phase showing less fluctuation, and less deviation from zero, than in the ascending aorta. As with ascending aortic impedance patterns, these findings are unexpected, given the larger body size and longer distance to presumed reflecting sites in human compared

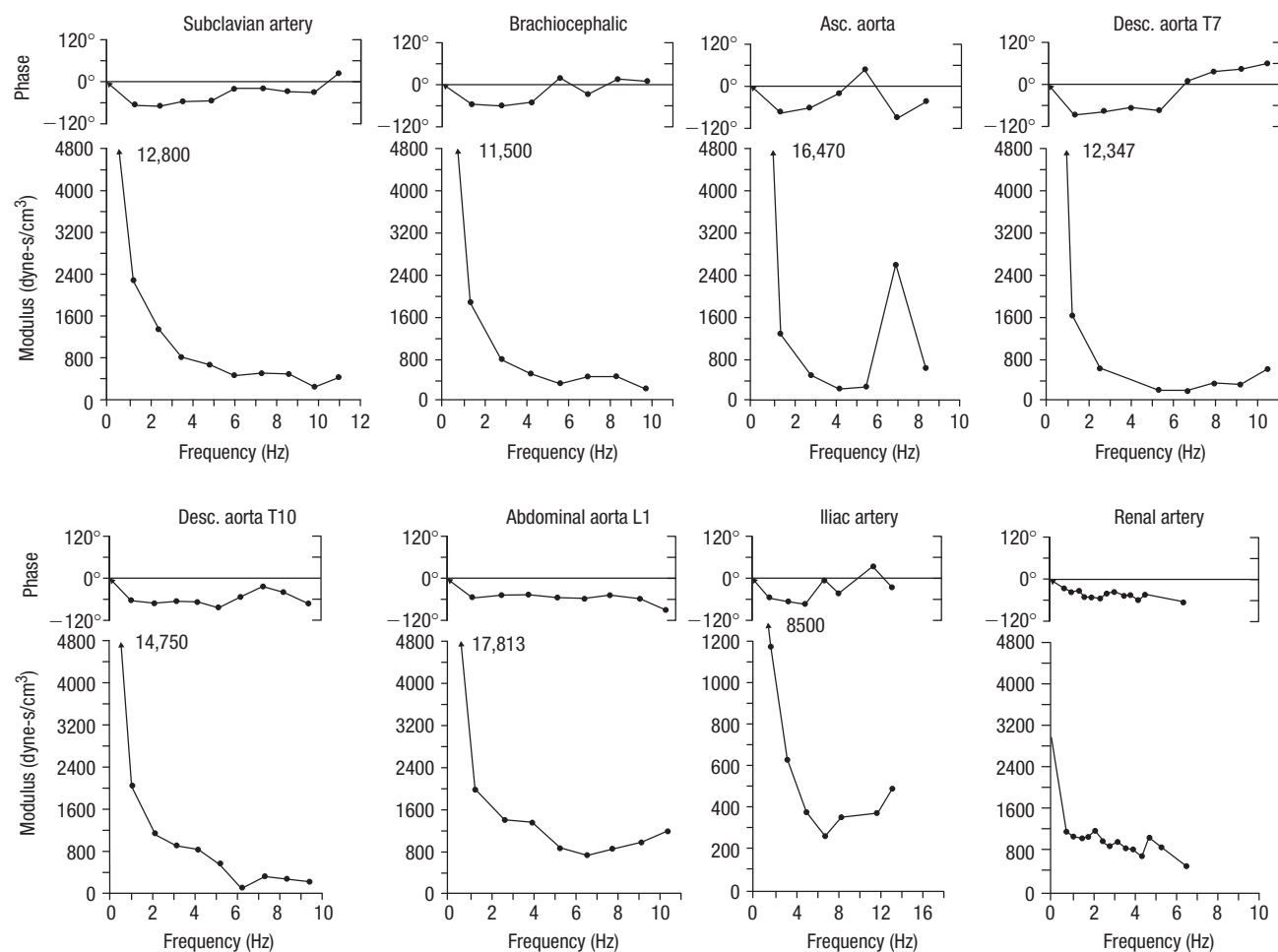


Figure 12.38 Vascular impedance in various arteries of human adults studied during diagnostic cardiac catheterization. The modulus is expressed in terms of linear flow velocity (dyne-s/cm^3). Reproduced from Mills *et al.* (1970)

with canine lungs. Definite differences in pulse wave velocity of the pulmonary arteries have not been reported between humans and dogs but could explain observed findings.

Ascending aortic and pulmonary input impedance have been determined in human disease, under control conditions and during the administration of therapeutic agents. Data have also been obtained in experimental models of human disease. Such determinations are very relevant to ventricular hydraulic load, to modification of such load by disease and to strategies whereby ill effects of disease may be offset. This subject is addressed in Chapters 19, 20, 21, 22, 23, 24, 25, and 26, after consideration of impedance as cardiac load (Chapters 13 and 14).

THEORETICAL AND TECHNICAL PROBLEMS IN THE DETERMINATION AND INTERPRETATION OF VASCULAR IMPEDANCE

The concept of vascular impedance is an extremely useful one for the study of the circulation, and particularly for

interpretation of the properties of large arteries. A number of points and reservations must, however, be borne in mind when applying theory and concept to practical application. These have been discussed by O'Rourke (O'Rourke, 1982a; O'Rourke, 1982b; O'Rourke, 1982c) as follows.

Terminal impedance

The concept of input impedance assumes that blood enters a vascular bed in a pulsatile fashion and at high mean pressure, and leaves this bed through its draining vein in a steady stream and at negligible mean pressure. In such a system, both resistance and impedance are calculated by relating pressure to flow at the input, neglecting venous pressure. This may be a reasonable assumption for the systemic circulation under normal circumstances, because right atrial pressure is normally close to zero and because venous pressure and flow waves are transmitted back from the heart (McDonald and Taylor, 1959; Attinger, 1963; Attinger *et al.*, 1966c) not through the arterioles, and do not normally alter pressure or flow

patterns in the capillaries, having been damped out in the smaller vessels. In the pulmonary circulation, however, the situation is not so clear-cut. Under control conditions, left atrial pressure is considerably higher than right atrial pressure and is certainly not negligible in relation to mean pulmonary arterial (PA) pressure. In normal humans, the mean pulmonary arterial pressure is 10–20 mmHg, and the mean left atrial pressure is 4.0–12 mmHg. It is certainly not reasonable to neglect mean pulmonary venous pressure in the calculation of pulmonary vascular resistance; this should be determined as (PA – left atrial mean pressure) divided by cardiac output. Also, because the pulmonary arteriolar caliber is relatively high (and the resistance is low), some arterial pulsations are transmitted through the pulmonary capillaries into the pulmonary veins (Kaplan *et al.*, 1951; Reuben *et al.*, 1971). Venous pulsation is, however, only some 5.0 per cent of pulmonary arterial pulsation (Bergel and Milnor, 1965) so can reasonably be ignored in the calculation of input impedance. In determining input impedance, therefore, pulsatile arterial pressure and flow are related directly in both the systemic and pulmonary circulations and venous pulsations are ignored. In calculating the resistive component of pulmonary impedance, however, and from this estimating terminal impedance and reflection coefficient, mean left atrial pressure must always be considered. In calculating the resistive component of systemic impedance, the right atrial mean pressure can reasonably be ignored under normal circumstances, but must be considered when elevated (as in right heart failure).

In his theoretic studies of vascular impedance, Taylor and co-workers (McDonald and Taylor, 1959; Taylor, 1957a; Taylor, 1957b; Taylor, 1959a; Taylor, 1964) showed that terminal impedance at zero frequency is equivalent to peripheral resistance. He demonstrated this experimentally (Taylor, 1966d) by showing that the frequency-dependent components of impedance modulus increase gradually and progressively as frequency is reduced between 1.00 and 0.01 Hz. Using resistance as equivalent to determined impedance, Taylor calculated the relatively high values of reflection coefficient referred to above. McDonald, who had previously taken a different view in the first edition of this book, conceded the point in the second edition.

Characteristic impedance

As previously discussed, characteristic impedance is input impedance in the absence of wave reflection. Since some degree of wave reflection is always present under normal circumstances, characteristic impedance cannot, except under special circumstances and with special techniques (see above), be measured directly. With the use of power spectrum analysis applied to data obtained with the heart beating irregularly (Taylor, 1966d; Alexander *et al.*, 1989; Zuckerman and Yin, 1989a), it is possible to obtain

impedance values up to 25 Hz. At this frequency, fluctuations due to wave reflection are almost completely attenuated, at least in the major arteries (Taylor, 1966b; O'Rourke and Taylor, 1967). Using this technique in Taylor's laboratory in Sydney, characteristic impedance was calculated as the average value of modulus between 15 and 25 Hz (O'Rourke, 1965a; Taylor, 1966b; O'Rourke and Taylor, 1967). In other studies, where the impedance modulus has been determined from Fourier analysis of waves recorded with the heart beating regularly, the characteristic impedance has been variously estimated as the average value of modulus between 2.0 and 12 Hz (Bergel and Milnor, 1965), above 2.0 Hz (Nichols *et al.*, 1977a; Murgo *et al.*, 1980a; Murgo *et al.*, 1981a; Murgo *et al.*, 1981b; Nichols *et al.*, 1985a), or above the frequency of the first minimum, as in the second edition of this book. One must appreciate that these determinations are arbitrary and provide estimates only. These estimates are likely to be reasonably accurate when applied to data when the first impedance minimum is reached at a low frequency, but quite inaccurate when applied to data (such as derived from a peripheral bed or from a small animal) where the first impedance minimum is reached well above 2.0 Hz. In averaging impedance moduli through a band of frequencies to calculate characteristic impedance, one hopes to include as much fluctuation above the true characteristic impedance as below. Success in this obviously will depend on the band of frequencies selected in relation to the maximal and minimal values of modulus. Using these arbitrary averaging techniques, different values of characteristic impedance have been obtained from the same impedance data (O'Rourke and Taylor, 1967).

The characteristic impedance of a vascular bed is determined principally by the distensibility of its major artery immediately distal to the site of measurement (see above). Determinations of characteristic impedance have been used as indices of arterial distensibility and to study the effects of different interventions, drugs, and diseases on arterial distensibility (see Chapters 19, 20, and 26). It is clear that one must make comparisons and draw conclusions with extreme caution in view of the technical difficulties in estimating characteristic impedance. Further problems in the interpretation of characteristic impedance under different conditions arise from the effects of alterations in arterial diameter, as these alter the relationship between velocity flow and volume flow (and so impedance expressed as dyne-s/cm³ or dyne-s/cm⁵) (Bargainer, 1967; O'Rourke, 1982c; Avolio, 1994; Eaton *et al.*, 1994), and from the effects of tapering with a Venturi effect on lateral pressure and high flow velocity (O'Rourke, 1968; O'Rourke, 1982c). These problems are discussed on the following pages.

Characteristic impedance is, in practice, difficult to calculate and to interpret. Although it is an index of arterial elasticity, it is probably less accurate and less reliable than simpler and better-established indices such as

foot-to-foot pulse wave velocity (see Chapters 4, 19, and 20) (Mitchell *et al.*, 2002; Mitchell *et al.*, 2003; O'Rourke *et al.*, 2004a; O'Rourke and Hashimoto, 2007; Mitchell *et al.*, 2010).

Units for the expression of impedance modulus

Impedance modulus, peripheral resistance, terminal impedance, and characteristic impedance are determined as pressure divided by flow. Flow can be expressed in volumetric terms (i.e. as cm^3/s) or in terms of velocity (i.e. as cm/s). Impedance can be expressed accordingly in units of volumetric flow ($\text{dyne}\cdot\text{s}/\text{cm}^5$) or units of velocity flow ($\text{dyne}\cdot\text{s}/\text{cm}^3$). There is no accepted convention for expression of impedance modulus. Some authors have not used either of the units but have expressed impedance as normalized for weight (Bergel and Milnor, 1965; Milnor *et al.*, 1966) or in relation to peripheral resistance (Patel *et al.*, 1965b; Patel and Vaishnav, 1980) or characteristic impedance (Murgo *et al.*, 1980a). Some standardization is clearly desirable.

Use of volume flow (cm^3/s) has the benefit of familiarity and appears to make more sense in the calculation of peripheral resistance than does flow velocity (cm/s). However, volume flow varies by several orders of magnitude in different arteries of the same animal and in the same artery of different animals, whereas blood flow velocity, like pressure, is similar in different arteries and in different species. The impedance modulus expressed in terms of linear flow velocity (i.e. as $\text{dyne}\cdot\text{s}/\text{cm}^3$) can be compared on the same axes of a graph for different arteries of the same animal (see Fig. 12.17) or for corresponding arteries of different animals (Fig. 12.39). This is not possible when impedance modulus is expressed as volume flow unless a logarithmic scale is used (Fig. 12.19). Thus, comparisons of impedance between different arteries and between different animals can best be made when the modulus is expressed in terms of linear flow velocity.

The same argument also applies when comparisons are made in the same artery in different circumstances where the artery dilates or constricts. In studies of nor-epinephrine on ascending aortic impedance (see above) and of serotonin on pulmonary artery impedance (see above), characteristic impedance expressed in terms of volume flow (as $\text{dyne}\cdot\text{s}/\text{cm}^5$) did not appear to increase, whereas arterial stiffness and pulse wave velocity were known to increase markedly. The use of volume flow under these circumstances fails to take into account the marked change in arterial caliber that accompanies change in mean pressure. When characteristic impedance is expressed as velocity flow in the unconstrained artery beyond the cuff-type flow transducer used in these experiments, the characteristic impedance increases to a similar degree as pulse wave velocity when mean arterial pressure and arterial diameter are increased.

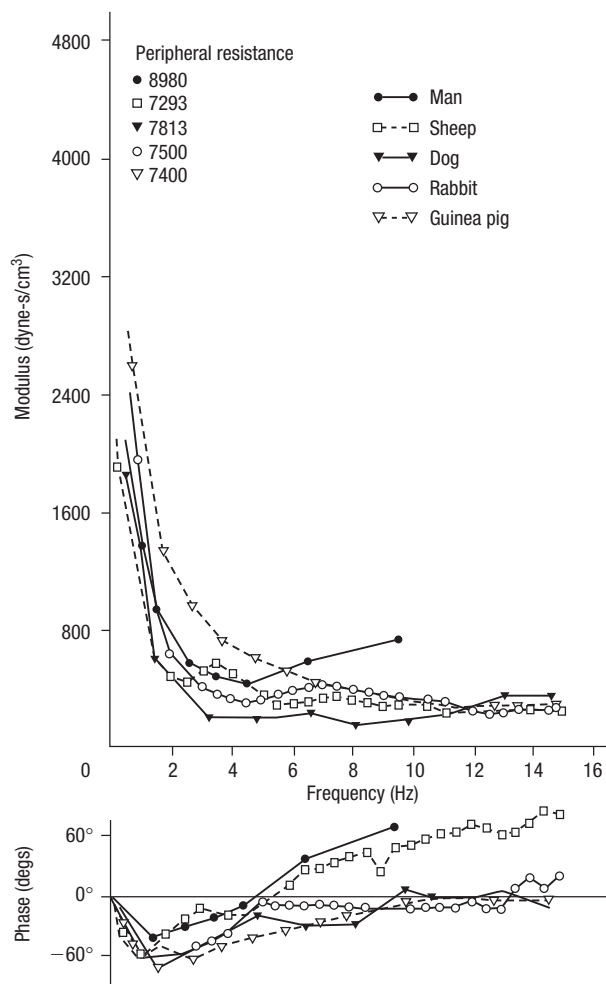


Figure 12.39 Vascular impedance in the ascending aorta of five different mammals. The modulus is expressed as $\text{dyne}\cdot\text{s}/\text{cm}^3$, as determined from linear flow velocity. Reproduced from Avolio (1976)

In the second edition of this book, McDonald pointed out that the correct expression of acoustic and hydraulic impedance is as $\text{dyne}\cdot\text{s}/\text{cm}^3$ (i.e. in terms of linear flow velocity) and suggested that vascular impedance be expressed in the same way. Information obtained subsequently reinforces this view.

Taylor (Taylor, 1964; Taylor, 1965), Bargainer (1967), and Murgo (1980a) discussed the relationship between characteristic impedance and pulse wave velocity in terms of the 'waterhammer formula' where $P = \rho V c_0$, P being pressure (in dyne/cm^2), ρ being density (g/cm^3), c_0 wave velocity (cm/s), and V linear velocity (cm/s). This can be rearranged as $P/V = \rho c_0$, where P/V is characteristic impedance in $\text{dyne}\cdot\text{s}/\text{cm}^3$; i.e. $Z_0 = P/V = \rho c_0$. Hence characteristic impedance, Z_0 , is directly related to the pulse wave velocity and is therefore, through this relationship, a measure of arterial distensibility. Since the blood density is close to unity (usually taken as approximately $1.05 \text{ g}/\text{cm}^3$), the two values are numerically almost

identical when pulse wave velocity is expressed as cm/s and Z_0 is dyne-s/cm³.

Normalization of impedance modulus expressed as volume flow to body weight, surface area, peripheral resistance, or to characteristic impedance, represents an attempt to allow for differences in body size. By expressing the modulus in terms of linear velocity, one accomplishes the same objective through normalization of volume flow to arterial cross-sectional area. In so doing, one has a term that has the same units of expression as acoustic and hydraulic impedance in the physical sciences; the term is directly and numerically comparable to pulse wave velocity and is an index of arterial distensibility. Expression of the impedance modulus as dyne-s/cm³ – in terms of linear flow velocity – appears to have advantages over other forms of expression. A final reason for expressing impedance in terms of flow velocity is that flow measurements are now usually measured from ultrasound which provides velocity directly. Converting this to volume flow requires measurement or assumption of arterial cross-sectional area.

Pressure measurement for the determination of vascular impedance

In the theoretic development, described earlier, of the concept of vascular impedance, no consideration was given to an important practical problem: the conversion of pressure energy to kinetic energy at high flow velocity. The theory applies strictly only to circumstances in which non-linearities are small, and hence to arteries in which the peak flow velocity is so low that kinetic energy is insignificant. In these circumstances, the pressure measured lateral to the direction of flow represents the total energy of blood in the artery (see Chapter 2), and the energy expended when this blood is forced through arterioles and capillaries into veins where lateral pressure is negligible.

As the cardiac ventricle contracts, energy is imparted to the blood within, which is moved into the aorta or pulmonary artery. Within these vessels, blood gains high velocity at the peak of ejection, so that some pressure energy is converted to kinetic energy. The amount of energy conversion depends on the velocity achieved. In a frictionless system, to which the large vessels approximate (O'Rourke, 1968; O'Rourke and Milnor, 1971), conversion can be calculated from Bernoulli's theorem (see Chapter 2). Kinetic energy has been calculated in the ascending aorta and pulmonary artery and shown to be relatively low in systemic arteries, but in the pulmonary system to be a significant fraction (some 25 percent) of total pulsatile energy under basal conditions (Milnor *et al.*, 1966; O'Rourke, 1967b; Porjé, 1967; Milnor *et al.*, 1969; Nichols *et al.*, 1977a). The kinetic component of pulsatile energy is substantially greater when flow velocity is increased during exercise (Elkins and Milnor, 1971), in arteriovenous shunts or when an artery is narrowed by a cuff-type flow transducer (O'Rourke, 1968).

When the arterial kinetic energy is high, lateral pressure underestimates the total energy present and thus the energy lost in vessels downstream. The question thus arises of how the vascular input impedance should be calculated: should it be determined as the relationship between total energy (expressed as pressure) and flow, or, indeed, should the concept of impedance be used at all?

Burton (1965) first raised this issue in relation to the calculation of peripheral resistance. He stressed that flow should be related to the total energy difference between artery and vein, and that the pressure equivalent of kinetic energy should be added to the recorded lateral pressure in the artery. If proper for calculation of resistance, the reasoning should apply to impedance; indeed it should be more relevant since the pulsatile flow velocity (and thus the pulsatile kinetic energy) is far greater than the mean flow velocity (and mean kinetic energy). O'Rourke and Taylor (O'Rourke and Taylor, 1966; O'Rourke and Taylor, 1967) took this line in their early studies and have followed it consistently since, but most other workers have continued to calculate impedance as the relationship between flow and lateral pressure, thus ignoring the contribution of kinetic energy to pressure. Some determinations of impedance in the presence of high flow velocities (Hopkins *et al.*, 1979; Yin *et al.*, 1981) may have given misleading information on this account (O'Rourke, 1980). The use of lateral pressure for impedance determination in the ascending aorta and main pulmonary artery excludes a component of pressure caused by acceleration of blood from the ventricle. In determining ventricular hydraulic load and performance from impedance, erroneous assumptions are likely to be made. Impedance and ventricular load may appear to decrease when contraction and ejection velocity increase, and to increase when contractility decreases!

The usual way to include kinetic with potential energy is to measure pressure end-on to the direction of flow. Because flow velocity profiles are generally flat in the large blood vessels, and because only forward flow velocity is high (and back flow velocity low), end-on or impact pressure represents the total energy of blood in the artery throughout the cardiac cycle (O'Rourke and Taylor, 1966; O'Rourke, 1967b; O'Rourke and Taylor, 1967; O'Rourke, 1968). End-on or impact pressure, of course, also represents what lateral pressure would be if blood had no kinetic energy.

Our recommendation is that vascular impedance be calculated as the relationship of flow to impact pressure. We acknowledge that small and quite tolerable errors will occur when flow is recorded under usual conditions with a catheter-mounted pressure transducer that measures lateral pressure. Our real concern is with data obtained when flow velocities are very high, as when an artery is constricted – if only slightly – with a cuff-type flow transducer, during exercise or in the presence of arteriovenous shunts.

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Aortic input impedance as ventricular load

'Afterload' and impedance	311	Pulmonary artery impedance and right ventricular	
Ascending aortic impedance and left ventricular hydraulic		hydraulic load	316
load	313	The relationship between external ventricular power and	
Ascending aortic and left ventricular pressure	315	input impedance	316

Arterial input impedance describes the relationship between pressure and flow in an artery and characterizes the properties of the arterial bed downstream. Vascular input impedance is completely independent of the flow wave into the vascular bed, so determines the amplitude and contour of the pressure wave generated by any flow wave into the bed. These features make input impedance in the ascending aorta and in the main pulmonary artery of particular value in the study of the whole systemic and pulmonary vascular circulations, and in the interaction between systemic and pulmonary circulations with the left and right ventricle, respectively (O'Rourke and Taylor, 1967; Milnor, 1975; Nichols *et al.*, 1977a; Noble, 1979a; Noble, 1979b; Elzinga *et al.*, 1980; Nichols *et al.*, 1980b; Nichols and Pepine, 1982; Brin and Yin, 1984; O'Rourke *et al.*, 1984; Laskey *et al.*, 1985; Nichols *et al.*, 1985a; Yaginuma *et al.*, 1985; Piene, 1986; O'Rourke *et al.*, 1986a; Yin, 1986; Laskey and Kussmaul, 1987; Latham *et al.*, 1988b; Milnor, 1989; Zuckerman and Yin, 1989a; Chen *et al.*, 1990; Ting *et al.*, 1991; Kelly and Fitchett, 1992; Ting *et al.*, 1993; Ting *et al.*, 1995a; Ting *et al.*, 1995b; Mitchell *et al.*, 1996; Mitchell *et al.*, 2003; O'Rourke *et al.*, 2004a; Jegger *et al.*, 2006; Segers *et al.*, 2007b; Mitchell *et al.*, 2008; Chirinos *et al.*, 2009a; Chirinos *et al.*, 2009b; Chirinos and Segers, 2010a; Chirinos and Segers, 2010b).

The contour and amplitude of the ascending aortic flow and pressure waves depend on the pattern of ventricular ejection (i.e. on cardiac action) and on the properties of the systemic arterial system. The properties of the systemic arterial system are characterized as the ascending aortic impedance (Patel *et al.*, 1963a; O'Rourke and Taylor, 1967; Nichols *et al.*, 1977a; Noble, 1979b; Nichols *et al.*, 1980b; Nichols *et al.*, 1985a; Nichols *et al.*, 1987; Elzinga and Westerhof, 1991; Binkley *et al.*, 1996). When these arterial properties remain constant, the ascending aortic impedance remains constant, so that there is a unique relationship between pressure and flow

in the ascending aorta. For any flow ejection pattern, the contour of the resulting pressure waves can be precisely determined. Hence, ascending aortic impedance is regarded, quite properly, as an expression of left ventricular hydraulic load (Milnor, 1975; Hunter and Noordergraaf, 1976; Nichols *et al.*, 1980b), and, for the same reasons, main pulmonary artery impedance is the appropriate expression of right ventricular hydraulic load (Bergel and Milnor, 1965; Piene, 1986). Studies of arterial impedance in these vessels give valuable information not only on arterial properties but also on the interaction between the heart and vascular system and on the adaptation of one to the other (O'Rourke and Taylor, 1967; Yin, 1986).

'AFTERLOAD' AND IMPEDANCE

The term ventricular 'afterload' is used by clinicians to describe the external vascular factors that oppose ventricular systolic ejection. There is no general agreement about how this can best be characterized. Milnor (1975) has suggested that ventricular 'afterload' is the hydraulic input impedance presented to the ventricle (and thus measurable as ascending aortic impedance for the left ventricle and main pulmonary artery impedance for the right). A good argument can be made for this concept of 'afterload' since the input impedance is completely independent of cardiac properties, and thus determines what pressure is generated by any given ventricular ejection wave. However, there are, of course, difficulties (Noble, 1979b; O'Rourke, 1982a; O'Rourke, 1982b), and the concept has not yet gained popular appeal among cardiac physiologists or cardiologists, however, it is gaining popularity since it can now be measured non-invasively (Kelly and Fitchett, 1992; Chirinos and Segers, 2010a; Chirinos and Segers, 2010b). Impedance is a complex quantity that comprises modulus and phase,

both of which vary with frequency; therefore one cannot say that impedance is increased or decreased with different interventions without qualifying the answer by specifying phase and frequency. Furthermore, impedance modulus has the units of pressure divided by flow ($\text{dyne}\cdot\text{s}/\text{cm}^5$ or $\text{dyne}\cdot\text{s}/\text{cm}^3$), whereas afterload as applied to isolated papillary muscle preparations has the units of force (dyne) or pressure (dyne/cm^2). The concept of input impedance and ventricular 'afterload' would be more useful if ventricular contraction could be expressed (like impedance) in frequency-dependent terms. This has been attempted (Abel, 1971; Westerhof and Elzinga, 1978; Piene, 1986) with the heart considered as a pressure or flow generator in series with its own source impedance. This concept has not proved helpful, however, and there has been considerable debate about whether the heart is a 'pressure source' or a 'flow source', and about the degree to which generator and source impedance are altered by change in myocardial contractility (O'Rourke, 1982a; O'Rourke, 1982b; Milnor, 1989) (see Chapter 14). In contrast to the passive arterial system, the heart is an active pump whose properties, in systole and diastole, vary with time. It cannot be characterized in the frequency domain, and attempts to do so have been largely abandoned (Piene, 1986).

The term 'afterload' has been applied to the ventricle through application of studies on isolated muscle whose contraction was initiated at a given resting stretch but then had to lift another weight as contraction proceeded (Ross *et al.*, 1966; Ross, 1976; Pouleur *et al.*, 1979). In these isolated muscle studies, afterload remained constant throughout contraction.

Afterload is probably an unduly simplistic concept to apply to contracting ventricles. Yet there can be no argument that input impedance represents the hydraulic load presented to the ventricle by the systemic or pulmonary circulation. The problem arises of how this is used when describing ventricular function in different circumstances. Some have used resistance as afterload (Sonnenblick and Strobeck, 1971; Ross, 1976; Meredith *et al.*, 2008; Mullens *et al.*, 2008; Yancy, 2008), while some have suggested characteristic impedance as afterload (Milnor, 1975; Milnor, 1989; Chirinos *et al.*, 2009a). Clearly, neither is adequate and completely appropriate (O'Rourke and Nichols, 2009a; O'Rourke, 2010). Each is only a component of impedance and takes no account of the fluctuations in impedance modulus and phase between zero frequency (where the modulus equals peripheral resistance) and high frequencies (where the modulus equals characteristic impedance). Wave reflection is a major determinant of the values of modulus and phase between very low and very high frequencies, and indeed over the frequency range that contains the major harmonic components of the ascending aortic flow wave when the heart is beating normally (Chapter 9). Therefore, total ventricular external hydraulic load should include all three components – resistance, compliance,

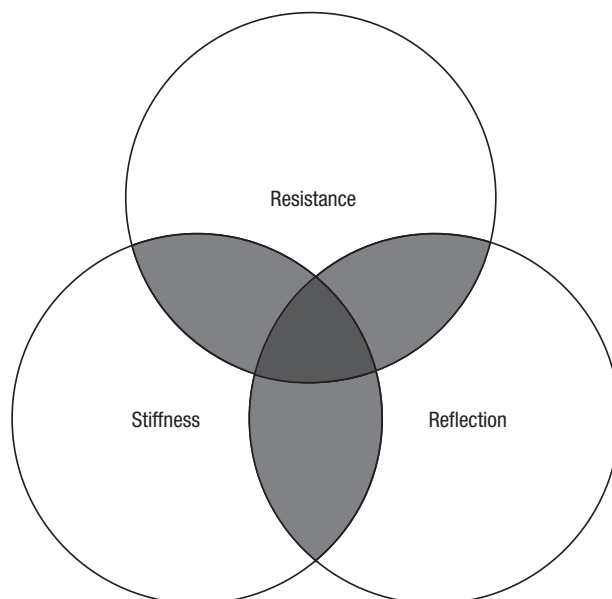


Figure 13.1 Venn diagram illustrating the different components of ventricular 'afterload'. The nonpulsatile or static component is total peripheral resistance (mean pressure/mean flow) and the pulsatile or dynamic components are stiffness and reflection. Changes in any (or all) of these load components can alter ventricular performance. Reproduced from Nichols *et al.* (1987)

and reflections – which can be obtained from the input impedance spectra (Fig. 13.1) (Nichols *et al.*, 1980b; Nichols and Pepine, 1982; Nichols *et al.*, 1987).

The best analogy for afterload of a papillary muscle preparation in the contracting ventricle is the ventricular pressure itself. This can be estimated as ascending aortic pressure. At least this is expressed in the same units and is the parameter that opposes ejection of blood flow from the ventricle. The problems here are, however, that pressure varies during ejection: it is not constant as in the papillary muscle preparation, and the pressure generated depends on ventricular contraction; that is, the load is largely determined by the contracting muscle and is not independent of the contracting ventricular muscle (Milnor, 1975; Noble, 1979b; Milnor, 1989). There are other problems as well – the load borne by individual cardiac muscle fibers within the heart depends not only on ventricular pressure but also on the orientation of muscle fibers, the thickness of the ventricular wall, the degree of ventricular dilatation, and myocardial elasticity and viscosity (Buckberg *et al.*, 2008). This subject is pursued in greater depth in Chapter 14. The point that can be made here is that input impedance of the systemic and pulmonary circulations is the hydraulic load presented to the respective ventricle by that circulatory system. Impedance is a major determinant of aortic and thus left ventricular pressure while the aortic valve is open, and is the only determinant that is independent of ventricular properties and ventricular filling characteristics. It also has

the potential for modification to the heart's advantage. If pressure in systole is afterload, impedance is the principal determinant of this.

However conceived or expressed, ventricular afterload is largely determined by input impedance, so that impedance must be considered carefully when interpreting cardiac function under different conditions.

When considering input impedance as hydraulic ventricular load, it is necessary to consider the fundamental role of the heart and the circulation. This is, of course, to pump blood through the capillaries of the body so that cells may function effectively. The need of the body is for steady flow, and the demand on the heart is to generate this and on the arteries and arterioles to distribute it to the capillaries where exchange of oxygen and nutrients takes place. However, the heart is not a continuous pump; its action is intermittent. Pulsations of pressure and flow appear to be a consequence of organic heart design, of its inability to generate steady continuous flow. Seen this way, pulsations are unnecessary and wasteful but result from the heart's design limitation (O'Rourke, 1965a; O'Rourke, 1982a; Taylor, 1994). The role of the arterial system as a type of Windkessel is to minimize energy wastage as pulsations and to maintain as low as possible that extra component of hydraulic load which is a consequence of the heart's intermittency (O'Rourke, 1967b; Taylor, 1969; Maeta and Hori, 1985; Hashimoto *et al.*, 2008a). This type of basic reasoning leads to a separation of input impedance into a functionally useful or resistive component and another pulsatile component which is wasteful of cardiac energy and which should be maintained as low as possible in relation to the resistive component (see Fig. 13.2). Such an approach leads to comparison of resistive against reactive components of load and separation of external heart work into steady (useful) and pulsatile (wasteful) components. Since the function of the arterial system is to cushion cardiac pulsations as well as to distribute blood, its efficiency will be seen through comparison of wasteful (pulsatile) to useful (steady) components of external heart work and of the reactive and resistive components of impedance.

ASCENDING AORTIC IMPEDANCE AND LEFT VENTRICULAR HYDRAULIC LOAD

The principal features of ascending aortic impedance patterns are described in Chapter 12 for humans, dogs, and a variety of experimental animals. Particular attention is directed at dogs, against which other species are compared.

The modulus of ascending aortic impedance falls steeply from its value at zero frequency (the peripheral resistance) to the value at high frequencies (the characteristic impedance), which is approximately 5 percent of peripheral resistance. The impedance modulus is even lower over a band of frequencies (around 2–8 Hz in dogs) where the average distance to peripheral reflecting sites

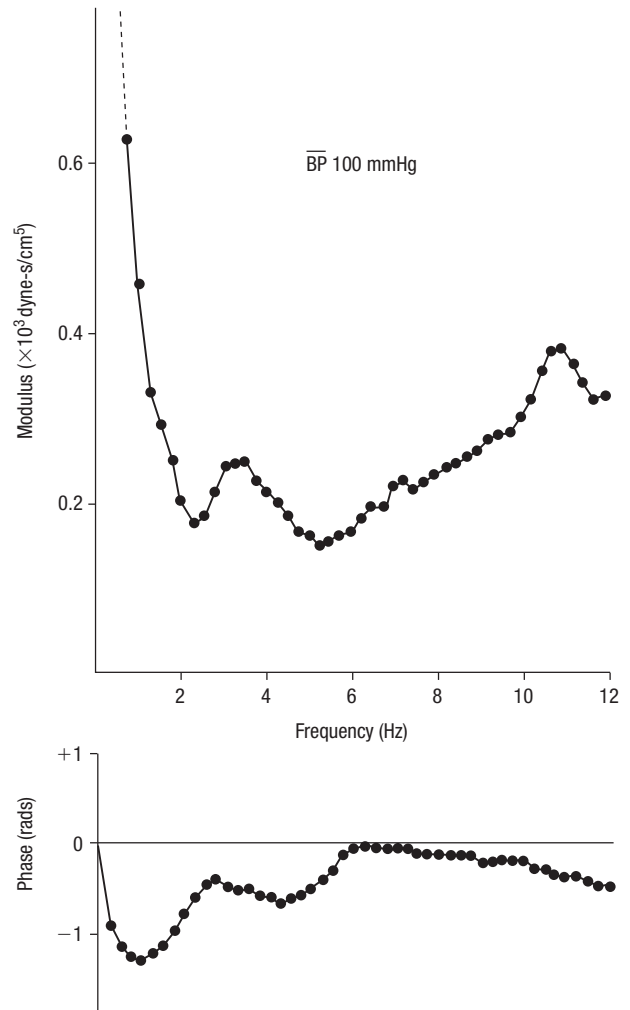


Figure 13.2 Modulus and phase of ascending aortic impedance in a 23 kg dog. Peripheral resistance (7.7×10^3 dyne-s/cm⁵) is approximately 20 times the value of characteristic impedance. Reproduced from O'Rourke (1970a)

represents one-quarter wavelength. Over this band, the impedance modulus may be 2.5 percent or less of peripheral resistance. However, the in-phase impedance (the component of pressure which is in phase with flow and determined at each frequency as the modulus multiplied by the cosine of phase) is even lower still, being 2.0 percent or less of peripheral resistance over this frequency band (Figs 13.2, 13.3, and 13.4).

These features of ascending aortic impedance are ideally suited for left ventricular ejection; this favorable interaction is best seen when ascending aortic impedance or the in-phase component of impedance is compared against the frequency components of the left ventricular ejection (or ascending aortic flow) wave (Figs 13.3 and 13.5). When so plotted, it is apparent that the lowest values of moduli (and of in-phase moduli) correspond to the highest values (the first three harmonic components) of the ventricular ejection (flow) wave in animals and humans with a compliant ascending aorta. In other words,

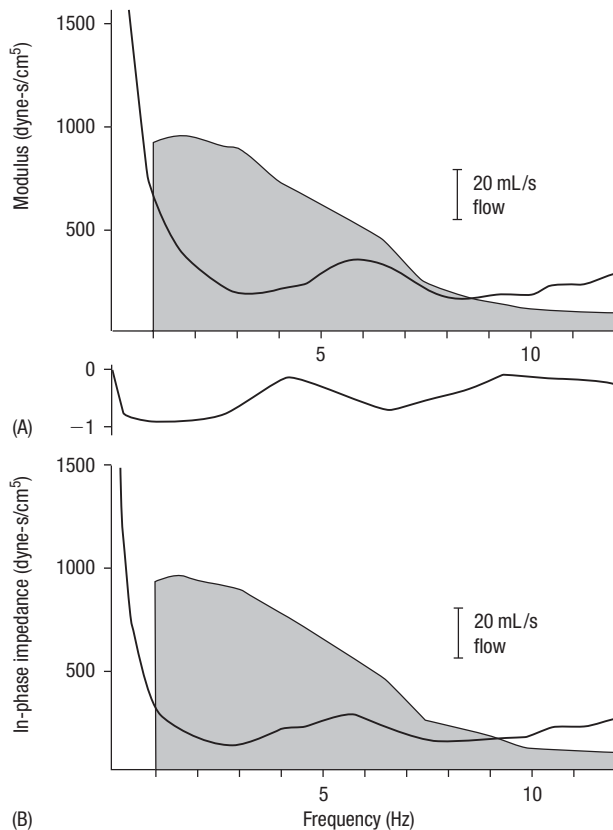


Figure 13.3 (A) Relationship between impedance modulus in the ascending aorta of a dog and the amplitude of harmonics of ventricular ejection (flow) waves (stippled area, as determined in a dog of similar size). (B) Relationship between in-phase impedance ($Z \cos \phi$) in the ascending aorta of the same dog and the amplitude of harmonics of the ventricular ejection wave. In-phase impedance is lower than impedance modulus at the frequencies where flow harmonics are highest. Reproduced from O'Rourke (1982a)

the impedance modulus is least over the frequency band that contains most of the pulsatile components of the left ventricular ejection wave. The pulsatile pressure generated by pulsatile flow at these frequencies is extremely low in relation to the mean pressure generated by the mean component of flow.

Taylor (Taylor, 1965; Taylor, 1966a; Taylor, 1966b; Taylor, 1966c; Taylor, 1969) examined the properties of arteries that are responsible for the highly desirable input impedance characteristics of the systemic circulation. The main factors were:

1. High distensibility of the proximal aorta, and lower distensibility of peripheral arteries.
2. Peripheral wave reflection.
3. Spatial dispersion of peripheral reflecting sites.
4. Asymmetry of the heart's location with respect to peripheral sites.
5. Appropriate relationship between body length and heart rate.

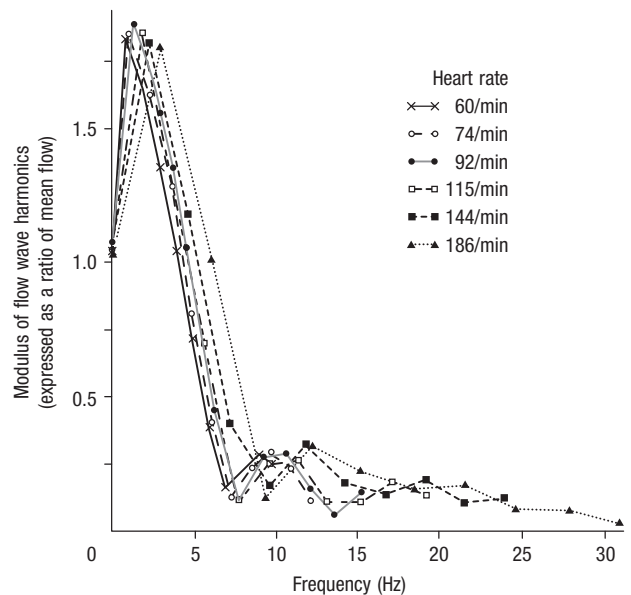


Figure 13.4 Moduli of the first ten harmonics of ascending aortic flow waves at heart rates between 60 and 186 beats per minute. Data were obtained from a dog with surgically induced heart block and electronic pacing. Despite changes in heart rate, the highest harmonic flow content remained in the frequency band of 1.0–7.0 Hz. Reproduced from O'Rourke and Taylor (1967)

High distensibility of the proximal aorta is responsible for the low value of characteristic impedance in the ascending aorta – lower than in any other systemic artery (see Chapter 12). The left ventricle 'sees' the low characteristic impedance of this vessel, rather than the higher characteristic impedance of arteries downstream. As far as the heart is concerned, for initial ejection of blood which is not influenced by wave reflection, the arterial system behaves as though it were all as distensible as the ascending aorta, yet the arterial system avoids the disadvantages of high overall distensibility by having progressively increased stiffness (decreased distensibility) of more distal arteries (Taylor, 1965). Wave reflection is responsible for the early minimal value of impedance modulus, where it dips below the value of characteristic impedance. For this to be of value to the heart, however, there must be a type of 'tuning', with the minimal value of impedance corresponding to maximal values of flow harmonics. As described (Figs 13.3, 13.4, and 13.5), such a relationship does apply in most mammals, but the system would be very unstable if the modulus rose to a high value at twice the frequency of that first minimum as would be expected in a system with a single peripheral reflecting site (see eqn 9.31). This instability is avoided by the spatial dispersion of peripheral reflecting sites, including the relative proximity to upper body sites and the relative distance to sites in the trunk and lower limbs (Taylor, 1966a; O'Rourke and Taylor, 1967). This arrangement appears to preserve the advantages of wave reflection at low frequencies (minimal value of modulus) while

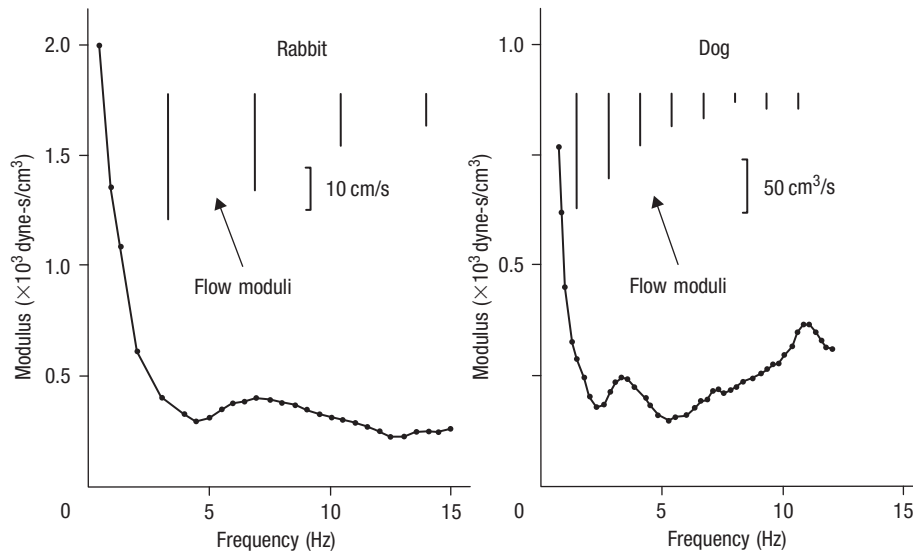


Figure 13.5 Relationship between modulus of ascending aortic impedance (below) and modulus of the first four to eight harmonics of ascending aortic flow (above) for a rabbit and a large dog. The impedance modulus is low at those frequencies where flow modulus is greatest. Reproduced from O'Rourke *et al.* (1984)

avoiding the disadvantage of a subsequent large rise in impedance modulus.

Pulse wave velocity is similar in corresponding arteries of different mammals (O'Rourke, 1982b), irrespective of body size. Because of this, and because arterial architecture is proportional to body dimensions, both the timing of wave reflection as seen at the heart, and the frequency of the first minimal value of ascending aortic impedance, are related to body length, being earlier (and at higher frequency) in small animals, and later (and at lower frequency) in large animals (Fig. 13.5). The favorable correspondence between the first minimal value of the impedance modulus in the ascending aorta and harmonic components of the left ventricular ejection wave will only be preserved if ventricular ejection changes in an appropriate fashion with body size. This, indeed, is seen as the difference in duration of ejection in animals of different sizes. The flow ejection pattern is similar in all animals studied, but the heart rate varies considerably with body size. The well-known inverse relationship between heart rate and body length (Fig. 13.6) is thus readily explained on the basis of physical principles and constitutes a survival advantage related to cardiac efficiency.

There are notable exceptions to the general principle of correspondence between the highest harmonics of the ascending aortic flow wave and minimal values of the ascending aortic impedance. In humans, most studies have shown a higher value of frequency for the minimal value than would be deemed desirable, and a mismatch between this and the highest harmonic values of flow. This is discussed fully in Chapter 19, where the mismatch is attributed to asymptomatic arterial degeneration with age. Published values of impedance are from adults, with an average age in the sixth decade. A similar mismatch, however, is seen in guinea pigs (Fig. 13.7B) (Avolio *et al.*, 1976)

and probably also in rats and mice. The reason for such a mismatch in these species is not clear. All, however, show an ascending aortic pressure wave contour similar to that in elderly humans (Fig. 13.7A).

ASCENDING AORTIC AND LEFT VENTRICULAR PRESSURE

In normal circumstances, there is little difference between pressures recorded in the left ventricle and the ascending aorta during systole, while the aortic valve is open. The tracings are almost superimposable (see Fig. 7.3). Ascending aortic impedance clearly determines left ventricular, as well as aortic systolic pressure. In most mammals, the peak aortic pressure is reached in early to mid-systole, but in adult humans and horses, guinea pigs, rats, and mice, the peak pressure is reached in later systole (Figs 10.2 and 13.7A). This late, augmented, pressure peak is attributable to relatively early wave reflection returning before ventricular ejection has been completed and is manifest on the impedance moduli spectrum as enhanced amplitude of the fundamental harmonic. Its presence is a manifestation in the time domain of the mismatch referred to above in the frequency domain. The mismatch between cardiac ejection and vascular properties leads to a higher pressure being achieved during systole than would be the case if the ejection duration were shorter, or the wave velocity lower, or the body length greater. This extra effort generated by the ventricle is wasted since it does not contribute to forward flow (Nichols and Singh, 2002; Nichols *et al.*, 2008; Hashimoto *et al.*, 2008a; Nichols and Epstein, 2009). However, in the time domain, another aspect of mismatch is apparent as well in the near-exponential decline of pressure during diastole in adult

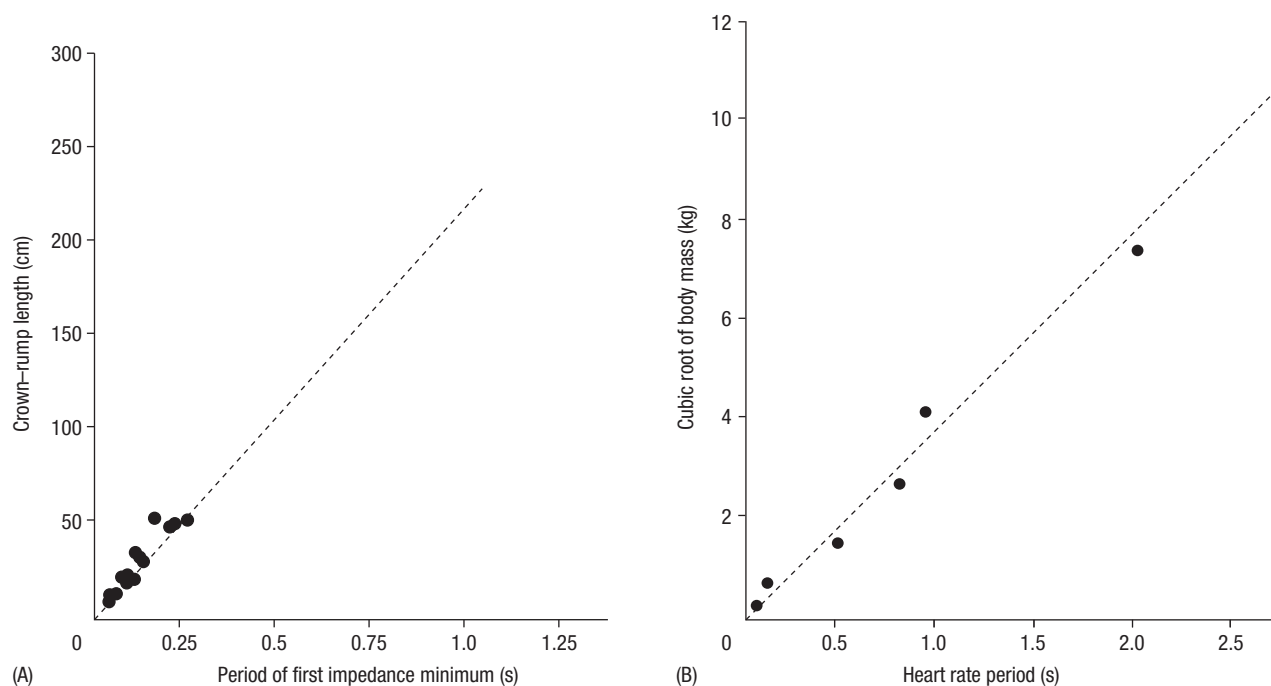


Figure 13.6 (A) Relationship between crown-rump length and period of first ascending aortic impedance minimum in four animal species (guinea pig, rabbit, dog, and sheep). (B) Relationship between the cubic root of body mass (which is proportional to crown-rump length) and resting heart rate period in a series of animals from mouse to elephant from data of Kenner (1972). Reproduced from O'Rourke *et al.* (1984)

humans and in these species. In young human adults and in other mammals, pressure during diastole is augmented by a broad and quite discrete reflected wave (Chapter 9). Coronary blood flow is generated by the pressure maintained during diastole, while ventricular myocardium is relaxed and the smallest vessels patent (see Chapter 17). Appropriately timed wave reflection in diastolic aids coronary flow as well as helping to minimize ventricular hydraulic load (O'Rourke *et al.*, 1986a).

Reference has been made to the efficiency of the arterial system in accepting pulsatile flow from the heart. This may be quantified in terms of the proportion of pulsatile to steady external left ventricular work (see below). It can also be described more simply in terms of the difference between mean ascending aortic pressure during systole and mean ascending aortic pressure during the period of diastole (O'Rourke, 1967b). Chirinos *et al.* (2009c) have confirmed that components of ascending aortic impedance – peripheral resistance, characteristic impedance (Z_0), and measures of early wave reflection predict left ventricular (LV) mass, but that change in Z_0 and early wave reflection markedly improved prediction.

PULMONARY ARTERY IMPEDANCE AND RIGHT VENTRICULAR HYDRAULIC LOAD

The principles applied to the left ventricle and systemic circulation can be applied *mutatis mutandis* to the right

ventricle and pulmonary circulation. The pulmonary circulation is at a lower mean pressure through shorter but more distensible arteries, and with lower peripheral resistance. In the pulmonary circulation, wave reflection is less apparent than in the systemic, and there is no clear asymmetry between the two lungs as there is between the upper and lower parts of the body. Features of pulmonary vascular impedance are described below. There is a less steep fall in modulus from its value at zero frequency, and all values of modulus are lower than in the systemic circulation. The impedance phase shows less fluctuation than in the ascending aorta. However, shorter lengths to peripheral reflecting sites in the lung are compensated by lower pulse wave velocity in pulmonary arteries, so that the minimal values of the impedance modulus and phase crossover occur at similar frequencies to those in the ascending aorta (see Fig. 16.11). A heart rate that is appropriate to the systemic circulation for an ideal 'match' is thus appropriate to the pulmonary circulation as well.

THE RELATIONSHIP BETWEEN EXTERNAL VENTRICULAR POWER AND INPUT IMPEDANCE

External ventricular work (or power) is the energy generated by the ventricle that is lost as blood flows through the circulation (Katz, 1931; Remington and Hamilton, 1947; Burton, 1965; O'Rourke, 1967b). There are two components to this energy: the steady component W_m ,

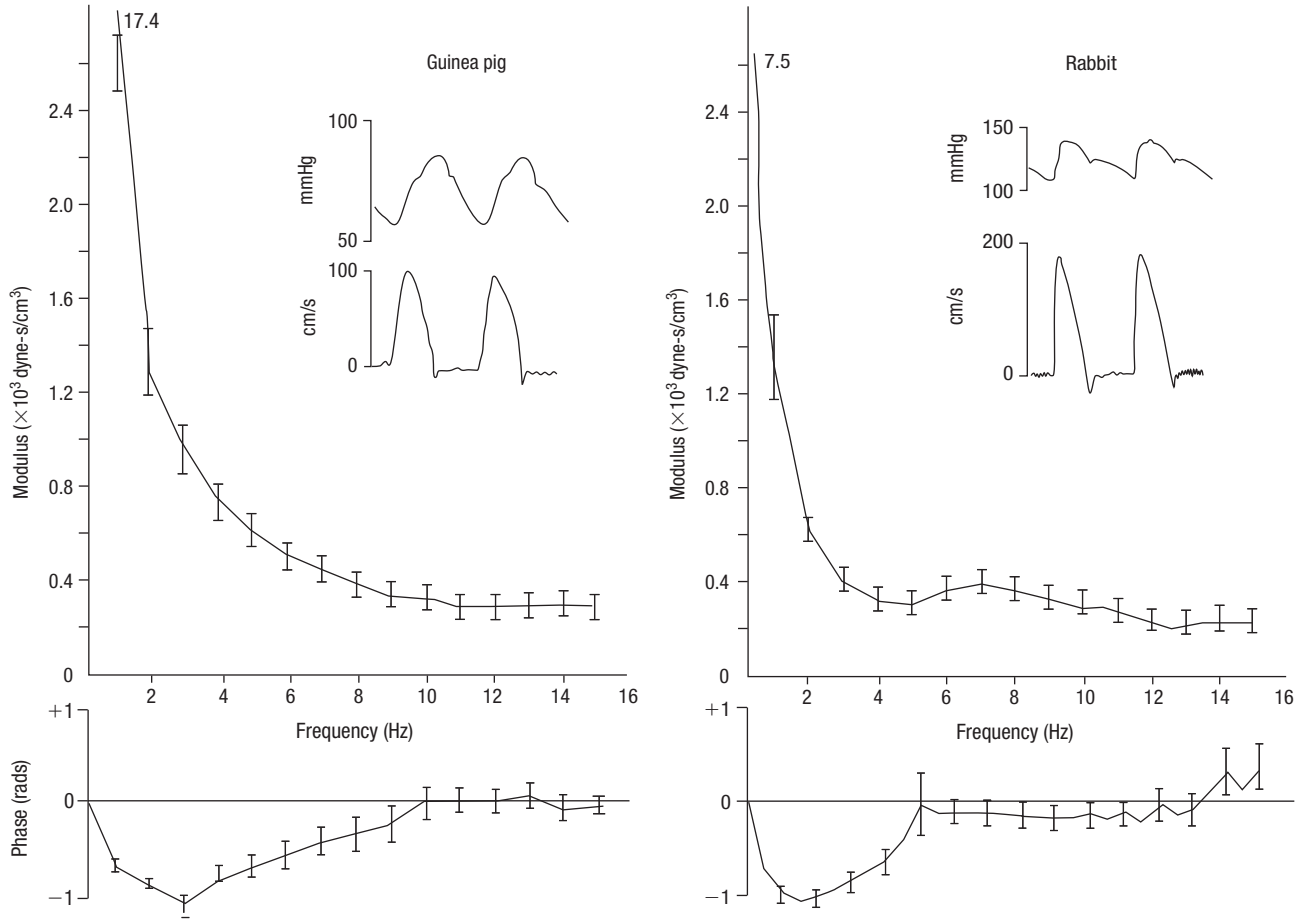


Figure 13.7 Typical flow and pressure waves in the ascending aorta of a rabbit and guinea pig (inset). The time calibration is the same for both tracings (below). Ascending aortic impedance of a series of 16 guinea pigs and 22 rabbits. Values are mean $\pm$ standard error of the mean. The modulus is expressed in terms of linear flow velocity. With similar heart rates (3–4/sec) and flow waves in guinea pigs as in rabbits, there is a mismatch in guinea pigs between impedance modulus and flow harmonics. This is responsible for the late systolic peak of pressure in the ascending aortic pressure wave. Reproduced from Avolio *et al.* (1976)

which is calculated as the product of cardiac output and mean arteriovenous pressure gradient, and the pulsatile component $\dot{W}_0$, which is calculated from input impedance to the circulation as

$$W_0 = \frac{1}{2} \sum_{n=1}^N (Q_n)^2 Z_n \cos \theta_n \quad 13.1$$

from harmonic components Q_n of the ventricular ejection wave and input impedance modulus Z_n and phase θ_n (Morris and O'Rourke, 1965; Milnor *et al.*, 1966; Porjé, 1967). The total external ventricular power or energy can be calculated as

$$W_T = P_0 Q_0 + W_0 = PQ, \quad 13.2$$

where P_0 and Q_0 are the mean values and P and Q are the instantaneous values of pressure and volume flow in the ascending aorta (or pulmonary artery). The pulsatile

component of external ventricular power can be obtained separately by subtraction of mean power ($W_m = P_0 Q_0$) from total power W_T . In normal circumstances, the pulsatile component of external ventricular power is only some 10 percent of total external left ventricular power (O'Rourke, 1967b; Nichols *et al.*, 1977a) and some 25 percent of total external right ventricular power (Milnor *et al.*, 1966).

The concept of steady and pulsatile components of external ventricular power was introduced by Bramwell and Hill (1922b) and subsequently used by Remington and Hamilton (1947). The pulsatile component was identified as representing energy lost in arterial pulsations and the steady component as representing energy lost in steady flow through the peripheral resistance. This and later work used the ratio of pulsatile to total external power as an inverse index of arterial efficiency, with the implication that steady power is physiologically useful in pumping blood forward, whereas pulsatile power is

wasted in arterial pulsations (Morris and O'Rourke, 1965; Milnor *et al.*, 1966; Porjé, 1967; O'Rourke, 1967b; Milnor, 1989).

In discussing the components of external ventricular power, two points should be made: the first regarding ventricular function and the second regarding kinetic energy and potential energy as components of ventricular power.

First, external ventricular power is the hydraulic energy imparted to blood by the ventricle but lost as the blood flows through the systemic or pulmonary circulations. This power bears no constant relationship to total ventricular power nor to myocardial oxygen needs, which are determined very largely by the pressure each ventricle must generate, and thus by the tension developed and sustained by the ventricular muscle during systole (Braunwald, 1969; Braunwald, 2009). Because external ventricular power represents energy lost within the vascular system, it is more relevant to the circulation than to the ventricle. The ratio of pulsatile to total external ventricular power is thus a useful index of arterial (in)efficiency, although perhaps no more useful than the difference between mean systolic and mean diastolic pressure in the ascending aorta (Fig. 13.8). The second point concerns the kinetic and potential components of external ventricular power. This is a complex matter, which can be simplified (O'Rourke, 1967b; O'Rourke, 1968). It has been the practice to separate the pulsatile and steady components of external ventricular power into their kinetic and potential elements (Evans, 1918; Prec *et al.*, 1949; Milnor *et al.*, 1966; Milnor, 1989). This becomes confusing because the relative magnitude of the kinetic component depends on the square of the blood velocity, and thus is likely to be markedly altered by relatively minor degrees of arterial obstruction caused by a cuff-type flow probe used to obtain experimental data (Fig. 2.10). The magnitude of the kinetic element thus depends to some extent on the technique of flow

measurement. Furthermore, the kinetic and potential components of ventricular power have no physiologic significance. The kinetic component is included with potential power when pressure is measured end-on to the direction of flow from the ventricle (O'Rourke and Taylor, 1967; O'Rourke, 1967b; O'Rourke, 1968).

Figure 13.9 shows the calculations by Milnor *et al.* (1966) of the steady and pulsatile components of right ventricular external power as measured from pressure and flow in the pulmonary artery of a dog (see also

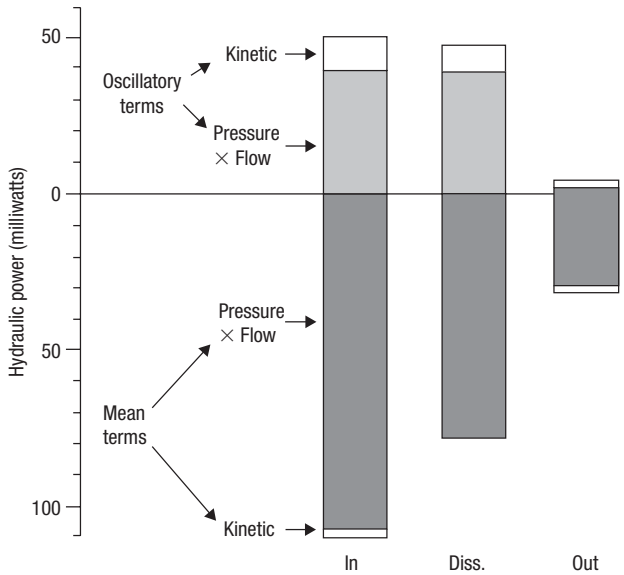


Figure 13.9 Average hydraulic power at the inlet (pulmonary artery, left) and outlet (pulmonary veins, right) of the pulmonary vascular bed of dogs. The difference between the two (center) is power dissipated (Diss.) by flow through the pulmonary circulation. Power associated with pulsatile terms is shown above the line and power associated with mean terms below the line. Kinetic and 'potential' components of energy are shown separately. Reproduced from Milnor *et al.* (1966)

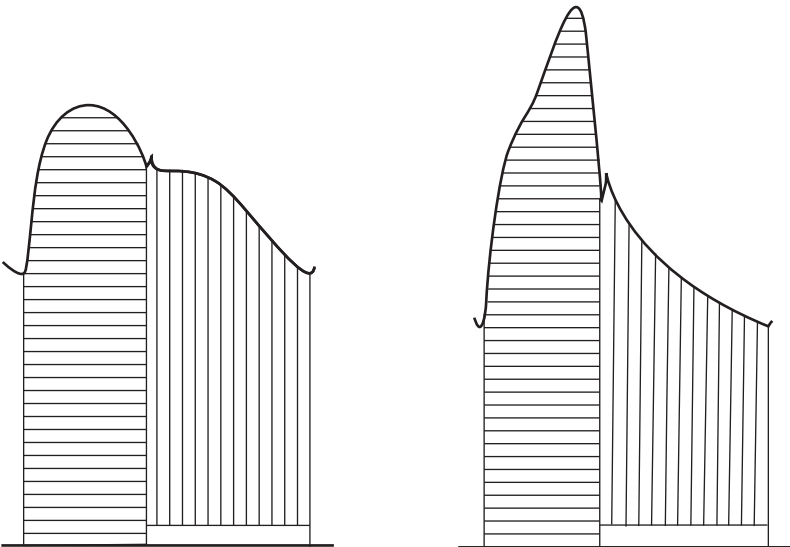


Figure 13.8 'Efficiency' of the arterial system in accepting pulsatile flow from the heart is manifest as a small difference between mean pressure during systole (horizontally hatched) and mean pressure during diastole (vertically hatched). Pressure during systole is a major determinant of myocardial oxygen requirement, while pressure during diastole is a major determinant of coronary blood flow. An 'efficient' arterial system generates the type of pressure wave seen on the left, while an inefficient system generates the type of wave seen on the right. Reproduced from O'Rourke (1982b)

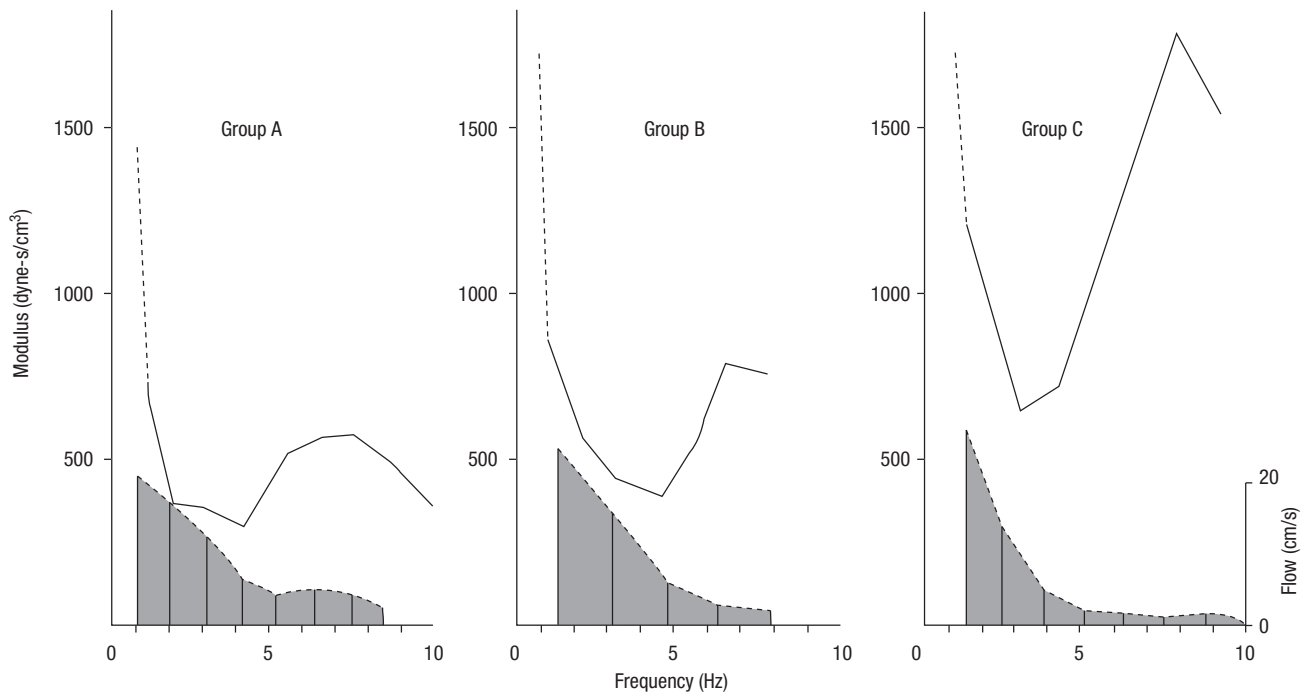


Figure 13.10 Impedance modulus in the ascending aorta (top) and harmonics of ascending aorta flow waves (bottom) in three groups of human subjects. Group A had no known cardiac or vascular disease, group B were subjects with known atherosclerosis, and group C were subjects with hypertension and known atherosclerosis. Data from Nichols *et al.* (1977a) as published in O'Rourke (1982a)

Chapter 16). The figure includes the kinetic and potential terms and shows pulsatile and steady energy at the input of the pulmonary circulation, and in the pulmonary veins, with energy dissipated shown separately. In terms of percentages, the oscillatory power was approximately 37 percent of mean power (25 percent of total power). In the systemic circulation, the pulsatile component of external ventricular power is far less than the steady component, as one would expect from the far higher mean pressure in the systemic circulation. In a group of normal dogs (O'Rourke, 1967b), the ratio of pulsatile to total external left ventricular power was found to be 10 percent. This value, however, was found to be markedly increased with artificial stiffening of the aorta, with vasodilatation, and with bradycardia (O'Rourke, 1967b). The ratio of pulsatile to total external left ventricular power was higher, too, around 20 percent in humans than in experimental animals (Porjé, 1967; Nichols *et al.*, 1977a; Kelly and Fitchett, 1992), and this can be attributed to unsuspected arterial degeneration. The ratio is further increased in patients with known arterial disease (Milnor, 1989), as would be expected with the increase in frequency-dependent values of impedance, and mismatch between ventricle and vascular system seen in disease (see also Chapter 19) (Fig. 13.10).

In both the systemic and pulmonary circulations, the ratio of pulsatile to total external ventricular power is markedly altered by changes in heart rate. This is illustrated in Fig. 13.11. Here as seen that in a typical dog, at a heart rate of 100 beats/min, the steady and

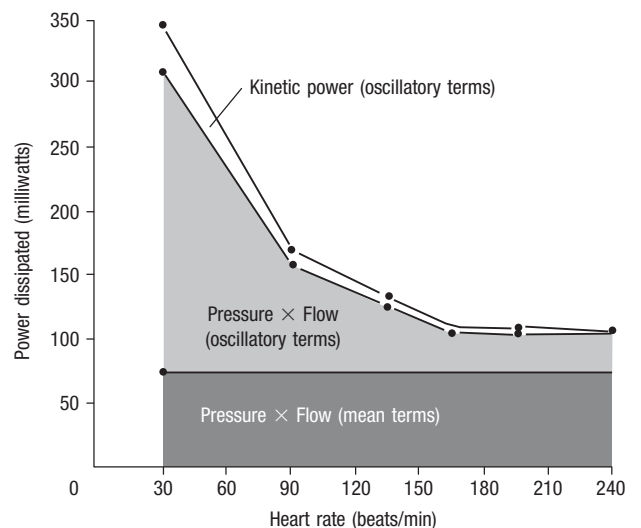


Figure 13.11 A composite diagram of the relation between heart rate (abscissa) and power dissipated in the pulmonary bed (ordinate) for a constant mean pulmonary blood flow of 42 mL/s. It is assumed that output power is not changed with heart rate. It can be seen that pulsatile and steady components of power dissipated are roughly equal at a heart rate of 100 beats/min, but that pulsatile power increases markedly below this rate and falls slightly above 100/min. Reproduced from Milnor *et al.* (1966)

pulsatile components of right ventricular external power dissipated in the lungs are approximately equal. At heart rates below this, the pulsatile component increases markedly, while at heart rates above this the pulsatile

component decreases slightly; the steady component remains constant at all frequencies.

There is now general agreement that ventricular metabolic demands are determined more by pressure generated than by flow ejected (see Chapter 14). In consequence, there is less interest now in external ventricular power as a measure of cardiovascular performance and

more on the determinants of pressure in the ascending aorta against which the ventricle must pump. While ventricular power is now downplayed, the importance of impedance remains as the major factor determining the aortic pressure, and thus the ventricular pressure during systole and hence total myocardial oxygen requirements.

Coupling of the left ventricle with the systemic circulation: implications to cardiac failure

Perspective	322	Power–afterload relationship	331
Relationship of ventricular pressure to volume: the 'law of the heart'	323	Pump function curves	331
Ventricular energy expenditure	323	Summary	333
Force–velocity relationships	324	Cardiac failure	334
End-systolic pressure–volume relationship	325	External heart work	341
Relationship of instantaneous pressure to instantaneous flow	328	Qualification	342

Science like life feeds on its own decay. New facts burn old rules; then newly developed concepts bind old and new together into a reconciling law.

William James, quoted by Fishman and Richards (1964)

The physiologist is at the threshold of an unprecedented scientific era. To him or her will fall the task of making sense of the bits and pieces of new knowledge that are so rapidly accumulating. The whole is more than the sum of its parts, but an intimate appreciation of the parts is needed to assemble a coherent whole. This is the challenge – and the privilege – of today's integrative physiologist.

Lenfant (1995)

Claude Lenfant, when Director of the National Heart Lung and Blood Institute, issued an inspirational challenge to those who, like the authors, are firm believers in 'integrative physiology' and who fear that attention to fine detail has detracted attention from the 'big picture'. Nowhere is this more relevant than to the issue of ventricular–vascular interaction. This issue is of supreme importance in the study of cardiac function and cardiac failure. However, the issue was virtually ignored in the report of the Task Force on Research in Heart Failure, presented by Dr Lenfant at the National Heart, Lung and Blood Institute (NHLBI) conference in 1994 (Lenfant,

1994; Lenfant, 1995) and in more recent reports of the NHLBI (Cohn *et al.*, 1997). A new updated report is expected to be published under editorship of Suzanne Oparil in early 2012. In recent issues of the *Journal of American College of Cardiology*, new approaches are foreshadowed (Nabel and Lauer, 2009) by the NHLBI, but these appear to emphasize narrow new technology related to myocardial subcellular processes. In contrast, cardiac load, described as impedance, is considered synonymous with peripheral resistance, and no account is taken of the gross changes in arterial stiffness and wave reflection that add to resistive hydraulic load in aging humans with cardiac failure (Meredith *et al.*, 2008; Mullens *et al.*, 2008; Yancy, 2008; O'Rourke and Nichols, 2009a; O'Rourke, 2010). The broad topic of ventricular–vascular interaction was not perceived as important to present problems or future research funding 15 years ago. This subject (aging, vascular stiffness, and cardiovascular function) was, however, identified as a priority area for research funding (NIH Guide 24: 24, 30 June 1995), and is emphasized by the most recent work from Framingham and Boston, Mayo Clinic, Wake Forest, Johns Hopkins, and elsewhere on cardiac failure in older people (Borlaug *et al.*, 2010; Chirinos and Segers, 2010a; Chirinos and Segers, 2010b; Derval and Jais, 2010; Maeder *et al.*, 2010; Marwick and Grimm, 2010; Paulus, 2010; Solomon *et al.*, 2010; Spragg *et al.*, 2010; Van Bommel *et al.*, 2010) (see Chapter 15).

The challenge is taken up in this chapter, which aims to collate presently available information into a unifying hypothesis. Our initiative carries a danger that we might

oversimplify or overlook important pieces of the puzzle. We hope that readers will tolerate this if they see some value to the incomplete ‘big picture’ that is presented here. If it is close to correct, it might serve as a stimulus and as a framework for others to complete, and we will have served a useful purpose.

The position we take is as follows: in description of vascular–ventricular interaction, consideration of pressure is dominant, and all-important details of arterial function can only be expressed in the frequency domain. However, interpretation can be made for the time domain, so that conventional approaches can be applied to ventricular function in the presence of varying arterial and vascular properties. Left ventricular pressure during systole is determined by ventricular volume (‘preload’) in accordance with the principles first enunciated by Frank and Starling. Left ventricular energy requirements and oxygen demand are determined by pressure in accordance with the work of Hill (Hartree and Hill, 1921; Hill, 1965), Rhode (1912), and others of the same era. Left ventricular output is related to pressure according to principles established by Sarnoff (Sarnoff *et al.*, 1958), Braunwald (Braunwald, 1997b; Braunwald, 2009), Sagawa (Sagawa *et al.*, 1988), and others in more recent times (Buckberg *et al.*, 2008; Maruyama, 2009). In cardiac failure with impaired myocardial contractility, the altered relationship between pressure and capacity of muscle to shorten results in ventricular ejection being absolutely dominated by pressure, so that the heart behaves as a ‘pressure source’. Ejection is limited by pressure which is determined by the physical properties of the vascular system. Such a property is apparent irrespective of whether relationships with pressure are expressed either instantaneously, per beat, per minute, or in the frequency domain. Such behavior of the failing heart explains efficacy of afterload reduction and gives new insight into how arterial properties and hydraulic load can be manipulated to the heart’s advantage. The same principles are relevant to prevention of heart failure and treatment of hypertension, as set out in subsequent chapters.

PERSPECTIVE

The function of the left ventricle is to pump blood, so that the tissues of the body may be perfused with nutrients according to their need. Whereas blood flow through the tissues is continuous, or nearly so, the heart’s output is intermittent. The arterial system links the left ventricle to the capillary circulation (Fig. 14.1) and thus subserves two functions – first, as a low-resistance conduit, and second as a cushion (or buffer) to flow pulsations at its input (O’Rourke, 1982a; O’Rourke, 1982b; O’Rourke, 2009a). Efficiency of the arterial system is apparent in the tiny mean pressure gradient (normally 1.0 to 3.0 mmHg) between central and peripheral arteries, and in the relatively small amount of extra energy (normally around 10

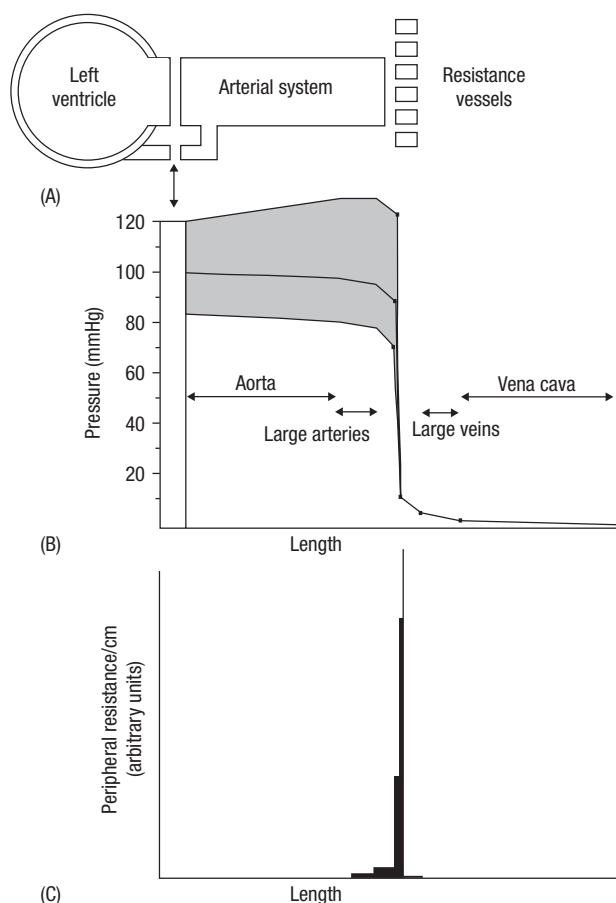


Figure 14.1 (A) Diagrammatic representation of the left ventricle, arterial system, and resistance vessels. (B) Change in mean pressure and pulsatile pressure between the central aorta and vena cava. (C) Resistance in corresponding segments of the vascular bed. Adapted from O’Rourke (1982b)

percent of total external work) dissipated in the arterial tree on account of flow pulsations (O’Rourke, 1968; Nichols *et al.*, 1977a; Kelly and Fitchett, 1992) (Chapter 13). However, both conduit and cushioning functions can be affected by disease – the former by narrowing and the latter predominantly by stiffening of arteries.

The purpose of this chapter is to provide as comprehensive and as logical an explanation as possible for interaction between heart and systemic arterial system under different conditions, not only in healthy experimental animals, but in simulated cardiovascular disease as well, and in humans when young and healthy, when old, with elevation in arterial pressure, and in the presence of cardiac failure.

There is one fundamental problem in tackling this subject which has long exasperated physiologists and bioengineers alike. In characterizing the arterial system, the most appropriate method of analysis is in the frequency domain, with pressure–flow relationships expressed as vascular impedance (O’Rourke and Taylor, 1967; O’Rourke, 1982a; O’Rourke, 1982b; Nichols *et al.*, 1985a; Milnor, 1989; Segers *et al.*, 2007b; Mitchell, 2009b; O’Rourke,

2009a; Chirinos and Segers, 2010b). The previous two chapters deal extensively with this approach, while elsewhere the effects of aging, of hypertension, of vascular disease, and of different therapies are described in terms of their effect on ascending aortic impedance. This (ascending aortic impedance) is acknowledged to be the most exact description possible of left ventricular hydraulic load. The problem is that the heart cannot be conveniently characterized in the same way (Abel, 1971; Hunter and Noordergraaf, 1976; Westerhof and Elzinga, 1978). This is due to the fact that the left ventricle is an active time-varying system, so that its properties cannot be quantified in the frequency domain using impedance. Attempts to do so have failed (Abel, 1971; Westerhof and Elzinga, 1974; Reichel and Baumann, 1979). Hence, cardiac function has been characterized in the time domain, from studies of the relationships of generated pressure and flow versus time, or through relationship of pressure versus flow or versus volume. Indeed, a variety of methods have been used to characterize cardiac function in these ways. Since all describe the same fundamental mechanical properties of the heart, all should be capable of reconciliation with each other. This is a principal goal of our chapter, but to tackle it we should trace the development of the field.

RELATIONSHIP OF VENTRICULAR PRESSURE TO VOLUME: THE 'LAW OF THE HEART'

The earliest relevant studies of cardiac function were conducted by Otto Frank in the late nineteenth century, using the slowly beating turtle heart enclosed in a cardiometer to

measure volume change, and his newly developed mechanical and optical manometers (Frank, 1895). In this non-ejecting preparation, Frank showed a relationship between pressure generated and initial volume, with pressure increasing up to a certain level and then decreasing with further increase in volume (Fig. 14.2A). He pointed out similarities with data obtained by Blix on the relationship between force and initial length of skeletal muscle (Fig. 14.2B). Ernest Starling (Patterson *et al.*, 1914) reproduced Frank's work in the mammalian heart (Fig. 14.2C) and coined the term 'law of the heart' to describe the relationship between pressure and volume (Katz, 2002; Konhilas *et al.*, 2002). Starling considered that the declining limb of the pressure–volume graph may be related to heart failure, but both he and Frank noted that in the weakened heart, the initial slope of the pressure–volume relationship is less steep, although it can be made steeper by inotropic stimulation. Subsequent work has confirmed that of Frank and Starling and has shown that the initial slope is related to contractility of the ventricle and a result of interaction between actin and myosin filaments in the sarcomere (Braunwald and Ross, 1979; Braunwald *et al.*, 1992; Braunwald, 2009). The falling limb at very high pressures is now known to be an artifact and irrelevant to physiologic and clinical problems (Braunwald *et al.*, 1992; Elzinga, 1992).

VENTRICULAR ENERGY EXPENDITURE

Starling's law of the heart was related by him to cardiac output. Around the same time, Rhode in Germany and the Nobel Laureate AV Hill and colleagues in the United

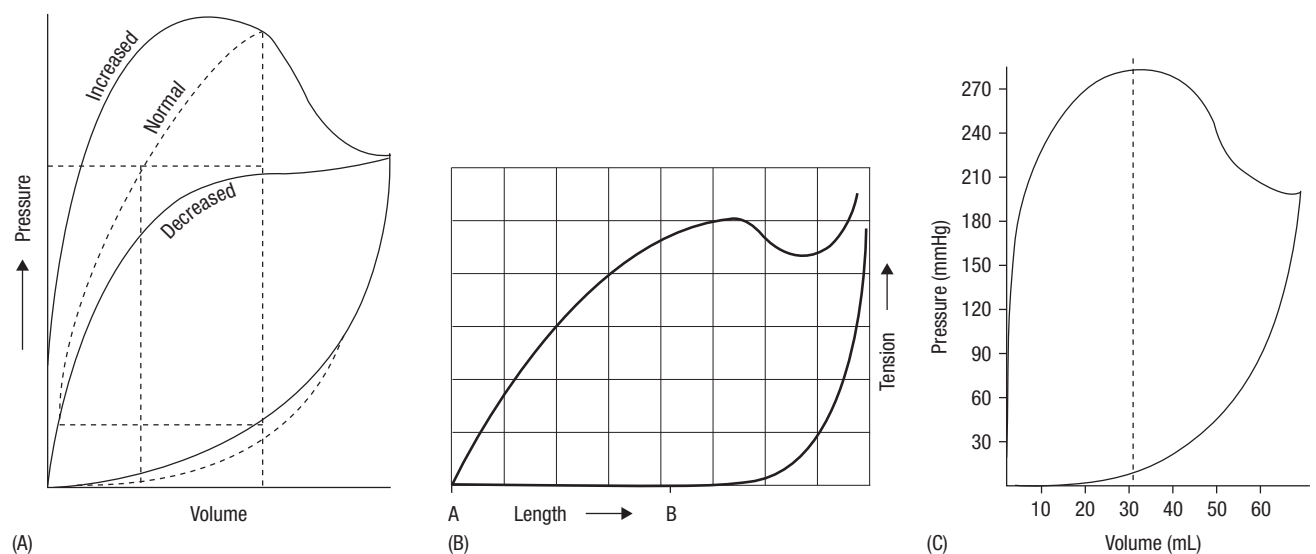


Figure 14.2 (A) Relationship between ventricular pressure (ordinate) and ventricular volume (abscissa) for normal heart (center, dashed line), and with increased (above) and decreased (below) contractility. The lowest line is the passive relationship of pressure and volume in diastole. The upper three lines describe the relationship of pressure and volume during isometric contraction. Reproduced from Frank (1895, 1959). (B) Relationship between developed tension (above) and passive tension (below) and length in skeletal muscle. Data of Blix, reproduced by Patterson *et al.* (1914). (C) Description of the 'law of the heart' by Patterson *et al.* (1914). Data were acknowledged as from Frank, with axes adjusted to conform to the dimensions of a dog's heart

Kingdom conducted studies on the energetics of contracting skeletal and cardiac muscle, establishing that heat production and energy expenditure are related to pressure generated rather than to muscle shortening or cardiac ejection (Rhode, 1912; Hartree and Hill, 1921; Fenn, 1923; Hill, 1965). This was subsequently confirmed by Sarnoff *et al.* (1958), who showed that myocardial oxygen consumption is closely related to the tension–time index (the product of mean aortic systolic pressure and time of ejection; see Fig. 13.8), and not to stroke work. This view remains current, utilizing the most modern techniques of measurement (Braunwald, 1969; Suga, 1979, Suga, 1990; Gibbs, 1995; Braunwald, 2001b; Chen *et al.*, 2001; Shapiro *et al.*, 2007; Braunwald, 2009). In referring to ventricular–vascular interaction, Hill had pointed out the importance of arterial stiffness, measuring this as pulse wave velocity and stating that ‘the amount of energy which the heart has to expend per beat, other things being equal, varies inversely with the elasticity of the arterial system’ (Bramwell and Hill, 1922b).

FORCE–VELOCITY RELATIONSHIPS

During the 1950s to 1970s, detailed studies were conducted on ventricular contraction using cardiac papillary muscles and focusing on the relationship between force and velocity of shortening (Fig. 14.3) (Abbott and Mommaerts, 1959; Sonnenblick, 1962; Braunwald, 1976). These were translated to the whole ventricle so as to

characterize contraction of the ventricle against different loads (Brutsaert and Sonnenblick, 1973; Braunwald, 1976). For the whole ventricle, as for the papillary muscle, a reciprocal relationship was established between stroke volume and measures of ventricular contraction (as peak systolic pressure, mean systolic pressure, or end-systolic pressure) (Weber *et al.*, 1974; Braunwald, 1976; Nichols *et al.*, 1977b; Noordergraaf and Melbin, 1978; Westerhof and Elzinga, 1978; Noble, 1979a; Noble, 1979b; Ford, 1980; Braunwald, 1992). Different effects of altering ventricular volume and contractility were demonstrable. It was shown that the normal ventricle is similar to normal papillary muscle, demonstrating an inverse relationship between pressure and output, showing a bell-shaped relationship between developed power (as pressure $\times$ flow) against generated pressure, with peak power generated at normal operating pressures (Braunwald and Ross, 1979; Sunagawa *et al.*, 1985a; Braunwald *et al.*, 1992; Chiu *et al.*, 1992). These findings were taken to bespeak efficient operation of the heart at normal operating pressures. The detailed studies explain findings in the intact heart and the relationship between ejection and load under different conditions (Elzinga and Westerhof, 1973; Elzinga and Westerhof, 1974; Abbott and Gordon, 1975; Elzinga and Westerhof, 1976; Brutsaert and Paulus, 1977; Elzinga and Westerhof, 1978; Paulus *et al.*, 1979; Noble, 1979a; Noble, 1979b; Elzinga and Westerhof, 1980a; Elzinga and Westerhof, 1980b; Kelly *et al.*, 1992a; Chen *et al.*, 2001; Shapiro *et al.*, 2007; Borlaug *et al.*, 2009; Braunwald, 2009; Najjar, 2009).

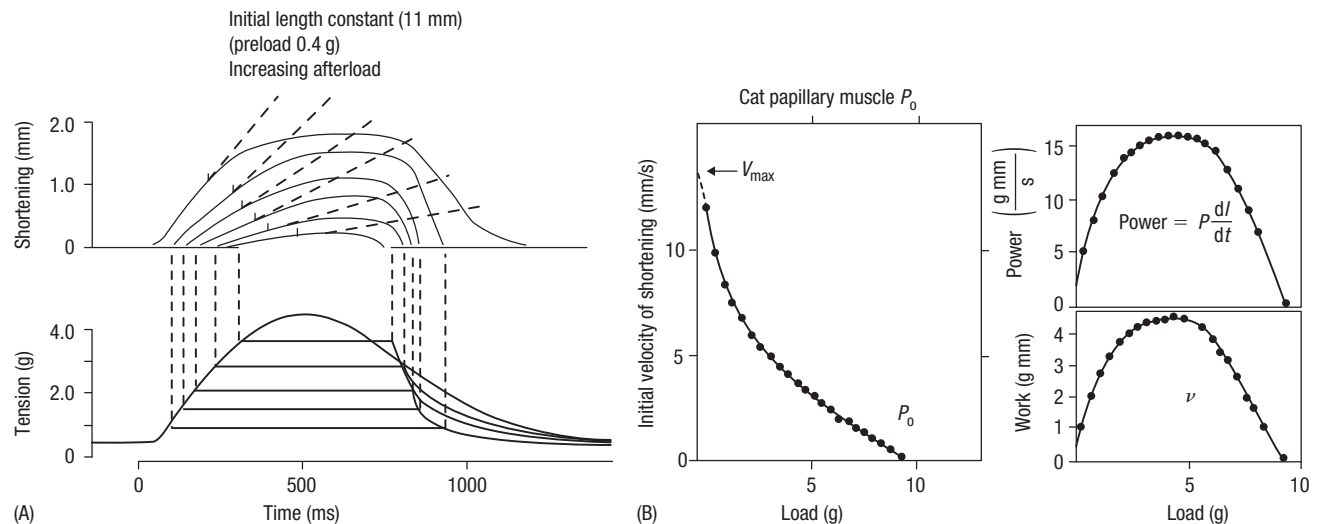


Figure 14.3 Use of afterloaded isotonic contractions to obtain force–velocity relations in cat papillary muscle. (A) Effects of increasing afterloads on the course of tension development and subsequent shortening. Several superimposed contractions are displayed. The muscle develops a force equal to the afterload and thereafter shortens. As the afterload is increased, the velocity of shortening (dashed lines) and the extent of shortening decline. (B) Velocity of shortening plotted as a function of load: the force–velocity relation. As the load is increased, the velocity of shortening decreases. When the load is so high that no external shortening is recorded, velocity is zero and the force is equivalent to the isometric contraction (P_0). When the curve is extrapolated back to zero load, V_{max} is obtained. Also shown in (B) are power (top) and work (bottom) curves as a function of increasing afterloads. Both power and work are zero when the load is zero or with isometric contractions, and both curves peak at an intermediate load. Reproduced from Braunwald (1976)

END-SYSTOLIC PRESSURE–VOLUME RELATIONSHIP

The classic studies of Starling, Frank, and others were taken one step further by Sagawa and colleagues in Baltimore, and by Weber and colleagues in Birmingham during the early 1970s (Suga *et al.*, 1973; Weber *et al.*, 1974; Sagawa *et al.*, 1988; Suga, 1990), by measuring pressure–volume relationships in a heart preparation that was capable of ejecting (changing volume) while being adequately perfused through the coronary circulation. These studies focused on the rising limb of the pressure–volume relationship, showing that this is nearly linear and showing a unique relationship between ventricular pressure and volume that was independent of absolute volume ejected (Figs 14.4 and 14.5). These studies established the concept of the ventricle as a time-varying elastance, where properties could be expressed in terms of the slope of the pressure–volume relationship and of the intercept of this line on the volume axis. The slope was shown to depend on ventricular contractility. Special attention was given to the relationship of pressure and volume at end systole.

These studies on end-systolic pressure–volume relationships have dominated the approach to measurement of cardiac performance and ventricular energetics right through the 1990s until the present. The slope of the end-systolic pressure–volume relationship has been taken as the measure of myocardial contractility, and its extrapolated intercept with the volume axis has been used to measure ventricular energy release and oxygen demand (Suga, 1979; Suga, 1990; Gibbs, 1995). A further development has been determination of ventricular load as arterial elastance, with this taken as the slope of the line joining the end-diastolic pressure–volume point and the end-systolic pressure–volume point (Fig. 14.6) (Sunagawa *et al.*, 1983; Sunagawa *et al.*, 1985a; Burkhoff and Sagawa, 1986; Kelly and Kass, 1992a; Kelly and Kass, 1992b; Starling, 1993; Chen *et al.*, 2001; Shapiro *et al.*, 2007; Borlaug *et al.*, 2009).

A healthy degree of skepticism has crept into assessment of all this work and of the conclusions which have arisen therefrom (Sasayama and Asanoi, 1991; Kass and Kelly, 1992; Burkhoff *et al.*, 1993; Noda *et al.*, 1993; Gibbs, 1995). These are set out below but should not detract from the fundamental value of this work, which gives another

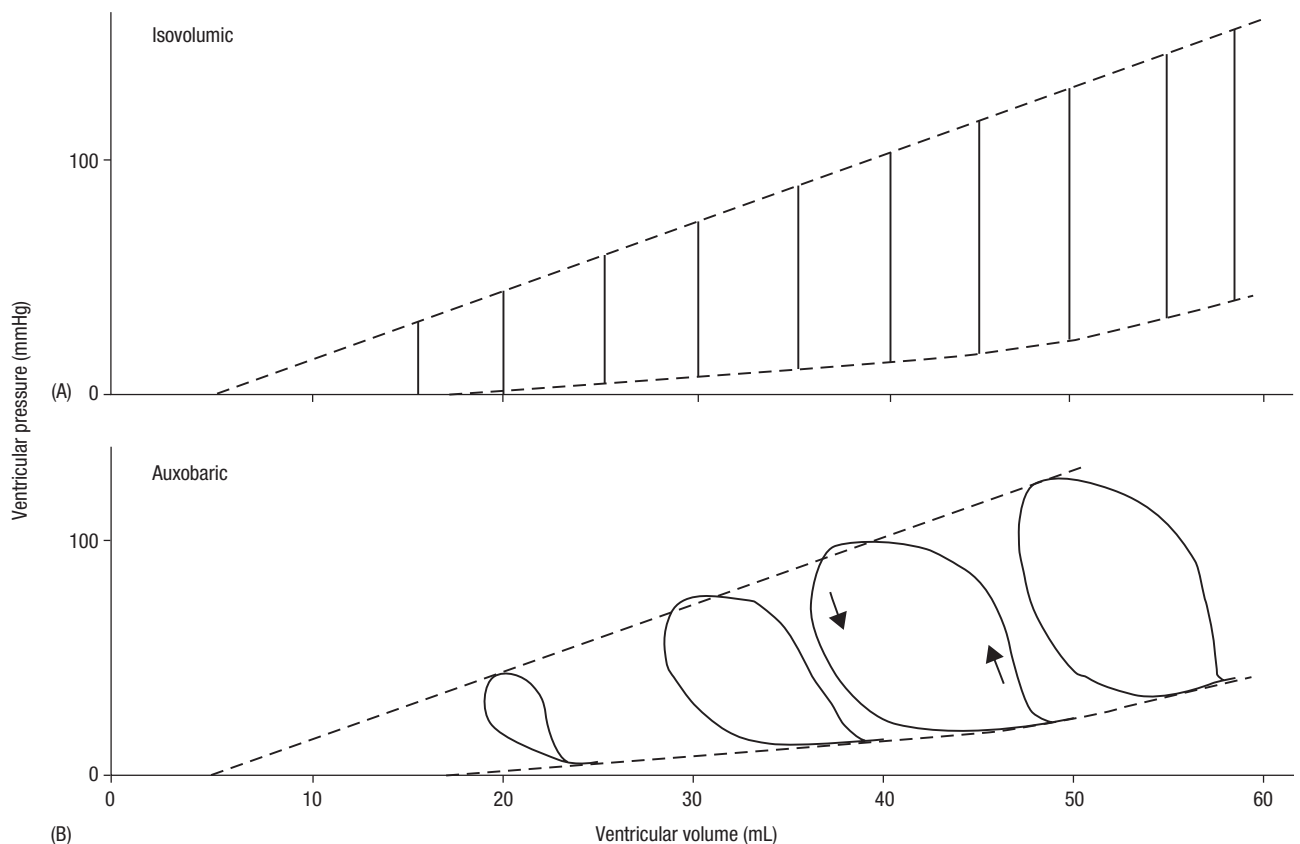


Figure 14.4 (A) Tracings of left ventricular pressure–volume relations during isovolumetric contractions from different end-diastolic volumes; with no change in systolic or diastolic volume, the loops are completely closed. The dotted line at top traces the peak of the relations, i.e. the end-systolic pressure–volume relationship. (B) Tracings of left ventricular pressure–volume loops in the same heart during isotonic contraction, i.e. as the heart was ejecting blood. The dotted line at the top is transcribed from the upper figure. For explanation, see text. Reproduced from Suga and Sagawa (1974)

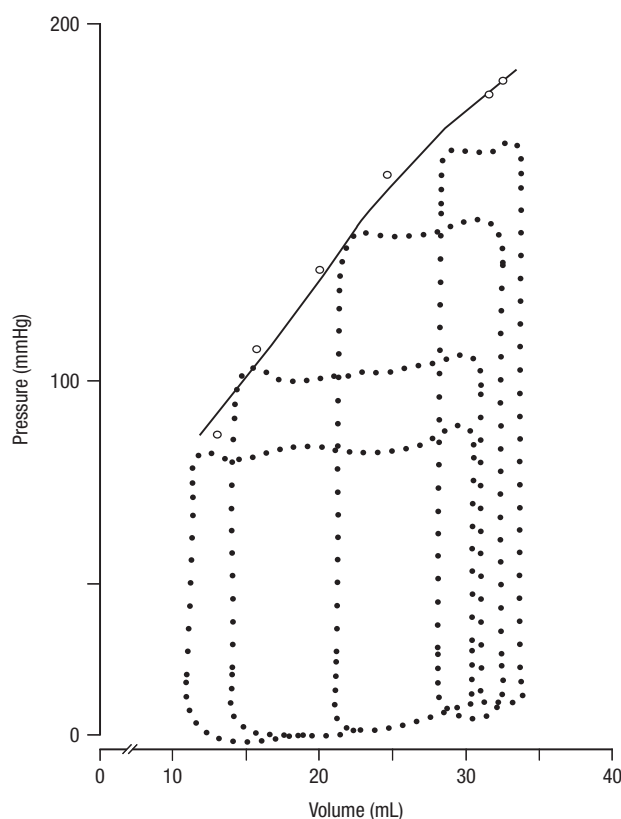


Figure 14.5 Relationship between ventricular pressure and ventricular volume in an isolated heart preparation. For a given end-diastolic volume, volume ejected at different diastolic pressures is determined by the Frank–Starling curve, ejection ceasing when pressure reaches the isovolumic pressure normally generated at that given volume. Reproduced from Weber *et al.* (1974)

dimension (change in volume during contraction) to the original isometric Frank and Starling view on the ‘law of the heart’ and confirms the value of the rising limb of the pressure–volume relationship to describe ventricular contractility and changes thereof. More importantly, this demonstrates the difference between change in volume ejection achieved with change in pressure load for different degrees of ventricular contractility (Fig. 14.7). The inverse relationship between volume output (or shortening) and pressure has been described (Fig. 14.3), but not the exquisite quantitative change in this relationship in the presence of impaired myocardial contractility. This is evident in the relationships between pressure and flow (Miyashita *et al.*, 1993) and power and pressure (Chiu *et al.*, 1992), and in pump function graphs (Elzinga and Westerhof, 1973; Elzinga and Westerhof, 1974; Elzinga and Westerhof, 1976; Elzinga and Westerhof, 1978; Elzinga and Westerhof, 1979; Elzinga and Westerhof, 1980a; Elzinga and Westerhof, 1980b) as described later, but it is well illustrated by the pressure–volume relationships. The capacity of the ventricle to expel more blood for a given reduction in ‘afterload’ when the ventricle is failing as compared to when it is not (Fig. 14.7) is explicable on the basis of the slope of the end-systolic pressure–volume relationship (ESPVR) and on the concept of changing pressure–volume loops during contraction. Analogous behavior, expressed in a different way, is discussed below. Effects on the ejection of the right ventricle are described in Chapter 16. In pulmonary hypertension, pulmonary pressure can rise to (relative) much higher pressure than in systemic hypertension, and is associated with a distinctive pulmonary flow pattern, with markedly reduced ejection from the right ventricle (Fig. 16.13).

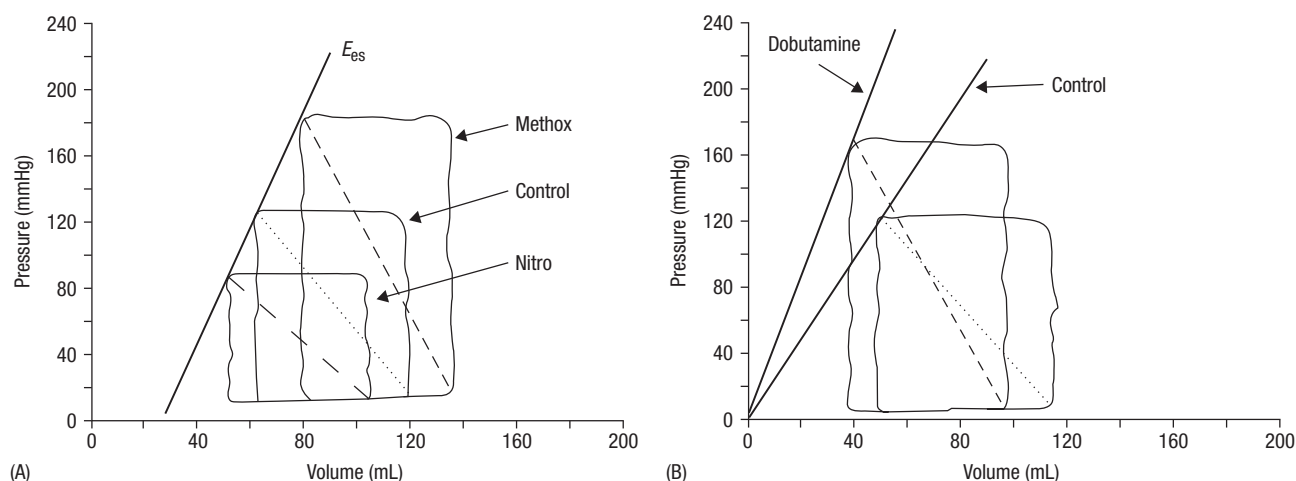


Figure 14.6 (A) Schematic representation of the effects of altered left ventricular (LV) load on effective arterial elastance and the relationship with LV chamber elastance. (B) Effects of enhanced inotropy on the relationship between LV chamber elastance and effective arterial elastance. E_{es} , LV chamber elastance; Methox, methoxamine; Nitro, nitroprusside. Reproduced from Starling (1993)

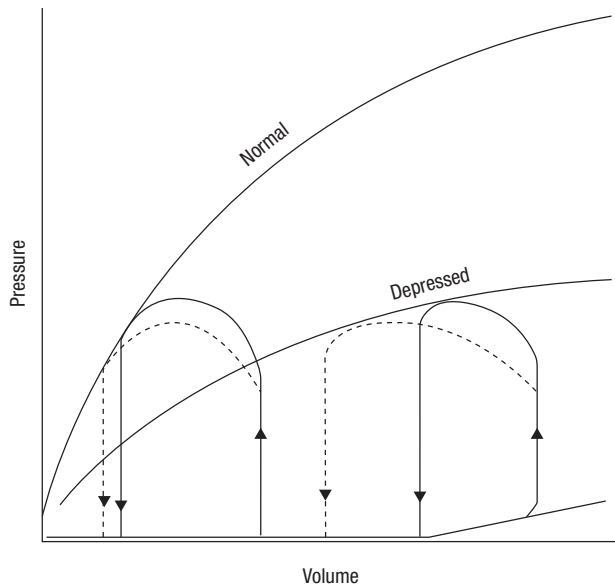


Figure 14.7 Diagrammatic ventricular pressure–volume loops with normal and depressed ventricular contractility, illustrating the greater effect of arterial pressure reduction (dashed line) on volume ejection in the presence of impaired contractility. Reproduced from O'Rourke (1982a)

The re-evaluation of enthusiasm for pressure–volume loops relates principally to technical issues and the precision of measuring ventricular volume and the end-systolic pressure–volume point accurately.

The earliest studies of pressure–volume loops were conducted in a perfused isolated heart, whose afterload could be controlled precisely, and in a similar way to afterload of a loaded papillary muscle preparation. The beginning and end of ejection were readily measurable, and pressure was kept at the same level during systole, so that ejection was 'isotonic'. Even with this type of preparation, however, doubts began to emerge about whether the ESPVR was linear over a wide range of pressure (and volume) and was truly independent of other variables (Fig. 14.8). The big problems emerged when the concept was taken to the intact animal and (more so) to human studies (Sasayama and Asanoi, 1991; Kass and Kelly, 1992; Burkhoff *et al.*, 1993; Noda *et al.*, 1993; Gibbs, 1995; Jegger *et al.*, 2006; Jegger *et al.*, 2007; Osranek *et al.*, 2008; Delnoy *et al.*, 2009; Michaels *et al.*, 2009). In all these studies, the queries raised in carefully controlled isolated heart preparations were even more apparent, and other issues emerged. Ventricular volume is notoriously difficult to measure accurately. That this is not accurately measurable is apparent in virtually all published studies (which are assumed to show the best data); these show variations, often extreme, during isometric ventricular contraction and isometric ventricular relaxation, when aortic and mitral valves have shut, and when ventricular volume must be constant (Figs 14.4 and Fig. 14.9) (Suga and Sagawa, 1974; Kelly *et al.*, 1992a; Kelly *et al.*, 1992b; Burkhoff *et al.*, 1993; Schreuder *et al.*, 1995).

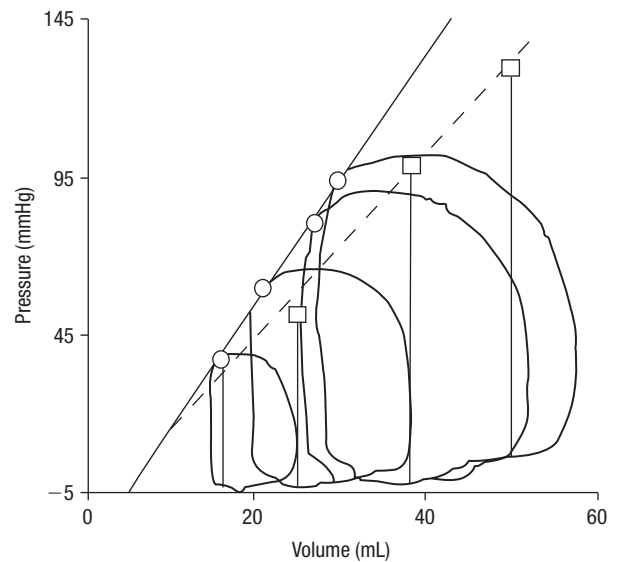


Figure 14.8 The effect of physiologic afterload on the end-systolic pressure–volume relation in isolated canine heart. Switching from isovolumic contraction (□--□) to ejection at an effective ejection fraction EF_{eff} of 58 percent ($EF_{eff} = SV / (V_{ed} - V_0)$), (○--○) produced an increase in the slope of the ESPVR. Thus ejection favorably influenced contractile performance. Reproduced from Kass and Kelly (1992)

However, there is a problem with ventricular pressure measurement as well. Most published data for intact animals and humans show pressure fluctuations far smaller than measurable in the aorta for pressure between aortic valve opening and closure, and to the peak of pressure (Fig. 14.6) (Kelly *et al.*, 1992a; Kelly *et al.*, 1992b; Starling, 1993). One expects that the pressure variation during ejection would follow the same course as seen for the left ventricular and ascending aortic pressure wave. However, data displayed often show a flat ventricular pressure during change in ventricular volume or, even more surprisingly, a fall in ventricular pressure from aortic valve opening to aortic valve closure. When volume data are suspect, and pressure data also suspect, how accurately can one measure the ESPV point for a given loop? In many published figures, this appears as the peak of pressure rather than the point that should correspond to end systole (Figs 14.6, 14.8, and 14.9). Add to these problems with measurement of pressure and volume, the need to repeat measurements at different filling pressures so as to establish the linearity of the ESPVR, and one has a very serious technical problem indeed. All these issues reflect on the data that have been obtained from intact animals and in man, and on the precision claimed by many. The ESPVR appears to be fraught with technical hazard and is possibly far less precise than information that can come through investigation of other relationships that pertain to ventricular performance.

Recently conducted studies of ESPVR in human adults at cardiac catheterization have shown evidence (in the majority) of a substantial increase in pressure during the

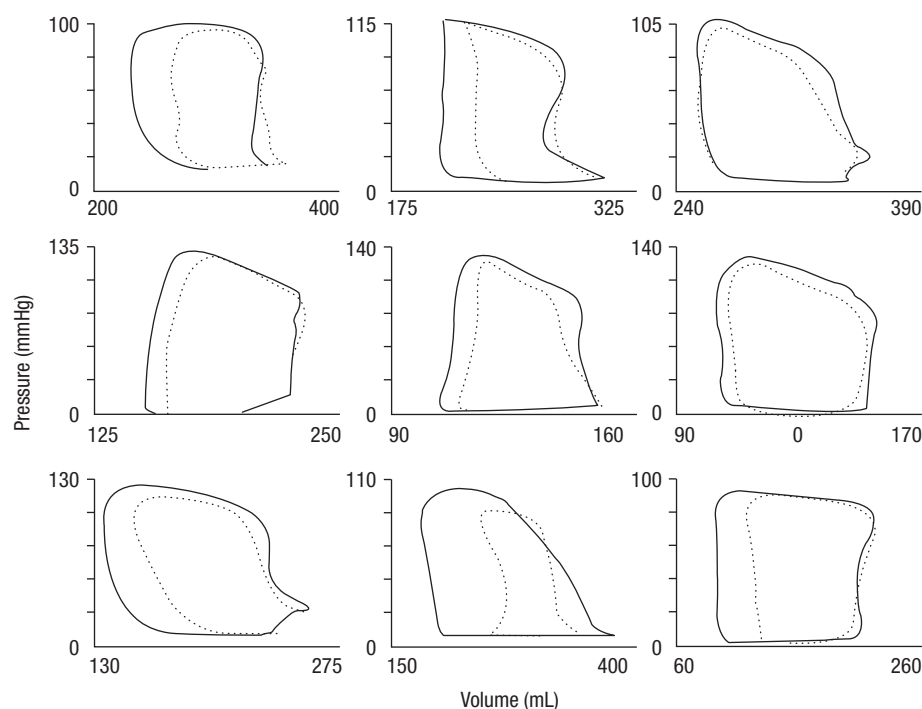


Figure 14.9 Left ventricular pressure–volume loops of nine patients after cardiomyoplasty at the best setting. Dotted lines represent the unassisted beats; solid lines represent the latissimus dorsi assisted beats. Reproduced from Schreuder *et al.* (1995)

latter part of systole (Fig. 14.8) (Kass and Kelly, 1992). This is the pressure pattern one would predict in the presence of stiffened arteries and early return of wave reflection. However, in these, the rise in pressure between the beginning of ejection and the systolic peak was often lower than one might predict (averaging 35 mmHg in the data of Kelly *et al.* (1992a)), and the end-systolic pressure–volume point still appears close to the top of the loop (near peak systolic pressure) rather than on its descending limb. These issues can be reconciled at least in part by considering mid-wall stress and strain during systole as the ventricular volume decreases and wall thickness increases (de Simone and Devereux, 2008; Chirinos *et al.*, 2009a; Najjar, 2009). In some studies, the ordinate is measured as mid-wall stress rather than pressure (Chirinos and Segers, 2010a). When this is done, one can explain stress/strain loops more readily, since the stress will fall as the ventricle ejects blood and becomes smaller while its wall thickness increases.

Three other approaches have been used to describe ventricular–vascular coupling, and all are useful in describing myocardial failure and the logical hemodynamic approach to this. Such approaches have been championed by Yaginuma and colleagues in Japan, by Carroll and colleagues initially in Chicago, and by Westerhof and colleagues in Amsterdam. More results will, no doubt, be forthcoming since simultaneous high fidelity left ventricular (LV) pressure and volume can now be measured continuously in humans (Delnoy *et al.*, 2009; Michaels *et al.*, 2009) using a conductance catheter that

has been tested in animals (Jegger *et al.*, 2006; Jegger *et al.*, 2007).

RELATIONSHIP OF INSTANTANEOUS PRESSURE TO INSTANTANEOUS FLOW

Yaginuma drew attention to the time course of pressure and flow in the ascending aorta during ventricular systole (Yaginuma *et al.*, 1985; Yaginuma *et al.*, 1986). He confirmed that aortic and ventricular pressure are essentially identical during systole except for the small pressure gradient attributable to acceleration of blood during the early part of systole (see Chapter 7). Yaginuma's early work was directed at the measurement of ascending aortic impedance from pressure–flow relationships in the frequency domain (O'Rourke and Yaginuma, 1984; Yaginuma *et al.*, 1986). However, he became intrigued with the boost to pressure in late systole that had first been stressed by Murgo *et al.* (1980a) and is attributable to the early return of wave reflection from peripheral sites in adult humans and hypertensive subjects (O'Rourke, 1982a; O'Rourke, 1982b; Nichols *et al.*, 1993).

Yaginuma *et al.* (1985) referred to this as augmentation, stressed how it added to left ventricular load and to oxygen requirement, and noted that in older individuals and in hypertensive individuals its magnitude could exceed 50 mmHg and constitute almost half of aortic pulse pressure (Fig. 20.12). Further work from this group showed how aortic augmentation could be assessed

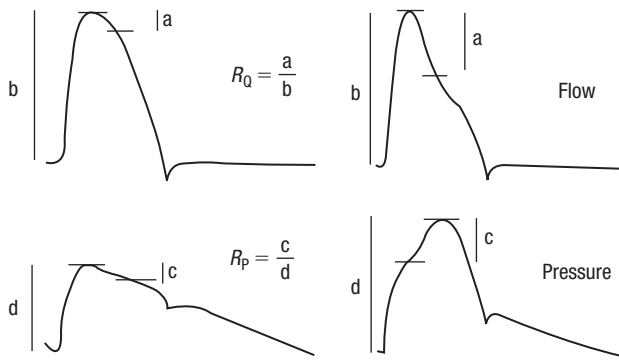


Figure 14.10 Schematic representation of ascending aortic flow (above) and pressure (below) in a young human adult (left) and an older human subject (right). The effects of wave reflection on flow are indicated by the decrease in flow at a point one-third in time from peak flow (a) to the incisura and expressed in relation to peak flow (b) as $R_Q = a/b$. The effects of wave reflection on pressure are indicated by the boost to late systolic pressure peak after the initial shoulder (c) relative to pulse pressure (d) as $R_P = c/d$. This is defined as pressure augmentation. R_Q and R_P are indices of the effects of wave reflections on flow and pressure waves, respectively. Modified from Miyashita *et al.* (1994)

non-invasively from the carotid pressure wave recorded with high-fidelity instrumentation (Fujii, 1989). This work has been extended by Takazawa, Miyashita, and others in Japan and by our own groups in Sydney, Gainesville, and Athens. Takazawa and colleagues related systolic pressure augmentation in humans to impaired diastolic relaxation and to heart failure with diastolic dysfunction (Yabe *et al.*, 1991). This was later confirmed by Weber *et al.* (2006) and Abhayaratna *et al.* (2008). Sugishita and colleagues from Tsukuba have related the relative increase in systolic pressure and decrease in diastolic aortic pressure in models of human arteriosclerosis to the development of myocardial ischemia (Kelly *et al.*, 1992a; Watanabe *et al.*, 1992; Watanabe *et al.*, 1993; Ohtsuka *et al.*, 1994; Kass *et al.*, 1996). Work from Gainesville (Nichols *et al.*, 1992; Nichols *et al.*, 1993; Nichols, 1994; Nichols and Edwards, 2001) and Tokyo (Sakai *et al.*, 1992; Miyashita *et al.*, 1993) expanded Yaginuma's observation that extreme pressure augmentation in late systole is accompanied by deceleration of flow during this period. This feature had been overlooked before but is predictable on the basis of the known fundamental inverse relationship between pressure and ventricular muscle shortening referred to earlier (Fig. 14.3). Miyashita coined an index for the aortic flow wave, relating flow at a point one-third between the peak and nadir to peak flow (Fig. 14.10). He showed that this ratio has an inverse relationship to pressure wave augmentation (Fig. 14.11). Again, this finding is predictable on the basis of known myocardial behavior. Nichols and colleagues and Miyashita, Yaginuma and colleagues describe the change in pressure wave contour and flow wave contour as resulting from the early return of wave reflection from the

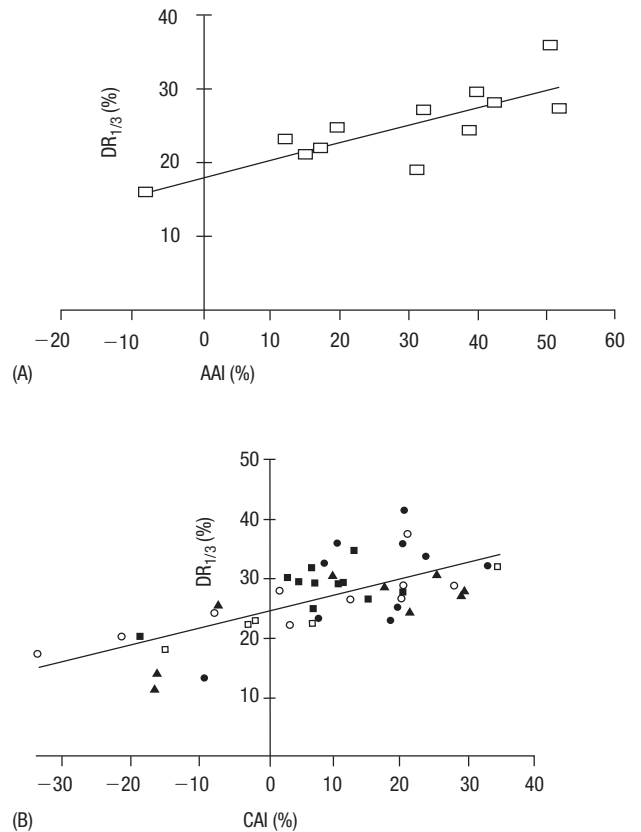


Figure 14.11 (A) Relationship between an index ($DR_{1/3}$) of wave reflection on the ascending aortic flow wave (ordinate), plotted against an index (AAI) of wave reflection on the ascending aortic pressure wave ($DR_{1/3}$ percent corresponds to R_Q in Fig. 14.10). Aortic augmentation index (AAI) in percent corresponds to R_P in Fig. 14.10. Data recorded at cardiac catheterization. (B) Relationship between similar indices determined non-invasively from Doppler flow in the ascending aorta and the pressure wave recorded from the carotid artery (CAI). Reproduced from Miyashita *et al.* (1994)

periphery. Their findings were that such early wave reflection may either add to pressure and/or subtract from flow (see Chapter 10) or cause a combination of the two (Miyashita *et al.*, 1993). We have taken this one step further by applying Westerhof's concept of wave reflection as adding to pressure and subtracting from flow (Nichols *et al.*, 1992; Nichols *et al.*, 1993; Nichols, 1994; O'Rourke, 1994; Westerhof and O'Rourke, 1995; Nichols and Edwards, 2001; O'Rourke and Hashimoto, 2007; O'Rourke, 2009a; O'Rourke, 2009b). Nichols' and Miyashita's findings are completely logical and predictable on both the basis of muscle mechanics and physical principles of wave reflection. They also help explain pressure and flow contour in the pulmonary artery in pulmonary hypertension and right heart failure (Fig. 16.14).

Miyashita *et al.* showed the effects of reducing wave reflection on the contour of the ascending aortic pressure and flow waves. Augmentation of pressure was reduced, but so was the deceleration to flow restored (Fig. 14.12).

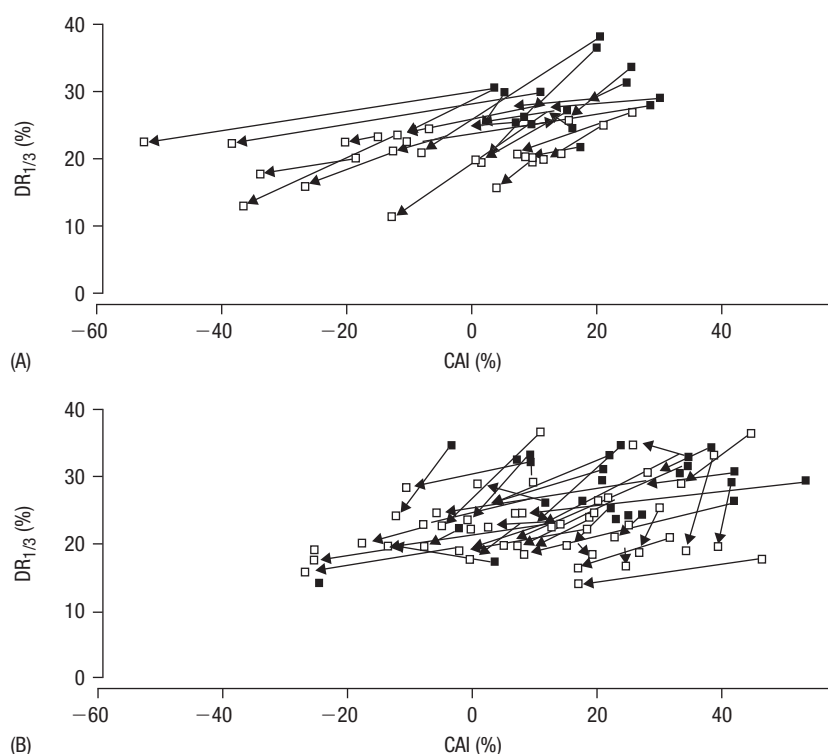


Figure 14.12 Effects of nitroglycerin (A) and nifedipine (B) on indices of wave reflection determined non-invasively as in Fig. 14.11. (■) Before drug; (□) after drug. Reproduced from Miyashita *et al.* (1994)

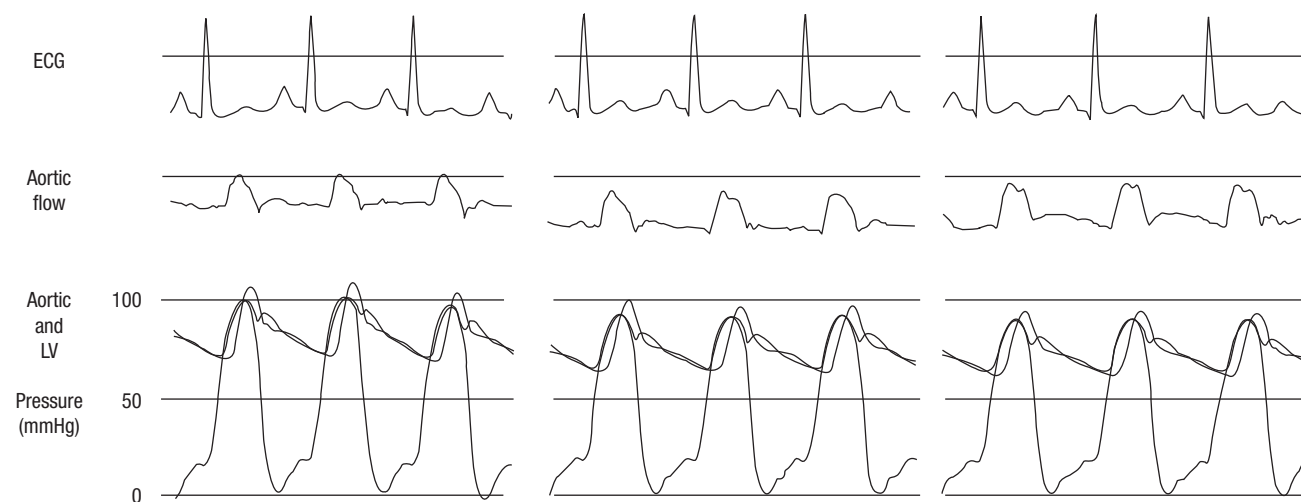


Figure 14.13 Typical example of ascending aortic pressure and flow data acquired at cardiac catheterization from a patient with cardiac failure before and during nitroprusside infusion. The tracings in each panel are, from top to bottom, electrocardiogram (ECG), aortic blood flow velocity, and ascending, descending, and left ventricular (LV) pressure. Left: The control state. Middle and right: During nitroprusside infusion at 0.5 and 0.75 mg/kg/min, respectively. Flow from the heart is increased substantially with trivial decrease in LV systolic pressure. Reproduced from Chiu *et al.* (1992)

One might expect that this phenomenon would be most marked in patients with heart failure, where further augmentation to pressure is impossible and reduction to flow is the major predicted effect of wave reflection. Such an increase in flow is apparent in data presented for nitroprusside by Yin *et al.* (1983b) and by Chiu *et al.* (1992) (Fig. 14.13) and also by Sakai *et al.* (1993) for nitroglycerin.

This work by Japanese workers shows application of the inverse pressure–flow characteristics of cardiac performance but illustrates that they are apparent as an instantaneous relationship and that reduction in pressure through systole causes a simultaneous increase in flow. Surprisingly, this approach has not been taken up by cardiologists in the Western world despite widespread availability of echocardiography. This may change with

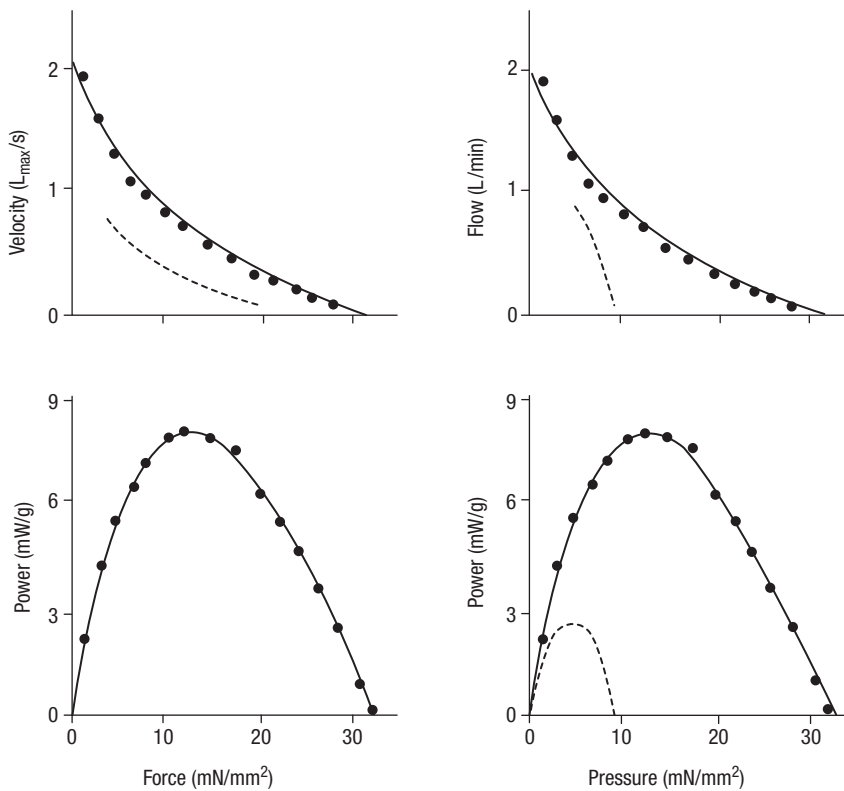


Figure 14.14 Left: Force–velocity curve as obtained in a papillary muscle preparation (top) together with power–force relationships derived therefrom (bottom). Reproduced from Chiu *et al.* (1992). Right: Flow–pressure relationships predicted for a normal heart and a heart with impaired myocardial contractility (top), together with power–pressure relationships derived from these data (bottom). In the presence of heart failure, the curve is shifted to the left (broken line)

introduction of magnetic resonance imaging which provides more accurate waveforms than echocardiography (Redheuil *et al.*, 2010).

POWER–AFTERLOAD RELATIONSHIP

Reference has been made to the early work of Braunwald and colleagues who drew attention to force–velocity relationship and to the relationship between ventricular (after)load and power in isolated papillary muscle. It was shown that there is a bell-shaped curve describing power and load, and that the papillary muscle normally operates about an applied load where power generation is maximal. This work has been applied to the intact heart, and the same relation has been shown to hold, with maximal power generated at near-normal ventricular dimensions and aortic pressure (Fig. 14.14). With the induction of cardiac failure, the curve is shifted to the left and a mismatch develops (Myhre *et al.*, 1986). Chiu *et al.* (1992) have applied this concept to patients with cardiac failure and dilated cardiomyopathy. They have confirmed that the ventricle operates about the right-hand end of the bell-shaped curve, such that power falls as pressure increases, but increases when pressure decreases. Since power is the product of pressure and flow, these findings represent substantial and excessive decrease in flow with further increase in pressure, but substantial increase in flow with decrease in pressure (Fig. 14.15, page 332). These data

from humans are completely consistent with observations in experimental animals with cardiac failure and with the human data of Yaginuma, Miyashita, Takazawa, and others described above. They imply that the inverse relationship between pressure and flow described repeatedly by others is accurately depicted as the strength of myocardial contractility, either as measured instantaneously (by Miyashita, Yaginuma, Sakai, and Yin) or on a beat-to-beat basis (as described by Chiu *et al.*, 1992).

PUMP FUNCTION CURVES

The key part of this puzzle is provided by the work of Elzinga and Westerhof from Amsterdam (Elzinga and Westerhof, 1973; Elzinga and Westerhof, 1974; Elzinga and Westerhof, 1976; Elzinga and Westerhof, 1978; Elzinga and Westerhof, 1980a; Elzinga and Westerhof, 1980b) who describe ventricular function in terms of pump function curves, relating mean ventricular pressure to cardiac output. Their work in experimental animals was preceded by work in mechanical pumps which established that this is an appropriate way to characterize the behavior of any pump – mechanical or biological – and its innate strength and response to load. These workers showed that, over a wide range of operation, the left ventricle exhibits an ellipsoid relationship between these two parameters (Fig. 14.16, page 333). At high developed pressures, flow is low and the ventricle exhibits a relatively large change in flow

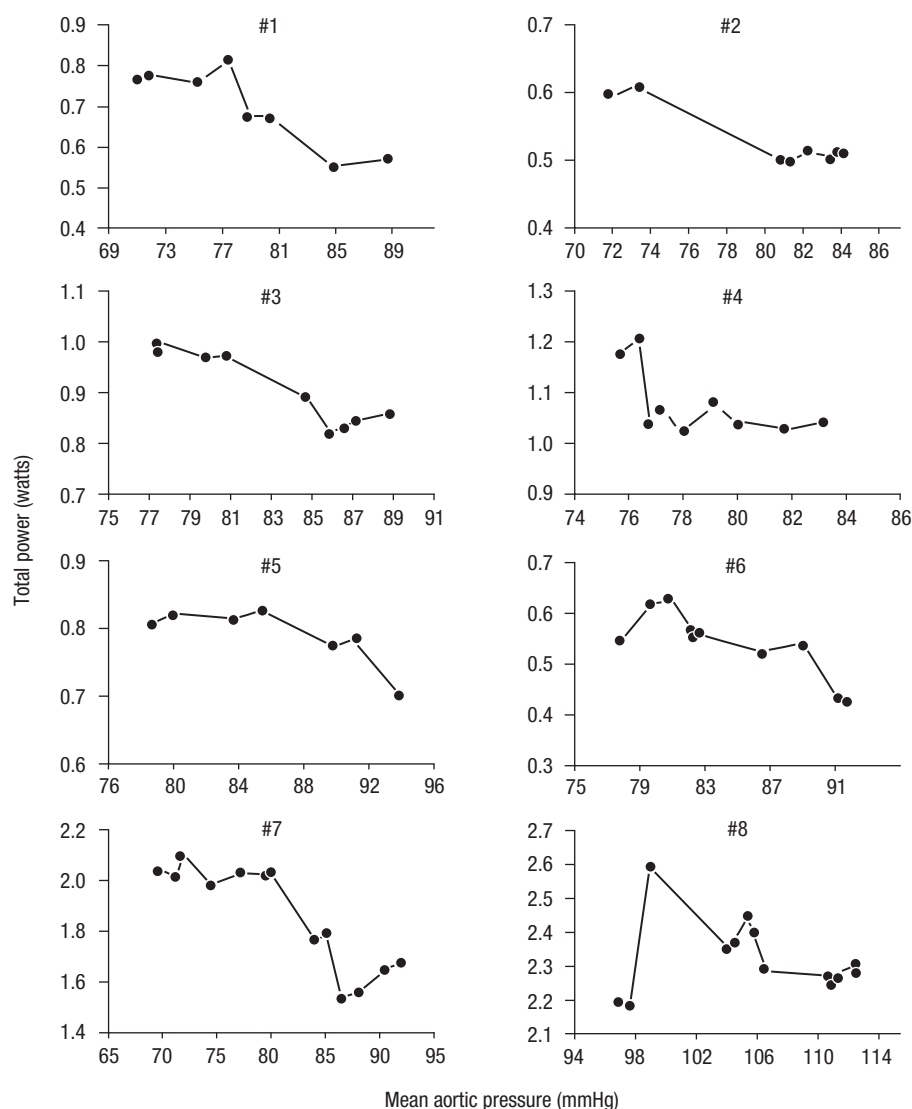


Figure 14.15 Power–afterload relations in eight patients with cardiomyopathy and severe heart failure. In all the plots, the control state is the point on the right-hand end of the graph with the highest blood pressure. As pressure is decreased with nitroprusside, power increases and then plateaus. Since power is pressure $\times$ flow, pressure reduction is associated with an increase in flow.

Reproduced from Chiu *et al.* (1992)

to a small change in pressure. Here, the graph approaches the pressure axis at an angle of 90° . At low pressures, flow is high and small changes in flow only are achieved with large changes in pressure. Here, the graph approaches the flow axis at an angle of 90° . The data indicate that when the heart is beating against a high load (pressure), it acts as a ‘pressure source’ with pressure limited but with large increases in flow achieved for small decreases in pressure. When the heart is beating against low pressure, it behaves as a flow source, with flow already high, and small changes in flow are achieved with large changes in pressure.

Elzinga and Westerhof went on to study the effects of altered myocardial contractility on pump function graphs and showed that when contractility is impaired, the graph

is lower (Fig. 14.17, page 333), so that the ellipsoid axis meets the pressure axis at a lower value, whereas when contractility is increased, the ellipsoidal curve meets the pressure axis at a higher level. The concept that arises from this is that the slope of the curve relating pressure to pump output is dependent on ventricular contractility. When contractility is impaired (as in heart failure), the heart pumps as though it were a ‘pressure source’, with flow exquisitely sensitive to pressure – falling off substantially if pressure is increased, but increasing dramatically when pressure is decreased (Fig. 14.18, page 334) (Ross *et al.*, 1966; O’Rourke, 1994; Westerhof and O’Rourke, 1995; Denardo *et al.*, 2010). This is the most fundamental principle in treatment of cardiac failure

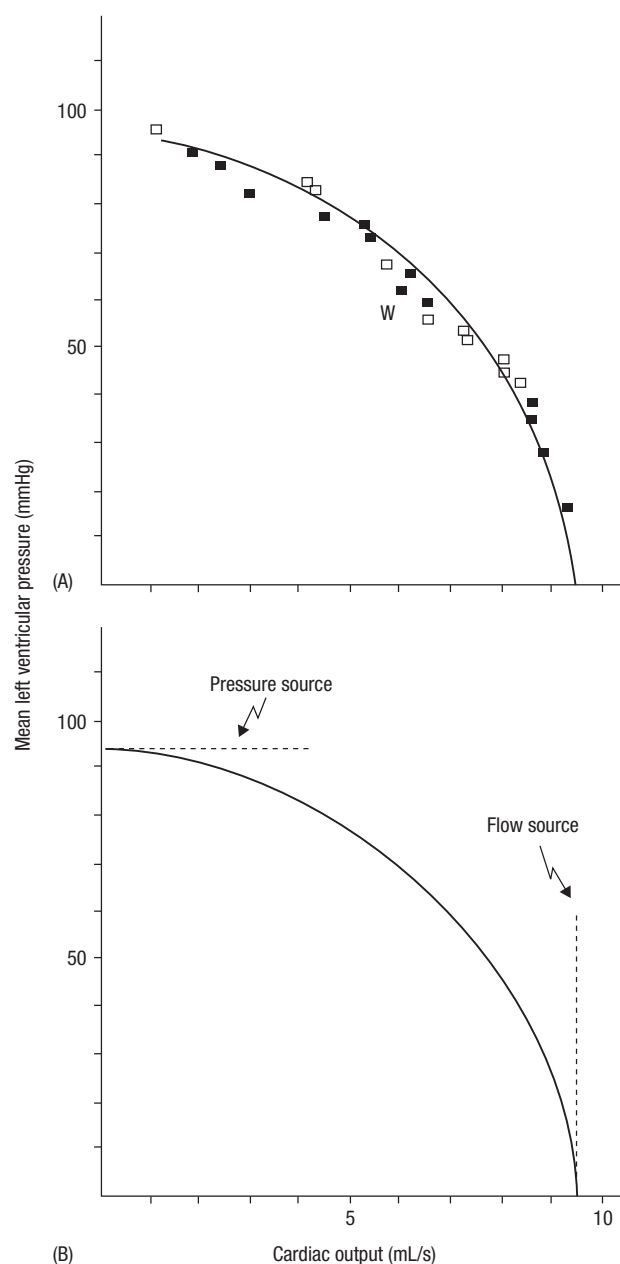


Figure 14.16 The pump function graph. (A) The pump function graph is the inverse relation between mean left ventricular pressure and cardiac output. This pump function graph was determined in the isolated cat heart. Adapted from Elzinga and Westerhof (1978). W is the working point where in baseline conditions pressure and flow are found. (B) The heart approaches a pressure source at low output and high pressure, and approaches a flow source at high output and low pressure

(De Keulenaer and Brutsaert, 2009; Jessup *et al.*, 2009). Control of LV and aortic systolic pressure by drugs or by counterpulsation can lead to recovery of myocardium and removal of a potential recipient from the cardiac transplantation list (Desai and Bhatt, 2009; Wilson *et al.*, 2009).

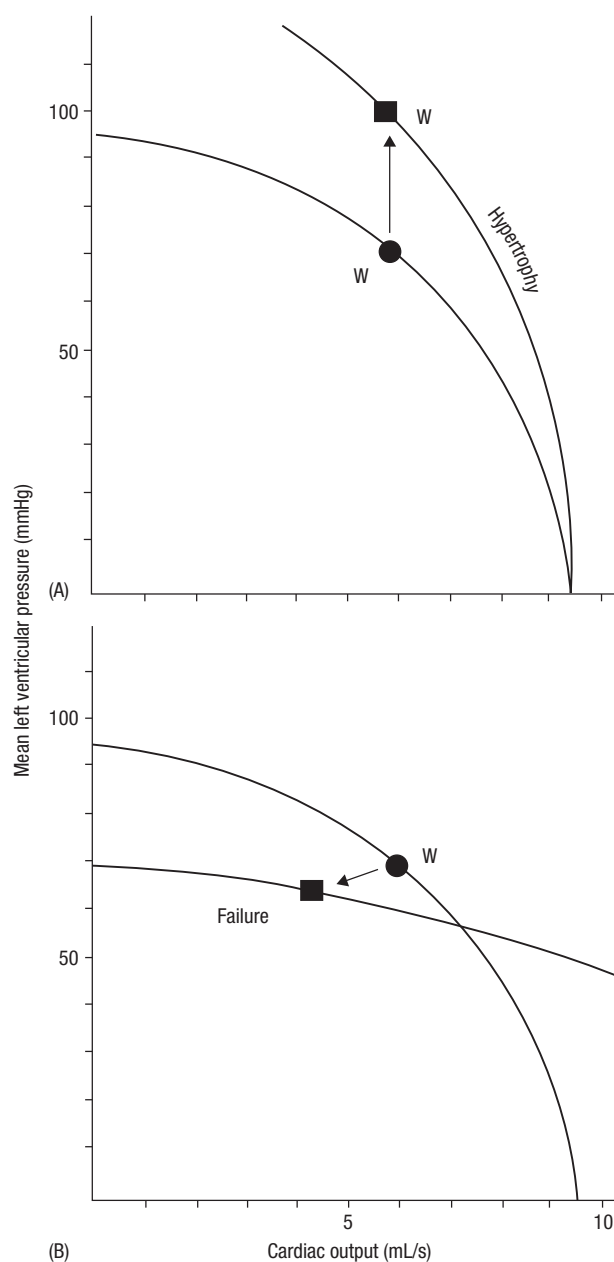


Figure 14.17 (A) The pump function graph in control conditions and during compensated hypertension (concentric hypertrophy). The working point (the interaction of heart and arterial load) is indicated with the filled circle and square, respectively. (B) The pump function graph in control conditions and failure

SUMMARY

The data presented here describe different ways of describing cardiac contraction and of characterizing mechanical properties of the heart. General principles are:

1. The capacity to generate pressure is dependent on initial volume at the beginning of contraction. Differences in contractility are manifest as



Figure 14.18 Change in left ventricular ejection caused by a stepwise increase in aortic (and thus left ventricular) pressure, with constant contractility and end-diastolic length in a dog with cardiac failure. Reproduced from O'Rourke (1982b), the diagram was redrawn from data published by Ross *et al.* (1966)

differences in the slope of the pressure–volume relationship at the end of contraction (for isometric activity – Frank–Starling) or at end-systole (for the ejecting heart – Suga and Sagawa).

2. Capacity to generate flow is dependent on the pressure against which ejection occurs. Little distinction can be drawn between peak, mean, and end-systolic ventricular pressure, but instantaneous mid-wall tension may be the most appropriate measure.
3. When ventricular contractility is depressed, the ventricle acts as a 'pressure source', with the capacity to eject being dependent exquisitely on pressure generated, and variable substantially through small changes in such pressure.
4. All these properties of the heart depend on fundamental properties of the ventricular myocardium and on the inverse relationship between the capacity of muscle fibers to shorten and generate pressure. They are apparent under different conditions, when indices of pressure and output are expressed instantaneously (Yaginuma, Nichols, Miyashita), on a beat-by-beat basis (Suga and Sagawa, Chiu and Carroll), or over a period of time (Elzinga and Westerhof).
5. Myocardial energy requirement is dominated by generation of tension and requires consideration of the pressure generated, the time the pressure is maintained and the dimensions of the ventricle.

CARDIAC FAILURE

In virtually all cases of 'systolic' cardiac failure, impaired left ventricular contractility is a major or the sole cardiac factor (Kannel and Belanger, 1991; Braunwald *et al.*, 1992; Levy *et al.*, 2002; Braunwald, 2009). When contractility is

impaired, the ventricle dilates through use of the Frank–Starling mechanism to maintain pressure and output. In discussing ventricular–vascular interaction in a book on arterial function, venous return and ventricular 'preload', important as they are, need be set aside. The focus must be on the load presented by the arterial system at the ascending aorta and against which the heart must contract (Chirinos and Segers, 2010a; Chirinos and Segers, 2010b).

Effects of this 'after' load can be dealt with in different ways, but the main feature of this condition is the exquisite dependence of ejection on pressure (Wilcken *et al.*, 1964; Ross *et al.*, 1966; Maruyama *et al.*, 1993) (Fig. 14.18) and the large (inverse) changes in output that can be achieved with small changes in pressure.

Figure 14.19 (page 335) illustrates this property of the failing heart, as described by the four different approaches already presented. When impaired contractility is described as (A), a lowered slope of the ESPVR, ventricular pressure, and output are maintained at a high resting ventricular volume. When ejection occurs against a lower pressure, volume decreases to a greater degree than in normal circumstances before the ESPVR point is reached. Stroke volume is increased to a far greater degree than for a similar reduction of pressure in the normal heart. The slope of the ESPVR accounts for the difference.

When impaired contractility is apparent as (B), depressed ejection during late systole in the ascending aortic flow wave form, a reduction in pressure leads to a higher maintained flow velocity and longer duration of ejection (see also Figs 14.13 and 14.18, pages 330 and 334). When seen in relation to power/afterload curves, (C) reduction in pressure is associated with increase in power. Since power is pressure $\times$ flow, such an increase (with pressure falling) represents a disproportionate increase in flow. When expressed in terms of pump function curves, (D), the failing ventricle operates in a region where the curve is almost parallel to the flow axis and at a high angle with the pressure axis. This is probably the best expression of the failing heart acting as a pressure source and shows how small changes in pressure can lead to large changes in flow output. The therapeutic goal is, of course, to decrease pressure, so as to increase cardiac output.

Strategies for reducing ventricular hydraulic load

The principles described above relate to reduction in left ventricular pressure during systole and apply irrespective of how such pressure is decreased. Possible ways of decreasing pressure are shown in Fig. 14.20 (page 336) and entail a reduction in mean arterial pressure (through reduction in peripheral resistance), a decrease in wave reflection (through the use of purely 'arterial' dilators), a delay in early wave reflection, with intra-aortic balloon counterpulsation (O'Rourke *et al.*, 1993a) or with enhanced external counterpulsation (Michaels *et al.*, 2002;

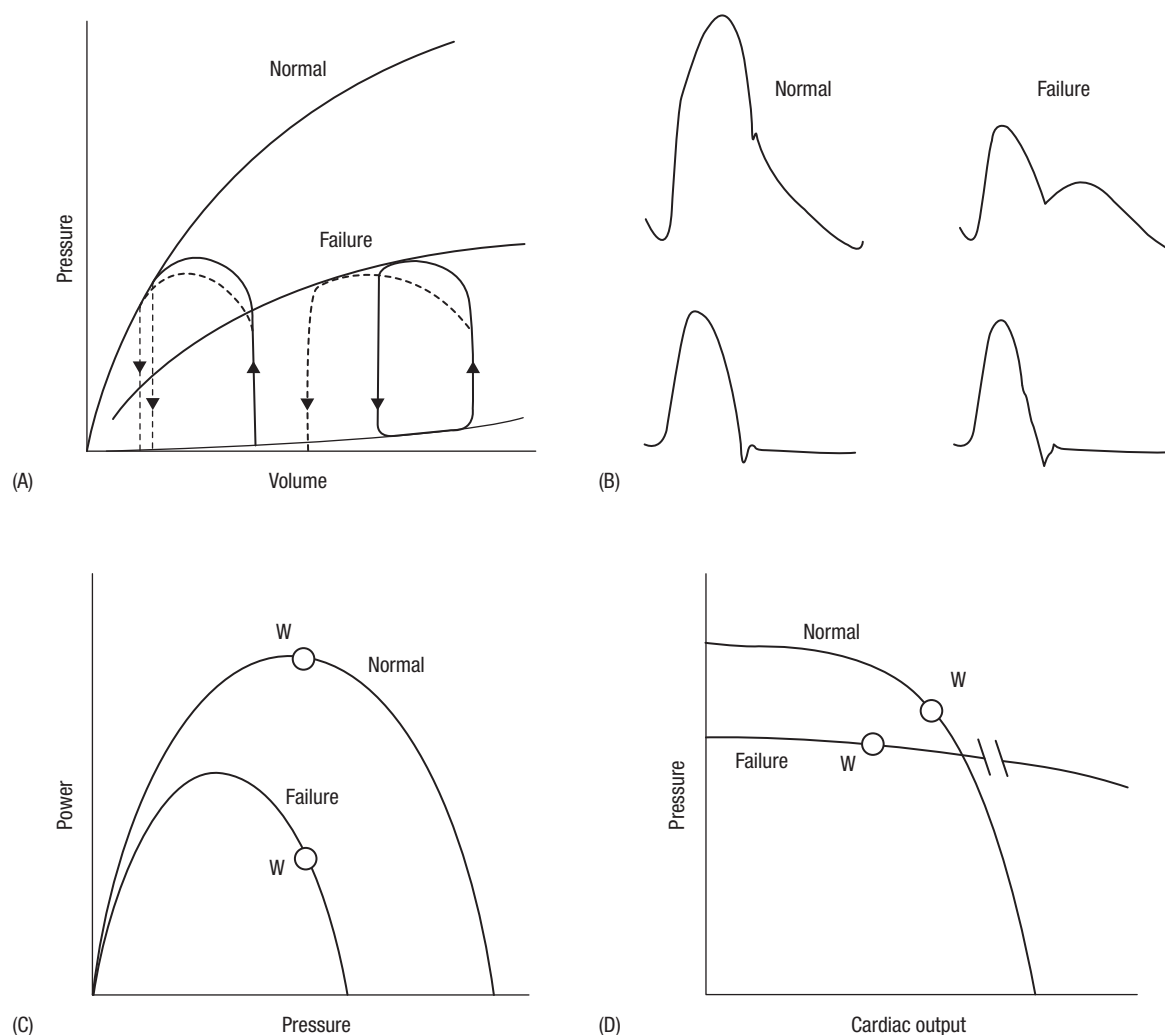


Figure 14.19 Fundamental differences between properties of the normal and failing heart as seen in different ways, but all showing the extreme dependence of the failing heart on pressure. (A) Pressure–volume loops. In the failing heart, the slope of the ESPVR is less steep than in the normal heart. When pressure is reduced, the lower slope permits more blood to be ejected before the ESPVR is reached. (B) Aortic pressure and flow waves. In the failing heart, the flow curve is concave to the right, and systole is abbreviated. (When pressure is lowered during late systole, the flow curve regains its normal convex shape, ejection is prolonged and stroke volume is increased.) (C) Power curves. Normally, the ventricle operates near the apex of the power curve. In failure, the curve is displaced to the left and the ventricle operates on the falling limb of the curve, where small changes in pressure cause large changes in power (and hence in stroke volume). (D) Pump function curve. In heart failure, the pump function curve's intercept on the pressure axis is lowered. While the normal ventricle operates at a point where the ventricle is neither a 'pressure' nor a 'flow' source, the failing ventricle operates at a point where the ventricle acts like a pressure source with large changes in cardiac output possible with small changes in pressure

Nichols *et al.*, 2006). With many drug therapies, reduction in aortic pressure during systole is attributable to a combination of peripheral vasodilatation and reduced and delayed wave reflection.

The earliest studies to show benefit of drug therapy in hypertensive heart failure were conducted with ganglion-blocking agents (Smirk and Alstad, 1951; Smirk, 1957). Improvement in cardiac performance and output with these agents is attributable to a reduction in peripheral resistance and mean arterial pressure. Aortic systolic

pressure may have been reduced further by delay of wave reflection. Hydralazine and prazosin were employed early in the 1970s for the relief of cardiac failure (Packer *et al.*, 1979; Massie *et al.*, 1981; Smith *et al.*, 1992). The acute benefits were not maintained; such a problem was attributable to tolerance (Packer *et al.*, 1979). These drugs are pure arteriolar dilators and work in much the same way as the ganglion-blocking agents (O'Rourke *et al.*, 1993a). Terazosin, an analog of prazosin, has been studied in acute experiments on humans and not found to have any

significant effect on wave reflection (O'Rourke *et al.*, 1985). In contrast, nitroglycerin causes a substantial decrease in wave reflection, as manifest in the reduction of late systolic pressure augmentation, and in the decrease in modulus and phase of the low-frequency components of ascending aortic impedance (Yaginuma *et al.*, 1986). This drug in conventional clinical doses has little effect on peripheral resistance or on aortic pulse wave velocity

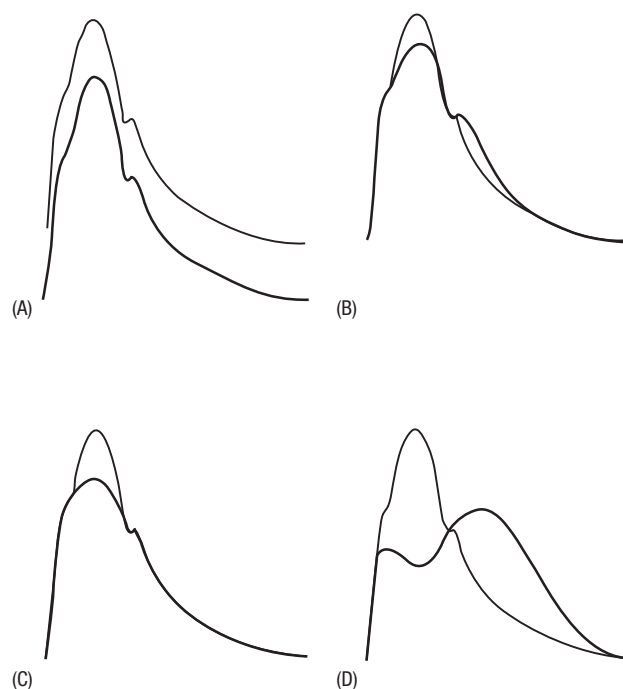


Figure 14.20 Four different strategies for reducing left ventricular hydraulic load, with expected benefit for increase in stroke volume and decrease in myocardial oxygen demand. (A) Reduction in mean arterial pressure. (B) Delay in wave reflection. (C) Decrease in reflected wave amplitude. (D) Arterial counterpulsation. Thick line: after intervention

(Yaginuma *et al.*, 1986; Fitchett *et al.*, 1988a; Fitchett *et al.*, 1988b; Latson *et al.*, 1988). Nitroglycerin and isosorbide mononitrate are extremely effective drugs for the relief of acute and chronic heart failure (Cohn *et al.*, 1991; Goldman *et al.*, 1993) (as well as, and independent of, their effects on angina), and are very effective for increasing exercise tolerance acutely and long term in patients with cardiac failure (Leier *et al.*, 1983; Cohn *et al.*, 1991). As described in Chapters 9, 10, and 26, the beneficial effects of this drug in decreasing aortic and left ventricular pressure are underestimated from the measurement of brachial artery pressure (Kelly *et al.*, 1990a; Simkus and Fitchett, 1990; Devereux, 1991; Takazawa *et al.*, 1993b). The beneficial effects on 'afterload' reduction have been overlooked in the past, since this was gauged from the brachial systolic pressure or peripheral resistance. Nitrates are among the most effective drugs available for the management of cardiac failure (Taylor *et al.*, 2004). Apart from their effects on veins and venous return (Smith *et al.*, 1992; Smith *et al.*, 1997), the beneficial actions of oral and transdermal nitrates in conventional dosage are entirely attributable to the reduction in wave reflection and in the peak of pressure in the left ventricle and ascending aorta (Fig. 14.21) (O'Rourke *et al.*, 1993a).

The ill effects of aging and arterial stiffening on left ventricular load, first pointed out by Bramwell and Hill (Bramwell and Hill, 1922a; Bramwell and Hill, 1922b), could be reversed by a delay in wave reflection. This is accomplished to a degree by any drug that lowers arterial pressure, since the less tense aorta transmits the arterial pulse more slowly from, as well as to, the periphery. This passive effect appears to be relatively minor under most conditions (Latson *et al.*, 1988; Carroll *et al.*, 1991) and amounts to approximately 5–10 cm/s reduction in carotid-femoral pulse wave velocity (PWV) per 1.0 mmHg decrease in mean arterial pressure (Asmar *et al.*, 2001b). Dilator drugs, even nitrates, appear to have little direct

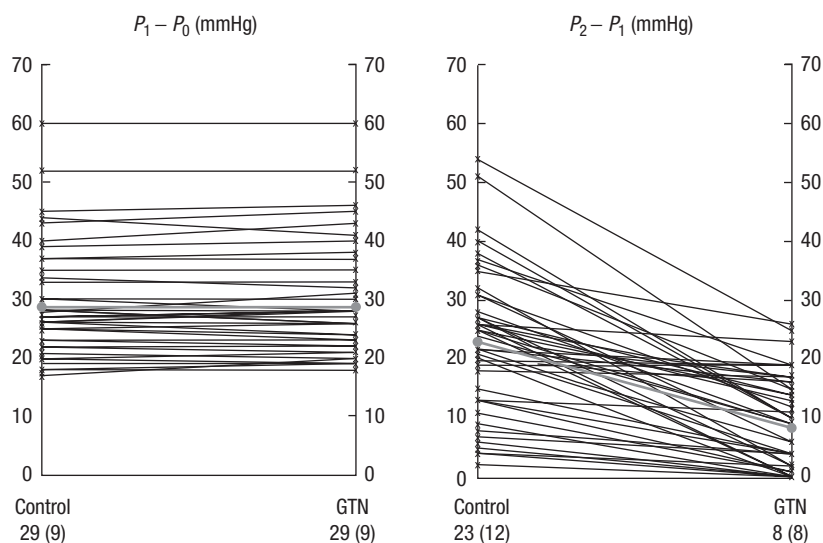


Figure 14.21 Effects of an arterial dilator (nitroglycerine intravenous infusion) on the incident and reflected components of the aortic pressure wave in 50 consecutive patients undergoing coronary artery bypass, and prior to initiation of cardiac pulmonary bypass. Reproduced from Pauca *et al.* (2005). Left: Incident wave amplitude under control conditions and during nitroglycerine infusion. Wave amplitude 29 ± 9.9 mmHg before infusion, 29 ± 9.0 mmHg during infusion, $p = \text{not significant}$. Right: Reflected wave amplitude under control conditions 23 ± 12 mmHg, 8 ± 8.0 mmHg during infusion, $P < 0.001$

effect on the aorta and large arteries, so do not appreciably decrease wave velocity directly; hence, they have little direct pharmacological effect in the delay of wave reflection (Fig. 14.21) (O'Rourke *et al.*, 1993a).

The angiotensin-converting enzyme (ACE) inhibitors have proven to be the most effective of all agents for the treatment of chronic cardiac failure and reduction in mortality (Cohn *et al.*, 1991; Pfeffer *et al.*, 1992; SOLVD Investigators, 1992; Smith *et al.*, 1997). Although there are undoubtedly direct effects on the heart, a major part of the benefit must come about through a reduction in left ventricular and aortic pressure (Devereux, 1991; Devereux, 1997). Reduction in wave reflection has been confirmed in acute studies (Ting *et al.*, 1993; Chen *et al.*, 1995; Asmar *et al.*, 2001b; London *et al.*, 2004a); this, together with reduction in arterial mean pressure and delay in wave reflection, must constitute a major part of their therapeutic role. Calcium antagonists have similar effects (Packer, 1989; Packer *et al.*, 1996; Noda *et al.*, 2006), although the possibility of adverse long-term effects in patients with ischemic heart disease has not been excluded (Messerli, 1995; Opie, 1995). Long-term effects were investigated in the NHLBI-sponsored ALLHAT trial (ALLHAT, 2002). In the short and intermediate term, these agents have been shown to improve cardiac failure and decrease symptoms (Packer *et al.*, 1996), and to delay the need for valve replacement in patients with aortic valve incompetence (Bonow, 1994; Scognamiglio *et al.*, 1994). Beneficial effects of dobutamine in cardiac failure are largely attributable to reduction in wave reflection (Binkley *et al.*, 1990; Binkley *et al.*, 1996).

With intra-aortic balloon counterpulsation (Weber and Janicki, 1974; O'Rourke, 1986; Pennington and Swartz,

1992), a sausage-shaped balloon is placed in the descending thoracic aorta and rhythmically inflated and deflated synchronously with the heart beat, such that deflation occurs during systole and inflation during diastole. Balloon deflation during systole decreases left ventricular and aortic systolic pressure, thus aiding left ventricular ejection, especially in the presence of cardiac failure, and decreases myocardial oxygen demand. Inflation during diastole increases the aortic pressure throughout the time when the coronary artery radicles in the left ventricle wall are capable of being perfused (Fig. 17.5); this improves myocardial blood flow and oxygen supply. The timing of balloon inflation and deflation is adjusted according to the effect on the measured aortic pressure wave. The optimal effects are seen when deflation reduces pressure throughout systole as much as possible, and increases pressure throughout diastole as much as possible (Fig. 14.22).

Intra-aortic balloon counterpulsation was originally introduced for the relief of myocardial ischemia accompanying coronary artery occlusion and cardiogenic shock (O'Rourke, 1986; Pennington and Swartz, 1992). The major role for this device now is in supporting the left ventricle during transient left ventricular functional derangement following cardiac surgery, and in severely ill patients with cardiac failure awaiting cardiac transplantation. Although other mechanical devices have been introduced for circulatory support, this device still remains the most frequently used, simplest, and most effective nearly 50 years after its introduction. The theoretic basis of its benefit is impeccable. Its problems are technical and include balloon fatigue and rupture. Another problem is that it can only support the failing

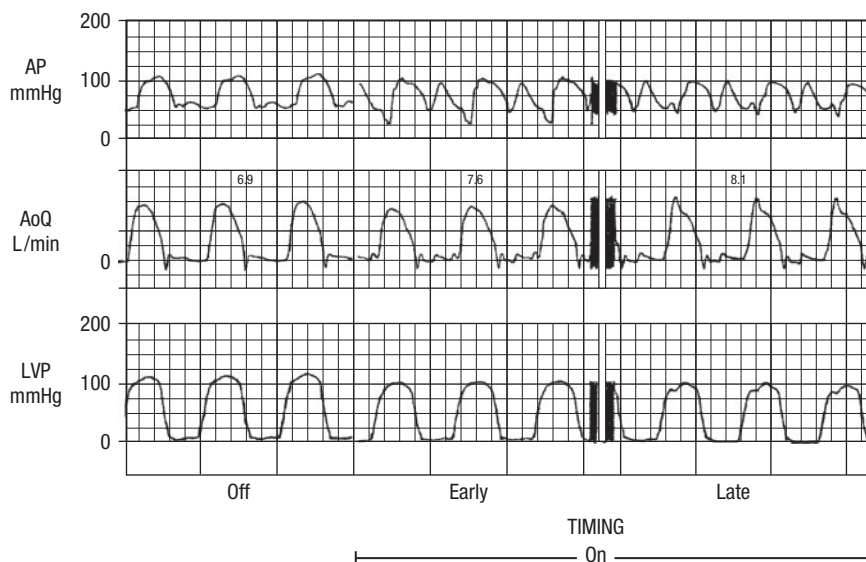


Figure 14.22 Basal and intra-aortic balloon counterpulsation (IABP) demonstrating the influence of early and late timing of deflation on aortic pressure (AP), aortic flow (AoQ), and left ventricular pressure (LVP) in a conscious calf. Note that both IABP modes increase cardiac output with respect to the basal state (6.9 to 7.6 and 8.1 L/min), and late deflation provides an additional increment in cardiac output with respect to that with early deflation (7.6 to 8.1 L/min). Reproduced from Weber *et al.* (1972)

heart; it is not a flow source. It merely optimizes a hemodynamic condition for ventricular ejection and myocardial perfusion.

Ascending aortic impedance in systolic heart failure

Ascending aortic impedance is a measure of LV afterload and has been determined from recordings of pressure and flow taken invasively at cardiac catheterization (Merillon *et al.*, 1978; Pepine *et al.*, 1978; Gundel *et al.*, 1981; Yin *et al.*, 1983b; Laskey *et al.*, 1985; Kromer *et al.*, 1992). Recordings have also been taken during limited exercise (Laskey and Kussmaul, 1987), during vasodilator drug therapy (Merillon *et al.*, 1978; Pepine *et al.*, 1979; Gundel *et al.*, 1981; Yin *et al.*, 1983b; Laskey *et al.*, 1985; Laskey and Kussmaul, 1987) and with inotropic therapy (Binkley *et al.*, 1990; Binkley *et al.*, 1996). Further details of these are given in Chapters 26 and 27 on drug therapy and exercise, respectively. These studies are supplemented by other determinations of peripheral blood flow and large artery properties (Arnold *et al.*, 1991; O'Rourke *et al.*, 1993a).

These studies show surprisingly little difference to features of ascending aortic impedance in normal subjects (Fig. 14.23); surprisingly so, because there are very dramatic differences between patterns of pressure (and flow)

in normal subjects and in patients with cardiac failure (Fig. 14.13). In cardiac failure due to impaired systolic ejection, there is a progressive decrease in ventricular ejection duration (even when allowance is made for any increase in heart rate) (Weissler *et al.*, 1968; Kromer *et al.*, 1992; O'Rourke *et al.*, 1992a), and the aortic flow wave tends to fall linearly from its peak to the incisura, whereas the flow wave is normally convex to the right (Fig. 14.10) (Yaginuma *et al.*, 1985; Sakai *et al.*, 1992; Miyashita *et al.*, 1993; Nichols *et al.*, 1993). The greatest difference is in the aortic and left ventricular pressure wave. Instead of showing the characteristic late systolic peak of mature and older humans, this shows the single rounded early systolic peak seen in young healthy adolescents (Lewis, 1906; Hamilton, 1944; Ewy *et al.*, 1969; Orchard and Craigie, 1980; Chiu *et al.*, 1992; Kromer *et al.*, 1992; O'Rourke *et al.*, 1992a; Denardo *et al.*, 2010). The arterial pressure wave also shows a prominent diastolic wave (as normally in adolescence), although this may be markedly exaggerated in the brachial or radial arteries (see Fig. 10.10). The differences that have been shown in aortic impedance with heart failure, while relatively modest (Avolio, 1994), include increased resistive component (consistent with increased arteriolar tone) and increased characteristic impedance (Pepine *et al.*, 1978; Pepine *et al.*, 1979; Eaton *et al.*, 1993). Evidence of early return of wave reflection is usually seen despite the 'young' aortic pressure waves (Merillon *et al.*, 1978; Pepine *et al.*, 1978; Pepine *et al.*,

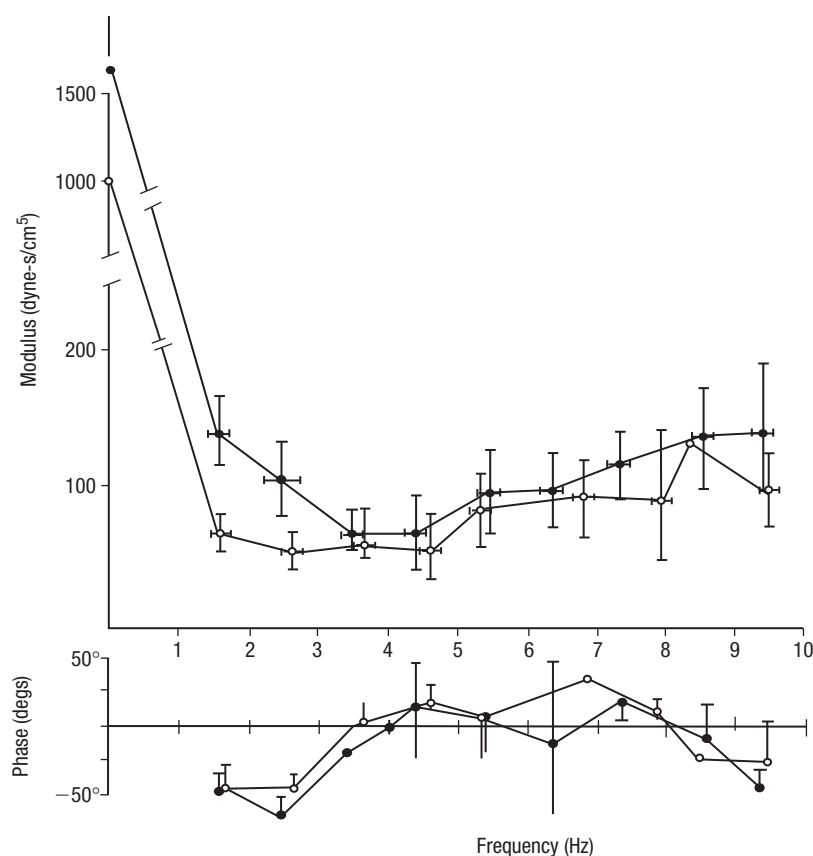


Figure 14.23 Changes in ascending aortic impedance in six patients with heart failure under control conditions (closed circles) and after infusion of nitroprusside (open circles). Reproduced from Merillon *et al.* (1984)

1979; Yin *et al.*, 1983b; Laskey *et al.*, 1985; Laskey and Kussmaul, 1987; Kromer *et al.*, 1992). Arterial pulse wave velocity may be increased (Arnold *et al.*, 1991; Chiu *et al.*, 1992) despite fall in mean pressure. These changes are all attributable to increased circulating levels of angiotensin II and of catecholamines (Kromer *et al.*, 1992), together with increased sympathetic tone. In patients with severe LV dysfunction and reduced mean arterial pressure, pulse wave velocity and the resulting late systolic pressure boost may be decreased (Denardo *et al.*, 2010).

Impedance studies have shown the same evidence of wave reflection as seen in normal subjects. As in normal subjects, and in those with hypertension, wave reflection and lower frequency moduli are attenuated by vasodilators (Fig. 14.23), including nitroglycerin, isosorbide dinitrate, nitroprusside, nifedipine, dobutamine, and enalapril (Chapters 9, 12, and 26). However, wave reflection appears to be maintained during leg exercise, despite vasodilatation in exercising muscle (Laskey and Kussmaul, 1987). Such a finding may be due to greater vasoconstriction elsewhere and in arteries proximal to the arterioles where vasomotor tone is normally relatively low.

Wave reflection in systolic heart failure – modification of flow, rather than pressure

Maintenance of ascending aortic impedance patterns in cardiac failure mandates that there is no major change in the properties of the systemic circulation – specifically no major change in wave reflection. Yet there are very substantial changes in ascending aortic pressure waves, with disappearance of the pattern of augmentation, which is generally taken to characterize the effects of early wave reflection. This has been referred to earlier and follows from the studies of Yaginuma *et al.* (1985), Sakai *et al.* (1992), and Miyashita *et al.* (1993) from Japan. The explanation is that wave reflection remains, and is strong, but that it cannot be manifest on the pressure wave since the ventricle is incapable of generating such a pressure boost (Fig. 14.19) (Maruyama *et al.*, 1993; O'Rourke, 1994; Westerhof and O'Rourke, 1995; Denardo *et al.*, 2010). The ventricle, acting as a pressure source in severe cardiac failure, shows the same behavior instantaneously as previously described for weakened papillary muscle and myocardium. Instead of generating pressure, its shortening is decreased by wave reflection. Reflection is still present, but as a subtraction to the flow wave rather than an addition to the pressure wave (see Chapter 10). Such an effect on reflection (adding to pressure or subtracting from flow) is precisely what one would expect also from physical principles (Westerhof *et al.*, 1972). The subtraction from flow is the change in shape of the flow wave, with deviation from the convex (to right) contour during late systole and decrease in ejection duration. The effect of wave reflection on flow is responsible for the short, sharp contour of the aortic flow wave, the early aortic valve

closure and the decreased stroke volume. Seen thus, early wave reflection is a cause of reduced cardiac output and the problem of cardiac failure itself. This argument is summarized in Fig. 14.24, which also shows how effects on aortic (and left ventricular) pressure may not be apparent when systolic pressure is taken only from the brachial artery by cuff sphygmomanometer, or even from the peak of the recorded wave (see Kelly *et al.*, 1990a; Simkus and Fitchett, 1990; Devereux, 1991; Takazawa *et al.*, 1993b; Denardo *et al.*, 2010). This figure also shows another characteristic feature of the pressure pulse in cardiac failure – its dicrotic appearance. This, as described above, is a consequence of early aortic valve closure in the presence of unchanged wave reflection. Since the valve is closed, reflection cannot alter aortic flow; it can only alter the pressure, and it does so by creating a prominent wave in early diastole. Such diastolic waves are particularly prominent in severe heart failure and may interfere with identification of the cardiac incisura and with the timing of arterial counterpulsation (Lewis, 1906; Ewy *et al.*, 1969; Orchard and Craigie, 1980; O'Rourke *et al.*, 1992a; Denardo *et al.*, 2010).

Further evidence to support the above proposition comes from the administration of drugs, such as nitroprusside or dobutamine, which decrease wave reflection (Merillon *et al.*, 1978; Pepine *et al.*, 1979; Gundel *et al.*, 1981; Laskey *et al.*, 1985; Binkley *et al.*, 1990; Chiu *et al.*, 1992). These drugs have little effect on the contour of the systolic part of the pressure wave in patients with severe heart failure, but alter the contour of the aortic flow wave, increasing late systolic flow and increasing ejection duration (see Fig. 14.13) (Yin *et al.*, 1983b; Chiu *et al.*, 1992; Miyashita *et al.*, 1993; Sakai *et al.*, 1993). In people without heart failure, these same drugs have little effect on the flow wave but cause a marked reduction in late systolic augmentation of the pressure wave (see Figs 10.26 and 10.27, and Chapter 26) (Yaginuma *et al.*, 1986; O'Rourke *et al.*, 1993a).

The proposition put here continues the same logical argument presented elsewhere for the management of hypertension through the asymptomatic phase (O'Rourke and Kelly, 1993; Safar and O'Rourke, 1993; Denardo *et al.*, 2010) into the period when cardiac failure develops. Early wave reflection is seen as an undesirable feature, responsible for increased left ventricular load. The same vasodilator drugs which decrease wave reflection and pressure augmentation at this stage (Cohn *et al.*, 1991; SOLVD Investigators, 1992) are also beneficial when failure develops (Jessup *et al.*, 2009). The basic problem is early wave reflection; this is disadvantageous to the heart. However, when the heart is normal (or even hypertrophied) and capable of pumping strongly, it is manifest as a boost to systolic pressure, and as the heart weakens, it becomes manifest as a reduction in flow (O'Rourke, 1994; Westerhof and O'Rourke, 1995). Early wave reflection is the bogey under all circumstances. The major therapeutic strategy is to reduce wave reflection when this occurs early

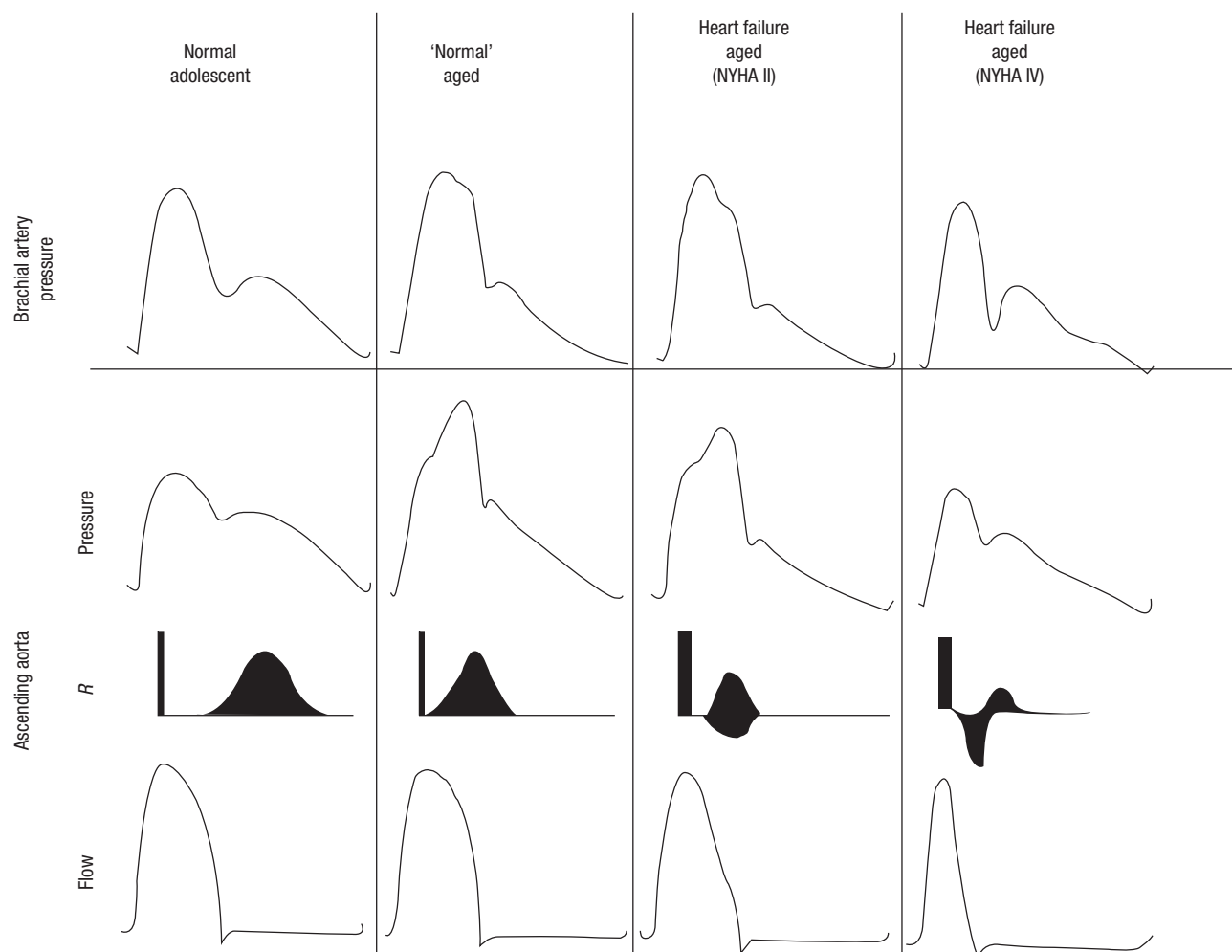


Figure 14.24 Effects of wave reflection on aortic pressure and left ventricular output with the development of heart failure in a patient with isolated systolic hypertension, all shown schematically. From above: brachial artery pressure, aortic pressure, wave reflection (R) (with effect on pressure shown as upward deflection, and effect on flow as downward deflection) and ascending aortic flow (bottom tracing)

as a consequence of aging or hypertensive change. Such a strategy then appears to be appropriate when arteries are stiffened and wave reflection occurs early, i.e. in any adult beyond the age of 30 years with hypertension or cardiac failure (O'Rourke and Hashimoto, 2007; O'Rourke and Nichols, 2009a; O'Rourke, 2010).

There are fewer data on the treatment of children and adolescents, since cardiac failure in this age group is rare (Towbin *et al.*, 2006; Shaddy *et al.*, 2007). In children (at least in those over the age of 12), and in adolescents, aortic PWV is low, but LV ejection similar to that in adults. Wave reflection from the lower body is ideally timed and does not (except in aortic coarctation) return during systole to boost aortic and LV pressure. There is considerable doubt on the benefit of β -blockers, such as carvedilol (Shaddy *et al.*, 2007) and of ACE inhibitor or angiotensin receptor blocker (ARB) drugs in children and adolescents (Kantor *et al.*, 2010; Singh *et al.*, 2010). One would not expect reduction in wave reflection from peripheral sites to have the same benefit as in older cohorts.

Arterial and ventricular elastance

In a series of articles emanating from the group of Kass *et al.* from Johns Hopkins, the concepts of arterial and ventricular elastance have been combined to determine invasively and non-invasively the interaction between heart and arterial system with aging, with change in venous return, with exercise and with drugs (Kass and Kelly, 1992; Kelly and Fitchett, 1992; Kelly *et al.*, 1992a; Kelly *et al.*, 1992b; Kass *et al.*, 1996; Chen *et al.*, 1998; Chen *et al.*, 1999; Chen *et al.*, 2001; Kawaguchi *et al.*, 2003; Borlaug *et al.*, 2009). While arterial stiffening and drug effect were described in terms of aortic impedance and wave reflection, the simplified concept of aortic compliance (which necessarily excludes wave reflection) was used to relate arterial to ventricular performance under different conditions. What emerged from these studies is an endorsement of the concept previously presented together with a better understanding of the problem that the aging vasculature presents to the heart, which limits exercise

capacity and predisposes to faints and 'flash' pulmonary edema (Chen *et al.*, 1998; Gandhi *et al.*, 2001; O'Rourke and Hashimoto, 2007; Shapiro *et al.*, 2007; Abhayaratna *et al.*, 2008; Weber *et al.*, 2008).

Ventricular-vascular interaction in diastole: diastolic heart failure

There is relevance of the principles enunciated here to diastolic, as well as systolic heart failure. In diastolic failure, ventricular filling is limited by slow ventricular relaxation (Fig. 14.25) (Brutsaert *et al.*, 1993; Federmann and Hess, 1994; Wachter *et al.*, 2009; Borlaug *et al.*, 2010; Paulus, 2010). This may cause a decrease in cardiac output, but always causes an increase in left ventricular diastolic pressure, and thus creates congestive symptoms (Brutsaert *et al.*, 1984; Le Carpentier *et al.*, 1991; Redfield, 2004; Zile *et al.*, 2004; Paulus *et al.*, 2007; de Keulenaer and Brutsaert, 2009; Lee *et al.*, 2009; Borlaug *et al.*, 2009; Najjar, 2009). Early return of wave reflection augments systolic pressure and delays the systolic peak of the pressure wave. Both the delay in the systolic peak and the degree of augmentation have been shown to correlate with delayed diastolic ventricular relaxation (Hori *et al.*, 1985; Yabe *et al.*, 1991; Gillebert *et al.*, 1997). Reduction in late systolic augmentation is associated with accelerated diastolic relaxation (Yabe *et al.*, 1991). Ventricular hypertrophy and myocardial ischemia are important factors also in diastolic dysfunction. Both are aggravated by high aortic systolic pressure, caused by arterial stiffening (Hundley *et al.*, 2001; O'Rourke, 2001; Borlaug *et al.*, 2009; Najjar, 2009).

In diastolic dysfunction, duration of left ventricular ejection is prolonged (Fig. 14.25). This provides a useful distinction from systolic dysfunction where (as discussed above) left ventricular ejection is abbreviated. As underlying disease progresses, systolic dysfunction may complicate diastolic dysfunction (Fig. 14.25). Sphygmocardiography (Chapter 30) may provide a means of separating predominantly 'systolic' from predominantly 'diastolic' heart failure (O'Rourke and Lei, 1997). In patients with 'diastolic' heart failure (more properly described clinically as heart failure with preserved ejection fraction), dyspnea occurs readily with exertion. Increase in LV and diastolic and pulmonary venous pressures are undoubtedly the cause of dyspnea (Paulus, 2010), but non-invasive and invasive indices have not consistently been abnormal during exercise (Borlaug *et al.*, 2010; Maeder *et al.*, 2010). It is surprising that such a simple index, such as ejection duration, has not been studied in such patients. We have found that the diastolic time is substantially reduced during exercise in such patients, and systolic ejection duration preserved, despite tachycardia and reduction of cycle length (Brutsaert *et al.*, 1993; Federmann and Hess, 1994; O'Rourke and Lei, 1997).

EXTERNAL HEART WORK

External heart work is referred to in Chapter 13. This is a measure of energy lost in the circulation. It can be measured (for the systemic circulation) from pulsatile pressure and flow waves in the ascending aorta (and assuming right atrial pressure to be negligible); for the pulmonary

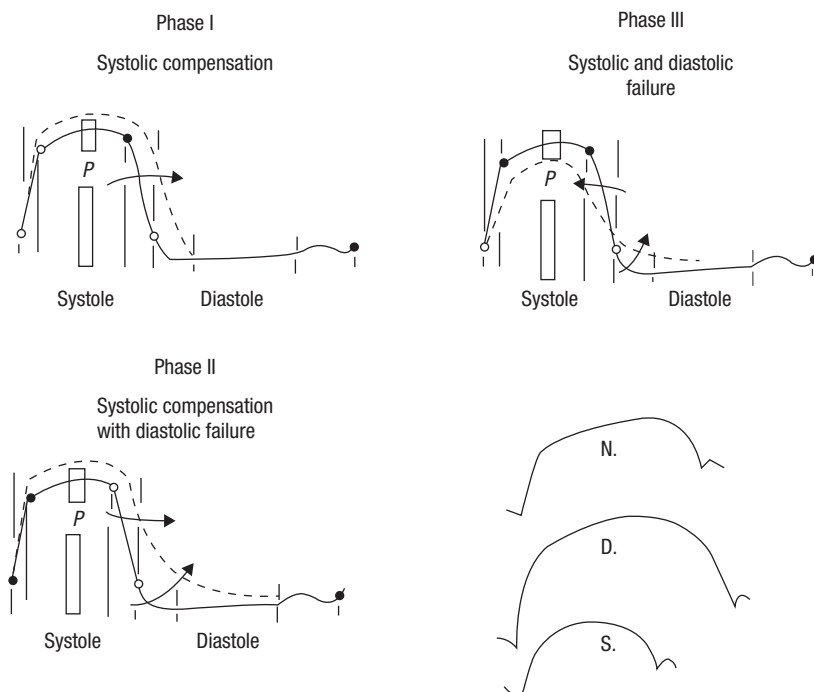


Figure 14.25 Successive phases of left ventricular malfunction, associated with left ventricular hypertrophy from chronic increase in ventricular load. Solid line is the normal ventricular pressure tracing. With hypertrophy, there is prolongation of ventricular contraction and increase in ejection duration (phase I). Delay in relaxation follows with increased left ventricular pressure throughout most of diastole (phase II). With impairment of ventricular contractility (phase III), peak systolic pressure is reduced and ejection duration is decreased, while relaxation remains prolonged, and pressure raised through diastole. On the right are the systolic portions of the aortic pressure waves taken from those data and drawn on an expanded scale for phase I (normal, N), phase II (diastolic failure, D), and phase III (with onset of systolic failure, S). From Brutsaert *et al.* (1993), as modified by Federmann and Hess (1994), and by ourselves

circulation, it can be measured from pressure and flow waves in the pulmonary artery.

External heart work comprises just a small fraction of ventricular work (for the systemic circulation normally some 10–20 percent of the total). In describing ventricular work, energy requirements and function as to this point, we have been dealing with total ventricular work. This is of overwhelming importance when describing cardiac function.

The concept of external heart work assumes importance in describing the function of the circulation in which the energy is lost (Milnor *et al.*, 1966; O'Rourke, 1968; Nichols *et al.*, 1977a). It has particular importance in describing the efficiency of the arterial system in accepting pulsatile flow from the ventricle. One can separate total external work into two components – pulsatile and steady. Pulsatile external work is lost in the large arteries where flow is pulsatile, whereas steady external work (mean pressure $\times$ mean flow) is principally lost in the arterioles and capillaries where flow is steady or almost so. Since steady external work is useful (for perfusion of the tissues) and pulsatile work useless, the ratio of pulsatile to total external work is sometimes used as an (inverse) index of the efficiency at which the circulation accepts pulsatile flow from the heart. In normal circumstances, the ratio is around 10–20 percent for the systemic circulation (O'Rourke, 1968; Nichols *et al.*, 1977a) and somewhat higher for the pulmonary circulation (Milnor *et al.*, 1966). In disease states, the ratio may approach or even exceed 50 percent – where pulsatile energy losses are of the same magnitude as energy losses in maintaining steady flow. In these circumstances, the circulation is very inefficient. Factors contributing to minimization of pulsatile external work include appropriate arterial distensibility and such timing of wave reflection that augmentation to aortic pressure from this occurs during diastole rather than systole (Taylor, 1964; Taylor, 1994).

The concept of external heart work is of less interest than in earlier years. Similar information on arterial function can be gained from examination and analysis of the pressure wave in the ascending aorta and through measurement of such features as systolic tension time index, diastolic pressure time index, or mean aortic pressure during systole and diastole directly from the aortic pressure wave. Such dispenses with need to measure load.

Pulsatile external heart work is relatively higher in the pulmonary circulation than in the systemic circulation. This is attributable to lesser cushioning of flow pulsations in the shorter elastic pulmonary arteries (Bergel and Milnor, 1965) than in the aorta. In the systemic circulation, flow pulsations are almost completely absorbed in the large elastic arteries, so that flow is almost steady in the smallest blood vessels. In the pulmonary circulation on the other hand, some 20–30 percent of flow pulsations extend through to the venous side so that flow in pulmonary capillaries is pulsatile (see Chapter 16).

With aging in humans, the aorta and elastic systemic arteries become progressively stiffer, and so less able to absorb flow pulsations. These extend down into the microvasculature, and more so into the vasodilated organs – the brain and kidneys. The small blood vessels in these organs absorb the pulsations, and the energy associated with them (O'Rourke and Safar, 2005), damages their walls. This topic is taken up in Chapter 18. Pulsatile energy loss in these organs can be calculated in the same way as in the ascending aorta (for the systemic circulation) and main pulmonary artery for pulmonary circulation, from the difference in pulsatile pressure and pressure at input and output. Measurement of pulsatile energy lost in brain and kidneys may permit organ damage to be quantified. To date, this has not been done, but it has been shown that artificial stiffening of the aorta in experimental animals leads to increase in external heart work from around 10 percent of total external work up to 30 percent (O'Rourke, 1967b). All studies on simulated arterial stiffening (Kelly *et al.*, 1992a; Kelly *et al.*, 1992b; Ohtsuka *et al.*, 1994; Kass *et al.*, 1996) have to date been directed more at understanding left ventricular load and function than at potential sites of organ damage in the systemic circulation.

QUALIFICATION

The arguments presented here refer to pressure and its relationships to ventricular ejection and to energy expenditure (and oxygen demands). Most published data refer to pressure since this is easily measurable. However, the really important parameter is tension within myocardial cells, and this is modified by dimensions – by diameter of the ventricular chamber in accordance with the law of Laplace, and by wall thickness, which relates force to stress. When diameter and wall thickness remain constant, pressure determines tension. When changes are small, pressure remains all important, but when changes are larger (as with wall thickening during ejection in a ventricle of normal size) (Chirinos *et al.*, 2009a; Chirinos *et al.*, 2009b; Chirinos *et al.*, 2009c), dimensional change must be considered. Thickening of the wall during systole may account for decreasing wall tension during ejection despite an increase in pressure.

Consideration of ventricular dimension could explain an apparent anomaly in the arguments presented here for the failing heart acting like a pressure source, with ejection utterly dependent on load. Data for force–velocity relationships of isolated papillary muscles show no evidence of such behavior, and the curve for muscle with impaired contractility is simply displaced below the normal curve. However, when allowance is made for the ventricular dilatation that necessarily accompanies impaired myocardial contractility and heart failure, another picture emerges (see Fig. 14.14). In the dilated failing heart, pressure is lower for any given wall tension; hence, the abscissa is foreshortened. On the other hand, in the

dilated heart, any degree of shortening will create greater volume change and greater flow from the ventricle. Hence, the ordinate is elongated. Thus, the relationship for aortic flow/ventricular pressure in a failing heart will be much steeper than the normal relationship and will approach the pressure axis at close to 90° consistent with behavior as a pressure source.

The work of Sarnoff *et al.* (1958) and of his predecessors was done in normal, non-failing hearts, and here a close relationship was shown between pressure-time integral and oxygen requirement. This relationship is changed in the dilated, failing heart, where wall tension is greater for any given ventricular pressure. Suga (1990) has confirmed the influence of ventricular volume on oxygen demands and has shown that the mechanical energy of contraction per beat can be determined from the ventricular pressure-volume loop (see Fig. 17.9).

Recent work on left ventricular dyssynchrony (Kass *et al.*, 1999; Kass, 2008; De Keulenaer and Brutsaert, 2009) explains the sinister implications of left ventricular bundle branch block (LBBB) and suggests another way of treating LBBB and other electrical conduction blocks with biventricular pacing (Jessup *et al.*, 2009). It is possible that timing may be optimized by pulse wave analysis, with attention directed at maximizing aortic pulse pressure and

ejection duration of paced beats. One may expect from principles discussed here that pulse waveform analysis may help select optimal candidates and optimal pacing site location, for biventricular pacing on the ability to increase pulse pressure and ejection duration with biventricular or left ventricular pacing. Indications for such pacing to restore left ventricular synchrony are increasing in patients with mechanical dyssynchrony caused by ventricular conduction disease. It is now accepted that mechanical indices of improved contraction and ventricular ejection should be measured (Delnoy *et al.*, 2009; Marwick and Grimm, 2010; Solomon *et al.*, 2010; Spragg *et al.*, 2010; van Bommel *et al.*, 2010). These include LV ejection duration, aortic pulse pressure, pressure-volume loops and LV max dP/dt. Radial arterial tonometry has been used in small numbers of patients, and may prove a very useful tool (Naqvi and Rafique, 2008).

This chapter has taken a viewpoint on ventricular-vascular interaction which appears to explain observed behavior of the heart under different conditions of altered vascular load and in the presence of heart failure. While presently acceptable to the authors as a tentative solution, its test will be in explaining other findings of which we are not aware and data yet to be published.

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Cardiac failure: clinical implications

The epidemic of cardiac failure	345	Mechanical ventricular unloading in treatment of cardiac failure	352
The cause of cardiac failure	347	Drug trials in cardiac failure	353
Changing concepts of cardiac failure	347	Drug trials for prevention of cardiac failure	354
'Diastolic' heart failure	349	Other therapies	354
Early wave reflection: a unifying theory for cause of systolic and diastolic heart failure?	351		

Why have a chapter on cardiac failure in a book on blood flow in arteries? There are two good reasons – first that arterial disease (of coronaries or aorta) is responsible for most cases of cardiac failure now seen by physicians around the world, and second, that modification of arterial properties and of left ventricular afterload is the most effective method for treating and preventing cardiac failure. If one wants to manage cardiac failure most effectively, one needs to know how arterial properties and wave reflection characteristics impact on vascular–ventricular interaction, and how these properties can be modified.

The previous chapters set out the basic concept on vascular–ventricular interaction, and the mechanisms whereby chronic increase in left ventricular (LV) afterload adversely affects cardiac performance. While the bulk of this chapter was originally written in 1997, there has been little need for change, since we were able to anticipate many subsequent developments. Perhaps we did underestimate their importance, and the magnitude of benefit which has now been established from use of mechanical therapies, low dose β -blocking and vasoactive drugs for reducing left ventricular afterload. Certainly, we did not anticipate the thrust of government and professional bodies to identify and treat even modest increase in arterial pressure so as to reduce cardiovascular events, and especially to delay onset of cardiac failure and reduce disability in our aging community.

Elsewhere in this book, and especially in Chapters 9, 14, 19, and 21, we emphasize the effect of aging on arterial stiffness, and stress that ill effects are substantially underestimated from the modest (perhaps 20 percent) increase in brachial systolic pressure seen in population studies between age 20 and 80 years. We pointed to a two-fold increase in aortic characteristic impedance (and pulse wave velocity) over this period, and in consequence

of early wave reflection, a four-fold increase, on average, in ascending aortic impedance at resting heart rate (O'Rourke and Hashimoto, 2007). Recent magnetic resonance imaging (MRI) studies suggest that ascending aortic stiffness and impedance increase to an even greater degree than this (Redheuil *et al.*, 2010). We pointed out too that wave reflection can be modified to the heart's advantage by drugs and other measures, even if structural properties of the arteries themselves cannot. These issues are important to modern management of heart failure, but where do we see them mentioned in texts or professional journals? In 2002, the *New England Journal of Medicine* published a perspective on cardiac load, equating this only with peripheral resistance (Baughman, 2002), and declined a correction which referred to arterial stiffness and wave reflections, which was acknowledged by the author. In 2008, the *Journal of the American College of Cardiology* published and editorialized similar views on LV load (Yancy, 2008; Mullens *et al.*, 2008), but did accept correspondence on the subject (O'Rourke and Nichols, 2009a). Major current articles, editorials and guidelines on cardiac failure make scant or no reference to wave reflection or pulsatile hemodynamics (Velagaleti *et al.*, 2008; Phan *et al.*, 2009; de Keulenaer and Brutsaert, 2009; Jessup *et al.*, 2009; Lee *et al.*, 2009) despite wave reflection being the principal reversible cause of failure (Denardo *et al.*, 2010). Such is the inertia of the medical profession to fundamental new views which challenge entrenched dogma.

THE EPIDEMIC OF CARDIAC FAILURE

Cardiac failure is increasing progressively, and is becoming the major cause of morbidity and mortality in

Western and Eastern societies (Fig. 15.1A) (Bristow, 2001; Levy *et al.*, 2002; Jessup and Brozena, 2003; Fang *et al.*, 2008; Jhund *et al.*, 2009). Cardiac failure cases increase as coronary disease lessens in the Western world (Fig. 15.1B) (Levy and Thom, 1998; Kesteloot, 1999; Levy *et al.*, 2002). The phenomenon is largely one of aging (Fig. 15.1C) (Redfield, 2002). Incidence of cardiac failure increases markedly with increased age (Fig. 15.1D) (Kannel *et al.*, 1972; Kannel and Belanger, 1991; Petrie *et al.*, 2001; Lloyd-

Jones *et al.*, 2002). With an increasing number of older people in the community, prevalence of cardiac failure – of symptomatic people – increased up to around 4.5 million in the United States (Fig. 15.1E) (Bristow, 2001). Death from coronary disease has decreased in recent years, in consequence of better prevention and better treatment (Levy and Thom, 1998; Tavazzi, 2001), but this is usually shown, as in Fig. 15.1B, in age-adjusted terms (Table 15.1) (Levy *et al.*, 2002). The epidemic of cardiac failure is a

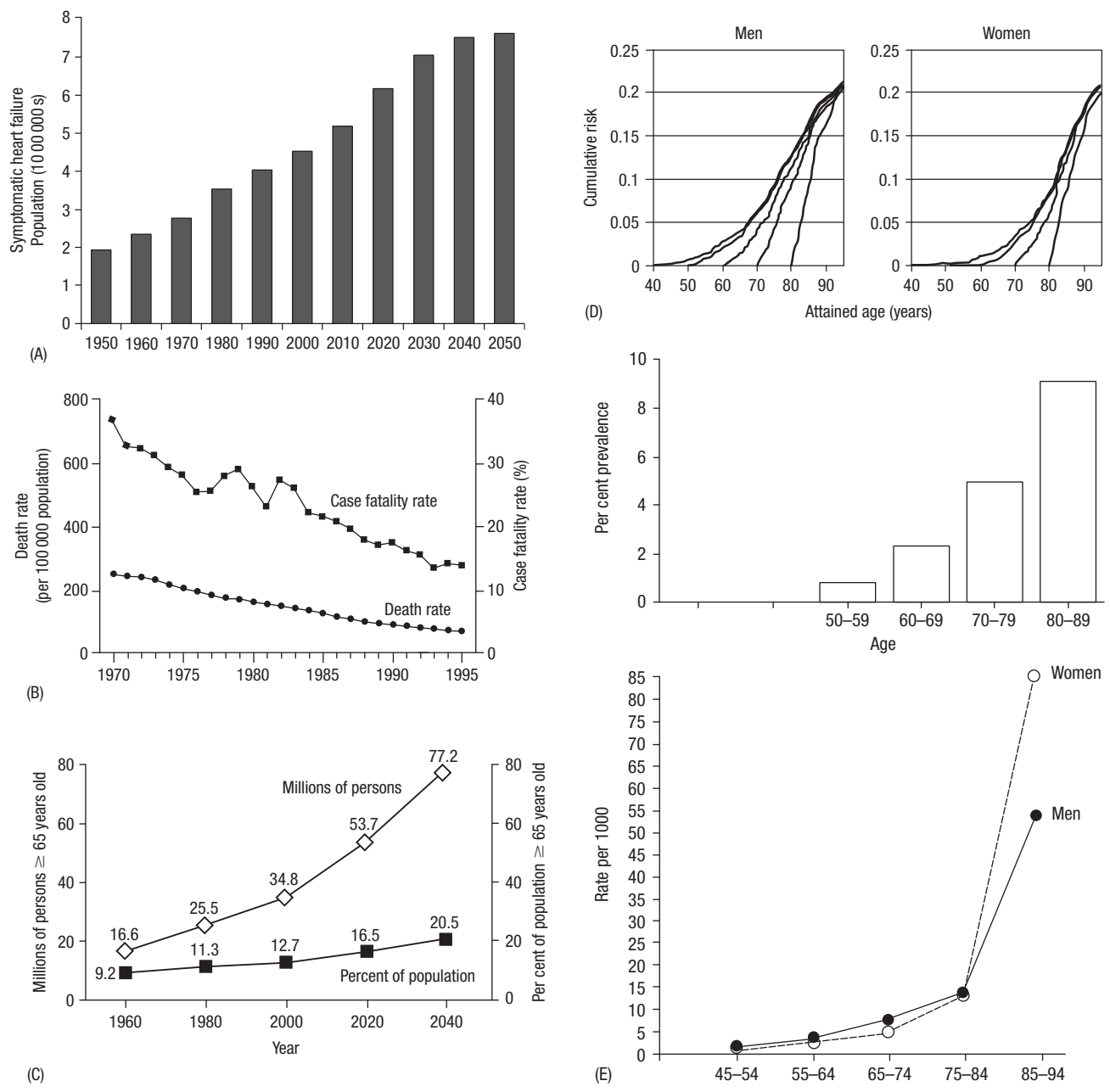


Figure 15.1 (A) Prevalence of symptomatic heart failure in the United States 1950–2050 (predicted), from Bristow (2001) after Hayflick (1994). (B) Death rate and case fatality rate from acute myocardial infarction among people 45–64 years in the United States. From Levy and Thom (1998). (C) Projected increases in the US population 65 years of age and older from the US Census Bureau and Redfield (2002). (D) Cumulative risk for congestive heart failure at selected index ages for men and women in the Framingham study. From Lloyd-Jones *et al.* (2002). (E) Prevalence (above) and incidence (below) of heart failure by age in the Framingham study. From Kannel and Belanger (1991)

Table 15.1 Temporal trends in the age-adjusted incidence of heart failure in the United States

Period	Men		Women	
	Incidence of heart failure	Rate ratio	Incidence of heart failure	Rate ratio
1950–69	627 (475–779)	1.00	420 (336–504)	1.00
1970–79	563 (437–689)	0.87 (0.67–1.14)	311 (249–373)	0.63 (0.47–0.84)
1980–89	536 (448–623)	0.87 (0.67–1.13)	298 (247–350)	0.60 (0.45–0.79)
1990–99	564 (463–665)	0.93 (0.71–1.23)	327 (266–388)	0.69 (0.51–0.93)

All values were adapted for age. Values in parentheses are 95 percent confidence intervals. From Levy *et al.* (2002).

consequence of success in managing coronary disease and other conditions, with expectation of life increasing up to ten years over the past three decades (Tavazzi, 2001) in the Western world. The age-adjusted incidence of cardiac failure appears to have changed little over recent decades – at least in men (Levy *et al.*, 2002; Redfield, 2002; Fang *et al.*, 2008). The good news is that survival after onset of heart failure has improved, both in men and women (Levy *et al.*, 2002; Jhund *et al.*, 2009).

THE CAUSE OF CARDIAC FAILURE

The two major causes (antecedents) of cardiac failure are agreed as hypertension and coronary artery disease, although the relative importance of each has been disputed, and since both conditions often coexist. Yusuf and Pitt (2002) attributed 75 percent of cardiac failure cases to these two conditions. Experience from specialized centers gives a different view on etiology, depending on the expertise of a given center for a particular condition – which can explain the high prevalence of idiopathic cardiomyopathy in some (Givertz, 2000) and of coronary disease (Gheorghiade and Bonow, 1997) in others. The non-significant fall over years in incidence of cardiac failure in males despite the fall in coronary disease, implies that coronary disease is not the major cause (Levy *et al.*, 2002). The Framingham and other community studies (Fox *et al.*, 2001; Lee *et al.*, 2009) have given a lower prominence to coronary disease, and higher prominence to hypertension than hospital-based studies, one of which estimated that 68 percent of all cardiac failure cases were due to coronary atherosclerosis (Gheorghiade and Bonow, 1997; Tavazzi, 2001). Hospital-based cardiologists until recently largely overlooked the problem of diastolic left ventricular dysfunction in older people, and probably have often ascribed breathlessness to another cause. The large pool of elderly patients with cardiac failure due to arterial stiffening and systolic hypertension has been overlooked (see below). The Framingham group (Levy *et al.*, 1996) has found that hypertension was present in 91 percent of people who subsequently developed cardiac failure, with approximately half of these having diastolic LV dysfunction as the principal cause of the condition, and the other half having systolic LV

dysfunction subsequent to ventricular scarring and coronary atherosclerosis (Fig. 15.2) (Vasan *et al.*, 1996; Vasan and Levy, 2000; Vasan and Benjamin, 2001; de Keulenaer and Brutsaert, 2009; Lee *et al.*, 2009). In young men up to the age of 35, a prospective study showed that development of cardiac failure in the next 20 years was related to elevated arterial pressure, and that 26 of 27 of those affected were black (Bibbins-Domingo *et al.*, 2009). Coronary artery disease was not the cause in the vast majority. Independently, Heffernan *et al.* (2008) had shown that while brachial pressures are similar in young black and white men in the United States, central systolic pressure is consistently higher in blacks than in whites, while blacks in South Africa were shown to have higher aortic stiffening than whites (Shiburi *et al.*, 2006). In the elderly, females have higher prevalence of cardiac failure than males despite lower prevalence of coronary disease and similar brachial systolic pressure. As with blacks (Heffernan *et al.*, 2008), this is probably due to higher aortic systolic pressure (McEniery *et al.*, 2008) as a consequence of shorter body length, greater arterial taper (Cecelja *et al.*, 2009) and earlier wave reflection. This is discussed further in Chapter 29. Findings support those from Framingham (Levy *et al.*, 2002) and the views of Katz (Katz, 1990; Katz, 1998) on ‘Cardiomyopathy of Overload’ as the most common cause of cardiac failure at all ages.

CHANGING CONCEPTS OF CARDIAC FAILURE

A new paradigm is emerging to explain development of cardiac failure in elderly people. The view at the turn of the century was elegantly set out by Eugene Braunwald in his Shattuck, Denolin, and AHA Distinguished Scientist lectures (Braunwald, 1997a; Braunwald, 1997b; Braunwald, 2001a; Braunwald, 2009). This was based on the concept that cardiac failure was a consequence of systolic left ventricular dysfunction, and was predominantly caused by coronary disease and myocardial infarction (Gheorghiade and Bonow, 1998). Indeed, up until this time, a diagnosis of cardiac failure was not considered substantiated unless LV ejection was impaired, and trials of cardiac failure (with the exception of the DIG trial (Digitalis Investigation Group, 1997)) required ejection fraction of <35 percent. The emerging paradigm is

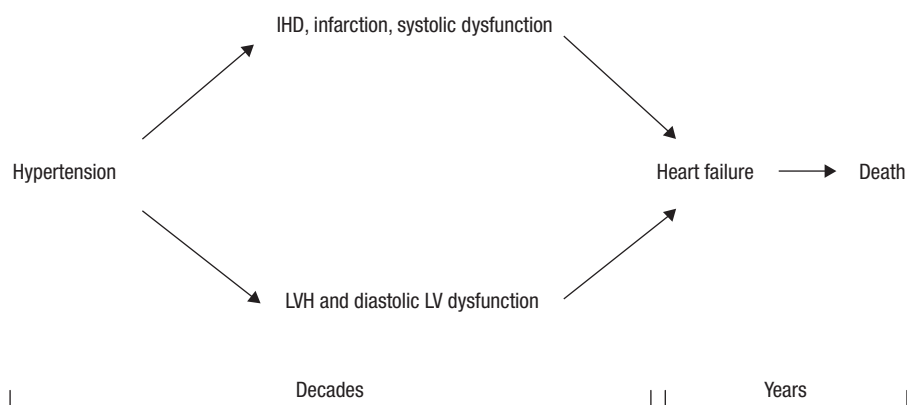


Figure 15.2 Schematic description of progression to heart failure over decades, after Vasan *et al.* (1996). Continued elevation of arterial pressure leads ultimately to systolic dysfunction from coronary disease and myocardial scarring (top), to diastolic dysfunction from left ventricular hypertrophy (LVH) (bottom) or to a combination. The problems develop slowly over decades, but once symptomatic from heart failure, further deterioration is rapid. IHD, ischemic heart disease. After Vasan *et al.* (1996)

necessary to explain the epidemic of cardiac failure in an aging society (Eastern and Western) in which coronary disease is lessening, but arterial stiffening continues to progress.

This paradigm follows the concept of ‘Cardiomyopathy of Overload’ set out by Katz (Katz, 1990; Katz, 1998; Katz, 2002) to explain the finding of Meerson (1961), and the observation of cardiologists including Mackenzie (1902) who was quoted by Katz: ‘The more I study the symptoms of heart failure, and the more I reflect on the part played by the heart muscle, the more convinced I am that ... heart failure is due to exhaustion of the reserve forces of the heart muscle’. Katz went on to describe left ventricular hypertrophy as a consequence of chronically increased LV load (increased systolic pressure), and what was initially compensatory to maintain left ventricular tension, but ultimately became pathological, as the hypertrophied myocytes outgrew their capacity to gain nutrients and discharge metabolites. These myocytes contracted less forcefully and relaxed more slowly (Grossman *et al.*, 1975). Additional problems subsequently observed were apoptosis of some myocytes and necrosis of others (Olivetti *et al.*, 1991; Narula *et al.*, 1996; Anversa *et al.*, 1997; Pons-Llado *et al.*, 2000) and extracellular matrix remodeling (Braunwald, 1997a; Weber, 1997; Braunwald, 2009).

The concept of ‘Cardiomyopathy of Overload’ readily explains the genesis of cardiac failure in older people on the basis of increasing aortic stiffness and early wave reflection causing chronic elevation of aortic and LV systolic pressure (Fig. 15.3) (Denardo *et al.*, 2010). The concept also explains development of cardiac failure in young adults, with Bibbins-Domingo *et al.* (2009) noting that in a prospective study most (75 percent) of the subjects developing cardiac failure had earlier elevation of arterial pressure. The Framingham study has repeatedly emphasized the importance of elevated systolic pressure and LV load as a cause of later cardiac events, and

particularly cardiac failure (Kannel *et al.*, 1972; Levy *et al.*, 1988a; Levy *et al.*, 1990; Ho *et al.*, 1993; Levy *et al.*, 1996; Levy *et al.*, 1999; Levy *et al.*, 2002) with or without previous evidence of LV hypertrophy. This work, confirmed elsewhere for systolic and pulse pressure (Kostis *et al.*, 2001; Petrie *et al.*, 2001), is the basis for exhorting better control of systolic pressure in order to prevent or delay development of cardiac failure (Levy *et al.*, 1999; Lloyd-Jones *et al.*, 2002; Yusuf and Pitt, 2002; Jessup *et al.*, 2009) (see also Chapter 21). All the studies identify increased load as increased brachial artery systolic (or pulse) pressure. This is an imprecise measure of left ventricular load since it underestimates aortic systolic pressure in younger people and does not show differences in aortic systolic pressure between blacks and whites (Heffernan *et al.*, 2008), or women and men (McEniery *et al.*, 2008). Aortic systolic pressure increases far more with age than does brachial pressure (Vlachopoulos and O’Rourke, 2000b; Wilkinson *et al.*, 2001; O’Rourke and Hashimoto, 2007). Systolic pressure also may fall with the onset of complications, such as myocardial infarction and/or frank systolic heart failure itself (see Fig. 15.3).

The emphasis on ‘Cardiomyopathy of Overload’ (Katz, 1990; Katz, 1998) is not inconsistent with the current views on heart failure championed by Braunwald (Braunwald, 1997a; Braunwald, 1997b; Braunwald, 2001a; Braunwald, 2009) and others. It does emphasize, however, the importance of afterload, and the appropriate expression of afterload, and it better explains the relentless progression of heart failure with time. It better explains the benefits of mechanical heart assistance, as well as of drug therapy, and it better embraces the concept of diastolic LV dysfunction as a cause of heart failure in ‘diastolic heart failure’ or more specifically ‘heart failure with preserved LV systolic function’ (Yamamoto *et al.*, 2009; Lam, 2010; Tapp *et al.*, 2010). It better explains the results of the ALLHAT (2002) study wherein an angiotensin-converting enzyme inhibitor (ACEI) appeared less

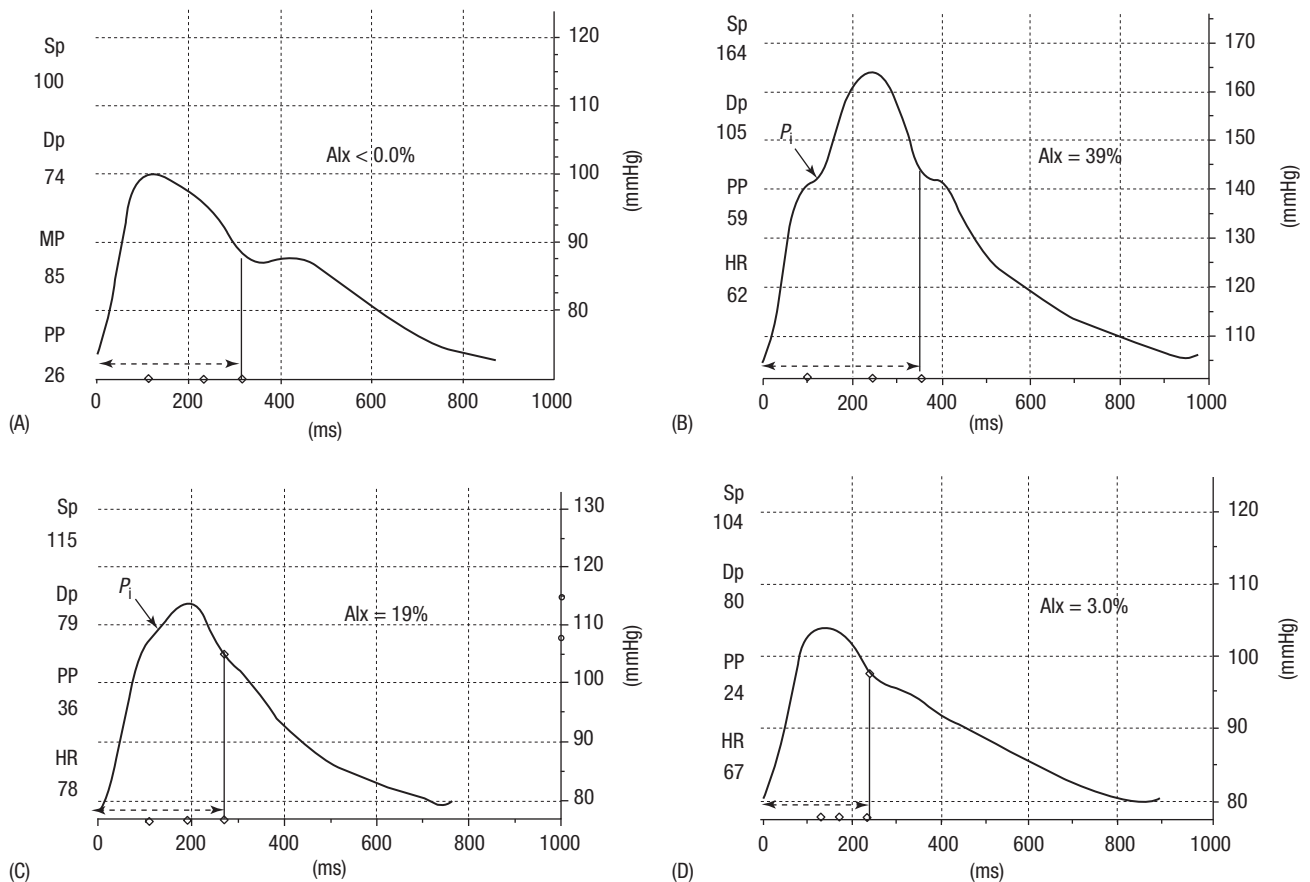


Figure 15.3 Illustration showing changes in the central aortic pressure wave that occur during the progression or development of left ventricular dysfunction (LVD). (A) Young normal healthy subject with reflected wave occurring predominately in diastole; ejection duration (ED) (broken line) is 316 msec, central pulse pressure (cPP) is 26 mmHg, and augmentation index (Aix) is <0.0 percent. (B) During aging and development of hypertension, the reflected wave (amplitude 23 mmHg in this example) occurs almost entirely in systole resulting in an increase in Aix (39 percent) and cPP (59 mmHg), probably causing LV hypertrophy which begins the process of systolic heart failure (HF). (C) As HF proceeds reflected wave amplitude (7 mmHg), Aix (19 percent), cPP (36 mmHg) and ED (269 msec) begin to decrease. (D) In LVD, the augmented systolic pressure becomes almost zero and LV ejection is shortened further (235 msec). Reproduced from Denardo *et al.* (2010)

effective than a diuretic in preventing onset of heart failure in a high-risk group, but where apparent overall benefit was related to actual systolic pressure fall.

'DIASTOLIC' HEART FAILURE

The Framingham (Vasan *et al.*, 1995) and Olmsted County (Senni *et al.*, 1998; Senni and Redfield, 2001) studies have confirmed that some 50 percent of people who develop cardiac failure in the community do not have impaired LV contractility to account for this. Furthermore, they showed that the condition was often associated with LV hypertrophy and was usually preceded by elevation in arterial pressure (Fig. 15.3). Most of the patients were elderly and the majority were female. In the Framingham study (Levy *et al.*, 1996), 91 percent of patients who developed congestive cardiac failure had pre-existing hypertension (blood pressure (BP) > 140/90). Some 50

percent were considered to have predominant diastolic LV dysfunction as the cause of cardiac failure (Vasan *et al.*, 1996). The initial process (of elevated LV load leading on to coronary disease or to myocyte hypertrophy and dysfunction) took decades (Fig. 15.2), whereas deterioration was swift after onset of symptomatic heart failure with some 50 percent of patients dying within five years (Fig. 15.4).

It is difficult to understand why cardiologists had overlooked the problem of diastolic LV dysfunction and the all-important influence of elevated arterial pressure, and why it was left to general practitioners outside the hospital, and to epidemiologists, for this issue to gain appropriate attention. The problem of diastolic LV dysfunction in hypertrophic cardiomyopathy was well known in the 1970s (with attention having shifted from an obstructive to a relaxation problem in the previous decade (Doi *et al.*, 1980)). Further, the problem of hypertensive hypertrophic cardiomyopathy had been described in 1977

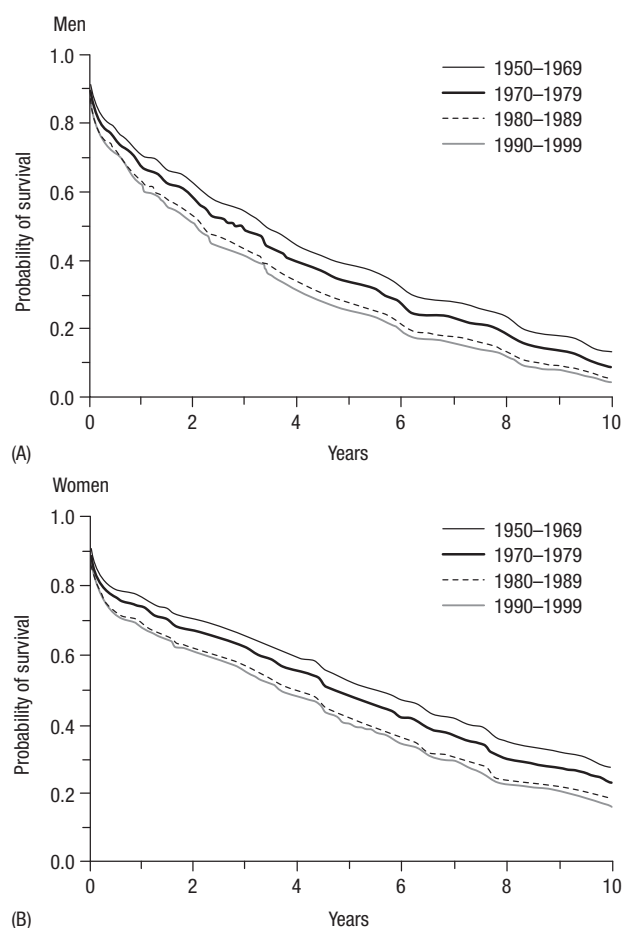


Figure 15.4 Temporal trends in age-adjusted survival (patients 65–74 years) after onset of symptomatic cardiac failure. From Levy *et al.* (2002)

(Come *et al.*, 1977; Murgo *et al.*, 1980b; Topol, 1988) with emphasis also on diastolic LV dysfunction. It was not appreciated at this time that the condition described – usually in older hypertensive black women – was an extreme, or the terminal phase of a much wider problem.

The issue of diastolic dysfunction in heart failure was considered sufficiently important to warrant a full supplement of *Circulation* (Vol. 81, No. 2) in 1990, but this was not followed up, at least in clinical trials, for a further decade. The pressure for consideration of diastolic dysfunction in heart disease came principally from Europe. In the United States, the focus of cardiologists was on LV systolic dysfunction. The Framingham and Olmsted (Mayo) data on this subject (Tsang *et al.*, 2004; Abhayaratna *et al.*, 2006a; Weber *et al.*, 2008) have been confirmed and are now generally accepted, but agreement on the importance of diastolic LV dysfunction as a cause of heart failure is not total (Caruana *et al.*, 2000; Banerjee *et al.*, 2002; Redfield *et al.*, 2003). Objections are based on the clinical diagnosis of congestive cardiac failure as including patients whose breathlessness may be due to age, obesity, pulmonary disease, or myocardial ischemia,

and the lack of precise methods to characterize diastolic LV dysfunction (Vasan and Levy, 2000; Cheitlin, 2001; Zile *et al.*, 2001; Zile and Brutsaert, 2002a; Zile and Brutsaert, 2002b) even at cardiac catheterization. The accepted non-invasive features of impaired diastolic function (elevated time constant of LV pressure fall during early diastole, reduced E/A ratio in echocardiographic LV filling, and reduced E wave deceleration time), and the ratio of E/E' are not all present in people with LV hypertrophy and diagnosed 'diastolic heart failure' (De Simone *et al.*, 2000; Zile *et al.*, 2001; Redfield, 2004; Zile *et al.*, 2004; Abhayaratna *et al.*, 2006b; Hatle, 2007; Paulus *et al.*, 2007; Weber *et al.*, 2008), and may be seen in elderly people without evidence of heart failure. Measurement of B-type natriuretic peptide (Hobbs, 2000; Maisel *et al.*, 2002; Massie, 2003; Maisel *et al.*, 2003; Weber *et al.*, 2006; Jourdain *et al.*, 2007; Latini *et al.*, 2007; Wang, 2007; Weber *et al.*, 2006; Weber *et al.*, 2008) may prove decisive in establishing the diagnosis of congestive cardiac failure and in therapy. The best accepted echocardiographic index of diastolic LV dysfunction is the ratio of E to E', where E is maximal early systolic flow and E' is maximal tissue relaxation velocity at the mitral valve (Gaasch and Little, 2007; Kasner *et al.*, 2007).

Hundley *et al.* (2001) confirmed the association between aortic stiffening with age and development of systolic hypertension and diastolic heart failure, but showed that aortic stiffening was not inevitable in older people. (Their controls were selected fit older people with systolic pressure (124 mmHg) identical to the younger controls.) Aortic stiffening is progressive and relentless with aging in all societies, irrespective of atherosclerosis prevalence (Cecelja *et al.*, 2011, see Chapter 19). Such progressive degeneration has been recognized for well over 100 years, and was credited by Mackenzie (1902) for the progressive deterioration in the heart's response to effort: 'during the fourth decade of life the ease with which violent efforts are made is gradually lessening and violent exertions are as a rule carefully avoided. Running after trains is not to be done in comfort and the ascent of hills is undertaken with more deliberation. All this proceeds *pari passu* with diminished resiliency of the arterial wall'. The problem with diastolic dysfunction is its ubiquity in the older population – where cardiac failure may only become clinically apparent with 'flash' pulmonary edema occurring in consequence of acute tachycardia (Gandhi *et al.*, 2001).

Diastolic dysfunction of the heart is characterized by delay in relaxation during diastole (Le Carpentier *et al.*, 1991; Brutsaert *et al.*, 1993; Federmann and Hess, 1994; Zile and Brutsaert, 2002a; Zile and Brutsaert, 2002b; Zile *et al.*, 2004; Gaasch and Little, 2007; Paulus *et al.*, 2007; Paulus, 2010; Borlaug *et al.*, 2010; Maeder *et al.*, 2010). Relaxation is an active process, requiring energy, and is slowed in the aged heart with hypertrophy and in ischemic heart disease. In ischemic heart disease, this process may be slowed to such a degree that 'post-systolic shortening'

may become apparent (Brown *et al.*, 1987; Skulstad *et al.*, 2002), at ventriculography or echocardiography.

Delayed diastolic relaxation is part of prolonged contraction, and this is a characteristic of people with cardiac failure and preserved LV contraction (Brutsaert *et al.*, 1993; Federmann and Hess, 1994; Chemla *et al.*, 2000; Grossman, 2000; Vasan and Benjamin, 2001; Burkhoff *et al.*, 2003; Jessup, 2003; Kawaguchi *et al.*, 2003; Masoudi *et al.*, 2003; Smith *et al.*, 2003; Zile, 2003; Weber *et al.*, 2006; Weber *et al.*, 2008). The prolonged period of systole seen in this condition contrasts with the shortened period of systole seen in patients with cardiac failure due to systolic LV dysfunction (see Fig. 15.3 and Chapter 14) (Lewis, 1906; Ewy *et al.*, 1969; O'Rourke *et al.*, 1992a; Denardo *et al.*, 2010). We have exploited this (O'Rourke and Lei, 1997; Abhayaratna *et al.*, 2006b; Weber *et al.*, 2006; Weber *et al.*, 2008; Shim *et al.*, 2011) in using the period of systole to effect a separation of patients with predominantly systolic LV dysfunction from those with predominantly diastolic LV dysfunction (Fig. 15.5). It comes as no surprise from what has been set out above, that the majority of patients seen in clinical practice appeared to have a mixture of systolic and diastolic LV dysfunction.

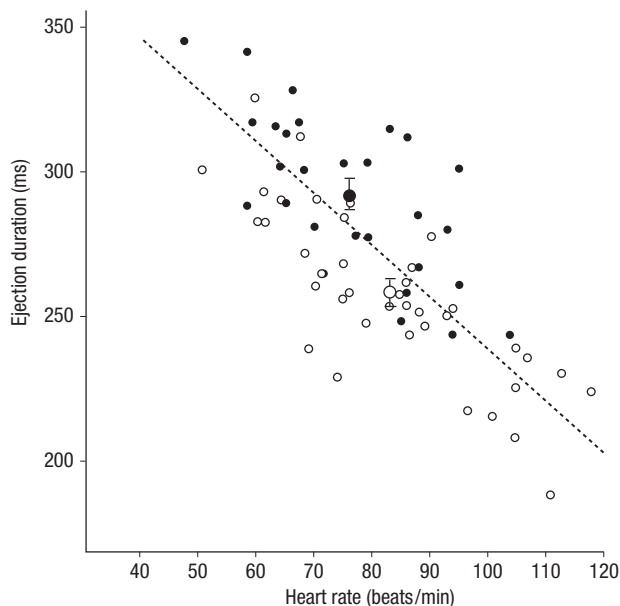


Figure 15.5 Relationship between left ventricular ejection duration and heart rate in 67 patients with cardiac failure attributable to myocardial disease. Predominant diastolic dysfunction was considered the cause of heart failure in 28, on the basis of left ventricular ejection fraction ≥ 35 percent (echocardiogram or gated blood pool scan) (closed circles). Predominant systolic dysfunction was considered the cause of heart failure in 39, on the basis of left ventricular (LV) ejection fraction < 35 percent (open circles). From O'Rourke and Lei (1997). Graph also shows mean $\pm$ SEM, and regression line from Weissler *et al.* (1963)

We have used measurement of ejection period not only to effect a separation of cardiac failure patients into those with predominantly systolic and diastolic dysfunction as the cause, but also to quantify the severity of systolic LV dysfunction (Fig. 15.6). In patients with apparent systolic LV dysfunction as the cause of heart failure, we found a difference in ejection duration when patients were separated into those with cardiac index > 2.8 L/min/m² and those with lower cardiac index (Fig. 15.6) (Uebaba *et al.*, 2002). Furthermore, we found that the patients with lowest cardiac index could be further characterized by the response to intravenous nitroglycerin, which reduces wave reflection, but which as predicted, provoked the greatest increase of cardiac index and of ejection duration in those with the most severe degree of cardiac failure (Fig. 21.11).

EARLY WAVE REFLECTION: A UNIFYING THEORY FOR CAUSE OF SYSTOLIC AND DIASTOLIC HEART FAILURE?

A ready explanation is available on the basis of the argument given above on LV afterload, and the views of Westerhof *et al.* on LV function (Chapter 14) (Figs 14.16, 14.24, and 15.3) (see Westerhof *et al.*, 1972; Westerhof and O'Rourke, 1995). The basic problem is seen as early return of wave reflection, caused by stiffening of the aorta and conduit elastic arteries with age. The consequence of this

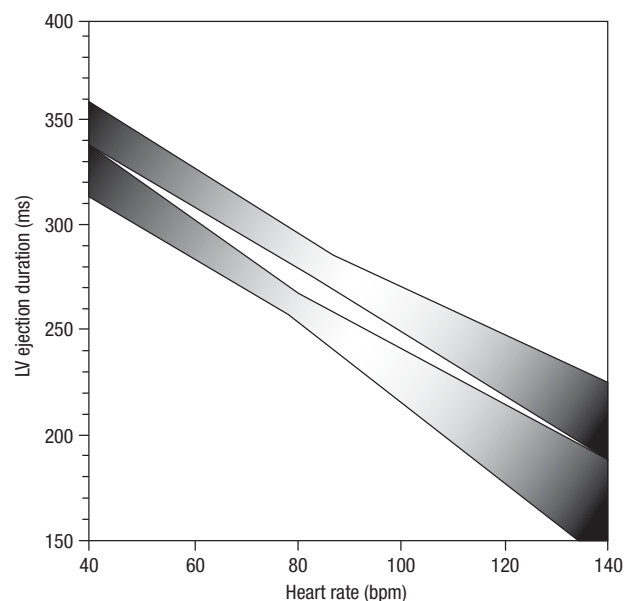


Figure 15.6 Relationship between left ventricular (LV) ejection duration and heart rate in a group of 159 patients with cardiac failure and dilated cardiomyopathy. Patients were separated into two groups with normal cardiac index (≥ 2.8 L/min/m², $n = 69$) (top), and low cardiac index (< 2.8 L/min/m², $n = 90$) (bottom). Regression lines with 95 percent confidence limits are shown. From Uebaba *et al.* (2002)

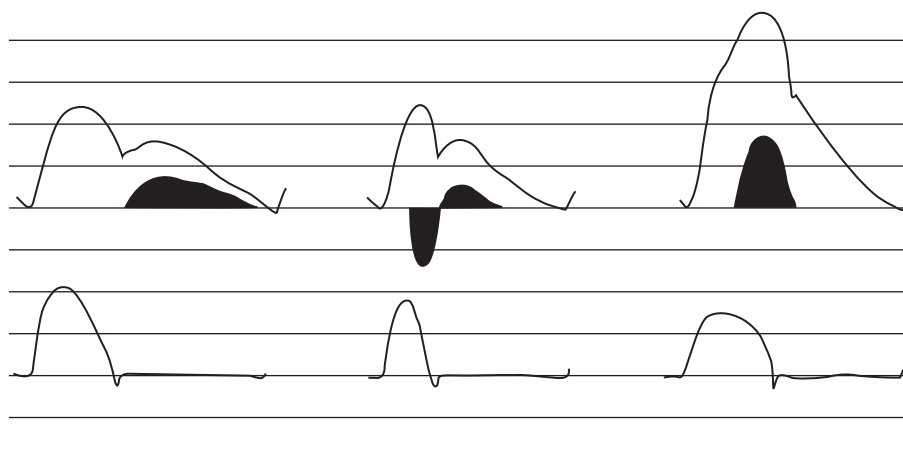


Figure 15.7 Schema illustrating the effects of early return of wave reflection caused by arterial stiffening in a patient with LV systolic dysfunction (center) and another with diastolic LV dysfunction (right), compared to a normal young subject without aortic stiffening (left). The aortic pressure wave shown above and flow wave below, with effects of wave reflection indicated (shaded area, center) on pressure as positive, and on flow as negative. In the young normal subject, reflection occurs in diastole only. With severe systolic dysfunction, effects in systole are exclusively on flow with deceleration and premature termination of systole. With diastolic dysfunction only, contraction is powerful and aortic flow is not altered; effects of reflection are seen as a boost to systolic pressure only with no influence on flow. After Westerhof and O'Rourke (1995) and O'Rourke (2001)

is different when the heart is able to eject normally, and when ventricular ejection is weakened from infarction, scarring, cardiomyopathy, hypertrophy, or any other cause. When the heart is able to contract normally and powerfully (preserved LV function), the effects of wave reflection are seen as a boost to the LV and aortic pressure waveforms and with the late systolic peak augmented (Figs 15.3 and 15.7). Such augmentation is exaggerated in the aged heart, or in LV hypertrophy when the period of ventricular ejection is prolonged (Fig. 15.3B). Such augmentation further delays the process of ventricular relaxation (Yabe *et al.*, 1991; Gillebert *et al.*, 1997). In systolic heart failure, when ventricular contractility is impaired, wave reflection does not boost systolic pressure (Fig. 15.3D), since the heart is incapable of responding, so that systole is terminated prematurely, and wave reflection is seen to have had a negative influence on flow (rather than a positive influence on pressure) (Fig. 14.24) (Westerhof and O'Rourke, 1995; Denardo *et al.*, 2010). This approach has been effectively used to characterize mechanical pumps, with these seen to act as a flow source (powerful ejection) or pressure source (ejection limited by pressure achieved), and is very usefully applied to the heart (Westerhof and O'Rourke, 1995; Denardo *et al.*, 2010). The concept is consistent with the epidemiological, pathophysiological, and clinical data that we have, and throws early wave reflection into the spotlight as the bogey, and the common thread between systolic and diastolic heart failure. If this is the case, the most effective treatment of both is afterload reduction. Such indeed is current clinical practice. Afterload reduction with ACEI, nitrates, and hydralazine have proved effective in treating systolic LV dysfunction (Cohn *et al.*, 1986; Consensus Trial Study Group, 1987; Cohn *et al.*, 1991; SOLVD

Investigators, 1991; SOLVD Investigators, 1992; Pfeffer *et al.*, 1992; Taylor *et al.*, 2004; Jessup *et al.*, 2009). Similar therapies and angiotensin receptor blockers (ARB) have been trialed for the treatment of cardiac failure with preserved LV function, and appear to be equally effective (Cleland *et al.*, 1999; Swedberg *et al.*, 1999; Packer *et al.*, 2001; Granger *et al.*, 2003; Khot *et al.*, 2003; Pitt *et al.*, 2003; Shekelle *et al.*, 2003; Cleland *et al.*, 2006; Solomon *et al.*, 2007; Borlaug *et al.*, 2009; van Veldhuisen *et al.*, 2009; Wachter *et al.*, 2009; Shah *et al.*, 2010). Precisely the same strategy is used for the treatment of hypertension, and for prevention of cardiac failure (Chapters 20 and 21). Many patients have a mixture of systolic and diastolic dysfunction as the cause of heart failure. Separation has been on the basis of ejection fraction, but this is viewed increasingly as a crude index (Borlaug *et al.*, 2009; Cho *et al.*, 2009). Echocardiographic indices of LV filling velocity such as early mitral flow velocity (E), of early divided by late mitral flow velocity (E/EA), or of tissue velocity (EM), or relative flow/tissue velocity (E/EM) are usually used clinically rather than ejection duration. The 'gold standard' for diastolic dysfunction is the LV diastolic pressure-volume relation and the earlier diastolic decay, but these are not routinely measured at cardiac catheterization, and require higher frequency response than the monitoring systems that are routinely employed.

MECHANICAL VENTRICULAR UNLOADING IN TREATMENT OF CARDIAC FAILURE

Intra-aortic balloon counterpulsation was introduced in 1968 for the treatment of cardiogenic shock following myocardial infarction (Kantrowitz *et al.*, 1968). Benefits

were attributed to improvement of diastolic coronary perfusion pressure, as well as to reduction in LV load, and were apparent as improvement in LV function. Subsequent use in patients with acute coronary disease (O'Rourke, 1986) was followed by use in patients with cardiomyopathy, and normal coronary arteries, and where benefits were attributable only to reduction in LV load. Such benefits were substantial (Richenbacher and Pierce, 2001; Frazier and Delgado, 2003; Stevenson and Rose, 2003; Willerson and Kereiakes, 2003; Cheng *et al.*, 2009b; Wilson *et al.*, 2009), and counterpulsation was widely used as a bridge to cardiac transplantation. With donor hearts in short supply, and counterpulsation continued for longer periods of time, it became apparent that substantial improvement was often maintained and patients previously dependent on the device were able to be weaned without transplantation. Similar results were seen with other mechanical support devices which unloaded the left ventricle, and histological studies showed recovery of damaged myocytes and reduction of intercellular fibrous tissue (Levin *et al.*, 1995; Muller *et al.*, 1997; Dipla *et al.*, 1998). Results were initially described in terms of 'resurrection after mechanical circulatory support' (Rose and Frazier, 1997) and were completely unexpected in patients who initially were completely dependent on the mechanical device. They could only be explained on the basis of chronic reduction in LV load (Parmacek and Epstein, 2009). They occurred in patients without coronary disease as well as those with, so that benefit could not just be attributed to regrowth of collateral vessels and recovery of stunned or hibernating myocardium (Braunwald, 2001a). Implantable LV assist devices are widely used for definitive long-term treatment of end-stage heart failure, as well as for short-term use, and as a bridge to cardiac transplantation (Jessup, 2001; Rose *et al.*, 2001; Frazier and Delgado, 2003; Stevenson and Rose, 2003; Willerson and Kereiakes, 2003).

DRUG TRIALS IN CARDIAC FAILURE

Apart from digoxin, which has other effects, drugs which increase myocardial contractility have been abandoned for long-term treatment of cardiac failure. They hasten rather than delay death (Packer *et al.*, 1991; Bristow, 2001). In contrast, drugs which reduce LV load have been shown in acute studies (Kameyama *et al.*, 1991) and long-term trials to improve cardiac performance and reduce mortality (Cohn *et al.*, 1986; Consensus Trial Study Group, 1987; SOLVD Investigators, 1991; Cohn and Finkelstein, 1992; Pfeffer *et al.*, 1992; SOLVD Investigators, 1992; Taylor *et al.*, 2004; Jamerson *et al.*, 2008; Jessup *et al.*, 2009). The initial long-term trial (Cohn *et al.*, 1986) used a nitroglycerin/hydralazine combination, while the later trials used predominantly ACEI for reduction in LV load. Benefits were substantial, and led to ACEI drugs being considered as mandatory for the treatment of cardiac

failure. The usual explanation (Braunwald *et al.*, 2001a; Braunwald 2009; Jessup *et al.*, 2009) is that benefit is due to ACEI action (Khalil *et al.*, 2001; Brunner and Gavras, 2002), whereas all these drugs and the nitrates also reduce wave reflection, and cause greater reduction in aortic and LV systolic pressure than apparent from measurement of brachial pressure (Kelly *et al.*, 1989f; Simkus and Fitchett, 1990; Takazawa *et al.*, 1993b; Ting *et al.*, 1995a; Mitchell *et al.*, 1996; Hirata *et al.*, 2005b; Griffin *et al.*, 2003; London *et al.*, 2004a). The second SOLVD trial with enalapril in patients with impaired LV systolic function (LVEF < 35 percent), but without symptoms of cardiac failure, likewise showed reduction in morbidity and mortality for the active agent compared to placebo.

The ill effects of adrenergic stimulation on the heart are described above. The sympathetic nervous system is activated in heart failure and contributes to deterioration. Prolongation of survival with use of β -blocking drugs was first described in 1979 (Swedberg *et al.*, 1999) and confirmed in trials (Packer *et al.*, 1996; CIBIS-II Investigators, 1999; MERIT-HF Study Group, 1999; Braunwald, 2001b; Packer *et al.*, 2001; Barry and Gilbert, 2003; Reiken *et al.*, 2003; Shekelle *et al.*, 2003). Benefits are additional to those achieved with ACEI drugs or nitrates, and are associated with the same evidence of myocardial recovery as seen with LV assist devices (Azeka *et al.*, 2002). As with mechanical support systems, addition of a β -blocking agent often enabled a patient to recover function sufficiently to be taken off a heart transplantation list (Azeka *et al.*, 2002). As with mechanical or drug-induced unloading of the ventricle, benefits can be attributed to reduction in myocardial work and oxygen requirement. Benefit often takes weeks or months to become apparent, and careful titration of β -blocking agents is required on initiation of treatment (Swedberg, 2009).

To date, no treatment has proved benefit in patients with cardiac failure due to diastolic LV dysfunction, so that placebo-controlled trials are still considered to be ethical. Such trials have tested perindopril (Cleland *et al.*, 1999), valsartan (Swedberg *et al.*, 1999), and irbesartan (Devereux *et al.*, 2001). Since these drugs reduce wave reflection (Hirata *et al.*, 2005b; Klingbeil *et al.*, 2002), definite benefits were anticipated but have not yet been confirmed (Chaudhry *et al.*, 2010; Stevenson, 2010; Tapp *et al.*, 2010). In a meta-analysis of these trials, Shah *et al.* (2010) showed no consistent reduction in 'hard' end-points of heart failure hospitalization or mortality. For valsartan, Solomon *et al.* (2007) showed no significant improvement in diastolic relaxation velocity compared to the placebo group whose blood pressure had been controlled to the same level with other drugs (usually calcium channel blockers (CCB)) which had no direct action on the renin-angiotensin system. Cleland *et al.* (2006) compared perindopril against placebo in the PEP-CHF trial (Perindopril in Elderly People with Chronic Heart Failure). The trial was underpowered to reach the primary end point, but did show improved symptoms and exercise

capacity, and fewer hospitalizations for heart failure in the relatively small number of patients who completed the trial. In the SENIORS trial, van Veldhuisen *et al.* (2009) showed no difference in outcome for patients with systolic dysfunction ($EF \leq 35$ percent) compared to those with diastolic dysfunction ($EF > 35$ percent) as the cause of heart failure, using the vasodilating β -blocker nebivolol.

All of these trials are consistent with the view that benefits of most new therapies for cardiac failure are a consequence of reduction in early wave reflection, and are manifest equally in patients with 'systolic' heart failure and in 'diastolic' heart failure. Benefit is manifest in the former as increase of flow from the heart (stroke volume) and in the latter as decrease in aortic and LV late systolic pressure augmentation. Views were predicted on the basis of 'systolic' and 'diastolic' heart failure being extremes of myocardial dysfunction, with ejection against load being predominantly limited in the former, and filling during diastole being predominantly limited in the latter – and with most older subjects with heart failure falling in between (O'Rourke, 1994; Westerhof and O'Rourke, 1995).

DRUG TRIALS FOR PREVENTION OF CARDIAC FAILURE

These trials in hypertensive and high-risk subjects are described in Chapter 21. In comparison to placebo, active therapy has reduced incidence of cardiac failure by almost 50 percent. ACEI drugs have shown benefit as have calcium channel antagonists, diuretics, and β -blockers. Again, with respect to cardiac failure prevention, benefits are attributable to reduction in LV afterload, from a combination of arterial and arteriolar dilation. Strong support for a specific ACEI action independent of blood pressure reduction fails to consider 24-hour change in blood pressure (in the HOPE trial) (O'Rourke, 2000b; Yusuf *et al.*, 2000), or the greater reduction in central then brachial pressure with these agents (Griffin *et al.*, 2003; Kurtz, 2003; London *et al.*, 2004a; Hirata *et al.*, 2005b; Williams *et al.*, 2006; Matsui *et al.*, 2009). The trials are further thrown into question with the recent publication of the ALLHAT trial which was funded independently by the US National Institutes of Health. This showed no benefit for ACEI drugs in prevention of cardiac failure over the diuretic or CCB, despite similar reduction in brachial artery pressure. In the HOPE (Yusuf *et al.*, 2000), LIFE (Dahlöf *et al.*, 2002), ACCOMPLISH (Jamerson *et al.*, 2008), and ASCOT (Dahlof *et al.*, 2005) trials, cardiac failure was reduced by adding an ACEI, ARB, or CCB drug to the treatment of high risk patients, and was independent of reduction in brachial pressure. CAFE (Williams *et al.*, 2006), a substudy of ASCOT, showed that benefit of amlodipine against atenolol was attributable to reduction in aortic systolic pressure. Other smaller studies showed that a CCB was superior to a diuretic in reducing

aortic systolic pressure in patients on an ARB with no difference in brachial cuff pressures (Matsui *et al.*, 2009) providing an explanation for ACCOMPLISH on the same basis as CAFE. Jiang *et al.* (2007) showed superiority of an ACEI over a diuretic in reducing aortic systolic pressure. This provides a challenge to ALLHAT, suggesting that the diuretic may have suppressed congestive features, and so delayed the diagnosis of cardiac failure.

Comments made here are on mechanism of action, not on the value of modern drugs. Most patients treated for hypertension require a combination of drugs, and the ACEI and ARB drugs, and CCB are very effective and very well tolerated. With respect to ALLHAT, it must be said that results might have been anticipated from the theoretic principles exposed here, and from Staessen's classic meta-analyses (Staessen *et al.*, 2001b; Staessen *et al.*, 2003; Staessen *et al.*, 2010).

OTHER THERAPIES

Surgical treatment to reduce LV load is at an early stage. This is aimed at reducing the stiffness of dilated degenerate aorta in elderly patients with an elastic wrap. The aim is to simultaneously decrease diameter of the dilated aorta while converting its elastic properties to those of the material used to encircle the aorta (Avolio *et al.*, 2002; Byrom *et al.*, 2010). This represents the reverse of a procedure undertaken in dogs, wherein rigid Lucite ferrules were inserted over the brachiocephalic artery and descending thoracic aorta, between the intercostal arteries (O'Rourke, 1968). The procedure resulted in marked increase of ascending aortic characteristic impedance and external LV work (O'Rourke, 1968). We are currently actively pursuing this topic with our bioengineering and cardiac surgical colleagues in Sydney and the United States.

Surgical therapy to decrease LV tension is applied in the excision of LV aneurysms, and in the Batista operative procedure. These procedures are finding their place in clinical practice (Ratcliffe, 2000; Starling *et al.*, 2000). It makes good sense in the treatment of cardiac failure to reduce ventricular dimensions, and thereby reduce ventricular tension, but it never makes good sense to remove normally contracting myocardium. Best results are expected where maximal reduction is achieved in ventricular dimensions with minimal myocardial loss – as in ventricular aneurysmectomy.

In patients with cardiac failure, left bundle branch block can markedly impair cardiac efficiency, since the last part of the LV to contract does so as the first relaxes, and there is a component of side to side 'sloshing'. With normal ventricular activation increased from 0.10 msec to perhaps 0.20 msec, the last part to contract (usually the LV free wall) does so just as the reflected wave returns from the periphery – so predisposing to premature termination of contraction (see above and Figs 14.24 and 15.7).

Biventricular or just LV pacing can improve mechanical synchronization and markedly improve LV function, prognosis, and symptomatic status (Kass *et al.*, 1999; Leclercq *et al.*, 2002; Kass, 2007a; Kass, 2008; Derval and Jais, 2010; Marwick and Grimm, 2010; Spragg *et al.*, 2010; van Bommel *et al.*, 2010).

Specific therapies for the heart are suggested from the findings that mitotic activity has been observed following infarction in surviving myocytes, and that myocytes may be stimulated to reproduce and form more contracting cells (Rosenthal, 2001). Clinical trials are under way with blood-derived and bone marrow progenitor cells in the hope of regenerating lost myocardial cells (Assmus *et al.*, 2002; Parmacek and Epstein, 2009).

Cardiac transplantation remains the ultimate surgical procedure for treatment of end-stage cardiac failure and long-term results at our own hospitals are excellent, with five-year survival in the region of 70 percent and long-term mortality approaching 3 percent per annum. An example of the observed hemodynamic changes associated with heart transplantation is shown in Fig. 15.8 (Nichols *et al.*, 2008). The new pump is now able to generate the force required to overcome the late systolic pressure boost seen in the central aortic pressure wave. Mechanical heart devices are also frequently used on account of limited clinical availability of donor hearts. Results of the REMATCH trial (Rose *et al.*, 2001) showed superiority over intensive medical therapy in end-stage cardiac failure

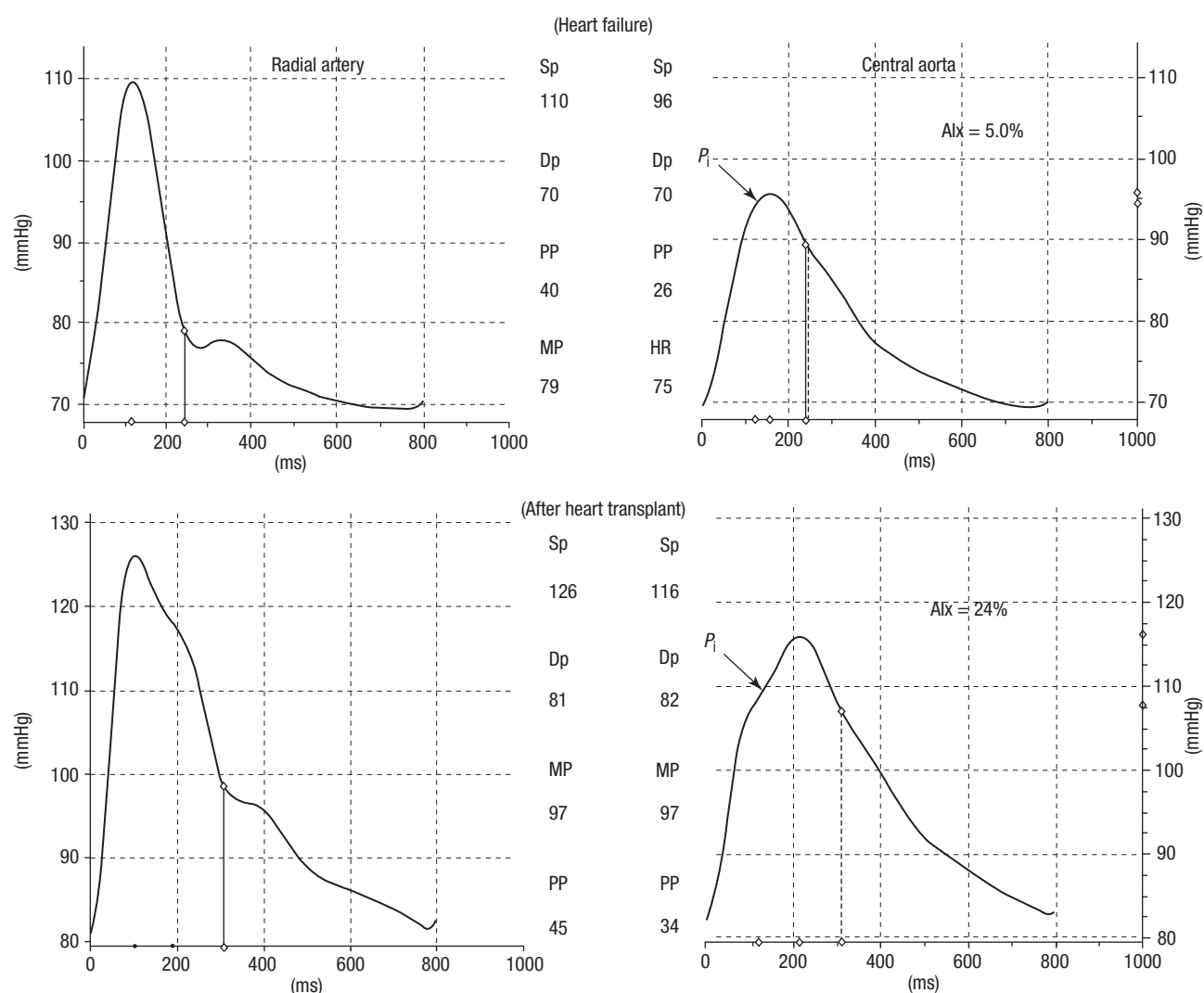


Figure 15.8 Measured radial artery (left) and synthesized aortic (right) pressure waves recorded in a heart failure patient (53 years of age) before (top) and after heart transplantation (bottom). The failing LV could not generate enough force to overcome the late systolic pressure boost usually seen at this age. This reduced contractile force resulted in an abbreviated ejection duration (ED) (240 msec) and a low augmentation index (Alx). After heart transplant, the LV could again generate the necessary force to overcome the late systolic pressure boost which caused an increase in ED (309 msec) and Alx. Heart rate was similar before and after transplant (75 beats/min). Dp, diastolic pressure; HR, heart rate; MP, mean pressure; PP, pulse pressure; Sp, systolic pressure. Reproduced from Nichols *et al.* (2008)

as did Deng (2004) and Lewis *et al.* (2004). Even with such good results of heart transplantation, prevention of ventricular damage is still seen increasingly as a goal in management of coronary disease. As well as prevention of coronary disease or control of its progression, this entails better management of the 'infarcting' stage with prompt revascularization by lytic agents or angioplasty and better

late management of myocardial infarction, with limitation of infarct expansion and progressive ventricular dilation through judicious rehabilitation (White *et al.*, 1987; Jugdutt *et al.*, 1988; Jessup *et al.*, 2009) and use of drugs which slow ventricular expansion, and ventricular hypertrophy, including ACEI, ARB, CCB β -blockers, and nitrates, as previously described.

The pulmonary circulation

Pressure and flow waves in pulmonary arteries	359	Pulmonary artery pressure and flow waves in pulmonary disease	367
Pulse wave velocity and arterial stiffness	360	Simulation of disease in experimental animals	369
Pulmonary artery impedance in experimental animals	361	Pulmonary vascular impedance and right ventricular load	371
Pulmonary artery impedance in humans	365	Primary pulmonary hypertension	372

Like the systemic circulation, the pulmonary circulation receives the same blood flow from the heart with the same periodicity. There the similarity ends. The main functional difference was stressed by Harvey in 1628, that the pulmonary arteries carry venous blood and the ascending aorta carries red oxygenated blood. From the hemodynamic point of view, the main differences between the right and left heart circulations are as follows.

Anatomically:

1. Path lengths along vascular segments are much shorter.
2. Path lengths to corresponding segments of the right and left lungs are similar; there is not the same asymmetry as between the upper and lower parts of the body as in the systemic circulation (Fig. 16.1) (Singhal *et al.*, 1973; Horsfield, 1978; Piene, 1986; Milnor, 1989).
3. The main pulmonary artery exhibits tapering to its point of bifurcation (Attinger, 1963; O'Rourke and Milnor, 1971).
4. The major pulmonary arteries are elliptical in cross-section rather than round (Attinger, 1963).
5. Wall thicknesses of pulmonary arteries are less than those of systemic arteries (Patel *et al.*, 1960; Milnor, 1989).
6. There are no well-developed arterioles in the pulmonary circulation (Grover *et al.*, 1983; Milnor, 1989).

Functionally (Fig. 16.2):

1. The mean pulmonary arterial pressure is much lower – about one-sixth of systemic and increases with age (Davidson and Fee, 1990; Ghali *et al.*, 1992). Also, mean pulmonary artery is an independent predictor of mortality (Cooper *et al.*, 1991).

2. The pulmonary vascular resistance is also lower – about one-sixth of systemic and also increases with age (Davidson and Fee, 1990; Ghali *et al.*, 1992; Lankhaar *et al.*, 2006).
3. The pulse pressure is lower (Fig. 16.3) (Kussmaul *et al.*, 1988; Lankhaar *et al.*, 2006; Syed *et al.*, 2008; Chemla *et al.*, 2009).
4. The main pulmonary arterial distensibility is higher (Patel *et al.*, 1960; Gozna *et al.*, 1974; Piene, 1986;

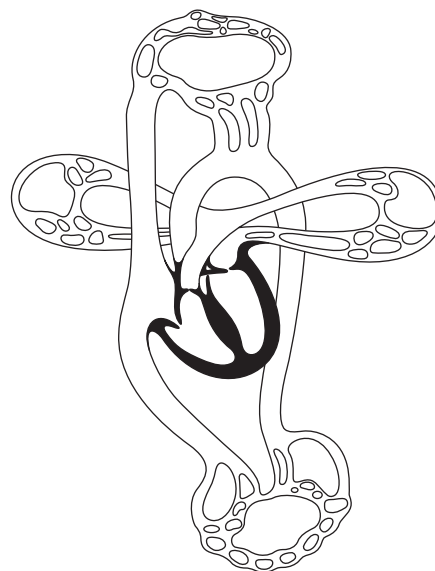


Figure 16.1 This logo of the Cardiovascular Systems Dynamics Society illustrates the point that path lengths to vessel terminations in the upper and lower parts of the body in the systemic circulation are quite dissimilar, whereas path lengths to vessel terminations and the capillary network in right and left lungs are similar. Logo shown by permission of the Secretary of the CSDS

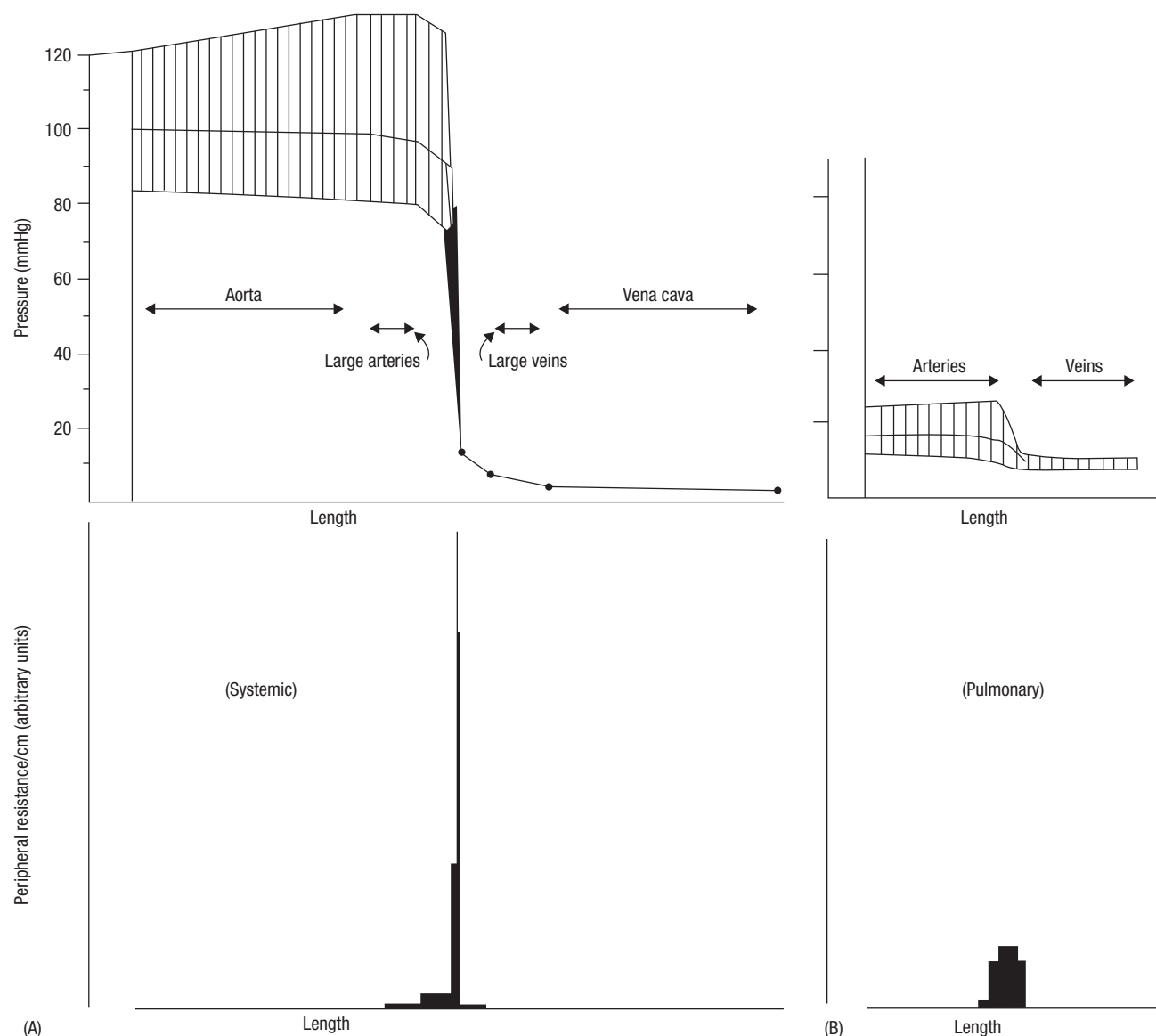


Figure 16.2 Top: Change in mean pressure and in pulsatile pressure generated by the left (A) and right (B) ventricles between the ascending aorta and vena cava, and between main pulmonary artery and pulmonary veins, respectively. Mean pressure falls only slightly in the large arteries, whereas pulsatile pressure is augmented peripherally (less so in pulmonary arteries) as a result of wave reflection. Mean pressure and pulse pressure fall steeply over a short length in the smallest arteries and arterioles as a result of the relatively high resistance (lower in the lungs) that these vessels present. Bottom: Resistance in corresponding segments of the systemic (A) and pulmonary (B) vascular bed. Reproduced from O'Rourke (1982b) and prepared from data of Schleier (1918)

- Slife *et al.*, 1990; Grant *et al.*, 1991; Fourie *et al.*, 1992; Sanz *et al.*, 2009a) and characteristic impedance (or stiffness) is lower (Lankhaar *et al.*, 2006) than in the ascending aorta.
5. The mean pressure in pulmonary veins and the left atrium is relatively high (see Fig. 16.3) (Grover *et al.*, 1983; Milnor, 1989).
 6. Pulmonary arteries do not exhibit increasing stiffness between central and peripheral sites, as do the systemic arteries (Patel *et al.*, 1960; Attinger, 1963).
 7. The peripheral reflection coefficient in the pulmonary arterial system is lower (Bergel and

Milnor, 1965; Bergel *et al.*, 1967; Reuben *et al.*, 1971; Van den Bos *et al.*, 1982; Murgo and Westerhof, 1984; Kussmaul *et al.*, 1992).

8. Low perfusion pressure and hydrostatic effects result in differential perfusion of the lung, with little or no perfusion of the apex in the erect position at normal pressure, and pulmonary vascular resistance that is dependent on arterial pressure (Grover *et al.*, 1983).
9. A substantial (20–30 percent) proportion of pressure and flow pulsation extends through normal lungs to the left atrium (Bergel *et al.*, 1967).

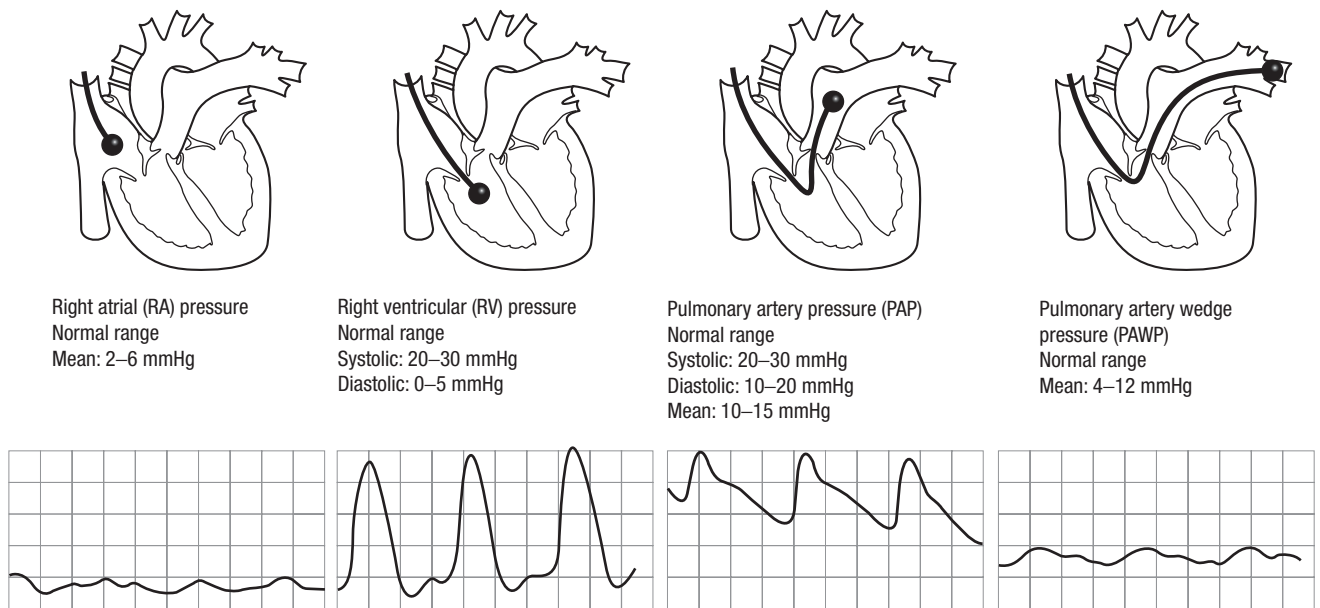


Figure 16.3 Pressure recordings in various chambers of the right heart circulation

From a clinical point of view, the pulmonary circulation has created great interest among pediatricians because of the frequency and significance of pulmonary vascular changes in congenital heart disease (Lucas *et al.*, 1975; Wilcox and Lucas, 1980; Radke *et al.*, 1985), and among adult cardiologists dealing with pulmonary hypertension in chronic rheumatic and thromboembolic disease (Milnor *et al.*, 1969; Reuben, 1970; Haneda *et al.*, 1983; Kitabatake *et al.*, 1983; Okamoto *et al.*, 1984; Chen *et al.*, 1990; Kussmaul *et al.*, 1992; Galie and Rubin, 2004; Gan *et al.*, 2007; Lankhaar *et al.*, 2008; Lam *et al.*, 2009) (see also *Journal of the American College of Cardiology* Supplement on the Fourth World Symposium 2009; 54 (Suppl. S): S9–S116, edited by Humbert M and McLaughlin V). The systemic circulation has held more interest for adult cardiologists and physicians dealing with hypertension, atherosclerosis, and other degenerative disease. This is beginning to change, with increasing interest in pulmonary hypertension and its management with prostacyclin and its analogs, with inhaled nitric oxide, and orally administered endothelin antagonists and phosphodiesterase inhibitors (Castlelain *et al.*, 2001; Nakayama *et al.*, 2001; Wilkens *et al.*, 2001; Zhao *et al.*, 2001). In contrast to the pulmonary circulation, less has been published on pulsatile pressure and flow phenomena in the systemic circulation of children (Endo *et al.*, 1994).

From a technical viewpoint, there are some peculiar difficulties in the interpretation of data obtained from the pulmonary circulation, especially during interventions designed to simulate arterial disease. These are less of a problem in the systemic circulation where the relative changes induced are far smaller. The mean arterial pressure can increase four-fold in pulmonary hypertension, whereas systemic hypertension is considered severe with a

25 percent increase in mean arterial pressure. Because the pulmonary artery is so distensible, a two-fold rise in mean pressure can cause marked dilatation of that artery; if the artery is constrained within a cuff-type flow transducer, as in chronic dog experiments, the mean flow velocity downstream will be considerably overestimated (O'Rourke, 1982b). In addition, high flow velocity and a disturbed velocity profile may produce Venturi effects with distortion of the pressure pulse downstream (see Chapter 10) (O'Rourke, 1968; Hopkins *et al.*, 1979; O'Rourke, 1980). Because the pressure pulse is of lower amplitude in the pulmonary than in the systemic circulation, it is more subject to distortion by Venturi effects and to artifacts. These points aside, there is general agreement on the patterns of pressure and flow waves and of vascular impedance in the pulmonary vessels of experimental animals and of humans under normal conditions and in human disease, and there is agreement that the pulmonary system behaves to a first approximation in a linear fashion with respect to pulsatile pressure–flow relationships (Bergel and Milnor, 1965; Reuben *et al.*, 1971; Piene, 1986), just as does the systemic system.

PRESSURE AND FLOW WAVES IN PULMONARY ARTERIES

Compared with the ascending aortic flow wave (Fig. 16.4), the flow wave in the main pulmonary artery is usually of lower amplitude with a longer duration of forward flow and later incisura. The amplitude of the first harmonic of the flow wave is usually greater than of all other harmonics combined, whereas in the ascending aorta the first harmonic is relatively

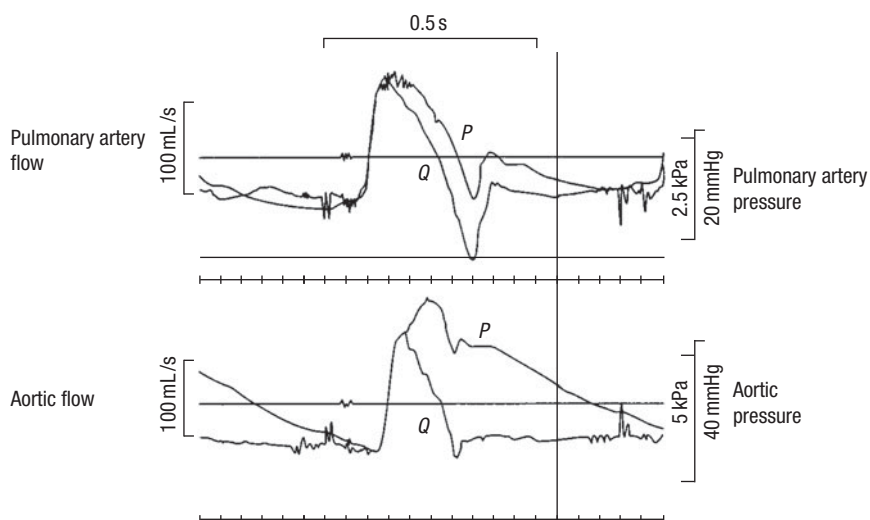


Figure 16.4 Simultaneously recorded pressure (P) and flow (Q) waves in the main pulmonary artery and in the ascending aorta of a dog. Reproduced from Van den Bos *et al.* (1982)

smaller and the higher harmonics are of relatively greater amplitude than in the pulmonary artery (Patel *et al.*, 1963a).

The pressure wave generated in the main pulmonary artery is likewise more rounded than in the ascending aorta and often looks surprisingly similar to the flow wave. Unlike flow, of course, the incisura is higher than the foot of the wave and pressure falls during diastole, but usually without any definite diastolic wave as usually seen in the ascending aorta. Both pressure and flow waves are more rounded due to minimal effect of wave reflections. These features explain why the impedance phase of the pulmonary artery is very much less than in systemic arteries.

In the more peripheral pulmonary arteries, the amplitude of the flow wave decreases as in systemic vessels, but there are no reciprocal flow oscillations or diastolic flow waves as usually seen in the brachiocephalic artery and descending aorta. The pressure wave appears to increase in amplitude between the central and peripheral arteries (Attinger, 1963; Reuben, 1970; Milnor, 1989), but to a far lesser degree than in the systemic circulation (see Chapter 10). The pressure wave is attenuated in the pulmonary capillaries, but flow and pressure are still pulsatile here. Flow pulsation can be measured in humans with the body plethysmograph (Reuben, 1970; Grover *et al.*, 1983). Some pulsation (believed to be around 5.0–10 percent of that in the pulmonary artery) passes through the capillaries into the pulmonary veins and contributes to pulsations in these vessels (Bergel and Milnor, 1965; Milnor *et al.*, 1966; Maloney *et al.*, 1968).

PULSE WAVE VELOCITY AND ARTERIAL STIFFNESS

The pulse wave velocity has been measured as foot-to-foot velocity or as apparent phase velocity (at high frequencies) from pressures recorded at central and peripheral sites (Attinger, 1963; Bargainer, 1967; O'Rourke and Milnor, 1971; Reuben *et al.*, 1971; Peng *et al.*, 2006; Bradlow *et al.*, 2007) or calculated from determinations of arterial

distensibility (Patel *et al.*, 1960; Milnor *et al.*, 1969; Gozna *et al.*, 1974). Recent studies by Peng *et al.* (2006) and Bradlow *et al.* (2007) used magnetic resonance imaging (MRI) to measure pulmonary artery pulse wave velocity in normal human subjects. Results are in general agreement, with values around 250 cm/s in dogs and humans at normal distending pressure. Such a value is about half that noted in the ascending aorta of experimental animals and young humans. The pulse wave velocity does not appear to increase in the periphery of the pulmonary arterial tree (as it does in the systemic arteries) (see Attinger, 1963) and in humans it appears to increase less with age than in the systemic circulation (Gozna *et al.*, 1974; O'Rourke, 1982b). The pulse wave velocity (or apparent phase velocity which is impedance in the absence of wave reflections and correlates directly with arterial stiffness) has been measured at different pulmonary distending pressures in animals and shown to change markedly with pressure. Bargainer (1967) reported a 56 percent increase in apparent phase velocity (from 267 to 417 cm/s) at high frequencies (where 'true' wave velocity is approximated), with doubling of the mean arterial pressure (from 12.4 to 29.6 mmHg) through infusion of serotonin in dogs. This change is similar to that calculated from changes in mechanical properties of the main pulmonary artery (Patel *et al.*, 1963a) with passive distension. Bargainer considered the change to be partly due to the effects of serotonin on the large arteries, and this view was supported by Reuben *et al.* (1971). An increase in the apparent phase velocity of similar degree has also been noted in patients with pulmonary hypertension (Milnor *et al.*, 1969; Chen *et al.*, 1990; Kussmaul *et al.*, 1992). Milnor considered this to be partially due to a direct effect on the large arteries, but it could have been purely a passive phenomenon resulting from greater stretch of the vessel wall with transfer of tension from elastic to collagenous components of the wall (see Chapter 4). More recently, Santana *et al.* (2005) measured pressure and diameter in sheep and calculated pressure-strain elastic

modulus (E_p) during passively induced acute pulmonary hypertension (mechanical occlusion). This maneuver produced an increase in mean pulmonary artery pressure of 41 percent (from 14.0 to 19.8 mmHg) and an increase in E_p of 104 percent (from 2.31 to 4.69×10^5 dyne/cm²).

There is some conflict in the literature (see Piene, 1986) on the relationship between wave velocity and characteristic impedance with change in arterial pressure. This only applied when the flow used for calculation of impedance was measured with a cuff-type flow probe applied to the pulmonary artery, and can (see below) be attributed to overestimation of the pulmonary flow velocity through failure to consider dilatation of the pulmonary artery at higher distending pressures. When characteristic impedance is expressed in terms of pressure divided by flow velocity in the unconstrained vessel (as dyne-s/cm³), results are completely congruous (O'Rourke, 1982a). (This subject is also discussed in Chapter 12).

PULMONARY ARTERY IMPEDANCE IN EXPERIMENTAL ANIMALS

Pioneering studies of pulmonary vascular impedance were performed by Caro and McDonald (1961) and by Patel *et al.* (1963a). These studies established the principal features of pulmonary vascular impedance. In the only study of its type ever reported, Caro and McDonald studied the perfused isolated pulmonary bed of rabbits, using a sinusoidal pump of variable frequency imposed on steady flow. Only the modulus was recorded at the time (Fig. 16.5). The results of Patel *et al.* (1963a) from dogs

were more precise and were determined in the conventional way from harmonic analysis of pressure and flow waves reported in the pulmonary artery of living animals. The most detailed study of pulmonary vascular impedance in normal dogs was conducted by Bergel and Milnor (1965) (Fig. 16.6), who also documented the pulmonary effects of serotonin. The same workers showed that inflation and deflation of the lung caused little change in pulmonary vascular impedance and that values in the unanesthetized animal were similar to those in anesthetized animals of similar size (Milnor *et al.*, 1966). They also studied the implications of impedance for the pulsatile and steady components of energy loss in the lungs (Milnor *et al.*, 1966) and pulmonary artery pressure wave transmission (Bargainer, 1967).

Patel *et al.* (1963a) and Bergel and Milnor (1965) noted differences between impedance patterns in the main pulmonary artery and ascending aorta. In the pulmonary artery, the resistive component was about one-sixth of that in the aorta, while the characteristic impedance was about one-half, with the result that the impedance modulus fell less steeply from its value at zero frequency to the next value at heart frequency. The phase was lower and fluctuated less than in the aorta. The first minimum of modulus was reached at about the same frequency as in the aorta of a similar-sized dog (around 3.0 Hz), but was followed by a definite maximum at twice this frequency, unlike in the ascending aorta. The differences were attributed to less wave reflection than in the ascending aorta and the presence of a single, functionally discrete reflecting site in the pulmonary vascular bed. This reflecting site was considered to represent the myriad of

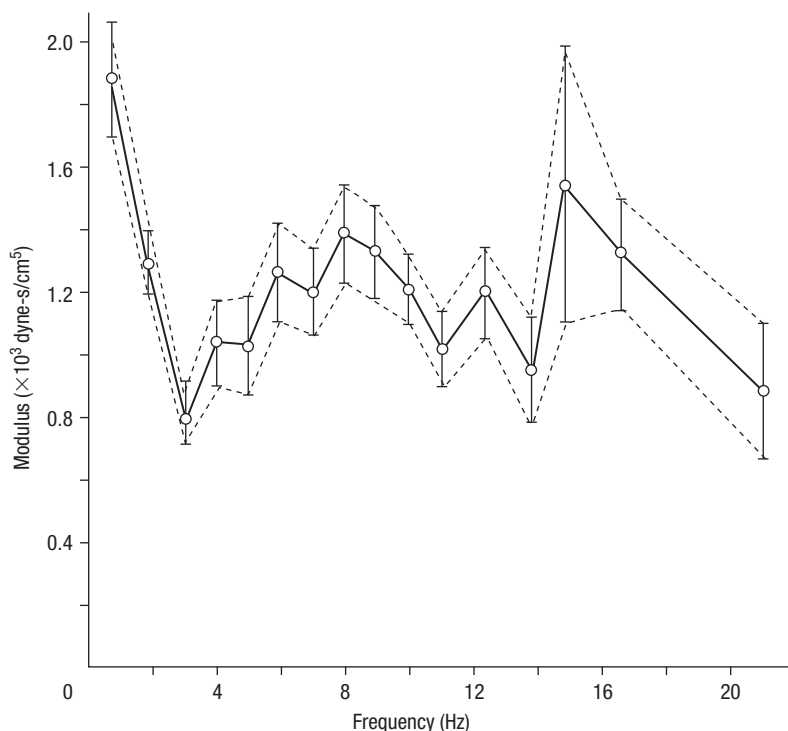


Figure 16.5 Modulus of vascular impedance in the main pulmonary artery of 24 isolated rabbit lungs, perfused with a sine wave oscillatory pump of variable frequency. Vertical bars show standard error of the mean. Reproduced from Caro and McDonald (1961)

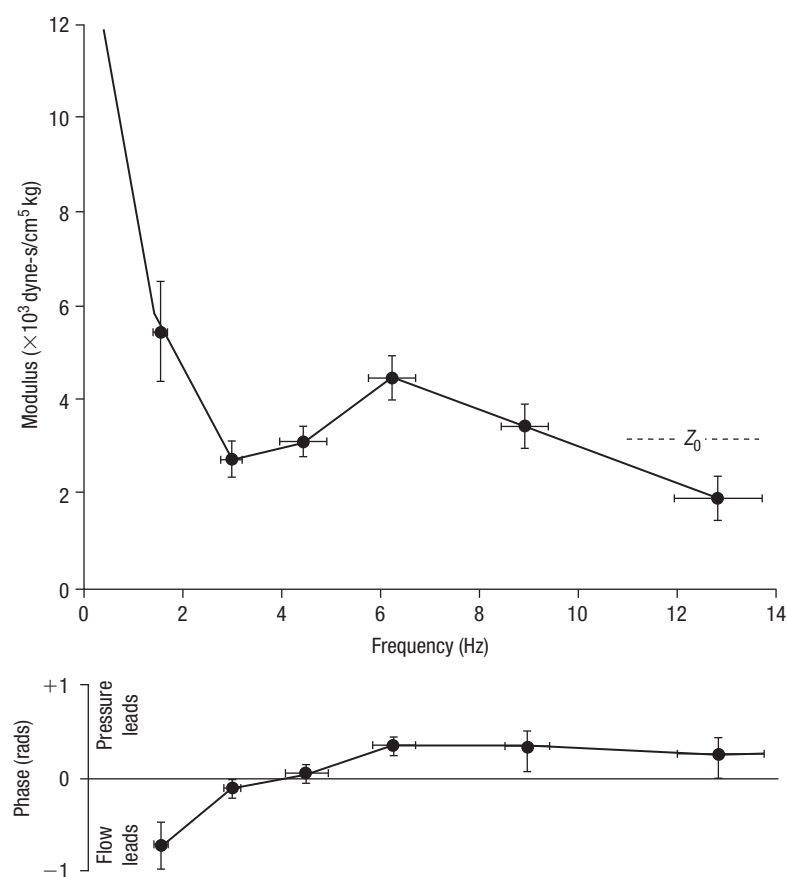


Figure 16.6 Modulus and phase of pulmonary vascular impedance in 13 anesthetized open-chest dogs, determined by Fourier analysis of pressure and flow waves recorded in the main pulmonary artery. Values are mean $\pm$ standard errors of the mean. Input impedance at zero frequency (mean pulmonary artery pressure divided by mean flow) was 14.2×10^3 dyne-s/cm⁵ kg, and estimated characteristic impedance (Z_0) 3.1×10^3 dyne-s/cm⁵ kg. Reproduced from Bergel and Milnor (1965)

individual reflecting sites in the lungs and principally those in the lung periphery. Bergel and Milnor (1965) and Reuben *et al.* (1970) calculated the global reflection coefficient as approximately 0.3, which is approximately one-third that of the systemic circulation (see Chapter 12). The similarity of the impedance minimum in the ascending aorta and pulmonary artery indicated a similar timing of wave reflection from peripheral sites in the systemic and pulmonary circulations. It appears that the shorter distances to reflecting sites in the lungs are completely offset by the lower pulse wave velocity in the pulmonary arterial tree (see Chapter 13). Bergel and Milnor (1965) studied the effects of pulmonary vasoconstriction on pulmonary vascular impedance. Vasoconstriction was induced by intravenous infusion of serotonin and was associated with a marked increase in the mean pulmonary artery pressure. Serotonin infusion was associated with a shift of impedance curves to the right (Fig. 16.7), with an increase in the frequency at which the modulus reached its minimal value and the phase crossed zero; there was an increase in the resistive value, but no increase in the characteristic impedance. Bergel and Milnor (1965) attributed this change to early return of wave reflection, as well as peripheral vasoconstriction. Equating characteristic impedance with pulse wave velocity, and considering this not to have changed, these workers suggested that reflecting sites had shifted closer to the heart (i.e. that vasoconstriction was

occurring in the larger pulmonary arteries). Further studies in the same laboratory, reported by Bargainer in 1967, suggest a different interpretation. Bargainer showed that serotonin infusion causes a substantial (56 percent) increase in pulmonary arterial wave velocity; this by itself was almost sufficient to explain the observed findings (i.e. that they arose from more rapid return of reflected waves from the same sites); constriction of more central arteries may, however, have contributed. Investigating the same phenomenon, Reuben *et al.* (1971) considered the effects of serotonin to arise from more proximal reflection and more rapid pulse wave velocity; the latter was considered to be the dominant mechanism in pulmonary hypertension, however caused (Reuben and Kitchen, 1975; Piene, 1986). Recent studies in sheep indicate that acute pulmonary hypertension induced by mechanical occlusion of the left pulmonary artery increases arterial stiffness and produces a rightward shift in the impedance spectra. This modification in large artery elastic properties caused an increase in characteristic impedance (Z_0) and reflected wave amplitude (Santana *et al.*, 2005; Grignola *et al.*, 2007).

Reference has been made to the apparent paradox of increases in the pulmonary artery pressure, in the pulse wave velocity, and in the impedance minimum, without a change in the characteristic impedance. This is also seen in the systemic circulation with norepinephrine infusion and has been reported for the pulmonary artery during

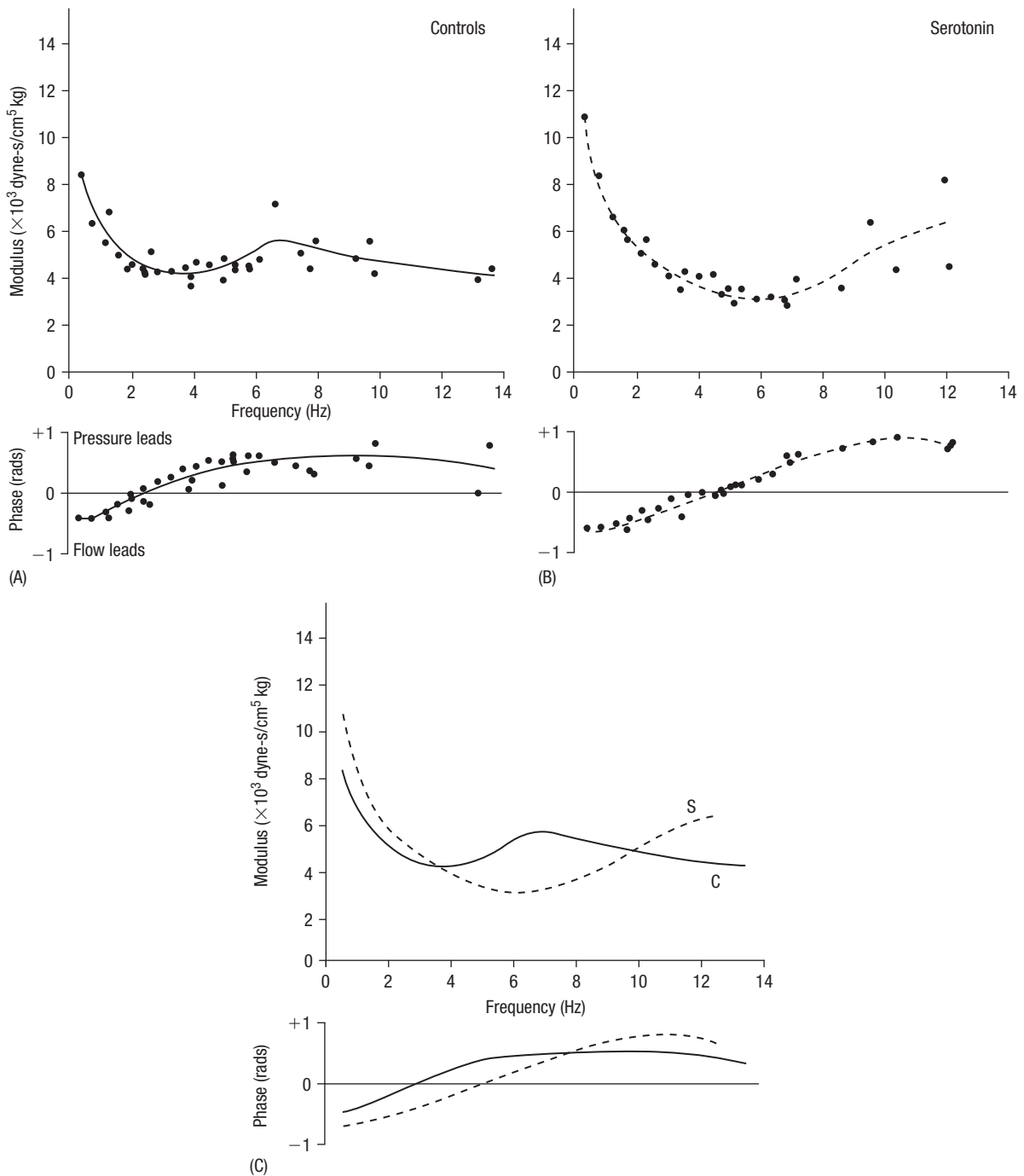


Figure 16.7 Modulus and phase of pulmonary vascular impedance in a dog with complete heart block and with the heart paced at different frequencies (A) under control conditions and (B) during intravenous infusion of serotonin (0.5 mg/kg/min). Lines are drawn to approximate the average impedance values (solid for control (C), dotted for serotonin (S)) and are superimposed in (C). Reproduced from Bergel and Milnor (1965)

infusion of serotonin (Bergel and Milnor, 1965; Reuben *et al.*, 1971; Van den Bos *et al.*, 1982) and with injection of microspheres (Elkins *et al.*, 1974; Calvin *et al.*, 1985). Womersley's relationship (see eqn 7.28), which simplifies to $Z_0 = \rho c_0$ (the Water-Hammer equation) in large arteries at high values of α , applies to Z_0 measured in terms of linear flow velocity. When (as in human studies) an electromagnetic flow catheter is used and the artery is free

to dilate, the similarity is obvious, but it is not, if one utilizes the cross-sectional area of the artery enclosed within a rigid cuff transducer and calculates flow velocity from this rather than in the unconstrained artery downstream. When allowance is made for arterial dilatation with increased distending pressure, and for a relative decrease in the flow velocity within a distended vessel immediately downstream from a flow transducer, the

characteristic impedance can be shown to increase to a similar degree as pulse wave velocity (O'Rourke, 1982a; O'Rourke, 1982b).

Van den Bos *et al.* (1982) determined the vascular impedance in the main pulmonary artery and ascending aorta of chronically instrumented dogs and examined the concept that major differences between the patterns of each, and of the contour of ascending aortic and pulmonary arterial pressure waves, were attributable to differences in the intensity of wave reflection. When nitroprusside was infused intravenously to reduce the arterial pressure and cause peripheral vasodilatation, aortic impedance patterns and pressure waves came to resemble those normally seen in the pulmonary arteries; this intervention markedly reduced the measured wave reflection. When serotonin was infused intravenously to increase the pulmonary artery pressure and cause pulmonary vasoconstriction, pulmonary impedance patterns and pressure waves came to resemble those normally recorded in the ascending aorta; this intervention markedly increased measured pulmonary wave reflection. Findings are illustrated in Fig. 16.8.

Impedance patterns in the ascending aorta and pulmonary artery of the same animals (dogs) (Patel *et al.*, 1963a; O'Rourke, 1965a; Van den Bos *et al.*, 1982) show that the first minimum occurs at almost the same frequency (around 3.0 Hz) and just above the normal heart

rate frequency. The similarity of impedance frequencies implies that wave reflection returns with the same timing in both, indicating that differences in arterial lengths and reflection site distances in the pulmonary and systemic circulations are balanced by corresponding differences in pulse wave velocity, suggesting that this balance is of benefit to normal cardiac function. Milnor *et al.* (1966) showed that patterns of impedance in conscious, unanesthetized dogs are similar to those in the anesthetized open-chest preparation.

Elkins and Milnor (1971) studied the effects of exercise on pulmonary vascular impedance. The changes seen were explicable on the basis of passive distension of pulmonary vessels, with increasing arterial pressure reducing compliance of the large pulmonary arteries. Greater fluctuation of phase during exercise was difficult to explain, especially in the presence of decreased fluctuations in the impedance modulus.

The effect of a change in body size on pulmonary vascular impedance was determined by Greenwald *et al.* (1985) (Fig. 16.9). Measurements of pulmonary artery pressure and flow were taken in 27 anesthetized, open-chest pigs ranging in age from 1.0 hour to 4.0 weeks. Over this period, there had been a 7.4-fold increase in body weight (from mean 1.45 to 10.8 kg). The impedance modulus (expressed in terms of volume flow) decreased substantially, although to a greater degree than would have

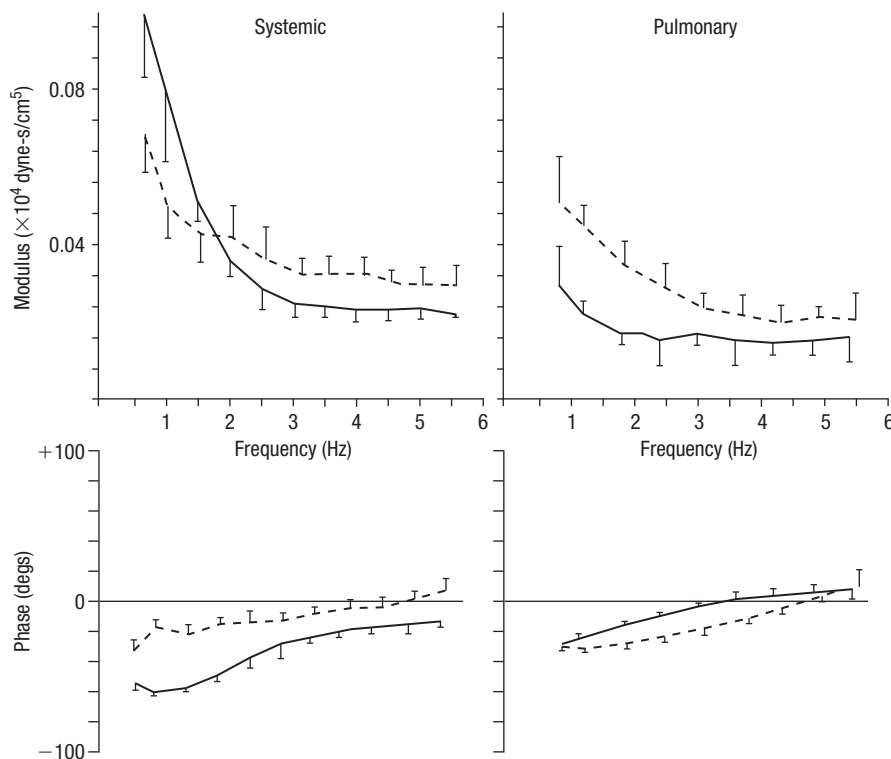


Figure 16.8 Input impedance of the systemic (left) and pulmonary (right) circulations before (solid lines) and during the infusion of the vasoactive agent – nitroprusside (50–200 mg/min) for the systemic data, and serotonin (0.5–0.75 mg/min) for the pulmonary data (broken lines). For an explanation, see the text. Reproduced from Van den Bos *et al.* (1982)

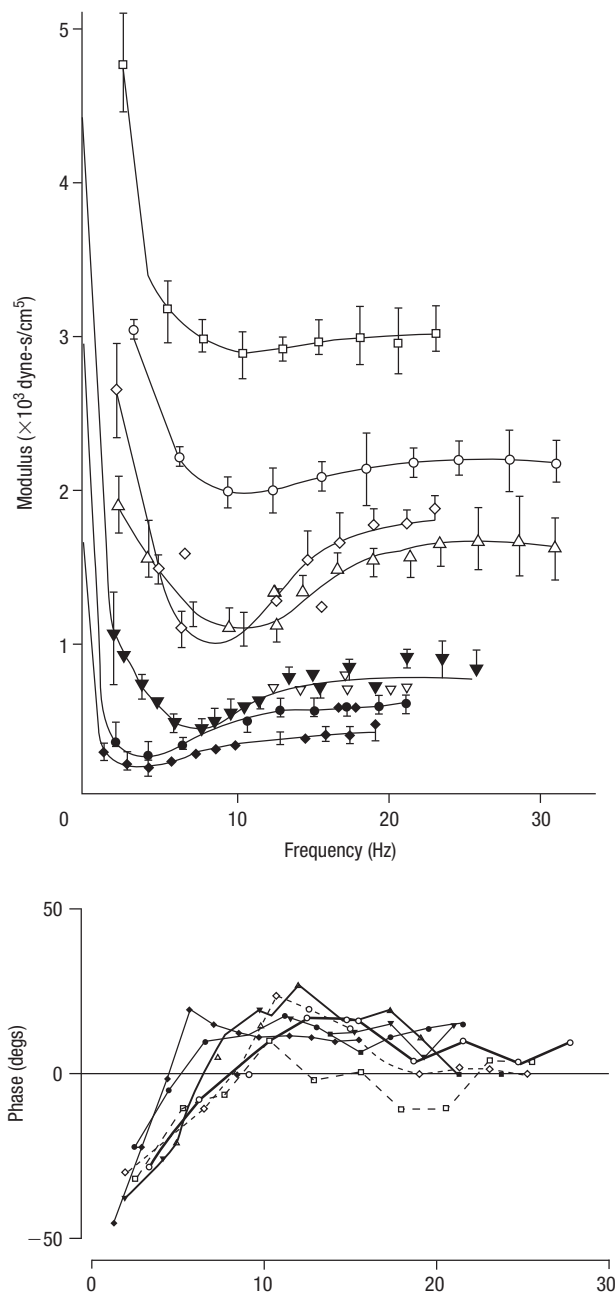


Figure 16.9 Changes in pulmonary vascular impedance modulus and phase with growth in a series of pigs reported by Greenwald *et al.* (1985). 0 H Pigs ($\diamond$) were studied at birth, 4 W pigs ($\blacklozenge$) at 4 weeks after an average weight gain of from 1.45 to 10.8 kg. Other studies were conducted at intermediate periods: ($\circ$) 12 H; ($\square$) 24 H; ($\triangle$) 48 H; ($\blacktriangledown$) 1 W; ($\bullet$) 2 W. The modulus units are newton meters; the phase units are degrees. H, hour; W, week

been predicted from the increase in body size and pulmonary blood flow, but changes in the frequency of the modulus minimum and phase crossover were of the order predicted (10–4.0 Hz and 9.0–4.0 Hz, respectively). At all ages, the calculated reflecting site corresponded to the lung periphery. Radke *et al.* (1984) pointed out that in

humans the frequency of the impedance minimum and phase crossover is dependent on age and body size.

PULMONARY ARTERY IMPEDANCE IN HUMANS

Studies in 'normal' adult humans have been reported by Milnor *et al.* (1969), Mills *et al.* (1970), Murgo and Westerhof (1984), Chen *et al.* (1990), Slife *et al.* (1990), Huez *et al.* (2004) and by Noda *et al.* (2006). Data were obtained from patients undergoing diagnostic cardiac catheterization who were found to have no pulmonary vascular disease, and no or trivial heart disease. Milnor *et al.* (1969) measured flow with Womersley's differential pressure technique (see Chapter 6), while other workers used a catheter-mounted electromagnetic flow transducer. Impedance patterns in humans are similar to those in dogs (Fig. 16.10). The minimal value of impedance modulus and of phase crossover in humans occurs at a similar frequency to that in dogs, despite the considerable difference in size of body and of lungs. These features are different from those seen in pigs of different size (see Fig. 16.9) and suggest that in adult humans the pulse wave velocity in the pulmonary artery is higher than that in dogs. Similar findings have been noted for the human systemic circulation, where the inappropriately high value of impedance minimum and phase crossover in the ascending aorta has been attributed to asymptomatic arterial degeneration with age (see Fig. 16.11). Murgo and Westerhof (1984) were able to determine pulmonary and aortic input impedance in the same subjects (Fig. 16.10), as have Van den Bos *et al.* (1982) in dogs. As in dogs, the first minimum of the impedance modulus and of phase crossover occurred at similar frequency in the ascending aorta and pulmonary artery, indicating similar timing of wave reflection, with shorter path lengths and reflection site distance in the pulmonary circulation compensated for by relatively lower pulse wave velocity.

Murgo and Westerhof (1984) settled a question raised by other authors, that pulmonary vascular impedance may be altered significantly by lung inflation or deflation; they were unable to show any significant change with normal breathing.

Milnor *et al.* (1969), Yin *et al.* (1983b), Chen *et al.* (1990), and Kussmaul *et al.* (1992) determined the pulmonary vascular impedance in patients with pulmonary hypertension secondary to mitral stenosis or left ventricular (LV) failure. Patterns of vascular impedance in pulmonary hypertension were similar to those seen with serotonin infusion or pulmonary embolization in experimental animals, there being an increase in the resistive component of modulus, an increase in the frequency of the first minimum of the modulus and of phase crossover, an increase in characteristic impedance and, where measured, an increase in phase velocity as well. In these studies, changes in wave velocity and characteristic

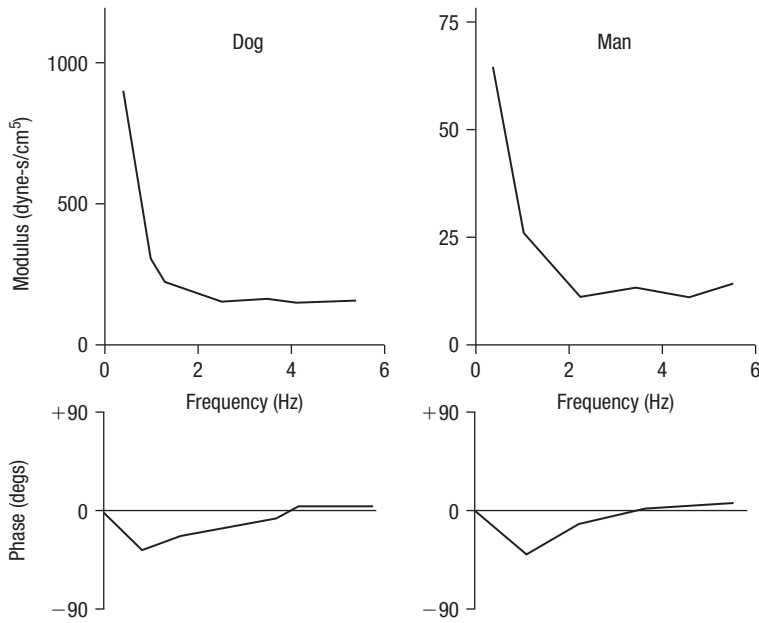


Figure 16.10 Average values of modulus and phase of pulmonary vascular impedance in a group of six dogs weighing 21–43 kg (left) and in a group of ten normal adult humans weighing 69–94 kg (right). Data are from Van den Bos *et al.* (1982) for dogs, and from Murgu and Westerhof (1984) for humans

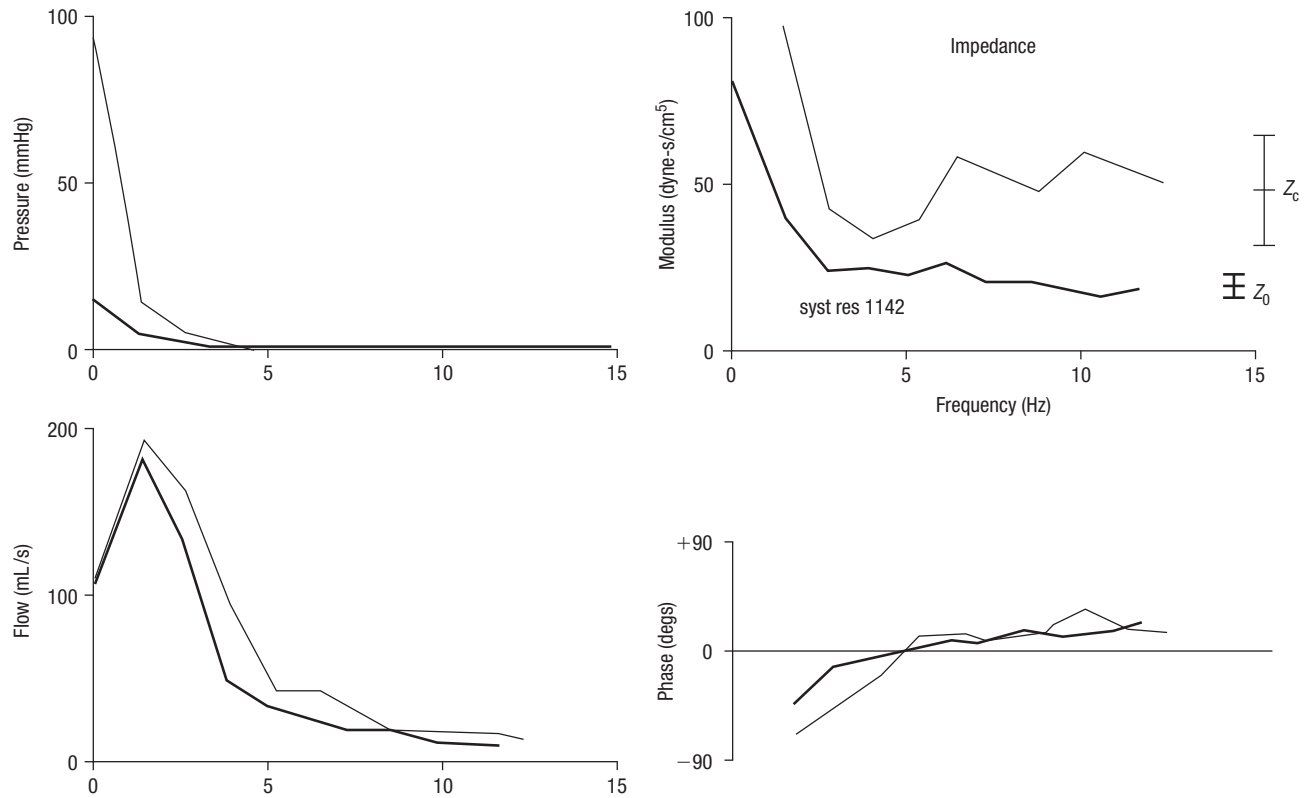


Figure 16.11 Comparison of systemic and pulmonary vascular impedance in humans. The left panels show average modulus values of the first eight pressure (top) and flow (bottom) harmonics in the pulmonary artery and ascending aorta, from which impedance was calculated. The right panels compare modulus (top) and phase (bottom) of impedance in the ascending aorta (thin line) and pulmonary artery (thick line). Z_c , characteristic impedance. Reproduced from Murgu and Westerhof (1984)

impedance were completely congruous, probably because flow was measured with a catheter and not with a device that constrained the vessel wall (see above). Kussmaul *et al.* (1988) determined pulmonary vascular impedance at rest and during exercise in a group of

patients with ischemic heart disease and compared the results with a group of subjects free of detectable heart, lung, or vascular disease. There was no significant difference in characteristic impedance at rest between the two groups, but mean pulmonary pressure and characteristic

impedance increased significantly only in the group with ischemic heart disease during exercise; this was not accompanied by any change in pulmonary vascular resistance. Exercise led to a substantial increase in pulsatile and total hydraulic power in the group of patients with ischemic heart disease. Results showed that exercise-induced ischemia increased pulsatile hydraulic load to the right ventricle and impaired the hydraulic efficiency of right ventricular–pulmonary artery coupling.

Lucas, Wilcox, Radke, and co-workers from North Carolina determined pulmonary vascular impedance from measurements of pressure and flow taken during operation for atrial or ventricular septal defects in infants and children. Interpretation was aided by computer simulation studies using Milnor's (1989) model of the pulmonary arterial tree. In children with normal pulmonary arterial pressure, impedance patterns were similar to those in dogs; they were similar, too, to those determined in human adults despite the difference in size. In a group of five six-year-old children (Lucas *et al.*, 1975), the minimum of the impedance modulus was seen at 2.0–4.0 Hz. In infants less than 18 months (Lucas *et al.*, 1975), the impedance minimum and phase crossover was around 5.0 Hz. Assuming a similar position for reflecting sites in infants, children, and adults, these findings indicate that the pulse wave velocity must be markedly lower in infants and children than in adult humans. The changes in pulmonary arterial distensibility noted by Gozna *et al.* (1974) appear insufficient to account for these findings. In the study by Lucas *et al.* (1975), impedance patterns were similar before and after closure of an atrial septal defect, indicating that the properties of large arteries did not change at the time of surgery, when mean pressure remained normal.

These workers explored the possibility that impedance patterns might be of value in predicting the presence of secondary structural lung disease in infants and children with pulmonary hypertension due to left-to-right shunt. This was not confirmed. The major determinants of pulmonary vascular impedance patterns appear to be body size and age (Radke *et al.*, 1984) and the level of pulmonary arterial pressure (Wilcox and Lucas, 1980; Radke *et al.*, 1985).

In children with pulmonary hypertension, there are marked increases in characteristic impedance and in the frequency of the impedance minimum and phase crossover, as well as in pulmonary vascular resistance and pulmonary artery pressure (Fig. 16.12).

PULMONARY ARTERY PRESSURE AND FLOW WAVES IN PULMONARY DISEASE

Using a pulsed-wave Doppler (Kitabatake *et al.*, 1983; Huez *et al.*, 2004; Weinberg *et al.*, 2004; Hunter *et al.*, 2008) or MRI technique (Lankhaar *et al.*, 2006), pulmonary artery flow velocity has been recorded non-

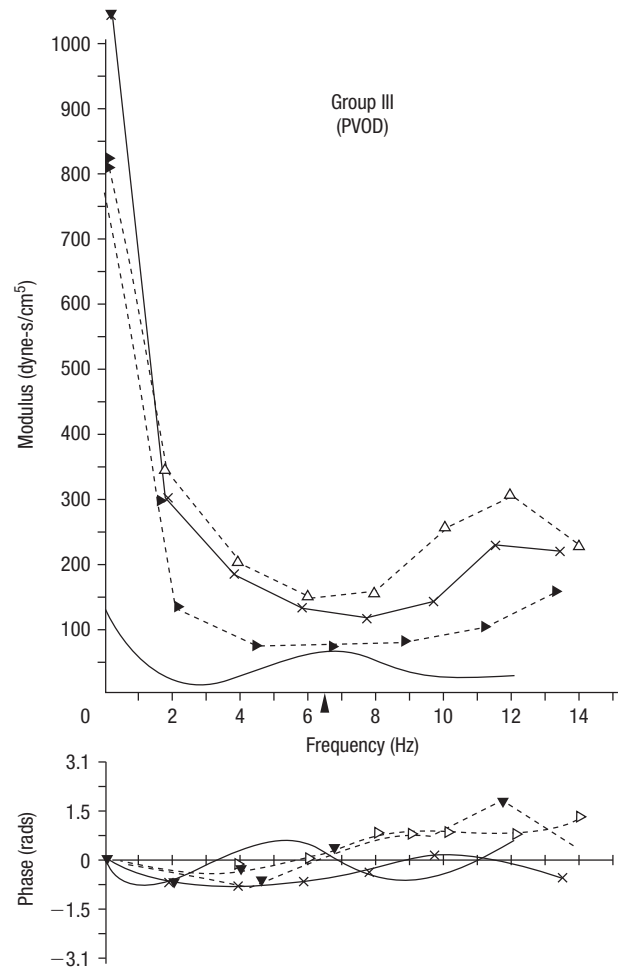


Figure 16.12 Impedance modulus and phase in three children aged two to six years with ventricular septal defect and pulmonary hypertension (mean pulmonary artery pressure 54 mmHg), compared with (solid lines) impedance in a model of the pulmonary arterial tree of normal children based on impedance data of 25 children with atrial or ventricular septal defect but without pulmonary hypertension. PVOD, pulmonary vascular obstructive disease. Reproduced from Wilcox and Lucas (1980)

invasively in adults and children with pulmonary hypertension. Both techniques have shown characteristic changes in the pulmonary artery flow velocity pattern in this condition, a decrease in time to peak flow, and either a concavity to the right or a notch in late systole during deceleration (Lankhaar *et al.*, 2006), with duration of systole often abbreviated (Kitabatake *et al.*, 1983) (Figs 16.13 and 16.14). The flow pattern described is similar to that in the aorta of patients with heart failure caused by systolic LV dysfunction (see Chapters 14 and 15). Kitabatake *et al.* (1983) showed a close correlation between time to peak velocity and mean pulmonary artery pressure (Fig. 16.15) and suggested their method as a means of determining pulmonary artery pressure non-invasively. This has not been taken up in routine echocardiographic

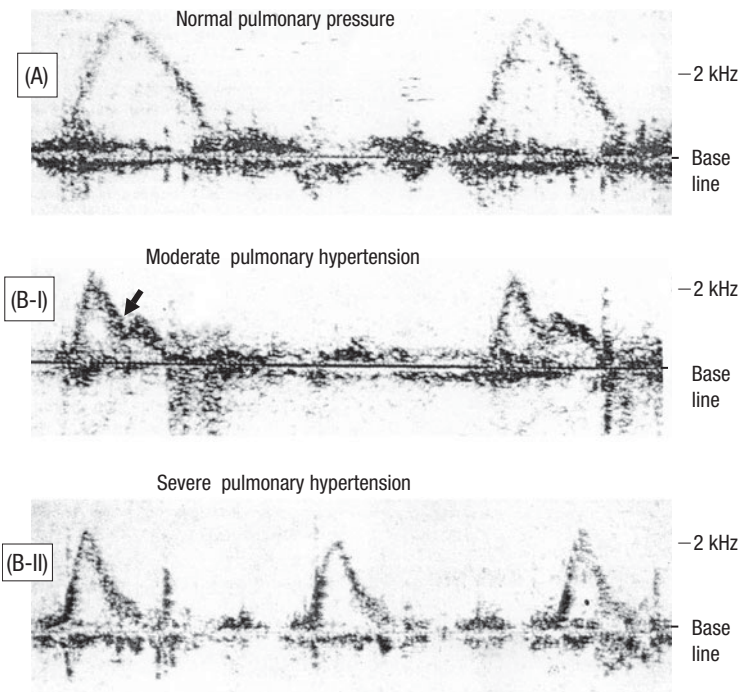


Figure 16.13 (A) Representative flow velocity pattern in the right ventricular outflow tract in a patient without pulmonary hypertension, demonstrating the dome-like contour in systole, with a peak level in mid-systole. (B) Flow velocity patterns in the right ventricular outflow tract in pulmonary hypertension. I. Flow velocity pattern in a patient with moderate pulmonary hypertension (mean pressure 38 mmHg). Note the mid-systolic notching in the deceleration phase of the flow velocity wave during ejection. II. Flow velocity pattern in a patient with severe pulmonary hypertension (mean pressure 68 mmHg), showing a rapid acceleration and deceleration without mid-systolic notching. From Kitabatake *et al.* (1983)

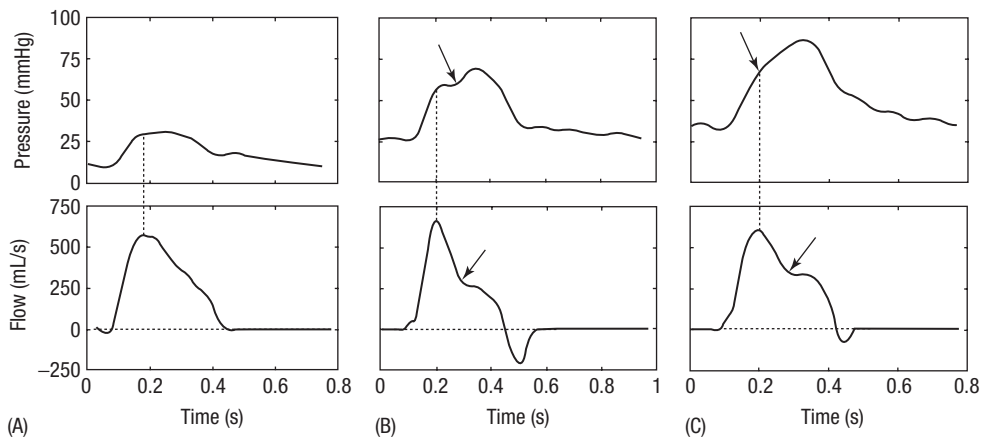


Figure 16.14 Pressure and flow velocity (magnetic resonance imaging (MRI)) waves recorded in the main pulmonary artery. Note that diastolic flow velocity was set to zero. The arrows represent the inflection point on the pressure wave and notching on the deceleration phase of the flow velocity wave. (A) No pulmonary hypertension; (B) chronic thromboembolic pulmonary hypertension; (C) idiopathic pulmonary hypertension. Reproduced from Lankhaar *et al.* (2006)

studies. Notching of the pulmonary flow velocity wave during deceleration in late systole has also been observed by those who have recorded phasic flow velocity directly from the pulmonary artery in pulmonary hypertension with the electromagnetic catheter technique (Fig. 16.16). These findings are intriguing since they raise the suggestion that the phenomenon is likely an expression of early wave reflection, with this causing deceleration of flow in the presence of relative or real right ventricular myocardial weakening (Figs 14.10, 14.11, and 14.12) (Miyashita *et al.*, 1994). In these studies, notching or early deceleration of flow corresponded to augmentation of pulmonary artery pressure in late systole, which supports a suspicion that

early wave reflection is the cause of this phenomenon (Figs. 16.14, 16.16, and 16.17).

Studies in experimental animals (dogs, goats, and pigs) have shown that acute pulmonary artery embolization increases pulmonary pressure and causes an increase in wave reflection amplitude (Maggiorini *et al.*, 1998; Wau-thy *et al.*, 2004). Subsequent studies in pulmonary hypertension have focused on contour of the pulmonary artery pressure waveform (Fig. 16.18) (Castlelain *et al.*, 2001; Nakayama *et al.*, 2001). Both of these human studies showed that time to the inflection point was shortened, and augmentation index was greater in patients with chronic pulmonary thromboembolism (CPTe) compared

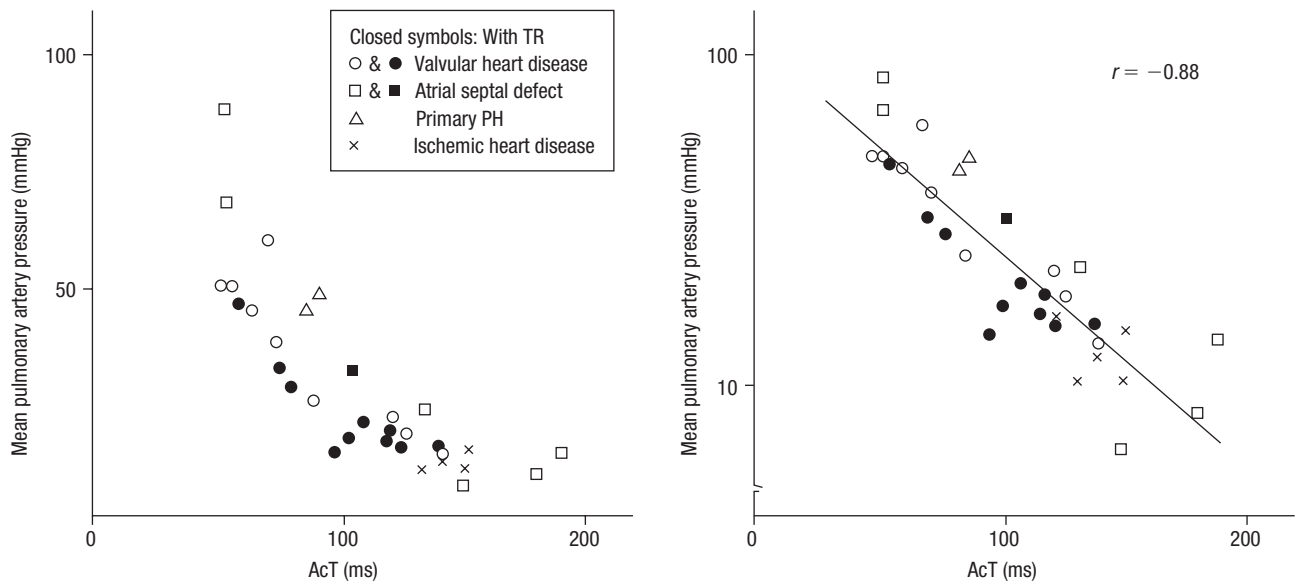


Figure 16.15 Time to peak flow velocity in the pulmonary artery (AcT) plotted against mean pulmonary artery pressure (on a linear scale at left, logarithmic scale at right) in 33 patients with a variety of cardiac and pulmonary diseases (see inset). Flow was measured non-invasively with the pulsed Doppler technique. PH, pulmonary hypertension; TR, tricuspid regurgitation. Reproduced from Kitabatake *et al.* (1983)

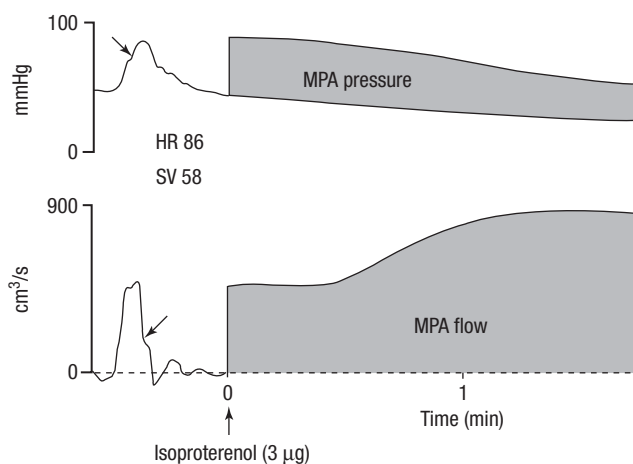


Figure 16.16 Main pulmonary artery (MPA) pressure and flow waves recorded with an electromagnetic velocity catheter in a patient with primary pulmonary hypertension (PHT) during isoproterenol infusion. Arrows indicate inflection point on the pressure wave and notching on the deceleration phase of the flow velocity wave

to those with primary pulmonary hypertension (Fig. 16.19).

These findings in both experimental animals and humans are readily explained on the basis of earlier wave reflection in the presence of thromboembolism as a consequence of obstructive lesions (major reflection site) being present in the major arteries closer to the heart rather than in the distal small peripheral vessels as in primary pulmonary hypertension. More recently, Lam *et al.* (2009) showed that heart failure with preserved LV systolic function (HFpEF) is associated with pulmonary hypertension, and that mortality in this condition can be predicted from the level of pulmonary artery systolic pressure. Lankhaar *et al.* (2008) showed that pulmonary vascular resistance and pulmonary vascular compliance (measured as cardiac output $\div$ pulmonary pulse

pressure) are inversely related at rest and during drug therapy, reinforcing the view of Milnor *et al.* (1969) that components of impedance apart from resistance need to be considered in management of pulmonary hypertension (see also Huez *et al.*, 2004; Weinberg *et al.*, 2004; Hunter *et al.*, 2008; Souza, 2008; Chesler *et al.*, 2009).

SIMULATION OF DISEASE IN EXPERIMENTAL ANIMALS

Pulmonary hypertension in humans has been simulated in experimental animals by infusion of serotonin (Bergel and Milnor, 1965; Reuben and Kitchen, 1975) and a stable thromboxane analog U46619 (Leather *et al.*, 2002), by embolization of the pulmonary artery with microspheres

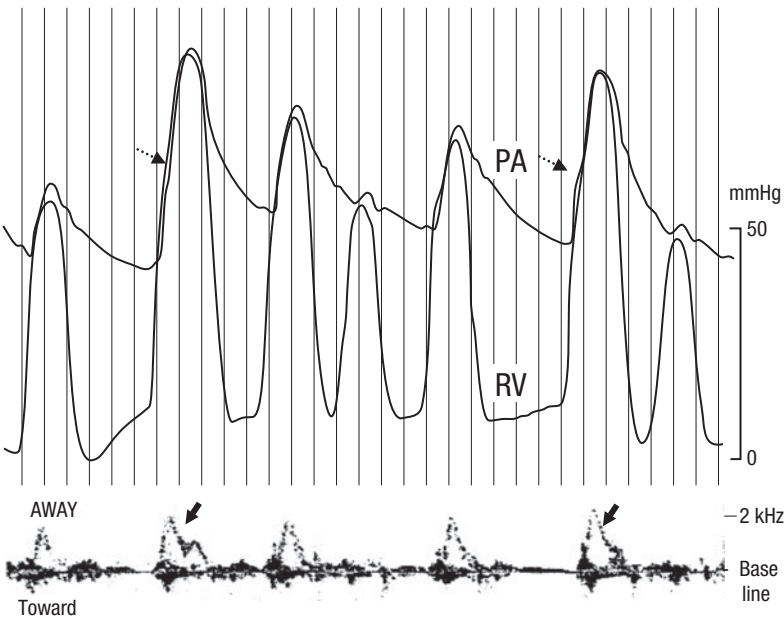


Figure 16.17 Simultaneous recordings of the flow velocity wave in the right ventricular outflow tract and the pressures of the pulmonary arterial trunk and the right ventricle in a patient with pulmonary hypertension and atrial fibrillation. The flow velocity pattern changes in its configuration from beat to beat. In the fourth and seventh beats, significant ejection flow is not observed. In the second and sixth beats, mid-systolic notching (solid arrows) is observed in the flow velocity pattern during deceleration, and in the other beats, mid-systolic notching is absent. The broken arrows indicate inflection points on the pressure waves associated with mid-systolic notching of the velocity waves. PA, pulmonary artery pressure; RV, right ventricular pressure. From Kitabatake *et al.* (1983)

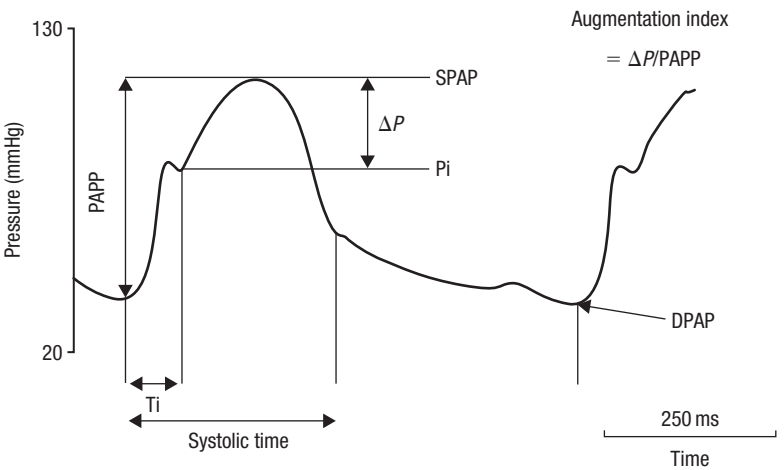


Figure 16.18 Typical pulmonary artery pressure tracing in a patient with pulmonary hypertension showing an early systolic shoulder, attributed to early return of wave reflection. Heart rate 89/min. DPAP, diastolic pulmonary artery pressure; PAPP, pulmonary artery pulse pressure; SPAP, systolic pulmonary artery pressure. From Castelain *et al.* (2001)

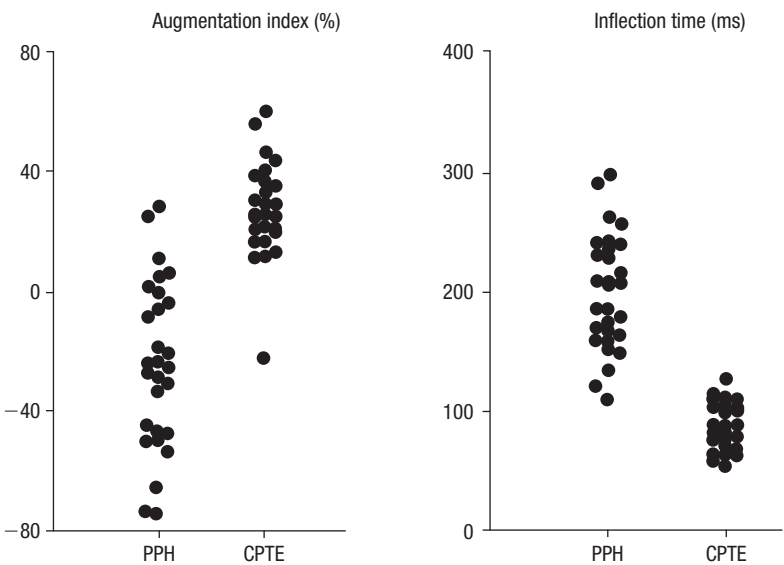


Figure 16.19 Comparison of pulmonary artery reflection between primary pulmonary hypertension (PPH) and chronic pulmonary thromboembolism (CPTe). From Nakayama *et al.* (2001)

(Elkins *et al.*, 1974), blood clots (Maggiorini *et al.*, 1998), or glass beads (Wauthy *et al.*, 2004), through direct pulmonary artery constriction (Calvin *et al.*, 1985; Santana *et al.*, 2005), through elevation of left atrial pressure with a left atrial balloon (Hopkins *et al.*, 1980) or through an arteriovenous fistula (Hopkins *et al.*, 1979). Such simulations have been useful in understanding the effects of disease in humans, but are subject to technical difficulties and artifacts and to problems in scaling. None of the measures designed to produce pulmonary hypertension in experimental animals has shown the dramatic increase in characteristic impedance seen in children (Fig. 16.12), even though they show similar changes in impedance shift of minimal values to higher frequencies. Many of these studies appear to have been 'over-interpreted'. For instance, Hopkins *et al.* (1979) showed an apparent decrease in the modulus of pulmonary vascular impedance and in pulmonary artery pressure wave amplitude with opening of an arteriovenous shunt. They attributed these changes to improvement in pulmonary artery distensibility in the presence of high pulmonary artery blood flow. The changes reverted on closure of the shunt. A more likely explanation was that high flow velocity in the pulmonary artery created a Venturi effect at the site of pressure measurement downstream from the implanted cuff flow probe. These studies have been challenged (O'Rourke, 1980) and have not, to our knowledge, been confirmed. The inconsistencies are discussed by Piene (1986).

Dujardin *et al.* (1982) inferred changes of pulmonary arterial distensibility from characteristic impedance determinations in experimental animals during shock and volume expansion. They measured characteristic impedance from both the frequency and time domain; the latter method permitted comparisons of impedance at the same level of mean arterial pressure. The results indicate increased characteristic impedance and arterial stiffness in hemorrhage and decreased characteristic impedance and arterial stiffness with volume expansion. It is difficult to understand the mechanism underlying such a change. Piene (1986) noted an increase in characteristic impedance with sympathetic stimulation and the use of vasoconstrictor agents. The mechanism may be similar to that described in peripheral systemic arteries. As for systemic arteries, interpretation of data is confounded by the accompanying changes in transmural pressure and in arterial diameter.

PULMONARY VASCULAR IMPEDANCE AND RIGHT VENTRICULAR LOAD

As with ascending aortic impedance and the left ventricle, pulmonary vascular impedance may be used to characterize right ventricular load. Indeed, pulmonary artery impedance has been simulated in three-element

(Lankhaar *et al.*, 2004; Westerhof *et al.*, 2009) and hydraulic models for studies of right ventricular performance (Piene and Sund, 1979; Piene and Sund, 1982; Piene, 1986; Weinberg *et al.*, 2004). This whole subject is discussed in Chapters 13 and 14. One major difference between the impedance spectra of the systemic and pulmonary circulations is the relatively greater magnitude of the oscillatory components of modulus compared with the resistive component in the pulmonary circulation. There are, in consequence, relatively greater pulsatile energy losses within the pulmonary circulation. In the normal dogs reported by Milnor *et al.* (1966) and more recently by Maggiorini *et al.* (1998), pulsatile external right ventricular power was approximately 25 percent of total external right ventricular power. The magnitude of the pulsatile component increased with serotonin infusion and with slowing of the heart rate and decreased with pulmonary embolism. In patients with pulmonary hypertension, pulsatile hydraulic power dissipated in the lungs was of similar magnitude to steady power (Fig. 16.20) and oscillatory power was found by Saouti *et al.* (2010) to be a constant fraction of total power irrespective of pulmonary artery pressure. This finding was discussed by Milnor *et al.* (1969), who pointed out that, in pulmonary hypertension, decreased arterial distensibility is of similar importance to increased resistance in the elevation of right ventricular load. Souza (2008) and Lankhaar *et al.* (2008) concur.

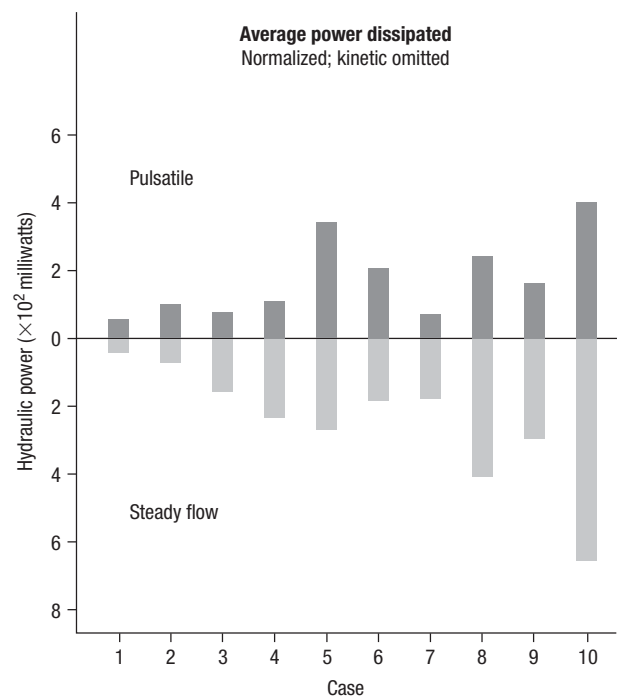


Figure 16.20 Magnitude of pulsatile (top) and steady (bottom) power dissipation in the pulmonary circulation of ten patients ranked in relation to mean pulmonary pressure (case 1 lowest, 10 highest). Reproduced from Milnor *et al.* (1969)

The separate influence of change in resistance and compliance on right ventricular load has been studied by Elzinga *et al.* (1980), Piene (1986), and Fourie *et al.* (1992). Changes in resistance and in compliance alter the ventricular load and ventricular ejection in a completely predictable fashion (Piene, 1986) (Fig. 16.21) and pulmonary input impedance predicts clinical outcomes better than pulmonary resistance alone in pediatric patients with pulmonary hypertension (Hunter *et al.*, 2008).

The relationship of vascular impedance to total ventricular load is misunderstood by many. It is sometimes stated that the frequency-dependent components of impedance are an indication of the pulsatile external energy losses in the circulation. This is true, but the argument is sometimes taken further: first, that the frequency-dependent components of impedance are a small fraction of the resistive component; and second, that oscillatory ventricular power is a small fraction of total ventricular power, so that the frequency components of impedance are indeed a very small fraction of total ventricular power. This is not realistic. External ventricular power is a convenient way of representing energy losses in the circulation. While the frequency-dependent components of impedance can be used to quantify what fraction of these energy losses is pulsatile, the impedance is directly

related to total ventricular power and total myocardial oxygen demand, because the frequency-dependent components of impedance determine directly what pressure is generated in the aorta and in the left ventricle by LV contraction, and in the pulmonary artery and right ventricle by right ventricular contraction.

PRIMARY PULMONARY HYPERTENSION

Primary pulmonary hypertension is a rare but lethal disease characterized by obstructive disease in the smallest pulmonary arteries which causes elevation in pulmonary vascular resistance and stiffness, elevated pulmonary artery pressure, and early and increased wave reflection from the peripheral arteries (Castlelain *et al.*, 2001; Nakayama *et al.*, 2001). The associated increase in input impedance (resistance and stiffness) causes a right-ward shift in the moduli minimum and phase zero crossing and an elevation in the impedance fundamental modulus (Fig. 16.22).

Interest is focusing on this condition because it sometimes responds favorably to drugs, and responds differently from systemic hypertension. Nitrates and

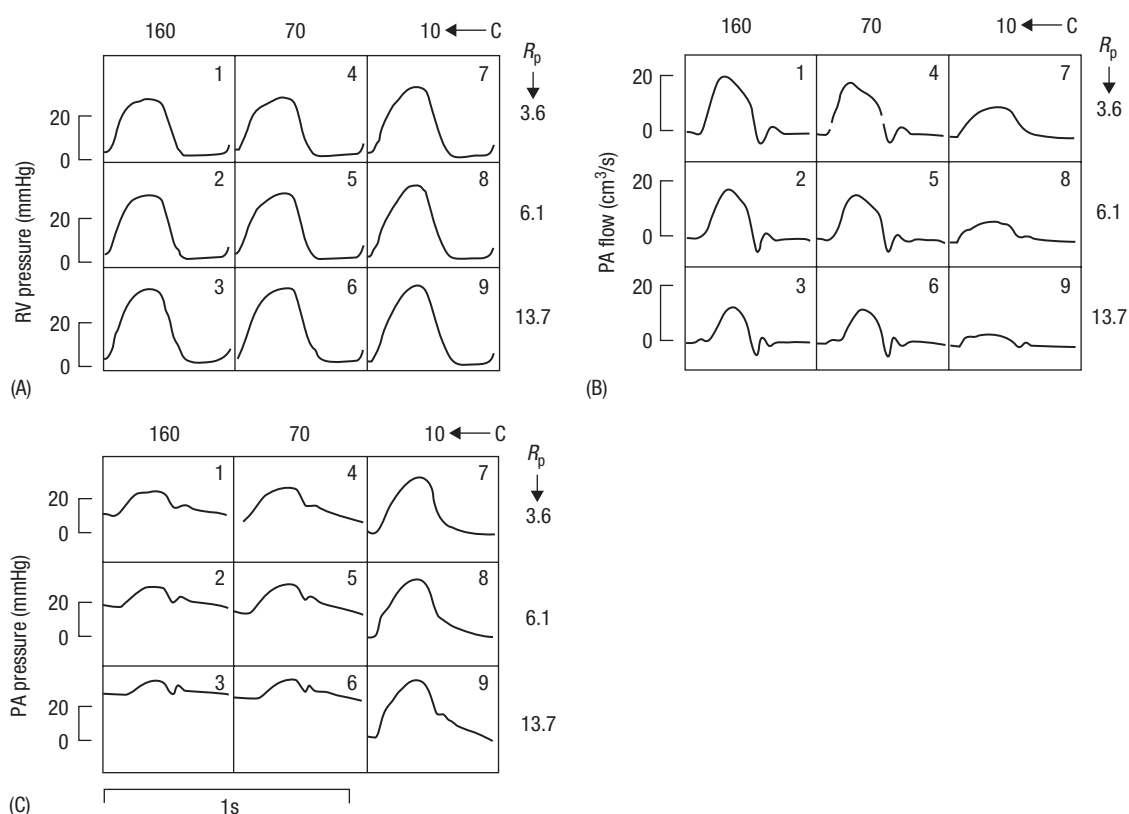


Figure 16.21 Dependence of right ventricular (RV) pressure (A), pulmonary arterial (PA) flow (B) and PA pressure (C) of isolated cat hearts paced at 2.0 Hz on independently variable resistance (R_p), which was increased four-fold, and on compliance (C), which was increased 16-fold. Reproduced from Elzinga *et al.* (1980)

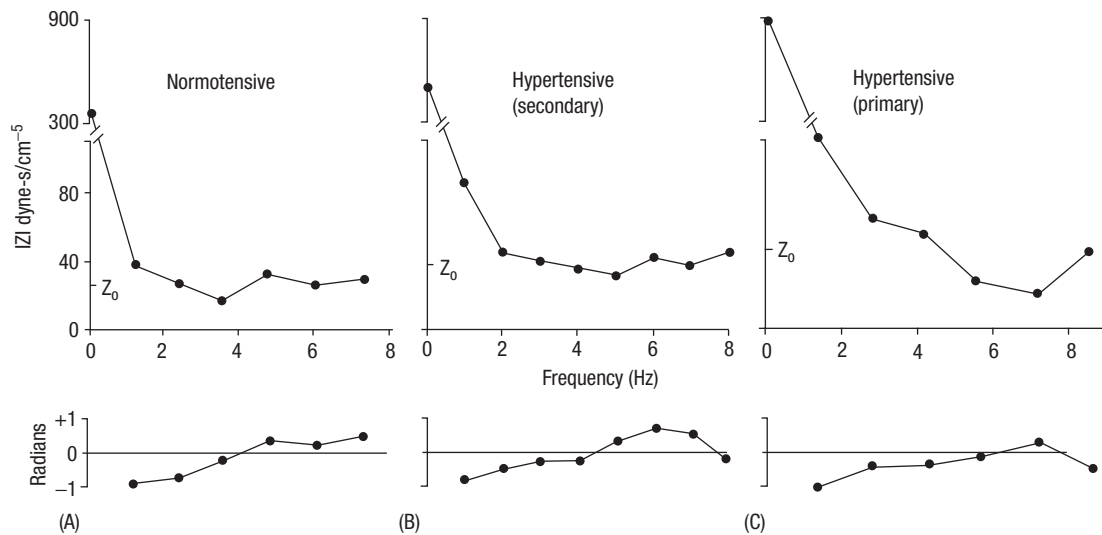


Figure 16.22 Modulus (above) and phase (below) of pulmonary vascular input impedance in three humans obtained from Fourier analysis of pressure and flow waves measured with an electromagnetic velocity catheter. (A) Normal pulmonary artery pressure, (B) pulmonary hypertension secondary to mitral stenosis and (C) primary pulmonary hypertension. Z_0 , characteristic impedance (average of impedance moduli > 2 Hz)

calcium channel antagonists had shown some benefit in the past, but were shown inferior to continuously infused prostacyclin. While effective in lowering pulmonary artery pressure and decreasing mortality, this treatment was extremely expensive and had a number of side effects (Barst *et al.*, 1996). Prostacyclin analogs taken orally were not effective, but inhaled iloprost has recently shown benefit (Olschewski *et al.*, 2002), as has sildenafil (a phosphodiesterase-5 (PDE5) inhibitor), bosentan, an endothelin-1 antagonist (Rubin *et al.*, 2002) and its analog, sitaxsentan (Barst *et al.*, 2006; Benza *et al.*, 2007).

Nitric oxide, inhaled continuously, has been shown effective as a pulmonary vasodilator in primary and other forms of pulmonary hypertension (Krasuski *et al.*, 2000) either alone or in association with sildenafil. Sildenafil alone has also proven effective as a pulmonary vasodilator (Zhao *et al.*, 2001) and is currently being evaluated in acute and long-term studies. Other PDE5 inhibitors, including tadalafil and vardenafil, are also under study. Another vasodilator, tolazoline hydrochloride, an α -adrenergic receptor antagonist, has also been used in some patients to treat primary pulmonary hypertension. This agent decreases pulmonary artery pressure, wave reflection amplitude, and characteristic impedance (Fig. 16.23).

None of these agents are effective as treatment for systemic hypertension. The reason for inefficacy of drug treatment in primary pulmonary hypertension (and in many other forms, particularly that associated with left to right shunts) may well be the structural nature of the obstructive defect in small blood vessels. When fully established (Edwards, 1957; Sheehan *et al.*, 2005), one sees plexiform lesions with thickened and thinned walls and a mixture of microaneurysms, microhemorrhage, and thrombosis, similar to what are seen in the brain of elderly

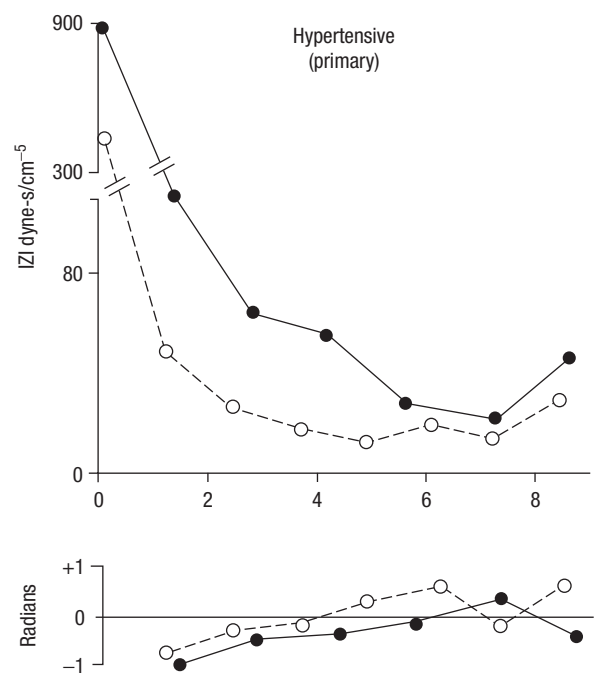


Figure 16.23 Modulus (above) and phase (below) of pulmonary vascular input impedance in a patient with primary pulmonary hypertension under control conditions ($\bullet$ — $\bullet$) and during the administration of the α -adrenergic receptor antagonist tolazoline hydrochloride ($\circ$ — $\circ$). The major effects of the drug were on pulmonary resistance and the lower impedance moduli (indicating reduced wave reflection)

persons with dementia and stiffened arteries. This is discussed elsewhere (Chapter 18) and is attributed to high pulsatile energy loss in small blood vessels causing endothelial and medial damage. Endothelial damage is

manifest as shedding of endothelial cells into the bloodstream. Severity and irreversibility of pulmonary hypertension in congenital left-to-right shunt (in ventricular septal defect, patent ductus arteriosus, etc.) can be gauged by count of circulating endothelial cells (Smadja *et al.*, 2009). Identification and early treatment of pulmonary

hypertension may be of more importance than early treatment of systemic hypertension. Early surgical treatment of congenital left to right shunts is well known and the accepted procedure to prevent development of pulmonary hypertension.

The coronary circulation

Coronary flow reserve	376	Myocardial ischemia	386
Coronary artery stenosis	378	Factors limiting coronary blood flow	388
Coronary 'steal'	380	Therapeutic implications	390
Specific principles of blood flow to the normally contracting ventricle	382	Arterial counterpulsation	390
		New diagnostic methods	391

More is known about the coronary than about any other circulation in humans. This is a consequence of clinical importance and diagnostic need. Disease of the coronary arteries is responsible for more morbidity and mortality than any other disease in Western society (Assmann *et al.*, 1999; Brown and O'Connor, 2010; Yeh *et al.*, 2010). Invasive and non-invasive methods, including mathematical and computer reconstruction models, have provided precise information on the coronary arteries – especially on their anatomy and involvement by atherosclerotic disease (Falk *et al.*, 1995; Mittal *et al.*, 2005; Zamir, 2005; Huo and Kassab, 2006; Achenbach and Raggi, 2010; Applegate, 2010; Kern and Samady, 2010; Mark *et al.*, 2010).

Epidemiological studies continue to reinforce the importance of conventional risk factors in the development of coronary artery disease, and tables from Framingham and other sources are readily available on the internet and from national societies to calculate the risk in individuals without prior heart disease. The conventional risk factors remain as age, male gender, blood pressure, cholesterol level, presence of diabetes mellitus, smoking, and obesity. Attention to lifestyle and risk factors is largely responsible for the steady decrease in age-adjusted mortality from coronary artery disease over the last few decades (Ford and Capewell, 2007; Goldberg, 2009). A present focus on obesity and glucose intolerance is a reason for the slowing in the rate of decline – and attributable to unhealthy diet and inactivity. The major Framingham risk factors have shown their predictive value, and this is only slightly improved by inclusion of newer biomarkers, except from measurement of pulse wave velocity as an index of aortic stiffness, and of pulse waveform indices in males under age 60 (Weber *et al.*, 2004; Galderisi, 2008; Guelen *et al.*, 2008; Ikonomidis

et al., 2008; Fischer-Rasokat *et al.*, 2009; Vlachopoulos *et al.*, 2010a; Vlachopoulos *et al.*, 2010b; Vlachopoulos *et al.*, 2010c). A wonderful personal perspective on the Framingham study is available in Levy and Brink's *A Change of Heart* (Levy and Brink, 2005).

The information from Framingham and other studies is principally directed at the coronary arteries that, like a corona, run on the surface of the heart. Arteries within the heart muscle show little evidence of disease, and presumably are spared from the shocks and movements by the protection of surrounding myocardial fibers. They appear to be spared from high systolic pressures through systolic compression of the myocardium – and do not show the abnormalities seen with aging in small vessels of the equally highly perfused brain and kidney. These are described in Chapter 28.

Little is known on the function of the smallest arteries, arterioles, and capillaries in the heart. Dilation appears to be caused by metabolites during activity, by adenosine, and by endothelially released nitric oxide. Throughout life, these remain youthful, but clinicians presume that ischemic events, 'microvascular angina', in the heart must be caused by narrowing of such small vessels through endothelial dysfunction just as the more common ischemic events are caused by narrowing of the coronary arteries on the heart's surface (Cannon and Epstein, 1988; Chilian, 1997; Nisanci *et al.*, 2002; Bugiardini and Merz, 2005; Camici and Crea, 2007; Pries *et al.*, 2008; Cannon, 2009; Phan *et al.*, 2009). There are other explanations, and they are discussed later in this chapter.

A separate chapter on coronary hemodynamics is expected in a book of this type. Such a chapter permits discussion on general principles that pertain also to perfusion of other vascular beds. However, the text must also consider the unique characteristics of coronary blood

flow – that this occurs into an organ which is alternately contracting and relaxing, thus altering the phasic nature of blood flow into and through its muscular wall (Gregg and Shipley, 1944; Khouri and Gregg, 1963; Feldman *et al.*, 1978a; Feldman *et al.*, 1978b; Hoffman, 1984; Hoffman, 1987; Olsson and Bungert, 1987; Nichols *et al.*, 1988a; Nichols *et al.*, 1988b, Nichols *et al.*, 1994a; Nichols *et al.*, 1994b; Gould, 1997; Gould, 1998a; Gould, 1998b; Boudoulas and Leier, 2000; Ootaki *et al.*, 2005; Ootaki *et al.*, 2008). The heart is unique in that it generates its own blood flow, therefore, flow is altered by the process of its own generation (Gielen *et al.*, 1999) with the large majority of flow occurring in diastole (about 80 percent) when the heart is relaxed.

Resting coronary blood flow per gram of tissue is high, as it is to the brain and kidneys, and exhibits an age-related decrease (Czernin *et al.*, 1993). Like the brain, but unlike the kidney, oxygen extraction from blood perfusing the myocardium is high, so that venous blood exiting the heart muscle has a very low oxygen content. In consequence, an increase in myocardial oxygen supply can only be met by an increase in coronary blood flow. There is thus normally a very close relationship between oxygen requirement by the heart muscle and coronary blood flow (Fig. 17.1) (Khouri *et al.*, 1965). Coronary blood flow exhibits autoregulation, as in the brain, with blood flow maintained relatively constant over the physiologic range of arterial pressure, showing substantial reserve over the physiologic range but limited reserve below (Fig. 17.2), and a positive intercept on the pressure axis (Hoffman, 1984; Hoffman, 1987; Boudoulas and Leier, 2000; Kern, 2000).

CORONARY FLOW RESERVE

Coronary blood flow (or vascular) reserve is measured as the difference between basal coronary blood flow and flow at peak (or near peak) vasodilatation (see Figs 17.2, 17.3 and 17.4). The normal heart with normal coronary arteries can increase its coronary blood flow up to four or five times resting values during exercise, reactive hyperemia, or drug infusion (Von Restorff *et al.*, 1975; Bourdarias, 1995; Duncker and Bache, 2008). Since coronary flow reserve (CFR) varies with heart rate and mean perfusion pressure (Fig. 17.2), it must be expressed and comparisons must be made at a given heart rate and mean arterial pressure. CFR is low at low perfusion pressure where the effects of autoregulation cause arteriolar dilatation and where the effects of anomalous blood viscosity cause the pressure–flow relationship to have an apparent zero flow at 20–40 mmHg (Klocke *et al.*, 1981; Hoffman, 1984; Hiramatsu *et al.*, 1994). At perfusion pressures below about 50 mmHg (i.e. below the autoregulatory range), a prolonged diastolic duration is required in order to maintain coronary flow and protect the myocardium (Merkus *et al.*, 1999). This modification in diastolic

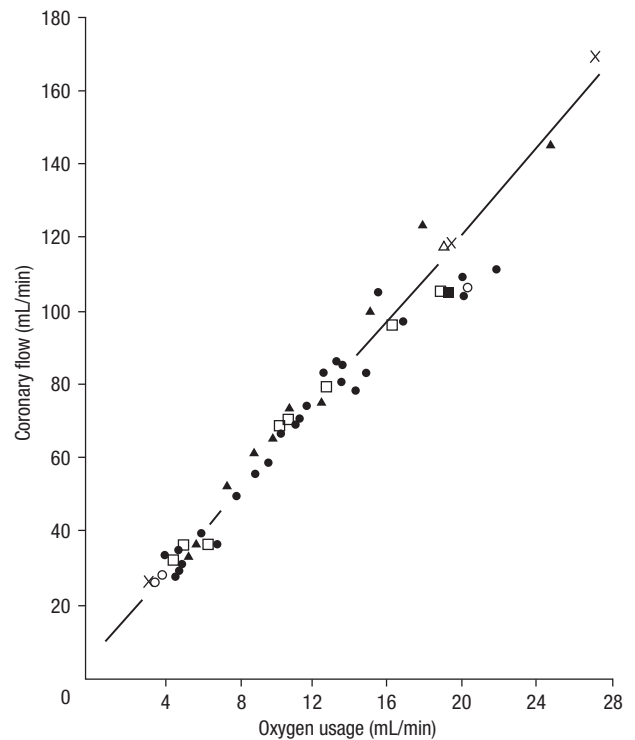


Figure 17.1 Relationship between coronary blood flow and myocardial oxygen usage in conscious dogs during treadmill exercise. Each symbol represents a different dog. Reproduced from Khouri *et al.* (1965)

pressure duration is blunted with the nitric oxide synthase inhibitor IV(G)-monomethyl-L-arginine (L-NMMA) suggesting nitric oxide-mediated mechanisms. CFR is also decreased in the presence of coronary stenosis, where limitation to flow is apparent during vasodilatation before being apparent at rest (Gould *et al.*, 1974; Feldman *et al.*, 1978a; Feldman *et al.*, 1979a; Feldman *et al.*, 1979b; Feldman *et al.*, 1983; Hoffman, 1984; Bache, 1998) (Figs 17.3 and 17.4), and in the ischemic myocardium because of the ‘no reflow’ phenomenon (Nichols *et al.*, 1988a; Nichols *et al.*, 1988b; Mehta *et al.*, 1989).

CFR is also limited with aging and with syndrome X, even when the large epicardial coronary arteries are completely normal (Weisfeldt *et al.*, 1992; Monti and Piatti, 2005; Galiuto *et al.*, 2007; Cemin *et al.*, 2008). This has been attributed to changes in the microcirculation, but recent studies have emphasized the importance of increased basal flow (attributable to increased coronary blood flow consequent on ventricular hypertrophy and increased arterial stiffness), with the peak flow at a given arterial pressure well preserved (see Chapter 19) (Czernin *et al.*, 1993). Reduction of CFR in hypertension and aortic valve disease can also largely be explained on the basis of increased ventricular mass and myocardial oxygen need, increased basal blood flow (Braunwald, 1992; Gould, 1997; Galderisi *et al.*, 2003; Braunwald, 2009), and endothelial dysfunction (Frohlich, 2001; Palmieri *et al.*,

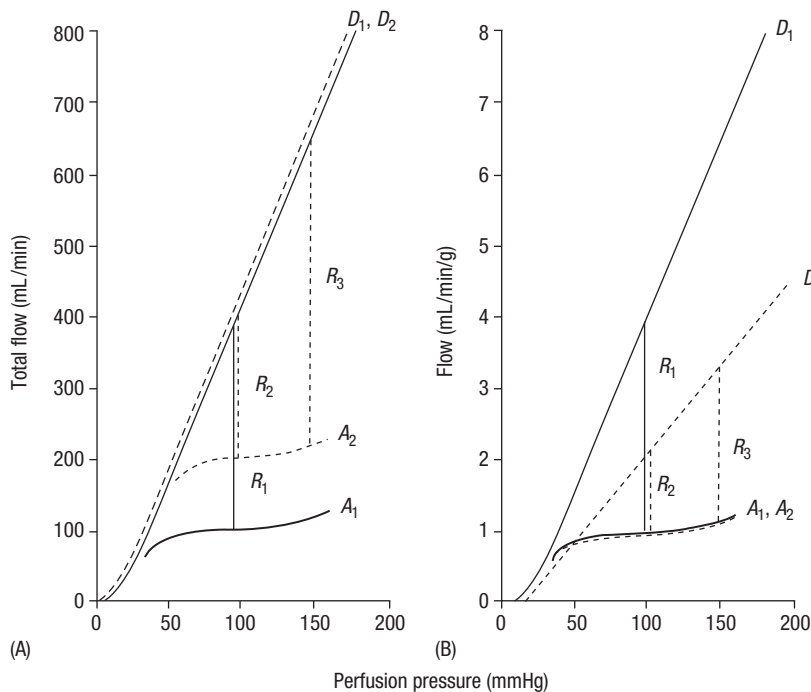


Figure 17.2 Diagram to illustrate measurement of coronary blood flow reserve. Pressure–flow relationships for normal left ventricle (100 g) and hypertrophied left ventricle (200 g) during autoregulation (A) and maximal vasodilatation (D) are shown. The difference (D minus A) between the two curves represents blood flow reserve (R) of the distal coronary bed at a given perfusion pressure. Normal: A_1 , D_1 , R_1 , and solid lines; hypertrophied: A_2 , D_2 , R_2 , R_3 , and dashed lines. D_1 and D_2 are superimposed. Coronary flow reserve for hypertrophied ventricles is shown at normal (R_2) and elevated (R_3) perfusion pressures. (A) Total flow per ventricle; (B) flow per unit mass. Reproduced from Hoffman (1987)

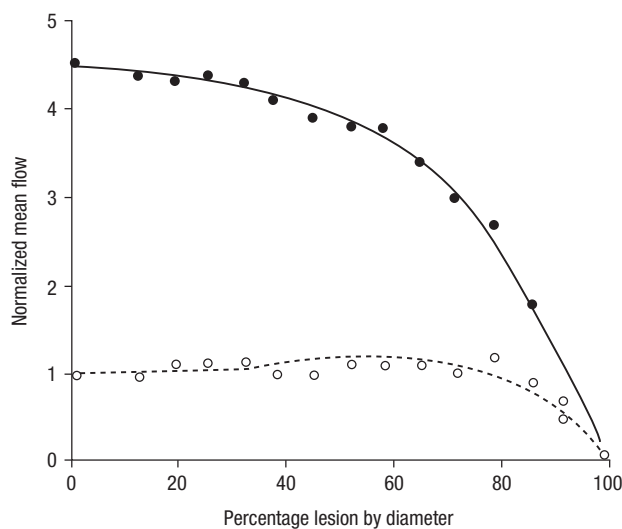


Figure 17.3 Effect of progressive constriction on resting blood flow in the circumflex coronary artery (○--○) and reactive hyperemic response (●--●) in a dog. Percentage constriction by diameter is displayed on the horizontal axis. Reproduced from Gould *et al.* (1974)

2005; Ikonomidis *et al.*, 2008). Depressed CFR in dilated cardiomyopathy cannot (Tsagalou *et al.*, 2008).

Studies have shown a decrease in CFR in patients with hypertension, diabetes, and aortic valve disease, and in dogs with left ventricular hypertrophy resulting from aortic banding (Marcus *et al.*, 1983; Hoffman, 1987; Nahser *et al.*, 1995; Palmieri *et al.*, 2005; Ikonomidis *et al.*, 2008). This decrease in CFR has been attributed to changes in the microcirculation (Camici and Crea, 2007). In the absence of change in myocardial oxygen demands

or in coronary stenosis, this may be the appropriate inference. However, since coronary flow is also dependent on other factors, including duration of diastole, arteriovenous pressure gradient in diastole, and rate of ventricular pressure decay during early diastole (see below), such an inference cannot be made with the same confidence that it can for other vascular beds. Changes in cardiac variables and endothelial function may influence CFR (Nichols *et al.*, 1988a; Nichols *et al.*, 1994b) and, as discussed below and in Chapter 5, may indeed induce myocardial ischemia and anginal symptoms in the presence of minor coronary atherosclerosis (Buckberg *et al.*, 1972a; Buckberg *et al.*, 1972b; Hoffman and Buckberg, 1978; Ferro *et al.*, 1995; Gould, 1997; Gould *et al.*, 2000; Hoffman and Chilian, 2000; De Bruyne, 2001a; Cai *et al.*, 2002; Guyton, 2002; Kern *et al.*, 2006; Braunwald, 2009).

Several methods have been used to study CFR. These include invasive methods (intracoronary Doppler flow wire), semi-invasive methods (transesophageal echocardiography), and non-invasive methods (positron emission tomography (PET) and transthoracic echocardiography). Invasive CFR (iCFR) can be measured using a 0.014-in. Doppler wire. The tip of the wire is advanced distal to the lesion, and baseline measurement of flow velocity is obtained. Intracoronary or intravenous adenosine is injected to induce hyperemia, and then the hyperemic flow velocity is measured. Assuming that the vessel cross-sectional area remains constant, iCFR is calculated as the hyperemic to basal flow velocity. iCFR begins to decrease at a diameter stenosis of 60 percent or greater, whereas resting flow decreases when the diameter stenosis is 90 percent or greater. iCFR depends on heart rate and blood pressure, as tachycardia and increased blood pressure decrease iCFR. iCFR < 2.5 is considered

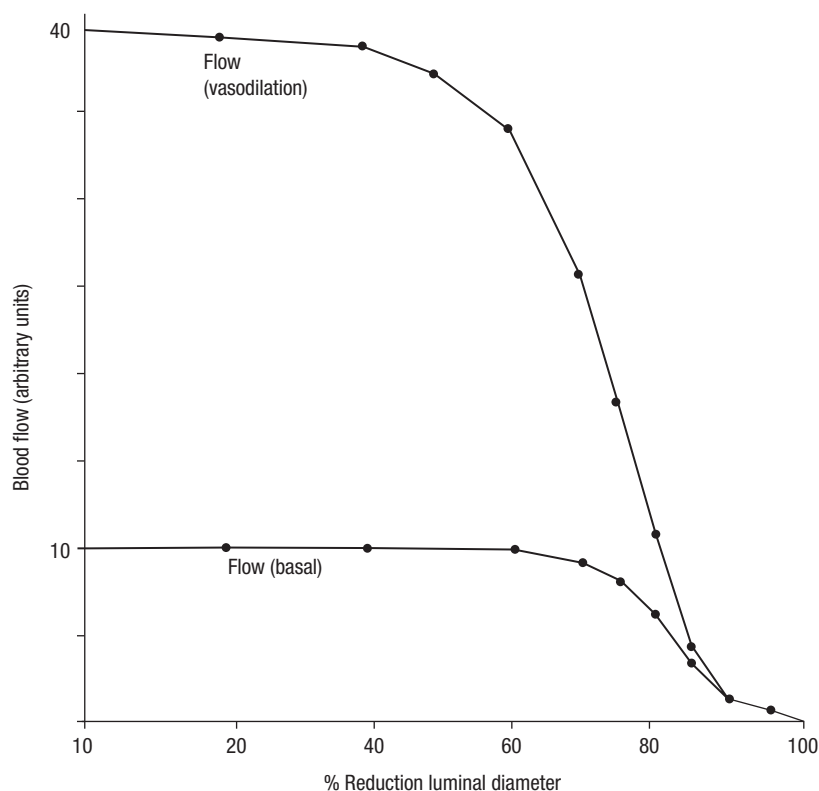


Figure 17.4 Calculation of blood flow under control conditions and during vasodilatation in a model of the coronary vascular bed, with progressive reduction in caliber of a 0.5 cm arterial segment. Change in resistance of this segment is calculated from Poiseuille's formula. Findings approximate those found in life (see Fig. 17.3). Reproduced from O'Rourke (1982b)

abnormal and may be used to evaluate whether a moderate epicardial stenosis is hemodynamically significant or not. However, iCFR interprets a two-component system, namely the epicardial coronary artery and the microvasculature. Thus, an abnormal iCFR can be due to epicardial artery obstruction or microvasculature dysfunction. Relative iCFR (rCFR), which is the iCFR of the target vessel divided by the iCFR or an unobstructed reference vessel, partly addresses this issue, assuming that the microvascular function is the same. An rCFR < 0.8 is considered abnormal. The disadvantage is that iCFR must be measured in two coronary arteries to calculate rCFR.

Fractional flow reserve (FFR) is a valuable tool that evaluates the physiologic significance of a coronary artery stenosis (De Bruyne *et al.*, 1995; Pijls *et al.*, 1996). FFR is defined as the ratio of the flow in an obstructed artery at maximal vasodilation divided by the flow in the artery in the theoretical absence of epicardial obstruction at maximal vasodilation. FFR is calculated as the ratio of the pressure distal to the stenosis divided by the aortic pressure at maximal vasodilation, assuming that at maximal vasodilation the resistance distal to a normal and a diseased artery is minimal and the same. FFR is measured using a pressure wire. The tip of the wire is advanced distal to the lesion, and intracoronary or intravenous adenosine is injected to induce hyperemia. FFR is calculated as the pressure distal to the stenosis divided by the aortic pressure obtained by the guiding catheter, as aortic pressure is equal to coronary artery pressure in the absence of stenosis. A cut-off value of 0.75–0.80 has been shown to correlate well with non-invasive indices of

reversible ischemia. FFR is a very useful tool to decide whether an angiographically moderate stenosis is hemodynamically significant or not. Recently, the randomized FAME (Fractional Flow Reserve Versus Angiography for Multivessel Evaluation) trial showed that use of FFR to guide stenting in patients with multivessel coronary artery disease significantly reduces the rate of the composite end point of death, non-fatal myocardial infarction, and repeat revascularization at one year (Tonino *et al.*, 2010). FFR is free of the limitations that compromise iCFR. In particular, FFR is more reproducible, is lesion-specific (microvasculature does not affect FFR), and importantly, it does not depend on heart rate, blood pressure, or cardiac contractility (De Bruyne *et al.*, 1996; Lanza *et al.*, 2008; Pennell, 2008). Furthermore, with use of intravenous adenosine, serial coronary stenoses can be evaluated by pulling the pressure wire back.

CORONARY ARTERY STENOSIS

Since the high incidence of atherosclerotic coronary artery disease has provided a major impetus for the study of the coronary circulation, it is not surprising that much attention has been devoted to the study of the effects of stenosis on coronary blood flow (Gould *et al.*, 1974; Feldman *et al.*, 1978a; Feldman *et al.*, 1978b; Gould, 1978b; Feldman *et al.*, 1979a; Feldman *et al.*, 1979b; Feldman *et al.*, 1980; Brown *et al.*, 1982; Gould and Kelley, 1982; Sabbah and Stein, 1982; Feldman *et al.*, 1983; Brown *et al.*, 1986; Gould, 1988; Brown *et al.*, 1990; Gould, 1990;

Gould, 1992; Brown *et al.*, 1993; Boudoulas and Leier, 2000; Gould *et al.*, 2000; Kern, 2000; Magnus and Beaglehole, 2001; Kern *et al.*, 2006; Braunwald, 2009). Early attempts to quantify the effects of stenosis utilized an external coronary artery snare, adjustable with a micrometer to achieve a known reproducible degree of stenosis (Gould *et al.*, 1974; Feldman *et al.*, 1978a; Feldman *et al.*, 1980; Gould *et al.*, 2000). The quantitation of stenosis severity in such studies has been variably expressed as percentage reduction in either lumen diameter or cross-sectional area. These studies have emphasized that the physiologic significance of stenosis depends on the level of coronary flow and consequently on flow velocity. In the presence of autoregulation in the normal ventricle, an external snare stenosis resulting in approximately an 80–85 percent diameter reduction, or a 60–80 percent reduction in cross-sectional area, is necessary before a reduction in resting flow is noted (see Fig. 17.3) (Gould *et al.*, 1974; Feldman *et al.*, 1978a; O'Rourke, 1982b; Gould *et al.*, 2000). However, in the absence of autoregulation, reductions in maximal coronary flow levels can be observed with a diameter reduction as small as 30 percent (Gould *et al.*, 1974; Feldman *et al.*, 1978a). Consequently, assessment of coronary flow through a stenotic vessel segment during reactive hyperemia has been proposed as a useful indicator of the significance of a stenosis (Hillis and Friesinger, 1976). These studies have resulted in the concept of a 'critical stenosis', defined as a stenosis sufficient to prevent an increase in coronary blood flow above resting values in response to increased oxygen demands, while not significantly attenuating resting flow (Gould *et al.*, 1974; Feldman *et al.*, 1978a).

The movement of coronary blood flow through a stenotic segment is associated with a reduction in perfusion pressure, or the pressure gradient, through the stenosis. The pressure gradient is related to stenosis geometry, flow velocity, blood viscosity and density, and flow waveform (Young, 1979). However, the most significant factors are found to be flow velocity and the ratio of the area of the vessel at the site of stenosis to the unobstructed vessel area (Young, 1979). The pressure gradient over a stenotic segment and geometric descriptors of the stenosis have been used to predict coronary flow, using a non-linear, quadratic model (Sun *et al.*, 1983). Despite the major influence of stenosis severity on flow velocity, the remaining factors cannot be overlooked. For example, lengthening of coronary stenoses, particularly beyond 5 mm, has been shown to reduce the reactive hyperemia response significantly (Feldman *et al.*, 1978a; Feldman *et al.*, 1978b).

Since the diseased coronary artery may often have multiple sites of narrowing, the influence of multiple stenoses on coronary flow have been examined. Multiple stenoses would be expected to have a greater impact on flow than a single stenosis of similar total length, owing to multiple entrance and exit points with increased propensities for flow turbulence. However, data from

experiments in the coronary circulation, as in the peripheral circulation, have been inconsistent. Some studies have concluded that multiple stenoses produce a resistance to flow more severe than a single stenosis of similar length (Gould *et al.*, 1974; Sabbah and Stein, 1982), while others have observed that the number of stenoses was less critical than the lumen diameter and total length in determining the flow response (Feldman *et al.*, 1978a; Feldman *et al.*, 1978b). Many of the apparent differences between studies in this field undoubtedly relate to the widely differing experimental preparations employed to approach the problem. When considering the application of experimental data to the clinical situation, it should be noted that the experimental studies have generally employed a concentric stenosis, produced by a variety of means. However, an eccentric stenosis with a normally elastic segment opposite the plaque, a situation often observed in diseased coronary arteries (Brown *et al.*, 1984), produces a greater pressure gradient and resistance than does a fixed concentric stenosis of comparable dimensions (Logan, 1975; Wong *et al.*, 1984). This phenomenon is thought to be due to a collapse of the elastic segment, resulting in additional narrowing of the arterial lumen during high flow velocity through the stenosis (Logan, 1975; Wong *et al.*, 1984). Consequently, caution must be used in directly extrapolating data which attempt to predict the significance of an experimental stenosis to the clinical evaluation of an abnormal coronary artery.

Experimental flow data from coronary arteries with snare-induced narrowings have indicated the importance of the peripheral vascular bed in determining the calculated stenosis resistance. With a critical stenosis, the release of a total coronary occlusion is not accompanied by an increase in coronary flow above control levels, since the decrease in peripheral coronary resistance is balanced by an increase in stenosis resistance. However, a more severe stenosis results in a decrease in coronary flow after release of a coronary occlusion. This reduction in flow below control levels is associated with a large increase in stenosis resistance. This led to the concept of a 'compliant' or 'dynamic' stenosis (Gould, 1980; Gould and Kelley, 1982; Santamore *et al.*, 1982; Schwartz, 1983). One explanation for this phenomenon is that the coronary artery within or distal to the stenosis collapses or narrows passively owing to the vasodilatation of the peripheral coronary bed and the reduction in peripheral coronary pressure (Walinsky *et al.*, 1979). Other possible explanations include dilatation of the normal coronary artery adjacent to the stenosis, with a resulting increase in relative stenosis severity, change in arterial smooth muscle tone within the stenotic segment itself or the appearance of fully developed turbulence within the stenotic segment rather than downstream of the stenosis site (Gould, 1980; Gould, 1990). Consequently, the mechanisms leading to apparent changes in stenosis remain to be conclusively identified.

While this discussion has thus far dealt with flow through a stenosed coronary artery, the transmural distribution of coronary flow under these conditions is an important additional determinant of the survival of myocardium. In the absence of a stenosis, the transmural distribution of coronary flow is relatively uniform, with endocardial/epicardial ratios equal to or slightly greater than 1.0. This ratio is not altered by vasodilatation of the coronary bed. In the presence of a coronary stenosis, the transmural distribution of coronary flow during low myocardial flow demand is not affected until the degree of stenosis exceeds a 90 percent diameter reduction (Buss *et al.*, 1978). With more severe stenoses, the endocardial/epicardial ratio falls, primarily owing to a reduction in flow to the subendocardial layer. Coronary vasodilatation in the presence of a stenosis results in a reduction in the endocardial/epicardial flow ratio at a much more moderate degree of stenosis (60 percent diameter reduction) (Buss *et al.*, 1978). Once again, the fall in the endocardial/epicardial ratio is primarily the result of a marked reduction of flow to the subendocardium.

Transmural flow distribution in the presence of multiple stenoses has been examined by creating two sites of external coronary compression (Warltier *et al.*, 1983). Each stenosis, applied alone, did not alter the endocardial/epicardial ratio, despite a reduction of 70 percent in the reactive hyperemia response. However, the combination of two stenoses resulted in a significant fall in the endocardial/epicardial ratio, once again due primarily to a fall in blood flow to the subendocardium (Warltier *et al.*, 1983).

Numerous efforts, both pharmacological and mechanical, have been oriented at improving coronary flow to the ischemic myocardium. The advent of intra-aortic balloon counterpulsation and enhanced external counterpulsation (EECP) originated from experiments that produced a delay in the arterial pressure pulse perfusing

the coronary arteries (Kantrowitz and Kantrowitz, 1953). A significant increase in coronary blood flow can be produced by delaying the coronary arterial pressure pulse so that it occurs during ventricular diastole (Fig. 17.5) (Michaels *et al.*, 2002; Wu *et al.*, 2006). These observations form the basis of the intra-aortic balloon pump and EECP, which act by inflation of an intra-aortic balloon or external cuffs on the legs at the end of ventricular systole (see below and Chapter 14).

CORONARY 'STEAL'

The concept of 'steal' was first applied to the cerebral circulation as a means of explaining brainstem ischemia during activity of an upper limb, when an obstruction existed in the subclavian artery proximal to the vertebral artery origin (Weatherall *et al.*, 1983). The concept explained cerebral ischemia on the basis that vasodilatation in the arm led to hypotension distal to the subclavian obstruction, causing reversal of flow in the vertebral artery away from the brain and towards the arm.

The possibility of steal exists in other regions when arteriolar dilatation increases blood flow beyond the capacity of conduit arteries for its supply. That 'steal' can occur in the heart has been known for many years (Fam and McGregor, 1964; Schaper *et al.*, 1973; Becker, 1976; Cohen *et al.*, 1976; Becker, 1978). In the heart, it appears that steal can only occur in the presence of vasodilatation when a major coronary artery is totally occluded and when collateral blood supply from adjoining regions is limited by stenoses in their arteries of supply (Fig. 17.6) (Becker, 1976; Becker, 1978). Thus, in the heart, steal requires three factors: total occlusion of a coronary artery, stenosis in the adjoining artery that feeds the collaterals, and arteriolar dilation which decreases the collateral perfusion pressure and ultimately collateral flow (Becker,

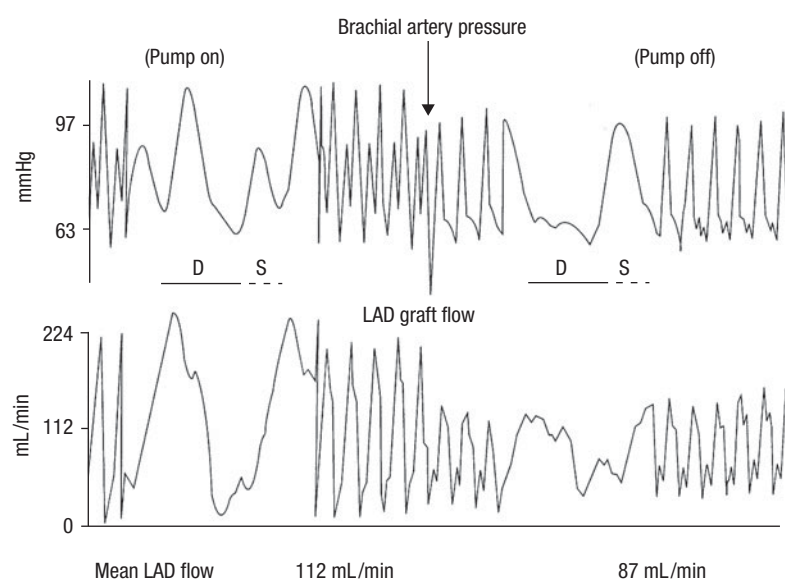


Figure 17.5 The effect of intra-aortic balloon counterpulsation on blood flow through a vein graft from the ascending aorta to the left anterior descending (LAD) coronary artery of a patient during surgery. Shown at the top is a tracing of brachial arterial pressure recorded with a liquid-filled catheter–manometer system. When the pump is turned off (at arrow), the diastolic pulse disappears and blood flow through the graft decreases. The duration of diastole (D) is indicated by the solid line (—) and the duration of systole (S) by the broken line (---). Note the delay in the pressure record

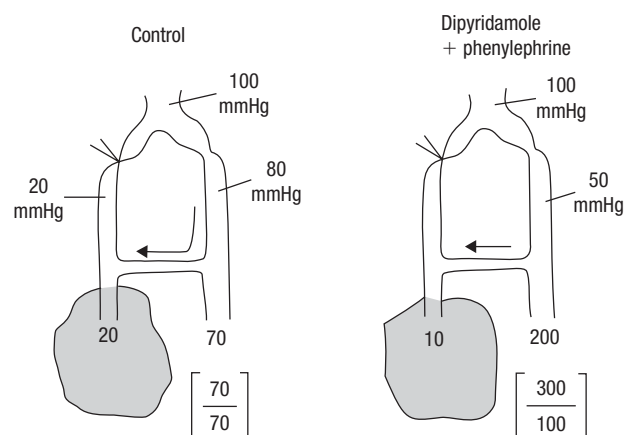


Figure 17.6 Schematic to show the mechanism for dipyridamole-induced coronary 'steal' (with perfusion pressure maintained with phenylephrine). The coronary artery divides into two branches, the first completely occluded and the second stenosed but providing collaterals to the first. Under control conditions, coronary flow is normal (70 units) through the bed supplied by the second branch but reduced (20 units) in the bed supplied by the first. With dipyridamole, and at the same perfusion pressure, blood flow to the second bed is increased to 200 units while flow to the first is reduced to 10. The second bed has 'stolen' from the first. The phenomenon can be explained on the basis of arteriolar vasodilatation, increased flow, and altered pressure gradients across the unchanged stenosis and collateral vessel (see text). Reproduced from Becker (1978)

1978; Schulz *et al.*, 1985). This scenario is often termed 'lateral' coronary steal. However, in a broader sense, steal occurs also when flow is redistributed from the subendocardium to the subepicardium in the case of a severely stenosed epicardial artery and this is termed 'transmural' steal. In this scenario, when an epicardial artery is severely stenosed, subendocardial arterioles of the area perfused by the stenosed artery are maximally dilated, even at resting conditions, while subepicardial ones may not be dilated to the same extent. In case of additional dilation of the subepicardial arterioles, blood is redistributed from the subendocardium to the subepicardium. Steal may be absolute (when blood flow to an ischemic region is further reduced, as in Fig. 17.6) or relative, when blood flow is not increased to the degree required by increased metabolic activity. The former can be demonstrated experimentally (Becker, 1978), while the latter may be manifested through clinical and electrocardiographic evidence of myocardial ischemia during exercise (Schulz *et al.*, 1985) or chemically simulated exercise. Steal may be induced by metabolite-induced arteriolar vasodilatation such as in exercise (including exercise evoked by dobutamine in stress echocardiography), or by pharmacologic arteriolar vasodilatation with adenosine or dipyridamole (Gould, 1978a; Schulz *et al.*, 1985; Klocke *et al.*, 1987). Steal is only induced by arteriolar dilatation. It is readily seen with adenosine-like drugs, such as dipyridamole,

which are strong arteriolar dilators, but which have no significant effects on larger arteries (Becker, 1978). This fact is utilized for the performance of radionuclide myocardial perfusion studies (Gould *et al.*, 1978b) or stress echocardiography (Oh *et al.*, 1999). In these studies, exercise is simulated and abnormal myocardial perfusion is demonstrated in patients with coronary disease, through an intravenous infusion of adenosine, dipyridamole, or dobutamine. Dipyridamole stress tests are frequently performed in patients with peripheral atherosclerotic arterial disease who, because of leg claudication, cannot exercise on a treadmill or bicycle.

Drugs that dilate arteries do not cause steal. This has never been demonstrated with nitroglycerin, which markedly dilates small conduit arteries in conventional doses (Feldman *et al.*, 1981; Wheatley *et al.*, 1994), but has little effect on peripheral arterioles (Fig. 17.7) unless administered intra-arterially (Nichols *et al.*, 1988a; Nichols *et al.*, 1988b). In patients with angina, nitroglycerin preferentially redistributes myocardial blood flow to ischemic regions (Fallen *et al.*, 1995). In exercise-induced steal, nitroglycerin or isosorbide dinitrate may have a favorable effect through dilatation of collateral vessels (Becker, 1978; Schulz *et al.*, 1985; Klocke *et al.*, 1987; Fallen *et al.*, 1995) or through dilatation of coronary stenoses (Brown, 1985). In Becker's animal studies (Becker, 1978), steal was not improved with nitroglycerin, whereas in clinical studies (Schulz *et al.*, 1985), nitrates decreased evidence of steal. The difference between animal and clinical studies can be explained by the nature of the stenosis – with the eccentric type so common in humans having a lumen that can be increased by relaxation of smooth muscle in its wall (Brown, 1985), but with this phenomenon impossible in the clamped or tied, symmetrically constricted artery of the experimental preparation.

The phenomenon of steal focuses attention on the differential effects of different drugs on arteries and on arterioles. Other implications are discussed in Chapter 26 and elsewhere. Many current texts make no distinction between the two different actions. It is clear, however, that they are important with respect to pulsatile phenomenon (arterial distensibility and wave reflection), as well as to the steady flow principles of steal discussed here.

Drugs, such as nitroprusside and nifedipine, dilate both arteries and arterioles in the systemic circulation and in the heart (Vatner and Hintze, 1982). One action (on arterioles) might tend to promote steal while the other would tend to offset it. There is no convincing evidence of steal occurring with nifedipine or other dihydropyridines. These drugs are widely used in patients with angina pectoris; they appear, like nitrates, to dilate arterial stenoses and collateral vessels (Engel and Lichtlen, 1981), as well as the conduit arteries (Stone *et al.*, 1983). The possibility of an adverse effect, however, exists for any drug that dilates arterioles, as well as arteries (Klocke *et al.*, 1987; Furberg *et al.*, 1995), should the former effect predominate over the latter.

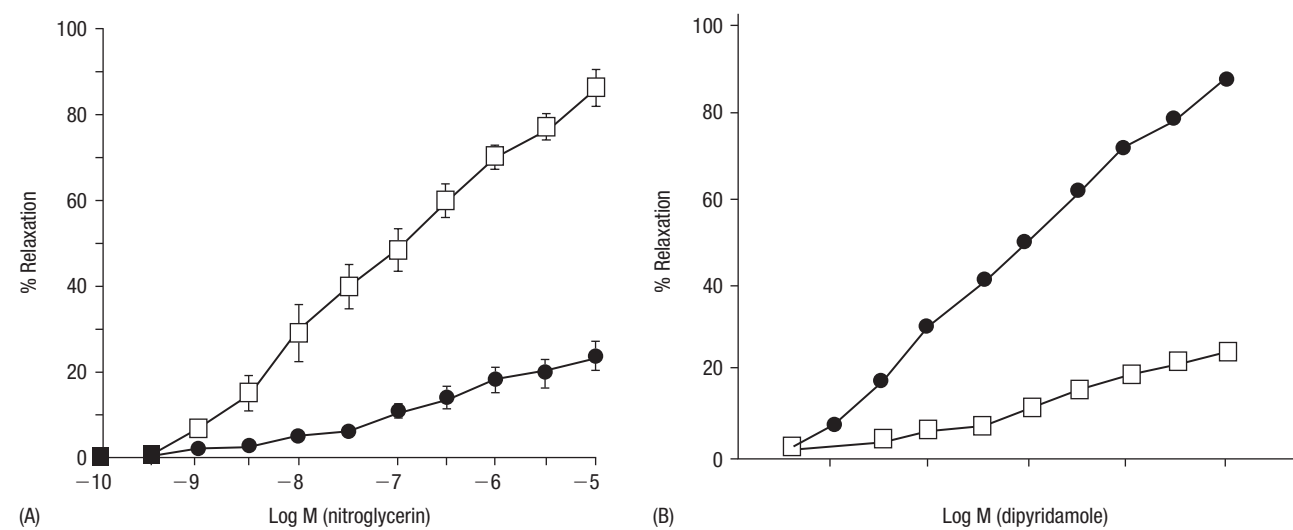


Figure 17.7 (A) Percentage increase in internal diameter (vasodilatation) of coronary arteries of different size during nitroglycerin administration. Diameter increase is much less in vessels the size of arterioles (< 100 mm; •–•) than in vessels the size of small conduit arteries (200–300 mm; □–□). After Wheatley *et al.* (1994). (B) Schematic diagram to show effects of dipyridamole on the caliber of the same-sized vessels. Dipyridamole markedly dilates arterioles, but has little effect on conduit arteries

SPECIFIC PRINCIPLES OF BLOOD FLOW TO THE NORMALLY CONTRACTING VENTRICLE

Among bodily organs, the heart is unique in that its cyclic beating activity alters phasic flow through the coronary arteries. Flow cannot readily be measured in the minute intramyocardial arteries (Hoffman, 1995; Iwanaga *et al.*, 1995) since these are squeezed shut by contraction of surrounding myocardium. Intramyocardial pressure is greater in subendocardial than in subepicardial regions (Stein *et al.*, 1985), and there is some evidence that flow is reversed from subendocardium to subepicardial regions during systole. In the epicardial coronary arteries where flow can readily be measured, repetitive contraction and relaxation of the heart causes pronounced changes in phasic blood flow (Gregg and Green, 1940; Gregg and Shipley, 1944; Berne and Levy, 1977; Ootaki *et al.*, 2005; Hartley *et al.*, 2008; Ootaki *et al.*, 2008). These cyclic changes are less marked in the right coronary arteries supplying predominantly the right ventricle (Lowensohn *et al.*, 1976; Ootaki *et al.*, 2008), but are very marked in the left coronary artery, which supplies the left ventricle (see Figs 9.19 and 17.8). In the left coronary artery, flow during ventricular systole is very small, then increases markedly in early diastole whereafter it falls off in parallel with aortic pressure during mid-to-late diastole, and then falls abruptly during isometric ventricular contraction, prior to aortic valve opening and blood flow acceleration. These coronary flow patterns are readily explained (Spaan *et al.*, 1981; Spaan *et al.*, 2008). Left ventricular contraction appears to ‘throttle’ branches of the coronary arteries entering the myocardium so that no flow can occur through these vessels during systole. The small amount of blood flowing into the left coronary artery during systole

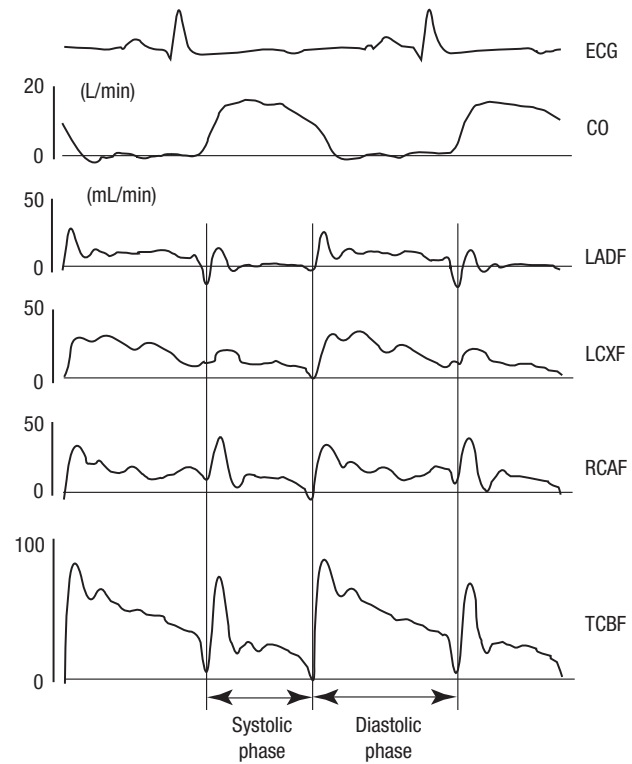


Figure 17.8 Sample cardiovascular data. CO, cardiac output (phasic bold flow); ECG, electrocardiogram; LADF, left anterior descending coronary artery flow; LCXF, left circumflex coronary artery flow; RCAF, right coronary artery flow; TCBF, total coronary blood flow. Reproduced from Ootaki *et al.* (2005)

probably just dilates the epicardial branches of this artery (Douglas and Greenfield, 1970; Reneman and Arts, 1985; Hoffman and Chilian, 2000). During diastole, throttling ceases with myocardial relaxation, and the

coronary system of the left ventricle is perfused by the pressure gradient between origin (aorta) and exit (right atrium) (Gould, 1997).

These hemodynamic factors provide complexities to the measurement and interpretation of coronary blood flow under different conditions. As pointed out above, it is still possible to discuss mean pressure and mean flow as coronary resistance, 'steal' and coronary reserve, but one must be more circumspect in interpretation of pressure–flow relationships with respect to microvascular change. These principles are of crucial importance for the interpretation of coronary flow in disease states and under conditions where cardiac function is altered.

Myocardial oxygen requirements

Since there is a close relationship between coronary blood flow and myocardial oxygen requirements (see Fig. 17.1) (Khoury *et al.*, 1965), consideration of the former must entail understanding of the latter.

There are three components of myocardial oxygen requirements: basal need to sustain viability in the absence of activity, energy required to maintain electrical activation of muscle cells, and energy expended in contraction (Gibbs, 1978; Hoffman and Buckberg, 1978; Gibbs and Chapman, 1979; Suga, 1990; Gibbs, 1995; Hashimoto *et al.*, 2008a). Of these, the third is dominant, amounting to some 60 percent of the total under basal resting conditions. Basal myocardial oxygen need is increased when the ventricle is hypertrophied (Czernin *et al.*, 1993; Gould, 1997); energy expended in contraction is increased when mechanical activity is increased (Gibbs and Chapman, 1979; Suga, 1990) and when afterload (i.e. arterial stiffness, wave reflections, and peripheral resistance) is increased.

This field has created keen interest in cardiac physiology from the earliest part of the twentieth century, when debate arose as to whether most energy was associated with pressure generation or with ejection of blood (Rhode, 1912; Hartree and Hill, 1921; Fenn, 1923). In more recent times, Suga and others (Suga, 1990; Gibbs, 1995) have shown how the basal plus electrical energy requirements can be separated from mechanical energy requirements by study of pressure–volume relationships (Fig. 17.9). This work has shown the importance of pressure generation and of ventricular volume in determining ventricular oxygen requirement associated with mechanical activity. This can also be expressed in terms of tension development, with ventricular wall tension dependent on pressure generated and on ventricular dimensions through the law of Laplace.

At any given dimension, and in the absence of cardiac failure, Sarnoff *et al.* (1958), in their classical studies, confirmed the utter dependence of oxygen requirement on left ventricular pressure generation, but showed that this depends too on the duration that pressure is maintained, i.e. on the tension–time index (TTI) or systolic

pressure–time index (SPTI) (Hoffman and Buckberg, 1978; Braunwald, 1992; Braunwald, 2009). Under a variety of experimental conditions, with an isolated heart preparation, oxygen consumption associated with mechanical activity was linearly related to the integral of ventricular pressure and time throughout systole (and when pressure was atmospheric during diastole, throughout the whole cardiac cycle). The Suga and Sarnoff approaches are complementary. Time is not represented in the former; ventricular dimensions are not fully represented in the latter.

The Sarnoff experiments have been repeatedly confirmed. Approximations have been introduced, with aortic pressure substituting for ventricular pressure. This omits the normally small isovolumic pressure increase preceding aortic valve opening, and the isovolumic pressure decrease following aortic valve closure prior to mitral valve opening. The former is increased in cardiac failure with systolic dysfunction (Boudoulas, 1990), the latter in the presence of diastolic dysfunction (Brutsaert *et al.*, 1993). In most circumstances, the aortic pressure–time index is an acceptable substitute for the ventricular pressure–time index (Buckberg *et al.*, 1972a; Buckberg *et al.*, 1972b; Braunwald, 1992; Braunwald, 2009). The same cannot be said, however, for the pressure–time index determined from a peripheral artery on account of amplification of the pressure wave and uncertainty in identification of the cardiac incisura, which denotes aortic valve closure. Other substitutes have been used for estimation of myocardial oxygen consumption and blood requirement. These include the double product (of peak brachial artery pressure and heart rate). This assumes a constant relationship between peak systolic pressure and integrated aortic systolic pressure, and between systolic time and heart rate. Although frequently used in clinical studies, such an index has proved unreliable in estimating myocardial oxygen requirements (Moffitt *et al.*, 1984), as one might expect.

The ability to generate the ascending aortic pressure wave from brachial, carotid, or radial pressure wave, and to separate systole from diastole (Chapter 30), provides measures of the aortic pressure–time indices (systolic and diastolic), which should be more useful than the currently used 'double product' or other presently used estimates (Sharman *et al.*, 2005).

Determinants of myocardial blood flow

As previously discussed, the normal coronary circulation has the capacity to supply adequate blood flow for maximal myocardial needs. However, an inability to supply adequate flow is the cause of more death and disability in Western society than any other disease (Assmann *et al.*, 1999). To understand this, one must understand the factors that normally determine, and which may limit, coronary blood flow.

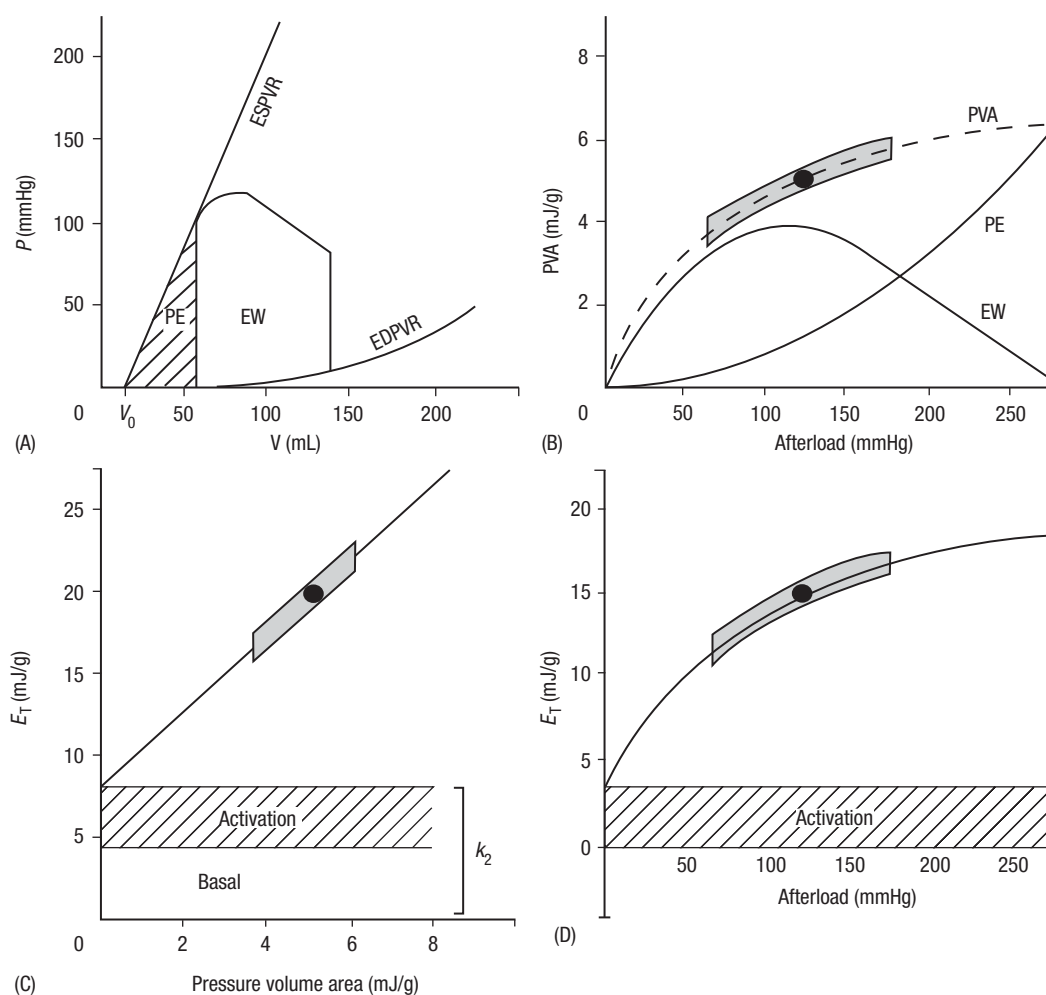


Figure 17.9 (A) Schematic view of a ventricular work diagram for humans, showing the end-systolic pressure-volume relationship (ESPVR) line and the end-diastolic pressure-volume relationship (EDPVR). Pressure-volume area (PVA) is given by the sum of the external work (EW) and the potential energy (PE) terms and V_0 is the volume axis intercept of the ESPVR. (B) Schematic drawing showing the relationship between PVA and afterload (pressure); the solid circle denotes a likely normal value for humans and the boxed region of the PVA line shows the physiological limits at this particular level of contractility. (C) Schematic of the energy E_T or mVO_2 -PVA relation. The energy intercept *in vivo* is made up of energetic contributions from basal and activation metabolism. The solid line represents a likely normal energy per beat value for humans and the shaded part of the relationship illustrates likely physiologic limits for this particular level of contractility. The slope of the relationship is drawn at 40 percent contractile efficiency. (D) Schematic diagram of the E_T or mVO_2 -afterload relationship, showing physiologic limits and a likely normal value. The basal metabolism has been subtracted. mVO_2 is myocardial oxygen consumption. Reproduced from Gibbs (1995)

The most important determinant of flow is the caliber of the epicardial coronary arteries. Normally, this is very high, such that resistance of the epicardial coronary arteries is a tiny, insignificant component of total coronary resistance. Variations in arteriolar tone normally control coronary blood flow, as in other organs. Effects of coronary stenoses are considered above and in Chapter 23. Atherosclerosis is the usual cause of coronary narrowing. Coronary spasm is rare, but may narrow or completely occlude a coronary artery and impede flow (Maseri *et al.*, 1978; Maseri *et al.*, 1991; Gianni *et al.*, 2006; Lanza *et al.*, 2008; Pennell, 2008). Combinations of atherosclerosis and increased smooth muscle tone may contribute to coronary

syndromes (Brown, 1985; Bogaty *et al.*, 1994), as may combinations of atherosclerosis and thrombus formation (see Chapter 23) (Fuster *et al.*, 1992a; Fuster *et al.*, 1992b). Clinicians dealing with angina pectoris usually show greatest concern for such coronary artery lesions as demonstrated during coronary angiography, with far less concern for the other factors that may alter coronary blood flow (Merkus *et al.*, 1999; Fokkema *et al.*, 2005; Kolyva *et al.*, 2008). Such an approach is challenged by the work of Ferro *et al.* (1995), which showed that a minor change in just one factor (duration of diastole) could have as much effect on coronary blood flow as could a substantial increase in coronary stenosis (Fig. 17.10). Here, a

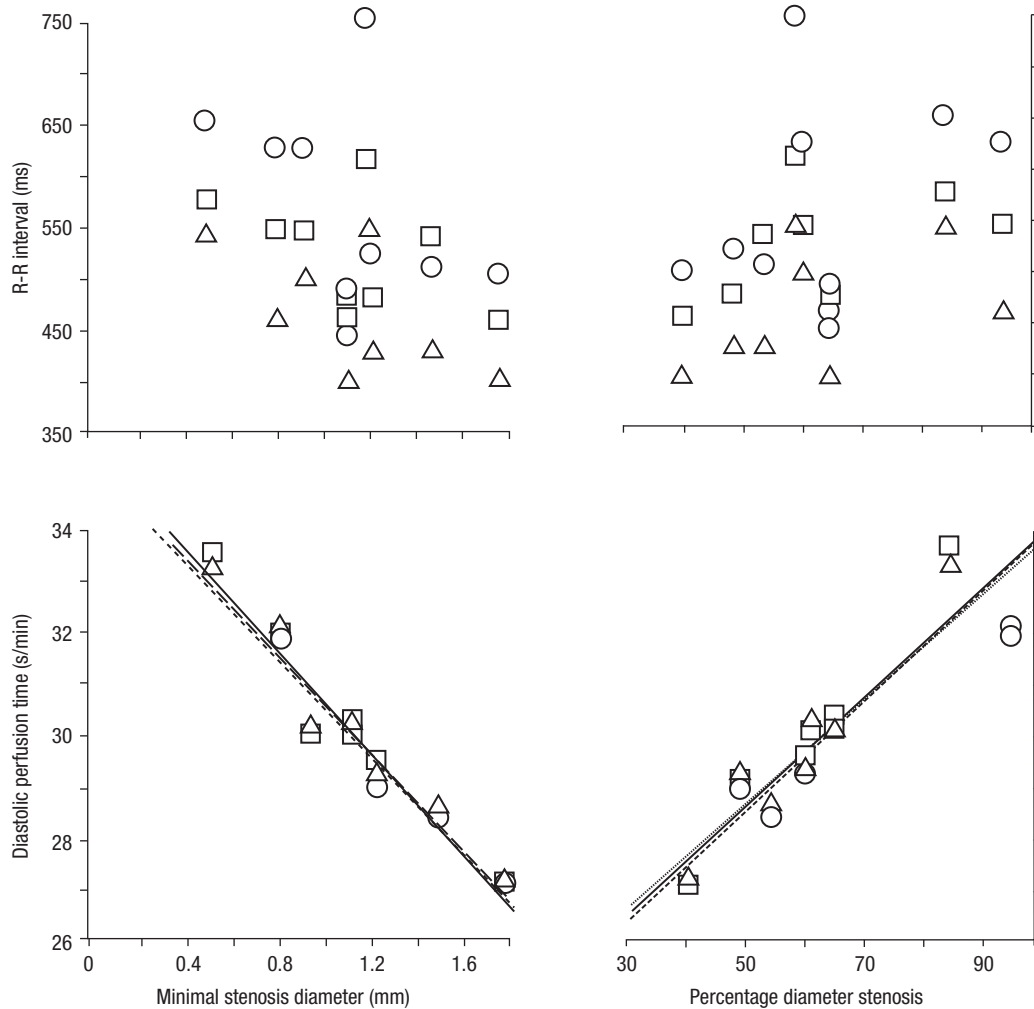


Figure 17.10 Scatterplots showing the relation of diastolic perfusion time (below) and R-R interval (above) to minimal stenosis diameter and percentage diameter stenosis in patients during upright exercise ($\square$), supine exercise ($\circ$) and atrial pacing ($\triangle$). Reproduced from Ferro *et al.* (1995)

decrease in diastolic period from 54 to 45 percent of the cardiac cycle has the same effect in limiting coronary blood flow as an increase in coronary narrowing from 40 to 90 percent of lumen diameter, i.e. a relatively small change in diastolic perfusion time has the same effect as a very large change in coronary narrowing.

Coronary blood flow to the left ventricular myocardium can only occur during diastole when the 'throttling' effect of myocardial compression on intramyocardial radicles of the coronary arteries has been reduced (Douglas and Greenfield, 1970) as described above. Determinants of coronary flow thus comprise duration of diastole and the pressure gradient between aorta and right atrium over this period (Fig. 17.11). If left ventricular pressure throughout the period of diastole is elevated, as by left ventricular failure, this gradient may be reduced by a 'waterfall' effect, such that the effect of the perfusing gradient is between the aorta and left ventricle. This pressure gradient may be measured directly from

catheterization of the aorta and left ventricle (Fig. 17.11), or it may be estimated from the aortic pressure-time index (DPTI), which assumes that the pressure falls instantaneously from the incisura and rises instantaneously to the aortic wave foot, as well as approximating zero in the left ventricle during diastole (Buckberg *et al.*, 1972a; Buckberg *et al.*, 1972b). The aortic DPTI provides a useful guide to the hemodynamic gradient that generates coronary perfusion but will overestimate this, particularly in the presence of cardiac failure with systolic (or diastolic) dysfunction, when diastolic ventricular pressure is high and when pressure rises (or falls) more gradually to or from its levels during systole. Even under normal conditions, the DPTI overestimates the hemodynamic gradient for perfusion, since perfusion of the coronary system effectively stops during long periods of diastole when aortic pressure is still in the region of 20–30 mmHg. This is not unique to the coronary circulation, but pertains to other beds as well. The

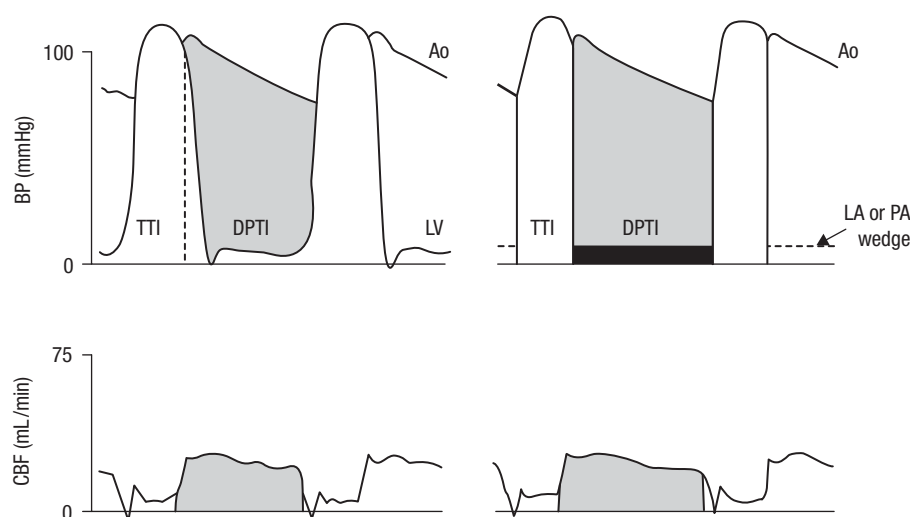


Figure 17.11 Left panel: Superimposed phasic recordings of aortic (Ao) and left ventricular (LV) blood pressure and phasic left coronary blood flow (CBF). The stippled area in the pressure recording was used to calculate the diastolic pressure–time index (DPTI), and the stippled area under the flow recording was used to calculate diastolic CBF. Right panel: How DPTI is calculated from aortic and mean left atrial (LA) or pulmonary artery (PA) wedge pressure. (Right atrial pressure is used if it exceeds these.) The blackened area shows how these pressures affect DPTI. TTI, tension–time index. Reproduced from Buckberg *et al.* (1972b)

phenomenon is due to anomalous viscosity of blood (see Chapter 2), with very high values achieved at very low flow rates (Whitmore, 1968; Hoffman, 1984). It accounts for the inability to describe precisely pressure–flow relationships as resistance, and the low coronary reserve at low diastolic pressures.

MYOCARDIAL ISCHEMIA

Myocardial ischemia (insufficient blood supply to the myocardium) results when there is an imbalance between supply and demand, with blood supply insufficient to meet demand. This may result from increased demand, as when heart rate and systolic pressure increase during exercise, or when supply is limited by coronary narrowing, endothelial dysfunction, or decreased coronary perfusion time. If severe and prolonged, ischemia results in irreversible myocardial damage (myocardial infarction). The classic symptom of myocardial ischemia is oppressive chest pain or discomfort and is termed ‘angina pectoris’; myocardial ischemia, however, may infrequently occur without angina.

Myocardial ischemia is associated with abnormal function of the affected myocardial cells. Relaxation is delayed and contraction impaired. If sufficient myocardium is affected, this may alter ventricular performance and further accentuate angina by causing decrease in cardiac output and blood pressure where reflex tachycardia cause relative shortening of the diastolic period and increasing the ventricular diastolic pressure, creating a vicious circle where ischemia begets ischemia. In left

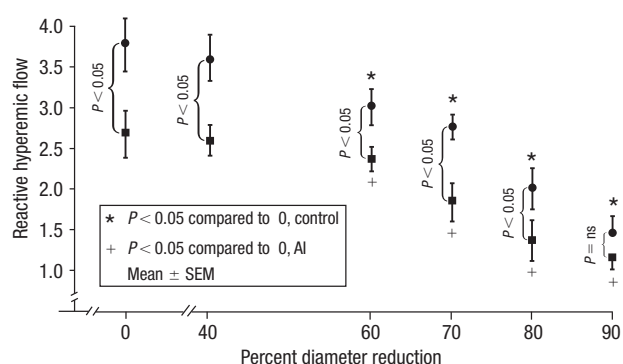


Figure 17.12 Normalized peak reactive hyperemic flow with control (●) and aortic insufficiency (AI) (■) at different degrees of coronary artery diameter reduction. At all degrees of narrowing, < 90 percent AI responses were less than control responses. Compared with their respective responses at 0 percent narrowing, both control and AI responses decreased with narrowing > 40 percent. Reproduced from Feldman *et al.* (1979b)

ventricular hypertrophy resulting, for example, from aging, systemic hypertension, aortic coarctation, or valvular heart disease, coronary artery blood flow (i.e. supply) begins to decrease and myocardial ischemia appears at a coronary narrowing less severe than would be expected (Fig. 17.12) (Feldman *et al.*, 1979b; Feldman *et al.*, 1983; Czernin *et al.*, 1993; Gould, 1997). Also, in severe coronary artery disease, the autoregulatory mechanism is exhausted and coronary blood flow is totally dependent upon perfusion pressure (Westerhof *et al.*, 2006; Westerhof *et al.*, 2010; (Fig. 17.13).

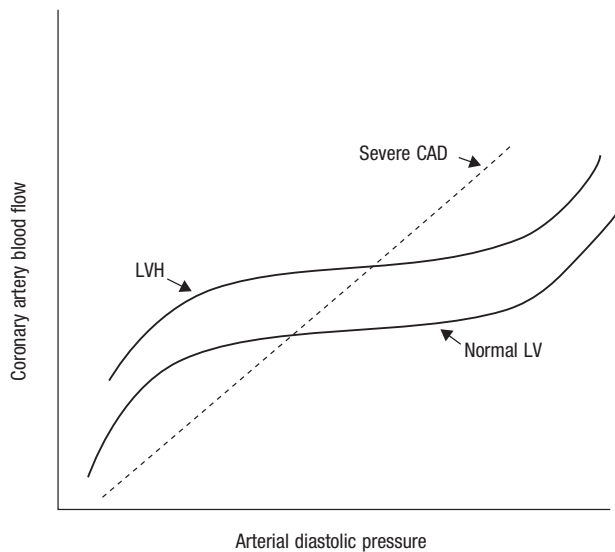


Figure 17.13 Diagram illustrating the relationship between coronary artery blood flow and arterial diastolic blood pressure (i.e. perfusion pressure). Pressure–flow relationships for a normal (LV) and a hypertrophied left ventricle (LVH) are shown. If the coronary arteries have minor or no disease autoregulation maintains flow at a constant level for pressures between about 50 and 130 mmHg. With severe coronary artery disease (CAD), blood flow is totally dependent upon diastolic arterial pressure (i.e. there is a linear relationship between arterial diastolic pressure and coronary blood flow, broken line). Therefore, when diastolic pressure falls below about 70–80 mmHg in ventricles of normal mass it becomes jeopardized. This ill effect occurs at a higher pressure in LVH. Reproduced from Nichols and Edwards (2001)

Subepicardium versus subendocardium

All the points made above on pulsatile hemodynamic factors refer to the myocardium as a whole. However, the subendocardial region is more susceptible to impaired blood flow and to myocardial ischemia than is the subepicardial region of the myocardium (Buckberg *et al.*, 1972a; Buckberg *et al.*, 1972b). The classic evidence of a positive exercise stress test is depression of the electrocardiographic ST segment; this is due to reduced subendocardial flow (Fig 17.14) resulting in ischemia in the presence of normal or near-normal subepicardial perfusion. The regulation of coronary blood flow during exercise has been published recently in an outstanding detailed review by Duncker and Bache (2008).

There are two reasons for special susceptibility of the subendocardium, one anatomical and the other functional, but both act to compromise blood flow to this region. The major coronary arteries run on the epicardial surface of the heart, with the branches diving into the muscular wall and running perpendicularly from the epicardial to endocardial regions (Fulton, 1965). The epicardial region is closer to the source of supply, the subendocardial region further away with a pressure gradient in the small

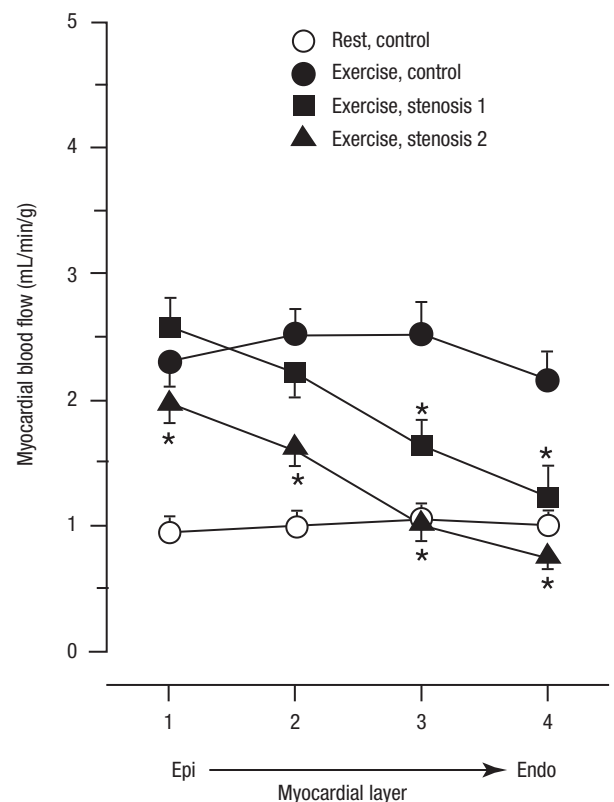


Figure 17.14 Effect of a coronary artery stenosis on the transmural distribution of myocardial blood flow during treadmill exercise in dogs. Blood flows are shown at rest and during exercise with normal arterial inflow and in the presence of a modest (stenosis 1) and severe (stenosis 2) degree of stenosis. Note the progressive hypoperfusion from the outer (subepicardial, epi) to the innermost (subendocardial, endo) myocardial layer. Reproduced from Duncker and Bache (2008); Ball and Bache (1976)

perforating arteries. The second reason relates to intramyocardial pressure that ‘throttles’ the coronary artery branches within the myocardium and impairs coronary blood flow during muscular contraction. This pressure is greatest deep in the subendocardial layers of the heart and least in the subepicardial region (Stein *et al.*, 1985; Hoffman, 1995).

As might be expected, the potential for subendocardial ischemia is increased in the presence of ventricular hypertrophy (Frohlich *et al.*, 1992b; Gould, 1997). Here, not only is oxygen (and blood) demand increased, but also the wall is thicker with greater distance from epicardium to endocardium, and (when pressure is high or the ventricle dilated) intramyocardial pressure is increased.

The different susceptibility of subepicardium and subendocardium to ischemia has been utilized to study the effect of factors that increase myocardial demand or restrict supply. In experimental circumstances, the difference between subepicardial and subendocardial supply can be measured with radioactive microspheres. Buckberg

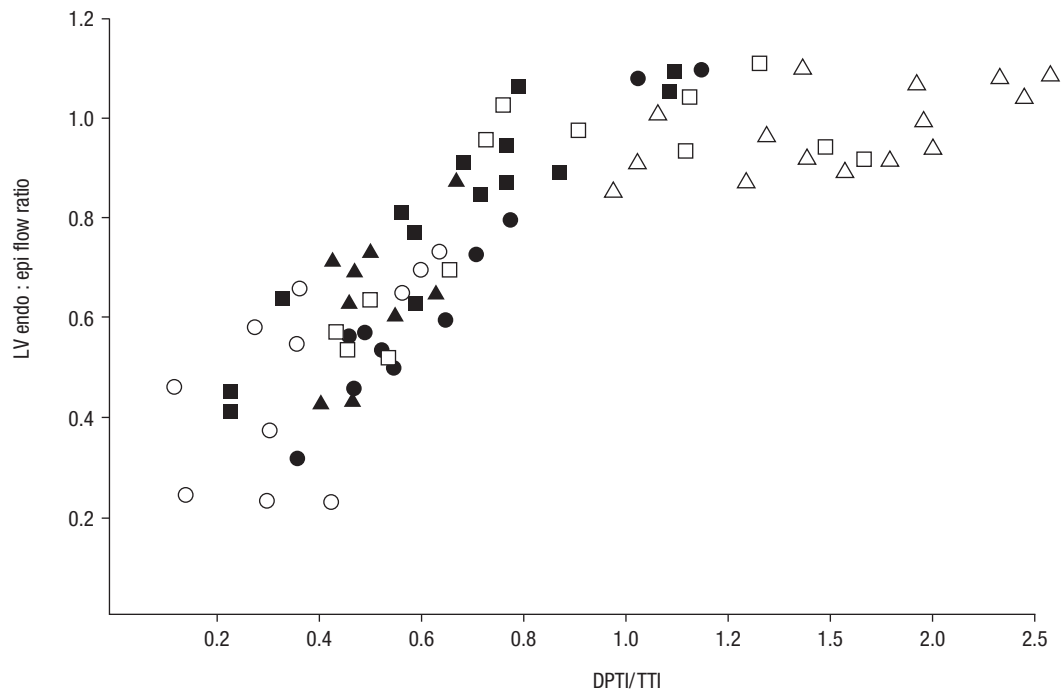


Figure 17.15 Diastolic pressure–time index (DPTI)/tension–time index (TTI) is plotted against left ventricular (LV) flow distribution measured with radionuclide microspheres. Note that subendocardial (endo) muscle is underperfused relative to subepicardial (epi) muscle when the ratio DPTI/TTI is below 0.7. (Δ) Control; ($\blacktriangle$) atrioventricular fistula; ($\circ$) SVAS; ($\bullet$) Isuprel (intravenous isoproterenol infusion); ($\blacksquare$) patent ductus arteriosus; ($\square$) pacing. SVAS, supraaortic stenosis. Reproduced from Buckberg *et al.* (1972b)

et al. (Buckberg *et al.* 1972a; Buckberg *et al.* 1972b) have confirmed the extreme susceptibility of the subendocardium to factors that increase myocardial oxygen demand and/or restrict blood supply (Fig. 17.15). They have coined the term ‘subendocardial viability ratio’ as the ratio of aortic diastolic pressure time integral and aortic systolic pressure time integral (SPTI or TTI), i.e. as a measure of hemodynamic capacity for supply divided by myocardial oxygen demand. They have shown that when this exceeds unity, there is no significant difference between subepicardial and subendocardial blood flow, with both perfused adequately, but when the value falls below 0.5, the ratio of subendocardial to subepicardial flow falls precipitously, indicating subendocardial ischemia. This has been used clinically by Buckberg *et al.* (1972b) and has been adapted by ourselves as a useful measure in sphygmocardiography (Chapter 30).

FACTORS LIMITING CORONARY BLOOD FLOW

Coronary narrowing

Coronary atherosclerosis (or the formation of plaque in the artery) is the most frequent and the most obvious cause of coronary flow limitation and myocardial ischemia. Symptoms of angina pectoris usually occur when there is substantial epicardial coronary artery narrowing,

however, angina occurs quite often in individuals (especially women) with normal coronary arteries (see below). Coronary narrowing is usually due to atherosclerosis (see Chapter 23), but may rarely occur on account of localized muscular contraction (spasm) in the artery wall (Maseri *et al.*, 1978; Braunwald, 1992; Braunwald, 2009). The severity of atherosclerotic narrowing may be potentiated by a generalized or localized increase in muscular tone in the coronary arterial wall consequent on increased autonomic nerve activity (Vatner *et al.*, 1980; Vatner and Hintze, 1982; Brown, 1985; Bogaty *et al.*, 1994), endothelial dysfunction (Egashira *et al.*, 1993b; Egashira *et al.*, 1993c), or perverted response to therapy with such drugs as ergotamine, serotonin, or sumatriptan (Golino *et al.*, 1991; Hillis and Lange, 1991; McFadden *et al.*, 1991). Also, plaque disruption with superimposed thrombosis is the main cause of acute coronary syndromes of unstable angina, myocardial infarction, and sudden death (Fuster *et al.*, 1992a; Fuster *et al.*, 1992b; Falk *et al.*, 1995; Libby, 2002; Libby *et al.*, 2002a; Libby *et al.*, 2002b; Finn *et al.*, 2007; Tanaka *et al.*, 2008). These matters are discussed further in Chapter 23.

Clinicians are most concerned – even preoccupied – with coronary artery narrowing being the cause of angina pectoris, with other causes of this condition not receiving appropriate consideration. The term ‘syndrome X’ has been coined for angina pectoris with normal coronary arteriograms (Kemp, 1973; Cannon *et al.*, 1992; Conti, 1993; Panting *et al.*, 2002; Cannon, 2009). This syndrome

clearly includes angina caused by all other mechanisms; unfortunately, it may also include many people without myocardial ischemia, but with non-cardiac chest pain simulating myocardial ischemia.

Narrowing of small intramyocardial coronary arteries

'Microvascular angina' is implicated as a cause of syndrome X and a factor in hypertension and diabetic patients, and in patients with connective tissue disease (Nahser *et al.*, 1995; Cannon, 2009). Histological studies in these conditions have not been decisive, and evidence of small vessel coronary narrowing is usually taken by implication on finding a reduced CFR (endothelial dysfunction) in the presence of normal epicardial arteries (Tillmanns *et al.*, 1991; Suzuki *et al.*, 1994; Chilian, 1997; Bugiardini and Merz, 2005; Monti and Piatti, 2005; Galiuto *et al.*, 2007; Cemin *et al.*, 2008; Phan *et al.*, 2009). As discussed above, this may be a reasonable implication for other vascular beds, but cannot be applied with confidence to the coronary system, where factors associated with contraction and diastolic perfusion gradient and duration cannot be controlled. Small vessel disease appears to be very rare. 'Microvascular angina' may be due to the factors related to the perfusing pressure gradient. A subgroup of women in the Women's Ischemia Syndrome Evaluation (WISE) study were found to have increased aortic wave reflection, central pulse pressure, SPTI, and abbreviated diastolic pressure time. Cannon (2009), in writing a state-of-the-art paper entitled 'The continuing dilemma of chest pain with normal coronary arteriogram', made no mention at all of perfusion pressure gradient, and referred only to arterial narrowing as the cause of anginal pain in this common condition, especially of older women. A challenge to the author (O'Rourke and Nichols, 2009b) was published, but no response was forthcoming. A new 'syndrome Y' or 'coronary slow-flow phenomenon' has been proposed to differentiate this from Cannon's 'syndrome X' (Fineschi *et al.*, 2010). Coronary slow-flow phenomenon has been associated with increased aortic stiffness and altered central hemodynamics (Tanriverdi *et al.*, 2010).

Diastolic perfusion pressure

Decreased diastolic perfusion pressure may result from arterial hypotension (as in cardiogenic shock, where it is a major factor in the vicious circle of cardiac decompensation), from increased left ventricular diastolic pressure (in left ventricular failure), from increased right atrial pressure (in right ventricular failure), or from left ventricular dysfunction with slowed ventricular relaxation (Frohlich *et al.*, 1992a). These factors are rarely the cause of myocardial ischemia entirely by themselves, but usually

act in combination with coronary narrowing or with ventricular hypertrophy (Frohlich *et al.*, 1992b; Gould, 1997). Aortic stiffening causes a reduction in coronary perfusion pressure and coronary flow reserve and predisposes to myocardial ischemia (Weber *et al.*, 2006; Galderisi, 2008; Guelen *et al.*, 2008; Ikonomidis *et al.*, 2008; Nemes *et al.*, 2008), but this usually requires other factors, including increased myocardial demand (from increased left ventricular systolic pressure or relatively increased systolic period from tachycardia) or a minor degree of coronary narrowing (Watanabe *et al.*, 1992; Watanabe *et al.*, 1993; Kass *et al.*, 1996).

Diastolic time

Diastolic time as a factor in the induction of myocardial ischemia appears to have been underestimated to date, but has been highlighted by the work of Ferro *et al.* (1995). These workers examined patients with single proximal left anterior descending coronary artery lesions and related the severity of stenosis to the diastolic period at a pre-determined stage of myocardial ischemia during exercise stress testing. There was a very close association between stenosis severity and duration of diastole (see Fig. 17.10). Indeed, in percentage terms, change in diastolic duration was more relevant than was severity of coronary stenosis. This fascinating work will need to be confirmed. It is consistent with the findings of Buckberg, with pathophysiologic principles previously enumerated in this chapter and the long-standing work of Boudoulas and colleagues (Boudoulas *et al.*, 1979; Boudoulas *et al.*, 1982; Boudoulas, 1990; Boudoulas, 1991; Boudoulas *et al.*, 1995b; Boudoulas and Leier, 2000) and more recently by Merkus *et al.* (1999), Fokkema *et al.* (2005) and Kolyva *et al.* (2008) and by our group here at the University of Florida. The ability to estimate perfusion pressure, as well as diastolic time, non-invasively from peripheral pressure measurements, and to estimate myocardial oxygen demand as well, should enhance further studies of this subject (see Chapter 30).

Increased aortic stiffness

As central arterial stiffness and wave reflection amplitude increase, aortic systolic pressure rises, pulse pressure widens, and ventricular wall stress and myocardial oxygen demand increase, while aortic diastolic pressure and, therefore, coronary artery perfusion pressure decreases. These alterations in arterial wall properties can occur without a change in peripheral resistance (Kass *et al.*, 1996) and indeed without increase in brachial systolic or pulse pressure (Kingwell, 2000). Since the majority of coronary arterial and left ventricular myocardial blood flow (>80 percent) occurs during diastole (Douglas and Greenfield, 1970), such abnormalities in ventricular/

vascular coupling unbalance the favorable supply/demand ratio (Pepine and Nichols, 2007; Guelen *et al.*, 2008) and promote myocardial ischemia, angina pectoris, and contractile dysfunction in coronary artery disease. In the normal coronary circulation, however, blood flow is maintained over a wide range of perfusion pressures by the process of autoregulation (see Figs 17.2 and 17.13) – as perfusion pressure falls, vasodilation maintains a normal blood flow (Hoffman, 1987). Also, in left ventricular hypertrophy and other conditions of increased myocardial oxygen demand, coronary blood flow increases to meet the demand. Indeed, in the study by Watanabe *et al.* (1993), when the heart ejected into a stiff or non-compliant aorta, systolic pressure and myocardial oxygen demand increased and diastolic pressure decreased, but coronary blood flow also increased in response to the increased demand, and myocardial contractile function was maintained at rest. However, the increased arterial stiffness caused a decrease in coronary blood flow reserve (Leung *et al.*, 2005; Wang and Hu 2005; Fukuda *et al.*, 2006; Ikonomidis *et al.*, 2008; Nemes *et al.*, 2008) so that during increased myocardial contractility, endocardial blood flow was impaired and the subendocardial electrocardiogram showed signs of ischemia (Ohtsuka *et al.*, 1994). These undesirable alterations were enhanced in the presence of a high-grade coronary artery stenosis (Watanabe *et al.*, 1993) and during reductions in diastolic blood pressure (Isoyama *et al.*, 1983). During total coronary artery occlusion and myocardial ischemia, increased aortic stiffness caused marked enhancement in cardiac dysfunction (Kass *et al.*, 1996). When the duration of coronary artery occlusion was increased to produce myocardial infarction, increased coronary perfusion pressure caused a decrease in infarct size, while lowering perfusion pressure had the opposite effect (Maroko *et al.*, 1971).

The most recent publications that investigated the various blood pressure components (systolic, diastolic, and pulse) as predictors of risk for coronary heart disease involved the Framingham (Franklin *et al.*, 1999; Franklin *et al.*, 2001; Haider *et al.*, 2003) and other (Boutitie *et al.*, 2002; Gasowski *et al.*, 2002; Protogerou *et al.*, 2007) populations. The results from these studies showed that, in middle-aged and elderly individuals, coronary heart disease risk was significantly increased with lower diastolic pressure. Others have also shown an inverse (or negative) relationship between diastolic blood pressure and both myocardial infarction (Madhavan *et al.*, 1994; Mitchell *et al.*, 1997a) and cardiovascular mortality (Benetos *et al.*, 1997; Protogerou *et al.*, 2007). Also, in patients with left ventricular hypertrophy and coronary artery disease resulting from increased aortic stiffness and wave reflection amplitude, acute lowering of diastolic blood pressure to less than 85 mmHg increases ischemic heart disease events (Farnett *et al.*, 1991). These investigators warned against excessive lowering of diastolic blood pressure in high-risk patients.

THERAPEUTIC IMPLICATIONS

The above considerations emphasize that there are many strategies for optimal treatment of myocardial ischemia in addition to reduction of coronary artery narrowing by angioplasty, stent insertion, or bypass grafting (see Chapter 26) (Boden *et al.*, 2007). Strategies depend upon a complete understanding of myocardial oxygen supply and demand and the implications of cardiac contraction and relaxation.

Reduction in oxygen demand is accomplished by reduction in left ventricular mass and left ventricular pressure throughout systole, reduction in systolic duration, and reduction in left ventricular dilatation. Therapies include the use of drugs (angiotensin-converting enzyme inhibitors (ACEI), calcium channel antagonists, angiotensin receptor blockers (ARB), nitrates, phosphodiesterase type-5 inhibitors, alpha-blockers, diuretics, and β -blockers) for hypertension, and reduction in left ventricular systolic pressure and left ventricular mass, together with reduction in systolic time through reduction of heart rate with β -blocking agents and calcium channel antagonists. Reduction in ventricular dilatation may result from appropriate treatment of hypertension and of cardiac failure – with ACEIs, nitrates, diuretics, or specific therapy directed at the cause of cardiac failure (see Chapters 15 and 26).

An increase in coronary blood (and oxygen) supply is achieved by improving coronary perfusion pressure through treatment of left and right ventricular failure if present, or of impaired diastolic relaxation if possible, and through prolongation of diastolic duration by a reduction in heart rate (with β -blockers or non-dihydropyridine calcium channel antagonists) or through prevention of undue tachycardia as may occur with short-acting dihydropyridines. Since these factors are interrelated, compromise is required. Undue reduction in heart rate may cause cardiac failure; undue reduction in arterial pressure may impair perfusion more than myocardial requirement. Clinical assessment may be aided by non-invasive pressure wave analysis (see Chapter 30). Knowledge of coronary ‘steal’ endorses the use of vasodilator agents that dilate coronary arteries (especially nitrates) rather than those whose predominant effect is on arterioles (dipyridamole, hydralazine, prazosin, and terazosin). Nitrates relieve the type of ‘steal’ that can occur with these drugs and with exercise, as well as causing dilatation of eccentric coronary stenoses. The effect of reducing wave reflection and decreasing aortic and left ventricular systolic pressure and myocardial oxygen demand is shared by ACEIs, ARBs, and calcium channel antagonists (Chapters 12 and 26).

ARTERIAL COUNTERPULSATION

Arterial counterpulsation is a well-recognized and highly effective method for left ventricular mechanical support,

especially in the presence of myocardial ischemia (O'Rourke, 1977; Braunwald, 1992; Waksman *et al.*, 1993) and refractory angina (Lawson *et al.*, 1992). This is discussed further in Chapter 24. Counterpulsation illustrates the principles discussed previously in this chapter. By rhythmically inflating an intra-aortic balloon (Fig. 17.16) or external leg cuffs (Fig. 17.17) in phase with the heart, counterpulsation increases aortic pressure throughout diastole, thus increasing coronary perfusion pressure and blood flow (Kern *et al.*, 1993a; Kern *et al.*, 1993b), while decreasing aortic and left ventricular pressure during systole, and hence decreasing myocardial oxygen need and promoting more complete left ventricular ejection. Intra-aortic balloon counterpulsation is widely used for left ventricular support postoperatively, for acute coronary syndromes or as a bridge to transplantation. Also, the counterpulsation technique has been employed non-invasively in patients with refractory angina. Three pairs of pneumatic cuffs are wrapped around the calves, lower thighs, and upper thighs. During early diastole, these cuffs inflate sequentially from distal to proximal, resulting in a 'milking effect' of blood from the lower extremities and an augmentation in diastolic pressure (Werner *et al.*, 1999; Bonetti *et al.*, 2003b; Bonetti *et al.*, 2003c). This diastolic pressure wave is transmitted backwards toward the heart (Fig. 17.17) thereby increasing perfusion pressure and coronary blood flow (see Fig. 17.5) (Werner *et al.*, 1999; Michaels *et al.*, 2002; Wu *et al.*, 2006). At the onset of systole, all cuffs are deflated, resulting in acute afterload reduction (Michaels *et al.*, 2002). Several small trials have

demonstrated a consistent and prolonged positive response in patients treated with EECP after 35×1 hour per day sessions. Benefits associated with EECP include reduction of angina (Stys *et al.*, 2001; Nichols *et al.*, 2006; Casey *et al.*, 2008a), and nitrate use (Lawson *et al.*, 2000), increased exercise tolerance (Stys *et al.*, 2001), favorable psychosocial effects, and enhanced quality of life (Springer *et al.*, 2001). These long-term beneficial effects of EECP may be due to improvement in endothelial function through increased release of nitric oxide (Bonetti *et al.*, 2003a; Bonetti *et al.*, 2003b; Bonetti *et al.*, 2003c; Braith *et al.*, 2010; Casey *et al.*, 2010).

NEW DIAGNOSTIC METHODS

Intravascular ultrasonography

Intravascular ultrasonography (or IVUS) is a relatively new imaging modality in which ultrasonic crystals are mounted on the distal tip of a flexible catheter small enough to allow percutaneous insertion by standard catheterization techniques and advanced into a vascular structure, usually a coronary artery. These catheters are inserted over a guide wire and used to obtain vessel wall structure. Great interest has been generated in the potential use of this device to determine the severity, composition, and physiologic significance of coronary artery atherosclerotic lesions or plaque volume (Nissen *et al.*, 1991; Nissen *et al.*, 2000; Schoenhagen *et al.*, 2000).

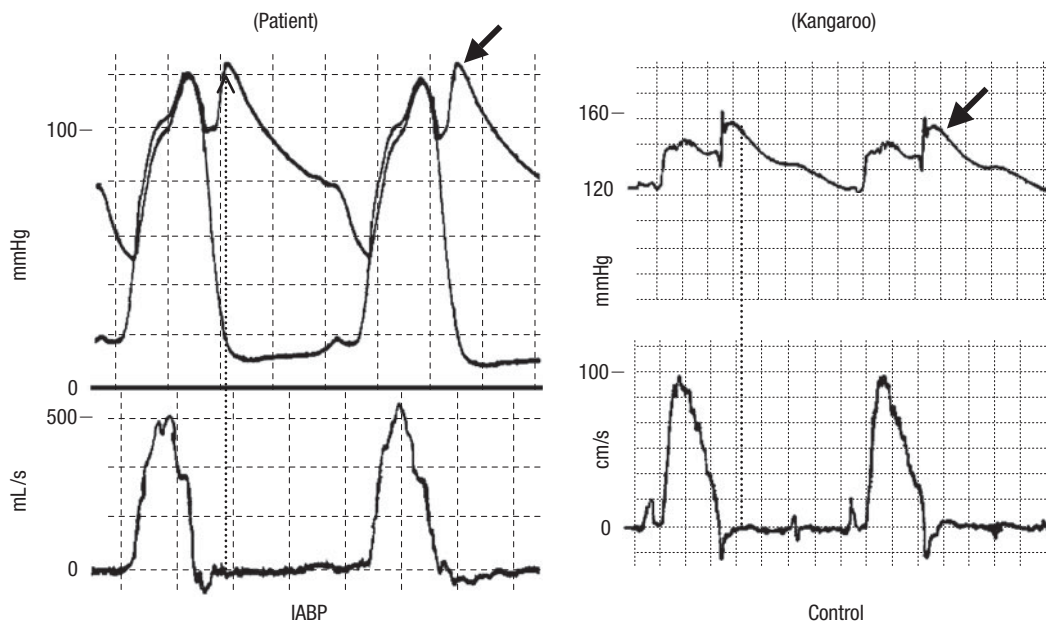


Figure 17.16 Left: High-fidelity ascending aortic and left ventricular pressures and aortic blood flow waves measured in a patient during intra-aortic balloon counterpulsation. Right: Ascending aortic pressure and flow waves measured in a kangaroo. Arrows denote the peak of the augmented diastole pressure wave. In the human, the wave is caused by balloon inflation in the descending aorta while in the kangaroo the wave is caused by intense reflection from the lower body. Neither generated any measurable flow (broken lines). IABP, intra-aortic blood pressure

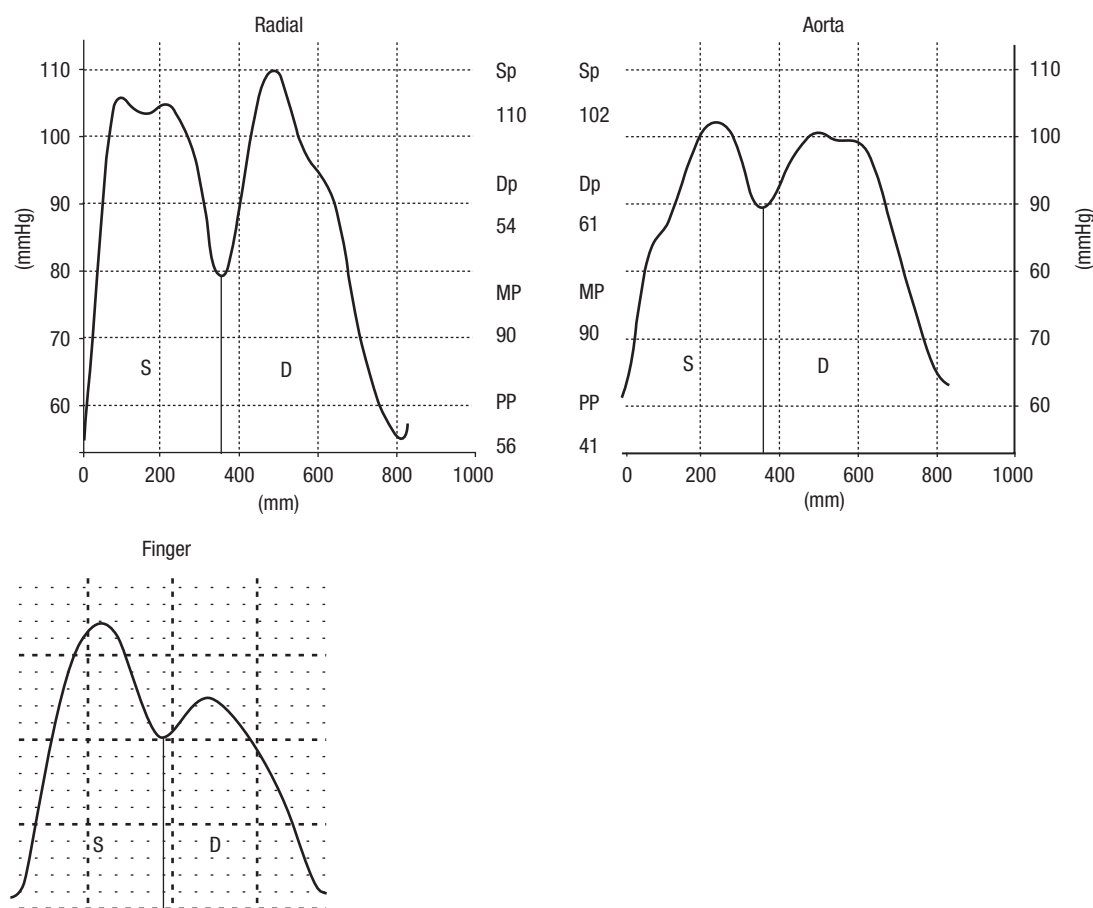


Figure 17.17 Simultaneous pressure waves obtained during enhanced external counterpulsation (EECP) in a patient with refractory angina. The finger pulse was recorded with photoplethysmography while the radial artery pressure wave was recorded using applanation tonometry and the aortic pressure wave was synthesized from the radial wave using a mathematical transfer function

For excellent reviews of intravascular ultrasonography, see Hibbard *et al.* (1996), (Nissen, 2001a; Nissen, 2001b; Nissen, 2002), Mintz *et al.* (2001), Bocksch *et al.* (2005), Nicholls *et al.* (2007b) and Garcia-Garcia *et al.* (2010). IVUS, developed over the past two decades, has many potential advantages in characterizing the atherosclerotic disease process and effects of intervention on plaque volume and regression (Nissen *et al.*, 1991). The catheter uses high frequency ultrasound to image not only the coronary lumen, but also the structure of the vessel wall, including the atherosclerotic plaque. The catheter is approximately 1 mm in diameter, operates typically at 20–45 MHz, and generates full-motion, cross-sectional images at 30 frames per second. With modern devices, axial resolution is approximately 50–80 μm , which is approximately one order of magnitude greater resolution than available by angiography. Refinements in technology have permitted the morphologic characterization of arterial lesions before, during and after pharmacologic and catheter-based vascular interventions. Preliminary designs have combined ultrasound imaging with atherectomy (Yock *et al.*, 1990), laser (Aretz *et al.*, 1989), and balloon (Isner

et al., 1991) devices to provide on-line feedback in guiding catheter-based interventional procedures. Advances in image processing have permitted three-dimensional reconstruction of arterial structures (Rosenfield *et al.*, 1991). When dynamic two-dimensional information is combined with Doppler assessment of blood flow velocity, physiologic pressure–volume relations can be established. At present, two different catheters must be used to obtain blood flow velocity and vessel cross-sectional area, but future designs will, no doubt, incorporate transducers in one catheter for both measurements.

The traditional model of arterial disease, in which the plaque develops in the vessel wall over many years, gradually narrowing the lumen to produce symptoms, is not completely accurate. A more accurate model of atherosclerosis was originally described by Glagov *et al.* (1987) who showed that coronary ‘remodeling’ enables patients to develop large atherosclerotic plaques without reduction in lumen size (Fig. 17.18). In essence, during the early progression of disease, atherosclerotic material grows into the vessel wall, outwardly displacing the external elastic membrane. Accordingly, patients can develop a large

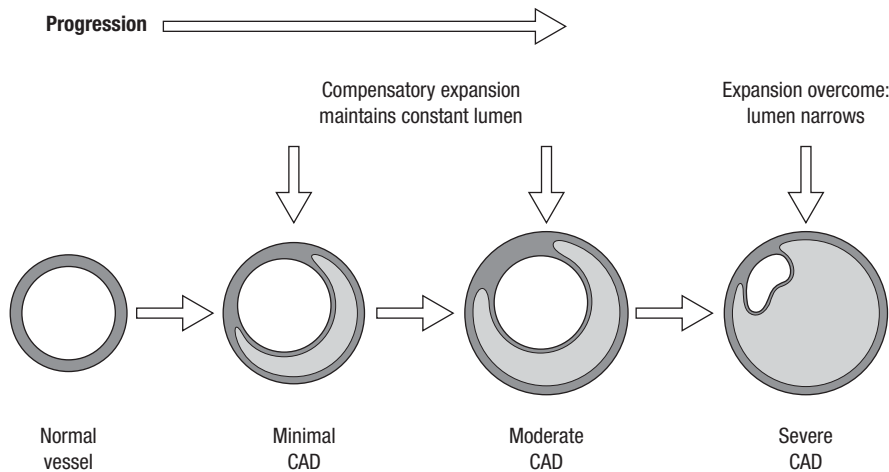


Figure 17.18 Illustration of the coronary artery remodeling hypothesis: in coronary artery disease (CAD) compensatory expansion maintains the vessel lumen diameter despite increasing plaque size. Reproduced from Glagov *et al.* (1987)

atherosclerotic plaque with no luminal narrowing. It is suggested that this 'compensatory' expansion through maintaining a constant lumen, would protect patients against ischemic symptoms.

Great interest has been generated in the use of IVUS to determine the severity and composition of atherosclerotic lesions (Nissen *et al.*, 1991; Nissen *et al.*, 2000; Schoenhagen *et al.*, 2000; Taki *et al.*, 2010). IVUS can provide valuable information regarding plaque characteristics (plaque volume, calcification, remodeling, encroachment into the arterial lumen, cross-sectional lumen area, etc). This technique is commonly employed as a tool for tracking changes of atherosclerotic plaque volume in the setting of clinical studies. A major clinical trial using IVUS to compare aggressive to moderate lipid-lowering therapy in patients receiving either 80 mg atorvastatin or 40 mg pravastatin (REVERSAL study) showed superiority of the more aggressive drug therapy with respect to the primary outcome – plaque volume in the coronary artery wall (Nissen, 2002; Schoenhagen *et al.*, 2006; Sipahi *et al.*, 2006). Other clinical trials (e.g. ASTEROID, ACTIVATE and CAMELOT) using similar methodology have been designed and are in progress or finished (Nissen *et al.*, 2004; Nissen *et al.*, 2006; Bayturan *et al.*, 2009). Analysis of such IVUS data revealed that there was a direct relationship between the level of LDL-C achieved and the rate of disease progression, with regression observed when LDL-C was less than 70 mg/dL (Nissen *et al.*, 2006). A schematic drawing of the methodology for remodeling measurements used in the REVERSAL study is shown in Fig. 17.19. Contrast-enhanced IVUS after intracoronary injection of microbubbles has been recently used as a novel tool for characterization of neovascularization and detection of vulnerable coronary atherosclerotic plaques (Vavuranakis *et al.*, 2008). IVUS has also been used to evaluate coronary artery distensibility for research purposes (Vavuranakis *et al.*, 1999). For pure

clinical purposes, in the current stent era, IVUS is used to evaluate left main stenosis, to size stent, and detect stent under-expansion or stent fracture, especially in patients with stent restenosis or thrombosis (Doi *et al.*, 2010). Although certain arterial cross-sectional area cut-offs to define a significant stenosis have been proposed, functional assessment with FFR is a more reliable means to determine whether an angiographically moderate stenosis needs revascularization. Accordingly, with the exception of evaluation of left main stenosis, in the clinical setting IVUS does not weigh on the decision to revascularize a specific lesion, but greatly determines the strategy of the percutaneous interventional procedure should the decision be taken.

Quantitative coronary angiography

Quantitative coronary angiography (QCA) consists of the application of computer-derived measurements to determine the severity of coronary artery stenoses. This approach was first proposed and validated in the late 1970s and early 1980s (Brown *et al.*, 1977; Brown, 1985; Brown *et al.*, 1986) prior to the development of digital angiography. For an excellent review of the principles and applications of digital imaging in coronary angiography, see Nissen (1994), Garrone *et al.* (2009), and Pantos *et al.* (2009). Traditional QCA requires digitalization of cine film, usually via a projector equipped with a high-quality video camera. The digitized image is computer-analyzed, typically employing automated border detection and precise correction for radiographic magnification. This technique yields measurements of both the minimum luminal diameter and the percentage diameter stenosis. Under controlled conditions, several investigators have demonstrated excellent precision and interobserver variability, and shown good reproducibility for stenosis

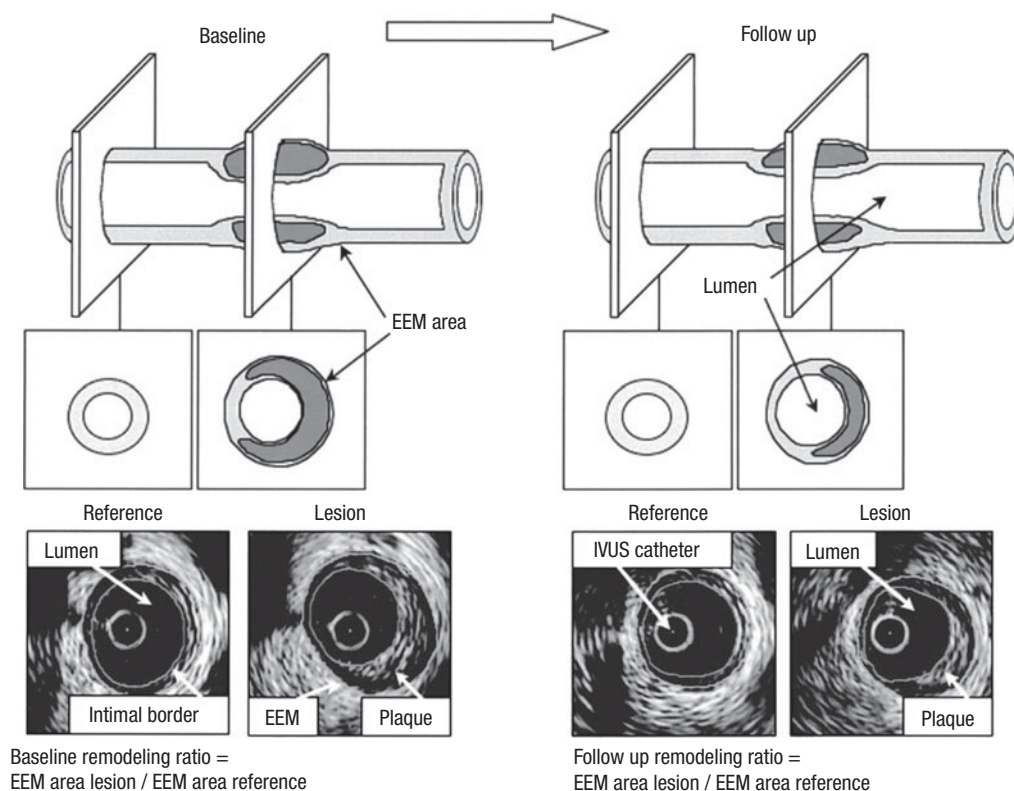


Figure 17.19 Schematic drawing of the methodology for remodeling measurements using intravascular ultrasound (IVUS). EEM: External elastic membrane. Reproduced from Schoenhagen *et al.* (2006)

sizing using QCA (Brown *et al.*, 1977; Kirkeeide *et al.*, 1986).

Traditional QCA employs geometric calculations to determine stenosis severity. An alternative method employs computer-derived density measurements obtained by analysis of digital angiographic images (Nichols *et al.*, 1984). This densitometric approach to stenosis sizing calculates percentage luminal area reduction by measuring summated density for two regions of interest, one placed over the lesion and the other over the nearest adjacent normal site. Theoretically, densitometric stenosis measurements should be independent of the radiographic projection. Although superficially attractive, the densitometric method is seriously flawed, primarily because of an inability to image both the stenosis and reference sites in a single orthogonal projection.

Quantitative measurements were only available 'off-line' until recently – a constraint that was acceptable for research but not immediate clinical decision-making. Improvements in computer technology have enabled some newer digital systems to provide nearly instantaneous quantitative angiography – a capability that enables precise measurement of stenosis severity during the procedure. The ability to precisely measure coronary diameter during interventional catheterization facilitates balloon sizing. However, it is rarely used for this purpose today, as intracoronary ultrasound provides more precise

measurements. Data show that variables obtained with quantitative angiography may have predictive value. For example, progressive luminal obstruction as assessed with QCA has been shown to predict cardiovascular events in longitudinal trials (Brown *et al.*, 1993; Waters *et al.*, 1993; Brown, 2007). Furthermore, QCA was used during the 1990s to evaluate the effect of treatment on the coronary atherosclerotic burden. A recent meta-analysis of QCA studies including patients with coronary artery disease who were treated with different hypolipidemic drugs showed that a 1.0 percent reduction in diameter stenosis as assessed with QCA reduces event rate by almost 20 percent (Brown *et al.*, 2006; Brown, 2007). Although intuitively useful, there exist no randomized studies documenting the improvement in outcome when interventional procedures are guided using QCA. Accordingly, stenosis quantification by QCA is currently not a necessary feature for clinical decision-making.

Following the development of the first automated methods in the 1980s, QCA has undergone substantial evolution from simplistic computer-assisted methods of visual interpretation of coronary angiograms to recently introduced three-dimensional reconstruction and quantitative analysis algorithms. The ability of QCA to provide objective dimensional assessment of coronary lesions is still useful and has led to its widespread application in both scientific research and clinical practice. Most of the

systems available nowadays are based on vessel edge detection algorithms to designate the arterial lumen on conventional two-dimensional coronary angiograms. Three generations of these systems have been developed with improved accuracy and precision of obtained measurements. Techniques for reconstruction of the two-dimensional coronary angiogram in space have allowed the implementation of three-dimensional QCA. This new technology may solve many of the limitations inherent in two-dimensional QCA and provide stenosis assessment capabilities not available with two-dimensional imaging (Pantos *et al.*, 2009).

Optical coherence tomography

Optical coherence tomography (OCT) is a catheter-using back-based imaging modality using reflected infrared light, and currently it provides the highest resolution of all invasive modalities (5–20 μm). In the last few years, there has been a significant and continuous technical development that has improved the quality of image. The clinical applicability of the technique continues to increase. Due to its high resolution, OCT is the only modality so far that can identify thin-cap fibroatheroma with a cap thickness of 65 μm or less, which corresponds to vulnerable plaques (Gonzalo *et al.*, 2008). OCT is a useful tool for the evaluation of the coronary vessel wall, and the impact of coronary interventions on plaque structure. Furthermore, OCT can evaluate stent apposition in great detail and can also identify the presence of vessel wall injury after implantation of a stent, or in-stent restenosis due to tissue hyperplasia (Gonzalo *et al.*, 2008; Guagliumi and Sirbu, 2008; Guagliumi *et al.*, 2010). Currently, the second generation Fourier domain OCT is used; this employs a non-occlusive technique that does not require temporary occlusion of the vessel, a limitation of the previous generation. However, the limited perpendicular imaging penetration of the vessel wall, inherent to the high resolution of the technique, remains.

Other invasive techniques

Several new catheter-based modalities for coronary plaque imaging have emerged during the last decade. ‘Virtual histology’ uses the ultrasound scattered reflection wave in order to improve tissue characterization by IVUS. ‘Palpography’ or ‘elastography’ is a new IVUS-derived technique used to evaluate the stress–strain relationship on a coronary lesion and to identify thin cap fibroatheroma. Intravascular magnetic resonance imaging (MRI) is a new modality to assess the atherosclerotic burden and the composition of coronary plaques. ‘Angioscopy’ classifies coronary lesions and evaluates their composition by direct visualization of the plaques’ color. ‘Spectroscopy’

(near-infrared and Raman spectroscopy) evaluates the chemical composition of plaque components. ‘Coronary thermography’, although not strictly an imaging technique, may greatly aid identification of the vulnerable plaque (Stefanadis *et al.*, 1999a).

New non-invasive methods

Advances in computed tomography (CT) technology allow images to be obtained with high spatial and temporal resolution. These features now permit non-invasive coronary CT angiography (CCTA). Many studies addressing proof of concept, feasibility, and clinical robustness have been published since CCTA was first described. More recently, the scientific evaluation of CCTA has rightly focused less on technical aspects and more on multicenter trials of the diagnostic value of CCTA and on head-to-head comparisons with other non-invasive (stress myocardial perfusion imaging with radionuclides) and invasive (QCA) modalities for the detection of coronary artery disease (CAD). Overall, there is high agreement between CCTA and both stress myocardial perfusion imaging with radionuclides and QCA for the presence of CAD. However, CCTA can over- or underestimate the severity of CAD compared to QCA as a reference standard. Initial studies that compared CCTA to stress myocardial perfusion imaging with radionuclides found their accuracies for determining the presence of high-grade luminal obstructions comparable. Limitations of CCTA include the inability to reliably assess the coronary artery lumen dimensions in patients with large amounts of coronary artery calcium, artifacts caused by coronary and respiratory motion, and the need for ionizing radiation and intravenous administration of iodinated contrast material. Although current expert consensus does not call for CCTA to be a first-line test for CAD, particularly for screening in asymptomatic individuals, current data suggest a promising role in the evaluation of symptomatic patients for possible CAD (Bowman *et al.*, 2009).

Currently particularly controversial in Australia is government-proposed rebate for use of non-invasive CT scanning of coronary arteries. Three indications will be approved for reimbursement on referral from a cardiologist or cardiac surgeon: (1) anatomic anomalies or coronary shunts; (2) patients awaiting non-coronary cardiac surgery (in whom angiography would otherwise be done); (3) patients with stable angina and low–medium risk, where stratification may be improved and medical therapy optimized. These indications for reimbursement of non-invasive coronary CT appear to be justified.

There has been a veritable explosion of new methods to image and assess coronary anatomy and function. New journals (including an imaging edition of the *Journal of the American College of Cardiology*) have been formed to report advances. At present, it is fair to say that coronary

anatomy and variations can be accurately displayed, but severity of stenoses cannot. There are many recent reviews and position statements that may lose currency within months – e.g. Gibbons *et al.*, 2007; Bluemke *et al.*, 2008; Choi *et al.*, 2008. This book cannot keep up with changes

in the field, but can direct interested readers to imaging journals, the traditional cardiac journals, and *New England Journal of Medicine*, which summarizes advances and applications at approximately yearly intervals. Manufacturers post new information on their websites.

Special circulations

The cerebral circulation	398	The liver circulation	408
The renal circulation	408	Summary	408

The pulmonary and coronary circulations are discussed in separate chapters (Chapters 16 and 17), as in previous editions. The lungs (Chapter 16) warrant separate attention because of their high blood flow at low pressure; this confirms the principles that were first established for the systemic circulation. Coronary arteries warrant separate attention because of the unusual (predominantly diastolic) nature of flow through the left ventricular (LV) myocardium, and their relevance to atherosclerotic and myocardial disease.

In this edition, we introduce a chapter on special circulations in order to highlight new advances, where attention to pulsatile flow, and its change with age and disease, has provided new insight into the mechanisms of disease, especially dementia. This area is likely to expand widely as a consequence of new techniques, especially magnetic resonance imaging (MRI) (Vermeer *et al.*, 2003; Bateman, 2004; Kuehne *et al.*, 2004; van Dijk *et al.*, 2004; Henry-Feugeas *et al.*, 2005; Greenberg, 2006; Heckbert *et al.*, 2006; Henry-Feugeas, 2007; Vernooij *et al.*, 2007; Mauritz *et al.*, 2008), and was addressed dramatically at the 2010 World Congress of Cardiology Meeting in Beijing by Valentin Fuster, one of the world leaders in the field (Fuster and Vahl, 2010). He pointed out how new imaging techniques are poised to explain dementia as a preventable microvascular disease, related strongly to aging and hypertension, but not to atherosclerosis. New advances have also followed a consideration of the inevitable effects of aortic and elastic arterial stiffening with age. Our initial focus was on large artery lesions and LV load, as described in previous chapters. However, we had to explain another consequence: what happens when the pulsations generated by the heart cannot be absorbed (or cushioned) in these elastic arteries? The obvious answer was that they must pass down into the smaller arterial resistance vessels, particularly those in highly perfused organs, such as the brain and kidney. Pulsatile energy must be absorbed by these fragile vessels. What are the consequences? Explanation appears to draw together many lines of research

which has the potential to explain the development of dementia and kidney failure in the aging population, to link large arterial disease with small vessel disease, to provide a strategy for delaying or preventing these conditions, and for treating cerebral edema as a consequence of head injury or stroke (Berge, 2007; Vernooij *et al.*, 2007; Hirata *et al.*, 2007; Wang *et al.*, 2007; Bateman *et al.*, 2008; Elias *et al.*, 2008; Hashimoto *et al.*, 2008b; Henskens *et al.*, 2008a; Hirata *et al.*, 2008; Knecht *et al.*, 2008; Pickering, 2008; Waldstein *et al.*, 2008; Elias *et al.*, 2009; Henskens *et al.*, 2009; Buyck *et al.*, 2009; Guo *et al.*, 2009; Ito *et al.*, 2009; Laurent *et al.*, 2009; Stewart *et al.*, 2009; Hirata and O'Rourke, 2010; Soiza *et al.*, 2010; Weber, 2010; White *et al.*, 2010; Cooper *et al.*, 2011). The field is advancing quickly and many aspects are controversial, so that just a broad introduction is possible.

Until recent times, textbooks on physiology have been prepared to discuss blood flow to organs in terms of steady blood flow, and to describe changes in mean flow with activity and posture in terms of autoregulation and vasomotor control through metabolites, hormones, and autonomic nerve control. Endothelial-derived nitric oxide release was the most recent advance in this field (Chapter 5). Resting blood flow to bodily organs varies markedly (Bard, 1961) (Table 18.1) and is dependent on their metabolism and function. The most highly perfused organs (brain and kidneys) have low resting vasomotor tone. As well as high mean blood flow velocity in the arteries of supply, the flow patterns in these arteries are different to those in other arteries, whereas the pressure waveforms (where measurable) are similar. Such differences in flow patterns can be used to describe properties of the circulation downstream, though indices such as the pulsatility index (PI), or through description of pulsatile pressure/flow relationship as vascular impedance. Changes in blood flow patterns can also be used as flow augmentation index (FAI) to describe change in arterial properties upstream which provide the 'vis a tergo', such as wave reflection, and which in older

Table 18.1 Resting blood flow and O₂ consumption of various organs in a 63 kg adult man with a mean arterial blood pressure of 90 mmHg and an O₂ consumption of 250 mL/min

Region	Mass (kg)	Blood flow			Oxygen consumption		Resistance (R units) ^a		Percentage of total	
		mL/min	mL/100 g/min	Arteriovenous oxygen difference (mL/L)	mL/min	mL/100 g/min	Absolute	Per kg	Cardiac output	Oxygen consumption
Liver	2.6	1500	57.7	34	51	2.0	3.6	9.4	27.8	20.4
Kidneys	0.3	1260	420.0	14	18	6.0	4.3	1.3	23.3	7.2
Brain	1.4	750	54.0	62	46	3.3	7.2	10.1	13.9	18.4
Skin	3.6	462	12.8	25	12	0.3	11.7	42.1	8.6	4.8
Skeletal muscle	31.0	840	2.7	60	50	0.2	6.4	198.4	15.6	20.0
Heart muscle	0.3	250	84.0	114	29	9.7	21.4	6.4	4.7	11.6
Rest of body	23.8	336	1.4	129	44	0.2	16.1	383.2	6.2	17.6
Whole body	63.0	5400	8.6	46	250	0.4	1.0	63.0	100.0	100.0

^aR units are pressure (mmHg) divided by blood flow (mL/s).
Reproduced from Bard.

humans, adds to the pulsatile perfusion pressure in the artery of supply. The difference between flow augmentation index and flow PI calculation is shown in Fig. 18.1. The distinction is important, particularly as it relates to cerebral and renal perfusion in older people, as will become apparent in the following pages.

THE CEREBRAL CIRCULATION

Steady flow

The brain and kidneys are the most highly perfused organs supplied by the systemic circulation, together receiving some 40 percent of cardiac output though being < 3 percent of total body mass (Table 18.1). The brain has a high metabolic rate and requires high and continuous flow for normal instantaneous bodily control. This is achieved through autoregulation (Berne and Levy, 1983; Baumbach and Heistad, 1985; Faraci and Heistad, 1990) and the presence of baroreceptors in the proximal part (carotid bulb and aorta) of the principal arteries of blood supply to the brain.

The unique feature of the cerebral circulation of humans is its location within a rigid box (the skull) with blood entering it through the carotid arteries, which pass through a long bony canal or through vertebral arteries passing through cervical vertebrae and vertebral canal. The traditional view is that intracranial contents are incompressible such that any increase in arterial inflow must be associated with a corresponding increase in venous outflow (Berne and Levy, 1983). Views have now changed dramatically on the basis of pulsatile arterial

blood flow measurements in the carotid arteries at the base of the brain, and cerebral arteries and veins within the brain, and in consequence of pathological and MRI studies of normal, aged, and diseased brains (Vermeer *et al.*, 2003; van Dijk *et al.*, 2004; Greenberg, 2006; Vernooij *et al.*, 2007; Waldstein *et al.*, 2008; Kearney-Schwartz *et al.*, 2009; White *et al.*, 2010). Yet another challenge is to identify and apply relevant principles to the problems of cerebral edema and arterial 'spasm' which follow head injury and cerebral hemorrhage (Martí-Fàbregas *et al.*, 2003; Brisman *et al.*, 2006; Broderick *et al.*, 2007; Barbosa *et al.*, 2010; Nagai *et al.*, 2010), and may be seen in cerebral tumors and the malignant phase of hypertension (Pickering, 1968; Byrom, 1969; Broderick *et al.*, 2007).

Pulsatile blood flow

CAROTID ARTERIES

Patterns of carotid pressure and flow waveforms have been measured in clinical practice for many years. Pressure waveforms are often used as a surrogate of aortic pressure waveforms (Chen *et al.*, 1997; Kelly *et al.*, 1989b; Mitchell *et al.*, 2004a; Segers *et al.*, 2007b; Mitchell *et al.*, 2008a; Mitchell, 2009a; Mitchell, 2009b; Segers *et al.*, 2009) and are described in Chapter 28. Flow waveforms are utilized in the diagnosis of presence and severity of carotid stenosis, on the basis of high flow velocity during systole, persisting into diastole.

Only in recent years has the relationship between carotid artery pressure and flow waves been examined in any detail. Hirata *et al.* (2006) measured pressure

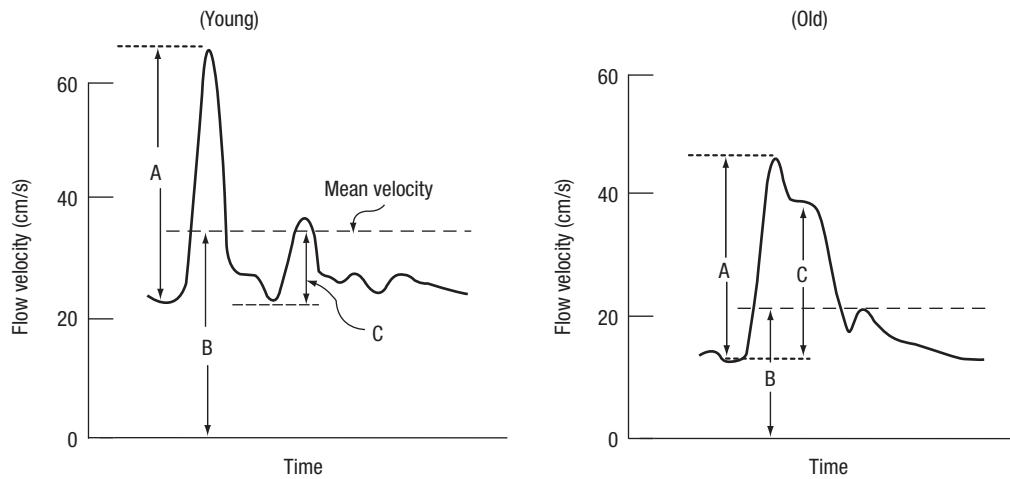


Figure 18.1 Carotid blood flow velocity waveforms in a young (left) and an old (right) human subject, showing calculation of flow pulsatility index (PI) and flow augmentation index (FAI). PI is amplitude (or peak) of the flow waveform (A) divided by mean flow (B) ($A \div B$). This is determined principally by the resistive properties of the vascular bed downstream, and is similar in young and old subjects. FAI is the boost to late systolic blood flow caused by return of wave reflection from the lower part of the body. It is calculated as amplitude of the late systolic peak (C) divided by amplitude (A) of the wave ($C \div A$). FAI is low in young adult humans and increases progressively with age

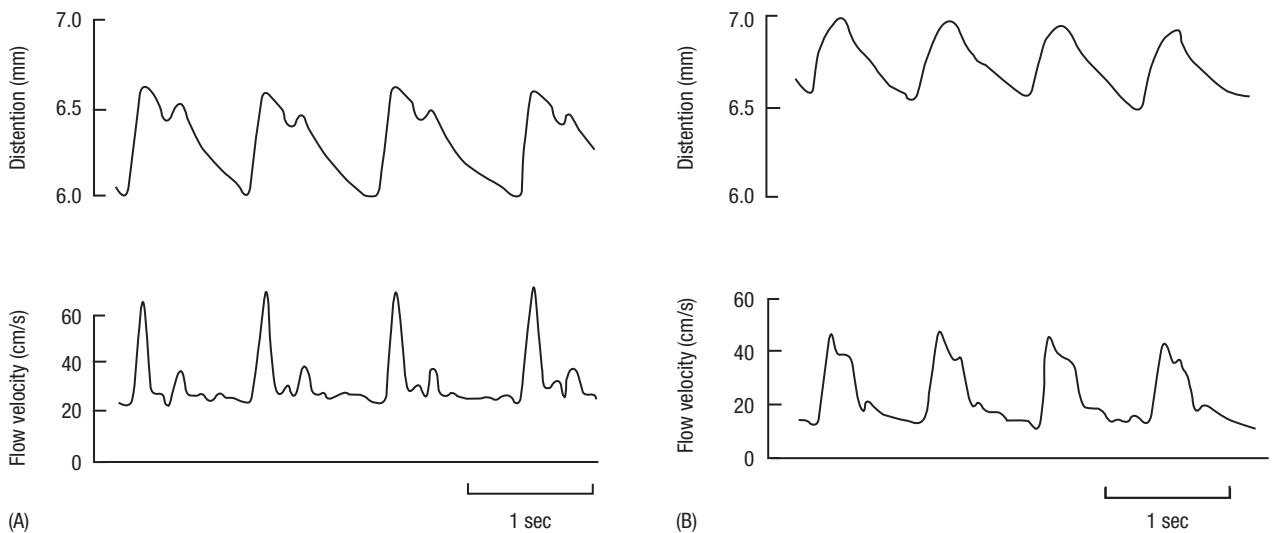


Figure 18.2 Representative pressure (top) and blood flow velocity (bottom) waveforms in the common carotid artery of a 23-year-old subject (A) and in a healthy 65-year-old subject (B). The distention wave was used as a surrogate of the pressure wave. Reproduced from Hirata *et al.* (2006)

(as distention) and flow velocity waves in the common carotid artery of 56 healthy people aged 20 to 72 years. Pressure (distention) waves were similar to those described by others (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b), and showed the same aging change – a progressive increase in late systolic pressure augmentation. Carotid artery flow waves were, also, similar to those recorded by others (Mills *et al.*, 1970), but showed a distinctive change with age that corresponded to the change in carotid pressure patterns (Fig. 18.2). Hirata's flow waveforms were recorded in the common carotid artery, but are almost identical

to flow patterns recorded by others in the internal carotid artery which supplies exclusively the brain and eyes. They are also similar to those recorded at different ages by transcranial Doppler techniques in the optic arteries (Michelson *et al.*, 2007). Surprisingly, in normal subjects, these cerebral flow waves are quite similar to pressure waves recorded in the radial artery of people at similar age.

A common feature at all ages is that carotid blood flow is high, and persistent through systole and diastole (Figs 18.1 and 18.2). The absolute value of mean blood flow velocity and flow pulsation (systolic peak – diastolic

nadir) is comparable, so that PI is normally near unity. The principal change with age is in late systolic flow which is relatively low in youth and increases progressively with age (Hirata *et al.*, 2006). In young adults, there is a sharp narrow peak of flow in early systole whereafter flow remains relatively constant through late systole and diastole. The late systolic flow peak increases with age and corresponds to the progressive increase in carotid pressure augmentation (Fig. 18.2). Just as pressure augmentation can be expressed as an augmentation index (pressure augmentation/pulse pressure, AIx), so this change in the flow wave can be expressed as a FAI – increase in late systolic flow shoulder from wave foot/amplitude of the flow wave (Fig. 18.1). Both carotid flow augmentation and pressure augmentation increase with age (Figs 18.1 and 18.2), and show a close relationship with each other (Fig. 18.3). The changes in carotid flow augmentation with age are similar to those seen in the ophthalmic artery which supplies the eye (Michelson *et al.*, 2007; Schmieder, 2008), and also similar to changes in the radial pressure waveform with age (Fig. 10.34).

Increase in carotid pressure augmentation with age is caused by early return of wave reflection from the lower body (Chapters 19 and 20). Progressive increase in carotid flow augmentation is attributable to the same phenomenon, as are the changes in the radial pressure pulse. This phenomenon can explain age-related pathology in the brain and a strategy for its modification. This is detailed below. There is a relationship between carotid flow augmentation and cerebral lacunar infarction and white matter hyperintensities (Hirata *et al.*, 2008; Hirata and O'Rourke, 2010). Both carotid flow augmentation and

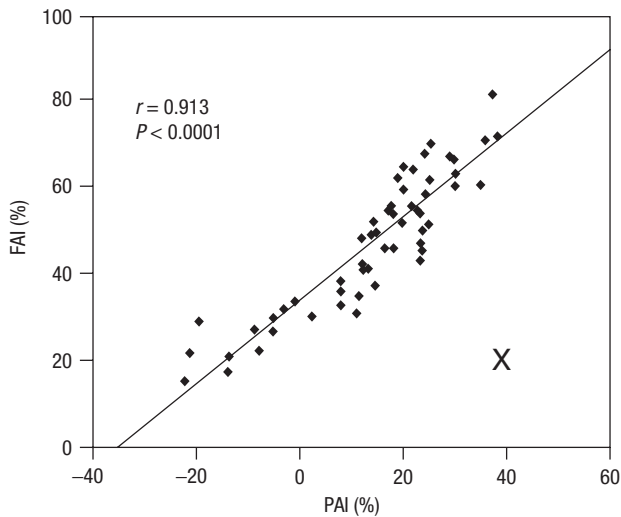


Figure 18.3 Relationship between flow augmentation index (FAI) and pressure augmentation index (PAI) in the common carotid artery of 56 normal subjects. The point inset is the relationship that would be expected in a patient with intracranial hemorrhage and cerebral edema with waveforms as in Fig. 18.6. Reproduced from Hirata *et al.* (2006)

pressure augmentation are reduced by drugs which decrease wave reflection (Fig. 18.4) (Hirata *et al.*, 2007).

The carotid arteries are the last possible site for non-invasive pressure estimation before blood enters the skull to perfuse the brain. Blood flow waveforms can, however, be measured by transcranial Doppler and MRI techniques in large arteries within the brain, and in the cerebral veins and venous sinuses (see Chapter 6) (Bateman, 2002a; Bateman *et al.*, 2008). Pressure cannot be measured in these vessels, although its level is often inferred from the flow patterns. Many assume that pressure drops between carotid and peripheral cerebral arteries in a similar manner to that in the whole systemic circulation, i.e. with little fall in mean pressure but substantial fall in pulsatile pressure between large and small arteries (Struijker-Boudier *et al.*, 2006). However, Heistad and colleagues (Baumbach and Heistad, 1985; Faraci and Heistad, 1990) have shown that consequent on high cerebral blood flow and low arteriolar resistance, pressure drop along the large cerebral arteries is greater, and so that the situation has some analogies to the lung (Fig. 18.5) (Milnor *et al.*, 1966), where flow and pressure pulsations extend into the capillary network, and through capillaries into the pulmonary veins. Other implications of this phenomenon are referred to below. As pointed out above, there is limited information available on intravascular pressure in the brain, so this is often inferred from the flow patterns.

We found only a limited number of studies on cerebrovascular impedance and these were performed in dogs

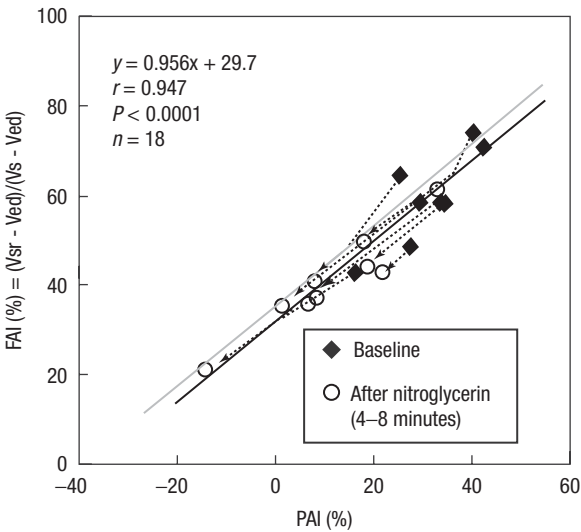


Figure 18.4 Relationship between carotid pressure augmentation index (PAI) and flow augmentation index (FAI) in nine patients with coronary artery disease (CAD) before and after (arrows) administration of nitroglycerine sublingually. The dotted line is the regression line from the normal subjects shown in Fig. 18.3. The solid line is the regression line for the nine CAD patients (...., regression line for normal subjects; —, regression line for present study). Reproduced from Hirata *et al.* (2007)

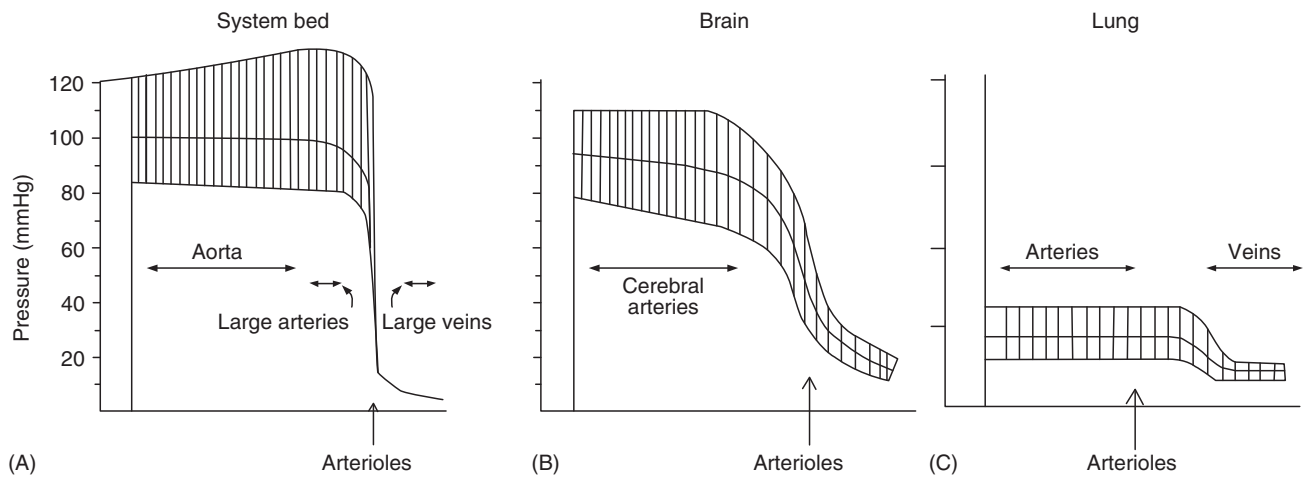


Figure 18.5 Schematic diagram of mean and pulsatile pressure changes between carotid artery and venous sinuses (B), with comparison with the systemic vascular bed (A) and the pulmonary circulation (C). See also Fig. 16.2

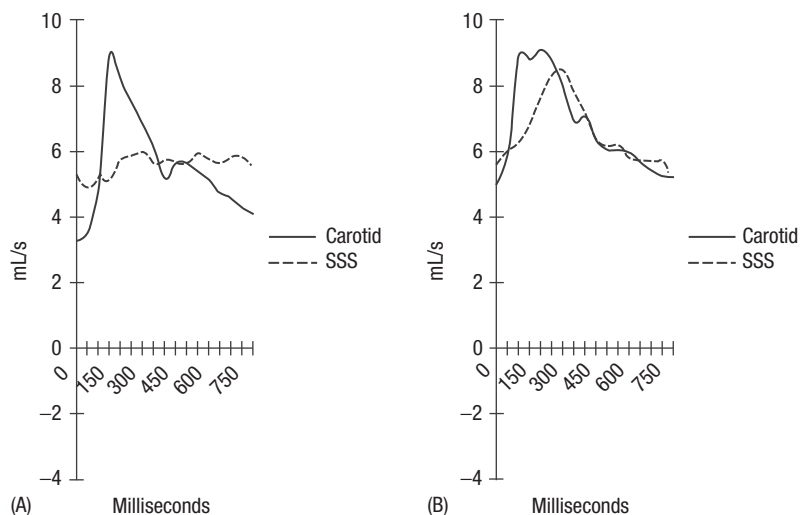


Figure 18.6 (A) Blood flow waveforms (as volume flow) in the internal carotid artery and superior sagittal sinus (SSS) of a patient with secondary intracranial hypertension (SIH). Note low pulsatility in the SSS and single peak of flow in the carotid artery. (B) Blood flow waveforms (as volume flow) in the internal carotid artery and SSS of a normal healthy subject. Note relatively high flow pulsation in the SSS and delay between dual peak in the carotid artery (high FAI) and single peak in the SSS. Reproduced from Bateman (2002b)

(Taylor and Tukmachi, 1985; Taylor and Tukmachi, 1987). Such studies are crucial if we are to understand normal cerebral perfusion and the effects of disease. These authors measured pulsatile pressure and flow in the internal and external carotid and vertebral arteries and calculated input impedance spectra. The general form of the impedance spectra of the internal carotid and vertebral arteries suggested vasculatures with little wave reflection. In contrast, the external carotid bed had a higher input impedance and greater wave reflection. The studies of Hirata *et al.* (Hirata *et al.*, 2006; Hirata *et al.*, 2008) in humans, with low PI and constant relationship between common carotid flow and pressure augmentation, suggest a vascular bed reflection coefficient around 0.4 (compared to around 0.8 in a systemic vascular bed and 0.3 in the pulmonary circulation). The changes in patients with cerebral edema and secondary intracranial hypertension (SIH), whose carotid PI increases substantially (Fig. 18.6A) (Bateman, 2002a; Bateman, 2002b; Martí-Fàbregas

et al., 2003), imply marked increase in cerebral vascular resistance and reflection coefficient. In such conditions, one would expect a marked deviation of flow augmentation from pressure augmentation – with carotid FAI markedly decreased, but carotid pressure augmentation index maintained, and with a corresponding change in cerebral vascular impedance (and increase in wave reflection from the cerebral bed).

INTRACEREBRAL ARTERIES, CAPILLARIES, AND VEINS

In normal subjects, flow patterns in the smaller cerebral arteries and ophthalmic arteries are similar to those in the carotid arteries and show the same changes with aging – increase in late systolic flow velocity with broadening of the systolic peak (Stolz *et al.*, 1999; Michelson *et al.*, 2007). Michelson *et al.* (2007) have drawn attention to the accompanying increase in the first compared to

higher harmonics of the flow wave, as also seen for the carotid artery pressure waveform (Gallagher *et al.*, 2004).

Transcranial Doppler has also been used to measure flow waveforms in the cerebral veins and venous sinuses (Valdúeza *et al.*, 1996; Baumgartner *et al.*, 1997; Stolz *et al.*, 1999). These venous studies, supplemented more recently by MRI studies (Bateman, 2002a; Bateman, 2002b; Bateman, 2003; Henry-Feugeas *et al.*, 2005; Bateman, 2008) show persistence of pulsatile flow, and with PI (or resistance index) almost comparable to indices in the arterial tree, although with the venous flow wave delayed compared to that in the arteries upstream (Fig. 18.6B). Bateman (Bateman, 2002a; Bateman, 2002b; Bateman, 2003) described a PI of 0.72–0.75 in the carotid artery and 0.41–0.47 in cortical veins and superior sagittal sinuses of normal subjects, with higher values (1.09 and 0.49–0.67), respectively, for arteries and veins in people with dementia. Bateman considered that the high flow pulsations had been transmitted indirectly from artery to vein via extravascular pathways rather than through the capillaries, and in the process had caused edema and white matter lesions. In the lung, similar flow patterns are seen, but the relatively high pulmonary vein pulsations are known to have been transmitted directly through the pulmonary capillaries (Milnor *et al.*, 1966).

MRI studies have revolutionized the investigation and management of cerebral disease, principally by drawing attention to the relatively high prevalence of white matter changes that are attributable to cerebral microvascular disease (Vermeer *et al.*, 2003; Greenberg, 2006; Vernooij *et al.*, 2007) and reversible edema that can be seen acutely and which disappears promptly with treatment (Hinchey *et al.*, 1996). Bateman (2002a) linked such microvascular disease to the increase in pulsatile flow within the brain and coined the term ‘pulse wave encephalopathy’ to explain leukoaraiosis (and white matter changes). He later extended this (Bateman, 2004) to explain other cerebral lesions as seen in various types of dementia. Henry-Feugeas *et al.* (2005) broadened this view on pulse wave encephalopathy to explain aging changes seen using MRI with emphasis on the effects of pulsatile flow in the smallest arterioles and capillaries as causing disturbance in capillary exchange. There was less emphasis on venous pulsations and apparent assumption that pulsatile flow had passed through arteries and capillaries into veins, causing functional and structural damage in the process. Henry-Feugeas attributed the high flow pulsations to arterial stiffening and an impaired ‘Windkessel effect’ (failure of the aorta and large elastic arteries to cushion and absorb flow pulsations). Henry-Feugeas (2007) described Alzheimer’s dementia as due to ‘devastating cerebrovascular dysfunction’ from microvascular disease through this mechanism. Bateman’s earlier hypothesis attempted to account for the localization of lesions in relation to small arteries and arterioles. A meta-analysis of 22 longitudinal studies by DeBette and Markus (2010) has

confirmed white matter hyperintensities as a marker for incident dementia, stroke, and death.

Link between large systemic arteries and cerebral microvessels

Elsewhere in this book (Chapters 19, 20, and 30), we describe the effects of aging on the aorta and large elastic arteries, with stiffening, increased pulse wave velocity and early return of wave reflection. The major ill effects are progressive increase in LV load and a growing imbalance between myocardial oxygen and blood requirements and capacity for myocardial perfusion. These effects largely explain development of cardiac failure and myocardial ischemia in the elderly, even independent of associated atherosclerotic disease (O’Rourke, 2008). We could not, however, explain the other vascular disabilities which befall older people, particularly intellectual deterioration leading to dementia and cerebral abnormalities, both hemorrhagic and thrombotic, which are not caused by atherosclerotic disease. Nor could we explain increasing incidence of renal failure.

Description of high pulsatile flow patterns in cerebral arteries and veins, as previously described and with progressive increase with age, provided a ready explanation. With aging, there is evidence of microvascular disease attributable to medial and endothelial damage. This is outlined below. The first task was to explain how macrovascular disease in systemic arteries may result in microvascular disease in the brain.

The connection is expected and inevitable (Fig. 18.7) (Mitchell *et al.*, 2005a; O’Rourke and Safar, 2005; Nichols and O’Rourke, 2006b). The role of the normal arterial system is to cushion flow pulsations generated by the intermittently beating heart. Such flow pulsations into the ascending aorta continue throughout life, but the aorta loses its capacity to cushion these, so that they must pass peripherally into smaller vessels, and preferentially into vasodilated organs, such as the brain and kidney. The high flow pulsations in small cerebral arteries described above are an inevitable consequence of reduced pulsatile aortic expansion. The next step is to explain what ill effects such high flow (and pressure) pulsations may have. This is described below under Pathological changes in cerebral microvessels, p. 403. Pathological changes are attributable to excessive energy losses in the microcirculation, caused by increased pulsations of pressure and flow within these. Increased pulsatile pressure pulsations are likely to cause medial damage; increased pulsatile flow pulsations are likely to cause endothelial dysfunction and damage. The changes one looks for are hemorrhage and thrombosis in small arteries, arterioles, and capillaries. Such is the pathology observed (see below). It is associated with and can be attributed to increased pulsatile energy losses in the systemic circulation caused by aortic stiffening. O’Rourke (1970a) previously showed that pulsatile energy losses in

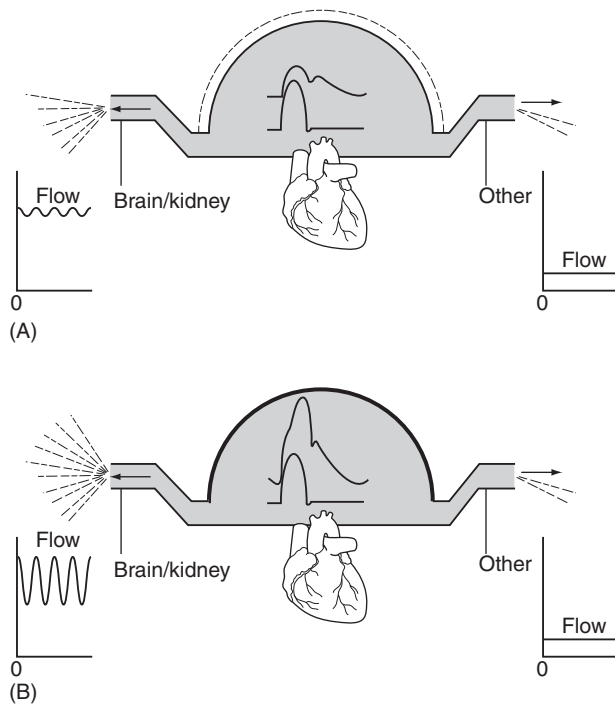


Figure 18.7 A simplified 'Windkessel' model of the arterial system explains the fundamental relationship between large and small artery blood flow. Under normal circumstances (A), pulsatile flow ejection from the left ventricle creates pulsations of pressure which are cushioned in the elastic arteries, so that flow through most bodily organs is steady and non-pulsatile. When the elastic arteries are stiffened (B), pressure fluctuations are larger, and cannot be well cushioned, so they enter down into small vessels of low resistance organs (such as brain and kidney) where they are absorbed. Reproduced from Adji *et al.* (2011)

the systemic circulation of normal young dogs were around 10 percent of energy losses from steady flow. With aortic stiffening, pulsatile energy losses increased to over 20 percent. In middle-aged humans, undergoing diagnostic cardiac catheterization, pulsatile arterial energy losses were around 20 percent (Nichols *et al.*, 1977a; Kelly and Fitchett, 1992). In young people, most pulsatile energy losses are confined to the large arteries; with age, there must be increased pulsatile energy loss in the microcirculation, especially in the brain and kidneys.

It is easiest to explain this association in terms of a Windkessel model (Fig. 18.7). The better and more comprehensive explanation requires a distributed model, since the major effect of aortic stiffening is through increasing flow in late systole up the carotid arteries. This is caused by early wave reflection from the lower body, as well as continuing ejection from the heart. Support for a distributed model also stems from the carotid pressure and flow studies; reduction in wave reflection by nitroglycerine has a potent effect in reducing late systolic flow

and pressure at doses that do not cause reduction in mean pressure (Fig. 18.4) (Hirata *et al.*, 2007).

Pathological changes in cerebral microvessels

Pathological studies of the brain, beginning with the work of Charcot and Bouchard (Pickering, 1968), showed that small aneurysms develop in the brain with age around branching points, that they increase in size and number with age, and that cerebral hemorrhage frequently arises through their rupture. As with elastin fibers in the aortic wall media, such small arterial damage can be attributed to the fatiguing effects of cyclic stress, with catastrophic rupture being the ultimate manifestation of an aging change, hastened by increase in pulsatile stress. Weakening of the wall of small arteries can occur at other sites in consequence of aging change and high pulsatile stress and can account for microbleeds (Fisher, 1969; Fisher, 1982; Buee *et al.*, 1997; Henskens *et al.*, 2008a; Henskens *et al.*, 2008b). Microbleeds are associated with cerebral damage and intellectual deterioration. Amyloid plaques, as seen in Alzheimer's disease can now be considered as the end result of microbleeds (Cullen *et al.*, 2005; Cullen *et al.*, 2006). The pathological studies of Cullen and Stone challenge the traditional view that is still promoted without qualification (Mayeux, 2010; Querfurth and LaFerla, 2010) which assumes that amyloid plaque deposition is the primary event in causation of Alzheimer's disease.

Lacunar infarcts are small cerebral infarcts, often asymptomatic, and are often seen in association with microbleeds and with Charcot–Bouchard aneurysms in older people. They are caused by damage to and thrombosis in small cerebral perforating arteries. Lammie *et al.* (2000) described the lesion causing lacunar infarcts as follows: 'In its acute form, this lesion is characterized by fibrinoid vessel wall destruction; in its healed form by a complex disorganized lesion with or without aneurysmal dilation; Fisher (Fisher, 1969; Fisher, 1982) called this "lipohyalinosis". Lesions occur independently of atherosclerosis, but their frequency increases in older people and in the presence of hypertension. Pathological studies (Fisher, 1969; Fisher, 1982; Lammie *et al.*, 1997; Lammie *et al.*, 2000) have commented on the association between the intimal obstructive lesions (causing thrombosis and infarction) with medial damage and disruption (causing hemorrhage), as suggesting a common etiological mechanism of both. Abnormally high pulsatile pressure and flow can account for both – high pulsatile pressure by tearing smooth muscle cells in the vessel wall, and high pulsatile flow by ripping endothelial cells off the wall. It is easy to envisage the yield stress for endothelial cells (around 400 dyne/cm²) being exceeded in the presence of high pulsatile shear stress. High levels of endothelial cell fragments in the peripheral blood of people with stiffened aortae can be explained on this basis (Wang *et al.*, 2007),

as can high levels of endothelial cells in the blood of children with pulmonary hypertension caused by congenital left-to-right shunts (Smadja *et al.*, 2009) where a similar mechanism prevails (Chapter 16) (Edwards, 1957). Genesis of small vascular lesions was studied carefully by Donald Fry and colleagues at the National Institutes of Health almost 50 years ago (Fry, 1969a; Fry, 1969b); their views are supported by recent studies on the micro-circulation (see Chapters 2 and 5) (Phelps and DePaola, 2000; Kim *et al.*, 2005).

Fry (1969a) stated

When the endothelial surface is exposed to shearing stresses in excess of some critical value ... the cell appears to become mechanically unstable and is washed away from its moorings to the basement membrane ... Continued exposure appears to produce erosion of the subjacent intrafibrillar matrix ... and deposition of thrombotic material.

Fry (1969a) calculated yield stress for a series of animals as between 200 and 1320 dyne/cm² and erosion stress as between 430 and 1610 dyne/cm², but noted that impaired endothelial function with greater permeability to protein and fat was apparent at lower levels of shear. Fry stressed the particular problems with pulsatile compared to steady shear. He compared the vascular endothelium to a thatched roof, which could leak, and be blown away by intermittent winds in a storm, but could be completely secure and waterproof when the wind was steady (Fry, 1973). It is refreshing to read the old accounts by Fry (1969a,b), Byrom and Dobson (1949), Byrom (1969), and George Pickering (1968), and delight in their clarity, logic, and relevance.

Lesions in the brain with regions of edema, hemorrhage, and thrombosis occur in hypertensive encephalopathy complicating malignant hypertension and eclampsia; these are reversible with blood pressure lowering (Pickering, 1968; Byrom, 1969; Ledingham, 1995; Cipolla, 2007). They are attributable to acute damage to the walls of small arteries with tearing of smooth muscle cells and incomplete or complete disruption of the vessel wall (Pickering, 1968; Byrom, 1969). Similar lesions are seen in the kidney of patients with malignant hypertension, in experimental animals with acute hypertension, and in patients with relatively atrophic arteries where blood pressure is raised from subnormal to normal (as for lower body arteries following repair of aortic coarctation) (Benson and Sealy, 1956; Pickering, 1968; Byrom, 1969).

Some of the most interesting pathological and pathophysiological studies on this subject were done by Byrom in Sydney through experimental studies in rats with lucite 'windows' in the skull, and by correlation with human pathology (Fig. 18.8) (Byrom, 1954; Byrom, 1969). Byrom is credited with establishing the vicious cycle theory for progression of hypertension and for the mechanism of arterial damage in malignant hypertension (Ledingham, 1995).

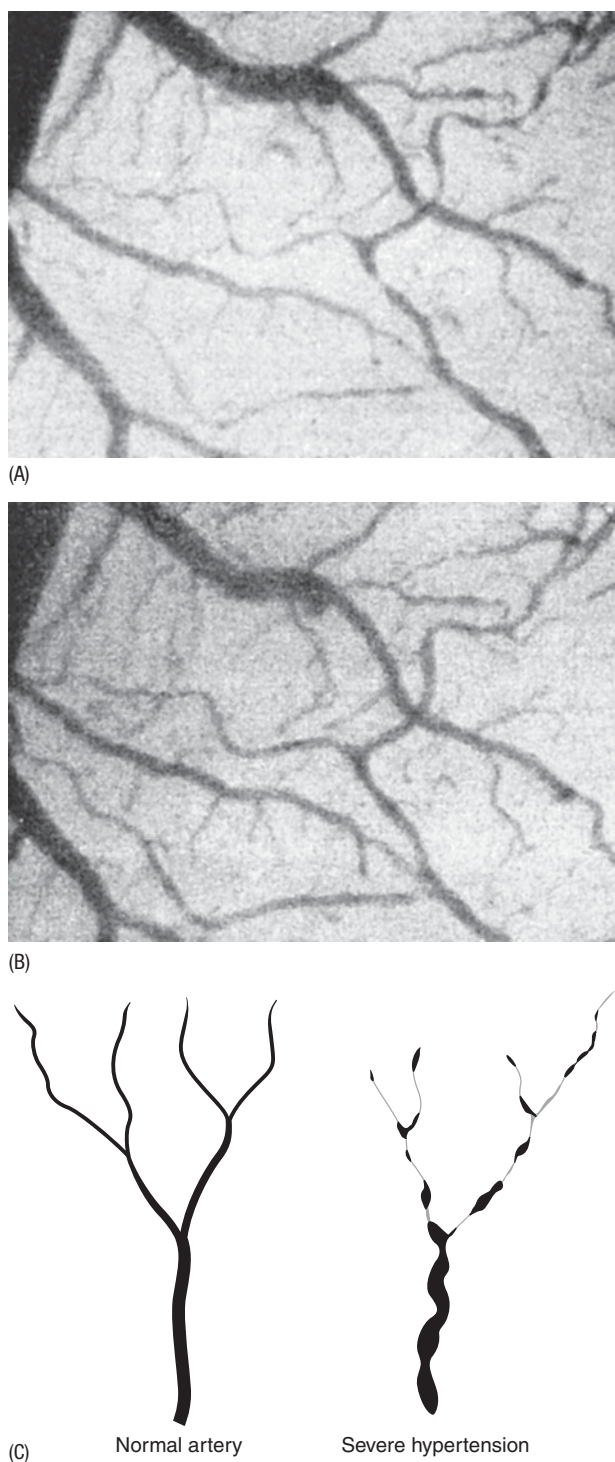


Figure 18.8 Small pial arteries photographed through lucite 'windows' in the skull of rats. (A) Light ether anesthesia, BP 240. Gross irregularity in caliber of small arteries, veins normal. (B) Five minutes later, deep ether anesthesia, BP 180. Arteries almost uniform in caliber. (C) Schematic diagram: Caliber of the cerebral arteries of the rat in uncomplicated renal hypertension as seen through permanent acrylic windows. Left: normal artery. Right: the typical picture of severe hypertension (BP > 240 mmHg systolic) – dilation, tortuosity, irregularity of largest arteries, tight uniform constriction of smallest arteries, and alternating constriction and dilation of intermediate arteries. Reproduced from Byrom (1969)

He emphasized the importance of physical stresses and of arterial weakness, and the relative unimportance of neurohumoral influences for causing acute damage. He stressed that inflammatory change in small arteries was a consequence of damage, not the cause. He noted the association between arterial dilation and narrowing in the same segment of a cerebral artery damaged by hypertension, and the greater evidence of mural damage in the dilated rather than the constricted segment. Byrom's work downplayed views on autoregulation, defects of which had hitherto been invoked to explain acute hypertensive emergencies, as had paracrine and endocrine mechanisms. His views supported the contemporary approach to treatment, emerging at the time (1950), that the appropriate treatment of hypertensive encephalopathy was to reduce arterial pressure, by whatever means was available (Smirk, 1957; Levy and Brink, 2005; Messerli, 2007). George Pickering (1968) referred to this (among other issues) as 'a blinding glimpse of the obvious'.

The 'pulse wave encephalopathy' described by Bateman (Bateman, 2002a; Bateman, 2004), Henry-Feugeas *et al.* (Henry-Feugeas *et al.*, 2005; Henry-Feugeas, 2007) and others (O'Rourke and Safar 2005; O'Rourke and Hashimoto, 2007) as the cause of cerebral damage and dementia appears to be a chronic representation of the acute cerebral arterial damage described by Byrom, Fisher, Lammie, and others. It is attributable to the same factors – of physical damage to weakened arterial walls as a consequence of aging, with a combination of thrombosis and lacunar infarction caused by increased pulsatile shear stress, endothelial damage, fluid exudation, and thromboses, and of increased pulsatile tensile stress with microaneurysm formation and microbleeds, with these turning into amyloid plaques (Cullen *et al.*, 2005; Cullen *et al.*, 2006). All are explicable on the basis of aortic stiffening with high pulse pressure and flow caused by early return of wave reflection from the lower body (O'Rourke and Safar, 2005; Nichols and O'Rourke, 2006b; Hirata *et al.*, 2007; Wang *et al.*, 2007; Abbatecola *et al.*, 2008; Hashimoto *et al.*, 2008b; Henskens *et al.*, 2008a; Henskens *et al.*, 2008b; Hirata *et al.*, 2008; Waldstein *et al.*, 2008; Buyck *et al.*, 2009; Elias *et al.*, 2009; Guo *et al.*, 2009; Henskens *et al.*, 2009; Ito *et al.*, 2009; Stewart *et al.*, 2009; White *et al.*, 2010). The predilection of the brain to such lesions results from its low vascular resistance (Baumbach and Heistad, 1985; Taylor and Tukumachi, 1985; Taylor and Tukumachi, 1987; Faraci and Heistad, 1990), which permits arterial pressure and flow waves to extend further into the cerebral microvessels than in any organ other than the kidney, where similar lesions are also seen (Ford *et al.*, 2010).

Despite what is set out above, neuroradiologists are reluctant to report cerebral white matter hyperintensities as abnormalities, mainly because they are seen commonly in apparently normal older people and because their presence may generate alarm and impose restrictions (in gaining travel and other insurance) when their natural

history is at present not known. An excellent *New England Journal of Medicine* review on this subject is by Vernooij *et al.* (2007) from Rotterdam. These authors found quantifiable (median 1.8, 3.1, and 7.7 mL) white matter intensities in groups of normal subjects aged 45–49, 60–74, and 75–79 years, respectively. The proportion of people without any white matter lesions fell from 5.4 percent in 45–49 year olds to 2 percent in people > 75 years. It is disappointing that the same journal, three years later, contained two articles on Alzheimer's disease which made no mention of this issue. Dettie and Markus (2010) reported a meta-analysis of 22 longitudinal studies; white matter hyperintensities predicted dementia, stroke, and death. An association with faster decline in global cognitive performance executive function and processing speed was also suggested.

Similarity of cerebral with pulmonary lesions

At the beginning of this chapter, we described the cerebral circulation as having similar features to the pulmonary circulation, in that pulsations normally pass through the capillaries into the draining veins (Fig. 18.5). This appears to have no long-term implications for the pulmonary circulation, such as that seen in the brain, since pulmonary arteries appear to remain relatively distensible throughout life. It may be relevant to the development of pulmonary damage in adults when small arterial vessels are particularly friable, as in connective tissue disease. However, pulmonary flow pulsations are increased markedly in congenital left-to-right shunts such as ventricular septal defect (VSD) and patent ductus arteriosus (PAD) (Edwards, 1957). When this occurs, there are usually no immediate problems, and mean pulmonary artery pressure initially remains within normal limits and the pulmonary microvasculature remains normal. Over years, however, progressive changes develop in the small pulmonary arteries. These appear to be caused only by the high flow pulsations, since they are not initially accompanied by corresponding increases in mean pressure. The lesions which develop were described by Jesse Edwards in his Lewis Connor lecture to the American Heart Association in 1956 (Edwards, 1957). They are predominantly occlusive and ultimately lead to fixed pulmonary hypertension (Eisenmenger's disease) which persists even if the shunt is repaired. However, they do not develop at all if the shunt is detected early and corrected. The pathology described by Edwards is complex, comprising a mixture of medial and intimal damage with aneurysm-like formation and hemorrhage, together with predominant intimal damage and thrombosis. They appear to have many of the same features as described for aging change in the brain. It is possible that both 'pulse wave encephalopathy' and such a form of pulmonary hypertension have a similar mechanism. It is possible too that some of the vascular abnormalities seen in computed tomography (CT) scans

of the lungs correspond to cerebral lesions seen in MRI scans (Sheehan *et al.*, 2005); furthermore, as mentioned earlier, high counts of shed endothelial cell fragments are seen in blood in both situations (Wang *et al.*, 2007; Smadja *et al.*, 2009) and can be attributed to the same mechanism. In pulmonary hypertension caused by left-to-right shunt, the number of cell fragments is related to the severity of disease and so is of diagnostic and prognostic value (Smadja *et al.*, 2009).

Implications to therapy

Consideration of arterial flow pulsations in the brain provides a new approach to Alzheimer's disease and other forms of dementia. We support the view that many forms of dementia are caused by microvascular damage, and that this is an inevitable consequence of arterial stiffening with early return of wave reflection from the lower body, and with this boosting the pressure and flow pulse which surges through the brain in consequence of its high blood supply and low vascular resistance. If this is the case, one would expect to reduce development of cerebral microvascular disease through the use of drugs, such as calcium channel blockers, angiotensin-converting enzyme inhibitors (ACEIs), and angiotensin receptor blockers (ARBs) which reduce wave reflection from the lower body. Such drugs do reduce carotid pressure augmentation (Chapters 19, 20, and 26), and as with nitrates (Fig. 18.4) (Hirata *et al.*, 2007; Hirata and O'Rourke, 2010) are associated with reduced carotid flow augmentation as well. Such drugs do reduce long-term progression to dementia, as well as cerebral vascular events when used in treatment of hypertension (Forette *et al.*, 2002; The PROGRESS Collaborative Group, 2003; Dufouil *et al.*, 2005; Nagai *et al.*, 2010). Material presented in this chapter provides further support for aggressive treatment

of systolic hypertension in older people, and from an earlier age (Eze-Nliam *et al.*, 2010; Pase *et al.*, 2010; Soiza *et al.*, 2010; Weber, 2010; White *et al.*, 2010; Adji *et al.*, 2011).

Intracranial hypertension

Characteristic features of secondary intracranial hypertension are decreased pulsatility of cerebral veins and decrease in the late systolic component of carotid flow waveform (Figs 18.6 and 18.9) (Bateman, 2002a; Bateman, 2002b; Martí-Fàbregas *et al.*, 2003). Both are attributable to cerebral swelling and secondary intracranial hypertension. There is a dearth of hemodynamic information on this subject. Opinions on treatment vary considerably, with artificially induced hypertension recommended under some conditions (Brisman *et al.*, 2006), but rejected by some (Messerli, 2007); different views are acknowledged in official guidelines (Broderick *et al.*, 2007).

In this situation, issues of mean flow and the 'brain in a box' assume supreme importance. Elevated intracranial pressures by compressing blood vessels will increase resistance and reduce mean cerebral blood flow unless mean blood pressure is elevated. Harvey Cushing (1901) was the first to study this. He described how acute major cerebral lesions cause elevated arterial pressure and regarded this as compensatory to maintain cerebral blood flow. Cushing's work formed the basis of views still current in many places that elevated blood pressure in patients with stroke should not be treated. Compression of vessels is inevitable when the brain cannot expand. Compression of blood vessels in the brain will only intensify the problems already set out in the cerebral microvasculature, since through narrowing of these vessels they will contribute to increased pulsatile flow velocity as well as limited mean cerebral flow.



Figure 18.9 Carotid flow waveforms from the left (L) and right (R) sides in a patient with a large R-sided cerebral hemorrhage and secondary intracranial hypertension. Pulsatility index (PI) is 3.05 for the left and 4.9 for the right. FAI is approximately 30 percent for both sides. Reproduced from Martí-Fàbregas *et al.* (2003)

With limited competence and information, one can only suggest that improved monitoring might include measurement of carotid blood flow and pressure waveforms – where a deviation from the normal pattern of flow and pressure augmentation is expected (Figs 18.6 and 18.9), and that treatment might comprise use of a drug that has least effect on mean pressure and greatest effect on wave reflection, so as to reduce the effects of wave reflection on late systolic pressure and flow extending into the brain. Nitroglycerine is likely to be the most effective.

If, after cerebral damage, by stroke or trauma, high pulsatile pressure and flow pulsations enter the brain, and do not exit thereafter, they must be absorbed by the brain with loss of pulsatile energy therein. Simple modeling studies show that if the cerebral arteries are compressed and narrowed, flow pulsations and shear stress at the wall will be increased. Compression from outside the artery will limit pulsatile expansion and favor increased pulsation in the longitudinal direction, i.e. as flow. Shear stresses under these conditions are likely to exceed the yield limit (around 400 dyne/cm²) and dislodge endothelial cells, favoring exudation of protein and fluid, then vascular disruption and thrombosis within the artery, as described by Fry (Fry, 1969a; Fry, 1969b; Fry, 1973). Such a mechanism can explain the arterial narrowing and obstruction which is often attributed to spasm in cerebral blood vessels at a distance following subarachnoid hemorrhage. Favorable effects of the calcium-channel blocker nisoldipine, the drug usually employed under these circumstances (Brisman *et al.*, 2006), may be a consequence of decreased wave reflection from the lower body and a decrease in carotid pressure augmentation

rather than to any local effect of the drug on cerebral arteries (Noda *et al.*, 2006). Nitroglycerine may be better.

Therapeutic views: old, new, and emerging

There are no conflicts between these views on treatment of cerebral edema and raised intracranial pressure and the views which are currently being put into practice in intensive care wards for the treatment of stroke, head injury, and hypertensive encephalopathy (Adams *et al.*, 2007; Berge, 2007; Broderick *et al.*, 2007; Vahedi *et al.*, 2007; Anderson *et al.*, 2008; Soiza *et al.*, 2010; Weber, 2010). Indeed, our views complement these and are similar to those applied to malignant hypertension. The old views were based exclusively on mean pressure/flow considerations, and followed the classic work of Harvey Cushing nearly 100 years ago (Cushing, 1901). Sixty years ago, Cushing's (1901) logic was routinely applied to the management of malignant hypertension. However, such views on the management of malignant hypertension were overturned 50 years ago. Byrom's work (Byrom, 1969) encouraged Smirk (1957) and others to reduce arterial pressure by all means available, so that the vicious circle would be broken and arterial lesions could heal. This is now standard practice, but was revolutionary at the time. For his work, Byrom won the Volhard Award of the International Society of Hypertension in 1976.

The logic of treatment entails maintenance of normal mean blood pressure and volume flow, while decreasing pulsatile pressure and flow. Practical therapies include cerebral decompression (Berge, 2007; Vahedi *et al.*, 2007; Cooper *et al.*, 2011) and nitrates (Wilmot *et al.*, 2006; Hirata *et al.*, 2007). Treatment for non-hemorrhagic stroke might include consideration of non-pulsatile cardiac bypass. Monitoring might entail interpretation of carotid flow or pressure contour or of pressure and flow relationships as impedance (Hirata *et al.*, 2008; Soiza *et al.*, 2010).

What is normal and what is abnormal?

Reference has been made to the ubiquity of white matter intensities in MRI scans of apparently normal subjects, and their relationship with age-related aortic stiffening, and with microvascular disease in the brain and decline in intellectual function. This brings attention to the issue of what is normal and what is abnormal, and when does aging change constitute a disease (O'Rourke and Hashimoto, 2007)? We do not wish to confront these issues, but they were discussed by William Osler in his textbook (Osler, 1892), as was the related subject of the 'Fixed Period' in his last graduation address before leaving Johns Hopkins Hospital for Oxford (Osler, 1906). He believed that a scholar's best years were before 40 years of age and after age 60 it was best that they be replaced.

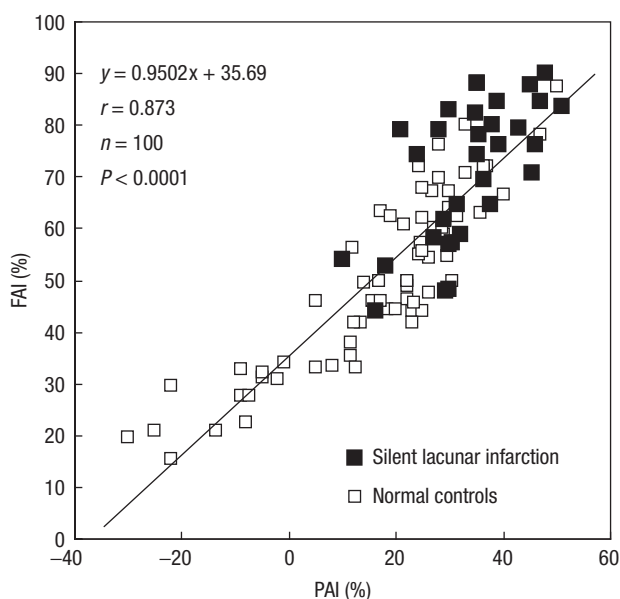


Figure 18.10 Relationship between carotid pressure (PAI) and flow (FAI) in 32 patients with silent lacunar infarcts who visited a brain health screening unit at a medical center and 68 normal subjects. Reproduced from Hirata *et al.* (2008)

The most recent contribution on this issue comes from Melbourne, Australia, where it was shown that 'healthy middle-aged individuals are vulnerable to cognitive defects as a result of increased arterial stiffness' (Pase *et al.*, 2010). Such observations fit with Osler's opinion that,

as it can be maintained that all the great advances have come from men under forty, so the history of the world shows that a very large portion of the evils may be traced to the sexagenarians – nearly all the great mistakes politically and socially, all of the worst poems, most of the bad pictures, a majority of the bad novels, not a few of the bad sermons and speeches ...

THE RENAL CIRCULATION

While the kidneys have specialized cells with high metabolic rate, they are few and require a relatively low flow rate to remain viable. Kidney perfusion however is high – taking some 25 percent of cardiac output (420 mL/100 g/min) (Table 18.1), since kidney function is to filter the blood and maintain constancy of the body's chemical internal environment (Bard, 1961; Boron and Boulpaep, 2005). They have two circulations in series: to the glomerular capillaries through the afferent arterioles, and to the tubular network through the efferent arterioles. Local, hormonal, and neural control of afferent and efferent arteries maintains appropriate pressure in glomerular and distal capillaries so as to control kidney function, while the vasa recta secrete renin in response to change in afferent arteriolar pressure and electrolyte content.

Sympathetic tone in the kidney is normally low, accounting for high resting blood flow and low resistance. Perfusion to the kidney per unit mass is the highest in the systemic circulation. Flow waveforms in the renal arteries show high diastolic and mean flow velocity and low PI and evidence of low peripheral reflection, as in the brain.

In consequence of its low resistance, impedance plots in the renal artery show little evidence of wave reflection, so that renal flow wave contour is similar to pressure wave contour. Flow pulsations and pressure pulsations extend down through the low resistance afferent arteriole to the glomerular capillaries (Loutzenhizer *et al.*, 2004; Struijker-Boudier *et al.*, 2006). It follows that these vessels – and to a lesser degree the efferent arterioles and tubular capillaries – are exposed to high pulsatile pressure and flow, and to the consequences of aortic stiffening and early wave reflection. The processes that have been described for the brain are expected to occur in the kidneys, and indeed have been so described in the presence of hypertension, especially malignant hypertension (Pickering, 1968; Byrom, 1969; Ford *et al.*, 2010).

Because of their low vascular resistance, one expects that the kidneys would be more vulnerable than any other organ to the ill effects of aging, arterial stiffness, and hypertension. This is the case and explains in part why

high arterial pressure and renal disease go hand in hand (Bakris, 2005; Verhave *et al.*, 2005). Proteinuria (Bakris, 2005; Ford *et al.*, 2010), and its increase in the malignant phase of hypertension, can be explained on the same basis as increased capillary permeability in the brain microvasculature consequent on increase in flow pulsations, i.e. on damage to and shedding of endothelial cells. Hematuria in severe hypertension can be explained at least in part on the same basis as cerebral microbleeds. Treatment of hypertension causes reduction in proteinuria, and this is associated with maintained renal function, and reduction in cardiovascular events (Safar *et al.*, 2004; Ibsen *et al.*, 2005; Lea *et al.*, 2005; Matsui *et al.*, 2010).

Renal artery flow patterns are used for the diagnosis of renovascular hypertension through demonstration of high flow pulsation within a stenosis and low PI above and below a localized stenosis. In the presence of distal renal vascular disease, flow pulsation is high relative to mean flow, so that PI is increased. Progressive increase in flow pulsation, as seen in the brain, has not to our knowledge been a feature of aging studies in the kidney.

THE LIVER CIRCULATION

The liver is a very vascular organ which receives some 25 percent of cardiac output. Three-quarters of this comes through the portal vein and just one-quarter (6 percent) through the hepatic artery (Table 18.1). Arterial and venous blood mix at the center of the hepatic acinus then flows through sinusoids to the periphery of the acinus whence blood flows into the hepatic vein to the inferior vena cava.

The liver is not exposed to direct arterial pulsations. Hence it is not exposed to the same damaging effects of arterial stiffening with age as the brain and kidney, despite its vascularity. As for the myocardium (see Chapter 17), one does not expect to see microvascular damage in the liver of elderly people with arterial stiffening. The smallest myocardial arteries show no such lesions either. These are protected from high systolic pressure by contraction of the myocardium around them.

SUMMARY

The new views on ill effects of flow pulsations on the microcirculation emerged in the early part of the twenty-first century as a logical extension of earlier work by authorities in hypertension research, such as Byrom and Pickering (as acknowledged by Laragh and Brenner (1995)), by histopathologists such as Fisher, clinical researchers in neurology such as Heistad, Schiffrin, and Donnan, and hemodynamic bioengineers such as Fry. These have been pressed by members of the Working Groups on Vascular Structure and Function of the European Society of Hypertension, such as Safar, Laurent,

Boutouyrie, and Struijker-Boudier. They have brought together researchers whose emphasis is on the macro-circulation with those whose emphasis is on the micro-circulation; neither can now see their own field in isolation as was the tendency in the past where expertise was dictated by availability and interest in specific techniques. We need to remember that our patients, and we ourselves, are aging, and that a macrovascular arterial system that progressively becomes inefficient to the heart, causing disease of the heart, also becomes inefficient to the microcirculation, causing disease of the brain and kidneys, which becomes evident insidiously with cognitive decline and with renal insufficiency that hastens degeneration of the macrocirculation. Byrom (1969) made his major contribution through introducing the concept of a vicious cycle causing malignant hypertension – which we hardly see now with prompt diagnosis and treatment (Ledingham, 1995). However, there are other vicious circles which accelerate disability and death through vascular mechanisms affecting heart, kidney, brain, and other organs. Just one is the interaction of small and large arterial vessels (Laurent *et al.*, 2009; Adji *et al.*, 2011).

Recent studies have endorsed the views stated by Safar and O'Rourke in 2006, and expanded in this chapter. Poor cognitive performance is seen as the first stage of dementia (Staessen *et al.*, 2007) and is associated with arterial stiffness, independent of age, and with risk factors of arterial disease, including obesity, cigarette smoking, and diabetes, but not cholesterol level. It is strongly associated with elevated arterial pressure (Waldstein *et al.*, 2008) and even with high-normal blood pressure (Elias *et al.*, 2008; Knecht *et al.*, 2008; Elias *et al.*, 2009; Nagai *et al.*, 2010). There must be an age limit to this relationship since elderly people usually show a progressive fall in systolic and pulse pressure in the final years of life, in association with debility (Staessen *et al.*, 2007; Axelsson *et al.*, 2008).

Staessen *et al.* (2010) presented strong data for the view that Alzheimer's dementia, as well as vascular dementia, was caused by arterial disease, and in mid-life at least, was delayed or prevented by treatment of hypertension. Other recently reported studies support this view (Hanon *et al.*, 2006; Stewart *et al.*, 2009), although the view is not universal (Pickering, 2008) and is not seriously considered by many neurologists.

Recent studies have confirmed the relationship between arterial stiffness and cerebral microvascular disease (Henskens *et al.*, 2008a; Henskens *et al.*, 2008b; Henskens *et al.*, 2009) and with renal microvascular disease (Abbatecola *et al.*, 2008; Ishikawa *et al.*, 2008). Arterial stiffening is seen as a link between cerebrovascular and renovascular disease (Hashimoto *et al.*, 2008b; Ito *et al.*, 2009; Laurent *et al.*, 2009) and with micro-albuminuria corresponding with patchy cerebral vascular edema seen in MRI studies. Cerebral white matter lesions remain a mystery, because of their frequency in apparently normal older people (Vernooij *et al.*, 2007). Perhaps they are as 'normal' as isolated systolic hypertension was considered 'normal' in older people just 25 years ago. They are associated with a risk of stroke (Debette and Markus, 2010), but not with vascular events elsewhere in the body (Buyck *et al.*, 2009). They are related to all blood pressure components (Guo *et al.*, 2009). While these views may be considered as nihilistic and depressing, they need not be. Byrom (1969) stressed that the basic problem in malignant hypertension was reversible. So it should be with the microvascular lesions caused by aortic stiffening. The logical approach is to limit their development by healthy lifestyle, and early treatment of elevated arterial pressure. We will not live for ever, but we can live better. It is pleasing to have a distinguished medical scientist, with the expertise, standing, and foresight of Valentin Fuster, taking on this new challenge (Fuster and Vahl, 2010).

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Aging

Principal cardiac changes with aging	415	Ventricular–vascular interaction	432
Cardiovascular function in normal aging humans	416	Arterial pressure and flow waves	433
Structural changes in arteries with age	417	Amplification	438
Functional change in arteries	423	Growth	441
Pulse wave velocity	426	Mechanism of arterial degeneration with age	442
Ascending aortic impedance	430	Perspective	446

Only in the case of young children do we find that the elasticity of arteries is so perfectly adapted to the requirements of the organism as it is in the case of the lower animals.

Roy, 1880

The amount of energy expended by the heart as measured by its oxygen consumption or CO₂ output has been shown to be proportional to the pressure developed (Rohde, 1910; Rohde, 1914); hence the amount of energy which the heart has to expend per beat, other things being equal, varies inversely with the elasticity of the arterial system.

Bramwell and Hill, 1922b

Pioneers of vascular and cardiac physiology, including the Nobel Laureate AV Hill, were well aware of the progressive stiffening of arteries with age, and the ill effects that such stiffening has on cardiac metabolism and function (Roy, 1880; Bramwell and Hill, 1922a; Bramwell and Hill, 1922b). Relevance to deterioration in physical fitness from early adulthood was acknowledged by Mackenzie (1902). Regrettably, studies on this subject lapsed in the 1920s with clinical acceptance of the cuff sphygmomanometer, perhaps on account of preoccupation with diastolic pressure, and with the emergence of the view that peripheral resistance provided a complete picture of left ventricular (LV) load (Mackenzie and Orr, 1926). Such studies have only recently been rekindled by epidemiological evidence that has confirmed the sinister implications of rising systolic blood pressure with age (see Chapter 21) (Amery *et al.*, 1985; Nichols *et al.*, 1985a; Rutan *et al.*, 1986; Kannel and Belanger, 1991; SHEP, 1991; Folkow and

Svanborg, 1993; Franklin *et al.*, 2001; Nichols and Singh, 2002; O'Rourke 2002a; Lakatta and Levy 2003a; Lakatta and Levy 2003b; O'Rourke and Hashimoto, 2007; McEniery *et al.*, 2010b; Mitchell *et al.*, 2010).

Current physiology texts do not deal with cardiovascular aging, and neither do undergraduate teaching programs, perhaps on the premise that the effects of aging constitute abnormality and should consequently be dealt with in medical texts and programs (Parmley, 1997). Up until recently, only one of the major texts on cardiovascular disease (*Braunwald's Heart Disease*) contained a separate chapter on aging and in this way may have influenced specific disease processes (Weisfeldt *et al.*, 1992; Lakatta *et al.*, 1997). Regrettably the sixth to ninth editions (Braunwald *et al.*, 2001; Braunwald *et al.*, 2004) of Braunwald's text substituted the chapter on aging with one on disease in the elderly. While medical and physiological texts have in general avoided discussion of the aging process, this has been featured in a number of general scientific journals over the last two decades (Rusting, 1992; Ricklefs and Finch, 1995; Black, 1999; Olshansky *et al.*, 2001; Olshansky *et al.*, 2002; Martin *et al.*, 2003 and the Science and SAGE KE website: <http://sageke.sciencemag.org/>).

There is no shortage of research on vascular aging (see Chapter 3), but much of this may be irrelevant or misleading. Animal research has largely been conducted on rats, whose normal life span of 30 months is less than 5 percent of Western humans and whose cumulative exposure to cyclic stress is approximately 25 percent that of humans (Wolinsky and Glagov, 1964; Wolinsky, 1972; Cox, 1978). There is good reason to believe that the accelerated lifespan of rats and of other animals does not fully simulate that of humans with respect to cardiovascular aging (Weisfeldt, 1980; Bulliman *et al.*, 1981;

Yin *et al.*, 1981; Yin *et al.*, 1983a; Nichols *et al.*, 1985a; Nichols *et al.*, 1987; O'Rourke, 1990a; Folkow and Svanborg, 1993; Nichols *et al.*, 1993; Nichols, 1994; Lakatta, 1995; Nichols and Edwards, 2001; Olshansky *et al.*, 2001; Olshansky *et al.*, 2002) since humans live so much longer than experimental animals. Most studies of cardiovascular aging in humans have concentrated on arterial pressure and particularly on diastolic pressure (US National Health Survey, 1977; Kohn, 1978; Fries, 1980; Weisfeldt, 1980; Roy *et al.*, 1984; Lakatta, 1989a; Lakatta, 1989b; Risk Factor Prevalence Study Management Committee, 1989; Lohman *et al.*, 1992). Diastolic pressure changes little with age on account of progressive arterial stiffening (Nichols *et al.*, 1985a; Berger and Li, 1990; O'Rourke, 1990a; Nichols *et al.*, 1992) at least up to age 60. Beyond 60 years, the Framingham and other studies have documented a definite and substantial fall in diastolic pressure, even as systolic pressure continues to rise (Kannel, 1985; Burt *et al.*, 1995; Franklin *et al.*, 1997; Franklin *et al.*, 1999; Franklin *et al.*, 2001; McEniery *et al.*, 2005a). Implications of this are discussed in Chapter 21 and by Najjar *et al.* (2005), Wang and Lakatta (2006), and O'Rourke and Hashimoto (2007). The rise in LV and aortic systolic blood pressure with age during adulthood is underestimated by the cuff sphygmomanometer (Fig. 19.1). The apparent plateau in systolic pressure in the (1977) US population between the ages of 20 and 50 years is also seen in the Australian population (Risk Factor Prevalence Study Management Committee, 1989). This appears to be artifactual and attributable to extreme amplification of the brachial systolic pressure peak in young adults (Fig. 19.2, and see Chapter 21; Figs 21.2 and 21.3) (Vlachopoulos and O'Rourke, 2000b; O'Rourke and Hashimoto, 2007; Avolio *et al.*, 2009b). A similar plateau is apparent for white males and females in the later US survey (Burt *et al.*, 1995) and from all data pooled from the INTERSALT study (Rodriguez *et al.*, 1994). The dotted line in Fig. 19.1 is the regression line for directly measured aortic systolic pressure in a group of apparently normal people undergoing cardiac catheterization for chest pain syndrome (Nichols *et al.*, 1985a). There was a steady increase in aortic systolic pressure from young adulthood, as there was in aortic pulse wave velocity (PWV) and characteristic impedance. Increase in systolic pressure with age is even greater in Chinese populations with high prevalence of hypertension (Wu *et al.*, 1996).

Many publications have tackled these issues, specifically Weisfeldt and Lakatta from Johns Hopkins Hospital and the US National Institutes of Aging (Weisfeldt, 1980; Lakatta, 1989a; Lakatta, 1989b; Weisfeldt *et al.*, 1992; Lakatta, 1993a; Lakatta, 1993b; Lakatta, 1995; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b; Lakatta, 2003), Folkow and Svanborg (1993) in *Physiological Reviews*, and our own reviews on the topic (O'Rourke, 1982a; O'Rourke, 1982b; Nichols *et al.*, 1985a; Nichols *et al.*, 1987; O'Rourke, 1990a; Nichols *et al.*, 1993; O'Rourke and



Figure 19.1 Change in arterial blood pressure with age in (A) USA (US National Health Survey, 1977) and (B) Australian National Heart Foundation (1989) populations. Solid lines represent published epidemiological data; thin lines indicate pressures recorded in the ascending aorta at diagnostic cardiac catheterization by Nichols *et al.* (1985a)

Kelly, 1993; Nichols, 1994; Westerhof and O'Rourke, 1995; Vlachopoulos and O'Rourke, 2000b; O'Rourke, 2002a; Safar and O'Rourke, 2006; O'Rourke and Hashimoto, 2007; Nichols *et al.*, 2008; Hashimoto and Ito, 2009).

All are made highly relevant by findings of the Build and Blood Pressure, Framingham, MRFIT, SHEP, European Collaborative, and other studies, which stress the influence of systolic pressure on cardiovascular events and the lack of correlation of such events with diastolic pressure in older subjects (Kannel *et al.*, 1981; Amery *et al.*, 1985; Rutan *et al.*, 1988; Perry *et al.*, 1989; Rutan *et al.*, 1989; Franklin and Weber, 1994; Avanzini *et al.*, 1995), and the SHEP, STOP Hypertension, European Collaborative, MRC, CONSENSUS, ALLHAT, HOPE, REASON, LIFE, ALLHAT, ASCOT, CAFE, and HYVET studies. All show the benefit of reducing elevated systolic

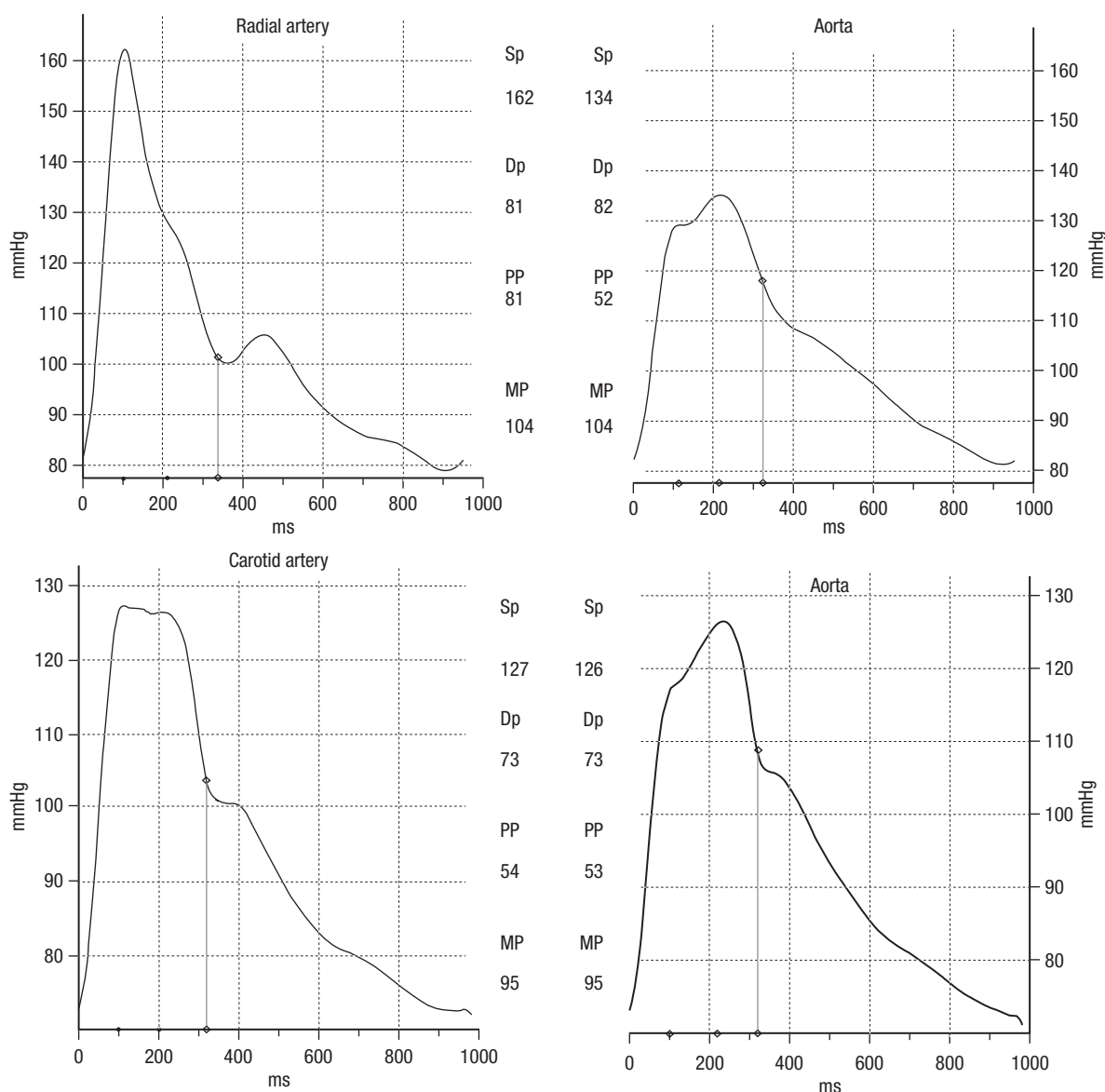


Figure 19.2 Spurious systolic hypertension in youth detected by applanation tonometry (see Chapter 30). Top: Pressure wave recorded non-invasively by applanation tonometry in the radial artery (left) and synthesized aortic pressure wave (right) of a young adult referred to the clinic with a diagnosis of systolic hypertension, but without any other abnormality (amplification 1.6). Bottom: Pressure waves recorded non-invasively by applanation tonometry in the same person from the carotid artery (left) and compared to the aortic pressure wave (right).

blood pressure in the elderly (Amery *et al.*, 1986; Consensus Trial Study Group, 1987; Dahlöf *et al.*, 1991; SHEP, 1991; Winker and Murphy, 1991; Medical Research Council Working Party, 1992; Yusuf *et al.*, 2000; Asmar *et al.*, 2001b; ALLHAT, 2002; Dahlöf *et al.*, 2002; London *et al.*, 2004a; Dahlöf *et al.*, 2005; Beckett *et al.*, 2008). Principles espoused are relevant also to the treatment of subclinical and clinically manifest cardiac failure (see Chapters 15 and 21), where the absolute cornerstone of treatment is reduction in LV load, even if this is not elevated by conventional criteria.

Studies of aging in humans are bedevilled by a number of specific problems. These include: (1) inadequacy of conventional non-invasive markers (especially diastolic

blood pressure); (2) difficulty in separating the aging process from concurrent disease; (3) difficulty in agreeing on what in the aging process is normal and what is abnormal; (4) allowance for the 'natural' death of cells and the atrophy of tissues; (5) the difficulty in separating cause from effect (i.e. is inflammation a cause of arterial disease or a consequence of physical damage with healing of the damaged media); (6) the presence of the reserve capacity of the heart (and of other organs); and (7) the fact that cross-sectional studies are flawed by the inclusion only of survivors from an earlier age – people who may be 'superfit' and relatively immune from the aging process itself (Kohn, 1978; Weisfeldt, 1980; Lohman *et al.*, 1992; Vaitkevicius *et al.*, 1993; Perls, 1995). Furthermore, studies

of aging in humans tend to follow the social and psychological profile of growth, maturity, stability, then decline – as set out in Shakespeare's *As You Like It* as the Seven Ages of Man. With aging, there is evidence of inflammation, and increased susceptibility to infections and neoplasm. This has been described as 'inflamm-aging', associated with immunosenescence (Larbi *et al.*, 2008). This issue was a point of contention with respect to the inflammation seen in cerebral and renal microcirculation of humans and experimental animals with malignant hypertension. Byrom and Dodson (1948) and Byrom (1969) had to tackle this and argued that inflammation was due to repair of over-distended damaged arteriolar walls, and Pickering (1968) agreed. Most others at the time considered that inflammation was the primary cause. The issue was solved when the lesions were seen to resolve with reduction in arterial pressure (Smirk, 1957; Pickering, 1968). The same issue arises anew with respect to cerebral microvascular lesions in people with dementia (Chapter 18).

The subject of aging does have important social connotations. Expectation of life has improved markedly from less than 40 years in the period before the twentieth century to around 80 years in Western countries today. Every new success in medical therapy – for cardiovascular disease, for cancer, for anything – extends this period and swells the numbers for whom society must care and whom physicians must attend. Median life expectancy for Australian men is 45, 35, 25, 16, and 9 years at 40, 50, 60, 70, and 80 years and some three years longer for women (Australian Bureau of Statistics in Population Projections, Australia, 2006–2101). It behoves carers and attenders to understand the difference between older people and those animals whose physiologic and pathophysiologic processes have been included in their training. It behoves them also to consider changes in mortality of the population (Fig. 19.3), where mortality is being compressed into a smaller range of age, but at a greater age, and in consequence there is more opportunity to put 'quality' into the life of older people before the onset of terminal disease (Fries, 1980; Lohman *et al.*, 1992; Lakatta *et al.*, 1997; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b; Najjar *et al.*, 2005; O'Rourke and Hashimoto, 2007).

There is thus a great deal of imbalance and controversy in the literature on the subject of cardiovascular aging. Such controversy relates to the nature of aging change, the magnitude of aging change, and the cause of aging change. In this chapter, we will argue:

1. that the effects of aging on arteries of humans are gross, are progressive throughout life, and are underestimated by conventional clinical indices of cardiovascular function;
2. that the effects of arterial aging are responsible for progressive increase in LV load, and so for LV hypertrophy, increased LV oxygen demand and ultimately the systolic and diastolic myocardial dysfunction that causes cardiac failure;

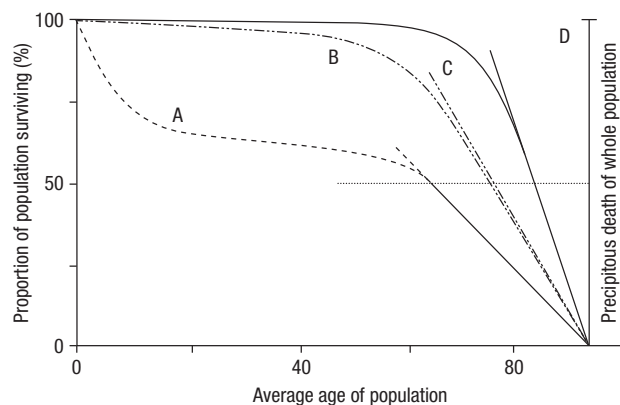


Figure 19.3 Hypothetical survival curves for four different population circumstances: (A) non-industrialized country, 1991; (B) industrialized country, 1991; (C) industrialized country, 2000; (D) 'drop dead' situation. All curves are linearly extrapolated to the estimated maximum average lifespan of 95 years for the year 2000. Reproduced from Lohman *et al.* (1992)

3. that while arterial degenerative changes are responsible for cardiac dysfunction and failure in the elderly, they are also responsible for deterioration in cardiovascular performance during exercise in younger subjects – beginning with competitive performance in the third decade, through social and recreational performance in subsequent decades;
4. that arterial aging is a degenerative process, primarily caused by elastin fiber fracture, consequent on repetitive pulsations, and explicable on the theory of material fatigue, with the secondary effects of dilatation, stiffening, and thickening, attributable to repair and fibrous remodeling;
5. that aging change with marked stiffening of the aorta and proximal elastic arteries, results in pulsatile flow extending further downstream into vasodilated vascular beds of brain and kidney, with exaggerated energy loss in small arterial vessels causing damage which results in microinfarcts and microbleeds which are the basis of dementia (brain) and renal failure (kidney);
6. that while damage to elastin fibers is expected from systolic pulsations, the damage to large arteries, small arteries, and heart can be reduced or delayed by 'trapping' reflected waves in the peripheral arteries of non-specialized tissues so that they do not return during the period of systole to elevate aortic and LV pressure and to increase cerebral/renal flow pulsation.

Our views on aging, set out in this chapter and elsewhere (Nichols and O'Rourke, 2006a; Nichols and O'Rourke, 2006b; Safar and O'Rourke, 2006; O'Rourke and Hashimoto, 2007; Nichols and O'Rourke, 2009; O'Rourke *et al.*, 2010b), concentrate on non-living, inert components

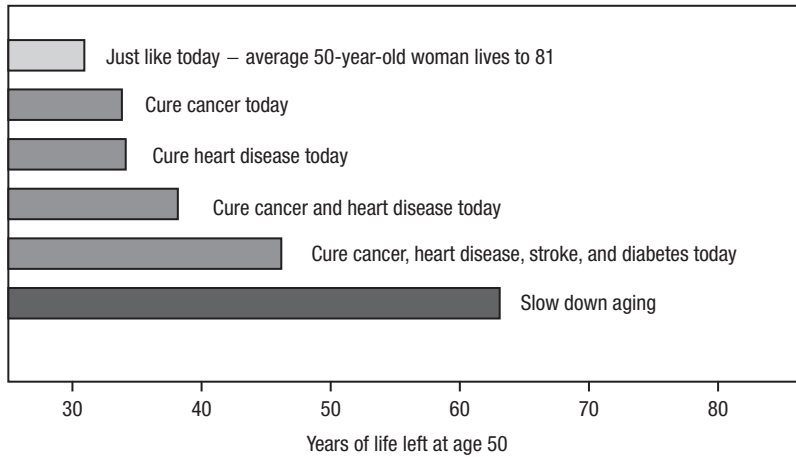


Figure 19.4 Projections on aging. Expectation of life for a normal US woman in 2003 (top), then for the average women if the most common causes of death were found to be treatable. Life expectancy would be highest (more bang for research buck) if aging could be slowed but these diseases remained. Reproduced from Martin *et al.* (2003)

of the body. These include aortic elastin, the most inert material in the body, but apply also to other components of the body which are not turned over regularly by cellular activity – the hair, teeth, joints, even the elastin in skin, which create such obvious problems in older people. We are subject to criticism by others whose techniques focus on genes, cells, enzymes, and drugs, but we do not exclude a role for such factors, although we see them more likely to be a lesser or later part of the degenerative effect, rather than the cause. It is not correct to portray such views as nihilistic and ‘no longer tenable’ (Najjar *et al.*, 2005), just because other mechanisms exist but have not yet been exploited. We do apply the principles we have espoused by treating systolic pressure elevation early with drugs which reduce aortic pressure augmentation – with angiotensin-converting enzyme inhibitors (ACEI), angiotensin receptor blockers (ARBs), calcium channel blockers (CCBs), nitrates, and vasodilating β -blockers, and with encouragement of weight loss and regular aerobic exercise – all aimed at improving endothelial function and reducing wave reflection. However, the need for more research is obvious from consideration of Fig. 19.4 from *Science* (Martin *et al.*, 2003), which argues that research on aging, even just on slowing down aging could add more years to life than if all forms of cancer, heart disease, stroke, and diabetes were cured today.

PRINCIPAL CARDIAC CHANGES WITH AGING

Primary aging effects on the heart are relatively minor and contrast with the gross effects of age on the arterial system. Most of the effects of aging on the heart are attributable to a change in arterial properties and the secondary effects of increased hydraulic load (Nichols *et al.*, 1985a; Nichols *et al.*, 1987; Lakatta, 1989a; Lakatta, 1989b; Katz, 1990; Weisfeldt *et al.*, 1992; Folkow and Svanborg, 1993; Nichols *et al.*, 1993; Lakatta, 1995; Lakatta *et al.*, 1997; Braunwald, 2001a; Lakatta, 2003; Lakatta and Levy 2003a; Lakatta and Levy 2003b; O’Rourke and Hashimoto, 2007).

After neonatal development, there is no increase in the number of myocardial cells. Those present in an octogenarian have been present and contracting rhythmically since infancy or earlier (Morley and Reese, 1989; Zuckerman and Yin, 1989b; Folkow and Svanborg, 1993). There are claims that under some conditions, heart muscle cells may be induced to multiply, but this view conflicts with earlier findings which showed that there is, in fact, a decrease in the number of myocardial cells (estimated at a decrease of 38 million myocardial nuclei in the left ventricle per year (Olivetti *et al.*, 1991)). This is in concordance with the known loss of cells in other organs, such as the brain, kidney, and liver (Kohn, 1978; Fries, 1980). In the heart, however, the remaining myocardial cells hypertrophy – by an average of $110 \mu\text{m}^3/\text{year}$ in one study (Olivetti *et al.*, 1991), such that ventricular muscle thickness is preserved. Some studies have shown an increase in LV mass with aging (Gerstenblith *et al.*, 1977; Gardin *et al.*, 1979; Gardin *et al.*, 1987; Pearson *et al.*, 1991a; Pearson *et al.*, 1991b; Folkow and Svanborg, 1993; Bikkina *et al.*, 1994; Cheng *et al.*, 2010; Scalia *et al.*, 2010), but some have shown little significant change except in the presence of disease (Lie and Hammond, 1988; Weisfeldt *et al.*, 1992). Since there is universal decrease in lean body mass with aging, all studies are consistent with an increase in myocardial mass per unit of lean body mass. These cardiac findings contrast with virtually all other organs where cellular loss is associated with a decrease in mass (Kohn, 1978; Fries, 1980) since there is no similar hypertrophy of the remaining cells. In the heart, preservation or increase in mass is largely attributable to hypertrophy of myocardial cells (Unverferth *et al.*, 1983), although there is some corresponding increase in the fibrous connective tissue matrix (Tan *et al.*, 1992; Weber *et al.*, 1992). The increase in myocardial and fibrous tissue is, however, not accompanied by a corresponding increase in capillary and vascular growth (Morley and Reese, 1989; Frohlich *et al.*, 1992a). In consequence, with aging, the stage is set for a decreased supply of nutrients and an impaired release of metabolites, both from the cell (on account of the greater volume

per unit of surface area) and from/into the bloodstream. When there is an increase in myocardial blood flow requirement consequent on increased LV load and mass, there is considerable decrease in myocardial blood flow reserve as well (Fig. 19.5, see Chapter 17) (Morita *et al.*, 1991; Watanabe *et al.*, 1992; Czernin *et al.*, 1993; Masuyama *et al.*, 1994; Ohtsuka *et al.*, 1994; McGuire *et al.*, 2001a; McGuire *et al.*, 2001b).

With respect to ventricular function, there is preservation of the ability of myocardial cells to develop tension during systole (Weisfeldt *et al.*, 1992). There is a modest decrease in the velocity of myocardial shortening with age, but the duration of contraction is prolonged (Lakatta *et al.*, 1975; Lakatta, 1989b; Weisfeldt *et al.*, 1992; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b). Diastolic changes are more marked and comprise delayed

relaxation with a consequent decrease in early LV filling and a potential for decrease in ventricular filling throughout diastole (Pasierski *et al.*, 1991; Pearson *et al.*, 1991a; Pearson *et al.*, 1991b; Schulman *et al.*, 1992; Weisfeldt *et al.*, 1992; Masuyama *et al.*, 1994; Villari *et al.*, 1997; Scalia *et al.*, 2010; Shim *et al.*, 2011). This phenomenon is accentuated by a concurrent increase in myocardial stiffness, caused by myocardial cell hypertrophy and fibrosis (Bonow and Udelson, 1992; Weber *et al.*, 1992; Hundley *et al.*, 2001; Weber *et al.*, 2006; Abhayaratna *et al.*, 2008; Weber *et al.*, 2008; Scalia *et al.*, 2010). All these factors – slowed ventricular relaxation, cellular hypertrophy, and fibrosis – predispose to increasing LV pressure during diastole and thus contribute to diastolic dysfunction (Redfield, 2004; Paulus, 2010), to increased venous filling pressure and, through decreasing coronary arteriovenous pressure gradient, to myocardial ischemia (see Fig. 13.7 and Chapter 15 which deals with the clinical problem of cardiac failure due to diastolic LV dysfunction). These problems are accentuated by tachycardia (Masuyama *et al.*, 1994; Ohtsuka *et al.*, 1994; Warltier *et al.*, 2000; Shammash *et al.*, 2001; Taylor and Pagliarello, 2003; Prichard and Vallance, 2004), and provide a good reason for using β -blocking drugs in the perioperative period for older people undergoing non-cardiac surgery.

The functional changes described above are largely attributable to cellular hypertrophy consequent on increased LV hydraulic load (Frohlich *et al.*, 1992a; Frohlich *et al.*, 1992b; Sagie *et al.*, 1993). Impaired diastolic relaxation, however, is partly due to increased late systolic pressure which is due to early wave reflection (Hori *et al.*, 1985; Takazawa *et al.*, 1993a; Takazawa *et al.*, 1993b; Villari *et al.*, 1997) and is partly reversible when pressure is reduced, as by ACE inhibitors (Yabe *et al.*, 1991; Kelly and Kass, 1992b; Lorell, 1995) or angiotensin-blocking drugs, CCB, or nitrates (see Chapters 15 and 26).

From a physiologic viewpoint, another major cardiac change with age is decreased responsiveness to the sympathetic nervous system and to circulating catecholamines, with decreased inotropic, chronotropic, and vasodilatory responsiveness (Weisfeldt *et al.*, 1992; Lakatta, 1993a; Lakatta, 1993b; Paulus, 2010). This appears to be a post-receptor phenomenon.

CARDIOVASCULAR FUNCTION IN NORMAL AGING HUMANS

Conventional measures of cardiovascular function show little change at rest. The only definite rest changes in 'Westernized' populations are an increase in arterial systolic (and pulse) pressure (Kannel *et al.*, 1971; Kannel and Gordon, 1978; Stamler *et al.*, 1978; Kannel *et al.*, 1980a; Kannel *et al.*, 1980b; Svardsudd and Wilhelmsen, 1980; Nichols *et al.*, 1985a; Franklin *et al.*, 1997), slowed and delayed LV filling during diastole (Gerstenblith *et al.*, 1977; Messerli *et al.*, 1984; Voutilainen *et al.*, 1991;

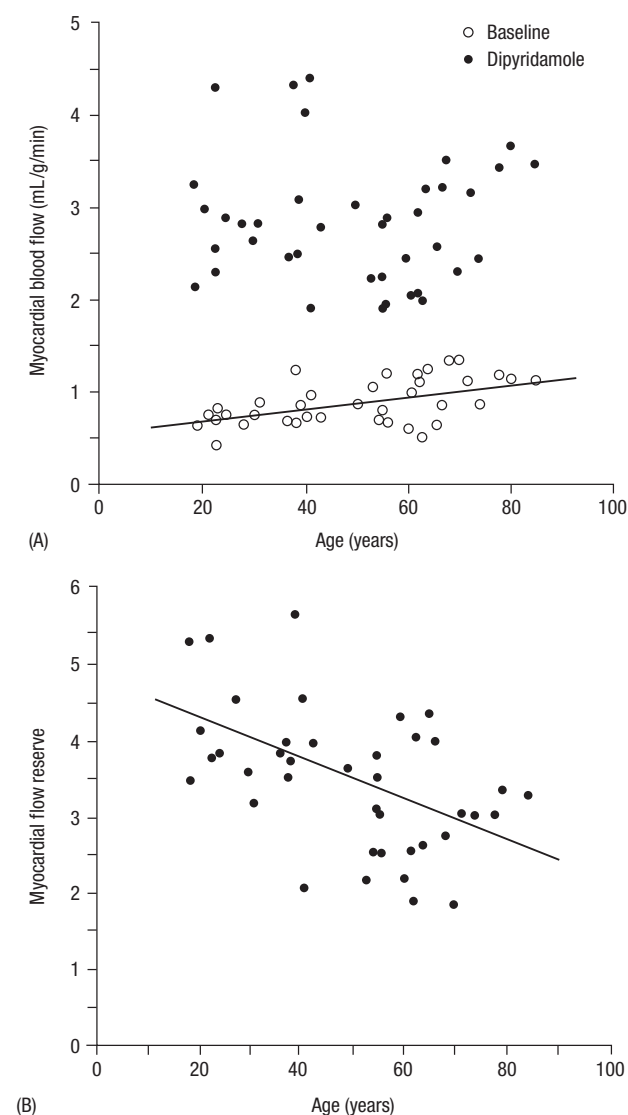


Figure 19.5 (A) Relationship between 'maximal' (●) and resting (○) myocardial blood flow and age. (B) Relationship between myocardial blood flow reserve (maximal – resting flow) and age in the same group. Reproduced from Czernin *et al.* (1993)

Villari *et al.*, 1997; Cheng *et al.*, 2010; Scalia *et al.*, 2010) and a decrease in peak ascending aortic blood flow velocity (Mowat *et al.*, 1983; Haites *et al.*, 1985). Arterial diastolic blood pressure increases from childhood until the fifth decade then after a plateau around age 50, decreases progressively above age 60 (Kannel, 1985; Burt *et al.*, 1995; Franklin *et al.*, 1997; McEniery *et al.*, 2005a) (Fig. 19.1). Systolic pressure increases by around 30 mmHg between age 20 and 80 years, although the greatest increase in systolic brachial pressure occurs before age 20 and after age 50 (see Fig. 19.1). Resting cardiac output (and cardiac index) and stroke volume (area underneath the volumetric flow curve) have been found to be either reduced (Lewis, 1934; Cournand *et al.*, 1945; Brandfonbrener *et al.*, 1955; Julius *et al.*, 1967; Messerli *et al.*, 1981; Nichols *et al.*, 1985a; Coggan *et al.*, 1990) or unchanged (Higginbotham *et al.*, 1984; Rodeheffer *et al.*, 1984; Rodeheffer *et al.*, 1986; Fleg *et al.*, 1990) in older versus younger individuals. We believe that stroke volume (and cardiac output) decreases slightly because of the age-related increase in amplitude of the reflected wave, which subtracts from the measured flow wave during deceleration (see Chapters 9 and 15). Peak ascending aortic volumetric blood flow (blood flow velocity $\times$ cross-sectional area) probably changes very little with age since ascending aortic cross-sectional area increases with age (Gerstenblith *et al.*, 1977; Nichols *et al.*, 1985a; Lakatta, 2000), while peak blood flow velocity decreases (Mowat *et al.*, 1983; Haites *et al.*, 1985); therefore, their product does not change (or decreases slightly).

In contrast to these minor changes at rest, aging is associated with quite marked deterioration in exercise ability (Lakatta *et al.*, 1997; McGuire *et al.*, 2001a; McGuire *et al.*, 2001b; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b) (Fig. 19.6) and in indices of cardiovascular efficiency during exercise. There is a decrease in maximal oxygen uptake, a decrease in maximal heart rate and stroke volume, a decrease in cardiac output and ejection fraction, with an increase in both end-systolic and end-diastolic volume (Fig. 27.6). In people with apparently preserved cardiovascular function, exercise uncovers deficiencies with age, just as exercise uncovers deficiencies and symptoms in people with cardiovascular diseases, such as coronary atherosclerosis and cardiac failure. The decline in cardiovascular function during exercise with age shows no lower limit. The aging process is probably responsible for the progressive decline in physical performance throughout life from the time in adolescence when full body height is attained (Mackenzie, 1902).

STRUCTURAL CHANGES IN ARTERIES WITH AGE

Concurrent arterial disease

The general problem of separating age-related changes from disease-related changes plagues research into aging.

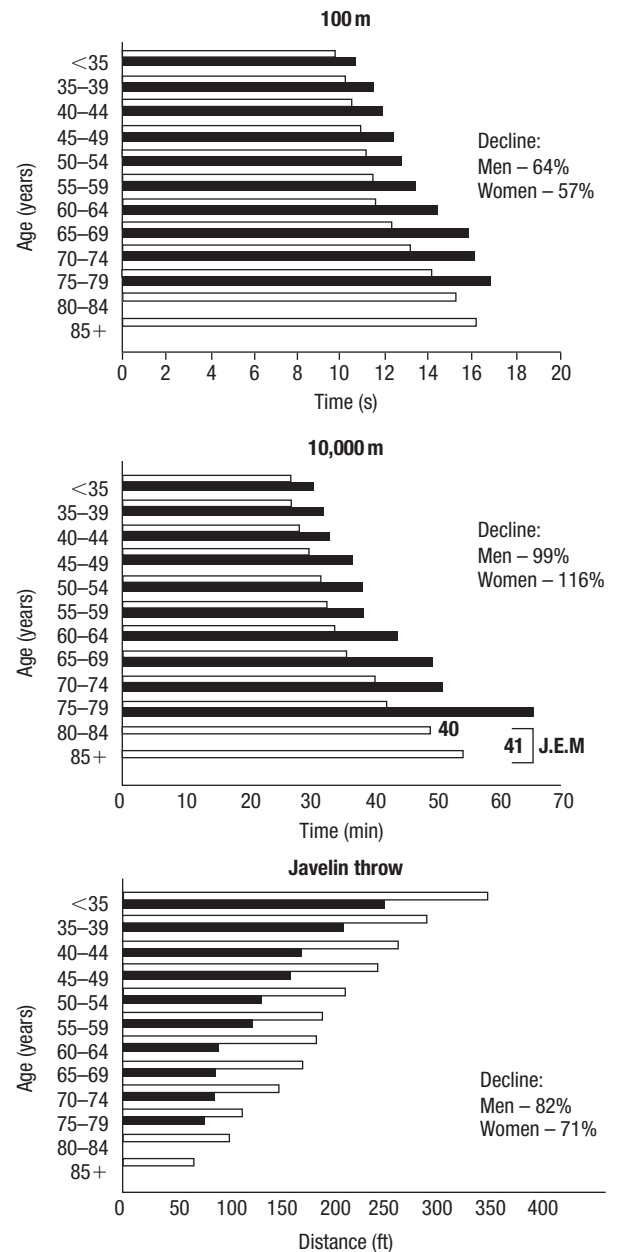


Figure 19.6 Representation of the decline in world records for the 100 and 10 000 meters and the javelin throw with advancing age. Although the decline with age appears inevitable, it should be noted that the best older athletes have performances that are superior to those of average middle-aged men. This is illustrated in the 10 000 meters where one of the authors' (JEM, age 40–41 years) best times are shown to be equivalent to the world record for an 85-year-old man. Males, solid bars; females, open bars. From Morley and Reese (1989)

This is especially true in studies of arterial structure in Western populations on account of the ubiquity and severity of atherosclerosis, even in asymptomatic subjects. When the subject of arterial aging arises at a clinical meeting, most Western physicians refer to atherosclerosis and its increasing prevalence with age. This is not a

normal aging change. Atherosclerosis is best considered as a disease and a condition quite separate and distinct from true aging change (Moschowitz, 1950; Weisfeldt, 1980; Kissane, 1985; Vlachopoulos and O'Rourke, 2000b; O'Rourke and Hashimoto, 2007). Councilman (1891) and Osler (1892), in their influential works, referred to aging change as 'senile arteriosclerosis' and to atherosclerosis as 'nodular arteriosclerosis', stressing that the former is diffuse and dilatory, while the latter is localized and obstructive. Many use the terms 'atherosclerosis' and 'arteriosclerosis' interchangeably. The American Heart Association (AHA) journal *Arteriosclerosis and Thrombosis* (now *ATVB*) deals almost exclusively with atherosclerosis (and thrombosis). The index to abstracts for all but the most recent AHA meetings comprised an identical range of subjects under the titles 'Arteriosclerosis' and 'Atherosclerosis'. In this book, we eschew the general term 'arteriosclerosis' and refer to the degenerative changes of aging and hypertension without providing a specific name, but we consider these to be separate from atherosclerosis, recognizing of course that all conditions may be seen together. The problem embraces not only the term 'arteriosclerosis' but the prefix 'senile' as well; as first recognized by Roy (1880), arterial aging begins in childhood, is obvious in the elderly and is apparent when sought from the contour of the radial, carotid, and aortic pressure pulse in most human adults (Marey, 1863; Mahomed, 1872; O'Rourke and Hashimoto, 2007).

In this chapter, we wish to concentrate on aging change itself. The distinction between this and atherosclerosis could hardly be more stark (Table 19.1). Atherosclerosis is initially and predominantly an intimal disease and localized to sites of predilection in major arteries. Aging change is diffuse, initially and predominantly medial. Atherosclerosis constricts while aging dilates. Atherosclerosis causes ill effects from ischemia downstream; aging causes ill effects from increasing LV load upstream. Atherosclerosis is very rare in some populations, whereas aging change is ubiquitous, although hastened by elevated arterial pressure. Atherosclerosis is a major cause of aneurysmal dilatation at only one site – the distal

abdominal aorta – whereas aging change weakens the arterial wall and predisposes to arterial dilatation, aneurysm formation, and rupture throughout the body, especially in the aorta itself and in cerebral arteries (see Chapter 20).

Atherogenesis and atherosclerosis are discussed in Chapter 23, while other arterial diseases are described in Chapters 24 and 25. Hypertensive arterial disease has many similarities to the aging process and can be regarded as a type of accelerated aging. The underlying mechanism of damage to large elastic arteries is similar. Osler (1892) saw little distinction between senile arteriosclerosis, and what he called 'diffuse arteriosclerosis' in young people. This probably was hypertensive disease (he had no means of measuring blood pressure). Hypertension as presently defined is discussed separately in Chapter 20.

Some ten years ago, Dzau *et al.* (Dzau *et al.*, 2006a; Dzau *et al.*, 2006b) introduced the view of a 'cardiovascular continuum' to explain the development of coronary atherosclerosis from pre-existing risk factors, and its development into myocardial infarction, cardiac failure, and other complications, and ultimately leading to death. The continuum described different stages in the process where interventions could delay or redirect progression and thus extend or improve life. This view can now be extended into an 'aging cardiovascular continuum' which traces the effects of aging change which, independently of atherosclerosis and conventional risk factors, can lead on to similar cardiovascular events, and ultimately lead on to cardiac failure and death. This 'continuum' appears to be more relevant to societies such as China where atherosclerosis is not a major problem at this time, but strokes from small vessel disease are more common and cardiac failure as common as in Western societies. The aging continuum will probably become more important in Western societies as coronary disease is better controlled by preventive and therapeutic measures. Figure 19.7 shows the two continua side by side (O'Rourke *et al.*, 2010b).

Intrinsic aging change

Little attention is given to the aging change of arteries of the systemic circulation in textbooks of histology, pathology, or cardiology, yet these changes are dramatic and gross in human subjects. The changes are not usually seen in experimental animals over the normal lifespan and are not apparent in the human systemic or pulmonary veins. Detailed studies of aging change on arterial wall structure and chemistry have been reported (Moritz and Oldt, 1937; Mitchell and Schwartz, 1965; Deyl *et al.*, 1971; Kohn, 1978; Stehbens, 1979; Cliff, 1981; Masoro, 1981; Harris, 1982; Yin *et al.*, 1983a). The histologic morphometric studies of Gerrity and Cliff (1975) and of Toda *et al.* (1980) have now been supplemented by our own studies with colleagues in Sydney (Avolio *et al.*, 1983;

Table 19.1 Contrasting features of aging change and atherosclerosis in human arteries

	Atherosclerosis	Aging change
Anatomic location	Focal	Diffuse in elastic arteries
Vascular location	Intima	Media
Vascular effect	Constriction	Dilation
Consequence (distal)	Ischemia	Microvascular damage in brain and kidney
Consequence (proximal)	Nil	Left ventricular load

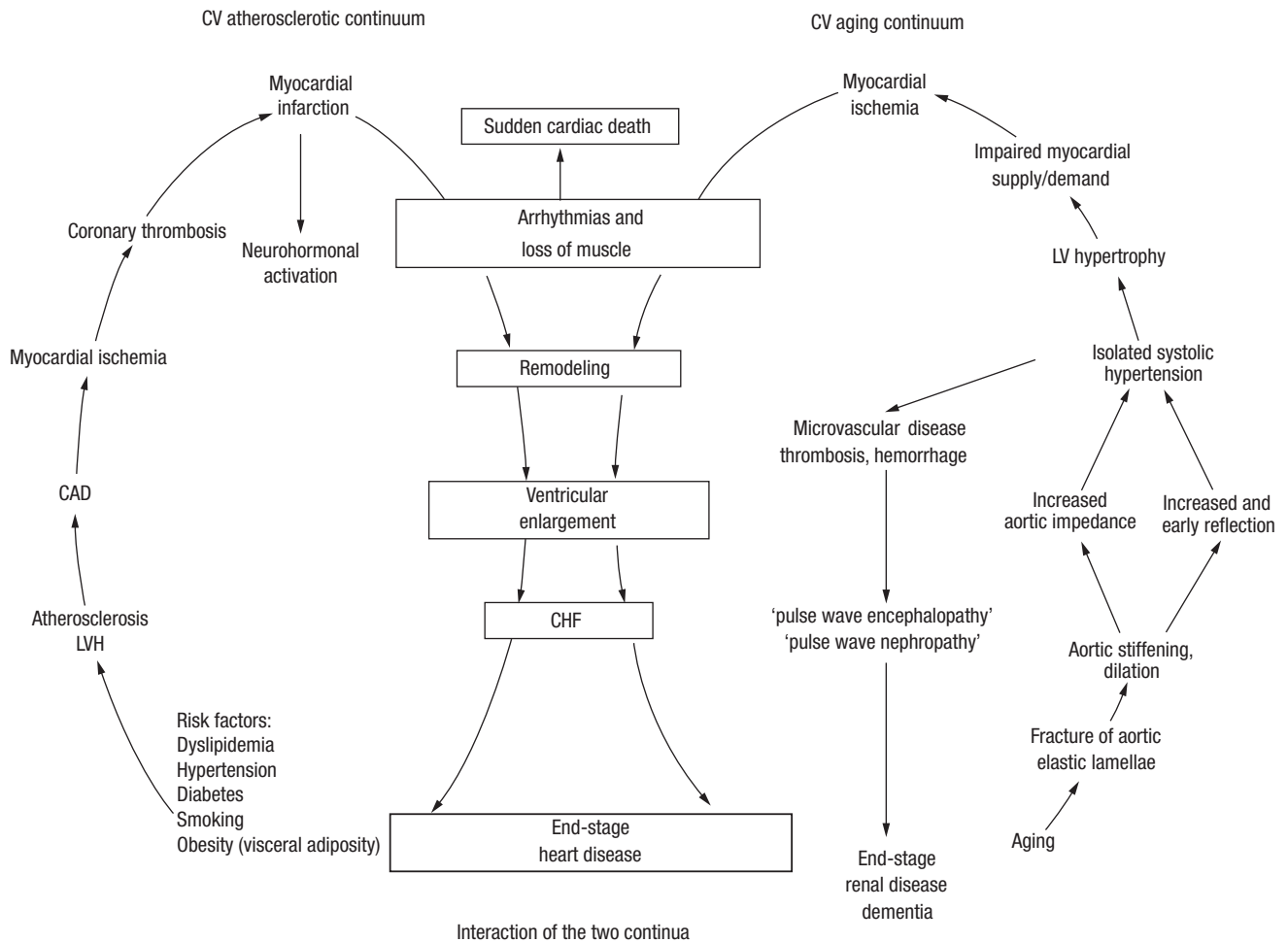


Figure 19.7 The cardiovascular continuum of Dzau *et al.* (2006a; 2006b) (left) compared to the aging continuum (O'Rourke *et al.*, 2010b) (right). The former is based on atherosclerosis and pre-existing risk factors. The latter is based on aortic aging and its ill effects which include macrovascular damage, cardiac failure, and angina, together with microvascular disease of brain and kidney. The final progression to end-stage heart disease and heart failure is common to both continua. Reproduced from O'Rourke *et al.* (2010b)

Avolio *et al.*, 1985a; O'Rourke *et al.*, 1987; Jiang *et al.*, 1993) and through collaboration with centers in China and the United States (Virmani *et al.*, 1991; Jiang *et al.*, 1993), and by fascinating recent work using magnetic resonance imaging (MRI) (Redheuil *et al.*, 2010). The Chinese studies were most valuable, since unlike earlier studies of arterial function, they included populations with virtually no atherosclerosis but with a different prevalence of hypertension, and thus helped to separate the different contribution of aging, of hypertension, and of atherosclerosis. We believe that such morphometric studies are crucial to the understanding of underlying mechanisms of aging since they provide the means for comparing the rate of degeneration between different human groups and different animal species, and for testing the hypothesis that the degeneration of the media with age is due to the fatiguing effects of cyclic stress on the elastic fibers.

Aging changes to the human arterial wall involve the intima (which undergoes hyperplasia – even in the

absence of atherosclerosis) but are most marked in the load-bearing media where there is progressive disorganization, with loss of the orderly arrangement of elastin fibers and laminae (see Chapter 4). These undergo thinning, splitting, fraying, and fragmentation (Fig. 19.8) (Lakatta *et al.*, 1987; O'Rourke *et al.*, 1987; Lakatta, 1989a; Lakatta 1989b; Virmani *et al.*, 1991; Lakatta *et al.*, 1997; Fonck *et al.*, 2009). Degeneration of elastin fibers is associated with an increase in collagenous material and in ground substance, and often with deposition of calcium in this and in degenerate elastin fibers (McEniery *et al.*, 2009). Either of these processes may occur to an extreme, with cystic medial degeneration (Carlson *et al.*, 1970) and pooling of mucoid material (medionecrosis) or extensive medial calcification (Monckeberg's sclerosis). Changes are most marked in the aorta and central elastic arteries, which dilate and become tortuous (Figs 19.9 and 19.10). Calcification may be patchy (Fig. 19.9B) or so diffuse that the aorta appears more radiodense than the thoracic vertebral bodies (Fig. 19.9A). Quantitative changes of

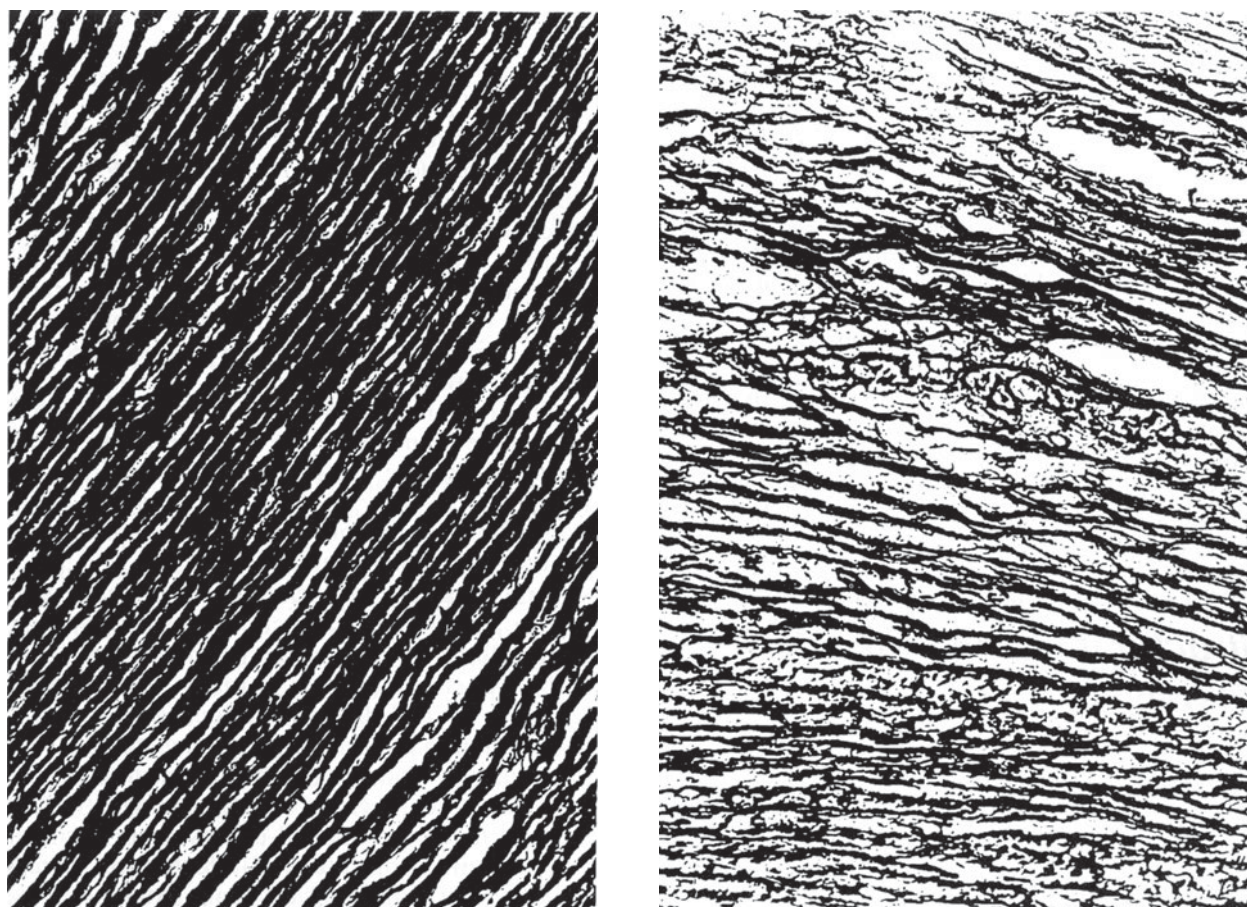


Figure 19.8 Comparison of the arterial wall elastin composition in a child (left) with that in an elderly adult (right). Stain: Orcein and Giesen, magnification $\times 450$. Prepared by P Lauren

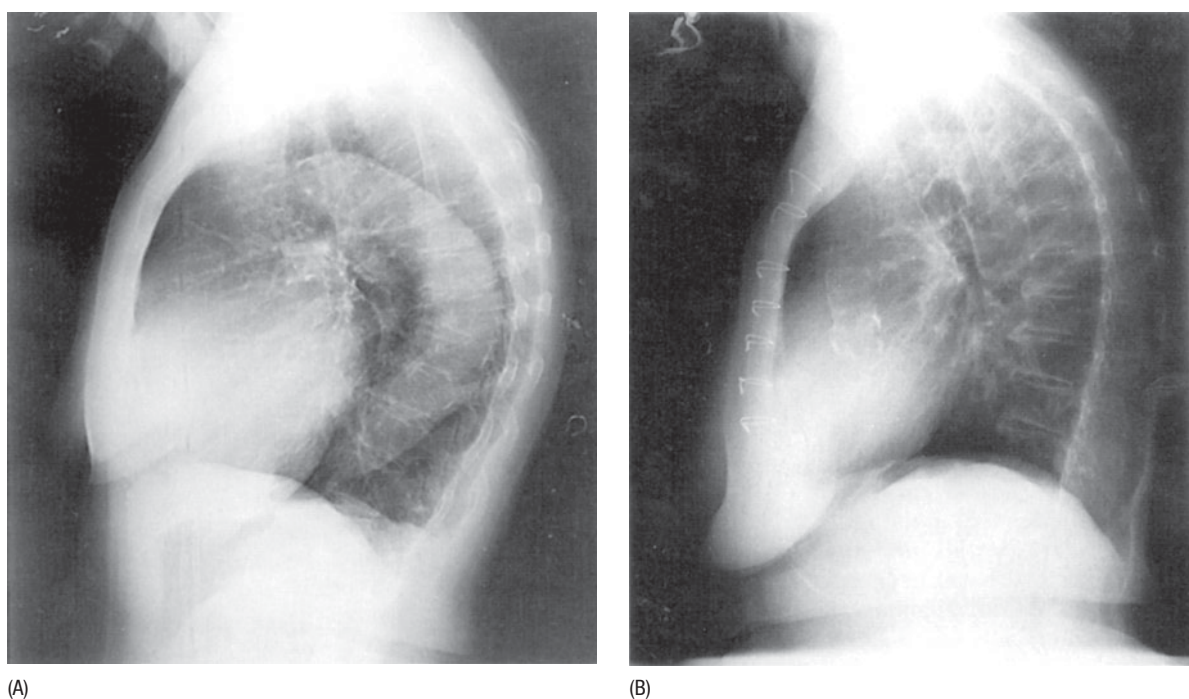


Figure 19.9 Lateral chest x-rays of two older women, showing diffuse (A) and localized (B) calcification of the thoracic aorta. The degree of calcification appears comparable to that of the ribs and spine

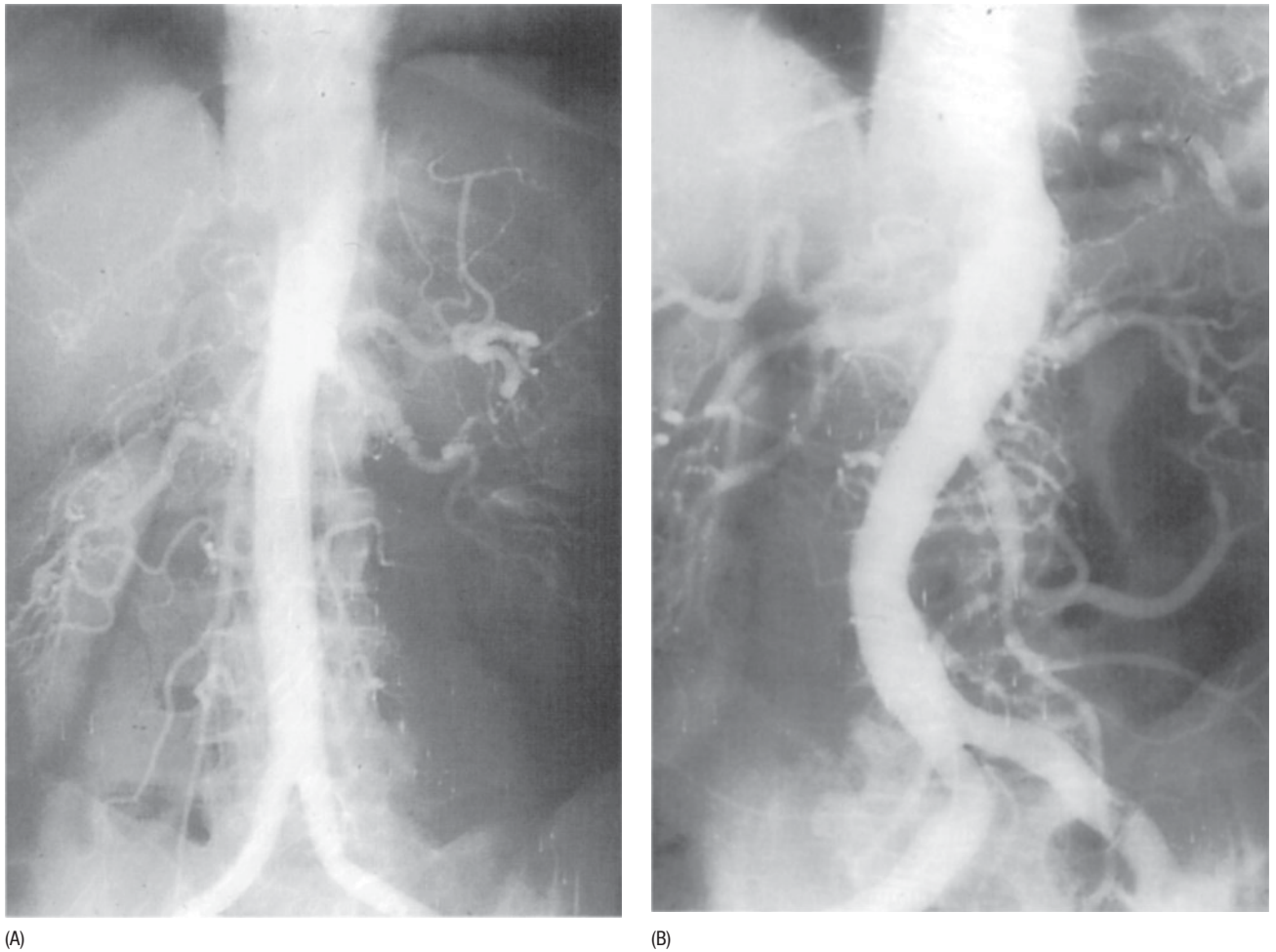


Figure 19.10 Typical examples of change in diameter and length of systemic arteries with age in humans: (A) young subject; (B) old subject

aortic cross-sectional area (Fig. 19.11A,B, page 422) show that these are not a strict senile phenomenon but, like the accompanying functional changes, are progressive throughout life (Table 19.2, page 423) (Mitchell and Schwartz, 1965; Nichols *et al.*, 1985a; Roman *et al.*, 1989; Ingelsson *et al.*, 2008).

With aging, there is progressive increase in carotid intima-media thickness, which is of similar degree in males and females (Fig. 19.11C, page 422). The autopsy studies we have undertaken in the United States, China, and Australia show that this is largely due to intimal hyperplasia; dilation of the aorta and elastic arteries (Fig. 19.11D, page 422) is associated with thickening of the intimal layer but no change in the load-bearing medial layer (Fig. 19.11E, page 422) (Virmani *et al.*, 1991). Individual load-bearing elastin fibers within the media are thinned (Fig. 19.8, right) (O'Rourke *et al.*, 1987).

Chemical studies of arterial aging change provide little insight into the extent or cause of aging change. Certainly, in the aorta there is an age-related increase in collagen and glycosaminoglycans, and a decrease in smooth muscle

and 'normal' elastin as well (Faber and Moller-Hou, 1952; Ahmed, 1967). Elastin, although present, is in a different structural form, and in a form that is not well suited to bear stresses applied to the arterial wall. Anatomic changes and their explanation are based on the earlier work of Burton and Roach (Burton, 1954; Roach and Burton, 1957; Roach and Burton, 1959; Roach, 1977a; Roach, 1977b; Roach, 1983) and the concept of medial lamellar units proposed by Wolinsky and Glagov (Wolinsky and Glagov, 1964; Wolinsky and Glagov, 1967; Wolinsky and Glagov, 1969) (see also Glagov (1979), Glagov *et al.* (1992) and the excellent review by Masoro (1981)). Clark and Glagov (1985) have modified this concept with a description of the media in terms of musculoelastic fascicles. All these studies have shown that aging changes do not simply develop in the elderly but are progressive throughout life and are well developed by early adulthood (see Figs 19.1 and 19.11, 19.12 (page 423), 19.13 (page 424), 19.14 (page 425), 19.15 (page 426), 19.16 (page 426)). One expects, then, for their effects to be apparent in data taken from most human adults, not just the elderly.

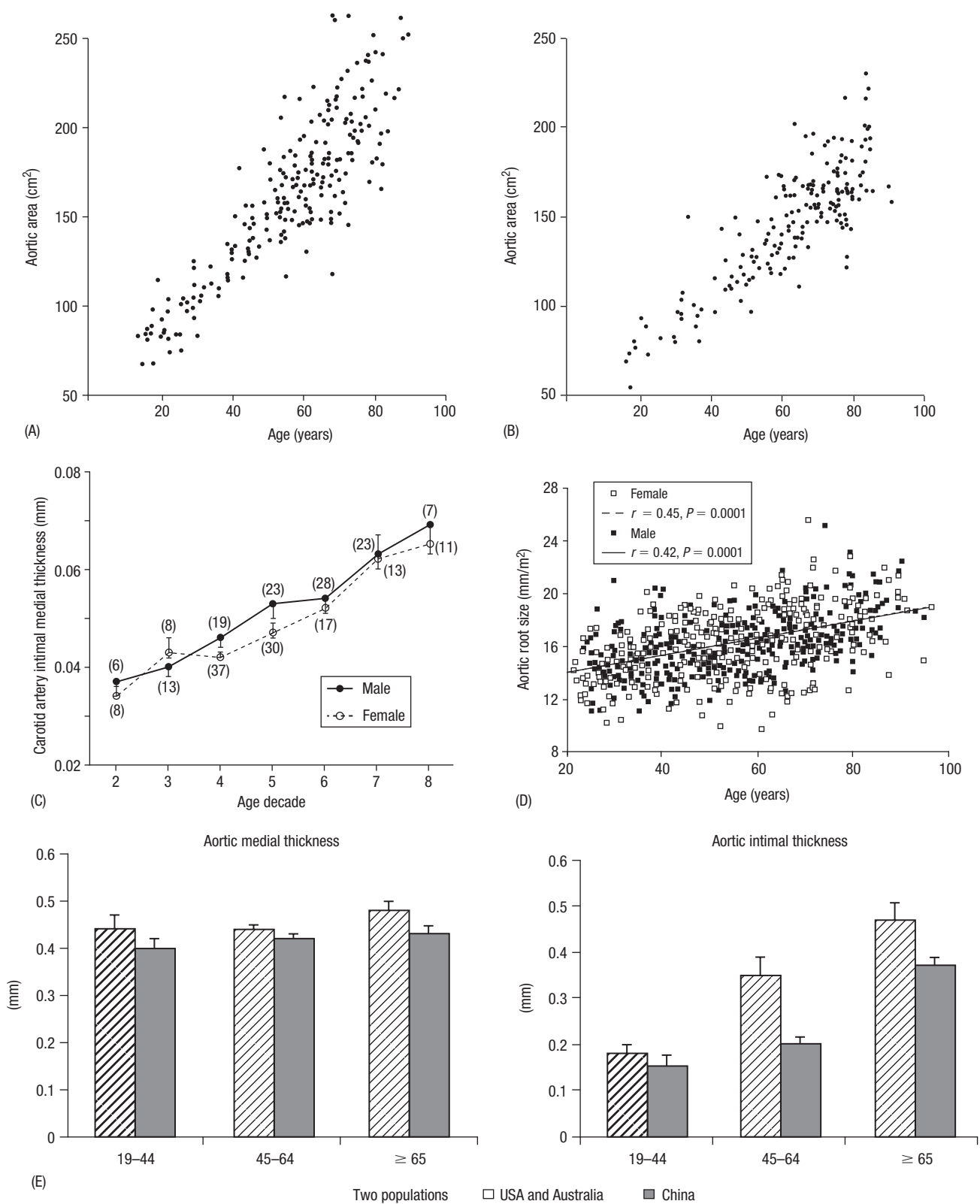


Figure 19.11 Increase in size of the thoracic aorta with age (A) in men and (B) in women, from the unselected autopsy study of Mitchell and Schwartz (1965). (C) Change in intima-media thickness of the carotid artery, measured by ultrasound, as a function of age in an apparently normal cohort of men and women. Courtesy of E Lakatta, US National Institute of Aging, NIH. (D) Change in aortic root diameter, measured by ultrasound and expressed in terms of body surface area, with age. The regression line for males and females was identical. Courtesy of E Lakatta, US National Institute of Aging, NIH. (E) Difference in aortic medial (left) and intimal (right) thickness with age in an autopsy series of people without known cardiovascular disease. From data of Virmani *et al.* (1991)

Table 19.2 Effects of age on arterial hemodynamics

Variable	Slope	Intercept	Correlation coefficient	<i>T</i>	<i>P</i> value
Aortic radius (cm)	0.0097	0.98	0.61	5.08	0.001
Heart rate (beats/min)	0.0486	73.64	0.05	0.30	NS
Systolic pressure (mmHg)	0.6852	95.00	0.63	5.36	0.001
Diastolic pressure (mmHg)	0.0702	76.12	0.11	0.73	NS
Mean pressure (mmHg)	0.1911	90.27	0.33	2.26	0.01
Pulse pressure (mmHg)	0.6150	18.89	0.70	6.38	0.001
Cardiac output (L/min)	−0.0365	7.85	−0.37	−2.63	0.005
Stroke volume (mL)	−0.5605	109.30	−0.30	−2.076	0.025
Peak flow (mL/s)	−3.2638	777.92	−0.31	−2.17	0.025
Characteristic impedance (dyne-s/cm ⁵)	1.2135	10.3	0.66	5.71	0.001
Peripheral resistance (dyne-s/cm ⁵)	9.7409	853.97	0.47	3.45	0.001
Frequency of $ Z _{\min}$ (Hz)	0.0364	2.41	0.65	5.59	0.001

$|Z|_{\min}$ = first minimum of impedance moduli; NS = not significant.
From Nichols *et al.* (1985a).

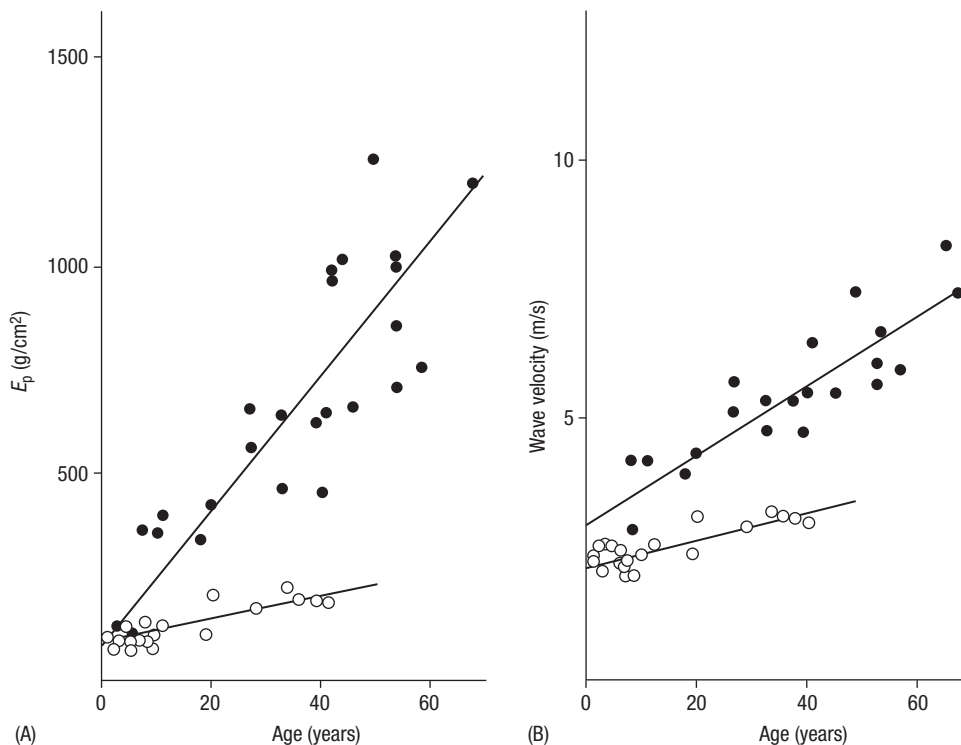


Figure 19.12 Increase in incremental elastic modulus E_p (A) and calculated pulse wave velocity (B) with age in the proximal thoracic aorta (●) and proximal pulmonary artery (○) of human subjects. From data published by Gozna *et al.* (1974)

Any hypotheses for the mechanism of arterial degeneration with age will have to embrace the above findings of progressive change with aging, severity related to absolute age rather than lifespan, and severity related to degree of mechanical stress (Ricklefs and Finch, 1995; Olshansky *et al.*, 2001; Olshansky *et al.*, 2002; Laurent and Boutouyrie, 2007). A logical explanation arises from the application of mechanical engineering principles of material fatigue to the non-living, extracellular components of the arterial wall (see Chapter 4).

FUNCTIONAL CHANGE IN ARTERIES

With increasing age, elastic arteries progressively stiffen as well as dilate (Fig. 19.12). Such a change has been noted in rats (Cox, 1978; Benetos *et al.*, 1993a), in mice (Reddy *et al.*, 2003), and in beagle dogs (Bullman *et al.*, 1981; Yin *et al.*, 1983a), but is most obvious in humans (Figs 19.11 and 19.12). CS Roy in 1880 first studied changes in the arterial elasticity of human and animal arteries with age and made the perceptive

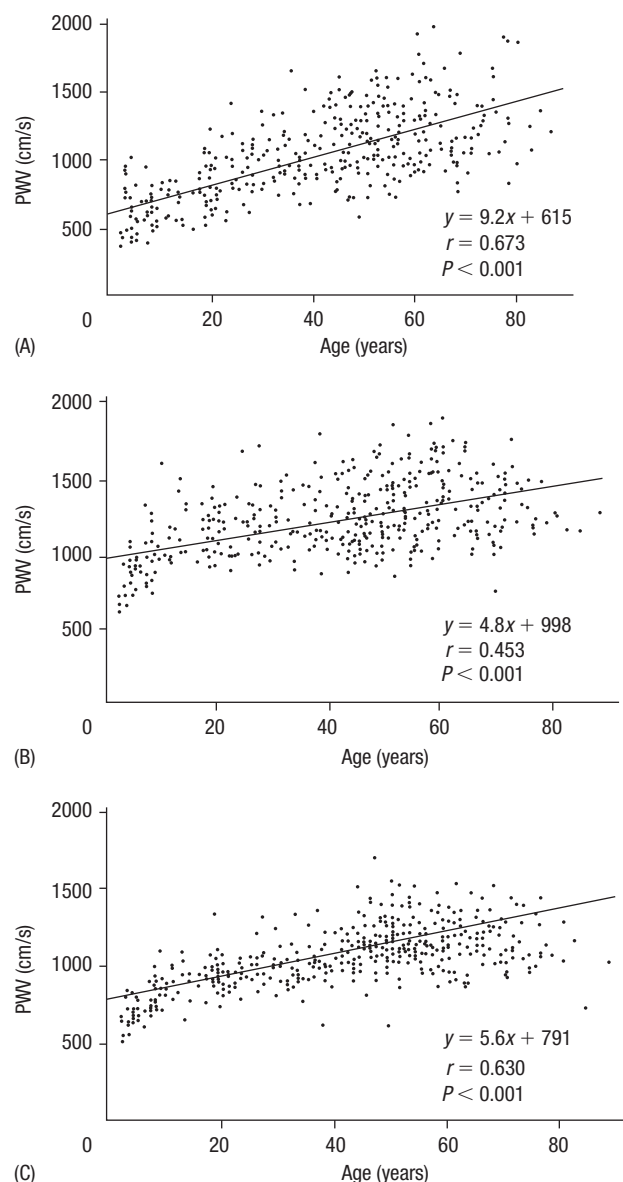


Figure 19.13 Changes in arterial pulse wave velocity (PWV) with age in a group of 480 human subjects with low prevalence of atherosclerosis in urban Beijing. (A) Wave velocity in the aorta and iliac arteries between the aortic root and the femoral artery at the groin. (B) Wave velocity between the brachial artery in the upper arm and the radial artery at the wrist. (C) Wave velocity below the femoral artery in groin and the dorsalis pedis or posterior tibial artery at the ankle. Reproduced from Avolio *et al.* (1983)

observation stated at the beginning of this chapter. Bramwell and Hill (Bramwell and Hill 1922a; Bramwell and Hill, 1922b) determined change in arterial elasticity from PWV in humans at different ages and stressed the functional effects of arterial stiffening (see Chapter 4), as did Mackenzie in 1902, and now well documented (Ioannou *et al.*, 2009).

The views of Bramwell and Hill are at variance with findings of Sunagawa *et al.* (1985b), which appeared to

show that arterial stiffening has no significant effect on cardiac performance and that resistive load is utterly dominant. This work, however, was subject to technical problems and is out of line with previous and subsequent investigations (O'Rourke, 1967b; Urshel *et al.*, 1968a; Elzinga and Westerhof, 1973; Ishide *et al.*, 1980; Yin *et al.*, 1981; Pien, 1986; Morita *et al.*, 1991; Watanabe *et al.*, 1992; Maruyama *et al.*, 1993; Ohtsuka *et al.*, 1994; Chen *et al.*, 1998; Chen *et al.*, 1999; Chen *et al.*, 2001; Cheng *et al.*, 2010; Scalia *et al.*, 2010). The Bramwell and Hill concept has indeed gained general support from most subsequent investigations.

The findings of Bramwell and Hill on arterial stiffening with age have been confirmed subsequently for human subjects in Western populations (Schimmler, 1965; Learoyd and Taylor, 1966; Gozna *et al.*, 1974; Schimmler, 1975; Ho, 1982; Langewouters *et al.*, 1984; Länne *et al.*, 1992a; Länne *et al.*, 1992b; Sonesson *et al.*, 1993; Pearson *et al.*, 1994) and in Asian, African, European, and US populations (Avolio *et al.*, 1983; Avolio *et al.*, 1985a; Mitchell *et al.*, 2004a; McEniery *et al.*, 2005a; Shiburi *et al.*, 2006; Mitchell *et al.*, 2007b; McEniery *et al.*, 2008; Vermeersch *et al.*, 2008; Lieb *et al.*, 2009; Redheuil *et al.*, 2010; Taviani *et al.*, 2010) (Figs 9.13, 19.14, and 19.17 (page 427)), using determinations of PWV *in vivo* or the extensibility of arterial segments *in vitro* and recently *in vivo* (see Chapter 4). The similarity of findings in Beijing, where atherosclerosis is uncommon, to those in Western societies, where atherosclerosis is common, add weight to the concept that aortic stiffening is a true aging change and not due to concomitant atherosclerosis (Figs 19.13 and 19.14) (Haynes *et al.*, 1936; Gerstenblith *et al.*, 1977; O'Rourke *et al.*, 1986a). In Western groups, arterial stiffness measured as PWV is generally similar in subjects with and without clinically apparent atherosclerosis (Megnien *et al.*, 1998) and even in many people with hypercholesterolemia and diabetes mellitus. Contrary findings have been reported (see Chapter 25) (Hirai *et al.*, 1989; Stefanadis *et al.*, 1990). It appears that atherosclerosis is often not sufficiently widespread to significantly alter arterial distensibility, and that age and arterial pressure are the major determinants of this. Recent reports, however, confirm the older studies by Remington *et al.* (1948) and the teachings of Osler (1892) that at any one age there is a wide range of arterial distensibility in humans. People with diabetes, atherosclerosis, renal insufficiency, etc., are usually within this range, although usually in the upper part – either as a consequence of the disease or through predisposition as in the EVA (early vascular aging) syndrome (see Chapters 24 and 25) (Nilsson *et al.*, 2008).

Our initial studies of aortic and brachial PWV showed no significant difference between males and females. Subsequent studies in Framingham (Mitchell *et al.*, 2004a), Baltimore (Fig. 19.14C), and Europe (McEniery *et al.*, 2005a; Vermeersch *et al.*, 2008), likewise show little difference between males and females, and the

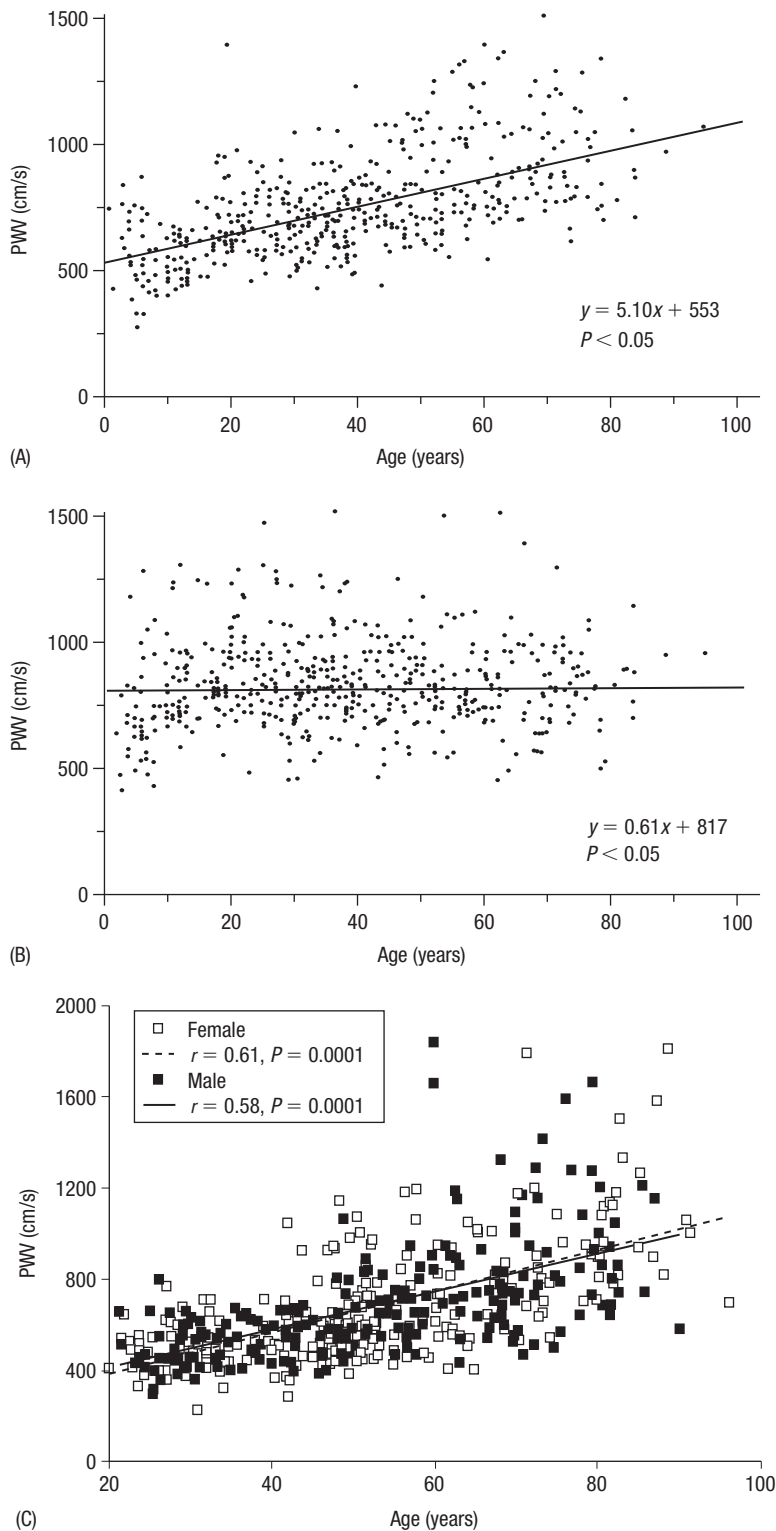


Figure 19.14 Relationship between aortic (A) and arm (B) pulse wave velocity (PWV), measured non-invasively, with age in the rural southern Chinese community of Guangzhou. Reproduced from Avolio *et al.* (1985a). (C) Change in aortic pulse wave velocity, measured non-invasively in normal males and females, as a function of age. Courtesy of E Lakatta, US National Institute of Aging, NIH

same progressive increase with aging, without any change at the time of menopause. Again, contrary reports exist.

An increase in stiffness with age is virtually confined to the aorta and central, predominantly elastic arteries (Boutouyrie *et al.*, 1992; Länne *et al.*, 1992a; Länne *et al.*, 1992b; Benetos *et al.*, 1993b; Sonesson *et al.*, 1993;

Redheuil *et al.*, 2010). There is little or no consistent change in distensibility of the muscular arteries, e.g. femoral, radial, or brachial, with age, in contrast to changes in central elastic arteries (Figs 4.24, 19.14B, 19.15 (page 426), and 19.18A (page 428)) (Ho, 1982; Avolio *et al.*, 1983; Avolio *et al.*, 1985a; Boutouyrie *et al.*, 1992;

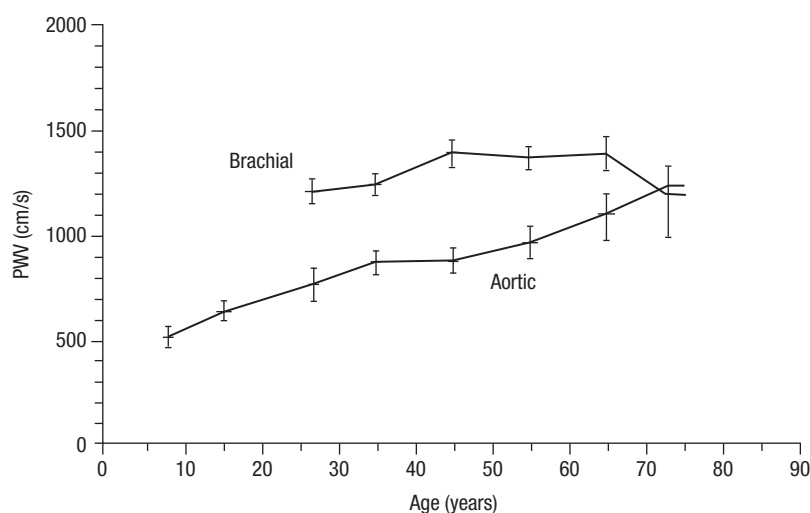


Figure 19.15 Age-related changes in aortic and brachial artery pulse wave velocity (PWV) in an Australian population. Aortic PWV increases with age, whereas brachial artery PWV does not. Reproduced from Ho (1982)

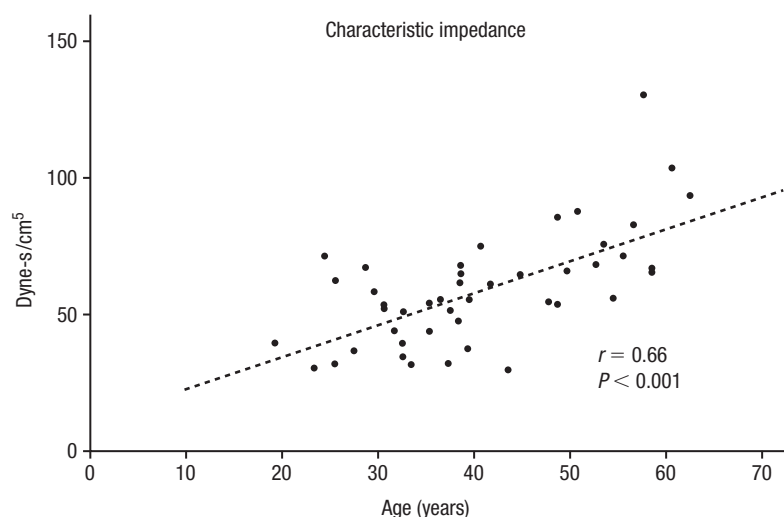


Figure 19.16 Relationship between ascending aortic characteristic impedance (in volumetric terms – dyne-s/cm⁵) and age. Reproduced from Nichols *et al.* (1985a)

Benetos *et al.*, 1993b; McEniery *et al.*, 2008; Nichols and O'Rourke, 2009).

Measures of total systemic vascular compliance estimated as the ratio of stroke volume (SV) and brachial pulse pressure (PP) (Fig. 19.18B, page 428) show considerable scatter and just a modest decrease with aging. This type of indirect measure is subject to a number of problems, and is not favored for studies of arterial stiffness (O'Rourke *et al.*, 2002a; Laurent *et al.*, 2006).

Learoyd and Taylor (1966), Cox (1978), and Yin *et al.* (1983a; 1983b) found no consistent difference in arterial wall viscosity with age in humans, rats, and dogs, respectively.

Findings of increased stiffness with age in the central elastic arteries but not the peripheral muscular arteries of humans are consistent with the theory previously proposed (O'Rourke, 1976) that fracture of elastic load-bearing elements of the arterial wall causes the wall to weaken and to stretch, so that stress is transferred to collagenous elements in the wall, just as when an artery is acutely distended by a greater pressure within (see Chapter 4) (Burton, 1954; Roach and Burton, 1957).

PULSE WAVE VELOCITY

As stated above, in the human systemic arterial circulation, PWV shows a greater increase with age in the aorta than in peripheral arteries. The increase with age is progressive from childhood and occurs independently of the increase in mean arterial pressure (Avolio *et al.*, 1983). The majority of recent studies show a curvilinear relationship of carotid-femoral PWV with age, whereas earlier studies showed a linear increase (Mitchell *et al.*, 2004a; McEniery *et al.*, 2005a; Najjar *et al.*, 2005; Shiburi *et al.*, 2006; Mitchell *et al.*, 2007b; McEniery *et al.*, 2008). It is, however, greater in populations with a greater salt intake and a relatively high prevalence of hypertension than in a population with a low salt intake and a relatively low prevalence of hypertension (see Figs 19.13 and 19.14) (Avolio *et al.*, 1983; Avolio *et al.*, 1985a; Safar *et al.*, 2009). Greater change in the aorta than in peripheral vascular beds is consistent with greater disorganization of the arterial media in central elastic arteries, caused by fatigue and fracture of elastin fibers as a consequence of greater

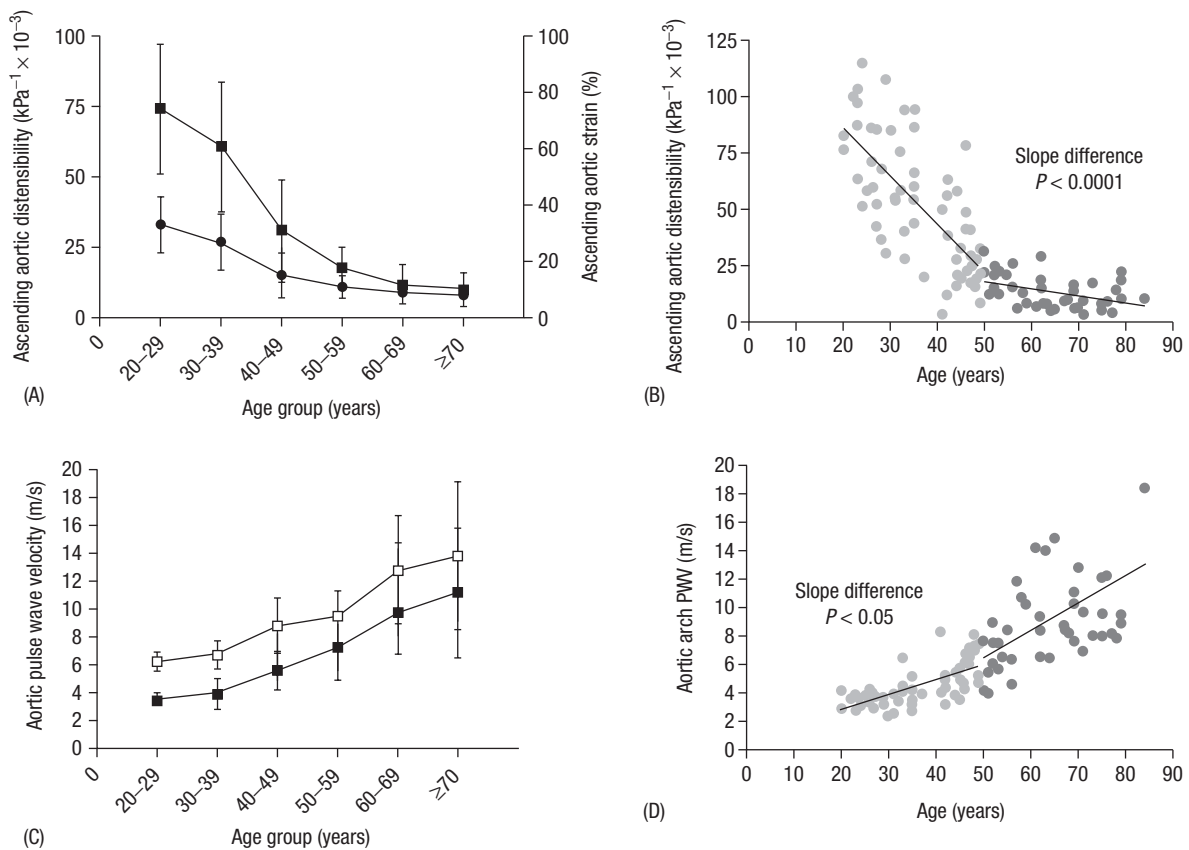


Figure 19.17 Age-related changes in central aortic function, measured by magnetic resonance imaging (MRI). (A) Ascending aortic strain (percentage change in diameter – black circles, right axis) and ascending aortic distensibility (black squares, left axis) plotted against age. (B) Ascending aortic distensibility (individual patient data) plotted against age from patients aged from 20 to 84 years. Separate linear regression lines are given for 20–50 years and over 50 years. (C) Aortic arch pulse wave velocity (PWV) (black squares) and carotid femoral PWV (open squares), measured by tonometry, plotted against age. (D) Aortic arch PWV with individual data points plotted against age and regression lines 20–50 years and over 50 years. Reproduced from Redheuil *et al.* (2010)

pulsatile strain (see below). In arteries of the upper limb, there is little increase in PWV with age (Figs 19.13B, 19.14B, and 19.15) (Hallock, 1934; Landowne, 1958; Ho, 1982; Avolio *et al.*, 1983; Avolio *et al.*, 1985a). This fact justifies the use of a simple generalized transfer function to calculate the central aortic pressure wave from the radial artery pressure wave in fully grown adult humans of all ages (see Chapter 30). The minor increase in PWV with age in the upper limb contrasts with the marked increase in the aorta, where PWV more than doubles between the second and eighth decade of life. This change of over 100 percent during a lifetime is greater than any other measured change in a vascular parameter with aging, except for aortic characteristic impedance with which it is closely correlated (Chapter 3) (see Fig. 19.16). However, its effect on the low-frequency components of the ascending aortic impedance modulus is likely to be far greater (Nichols *et al.*, 1993), since these components are increased directly by increased aortic stiffness and indirectly by an increase in PWV through early return of wave reflection (see Chapter 9). A two-fold increase in aortic PMV translates to a four-fold increase in impedance

modulus at frequencies of greatest interest (see Fig. 20.4 and Chapter 12).

Studies of PWV with age in humans show that, in infancy and adolescence, central elastic arteries are much more distensible than peripheral arteries. Human systemic arteries exhibit the same type of elastic non-uniformity previously demonstrated in experimental animals and shown to be highly desirable for optimal arterial function and ventricular/vascular interaction (Taylor, 1964; Taylor, 1994). Such elastic non-uniformity is lost as the aorta progressively stiffens, with values of aortic PWV coming to approach and even often exceed those of the peripheral arteries in people over the age of 60 years. These data are not completely conclusive, however, since wave velocity was not measured in the most proximal part of the aorta. Latham's invasive measurements (Latham *et al.*, 1985) showed that PWV does increase progressively along the aorta, with the value in the ascending aorta of adult humans at mean age 44 years (4.4 m/s) being about the same as that in the corresponding segment of a dog (see Fig. 4.22). These data have been recently confirmed by Taviani *et al.* (2010) using non-invasive techniques.

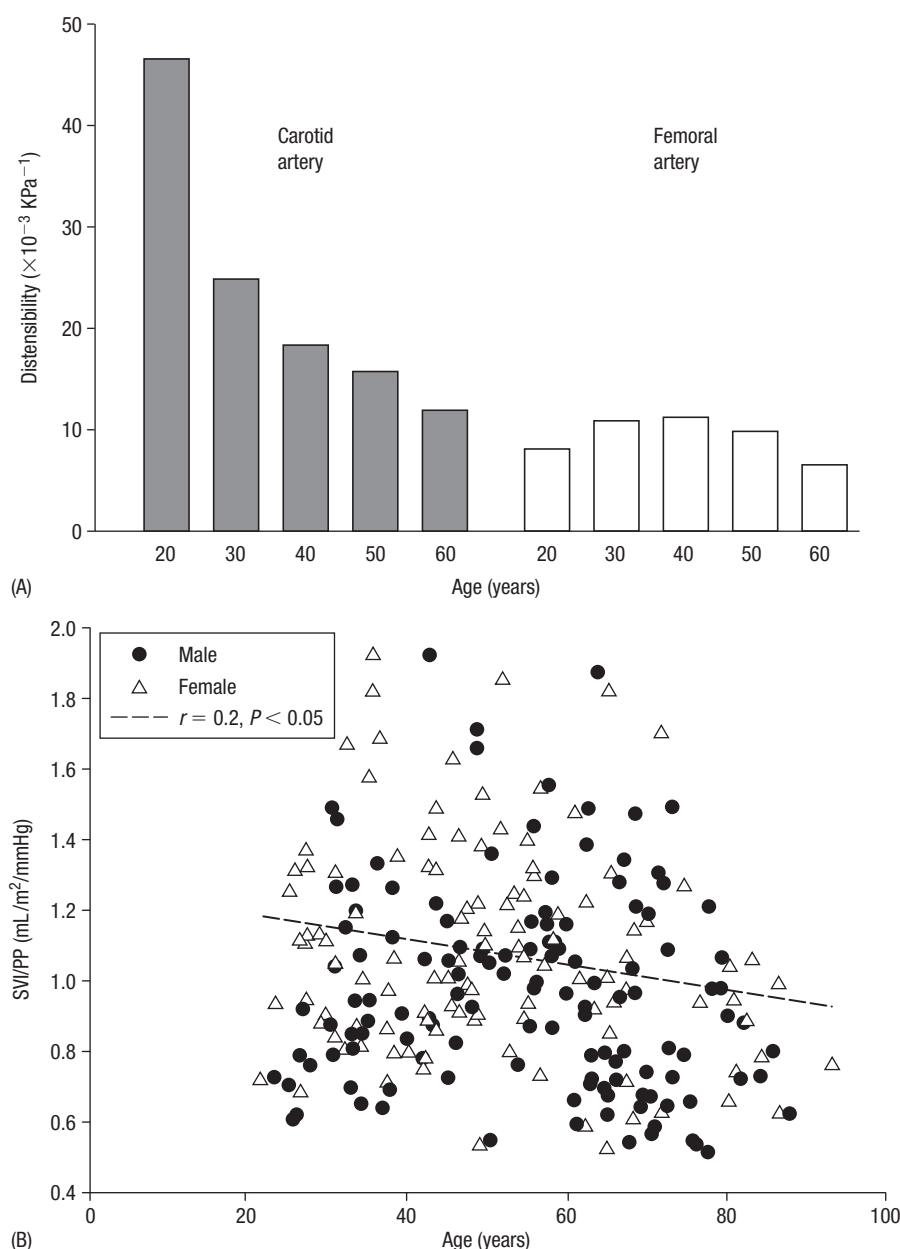


Figure 19.18 (A) Relationship between distensibility and age for a central, predominantly elastic artery (■), and for a peripheral, predominantly muscular artery (□). Reproduced from Benetos *et al.* (1993b). (B) Change in total systemic vascular compliance with age, expressed in relation to body surface area. Courtesy of E Lakatta, US National Institute of Aging, NIH

Merillon *et al.* (Merillon *et al.*, 1982a; Merillon *et al.*, 1982b) calculated PWV from pressure–radius changes in the ascending aorta and showed that this increased in normal subjects by an even greater amount than determined in our non-invasive studies, from about 4 m/s at age 25 to about 8 m/s at age 50 and about 10 m/s in the late sixties, an increase of some 150 percent over 40 years. Recent MRI studies on proximal aortic PWV and aortic stiffness support these larger changes in normal subjects with age (Fig. 19.17) (Redheuil *et al.*, 2010). In beagle dogs, Bulliman (1980) found an increase in the central aortic PWV of 40 percent over ten years. Differences in the lifespans of humans and dogs are quite different, but

the change, when expressed in terms of years, amounts to an increase of around 4.0 percent per year or around 1 m/s per decade. Similar values of 1 m/s per decade have been reported by Alecu *et al.* (2008). Values a bit less than this were reported by Sugawara *et al.* (2010b) and Taviani *et al.* (2010). Values around half of this were found by Khoshdel *et al.* (2006) in a meta-analysis of studies which used the COMPLIOR device. In the brachiocephalic artery of dogs, Bulliman (1980) found an increase in the PWV of approximately 24 percent over ten years. It is our contention that the increase in aortic wave velocity with age in humans is due to the degeneration of elastic fibers and is the major reason for differences in impedance

patterns and in pressure and flow wave contours between adult humans and experimental animals. Difference in the degree of dilatation with age, with the predominantly elastic thoracic aorta diameter increasing to a greater degree than the collagenous abdominal aorta (Langewouters, 1982) and the further decrease in area ratio at the aorta–iliac junctions (Gosling *et al.*, 1971; Greenwald *et al.*, 1990) (Fig. 9.10), may partly explain the apparent shorter distance to reflection sites in the lower body of adult human females compared to males (Latham *et al.*, 1985) as due to change in phase of reflection. This may explain in part the greater propensity for atherosclerosis in the infrarenal aorta than in the proximal aorta, on the basis of greater fluctuation in shearing stress at the aortic wall in such a region (see Chapters 4 and 23). Mitchell *et al.* (2004a) have argued against this view, postulating that a ‘lumped’ peripheral reflected site moves further distally with age, thus, reduces wave reflection amplitude. We have serious difficulties with Mitchell’s analysis (O’Rourke and Nichols, 2004b). These concerns are enhanced by recent publications (McEniery *et al.*, 2005a; O’Rourke and Adji, 2010b) (see also Chapter 9).

Changes in arterial distensibility with age have frequently been assessed from measurements of PWV in the arms and legs. It is clear that these underestimate changes in aortic properties. Smulyan *et al.* (1983) considered that apparent arterial stiffening with age is due entirely to a concomitant increase in arterial pressure. These data were determined from measurement of PWV in the upper limb. While this is probably true for the brachial artery, such findings cannot be extrapolated to central elastic arteries, nor to the arterial system as a whole. Other measures of arterial stiffness in humans have confirmed the progressive increase with age in central, predominantly elastic arteries, with little or no change in peripheral, predominantly muscular arteries (see above and Chapter 4).

Gender

In the largest population study on arterial stiffness that provided reference and normal values for aortic PWV based on a European population ($n = 11\,092$), gender differences were negligible (although still significant because of the size of the sample) (The Reference Values for Arterial Stiffness’ Collaboration, 2010). The difference was <0.1 m/s (higher aortic PWV in males), which is small when the precision of the technique is taken into account. In other large recent studies, aortic PWV was found to be similar in adult males and females (McEniery *et al.*, 2005a; Vermeersch *et al.*, 2008) (see Fig. 9.18). In accordance with these studies, recent studies on children and adolescents showed no significant differences in aortic PWV values between genders (Ayer *et al.*, 2010; Reusz *et al.*, 2010). Furthermore, Jarauta *et al.* (2010) found no age-related difference in carotid intima-media thickness in males and females (see Fig. 4.2). On the other hand,

studies have repeatedly demonstrated gender differences in central hemodynamics (Hayward *et al.*, 1997; Smulyan *et al.*, 2001; Mitchell *et al.*, 2004a; McEniery *et al.* 2005a; Cecelja *et al.*, 2009; Janner *et al.*, 2010; Lieber *et al.*, 2010). Augmented pressure and augmentation index are systematically higher in women. Aortic augmented pressure increases linearly in both sexes, while augmentation index tends to a plateau value above the age of 60 years (Fig. 9.18) (Kelly *et al.*, 1989a; McEniery *et al.*, 2005a; Avolio *et al.*, 2009b). Differences in wave reflection indices are largely attributed to differences in height between the two sexes and not to differences in pulse transit time. Due to shorter body stature, the reflecting site in women is closer to the central circulation than it is in men. Thus, with aortic PWV being equal, the pressure wave, reflected from a closer reflection site in the arterial tree, returns towards the proximal aorta earlier and tends to amplify the primary wave in systole rather than in diastole. However, wave reflection indices are still elevated in women even after adjustment for height (McEniery *et al.*, 2005a; Ayer *et al.*, 2010; Lieber *et al.*, 2010) implying contribution of differences in the magnitude of the reflected wave. The latter is dependent on vasomotor tone, which in turn is influenced by additional morphometric and physiologic factors including sex hormones (Smulyan *et al.*, 2001; Thompson *et al.*, 2003) and aortic tapering (Cecelja *et al.*, 2009). Tendency towards higher wave reflection indices in women due to the above-mentioned factors may be partly compensated by a faster heart rate in women that would make the reflected wave return closer to diastole. Indeed, significance of difference in such indices increases after adjustment for heart rate (Lieber *et al.*, 2010).

Race/ethnicity

Generally, aortic PWV has been shown to be higher in African American men compared with white men, despite comparable mean arterial pressure (Ferreira *et al.*, 1999; Chaturvedi *et al.*, 2004; Din-Dzietham *et al.*, 2004; Bhuiyan *et al.*, 2006; Heffernan *et al.*, 2008; Lewis *et al.*, 2010) PWV overall, but subgroup analysis indicated that the effect was significant only in Blacks despite the fact that Blacks achieved a smaller reduction in salt intake and a similar fall in blood pressure, compared with Whites and Asians (He *et al.*, 2009). Given the overall predictive role of aortic PWV for cardiovascular events, as well as that for developing hypertension, these findings are in accordance with the fact that the incidence of hypertension in adults is more common in non-Hispanic African-Americans, who also are more likely to develop hypertensive complications, particularly renal disease. Furthermore, these findings imply that reduction of arterial stiffness is a particularly appealing target in black hypertensive patients. Results regarding other ethnicities are not consistent. In our early studies involving large numbers of

subjects, Chinese people generally had higher aortic PWV values than Caucasians of similar age who were recruited from different localities around the globe (Fig. 19.13) (Avolio *et al.*, 1983). In other, smaller studies, white European men have similar aortic PWV values with Asian men (Tillin *et al.*, 2007). Regarding central hemodynamics, aortic augmentation index, as well as central systolic and pulse blood pressure, are shown to be higher in African-American men compared with white men (Heffernan *et al.*, 2008). On the contrary, white European men have no significant differences in aortic augmentation index with Asian men (Sugawara *et al.*, 2010c).

ASCENDING AORTIC IMPEDANCE

Studies of aging in beagle dogs have failed to show any difference in ascending aortic impedance over the normal lifespan (Bullman, 1980; Yin *et al.*, 1981; Yin, 1986). The patterns of impedance are surprisingly similar under resting conditions in young and senescent beagle dogs aged between 2.0 and 12 years. However, Reddy *et al.* (2003) found an age-related increase in aortic stiffness in mice which resulted in increased aortic input impedance, characteristic impedance, pulse pressure, and augmentation index.

Murgo *et al.* (Murgo *et al.*, 1980a; Murgo *et al.* 1980b), Merillon *et al.* (Merillon *et al.*, 1982b; Merillon *et al.* 1982c), and Nichols *et al.* (Nichols *et al.*, 1985a; Nichols *et al.* 1993) have shown a substantial change in human ascending aortic impedance with age, but over a range of 40 years. These changes include an increase in characteristic impedance, a shift of the curves to the right, with an increase in the frequency of the modulus minimum and phase crossover, and an increase in fluctuation of the impedance modulus (Fig. 19.19; Table 19.2). The combination of increased characteristic impedance and shift to higher frequencies caused very substantial increase of impedance modulus at low frequencies (i.e. up to 4.0 Hz).

The change in characteristic impedance in Nichols' data was surprisingly high (137 percent over 40 years though lower when expressed in terms of linear velocity, dyne-s/cm^3), while in the Merillon data this was approximately 81 percent and more in keeping with the increase in frequency of the impedance minimum (by approximately 57 percent, from 2.8 to 4.4 Hz over this period), and in our non-invasively determined PWV, which increased by some 60 percent over 40 years. As discussed, however (see above), Merillon *et al.* had found evidence of a substantial (150 percent) increase in the ascending aortic PWV over 40 years. These data suggest that there is, with age, a greater degree of degeneration in the proximal ascending aorta than in the arch and descending aorta. Further data are needed on this subject, especially from the extremes of age – in children and in the elderly; few are presently available (Redheuil *et al.*, 2010; Taviani *et al.*, 2010). Indirect evidence has been

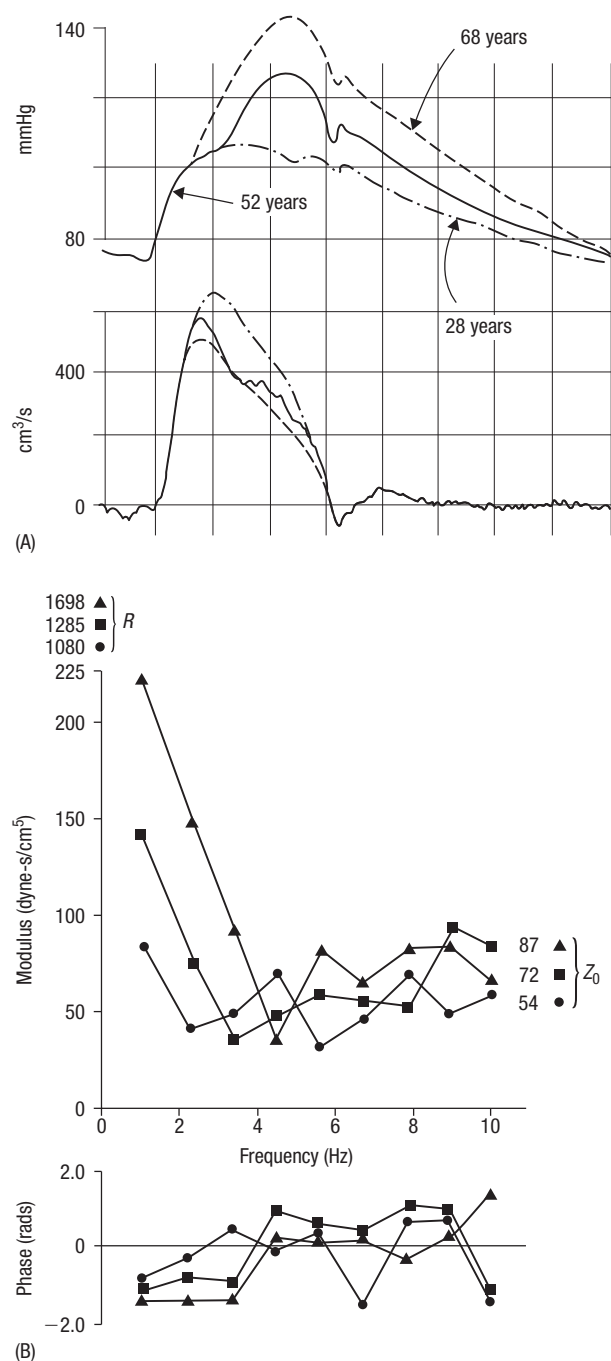


Figure 19.19 (A) Typical pressure and flow waves recorded in three normotensive subjects aged 28, 52, and 68 years. (B) Ascending aortic impedance spectra calculated from the above pressure and flow waves. R represents the peripheral resistance and Z_0 the characteristic impedance. (●-●) 28 years; (■-■) 52 years; (▲-▲) 68 years. Impedance modulus is expressed in volume flow terms (mL/s). When expressed in velocity flow (cm/s), assuming differences in aortic cross-sectional area from the same paper, values of characteristic impedance were significantly higher for the youngest to oldest group. Reproduced from Nichols *et al.* (1993)

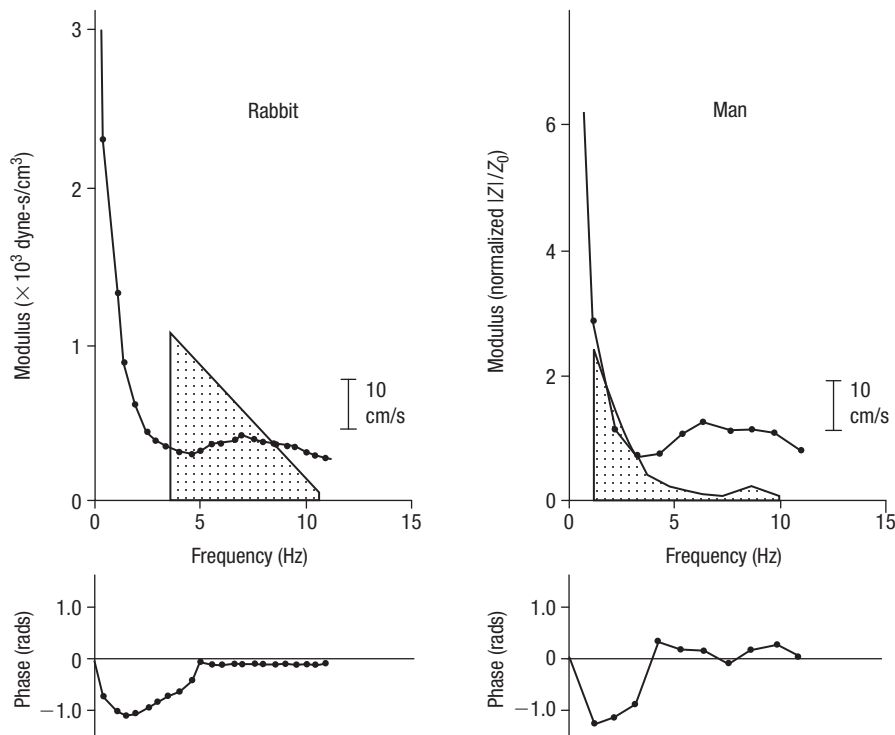


Figure 19.20 Ascending aortic impedance in a series of rabbits, from Avolio *et al.* (1976) and in a group of relatively young, apparently normal, human adult subjects reported by Murgo *et al.* (1980a), together with (inset) amplitude of the first 2–9 harmonics of a typical ascending aortic flow wave in each case. Despite an enormous difference in size, the impedance curves show similar features, indicating similar timing of wave reflection from peripheral sites. Impedance ‘match’ in the rabbit is not apparent in humans

available for some time, however, that the ascending aortic impedance in humans is markedly altered by aging change. Thus, despite enormous differences in body size, impedance patterns in the ascending aorta of adult humans are surprisingly similar to those in rabbits (Fig. 19.20). Such findings suggest that aging may cause a greater change in the PWV than is apparent from measurements taken to date and that the aging change in humans causes the reflected wave to arrive early at the heart from the lower body (as discussed previously and in Chapter 9).

Mills *et al.* (1970) drew attention to differences in the patterns of the impedance modulus in the ascending aorta of dogs and humans, with adult humans showing more fluctuation and less evidence of two peripheral reflecting sites at different distances from the heart. Murgo *et al.* (1980a) noted that these patterns were age dependent, with less fluctuation in the young and more in older subjects; this was confirmed by Nichols *et al.* (1985a). O’Rourke and Avolio (1980) suggested that the change with age in humans was due to a greater degree of degeneration in the descending aorta and arteries supplying the lower than upper body arteries with age, with relatively faster wave travel to normally placed peripheral reflecting sites in the lower body. This view was supported by Burattini *et al.* (1991). Using a multisensor pressure catheter technique, Latham *et al.* (1985) suggested another factor – that in adult humans an anatomically discrete

reflecting site is present in the abdominal aorta below the renal arteries where the aorta shows an abrupt anatomical taper. In an autopsy series, Langewouters (1982) confirmed that such a taper exists as had Gosling *et al.* (1971) and Greenwald *et al.* (1990) (Fig. 9.11); it presumably develops with aging as a consequence of greater dilatation above, than below, the renal arteries.

The implications of change in ascending aortic impedance with age have been discussed in a series of publications. Yin *et al.* (1981) have shown that, in dogs, differences in ascending aortic impedance do become manifest during exercise and act to impede ventricular ejection and cardiac performance. The effects of aging must be more important in humans and can certainly account for the decreased capacity for ventricular ejection during exercise (Fig. 27.5) (Port *et al.*, 1980; Weisfeldt, 1980; Yin, 1986; Weisfeldt *et al.*, 1992; Folkow and Svanborg, 1993; Lakatta, 1993a; Lakatta, 1993b; Nussbacher *et al.*, 1995; McGuire *et al.*, 2001a; McGuire *et al.*, 2001b). Clearly, considerations of ascending aortic impedance during exercise in humans are relevant to angina pectoris and cardiac failure. These points are discussed in Chapters 13, 14, 15, 17, and 27.

Invasive data on ascending aortic impedance have not been presented for people over the age of 70 years. Non-invasive data are, however, available and show continuing increase in systolic pressure, in pulse pressure (see Fig. 19.1) and in aortic PWV (see Figs 19.13A, 19.14A,

and 19.17C,D) beyond age 70 (Franklin *et al.*, 1997). These suggest truly dramatic increases in ascending aortic impedance in the very old (those over 80 years of age). If one assumes a two-fold increase in ascending aortic characteristic impedance, and a two-fold increase in the minimal value of impedance between age 20 and 80 years, consistent with published values of aortic PWV, values of ascending aortic impedance up to 5.0 Hz are increased at least four-fold (Fig. 19.21) when expressed in velocity terms (dyne-s/cm³). With such ascending aortic impedance change, a given ascending aortic flow wave would generate pressure waves of four times the amplitude found in a 20-year-old subject. Published values of brachial arterial blood pressure change with age show perhaps a two-fold increase in pulse pressure over this age span. When allowance is made for the substantial amplification of pressure waves in adolescence, which is not seen in the elderly (see Figs 19.1, 19.2 and 20.1), pulse pressure changes of this magnitude do indeed occur, and not infrequently in the elderly (see Fig. 30.28). An ascending aortic pulse pressure at rest of around 25 mmHg would be very frequent in adolescents, and of 80 mmHg in octogenarians (Nichols *et al.*, 1985a). Absolute changes in impedance are, of course, partially offset by dilatation of

the ascending aorta with age (Nichols *et al.*, 1985a; Nichols *et al.*, 1993). Segers *et al.* (2007b) and Mitchell *et al.* (2004a) have reported values of ascending aortic impedance, determined non-invasively in the large Asklepios and Framingham studies, respectively. Findings are similar to those found with invasive measurement of ascending aortic pressure and flow waves (see Fig. 12.34). Changes in impedance will be followed longitudinally. So far, only cross-sectional data are available, and in published studies differences of simple indices, such as AIx and PWV, appear to be more consistent than impedance.

VENTRICULAR-VASCULAR INTERACTION

The progressive deterioration in vascular-ventricular interaction with increasing age has been described by Merillon *et al.* (Merillon *et al.* 1982b; Merillon *et al.* 1982c), Nichols *et al.* (Nichols *et al.*, 1985a; Nichols *et al.*, 1987; Nichols *et al.*, 1993), O'Rourke *et al.* (O'Rourke *et al.*, 1986a; O'Rourke *et al.*, 1990), O'Rourke and Kelly (1993), Westerhof and O'Rourke (1995), and is discussed in Chapter 14. This is underestimated from brachial artery systolic pressure measurements. A progressive increase in cardiac load with age is related to and almost certainly responsible for the gradual increase in LV mass with age and in hypertension (Fig. 19.22) and for the progressive impairment in LV function with advancing years (Merillon *et al.*, 1982b; Merillon *et al.*, 1982c; Miyashita *et al.*, 1993; Westerhof and O'Rourke, 1995) (see Chapters 14 and 15). The implications for the development of cardiac failure are dealt with by Weisfeldt *et al.* (1992), Westerhof and O'Rourke (1995), Lakatta *et al.* (1997), and

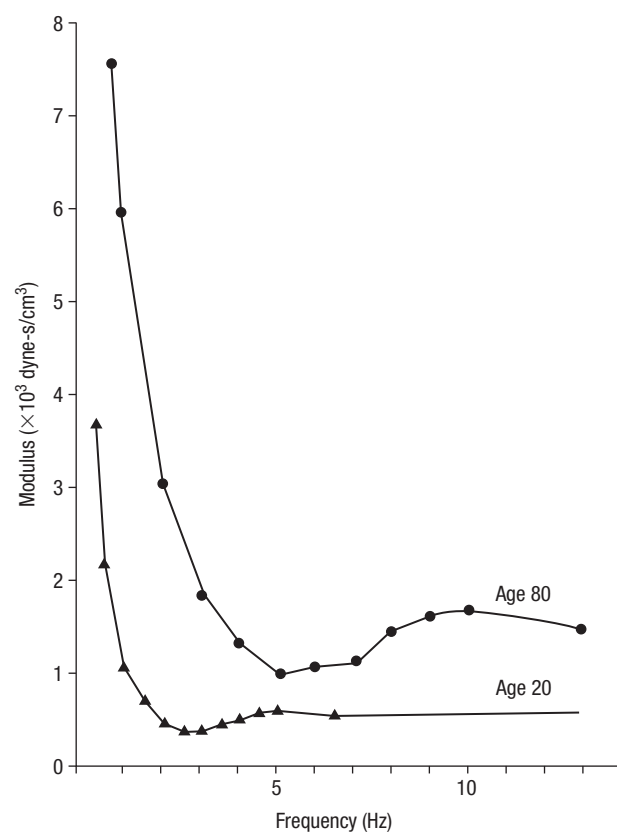


Figure 19.21 Estimated impedance modulus in a typical 20-year-old and a typical 80-year-old human subject, based on two-fold increase in characteristic impedance and halving of the time for wave reflection over this period. For explanation, see text

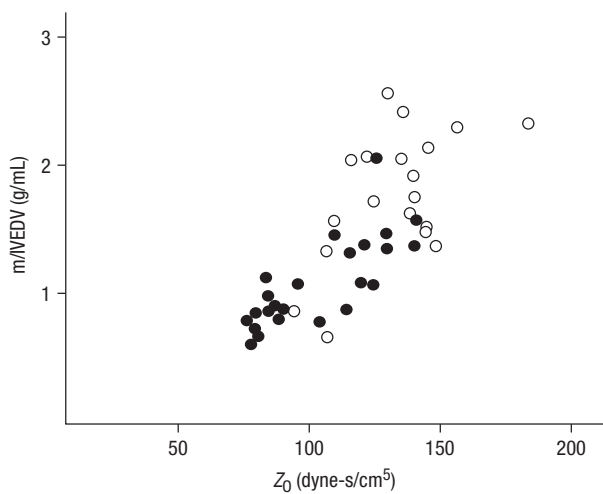


Figure 19.22 Relationship between an index of left ventricular (LV) mass (LV mass divided by LV end diastolic volume) (ordinate) and ascending aortic characteristic impedance (abscissa) in normal subjects (●) and in hypertensive patients (○). Reproduced from Merillon *et al.* (1982a)

Denardo *et al.* (2010); implications for impairment in exercise capacity are discussed by Lakatta (1993a), Folkow and Svanborg (1993), Lakatta and Levy (Lakatta and Levy, 2003a; Lakatta and Levy, 2003b), and in Chapter 27. The American Heart Association has published guidelines on exercise in heart failure (Arena *et al.*, 2007).

Considerations of ascending aortic impedance and its change with age (see Figs 19.19, 19.20, and 19.21) indicate that the increase in LV load – at rest, let alone with exercise – is far greater than ever conceived or seriously considered to date. Figure 19.19 shows that even by age 52, the arterial system of humans has deteriorated to the point of marked inefficiency, where there is marked ‘detuning’ and disparity between the minimal values of ascending aortic impedance and the harmonic components of the LV ejection (flow) wave. Figure 19.21 indicates substantial further progression by age 80, where there is a gross disparity between arterial load and ventricular ejection.

While studies of ascending aortic impedance provide the best information of a quantitative nature on ventricular–vascular interaction, this same information is available from studies of pressure and flow in the time domain. In older people, there is a gross augmentation of ascending aortic and LV pressure in late systole – with values of 20–50 mmHg often noted. Effects of aging change are probably responsible for the early deceleration of aortic flow in late systole of older people, even in the absence of cardiac failure – as first stressed by Nichols *et al.* (1987), Miyashita *et al.* (1993), and Yabe *et al.* (1991). The aging change clearly has an important influence on the development of LV hypertrophy and of LV failure (Katz, 1990; Frohlich *et al.*, 1992a; Frohlich *et al.*, 1992b; Westerhof and O’Rourke, 1995; Vasan and Levy, 2000; Hundley *et al.*, 2001; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b; Denardo *et al.*, 2010) in the elderly, as well as on exercise capacity at a younger age (Fig. 19.22).

Clearly, this subject has been overlooked in the past by those who use a lumped model for characterization of aortic properties (Sunagawa *et al.*, 1985a; Sunagawa *et al.*, 1985b). More recent studies of ventricular–vascular interaction (Morita *et al.*, 1991; Kelly and Kass, 1992a; Kelly and Kass, 1992b; Watanabe *et al.*, 1992; Ohtsuka *et al.*, 1994; Chen *et al.*, 1998; Chen *et al.*, 1999; Chen *et al.*, 2001; Kass *et al.*, 2001) began to address this subject realistically, with results in line with those of Roy (1880), Mackenzie (1902), Bramwell and Hill (Bramwell and Hill, 1922a; Bramwell and Hill 1922b), and more recently researchers (Segers *et al.*, 2007b; Chirinos *et al.*, 2009a; Chirinos *et al.*, 2009b). Chirinos and Segers (Chirinos and Segers, 2010a; Chirinos and Segers, 2010b) have published reviews of ventricular–vascular interaction with aging and assessment with non-invasive methods. Mixed with authoritative articles in this field are a number that appear quite erroneous. An example is the hybrid Windkessel plus transmission line utilized by a group from Imperial

College London (Davies *et al.*, 2007; Baksi *et al.*, 2009). This model is used to justify attribution of the late systolic shoulder in the aortic and LV pressure pulse to a ‘Windkessel’ effect rather than to early wave reflection from peripheral sites. With the ‘Windkessel’ effect removed, the adjusted pulses contain a higher proportion of high frequency components – and these are seen as wave reflection. Authors conclude that the largest component of LV load is generated by the proximal stiffened aorta, interacting with the LV, and that effects of wave reflection are trivial. Such a conclusion runs counter to findings of Namasivayam *et al.* (2009a) for a large group of normal subjects, and to Taylor’s (Taylor, 1966a; Taylor, 1966b) views from experimental and model studies, that wave reflection is high for low frequency components of the pulse, and low for high frequency components (see Table 9.1). The Imperial group claim support for their concept from the analysis of Mitchell *et al.* (2004a) in the Framingham cohort, but this was based on their view that wave reflection inferred solely on aortic augmentation index (AIx) change, decreases with age. As Namasivayam *et al.* (2010) pointed out, change in AIx with age is a mathematical phenomenon, where amplitude of the primary and secondary (reflected) components of the pulse both increase with age, but they have different intercepts on the pressure axis. This is part of the phenomenon wherein the augmented (reflected) component of the pulse increases towards amplitude of the primary wave, but cannot increase beyond (i.e. AIx cannot exceed 50 percent), even when aortic stiffness and wave reflection are extreme.

ARTERIAL PRESSURE AND FLOW WAVES

Attention has been directed to differences in arterial pressure and flow wave patterns in adult humans compared with experimental animals (see Chapter 10). These can be explained on the basis of changes in arterial impedance and PWV with age, both resulting from asymptomatic arterial degeneration. Murgo *et al.* (1980a) noted different contours of pressure waves in young and old subjects, and their relationship to different ascending aortic impedance patterns (Fig. 19.23). These can be explained on the basis of different timing of wave reflection from peripheral sites. In young subjects, a ‘type C’ pattern is usually observed, with a definite diastolic wave following the incisura (see Fig. 9.2, top); this is the type of pattern seen in large experimental animals (rabbits, dogs, and sheep) and is associated with an ascending aortic impedance showing two separate early minima and little modulus fluctuation. It is explained on the basis of the asymmetric T-model of the arterial system (see Chapter 9) and wave reflection from the lower body returning late after aortic valve closure. The ‘type A’ pattern, typical of older human subjects, is explained on the basis of faster wave travel and earlier wave reflection, so that the

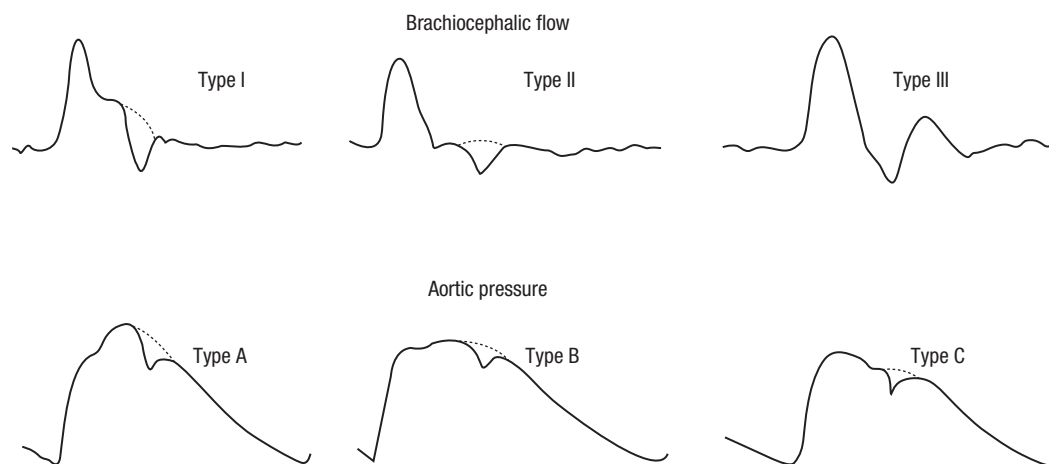


Figure 19.23 Comparison of types I, II, and III brachiocephalic flow waves as recorded by Mills *et al.* (1970) with corresponding ascending aortic pressure waves (designated type A, B, and C, respectively), recorded by Murgo *et al.* (1980a). All features can be explained on the basis of differences in the timing of peripheral wave reflection from the upper and lower body in relation to the duration of ventricular ejection. Reproduced from O'Rourke (1982b)

reflected wave appears in late systole and therefore increases pulse pressure (see Fig. 9.2, middle) (see Chapters 9 and 10). In the 'type D' pressure pattern typical of the very old, the reflected wave merges with the incident wave shortly after beginning of ejection so that no inflection point is visible (see Fig 9.2, bottom). The 'type A' and 'type D' pressure patterns are associated with the minima of impedance occurring at a higher frequency and greater fluctuation of the impedance modulus (see Fig. 19.19); this is caused by early wave reflection from the lower body, so that reflection returns almost in phase with upper body reflections, and the arterial system behaves as though there were one (not two) peripheral reflecting sites (O'Rourke and Avolio, 1980; O'Rourke, 1990a; O'Rourke, 1990b; O'Rourke, 1990c; Burattini *et al.*, 1991).

Changes in flow waves in the systemic arteries can be explained on the same basis. With increasing wave velocity and earlier wave reflection from the lower body, flow fluctuations are seen early in the brachiocephalic artery (Fig. 19.23), with the contour changing from type III to type I (Mills *et al.*, 1970; O'Rourke, 1982b), while in the lower body, fluctuations are lost, so that arterial flow contour comes to resemble that seen in the ascending aorta.

In older subjects, the thoracic aorta and peripheral and lower body arteries show none of the diastolic flow fluctuations characteristic of younger subjects and experimental animals, unless wave reflection is delayed relative to ventricular ejection, as in tachycardia, with hypotension, or during a Valsalva maneuver (see Chapter 10). However, fluctuations of flow during systole are quite marked in the brachial artery of older subjects, just as they are in the brachiocephalic artery, on account of the difference in the timing of wave reflection from sites in the arm itself and from the lower body (see Chapter 10).

As mentioned above and in Chapter 9, there is a progressive change in contour of the ascending aortic flow

wave with age (Nichols *et al.*, 1993). In late systole, the descending part of the wave is normally convex to the right, as it is in most experimental animals. With aging, this convexity gradually disappears, then becomes concave to the right (Figs 9.13, 9.14, and 19.19), and with this concavity corresponding in time to the late systolic augmentation of the pressure wave, caused by early return of wave reflection and consequent on increased aortic wave velocity (Miyashita *et al.*, 1993; Takazawa, 1987). We point out in Chapter 14 and elsewhere (Westerhof and O'Rourke, 1995; Denardo *et al.*, 2010) that this phenomenon is exaggerated when ventricular contractility is impaired and is the basis of shortened ventricular ejection duration and decreased stroke volume in cardiac failure in older people (see Chapter 15). Such changes are seen better in MRI than Doppler recordings, since the former better account for forward and backward flow.

New information on arterial pressure wave change with aging has come from the non-invasive studies initiated by O'Rourke, Kelly, Hayward, Avolio, and colleagues in their Sydney laboratory with high-fidelity sphygmographs (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; O'Rourke *et al.*, 1989b) (Fig. 19.24), which utilize the principle of applanation tonometry. These complement other invasive studies of aortic and LV pressure wave change with age. In a study of 1005 normal subjects ranging in age from 3 to 94 years, Kelly *et al.* (1989c) have shown characteristic aging change in the femoral, carotid, and radial artery pulses (see Fig. 10.34). All show a progressive increase in amplitude with age, attributable to progressive arterial stiffening, but also characteristic changes in wave contour, notably an increase in augmentation of the late systolic wave (Figs 19.23 and 19.24). This is most marked in the left ventricle and proximal aorta, less marked in the carotid, and least marked in the radial and finger arteries. Augmentation is more marked in females than in males

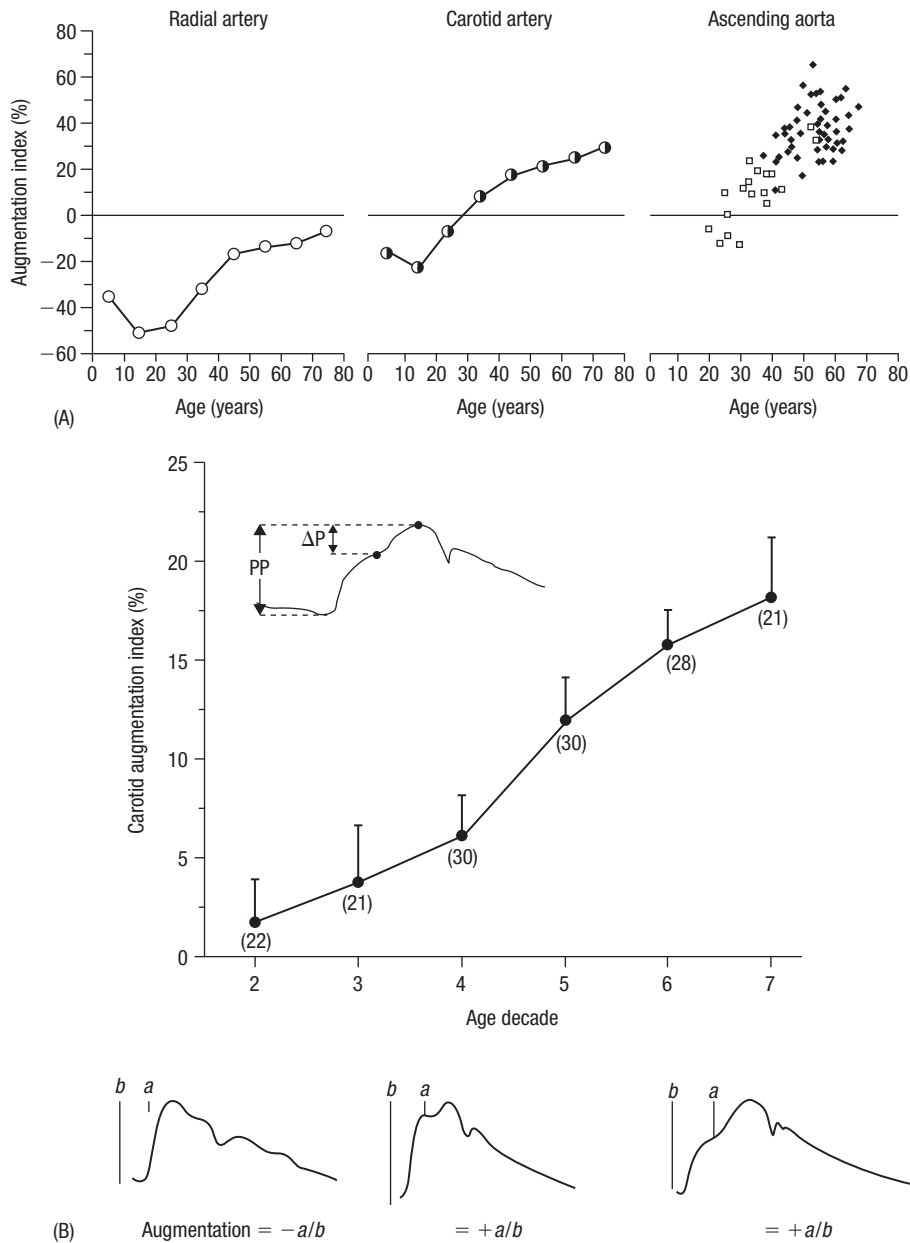


Figure 19.24 (A) Change of augmentation index (here shown as ratio of the amplitude of shoulder to peak pulse pressure (a/b), as in the pressure waves displayed below) with age in the radial artery (left), determined by Kelly *et al.* (1989e), supplemented with data from very elderly but healthy people by Marey (1863); in the carotid artery (center), determined by Kelly *et al.* (1989b); and in the ascending aorta (right) from Murgu *et al.* (1980a); ($\square$) and Takazawa (1987) ($\blacksquare$). High values in young children are attributable to short body length. (B) Change in carotid augmentation with age. The slight but significantly higher values in females than males cannot be attributed to higher aortic pulse wave velocity (Fig. 19.14C), but can be attributed to small body height and greater tapering of large arteries (Cecelja *et al.*, 2009). Courtesy of E Lakatta, US National Institute of Aging, NIH

(Smulyan *et al.*, 1998; Mitchell *et al.*, 2004a; McEniery *et al.*, 2005a; Lieber *et al.*, 2010). This is largely responsible for higher pulse pressure in older women than men (Smulyan *et al.*, 2001) and is attributable almost entirely to shorter stature and to greater ‘tapering’ of distal versus proximal aorta in women (Cecelja *et al.*, 2009). This is supported somewhat by the recent work of Ayer *et al.* (2010) who found that carotid augmentation index is greater in prepubertal girls than boys, independent of

height. Both short stature and aortic ‘tapering’ cause earlier return of wave reflection. Aortic PWV is virtually identical in men and women (Avolio *et al.*, 1983; Avolio *et al.*, 1985a; Vermeersch *et al.*, 2008; Wang *et al.*, 2010; Shim *et al.*, 2011), and shows no abrupt change at the time of menopause (Mitchell *et al.*, 2004a; O’Neill, 2010). Studies of both aortic and carotid augmentation index show that this increases progressively from youth up to age 60 whereafter it flattens (Kelly *et al.*, 1989b; Adji and

O'Rourke, 2004a; Filipovsky *et al.*, 2005; McEniery *et al.*, 2005a; McEniery *et al.*, 2008) (Figs 9.14, 9.18, and 19.24A,B) or may even decrease over age 60 if the ventricle begins to fail. This contrasts with progressive rise in aortic PWV throughout life, and the steady rise in aortic augmentation seen in older people undergoing cardiac catheterization but in whom no significant coronary disease was demonstrated (Murgo *et al.*, 1980a; Takazawa, 1987). The explanation we favor, and which covers all findings, is that aortic stiffness does increase progressively with age, as does aortic characteristic and input impedance. The apparent decrease in wave reflection, when measured as AIx in older people (> 60 years) is attributable at least in part to change in shape of the aortic flow wave, with lower velocity in the latter part of systole (see Figs 9.14 and 9.19), and caused by high late systolic pressure, with subtle impairment in late LV systolic ejection. Part of the flattening of aortic AIx with age is attributable to a purely mathematical phenomenon (Namasivayam *et al.*, 2009a; Namasivayam *et al.*, 2009b; Namasivayam *et al.*, 2010): the quotient of two values which increase linearly with age, but have a different age intercept, must be a curve. High AIx in older people with normal coronary arteries undergoing cardiac catheterization for chest pain syndrome (Murgo *et al.*, 1980b; Takazawa, 1987) is attributable to such a group having a substantial number with 'syndrome X', and powerful ejection throughout systole, and relative prolongation of ejection duration (p. 445) (Fig. 30.38) and shortened diastolic duration. The difference in amplitude of the late systolic wave in central and peripheral arteries was first noted by Marey in 1863 then by Mahomed in 1872 (see Fig. 10.34).

Mitchell *et al.* (2004a) have pointed out that stiffening of the aorta with age leads to carotid-femoral PWV approximating, and then exceeding, PWV in the peripheral muscular arteries. There can be no argument with this viewpoint, which was first pointed out by Learoyd and Taylor (1966), then established by Ho (1982) in a normal Australian cohort and Avolio *et al.* in two Chinese populations (Avolio *et al.*, 1983; Avolio *et al.*, 1985a) and McEniery *et al.* (2008) in the UK (see Fig. 4.24). Mitchell *et al.* (2004a) then went on to describe this as impedance matching between the central elastic aorta and peripheral muscular arteries, which shifts reflecting sites distally. Such a view is at odds with Taylor's view (Taylor, 1965; Taylor, 1966a; Taylor, 1966b) that non-uniform elasticity of the aorta (less stiff proximally, more stiff distally) is one of the factors which contributes to arterial efficiency in young animals and humans. Mitchell described such 'matching' as reducing wave reflection from the lower body, and explaining the flattening of carotid augmentation around age 60 years, and its reduction in women over the age 60 years. These changes in wave reflection characteristics reduce ventricular load and imply a beneficial effect to the heart. The measuring techniques of Mitchell have been questioned by many, but these interpretations have been questioned by our group through analysis of

our own data, and the group of normal subjects reported by McEniery *et al.* (2005a) and Namasivayam *et al.* (2009a).

In our own data, which supports that of McEniery *et al.*, we found that both proximal wave reflection (augmentation in mmHg) and pulse pressure increase with age, but with different intercepts on the pressure axis. As shown in Fig. 19.25, this results in a steep initial slope of AIx with age at around 20 years of age, but a steady flattening around age 60 years. This is not a manifestation of wave reflection, but a necessary result of the mathematical relationship of 'conservation of energy'. This is as necessary as is the fact that aortic augmentation index cannot exceed 50 percent, i.e. the reflected wave must be less than the incident wave.

Mitchell *et al.* (Mitchell, 2009b; Mitchell *et al.*, 2010) interpreted their data to show evidence that the initial wave generated by the heart increases with age while the reflected wave decreases relatively with age so that it becomes progressively less relevant as age advances. They concluded that their findings were consistent with the hypothesis that changes in central aortic stiffness and incident (or forward) wave amplitude, rather than wave reflection, are responsible for most of the increase in pulse pressure in the elderly. Namasivayam *et al.* (2009a) challenged this using the new technique of mediation analysis. This showed that below age 60 years, wave reflection dominates the initial wave as a cause of increase in aortic systolic and pulse pressure, while over age 60 there

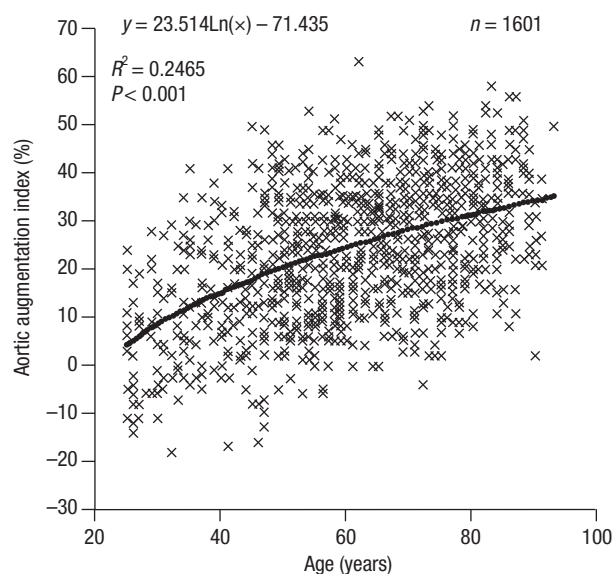


Figure 19.25 The observed pattern of increase in augmentation index with advancing age can be partly explained by the ratio between linear regression equations for augmented pressure and aortic pulse pressure. The ratio between these linear equations closely approximates the fitted logarithmic regression for AIx. Data are from patients attending a cardiovascular outpatient clinic. From Namasivayam *et al.* (Namasivayam *et al.*, 2009b; Namasivayam *et al.*, 2010)

is a similar change in the incident and reflected wave (Fig. 19.26).

It is desirable that these anomalies be settled before the Framingham waveform data, collected assiduously over

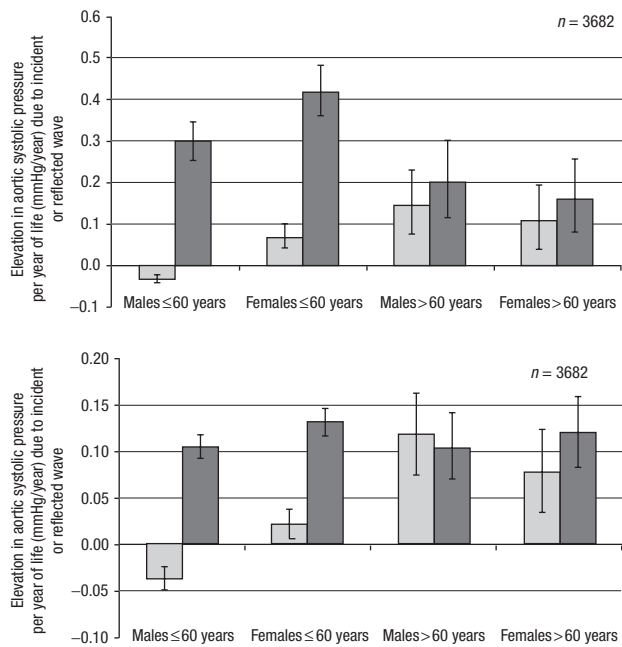


Figure 19.26 (Top) Contribution of incident and reflected waves to aortic systolic pressure elevation with advancing age (per year of life) in 3682 healthy subjects, before and after age 60 years. Vertical lines represent 95 percent confidence intervals. Light gray bars represent pressure at aortic initial peak, dark gray bars represent reflected wave. (Bottom) Corrected contribution of incident (light gray) and reflected (dark gray) waves with control for height, weight, heart rate, and mean arterial pressure. From Namasivayam *et al.* (2009a)

the past decade, undergo final analysis and publication. As presented up to now, the analysis discussed above has the capacity to taint this historic trial (O'Rourke and Adji, 2010b; O'Rourke *et al.*, 2010d).

These non-invasive studies have been extended by many others, as well as by our own groups. Attention has been focused on late systolic augmentation of the arterial pressure wave and on how this relates to LV load and function. Yaginuma, Takazawa, and colleagues in Japan have established the relationship between carotid and ascending aortic augmentation, pointing out that the former underestimates the latter (see Fig. 9.14) and provides highly useful information on LV load and function in different circumstances and with different therapies (Yaginuma *et al.*, 1985; Takazawa, 1987; Takazawa *et al.*, 1993a; Takazawa *et al.*, 1993b; Miyashita *et al.*, 1994; Miyashita *et al.*, 2010). The effect of aging on augmentation of ascending aorta, carotid, and radial pressure waves has been established (Fig. 19.24). The apparently paradoxical increase in augmentation in children can be explained on the basis of the short body length and the early return of wave reflection (Fig. 12.32) (Hsieh *et al.*, 1989; Gevers, 1994). Pressure wave patterns in infants and neonates are discussed later in this chapter, while detailed information of the use of arterial tonometry is presented in Chapter 30. Devereux, Roman, and colleagues in New York have pointed to the relationship between carotid pressure wave augmentation, carotid intima–medial hypertrophy and hyperplasia, and LV hypertrophy (Saba *et al.*, 1993; Roman *et al.*, 1995b), and have suggested that carotid augmentation may provide a non-invasive index of carotid disease and of LV hypertrophy. Shimada *et al.* (1994) have obtained similar results in a Japanese cohort (Fig. 19.27). Lakatta and colleagues in Baltimore have clarified the association between carotid augmentation

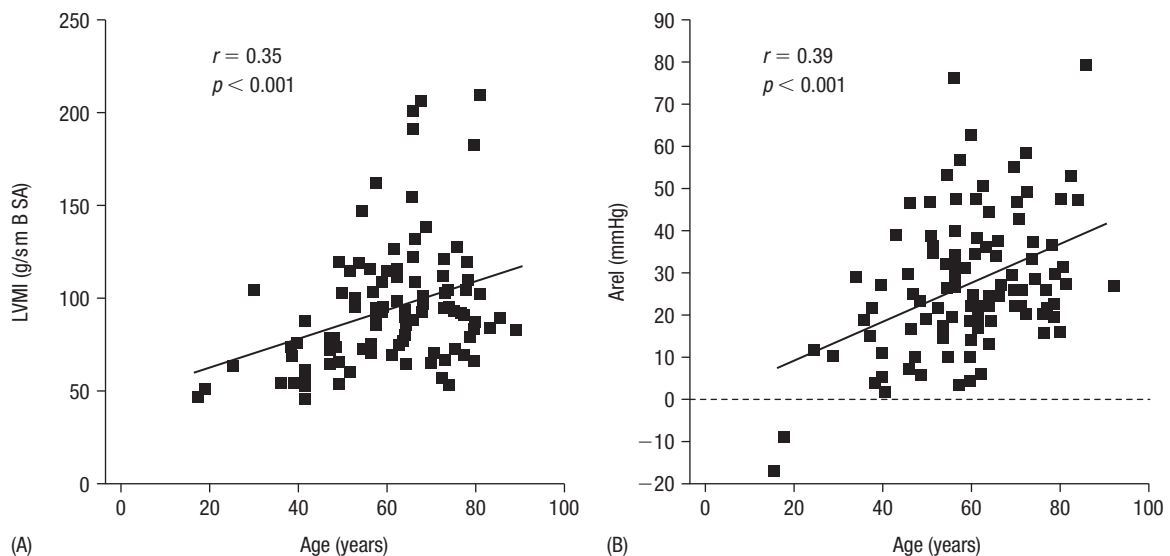


Figure 19.27 Relationship between left ventricular mass index (LVMI) and age (A) and between carotid artery augmentation index (Arel) and age (B). Reproduced from Shimada *et al.* (1994)

and age in their healthy aged cohort, confirming the work of Kelly *et al.* (Kelly *et al.*, 1989b; Kelly *et al.*, 1989f) on aging change and of Shimada *et al.* on ventricular hypertrophy, but showing in addition that the degree of augmentation is inversely related to exercise capacity (Vaitkevicius *et al.*, 1993) (Chapter 27). The strong implication in the title of the 1993 paper is that physical exercise modifies pressure wave augmentation, and the physical properties of the arteries are responsible for this phenomenon. One could take the opposite viewpoint and suggest that arterial properties, and thus augmentation, determine the capacity for exercise. There is support for both views (McGuire *et al.*, 2001a; McGuire *et al.*, 2001b). Kelly and Fitchett (1992) showed that aortic impedance and ventricular load could be estimated non-invasively from aortic Doppler flow and carotid tonometry. This has been used by others (Markus *et al.*, 1994; Chen *et al.*, 1999) and is the basis of the method used by Mitchell *et al.* (2002) to study effects of drugs on the heart and in the Framingham study (Mitchell *et al.*, 2004a; Mitchell *et al.*, 2005a; Mitchell *et al.*, 2005b; Mitchell *et al.*, 2007a; Mitchell *et al.*, 2007b; Mitchell *et al.*, 2010). Problems noted by Kelly and Fitchett and others, with carotid and particularly brachial tonometry (O'Rourke and Adji, 2010b), call into question the claimed precision of the Mitchell process to accurately measure characteristic impedance and wave reflection. Even when pressure and flow are measured directly in the ascending aorta with fully calibrated instruments, it is difficult to be precise on the calculation of characteristic impedance and frequency of fluctuations in modulus and phase (O'Rourke, 1982a).

Roman *et al.* (2009) have analyzed the results of the Strong Heart study with respect to relationship between central aortic systolic and pulse pressure with carotid intima-media thickness and LV mass. They found better correlation between central pulse pressure and arterial intima-media thickness, and better correlation between central systolic pressure and LV mass (Roman *et al.*, 2010). This supplements their previous study done with carotid tonometry and is what one would expect from the underlying principles.

AMPLIFICATION

Aorta and lower body vessels

Amplification of the arterial pressure wave is discussed in Chapters 9 and 10 and attributed principally to peripheral wave reflection at arterial–arteriolar junctions, and with elastic non-uniformity between central and peripheral arteries contributing also. Amplification is greatest when PWV is low and ejection duration short, and where wave reflection has the greatest positive effect on the peripheral pressure wave and the least positive effect on the central

pressure wave (type C pattern, Fig. 4.21) (i.e. where timing of reflection is such that the peripheral site is near an antinode and the central site near a pressure node, see Chapters 9 and 10). With this timing, and pressure at this frequency, the flow antinode will be highest centrally and lowest peripherally (i.e. flow pulsations are highest in the ascending aorta, and lowest in peripheral arterioles – just what an efficient arterial circulation requires (McDonald and Taylor, 1959; Taylor, 1964)).

Studies in humans have shown a progressive decrease in percentage amplification of the pressure wave between the ascending aorta and the femoral artery with age (Fig. 19.28) (O'Rourke *et al.*, 1968; Nichols *et al.*, 1993). In children around ten years of age, the femoral pulse pressure is some 50 percent greater than ascending aortic pulse pressure, but femoral pulse pressure is virtually the same as central pulse pressure at age 65. This aging change can readily be explained on the basis of increased aortic PWV, through explanations given for both the time and

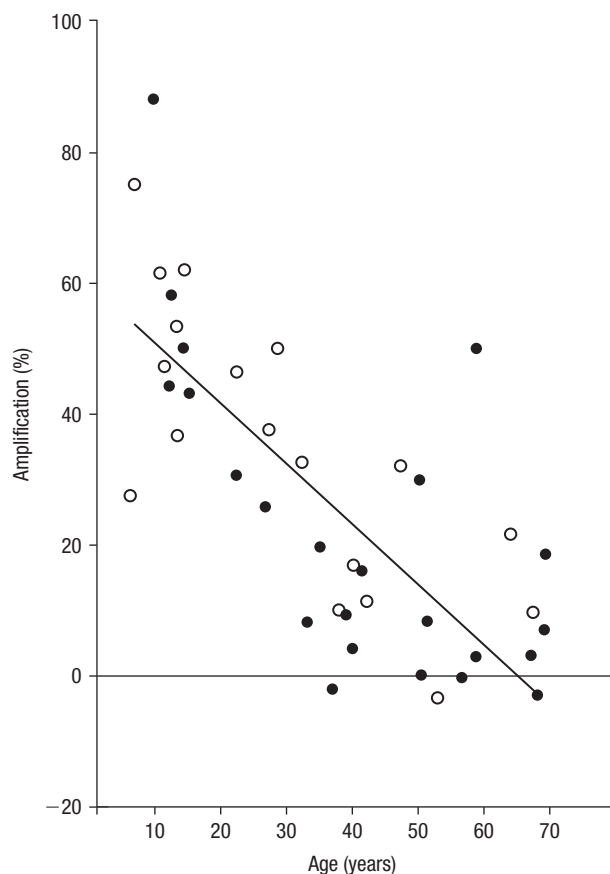


Figure 19.28 Amplification of the arterial pressure wave between the aortic arch and the iliac artery. Data represent the ratio of the pulse pressure increase in the iliac artery to the pulse pressure in the aortic arch, with data recorded following cardiac catheterization during withdrawal of the catheter from the central to the peripheral vessel. (○) Patients without clinical evidence of any vascular lesion; (●) patients with known vascular lesions. Reproduced from O'Rourke *et al.* (1968)

the frequency domain. The latter are more comprehensive. In the time domain, changes are attributable to a progressive increase in the amplitude of the central aortic pressure pulse, since with early return of wave reflection, this comes closer in time to the peripheral reflecting site and antinode, so that it is almost equally as affected by wave reflection as is the peripheral femoral pulse. In the frequency domain, one can perceive two separate effects, both related to increased aortic PWV: the first a consequence of wave transmission, and the second a consequence of a change in shape, and thus in harmonic content, of the central aortic pressure pulse. Increased aortic PWV causes a shift of the transfer function (Fig. 19.29) for pressure between the ascending aorta and the femoral artery; this has the effect of causing less amplification of the first three harmonics of the aortic pressure wave into the legs; these normally comprise the bulk of the wave and are responsible for most of its amplification. The second effect is a consequence of a

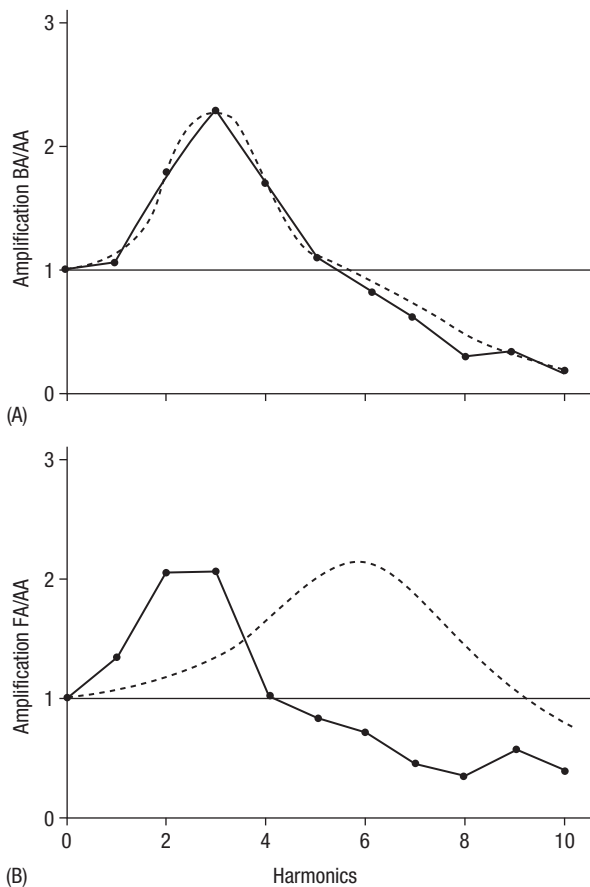


Figure 19.29 Amplification of the human arterial pressure wave as a function of frequency (A) between the aortic arch (AA) and brachial artery (BA), and (B) between the aortic arch and femoral artery (FA). Data were obtained from Fourier analysis of pressure waves recorded simultaneously in a young male subject by Kroeker and Wood (1955). The dotted lines indicate the expected changes in amplification in an older subject

change in the PWV on aortic impedance. The shift of impedance aortic curves to the right with age results in an increase of the lowest-frequency components of impedance modulus so that the first, second, and third harmonic components of the aortic pressure wave are markedly increased; there is, in consequence, a greater proportion of the aortic pressure wave that is amplified to the lowest degree (Fig. 19.30B). The two separate aging effects are thus: (1) a lower amplification of low-frequency components of the pressure wave, and (2) a greater amplitude of these low-frequency components.

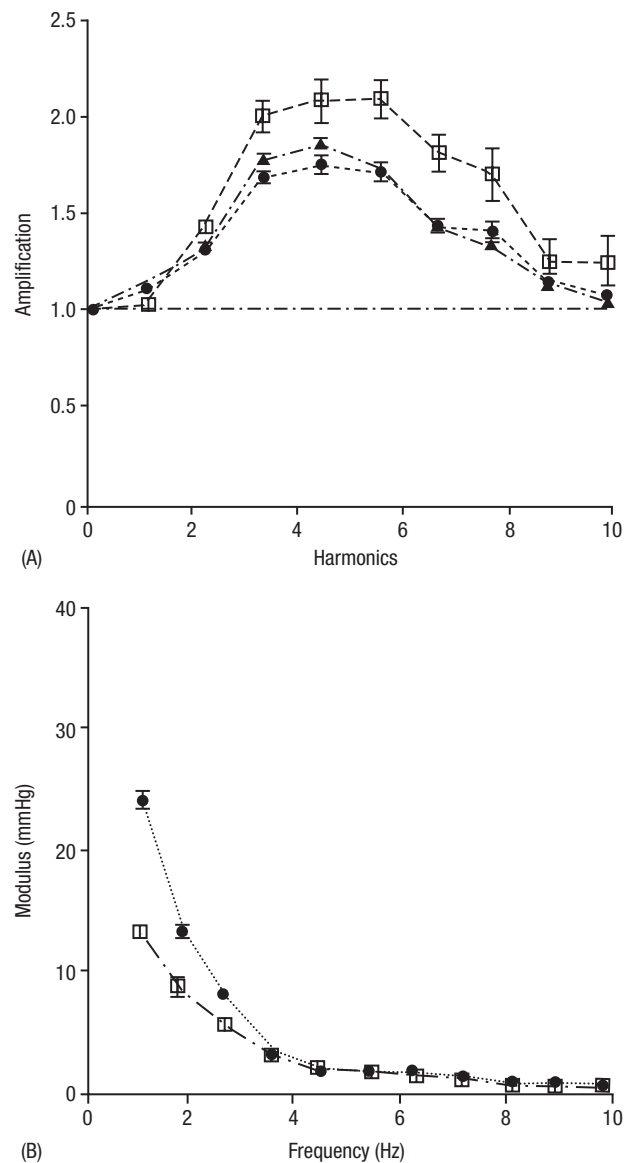


Figure 19.30 (A) Transfer function for pressure between the carotid artery and the radial artery for a group of 439 adult human subjects, subdivided into those over 65 years (●), those under 35 years (□) and the whole population (▲). (B) Amplitude of harmonic moduli of carotid pressure waves in the same group for subjects over 65 and those under 35 years of age; symbols as above. Reproduced from Gallagher (1993)

Upper limb

Amplification of the pressure pulse in the upper limb is of great clinical importance, since it bears on the interpretation of systolic and diastolic pressures measured in the brachial artery, with relation to systolic and diastolic pressures in the central arteries. This subject is ignored by most clinicians, who consider that 'systolic' pressure, measured at the brachial artery, is the same throughout the body, and 'diastolic' pressure likewise.

Amplification of pressure in the upper limb is different from that down the aorta to the femoral and more distal arteries. The reflected pressure wave returning from the lower body makes a substantial contribution to the contour of the brachial and radial pressure waves, being responsible for its secondary diastolic or late systolic surge (Kelly *et al.*, 1989a; Kelly *et al.*, 1989e; O'Rourke *et al.*, 1992a; Nichols and Edwards, 2001; Nichols, 2005) (see Chapter 30), whereas reflection from the upper limb has little effect on the lower body pulse (Karamanoglu *et al.*, 1994). Changes in the timing of wave reflection from the lower body with age are responsible for the progressive change in brachial and radial artery pressure wave contour, notably for the shift of the pressure surge from diastole to late systole, and the progressive increase in amplitude of the secondary systolic wave (see Figs 9.2, 10.34, and 19.24).

As with transmission to the lower body, pressure waves in the upper limb are progressively less amplified with age – at least from adolescence. In young adults, the amplitude of the brachial or radial pressure pulse may be 50 percent greater than in the ascending aorta, whereas all are of nearly equal amplitude in the elderly (see Figs 4.18 and 9.2, and Chapter 9) (O'Rourke *et al.*, 1992a; Nichols *et al.*, 1993). The difference affects predominantly the systolic pressure, and this may be 20 mmHg (or more) greater in the radial or brachial artery than in the ascending aorta; diastolic pressure is only 1.0–2.0 mmHg less (Pauca *et al.*, 1992; Pauca *et al.*, 2001). In contrast to wave amplification in the lower body, however, aging effects in the upper limb are almost entirely due to change in contour and harmonic content of the ascending aortic pressure wave itself, rather than to a change in properties of upper limb arteries. In contrast to the aorta, aging has little effect on PWV in the upper limb. Our own studies (Ho, 1982; Avolio *et al.*, 1983; Avolio *et al.*, 1985a) have shown no significant change with age in a southern Chinese population throughout life (see Fig. 19.14), no significant change above age 20 in a Northern Chinese population (see Fig. 19.13), and little change above age 30 in an Australian population (see Fig. 19.15). Similar findings have been noted for the Framingham cohort (Mitchell *et al.*, 2004a). Limited studies only have been conducted on transfer function in the upper limb from directly recorded pressure waves, but neither our studies (Karamanoglu *et al.*, 1993) (see Chapter 30), nor those of Lasance *et al.* (1976) or Chen *et al.* (1997), showed any apparent difference with age.

A single generalized transfer function was proposed to describe upper limb amplification at all ages in adults (O'Rourke *et al.*, 1992c; Karamanoglu *et al.*, 1993; Chen *et al.*, 1997; Fetics *et al.*, 1999). Figure 30.18 shows all data on upper limb transfer functions currently known to us in which appropriate manometric techniques were applied (Gallagher *et al.*, 2004). These are virtually identical except for the original paper of O'Rourke (1970b) in which distal pressure was measured close to the axilla. In a study by Hope *et al.* (2003), high and erratic values over 4 Hz are attributable to dynamic inaccuracy of the fluid-filled catheter system as used at routine cardiac catheterization. This study has been repeated using catheter-tip micromanometry; and as recalculated from the inverse published values (Hope *et al.*, 2004; Hope *et al.*, 2007), modulus of the transfer function is virtually identical to that originally described (Karamanoglu *et al.*, 1993) and in its initial validation by Chen *et al.* (1997). The Melbourne group (Hope *et al.*, 2007), however, had made an error with phase, as pointed out by Segers *et al.* (2007a), with the result that at high frequencies they had components of the distal wave preceding the proximal, and artifactual fluctuations in the calculated proximal pressure wave. This brings into question the veracity of the large ANBP2 study (Dart *et al.*, 2006) which found that cardiovascular outcomes were better predicted from brachial than central pressures (Vlachopoulos *et al.*, 2010b). Gallagher *et al.* (Gallagher, 1993; Gallagher *et al.*, 2004) have studied aging change in upper limb wave transmission through determining transfer function for pulses recorded non-invasively in the carotid artery and radial artery in 439 subjects at different ages. Differences in transfer function were minor (Fig. 19.30A); specifically, there was no shift in frequency with age. There was, however, an almost two-fold increase in amplitude of the first three harmonic components of the carotid pressure waves, attributable to an increase in ascending aortic impedance (Fig. 19.30B).

These principles can be applied to explain the curious increase in systolic blood pressure seen between the ages of 10 and 20 in major epidemiological studies such as the US National Health Survey of 1977 (see Fig. 19.1). This increase is greater than at any other time of life. Systolic blood pressure is often unusually high in the late teens and early twenties. Measurement of pressure waves in the radial or brachial artery shows a highly peaked pressure pulse with normal mean and diastolic pressure (see Figs 20.1 and 20.2). When the ascending aortic pressure wave is synthesized from the radial, using the generalized transfer function referred to above (O'Rourke *et al.*, 1992c; Karamanoglu *et al.*, 1993), this is seen to be of normal contour for age, and with aortic systolic peak pressure markedly less than in the brachial (or radial) artery (Fig. 20.2) (O'Rourke, 1996). It is clear that this finding is a consequence of pressure wave amplification in the upper limb. It appears that systolic pressure in the brachial artery is considerably higher than in the aorta

around this age (when full body height has been achieved, but little arterial degeneration has occurred), and that aortic systolic pressure actually rises steadily with age. This was shown in studies undertaken at cardiac catheterization by Nichols *et al.* (Fig. 19.1) (Nichols *et al.*, 1985a). The rapid rise in systolic pressure to age 20 and the subsequent plateau to age 50 (see Figs 19.1 and 19.2) appears to apply only to the brachial and radial arteries and not to the aorta (see dotted line in Fig. 19.1). Changes with age in amplification are shown in Fig. 19.31. Wilkinson *et al.* (2001) studied a group of normal subjects, and compared the peripheral/central ratio of two components of the pressure pulse – the pulse pressure itself, and the first component of the pulse from foot to initial peak or shoulder (i.e incident wave amplitude) (Fig. 19.31). With aging, the same change was seen as in Fig. 19.28, with progressive reduction in amplification with age. In contrast, amplification of the first component did not change with age.

We have shown a reciprocal relationship between aortic pressure wave augmentation and amplification of the

pulse between the ascending aorta and radial artery (Fig. 19.32) (O'Rourke and Adji, 2008). Given the inherent inaccuracy in identifying an early systolic inflection point on the aortic wave, and so in measuring pressure augmentation, we suggested that amplification may be a better measure of wave reflection than augmentation. This issue has not yet been settled. However, in seeking indices of pulse wave shape as surrogates of target organ damage, we found amplification better than augmentation in predicting reduction of LV mass after one year of antihypertensive therapy (Hashimoto *et al.*, 2007a; Hashimoto *et al.*, 2007b). Likewise, Wang *et al.* (2010) found pulse pressure amplification a potent predictor of cardiovascular outcomes in a large study extending over 15 years.

GROWTH

Studies of arterial pressure waves in the ascending aorta and upper limb arteries of human infants and children

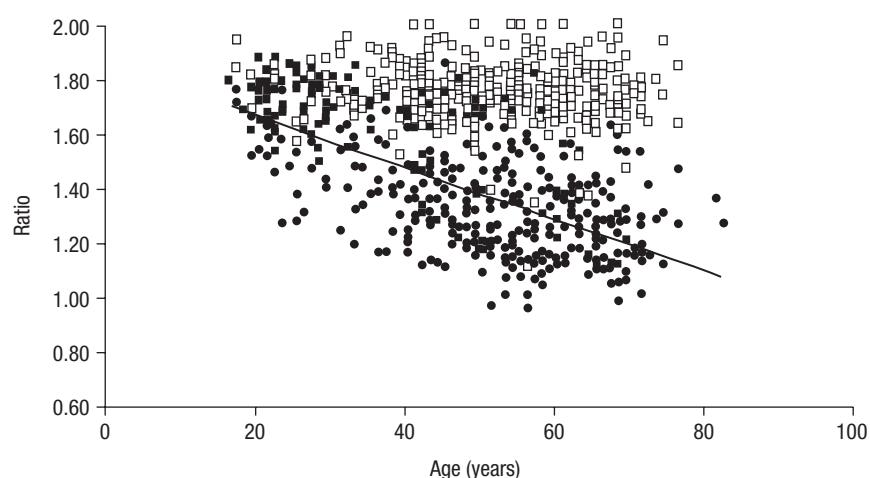


Figure 19.31 The relationship between amplification and age between the ascending aorta and radial artery for pulse pressure (●) and for the first component of the arterial pulse (□). Pulse pressure amplification fell with age, whereas amplification of the initial component did not. From Wilkinson *et al.* (2001)

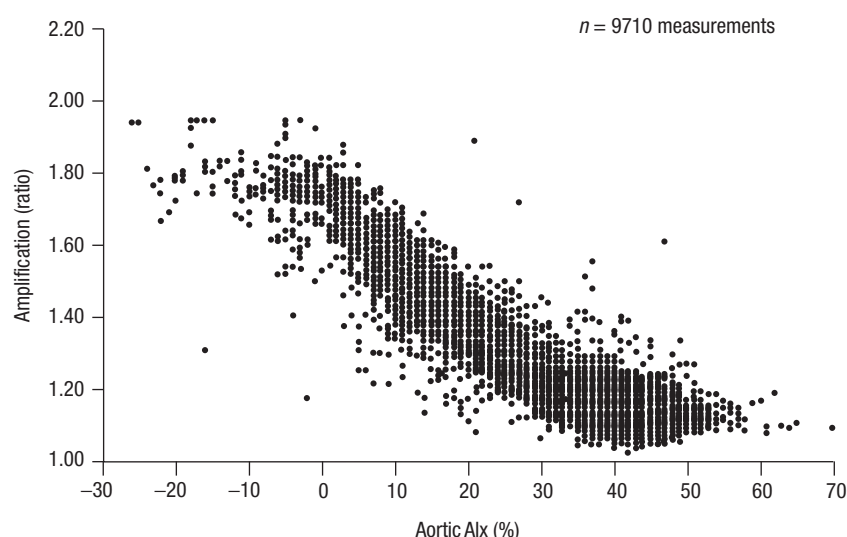


Figure 19.32 Amplification of the pressure pulse between ascending aorta and radial artery plotted against aortic augmentation index (Alx) in 1505 patients attending a cardiovascular outpatient clinic. For $Alx \ 0\% \leq 45\%$, there is an inverse linear relationship between amplification and Alx. Amplification is approximately 1.05 for $Alx > 45\%$; and approximately 1.80 for $Alx < 0\%$. From O'Rourke and Adji (2008)

have provided surprises. Pressure waves are quite unlike those seen in adolescents and paradoxically are similar to those recorded in older adults except that mean blood pressure is much lower (Hsieh *et al.*, 1989; Gevers *et al.*, 1993a; Gevers *et al.*, 1993b; Gevers, 1994; Gevers *et al.*, 1995a; Gevers *et al.*, 1995b; Gevers *et al.*, 1996; Murakami *et al.*, 2010). In young children, the aortic and radial pressure waves are like those recorded in elderly humans

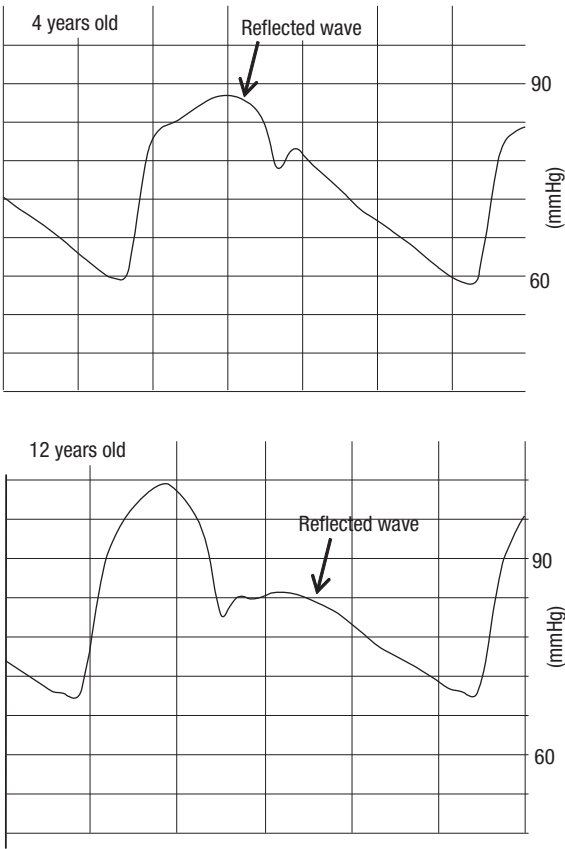


Figure 19.33 Central artery pressure waves in a four year old (body height 99.5 cm, upper) and a 12 year old (body height 163 cm, lower). Reproduced from Murakami *et al.* (2010)

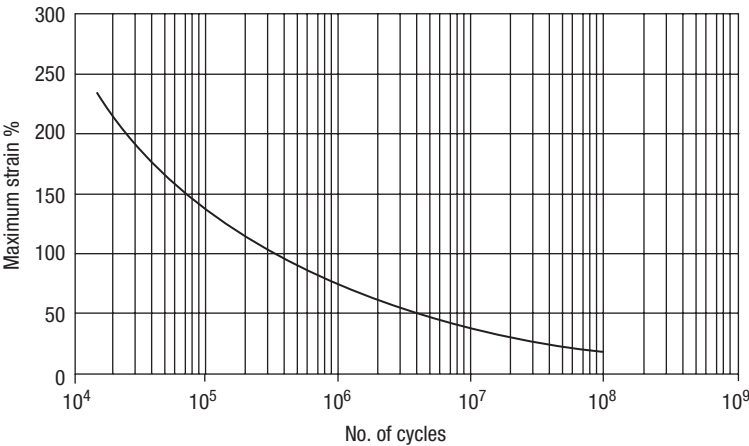


Figure 19.34 The relationship between strain (ordinate) and cycle number (abscissa, logarithmic scale) for the fracture of natural rubber

(Fig. 19.33). In children also, amplification of the pressure wave is decreased between the aorta and brachial artery, just as it is in the elderly.

These apparently paradoxical results can readily be explained on the basis of difference in body length and of the timing of wave reflection. The pressure waveforms of children appear similar to those of elderly subjects because reflected waves return earlier to central arteries and have their most pronounced effect on the systolic rather than the diastolic part of the pressure pulse. Reflection returns early, not because the aorta is stiff and aortic PWV high (as in the aged), but because body length is short, so reflection sites are anatomically closer to the heart. The same phenomenon largely if not entirely explains gender differences (Smulyan *et al.*, 1998; Smulyan *et al.*, 2001). Body height of women is substantially less than men, whereas aortic stiffness and PWV are similar (Mitchell *et al.*, 2004a; McEniery *et al.*, 2005a; Vermeersch *et al.*, 2008).

These findings in children should not be a surprise: they are predictable on the basis of short body length. The real surprise is that ejection duration in children is similar to that of adults for any given heart rate and not reduced, as in experimental animals of similar dimensions.

MECHANISM OF ARTERIAL DEGENERATION WITH AGE

Explanations for arterial degeneration need to account for all of the following:

1. The change itself – fracture of elastin lamellae and fibrous remodeling.
2. Progressive increase in severity with age.
3. Acceleration of the process in hypertension.
4. Severity of the process in the proximal aorta and elastic arteries with sparing of peripheral, predominantly muscular arteries.
5. Absence of comparable change in experimental animals.

6. Progressive dilatation, as well as stiffening of the aorta and elastic arteries.

In 1976, the theory was put forward that arterial degenerative changes are due to the fatiguing effects of cyclic stress on elastin fibers and lamellae within the arterial media, and that these become brittle and then

fracture as a consequence of repeated cycles of stress (O'Rourke, 1976). This theory was based on conventional engineering theory and practice as applied to aircraft, ships, bridges and, indeed, to any inanimate object that flexes or stretches (Gent, 1972; Sandor, 1972). Practicing cardiologists have become all too aware of this as the basic problem in fracture of struts in certain heart valves and

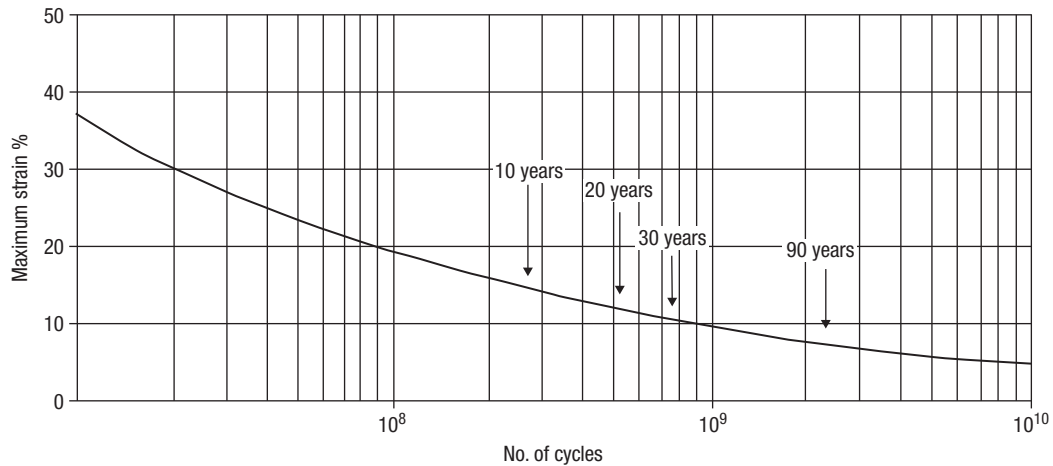


Figure 19.35 The relationship between strain and number of cycles predicted for elastin fiber fracture (assuming a similarity between rubber and elastin). At 10 percent stretch, fracture is predicted at 8×10^8 cycles, corresponding to 30 years of life at average heart rate 70 beats/minute. For a 5.0 percent stretch, fracture is not anticipated in 3×10^9 cycles, corresponding to a lifespan of 90 years

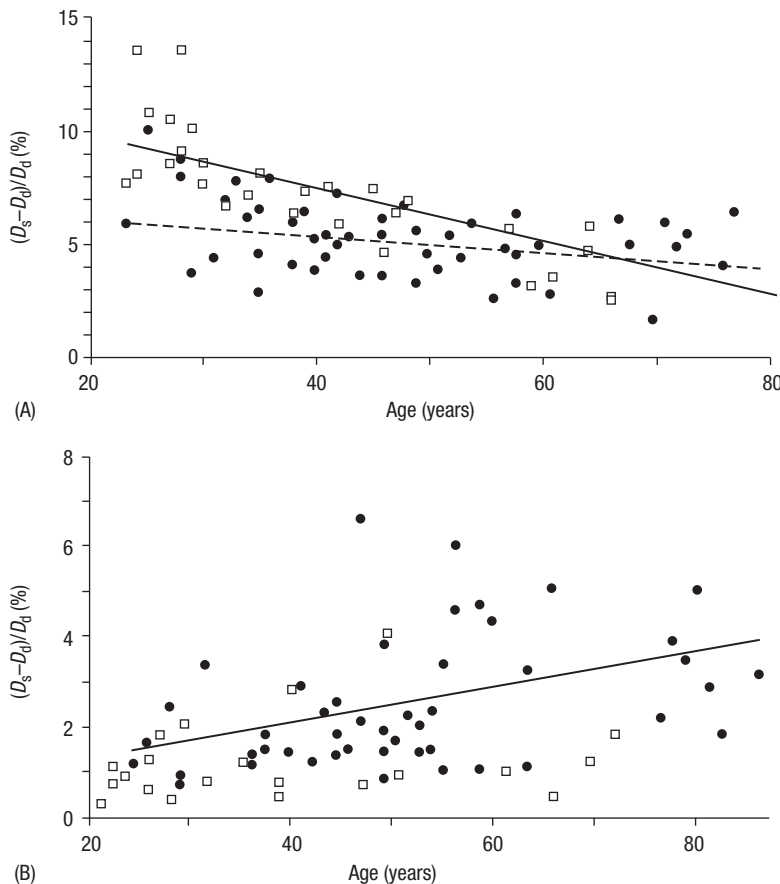


Figure 19.36 (A) The relationship between strain $(D_s - D_d)/D_d$ of the carotid artery and age for 32 normotensive adult subjects ($\square$) and 50 subjects with essential hypertension ($\bullet$). Strain in normotensive subjects averaged 10 percent under age 30 years, but decreased progressively with age, indicating increasing carotid artery stiffness. In hypertensive subjects, strain was lower, probably due to a smaller pulse change in diameter, but also decreased progressively with age. (B) The relationship between strain of the radial artery and age in normotensive and hypertensive subjects. Percentage strain increased with age (as did pulse pressure). There was no evidence of radial artery stiffening with age in either hypertensive or normotensive subjects. Reproduced from Boutouyrie *et al.* (1992)

pacemaker leads. Certainly, since 1976, the theory proposed has received support from many directions and can account for most if not all observations. As with other applications of physics and engineering principles, however, practicing physicians have been reluctant to embrace it.

Details of engineering principles are given in Caldwell *et al.* (1940), Lindley (1974), Rigby (1964), Broom (Broom, 1978a; Broom, 1978b; Broom, 1980), and elsewhere in engineering texts. The behavior of natural rubber is shown in Fig. 19.34; this plots strain as a percentage on the ordinate against the number of cycles plotted logarithmically on the abscissa. The line shows the relationship of strain and number of cycles at the point of fracture. At 100 percent stretch, fracture is probable after 200 000 cycles, while at 200 percent stretch, this is probable after just 20 000 cycles, and at 25 percent stretch, it is likely after 30 million cycles. There is no reason to believe that elastin behaves any differently from natural rubber, although, of course, elastin fibers are never stretched to the degree shown in this graph. If one extends the data

points shown on this graph (Fig. 19.35), one comes to values that are realistic and relevant. The carotid artery and other central elastic arteries, including the thoracic aorta, pulsate by around 10 percent with each heart beat (see Chapter 4), at least up until age 25 (Fig. 19.36) (Boutouyrie *et al.*, 1992), when the effects of arterial damage and stiffening become apparent. One would expect fracture to occur after around 800 million cycles. At an average heart rate of 70 beats per minute, this corresponds to around 25 years of life. Peripheral muscular arteries dilate to a far lesser degree, usually less than 5.0 percent per cycle (Fig. 19.36) (Boutouyrie *et al.*, 1992). At this degree of stretch, fracture would not be expected within a lifespan of 100 years.

The data in Figs 19.34 and 19.35 are given as a guide. There are no data available on the fatiguing properties of elastin. This may be different from that of natural rubber, but it would be expected to take the same general form. If elastin were to act like rubber as described by Osler (1892), the data in the figures show that the effects of repetitive stretch could explain the observations of

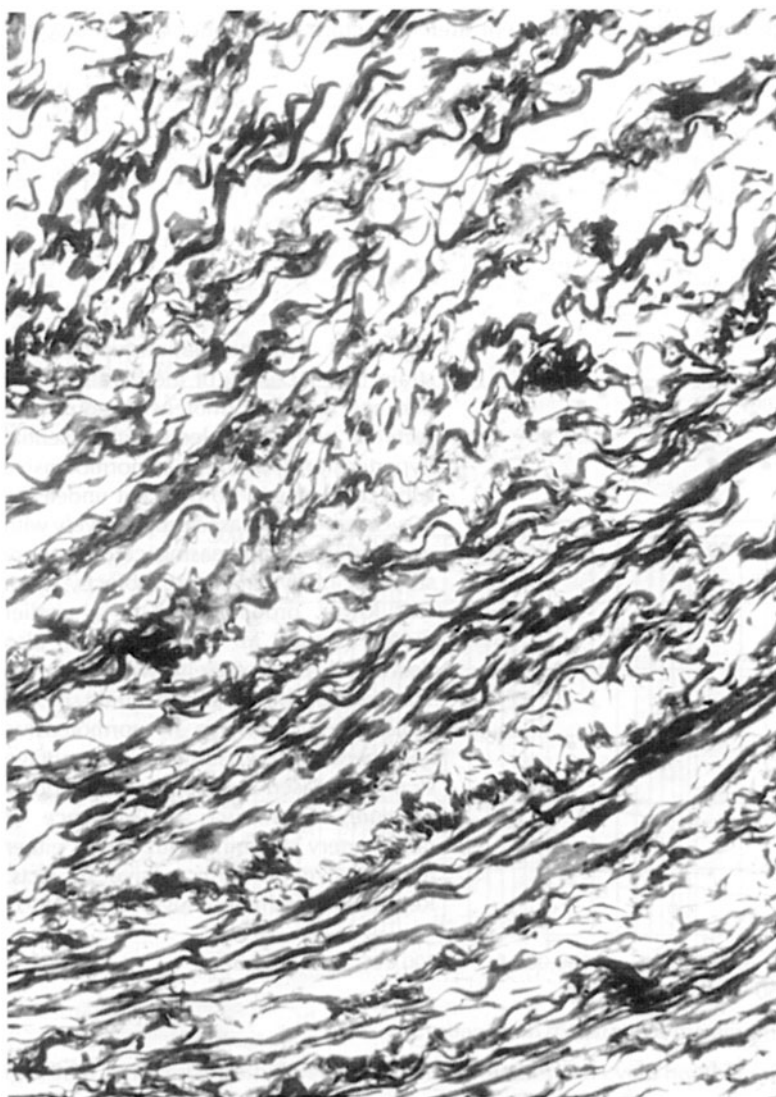


Figure 19.37 Elastin fiber fracture in the ascending aorta of a 27-year-old bird. Stain: Orcein and Giesen. Magnification $\times 415$. Prepared by J Yong. Compare this with Fig. 19.6, which shows similar changes in an elderly human

progressive fracture of elastin fibers from at least early adulthood in the central elastic, but not in other arteries. Data could also explain the increased prevalence of degeneration in hypertension, where (at least initially according to Boutouyrie *et al.* (1992)) greater pressure fluctuation could cause greater stretch, and the virtual absence in experimental animals, where in a lifespan nowhere near 1 billion cycles of stretch are ever achieved. Progressive dilatation of the affected arteries is an expected consequence of elastin fiber fracture, as is fibrous remodeling (Lam *et al.*, 2010). As expected, persistently elevated heart rate is associated with acceleration of aortic stiffening (Palatini and Parati, 2010).

These explanations, while logical and completely in accord with engineering principles, do not sit well with biologists, who are accustomed to dealing with continuous turnover in nature, with continuing repair and remodeling, and with a continuous influence of genetic processes on bodily activity throughout life. The question is often raised of how relevant these principles are to living organisms. We do not know the full answer, but it is fair to suggest that they cannot be irrelevant and must apply to the non-living components of the arterial wall. They would be less relevant if elastin had a high turnover rate, but available data indicate that elastin is very inert

and exists unchanged in the body for decades, if not the whole human lifespan (Fig. 19.36) (Rucker and Tinker, 1977; Kohn, 1978). Cardiac cells persist unchanged throughout a whole lifespan, albeit with turnover of chemical structure within. Arterial elastin persists unchanged chemically for decades, but must change physically in accordance with known physical laws related to applied stress. Elastin fracture is predictable on exposure to multiple cycles of stress and can account for arterial degeneration with age, and hence indirectly for the vast bulk of aging changes described in this chapter.

The theory presented is capable of testing. We predicted on the basis of theoretic principles that, while arterial medial degeneration may not be apparent in experimental animals, it must occur in birds, which have high arterial pressure, high arterial pulsation, and high heart rate, and which often live for decades. This process may be relevant to the propensity of turkeys to spontaneous aortic dissection and rupture (Simpson *et al.*, 1970). Figure 19.37 shows gross aortic medial degeneration with elastin fiber fracture in a 27-year-old bird with a predicted aortic pulsation of around 10 percent initially and a predicted cycle number around 3 billion. Figure 19.38 shows the ascending aorta in the same bird, dilated almost to the same diameter as the heart. The test

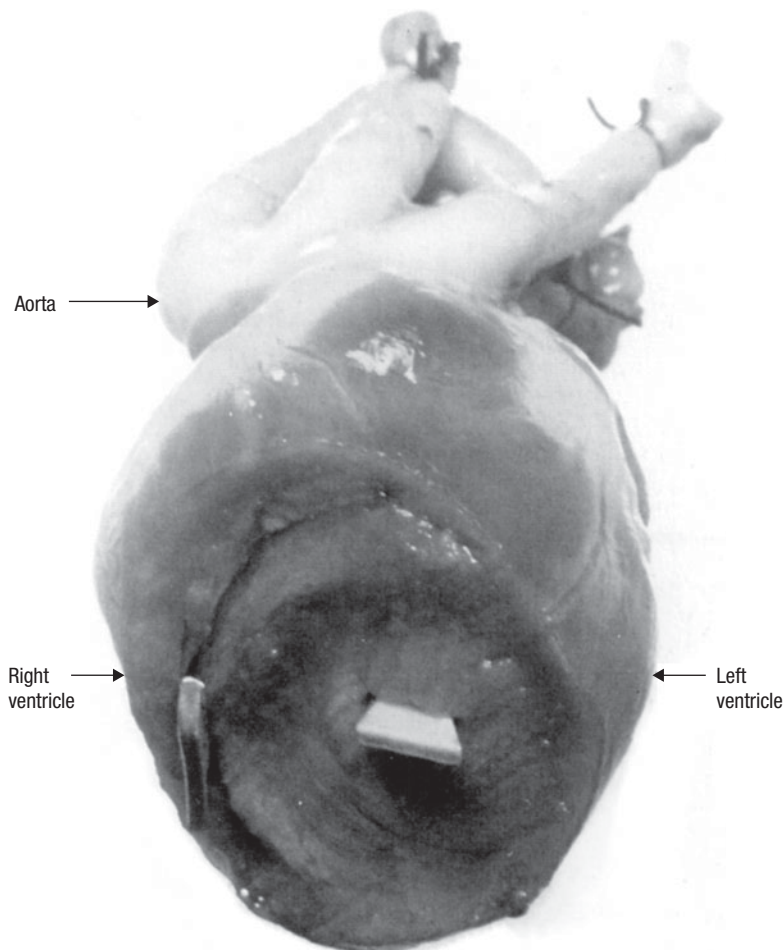


Figure 19.38 Dilated ascending aorta and central elastic arteries of the same 27-year-old bird seen in Fig. 19.34. Note severe left ventricular hypertrophy

will be extended to other long-living animals (crocodiles and turtles) with a low heart rate and low blood pressure. We predict that these will show no significant arterial elastin fiber change or aortic dilatation, in a lifespan of 100 years or more.

PERSPECTIVE

In this chapter, we began by quoting nineteenth century physiologists and physicians, and their views of arterial stiffening and its effects on the heart. In providing contemporary data, we have stressed the ill effects of fatigue and elastin fracture on the arteries over time as the cause of the aging change. This approach is not fashionable in an age which emphasizes genetic influences on biological phenomena, and on aging (Rusting, 1992; Ricklefs and Finch, 1995; Lakatta, 2003). While we have proposed a mechanism, we cannot account for the huge variation in elastic properties of arteries in the normal population at the one age, and we cannot fully explain why degeneration of arteries proceeds more quickly in some people than in others. Genetic factors must play a major role (Hanon *et al.*, 2001; Izzo, 2001; Lakatta, 2003), but the specifics have not yet been uncovered.

The great physician William Osler combined both views when describing arterial degeneration with age in

his famous textbook *The Principles and Practice of Medicine* (Osler, 1892; Osler, 1906). He wrote,

As an involution process, arteriosclerosis is an accompaniment of old age, and is the expression of the natural wear and tear to which the tubes are subjected. Longevity is a vascular question, which has been well expressed in the axiom that 'a man is only as old as his arteries'. To a majority of men, death comes primarily or secondarily through this portal. The onset of what may be called physiological arteriosclerosis depends, in the first place, upon the quality of arterial tissue (vital rubber) which the individual has inherited, and secondly upon the amount of wear and tear to which he has subjected it. That the former plays a most important role is shown in the cases in which arteriosclerosis sets in early in life in individuals in whom none of the recognized etiological factors can be found. Thus, for instance, a man of 28 or 29 may have the arteries of a man of 60, and a man of 40 may present vessels as much degenerated as they should be at 80. Entire families sometimes show this tendency to early arteriosclerosis – a tendency which cannot be explained in any other way than that in the make up of the machine ...

Hypertension

Basic mechanisms	449	Pressure wave contour and augmentation	459
Historical viewpoint	451	Left ventricular hypertrophy	461
Pathology	453	Therapy of hypertension	463
Studies of large artery stiffening in man	454	Epidemiology	466
Ascending aortic impedance	457		

These people appear to pass on through life pretty much as others do and generally do not suffer from their high blood pressures, except in their petty ailments upon which it imprints itself ... As age advances the enemy gains ascension of strength ... the individual has now passed 40 years, perhaps 50 years of age, his lungs begin to degenerate, he has a cough in the winter time, but by his pulse you will know him. Alternatively, headache, vertigo, epistaxis, a passing paralysis, a more severe apoplectic seizure, and then the final blow.

Mahomed, 1879

A separate chapter on hypertension in this book sits rather awkwardly with the authors. One has contributed an editorial review *Hypertension is a Myth* (O'Rourke, 1983a) to a national journal. We are strongly of the view, championed by Sir George Pickering (Pickering, 1968; Pickering, 1982), that hypertension is not a disease but a problem of magnitude, that high blood pressure, however measured or expressed, increases the load on the heart and the stresses on arteries, and accelerates cardiovascular degeneration and disease.

We have two major problems, one of which most clinicians are aware and the second that few recognize. The first problem is that classification of hypertension – systolic, diastolic, or systolic plus diastolic – is arbitrary and depends on criteria for the upper limit of 'normal' (Pickering, 1968; Burt *et al.*, 1995; Guidelines Subcommittee, 1999; Carretero and Oparil, 2000a; Carretero and Oparil, 2000b; Panza, 2001; Vasan *et al.*, 2001; Chobanian *et al.*, 2003; Kearney *et al.*, 2005; Mancia *et al.*, 2007b; Lloyd-Jones *et al.*, 2009). Our second problem is that this classification is set by the peak (maximum) and nadir (minimum) of the pressure wave in a peripheral

(brachial) artery into which the arterial pulse is amplified to a variable degree (Wilkinson *et al.*, 2001; McEniery *et al.*, 2005a; Laurent and Cockcroft, 2008; McEniery *et al.*, 2008). Implications of this phenomenon are discussed elsewhere with respect to exercise (Chapter 27), disease (Chapter 25), and therapy (Chapter 26). Aging effects are discussed in Chapter 19, where it is pointed out that in people over 70 years of age, and in small children, brachial (or radial) systolic pressure is a good guide to ascending aortic and left ventricular systolic pressure, whereas in adolescents and young adults, brachial (or radial) systolic pressure may overestimate central systolic pressure by 30–35 mmHg or more (Figs 20.1, 10.16, and 10.26) (Kroeker and Wood, 1955; Kroeker and Wood, 1956; Rowell *et al.*, 1968). Systolic pressure is recognized as an accurate index of cardiovascular risk in the elderly (Kannel *et al.*, 1981; Tuomilehto *et al.*, 1984; Dustan, 1989; MacMahon *et al.*, 1990), but a poor guide in the young (Black, 1999; Franklin *et al.*, 2001; Hyman and Pavlik, 2001). Differential amplification of the pulse (Figs 19.1, 19.2, and 20.1) readily explains why this is so. The apparent steep rise in arterial systolic pressure from age 10 to 20, then the plateau from 20 to 50 years, with the consistent rise thereafter (see Fig. 19.1) (US National Health Survey, 1977; Roy *et al.*, 1984; Risk Factor Prevalence Study Management Committee, 1989; Burt *et al.*, 1995; Uiterwaal *et al.*, 1997) is almost certainly due to the exaggeration of brachial and radial systolic pressure in relation to central systolic pressure, when the human body is fully grown, but aortic stiffness is still low (Vlachopoulos and O'Rourke, 2000b; Agabiti-Rosei *et al.*, 2007). An extreme of amplification can cause young (often tall) men to be diagnosed with 'isolated systolic hypertension'. Measurement of central aortic pressure can show that their aortic systolic and pulse pressure are normal, and they are considered to have 'spurious systolic

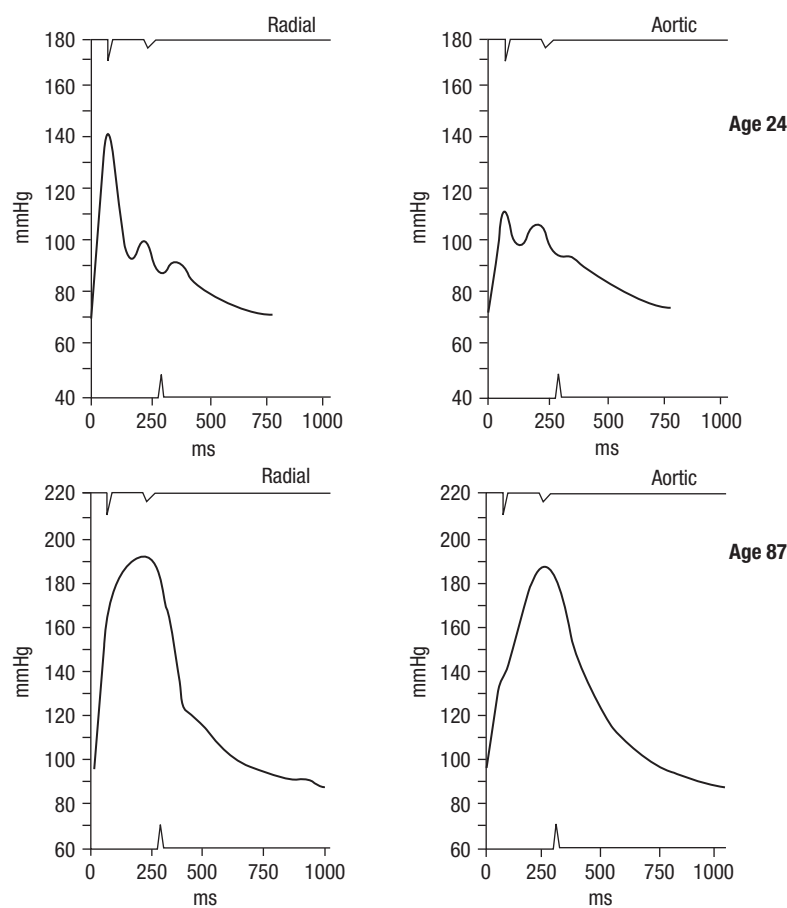


Figure 20.1 Ensemble-averaged pressure wave recorded in the radial artery (left) of a 24-year-old man (top) and an 87-year-old woman (bottom), and synthesized for the ascending aorta of the same subjects (right) using a generalized transfer function (O'Rourke *et al.*, 1993a)

hypertension' (O'Rourke *et al.*, 2000; Mahmud and Feely, 2003b; Hulsén *et al.*, 2006; Krzesinski and Saint-Remy, 2006). A similar condition of isolated systolic pressure elevation can be seen in young adults (often obese) with a hyperdynamic circulation and increased cardiac output (McEniery *et al.*, 2005b). Neither condition warrants pharmacological treatment as advised for elderly people, such as those studied in the SHEP trial (SHEP Cooperative Research Group, 1991).

These insights question seriously the current practice of diagnosing 'hypertension' on the basis of maximum systolic or minimum diastolic blood pressure in the brachial artery. In his seminal studies, conducted over 100 years ago, FA Mahomed (Mahomed, 1872; Mahomed, 1874; Mahomed, 1877; Mahomed, 1879; Mahomed, 1881) showed an awareness of this (Fig. 20.2). He diagnosed 'high blood pressure' not on the amplitude of the peak systolic pressure, but on the presence of the late systolic peak in the radial artery pressure wave: 'The tidal wave is prolonged and too much sustained' (Mahomed, 1874). Later nineteenth century physicians did the same, as did medical examiners for life insurance companies at the turn of the nineteenth century (Broadbent, 1890; Mackenzie, 1902; Postel-Vinay, 1996). Indeed, in the book *A Century of Arterial Hypertension*, published by the International Society of Hypertension to celebrate Riva-Rocci's centenary of the brachial cuff method to measure

systolic pressure, Postel-Vinay describes use of the pulse waveform by life insurance companies in France to accept or decline life insurance at the beginning of the twentieth century (Postel-Vinay, 1996).

We have a number of lesser problems. The first relates to minimum diastolic pressure, and the perception by many that this is directly related to peripheral resistance and arteriolar tone (Nichols and Edwards, 2001), and thus is a manifestation of hypertensive disease. This view is no longer tenable on the basis of the Framingham, MRFIT, SHEP, EWP, and other studies, which show the overwhelming importance of systolic pressure, particularly in the elderly in whom most cardiovascular events occur (Kannel *et al.*, 1981; Amery *et al.*, 1986; Rutan *et al.*, 1988; SHEP, 1991; SHEP, 1993; Franklin and Weber, 1994; He and Whelton, 1999; Kannel, 1999; MacMahon, 2000; Staessen *et al.*, 2000b; Franklin *et al.*, 2001; Vasan *et al.*, 2001; Pickering *et al.*, 2005a; Pickering *et al.*, 2005b; Safar and O'Rourke, 2006). However, while we are uncomfortable with the term 'hypertension', we are more uncomfortable still with other terms with official sanction – 'systolic' hypertension, 'diastolic' hypertension, 'labile' hypertension, 'borderline' hypertension – and by the view that distinct mechanisms prevail in these. The whole clinical approach is untidy, and the intellectual approach is lazy. The categorization is largely artificial and inaccurate. However, such is the strength of conviction among

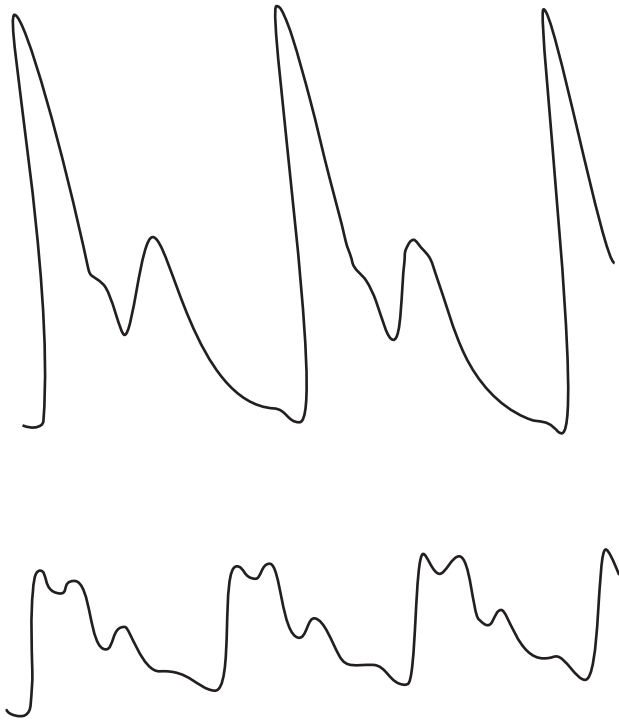


Figure 20.2 Sphygmographic tracings of radial artery pressure in a normal subject (top) and in a patient with hypertension and Bright's disease (bottom). Gain for the top tracing was set higher than for the bottom tracing. Reproduced from Mahomed (1874)

most clinicians that we are unlikely to achieve any desired change in approach unless we submit to such a degree as to accept, reluctantly, the status quo and the term 'hypertension' (O'Brien and Staessen, 1999; Chobanian *et al.*, 2003; Mancia *et al.*, 2007b; Vasan, 2009).

BASIC MECHANISMS

The basic problem in hypertension is the increase in peripheral resistance and decrease in arterial distensibility (Pickering, 1968; Kaplan, 1994; Lund-Johansen and Omvik, 1995; Ting *et al.*, 1995b; Safar, 2002; Simon, 2004). Resistance is also increased with aging, as a consequence of vascular rarefaction (fewer resistance vessels) and decreased arteriolar cross-sectional area (Hajdu *et al.*, 1990; Folkow and Svanborg, 1993; O'Rourke and Kelly, 1993); with low to normal cardiac output in the elderly. However, mean arterial pressure is usually only modestly elevated (see Chapter 19) (Lakatta and Levy, 2003a; Lakatta and Levy, 2003b) and tends to fall over the age of 60 (Franklin *et al.*, 1997).

In hypertension, cardiac output is normal or even initially high (Nichols *et al.*, 1986b; Izzard and Heagerty, 1995), so that mean arterial pressure is substantially increased (Lund-Johansen and Omvik, 1995) and peripheral resistance may be normal initially (Nichols *et al.*, 1986b;

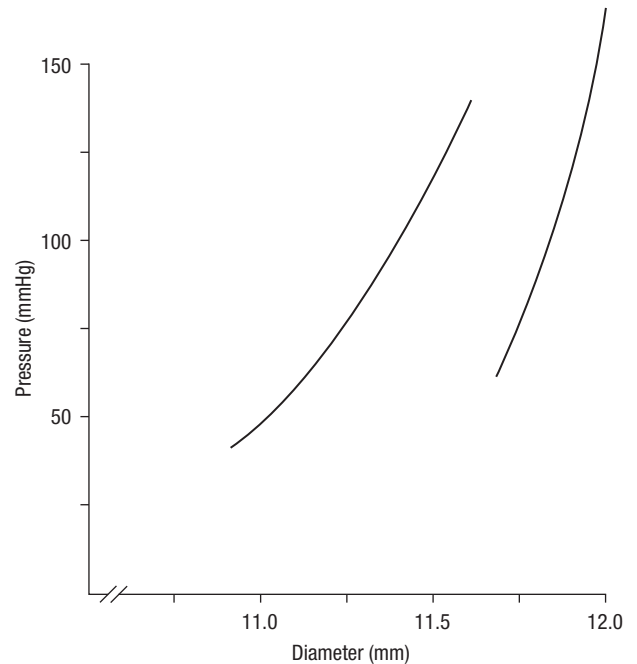


Figure 20.3 Pressure-diameter relationships shown schematically for an elastic artery of a normotensive subject (left) and from an age-matched patient with longstanding hypertension (right). After O'Rourke (1982b)

Carretero and Oparil, 2000a; Carretero and Oparil, 2000b; Vlachopoulos and O'Rourke, 2000b). Increase in mean arterial pressure causes a direct increase in diameter and stiffness of elastic arteries. As previously described in Chapter 4, the arterial wall behaves like a two-phase structure (Roach and Burton, 1957; Roach, 1977a; Armentano *et al.*, 1991; Armentano *et al.*, 1995). At low pressure, stresses are borne by the distensible elastin components, but as pressure increases, collagen fibers are progressively recruited (Figs 20.3 and 4.7). The pressure-diameter curve is non-linear, with convexity to the diameter axis (see Fig. 4.6). The artery is stiffer at higher rather than lower pressure. Thus, when blood pressure is elevated, as in hypertension, the central elastic arteries are larger and stiffer. This purely passive physical change causes elevation in ascending aortic impedance and in aortic pulse wave velocity (O'Rourke, 1970a; O'Rourke, 1976; Merillon *et al.*, 1982a; O'Rourke, 1982a; O'Rourke, 1982b; O'Rourke, 1985; Nichols *et al.*, 1986b; Ting *et al.*, 1986; Alexander *et al.*, 1989; O'Rourke, 1990a; O'Rourke, 1990b; Safar and O'Rourke, 2006; Laurent and Boutouyrie, 2007). The consequent changes are increase in amplitude of the initial (or incident) pressure wave generated by ventricular ejection and earlier return of the (reflected) pressure wave from peripheral sites, so that reflection returns to the proximal aorta in systole while the ventricle is still contracting and ejecting blood (see Chapter 9). Both these mechanisms contribute to increase the amplitude of the pressure pulse, to a disproportionate increase in systolic pressure (Figs 1.4, 13.8, and 20.4) and

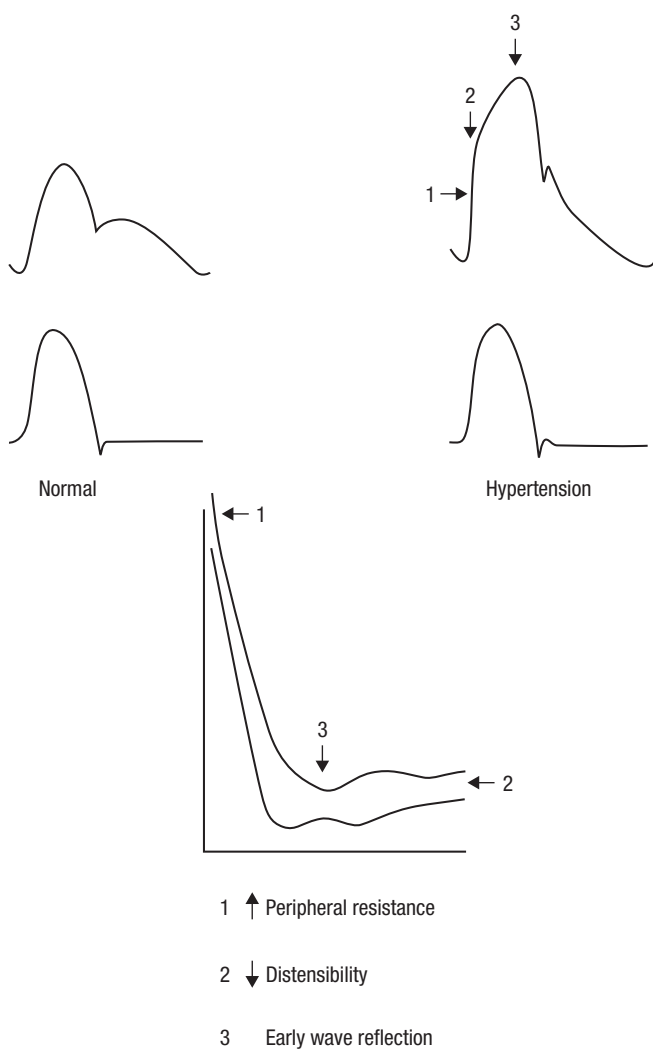


Figure 20.4 Effects of hypertension on aortic pressure wave contour (top) and on aortic impedance modulus (bottom). (1) Increased peripheral resistance causes increased mean pressure and an increased zero-frequency component of impedance. (2) Decreased arterial distensibility causes increased amplitude of the initial pressure peak and increased characteristic impedance. (3) Earlier return of reflected waves from arterial terminations causes the late systolic pressure peak and impedance curve to shift to the right. Reproduced from O'Rourke (1976)

to a relative reduction in pressure throughout diastole (O'Rourke, 1970a; Safar, 1989; O'Rourke, 1990a; O'Rourke, 1990b; O'Rourke and Kelly, 1993; Safar, 2002; O'Rourke and Nichols, 2005; Namasivayam *et al.*, 2009a). At low frequencies, the reflection coefficient also is increased (Fig. 20.5) (Alexander *et al.*, 1989; Chen *et al.*, 1995; Ting *et al.*, 1995a; Ting *et al.*, 1995b). These changes are acute and reversible and are readily seen in the ascending aorta of experimental animals during aortic occlusion and infusion of vasoconstrictor agents (Fig. 10.31). The characteristic feature of the arterial pressure pulse contour with hypertension is an increase in amplitude of the secondary late systolic or 'tidal' wave (see Figs 1.4, 20.1, 20.2, and 20.4) (Mahomed, 1877; Mahomed, 1881; O'Rourke, 1970a; Nichols *et al.*, 1977a; O'Rourke, 1982b; Nichols *et al.*, 1993; Safar *et al.*, 2003). A rise in mean pressure is necessarily associated with an increase in pulse pressure; this is exaggerated if, as in many young people with hypertension, cardiac output and stroke volume are increased as well (Nichols *et al.*, 1986b; Safar *et al.*, 1987a; Izzard and Heagerty, 1995; McEniery *et al.*, 2005b).

When hypertension is sustained, arterial degeneration is accelerated and changes similar to those seen with aging occur prematurely (O'Rourke, 1982a; O'Rourke, 1982b; Weisfeldt *et al.*, 1992; Nichols *et al.*, 1993; Benetos *et al.*, 2002; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b). From the point of view of the large arteries, hypertension can be looked upon as an accelerated form of aging (or aging as a premature form of hypertension). Similar pathologic changes are seen in the arterial walls with hypertension, but at an earlier age (Carlson *et al.*, 1970; Wolinsky, 1972; Gerstenblith *et al.*, 1977; Merillon *et al.*, 1982a; Folkow and Svanborg, 1993; Lakatta, 1993a; Lakatta, 1993b; Nichols *et al.*, 1993; Taddei *et al.*, 1997). The central elastic arteries stiffen and dilate to a greater degree than in normotensive people of similar age (O'Rourke, 1970a; Simon *et al.*, 1979a; Simon *et al.*, 1979b; Berger and Li, 1990; O'Rourke, 1990a; O'Rourke, 1990b; Simon *et al.*, 1991b; Nichols *et al.*, 1992). The more peripheral muscular arteries are spared from medial degenerative change, just as they are with aging (Boutouyrie *et al.*, 1992; Hayoz *et al.*, 1992; Benetos *et al.*, 1993a; Benetos *et al.*, 1993b; Laurent *et al.*, 1993; Mulvany, 1993;

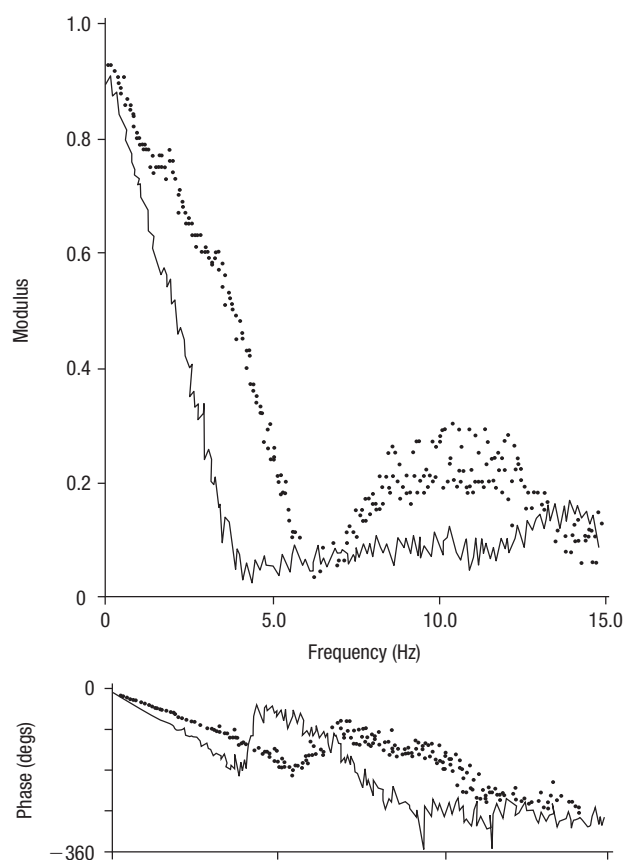


Figure 20.5 Reflection coefficient spectra measured at mean arterial pressures of 99 mmHg (solid line) and 168 mmHg (dotted line). Reproduced from Alexander *et al.* (1989)

McEniery *et al.*, 2005a; McEniery *et al.*, 2007), but these vessels suffer accelerated intimal change, with intimal hyperplasia, accelerated atherosclerosis (Roman *et al.*, 1992b; Weisfeldt *et al.*, 1992), and impaired endothelial function (Giannattasio *et al.*, 1997; Taddei *et al.*, 1997; Safar, 2002; Lind, 2006; Wallace *et al.*, 2007).

Hypertension begets hypertension (Byrom and Dodson, 1948; Pickering, 1968; Byrom, 1969; Ledingham, 1995). Elevated arterial pressure causes reactive vascular changes in the kidneys that have the effect of reducing renal blood flow and causing the release of renin, together with increased reabsorption of salt (Kaplan, 1992; Goldblatt, 1995). Renin catalyzes angiotensin I formation under the influence of angiotensin-converting enzyme (ACE). This forms angiotensin II, a potent vasoconstrictor and a stimulus for the production of aldosterone. The latter promotes further salt retention and increased blood volume.

Through these mechanisms, a rise in arterial blood pressure causes a further rise in arterial blood pressure, consequent on an increase in peripheral resistance from angiotensin II and increased cardiac output from volume overload. These are the principal mechanisms responsible for the progressive increase in arterial pressure in the absence of treatment and ultimately for its severe and

malignant phase (Byrom and Dodson, 1948; Pickering, 1968; Ledingham, 1995).

HISTORICAL VIEWPOINT

Many current controversies related to hypertension are set in perspective, and so are better understood, when considered in a historical context. There have been major changes in approach to this condition (with the modern approach revisiting the original) and, more significantly, a change in the presentation and progression of the disease in affluent countries. A worldwide consensus has emerged on the need to identify and treat people with elevation of arterial pressure before vascular or cardiac damage occurs (Chobanian *et al.*, 2003; ESH/ESC Guidelines Committee, 2003; ISH Writing Group, 2003; Mancia *et al.*, 2007a; Mancia *et al.*, 2007b; Vasan, 2009).

Richard Bright (1827) was the first to draw attention to elevated arterial blood pressure. His observations were based entirely on clinical and pathologic findings. He drew attention to 'hardness of the pulse' and its association with albuminuria and left ventricular hypertrophy, together with risk of stroke and heart failure. While most of Bright's cases were of glomerulonephritis, it is now recognized that some too were a consequence of essential hypertension (Pickering, 1982; O'Rourke, 1992). This view was first put forth by FA Mahomed, who in 1872 developed the first quantitative sphygmograph and described the natural history of essential hypertension and the characteristic features of the arterial pulse (see Figs 10.34, 20.1, 20.2, and 20.4) (Mahomed, 1872; Mahomed, 1874; Mahomed, 1877; Mahomed, 1879; Mahomed, 1881). He wrote:

what has been the cause in one case may be the result in another; thus general disorder may cause high arterial pressure, and this, in its turn kidney changes; while on the other hand kidney changes may be primary and acute and they may in their turn produce impurity of blood and this general (high) pressure. But whether we read the tale backwards or forwards it is the same tale in the end.

From the 1880s, sphygmographs were widely used in clinical practice and for illustrating disease processes and the therapeutic effects of drugs in journals and textbooks (see Figs 20.2 and 26.6). The influence of the cuff sphygmomanometer, introduced in 1896 and perfected in 1905, was not felt until the 1920s. One of the most prominent physicians of this period was Sir James Mackenzie, who wrote a number of highly influential books and used sphygmograms and cuff sphygmomanometers in his practice. In the first and second editions of his major text, Mackenzie cautioned against uncritical acceptance of numbers obtained with the sphygmomanometer, 'I write this so you may avoid some of the fallacies and short cuts

that at present are hindering the pursuit of a very important subject' (Mackenzie, 1917). The third edition of this text (Mackenzie, 1926) appeared after Mackenzie's death and contained the very type of simplistic advice that he had previously argued against: 'As regards the relative importance of systolic and diastolic pressures, it may be said that the systolic pressure represents the maximum force of the heart, while the diastolic pressure measures the resistance the heart has to overcome.' This statement must be attributed not to Mackenzie himself but to James Orr, Mackenzie's literary executor, and appears to have begun the legend that systolic pressure is benign (since it is a manifestation of a vigorous heart) and diastolic pressure is dangerous (since it is a manifestation of increased arteriolar tone).

Life insurance companies played a major role in popularizing the cuff sphygmomanometer. Follow up of screened and accepted clients was complete – the policy was redeemed – and the ill effects of raised diastolic and systolic pressure were confirmed. The first convincing evidence on the relationship between blood pressure and mortality was Fischer's presentation to the Society of Actuaries in 1917 (Table 20.1). This established a relationship between mortality and brachial systolic pressure. By the 1930s, elevated blood pressure was established as the major asymptomatic risk factor for premature death. The emphasis, however, was on diastolic pressure, and followed Mackenzie's perceived advice (rather than Fisher's data), together with the more ready determination of consistent diastolic than systolic pressures under conditions of clinical screening. An additional factor, apparent now, is the marked discrepancy between brachial and aortic systolic pressures in young adults, such as those taking up life insurance at the time (see Figs 19.1, 19.2, and 20.1) (Vlachopoulos and O'Rourke, 2000b). Nonetheless, mortality was found to be related to systolic as well as to diastolic pressure in the insurance publications

of the 1940s (see Pickering, 1968). Around this period, hypertension was a disease of middle-aged and young people and caused by increased peripheral resistance.

The seminal studies of Goldblatt in the 1930s drew attention to the importance of renal vascular disease in the development of some forms of hypertension, and the progression of all forms with time (Goldblatt, 1947; Byrom and Dodson, 1948; Ledingham, 1995; Vasan, 2009). Sir George Pickering initiated his studies in the 1930s and brought perspective and balance to the clinical and experimental arena. His classic book, properly and deliberately entitled *High Blood Pressure* (a companion simplified text was called *Hypertension*), first published in 1955, stressed pathophysiology, mechanisms, and a quantitative approach, as had Mackenzie 50 years before (Pickering, 1968; Pickering, 1982). Byrom, working in Sydney's St Vincent's and Prince Henry Hospitals, established the basis of malignant hypertension and the vicious circle of high blood pressure acting through the kidney (Byrom and Dodson, 1948; Byrom, 1954; Byrom, 1969). This work, acclaimed by Pickering in 1968 and by Laragh and Brenner (1995), confirmed the relevance of Goldblatt's work in the progression of hypertension. The studies of these three, Goldblatt, Pickering, and Byrom, were properly recognized in the only three chapters given to historical background in Laragh and Brenner's epic text (Goldblatt, 1995; Ledingham, 1995; Pickering, 1995).

Effective therapy for hypertension was first introduced in the 1940s, initially for hypertensive heart failure (Smirk and Alstad, 1951), then for malignant hypertension. Such therapy was not available to Franklin Delano Roosevelt, whose accelerated hypertension was not treated and led to his fatal stroke in 1945 (Calhoun and Oparil, 1995; Messerli, 1995). Demonstration that lowering blood pressure could resolve fibrinoid necrosis, confirmed Byrom's research and Pickering's views. Over the past 50 years, new, more effective, and better tolerated anti-hypertensive

Table 20.1 Summary of the mortality experience of the NW Mutual Life Insurance Co., with respect to systolic blood pressure

Period	No. risks		Ages (inc.)	BP (mmHg)		Other imp't	Mortality to Aug. 1st 1915. MA table			
	Acc'td.	Rej'td.		Range	Av.		100%	200%	300%	%
Aug. 1st 1907 to	2630		40–60	140–149	142	–				93
Aug. 1st 1910	521		40–60	150–160	152	–				127
		302	40–60		170	None				250
		288	40–60		171	One or more				302
Aug. 1st 1907 to	1274		40–60		160	None				220
Aug. 1st 1915	956		40–60		165	One or more				263
Nov. 1911 to Aug. 1915	495		16–39		150	None				142
Low BP										
1907–1910 (inc.)	200		40–60	105 and under	–					47
	427		40–60	106–110	–					65
Nov. 1911 to Aug. 1915	433		16–39	100 and under	–		2 deaths			
	60		40–60		–		No deaths			

Source: Abstracts of the Proceedings of the Association of Life Insurance Medical Directors of America (1917). From Postel-Vinay, 1996.

drugs have been successively introduced by a pharmaceutical industry that has gained and contributed immensely. The pharmaceutical industry has played a major role in changing the natural history of hypertension – through providing drugs which can break the vicious circle acting through renal damage – and has exerted considerable influence on the medical profession as well.

In recent years, the natural history of hypertension has changed. As a consequence of early recognition and effective therapy, hypertensive disease is rarely a problem in the young, and malignant hypertension, presenting as eclampsia, hypertensive encephalopathy, acute left ventricular failure, or acute renal failure, is virtually unknown in affluent Western countries. Essential hypertension and isolated systolic hypertension are principally problems of the aged, and the major etiological factors are capillary rarefaction and increased arterial stiffness, with a small decrease or no change in cardiac output; the major consequence is elevation in systolic blood pressure with little change in diastolic blood pressure. This is discussed in Chapter 19.

This change in natural history and clinical presentation over time is described by Harriet Dustan (1989) as follows:

It is really not surprising that up until this time the focus was on diastolic hypertension. From a 1989 perspective, it is hard to realize what a serious problem diastolic hypertension was before the era of anti-hypertensive drug therapy. This condition affected middle-aged people at the height of their productivity. It either killed them suddenly by cerebral haemorrhage or slowly and miserably by cardiac or renal failure. No one was concerned about elderly people with ISH (isolated systolic hypertension), particularly those who had lived longer than 70 years. However, when the effectiveness of anti-hypertensive drug treatment for diastolic hypertension was shown, the stage was set for a change in attitude towards ISH.

It is interesting to note that all the causes of death mentioned above are more robustly related to systolic blood pressure than to diastolic blood pressure (Port *et al.*, 2000; Franklin *et al.*, 2001; Safar, 2001; Safar, 2002; Chobanian *et al.*, 2003; Mancia *et al.*, 2007b).

PATHOLOGY

This book has no brief to deal with causes of hypertension. Aortic coarctation is dealt with in Chapter 24. The topics to be considered are those caused directly or indirectly by elevated arterial blood pressure.

Left ventricular hypertrophy is a consequence of volume or pressure overload (Katz, 1990; Frohlich *et al.*, 1992a; Frohlich *et al.*, 1992b; Levy *et al.*, 1996). There is no hyperplasia (increase in cell number), just an increase in

the size of existing myocytes. Such hypertrophy has the effect of maintaining stress. The heart does pump more vigorously, but it is set at a disadvantage, since the myocardial vascular network does not develop to the same extent, and ischemia is produced more readily (Weber *et al.*, 1992; Verdecchia *et al.*, 2001) because of decreased coronary blood flow reserve and increased myocardial oxygen demand. With the passage of time, myocardial cells degenerate and weaken so that contractile failure ultimately ensues (see Chapter 14) (Katz, 1990; Katz, 1998). A further problem with the hypertrophied heart is impaired diastolic relaxation. This is a major factor in hypertensive heart failure (Bonow and Udelson, 1992; Brutsaert *et al.*, 1993; Federmann and Hess, 1994; Mathew *et al.*, 2001; Wachtell *et al.*, 2002; Redfield, 2004; Abhayaratna *et al.*, 2006a; Weber *et al.*, 2006; Weber *et al.*, 2008; Borlaug *et al.*, 2010; Paulus, 2010). It is assumed that reduction in left ventricular hypertrophy by anti-hypertensive therapy is a desirable goal in therapy. Studies in progress are addressing this question; to date, it has not been fully established (Pfeffer and Pfeffer, 1990; Devereux, 1997; Gottdiener *et al.*, 1997; Lip, 2001; Mathew *et al.*, 2001; Wachtell *et al.*, 2002), but evidence is growing (Abhayaratna *et al.*, 2006a; Weber *et al.*, 2006; Chirinos *et al.*, 2009b).

Atherosclerosis is accelerated by hypertension, as are the complications arising therefrom. These are discussed in Chapter 23. Atherosclerosis can potentiate hypertension if it involves a renal artery; such a problem may have been responsible for President Franklin D Roosevelt's accelerated hypertension causing death from cerebral hemorrhage (Calhoun and Oparil, 1995).

Degenerative medial changes to the aorta and large predominantly elastic arteries are discussed in Chapter 19. These are most marked in hypertensive people at any age and are attributable to the fatiguing effects of cyclic stress (Prokop *et al.*, 1970; Gent, 1972; Sandor, 1972; O'Rourke, 1976; O'Rourke, 1982b; Kaplan, 1994). These changes can be considered as accelerated aging. As with aging, they are not apparent in the peripheral, predominantly muscular arteries (Hayoz *et al.*, 1992). The most extreme form of medial degeneration is cystic medial necrosis, where there is breakdown of elastin and collagen fibers in the aortic wall and collection of hyaline mucoid material. This is the basic cause of dissecting aortic aneurysm (Isselbacher, 2001) and is most common in elderly hypertensive subjects and in people with an inherited defect of elastin fibers (such as Marfan syndrome, see Chapter 24).

Hypertension causes hypertrophy of smooth muscle cells in the peripheral arteries and arterioles (Folkow, 1990; Mulvany, 1992; Mulvany, 1993; Izzard and Heagerty, 1995; Laurent and Boutouyrie, 2007). The significance of this is unclear. Were hypertrophy not to occur, these vessels would be more vulnerable to acute rises in pressure and to fibrinoid necrosis (see below). It is thought that such hypertrophy may contribute to increased vascular reactivity and to clinical syndromes associated with this.

It is also thought that this process may potentiate hypertension by decreasing arterial and arteriolar caliber. Work (Mulvany, 1992; Korsgaard *et al.*, 1993; Izzard and Heagerty, 1995) has shown that most of these mural changes are in the small conduit arteries upstream from the smallest vessels, which provide the greatest contribution to peripheral resistance and are responsible for the maintenance (and elevation) of mean arterial pressure.

Hypertension is a crucial etiological factor in cerebral hemorrhage and infarction – so much so that, in population studies, stroke incidence is taken as a measure of the prevalence of hypertension (Pickering, 1968; Kaplan, 1994). The basic cause of most cerebral hemorrhage, and of some (lacunar) infarcts, is degeneration in the media of small cerebral arteries (Fisher, 1969; Russell, 1975), attributable to hypertension. Hemorrhage usually arises from rupture of a small aneurysm of the type originally described by Charcot and Bouchard in 1868 (Russell, 1963; Pickering, 1968). These aneurysms are widespread in the brains of elderly hypertensive people. They are asymptomatic until they rupture. The aneurysmal wall is defective in collagen, elastin, and muscle fibers. Breakdown in these components appears to occur as a consequence of fatigue and fracture from repetitive cyclic expansion, and in regions of fiber weakness. Once developed, wall tension increases with increasing diameter according to the law of Laplace. Rupture is probably inevitable if life is sufficiently prolonged. Hypertension plays a major role in their genesis and ultimately in their rupture (Pickering, 1968; Kannel *et al.*, 1981; Kaplan, 1994). Increased aortic stiffness is a major factor (see Chapter 18) (O'Rourke and Safar, 2005; Payne *et al.*, 2010).

Lacunar infarcts are very common in the brains of older hypertensive subjects, and are responsible for minor cerebral events and for gradual intellectual deterioration (see Chapters 18 and 19) (Russell, 1975; Fisher, 1982; Lodder *et al.*, 1990; Ikeda *et al.*, 1994). These infarcts are due to thrombosis in small perforating arteries that show medial and intimal changes referred to as 'lipohyalinosis'. This appears to be primarily a medial degeneration of elastin fibers, collagen fibers, and smooth muscle. It is not atherosclerosis, and appears to be as common in Eastern as in Western populations with a different prevalence of atherosclerosis. It appears to be caused by hypertensive mechanical damage to the vascular wall, with secondary thrombosis in the lumen (Russell, 1975). The precise cause is not fully established, but its association with hypertension particularly, and with age, is very clear (see Chapter 18). Control of hypertension results in fewer lacunar infarcts. As with Charcot-Bouchard aneurysms, reduction of blood pressure markedly decreases the incidence of stroke from this cause (SHEP, 1991; Kaplan, 1992; Yusuf *et al.*, 2000; Klungel *et al.*, 2001; Dahlöf *et al.*, 2002; Spence, 2004). Most episodes of major cerebral infarction are caused by cerebral atherosclerosis or embolism from the carotid artery, aorta, or heart (Lodder *et al.*, 1990). In contrast to cerebral hemorrhage and

lacunar infarcts, such events are not markedly and immediately reduced by anti-hypertensive therapy, nor is embolism from the heart.

The one vascular event in hypertension which is usually associated with age is fibrinoid necrosis of small arteries. This is the basis of malignant hypertension. Byrom's work (Byrom and Dodson, 1948; Byrom, 1954; Byrom, 1969) clarified the mechanism and explained the form of the lesions themselves, their location in regions where an artery is dilated (rather than constricted), and their occurrence (apart from the malignant phase of hypertension) in weakened arteries (in connective tissue disease) and atrophic arteries (downstream from aortic coarctation) when arterial pressure is normal or only modestly elevated (Pickering, 1968; Ledingham, 1995). Byrom and Dodson (1948) described the mechanism as follows:

The sequence may be most readily explained by assuming that an intravascular tension bordering on the limits of tolerance leads directly to sudden local overstretching of the vascular wall, the overstretched muscle undergoing necrosis and lysis. Plasma and blood may then seep into the weakened zone to be converted into a fibrinoid plug by thromboplastin liberated from the dead muscle, and an inflammatory infiltration would be expected to follow in the adventitia as a first step towards repair.

Although overlooked initially, and ignored by many who perceived a chemical toxic injury to the wall, this elegantly simple explanation is now generally accepted (Pickering, 1968; Ledingham, 1995). Many still emphasize immunological, inflammatory and genetic factors in acute arterial damage, but these are probably a consequence rather than the primary cause (Verlohren *et al.*, 2009).

Fibrinoid necrosis of small arteries and arterioles is a consequence of acute elevation of blood pressure, with acute tear of smooth muscle. The artery may, but usually does not, rupture. This age-independent acute process contrasts with the chronic age-dependent process of elastin fiber fracture in the aorta, consequent on one billion or more cycles acting over decades (see Chapter 19). The lipohyalinosis of intermediate-size cerebral arteries may represent the effect of a similar phenomenon to that causing fibrinoid necrosis in the smaller vessels (see Chapter 18) (Russell, 1975). The thesis of Byrom is applicable to the cerebral and renal small arteries in the elderly, hence it is set out in full in Chapter 18.

STUDIES OF LARGE ARTERY STIFFENING IN MAN

The earliest *in vivo* studies by Bramwell and Hill (Bramwell and Hill, 1922a; Bramwell and Hill 1922b) were conducted through non-invasive determination of the

wave foot of sphygmograms, with calculation of pulse wave velocity. This technique of measuring arterial stiffness was resurrected in the late 1970s and 1980s (Gribbin *et al.*, 1976; Gribbin *et al.*, 1979; Avolio *et al.*, 1983; Avolio *et al.*, 1985a) and has indeed been shown to give very useful and highly reproducible results under different conditions (Asmar *et al.*, 1995b; Blacher *et al.*, 1999a; Asmar *et al.*, 2001b; Laurent and Boutouyrie, 2007; Mancina *et al.*, 2007b). Aortic stiffness has been shown to be an independent predictor of progression to hypertension in non-hypertensive subjects (Dernellis and Panaretou, 2005). Gribbin *et al.* (Gribbin *et al.*, 1976; Gribbin *et al.*, 1979) showed that increase in brachial artery pulse wave velocity in hypertension was reversible when blood pressure was normalized by the application of external pressure; there was no structural defect attributable to age or hypertensive disease, although there was some variability from person to person. Change in wave velocity in these studies was due to change in distending pressure. These findings were the basis for the introduction by the Casio Company of a wristwatch with the facility to measure arterial pressure. Findings of consistent wave velocity in the upper limb by Ho (1982), Avolio *et al.* (Avolio *et al.*, 1983; Avolio *et al.*, 1985a) and by Chen *et al.* (1997), and of consistent transmission characteristics with age (Gallagher, 1993), were the basis for using a generalized transfer function for the determination of the ascending aortic pressure from the radial pressure wave contour (see Chapter 30) (O'Rourke *et al.*, 1992b; Chen *et al.*, 1997; Pauca *et al.*, 2001). In the Casio watch, change in pressure from a preset individual value is calculated from the delay of the finger pulse, utilizing lead II of the ECG and finger photoplethysmograph through a pair of sensors, one electrical, the other optical, on the watch face.

These studies of pulse wave velocity complement data available from arteries studied *in vitro* after death (see Chapter 4). They have been complemented by non-invasive studies of arterial diameter change, utilizing ultrasound or techniques such as magnetic resonance imaging and computed tomography (Mao *et al.*, 2008; Lam *et al.*, 2010; Redheuil *et al.*, 2010). Results have not been entirely consistent or reproducible for determining arterial pressure–diameter relationships since arterial pressure has usually not been determined at the same site as the diameter. Such an oversight can bias results (O'Rourke, 2002a; Laurent and Boutouyrie, 2007). Direct invasive studies of pressure and diameter are more reliable and can be conducted in association with routine diagnostic cardiac catheterization (Merillon *et al.*, 1978; Stratos *et al.*, 1992). Similar non-invasive studies in humans have provided pulsatile pressure–flow relationships from which ascending aortic impedance could be determined (Kelly and Fitchett, 1992; Mitchell *et al.*, 2004a; Segers *et al.*, 2007b; Mitchell *et al.*, 2008; Mitchell, 2009b; Chirinos and Segers 2010a; Chirinos and Segers 2010b; Mitchell *et al.*, 2010). Results from all these studies, although sometimes giving conflicting results for other

conditions, have been reasonably consistent for patients with hypertension. They have, in the main, confirmed differences between central and peripheral arteries and the resulting pulse pressure amplification in hypertensive subjects (Avolio *et al.*, 2009b). Calibration of carotid pressure waves have, however, been a source of controversy (see Chapter 28) (O'Rourke and Takazawa, 2009; Segers *et al.*, 2009; O'Rourke and Adji, 2010b).

Aorta and central elastic arteries

In hypertensive subjects, the elastic modulus of the ascending aorta is greater than in age-matched controls (Fig. 20.6) (Gozna *et al.*, 1974; Merillon *et al.*, 1976; Stratos *et al.*, 1992; Nichols *et al.*, 1993). Similar data have been found for the arch and descending aorta and for the carotid arteries (Ting *et al.*, 1990; Boutouyrie *et al.*, 1992; Roman *et al.*, 1992; Benetos *et al.*, 1993b; van Merode *et al.*, 1993; Asmar *et al.*, 1995a; Asmar *et al.*, 1995b; Roman *et al.*, 1995a; Roman *et al.*, 1995b). Central elastic arteries of hypertensive patients are in general stiffer and dilated in comparison with age-matched controls. Treatment of hypertension leads to a passive decrease in diameter and stiffness, as one might predict (see Fig. 20.3), but the arteries may remain more stiff than those of age-matched controls (Asmar *et al.*, 1995b), indicating the presence of greater arterial medial degeneration in hypertensive subjects. It is usually not possible to attribute improvement in stiffness of the central arteries to the active effects of therapy (Stratos *et al.*, 1992), since

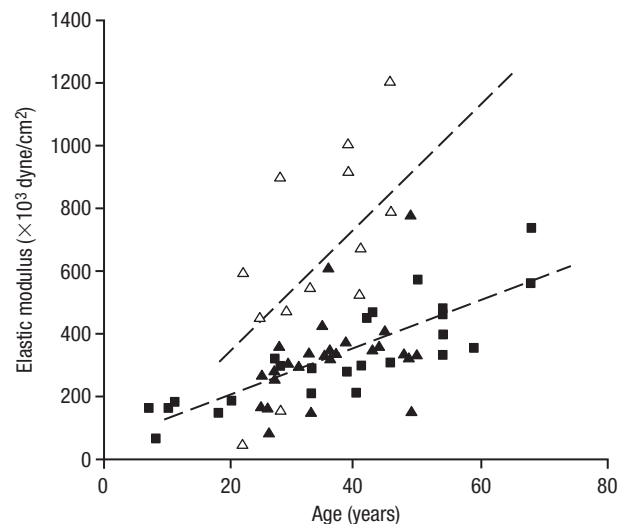


Figure 20.6 Age-related change in the pressure–strain elastic modulus of the ascending aorta of normotensive (■, ▲) and hypertensive (△) subjects. The elastic modulus increases with advancing age and is higher in hypertensive subjects compared with normotensive subjects of the same age. Broken lines are linear regression lines drawn through data points obtained from hypertensive and normotensive subjects. Adapted from Nichols *et al.* (1993)

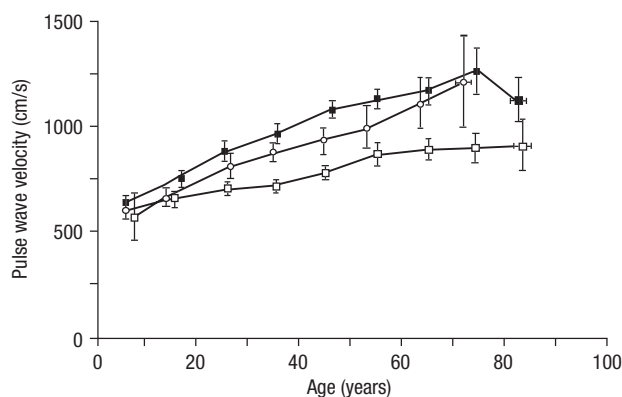


Figure 20.7 Comparison of age-related changes in pulse wave velocity measured in normal subjects of Beijing (■—■), Sydney (○—○), and Guangzhou (□—□). After the first decade, the pulse wave velocity of rural Guangzhou subjects is significantly lower than that of Beijing and Sydney subjects. Serum cholesterol levels were similar in Beijing and Guangzhou subjects. Urinary sodium excretion, however, was significantly higher in Beijing subjects (Avolio *et al.*, 1985a). Pulse wave velocities in Sydney subjects were similar to those measured in subjects with high and low serum cholesterol levels (Avolio *et al.*, 1986). These data indicate that arterial pulse wave velocity (and stiffness) increases with age and that sodium (rather than cholesterol) influences an increase in pulse wave velocity at a given age

diameter reduction usually accompanies a decrease in arterial pressure and stiffness (Asmar *et al.*, 2001b; Nichols and Edwards, 2001; London *et al.*, 2004b). Reduction in arterial pressure no matter how brought about, usually leads to decrease in carotid-femoral pulse wave velocity (PWV) of 0.5–1.0 m/s per 10 mmHg (Asmar *et al.*, 2001b; Hirata *et al.*, 2005b). In hypertensive patients whose arterial stiffness is not reduced by therapy, the artery may remain dilated, so that absolute volume change in any segment (compliance) may, in contrast to stiffness, normalize (Benetos *et al.*, 1991a; Asmar *et al.*, 1995b).

Similar data for pulse wave velocity along the aorta have been determined in hypertensive subjects by Merillon *et al.* (Merillon *et al.*, 1976; Merillon *et al.*, 1982a) and by Ting *et al.* (1990). Avolio *et al.* (1985a) showed lower aortic pulse wave velocity in a population with a low prevalence of hypertension compared with populations with a high prevalence of hypertension (Fig. 20.7).

Peripheral muscular arteries

Peripheral muscular arteries (e.g. femoral, brachial, radial) have been studied extensively by ultrasonic techniques, by Safar, Brunner, Hayoz, Hoeks, Mancia, Bank, and colleagues in Paris, Lausanne, Maastricht, Milan, and Minneapolis. Special problems, such as spontaneous arterial vasomotion (Hayoz *et al.*, 1995) and pressure measurement, have been identified and largely overcome. Data have been consistent and quite different from data from

the aorta, carotid, and other central, predominantly elastic arteries. Stiffness is directly related to change in blood pressure but is less marked than in central arteries. Any increase in stiffness in hypertensive subjects always returns to the normal range when pressure is reduced to the normal range; there is no residual increment in stiffness as often observed for central arteries. Furthermore, muscular arteries dilate in hypertensive subjects such that their compliance (absolute volume change with change in pressure) may appear to be normal or even increased in hypertensive subjects (Simon *et al.*, 1982; Safar *et al.*, 1983; Simon *et al.*, 1983a; Simon *et al.*, 1983b; Levenson *et al.*, 1984b; Simon *et al.*, 1984; Boutouyrie *et al.*, 1992; Hayoz *et al.*, 1992; Levenson *et al.*, 1992; Mulvany, 1992; Benetos *et al.*, 1993b; Laurent *et al.*, 1993; Boutouyrie *et al.*, 1994; Laurent *et al.*, 1994).

In contrast to the aorta and large central elastic arteries, many vasodilating drugs have a powerful effect on the peripheral muscular arteries, such that they dilate even though arterial pressure falls as a consequence of the peripheral effect of the drug on arterioles. This is a direct effect of the drug and not necessarily due to increased flow within the artery. Powerful arterial vasodilatation has been identified for nitrates, calcium antagonists, and ACE inhibitors, but not for pure arteriolar dilating drugs, such as hydralazine, dihydralazine, and prazosin (Simon *et al.*, 1982; Safar *et al.*, 1983; Simon *et al.*, 1983b; Levenson *et al.*, 1984a; Levenson *et al.*, 1984b; Simon *et al.*, 1984; Arcaro *et al.*, 1991a; Arcaro *et al.*, 1991b; Asmar *et al.*, 1993; Chaignon *et al.*, 1993; Bank *et al.*, 1995). Vasodilatation has been shown too for pindolol, nebivolol, and carvedilol, but not propranolol or atenolol (Maarek *et al.*, 1986; McEniery *et al.*, 2004; Polónia *et al.*, 2010). Conventional diuretics have no direct effect on these arteries (Smulyan *et al.*, 1984; Levenson *et al.*, 1992).

Vasodilatation in muscular arteries is associated with a decrease in arterial stiffness such that the wide-caliber artery shows even greater diameter fluctuations than when caliber is normal, for the same pulsatile pressure fluctuation within (see Fig. 26.2) (Simon *et al.*, 1982; Safar *et al.*, 1983; Simon *et al.*, 1983b; Levenson *et al.*, 1984a; Simon *et al.*, 1984; Asmar *et al.*, 1993; Chaignon *et al.*, 1993; Fitchett, 1993a). All the arterial dilating drugs – nitrates, calcium channel antagonists, ACE inhibitors, angiotensin receptor blockers (ARBs), and β -blockers with α -blocking activity have thus been shown to decrease stiffness of peripheral muscular arteries. This is in contrast to central elastic arteries, where such effects are not usually apparent and where any increase in diameter, as caused by increase in distending pressure, is associated with an increase in stiffness. An explanation has been offered (Fig. 20.8) that, in peripheral arteries, muscle fibers are in series with collagen and in parallel with elastin, such that a decrease in muscular tone transfers stress from collagen to elastin fibers. The peripheral dilated artery is less stiff, whereas any dilated central artery is more stiff (Chapter 4) (O'Rourke and Avolio, 1986a;

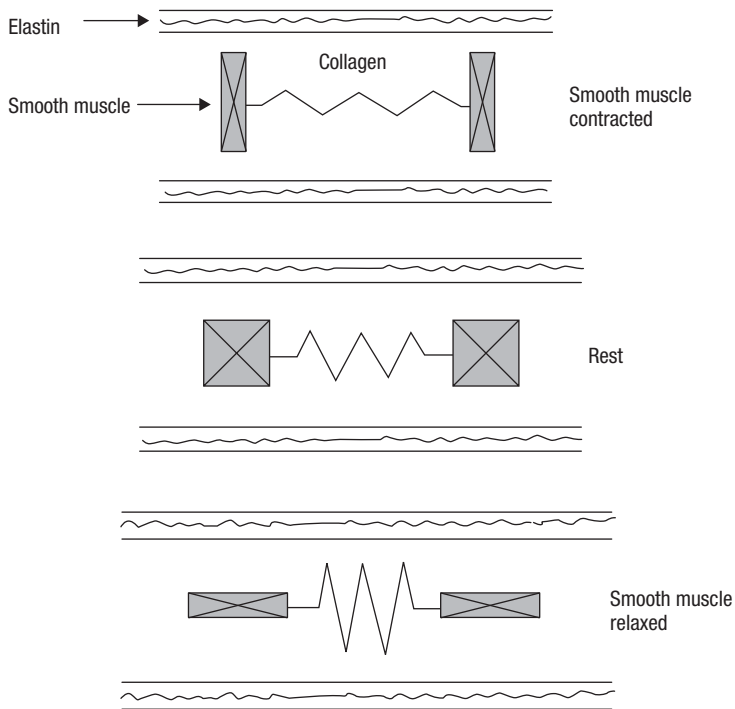


Figure 20.8 A proposed model for the arrangement of collagen, elastin, and smooth muscle in the arterial wall. The muscle is in series with the collagen elements, both are in parallel with the elastic lamellae. Contraction of muscle transfers stress to the collagenous elements in the wall and renders the wall stiffer, whereas muscular relaxation transfers stresses to the elastic lamellae, making the wall more distensible

O'Rourke and Avolio, 1986b). Usually, in the intact circulation, vasodilating drugs cause a decrease in peripheral resistance and mean arterial pressure. The decrease in arterial pressure, in turn, causes a passive decrease in diameter of large elastic arteries. If the decrease in diameter is great enough, arterial stiffness decreases. The opposite is true for vasoconstricting drugs. In large central arteries it appears, therefore, as though vasodilating drugs usually decrease stiffness and vasoconstricting drugs increase stiffness. These changes in arterial caliber and stiffness are, however, passive changes due to changes in arterial pressure.

These studies have drawn attention to the differential effects of vasodilator drugs at different sites. Differential effects on the caliber of coronary arteries have been described (see Fig. 17.7) (Feldman *et al.*, 1981; Wheatley *et al.*, 1994), as have differences in receptor responsiveness (Kutz *et al.*, 1991). Nitroglycerin has little or no direct action on the aorta (Latson *et al.*, 1988), but it has powerful dilator effects on the peripheral muscular arteries, such as the brachial and radial, and even more powerful effects on the smaller conduit arteries downstream (Fitchett, 1984; Smulyan *et al.*, 1986; Bank *et al.*, 1995), but no significant effect on arterioles unless given in high dose (Jiang *et al.*, 2002) or directly into the upstream artery. In contrast, dipyridamole, a powerful arteriolar dilator, has no significant effect on the larger conduit muscular arteries.

Smallest arteries

It is generally acknowledged that arterial vessels less than 150 μm in diameter are responsible for the major part of

peripheral resistance, and that arterial vessels of larger caliber are probably irrelevant to a change in peripheral resistance in hypertension (Mulvany, 1993; Izzard and Heagerty, 1995; Schofield *et al.*, 2002b). However, the larger small arteries appear to be a therapeutic target for anti-hypertensive therapy since, while hypertrophied in hypertension, they are sensitive to vasodilating drugs, such as nitrates, and can undergo substantial changes in caliber and distensibility in response to nitrate and other drug therapy. Such a change in caliber and distensibility is the only way to explain the type of digital throbbing, increased photoplethysmographic pulse signal, and increased distensibility of arterial vessels in an extremity when nitrates and other vasodilating drugs are administered (Avolio, 2002). These arterial changes, even with unaltered peripheral arteriolar tone and resistance, appear responsible for the attenuation of wave reflection from vessels downstream, such that the magnitude of the reflected wave to the aorta, left ventricle, and central arteries is reduced (Chapter 26) (Yaginuma *et al.*, 1986). This effect, attributed to Yaginuma (Yaginuma *et al.*, 1986), to greater dilation of daughter branches compared to parent vessel throughout an arterial bed has the effect of 'trapping' reflections in the peripheral circulation and impairing their return to the heart (Fig. 26.2).

ASCENDING AORTIC IMPEDANCE

Ascending aortic input impedance spectra in hypertensive patients show the changes predicted from theoretic principles and animal studies (see Figs 20.4 and 20.9) (Pepine *et al.*, 1978; Pepine *et al.*, 1979; Merillon *et al.*, 1982a;

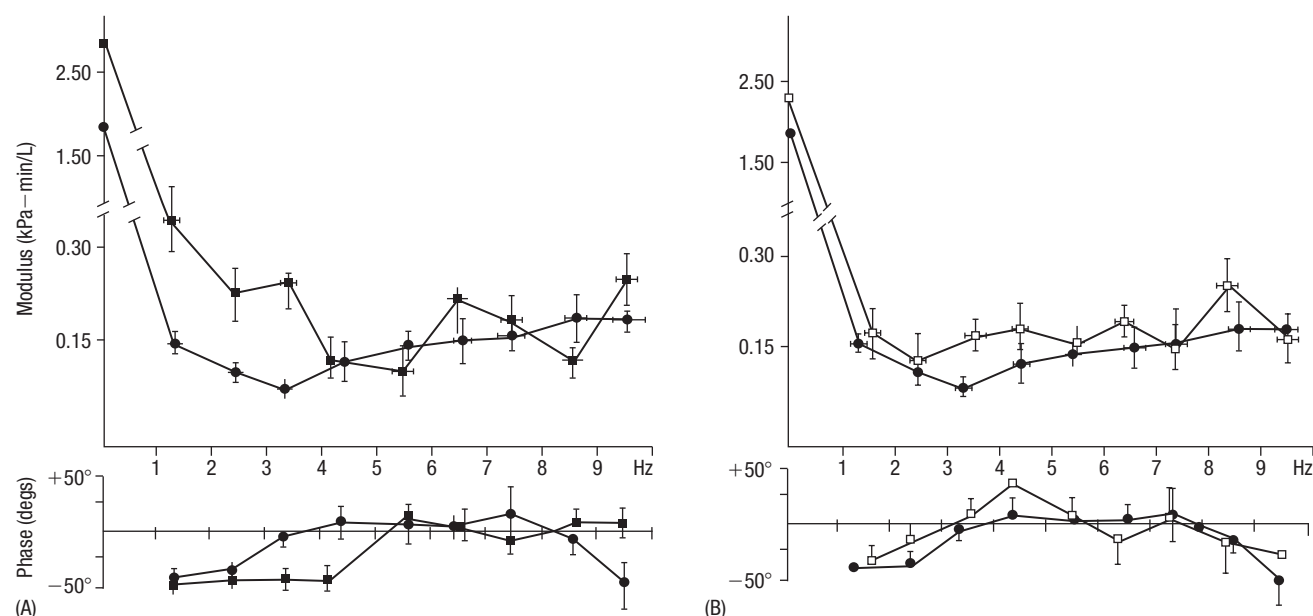


Figure 20.9 (A) Ascending aortic input impedance modulus and phase in a group of normal subjects (●-●) and in a group of hypertensive subjects of similar age (■-■). (B) Change in ascending aortic impedance of hypertensive patients with infusion of nitroprusside to restore normal mean arterial pressure (□-□). Closed circles represent data from normal subjects as in (A). Reproduced from Merillon *et al.* (1982a)

Merillon *et al.*, 1982b; Merillon *et al.*, 1982c; Nichols *et al.*, 1985a; Nichols *et al.*, 1986a; Ting *et al.*, 1986; Zuckerman and Yin, 1989a; Ting *et al.*, 1991; Nichols *et al.*, 1993; Ting *et al.*, 1993; Chen *et al.*, 1995; Ting *et al.*, 1995a; Ting *et al.*, 1995b). Spectra of impedance moduli are displaced upwards and to the right, as a consequence of decreased aortic compliance (increased characteristic impedance) and of early return of reflected waves from the periphery to the heart, consequent on increased aortic pulse wave velocity. Values of characteristic impedance in hypertensive patients are greater than those of age-matched controls (Fig. 20.10). Aortic input impedance data are similar to those in the elderly (see Figs 19.19 and 19.21). The shift in impedance spectra upwards and to the right with hypertension and with aging causes a disproportionate increase in the impedance modulus (Z_1) at the frequency of the first harmonic component of the pulse (Figs 19.21 and 20.9), such that this is increased far more than characteristic impedance or aortic pulse wave velocity. In Fig. 20.11, values of Z_1 in hypertensive patients over 50 years of age were five times greater than those of normotensive people under 30 years of age. Such values of Z_1 are in line with the impedance plots in Figs 20.9 and 19.19, and can explain the increase in amplitude of systolic augmentation and of pulse pressure seen in normal humans with age, and the gross increase that can be seen in older humans with hypertension (Fig. 20.12, page 460).

Merillon *et al.* (1982a) showed that ascending aortic impedance abnormalities in hypertension could be virtually normalized through treatment with intravenous nitroprusside (see Fig. 20.9). Decrease in peripheral

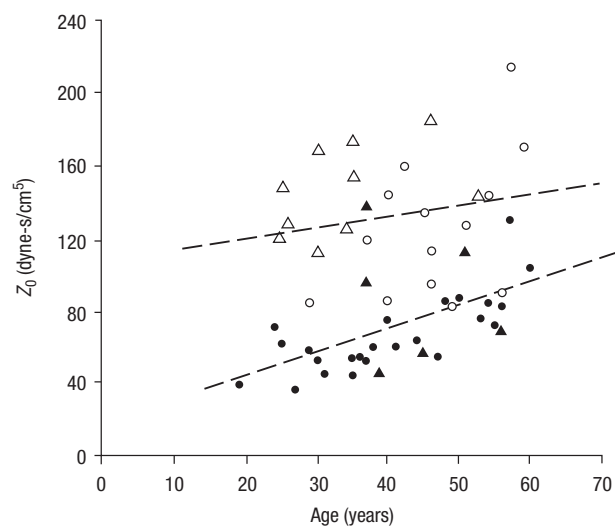


Figure 20.10 Age-related change in characteristic impedance (Z_0) of normotensive (▲, ●) and hypertensive (△, ○) subjects. Z_0 increases with age and is higher in hypertensive subjects than in normotensive subjects of the same age. From Nichols *et al.* (1993)

resistance and in mean arterial pressure were associated with a decrease in Z_1 and characteristic impedance, and a delay of travel and a decrease in amplitude of wave reflection, consequent on decreased aortic pulse wave velocity. In a series of reports from Taipei and Baltimore, Chen *et al.* (1995) and Ting *et al.* (1995a) confirmed Merillon's original work but have shown in addition that

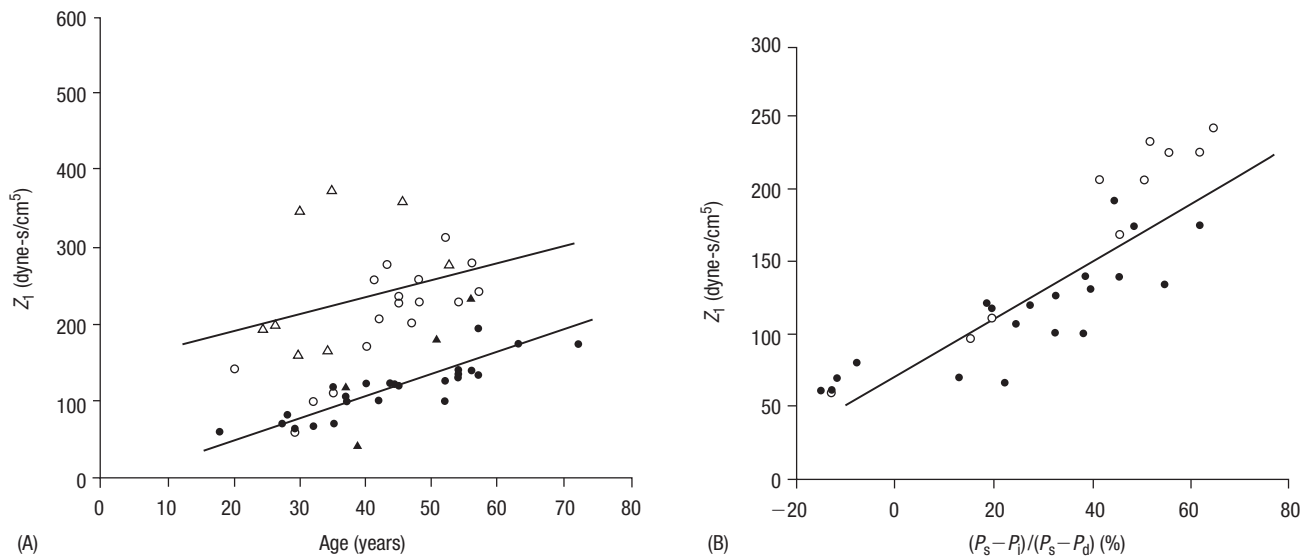


Figure 20.11 Age-related changes in the amplitude of the fundamental impedance modulus (Z_1) in normotensive ($\blacktriangle$, $\bullet$) and hypertensive ($\triangle$, $\circ$) subjects (A). Lines are linear regression lines drawn through data points obtained from hypertensive and normotensive subjects. Z_1 increases with age and is greater in hypertensive subjects compared with normotensive subjects of the same age. Z_1 is also directly related to the aortic augmentation index (B). From Nichols *et al.* (1993)

wave reflection intensity is increased in hypertension and is further increased by the β -blocking agent propranolol. These workers showed that the ill effects of hypertension on ascending aortic impedance are ameliorated but are not normalized by the α -blocking agent phentolamine, nor by an ACE inhibitor (Ting *et al.*, 1986; Ting *et al.*, 1991; Ting *et al.*, 1993; Ting *et al.*, 1995a). Ting *et al.* (1995b) and Yaginuma and colleagues (Noda *et al.*, 2006) have shown similar favorable effects with nifedipine.

PRESSURE WAVE CONTOUR AND AUGMENTATION

There has quite naturally been keen interest to apply the information obtained from invasive studies to the monitoring and treatment of patients with hypertension. Similarity of the grossly augmented ascending aortic pressure wave in Fig. 20.12 with that recorded non-invasively in the carotid artery (see Fig. 20.1) is apparent. Augmentation of the pressure wave ('the tidal wave is prolonged and too much sustained') was the principal characteristic of the arterial pressure pulse noted by Mahomed 140 years ago, and he was aware that the augmented or 'tidal' wave was of even greater amplitude in central arteries than were those conventionally recorded in the radial artery (Fig. 10.34) (Mahomed, 1874). Nichols *et al.* (1993) related the aortic impedance moduli Z_1 of the first harmonic of the pulse to the degree of augmentation and showed a close correspondence for people both with and without hypertension (see Fig. 20.11). People with high augmentation were usually older. When viewed in relation to age, augmentation of the ascending aortic

pressure wave was greater than that of normotensive controls when expressed in absolute or in relative terms (Fig. 20.13). As noted in Chapter 19 (see Fig. 19.22), Merillon *et al.* (1982a) have shown a close relationship between ascending aortic characteristic impedance and an invasively determined index of left ventricular hypertrophy; this relationship applied irrespective of age and irrespective of the absence or presence of hypertension.

Studies have been conducted in which the systolic pressure augmentation gauged from the carotid artery was used to assess the degree of reduction in central (left ventricular and aortic) systolic pressure. In a comparative study of atenolol and diltiazem (a vasodilating β -blocking agent), Kelly *et al.* (1989c) showed a benefit for the latter drug corresponding to decrease of 5–8 mmHg more than measured conventionally in the brachial artery. Pannier *et al.* (1994) and Chen *et al.* (1995) obtained similar results for the ACE inhibitor fosinopril and calcium antagonist lacidipine, respectively, while Asmar *et al.* (2001b) and Hirata *et al.* (2005b) showed similar results with an indapamide/perindopril combination and with ramipril, respectively.

The progressive increase in augmentation index with age, and left ventricular mass with age, has been noted (see Chapter 19) (Shimada *et al.*, 1994). It is to be expected that augmentation would be causally related to left ventricular hypertrophy. The relationship has been confirmed by Saba *et al.* (1993) using carotid augmentation and Hashimoto *et al.* (Hashimoto *et al.*, 2007a; Hashimoto *et al.*, 2007b; Hashimoto *et al.*, 2008a; Hashimoto *et al.*, 2008c) using ascending aortic augmentation, and by Roman *et al.* (2009) for aortic systolic and pulse pressure with carotid intima-medial thickness

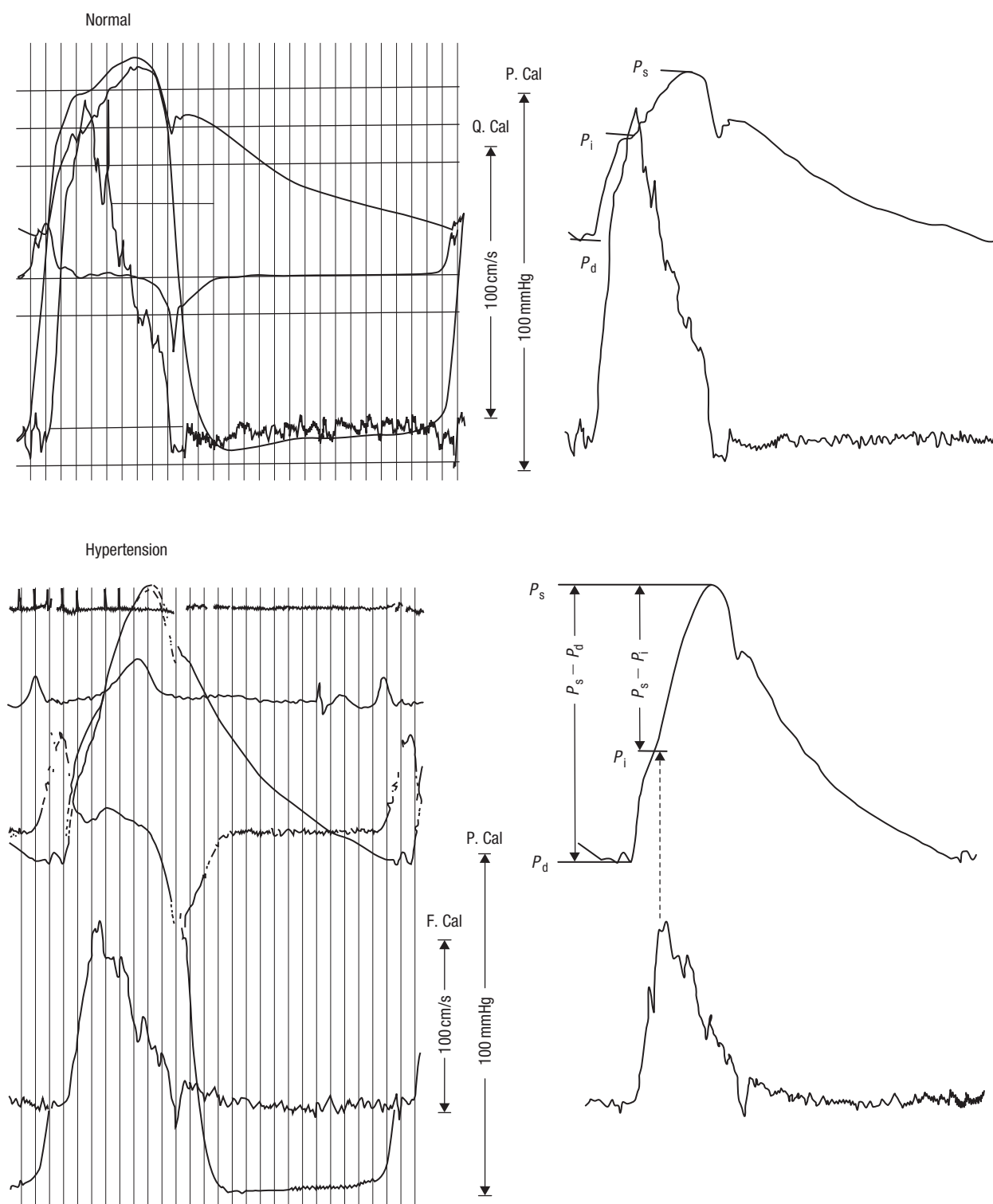


Figure 20.12 Aortic pressure and flow velocity wave shapes in patients with normal blood pressure (above) and hypertension (below). Reflection wave ratio is calculated as $(P_s - P_i)/(P_s - P_d)$. P_d , diastolic pressure; P_i , pressure at peak flow velocity; P_s , peak systolic pressure. Recorded data at left; aortic pressure and flow redrawn at right. Reproduced from Yaginuma *et al.* (1985)

and left ventricular hypertrophy, respectively, in the Strong Heart study. This group from Cornell Medical Centre in New York have also established a relationship between carotid augmentation and carotid intima-medial

thickness, and have suggested that carotid augmentation may play a useful role in detecting carotid disease, as well as left ventricular hypertrophy. A similar finding with respect to risk of stroke was made by Kannel in 1981 from

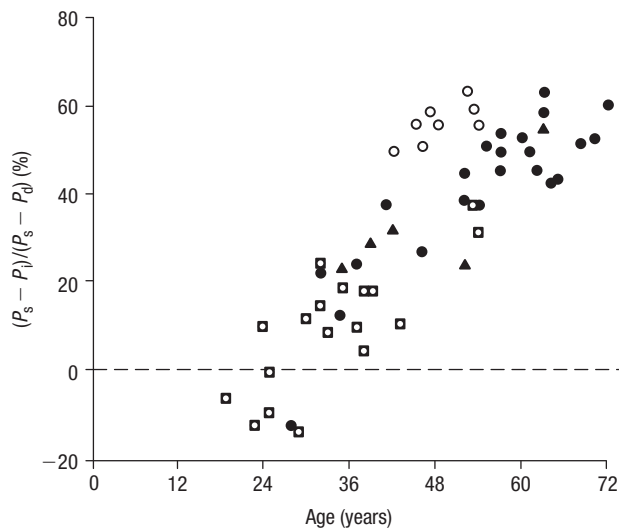


Figure 20.13 Age-related changes in augmentation (or reflection) index for the ascending aorta of normotensive ($\square$, $\blacktriangle$, $\bullet$) and hypertensive ($\circ$) subjects. The augmentation index increases with age and is higher in hypertensive subjects than in normotensives of the same age. Data from Murgo *et al.* (1980a), O'Rourke *et al.* (1968), and Nichols *et al.* (1993)

pulse contour data obtained in the Framingham study. Boutouyrie *et al.* (1995) have also established a close relationship between carotid artery stiffness and left ventricular hypertrophy. Vaitkevicius *et al.* (1993) have shown an inverse relationship between carotid augmentation and capacity for exercise, measured by maximum oxygen uptake (see Chapter 19).

This subject is pursued in Chapter 30 with respect to arterial tonometry. Here, the carotid augmentation index is shown to bear a close relationship to the ascending aortic augmentation index, although the former consistently underestimates the latter (Fig. 19.24). In Chapter 30, it is also shown that ascending aortic augmentation can be determined from the radial pressure pulse utilizing the relatively constant pulse wave velocity and transfer function for pressure in the upper limb (see Chapter 19) (Figs 19.29 and 19.30). This is certainly the most convenient way to estimate ascending aortic augmentation in relative and absolute terms. Another method, popularized in Japan (Takazawa, 1987; Takazawa, 1993a; Takazawa, 1993b; Takazawa *et al.*, 1998), utilizes the finger photoplethysmogram. This has shown a consistent relationship between ascending aortic augmentation and an index derived from the second derivative of the arterial pressure measured with the finger plethysmograph (Fig. 20.14, page 462).

These methods provide useful information on aortic properties, on left ventricular load and on the degree of left ventricular hypertrophy. The subject of augmentation is discussed elsewhere in relation to aging, drug therapy, and cardiac failure. Miyashita *et al.* (Miyashita *et al.*, 1994; Miyashita *et al.*, 1995) have stressed that augmentation of

the left ventricular, aortic, and carotid pulse, while due to early wave reflection, is dependent for its full expression on a powerfully contracting left ventricle. When ventricular function is impaired, the notch on the flow wave associated with augmentation of pressure is enhanced. Miyashita *et al.* have shown an inverse relationship between this boost to pressure and the corresponding reduction in aortic flow (see Chapter 14). Westerhof and O'Rourke (1995) and Denardo *et al.* (2010) have taken this further to show that pressure augmentation is completely absent when heart failure is severe, and where wave reflection causes premature termination of ventricular ejection, and thus decreases stroke volume (see also Chapter 14). Hence, studies of ventricular, aortic, and carotid augmentation, together with associated phenomena in the peripheral pressure pulse and photoplethysmogram, must take into account the presence or absence (and the severity if present) of cardiac failure, as well as other factors, including heart rate and ejection duration, when seeking to interpret data on arterial pressure wave augmentation.

LEFT VENTRICULAR HYPERTROPHY

Left ventricular hypertrophy in hypertension is attributable to prolonged increased left ventricular external arterial load over many years (Frohlich *et al.*, 1992a). In older subjects, it is better related to systolic than to diastolic pressure, and better related to ambulatory than to casual blood pressure recordings (Devereux *et al.*, 1995; Mancia *et al.*, 2007b). It is related to increased stiffness of the aorta and central elastic arteries (Bramwell and Hill 1922b; Milnor, 1975; Weisfeldt, 1980; Bouthier *et al.*, 1985; Roman *et al.*, 1993; Saba *et al.*, 1993; Boutouyrie *et al.*, 1994; Roman *et al.*, 1995a; Roman *et al.*, 1995b; Saba *et al.*, 1995b; Devereux, 1997; Roman *et al.*, 2009). The process of hypertrophy normalizes tension within myocardial fibers, which is desirable in the short term, but it also leads to greater increase in cell volume than cell surface area (which prejudices the capacity to gain nutrients and dispose of metabolites) and it is not associated with corresponding growth of vascular supply, potentiating a propensity to ischemia. Physiologic hypertrophy over years becomes pathologic hypertrophy where the cells have the capacity to develop less rather than more tension, and to maintain contraction longer, relax more slowly, and are susceptible to ischemia, especially with tachycardia, and even in the presence of normal epicardial coronary arteries (Katz, 1990; Kannel and Belanger, 1991; Frohlich *et al.*, 1992a; Gould, 1997; Verdecchia *et al.*, 2001; Verdecchia *et al.*, 2003; Weber *et al.*, 2004; Weber *et al.*, 2006; Weber *et al.*, 2008; Chirinos *et al.*, 2009b).

Left ventricular hypertrophy is usually measured clinically by echocardiography. Many studies have confirmed the presence of left ventricular hypertrophy as a predictor of future cardiovascular events, especially

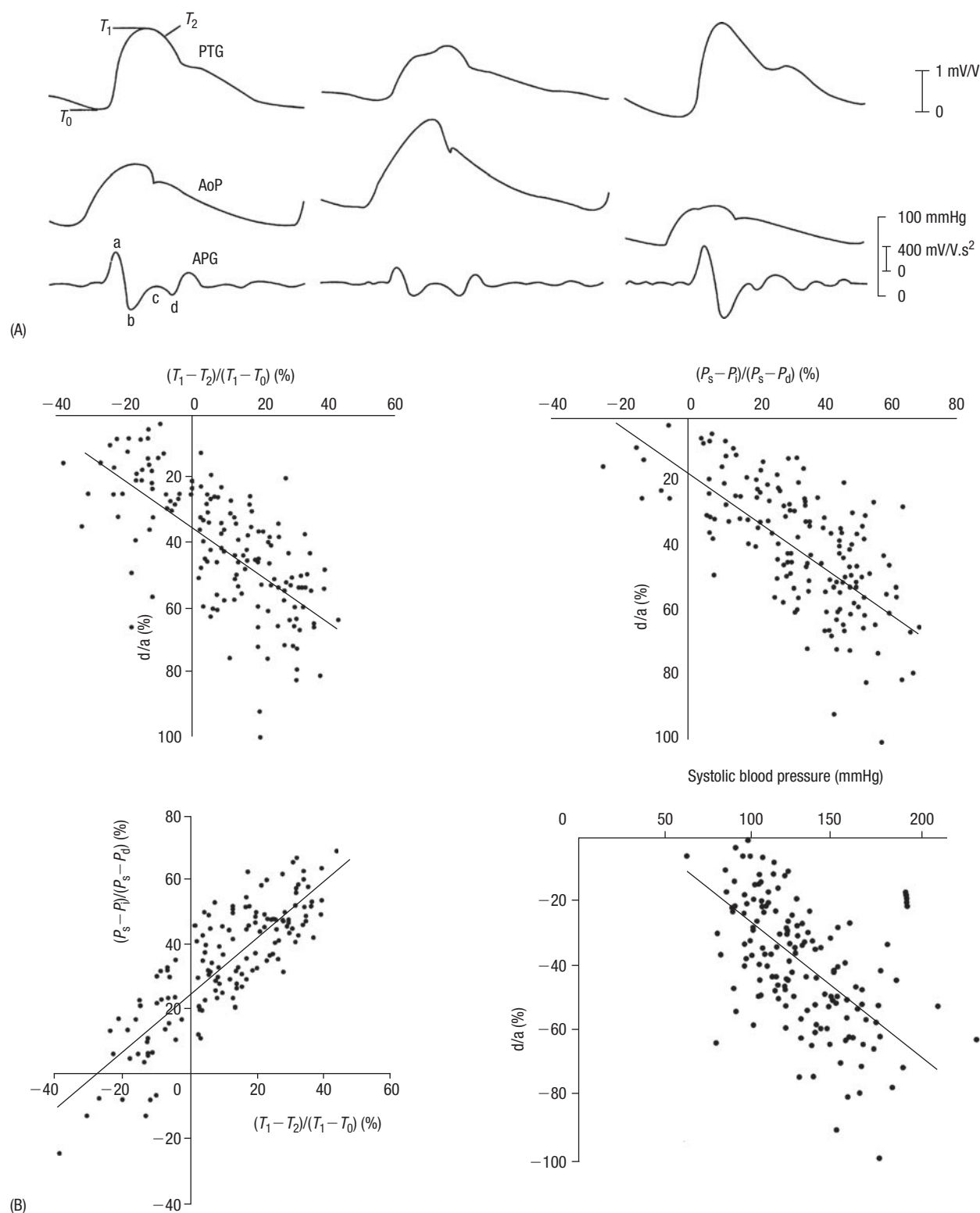


Figure 20.14 (A) Tracings of finger plethysmograph (PTG), aortic pressure (AoP), and 'acceleration plethysmograph' (APG – (second derivative of finger plethysmograph)) in a 64-year-old patient under control conditions, and with infusion of angiotensin (AGT, center) and nitroglycerin (NTG, right). Waves a, c and b, d are the discrete positive and negative systolic waves of the APG, respectively. (B) Relationship between indices derived from PTG, AoP, and APG in 56 patients with ischemic heart disease undergoing cardiac catheterization and infusion of angiotensin and nitroglycerin. The measure d/a is the ratio derived from the APG. Top left: photoplethysmograph augmentation index and d/a . Top right: aortic augmentation index and d/a . Bottom left: Aortic augmentation index and finger photoplethysmographic index. Bottom right: Ratio d/a and aortic systolic pressure. Reproduced from Takazawa *et al.* (1993a)

cardiac failure (Casale *et al.*, 1986; Levy *et al.*, 1990; Koren *et al.*, 1991; Bikkina *et al.*, 1994; Lip, 2001; Mathew *et al.*, 2001; Verdecchia *et al.*, 2001; Verdecchia *et al.*, 2003; Drazner *et al.*, 2004; Bibbins-Domingo *et al.*, 2009). In the absence of systolic heart failure, left ventricular hypertrophy is related to the degree (amplitude and duration) of late systolic augmentation of the central pressure wave resulting from wave reflection (Saba *et al.*, 1993; Hashimoto *et al.*, 2007a; Hashimoto *et al.*, 2007b; Hashimoto *et al.*, 2008c; Roman *et al.*, 2009), and with diastolic left ventricular (LV) dysfunction (see Chapter 15).

The degree of late systolic pressure augmentation is related to diastolic dysfunction. This was first demonstrated by Hori *et al.* (1985) in dogs, but has been confirmed in humans by Yabe *et al.* (1991). Part of the problem is structural and chronic, wherein hypertrophied muscle contracts and relaxes more slowly, but part is acute, reversible, and directly related to the amplitude and timing of the late systolic pressure peak (Yabe *et al.*, 1991). Reduction in this peak and/or delay of the peak, so that it occurs in early diastole, improves diastolic relaxation (Hori *et al.*, 1985; Zusman *et al.*, 1989; Yabe *et al.*, 1991; Kapuku *et al.*, 1993), and should ultimately be shown to improve symptoms of cardiac failure (Solomon *et al.*, 2007; van Veldhuisen *et al.*, 2009).

Reversal of left ventricular hypertrophy

Reversal of echocardiographically determined left ventricular hypertrophy has been demonstrated repeatedly in studies of anti-hypertensive therapy (Tarazi, 1982; Pfeffer and Pfeffer, 1990; Devereux, 1997; Gottdiener *et al.*, 1997; Devereux *et al.*, 2001; Devereux *et al.*, 2004; Schneider *et al.*, 2004; Mancia *et al.*, 2007b; Hashimoto *et al.*, 2008c; Fagard *et al.*, 2009). Whether this in itself translates into an improvement of life expectancy is yet to be determined, but it does appear to occur without any detriment in function and possibly with an improvement (Zusman *et al.*, 1989; Sagie *et al.*, 1993; Aurigemma *et al.*, 1995; Mathew *et al.*, 2001; Wachtell *et al.*, 2002; Frohlich, 2004; Solomon *et al.*, 2007).

There is some conflict in published data, doubtless reflecting the difficulties in measuring left ventricular hypertrophy, and in the relatively short duration (rarely greater than 1.0 year) that such studies were conducted. Comparative studies have shown greater benefit for calcium antagonists over β -blocking agents (Zusman *et al.*, 1989; Pfeffer and Pfeffer, 1990; Schulman *et al.*, 1990), and greater benefit of ACE inhibitors over diuretics (Dahlöf and Hansson, 1992), and ARBs and ACEIs over β -blockers (Asmar *et al.*, 2001b; Wachtell *et al.*, 2002; Schneider *et al.*, 2004). Where measured, reversal of left ventricular hypertrophy has been accompanied by a decrease in arterial stiffness and a decrease in augmentation of the arterial pressure pulse (Asmar *et al.*, 1988a; Asmar *et al.*, 1988b; Miyashita *et al.*, 1993; Boutouyrie *et al.*, 1994; Asmar *et al.*, 2001b; Asmar

et al., 2001d; London *et al.*, 2004a; Miyashita *et al.*, 2010). Meta-analyses of published data (Dahlöf and Hansson, 1992; Schmieder *et al.*, 1996; Fagard *et al.*, 2009) showed that while all anti-hypertensive agents could lead to the reversal of left ventricular hypertrophy, ACE inhibitors, calcium channel antagonists, and ARBs were the most effective.

All anti-hypertensive agents can reverse left ventricular hypertrophy, and this can be attributed to their action in reducing left ventricular systolic pressure (Yaginuma *et al.*, 1986; Fitchett *et al.*, 1988a; Fitchett *et al.*, 1988b; Latson *et al.*, 1988; Lip, 2001; Hashimoto *et al.*, 2007a; Hashimoto *et al.*, 2007b; Hashimoto *et al.*, 2008a). For ACE inhibitors, additional effects from a direct action of the drug on the heart are also possible (Tan *et al.*, 1992; Weber *et al.*, 1992) and are usually assumed. However, reversal of left ventricular hypertrophy in essential hypertension with ACE inhibition occurs without any concomitant change in right ventricular mass or structure, which argues against the direct effect of this drug on the heart (Aristizabal *et al.*, 1994; Griffin *et al.*, 2003; Kurtz, 2003), as does greater reduction of aortic than peripheral pressure in humans and experimental animals (Mitchell *et al.*, 1996; Asmar *et al.*, 2001b; Griffin *et al.*, 2003; Kurtz, 2003; Williams *et al.*, 2006; Nichols *et al.*, 2008; Matsui *et al.*, 2009; Miyashita *et al.*, 2010), and in other studies (Svensson *et al.*, 2001; Mancia *et al.*, 2007b) greater reduction in 24-hour than once daily pressure recording. It seems probable that a reduction in left ventricular afterload is the principal (and possibly the only factor) responsible for the reversal of left ventricular hypertrophy in hypertension (Bishop *et al.*, 2000), and that the superiority of ACE inhibitors and calcium antagonists stems from the fact that they, like nitrates and vasodilating β -blockers, cause greater reduction in aortic and left ventricular systolic pressure than is apparent from the measurement of brachial artery pressure (Kelly *et al.*, 1990a; Simkus and Fitchett, 1990; O'Rourke *et al.*, 1993a; Takazawa *et al.*, 1993a; Takazawa *et al.*, 1993b; Cholley *et al.*, 1995b; Nichols and Singh, 2002; Hirata *et al.*, 2005b; Williams *et al.*, 2006; Nichols *et al.*, 2008). Animal data (Mitchell *et al.*, 1996; Griffin *et al.*, 2003) support this view (Fig. 20.15), as do recent clinical studies (Asmar *et al.*, 2001b; De Luca *et al.*, 2004; London *et al.*, 2004a; Matsui *et al.*, 2009). Data from Hashimoto *et al.* (Hashimoto *et al.*, 2007a; Hashimoto *et al.*, 2008a) indicate that reduction in wave reflection is the most important mechanism.

THERAPY OF HYPERTENSION

Up until the late 1940s, no treatment except surgical sympathectomy was available for hypertension, even though the ill effects of this condition were widely known. The unhappy situation at this time was summarized by Weiss (1939):

What has been done in an effort to reduce the blood pressure? Because of an ill-founded idea that protein

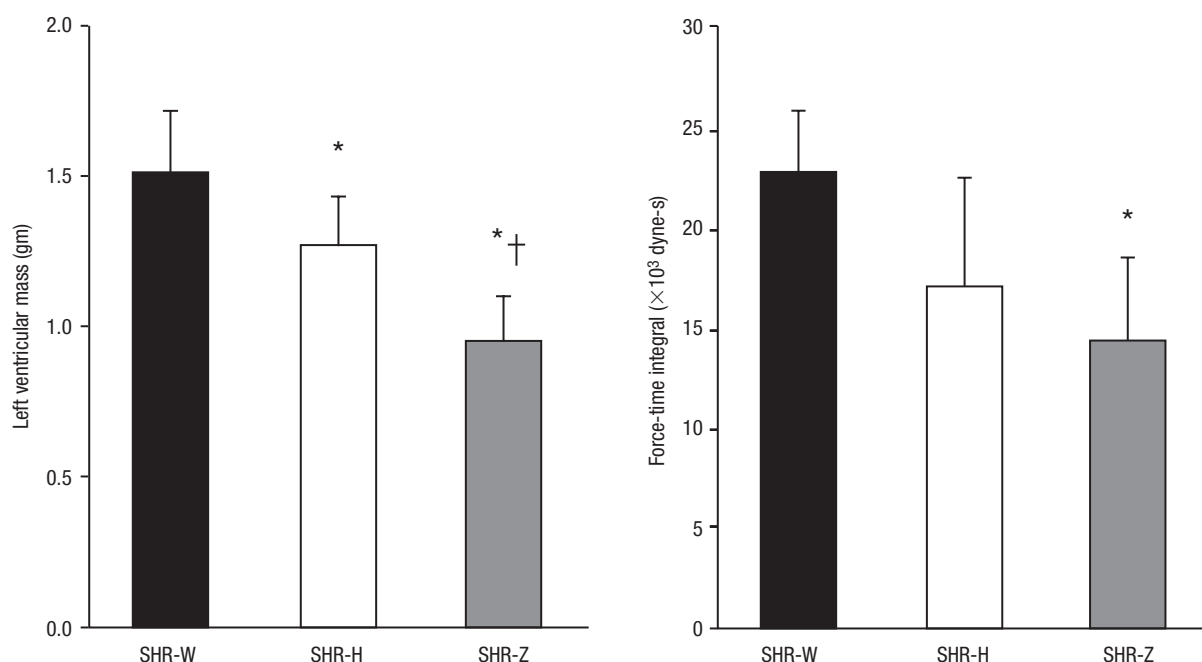


Figure 20.15 Relationships between left ventricular (LV) mass (left) and left ventricular systolic force-time integral (right) in hypertensive (SHR) rats treated with placebo (■), and with hydralazine (□) or an ace inhibitor (▒) to achieve an equivalent degree of blood pressure control, as determined by peripheral pressure measurements. There was a close relationship between regression of LV hypertension, and LV load, measured from pressure in the central aorta, despite no difference in peripheral systolic pressure. Reproduced from Mitchell *et al.* (1996)

was responsible for hypertension and kidney disease, the patient was denied meat and eggs, and especially red meat, which for some reason was looked upon with particular dread. His diet was rendered even more unpalatable by the withdrawal of salt. Sympathy would doubtless have been extended to this half-starved fellow except that he probably was not able to eat anyway, his teeth having been extracted on the theory that focal infection had something to do with hypertension. Even before this he had sacrificed his tonsils and had had sinuses punctured because of the same theory. In case some food had been consumed, the slight colonic residue was promptly washed out by numerous colonic irrigations, especially during the period when the theory of auto-intoxication was enjoying a wave of popularity. To add to his unhappiness he was often told to stop work and exercise. Of course he was denied alcohol and tobacco as well as coffee and tea, and as a climax to the difficulties of this unfortunate person, he may now fall into the clutches of the neurosurgeon who is prepared to separate him from his sympathetic nervous system.

Since the introduction of ganglion-blocking agents by Smirk and Alstad in the late 1940s (Smirk and Alstad, 1951), a variety of agents have been employed for the management of hypertension. Many, including diuretics, β -blocking agents, and calcium channel antagonists, were

not initially introduced for this purpose, and their value as anti-hypertensive agents was serendipitous and came as a surprise. It is now usual to employ these drugs in combination since the effects on arterial pressure are additive, but side effects are not, and often (as with β -blocking agents and calcium antagonists) tend to cancel. Modern therapy is convenient, and usually easily tolerable.

There is no consensus on ideal therapy for hypertension. Two general attitudes are presented, one based on the results of clinical trials, the other on pathophysiologic mechanisms.

The first approach (enshrined in official pronouncements) suggests that those agents be used first which have proved effective in large-scale clinical trials. Such agents are diuretics and β -blocking agents. Neither of these has any definite acknowledged specific effect on increased peripheral resistance as the cause of hypertension nor on abnormal arterial distensibility as the major important secondary effect of elevated pressure. Their main primary action is through reduction in blood volume, venous return, and cardiac output – and hence on arterial pressure. These drugs defy logic; they do not address the cause of hypertension, but they are nonetheless effective anti-hypertensive agents and they do reduce mortality. Through reducing arterial pressure, they passively reduce arterial stiffness and delay wave reflection, thus tending to reduce systolic augmentation of the aortic and left ventricular pressure. β -Blocking agents reduce heart rate and

stroke volume, and so decrease cumulative damaging cyclic pressure and diameter pulsations. They are logical therapy for delaying medial degeneration and for limiting the myocardial ischemia that can occur with heart rate increase when the left ventricle is hypertrophied (Stojanovic *et al.*, 2003; Prichard and Vallance, 2004).

The second approach is based on pathophysiologic mechanisms (Safar, 1980; Safar *et al.*, 1983; O'Rourke *et al.*, 1984; Safar *et al.*, 1984; O'Rourke *et al.*, 1985; Fitchett *et al.*, 1986; O'Rourke *et al.*, 1986a; O'Rourke and Avolio 1986a; Smulyan *et al.*, 1986; Yaginuma *et al.*, 1986; Dahlöf and Hansson, 1992; Devereux and Laragh, 1992; Kelly, 1992; Asmar *et al.*, 1993; Cholley *et al.*, 1995a; Cholley *et al.*, 1995b; Williams *et al.*, 2006; O'Rourke and Hashimoto, 2007; Williams, 2008; Matsui *et al.*, 2009). One would expect that the most logical therapy would be through the use of an agent that reduces peripheral resistance through arteriolar dilatation. Examples of such agents are prazosin, terazosin, and hydralazine. They are effective anti-hypertensive agents, but they have not been employed in studies that have shown mortality reduction.

Indeed, in the ALLHAT study, the α -blocking agent doxazosin was inferior to the diuretic chlorthalidone, and this arm of ALLHAT was terminated prematurely (ALLHAT Collaborators Research Group, 2002). The most popular anti-hypertensive agents are ACE inhibitors, ARBs, and calcium channel antagonists. These compounds dilate muscular conduit arteries as well as arterioles, and have been shown to reduce the compliance of these arteries by a direct effect on smooth muscle. These drugs, together with nitrates (Duchier *et al.*, 1987; Carroll *et al.*, 1991), have however little or no effect on the large, predominantly central elastic arteries (O'Rourke *et al.*, 1984; Yaginuma *et al.*, 1986; Fitchett *et al.*, 1988a; Fitchett *et al.*, 1988b; Asmar *et al.*, 2001b; Hirata *et al.*, 2005b), which are primarily concerned with cushioning pressure pulsations generated by intermittent ventricular contraction (Fig. 20.16). Their beneficial effects are indirect and are manifest through a reduction in wave reflection returning from peripheral reflecting sites (Yaginuma *et al.*, 1986; Nichols and Singh, 2002; Nichols, 2005; Nichols *et al.*, 2008) (see Chapter 26). This effect is very obvious

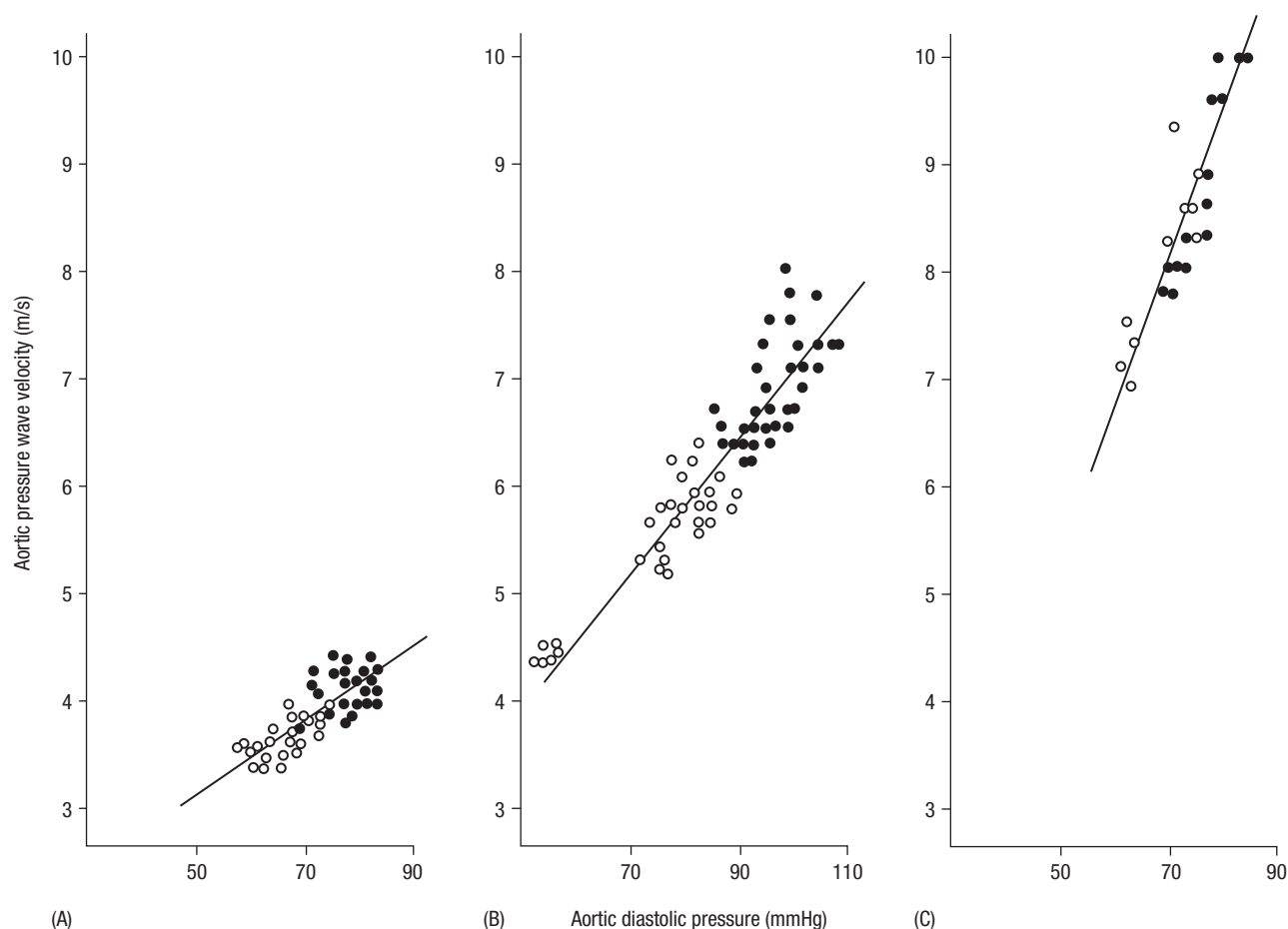


Figure 20.16 The relationship between aortic pulse wave velocity and aortic diastolic pressure shown for three patients with dilated cardiomyopathy but of different ages: (A) 25 years; (B) 43 years; and (C) 63 years. As pressure was lowered by nitroprusside, there was a fall in pulse wave velocity in all patients but the relationship was much steeper in the older compared with the younger patient. The vasodilator agent appeared to have no direct effect on pulse wave velocity. (●) Control beats; (○) nitroprusside beats. Reproduced from Carroll *et al.* (1991)

in pressure wave augmentation, which is substantially reduced by all these drugs – nitrates, calcium channel antagonists, ACE inhibitors and ARBs; ascending aortic impedance modulus at low frequencies is also decreased (see Fig. 20.9). From the theoretic standpoint, these drugs are a more logical treatment for hypertension than are diuretics and β -blocking agents, especially in the elderly, where early wave reflection provides a greater boost to aortic and left ventricular systolic pressure. They appear to be superior for reduction in left ventricular hypertrophy. As discussed earlier, their beneficial effects on left ventricular systolic pressure may not be manifest fully from conventional recording of pressure in the brachial artery. These views were tested and confirmed in the CAFE study (Williams *et al.*, 2006), in J-CORE (Matsui *et al.*, 2009), and by Miyashita *et al.* (2010).

Up until LIFE (Dahlöf *et al.*, 2002) and ALLHAT (ALLHAT Investigators, 2002), none of the arterial dilating drugs – ACE inhibitors, calcium channel antagonists, ARBs, or nitrates – had been utilized in trials designed to test their comparative ability to reduce morbid and lethal events in hypertensive patients, so none could be used with the same confidence as diuretics and β -blocking agents.

The LIFE study (Dahlöf *et al.*, 2002) compared losartan (ARB) against atenolol (β -blocker) in a four-year, double-blind, randomized trial of anti-hypertensive therapy in older patients with hypertension and ECG-diagnosed left ventricular hypertrophy. There was a modest reduction in the primary composite end point (death, myocardial infarction, stroke) in favor of losartan, although with no difference in all cause mortality or myocardial infarction. Brachial blood pressure reduction was identical (<1 mmHg difference for systolic or diastolic) suggesting that central systolic pressure had been reduced to a greater degree with losartan, and possibly sufficient to explain results (see above). There was no resounding benefit for the ARB and little reason for the enthusiastic response which followed, on a proposed direct cardiac effect of the ARB (Brunner and Gavras, 2002; Devereux *et al.*, 2004).

The ALLHAT (ALLHAT Collaborators Research Group, 2002) study results were even more sobering. This study compared a diuretic (chlorthalidone) against an α -blocker (doxazosin), and ACE inhibitor (lisinopril) and calcium channel antagonist (amlodipine). The α -blocker arm was discontinued early on account of superiority of chlorthalidone over doxazosin. At the planned study termination, there was no benefit, and probably disadvantage (for stroke with lisinopril, heart failure with amlodipine), shown with the newer anti-hypertensive agents. Over the early part of the study, brachial artery systolic pressure had been consistently higher with lisinopril and amlodipine compared to chlorthalidone (overall 2.5 mmHg greater with lisinopril, 1.25 mmHg greater with amlodipine). Central systolic pressure was not estimated, but was probably similar overall for the three therapies. Results are consistent with the view that the major mechanism of

benefit was blood pressure lowering rather than any specific benefit of the ACE inhibitor or calcium channel blocker (CCB). It must be stressed, however, that the one agent was not successful in controlling pressure in this study, and that it was a trial of initial therapy. The majority of patients required two or more drugs for control of arterial pressure.

A more recent trial comparing an ACE inhibitor with diuretic showed a marginal benefit to the ACE inhibitor (Wing *et al.*, 2003). This trial, whose findings appear to clash with ALLHAT (ALLHAT Collaborators Research Group, 2002), and an associated tonometric substudy (Gatzka *et al.*, 2001a; Gatzka *et al.*, 2001b), have created controversy, and are discussed elsewhere.

The ACCOMPLISH trial group showed outcome benefit for the combination of ACEI plus amlodipine over ACEI plus a diuretic in a three-year trial (Jamerson *et al.*, 2008). There was no measurement of central pressure in this study, but Matsui *et al.* (2009) in a smaller shorter study had shown that a combination of ARB (olmesartan) plus CCB (azalnidipine) led to a significant drop in central aortic systolic pressure of 5.2 mmHg compared to the ARB plus hydrochlorothiazide which led to a non-significant difference of 2.6 mmHg in brachial systolic pressure. Jiang *et al.* (2007) had previously shown that an ARB causes a greater drop of central pressure as compared to thiazide diuretic. Findings for the studies indicate that ACCOMPLISH results may be explained by greater benefit from the ACEI/CCB in reducing central aortic systolic pressure. More recently, Boutouyrie *et al.* (2010) published results of the Explor study, which compared amlodipine/valsartan with amlodipine/atenolol. There was no significant difference in brachial systolic pressure with the two therapies, but central aortic systolic and pulse pressure were 4 mmHg lower for the amlodipine/valsartan combination. A large multicenter study in premier Japanese institutions has confirmed the general benefit of modern arterial dilating drugs over conventional β -blocking and diuretic drugs (Fig. 20.17) (Miyashita *et al.*, 2010). These recent studies confirm prediction made by O'Rourke, Safar, and Dzau in their book, *Arterial Vasodilatation* (O'Rourke *et al.*, 1993a) which followed Kelly's first description of benefit for a vasodilating β -blocker over a conventional β -blocker in hypertension patients (Kelly *et al.*, 1989c).

EPIDEMIOLOGY

Epidemiological approaches to hypertension up until the 1980s were directed on the presumption that systolic and diastolic pressures represented specific physiologic and pathophysiologic phenomena, with elevated diastolic pressure necessary for a diagnosis of hypertension to be made and pathognomonic of this condition. A more mature approach has now evolved. All trials of hypertension up until 1985 entered patients on the basis of

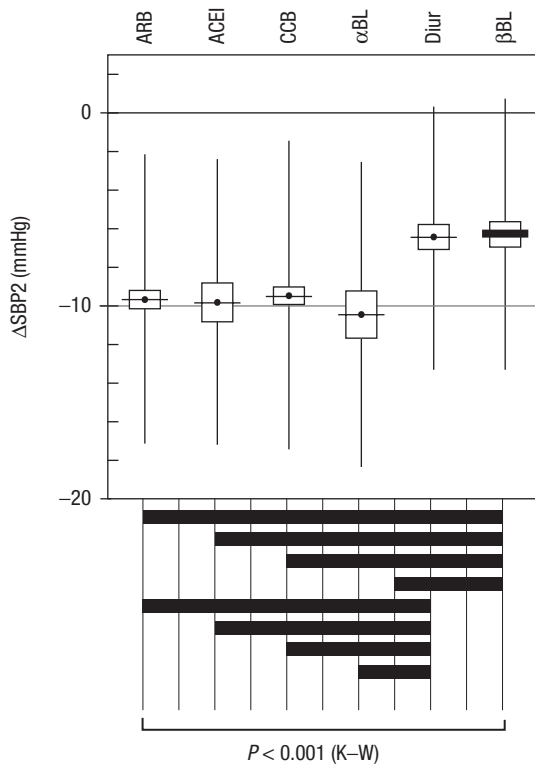


Figure 20.17 Difference between central and radial systolic pressure compared across different anti-hypertensive class. The newer generation of anti-hypertensives (angiotensin receptor blockers (ARBs); angiotensin-converting enzyme inhibitors (ACEIs), calcium channel blockers (CCBs), and α -blockers) showed significantly lower central systolic pressure than the older anti-hypertensives (diuretics and β -blockers). From Miyashita *et al.* (2010)

diastolic blood pressure and assessed response on the basis of a decrease in diastolic pressure.

The Systolic Hypertension in the Elderly Project (SHEP) (SHEP, 1991) was the first large trial to enter patients solely on the basis of systolic pressure. This was based on the findings of the Framingham studies (Kannel *et al.*, 1981), which had shown a more robust relationship between cardiovascular events and systolic pressure, and on the MRFIT trial (Rutan *et al.*, 1986) and the European

Collaborative Trial (Amery *et al.*, 1985), which had confirmed these. SHEP showed that a reduction in systolic pressure (even when diastolic pressure was normal) reduces cardiovascular events (SHEP, 1991; Winker and Murphy, 1991; Menard *et al.*, 1992; SHEP, 1993). This has caused a complete reassessment of hypertension and a search for new indices of vascular function apart from systolic and diastolic blood pressure levels themselves. Attempts are being made to separate steady (mean) and pulsatile components of blood pressure, and to seek separate influences and effects of each (Safar *et al.*, 1984; Darne *et al.*, 1989; Safar, 1989; Franklin and Weber, 1994; Madhavan *et al.*, 1994; Smith, 1994; Cavallini *et al.*, 1995; Scuteri *et al.*, 1995; Safar, 2001; Safar, 2002; Safar and Smulyan, 2004; Williams *et al.*, 2006; Matsui *et al.*, 2009).

Measurement of left ventricular hypertrophy has been useful and is incorporated in recent ongoing trials. Measurement of carotid or aortic augmentation from the arterial pressure pulse waveform has been related to adverse cardiovascular function, to left ventricular hypertrophy, and to morbidity and mortality (London *et al.*, 2001; Safar *et al.*, 2002a). Hashimoto *et al.* (Hashimoto *et al.*, 2007a; Hashimoto *et al.*, 2008a) have shown that indices determined from the pulse waveform can better predict reduction in LV mass with anti-hypertensive therapy than brachial artery cuff measurements in the clinic or at home. Williams *et al.* have shown that beneficial effects of amlodipine over atenolol in the ASCOT trial (Dahlof *et al.*, 2005) are explicable on the basis of greater reduction in aortic systolic and pulse pressure (Williams *et al.*, 2006). Roman *et al.* (Roman *et al.*, 2007; Roman *et al.*, 2009) have shown that central aortic blood pressure predicts better outcomes than brachial pressure in the Strong Heart study. These and other studies are considered in Chapter 21.

The time surely has passed when brachial artery blood pressure was considered a reason in itself to diagnose hypertension, to initiate therapy, and to assess the efficacy of therapy. The focus has moved to the heart and to the central arteries that are damaged by elevated pressure, to the ways by which stress on the heart and central arteries can be better measured, and to how drug and other therapies can best alleviate resulting ill effects.

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Interpretation of blood pressure in epidemiological studies and clinical trials

Explanation of brachial blood pressure change with age	469	Major therapeutic trials	480
Accuracy of conventional blood pressure measurement	470	Vasodilator therapy of cardiac failure	482
Cardiovascular events and brachial blood pressure	475	Current guidelines: JNC7, ESH/ESC, and WHO/ISH	483
Cardiovascular events and central systolic pressure	479	Wider implications	484
Cardiovascular events and indices of arterial stiffness	479		

The principles presented elsewhere in this book provide fresh insight into epidemiological studies and clinical trials. The principles provide an explanation as to why diastolic pressure is generally an unsatisfactory index of the hypertensive state and why systolic pressure is more useful. Indeed, this was a dominant theme in previous editions, and anticipated the present swing from diastolic to systolic pressure which followed the mass of evidence accumulating from Framingham (Kannel *et al.*, 1971; Kannel *et al.*, 1981), followed by the MRFIT (Rutan *et al.*, 1986; Domanski *et al.*, 2002) and other epidemiological studies and clinical trials (He and Whelton, 1999; Staessen *et al.*, 2000b; Staessen *et al.*, 2010).

However, events have moved further in the clinical and epidemiological field, from consideration of brachial diastolic to brachial systolic pressure and then on to consideration of pulse pressure and mean pressure in the brachial artery – and hence on to the concept of steady flow and pulsatile flow phenomena – i.e. the properties of small resistive arteriolar vessels on the one hand and large elastic vessels on the other. They have now moved further, in a series of studies which have considered pressure in the ascending aorta and large central arteries, where mean and pulse pressure are considered separately, and with emphasis on the conduit (steady flow) and cushioning (pulsatile flow) functions of the arterial tree. The theoretic approach of McDonald, Womersley, and Taylor, initiated in the 1950s, is now reaching into epidemiology and clinical trials, and is providing comprehensive explanations not previously available. These issues (which remain contentious (Prospective Studies Collaboration, 2002 and later *Lancet* correspondence)) will be addressed and summarized in this chapter. They add to the separate problem of accurate determination of cuff pressure in the

clinic, at home, or over a 24-hour period (O'Brien *et al.*, 2003; Pickering *et al.*, 2005a; Pickering *et al.*, 2005b; Mancia *et al.*, 2007a; Mancia *et al.*, 2007b).

EXPLANATION OF BRACHIAL BLOOD PRESSURE CHANGE WITH AGE

The detailed Framingham data from the original and off-spring cohort (Franklin *et al.*, 1997) supplement the data available from the NHANES and other cross-sectional studies (Fig. 21.1) (US National Health Survey 1977; Burt *et al.*, 1995; Benetos *et al.*, 2010; Mitchell *et al.*, 2010; Wang *et al.*, 2010). The Framingham longitudinal data show for the four original quartiles little change in mean arterial pressure from 40 to 80 years, a progressive increase in systolic pressure from age 30, and an increase in diastolic pressure to age 50, then a progressive decline in diastolic pressure thereafter. In contrast, there is a marked increase in pulse pressure from age 40 with advancing years. These data have been interpreted to show minimal increase in peripheral resistance with age (on the assumption that cardiac output remains relatively constant or decreases minimally with age at rest (Weisfeldt, 1980; Nichols *et al.*, 1985a; Lund-Johansen and Omvik, 1995), but progressive increase in large artery stiffness with age (Ho, 1982; Avolio *et al.*, 1983; Avolio *et al.*, 1985a; Asmar, 1999; O'Rourke, 2002a; Mitchell *et al.*, 2004a; McEniery *et al.*, 2005a; McEniery *et al.*, 2008). An additional interesting finding in the Framingham blood pressure longitudinal data was the relatively flat portion of pulse pressure at younger ages (30–40 years), especially in the lower two quartiles. These data can be supplemented with the longitudinal data published by Uiterwaal *et al.*

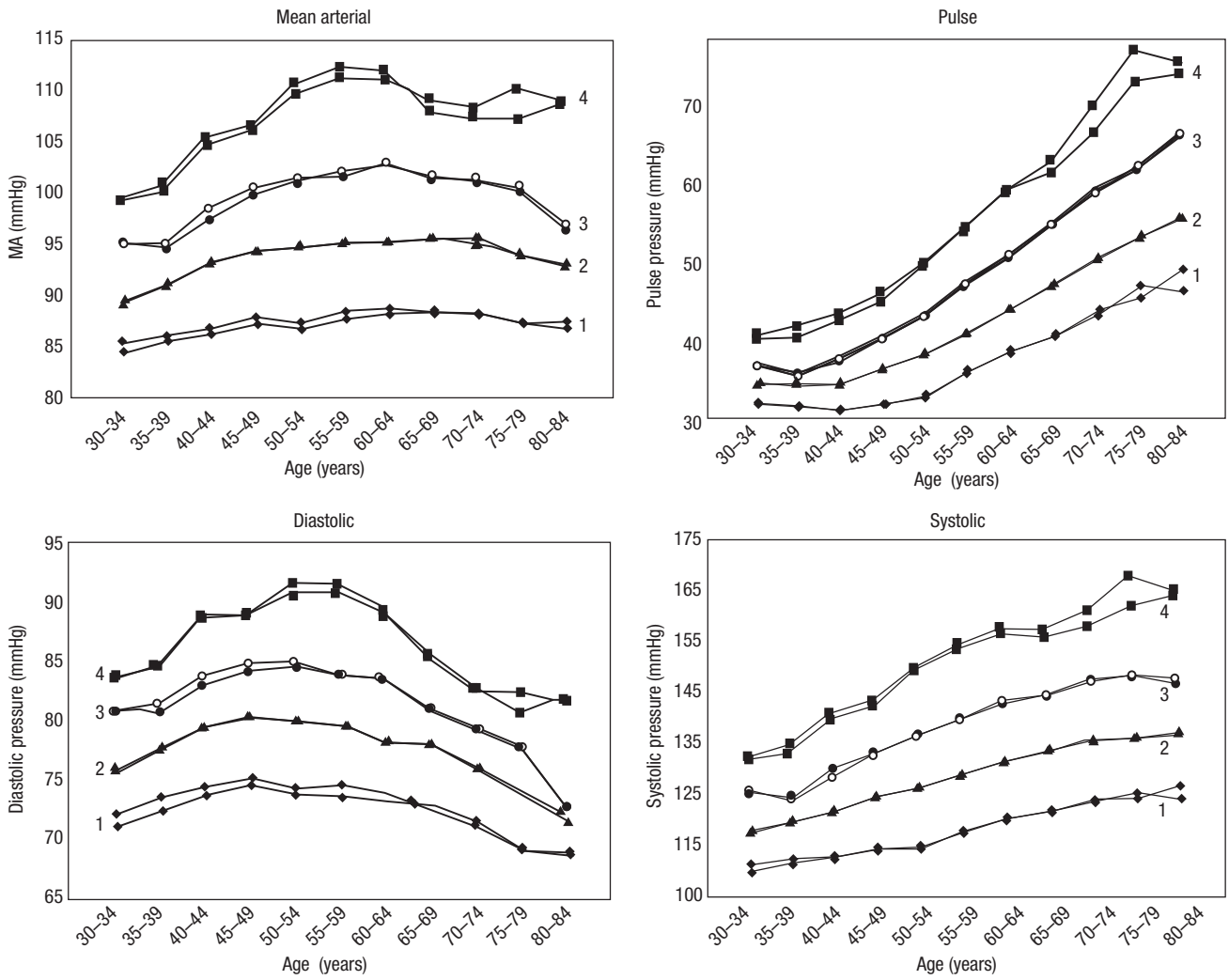


Figure 21.1 Longitudinal change with age of the original Framingham cohort in relation to the original quartile of systolic, diastolic, mean, and pulse pressure. After Franklin *et al.* (1997)

(1997) for change in blood pressure between infancy and age 32 years. These data show progressive rise in systolic pressure to age 18 years, when full body height is attained, then a plateau of systolic pressure to age 32 years (Fig. 21.2A). The plateau of systolic pressure between age 20 and 40 is apparent in virtually all epidemiological studies (Figs 21.2B and 21.3, see also Fig. 19.1) (Vlachopoulos and O'Rourke, 2000b), although it is most clear-cut in males (who are taller than females). The plateau in systolic pressure is associated with a trend for decline in brachial pulse pressure between age 20 and 40 years of age, as seen in the NHANES study for American Caucasians (Fig. 21.4A, page 473) (Burt *et al.*, 1995). Data from the US National Institute of Aging and NHANES III (Hajjar *et al.*, 2001) confirm the relative plateau in brachial pulse pressure between age 20 and 40, and progressive increase thereafter (Fig. 21.4B, page 473). These unusual changes in brachial, systolic, and pulse pressure between childhood and middle age can only be accounted for on the basis of pressure wave reflection and amplification in upper limb arteries

(Wilkinson *et al.*, 2001; Safar *et al.*, 2003) where pressures are measured in epidemiological studies. Amplification is greatest when maximal height is attained (O'Rourke *et al.*, 2000a), and is lowest when body height is short, in infancy (Gevers, 1994), or when the elastic arteries stiffen, for example with increasing age and hypertension. These issues are discussed further by Smulyan *et al.* (1998) and Wilkinson *et al.* (2001), and summarized in the *Handbook of Hypertension*, vol. 23 (Safar and O'Rourke, 2006).

ACCURACY OF CONVENTIONAL BLOOD PRESSURE MEASUREMENT

Up until recently, blood pressure in major trials has been measured with a cuff sphygmomanometer and the Korotkov auscultatory method, using phase I (appearance of sounds) to characterize systolic pressure, and phase V (disappearance of sounds) for diastolic pressure. There are known inaccuracies when such pressure values are

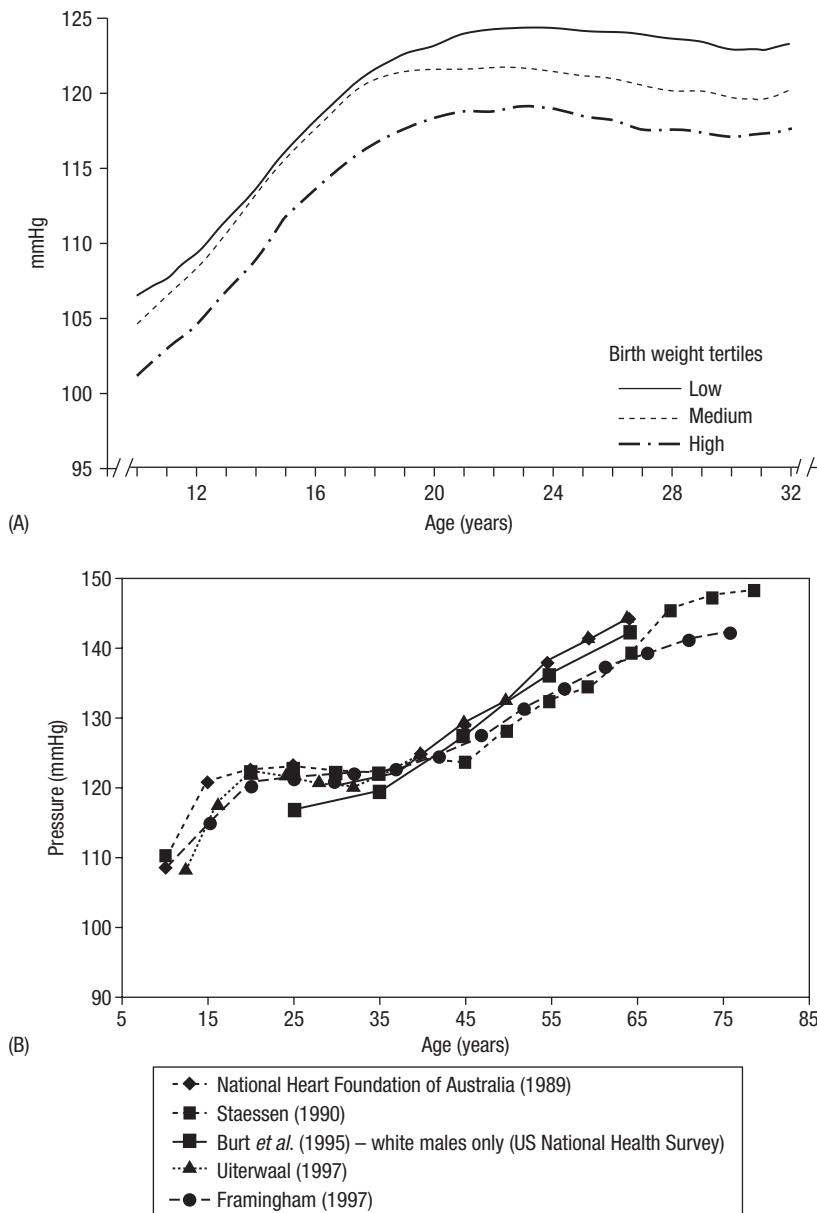


Figure 21.2 (A) Annual height- and weight-standardized mean systolic blood pressure by tertiles of birth weight (low, < 3200 g; medium, 3200–3600 g; high, > 3600 g). After Uiterwaal *et al.* (1997). (B) Changes in brachial systolic pressure with age show a steep rise from the age of ten years to a plateau when full body height is reached at age 18 years, then a subsequent rise after age 45 years. Data from National Heart Foundation of Australia 1989, US National Health Survey (white males) (Burt *et al.*, 1995), Framingham (Franklin *et al.*, 1997), and Uiterwaal *et al.* (1997). After O'Rourke *et al.* (2002)

compared against simultaneously recorded intra-arterial peak and nadir of the brachial artery pressure wave in the opposite arm (see Chapter 6) (Breit and O'Rourke, 1974; White *et al.*, 1990; Cohn *et al.*, 1995). Mean values are close, but the scatter is considerable (Fig. 21.5A–C), while coefficient of variation of paired measurements is much greater than seen with the better indices of arterial stiffness (see below).

In recent times, there has been a strong movement away from the Korotkov method to semi-automatic blood pressure recording devices. The Framingham study group has so moved, having been satisfied with correspondence between the two methods. However, there still are large differences between new methods and the Korotkov method, as taken to be the 'gold standard'. Such differences are tolerated by official bodies. The US AAMI (Association for the Advancement of Medical

Instrumentation) in its SP-10 criteria allows a mean difference of ≤ 5.0 mmHg for mean and ≤ 8.0 mmHg for standard deviation as indicating 'substantial equivalence', when a new method is compared against the cuff sphygmomanometer using the Korotkov sound technique (White *et al.*, 1993). The British Hypertension Society accepts as grade A a device showing 60 percent of measurement < 5.0 mmHg, and 85 percent < 10 mmHg, and 90 percent < 15 mmHg, when the device is compared against the standard Korotkov sound technique (O'Brien *et al.*, 2001c). The European Society of Hypertension accepts that a device be recommended if it satisfies AAMI SP-10 criteria and also is ranked as grade A or grade B as set out above; the society has noted, however, that a number of very popular devices currently used in hospitals and clinical trials do not satisfy these criteria (O'Brien *et al.*, 2001a; O'Brien *et al.*, 2001b).

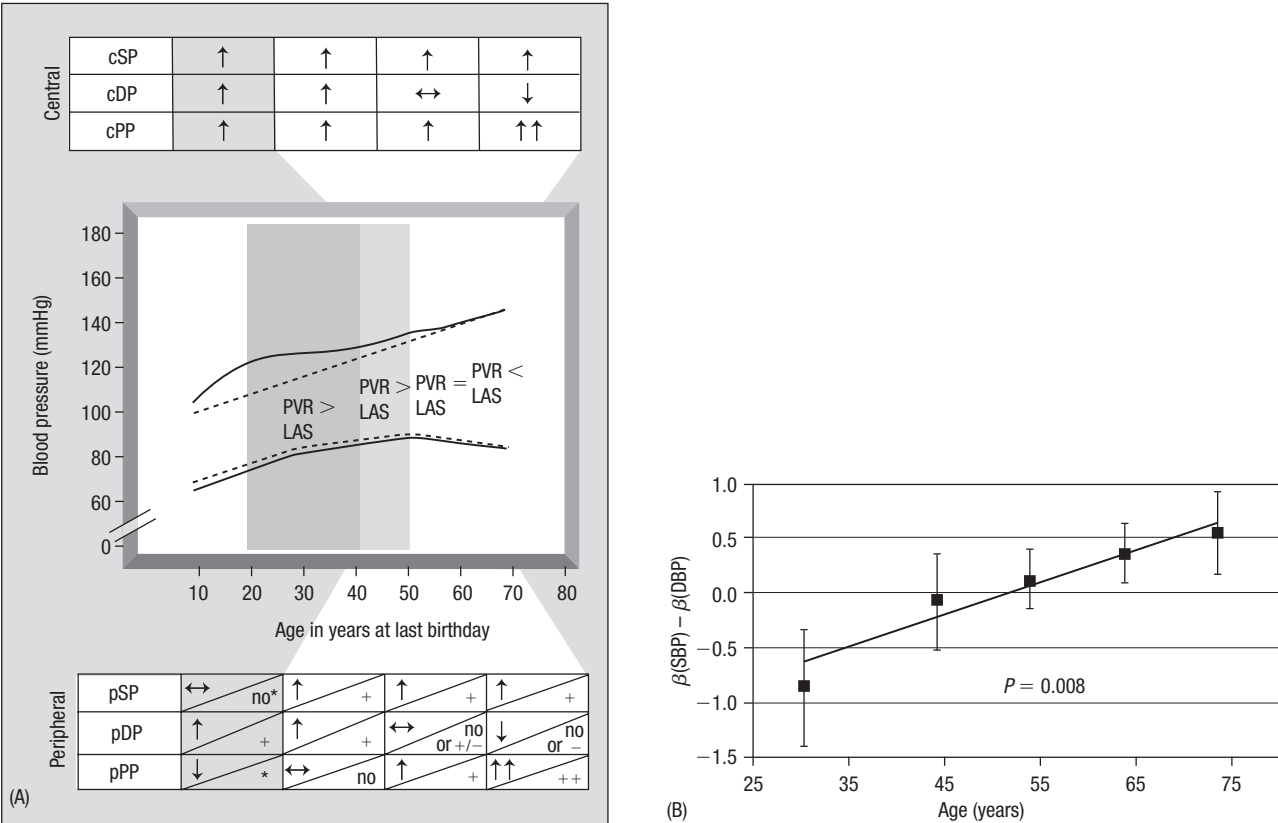


Figure 21.3 (A) Significance of change in central systolic cSP, diastolic cDP, and pulse pressure cPP (boxes above), and of change in peripheral (brachial) systolic pSP, diastolic pDP, and pulse pressure pPP (boxes below) in relation to changes in arterial pressure with age (in center), as determined from analysis of major epidemiological studies. The bottom segment shows in triangles the predictive value of peripheral blood pressure for events. In the graph, solid lines represent published data for brachial pressure; dotted lines indicate predicted pressures in the ascending aorta, based on known differences in brachial pressure amplification with age. The relative contribution of the underlying hemodynamic mechanisms (increase in peripheral vascular resistance (PVR) or in large artery stiffness (LAS)) is changing progressively from younger to older age. ↑ indicates increase; ↓ indicates decrease; ↔ indicates no change in blood pressure with age; +, positive; ++, strong positive; −, negative; +/-, weak positive; no, no predictive value. Adapted from Vlachopoulos and O'Rourke (2000b), Burt *et al.* (1995), and Franklin *et al.* (2000). *In males. (B) Difference in coronary heart disease prediction between systolic and diastolic pressure as a function of age. Difference in coefficients (from Cox proportional hazards regression) between systolic and diastolic pressure (ordinate) is plotted as a function of age. From Franklin *et al.* (2001)

These points refer to studies where all appropriate precautions are taken for accurate use of non-invasive sphygmomanometry, including positioning of the patients, use of appropriate cuff size, etc. Reason for random variation in systolic and diastolic values is unknown. When arterial pressure is measured at the same time in the two arms with identical automated devices, the same problem is seen, with standard deviation of the order of 8.5 mmHg (Fig. 21.5B) (Lane *et al.*, 2002).

All these comparative studies were performed under carefully controlled conditions where other sources of error were eliminated. Such is not the case in routine determination in the physician's office (Kaplan, 1998; Messerli *et al.*, 2002) or in the busy cardiac catheterization laboratory (O'Rourke, 2003). Messerli *et al.* repeated Kaplan's criticism of 'sloppiness' and noted that American Heart Association guidelines were regularly and systematically ignored in clinical practice.

The 'bottom line' of this issue is that assessment of an individual patient on the basis of arterial pressure alone is utterly unsatisfactory, and other information needs to be obtained for prognosis and therapy.

Current US (JNC7), ESH/ESC, and WHO criteria for staging and managing hypertension are based on sphygmomanometer cuff measurement of blood pressure. The guidelines, however, recognize the problems of sphygmomanometry by recommending multiple measurement of arterial pressure, and staging the grade of risk and need for therapy on the basis of other risk factors, evidence of target organ damage, and associated clinical conditions (Chobanian *et al.*, 2003; ESH/ESC Guidelines Committee, 2003; ISH Writing Group, 2003; Mancia *et al.*, 2007a; Mancia *et al.*, 2007b). We believe that measurements of arterial stiffness and degeneration, available from analysis of the arterial pulse waveform, provide further information to supplement the interpretation of blood pressure

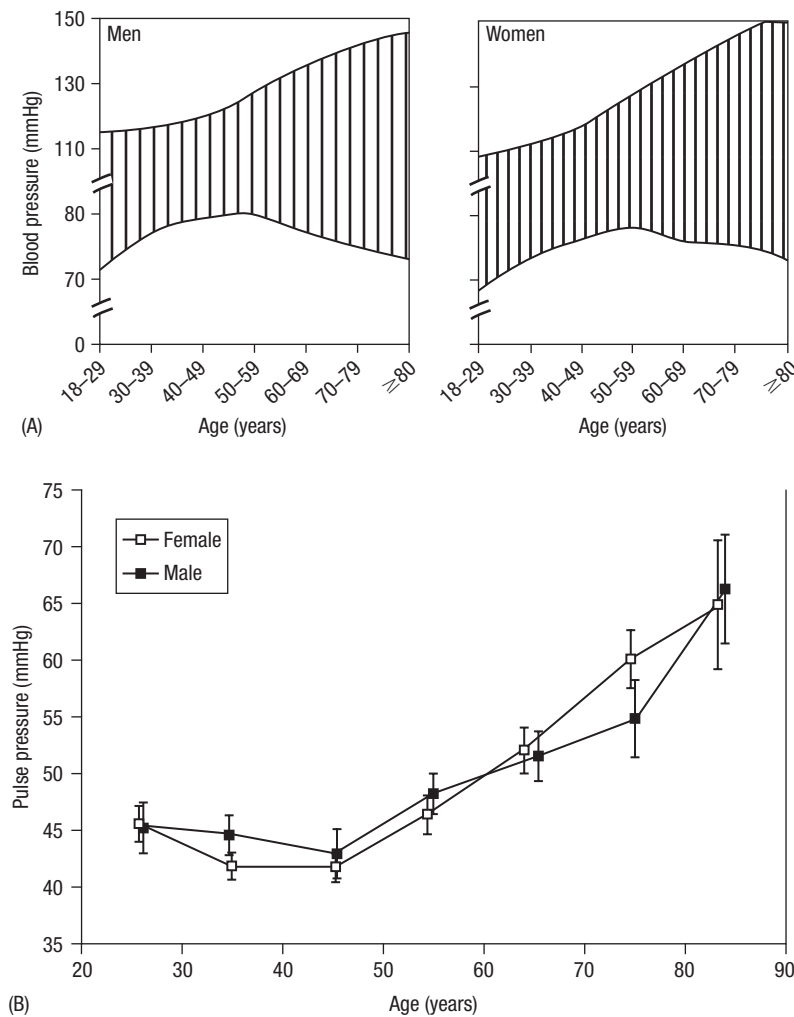


Figure 21.4 (A) Changes in brachial artery systolic, diastolic, and pulse pressure with age in white Caucasian males and females from the NHANES study. Brachial pulse pressure falls from the third to the fifth decade, then rises progressively from the seventh decade. Greater change in males than females is attributable to greater body height and lower arterial tapering in males. After Burt *et al.* (1995). (B) Changes in brachial pulse pressure with age in a normal cohort, studied at the US National Institute of Aging. Change in males and females was similar with values in the 20s and 30s apparently boosted by high amplification of the pressure wave between the ascending aorta and brachial artery. Courtesy of E Lakatta, NIA, NIH

recordings. This viewpoint, mentioned in the ESH/ESC report, has been supported by Vasan (2009) from Framingham and pressed by the working group on vascular structure and function of the European Society of Hypertension (Safar and London, 2000; Safar, 2001; Laurent *et al.*, 2006; Mancia *et al.*, 2007b). The US JNC8 guidelines are due for release in 2011.

The 1999 WHO/ISH document (Guidelines Subcommittee, 1999) referred to pulse pressure and arterial distensibility as among determinants of cardiovascular risk in hypertensive patients, together with previous stroke, coronary heart disease, heart failure, renal disease, and recurrent cardiovascular events, but stopped short of recommending any index of stiffness in stratification of risk for guidance of therapy. The document noted theoretic reasons for doing so, but found few data to support any independent predictive value. Such data have now been published for normal subjects (Mattace-Raso *et al.*, 2006; Willum-Hansen *et al.*, 2006; Benetos *et al.*, 2010; Wang *et al.*, 2010), patients with uncomplicated hypertension (Boutouyrie *et al.*, 1999; Boutouyrie *et al.*, 2000; Laurent *et al.*, 2001a; Boutouyrie *et al.*, 2002; Laurent *et al.*, 2003), end-stage renal disease at high vascular risk (Blacher *et al.*, 1999a; London *et al.*, 2001; Safar *et al.*,

2002a), diabetes mellitus and glucose intolerance (Cruickshank *et al.*, 2002), and in the elderly (Meaume *et al.*, 2001) (see also Table 29.1). These studies are discussed in *Journal of the American College of Cardiology* and *European Heart Journal* reviews, and subsequent correspondence (Vlachopoulos *et al.*, 2010a; Vlachopoulos *et al.*, 2010b; Vlachopoulos *et al.*, 2010c). In addition to data establishing independent predictive value, multiple studies (Table 21.1) have shown high reproducibility of the principal indices of arterial stiffness (aortic pulse wave velocity (PWV) and augmentation index) and lower coefficient of variation than seen for conventional measures of arterial pressure (comparing Fig. 21.5 with data in Table 21.1). These are included in the latest (2007) ESH/ESC Guidelines and are likely for the new US guidelines, which we are assured to be based on evidence.

Over the years, successive US JNC, European, and WHO/ISH committees have moved from consideration of arterial pressure alone as a guide of when and how to initiate anti-hypertensive drug therapy to consideration of total risk, and now emphasize consideration of associated cardiovascular disease, evidence of target organ damage and presence of risk factors, with consequent muting of emphasis on brachial arterial pressure numbers. Such an

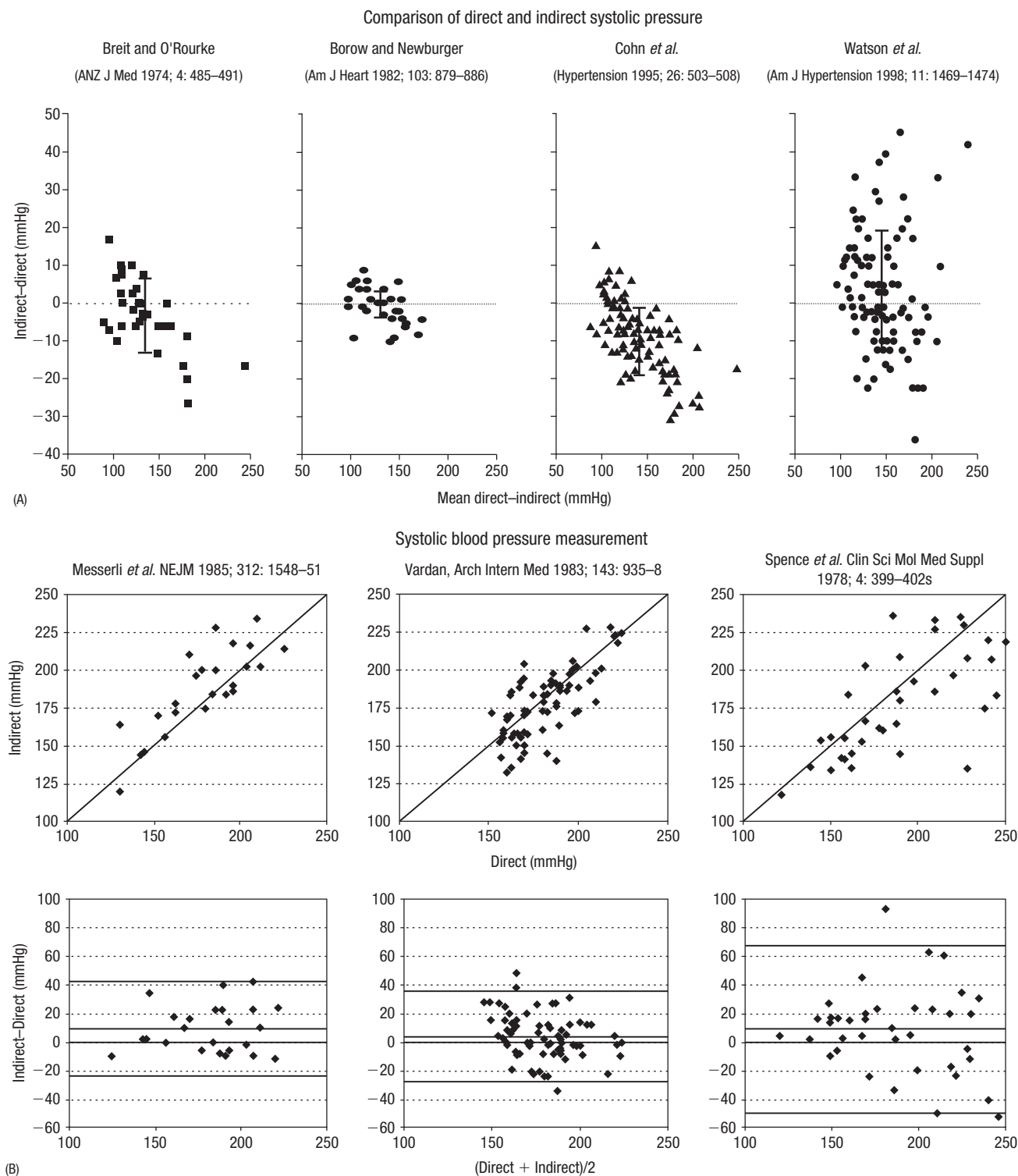


Figure 21.5 (A) Bland–Altman plots of direct and indirect systolic pressure in studies by Breit and O'Rourke (1974), Borow and Newburger (1982), Cohn *et al.* (1995), and Watson *et al.* (1998) showing mean $\pm$ s.d. (B) Comparison of directly measured brachial with indirectly measured brachial (cuff method, Korotkov sound technique) in three studies by Messerli *et al.* (1985b), Vardan *et al.* (1983), and Spence *et al.* (1978). Above, data are plotted on abscissa and ordinate, respectively, and the line of identity is shown. Below, data are plotted as Bland–Altman graphs which highlight the degree of variability of individual measurements, and show mean offset $\pm$ 2 s.d. (C) Bland–Altman plot of individual differences between the right and left arm pressure measured simultaneously by two identical, accredited, and validated oscillometric devices in 400 different subjects and patients. Systolic pressure differences at left (1.81 ± 8.6 mmHg), diastolic pressure differences at right (0.23 ± 8.3 mmHg) against the mean blood pressure. From Lane *et al.* (2002)

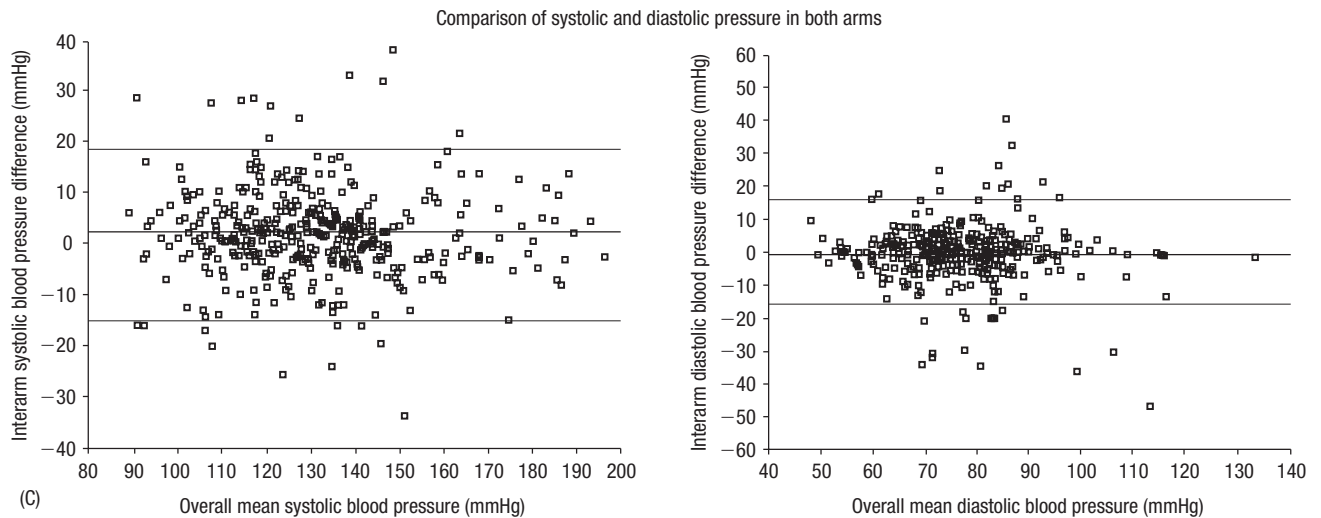


Figure 21.5 Continued

approach warrants strengthening in the future, and is being applied increasingly in drug development by pharmaceutical companies (Feig *et al.*, 2010) encouraged by the NIH, which held a special meeting on this subject in April 2011.

CARDIOVASCULAR EVENTS AND BRACHIAL BLOOD PRESSURE

Up until 1990, diastolic pressure was considered all important, not only in prediction of cardiovascular events, but also in the definition of hypertension. Up until the beginning of the SHEP trial through the US National Institutes of Health, entry into trials of anti-hypertensive therapy required elevation of diastolic pressure. A meta-analysis of clinical trials at this time (MacMahon *et al.*, 1990) showed a clear relationship between cerebrovascular and coronary events with brachial diastolic pressure (Fig. 21.6). Similar data had been shown for systolic pressure (which was preferred by the Framingham investigators), but the overwhelming clinical interest was in diastolic pressure. In considering the MacMahon meta-analysis of 1990, one has to concede that the results do not reflect present clinical experience. The nine trials considered entered patients as early as 1937 and were conducted through a period when effective anti-hypertensive therapy was not available, and when the vicious circle of the evolving malignant phase (Byrom and Dodson, 1948; Ledingham, 1995) could not always be broken, and when mortality from complications of elevated arterial pressure was very high. Criteria for initiating treatment for diastolic pressure elevation were generous (> 110 mmHg) and elevation of systolic pressure was considered irrelevant. Problems with high and ever-increasing peripheral resistance were the cause of blood pressure elevation, and far fewer people reached the age when increased arterial stiffness became the dominant cause of elevated pressure

and of cardiovascular events. The changes in clinical experience have been well summarized by Dustan (1989). Data in Fig. 21.6 are compelling, and many thousands of people were entered and followed over 6–25 years, with over a thousand events occurring. However, it must be conceded that the range of diastolic pressure was greater than in modern studies, and that people were younger than in modern studies. The data do not precisely define the risk of an individual presenting now with elevated arterial pressure.

The original Framingham cohort, whose change in blood pressure with age is shown in Fig. 21.1, has been analyzed with respect to coronary, cardiovascular, and all-cause mortality (Franklin *et al.*, 1999). A different picture emerged in this cohort compared to the 1990 meta-analysis in that diastolic pressure was not closely related to mortality, but systolic pressure was; and for any given level of systolic pressure, diastolic pressure was inversely related to mortality (Fig. 21.7, left). A corollary was that pulse pressure was directly related to all-cause mortality in this original Framingham cohort (Fig. 21.7, right). Similar findings have been noted for European populations (Fig. 21.8A, page 478) (Staessen *et al.*, 2000a) and indeed had been reported in earlier studies (Darne *et al.*, 1989; Madhavan *et al.*, 1994; Benetos *et al.*, 1997). Findings are explicable on the basis of arterial stiffening with age. Since pulse pressure is (other things being equal) a measure of arterial stiffness, and since arterial stiffening is both a manifestation of existing arterial degeneration, and a cause of future vascular events and of left ventricular hypertrophy and failure (Hundley *et al.*, 2001; Kostis *et al.*, 2001; O'Rourke, 2001; Domanski *et al.*, 2002; Franklin *et al.*, 2002), this relationship between brachial pulse pressure and cardiovascular events should come as no surprise. However, the earlier MacMahon *et al.* (1990) meta-analysis still needs to be acknowledged. Reanalysis of these data failed to confirm any direct relationship between pulse pressure and mortality. The Framingham study group pursued this

Table 21.1 Reproducibility of indices of arterial stiffness from multiple studies

(A) Reproducibility of stiffness indices			
(a) Aortic pulse wave velocity (PWV)			
	Intraobserver (m/sec)	Interobserver (m/sec)	Coefficient variation (%)
Liang <i>et al.</i> (1998)			3.2
Hirata <i>et al.</i> (2002)			3.3
Wilkinson <i>et al.</i> (1998)	0.1 (s.d. 1.2)	0.3 (s.d. 1.3)	
Sutton-Tyrrell <i>et al.</i> (2001a)		0.2 (s.d. 1.2)	
(b) Aortic augmentation index (calculated from radial artery using Sphygmocor)			
	Intraobserver (%)	Interobserver (%)	Coefficient variation (%)
Wilkinson <i>et al.</i> (1998)	0.5 (s.d. 5.4)	0.2 (s.d. 3.8)	
Siebenhofer <i>et al.</i> (1999)		0.4 (s.d. 6.4)	
Filipovsky <i>et al.</i> (2000)			7.9
Rietzschel <i>et al.</i> (2001b)			6.7
Savage <i>et al.</i> (2002a)	0.1 (s.d. 4)	0.1 (s.d. 3)	
Hirata <i>et al.</i> (2002)			2.3
(B) Reproducibility of indices on different days (same operators)			
	Coefficient of variation (%)		
Liang <i>et al.</i> (1998)			
Aortic PWV	3.2		
Leg PWV	5.0		
Augmentation index (carotid)	1.3		
Carotid intima media thickness	2.8		
Systemic arterial compliance	9.2		
Brachial flow-mediated dilation	10.3		
Carotid distensibility coefficient	10.0		
Hirata <i>et al.</i> (2002c)			
Brachial systolic BP	1.9		
Brachial diastolic BP	2.4		
Heart rate	4.0		
Aortic systolic BP	2.1		
Augmentation index (aortic from radial artery)	2.0		
Reflection time	2.0		
Ejection duration	2.3		
Aortic PWV	3.2		
Leg PWV	3.2		

Central augmentation index and aortic pulse wave velocity show acceptable reproducibility, with values at least as good as for more complex measures, and comparable with conventional indices of heart rate and arterial pressure.

issue by combining the original cohort with a younger offspring cohort who were entered some decades after the first and who were generally younger at entry. A more complicated mortality pattern emerged for the whole group (Franklin *et al.*, 2001). In younger subjects (40 years), diastolic pressure was the best predictor; in the middle-aged group age 40–60 years, diastolic pressure and systolic pressure were equally predictive; whereas in people > 60 years, pulse pressure was the best predictor and diastolic pressure was inversely related to outcome (Franklin *et al.*, 2001). This finding can only be explained on the basis of strong amplification of the brachial pressure wave in young men: a phenomenon whose extreme

manifestation is ‘spurious systolic hypertension of youth’ (O’Rourke *et al.*, 2000a; Franklin *et al.*, 2001; Wilkinson *et al.*, 2001; Mahmud and Feely, 2003a; Mahmud and Feely, 2003b). This was discussed in Chapter 20.

There have been a series of publications describing reanalysis of existing data with respect to pulse pressure and mortality. MacMahon (personal communication) has reanalyzed his original data and has found no definite association. This may have been a consequence of the relatively young age of subjects at entry. The Prospective Studies Collaboration (2002) found brachial pulse pressure to be unhelpful for prediction of events at any age, and systolic, diastolic, mean, and average $((\text{SBP} + \text{DBP})/2)$

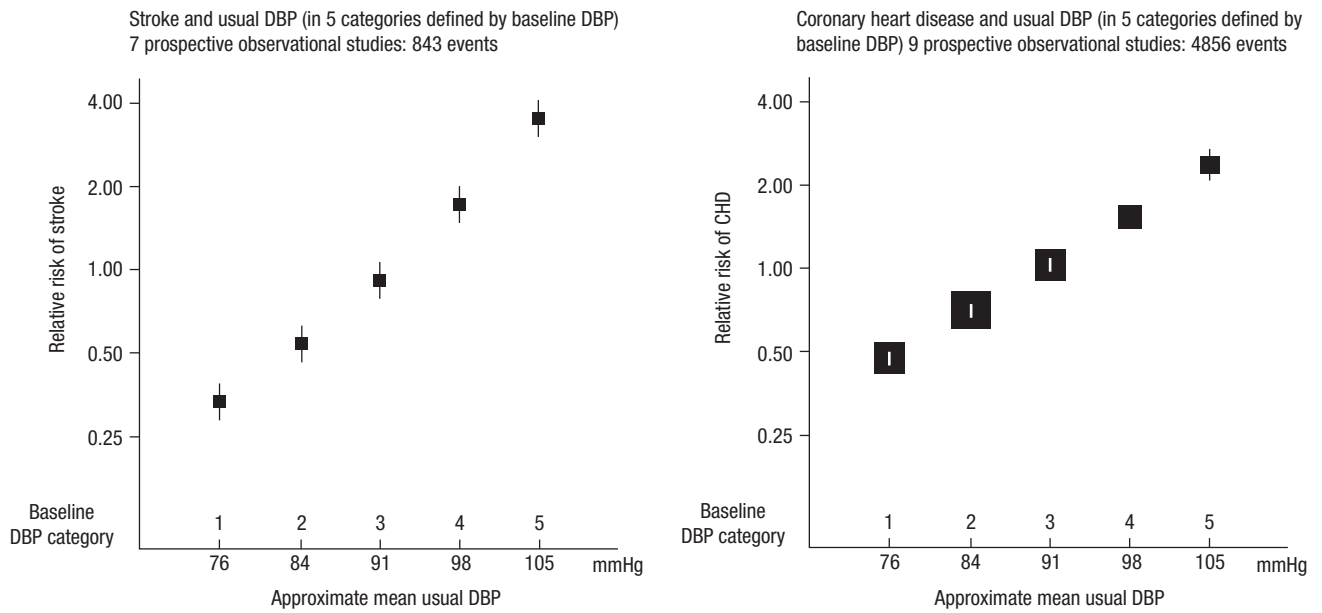


Figure 21.6 Relationship between stroke and usual diastolic blood pressure (DBP) (left), and between coronary artery disease (CHD) and usual diastolic blood pressure (right) from nine prospective observational studies. Size of squares proportional to number of events: 95 percent confidence intervals given by vertical lines. From MacMahon *et al.* (1990)

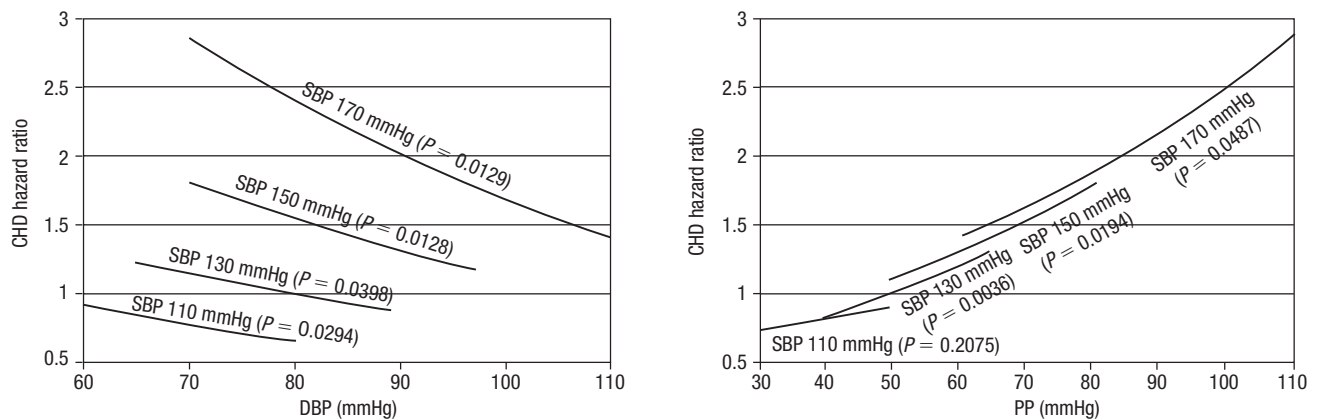


Figure 21.7 Coronary heart disease hazard ratio in relation to initial systolic (SBP) and pulse diastolic pressure (DBP) from the Framingham cohort. Left: hazard ratio is positively related to diastolic pressure, but for each level of initial systolic pressure, there is an inverse relationship to diastolic pressure. Right: In consequence of the above, there is a consistent relationship between hazard ratio and initial pulse pressure (PP). From Franklin *et al.* (1999)

to be most useful. This large meta-analysis has been challenged on a number of counts (see subsequent *Lancet* correspondence). Review of the large MRFIT database (Domanski *et al.*, 2002) has shown that only concordant elevation (of systolic and diastolic pressure) was predictive of events up to age 44, whereas both concordant and discordant (elevated systolic, but not diastolic pressure) elevation were predictive in older (age 45–57 years) subjects – implying support for the Framingham view. The vast majority of publications (see summary by Franklin *et al.* (2002)) have supported the Framingham observations. Comparisons have been made between influence of mean pressure and pulse pressure on

cardiovascular events. While most, conducted in older subjects, show a positive relationship between pulse pressure and events, mean pressure (calculated as diastolic pressure plus one-third pulse pressure) has not generally given additional predictive value (Blacher *et al.*, 1999a; Staessen, 2002). Again, the large but diffuse Prospective Studies Collaboration (2002) dissents.

As discussed above, indices of arterial stiffness as augmentation index and PWV have recently been included in the Framingham screening cycles. Results have recently been published (Mitchell *et al.*, 2010) and strongly support the use of carotid-femoral PWV as a measure of aortic stiffness, and a predictor of subsequent

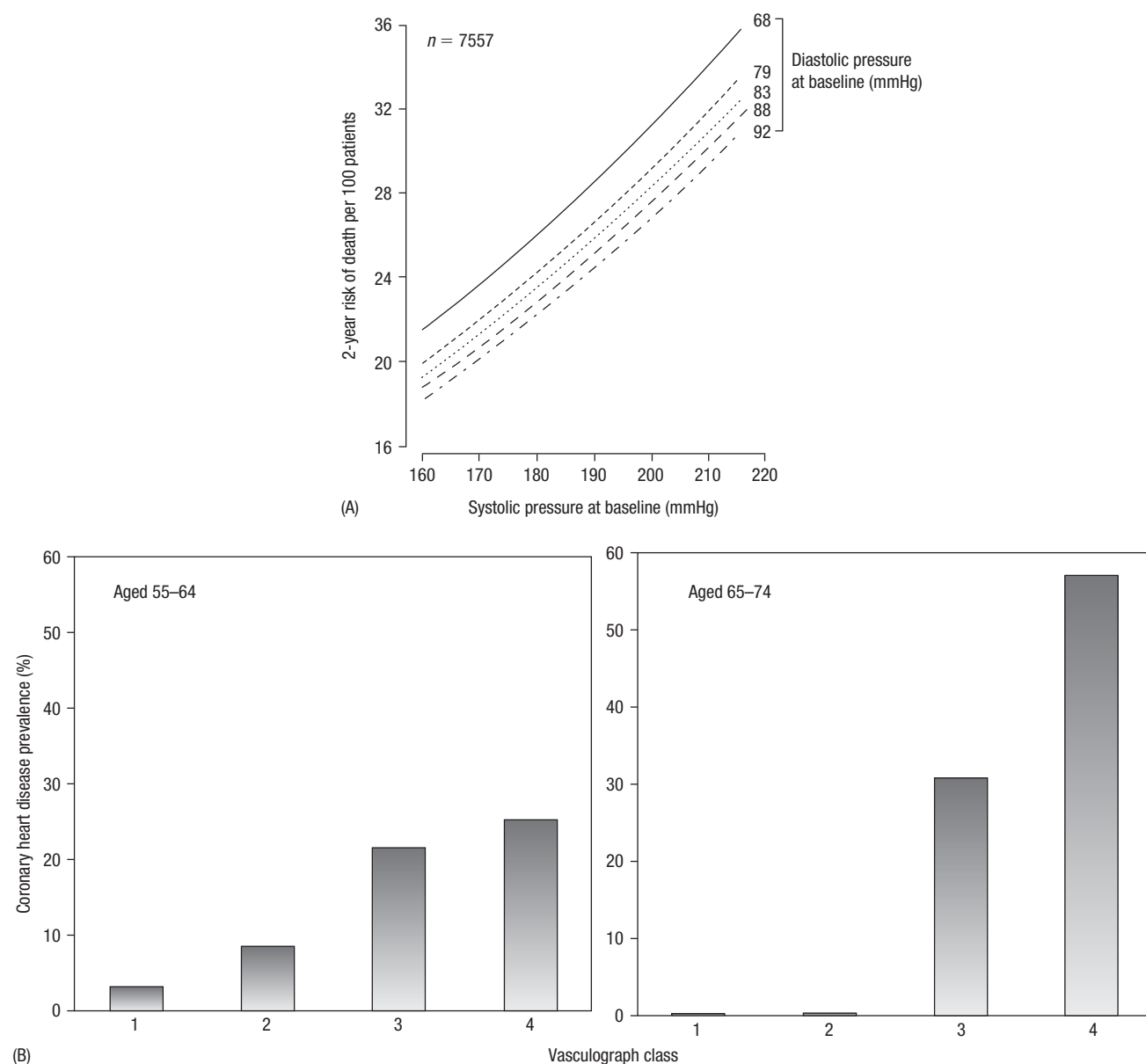


Figure 21.8 (A) Risk of death related to systolic pressure for different levels of diastolic pressure. Risk was greatest when diastolic pressure was low, and lowest when diastolic pressure was high, in agreement with the Framingham data in Fig. 20.7. From Staessen *et al.* (2000b). (B) Relationship between vasculograph class and prevalence of coronary artery disease in normotensive man aged 55–64 years (left) and 65–74 years (right) in the Framingham cohort of 1778 evaluable subjects. From data published by Dawber *et al.* (1973) and shown by Dr W Kannel in his 2003 William Harvey Annual Presentation to the American Society of Hypertension. Vasculograph class is based on the shape of the pressure wave recorded non-invasively in the finger. The relationship between class and coronary heart disease prevalence was significant ($P < 0.05$, $P < 0.01$) for the younger and older groups, respectively

events. Surprisingly, however, this study showed no value for indices of central pressure waveforms – as carotid systolic or pulse pressure, as augmentation index, or as carotid-femoral amplification, in prediction of cardiovascular events or mortality. A fundamental error is apparent in this study, in that Mitchell *et al.* depended on applanation tonometry to measure the brachial waveform. This has not been validated against intra-arterial pressure, as it has at radial and carotid sites (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; O'Rourke and Adji, 2010b), and the

method provides 'form factor' (FF) (mean pressure minus diastolic pressure pulse pressure) (Chemla *et al.*, 2002b; Segers *et al.*, 2007b; Segers *et al.*, 2009). Brachial FF is virtually identical to carotid (Segers *et al.*, 2009). In consequence (eqn 28.1), with brachial mean and diastolic assumed identical to carotid, systolic pressure in the carotid artery is not significantly different to brachial. Rather than central systolic and pulse pressure being 10–12 mmHg less than brachial, as in invasive, and radial, non-invasive studies (O'Rourke and Adji, 2010a), pressure

was not significantly different: +4 mmHg in males and –1 mmHg in females. Other large studies, conducted over long periods of time by Wang *et al.* (2010) and Benetos *et al.* (2010), show clear value of central pressure and indices of central pressure (central pulse pressure, AIx, amplification), independent of brachial values in prediction of outcomes. The recent results of Mitchell *et al.* (2010) also challenge the earlier Framingham studies of Dawber *et al.* (1973) over the value of pulse wave analysis (O'Rourke *et al.*, 2010d). In 1973, Dawber *et al.* measured the upper limb pulse with Henry Lax's finger plethysmogram in 1778 Framingham subjects and classified waveforms into vasculograph classes. These classes, 1–4, were based on the wave shape before and after the dicrotic notch, and roughly correspond to augmentation index as we describe this now (Takazawa *et al.*, 1993a). In normotensive subjects, and in the older age groups, there was a significant relationship between vasculograph class and prevalence of coronary disease (Fig. 21.8B). This index was referred to in just three later publications (Kannel *et al.*, 1980a; Kannel *et al.*, 1980b; Kannel *et al.*, 1981) where vasculograph class was shown to correlate with age and systolic blood pressure. Vasculograph class did correlate with development of stroke, but when adjusted for age and systolic pressure, the correlation was weak and not statistically significant.

CARDIOVASCULAR EVENTS AND CENTRAL SYSTOLIC PRESSURE

It is now clear that accurate measures of central pulse pressure and systolic pressure provide more precise predictive indices of cardiovascular events, and this indeed is the case. Ability to measure central systolic and pulse pressure eliminates the effect of pressure wave amplification in the upper limb, and so the differential effects of growth and arterial stiffening with aging, of heart rate, and of drugs on this relationship (Fig. 21.3) (Vlachopoulos and O'Rourke, 2000b; Wilkinson *et al.*, 2001; Franklin *et al.*, 2002). Central pressures have been measured directly, at cardiac catheterization, or have been measured indirectly from the carotid pressure or diameter waveform calibrated against brachial diastolic and mean pressure (Kelly and Fitchett, 1992; Van Bortel *et al.*, 2001b), or from use of a transfer function applied to the radial artery pressure waveform (Asmar *et al.*, 2001b; Williams *et al.*, 2006; Roman *et al.*, 2007; Roman *et al.*, 2009). This is discussed further in Chapter 28. These studies have shown a definite association between central systolic, pulse, and augmented pressure with extent of coronary atherosclerosis, likelihood of restenosis after angioplasty, presence of left ventricular hypertrophy and carotid intima-medial thickening and mortality (Agabiti-Rosei *et al.*, 2007; Vlachopoulos *et al.*, 2010a; Vlachopoulos *et al.*, 2010b; Weber *et al.*, 2010; Wang *et al.*, 2010). Details are given in Chapter 28.

CARDIOVASCULAR EVENTS AND INDICES OF ARTERIAL STIFFNESS

Measures of arterial stiffness including aortic pulse wave velocity and central pressure wave augmentation have been included in the Framingham and other major studies screening cycles. Information currently available comes largely from three groups in Paris – two large groups of hypertensive subjects studied by Benetos, Laurent, and Safar, and the other a group of patients with very high prevalence of cardiovascular events in end-stage renal failure studied by London *et al.* (2001). These studies were initiated 15–20 years ago and have largely been coordinated by Michel Safar. With a large number of events in the end-stage renal failure patients, definitive data were published first whereas outcome data from the hypertensive group were initially not quite so compelling (Boutouyrie *et al.*, 1999; Boutouyrie *et al.*, 2000; Laurent *et al.*, 2001a; Boutouyrie *et al.*, 2002; Laurent *et al.*, 2003).

The high risk end-stage renal failure group has been studied with respect to association of cardiovascular events with 'aortic' (carotid to femoral) PWV (Blacher *et al.*, 1999a), with carotid augmentation index (London *et al.*, 2001) and carotid pulse pressure (Safar *et al.*, 2002a) (see Figs 25.7, 25.8, 25.9, and 25.10). Results are striking. When divided into tertiles or quartiles, there was clear separation of mortality and of cardiovascular events at ten years, with as much as six-fold difference (600 percent) between the top and bottom quartile. While these patients were in end-stage renal failure and on dialysis, they did not die from renal failure, but almost always as a consequence of stroke, myocardial infarction, or cardiac failure. In recent years, data from a normal population have also been obtained from the radial artery waveform, with synthesis of the aortic waveform by the SphygmoCor process (see Chapter 30). Results are similar to those obtained from analysis of the carotid waveform, and have extrapolated to a group of 125 151 subjects in a French population (Benetos *et al.*, 2010).

In the high-risk renal failure group, it was possible to examine the effects of therapy on outcome. Guerin *et al.* (2001) showed that when aortic PWV was reduced by therapy, outcome was better than in patients where reduction in this index of stiffness did not occur (Fig. 25.8).

The hypertensive group studied by Benetos *et al.* (2002) included more patients, but they had fewer events. Therapy had doubtless influenced outcome, but this was administered in a conventional fashion, without targeting for any reduction in aortic stiffness or pressure wave reflection. Results to date are in line with those in high-risk end-stage renal failure patients (Figs 25.7 and 25.8) (Boutouyrie *et al.*, 2000; Boutouyrie *et al.*, 2002). Progression of aortic PWV over time has been demonstrated (Benetos *et al.*, 2002), and shown to be related to the level of systolic and pulse pressure, to heart rate, and to creatinine level, as well as to age. The relationship to

creatinine provides some clue to the association between renal failure and arterial stiffness. Influence of heart rate (Franklin *et al.*, 1999; Palatini and Parati, 2010) and pulse pressure can readily be explained on the basis of material fatigue, with greater likelihood of elastin fiber fracture when exposed to a greater number and greater extent of cycles of pulsatile stretch (see Chapter 20).

The REASON study of arterial hypertension was initiated by Michel Safar, to investigate the influence of an arterial dilator drug combination (perindopril/indapamide) against a β -blocker (atenolol) in the reduction of arterial stiffness and a surrogate of cardiovascular morbidity (left ventricular mass). Results (Asmar *et al.*, 2001b; De Luca *et al.*, 2004) were intriguing in that aortic PWV was reduced to a similar degree by both perindopril/indapamide, and by atenolol, and such reduction paralleled the fall in diastolic pressure. It appeared to be a passive effect, consequent on reduction of arterial distending pressure. In contrast, aortic and carotid augmentation and aortic and carotid systolic and pulse pressure fell to a greater degree with perindopril/indapamide than with atenolol. Part of this is attributable to the difference in heart rate between the two different therapies, but most was attributable to reduction in wave reflection with perindopril resulting from a greater decrease in muscular artery PWV. Whatever the underlying mechanism, left ventricular mass was reduced to a greater degree after 12 months with the vasodilator combination than with atenolol (Asmar *et al.*, 2001b; De Luca *et al.*, 2004; London *et al.*, 2004a; London *et al.*, 2004b). This subject was pursued by our group in Sydney who found, in comparison to atenolol, that the angiotensin-converting enzyme inhibitor (ACEI), ramipril, causes a similar decrease in aortic PWV, but a greater reduction in aortic augmentation index (Hirata *et al.*, 2005b) and in calculated aortic systolic and pulse pressure by 5.2 mmHg. Change in aortic augmentation index was associated with a decrease in PWV in the arm and leg and an increase in reflection site distance with ramipril, but not with atenolol. All findings can be explained on the basis of active dilation of muscular arteries by ramipril, but not by atenolol, and (as with perindopril/indapamide and atenolol in REASON) a passive effect only of blood pressure reduction in the degenerate aortic and proximal elastic arteries.

Subsequent trials have confirmed the REASON study and are discussed in Chapter 28. The first was the CAFE substudy of ASCOT (Sever *et al.*, 2001; Williams and O'Rourke, 2001). CAFE showed a systematic reduction of 4–5 mmHg in central systolic pressure for amlodipine compared with atenolol; authors attributed the benefit of amlodipine over atenolol in the CAFE substudy, and in the full ASCOT trial, to greater reduction in systolic central pressure with amlodipine (Williams *et al.*, 2006). Miyashita *et al.* (2010) have subsequently shown consistently higher aortic than brachial systolic blood pressure reduction with arterial dilating drugs (ACEI, angiotensin

receptor blocker (ARB), calcium channel blocker (CCB)) than with diuretic and β -blocker drugs (Fig. 20.17).

MAJOR THERAPEUTIC TRIALS

The modern era was revolutionized by the SHEP study, initiated by the US NHLBI. SHEP established the importance of systolic blood pressure lowering in older patients with isolated systolic hypertension (ISH), with greatest reduction in stroke (reduced by 37 percent) and cardiac failure (reduced by 54 percent), and lesser (27 percent) reduction in myocardial infarction and coronary death. Active treatment was chlorthalidone (diuretic), and with a β -blocker added if target blood pressure was not achieved. The comparator was placebo. SHEP's results were decisive, but led to another query: were newer more expensive anti-hypertensive drugs, i.e. ACEIs, CCBs, α -blockers, or ARBs, better than the older therapy with diuretics and β -blockers? The NHLBI ALLHAT trial was designed to answer this question and reported in late 2002. Findings (ALLHAT Collaborators, 2002) are consistent with the view that pressure reduction is all important and that the newer drugs (ACEIs, CCBs) have no special advantage for first-line treatment. Most ALLHAT patients did eventually require a cocktail of two or more drugs to achieve blood pressure control.

Many other trials have been conducted, and the pharmaceutical industry has had a major role in their design, conduct, and interpretation, and in the promulgation of results. At stake are billions of dollars in sales, and millions of dollars in research studies. There can be no doubt that patients and the profession have benefited by the introduction of the new drugs, since they are highly effective in reducing arterial pressure and decreasing cardiovascular events, and have very few side effects. However, the cost–benefit issue has not been fully settled, and is extremely important for poor communities and poor nations.

A series of meta-analyses have been published which establish the benefit of blood pressure reduction in people with hypertension, and with high likelihood of cardiovascular events (Blood Pressure Lowering Treatment Trialists' Collaboration, 2000; Staessen *et al.*, 2000b; Staessen *et al.*, 2001b; Staessen, 2002). These also showed little benefit of one drug class over another, and a general relationship between benefit (expressed in relation to stroke, coronary events, cardiovascular events, or cardiovascular death) and blood pressure reduction (Fig. 21.9) (Nissen *et al.*, 2004; PEACE Trial Investigators, 2004; Pitt, 2004). The relationship is enhanced, especially for stroke, when one makes allowance for the greater reduction in central than brachial pressure that occurs with arterial dilating drugs (ACEIs, ARBs, CCBs) (see Fig. 21.10).

Two recent studies (HOPE and LIFE) have created considerable controversy, as well as providing very useful information. HOPE (Yusuf *et al.*, 2000) studied the effect

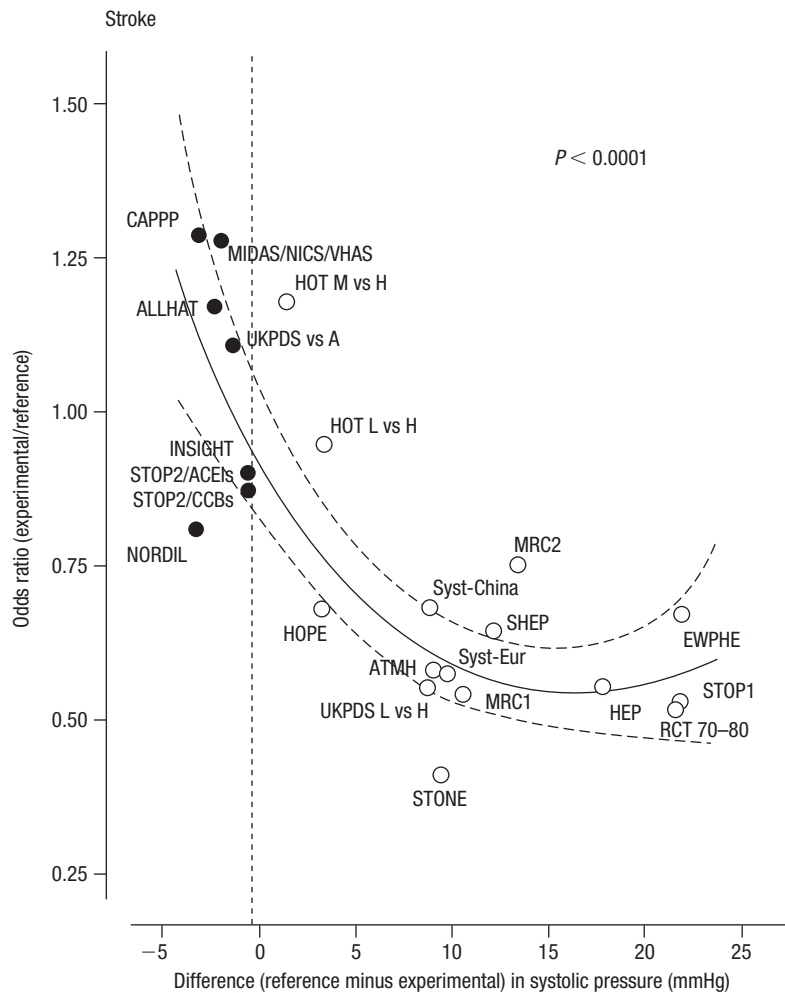


Figure 21.9 Odds ratio for stroke plotted against difference in achieved brachial systolic blood pressure for a series of clinical trials. There was a consistent relationship between risk and change in arterial pressure with treatment. From Staessen *et al.* (2001b)

of adding ramipril (10 mg) daily to the therapeutic regime of patients with high cardiovascular risk. Most patients were hypertensive and many were diabetic. HOPE showed that use of the ACEI ramipril caused reduction of cardiovascular events, especially stroke and cardiac failure. There was trivial reduction (averaging 3.3/1.4 mmHg) in brachial artery pressure. Comparing benefit to blood pressure reduction, authors considered the benefit could not be explained by the latter, and pressed the view of ramipril having a 'benefit beyond blood pressure reduction'. This view was encouraged by Aventis, who noted that the benefit appeared to be specific to ramipril. Other issues were ignored or set aside, including the greater 24-hour reduction in brachial blood pressure with ramipril compared to atenolol (Svennson *et al.*, 2001) and the expected greater reduction in central than brachial systolic pressure with this ACEI drug (Mitchell *et al.*, 1996; Griffin *et al.*, 2003; Hirata *et al.*, 2005b). The greatest benefit was seen in diabetic subjects – which is not surprising, given the greater apparent fragility of blood vessels in diabetes mellitus, and the known greater benefit of blood pressure reduction (see Chapters 19 and 25).

LIFE (Losartan Intervention For Endpoint reduction in hypertension study) (Dahlöf *et al.*, 2002)) compared the

ARB, losartan, against the β -blocker, atenolol, in a large group of hypertensive patients and came up with results similar to HOPE. There was a greater reduction in total cardiovascular events with losartan than atenolol. Results were 'driven' by a big difference in stroke which more than offset a marginal non-significant benefit of atenolol for coronary events. Losartan also caused greater reduction in left ventricular (LV) mass (Lip, 2001; Gerds *et al.*, 2002; Devereux *et al.*, 2004). The measured difference in brachial arterial pressure was trivial (about 1.0 mmHg systolic pressure). Beneficial effects, like those in HOPE, can be explained on the basis of greater reduction in central than brachial pressure (systolic and pulse) with losartan. This was promptly acknowledged by the investigators, who also suggested that any apparent difference in coronary events may be due to the anti-ischemic effect of atenolol through reducing heart rate and myocardial oxygen demand.

It might be added that on the other hand, any benefit of losartan over atenolol on renal function may be due to the action of the former in decreasing efferent arteriolar tone, and thereby reducing glomerular pressure. Such a mechanism was invoked to explain preservation of renal function with an ARB in diabetic patients (Brenner *et al.*, 2001; Palmer, 2002).

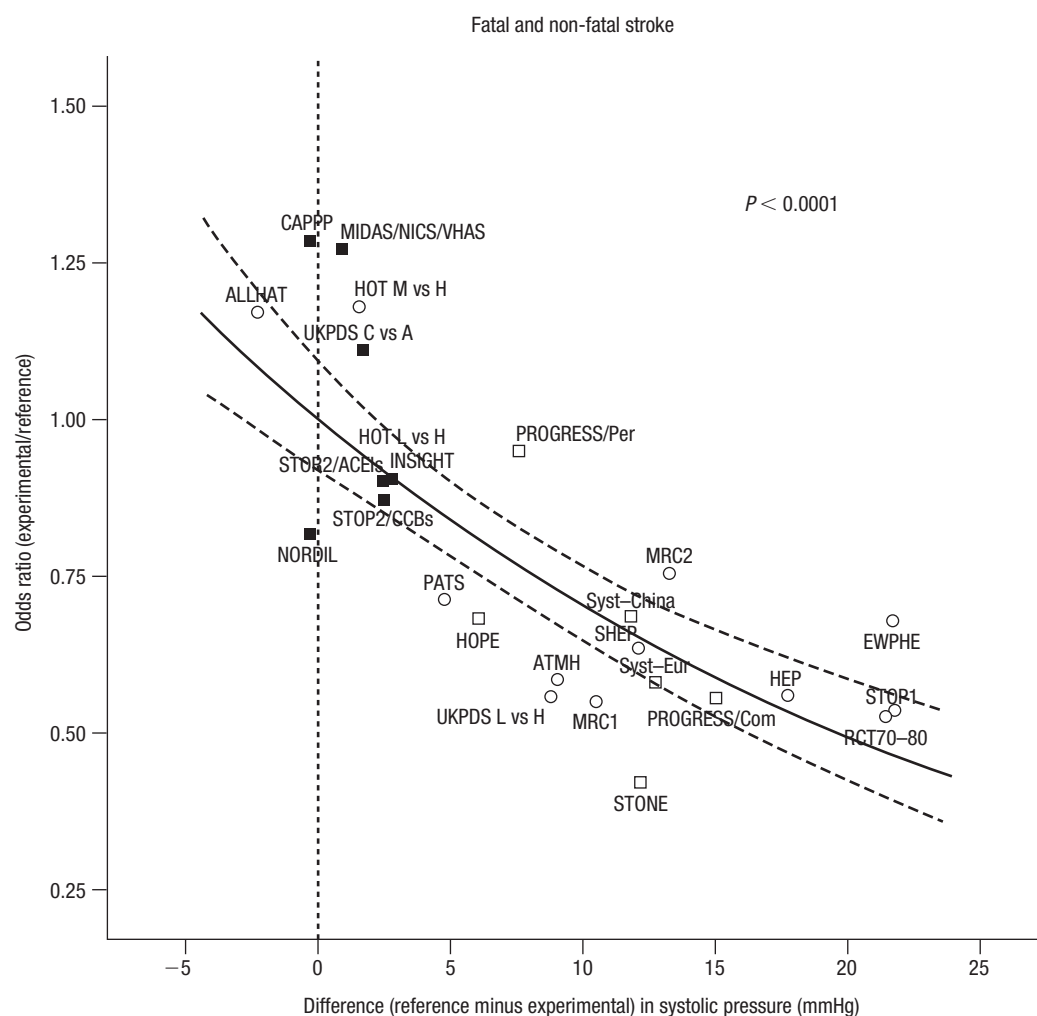


Figure 21.10 Odds ratio for stroke plotted against difference in estimated central systolic blood pressure for a group of clinical trials. Estimate of central systolic pressure allowed for a decrease over brachial pressure of 3.0 mmHg when the active therapy was a drug which reduces peripheral wave reflection (calcium channel blocker, ACE inhibitor or angiotensin receptor blocker). Data prepared by J Staessen and J Wang

New drugs are being developed to affect arterial properties without any effect on peripheral resistance or smooth muscle tone in conduit arteries. One such drug is ALT711 – an advanced glycation end product (AGE) crosslink breaker which it was hoped would improve aortic elasticity by cleaving collagen crosslinks in the wall. In older experimental animals (Vaitkevicius *et al.*, 2001) and in humans (Kass *et al.*, 2001), this has shown promise, although a lingering worry is that it may promote arterial dilation and aneurysm formation. The vasopeptidase inhibitor omapatrilat showed evidence of improving function of the proximal ascending aorta and aortic arch (Mitchell *et al.*, 2002), while Jiang *et al.* (2002) have confirmed that tiny doses of nitroglycerin (0.1 mg/hour) can markedly reduce aortic augmentation and central systolic pressure without any effect on other circulatory parameters.

VASODILATOR THERAPY OF CARDIAC FAILURE

Vasodilator therapy is accepted as the basis for treating patients with cardiac failure. There is better evidence on the importance of such drugs than there is on the traditional measures of diuretics and digoxin and more evidence of benefit in heart failure than in hypertension. The drugs used now for cardiac failure are the same as those used for hypertension, for diabetes mellitus, and for other patients with high cardiovascular risk. What escapes the attention of clinicians is that a common problem in all these conditions is early return of wave reflection, and a common feature of arterial vasodilator drugs is reduction in wave reflection (see Chapter 10). As described elsewhere, reduction of wave reflection in ISH decreases aortic systolic pressure, whereas in cardiac failure due to systolic LV dysfunction, reduction in wave reflection increases

stroke volume and cardiac output (Fig. 21.11 and Chapter 14) (Westerhof and O'Rourke, 1995; Uebaba *et al.*, 2002).

The first vasodilator drug trial compared a nitrate–hydralazine combination against placebo in patients with cardiac failure and low left ventricular ejection fraction (Cohn *et al.*, 1986). Benefit was substantial with 36 percent reduction in mortality at three years. This was followed by the CONSENSUS trial in Europe which used the ACEI, enalapril, as the vasodilator, then the second Veterans Administration Study (Cohn *et al.*, 1991), using enalapril, and the SAVE (captopril) and SOLVD (enalapril) trials in the United States. In all these, there were variable changes in arterial pressure and these did not explain the benefit of vasodilator therapy. Indeed, in the first VA study (Cohn *et al.*, 1986), there was no difference in systolic pressure, but an increase in ejection fraction with nitrate/hydralazine. In the second VA study (Cohn *et al.*, 1991), exercise capacity was improved by nitrate–hydralazine but not by enalapril after six months, whereas there was no difference in blood pressure at this time.

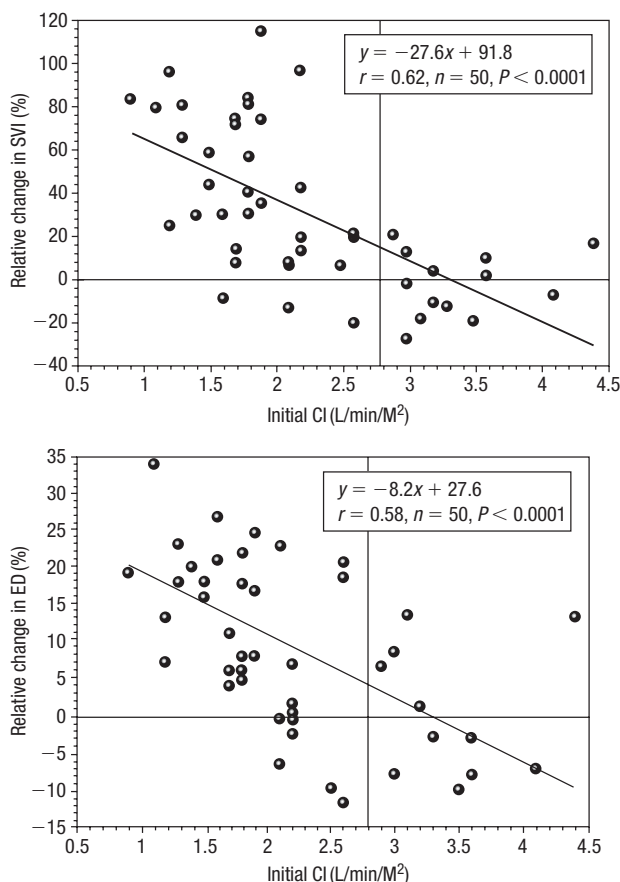


Figure 21.11 Effect of nitroglycerin infusion in increasing output from the heart in patients with cardiac failure is inversely related to the severity of failure, and is attributable to reduction in peripheral wave reflection with decreased afterload. Response measured in relation to change in stroke volume index (SVI) (top), and to change in ejection duration (ED) (bottom). From Uebaba *et al.* (2002)

Observers came to attribute benefit to direct effect on the renin angiotensin system, overlooking the point that the first such trial used a drug combination without any direct effect on this system (Cohn *et al.*, 1986), the point that in the second VA trial (Cohn *et al.*, 1991) the nitrate–hydralazine combination induced greater symptomatic benefit than enalapril, and the point that nitrate-based therapy of acute cardiac failure is very effective (Sharon *et al.*, 2000), and decreases plasma renin and angiotensin levels dramatically (Johnson *et al.*, 2002).

Investigators are reluctant to explain the benefit of ACEI or ARB therapy in cardiac failure on the basis of arterial vasodilatation. In most of the above-mentioned trials, benefit continued despite rise in brachial arterial pressure. The benefits of reducing wave reflection can be inferred from examination of the central (or peripheral) pressure waveform (see Chapters 9, 14, 15, and 26), but cannot be detected in these heart failure trials, nor in LIFE and HOPE from the simple measures of brachial systolic and diastolic pressure that were used. The concept that vasodilators require ACE inhibition or angiotensin antagonism for beneficial action on the heart remains unproven (Bishop *et al.*, 2000; Staessen, 2002; Griffin *et al.*, 2003; Kurtz, 2003; PEACE Trial Investigators, 2004; Pitt, 2004; Taylor *et al.*, 2004; O'Rourke and Hashimoto, 2007; Solomon *et al.*, 2007; van Veldhuisen *et al.*, 2009; Shah *et al.*, 2010; Staessen *et al.*, 2010).

CURRENT GUIDELINES: JNC7, ESH/ESC, AND WHO/ISH

Current guidelines on detection and management of hypertension have been provided by the US 7th National Committee on Prevention, Detection, Evaluation and Treatment of High Blood Pressure (JNC7) in 2003, by the European Societies of Hypertension and of Cardiology (Mancia *et al.*, 2007b), and by the World Health Organization/International Society of Hypertension (Guidelines Subcommittee, 1999; Chobanian *et al.*, 2003; ESH/ESC Guidelines Committee, 2003; ISH Writing Group, 2003). Guidelines are similar, although JNC7 regards systolic pressure levels from 120 to 140 mmHg to be abnormal, requiring lifestyle modification at least. ESH/ESC and WHO/ISH recognize levels of normality up to 140 mmHg systolic and/or 90 mmHg diastolic pressure, but stratify levels of hypertension at higher levels of pressure, and the need to initiate therapy according to risk, bearing in mind other risk factors, evidence of target organ damage, and associated clinical conditions, as well as the level of blood pressure. Authorities recommend anti-hypertensive therapy in patients with systolic pressure 130–139 mmHg in the presence of diabetes mellitus, cardiac failure, or renal insufficiency.

The importance of following guidelines has been pressed in a series of editorials and featured articles in the best medical journals (MacMahon, 2000; Wilkinson *et al.*,

2000a; Chobanian, 2001; Hyman and Pavlik, 2001; Vasan *et al.*, 2001; Frohlich, 2003; Franco *et al.*, 2004; Jones and Hall, 2004; Vlachopoulos *et al.*, 2010a; Vlachopoulos *et al.*, 2010b; Vlachopoulos *et al.*, 2010c), while attention is focused on the systolic rather than diastolic pressure as the target for therapy (Black, 1999; Lloyd-Jones *et al.*, 1999; Izzo *et al.*, 2000; Kostis *et al.*, 2001; Safar and O'Rourke, 2006). While concern has been raised about how low diastolic pressure may be permitted to fall, most agree that this is unlikely to be a problem if maintained at 60 mmHg or above (Somes *et al.*, 1999; Franklin *et al.*, 2002; Nissen *et al.*, 2004).

Doubtless, new data linking cardiovascular risk to arterial stiffness will be incorporated in future JNC, ESH/ESC, and WHO/ISH statements. This point has been pressed by the Working Group on Vascular Structure and Function of the European Society of Hypertension and others (Safar and London, 2000; Safar, 2001; Franklin *et al.*, 2002; Safar *et al.*, 2003; Safar *et al.*, 2004; Laurent *et al.*, 2006; Mancia *et al.*, 2007b; Staessen *et al.*, 2010; Vlachopoulos *et al.*, 2010a; Vlachopoulos *et al.*, 2010b).

WIDER IMPLICATIONS

What are the wider implications of these new moves? They add another dimension to the field where doubt has been cast on the generations-old fixation of diastolic blood pressure, where there are doubts on the ability of conventionally measured brachial pressure to show benefit of drug therapy (e.g. HOPE and LIFE studies together with VHEF (1 and 2), CONSENSUS, SAVE and SOLVD), where stratification of risk in hypertension is dominated by the presence or absence of target organ damage, associated clinical conditions, and risk factors, and where

anti-hypertensive therapy is recommended for people whose blood pressure is in the high normal range (when diabetes mellitus, cardiac failure, or renal insufficiency are present). Add to this the widening restrictions against conventional mercury manometers, their replacement with automatic and semi-automatic devices, the substantial inaccuracy of many of these devices (Fig. 21.5) (O'Brien *et al.*, 2001a; O'Brien *et al.*, 2001b; O'Brien *et al.*, 2001c; O'Brien *et al.*, 2003), and the high proportion of the normal elderly population who warrant use of anti-hypertensive therapy by current blood pressure criteria (Staessen, 2002; Staessen *et al.*, 2010), and one has a very difficult situation indeed, for individuals, for clinicians, and for public health authorities. What is likely to emerge, probably later rather than sooner, is a swing away from the tainted sphygmomanometer to other non-invasive measures of arterial degeneration, so that the sphygmomanometer might play the subsidiary rather than the dominant role in clinical practice. Should this occur, it would represent a return to the position before clinical acceptance of the cuff sphygmomanometer, when it was possible, even with primitive clinical screening methods, to recognize and treat patients with elevated arterial pressure (Mahomed, 1872; Mackenzie, 1902; O'Rourke, 1992; Swales, 1996a) and to decline for life insurance those with elevated cardiovascular risk (Postel-Vinay, 1996). Our minds need to be open to such possibilities. One already sees this among life insurance assessors, who despair of the value of conventional blood pressures in measuring risk (O'Rourke and Clubb, 2011). One also sees it in the modern practice of anesthesia where for all but major procedures, more notice is taken of digital pressure pulse waveforms and oxygen saturation values than of arterial pressure.

Arterial biomarkers

Arterial stiffness	487	Carotid intima-media thickness	495
Central pressure and wave reflection indices	490	Coronary artery calcium score	497
Endothelial dysfunction	493	Ankle-brachial index	497

... at the beginning a disease is easy to cure but difficult to diagnose; but as time passes, not having been treated or recognized at the outset, it becomes easy to diagnose but difficult to cure ...

Niccolo Machiavelli, *Il Principe*, 1513

A man is as old as his arteries.

Thomas Sydenham

The major issues in the natural history of cardiovascular disease have been brought together by Dzau *et al.* (Dzau *et al.*, 2006a; Dzau *et al.*, 2006b) in description of the cardiovascular continuum (Fig. 19.7). This concentrates on initial risk factors, and the development over time, of atherosclerotic disease with progression to coronary stenosis, myocardial ischemia, myocardial infarction, and scarring, with end-stage cardiac failure causing death if this does not occur before on account of a lethal arrhythmia. Such considerations led us to a new view on the link between macrovascular stiffening and the degeneration of microvasculature in the brain and kidney, which is a consequence of greater pulsatile flow entering these vessels and being cushioned therein rather than in the aorta and elastic arteries. These views are embraced as the cardiovascular aging continuum, which we see as the other side of the coin of the conventional continuum (or the other face of the cameo), and which explains how aging change even in the absence of atherosclerotic disease, can lead not only to myocardial ischemia, arrhythmia, and cardiac failure, but to cerebral and renal microvascular disease as well, with intellectual deterioration, dementia, and renal failure. We see both continua as applying in Western civilizations, but the aging process being of more importance now in developing nations with low prevalence of atherosclerotic disease and becoming more important in Western societies in the future as

atherosclerotic disease is controlled by better lifestyle, less smoking, and greater use of statins and anti-platelet agents. On the same wavelength are two terms that were recently coined, EVA which stands for early vascular aging and ADAM which stands for aggressive decrease of atherosclerosis modifiers (Fig. 22.1) (Nilsson *et al.*, 2009).

Prevention of cardiovascular disease (CVD) is greatly aided by the identification of proper biological markers or biomarkers. According to a standardized definition introduced by a National Institutes of Health (NIH) working group in 2001, a risk marker or biomarker is ‘a characteristic that is objectively measured and evaluated as an indicator of normal biological processes, pathogenic processes, or pharmacologic responses to a therapeutic intervention’ (Biomarkers Definitions Working Group, 2001; Vasan, 2006). Several large observational studies as well as randomized trials of the previous decades have identified age, gender, blood pressure, cholesterol, smoking, diabetes, and family history of coronary heart disease as the major risk markers for CVD. The levels or intensity of those risk markers predict the occurrence or development of CVD, whereas their modulation, when feasible, is associated with clinical benefit. However, newer data have identified other risk markers including obesity, metabolic syndrome, and physical inactivity. Among the recently recognized risk markers of CVD risk, accumulating evidence from several studies identified both ‘circulating biomarkers’, such as C-reactive protein and progenitor cells and endothelial molecular and cellular components, as well as ‘arterial biomarkers’, such as arterial stiffness, central blood pressures, wave reflection indices, ankle-brachial index, endothelial dysfunction, carotid intima-media thickness, and coronary artery calcium as risk markers for CVD morbidity and mortality, potentially independent of other traditional and currently validated risk factors.

Implementation of a biomarker into clinical practice is no easy task.

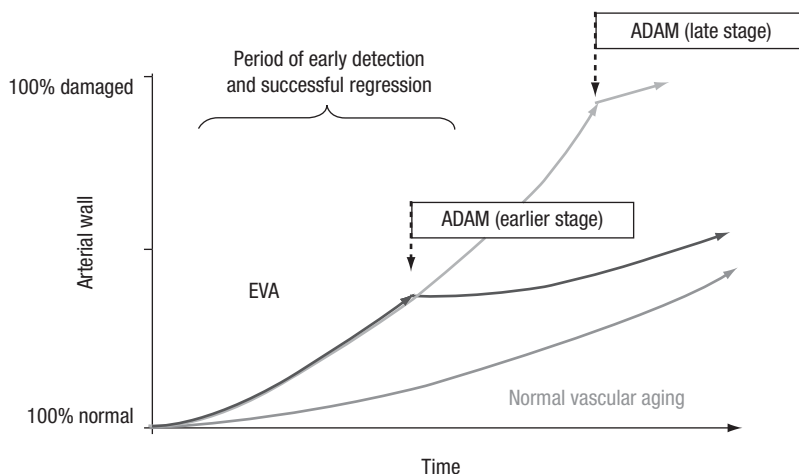


Figure 22.1 Early vascular aging (EVA) and the effect of aggressive decrease of atherosclerosis modifiers (ADAM) in high-risk patients. Adapted from Nilsson *et al.* (2009)

'Medicine is an art that ... has principles in action and reasoning in every case'

Plato, *Gorgias*

According to the Biomarkers Definitions Working Group, a surrogate endpoint or type 2 biomarker is 'a marker that is intended to substitute for a clinical endpoint, and is expected to predict clinical benefit or harm on the basis of epidemiological, therapeutic, pathophysiological, or other scientific evidence' (Biomarkers Definitions Working Group, 2001; Vasan, 2006). Such biomarkers serve as early, intermediate endpoints that identify progress of the disease at early stages before hard, irreversible endpoints occur; they also serve as endpoints of evaluation of a risk-reducing strategy (lifestyle modifying, drug, etc.), thus avoiding lengthy follow ups and increased numbers of participants that would otherwise be needed if reduction in hard endpoints was sought. To introduce a risk marker as a surrogate endpoint in everyday clinical practice, several important conditions/criteria must be fulfilled. The risk marker must be plausible, in a way that it is based on our current understanding of pathophysiology, which provides the background to conclude that the relation of the risk marker is consistent with the associated disease. The risk marker must be strong, so that the presence or a deviation of its value is associated with a meaningful and clinically significant change of the relative risk of developing an outcome versus the risk in the absence of this risk marker or when its value lies within the normal limits. The risk marker must exhibit a gradation of effect, in a way that a quantitative change in the magnitude of the risk factor or a change in the duration of exposure to the risk factor impacts on the outcome of interest. The clearer the gradation of effect, the more likely the factor is, indeed, a beneficially modifiable risk factor. The risk marker must be consistent, which means that the association between the risk factor and outcome must be demonstrated in a variety of populations with different age domains, in both genders, in different ethnic groups, and in several states of disease. Importantly, a new risk

marker (biomarker) will be clinically useful only if the methods/assays dedicated to its measurement are accurate and reproducible and the results are easy to interpret. Ideally, the risk marker/biomarker must explain a reasonable proportion of the outcome independent of established predictors (i.e. it must have an incremental predictive value beyond traditional risk markers and risk models). Furthermore, in order to be incorporated into clinical practice it must be supported by data suggesting that knowledge of the risk marker levels may affect clinical management. Finally, and perhaps most importantly, if the risk factor is modifiable by currently available strategies, favorable modulation of the risk factor should beneficially modify the outcome.

An enormous amount of data is published every year about new, innovative, and sophisticated cardiovascular biomarkers. Undoubtedly, apart from purely scientific motivations, an additional driving force to this data generation is the pressure from industry. Many of the studies are not founded on solid ground and as so they are hampered by methodological limitations affecting design, analysis, and reporting (Ioannidis, 2005). Further, in a recent comprehensive review of studies claiming improved risk prediction beyond the Framingham risk score most of them had flaws that cast doubt on the reliability of the claims for improved prediction (Tzoulaki *et al.*, 2009). Looking forward, there is room for improvement in risk assessment, and studies that include careful and meticulous study design and consideration of stringent statistical criteria and measures, as well as objective and unbiased reporting of data, will provide the most reliable information. When a new biomarker is evaluated, it is important to remember that the question of interest is not whether this new biomarker is a better predictor of disease than a previously known old biomarker. Rather, the pertinent question is whether the new biomarker improves the predictive accuracy of the best available model (representing the standard of care for that disease) that incorporates several known predictors of disease including the old biomarker, such as the Framingham risk score (Vasan, 2006). This comparison is

performed with four summary statistical measures (McGeechan *et al.*, 2008): (1) *Global measures of model fit*: how likely is it that the new model chosen would give rise to the data observed; (2) *Discrimination*: how well this new model separates individuals who develop the outcome from those who do not; (3) *Calibration*: how close the predicted risks are to the actual observed risks; and (4) *Reclassification*: whether the new model sufficiently changes a person's risk to move them into a different risk category and thus alter treatment decisions. These measures carry both strong points and limitations and always should be approached with extreme caution. The potential clinical use of a new biomarker lies in its ability to predict risk more accurately than existing models when inserted into a new model and also the risks predicted for individuals to be sufficiently different to warrant change in treatment decisions (McGeechan *et al.*, 2008).

Some traditional individual risk factors can fluctuate over time and their values, recorded at the time of risk assessment, may not reflect their true impact on the arterial wall. We have been particularly interested in arterial biomarkers because they have the advantage that they can better integrate both the cumulative effect of a traditional risk factor on the arterial wall over a long period of time, as well as any synergistic effects from risk factor combinations (Fig. 22.2, theoretic ROC curves). Of particular importance is the population that would benefit from the integration of arterial biomarkers in risk assessment. It appears that individuals who would benefit most are, apart from patients with hypertension or renal insufficiency those who, according to office-based evaluation with available risk charts (Framingham, European SCORE, Reynolds Risk Score) belong to the intermediate risk category, or to a low-risk category but with the presence of factors such as family history of early onset of

CVD, metabolic syndrome, substantially elevated single risk factor, chronic renal insufficiency, and novel risk factors (Kullo and Malik, 2007) (Fig. 22.3).

ARTERIAL STIFFNESS

A considerable number of studies have examined the ability of arterial stiffness to predict the risk of future fatal and non-fatal cardiovascular events (myocardial infarction, stroke, revascularization, aortic syndromes) and total mortality (Blacher *et al.*, 1999b; Laurent *et al.*, 2001a; Meaume *et al.*, 2001; Shoji *et al.*, 2001; Boutouyrie *et al.*, 2002; Cruickshank *et al.*, 2002; Blacher *et al.*, 2003;

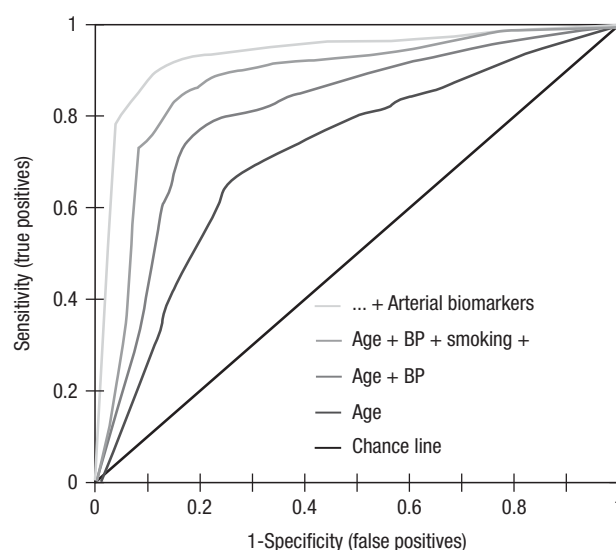


Figure 22.2 Theoretical ROC curves with the addition of biomarkers

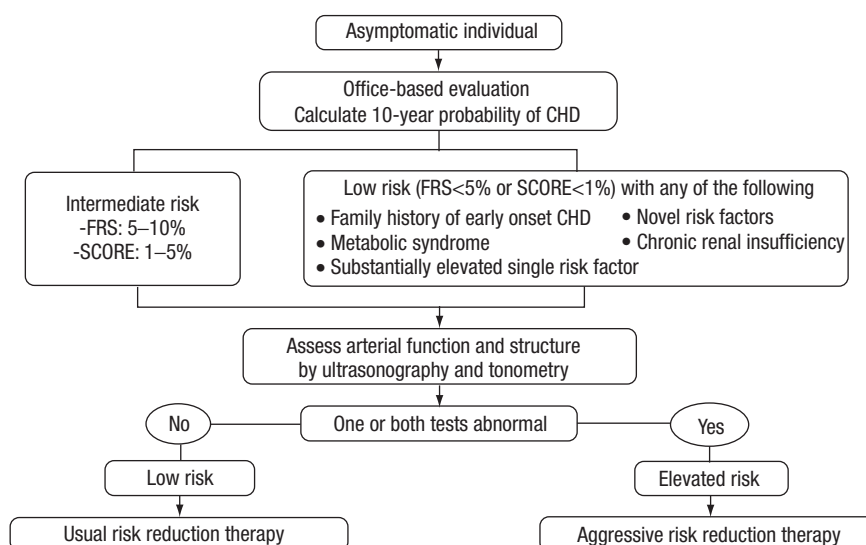


Figure 22.3 Algorithm for primary prevention with the use of 'arterial biomarkers'. FRS, Framingham risk score; SCORE, systemic coronary risk evaluation. Adapted and modified from Kullo and Malik (2007)

Laurent *et al.*, 2003; Gosse *et al.*, 2005; Kitahara *et al.*, 2005; Matsuoka *et al.*, 2005; Pannier *et al.*, 2005; Shokawa *et al.*, 2005; Sutton-Tyrrell *et al.*, 2005; Tomiyama *et al.*, 2005a; Mattace-Raso *et al.*, 2006; Willum-Hansen *et al.*, 2006; Choi *et al.*, 2007; Zoungas *et al.*, 2007; Terai *et al.*, 2008; Anderson *et al.*, 2009; Meguro *et al.*, 2009; Mitchell *et al.*, 2010; Rodondi *et al.*, 2010; Wang *et al.*, 2010). Historically, the first studies regarded patients belonging to high-risk groups, such as end-stage renal disease, while studies progressively extended to other patient groups, such as hypertensive or coronary artery disease patients, and from there to the general population. We recently performed a meta-analysis in which we pooled the aortic pulse wave velocity (PWV) (the 'gold standard' measure of aortic stiffness) data for 15 877 subjects from 17 available published studies who were followed up for a mean of 7.7 years (Vlachopoulos *et al.*, 2010a). Several populations, such as patients with hypertension, diabetes, end-stage renal disease (ESRD), and coronary artery disease, and subjects from the general population or ethnic minorities were included (Table 22.1). Aortic stiffness emerged as a strong predictor of future cardiovascular (CV) events and all-cause mortality. Within each patient group, the risk of CV events, CV mortality, and all-cause mortality in subjects with increased aortic PWV, was almost twice as high compared with the risk of subjects with lower aortic PWV. More specifically, the pooled relative risk (RR) of clinical events increased in a stepwise, linear-like fashion from the first to the third tertile of aortic PWV. The pooled RRs of total CV events, CV mortality, and all-cause mortality were 2.26, 2.02, and 1.90, respectively, for high versus low aortic PWV subjects. For total CV events and CV mortality, the RR was significantly higher in high baseline risk groups (coronary artery disease, renal disease, hypertension) compared with low-risk subjects (general population) (Fig. 22.4). An increase in aortic PWV by 1 m/s corresponded to an age-, sex-, and risk factor-adjusted risk increase of 14, 15, and 15 percent in total CV events, CV mortality, and all-cause mortality, respectively (Fig. 22.5). An increase in aortic PWV by 1 standard deviation was associated with respective increases of 47, 47, and 42 percent. Importantly, the predictive ability of arterial stiffness is higher in subjects with a higher baseline CV risk, such as renal disease (Fig. 22.6, page 491). The fact that arterial stiffness is predictive of CV events is largely expected since it is pathophysiologically important for both arterial and overall CV performance. However, arterial stiffness is predictive also for all-cause mortality with 45–50 percent of deaths attributed to non-cardiac causes. This important finding should be emphasized and further scrutinized. Although pathophysiological explanations are not readily identifiable, this could reflect the existence of common pathogenetic mechanisms, such as aging, inflammation, and oxidative stress, over a wide range of conditions. As pointed out earlier, an important measure to validate a new biomarker is its ability to change (reclassify) a

person's risk and move them into a different risk category. In the Framingham cohort, while addition of aortic PWV to the standard model yielded non-significant overall results, for individuals at intermediate CVD risk it resulted in a significant net reclassification of 15.7 percent ($P=0.03$) (Mitchell *et al.*, 2010). This indicates that individuals who belong in the intermediate risk group are those who would benefit most if PWV is included in their evaluation. All this wealth of evidence on the predictive value of aortic pulse wave velocity (Laurent *et al.*, 2006) justifies inclusion of arterial stiffness by European guidelines as a tool for assessment of subclinical target organ damage and suggests extension to other disease states or population groups (Mancia *et al.*, 2007a).

While assessing the risk of hard endpoints, such as mortality and coronary events, is the ultimate verification of a biomarker's predictive ability, assessing intermediate endpoints is also a valid step. The more pathophysiologically relevant the intermediate endpoint is, the more the chances are that the biomarker will predict hard endpoints. In patients with essential hypertension, indexes of large-artery stiffness indicate the presence and extent of atherosclerotic disease and correlate with the CV risk as assessed by the Framingham risk score (Blacher *et al.*, 1999a). Aortic PWV is associated with left ventricular hypertrophy (Toprak *et al.*, 2009), while it is an independent marker of cognitive impairment (Scuteri *et al.*, 2005) and correlates with silent small vessel cerebral disease (Henskens *et al.*, 2008b). Furthermore, PWV is an independent predictor of coronary flow reserve (Ikonomidis *et al.*, 2008) and renal function, as assessed by glomerular filtration ratio (Schillaci *et al.*, 2006) and microalbuminuria (Mulè *et al.*, 2009).

From the therapeutic standpoint, aortic stiffness appears as a worthwhile treatment target. However, to date, there is only one study showing that improvement in outcome was mediated through an improvement in aortic stiffness. Guerin *et al.* (2001) followed up patients with end-stage renal disease who were treated with reduction in dry weight and antihypertensive medication. Despite similar reduction in pressure, only survivors experienced a reduction in aortic stiffness, whereas non-survivors had an increase in PWV (Fig. 22.7, page 491).

Reference values and cut-off points are important elements for clinical integration of arterial stiffness in clinical practice. Such data were recently published for adults as well as for children and teenagers (The Reference Values for Arterial Stiffness Collaboration, 2010; Reusz *et al.*, 2010) (Table 22.2 (page 492), Figs 22.8 (page 492), 22.9 (page 493)).

It should be stressed that results supporting the role of arterial stiffness as an appealing biomarker are at this stage valid for aortic PWV (an index of regional arterial stiffness) only and are not necessarily expandable to indices of local arterial stiffness (aortic, carotid, femoral) or systemic arterial stiffness. Regarding other indices of regional arterial stiffness, brachial PWV is unlikely to have predictive value (Pannier *et al.*, 2005), while data pointing to

Table 22.1 Overview of studies on the association between aortic PWV and clinical events

Author, year	Population-sample size	Age (years)	Follow-up duration	Events	aPWV in m/sec (mean, s.d.)	aPWV cut-off (high versus low)
Anderson <i>et al.</i> (2009) ^b	Non-diabetic general population (<i>n</i> = 174)	60 ± 10	19.6 years	58 deaths	10.2 ± 2.1	> 10.6 m/s (upper tertile)
Blacher (1999a; 1999b) ^a	ESRD (<i>n</i> = 241)	51.5 ± 16.3	6 years	73 deaths, 48 CV deaths	11.1 ± 3.1	> 12.0 m/s (upper tertile)
Boutouyrie <i>et al.</i> (2002) ^a	Hypertension (<i>n</i> = 1045)	51 ± 12	5.7 years	53 coronary events, 97 CV events	11.5 ± 3.5	> 12.3 m/s (upper tertile)
Choi <i>et al.</i> (2007) ^a	Chest pain patients (<i>n</i> = 497)	57.7 ± 10.1	2.6 years	1 death, 0 CV death, 120 CV events	Not reported	> 12.5 m/s (upper tertile)
Cruickshank <i>et al.</i> (2002) ^a	Diabetes (<i>n</i> = 394)	60 ± 10	10.7 years	179 deaths	11.6 ± 3.8	3.8 m/s increase
Laurent (2001) ^a	Hypertension (<i>n</i> = 1980)	50 ± 13	9.3 years	107 deaths, 46 CV deaths	11.5 ± 3.4	5 ± m/s increase
Mattace-Raso (2006) ^a	Community-based adults (<i>n</i> = 2835)	71.7 ± 6.7	4–9 years	352 deaths, 156 CV events	13.3 ± 2.9	> 14.6 m/s in men; > 14.2 m/s in women (age-specific upper tertile)
Meaume <i>et al.</i> (2001) ^a	Subjects > 70 years (<i>n</i> = 141)	87.1 ± 6.6	2.5 years	27 CV deaths	14.2 ± 3.1	> 17.7 m/s (upper decile)
Pannier <i>et al.</i> (2005) ^a	ESRD (<i>n</i> = 305)	53.1 ± 16.2	5.8 years	96 CV deaths	11.1 ± 3.1	Upper tertile
Shoji <i>et al.</i> (2001) ^a	ESRD (<i>n</i> = 265)	55.4 ± 10.5	5.3 years	81 deaths, 36 CV deaths	8.6 ± 2.2	> 8.2 m/s (median)
Shokawa <i>et al.</i> (2005) ^a	Ethnic minority (<i>n</i> = 492)	63.7 ± 8.8	10 years	43 deaths, 14 CV deaths	9.7 ± 1.9	> 9.9 m/s (optimal cut-off by ROC curve)
Sutton-Tyrrell (2005) ^a	Community-based old adults (<i>n</i> = 2488)	73.7 ± 2.9	4.6 years	265 deaths, 111 CV deaths, 616 CV events	9.0 ± 3.9	> 8.4 m/s in men; > 7.9 m/s in women (age-specific median)
Terai <i>et al.</i> (2008) ^a	Hypertension (<i>n</i> = 676)	62 ± 12	4.8 years	22 deaths, 88 CV events	9.0 ± 0.6	> 8.8 m/s (median)
Willum-Hansen <i>et al.</i> (2006) ^a	General population (<i>n</i> = 1678)	40–70 years	9.4 years	62 CV deaths, 154 CV events	11.3 ± 3.4	> 13.1 m/s (upper quintile)
Zoungas <i>et al.</i> (2007) ^a	ESRD (<i>n</i> = 207)	55 ± 13	3.6 years	17 CV deaths, 65 CV events	s.d. 3.5	> 9.9 m/s (cut-off)
Mitchell <i>et al.</i> (2010) ^a	General population (<i>n</i> = 2232)	63 ± 12	7.8 years	151 CV events	9.3 (7.8–11.8) median (interquartile range)	> 9.3 m/s (median)
Wang <i>et al.</i> (2010) ^b	General population (<i>n</i> = 1272)	52 ± 13	15 years	225 deaths, 64 CV deaths	9.5 ± 2.3 (men), 9.5 ± 2.5 (women)	2.3 m/s increase (men); 2.5 m/s increase (women)

^aPWV is a predictor of events independent of peripheral pressures.

^bThese studies have not adjusted for pressure.

CV, cardiovascular; ESRD, end-stage renal disease; PWV, pulse wave velocity; ROC, receiver-operator characteristic.

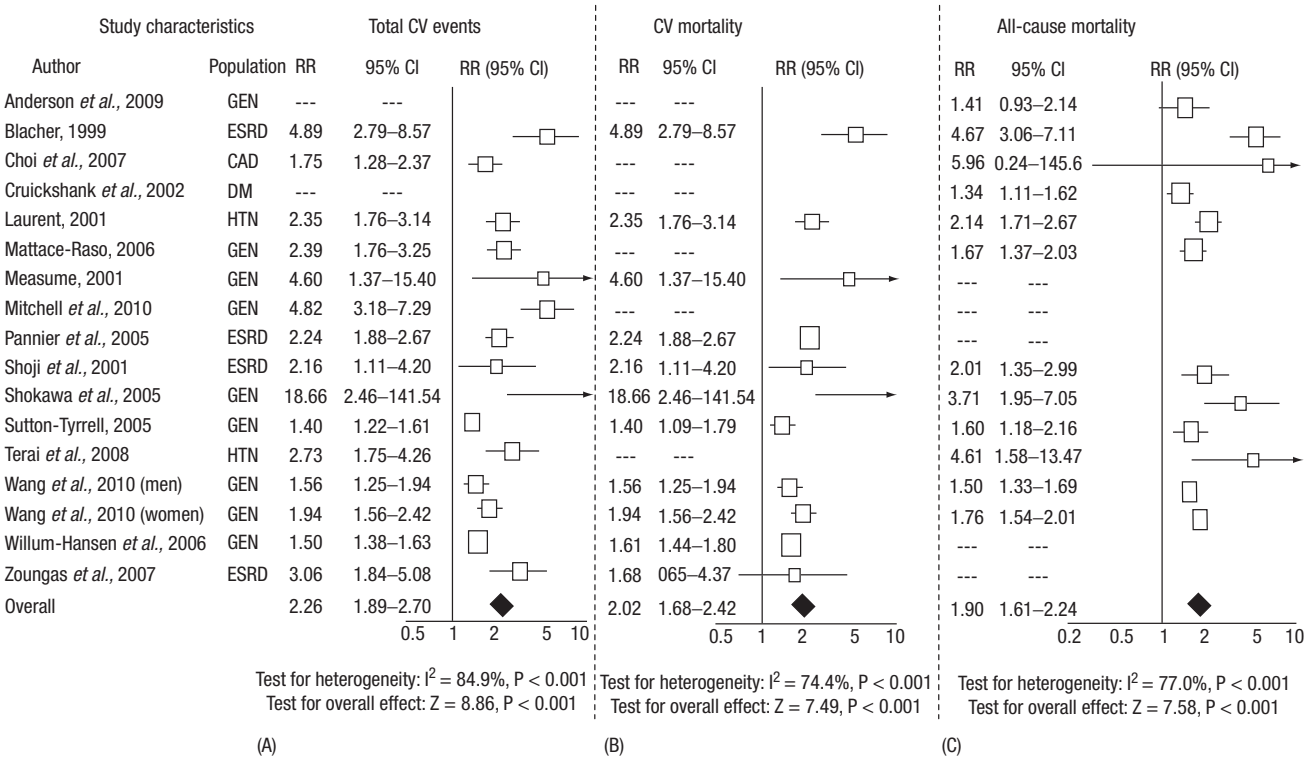


Figure 22.4 Relative risk (RR) and 95 percent confidence interval (CI) for high aortic pulse wave velocity (PWV) and clinical events. Adapted from Vlachopoulos *et al.* (2010a)

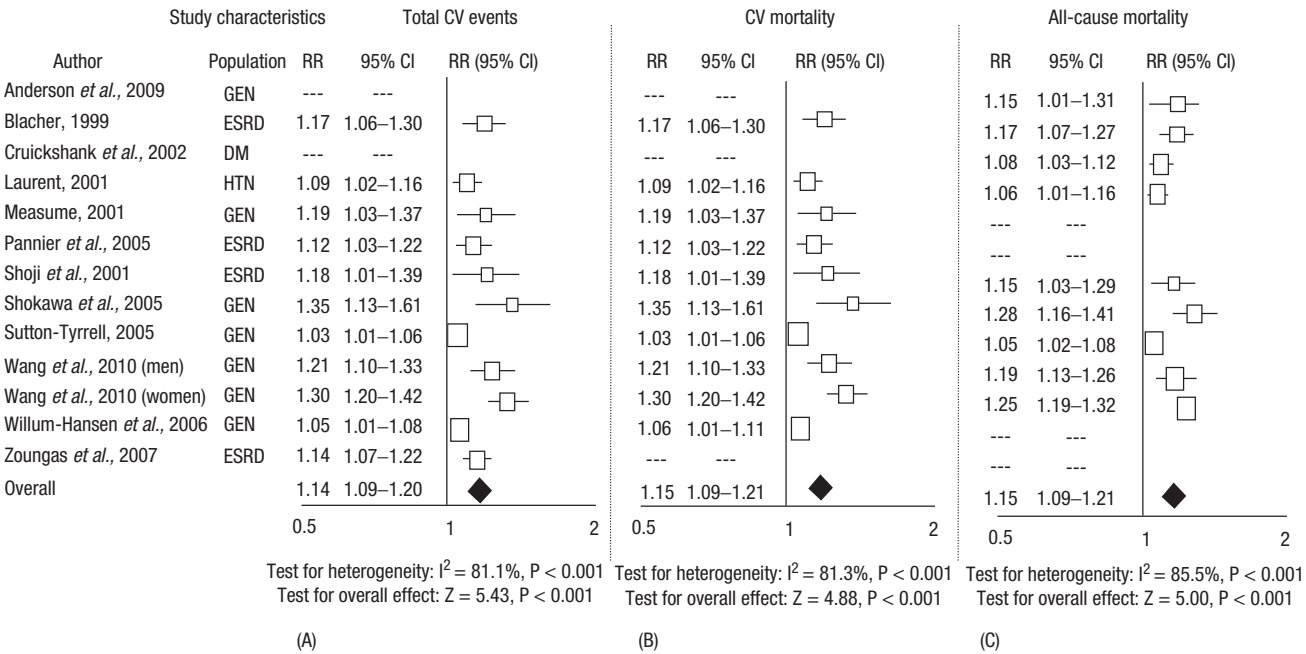


Figure 22.5 Relative risk (RR) and 95 percent confidence interval (CI) for a 1 m/s increase in aortic pulse wave velocity (PWV) and clinical events. Adapted from Vlachopoulos *et al.* (2010a)

an independent predictive role of brachial–ankle PWV are accumulating (Kitahara *et al.*, 2005; Matsuoka *et al.*, 2005; Tomiyama *et al.*, 2005a; Tomiyama *et al.*, 2005b; Meguro *et al.*, 2009; Miyano *et al.*, 2010) and await additional validation in order to be established.

CENTRAL PRESSURE AND WAVE REFLECTION INDICES

Central (aortic, carotid) pressures are pathophysiologically more relevant than peripheral pressures for the

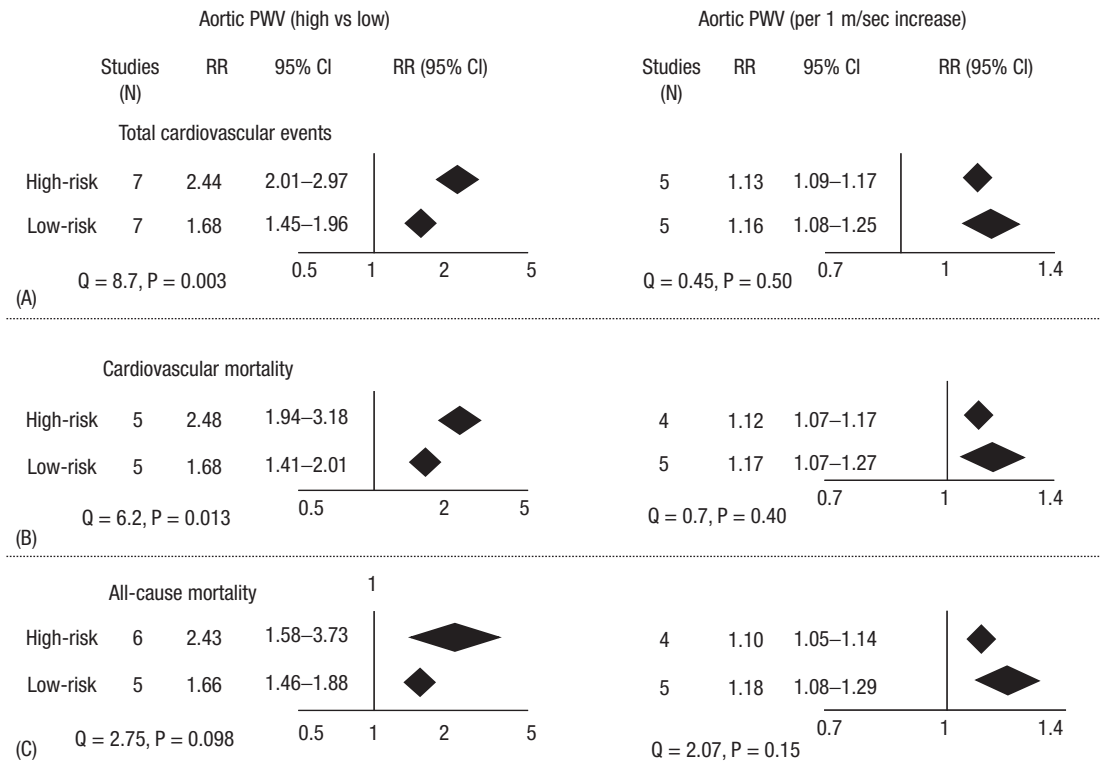


Figure 22.6 Pooled relative risk (RR) and 95 percent confidence interval (CI) aortic pulse wave velocity (PWV) and clinical events, according to baseline risk. Adapted from Vlachopoulos *et al.* (2010a)

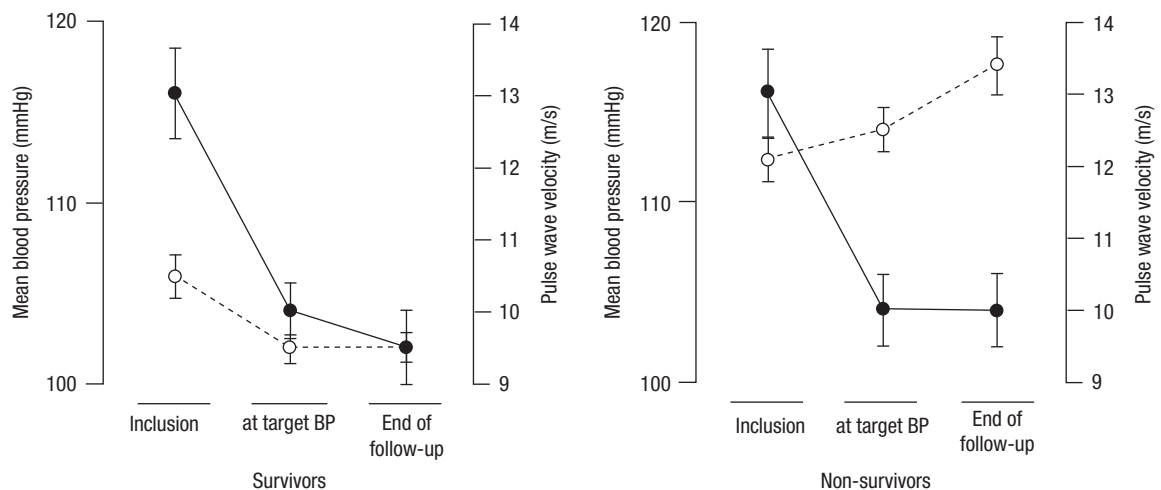


Figure 22.7 Changes of mean blood pressure (solid circle) and aortic pulse wave velocity (open circle) from inclusion until the end of follow up for survivors and non-survivors with end-stage renal disease. Adapted from Guerin *et al.* (2001)

pathogenesis of cardiovascular disease (Agabiti-Rossei *et al.*, 2007) and according to accumulating clinical evidence this is translated into a significant predictive role for future cardiovascular events. While several studies have shown an ability of central pressures and indices to predict future events, findings have not always been consistent. We meta-analyzed 11 longitudinal studies (Vlachopoulos *et al.*, 2010b) that had employed measures of central hemodynamics and had followed 5648 subjects for a mean

follow up of 45 months (Table 22.3) (London *et al.*, 2001; Lu *et al.*, 2001; Safar *et al.*, 2002b; Ueda *et al.*, 2004; Chirinos *et al.*, 2005; Weber *et al.*, 2005; Covic *et al.*, 2006; Dart *et al.*, 2006; Roman *et al.*, 2007; Jankowski *et al.*, 2008; Pini *et al.*, 2008; Roman *et al.*, 2009). The age- and risk factor-adjusted pooled relative risk of total CV events was 1.088 (95 percent CI 1.040–1.139) for a 10 mmHg increase of central systolic pressure, 1.137 (95 percent CI 1.063–1.215) for a 10 mmHg increase of central pulse

Table 22.2 Distribution of pulse wave velocity (PWV) values (m/s) in the reference value population according to age and blood pressure category (adapted and modified from The Reference Values for Arterial Stiffness' Collaboration, Eur Heart J 2010)

Age category (years)	Blood pressure category				
	Optimal	Normal	High normal	Grade I HT	Grade II/III HT
PWV as mean (± 2 s.d.)					
< 30	6.1 (4.6–7.5)	6.6 (4.9–8.2)	6.8 (5.1–8.5)	7.4 (4.6–10.1)	7.7 (4.4–11.0)
30–39	6.6 (4.4–8.9)	6.8 (4.2–9.4)	7.1 (4.5–9.7)	7.3 (4.0–10.7)	8.2 (3.3–13.0)
40–49	7.0 (4.5–9.6)	7.5 (5.1–10.0)	7.9 (5.2–10.7)	8.6 (5.1–12.0)	9.8 (3.8–15.7)
50–59	7.6 (4.8–10.5)	8.4 (5.1–11.7)	8.8 (4.8–12.8)	9.6 (4.9–14.3)	10.5 (4.1–16.8)
60–69	9.1 (5.2–12.9)	9.7 (5.7–13.6)	10.3 (5.5–15.1)	11.1 (6.1–16.2)	12.2 (5.7–18.6)
≥ 70	10.4 (5.2–15.6)	11.7 (6.0–17.5)	11.8 (5.7–17.9)	12.9 (6.9–18.9)	14.0 (7.4–20.6)
PWV as median (10–90 pc)					
< 30	6.0 (5.2–7.0)	6.4 (5.7–7.5)	6.7 (5.8–7.9)	7.2 (5.7–9.3)	7.6 (5.9–9.9)
30–39	6.5 (5.4–7.9)	6.7 (5.3–8.2)	7.0 (5.5–8.8)	7.2 (5.5–9.3)	7.6 (5.8–11.2)
40–49	6.8 (5.8–8.5)	7.4 (6.2–9.0)	7.7 (6.5–9.5)	8.1 (6.8–10.8)	9.2 (7.1–13.2)
50–59	7.5 (6.2–9.2)	8.1 (6.7–10.4)	8.4 (7.0–11.3)	9.2 (7.2–12.5)	9.7 (7.4–14.9)
60–69	8.7 (7.0–11.4)	9.3 (7.6–12.2)	9.8 (7.9–13.2)	10.7 (8.4–14.1)	12.0 (8.5–16.5)
≥ 70	10.1 (7.6–13.8)	11.1 (8.6–15.5)	11.2 (8.6–15.8)	12.7 (9.3–16.7)	13.5 (10.3–18.2)

s.d., standard deviation, 10 pc, the upper limit of the 10th percentile, 90 pc, the lower limit of the 90th percentile; HT, hypertension.

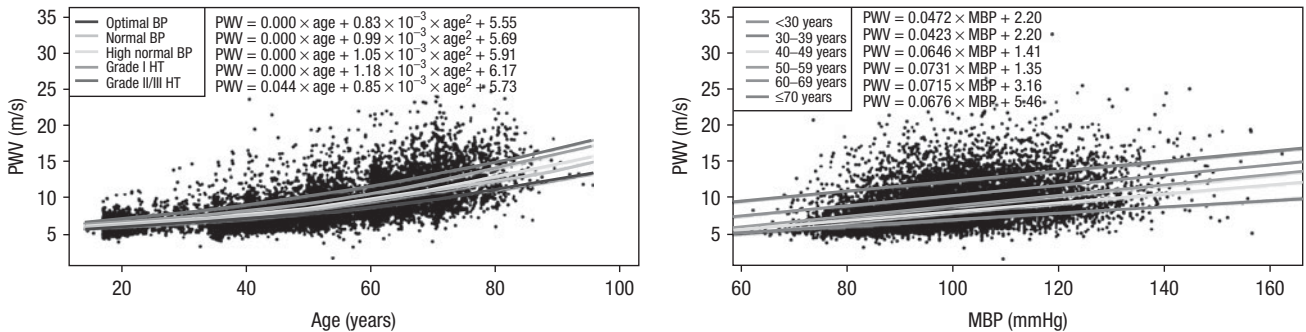


Figure 22.8 (Top) Pulse wave velocity (PWV) versus age in the reference value population along with regression equations. Regression lines denote the results of regression on age for different blood pressure (BP) categories. (Bottom) Pulse wave velocity versus mean blood pressure (MBP) in the reference value population along with regression equations. Regression lines denote the results of linear regression on mean blood pressure for different age categories. HT, hypertension; BP, blood pressure. Adapted and modified from The Reference Values for Arterial Stiffness' Collaboration (2010)

pressure, and 1.318 (95 percent CI 1.093–1.588) for a 10 percent absolute increase of central augmentation index. Furthermore, a 10 percent increase of central augmentation index was associated with a RR of 1.384 (95 percent CI 1.192–1.606) for all-cause mortality (Fig. 22.10). When three more studies (Mitchell *et al.*, 2010; Wang *et al.*, 2010; Weber *et al.*, 2010) that were published after the respective meta-analysis were included (a total of 9093 subjects and mean follow up 54.9 months) the RRs of total CV events were 1.115 (95 percent CI 1.029–1.209) for a 10 mmHg increase of central pulse pressure, and 1.303 (95 percent CI 1.098–1.546) for a 10 percent absolute increase of central augmentation index. Furthermore, a 10 percent increase of central augmentation index was associated with a RR of 1.328 (95 percent CI 1.167–1.511) for all-cause mortality.

A crucial issue to be addressed is whether central indices have a predictive ability over and above peripheral pressures. According to our meta-analysis, AIx has a predictive value independent of peripheral pressures and central pulse pressure has a marginally better predictive value compared to peripheral blood pressure ($P=0.057$). However, one study (Dart *et al.*, 2006) has been criticized on the basis of methodological flaws (O'Rourke *et al.*, 2007; O'Rourke and Adji, 2010b) that may account for the unexpected negative pulse pressure amplification (Avolio *et al.*, 2009b). When analysis was repeated excluding this study, central pulse pressure was associated with a higher RR of clinical events than brachial pulse pressure (1.338 versus 1.178, respectively, $P=0.017$). Apart from clinical events, central hemodynamic variables have been shown to be independently associated

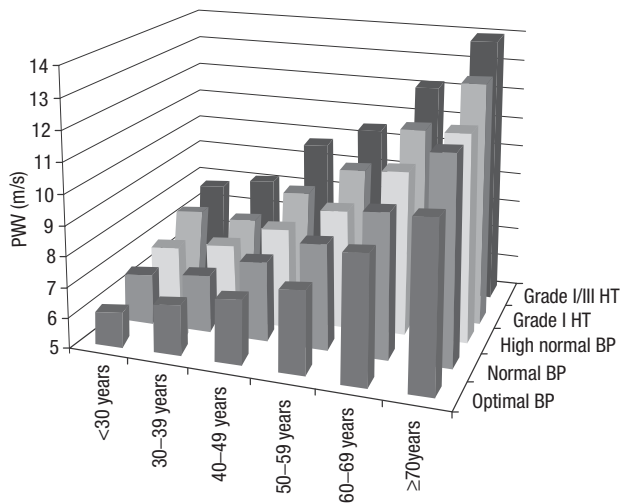


Figure 22.9 Reference values for pulse wave velocity (PWV): mean values according to age and blood pressure (BP) categories. HT, hypertension. Adapted from 'The Reference Values for Arterial Stiffness' Collaboration (2010)

with organ damage and incident cardiovascular disease in several populations (Table 22.4). Increased aortic augmentation index is associated with coronary artery disease (Weber *et al.*, 2004). In fact, in young patients with suspected coronary artery disease, increased wave reflections can separate patients with coronary artery disease from those without (Weber *et al.*, 2004). Central pressures also correlate with cardiovascular risk not only in patients with atherosclerotic disease, but also in apparently healthy subjects (Nurnberger *et al.*, 2002). The late systolic augmentation of the central pressure waveform is associated with an increase in left ventricular mass index independent of age and mean blood pressure (Saba *et al.*, 1993; Hashimoto *et al.*, 2007a), and carotid systolic pressure is an independent determinant of left ventricular wall thickness (Roman *et al.*, 2000). Moreover, central pressure is also more closely related to other important cardiovascular intermediate endpoints, such as vascular hypertrophy, extent of carotid atherosclerosis (Boutouyrie *et al.*, 1999; Roman *et al.*, 2007), and the ascending aorta diameter in patients with Marfan syndrome (Jondeau *et al.*, 1999) than brachial pressure. Recently, higher carotid pressure augmentation in older individuals has been linked to increased flow pulsations entering the cerebral circulation, which may increase the risk of cerebral microvascular damage (Hirata *et al.*, 2006). In inflammatory disorders, such as systemic vasculitis, augmentation index is a marker of disease activity, and it is independently associated with levels of C-reactive protein (Booth *et al.*, 2004b).

Despite similar effects on brachial pressure, antihypertensive drugs have differential effects on central pressure and this may explain the superiority of arterial vasodilating drugs in outcome trials. In the recent Conduit Artery Function Evaluation (CAFE) study, a substudy

of the Anglo-Scandinavian Cardiac Outcomes Trial (ASCOT), a higher treatment-related decrease of central pulse pressure, that was not evident in brachial pressure measurements, was independently associated with clinical benefit and reduced cardiovascular events (Williams *et al.*, 2006). Other studies have also shown that endpoints related to organ damage can be predicted by indices of central hemodynamics after therapy. Aortic and radial augmentation index, as well as amplification of the pulse between central and radial arteries, were superior to conventional cuff measures in predicting reduction in left ventricular mass during antihypertensive therapy (Hashimoto *et al.*, 2007b). Furthermore, the reduction in carotid wall diameter and hypertrophy with antihypertensive treatment was related to carotid PP, but not to mean BP (Boutouyrie *et al.*, 2000). The pREterax in regression of Arterial Stiffness in a controlled double-blind (REASON) trial, which compared a perindopril/indapamide combination against atenolol, showed that normalization of brachial systolic BP is achieved with a significantly greater reduction of carotid systolic BP after a 12-month treatment with the combination (London *et al.*, 2004a). In this study, compared with atenolol, the perindopril/indapamide combination was associated with a greater fall in left ventricular mass, and this was related to carotid but not brachial BP (de Luca *et al.*, 2004).

Undoubtedly, there is room for further refinement. The degree of pressure amplification towards the periphery is variable, depending on a number of factors including age, gender, and heart rate. Accordingly, extension of the existing data regarding the predictive ability of central hemodynamic indexes over and above brachial BP in a wider range of populations and disease states is mandatory. Furthermore, it should be noted, that evidence for a specific central pressure component or index does not necessarily apply to the others. Central pressures may be more easily adopted by the clinician compared to calculated indices, such as AIx and amplification, but, on the other hand, they may be more prone to error since they rely on the (in)accuracy of the cuff measurement. Amplification of the pulse between the aorta and the radial artery is inversely related to AIx in the aorta or radial artery. Despite limitations due to calibration issues, it has the potential to be measured as accurately as possible and may ultimately replace AIx in clinical studies (Avolio *et al.*, 2009b; O'Rourke and Adji, 2010b).

ENDOTHELIAL DYSFUNCTION

Given the important role of the endothelium for the pathogenesis of cardiovascular disease, assessment of endothelial function is, in principle, an appealing biomarker. To assess endothelial function, several methods have been utilized which generally may be identified as (1) those that measure the levels of progenitor cells and concentration of endothelial molecular and cellular

Table 22.3 Overview of studies on the association between central pressures and augmentation index and clinical endpoints

Author, year	Population-sample size	Age (years)	Follow-up duration (longitudinal studies)	Events	Index	Central indices vs peripheral BPs
Lu <i>et al.</i> (2001)	Stable CAD/angioplasty (<i>n</i> = 87)	72.5 ± 5.1	6.1 ± 4.1 m	Restenosis	Aortic PP, PPf, and PI	Not adjusted for peripheral BPs
London <i>et al.</i> (2001)	ESRD (<i>n</i> = 180)	54 ± 16	52 ± 36 m	Deaths and CV deaths	Carotid Alx	Central better
Safar (2002)	ESRD (<i>n</i> = 180)	54 ± 16	52 ± 36 m	Deaths and CV deaths	Carotid SBP, PP, brachial-carotid PP amplification	Central better
Ueda <i>et al.</i> (2004)	CAD/angioplasty (<i>n</i> = 103)	62 ± 9	6 m	Restenosis	Aortic Alx, aortic inflection time	Not adjusted for peripheral BPs
Chirinos <i>et al.</i> (2005)	CAD or non-obstructive coronary atherosclerosis (<i>n</i> = 297)	63.8 ± 10.3	40 ± 14 m	Deaths and CV events	Aortic PP, AP, Alx	Central better
Weber <i>et al.</i> (2005)	CAD/angioplasty (<i>n</i> = 262)	65 ± 10	24 m	Deaths and CV events	HR-corrected aortic Alx	Central better
Dart <i>et al.</i> (2006)	Elderly female hypertensives (<i>n</i> = 484)	72 ± 5	49 m (median)	CV events	Carotid SBP, PP, Alx	Peripheral better
Covic <i>et al.</i> (2006)	ESRD (<i>n</i> = 92)	42.6 ± 11.2	61 ± 25 m	Deaths	HR-corrected aortic, Alx	Central better
Roman <i>et al.</i> (2007)	American Indians free of CVD (<i>n</i> = 2403)	63.5 ± 7.5	58 ± 16 m	Deaths, CV deaths, and CV events	Aortic SBP, PP	Central better
Jankowski <i>et al.</i> (2008)	Subjects undergoing non-emergency coronary angiography (<i>n</i> = 1109)	52.7 ± 19.2	52.7 ± 19.2 m	Deaths, CV deaths, and CV events	Aortic PP, PPf	Central better
Pini <i>et al.</i> (2008)	Community-dwelling individuals ≥ 65 years (<i>n</i> = 398)	73 ± 6	94 ± 24 m	Deaths, CV deaths, and CV events	Aortic SBP, PP, Alx	Central better
Mitchell <i>et al.</i> (2010)	General population (<i>n</i> = 2232)	63 ± 12	7.8 years	CV events	Carotid Alx, brachial-carotid PP amplification, carotid PP	Central indices not anymore significant when adjusted for standard risk factors
Wang <i>et al.</i> (2010)	General population (<i>n</i> = 1272)	52 ± 13	15 years	Deaths, CV deaths	Carotid SBP, PP, Alx, RWTT, RI, Pi, Pa, Pf, and Pb	Central better
Weber (2010)	Males undergoing coronary angiography (<i>n</i> = 520)	64.0 (54–71.5)	49 m	Deaths and CV events	Aortic Alx, Aix at 75, pulse wave transit time	Central better

ESRD indicates end-stage renal disease; Alx, augmentation index; CAD, coronary artery disease; CVD, cardiovascular disease; FU, follow-up; BP, blood pressure; Pi, incident pressure wave height; Pa, augmented pressure; Pb, backward pressure amplitude; Pf, forward pressure amplitude; RWTT, reflected wave transit time; RI, reflection index; PP, pulse pressure; SBP, systolic blood pressure; AP, Augmented pressure.

components and (2) those that assess the endothelium-dependent vasomotion of an artery. Emerging techniques include laser Doppler flowmetry and orthogonal polarization spectral imaging of endothelial glycocalyx dimensions. Of these techniques, the most widely applied is the non-invasive brachial artery ultrasound imaging

during reactive hyperemia (flow-mediated dilatation) (Corretti *et al.*, 2002; Deanfield *et al.*, 2005; Roman *et al.*, 2006). Endothelial function is impaired in the presence of CV risk factors. Importantly, brachial conduit artery endothelial function correlates with endothelial function of the coronary arteries (Anderson *et al.*, 1995b). Endothelial dysfunction is related to the extent of coronary artery disease and studies have shown that it has a predictive role for clinical events both in the general population and in subjects with increased CV risk (Kullo and Malik, 2007; Yeboah *et al.*, 2009). A wealth of data shows that reversal of risk factors and pharmacological interventions reverse endothelial dysfunction (Fuentes *et al.*, 2001; Deanfield *et al.*, 2005; Liu *et al.*, 2009). However, it is difficult to determine whether the beneficial clinical outcome with these strategies is mediated, at least partly, through the improvement in endothelial function. Furthermore, strategies that have shown an improvement in endothelial function were not subsequently associated with clinical benefit, as was the case with administration of antioxidant vitamins. Practical problems include the multitude of techniques, the absence of a universal protocol even within a specific technique, and the lack of reference values. Assessment of endothelial function is a valuable research tool but the potential for clinical application is at this stage limited.

CAROTID INTIMA-MEDIA THICKNESS

Carotid intima-media thickness (IMT) is a measure of the composite thickness of both the tunica intima and the tunica media. IMT increases in diseases that mainly affect the media. It has a strong dependence on age and it correlates with atherosclerotic vascular disease in several arterial beds including the coronary arteries. Increased carotid IMT is associated with cardiovascular risk factors, prevalent cardiovascular disease, coronary artery calcification on computed tomography, presence and extent of angiographically determined coronary atherosclerosis, and plaque burden on intracoronary ultrasound (Smith *et al.*, 2000; Bots *et al.*, 2007; Kullo and Malik, 2007). Several studies show that increased carotid IMT is predictive of cardiovascular events in asymptomatic individuals and recurrent events in patients with known cardiovascular

Figure 22.10 Relative risk (RR) and 95 percent confidence interval (CI) for a 10 mmHg increase in central systolic and pulse pressure and total cardiovascular (CV) events, as well for an absolute 10 percent increase in central augmentation index and total cardiovascular (CV) events (top) or all-cause mortality (bottom). Boxes represent the RR and lines represent the 95 percent CI for individual studies. The diamonds and their width represent the pooled RR and the 95 percent CI, respectively. Studies appear in alphabetical order. CAD, coronary artery disease; ESRD, end-stage renal disease; GEN, general population; HTN, hypertension. Adapted from Vlachopoulos *et al.* (2010b)

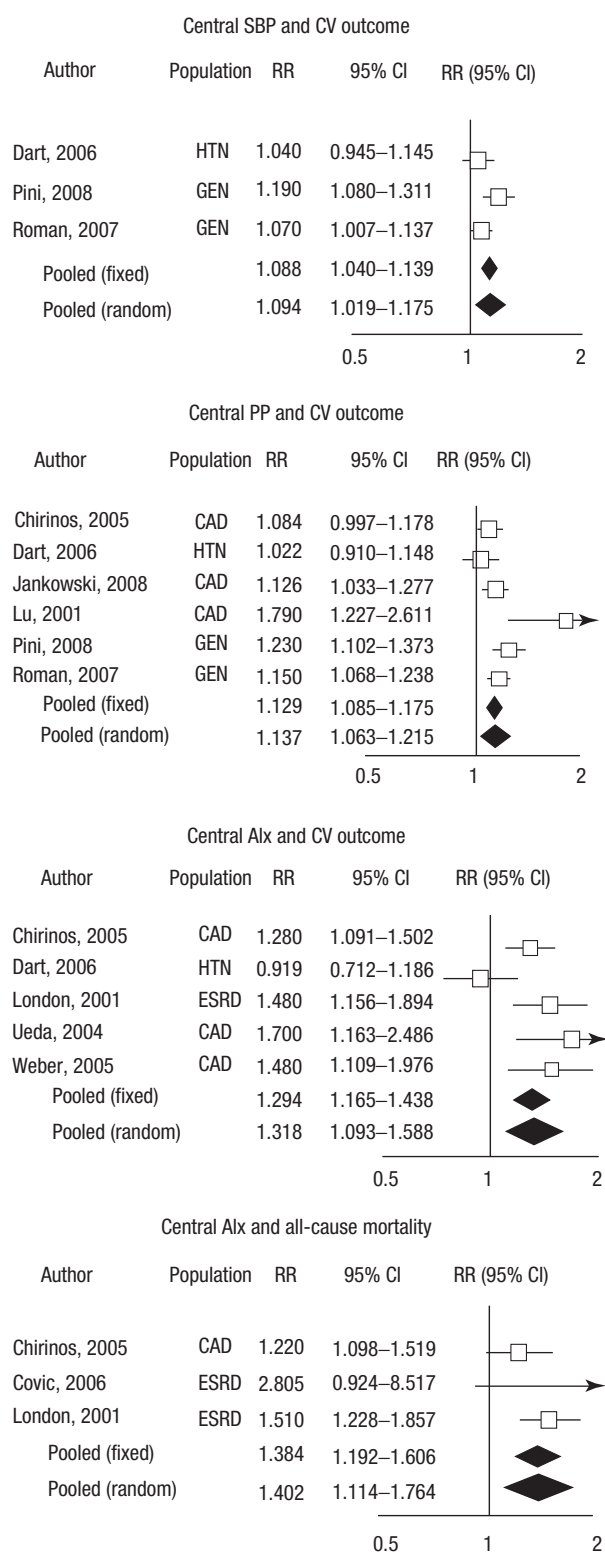


Table 22.4 CBPs/indices as markers of disease and predictors of surrogate endpoints

First author	Year, country	Population	Design	Parameter	Endpoint
Saba	1993, USA	Normotensives	Cross-sectional	Carotid Alx	LVMI ^a , carotid thickness
Boutouyrie	1999, France	Hypertensives	Cross-sectional	Carotid PP	Carotid thickness ^a
Jondeau	1999, France	Patients with Marfan syndrome	Cross-sectional	Carotid PP	Ascending aorta diameter ^a
Boutouyrie	2000, France	Hypertensives	Longitudinal (9-month FU)	Carotid PP	Carotid IMT reduction with treatment ^a
Roman	2000, USA	Normotensives, hypertensives	Cross-sectional	Carotid BP	Relative LV wall thickness
Waddell	2001, Australia	CAD	Cross-sectional	Carotid BP	Extent of CAD ^a
Nishijima	2001, Japan	Suspected CAD	Cross-sectional	Aortic fractional PP	Incident CAD
Nurnberger	2002, Germany	Healthy + CVD	Cross-sectional	Carotid Alx	CV risk scores
Philippe	2002, France	CAD	Cross-sectional	Aortic PP	Extent of CAD ^a
Hayashi	2002, Japan	Suspected CAD	Cross-sectional	Aortic Alx	Incident CAD
De Luca	2004, REASON study	Hypertensives	Longitudinal (1-year FU)	Carotid PP	LVMI reduction ^a
Weber	2004, Australia	Suspected CAD	Cross-sectional	Aortic BP, Alx	Incident CAD
Jankowski	2004, Poland	CAD	Cross-sectional	Aortic BP	Extent of CAD
Danchin	2004, France	Suspected CAD	Cross-sectional	Aortic PP	Incidence and extent of CAD
Booth	2004, UK	Systemic vasculitis	Cross-sectional	Aortic Alx	Disease activity
Roman	2007, USA	High-risk pts	Cross-sectional	Aortic PP	Carotid IMT and mass
Hashimoto	2007, Japan	Hypertensives	Longitudinal (1-year FU)	Aortic Alx	LVMI reduction with treatment ^a
Dart	2007, Australia	Elderly hypertensives	Longitudinal (4-year FU)	Carotid/Brachial BP	BP with treatment
Jiang	2007, China	Hypertensives	Longitudinal (2-mo FU)	Central SP	BP with treatment
Jankowski	2007, Poland	Normotensives and Hypertensives	Cross-sectional	Aortic BP	Extent of CAD ^a
Hope	2008, Australia	Suspected CAD	Cross sectional	Aortic Alx	Extent and severity of CAD
Wang	2009, China	Community population	Longitudinal (120 m)	Aortic SBP, PP, Alx	All-cause deaths, CV deaths, LVMI, IMT, eGFR ^a
Shrestha	2009, Japan	High-risk patients	Cross-sectional	Aortic SBP, Alx	White matter lesions, carotid IMT ^a
Stamatelopoulous	2010, Greece	Normotensives and Hypertensives	Cross-sectional	Aortic SBP, PP, Alx	Carotid IMT
Roman	2010, USA	American Indians	Cross-sectional	Aortic SBP, PP	LV hypertrophy and geometry

^aCBPs/indices were predictors of these surrogate endpoints independent of peripheral pressures.

Alx indicates augmentation index; BP, blood pressure; CAD, coronary artery disease; CV, cardiovascular; eGFR, estimated glomerular filtration rate; FU, follow up; IMT, intima-media thickness; LV, left ventricular; LVMI, left ventricular mass index; PP, pulse pressure; SBP, systolic blood pressure.

disease, independent of traditional risk factors (Kullo and Malik, 2007; Lorenz *et al.*, 2007). For an absolute carotid IMT increase of 0.1 mm, the future risk of myocardial infarction (MI) increases by 10–15 percent, and the risk of stroke increases by 13–18 percent (Lorenz *et al.*, 2007). Non-pharmacological and pharmacological interventions lead to either slowing of progression or even regression of increased IMT (Wang *et al.*, 2006; Crouse *et al.*, 2007; Shai *et al.*, 2010). Techniques are improving and protocols are being standardized (Stein *et al.*, 2008). Ranges according to age groups have been reported, however, proper reference values are lacking (Stein *et al.*, 2008). Despite the fact that the incremental value of IMT for risk prediction is challenged (Lorenz *et al.*, 2010), IMT measurement has been used in the evaluation of the efficacy of therapeutic interventions in large randomized studies (Kastelein *et al.*, 2007; Kastelein *et al.*, 2008; O'Leary and Bots, 2010) and the European Society of Hypertension/European Society of Cardiology recommended carotid ultrasound in the evaluation of hypertensive patients (cut-off value: IMT > 0.9 mm) (Mancia *et al.*, 2007a).

CORONARY ARTERY CALCIUM SCORE

Coronary artery calcium (CAC) is quantified with electron beam computed tomography and more recently with multidetector computed tomography scanners (Greenland *et al.*, 2007). CAC is assessed without contrast administration and, although doses of radiation are rather low (usually 1–2 mSv), exposure to radiation is a drawback of the technique. Calcium occupies approximately 20 percent of coronary plaques. CAC is rather an index of total coronary atherosclerotic burden and is not necessarily associated with the presence of obstructive coronary artery disease. Furthermore, although CAC may not directly identify plaques that are prone to rupture, areas of CAC and vulnerable or unstable plaques colocalize. Thus, patients with high CAC scores have a high likelihood of plaques that are prone to rupture. It should be noted that in patients with chest pain a zero CAC score cannot be

used to rule out the diagnosis of an acute coronary syndrome (Gottlieb *et al.*, 2010). An incremental prognostic value of CAC in the asymptomatic population has been shown (Polonsky *et al.*, 2010). A meta-analysis of six studies in more than 27 000 asymptomatic individuals showed that the presence of CAC quadruples the risk of a cardiovascular event in the coming three to five years (Greenland *et al.*, 2007). CAC also predicts risk in symptomatic patients (Pletcher *et al.*, 2004). The use of CAC scoring (with a cut-off value of 400) may be of value for risk stratification in asymptomatic individuals with intermediate cardiovascular risk (Greenland *et al.*, 2007). However, it has not been convincingly shown that management of an individual according to CAC classification alters prognosis.

ANKLE-BRACHIAL INDEX

The ankle-brachial index (ABI) is calculated as the ratio of the ankle to brachial systolic blood pressure and is a simple, non-invasive, and inexpensive test. Blood pressure in the lower limbs is normally higher than that in the upper limbs and normal ABI values range from 1.1 to 1.3. Values < 0.9 indicate with high specificity and positive predictive value the presence of flow-limiting arterial stenoses along the course of the arterial beds investigated. A supranormal ABI (i.e. > 1.4) is found in patients with generalized stiffening of blood vessels and advanced medial calcification most commonly seen in diabetic patients and patients with chronic kidney disease. There is a U-shaped association between ABI and CV events with a significantly increased risk in both low (< 0.9) and high (> 1.4) ABI groups (Ankle Brachial Index Collaboration, 2008). Clinically useful interpretation of ABI can be made only according to the fact that this index becomes abnormal in already progressed atherosclerotic disease. In this sense, it can be applied in the diagnosis of peripheral arterial disease in symptomatic patients and in the assessment of vascular risk in asymptomatic patients with CV risk factors (Mancia *et al.*, 2007a).

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Atherosclerosis

Atherosclerosis and inflammation	500	Atherogenesis and mechanical factors	502
Location of atherosclerosis	501	Effects of atherosclerosis	503
Risk factors for atherosclerosis	501	Embolization	504

Atherosclerosis is a disease of arteries which is widespread in the Western world and shows increasing prevalence in developing nations. In the West, atherosclerosis is the most common cardiovascular cause of death and disability, the major cause of heart attack and stroke, and is responsible for vast consumption of resources when it becomes symptomatic. Ironically, it is preventable.

Atherosclerosis is primarily an intimal disease of large- and medium-sized arteries. It is characterized by the presence of lipid in the intima, usually in localized form, and accompanied by inflammation and fibrosis, and by hyperplasia of the endothelium. The disease may spread into the surrounding media (Glagov *et al.*, 1987), but most commonly it extends into the arterial lumen.

The most common ill effect of atherosclerosis is to cause arterial narrowing (or stenosis) and reduction in blood flow to the tissues. The primary emphasis of this book is on arterial mechanisms, on pulsatile arterial blood flow, pressure, and diameter and on the cushioning function of the arterial system and on wave reflections. Other texts and reviews, surgical and cardiological, deal extensively with conduit function and steady flow (Libby, 2001c; Libby, 2002; Libby *et al.*, 2002; Russo *et al.*, 2002; Bonetti *et al.*, 2003a; Libby, 2003; Naghavi *et al.*, 2003a; Naghavi *et al.*, 2003b; Sary, 2003; Lloyd-Jones *et al.*, 2006; Serruys, 2006; Smith *et al.*, 2006; Jankowski *et al.*, 2008; He *et al.*, 2009; Libby *et al.*, 2009; Sanz *et al.*, 2009b) (see Chapter 2). Concentration on – even obsession with – arterial caliber has distracted attention away from other subtleties of disease and led to the phenomenon of ‘lumenology’ (Little *et al.*, 1995) and anecdotal accounts of the ‘oculodilatory reflex’ among procedural cardiologists and of the ‘audiodilatory reflex’ (‘if it is narrowed, I want it fixed’) among patients (Boden, 2004). Recent randomized studies in stable angina show comparable or even better results for a strategy of medical treatment directed at plaque stabilization than of revascularization by stenting or bypass surgery (Boden, 2004; Hambrecht

et al., 2004; Hueb *et al.*, 2004; Boden *et al.*, 2007; Boden *et al.*, 2009), although revascularization with urgent thrombolysis, stenting or surgery is the optimal strategy in most cases of acute occlusion of a central, peripheral, or coronary artery (Hueb *et al.*, 2007; Jacobs *et al.*, 2007; Bluemke *et al.*, 2008; DeMaria *et al.*, 2009; Sanz *et al.*, 2009b).

Since atherosclerosis is the most common serious disease affecting humans in Western and other affluent societies (Yusuf *et al.*, 2001; DeMaria *et al.*, 2009; Sanz *et al.*, 2009b), and since the major effect of this disease is to limit arterial blood flow to the tissues, it must be considered here. Emphasis will be directed towards mechanical principles as these relate to localization of the disease, to development of the disease, to complications including plaque rupture and thrombosis, and to effects of the disease on blood flow.

Confusion in terminology is described in Chapter 19. Many physicians, pathologists, and investigators still fail to distinguish between atherosclerosis and arteriosclerosis (Stehbens, 1994), and tend to assume that all arterial degenerative disease is caused by atherosclerosis. A specific distinction is made in Chapter 19, which deals with age-related changes in arterial wall properties. By arteriosclerosis as discussed here, we refer to the ‘nodular’ arteriosclerosis of Osler (1892) and Councilman (1891) – the focal, initially and predominantly intimal disease that is ‘characterized by formation of intimal plaques consisting of lipid accumulation, smooth muscle and inflammatory cells, connective tissue fibers, and calcium deposits’ (Zarins and Glagov, 1994) (Fig. 23.1), and which may limit blood flow to tissues downstream, weaken the wall locally, alter endothelial function, vascular tone, cell adhesion molecules and platelet adhesion, serve as a site for embolism or, when widespread, so disturb arterial distensibility as to increase left ventricular load and impair cardiac function (Falk, 1992; Amarenco *et al.*, 1994; Kupari *et al.*, 1994; Anderson *et al.*, 1995b; Gimbrone

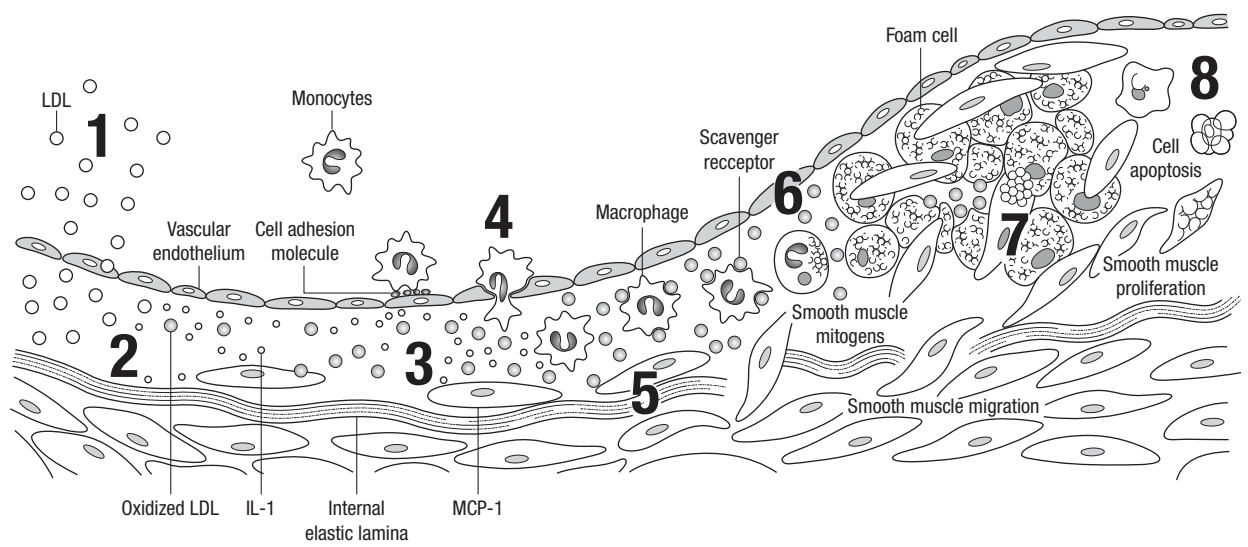


Figure 23.1 Schematic representing the evolution of atherosclerotic plaque as described by Libby (2001c)

Table 23.1 Classification of atherosclerotic lesions used in pathology

Type of lesion	Histological classification of lesions	Additional terms
I	Isolated macrophage foam cells	Fatty dot, fatty streak
II	Multiple foam cell layers	Fatty dot, fatty streak
III	Preatheroma, intermediate lesion	
IV	Atheroma	Fibrolipid plaque, fibrous plaque, plaque
V	Fibroatheroma	Fibrolipid plaque, fibrous plaque, plaque
VI	Fissured, ulcerated, hemorrhagic, thrombotic lesion	Complicated lesion
VII	Calcific lesion	Calcific lesion
VIII	Fibrotic lesion	

From Stary (2003).

et al., 2000; Huo and Ley, 2001; Smith *et al.*, 2006; DeMaria *et al.*, 2009). Definitions and classification of different atherosclerotic lesions have been agreed upon in an American Heart Association (AHA) document (Stary *et al.*, 1992; Stary *et al.*, 1995) and an atlas of atherosclerosis (Stary, 2003) (Table 23.1).

ATHEROSCLEROSIS AND INFLAMMATION

Over the past 20 years, there has been increasing attention to inflammation in atherosclerosis – as a cause of

progression of the disease with luminal narrowing, as a cause of weakening and rupture of the plaque, as a target for stabilization of the plaque, and possibly as the primary event in atherogenesis (Fuster *et al.*, 1996; Russell *et al.*, 2003; Virmani *et al.*, 2006; Cheruvu *et al.*, 2007; Epstein *et al.*, 2009; Libby *et al.*, 2009; Mathieu *et al.*, 2009; Libby and Crea, 2010). There is also attention to indices of inflammation such as C-reactive protein, as markers also of atherosclerosis. Research groups in Boston have been most active in this work (Libby, 2001a; Libby, 2001b; Libby, 2001c; Libby, 2002; Libby *et al.*, 2002; Cheruvu *et al.*, 2007; Libby *et al.*, 2009). This emphasis has been on monocyte infiltration in response to pro-atherogenic factors, such as oxidized lipoprotein, and the development of acute coronary syndromes or stroke (Fig. 23.1) (Ross, 1999; Libby, 2001a; Libby, 2001b; Bonetti *et al.*, 2003a; Itoh *et al.*, 2002; Libby *et al.*, 2002; Libby, 2003; Ridker *et al.*, 2003; Shimoni *et al.*, 2003; Suleiman *et al.*, 2003; Tracy, 2003; Danesh *et al.*, 2004; Mills and Bhatt, 2004). In the atherosclerotic plaque, a cholesterol-rich lipid core forms within the intimal layer and is infiltrated with cells, such as macrophages, macrophage-derived foam cells (laden with lipid drops), and smooth muscle cells. Plaque formation begins with accumulation of lipoprotein particles in the intima (Libby *et al.*, 2002). These particles aggregate and are susceptible to oxidation and glycation. Oxidative stress induces local cytokines that promote expression of adhesion molecules, which enable leukocytes to adhere to the endothelium (Huo and Ley, 2001; Epstein *et al.*, 2009). The cytokines also induce chemoattractant molecules (such as monocyte chemoattractant protein (MCP-1)) that facilitate migration of leukocytes. On entering the media, monocytes interact with stimuli (such as macrophage colony-stimulating factor) that promote expression of scavenger receptors. These receptors mediate uptake of modified lipid particles leading to formation of foam cells. The macrophage-derived foam cells are a storehouse of

atherogenic mediators and effector molecules, such as cytokines, superoxide anions, and matrix metalloproteinases. Smooth muscle cells migrate from the media to the intima and proliferate. This accumulation of smooth muscle cells leads to build up of extracellular matrix in plaque, and fatty streaks evolve to fatty fibrous lesions. Fibrosis continues in later stages, sometimes along with smooth muscle cell death. This leads to a relatively cell-free thin fibrous capsule surrounding a lipid-rich core. If the plaque ruptures, the lesion's thrombogenic core is exposed to blood in the lumen of the artery. Platelet adhesion and activation initiates formation of a thrombus and eventual cessation of blood flow to the tissues, thus causing ischemia or infarction if ischemia is prolonged (Libby, 2001c; Libby, 2002; Naghavi *et al.*, 2003a; Naghavi *et al.*, 2003b; Libby *et al.*, 2009).

LOCATION OF ATHEROSCLEROSIS

While fatty intimal plaques are seen at an early age in the aorta and other systemic arteries of Western humans (Wissler and Strong, 1998), the major sites of symptomatic arterial disease are in the carotid arteries (particularly in the carotid bulb), in the cerebral and epicardial coronary arteries, in the femoral and iliac arteries (particularly the superficial femoral arteries) and in the infrarenal part of the abdominal aorta (Wissler and Strong, 1998; Tuzcu *et al.*, 2001; Sary, 2003; Sanz *et al.*, 2009b). Any artery may be affected, but the disease usually spares arteries in the upper limb and the mesenteric, renal, internal mammary, and other muscular arteries (Roberts *et al.*, 1959a; Roberts *et al.*, 1959b; Sary *et al.*, 1992; Zarins and Glagov, 1994). Susceptible arteries are spared in certain locations – the internal carotid within the bony carotid canal of the petrous temporal bone, and the coronary arteries within the myocardium itself. Sites of predilection within susceptible arteries include the upstream rather than the downstream rim of aortic ostia, an artery immediately downstream from a bifurcation, and on its lateral side away from the flow divider (Fig. 2.28), and sites of dilatation (carotid bulb) and (paradoxically) relative constriction (the infrarenal aorta) (Roach, 1977b;

Glagov *et al.*, 1988a; Glagov *et al.*, 1988b; Nerem, 1988; Yoshida *et al.*, 1988; Jones *et al.*, 1990; Sary *et al.*, 1992; Giddens *et al.*, 1993; Bassiouny *et al.*, 1994; Caro, 1994; Sary *et al.*, 1995; Gimbrone *et al.*, 2000; Virmani *et al.*, 2006; Cheruvu *et al.*, 2007; Nakazawa *et al.*, 2010). The fact that such sites and such arteries show predilection emphasizes the relevance to local hemodynamic factors (see Chapter 2), since the known risk factors for atherosclerosis are general and should affect the blood and arteries equally at all sites. Sites of predilection provide clues – although to date few answers – as to why atherosclerosis develops and how it can be prevented or controlled.

RISK FACTORS FOR ATHEROSCLEROSIS

Risk factors for atherosclerosis include age and male sex, hypertension, hypercholesterolemia, cigarette smoking, diabetes mellitus, physical inactivity, adiposity, and race (Ross, 1992; Averbook and Wilson, 1994; McGill *et al.*, 1995a; McGill *et al.*, 1995b; McGill *et al.*, 1997; Wissler and Strong, 1998; Libby *et al.*, 2002; Richey Sharrett *et al.*, 2004; Smith *et al.*, 2006; Epstein *et al.*, 2009; Lloyd-Jones, 2010; Vasan, 2010). The long-term risk of coronary heart disease death in young men according to baseline total cholesterol level is shown in Fig. 23.2. Known risk factors cannot explain all features of the disease, although control of risk factors – especially hypertension, cigarette smoking, and hyperlipidemia – have led to a 70 percent decrease in the age-adjusted death rate from coronary atherosclerosis in Australia and the United States (National Heart Foundation of Australia Report, 1995; DeMaria *et al.*, 2009). New risk factors, including reduced HDL-cholesterol, elevated LDL-cholesterol, elevated lipoprotein (a), excess oxidative stress, elevated C-reactive protein, and elevated homocysteine level have been described (Stampfer and Malinow, 1995; Superko, 1998; Warnholtz *et al.*, 1999; Stamler *et al.*, 2000a; Itoh *et al.*, 2002; Ridker *et al.*, 2003; Shimoni *et al.*, 2003; Suleiman *et al.*, 2003; Pawlak *et al.*, 2004; LaRosa *et al.*, 2005; Wang *et al.*, 2006; Ware, 2006; Nicholls *et al.*, 2007a; Nicholls *et al.*, 2007b), together with air pollution and

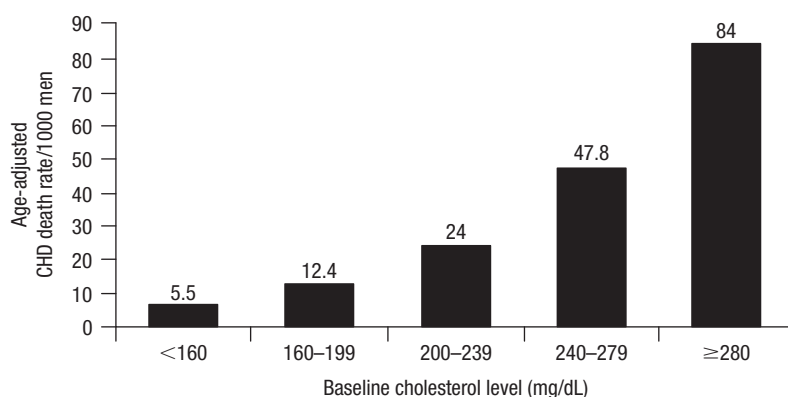


Figure 23.2 Relationship between risk of coronary heart disease and baseline total cholesterol level in 11 000 men aged 18–39 years followed up for 25 years. From Stamler *et al.* (2000a)

microparticles (Miller *et al.*, 2007). The principal recent focus is on the ‘epidemic’ of obesity, type 2 diabetes, insulin resistance, and metabolic syndrome (Smith *et al.*, 2006; Mathieu *et al.*, 2009). The prevalence of some forms of atherosclerotic lesion – such as abdominal aortic aneurysm – appears to be increasing, possibly as a consequence of aging of the community and/or lesser lower bodily exercise resulting from the greater use of motor vehicles for transport (Roach and MacLean, 1993; Zarins and Glagov, 1994). Abdominal aortic aneurysm of modern society has replaced the popliteal aneurysm of coach drivers – attributed to the pressure of the driver’s stiff leather boot on the popliteal artery behind the knee (Moore, 2005a). The US Center for Disease Control (CDC) has reported on the ‘actual’ causes of death in the United States and has attributed these to smoking (18 percent), poor diet, and physical inactivity (16 percent), with alcohol next in line and others – motor vehicle accidents and use of illicit drugs – far lower (McGinnis and Foege, 2004; Mokdad *et al.*, 2004).

ATHEROGENESIS AND MECHANICAL FACTORS

There is no agreement on what causes atherosclerosis to develop at particular sites. Theories include ‘encrustation’ (thrombus deposition and incorporation into the vessel wall) (von Rokitsansky, 1852; Duguid and Robertson, 1957; Berliner *et al.*, 1995; Grundy and Small, 1997; Libby, 2001c), ‘insudation’ (increased permeability of the endothelium to the entry of lipoproteins, particularly oxidized LDL) (Virchow, 1856; Fry, 1987; Friedman and Fry, 1993), ‘reaction to injury’ (an inflammatory proliferative response within the intima as a consequence of injury) (Ross, 1986; Collins and Fox, 1992; Fuster *et al.*, 1992a; Fuster *et al.*, 1992b; Libby, 2002; Libby *et al.*, 2002; Epstein *et al.*, 2009) or a combination of all under different circumstances (Fuster *et al.*, 1992a; Fuster *et al.*, 1992b; Ross, 1992; Davies, 1997; Naghavi *et al.*, 2003a; Naghavi *et al.*, 2003b). Common to all is some defect in the endothelial lining of the arterial wall such that the disease is initiated, develops, and progresses to a complicated lesion capable of causing a reduction in blood flow resulting in clinical events.

The endothelial lining of an artery is composed of endothelial cells that overlap like tiles on a roof (Fry, 1969a; Fry, 1969b; Zarins and Glagov, 1994), but are connected to each other and tend to be aligned in relation to the direction of blood flow. These cells are adherent to a basement membrane that separates them from the (usually) thin endothelium underneath, and with the intima separated from the load-bearing media by the internal elastic lamina. These endothelial cells are very susceptible to injury, but replicate quickly. Exhaustive research has been conducted over decades by some of the world’s most competent and resourceful bioengineers into the mechanical changes in arteries that occur at sites of

predilection for atherosclerosis (see Fry (1972); Caro (1977); Thubrikar *et al.* (1988); Yoshida *et al.* (1988); Nerem (1992); Friedman and Fry (1993); Caro (1994); Libby (Libby, 2001a; Libby, 2001b; Libby, 2002); Naghavi *et al.*, 2003a; Naghavi *et al.*, 2003b; Virmani *et al.* (2005); Reneman *et al.* (2006); Hoeks *et al.* (2008)). It has been extremely difficult to mimic and model the human arterial tree and to take into account not only the geometric patterns of the arteries but also the effects of arterial pulsation and of pulsatile blood flow. Attention has been directed at tensile stress in the arterial wall (which stretches the wall and causes the artery to dilate during systole) and shear stress at the vascular wall (which creates a drag on the endothelial cells) (Kornet *et al.*, 2000) and causes release of nitric oxide (see Chapters 2 and 5). It is clear that both are relevant in atherogenesis, the former in a diffuse and the latter in a localized way.

With respect to shear stress, it has been established that sites of predilection for atherosclerosis show either low average shear stress over a cardiac cycle, secondary flows or variable shear stress over the cardiac cycle or over longer periods of time (see Fig. 2.28) (Zarins *et al.*, 1983; Ku *et al.*, 1985b; Friedman and Fry 1993). These predilection areas are observed in artery bifurcations and are distinctly present in the carotid artery bulb opposite the flow divider (Bharadvaj *et al.*, 1982b; Reneman *et al.*, 1985; Rindt *et al.*, 1990; Rindt and Steenhoven, 1996). In these areas of disturbed flow, mean shear stress can be as low as 1–2 dyne/cm² (0.1–0.2 Pa) (Dai *et al.*, 2004). Initially, it was considered (Fry, 1969a) that endothelial damage might be due to high shear stress. However, it has been shown that this is not the case. Atherosclerosis is quite uncommon where shear stress is high, as at flow dividers, on the inner side of arteries downstream from a flow divider or on the outside of an arterial bend (see Fig. 2.28). The endothelial gene expression relative to the level of shear stress complies with these observations. Higher shear stress levels (10–15 dyne/cm²; 1.0–1.5 Pa) induce the expression of ‘atheroprotective’ genes of substances such as superoxide dismutase, cyclooxygenase-2, NO-synthase, and growth inhibitors and downregulate those of integrins, MCP-1, and endothelin-1 (Topper *et al.*, 1996; Malek *et al.*, 1999; Brooks *et al.* 2002). Lower shear stress levels <4.0 dyne/cm²; (<0.4 Pa) stimulate the expression of an ‘atherogenic phenotype’. At this level, genes of substances like endothelin-1, thrombomodulin, VCAM-1, and growth factors are upregulated (Malek *et al.*, 1999; Brooks *et al.*, 2002). Even more interesting is the observation that genes are differently expressed in areas of disturbed and undisturbed flow (Davies *et al.*, 2001; Brooks *et al.*, 2002; Dai *et al.*, 2004; Passerini *et al.*, 2004). In areas of disturbed flow with a mean shear stress of around 1.0 dyn/cm² (0.1 Pa), such genes as of chemokines and chemokine receptors (Brooks *et al.*, 2002; Dai *et al.*, 2004), and VCAM-1 are upregulated and genes of substances like superoxide dismutase-1 and superoxide dismutase-2 are downregulated (Brooks *et al.*, 2002).

Himburg *et al.* (2007) showed that in pulsatile shear stress, gene expression is frequency dependent. The repression of inflammatory transcription factors, including VCAM-1 and IL-8, and the induction of atheroprotective transcripts at 1 Hz was reversed into a pro-inflammatory response at 2.0 Hz. The latter was most pronounced under reversed and oscillatory shear. At arterial locations of high shear stress, endothelial cells align to the direction of blood flow and become more adherent to the underlying intima, while the intima shows more fibrous proliferation and becomes less permeable to the entry of monocytes and lipoproteins (Fry, 1969a; Fry, 1969b; Friedman and Fry, 1993; Caro, 1994; Zarins and Glagov, 1994). Problems occur at sites where fluctuations in blood flow prevent the cells from aligning, and where relative stasis causes relatively longer contact between the endothelial cells and the chemical and cellular elements of blood (Zarins and Glagov, 1994). Whether these changes in blood flow act through promoting entry of lipoproteins preferentially, through promoting adhesion of platelets or through physical damage and generation of an intimal proliferative response is not known. Perhaps all apply. However, an explanation is available for predilection at the lateral sites downstream from a bifurcation (stasis and secondary flows), in the carotid bulb (stasis and secondary flows), on the inner side of bends (stasis), and on the upper lip of intercostal arteries (stasis), the infrarenal aorta (greater blood flow pulsation including flow upwards into the renal arteries during diastole, see Chapter 18) and superficial femoral artery (relative stasis with sedentary lifestyle). These mechanisms of shear stress cannot explain all cases. The influence of tensile strain is apparent from protection of the carotid artery within the carotid canal (where it cannot expand) and the coronary arteries within the myocardium (ditto). The influence of other factors, such as hypercholesterolemia on atherosclerosis, is obvious from the fact that the disease is extremely rare in populations with a low cholesterol level and regresses or fails to progress when cholesterol level is reduced (Ornish *et al.*, 1990; Brown *et al.*, 1993; MAAS Investigators, 1994; Takagi *et al.*, 1997; Nissen *et al.*, 2004; Brown and Taylor, 2008; Kastelein *et al.*, 2008). The term 'regression' or 'reversal' of coronary artery disease (CAD) as used clinically incorporates the spectrum of beneficial changes in plaque composition and pathology, modest improvement in anatomic severity, endothelial healing, increased coronary flow and flow capacity, decreased symptoms, and improved survival. Standard coronary arteriography and standard non-invasive diagnostic tests (as commonly used) are inadequate for identifying or assessing severity of diffuse CAD. Newer technology or approaches using non-invasive positron emission tomography (PET), invasive intravascular ultrasound (IVUS) or pressure or flow velocity guide wires provide important new insights into the presence and severity of both segmental and diffuse coronary and other atherosclerotic disease (Gould *et al.*, 1998b; Kern *et al.*, 2006; Serruys,

2006; Kubo *et al.*, 2010). Non-invasive methods including magnetic resonance imaging (MRI) and computed tomography (CT) scanning are discussed elsewhere.

Pulsatile stress must be relevant, as well as pulsatile strain in the development of atherosclerosis (Thubrikar *et al.*, 1988; Kleinstreuer *et al.*, 2001; Himburg *et al.*, 2004; Reneman *et al.*, 2006; Fukumoto *et al.*, 2008; Hoeks *et al.*, 2008). Also, recent data show that increased circumferential tensile stress in the atherosclerotic plaque together with inflammatory change may play an important role in arterial remodeling and plaque rupture (Glagov *et al.*, 1987; Cheng *et al.*, 1993; Langille, 1996; Maseri, 1997; Ridker *et al.*, 1997; Heidland and Strauer, 2001; Epstein *et al.*, 2009). Atherosclerosis is uncommon in the pulmonary arteries, which expand more than systemic arteries with each beat of the heart, but with lesser mean and pulsatile luminal pressure and intramural stress.

EFFECTS OF ATHEROSCLEROSIS

Atherosclerosis may alter arterial blood flow and hemodynamics in six different ways:

1. by encroaching on the lumen and creating a localized stenosis;
2. by causing the formation of a localized thrombus that further narrows or occludes the arterial lumen;
3. by embolization of atheromatous material or associated thrombus to smaller peripheral arteries;
4. by encroaching on the media, causing destruction and atrophy of structural elements and leading to the formation of aneurysms;
5. by stiffening the artery and altering wave reflection in vessels upstream;
6. by creating an abnormality in vascular tone that favors vasoconstriction.

In assessing all these effects, a number of technical issues must be considered. Degree of stenosis is often assessed by angiography through comparing a narrowed segment with another segment (upstream or downstream) that appears not to be involved. This will underestimate the degree of stenosis if the artery is diffusely affected and will give results different from what one obtains with cardiovascular ultrasonic techniques, popularized by Nissen (Nissen *et al.*, 1991; Nissen and Yock, 2001) that relate the same stenosis to the external diameter of the arteries (Little *et al.*, 1995; De Franco and Nissen, 2001; Nissen, 2002). Pathologic (autopsy) studies examine arteries in a non-distended state and are apt to underestimate arterial diameter and wall thickness, and thus overestimate the degree of stenosis (Zarins and Glagov, 1994; Cheruvu *et al.*, 2007; Brown and Taylor, 2008). Such studies also show medial elastin lamellae as wavy structures rather than as straight sheets as they exist in life (see Chapter 19).

Arterial distensibility and stiffness in patients with atherosclerosis have been estimated at different sites from local diameter change and from pressure (usually brachial) at a distance (Mohiaddin and Longmore, 1989; Lehmann *et al.*, 1992a; Lehmann *et al.*, 1992b; Giannattasio *et al.*, 2001b). Results will be unreliable if there is a difference in pulsatile pressure between the diameter and pressure recording sites (see Chapter 4) (Caro, 1994).

Atherosclerosis and narrowing of conduit arteries

This subject is discussed in Chapter 17, with respect to coronary atherosclerosis and angina pectoris. The same principles apply to other sites.

Atherosclerosis and thrombosis

In his classic description of coronary artery occlusion as a clinical entity, Herrick (Herrick, 1912a; Herrick, 1912b) stressed the role of coronary thrombosis complicating coronary atherosclerosis as the cause of coronary occlusion, cessation of blood flow and myocardial infarction. This concept has been repeatedly confirmed, initially with the early angiographic studies of DeWood *et al.* (1980), and forms the basis of early experimental thrombolysis (Karsch *et al.*, 1983; Schumacher *et al.*, 1984; Tomaru *et al.*, 1989; Mehta *et al.*, 1991; Nichols *et al.*, 1991; Nicolini *et al.*, 1991) and the modern treatment of this condition in humans by fibrinolysis (Marder and Sherry, 1988a; Marder and Sherry, 1988b; O'Rourke *et al.*, 1988; Fuster *et al.*, 1992a; Fuster *et al.*, 1992b; Senen *et al.*, 2004). It is now recognized that while acute thrombus on an atherosclerotic lesion is responsible for coronary occlusion and 'transmural' (Q wave) myocardial infarction, non-occlusive thrombus can complicate coronary atherosclerosis and be responsible for other acute coronary syndromes, including sudden death, 'subendocardial' (non-Q wave) infarction and unstable angina, as well as contributing to the progression of the stenotic lesion without symptoms (Falk, 1992; Fuster *et al.*, 1992a; Fuster *et al.*, 1992b; Davies *et al.*, 1993; Davies, 1997; Libby, 2001c; Naghavi *et al.*, 2003a; Naghavi *et al.*, 2003b).

The first studies to draw attention to initiation of thrombosis were those of Davies in England (Davies and Thomas, 1984) and Falk in Denmark (Falk, 1985). Both showed that thrombosis was initiated by splitting or ulceration of the atherosclerotic plaque so that blood elements came into contact with collagen, lipid, or other components of the atherosclerotic intima. Recent advances support the current view of atherosclerosis as an inflammatory process that initiates and promotes lesion, development and disruption to the point of acute thrombotic complications and clinical events (Libby, 2002; Libby *et al.*, 2002; Serruys, 2006; Virmani *et al.*, 2006; Cheruvu *et al.*, 2007; Libby *et al.*, 2009).

The most recent studies on this phenomenon have been directed at understanding 'why do plaques rupture?' (Falk, 1992; Gotto, 1995; Hennerici, 2004; Tanaka *et al.*, 2004), with hopes 'towards the quiescent coronary plaque' (MacIsaac *et al.*, 1993). It has been shown that in both the coronary and carotid circulations, the plaques most likely to rupture are not chronic, fibrotic, stiff, or even highly obstructive lesions, but rather are more recently formed, soft, poorly supported, lipid rich, often less occlusive lesions (Falk, 1992; Taeymans *et al.*, 1992; Gotto, 1995; Milei *et al.*, 2003; Serruys, 2006; Cheruvu *et al.*, 2007). Also, plaque rupture and intraplaque hemorrhage may not correlate with symptoms in carotid artery stenosis (Milei *et al.*, 2003). Rupture has been shown to occur at the edge of such a plaque, where it is poorly supported, rather than in the region of its tough fibrous cap. Mechanical studies have shown that shearing and tearing forces at the plaque may cause rupture (Lee *et al.*, 1991a; Lee *et al.*, 1992; Cheng *et al.*, 1993; Davies *et al.*, 1993; Lee *et al.*, 1993; Arroyo and Lee, 1999; Clowes and Berceci, 2000). Such studies probably explain the increased incidence of coronary thrombosis during unaccustomed exertion (Fitzgerald, 1987; Willich *et al.*, 1993) and the success of such measures as cholesterol-lowering therapy, β -blocking agents, and possibly angiotensin-converting enzyme inhibitors (ACEIs) and antioxidants, for lowering the incidence of plaque rupture and coronary thrombosis (Gotto, 1995; Sukhova *et al.*, 2002). Apart from mechanical forces, plaque weakening by monocyte and/or macrophage activation undoubtedly also plays a role in plaque rupture (Davies *et al.*, 1993; Jude *et al.*, 1994; Moreno *et al.*, 1994; Davies, 1997; Maseri, 1997; Ridker *et al.*, 1997; Ganne *et al.*, 1999; Imanishi *et al.*, 2002; Stone, 2004).

EMBOLIZATION

It is generally accepted that cerebral embolism from carotid atherosclerosis is a common cause of cerebral vascular events and an indication for anti-platelet therapy, anti-coagulation or even carotid endarterectomy or stenting, as well as for cholesterol-lowering therapy. It is assumed that embolic material is usually a clot, dislodged from an ulcerated plaque, but atherosclerotic material with cholesterol has been observed to embolize into the cerebral and retinal vessels. A similar mechanism may operate at other sites, although less attention is usually given to this in the coronary circulation, possibly because collateral circulation is better in the peripheral coronary vasculature, so that a distal embolism is less likely to be symptomatic than the same material at its original site before embolization. Concern is expressed that over-expansion of a stent in acute evolving myocardial infarction may be one of the factors contributing to the 'no reflow' phenomenon through distal embolism (Nicolli *et al.*, 2009a; Nicolli *et al.*, 2009b). The wide caliber of the

carotid bulb may also be a factor for the cerebral circulation, with lesions here more likely to embolize than cause stenosis. This mechanism also appears to operate in the aorta itself. Transesophageal echocardiography has uncovered the extent and severity of asymptomatic atherosclerosis in the thoracic aorta during life; polypoid lesions are often seen, and their presence is associated with a greater risk of ischemic cerebral events (Amarenco *et al.*, 1994; Lehmann *et al.*, 1995a). Such patients are obviously at relatively high risk of stroke during or after aortic catheterization or bypass surgery. Demonstration of back flow from the descending aorta into carotid vessels (see Chapter 9) confirms that lesions in this vessel will not necessarily embolize only to the lower body.

Secondary medial effects

Atherosclerosis is often so severe as to encroach on the media and to damage the elastin and collagenous elements of the media and smooth muscle associated with these. It is not the usual cause of medial degeneration (see Chapter 19), but can account for localized patchy areas of damage and 'ectasia' associated with intimal plaque. It appears that atherosclerosis causes medial weakening with aneurysm formation predominantly in older males and at only one site with any frequency – the infrarenal abdominal aorta – although here also other factors may be operative (Tilson and Stansel, 1980; Reed *et al.*, 1992; Roach and MacLean, 1993; Zarins and Glagov, 1994; Lederle *et al.*, 2000; Lawrence-Brown *et al.*, 2001; Ashton *et al.*, 2002; Kent *et al.*, 2004; Song *et al.*, 2007; Golledge *et al.*, 2008; Kuivaniemi *et al.*, 2008; Diehm and Baumgartner, 2009). Ulceration of atherosclerotic aortic plaque can lead to aortic dissection or perforation and eventual rupture (Ganaha *et al.*, 2002).

Susceptibility of the infrarenal segment of the aorta to atherosclerosis and aneurysm formation is intriguing. Why is this region of the aorta so susceptible? No definite answers can be given, but a number of factors can be noted. The infrarenal aorta tapers down to a relatively narrow neck before expanding at the point where it bifurcates (Roach, 1977b; Langewouters *et al.*, 1984; Roach and MacLean, 1993). There is a higher steady flow downwards in the aorta above the renal arteries, since these normally receive some 25 percent of total cardiac output. However, since renal flow is almost continuous throughout the cardiac cycle, there is often some back flow during diastole from the lower body; hence flow in the infrarenal aorta is more pulsatile than that above. This is a factor cited as predisposing to atherosclerosis (Caro, 1994; Zarins and Glagov, 1994). An additional factor whose influence is unknown is the mechanical and chemical composition of the lower abdominal aorta. Whereas elastin is more abundant than collagen in the thoracic aorta, the opposite prevails in the lower abdominal aorta (see Chapter 4) (Cleary, 1963; Learoyd and Taylor, 1966).

Once established, abdominal aortic aneurysms continue to expand, and ultimately to dissect and rupture (Davies, 1993; Ernst, 1993; Ganaha *et al.*, 2002; Brady *et al.*, 2004; Song *et al.*, 2007; Golledge *et al.*, 2008). This is predictable. Tension in the wall increases as the aneurysm dilates (through the law of Laplace), and the wall progressively thins. Expansion rate accelerates as the aorta dilates (Brady *et al.*, 2004). Thrombus usually forms within the aneurysm, but this provides little mechanical support. A major commitment has been made by vascular surgical units to identify people with asymptomatic abdominal aortic aneurysms and to reconstruct the aorta before rupture occurs (Davies, 1993; Ernst, 1993). Etiology, diagnostic strategies, and therapeutic management of aortic dissection have been reviewed (Nienaber and Eagle, 2003a; Nienaber and Eagle, 2003b), with a more conservative approach applied than before (see Chapter 24).

Abnormalities in smooth muscle tone and in potentiation of thrombosis

Much recent work has come to focus on the influence of the endothelium on medial smooth muscle tone, on platelet function, and on fibrinolysis. This commenced with identification of plasminogen activator (t-PA) in the endothelium, identification of its action in dissolving spontaneously occurring clots, and its manufacture as a drug by genetic engineering techniques to treat coronary occlusion (TIMI, 1985; O'Rourke *et al.*, 1988; Collen, 1992; Collen and Van de Werf, 1993; Verstraete and Lijnen, 1994; Brzosko *et al.*, 2003; Rubenstein *et al.*, 2004).

Around the same time, it was found that other substances were produced by platelets and endothelial cells that could have a powerful action on platelet function in the blood and on the medial smooth muscle. Normal function was to discourage platelet adhesion and thrombosis, and to reduce contraction of smooth muscle. Abnormalities predispose to thrombosis and to increased smooth muscular tone with accentuation of stenosis. Messengers involved include prostacyclins, thromboxane, endothelin, and endothelial-derived relaxing factor (identified as nitric oxide) (Furchgott, 1988; Vanhoutte, 1988; Healy, 1990; Vane *et al.*, 1990; Moncada, 1991; Reneman, 1993). Research in this area has exploded to dominate many other fields of cardiology at this time (and is discussed in Chapter 5). The commercial success of recombinant t-PA foreshadowed other new products, including monoclonal antibodies against growth-stimulating factors to prevent restenosis after angioplasty (Serruys, 2006). The work already done provides possible answers to other previously unexplained phenomena, including the value of nitroglycerin in angina pectoris and the occurrence of increased vasomotor tone and sometimes frank spasm in what appear to be normal or minimally diseased coronary arteries.

Abnormalities of endothelial function have been demonstrated in atherosclerosis (Golino *et al.*, 1991; Williams *et al.*, 1992; Klemsdal *et al.*, 1994; Anderson *et al.*, 1995). Reviews have discussed the relationship between endothelial dysfunction and atherogenesis and atherosclerosis (Gimbrone *et al.*, 2000; Bonetti *et al.*, 2003a; Russo *et al.*, 2002; Sanz *et al.*, 2009b). Endothelial abnormalities may, however, be a consequence rather than a cause of the disease. Dysfunction may be caused by defective formation of active substances or from interference with the diffusion to active sites outside the intima. In the presence of coronary atherosclerosis, it has been shown that acetylcholine and serotonin constrict rather than dilate (indicating a defect in nitric oxide production) (Golino *et al.*, 1991; McFadden *et al.*, 1991), and that cholesterol-lowering and antioxidant agents reduce or abolish the vasoconstriction induced by acetylcholine on coronary arteries. These results appear to explain the greater than expected efficacy of cholesterol-lowering drugs in reducing acute coronary events (Brown and Taylor, 2008).

Arterial stiffening

Atherosclerosis appears to increase stiffening of the aorta and major arteries (Stefanadis *et al.*, 1990; Lehmann and Gosling, 1991; Lehmann *et al.*, 1995a; Stefanadis *et al.*, 1995a). Giannattasio *et al.* (2001b) found that arterial stiffness was increased in the internal carotid artery where there was an atherosclerotic plaque compared with the intact contralateral internal carotid artery. Also, stiffness was greater in the common carotid artery of the affected side in comparison with the contralateral common carotid artery. Beaussier *et al.* (2008) have studied carotid arterial

stiffness with a novel non-invasive echotracking system, and have shown that the affected artery can have greater or lesser stiffness depending on whether remodeling of the arterial segment was predominantly inward or outward. Evelien Hermeling, working with Reneman and Hoeks, has described a new method for measuring local arterial stiffness as pulse wave velocity of the carotid artery; this is expected to provide new insight on this subject (Hermeling *et al.*, 2007; Hermeling *et al.*, 2010). Many of the studies in this area have been flawed by failure to measure pressure at the same site as diameter (Caro, 1994) (see Chapter 19), so that effects of pressure wave amplification have been overlooked. Since atherosclerosis is associated with age, hypertension, diabetes, and smoking, and since it often affects a large part of the arterial system and causes some medial damage, it is not surprising to find reports of increased arterial stiffening in atherosclerosis. We note, however, that there are conflicts in reported data, that there have often been methodological problems (see above) and that we have not been able to distinguish people with from those without atherosclerosis on the basis of arterial stiffness indices. As discussed in Chapters 19 and 20, arterial stiffness is dominated by aging change and the effects of hypertension in populations with and without hypertension. Atherosclerosis must be a factor, but it is of lesser importance than arteriosclerosis due to aging and hypertension. We have little confidence that measures or indices of arterial stiffness will prove to be of specific diagnostic value in the identification of people with atherosclerosis. We acknowledge, however, that in symptomatic patients such indices have been useful in predicting severity and extent of coronary atherosclerosis (Weber *et al.*, 2004; Weber *et al.*, 2010).

Specific arterial disease

Marfan syndrome	507	Takayasu arteritis	521
Aortic aneurysm and dissection	510	Non-atheromatous coronary artery disease	521
Aortic coarctation	512	Disturbances of smooth muscle tone and caliber: specific	
Arteriovenous shunt	521	drug effects	521

Aging (Chapter 19), hypertension (Chapter 20), atherosclerosis (Chapter 23), diabetes (Chapter 25), and renal failure (Chapter 25) are the most common causes of altered arterial hemodynamics and of disturbed blood flow in the arteries of human subjects. There are a number of other conditions that can do likewise, but which are either less common or less potent in their effect. These are discussed here. As for the more common causes of arterial dysfunction, these act through either narrowing or occluding an artery and/or stiffening the artery, and/or creating abnormal reflection, and/or causing the artery to rupture. Effects can be manifest as ischemia of tissues downstream, as increased load on the heart and arteries upstream, or (through rupture) as disturbance of surrounding tissues or loss of blood.

As set out in Chapter 18, a new ill effect of arterial stiffening has emerged since the last edition of this book was published in 2005. This is the damage to the microvasculature in brain and kidney, and which can be attributed to funneling of flow pulsations into the beds of high flow organs, when such pulsations cannot be absorbed in the aorta and large arteries (O'Rourke and Safar, 2005).

MARFAN SYNDROME

This is a congenital defect of connective tissue, inherited as an autosomal dominant condition or occurring sporadically as a result of gene mutation. It is characterized by musculoskeletal features – tall stature with long, spindly limbs – visual problems, and cardiovascular abnormalities (Marfan, 1896; Murdoch *et al.*, 1972; Maron *et al.*, 1998; Pyeritz, 2001). The most serious abnormality, and that responsible for premature death, is a defect in formation of aortic elastin fibrils (now identified as the *FBN1* gene in chromosome 15 which encodes the connective tissue

protein fibrillin-1) (Dietz and Belz, 1991; Lee *et al.*, 1991; Milewicz, 1995). Patients with Marfan syndrome show structural defects and a decrease in the content of aortic elastin, which cause the aorta to degenerate prematurely, to dilate excessively with age (Fig. 24.1) and ultimately to rupture (Hwa *et al.*, 1993; Jeremy *et al.*, 1994; Finkbohner *et al.*, 1995; Legget *et al.*, 1995). Prior to the availability of surgical aortic repair, life expectancy was 47 years; it is now over 60 years (Finkbohner *et al.*, 1995). Pearson *et al.* (2008) give median probability of survival now as 72 years. Aortic root dilatation in Marfan syndrome is progressive and rapid, averaging 0.38–1.9 mm/year (Hwa *et al.*, 1993; Meijboom *et al.*, 2005; Lazarevic *et al.*, 2006); this contrasts with less than 0.1 mm/year in normal subjects (Vasan *et al.*, 1995). Surgery is recommended when aortic root diameter exceeds 50 mm or at a smaller diameter in specific conditions in order to prevent aortic dissection or rupture (Gott *et al.*, 1986; Hwa *et al.*, 1993; Devereux and Roman, 1999; Gott *et al.*, 1999; Nienaber and Eagle, 2003a; Hiratzka *et al.*, 2010).

The diagnosis of Marfan syndrome is usually made on the basis of clinical features using the Ghent diagnostic criteria (De Paepe *et al.*, 1996; Dean, 2007). Identification of genetic markers and of abnormal aortic wall properties with ultrasound and magnetic resonance imaging (Milewicz, 1995; Recchia *et al.*, 1995; Maron *et al.*, 1998) now assist in diagnosis. Over the past five years, Marfan syndrome has been diagnosed more frequently. Other mechanisms have been detected in Marfan syndrome apart from the defect in aortic elastin related to abnormalities in synthesis of fibrillin-1 (Pearson *et al.*, 2008). These better explain other features of Marfan syndrome, including mitral valve degeneration and non-cardiac manifestations (Gelb, 2006; Matt *et al.*, 2009). This was shown to be a defect in production of the latent transforming growth factor- β (TGF- β), caused by the fibrillin-1. Concurrently, it was recognized that there are

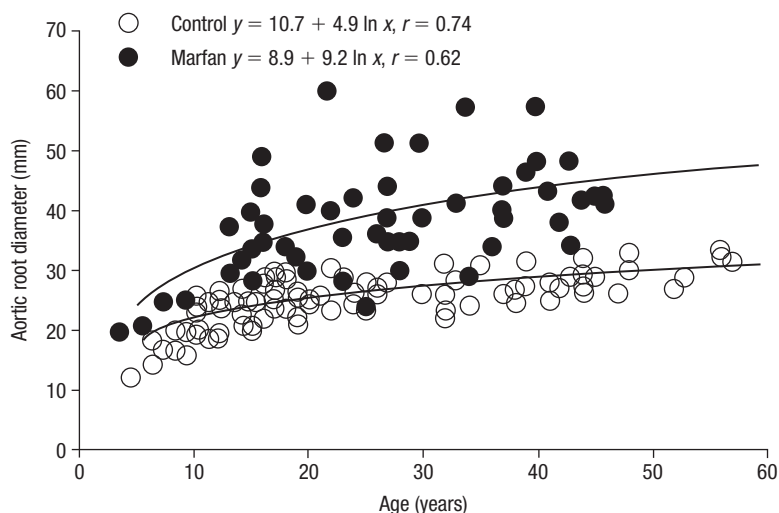


Figure 24.1 Comparison of the relationship between age and aortic root diameter in patients with Marfan syndrome and age- and gender-matched controls. The regression relations shown for each differ significantly, but there is considerable variation in the degree of aortic dilatation observed in patients with Marfan syndrome. Reproduced from Jeremy *et al.* (1994)

some patients in whom the TGF- β problems dominated over fibrillin-1, and other conditions were described which overlapped with Marfan syndrome. Two of these, Loeys Dietz syndromes 1 and 2, were named after the researchers who had dominated this field (Loeys *et al.*, 2006). Among other things, this led to change in the approach to medical treatment of Marfan and related syndromes (Yetman *et al.*, 2005; Habashi *et al.*, 2006; Lacro *et al.*, 2007; Yetman, 2007; Keane and Pyeritz, 2008; Williams *et al.*, 2008; Lindeman *et al.*, 2009). Matt *et al.* (2009) noted that TGF- β levels are increased in serum of patients with Marfan and related syndromes, and could be used as a diagnostic and prognostic marker, and as a guide to therapy.

Studies of aortic mechanics have shown increased stiffness (and decreased distensibility), as well as dilatation of the aorta (Fig. 24.2) (Yin *et al.*, 1989; Hirata *et al.*, 1991; Jeremy *et al.*, 1994; Adams *et al.*, 1995; Groenink *et al.*, 1998; Nollen *et al.*, 2004; Baumgartner *et al.*, 2006; Segers *et al.*, 2006). Hirata *et al.* (1991) demonstrated greater stiffness in the abdominal aorta even when undilated. Greater stiffness of the dilated aorta is attributable to the two-phase composition of the aorta (see Chapter 3), with stress borne at greater diameter by the collagenous components of the wall. The detailed studies of Yin *et al.* (1989) also included determination of ascending aortic impedance and the effects of vasodilator (nitroprusside) and β -blocking drugs used in the long-term treatment of this condition. Results show beneficial effects of nitroprusside attributable to a reduction of wave reflections (Fig. 24.3) and of systolic pressure; the β -blocking agent (propranolol) increased wave reflections (which may have been acutely deleterious), although the combination had an effect similar to nitroprusside alone. The use of β -blocking agents has been shown to delay aortic dilatation in Marfan syndrome (Shores *et al.*, 1994). The benefits of β -blocking drugs are attributable to a decrease in the number of fatiguing cycles, as well as in peak stress and maximum rate of change of aortic pressure (dP/dt)

(see Chapter 19) (Simpson *et al.*, 1970); the benefits of vasodilating agents (also employed in long-term therapy) are attributable to a reduction in early wave reflections and thus a decrease in peak systolic pressure, pulse pressure, and arterial wall stress. Jondeau *et al.* (1999) have shown that progressive aortic dilation in Marfan syndrome is related to central pulse pressure, but not to brachial pulse pressure or to mean arterial pressure. Groenink *et al.* (1998) have shown that use of β -blocking agents leads to greater reduction of aortic pulse wave velocity in Marfan syndrome than in normal subjects.

The effects of Marfan syndrome are to cause premature degeneration of the aortic media, with progressive dilatation and ultimate dissection or complete rupture. The same problem is seen in the elderly (see Chapter 19), and the same principles of therapy appear to apply – to decrease excessive wall strain by decreasing early wave reflections, to decrease the number of fatiguing cycles, decrease maximum aortic dP/dt and with consideration of surgical replacement of the proximal aorta when dilatation is extreme (> 50 mm) or progressive (> 10 mm/year) despite medical therapy, and when dissection or rupture appears imminent, or when aortic valve incompetence compromises cardiac function.

Medical management of Marfan and related syndromes still focuses on β -blocker (to reduce heart rate and number of strain cycles) and arterial dilator (to reduce aortic systolic and pulse pressure) through reduction in wave reflections (Yin *et al.*, 1989; Keane and Pyeritz, 2008). For excellent reviews on medical treatment of Marfan syndrome, see Yetman (2007) and Williams *et al.* (2008). A strong case exists now for using an angiotension blocking drug as the arterial dilator. As a consequence of recent research, reduction in TGF- β has become a target for therapy (Keane and Pyeritz, 2008). In experimental animals (a mouse model of Marfan syndrome), antibodies against TGF- β prevented development of mitral valve disease and aortic aneurysm. Losartan, an angiotensin II type 1 receptor blocking drug (ARB) had

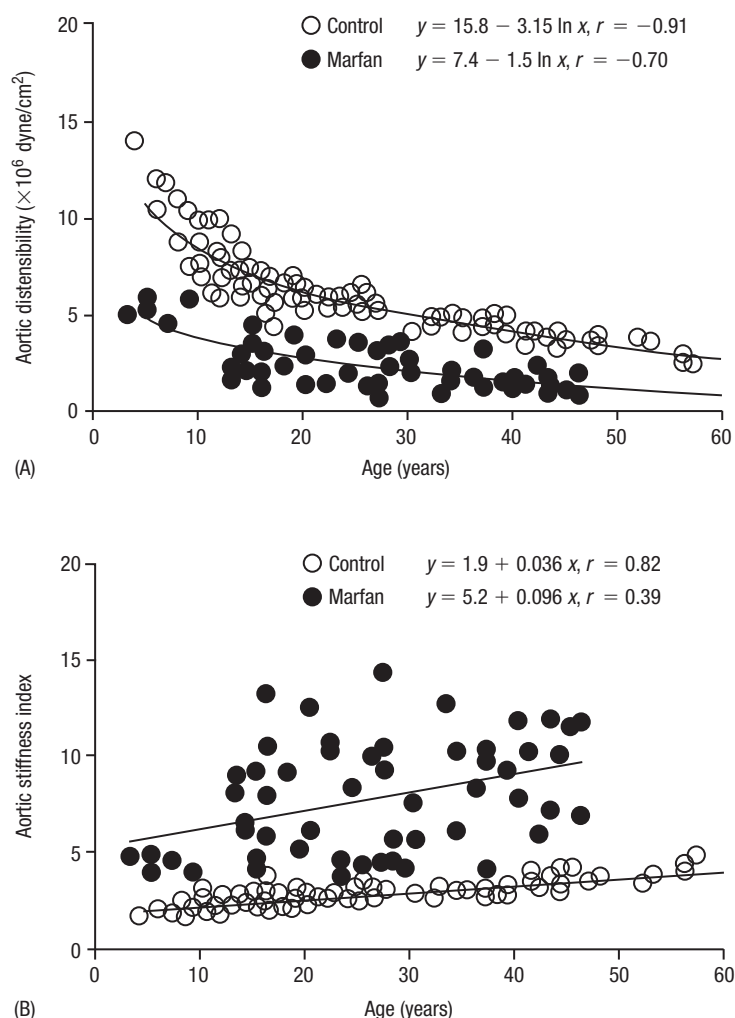


Figure 24.2 (A) Comparison of the relationship between age and aortic distensibility in patients with Marfan syndrome and age- and gender-matched controls. At all ages, patients with Marfan syndrome exhibit reduced aortic distensibility compared with controls. (B) Comparison of relationship between aortic stiffness index and age in the Marfan syndrome and control groups. Aortic stiffness increases with age in both groups, but is greater at all ages in the Marfan group. Reproduced from Jeremy *et al.* (1994)

similar beneficial effect in these mice with and was superior to the β -blocker propranolol (Habashi *et al.*, 2006) and to the angiotensin-converting enzyme inhibitor (ACEI) enalapril (Habashi *et al.*, 2011). Angiotensin II promotes vasoconstriction, cellular proliferation, and fibrosis and suppresses apoptosis when binding to its type 1 receptor, whereas binding to its type 2 receptor has opposite (and beneficial) effects. Beneficial effect of losartan is explained by the exclusive inhibition of type 1 receptors, whereas an ACEI, which blocks formation of angiotensin II, inhibits signaling through both receptors and may not be as effective.

Losartan therapy has been used in a small cohort of Marfan syndrome children with moderate–severe aortic dilation, as an add-on to β -blockade over two years (Brooke *et al.*, 2008). Despite the encouraging results of this observational study, equipoise is maintained regarding a role for ARB therapy in the treatment of patients with Marfan syndrome; the findings of this study must be confirmed by a prospective, randomized trial. A trial coordinated by the Pediatric Heart Network of the National Heart, Lung, and Blood Institute, comparing losartan with atenolol in patients with Marfan syndrome, began enrolling patients in the winter of 2007 (Lacro

et al., 2007). A more recent multicenter trial (COMPARE) using losartan therapy in adults with Marfan syndrome is in progress (Radonic *et al.*, 2010). Another clinical trial evaluating the effects of losartan versus nebivolol, a β -blocker with vasodilator activity, versus the association of both drugs on the progression of aortic root dilation in Marfan syndrome with *FBN1* gene mutations is also in progress (Gambarin *et al.*, 2009). Until data from these trials are available, evidence for the potential efficacy of ARB therapy in this setting should be considered preliminary.

Effects of ACEI on aortic wall mechanics and dilatation have also been studied. An open-label, non-randomized trial of ACEI versus β -blockers showed a superiority of ACEI in terms of increased aortic distensibility and smaller increases in aortic root diameter (Yetman *et al.*, 2005). A recent randomized controlled trial of the ACEI perindopril against placebo in a small group of patients with Marfan syndrome treated with standard β -blocker therapy showed reduced aortic stiffness, slower aortic root growth and decreased circulating TGF- β levels (Ahimastos *et al.*, 2007). Results of this trial were surprising given the potential lack of benefit expected with blockade of both type 1 and type 2 receptors by ACEI (Keane and Pyeritz,

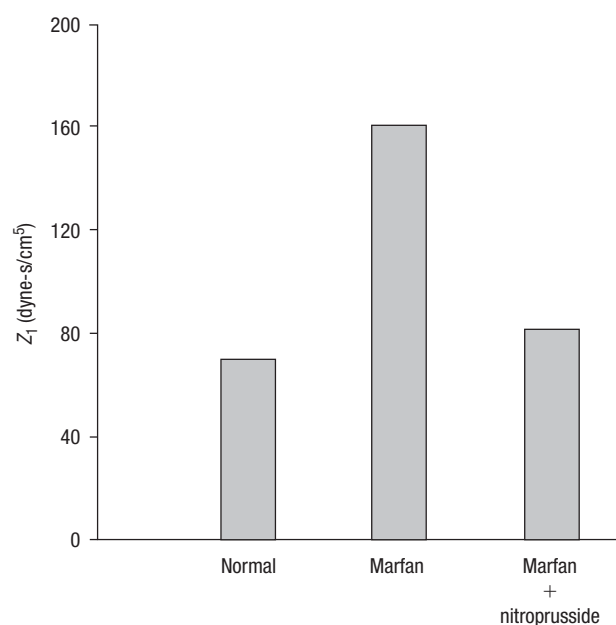


Figure 24.3 Ascending aortic impedance modulus of the first harmonic component (Z_1) in patients with Marfan syndrome and the effects of nitroprusside infusion. Nitroprusside returned Z_1 to a level close to that of age-matched normal subjects. Data from Nichols *et al.* (1985a) and Yin *et al.* (1989)

2008; Pearson *et al.*, 2008; Habashi *et al.*, 2011). Authors proposed an anti-apoptotic effect seen with blocking or receptor type 2 to explain the theoretically unfavorable inhibition of this receptor pathway, although apoptosis has not been observed in the ascending aorta of mouse models of Marfan syndrome (Pearson *et al.*, 2008). Furthermore, increase of aortic systolic diameter at the sinuses of Valsalva in controls over 24 weeks was surprisingly large (2.3 mm/m²), while reduction in central aortic stiffness in the treated group was also surprisingly large and uniform. Undoubtedly, larger trials of ACE inhibition are warranted as the authors admit.

Surgical management (usually involving replacement of the ascending aorta ± aortic valve) is recommended in adults before aortic diameter exceeds 50 mm, or at even a smaller diameter in cases with rapid increase in diameter (5 mm per year), large aortic root area-to-height ratio (> 10), significant aortic regurgitation, family history of dissection at diameter < 50 mm, and in women contemplating pregnancy. Thoracic aneurysms may involve one or more aortic segments (aortic root, ascending aorta, arch, or descending aorta) and are classified accordingly. Sixty percent of thoracic aortic aneurysms involve the aortic root and/or ascending aorta, 40 percent involve the descending aorta, 10 percent involve the arch, and 10 percent involve the thoracoabdominal aorta (with some involving > 1 segment) (Isselbacher, 2005). Histologically, ascending aortic aneurysms are characterized by fragmentation of the elastin fibers, an increase of collagen fibers, loss of smooth muscle cells, and replacement of

degenerated tissue with interstitial pools of basophilic mucopolysaccharide material inadequately termed ‘cysts’ (Klima *et al.*, 1983; Okamoto *et al.*, 2003). However, similar observations have been made in the aging but non-dilated thoracic aorta (Schlatmann and Becker, 1977). Therefore, none of these histological changes can be regarded as a specific structural alteration responsible for aneurysm formation (Kirsch *et al.*, 2006). According to Laplace’s law, wall stress is directly related to the radius and intravascular pressure, and inversely to wall thickness. Thus, once an aneurysm has developed, it is likely that increased wall stress is an important contributor to accelerating dilation and increasing the risk of rupture. Recently, using patient-specific data in a cylindrical model of the aorta, Okamoto *et al.* (2003) confirmed that mean circumferential wall stress increases linearly with aortic diameter.

AORTIC ANEURYSM AND DISSECTION

The vast database of the Yale University Center for Thoracic Aortic Disease – which includes information on 3000 patients with thoracic aortic aneurysm or dissection – has, over the last decade, permitted multiple glimpses into the ‘playbook’ of this virulent disease (Elefteriades, 2008). Understanding the precise behavioral features of thoracic aortic aneurysm and dissection permits more effective combat of this disease. Thoracic aortic aneurysm, while lethal, is indolent. Mortality usually does not occur until after years of growth. Sixty percent of thoracic aortic aneurysms involve the ascending aorta and grows slowly. The Yale group estimates this to be about 1 mm per year (the descending aorta grows somewhat faster) which is much less than that reported earlier (see below). Prior estimates had failed to account for negative observed growth (measurements of any object will vary about a mean) and other statistical complexities, leading to overestimates of actual growth. This group developed exponential equations specifically for the purpose of accurately estimating growth rates (Rizzo *et al.*, 1998). In fact, reports of rapid growth of the thoracic aorta are usually reflective of measurement error – specifically, measuring across an oblique portion of the aorta, especially the aortic arch. Over a patient’s lifetime, ‘hinge points’ at which the likelihood of rupture or dissection skyrockets are seen at 5.5 cm for the ascending and 6.5 cm for the descending aorta. Intervening at 5.5 cm for the thoracic aorta (ascending or descending) prevents most adverse events. Symptomatic aneurysms require resection regardless of size. The yearly rate of rupture, dissection, or death is 14.1 percent for a patient with a thoracic aorta of 6.0 cm diameter. The mechanical properties of the aorta deteriorate markedly at 6.0 cm diameter (distensibility falls, and wall stress rises) (Fig. 24.4). Thoracic aortic aneurysm and dissection are largely inherited diseases, with a predominantly autosomal-dominant pattern.

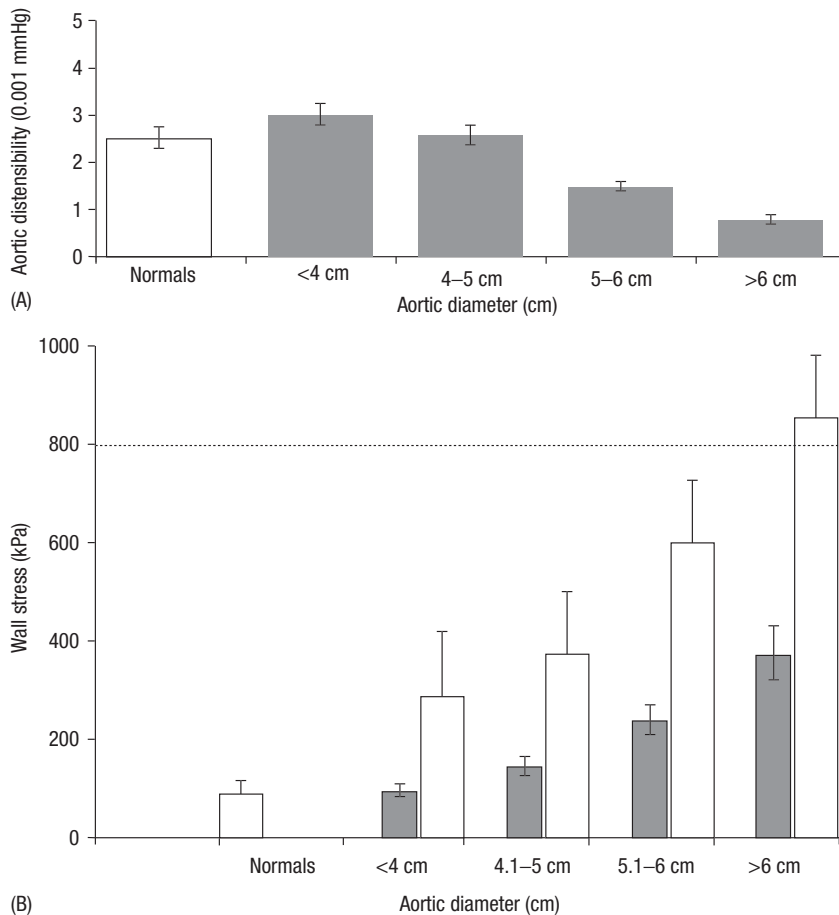


Figure 24.4 (A) Distensibility values in normal aortas and aortic aneurysms of different diameters. Distensibility of ascending aortic aneurysms decreases rapidly as diameter increases, to very low values at dimensions greater than 6 cm. (B) Exponential relationship between wall stress and aneurysm size in ascending aortic aneurysms. The dark bars represent a blood pressure of 100 mmHg, and the light bars represent a blood pressure of 200 mmHg. The line at 800 kPa represents the maximum tensile strength of the human aorta. Reproduced from Elefteriades (2008)

The specific genetics are being elucidated at the molecular level. Matrix metalloproteinase (MMP) overactivity participates in the destructive processes that degrade an aorta in individuals genetically preprogrammed to develop aneurysms. Most dissections are brought on via presumed momentary hypertensive crises by severe exertion or emotion (Hatzaras *et al.*, 2007).

It is interesting that aortic dissection has long been known to occur in circadian and diurnal patterns, with a preponderance of instances in the winter months and in the early morning hours. Acute aortic dissection also can occur in pregnancy, threatening both mother and fetus. The dissection usually occurs peripartum, during delivery or shortly thereafter. These are times when blood volume, cardiac output, blood pressure, and, of course, physical straining are high.

The prevalence of aortic aneurysmal disease has increased during the last three decades and is age- and gender-dependent (Bengtsson *et al.*, 1992). In this study, the prevalence among males increased rapidly after age 55 and reached a peak of 5.9 percent at age 80; that among females increased after age 70 and reached a peak of 4.5 percent above age 90.

In the proximal aorta, progressive dilatation to aneurysmal proportions is usually due to medial degeneration, caused by intrinsic weakness (Marfan or Marfan-related syndrome) or excessive wall stress (hypertension)

(Fig. 24.4) (Elefteriades, 2008). Conditions other than Marfan syndrome can cause aortic weakness (Schievink *et al.*, 1995; Devereux and Roman, 1999; Isselbacher, 2001; Pearson *et al.*, 2008; Biner *et al.*, 2009; Tadros *et al.*, 2009; Yetman and Graham, 2009) and predispose to progressive dilatation as previously discussed. Dilatation in elastic arteries proceeds with increasing age, as does dilatation of the normal aorta (see Chapter 18) (Stefanadis *et al.*, 1989a; Länne *et al.*, 1992a; Länne *et al.*, 1992b; Sonesson *et al.*, 1993; Sonesson *et al.*, 1994; Vasan *et al.*, 1995; Agmon *et al.*, 2003; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b; O'Rourke and Hashimoto, 2007) as one would expect on the basis of elastin fiber fatigue and fracture with cumulative cycles of pulsation. In the abdominal aorta, aneurysmal dilatation usually occurs distal to the renal arteries (Länne *et al.*, 1992a; Länne *et al.*, 1992b; Ernst, 1993) just above the bifurcation and is associated with, and usually attributed to, atherosclerosis and decreased elastin content (Davies, 1998; Zarins *et al.*, 2001; Lederle *et al.*, 2002) and prominent signs of diffuse inflammation of all layers of the vessel (intima, media, adventitia) (see Chapter 20) (Kuivaniemi *et al.*, 2008; Forsdahl *et al.*, 2009). Here too, dilatation occurs progressively with age and at a greater rate when the aorta becomes aneurysmal (Ures *et al.*, 1988; Jeremy *et al.*, 1994). In the study of Hirose *et al.* (1992), aneurysmal expansion averaged 4.2 mm/year for the thoracic aorta

and 2.8 mm/year for the abdominal aorta (Hirose *et al.*, 1992; Ernst, 1993; Brady *et al.*, 2004) and 3.3 mm/year in the UK study (UK Small Aneurysm Trial Participants, 1999) for abdominal aortic aneurysms. Surgery is warranted when dilatation exceeds 6 cm, so as to forestall aortic rupture (Davies, 1993; Ernst, 1993; Dake *et al.*, 1999; UK Small Aneurysm Trial Participants, 1999; Isselbacher, 2001; UK Small Aneurysm Trial Participants, 2002; Lederle *et al.*, 2002; Thompson, 2002; Hiratzka *et al.*, 2010). A more conservative approach is now being taken to management of aortic aneurysms, as it is to intracranial aneurysms (The International Study of Unruptured Intracranial Investigators, 1998; Caplan, 1998), but other factors including rate of expansion, level of arterial pressure, etc., also need be considered (see the *New England Journal of Medicine* correspondence and Lederle (2002), Powell *et al.* (2002), Brady *et al.* (2004), Song *et al.* (2007), and Isselbacher (2007)). Influence of hemodynamic factors in development and growth of intracranial aneurysms is described by Turner *et al.* (2001) and by Boussel *et al.* (2008).

Dissecting aneurysm of the aorta is partial disruption of the wall where blood tracks along the degenerate media, often over a considerable distance. The predisposition is aortic dilatation and medial degeneration (often to the stage of mucoid degeneration) as caused by Marfan syndrome, in other congenital defects, in long-standing hypertension, or with age. The initiation is usually a split in the intima, leading into the media. Spontaneous intramural hematoma of the aorta may proceed to dissection (von Kodolitsch *et al.*, 2003). The effect is further weakening of the aortic wall, predisposing to complete rupture, displacement of the aortic valve leading to valvular incompetence, and compression of the true lumen of the aorta and of its major branches. Aortic compliance is further impaired from narrowing of its true lumen, and systolic pressure is apt to be increased, favoring progression of dissection (Nakashima and Shiokawa, 1990; Nakashima *et al.*, 1990; Sutsch *et al.*, 1991; Isselbacher, 2001; Mehta *et al.*, 2002; Isselbacher, 2007).

Aortic dissection is a medical emergency, requiring prompt and aggressive reduction in disruptive forces within the aorta, complemented by surgery when the ascending aorta is involved (with the threat of aortic valve incompetence and rupture into the pericardial sac) (Wheat, 1980; Dake *et al.*, 1999; Vlahakes, 1999; Isselbacher, 2001; Nienaber and Eagle, 2003a; Nienaber and Eagle, 2003b; Isselbacher, 2007). Studies of aortic dissection have emphasized the importance of pulsatile phenomena and the rate of rise (dP/dt) of aortic pressure in the initiation and progression of dissection (Wheat, 1980; Isselbacher, 2001; Beller *et al.*, 2004). However, emphasis on shear stress at the aortic wall and on blood flow acceleration (dV/dt) appears to be inappropriate. Yin *et al.* (1989) found this (dV/dt) not to be reduced by β -blocking agents or by nitroprusside in patients with Marfan syndrome, although such therapy did reduce mean, pulsatile, and maximal tensile stress in the wall.

Medical treatment entails the use of anti-hypertensive therapies, initially administered intravenously, and monitored invasively, to reduce peak aortic systolic pressure and to prevent extension of dissection, in the expectation that blood will clot in the false lumen and that fibrous repair will occur. A combination of vasodilator and β -blocking agents is usually recommended, the former to reduce peripheral resistance and wave reflections, the latter in the hope of reducing maximal flow acceleration and maximum dP/dt in the aorta, as well as heart rate (see above). Sometimes re-entry occurs in the aorta, with a second intimal tear, and a 'double barrel' aorta is created. Surgical treatment entails repair of the intimal split and of secondary damage. Often, replacement of the whole ascending aorta and the aortic valve is necessary (Mehta *et al.*, 2002). Surgery to the descending aorta when required, as by progressive enlargement of the false aneurysm, especially beyond 22 mm in diameter (Isselbacher, 2007; Song *et al.*, 2007) may be undertaken with endovascular stent-grafts (Vlahakes, 1999; Chuter, 2009). Enlargement of an aneurysm, once developed, tends to be progressive, be this a thoracic or abdominal aneurysm, and caused by age, trauma, hypertension, or Marfan syndrome. All carry the risk of rupture; this is greater when the aneurysm is large or risk factors are uncontrolled. Surgical treatment is best undertaken with endovascular repair (Mani *et al.*, 2009; Sicard, 2009). Endovascular repair has become the favored treatment of aortic rupture caused by sudden deceleration as in motor vehicle accidents (Neschis *et al.*, 2008). Late development of aneurysm at the site of repair can occur months or years later, and is managed in the same way as after aortic dissection (Song *et al.*, 2007).

Spontaneous dissection can occur, although rarely, in the carotid and vertebral arteries (Schievink, 2001). Such a condition is virtually unknown in muscular arteries. The underlying problem is usually medial degeneration, as in the aorta, and precipitated by an intimal tear. Spontaneous dissection usually causes arterial obstruction, and is not uncommon as a cause of ischemic stroke in young people. Such people have high (and apparently inhomogeneous) carotid wall stiffness (Calvet *et al.*, 2004). Coronary artery dissection is a well-known hazard of coronary angiography.

AORTIC COARCTATION

In simple postductal aortic coarctation, the descending thoracic aorta is narrowed or occluded immediately beyond the origin of the left subclavian artery, and blood flow is channeled through large collateral branches of the subclavian and intercostal arteries into the aorta beyond this obstruction. Although most obviously an obstructive lesion, the clinical features of aortic coarctation are primarily a consequence of altered arterial hemodynamics in the upper part of the body, rather than of reduced blood

flow to the lower part. It has been standard practice in the past to relate these hemodynamic abnormalities in the upper part of the body to 'hypertension', although it is recognized that this condition is more sinister in its prognosis than is essential hypertension of equal severity (Pickering, 1968; Wood, 1968; Campbell, 1970).

Aortic coarctation is best considered in terms of its effects on conduit and cushioning function of the arterial system, through separate consideration of steady pressure and flow phenomena, and of pulsatile pressure and flow phenomena. When this is done, a relatively simple picture emerges that explains changes in arterial pressure on each side of the coarctation, and that provides a logical view of the mechanisms determining complications, and the principles of optimal management.

Hemodynamic alterations

Figure 24.5 shows changes in arterial blood pressure in the ascending aorta and femoral artery when, under acute conditions, the proximal descending aorta is progressively narrowed. These data, taken from Gupta and Wiggers (1951), show a fall in mean arterial pressure and in pulse pressure beyond the obstruction when the aorta is progressively narrowed. This is, however, not a good simulation of the clinical condition, where blood flow to the lower part of the body is maintained at near-normal levels (Lewis, 1933; Patterson *et al.*, 1957; Taylor and Donald, 1960; Frydrychowicz *et al.*, 2007; Hom *et al.*, 2008) at rest and during exercise through many large dilated collateral vessels. Since it enables precise hemodynamic characterization, velocity-encoded cine magnetic resonance imaging (MRI) is used for evaluating the severity of aortic coarctation and the qualitative assessment of the presence and direction of collateral vessels. It also allows accurate quantitation of key hemodynamic parameters, such as flow velocity, flow volume, and pressure gradients across the coarctation – functional information that is clinically useful for both preoperative planning and postinterventional monitoring (Hom *et al.*, 2008). Because of the presence of these collateral vessels, mean blood pressure below the coarctation (even when the aorta is totally obstructed) is far greater than that seen under acute conditions and is usually within normal limits (Fig. 24.6) (Bing *et al.*, 1948; Fuller *et al.*, 1952; Pickering, 1968; Hom *et al.*, 2008). Therefore, the arm–leg pressure gradient is an unreliable indicator of the severity of hemodynamic obstruction in aortic coarctation. While mean blood pressure is maintained at or near normal levels, pulse pressure is markedly reduced below the coarctation, as a result of attenuation of the pulse in travel through the long, tortuous and narrow collateral vessels, and the aorta when still patent. The influence of abundant collateral vessels in aortic coarctation on mean pressure and on pulse pressure is illustrated diagrammatically in the right panel of Fig. 24.5. The presence of collateral vessels

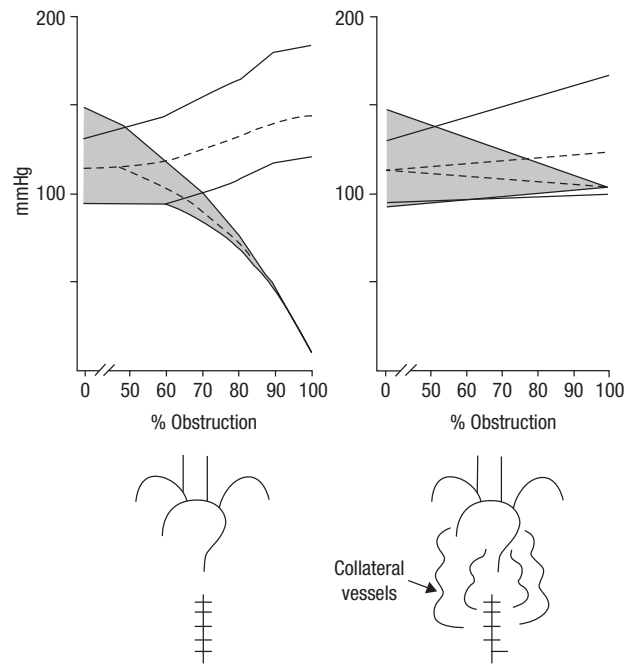


Figure 24.5 The effects of graded occlusion of the descending thoracic aorta on systolic, diastolic, and mean blood pressure immediately above and below (shaded) the obstruction. —, Mean pressure; left, no collateral vessels; right, plentiful collaterals. Reproduced from O'Rourke (1982b)

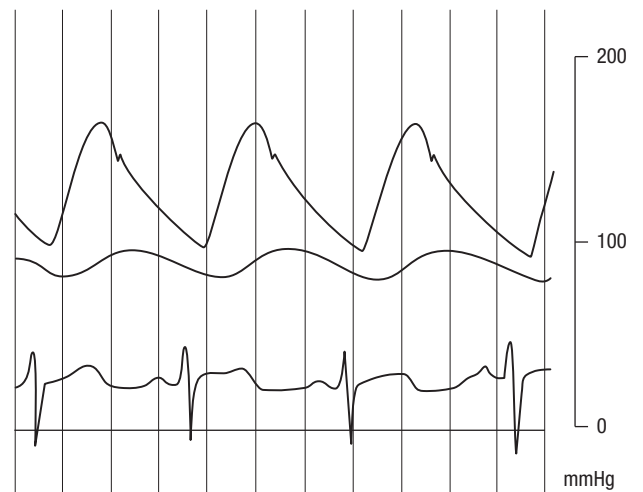


Figure 24.6 Pressure waves recorded simultaneously in the aortic arch (above) and lower thoracic aorta (below) in a nine-year-old boy with uncomplicated post-ductal aortic coarctation. Reproduced from O'Rourke and Cartmill (1971)

increases blood flow and causes mean pressure to fall only slightly in arteries below the coarctation and to be only slightly greater in the arteries above (Hom *et al.*, 2008). The collaterals, however, have a great effect on pulsatile phenomena, so that pulsatile pressure is markedly attenuated below the coarctation (in the same way as when

this is induced acutely) and augmented above. In terms of conduit and cushioning function of the arterial system, one can say that at rest, the collateral vessels provide almost complete compensation for abnormal aortic conduit function, but little or no compensation for altered cushioning function.

Figure 24.5 illustrates the changes in mean and pulsatile pressure that would be expected in the presence of abundant collateral vessels. Fig. 24.7 shows the pressure waves one would expect to record above and below the coarctation in patients with different degrees of collateral development. The adequacy of collaterals determines the extent of the fall in mean pressure, while the presence of coarctation and of collateral vessels appears to reduce pulse pressure to much the same extent in all; such an explanation accounts easily for the change in mean level and amplitude of the pressure wave below the coarctation, and thus for the systolic and diastolic pressures. The explanation is not so obvious when one considers systolic and diastolic pressures only and seeks to account separately for the differences in systolic pressure above and below the coarctation and of diastolic pressure above and below. It is clear that, in the presence of abundant collaterals, diastolic pressure may be higher below the coarctation than above because of increased collateral flow (Fig. 24.7). In the past, this occasional finding has been a source of confusion to clinicians, shackled as they are (or were) to the sphygmomanometer and to the notion that diastolic pressure is the sole indicator of arterial

hypertension. The finding of a higher than normal diastolic pressure below the coarctation was the source of the theory that there is generalized arteriolar vasoconstriction in aortic coarctation (Steele, 1941) and is still frequently quoted to support the theory that a renal mechanism is partly responsible for hypertension in this condition (Goldblatt and Kahn, 1938; Habib and Nanson, 1968; Pickering, 1968).

While there is some evidence that hypertension in coarctation is partly based on a renal mechanism, and that such a mechanism may be responsible for the frequency of unexplained residual hypertension following surgical correction, this has been disputed (Patterson *et al.*, 1957; Pickering, 1968) and in any case is not the major issue in determining the complications of coarctation. Preoccupation with diastolic pressure, hypertension and rival theories of hypertension in aortic coarctation (Goldblatt and Kahn, 1938; Steele, 1941; Bing *et al.*, 1948; Gupta and Wiggers, 1951; Pickering, 1968) appears to have diverted attention from the real problem – that only one-third of the whole arterial system is able to act as the cushion to intermittent left ventricular ejection (Fig. 24.8) (Taylor and Donald, 1960; O'Rourke and Cartmill, 1971).

Pressure pulse contour

When the proximal descending aorta is acutely occluded, the contour of the ascending aortic pressure pulse is considerably altered, with an increase in pressure during the latter part of systole and a disappearance of the diastolic pressure wave (Gupta and Wiggers, 1951; Nichols and Pepine, 1992). The pressure wave in acute aortic obstruction resembles that seen in hypertension (see Chapter 20) and arterial degeneration (see Chapter 19) and can be explained on the same basis – early return of wave reflection summing with pressure during systole and so augmenting the systolic peak. In contrast to hypertension and arterial degeneration, however, early wave reflection arises from an abnormal site (the coarctation itself) close to the heart, whereas in the other two conditions, reflected waves return early and predominantly from normally placed sites (the peripheral arterioles), because of increased pulse wave velocity (Fig. 24.8).

Because of the absence of collateral vessels, acute aortic obstruction is not strictly comparable with the clinical condition of aortic coarctation. Nonetheless, the proximal aortic pressure pulse wave in chronic experimental coarctation (Fig. 24.9) and in patients with coarctation (see Fig. 24.6) is exactly the same as in acute aortic constriction. The only difference is that mean pressure downstream is higher than when the aorta is acutely obstructed. These findings support the view that collateral vessels restore steady flow and thus maintain adequate conduit function, while the coarctation, and to a lesser degree the collaterals, are responsible for alterations in pulsatile phenomena in upper body arteries.

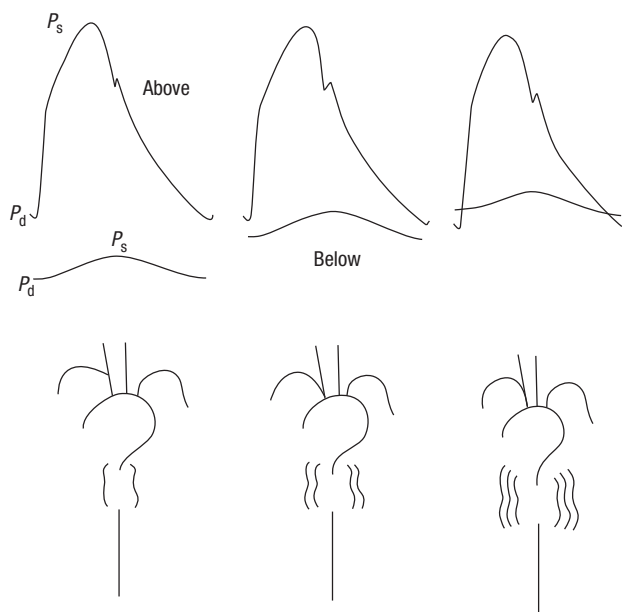


Figure 24.7 Diagrammatic representation of different degrees of collateral development on arterial pressure waves above and below an obstruction in the descending thoracic aorta. When collaterals are poorly developed, both systolic P_s and diastolic P_d pressures are lower beyond the obstruction. When collaterals are well developed, diastolic pressure may be higher beyond the obstruction. Reproduced from O'Rourke (1982b)

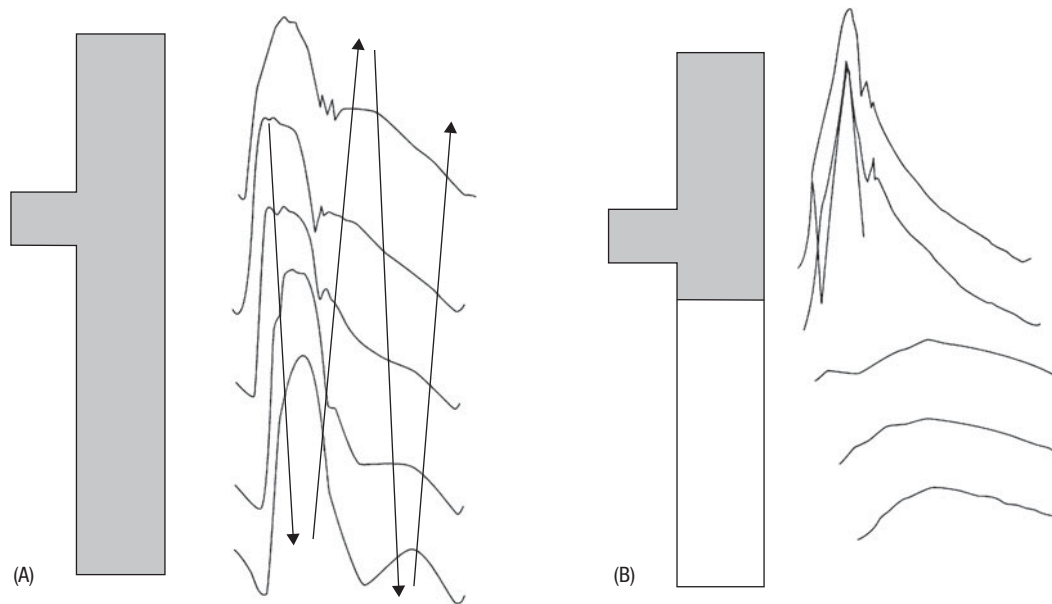


Figure 24.8 Model of the normal arterial system (A) and the arterial system in aortic coarctation (B), together with pressure waves recorded in a normal dog (A) and in a dog with aortic coarctation (B) at points corresponding to their position in the respective model. Superimposed on the waves are lines with arrows, which show the impulse generated by ventricular ejection passing back and forth over the arterial system at the velocity of the pulse (approximately 6.0 m/s). It is shown that the contour of the waves is attributable to summation of the incident and reflected components. Reproduced from O'Rourke and Cartmill (1971)

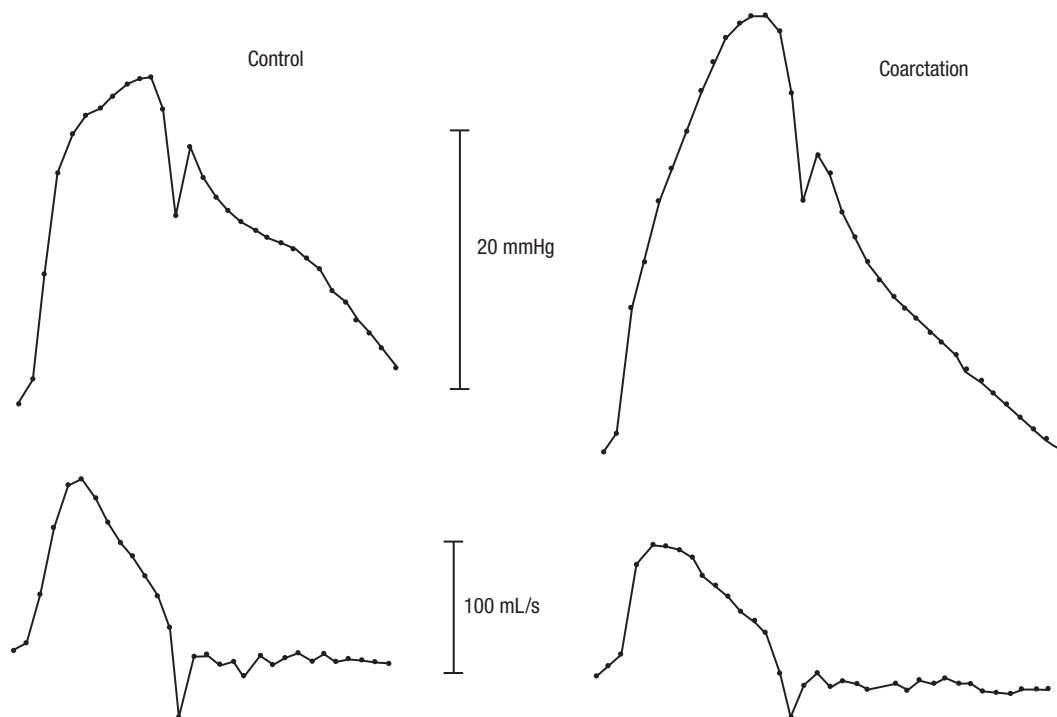


Figure 24.9 Pressure waves (above) and flow waves (below) recorded simultaneously in the ascending aorta of a normal dog (left) and in a littermate with chronic surgically induced aortic coarctation (right). Reproduced from O'Rourke and Cartmill (1971)

The pressure wave in Fig. 24.6 is typical of the pressure waves recorded in the ascending aorta and aortic arch of patients (including children) with aortic coarctation. O'Rourke and Cartmill (1971) found evidence of a

diastolic pressure wave in only one of 27 young patients with aortic coarctation. Young adults with no coarctation almost always exhibit a diastolic pressure wave in the proximal aorta (see Chapter 19) (O'Rourke *et al.*, 1968;

O'Rourke *et al.*, 1992a). Absence of a diastolic pressure wave in the ascending aorta or in the peripheral arteries of older children and young adults is quite abnormal and is suggestive of coarctation, hypertension, or premature arterial degeneration.

Change in contour of the arterial pressure wave in aortic coarctation can be explained on the basis of early return of wave reflections, with summation of primary and subsequent echoes with the incident wave (see Fig. 24.8). Alternatively, one can look at these phenomena in terms of smaller volume distensibility of the arterial system and by likening this to a Windkessel of reduced dimensions. The two explanations amount to the same thing, although the former is more precise and comprehensive.

Vascular impedance

Figure 24.10 shows ascending aortic impedance in three control dogs (Fig. 24.10A) and in their littermates in whom aortic coarctation had been induced three months previously. The first minimum of impedance modulus has been attributed to wave reflections from the lower part of the body (see Chapter 12) (O'Rourke and Taylor, 1967). This minimum is, predictably, lost in the dogs and rats with coarctation, so that the impedance modulus is considerably higher than under normal conditions (O'Rourke and Cartmill, 1971; Kobayashi *et al.*, 1996) and leads to left ventricular hypertrophy. Alteration of impedance spectra in coarctation is responsible for the higher amplitude and altered contour of the ascending aortic and left ventricular pressure waves, and for the mismatch

between aortic impedance and ventricular ejection. Figure 24.11 shows impedance spectra in two of the dogs studied by O'Rourke and Cartmill (1971) together with moduli of the first five or six harmonics of the ventricular ejection wave.

The favorable relationship between impedance minima and harmonic content of the left ventricular ejection wave (see Chapter 13) is lost in dogs and rats with aortic coarctation. No data on ascending aortic impedance in patients with aortic coarctation are presently available, but the ventricular ejection wave and ascending aortic pressure wave are known to be similar to those in dogs with chronically induced aortic coarctation (Porjé, 1967; O'Rourke and Cartmill, 1971). One would expect ascending aortic impedance spectra in humans with coarctation to show the same features as those in dogs.

Changes of arterial pressure with exercise

Murgo *et al.* (1981b) have studied the effects of exercise on arterial hemodynamics in the ascending aorta of patients with apparently normal vascular trees. They have shown that mild supine exercise causes no significant alteration in the frequency-dependent components of impedance, even though it leads to a reduction in the resistive term. There is no reason to believe that ascending aortic impedance in aortic coarctation behaves differently. Murgo *et al.* found that the first five harmonics of the ventricular ejection wave were approximately doubled in mild supine exercise when cardiac output was doubled, and that this led to a similar increase in aortic pressure

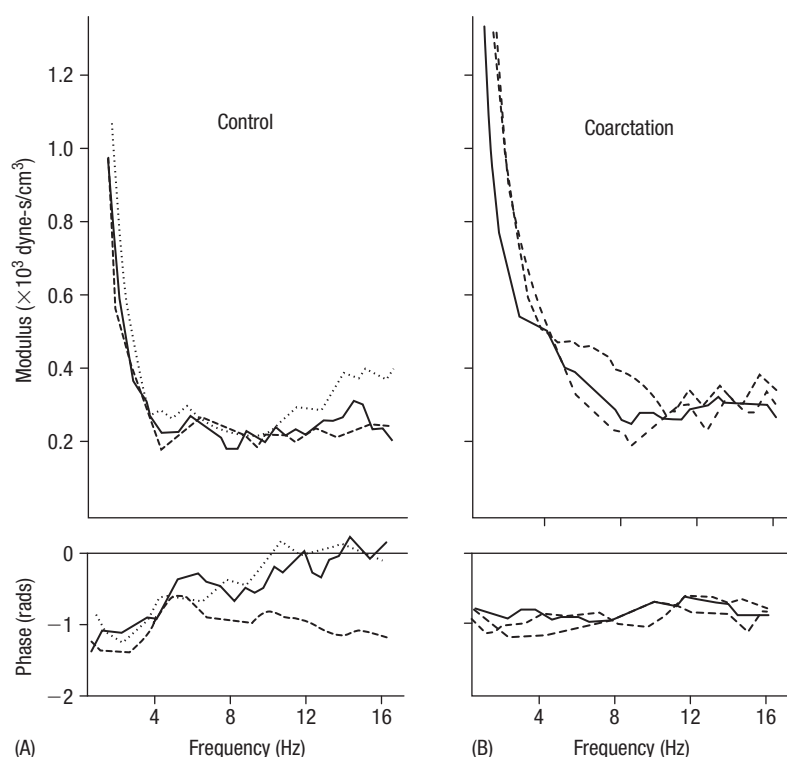


Figure 24.10 Input impedance to the systemic circulation in three normal dogs (A) and in three littermates with chronic aortic coarctation (B). Reproduced from O'Rourke and Cartmill (1971)

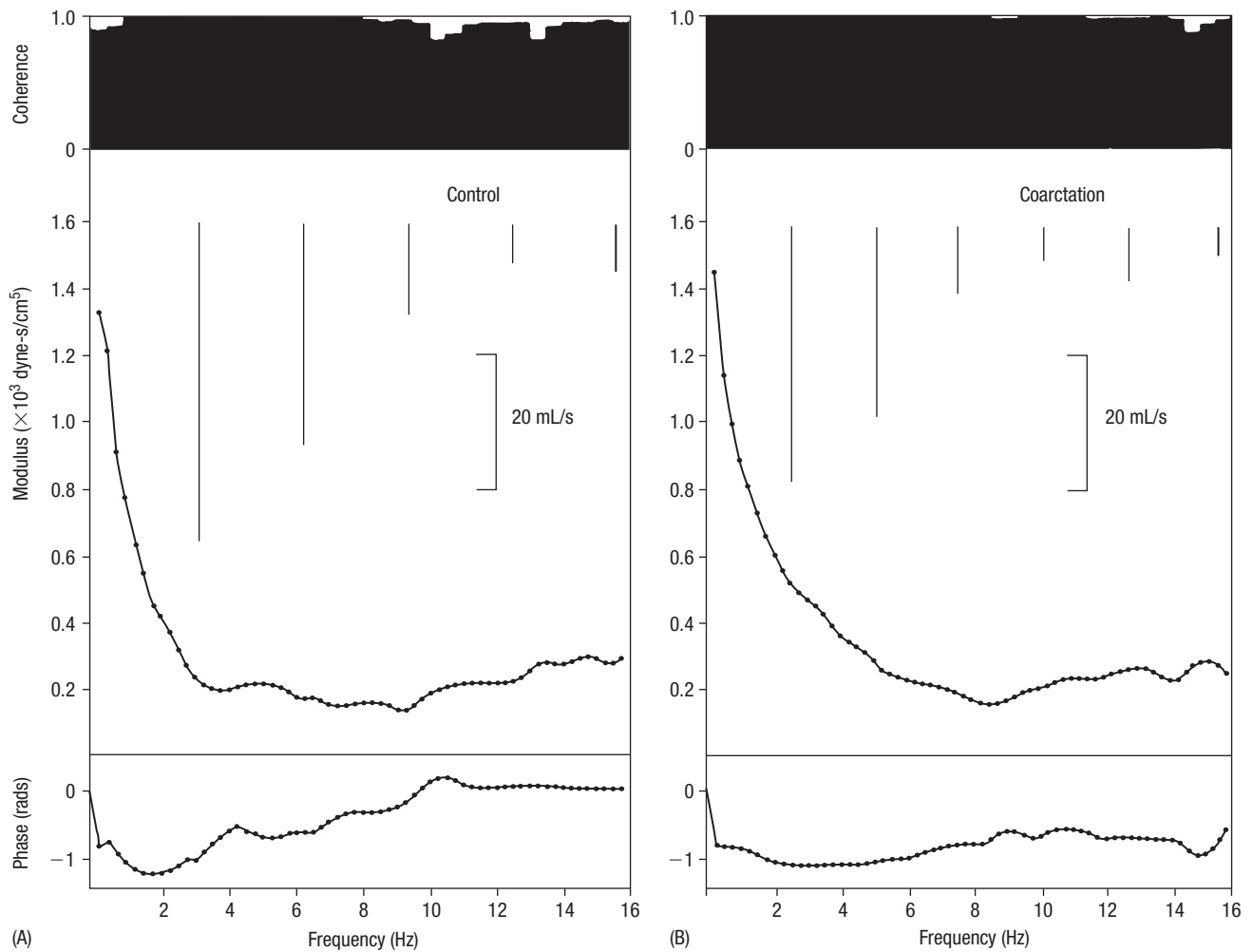


Figure 24.11 Input impedance to the systemic circulation (ascending aortic impedance) in (A) a normal dog and (B) a littermate with chronic, surgically induced aortic coarctation. The bars indicate the amplitude of the first five or six harmonics of the ascending aortic flow wave. Reproduced from O'Rourke and Cartmill (1971)

moduli and thus near doubling of the aortic pressure pulse. If one applies the same considerations to the nine-year-old boy whose pressure wave is shown in Fig. 24.6, and if one allows the same increase in mean pressure as found by Murgo *et al.* in normal subjects of around 14 mmHg (which probably underestimates the rise in coarctation), one can calculate that, with mild supine exercise, this child would generate a pressure wave of amplitude 135 mmHg and a peak systolic pressure of over 230 mmHg in the ascending aorta. Strenuous exercise would be expected to cause correspondingly greater pressure fluctuations and peaks.

Taylor and Donald (1960) drew attention to the dramatic increases in brachial artery pressure that occur during exercise in patients with aortic coarctation, and showed that these are far greater than seen in patients with severe essential hypertension who would be considered to require urgent medical therapy and restriction of activity (Fig. 24.12). These authors also pointed out that such changes in pressure usually occur without symptoms

(although they can induce cardiac failure), with moderate exertion only and in patients, otherwise asymptomatic who usually have had no advice to reduce their activities. These pressure increases are precisely what one would expect from a reduction in cushioning capacity of the arterial system. Their potentially damaging effects on the heart and arteries were stressed by Taylor and Donald and by others (O'Rourke and Cartmill, 1971; Maron *et al.*, 1973; Nanton and Olley, 1976; Cohen *et al.*, 1989), and form the background to the American Heart Association (AHA) Scientific Statement on exercise in the young patient with congenital cardiac disease (Maron *et al.*, 2004).

Pathophysiology

In aortic coarctation, conduit function of the arterial system is preserved, while cushioning function is markedly disturbed, with unusually high-pressure fluctuations

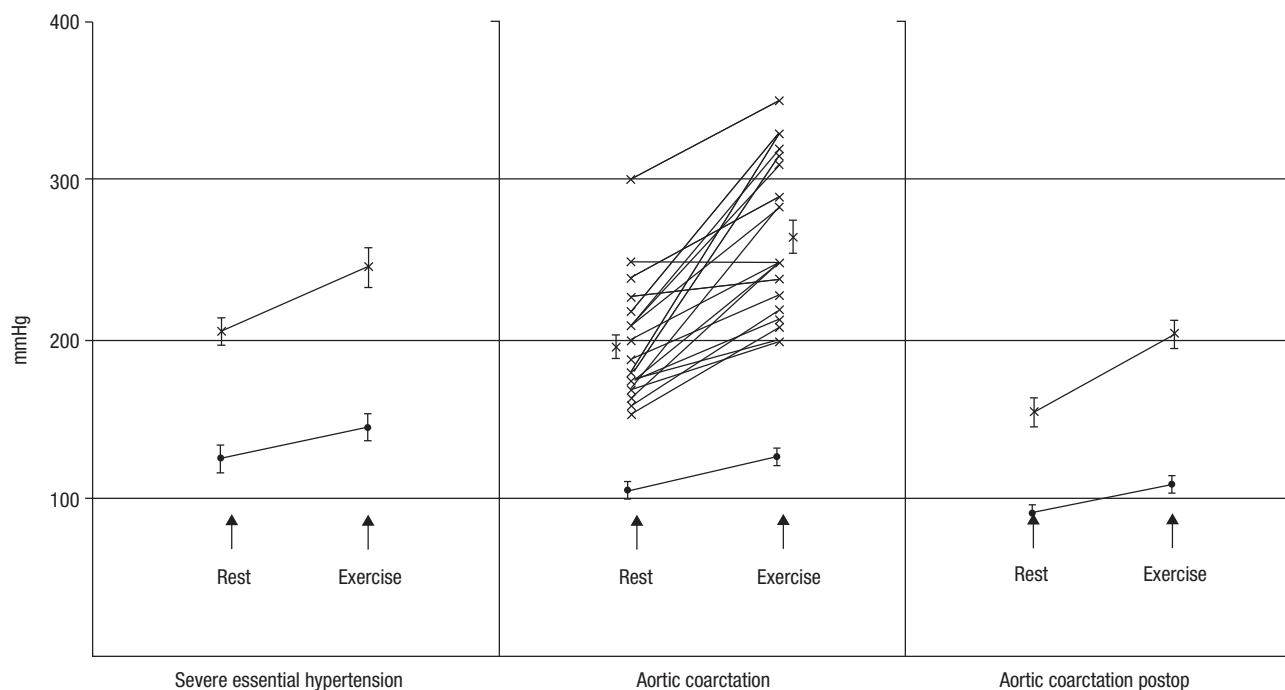


Figure 24.12 Systolic and diastolic pressures in the brachial artery at rest and during exercise of similar intensity in a group of patients with severe essential hypertension (left), patients with aortic coarctation (center), and patients with surgically corrected aortic coarctation (right). Individual values are shown for systolic pressure in all patients with coarctation. Other values are mean $\pm$ 1 SEM. Figure prepared from data published by Taylor and Donald (1960). Reproduced from O'Rourke (1982b)

being generated in the proximal aorta and upper body arteries by normal left ventricular ejection. The most apparent pathologic features of this condition (apart from the coarctation itself) are cardiac hypertrophy and pronounced degenerative changes in the proximal aorta and upper body arteries (Kobayashi *et al.*, 1996; Ou *et al.*, 2008a; Ou *et al.*, 2008b). These degenerative changes include thinning, splitting, and fragmentation of elastic fibers and laminae, mucoid degeneration of the aortic media and the development of aneurysms in the circle of Willis and within the brain substance itself (Boussel *et al.*, 2008). These pathologic features are similar to those seen with aging and in hypertension (Glynn, 1940; Davies and Fisher, 1943; Reifstein *et al.*, 1947; Blumenthal *et al.*, 1954; Pickering, 1968; Stehbens, 1979), but occur at a much earlier age, and are attributable – as are those of aging (see Chapter 19) and hypertension (see Chapter 20) – to fatigue and damage of structural components of the arterial wall as a result of repeated high pulsatile stresses over a long period of time. These pathologic changes in the heart and arteries are the basis of the lethal complications of aortic coarctation – subarachnoid hemorrhage, cerebral hemorrhage, aortic rupture, and cardiac failure; abnormally high stresses on the (often bicuspid) aortic valves and at the coarctation itself probably also contribute to the development of endocarditis and endarteritis at these sites. These complications of aortic coarctation are the most common causes of death, which in ‘simple’ uncorrected aortic coarctation usually occurred

before age 40 (Fig. 24.13) (Wood, 1968; Campbell, 1970). Onset of these complications was often abrupt in a patient previously asymptomatic, but autopsy almost invariably showed the presence of gross arterial degeneration and cardiac hypertrophy (Pickering, 1968; Stehbens, 1979).

Aortic coarctation is a sinister condition with an ominous prognosis. Patients usually appear quite normal until disaster strikes, and the main manifestation of the disorder is elevation of blood pressure in the upper part of the body. Since, in the past, the ill effects of coarctation were deemed to result from ‘hypertension’, and ‘hypertension’ was gauged on the basis of diastolic blood pressure level (which is less altered in aortic coarctation than is systolic or pulse pressure), operation was often deferred while the arterial system was literally being ‘ripped to pieces’. Misunderstanding of the pathophysiology of this condition is evident in textbooks published as recently as 1968 when Paul Wood stated that ‘surgical repair should only be advised when the blood pressure is at least moderately elevated (a moderate elevation being defined by Wood as diastolic pressure 110–125 mmHg or systolic pressure 180–230 mmHg) since nothing is gained by restoring the course of the blood flow *ipso facto*’. The presence of aortic coarctation is now generally agreed as an indication for surgery (Keith, 1978; Kirklin and Pacifico, 1978; Cohen *et al.*, 1989), or stenting (Hamdan *et al.*, 2001) and the only point of contention seems to be how early in childhood this should be performed (Cohen *et al.*, 1989; McCrindle, 1999; O’Sullivan *et al.*, 2002;

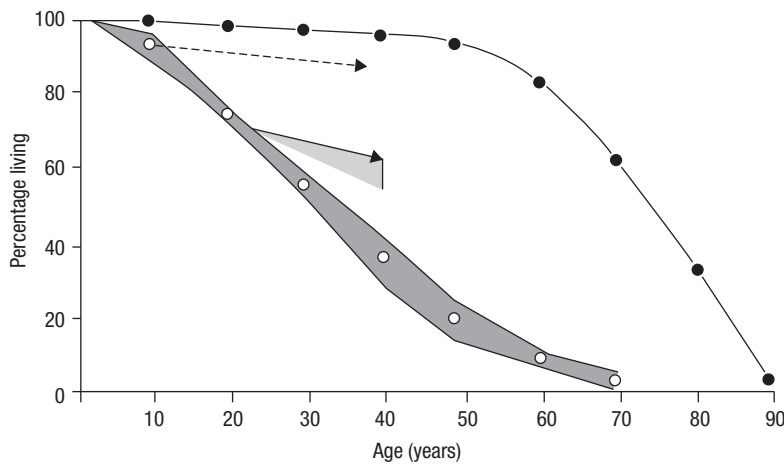


Figure 24.13 Life tables in normal subjects (solid line, closed circles) and in patients with aortic coarctation (open circles, stippled area), after Campbell (1970). The solid line with an arrow indicates improvement in survival after surgical correction of coarctation, as described by Maron *et al.* (1973). The shaded area below this line indicates the possible lower survival if patients lost to follow up in this survey had died. The dashed line with an arrow indicates the hopefully better prognosis in patients operated on electively at an earlier age. Reproduced from O'Rourke (1982b)

Celermajer and Greaves, 2002). Improvement in operative mortality, and appreciation that degenerative arterial lesions are a consequence of long-standing high arterial stresses, has led to a progressive decrease in the recommended age at which the defect is to be repaired. In addition, it is now appreciated that the residual hypertension after surgical repair and the late complications after surgery depend on the age at which this is performed (Keith, 1978; Clarkson *et al.*, 1983; Cohen *et al.*, 1989; O'Sullivan *et al.*, 2002). From the pathophysiologic viewpoint alone, one would think that the earlier corrective surgery is performed the better. The arguments presented here extend Keith's recommendation 32 years ago that elective surgery be undertaken at between four and seven years of age. Celermajer and Greaves (2002) have pointed to evidence of persistent arterial damage and dysfunction, even when repair is undertaken in infancy, and have suggested that coarctation may be just one manifestation of a generalized arterial defect.

Effects of surgery in aortic coarctation

Repair of aortic coarctation involves resection and end-to-end anastomosis, or resection with patch or conduit grafting. The operation carries an appreciable risk, even now, because of the abundant collateral vessels in the chest wall and about the anastomosis site, and because of the friable nature of the degenerate aorta when the condition has persisted into adolescence. The postoperative course of this condition is sometimes characterized by two interesting complications. The first of these is reactive hypertension, which is probably due to altered baroreceptor activity (March *et al.*, 1960; Keith, 1978) and frequently requires hypotensive therapy for several days. The second, rarely seen now that blood pressure is well-controlled postoperatively, is acute arterial and arteriolar damage and necrosis, which are attributable to excessive stresses on and fracture of the elastic fibers, laminae, and muscle fibers in atrophic lower body arteries and

arterioles (Byrom, 1954; Pickering, 1968; Keith, 1978). The relationship of this condition to fibrinoid necrosis in malignant hypertension is discussed in Chapter 20. Like the arterial degeneration in upper body arteries, this condition is attributable to excessive and damaging stresses on the arterial wall, but in this case applied acutely and in vessels structurally weak and so more easily damaged, even by normal or only slightly elevated pressures.

Repair of aortic coarctation is usually curative, with a reduction of arterial pressure in the upper body (Fig. 24.14), return of normal femoral pulses, regression of collateral vessels, and decrease or disappearance of electrocardiographic evidence of left ventricular hypertrophy (Taylor and Donald, 1960; Simon and Zloto, 1974; Keith, 1978; Clarkson *et al.*, 1983; Cohen *et al.*, 1989). However, residual problems often remain (Celermajer and Greaves, 2002; O'Sullivan *et al.*, 2002), especially in patients undergoing surgery after childhood. These problems are largely attributable to aortic narrowing at the site of the original coarctation due to problems at surgery or to growth of the child with non-growth of the anastomosis, and to persistence of degenerative changes caused by the coarctation before operative repair. Systemic hypertension (systolic rather than diastolic) is more common than in the normal population, and was found by Nanton and Olley (1976) in 24 percent of patients operated on at the Toronto Hospital for Sick Children, and by Maron *et al.* (1973) in approximately 40 percent of patients undergoing surgery (usually at an older age) at the Johns Hopkins Hospital. An unusually high increase in systolic pressure during exercise was noted by Taylor and Donald (1960) (see Fig. 24.12), and by James and Kaplan (1974), even in patients whose resting arterial pressures were normal. In their detailed long-term follow up, Maron *et al.* (1973) noted a very high incidence of premature cardiovascular disease and cardiovascular death in patients who had undergone aortic coarctectomy. Abnormalities of the precoarctation vasculature are known to occur after coarctation of the aorta repair and

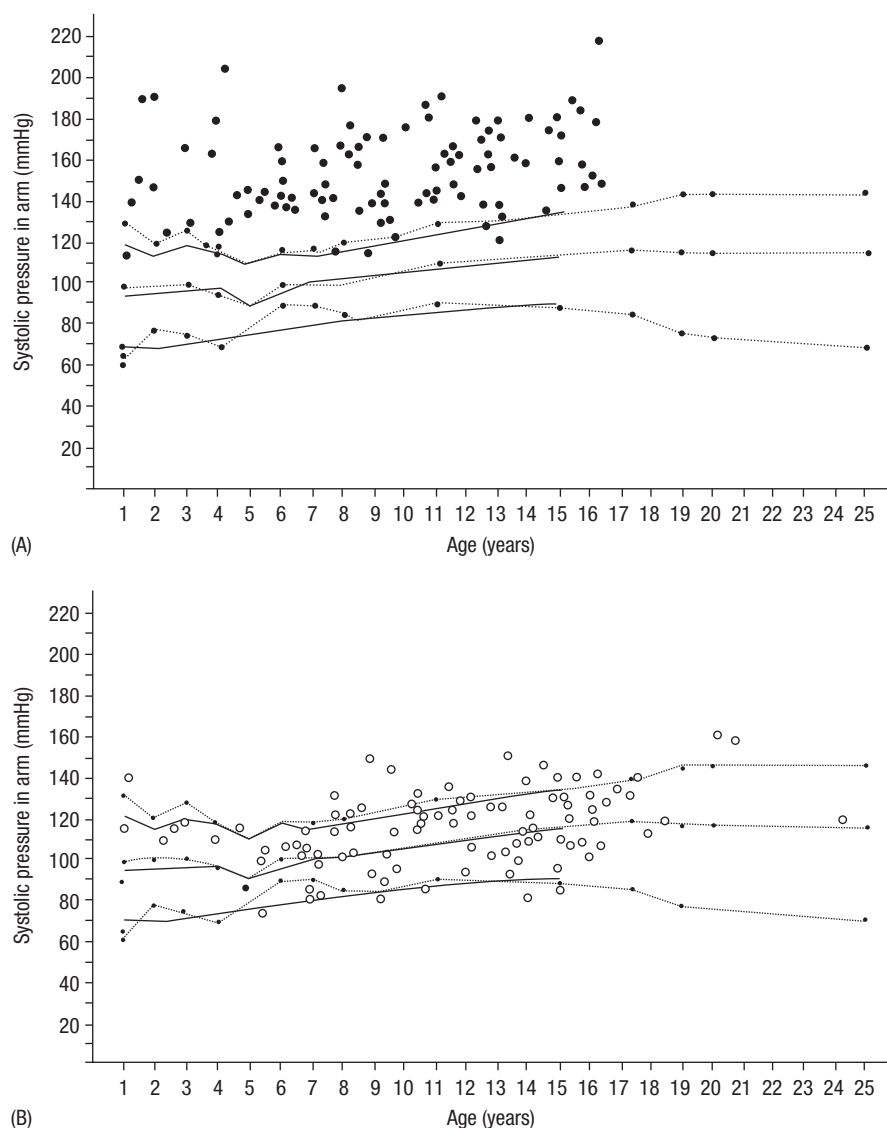


Figure 24.14 Systolic pressures in the arm before surgery (A) and at the last follow up (B) in 95 young patients after correction of aortic coarctation. The mean $\pm$ standard deviation lines represent values from a normal population. Eighty percent of patients after operation have pressures within the normal limits. Reproduced from Keith (1978)

appear related to adverse outcomes (Ou *et al.*, 2007; Ou *et al.*, 2008a; Ou *et al.*, 2008b; Swan *et al.*, 2010). Carotid artery intima-media thickness (IMT), aortic pulse velocity, stiffness index, tension–time index, pulse pressure, and wave reflections are all increased, while sub-endothelial viability index and brachial artery flow-mediated dilation (FMD) are reduced.

It is likely that with use of endovascular stents (Hamdan *et al.*, 2001), with improved surgical techniques, early diagnosis, and surgery, more careful postoperative surveillance and with reoperation where necessary, there will be further improvement in the results of corrective surgery for aortic coarctation. Since changes in attitudes to surgery have only come about in the last 25–30 years, such long-term benefits, if present, may not become evident for some years to come.

Clinical implications in management of aortic coarctation

Attention has been directed here to the abnormal cushioning function of the arterial system in aortic coarctation as the cause of asymptomatic arterial degeneration and cardiac hypertrophy, and of early death from the effects of these. It has been pointed out that abnormal cushioning function may accompany quite modest gradients (<10 mmHg) of mean pressure between proximal and distal aorta, and that diastolic pressure in upper body arteries and the arm–leg pressure gradient are poor guides to the significance of coarctation. The importance of assessing pressure change during exercise has been stressed, as has the importance of restricting exertion when pressure changes are great. Early operative surgical correction (or

stenting) of aortic coarctation has been advised (even if reoperation has to be undertaken during adolescence) in order to prevent the relentless progression of arterial degeneration which is the basis of early death in this condition, and of late complications, even after corrective surgery.

ARTERIOVENOUS SHUNT

Creation of a proximal arteriovenous shunt in experimental animals has very little effect on the frequency-dependent components of ascending aortic impedance or on the contour of the aortic pressure wave, even though it may cause a considerable decrease in resistance and an increase in cardiac output (Mang, 1976). In humans, arteriovenous shunts are well tolerated by the heart, showing the extreme tolerance this organ possesses in dealing with a volume load, in contrast to a pressure load (see Chapter 14). Small arteriovenous shunts (as in atrial or ventricular septal defect) do not require surgery, except when they pose a threat of a jet lesion and of infective endocarditis (in ventricular septal defect). Shunts are actually created artificially in the arm for convenient dialysis in patients with end-stage renal disease (Savage *et al.*, 2002b).

TAKAYASU ARTERITIS

This is a chronic occlusive disease affecting the aorta and its major branches, and caused initially by acute inflammatory arteritis (Sharma *et al.*, 1992; Mandell and Hoffman, 2001). It is most common in Asian populations and in women. It may present, as does aortic coarctation, with asymptomatic elevation of arterial pressure in the upper part of the body, or become apparent through detection of a low-amplitude or absent pulse in an arm. Symptoms may arise through disturbance of arterial conduit function – ischemia of a limb or organ, or disturbance of cushioning function with early wave reflections as in aortic coarctation, elevated central aortic and LV pressure, leading on to cardiac decompensation. Medical treatment of the chronic stage depends on the specific defect; surgical treatment depends on the anatomic defect and the possibility of and need for correction. The same principles apply in the chronic obstructive stage of the disease as in aortic coarctation.

NON-ATHEROMATOUS CORONARY ARTERY DISEASE

Virtually all structural coronary occlusive disease is caused by atherosclerosis. This is dealt with in Chapters 17 and 23. Rare causes (Kawai, 1995) are congenital anomalies (Becker, 1995), Kawasaki disease (Kato *et al.*, 1995;

Mandell and Hoffman, 2001), and connective tissue disease (Gonzalez, 1995). In heart transplant recipients, proliferative concentric intimal hyperplasia often occurs and may be a manifestation of chronic rejection (Schroeder and Gao, 1995; Taylor *et al.*, 2006). This is a common cause of graft failure in cardiac transplant recipients. Since the heart is denervated, such patients do not experience anginal pain from this problem, and it may present as cardiac failure or arrhythmia. In cocaine abusers, coronary atherosclerosis appears to be accelerated and may occur at an early age. Cocaine causes other problems in the heart attributable to coronary spasm or catecholamine sensitivity, with focal areas of myocardial damage and infarction (Karch and Billingham, 1995; Kloner and Rezkalla, 2003). Consequences are anginal pain, arrhythmias, or cardiac failure. Restenosis following coronary angioplasty is usually a consequence of intimal cell proliferation (Gould, 1992; Rensing *et al.*, 1992). It was hoped that stent restenosis (which was usually gradual and clinically silent) would be overcome by the introduction of drug-eluting stents. Hyperplasia was overcome but was replaced by a greater and more serious complication of late stent thrombosis, occurring up to three years after stent deployment. This is attributable to failure of stents to re-endothelialize because of slow drug elution, and warrants continuation of dual antiplatelet therapy for 12–24 months (Wright *et al.*, 2011). Coated stents are gradually being improved to overcome this problem.

Other less frequently occurring diseases, such as Fabry disease (Barbey *et al.*, 2006; Kalliokoski *et al.*, 2006), Adamantiades-Behçet disease (Ikonomidis *et al.*, 2006; Protogerou *et al.*, 2006) and β -thalassemia major (Cheung *et al.*, 2002) alter arterial stiffness, wave reflections, and peripheral endothelial function.

DISTURBANCES OF SMOOTH MUSCLE TONE AND CALIBER: SPECIFIC DRUG EFFECTS

This subject, as it pertains to arterial wall stiffness, is dealt with in Chapters 4 and 26, and, as it is related to hypertension and atherosclerosis, in Chapters 20 and 23. Cyclic changes of smooth muscle tone have been demonstrated in peripheral arteries and are usually of no clinical consequence, although they contribute to variability in the measurement of mechanical properties (Hayoz *et al.*, 1995).

Disturbances in endothelial function and smooth muscle tone are the cause of clinical problems in a number of conditions including variant (Prinzmetal) angina, migraine, and Raynaud's phenomenon, and of toxicity to different drugs, including cocaine and ergot. Such disturbances may also play a major role in generalized disease, including diabetes mellitus (Chapter 25).

In Prinzmetal or variant angina, an epicardial coronary artery narrows to complete closure due to spasm, with

resultant pain and electrocardiographic evidence of transmural myocardial ischemia (Prinzmetal *et al.*, 1959; Maseri *et al.*, 1978; Maseri, 1991; Bogaty *et al.*, 1994; Kawano and Ogawa, 2005). Arrhythmias and conduction disturbances may occur as in coronary thrombosis, but myocardial infarction does not, because spasm eases spontaneously within 30 minutes. Spasm is alleviated promptly by glyceryl trinitrate and prevented by calcium channel antagonists. The condition is rare, although relatively more common in women and in Asian nations. The onset of spasm is spontaneous; the cause is uncertain, although it is surmised that spasm is a consequence of a temporary imbalance between the production of dilating and constricting factors within the arterial intima. Sometimes a minor atherosclerotic lesion is demonstrable at the site of spasm. Patients with Prinzmetal or variant angina are particularly sensitive to serotonin and readily develop occlusion of epicardial arteries, with pain and electrocardiographic changes at low doses (McFadden *et al.*, 1991).

Cocaine

Cocaine can cause coronary artery spasm, simulating Prinzmetal angina or coronary occlusion (Isner *et al.*, 1985; Karch and Billingham, 1995). In contrast to the former, spasm may be sufficiently prolonged as to cause myocardial infarction (Kossowsky and Lyon, 1984; Isner *et al.*, 1986). It may also affect smaller intramyocardial arteries, resulting in small and sometimes multiple infarcts (Kloner and Rezkalla, 2003). Cocaine also predisposes to coronary atherosclerosis (Dressler *et al.*, 1990; Kolodgie *et al.*, 1991) and to arterial thrombosis (Virmani *et al.*, 1988). By increasing myocardial oxygen demands in the presence of coronary narrowing, cocaine may induce angina pectoris; if heart rate and arterial pressure remain elevated, myocardial infarction may result; also acute elevation of arterial pressure may precipitate aortic dissection (Karch and Billingham, 1995). Cocaine also potentiates the toxic effects of catecholamines.

Ergot

Sensitivity to ergot alkaloids varies. People with a predilection to coronary spasm are unduly sensitive, and ergotamine challenge has been used to induce spasm and confirm diagnosis. Ergot alkaloids may, like cocaine, cause sufficiently prolonged spasm to produce myocardial necrosis or even gangrene of an extremity (when a peripheral artery is involved).

Raynaud's phenomenon

In this condition, peripheral arteries are extremely sensitive to cold and respond by constriction (Zamora *et al.*, 1990; Wollersheim and van Zwieten, 1993), so that there is marked reduction of blood flow to the fingers and hand. This can be alleviated by vasodilators, especially nitrates and calcium antagonists. This is an accentuation of the normal reaction to cold. It is seen to a degree in all people subject to cold, may affect large arteries in the upper limb, and may be apparent as reduction in mean and pulse pressure in the hand following hypothermic cardiopulmonary bypass (Gravlee *et al.*, 1989a).

Migraine-serotonin and serotonin analogs

Headache in migraine is associated with, and appears to be caused by, dilatation of cerebral and extracerebral cranial arteries, sometimes with evidence of associated edema and inflammation (Iversen *et al.*, 1990; Olesen, 1994; Raskin, 1994). In migraine with aura, headache with arterial vasodilatation follows a preliminary wave of 'spreading depression' over the cerebral cortex. The mechanism is complex and not completely understood, but it appears to involve the release of serotonin from platelets at the beginning of the attack when vasoconstriction initially occurs. Sensory neuropeptides appear also to be involved and may be responsible for inflammation and the stimulation of pain receptors.

Serotonin has no effect on normal coronary arteries at low concentrations, but causes constriction at high doses; in contrast, patients with stable angina pectoris show generalized constriction at all concentrations (Golino *et al.*, 1991; McFadden *et al.*, 1991), while patients with variant angina show more marked arterial narrowing with localized epicardial coronary spasm (McFadden *et al.*, 1991). Serotonin contracts also cerebral arteries. The effects are considered to be due to activation of the 5HT₂ receptor (Connor *et al.*, 1989; Hillis and Lange, 1991). The serotonin agonist sumatriptan affects 5HT₁ receptors, which are usually not present in the heart, but which are present in cerebral vessels (Olesen, 1994; Raskin, 1994). This drug is extremely effective in the treatment of migrainous headache (Brown *et al.*, 1991; Dechant and Clissold, 1992; Pilgrim, 1994; Bateman, 2000). While far less likely than serotonin to cause coronary constriction (Dechant and Clissold, 1992; McIntyre *et al.*, 1992), there is no doubt that this can occur in susceptible individuals (Willett *et al.*, 1992; Ottervanger *et al.*, 1993).

Generalized and metabolic disease

Diabetes mellitus	524	Hyperparathyroidism	533
Kidney failure	529	Connective tissue disease	533
Hypercholesterolemia	533		

When we initiated our non-invasive clinical studies in the late 1970s to complement earlier invasive human and animal experimental studies, we confirmed the very substantial changes with age in humans which were not seen in experimental animals (O'Rourke *et al.*, 1968; Nichols *et al.*, 1977b; O'Rourke, 1982b) and attributed them to arterial disease (atherosclerosis) which accompanies aging. The opportunity then arose in the early 1980s to repeat these studies on pulse wave velocity (PWV) in mainland Chinese populations in regions with known high and low prevalence of hypertension. We were surprised to find in the former group, increase in aortic PWV similar to what we and others had recorded in Western populations with high prevalence of atherosclerosis, despite the Chinese group having virtually no evidence of atherosclerosis, and having low blood cholesterol levels (Ho, 1982; Avolio *et al.*, 1983). The Chinese group with low prevalence of hypertension (in rural Guangdong province) also showed increased aortic stiffening with age, but this, while beginning at the same level in childhood, was less than in the other group from urban Beijing in later life (Avolio *et al.*, 1985a; Avolio *et al.*, 1986; Avolio, 2003). There was a substantial difference in salt intake between the two Chinese groups, and this was considered to be responsible for the different prevalence of hypertension (Wu *et al.*, 1996). It may primarily have been responsible also for increased aortic stiffness in the urban Beijing population. We concluded that the aorta and elastic arteries progressively stiffen with age, and that this is accelerated by hypertension and high salt intake (Avolio *et al.*, 1985a). Studies of aortic structure in the Chinese groups were subsequently undertaken and confirmed similar aging change in China, the United States, and Australia (Virmani *et al.*, 1991) and supported our previous morphometric studies (Nichols *et al.*, 1985a; O'Rourke *et al.*, 1987). We concluded that atherosclerosis has little effect

per se on aortic stiffening with age, but that salt had an important adverse effect.

Our conclusions, 25 years ago, were that we and others had overestimated the effects of atherosclerosis (primarily an intimal disease) and underestimated the effects of aging change (primarily a medial condition of elastic fiber degeneration). In our early studies in Sydney, we saw little difference in PWV and arterial wave contour in people at the same age with known atherosclerosis, diabetes, renal disease, or diabetes mellitus (O'Rourke *et al.*, 1986a; Kelly *et al.*, 1989a). We downplayed the importance of all these conditions and emphasized the overriding influence of age and arterial pressure. We had hoped originally to use these indices to diagnose severity of atherosclerosis, but could not. Any difference in pressure wave contour or PWV with atherosclerosis, renal disease, or diabetes mellitus was lost in the great deal of unexplained variability which was apparent even in normal subjects at the same age and with similar arterial pressure. We did not pursue this field, but many others did in larger, better conducted studies, and with different results, which are described in a number of reviews (O'Rourke *et al.*, 1987; Cockcroft *et al.*, 2000; Bots *et al.*, 2002; Oliver and Webb, 2003; Laurent and Boutouyrie, 2007; O'Rourke and Hashimoto, 2007), and are summarized here. There is increasing interest in this field, despite the conflicts in much of the published (and unpublished) work. What is emerging is an acceptance that there is as much or more variation in other non-invasive indices of arterial structure and function (arterial calcium score (Detrano *et al.*, 2008; Agarwal *et al.*, 2010), carotid intima-media thickness (O'Leary *et al.*, 1999), flow-mediated brachial dilation (Celermajer, 1997; Halcox *et al.*, 2002b), even systolic and diastolic blood pressure (Franklin *et al.*, 1997)) as with the indices of arterial stiffness, and that such stiffness indices in long-running population studies are beginning to show strong prognostic

value, beyond conventional indices and conventional risk factors.

DIABETES MELLITUS

Diabetes mellitus does not cause heart disease directly but it does favour arteriosclerosis and coronary artery disease ... At least 50 percent of all diabetics die as a result of cardiovascular complications, and the relative incidence of this cause of death is steadily increasing as other fatal complications are eliminated. Marked atherosclerosis of the aorta with considerable dilation is common. Hypertension plays an important role in this group and frequently precedes the onset of the diabetes, sometimes by several years. Congestive failure due to hypertension or to coronary disease is not unusual, but death comes most frequently from coronary occlusion.

Thus wrote Paul Dudley White 60 years ago (White, 1951a; White, 1951b). His views have been strongly endorsed by findings published in the last 20 years. In the twenty-first century, research into arterial function and ventricular-vascular interaction carries the highest priority in diabetes care. Vascular disease, especially hypertension, does appear to predispose to glucose intolerance and diabetes, as White (1951a; 1951b) suggested (Barzilay *et al.*, 2006; Conen *et al.*, 2007; Elliott and Meyer, 2007), and some 80 percent of the complications and deaths in diabetic patients are indeed due to vascular, especially arterial events (Sowers *et al.*, 2001; Mancia, 2010; Pfeffer, 2010; Riddle, 2010). In one carefully conducted follow-up study, cardiovascular death in diabetic subjects was high (>2 percent per annum), and similar to non-diabetic people of the same age who had already had a myocardial infarct (Fig. 25.1) (Haffner *et al.*, 1998). In the Whitehall study, death rate from people with diabetes or impaired glucose tolerance (fasting glucose 6.1–6.9 mmol/L) was

twice that in controls (Fuller *et al.*, 1983; Davies, 1999). Subsequent studies have confirmed this view with Booth *et al.* (2006) estimating that diabetes confers an equivalent risk to aging 15 years, and Pries *et al.* (2008) showing diabetes subjects to have a two- to three-fold risk for cardiovascular disease compared to those without diabetes.

Earlier studies of arterial properties have shown that arterial stiffness is increased in patients with diabetes mellitus – either type 1 (insulin deficient) or type 2 (primarily insulin resistant) (Woolam *et al.*, 1962; Eliakim *et al.*, 1971; Lehmann *et al.*, 1992a; Megnien *et al.*, 1992; Salomaa *et al.*, 1995; De Angelis *et al.*, 2004). In the Bogalusa Heart Study, children with a positive parental history of diabetes also had stiffer arteries than children without such a parental history (Riley *et al.*, 1986; Li *et al.*, 2004). Hopkins *et al.* (1996) have shown greater aortic stiffness in normal healthy adults with a family history of diabetes. Stiffening involves the muscular as well as the elastic arteries (Megnien *et al.*, 1992) and unlike essential hypertension, causes intrinsic mechanical change to the arterial wall. Many recent studies have confirmed the earlier findings through measures of aortic impedance, pulse wave velocity, and/or aortic augmentation (Fig. 25.2) (Brooks *et al.*, 1999; Brooks *et al.*, 2000; Cockcroft *et al.*, 2000; Westerbacka *et al.*, 2000; Wilkinson *et al.*, 2000b; Brooks *et al.*, 2001; Cockcroft and Wilkinson, 2002; Cruickshank *et al.*, 2002; Fukui *et al.*, 2003; Haller *et al.*, 2004; Lacy *et al.*, 2004; Westerbacka *et al.*, 2005; Smith *et al.*, 2006; Sweitzer *et al.*, 2007; Urbina *et al.*, 2010). There have been confounders in interpretation of some of these studies. The higher aortic augmentation index for type 1 diabetic patients of Brooks *et al.* (1999) and Haller *et al.* (2004) was only significant when adjusted for heart rate, because the diabetic patients had heart rates greater than controls. This is an unfavorable prognostic indicator in itself, and largely attributable to autonomic neuropathy. Wilkinson *et al.* (2000b), on the other hand, reported that type 1 diabetes was associated with an increase in augmentation index. In type 2 diabetes, augmentation index was greater than controls without any adjustment (Brooks *et al.*, 2001; Ravikumar *et al.*, 2002; Fukui *et al.*, 2003;

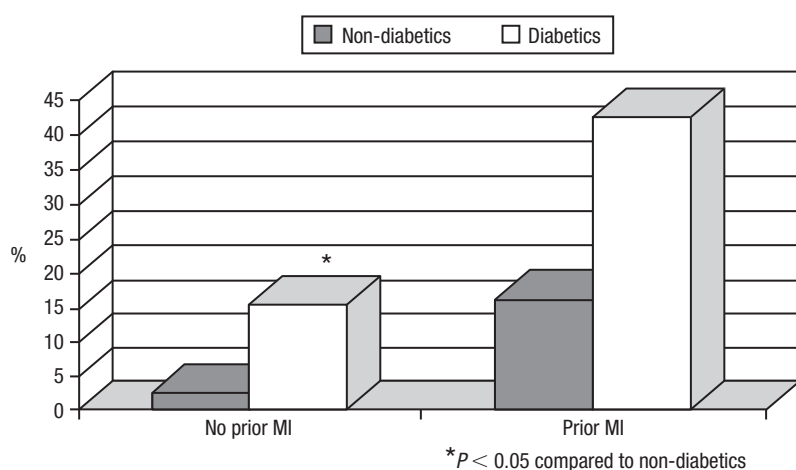


Figure 25.1 Incidence of death from cardiovascular causes after a seven-year follow up in a group of 1373 non-diabetic and 1059 diabetic Finnish subjects. Mortality rate was higher in the diabetic than non-diabetic subjects. In diabetic subjects, without previous myocardial infarction, death rate was as high as in the non-diabetic group who had suffered previous infarction. Data from Haffner *et al.* (1998), after Sowers *et al.* (2001)

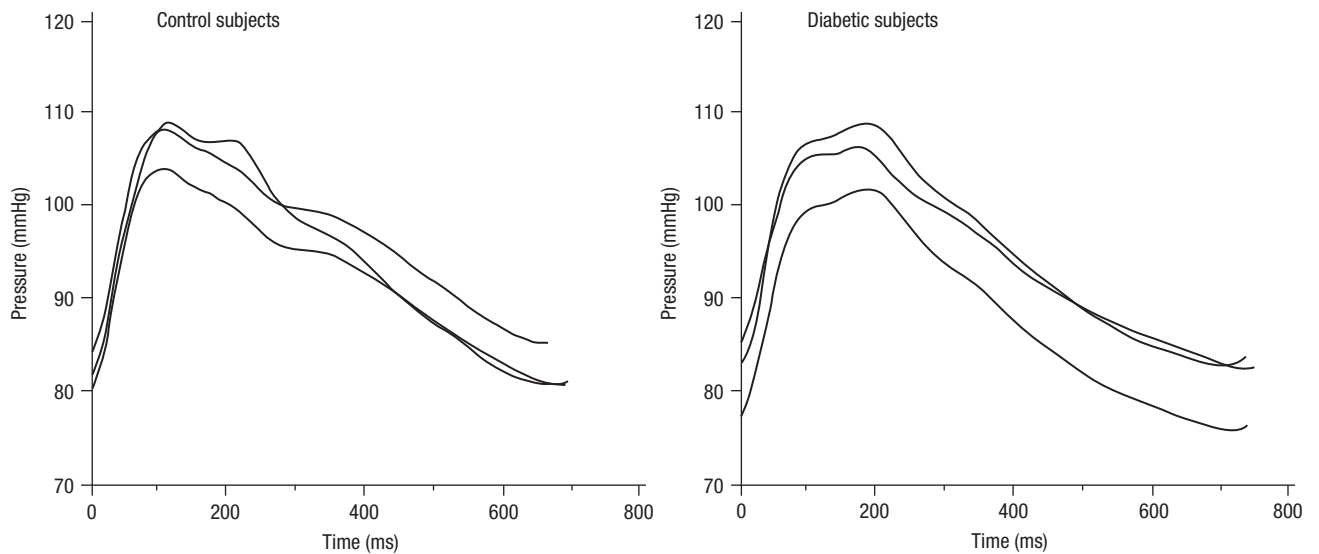


Figure 25.2 Comparison of typical synthesized aortic (from radial) pressure waves in diabetic subjects (right) and age- and sex-matched controls (left). In a group of 35 type 1 diabetic patients, mean age 30 years, aortic augmentation index was 7.1 percent (s.d. 1.6) in diabetics compared to 0.4 percent (s.d. 2.0) in controls ($P < 0.011$). From Wilkinson *et al.* (2000b)

Westerbacka *et al.*, 2005). Lacy *et al.* (2004) challenged the association between aortic augmentation index (AIx) and carotid-femoral PWV in diabetics. They found increase in PWV, but not in AIx. We (Maple-Brown *et al.*, 2005) found similar results in indigenous Australians with central obesity, insulin resistance, and diabetes mellitus. Initially, we attributed paradoxical findings to decrease in peripheral wave reflection, but subsequently considered the possibility of two other factors. The first of these is the decreasing rate of rise in AIx that occurs in older people and is itself an effect of aging (McEniery *et al.*, 2005a; Namasivayam *et al.*, 2009a). The second was a possible effect of concomitant cardiac steatosis (McGavock *et al.*, 2007; Ruberg, 2007) causing a subtle impairment in left ventricular (LV) contractility, impairing cardiac ejection in late systole and thereby causing decrease in flow and pressure augmentation. This is seen in the aged heart (see Chapter 14) (Miyashita *et al.*, 1994) and cardiac failure (Denardo *et al.*, 2010). Lacy's and Maple-Brown's findings may indicate an effect of diabetes on cardiac, as well as arterial aging. This is discussed below. Aging has a differential effect on aortic augmentation and aortic pulse wave velocity (McEniery *et al.*, 2005a) with AIx increasing more with age in the young, and aortic PWV more in the elderly. This is exaggerated in diabetics. Urbina *et al.* (2010) showed far greater difference in AIx in diabetic compared to normal children, and lesser relative difference in carotid femoral PWV, and only a slight difference between arm and leg PWV (Fig. 25.3).

The cause of greater arterial stiffness has not been determined with certainty (Franklin, 2002; Henry *et al.*, 2003; Schram *et al.*, 2004). The greater aortic stiffness confirmed by Brooks, Wilkinson, Westerbacka, and others corresponds to an aging change of 5–11 years (Table 25.1),

and this is how diabetic macrovascular disease is often explained – as accelerated aging. Booth *et al.* (2006) calculated aging difference between diabetics and non-diabetics as 14.7 years, from health claims of all adults in the province of Ontario (Canada). In the Framingham study, found that mortality rates of diabetics were two-fold higher than non-diabetics. Sung *et al.* (2009) presented data from the Korean National Health Insurance system which showed that there was a linear relationship between fasting glucose and risk of stroke and myocardial infarction which for ischemic stroke (the most common event) extended below the normal limit of 5.6 mm/L. Franklin (2002) has noted the increased formation of advanced glycosylation end-products (AGE) in diabetes (Schmidt *et al.*, 1999; Libby and Plutzky, 2002) and attributed increased stiffness to cross-linking of extravascular proteins to matrix components caused by AGE. Wilkinson *et al.* (2000b) noted this structural change, but emphasized a functional change as well, wherein endothelial dysfunction results in a defect in cGMP-mediated muscle relaxation, with reduction in caliber and distensibility of conduit arteries. Such a mechanism is supported by other studies of endothelial dysfunction (Cardillo *et al.*, 2002a; Ravikumar *et al.*, 2002) and highlights the endothelium as a target for therapy in diabetes mellitus. Ravikumar *et al.* (2002) compared various techniques for assessing vascular structure and function in diabetes, and showed differences in carotid intima-medial thickness (IMT), forearm blood flow response, and aortic augmentation index between diabetes and control subjects (Fig. 25.4). This was later confirmed by Westerbacka *et al.* (2005) who showed that augmented pressure correlated with longer duration of diabetes, urinary albumin excretion, and IMT.

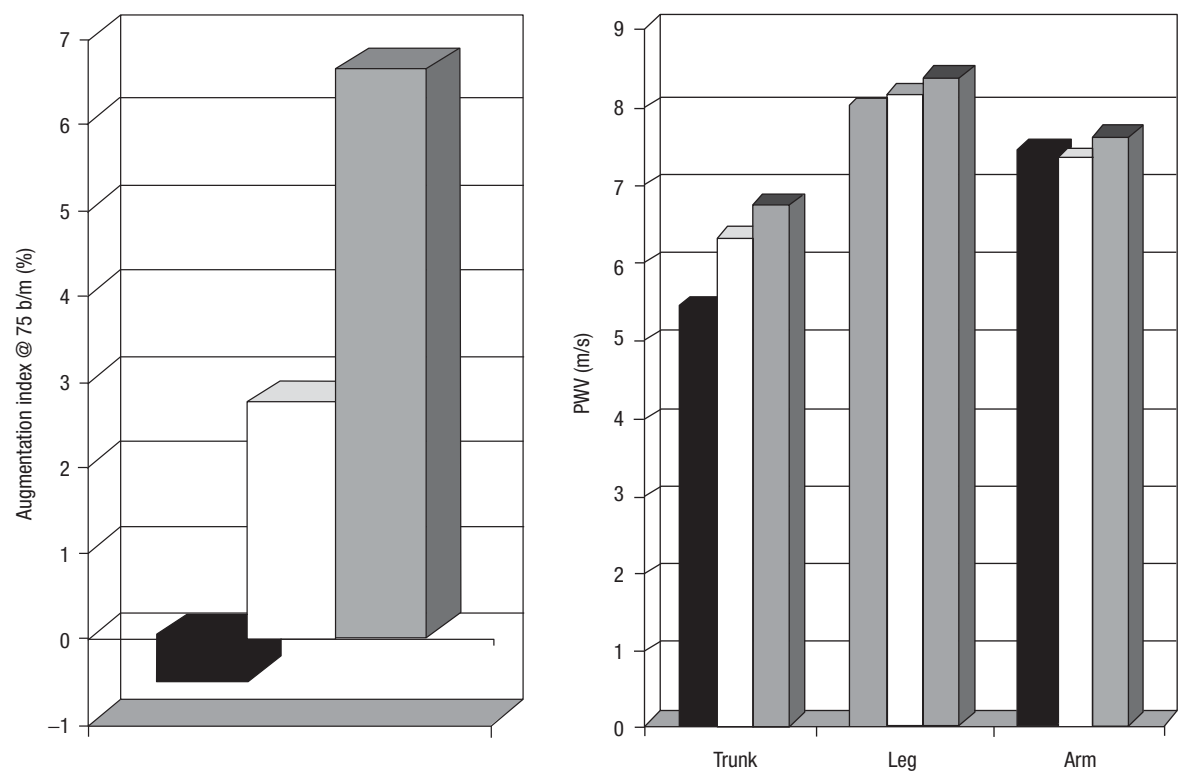


Figure 25.3 Calculated ascending aortic augmentation index, Alx (left) and pulse wave velocity, PWV (right). Alx is shown corrected for heart rate 75 b/min, in lean (black), obese (white), and type 2 diabetic adolescents (gray). At right, PWV is given in the same groups for the trunk, leg, and arm. There were substantial differences ($P < 0.001$) between lean and obese/diabetic subjects for Alx and carotid-femoral PWV, but lesser and non-significant differences between lean and obese subjects for arm and leg PWV. From Urbina *et al.* (2010)

Table 25.1 Effects of diabetes mellitus on calculated vascular age increase

Reference	Age
Type 1 Diabetes mellitus	
Brooks <i>et al.</i> (1999)	7.9 years (corrected for heart rate)
Cockcroft and Wilkinson (2002)	8.9 years
Type 2 Diabetes mellitus	
Brooks <i>et al.</i> (2000)	10.4 years
Ravikumar <i>et al.</i> (2002)	11.1 years
Fukui <i>et al.</i> (2003)	5.1 years
Booth <i>et al.</i> (2006)	14.7 years

Calculated from change in augmentation index with age in normal subjects (Kelly *et al.*, 1989a; Fujii, 1989; Wilkinson *et al.*, 2001; Nurnberger *et al.*, 2002).

In the studies of type 1 and type 2 diabetic subjects conducted by Wilkinson *et al.* (2000b) and by Brooks *et al.* (Brooks *et al.*, 1999; Brooks *et al.*, 2000; Brooks *et al.*, 2001), there was a significant decrease in the relationships of integrated aortic diastolic to integrated systolic pressure compared to control subjects. This, the subendocardial viability ratio (Buckberg *et al.*, 1972a; Buckberg *et al.*,

1972b; Hoffman and Buckberg, 1978), was decreased, partly because of increase in aortic augmented pressure, and partly because of a relatively longer period of systole and shorter period of diastole in the diabetic subjects. Decrease in subendocardial viability ratio and relative increase in ejection duration is associated with diastolic LV dysfunction (Abhayaratna *et al.*, 2006a; Weber *et al.*, 2006; Weber *et al.*, 2008), which was found in 60 percent of type 2 diabetic patients studied by Poirier *et al.* (2001). Decrease in subendocardial viability ratio also predisposes to myocardial ischemia and chest pain (angina) since it signifies relative increase in myocardial need and decrease in capacity to supply this need (coronary blood flow) during diastole – especially with any further increase in heart rate. Devereux *et al.* (2000a) and Liu *et al.* (2001) confirmed these adverse hemodynamic effects in type 2 diabetic patients and showed that they were related to arterial stiffening, and were associated with increased LV mass and wall thickness, independent of associated increases in body mass index and arterial pressure. High prevalence of diastolic LV dysfunction in diabetes explains the high prevalence of congestive heart failure, independent of the increased severity of coronary artery disease (Gustafsson and Hildebrandt, 2001). Abnormalities in cardiac metabolism in diabetics augment the ill effects of arterial stiffening on the heart, and their potentiation by

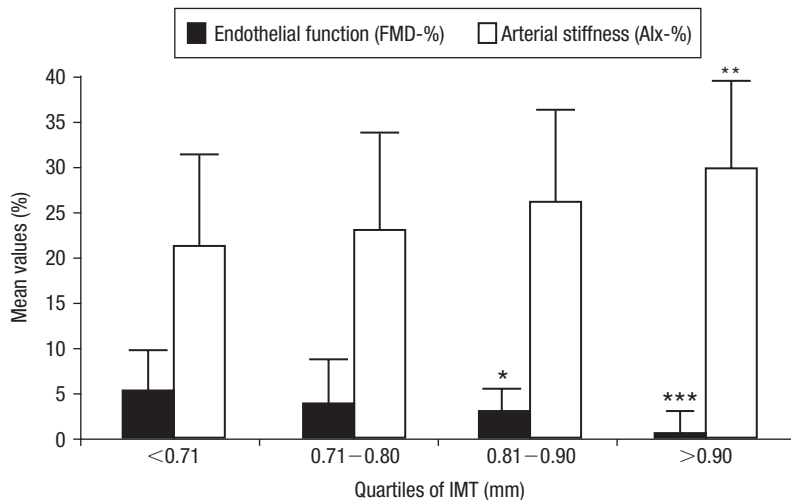


Figure 25.4 Relationship of calculated aortic pressure wave augmentation index Alx-percent (□) and brachial flow mediated dilation FMD-percent (■) to brachial intima-media thickness (IMT) in a group of 50 diabetic patients and 50 age-matched controls. From Ravikumar *et al.* (2002)

any elevation in arterial pressure (Taegtmeier *et al.*, 2002; Hatsuda *et al.*, 2006). Implications of these findings for systolic and diastolic reserve limitations in people with heart failure and preserved LV systolic function have been shown by Kawaguchi *et al.* (2003) and referred to by Fang and Marwick (2003) and Picano (2003).

Diabetes mellitus is inextricably linked with arterial stiffness, hypertension, and renal disease (Kimoto *et al.*, 2006). Hypertension often precedes development of diabetes (White, 1951b; Sowers and Bakris, 2000; Sowers *et al.*, 2001; Conen *et al.*, 2007) and is frequently associated with diabetes. Elliott and Meyer (2007) showed that anti-hypertensive therapy with different agents had different efficacy in limiting development of incident diabetes (Fig. 25.5). Diabetic blood vessels are uniquely vulnerable to hypertensive injury (Williams, 1999) and are more likely to rupture or occlude at any given blood pressure than non-diabetic vessels. Structural and functional changes in small diabetic arteries, especially an impaired myogenic response (Schofield *et al.*, 2002b), permits pulsatile flow and pressure to extend into the smaller vessels downstream. Damage to these vessels by a mechanism similar to that described by Byrom and Dodson (1948) may underlie the microvascular disease of diabetes (Safar *et al.*, 2003) with damage to retina, brain, and kidneys that is so common in this disease, and so amenable to blood pressure reduction (Eguchi *et al.*, 2003; Perkins *et al.*, 2003; Ritz, 2003; Frank, 2004; Safar *et al.*, 2004; Matsui *et al.*, 2010). In major trials of anti-hypertensive therapy, such as HOPE (Yusuf *et al.*, 2000), HOT (Hansson *et al.*, 1998), and LIFE (Dahlöf *et al.*, 2002), the greatest benefit for blood pressure lowering was seen in diabetic patients. Indeed in HOT (which was not powered to show overall benefit), there was a statistically significant reduction in cardiovascular mortality and major cardiovascular events in the diabetic subset in the comparison of vigorous against less vigorous lowering of blood pressure of hypertensives. Earlier studies had given similar results, and it was these which had led the US JNC6 and the WHO/ISH committees to recommend a target of

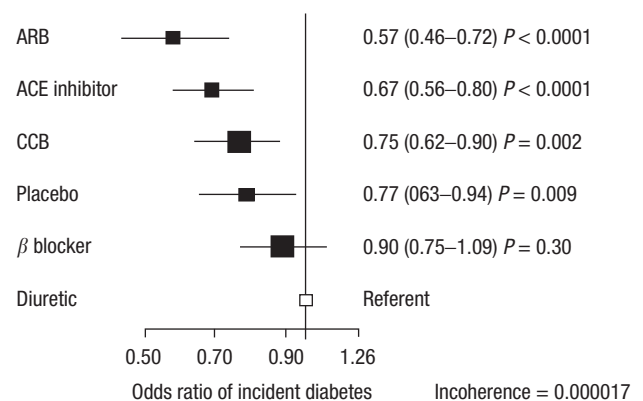


Figure 25.5 Incident diabetes according to principal drug used in a network meta-analysis of 143 153 patients within 22 clinical trials. Hazard ratios were lowest in the angiotensin receptor blocker (ARB), angiotensin-converting enzyme inhibitors (ACEI), and calcium channel blocker (CCB)-treated patients, and highest in those treated with diuretics. From Elliott and Meyer (2007). The hierarchy is similar to that found by Miyashita *et al.* (2010) for difference between aortic and upper limb systolic pressure for the same drug classes.

<130 mmHg systolic for patients with diabetes mellitus. Their recommendations have been further endorsed in statements from the American Heart Association, American Diabetes Association, and government bodies (Grundy *et al.*, 1999; Friesinger and Gavin 2000; Grundy *et al.*, 2002; Berlowitz *et al.*, 2003).

In detailed studies of diabetes care, the UK prospective diabetes study (UK Prospective Diabetes Study, 1998; Adler *et al.*, 2000), while showing benefits of glycemic control, have shown that the most important therapy for reducing vascular mortality was reduction of arterial pressure. Indeed, modest blood pressure control was more important than tight blood glucose control in a disease which was considered to be of glucose metabolism rather than of blood vessels (UK Prospective Diabetes Study, 1998; Laakso, 1999; Yudkin, 1999; Adler *et al.*, 2000;

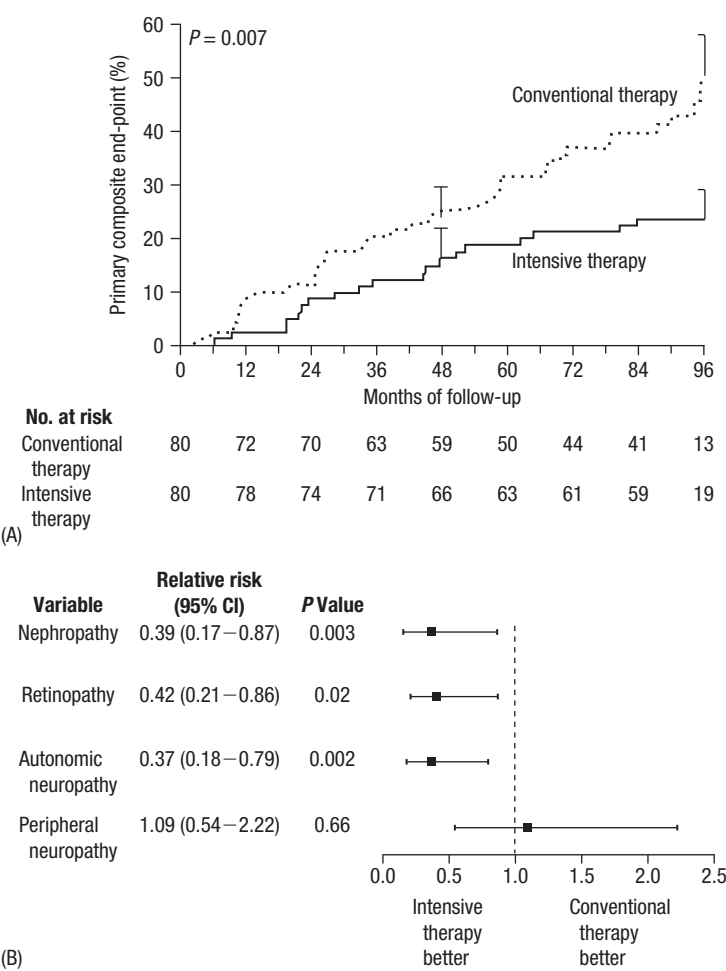


Figure 25.6 (A) Primary composite endpoint (cardiovascular death or cardiovascular event) on conventional or aggressive medical therapy of diabetes mellitus over eight years. (B) Relative risk of progression of nephropathy, retinopathy, and neuropathy for conventional and aggressive therapy over the eight-year period. From Gaede *et al.* (2003)

Libby and Plutzky, 2002). Benefits of aggressive therapy of diabetes, focused on blood pressure control, have now been established in a prospective study of type 2 diabetes mellitus (Fig. 25.6) (Gaede *et al.*, 2003; Solomon, 2003). Current evidence is set out by Snow *et al.* (2003), with guidelines by Vijan and Hayward (2003), Skyler *et al.* (2009), Marwick *et al.* (2009), and Mancia (2010).

The focus on diabetes as a major risk factor for cardiovascular disease has led to re-evaluation of criteria for diagnosis, and a downscaling of blood sugar levels at which the diagnosis is established. Even below this level (usually taken as 6.1 mm/L fasting), the condition of glucose intolerance is associated with high cardiovascular mortality (Barzilay *et al.*, 1999; Davris, 1999; Vinicor, 1999; Barzilay *et al.*, 2006; Sung *et al.*, 2009). Recent suggestions are that fasting blood sugar levels and glycosylated hemoglobin levels should be used like cholesterol to give a graded risk, not to pronounce a disease.

The problems of diabetes mellitus are interwoven with hypertension, renal disease, and all forms of cardiovascular disease, and increasingly are drawing together experts from different disciplines, including epidemiology (Sowers *et al.*, 2001; Cruickshank *et al.*, 2002; Pfeffer, 2010). Cruickshank *et al.* have used aortic pulse wave velocity as a measure of cardiovascular risk, and have

shown that this is a powerful independent predictor of risk, not only for frank diabetes mellitus, but for glucose intolerance as well. In Cox modeling, aortic PWV displaced systolic blood pressure as a risk factor, pointing to its value as a very useful integrated index of vascular status (Table 25.2).

Benefits of particular drug therapies are described in Chapter 26. The ALLHAT study (ALLHAT, 2002) has confirmed the benefits of blood pressure reduction against benefits of any particular drug class. Previously, angiotensin-converting enzyme inhibitors (ACEI) or angiotensin receptor blocker (ARB) drugs were considered preferable in diabetic patients, and they may still be in the presence of frank or impending nephropathy, over calcium channel antagonists (Brenner *et al.*, 2001; Viberti and Wheeldon, 2002; Frohlich and Sowers, 2003). The benefits of aggressive blood pressure lowering in the HOT study (Hansson *et al.*, 1998) for diabetics were achieved on calcium channel blocker (CCB)-based initial therapy. The Second Australian Blood Pressure Trial (Wing *et al.*, 2003) has shown marginal benefit for ACEI over diuretics, challenging the much larger ALLHAT results. The value of ACEI is, however, considered to have been generally overstated (Kurtz, 2003). The most recent ADVANCE, ACCORD, and VA Diabetes Trial (The ADVANCE

Table 25.2 Final Cox model of factors predicting mortality in a cohort of population-based control and type 2 diabetes participants

	Hazard ratio	95% CI	P
Age (per one year older)	1.07	1.04–1.09	<0.0001
Sex (female)	0.66	0.48–0.89	0.007
BP treated	1.10	0.82–1.48	0.54
Systolic BP (per 5 mmHg)	1.02	0.98–1.06	0.34
GTT status (normoglycemia)	1		
Glucose intolerant ^a	2.12	1.11–4.01	0.021
Known diabetes	2.34	1.50–3.74	<0.0001
PWV (per 1 m/s)	1.08	1.03–1.14	0.001
Smoking status (never)	1		
Current	1.57	1.10–2.23	0.012
Ex-smoker	1.16	0.81–1.66	0.42
Ethnicity (European)	1		
Gujerati	0.68	0.49–0.94	0.02
African-Caribbean	0.41	0.25–0.69	0.001
Other	1.34	0.80–2.33	0.25

^aBaseline fasting glucose > 6.1 mmol/L or 2-hour glucose > 11.1 mmol/L. *n* = 565. Six people without smoking data were omitted.

GTT, glucose tolerance test; PWV, pulse wave velocity.

From Cruickshank *et al.* (2002).

Collaboration Group, 2008; Dluhy and McMahon, 2008; Duckworth *et al.*, 2009) support views on anti-hypertensive drug therapy, but introduced concerns on aggressive use of anti-diabetic agents. Their concerns persist and have been extended to rosiglitazone and other new drugs (Drazen *et al.*, 2007; Nissen and Wolski, 2007). Elliott and Meyer (2007) showed a hierarchy of benefit for anti-hypertensive drugs in trials of diabetes (Fig. 25.5) which roughly corresponds to different effects of these drugs to lower central *c.f.* brachial systolic pressure (see Chapter 28) (Miyashita *et al.*, 2010).

KIDNEY FAILURE

Patients with end-stage renal disease are living longer than ever before, because of the availability of dialysis and renal transplantation. No longer is dialysis restricted to young adults, and in most Western centers the average age of initiating dialysis is over 65 years. No longer is diabetes mellitus an exclusion for entry into a dialysis program; indeed, diabetes is emerging as the most common cause of renal failure requiring dialysis (American Diabetes Association Position Statement, 2002). Patients with end-stage renal failure rarely die now from renal failure, but rather from the cardiovascular disease which accompanies it. Mortality rates are high, of the order of 10 percent per annum (London *et al.*, 2001).

Patients with renal failure show the same type of accelerated development of arterial disease as do diabetics, the same (or greater) likelihood of elevated arterial pressure as diabetics, the same sensitivity to elevated arterial pressure as diabetics, and the same benefit from anti-hypertensive therapy (Bergstrom *et al.*, 1986; Peterson

et al., 1995; Klang *et al.*, 1996; Palmer, 2002; Chue *et al.*, 2010). The ESH/ESC and JNC7 recommendations for therapy are the same as for diabetes, with a treatment goal of < 130 mmHg systolic pressure (see Chapter 21).

The cause of accelerated arterial degeneration in renal failure is not established. As with aging, fatigue and fracture of elastin fibers is a major cause of large artery degeneration (see Chapter 19), but as for diabetes, cross-linking of collagenous material from AGE accumulation may also be responsible, together with a greater degree of endothelial dysfunction. Medial degeneration and calcification are common in renal failure – even more so than in diabetes – and this is attributable to disturbance in calcium metabolism. Accelerated arterial degeneration has been quantified by Benetos *et al.* (2002), who found that in hypertensive subjects without overt renal failure, aortic pulse wave velocity increase with age is related to creatinine level (denoting degree of renal insufficiency), as well as arterial pressure and heart rate. This view has been questioned by Temmar *et al.* (2010b) who considered that the substantial increase in aortic stiffness is more related to age, systolic blood pressure, accompanying diabetes, and arterial calcification, than to uremic toxicity. Microvascular disease in the kidney is attributable to increased pulsatile flow in the small arterial vessels and glomerular capillaries, consequent on the inability of the aged aorta to cushion flow pulsations (see Chapter 18). In diabetic nephropathy, there is impaired autoregulation of the afferent arterioles as well, such that glomerular capillary hypertension occurs and capillary damage results, with subsequent albuminuria (Hostetter *et al.*, 1982; Giunti *et al.*, 2006). The kidney size increases and the organ pulsates. Pulsatile stress in the kidney is held responsible for secondary damage to renal capillaries and parenchyma

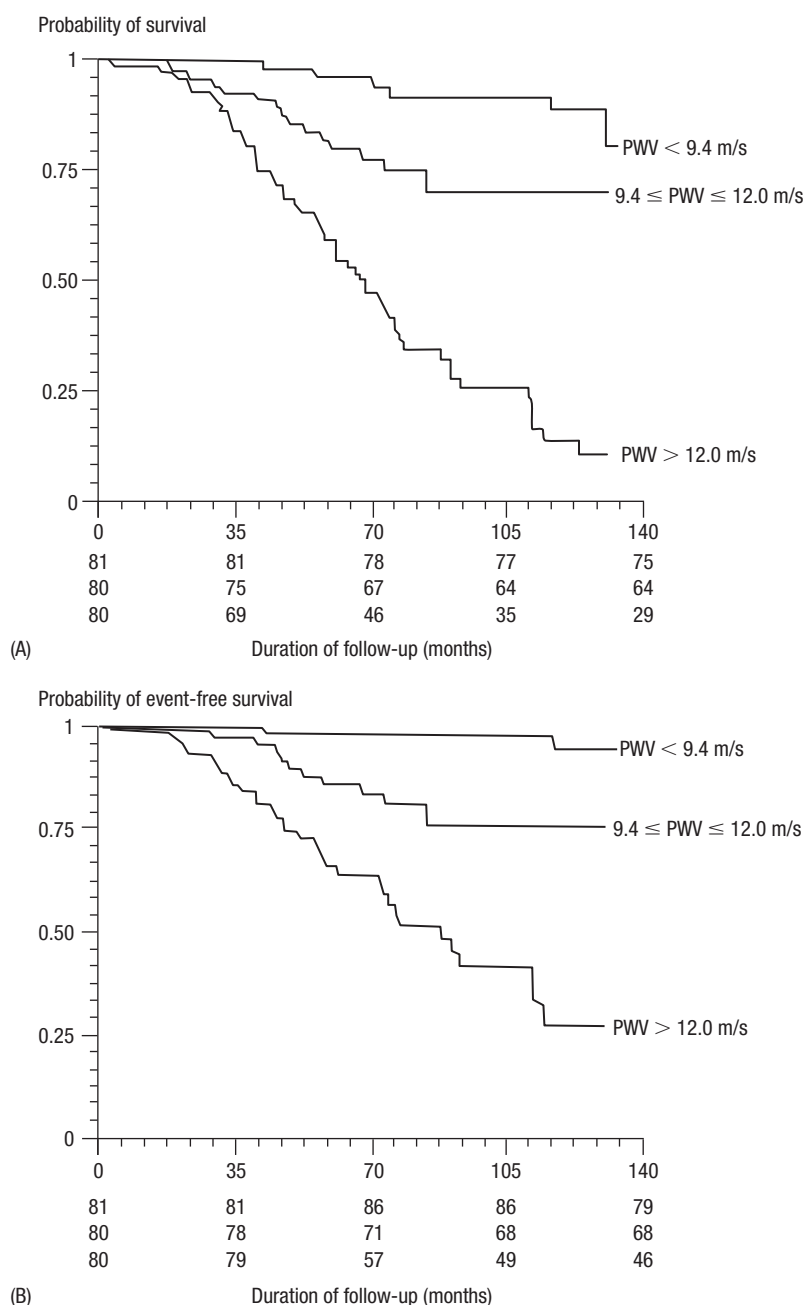


Figure 25.7 End-stage renal failure. Probability of survival (A) and probability of event-free survival (B) over a ten-year period in relation to aortic pulse wave velocity (PWV) at entry. There was a highly significant difference in both ($P < 0.0001$) for the upper compared to the middle tertile and for the lower component to the middle tertile. From Blacher *et al.* (1999a)

(Giunti *et al.*, 2006; Loutzenhiser *et al.*, 2006). Amelioration of capillary hypertension by ACE inhibitors is attributed to decrease in efferent arteriolar tone (Giunti *et al.*, 2006).

London and colleagues in Paris have conducted detailed studies in patients with end-stage renal disease over many years. They have determined measures of arterial stiffness, including aortic pulse wave velocity, central pressure wave augmentation, central pulse pressure, amplification of the pulse from central to radial arteries, and degree of arterial calcification. They have related these simple measures to outcome, and have shown strongly positive results for all, and with as much as an eight-fold difference in cardiovascular and total

mortality between the first and fourth quartiles of central pressure augmentation (Blacher *et al.* (1999a): Fig. 25.7; Guerin *et al.* (2001): Fig. 25.8; London *et al.* (2001): Fig. 25.9; Safar *et al.* (2002a): Fig. 25.10). Differences for PWV, AIx, aortic pulse pressure, and pulse pressure amplification were independent of conventional risk factors, including blood pressure and age. For pressure augmentation, published results to date are from the carotid pressure waveforms, but similar results are seen for the aortic pressure wave, synthesized from the radial artery pressure wave. (This technique was introduced at a later time than carotid tonometry.) Figures 25.7, 25.8, 25.9, and 25.10 are classic and were the first clear-cut evidence of using aortic stiffness measures in predicting outcome.

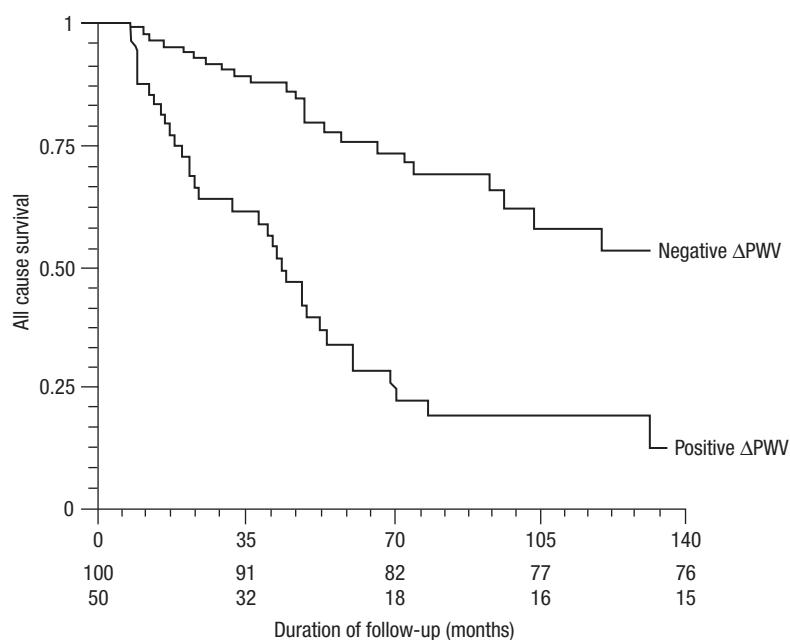


Figure 25.8 End-stage renal failure. Probability of all-cause survival according to ability of therapy to change aortic pulse wave velocity (PWV). Responders (top line) showed fall in PWV with anti-hypertensive therapy; non-responders (bottom line) showed no change or progressive increase in PWV. $P < 0.00001$. From Guerin *et al.* (2001)

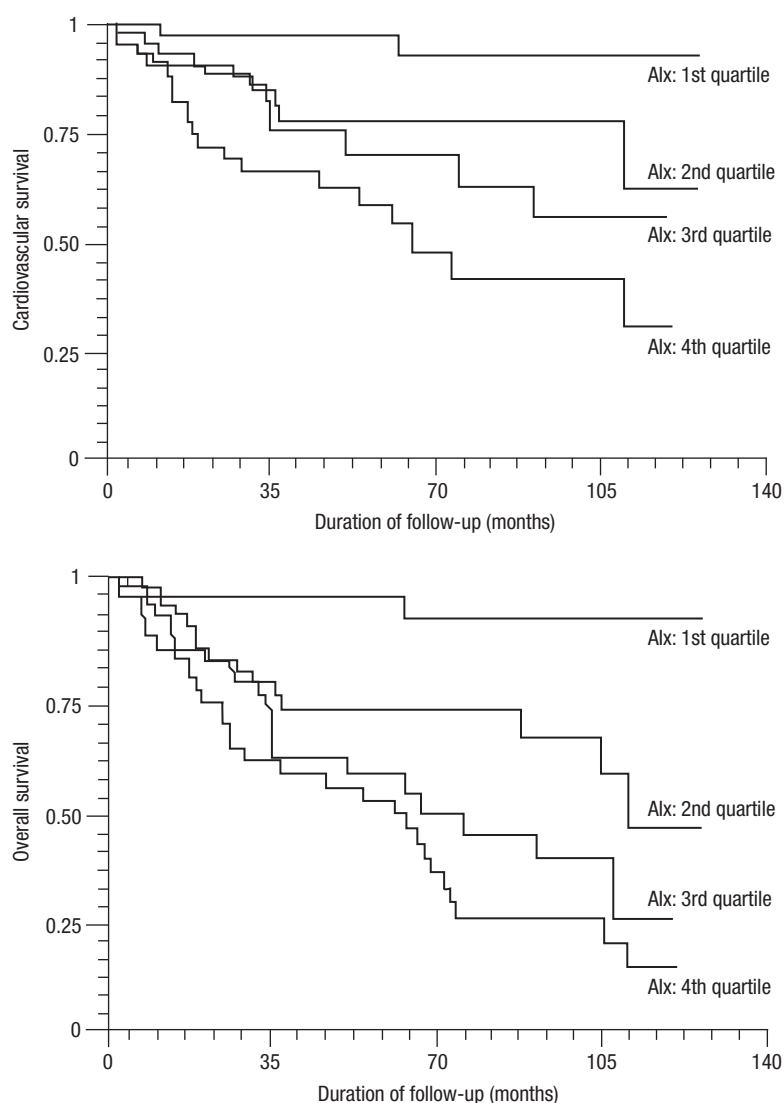


Figure 25.9 End-stage renal failure. Probability of cardiovascular (top) and overall survival (bottom) over ten years in relation to quartiles of carotid augmentation index (Alx). Log rank test for cardiovascular mortality $\chi^2 = 34.02$, $P < 0.0001$. Log-rank test for cardiovascular mortality $\chi^2 = 23.11$, $P < 0.00001$. From London *et al.* (2001)

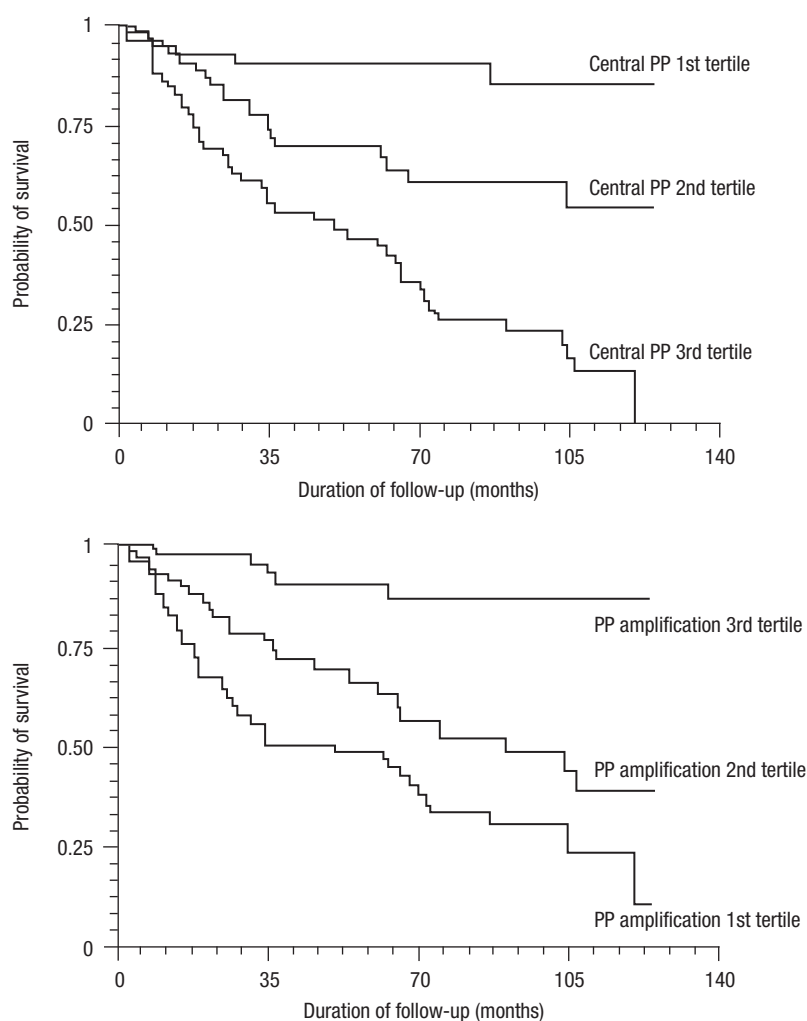


Figure 25.10 End-stage renal failure. Probability of overall survival over ten years in relation to tertiles of central pulse pressure (PP) (top) and PP amplification between the carotid and upper limb sites. Comparisons between the survival curves were highly significant ($P < 0.0001$ for both). Brachial blood pressure including brachial pulse pressure had no predictive value after adjustment in Cox analysis. From Safar *et al.* (2002a)

Apart from prediction of outcome, pressure pulse waveform analysis has been used to show the effects of volume overload (change before and after dialysis), of angiotensin II and of arteriovenous shunts (as used for hemodialysis) on central augmentation index and on ventricular–vascular interaction (Covic *et al.*, 2000; Ferro *et al.*, 2002; Savage *et al.*, 2002b; Vuurmans *et al.*, 2002). Presence of the arteriovenous shunt was associated with higher central pressure augmentation and lower sub-endocardial viability ratio. These factors predispose to myocardial ischemia and may help explain the high levels of cardiac troponins which are often found in patients with renal failure on dialysis (Freda *et al.*, 2002), as well as the high coronary event rate.

The outcome results for patients with renal failure are not specific for renal failure, and appear to be seen more readily and at an earlier time because of the severity of cardiovascular disease in this condition, and the high rate of cardiovascular events and of death. Similar results are seen in hypertensive patients (Laurent *et al.*, 2001a; Boutouyrie *et al.*, 2002; Laurent *et al.*, 2003), in elderly patients (Meaume *et al.*, 2001), and in diabetic patients and subjects with carbohydrate intolerance (Cruickshank

et al., 2002). Some problems with end-stage renal disease may be due to factors associated with dialysis. Covic *et al.* (2003) have shown that successful renal transplantation decreases aortic stiffness and increases vascular reactivity in dialysis patients.

Guerin *et al.* (2001) have investigated the value of drug therapy in reducing progression of renal failure and of cardiovascular complications and death. They showed that use of an ACE inhibitor could slow the steady increase in aortic pulse wave velocity and arterial stiffness with time. They also showed that in patients whose aortic pulse wave velocity decreased with therapy, event rate decreased, in comparison to those whose pulse wave velocity remained the same, or increased on therapy.

As discussed, JNC7 and ESH/ESC recommendation for therapy and for goals of therapy are similar for patients with diabetes mellitus and for renal insufficiency. It is generally held that inhibitors of the renin–angiotensin system are preferable for use in renal insufficiency in view of their greater ability to decrease albumen excretion in urine and delay progression to renal failure (Lewis *et al.*, 2001; Parving *et al.*, 2001). This view is based on their better effects on renal autoregulation, with more

pronounced dilation of the efferent than the afferent renal arterioles (Palmer, 2002). This issue remains under re-examination. Chapter 18 discusses the relationship between aortic stiffening and microvascular renal disease. As for the brain, whose white matter hyperintensities are shown on magnetic resonance imaging (MRI) in elderly people with aortic stiffening, renal damage is attributed to loss of pulsatile energy in the vasodilated renal arterioles, when the stiffened aorta fails to absorb pulsations imposed by the heart. Increased cellular apoptotic micro-particles are also seen in renal insufficiency and are a predictor of albumin excretion rate (Huang *et al.*, 2010). Matsui *et al.* (2010) have shown that, independent of home or office blood pressure reduction, a decrease in arterial stiffness by anti-hypertensive therapy (compared to control, over six months) reduced renal damage, assessed by urine albumin/creatinine ratio. Angiotensin receptor blockers also have this effect and remain a favored treatment of renal insufficiency due to hypertension, aging, or diabetes (Hollenberg *et al.*, 2007; Karalliedde *et al.*, 2008).

HYPERCHOLESTEROLEMIA

Dyslipidemia is a primary risk factor for the development of coronary atherosclerosis. But does it cause generalized arteriosclerosis? To assess the effect of dyslipidemia on the elastic properties of the aorta, Pitsavos *et al.* (1998) measured ascending aortic stiffness in young (< 40 years) patients with heterozygous familial hypercholesterolemia and compared the results with age-matched normolipidemic siblings. They found that aortic dimensions were similar in the two groups, but aortic stiffness was significantly increased in the hypercholesterolemia group.

Wilkinson *et al.* (2002b) investigated the relation between serum cholesterol, arterial stiffness, and central aortic blood pressure. Pressure waveforms were recorded from the radial artery in patients with hypercholesterolemia and age-matched control subjects, and corresponding central aortic pressure waveforms were generated using pulse wave analysis (see Chapter 30). There was no significant difference in peripheral blood pressure between the two groups, but central aortic pulse pressure was higher in the group with hypercholesterolemia. Aortic augmentation was also higher, as was the estimated aortic PWV, and pulse pressure amplification was less in the patients with hypercholesterolemia. In a multiple regression model, age, short stature, mean arterial pressure, smoking, and low-density lipoprotein cholesterol correlated positively with AIx, and there was an inverse correlation with heart rate. The increase in AIx and arterial stiffness in this and other reports is unclear. However, endothelial dysfunction and reduced nitric oxide bio-availability are associated with hypercholesterolemia and may contribute to the increase in arterial stiffness and AIx (Wilkinson *et al.*, 2002b). Administration of pravastatin

or atorvastatin for six months lowers both total cholesterol and elastic artery stiffness and increases stroke volume (Muramatsu *et al.*, 1997; Nichols and Singh, 2002), however, there is no change in the elastic properties of muscular arteries (Kool *et al.*, 1995). Aortic systolic and pulse pressure decrease while diastolic pressure and amplification increase (Nichols and Singh, 2002). These beneficial changes in wave reflection properties are associated with a decrease in AIx and wasted pressure effort and suggest a reduction in elastic artery stiffness, LV afterload, and myocardial oxygen demand. Also, later studies show that intensive cholesterol reduction with high-dose statins may be beneficial in the treatment of patients with isolated systolic hypertension and normal lipid levels, through a reduction in large artery stiffness (Ferrier *et al.*, 2002). In the CAFE substudy of ASCOT (Williams *et al.*, 2006), atorvastatin had no effect on aortic augmentation index or indices of aortic stiffness despite reduction in cholesterol level.

HYPERPARATHYROIDISM

This is an uncommon disease caused by excessive secretion of hormone by the parathyroid gland or glands. The usual cause is a functioning adenoma. The consequence is elevated blood and urine calcium. The complications include renal calculi and calcification of cardiac valves and blood vessels in association with premature arterial degeneration. Cardiovascular mortality is said to be increased, but this is not firmly established (Wermers *et al.*, 1998; Seely and Williams, 2001). Increased arterial stiffness has been reported by Smith *et al.* (2000) and Rubin *et al.* (2002), both of whom determined aortic augmentation from the radial tonometric pressure wave, using SphygmoCor.

Implications of the vascular findings remain under debate. There is no consensus about the management of mild primary hyperparathyroidism. The operative procedure is delicate and not without risk. Its benefit in extending life and health has not been established. Better understanding of the arterial disease, and ability to measure arterial stiffening in patients before and after surgery, may help in deciding on whom, and when to operate.

CONNECTIVE TISSUE DISEASE

Marfan syndrome is described in Chapter 24, as are other specific conditions which affect the large elastic arteries. There are a number of reports on studies of arterial stiffness in other conditions, including polyarteritis nodosa (Cheung *et al.*, 2000), systemic lupus erythematosus (Selzer *et al.*, 2001; Bjarnegard *et al.*, 2006; Knockaert, 2007; Pieretti *et al.*, 2007), and in rheumatoid arthritis (Maki-Petaja *et al.*, 2006; Roman and Salmon, 2007; Crilly *et al.*, 2009). Some rather

impressive results have been reported (Van Doornum *et al.*, 2003; Wong *et al.*, 2003) for rheumatoid arthritis, using a technique that has not been uniformly successful in detecting aging change or effects of drugs (see Chapters 19 and 26) (Manning *et al.*, 2002; Oliver and Webb, 2003). Using a universally accepted method for measuring aortic pulse wave velocity, Maki-Petaja *et al.* (Maki-Petaja *et al.*, 2006; Maki-Petaja *et al.*, 2007) found that rheumatoid arthritis was associated with increased aortic stiffness and endothelial dysfunction. Treatment with statins and anti-tumor necrosis factor- α reduced aortic stiffness and improved endothelial function. Bearing in mind the conflicting results for other conditions, for gender, and for hormonal action described elsewhere, one would have to be careful in accepting information in small studies until corroborated by different investigators using different methods and appropriate controls. Booth *et al.* (2004b)

have shown that arterial stiffness (gauged by carotid-femoral PWV and AIx is increased in patients with systemic vasculitis, and is related to C-reactive protein (CRP). A positive relationship between CRP and AIx, aortic PWV and pulse pressure has also been described for apparently normal subjects (Avolio, 2004; Kampus *et al.*, 2004; Yasmin *et al.*, 2004) however, CRP does not seem to have a causal role in the development of arterial stiffness (Schumacher *et al.*, 2009). Roman *et al.* (2003) and Roman and Salmon (2007) have described accelerated atherosclerosis and increased LV mass (Pieretti *et al.*, 2007) in patients with systemic lupus erythematosus. Constans *et al.* (2007) claim that an index of aortic stiffness (estimated using the QKd method from ambulatory blood pressure and electrocardiogram (Gosse *et al.*, 1996)) can predict progression in systemic sclerosis (see also Cecelja and Chowienczyk (2009)).

Therapeutic strategies

Pathophysiological principles	535	Arterial counterpulsation	549
Muscular arteries of different vascular beds	543	Monitoring and measurement of therapeutic effect	551
Medical therapies	544		

PATHOPHYSIOLOGICAL PRINCIPLES

Clinical and therapeutic implications are addressed throughout this book, particularly in its latter sections. We intend in this chapter to summarize these and to point out particular strategies that have either been shown, or which we expect to show, clinical relevance in the future. Medical, surgical, and mechanical issues will be addressed, and in a realistic way that makes the older concepts of arterial Windkessel completely inappropriate. Readers will have seen enough evidence of wave reflection in this book to appreciate that any conceptual model (including Windkessel) that excludes the possibility of wave reflection cannot be regarded as realistic and thus must show serious deficiencies with change in cardiac and vascular function. If these principles are to be useful for monitoring of sick patients as in an intensive care ward (which is our ultimate aim), they must apply widely with change in heart rate, blood pressure, cardiac output, and arteriolar tone. It is not enough to show association under basal conditions. In addition to therapeutic strategies, we will need to discuss how hemodynamic parameters can best be monitored and assessed.

Up until the late 1980s, clinicians showed little interest in the innate properties of arteries and regarded them simply as passive conduits for the transport of blood. With the single exception of arterial spasm, evaluated comprehensively by Maseri and others in the 1970s (Maseri *et al.*, 1978; Maseri, 1991), disease of conduit arteries was considered a matter for surgeons and procedural cardiologists, with opportunity for bypass, endarterectomy, angioplasty, and stenting. Activity of arterial vessels amenable to medical therapy was considered to reside in small peripheral vessels, especially in the most distal arterioles. Therapeutic attention was on arterioles and peripheral resistance, even as a full expression of left ventricular load (Lang *et al.*, 1986; Levenson and Simon,

1988; Baughman, 2002; Yancy, 2008). This view is no longer tenable (Healy, 1990; O'Rourke *et al.*, 1993a; Nichols and Singh, 2002; Meredith *et al.*, 2008). Even though it is still put forward in prestigious journals by eminent authorities and premier institutions, strong criticism is coming to light (O'Rourke, 2010).

The Framingham study, together with SHEP and later trials described in Chapters 20 and 21, have deflected attention from diastolic to systolic and pulse pressure, to the cushioning function of the arterial system, and thus to arterial stiffness and wave reflection and to ways in which stiffness and reflection can be modified by therapy. The work of Vane, Moncada, and others (Furchgott, 1988; Vane *et al.*, 1990; Moncada, 1991; Moncada *et al.*, 1991; Romero, 1991; Vane and Henderson, 1992) leading up to award of the Nobel Prize in 1998, set off a revolution in assessment of the vascular endothelium as a modulator of arterial tone and caliber and as a key factor in many disease states (Cooke, 1999). Disorders of endothelial function in disease cause not just quantitatively different, but also directionally different responses to naturally occurring chemicals and to drugs. The discovery of nitric oxide (NO) as the endothelium-derived relaxing factor (EDRF) (Hutchinson *et al.*, 1987; Ignarro *et al.*, 1987; Palmer *et al.*, 1987; Radomski *et al.*, 1987; Furchgott, 1988; Cooke, 1999) drew new attention to nitroglycerin and the possibility that 'the long used nitrates may be regarded as substitution therapy for a failing physiologic mechanism' (Vane *et al.*, 1990). The phosphodiesterase 5 (PDE5) inhibitors, sildenafil, tadalafil, and vardenafil can be seen as the most logical therapy for endothelial dysfunction (Vallance and Chan, 2001), since they delay the breakdown of cyclic GMP, whose production is limited as a consequence of reduced nitric oxide availability in people with endothelial dysfunction. There has been an explosion of interest in endothelial dysfunction, and this is addressed as a separate chapter (Chapter 5). Work on

serotonin, its receptors, and its analogs has led to the introduction of the specific and highly effective drug sumatriptan for the relief of cerebral arterial dilatation and vascular headache in migraine, bringing relief to its sufferers and prosperity to its manufacturer (Brown *et al.*, 1991; Olesen, 1994; Bateman, 2000). The same can be said for PDE5 and its inhibitors (Cheitlin *et al.*, 1999). The PDE5 inhibitors sildenafil, tadalafil, and vardenafil are widely used for erectile dysfunction and in patients with pulmonary hypertension. A new emerging use is in refractory hypertension (Oliver *et al.*, 2010).

The understanding of relevant therapeutic strategies in cardiovascular disease not only depends on the recognition of physical properties of the arterial system, as described throughout this book, but also requires the recognition of a number of specific differences. Such differences relate to the type of disease condition, specific vascular bed, type of blood vessel (artery, arteriole, or vein), type of artery (predominantly 'elastic' or predominantly 'muscular'), and size of artery (Safar *et al.*, 1983; Levenson and Simon, 1988; Daemen and De Mey, 1995; Boutouyrie *et al.*, 1999; Boutouyrie *et al.*, 2000; Van Bortel *et al.*, 2001a). This field is young and a comprehensive account cannot be given at this stage, but the following specific differences can be set out.

Different disease conditions

When an artery is de-endothelialized (or denuded), acetylcholine causes constriction rather than dilatation of the vessel (Furchgott and Zawadzki, 1980). EDRF in the normal vessel has now been identified as NO (Hutchinson *et al.*, 1987; Ignarro *et al.*, 1987; Palmer *et al.*, 1987; Radomski *et al.*, 1987). Defective release of NO, with constriction rather than dilatation of coronary vessels, has been described in patients with coronary atherosclerosis or with known propensity to vasospasm (Ludmer *et al.*, 1986; Golino *et al.*, 1991; Egashira *et al.*, 1993a; Egashira *et al.*, 1993b; Park *et al.*, 1995). Improvements have been demonstrated with cholesterol-lowering drugs (Anderson *et al.*, 1995; Treasure *et al.*, 1995). Abnormalities in response to acetylcholine have been shown throughout the coronary arterial tree in the myocardial ischemic dog preparation (Nichols *et al.*, 1988a; Mehta *et al.*, 1989), in patients with variant angina (Park *et al.*, 1995), and in heart transplant recipients (Mills *et al.*, 1992). In patients with coronary atherosclerosis, different grades of abnormal constrictor response to acetylcholine have been shown in different segments of the same coronary artery (Penny *et al.*, 1995). A defective (constrictor) response to acetylcholine has been described in other arteries of patients with coronary atherosclerosis (Liao *et al.*, 1991; Uehata *et al.*, 1993), with hypertension (Linde *et al.*, 1990; Panza *et al.*, 1990), with cardiac failure (Kubo *et al.*, 1991; Drexler *et al.*, 1992; Katz *et al.*, 1992; Lüscher and Noll, 1993), and with aging (Egashira *et al.*, 1993c; Chauhan

et al., 1996). A defective response has also been demonstrated in asymptomatic subjects with different risk factors for coronary atherosclerosis (Vita *et al.*, 1990; Seiler *et al.*, 1993), with the degree of response being related to the cumulative total of risk factors (Celermajer *et al.*, 1994b; Celermajer, 1998); this suggests that ability of endothelium to release NO may precede the development of coronary atherosclerosis and may be involved in the pathogenesis of the condition, rather than simply being an effect of the condition in one vascular bed. Abnormalities in endothelial function established at cardiac catheterization have been shown to predict subsequent outcome (Halcox *et al.*, 2002b). Patients with classical and variant angina behave differently from normal subjects in the response to ergot alkaloids (Maseri *et al.*, 1978; Maseri, 1991), to serotonin (Golino *et al.*, 1991; McFadden *et al.*, 1991; Vandenbrink *et al.*, 1998), and probably to other drugs as well (see Chapters 5, 23, and 24).

Different vessels – arteries, arterioles, veins

Attention to this subject was originally directed by Safar, Laurent, Simon, and colleagues in Paris in the early 1980s (Safar *et al.*, 1983; Simon *et al.*, 1983b) and has been pursued relentlessly since (Safar *et al.*, 1986a; Safar *et al.*, 1986b; Safar *et al.*, 1987a; Safar *et al.*, 1987b; Levy *et al.*, 1988a; Levy *et al.*, 1988b; Benetos *et al.*, 1993a; Benetos *et al.*, 1993b; O'Rourke *et al.*, 1993a; Safar and O'Rourke, 1993; Gierd *et al.*, 1994; O'Rourke *et al.*, 2001; Safar, 2001; Laurent *et al.*, 2006; Safar and O'Rourke, 2006). This is an area of particular confusion in contemporary texts, even the best in cardiology and pharmacology.

While a distinction is usually made between venous dilators and those which act predominantly on the arterial side of capillaries, rarely is any distinction made between arteriolar and arterial dilators. This distinction is of fundamental importance for organ perfusion (especially the brain and heart) in the presence of fixed organic arterial disease. Dipyridamole, dobutamine, and adenosine dilate only arterioles and not the conduit arteries upstream (see Fig. 17.7). Through such arteriolar dilatation, they can increase flow to normal myocardium (as occurs in exercise) and thus divert (steal) blood from myocardium normally supplied through collateral vessels (see Chapter 17) (Becker, 1978; Sudhir *et al.*, 1993). This is the basis for using such agents for precipitation and identification of regional ischemia in radionuclide or echocardiographic stress testing. This is also the probable basis of regional cerebral ischemia or 'watershed infarction' when other predominantly arteriolar dilating drugs, such as prazosin and hydralazine, are used for the treatment of hypertension in patients with localized cerebral atherosclerosis (Hankey and Gubbay, 1987; McLaren and Danta, 1987; O'Rourke, 1990a; Carey *et al.*, 2000).

The best anti-hypertensive vasodilating drugs for use in such patients are the angiotensin-converting enzyme

(ACE) inhibitors, angiotensin receptor blockers and calcium channel antagonists, which possess both arterial and arteriolar dilating effects (O'Rourke, 1990a; Fitchett, 1993a; O'Rourke *et al.*, 1993a; Bank and Kaiser, 1998; Safar, 2001). Even calcium channel antagonists may occasionally induce coronary 'steal' and myocardial ischemia if their arteriolar dilating effects predominate over their arterial dilating effects. Nitrates do not cause coronary 'steal' or 'watershed' cerebral infarction because they dilate arteries, but not arterioles (see below). However, they are not regarded as effective anti-hypertensive agents because they do not dilate arterioles (they do not decrease peripheral resistance), except in relatively high doses. They are, however, perfectly logical and quite effective for reducing systolic pressure in isolated systolic hypertension (Duchier *et al.*, 1987; Stokes *et al.*, 1999; Stokes *et al.*, 2003). This and other points relating to nitrates and hypertension, coronary disease, and cardiac failure are dealt with in Chapters 14, 17, and 21.

The response of the coronary arterial vessels to nitroglycerin is characterized by potent dilatation of larger coronary arteries with minimal and only transient effects on coronary perfusion (Winbury *et al.*, 1969; Feldman *et al.*, 1981). It is this property of nitroglycerin that minimizes the increases in blood flow to regions of the myocardium that are already adequately perfused under basal conditions and thus prevents the development of the coronary steal phenomenon. *In vitro* (Sellke *et al.*, 1990) and *in vivo* (Kutz *et al.*, 1991; Wheatley *et al.*, 1994) studies of the coronary microcirculation have provided insight into this differential effect of nitroglycerin. These studies have shown that coronary microvessels less than 100 μm in diameter are relatively insensitive to nitroglycerin, whereas those over 100 μm in diameter are potently dilated by the drug (Fig. 17.7). This phenomenon seems to be related to an inability of the smaller coronary microvessels to metabolize nitroglycerin to its vasoactive metabolite, because putative metabolites of nitroglycerin such as NO or S-nitrosocysteine are capable of producing marked dilatation of these smaller vessels (Sellke *et al.*, 1990; Sage *et al.*, 2000). These studies suggest that (1) sulfhydryl groups are required for the conversion of nitroglycerin to its vasoactive metabolite; (2) the interaction between nitroglycerin and sulfhydryl residues is a stereospecific process, indicating either an intracellular mechanism or a membrane-associated enzymatic reaction; and (3) a lack of available sulfhydryl groups may be responsible for the lack of response of the smallest coronary arterioles to nitroglycerin. As explained in Chapter 4, relaxation of smooth muscle in the wall of a small artery causes it to become more distensible (Fitchett *et al.*, 1984; Westling *et al.*, 1984; Bank *et al.*, 1996; Bank and Kaiser, 1998), as well as to dilate (see Chapter 4). The dilated conduit artery shows greater pulsation with each beat (Fig. 26.1). Such dilatation and increased pulsation appears to be the basis of headache and the sensation of throbbing that are commonly associated with glyceryl

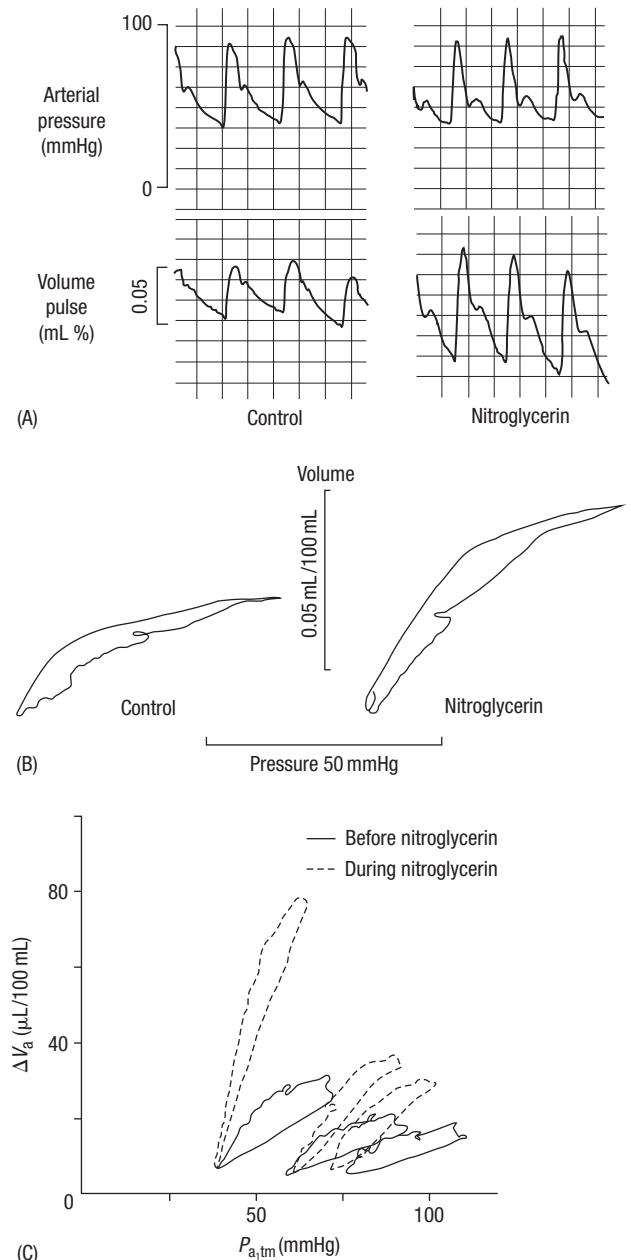


Figure 26.1 Effects of nitroglycerin on arterial distensibility in the human arm. (A) Arterial pressure (above) and plethysmographic volume pulse (below) before (left) and after (right) administration of 0.15 mg nitroglycerin sublingually. The volume pulse increases almost two-fold despite unchanged arterial pulse pressure. Reproduced from Fitchett *et al.* (1984). (B) Effects of nitroglycerin on the arterial volume-pulse loop. Reproduced from Fitchett *et al.* (1984). (C) Volume-pressure loops at three levels of transmural arterial pressure before and during nitroglycerin infusion in a normal subject. Reproduced from Westling *et al.* (1984)

trinitrate and other arterial dilating agents. These vascular changes have a considerable hemodynamic effect on the arterial pulse as well, through attenuation in amplitude of the reflected wave returning from peripheral arterioles

(Fig. 26.2) (Yaginuma *et al.*, 1986; Fitchett *et al.*, 1988a; Fitchett *et al.*, 1988b; Latson *et al.*, 1988; Kelly *et al.*, 1990a; Jiang *et al.*, 2002). Reduction in wave reflection is apparent in the decreased amplitude of the lowest-frequency components of impedance modulus in the ascending aorta, and in the decrease in amplitude of the secondary systolic peak of pressure in the ascending aorta and left ventricle (Figs 26.2, 26.3, 26.4, and 26.5). These effects on central systolic pressure are very substantial and, with nitroglycerin in low dose, are not accompanied by any decrease in peripheral resistance (Yaginuma *et al.*, 1986; Fitchett *et al.*, 1988a; Fitchett *et al.*, 1988b; Latson *et al.*, 1988; Jiang *et al.*, 2002), and little or no change in caliber or stiffness of the aorta and elastic arteries.

It is ironic that glyceryl trinitrate and isosorbide mono- and di-nitrate are classified as 'venous' dilators in most modern texts, even though their actions on conduit arteries have been known for over a century, as has the throbbing arterial headache attributable to arterial dilatation (Murrell, 1879). Such texts usually attribute anti-anginal action to venous dilatation or to coronary dilatation, yet generalized arterial dilatation with a reduction in wave reflection and left ventricular systolic pressure is a major and perhaps the predominant part of their action (Figs 26.1, 26.2, 26.3, 26.4, and 26.5) (Fitchett *et al.*, 1984; Westling *et al.*, 1984; Yaginuma *et al.*, 1986; Kelly, 1992; Jiang *et al.*, 2002). It appears that clinicians have been reluctant to describe a drug as an 'arterial' dilator unless it reduced mean or diastolic pressure in a peripheral artery or reduced peripheral resistance. (All these effects are manifestations of arteriolar rather than arterial dilatation.) Certainly, these drugs do not dilate arterioles in conventional clinical dosage, but their obvious beneficial effects on the heart are not due to, and cannot be gauged from, determinations of systolic and diastolic pressure in the brachial or radial artery or from peripheral resistance (see Figs 10.26 and 10.27) (Kelly *et al.*, 1990a; Simkus and Fitchett, 1990; Takazawa *et al.*, 1995b; Jiang *et al.*, 2002). However, in the first clinical description of nitroglycerin as an anti-anginal agent, Murrell (1879) (Fig. 26.6 page 541) showed the same characteristic changes in the radial pulse that can now be identified as due to decreased wave reflection from the lower body (see Figs 10.26 and 10.27).

This subject and its clinical implications are discussed further in the monograph *Arterial Vasodilation* (O'Rourke *et al.*, 1993a) and by Chen *et al.* (1995), Safar and Frohlich (1995), Grassi *et al.* (1995), Joannides *et al.* (1995), Delacrétaiz *et al.* (1995), Ting *et al.* (1995a), and van Bortel *et al.* (1995), and in proceedings of a Paris workshop on 'Structure and Function of Large Arteries' published in the October–December (2001) issues of *Hypertension* and by Miyashita *et al.* (2010).

Drugs are often referred to as 'vasodilators' without consideration of their major site of action, or the consequences of different actions on different vessels. The difference between arterial and arteriolar dilation is profound. Arterial dilators are used to treat angina, arteriolar

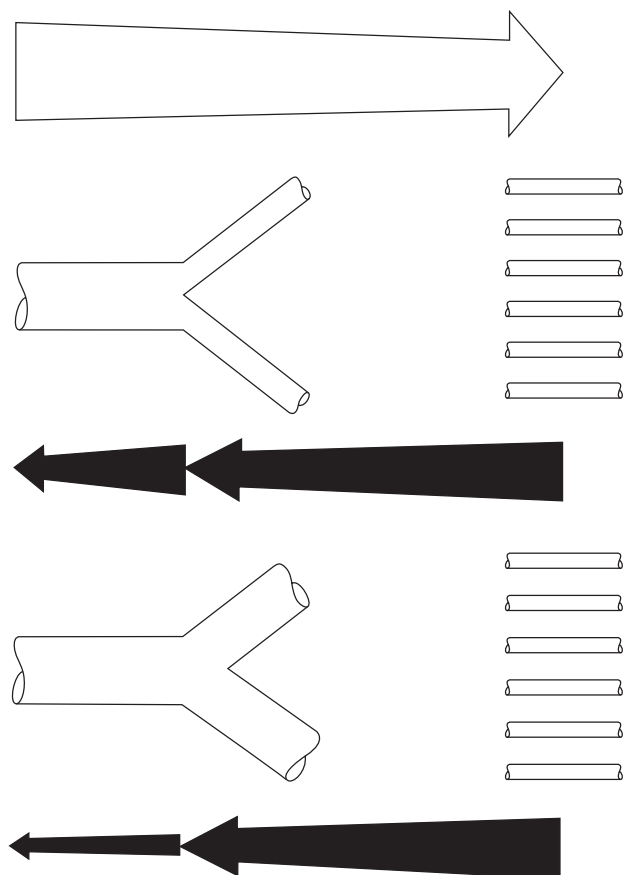


Figure 26.2 The proposed mechanism for reduction in peripheral reflection coefficient with glyceryl trinitrate. The forward-traveling wave is shown above, reflected at peripheral arterioles indicated at the right. Under control conditions (upper panel), the backward-traveling wave (dark arrow) is prominent on returning to the heart. With nitroglycerin, peripheral small arteries are dilated, causing a type of 'open-end' reflection at peripheral arterial branching points so that the amplitude of the backward-traveling wave (dark arrow, below) is markedly reduced on returning to the heart, despite unchanged reflection at the peripheral arterioles. Reproduced from Yaginuma *et al.* (1986)

dilators are used to provoke angina in simulated exercise stress testing. Nitroglycerin is an example of the former; dipyridamole is an example of the latter. Arterial dilation in the penis creates an erection; venous dilation causes penile flaccidity. The PDE5 inhibitors, such as sildenafil and tadalafil, promote arterial dilation without directly affecting the penile veins. Typically, anti-hypertensive agents such as ACE inhibitors, angiotensin receptor blockers (ARB), and calcium channel inhibitors have a mild effect only on arteriolar resistance vessels, and a greater effect on arteries; arterial dilation can explain the reduction in wave reflection that leads to reduction in systolic and pulse pressure. The propensity of calcium channel antagonists to cause peripheral edema is probably a consequence of their failure to dilate veins, so that with unopposed arterial and arteriolar dilation, capillary

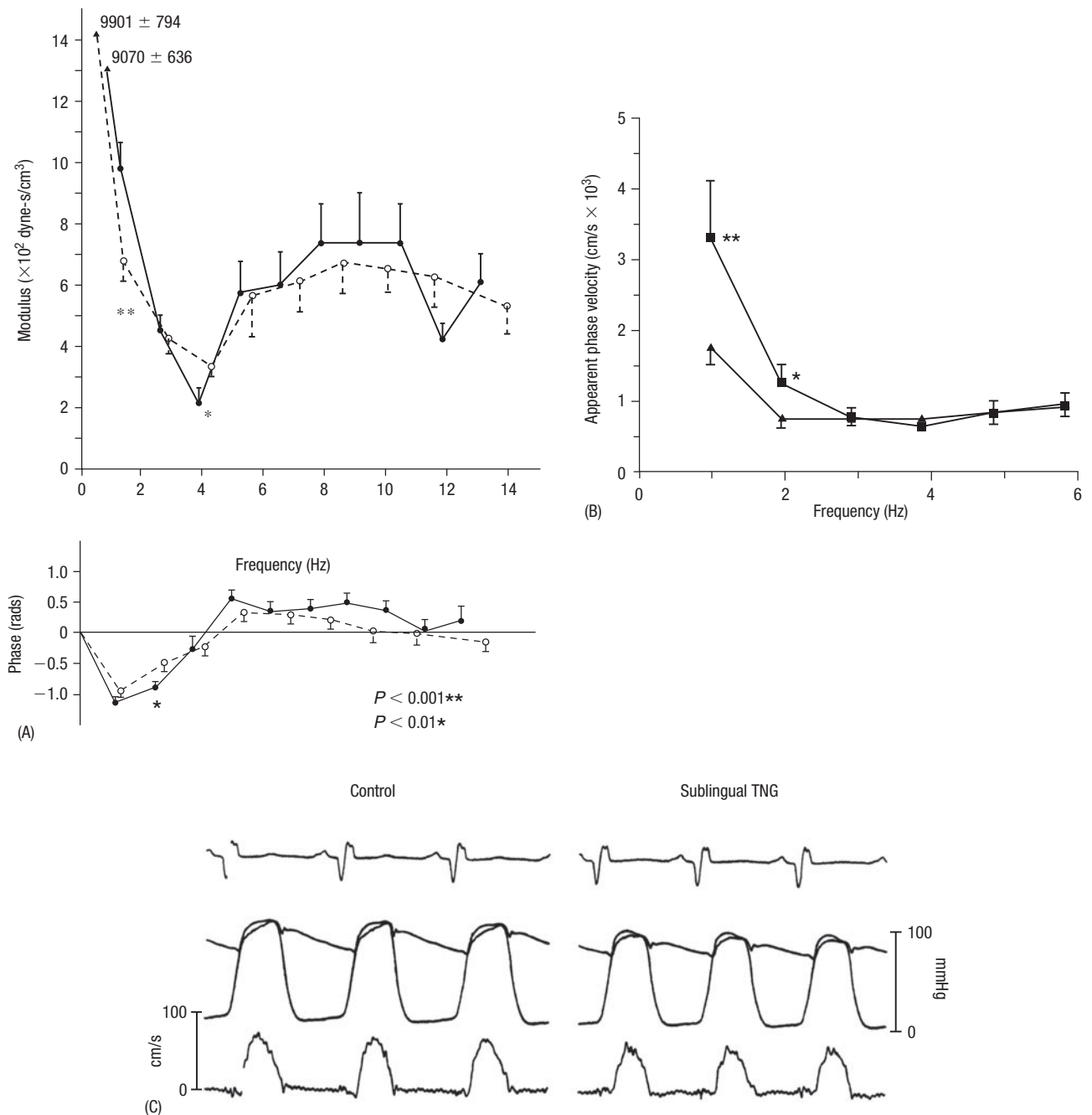


Figure 26.3 (A) Ascending aortic impedance in a group of patients before (●-●) and after (○-○) administration of sublingual nitroglycerin (TNG). The findings suggest a decrease in peripheral wave reflection without a change in arteriolar tone or in the properties of the proximal aorta. Reproduced from Yaginuma *et al.* (1986). (B) Effect of nitroglycerin on apparent phase velocity in the proximal aorta. The first and second harmonics were reduced by nitroglycerin. (■) Control; (▲) after nitroglycerin. * $P < 0.05$; ** $P < 0.01$. Reproduced from Fitchett *et al.* (1988a). (C) Representative left ventricular and aortic pressure tracings together with ascending aortic flow velocity in a patient with angina pectoris before (left) and after (right) administration of sublingual nitroglycerin 0.6 mg. Reproduced from Yaginuma *et al.* (1986)

hydrostatic pressure is increased. Benefits of lercanidipine over other dihydropyridines are attributable to a greater degree of venodilation, with maintenance of lower capillary pressure (Lacolley *et al.*, 1998; Messerli and Grossman, 2002). A simple overall schema for vasodilation (after Smith (1996)) is shown in Table 26.1.

Different conduit arteries – predominantly elastic and predominantly muscular

In Chapters 19 and 20, the point was made that the effects of aging and the long-term effects of hypertension are greatest on predominantly elastic vessels, such as the aorta

and carotid arteries, and least on muscular conduit arteries, such as the brachial, radial, and femoral. The very opposite holds for the effects of most clinically useful vasodilator agents. These have substantial and consistent effects on the large muscular arteries, with vasodilatation

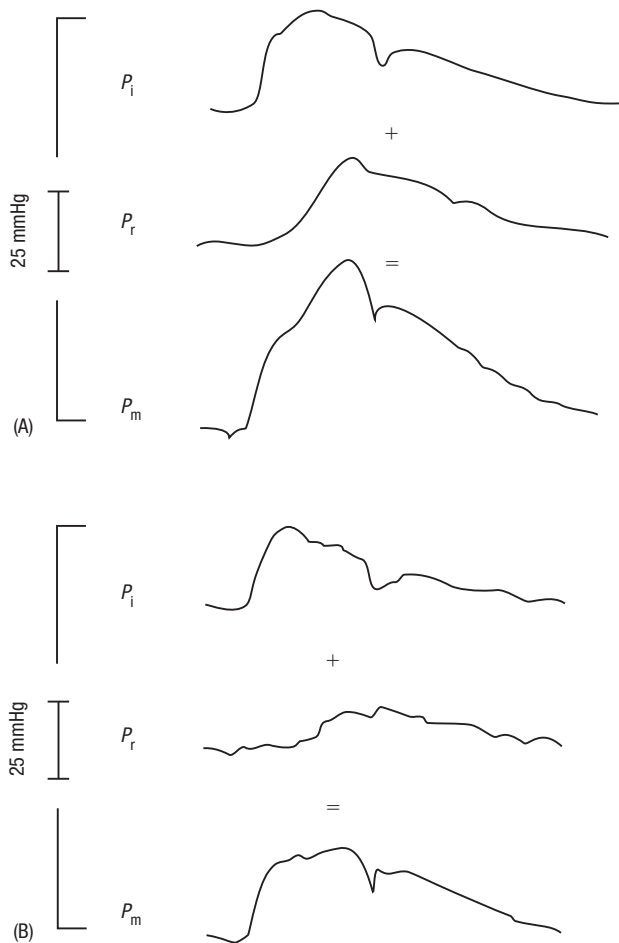


Figure 26.4 Explanation of aortic pressure wave contour. The measured aortic pressure wave (P_m) is made up of a forward-traveling (P_i) and backward-traveling (P_r) wave, the former generated by the heart and the latter caused by peripheral wave reflection. Nitroglycerin reduces the amplitude of (P_r) and thus offsets its most obvious effect – the secondary late systolic rise in measured aortic pressure. (A) Control; (B) after nitroglycerin. Reproduced from Yaginuma *et al.* (1986)

accompanied by a decrease in stiffness (see Fig. 26.1) (Safar *et al.*, 1983; Simon *et al.*, 1983b; Fitchett *et al.*, 1984; Westling *et al.*, 1984; Safar *et al.*, 1986a; Safar *et al.*, 1986b; Safar *et al.*, 1987a; Safar *et al.*, 1987b; Simon *et al.*, 1991b; Benetos *et al.*, 1993a; Benetos *et al.*, 1993b; Safar and

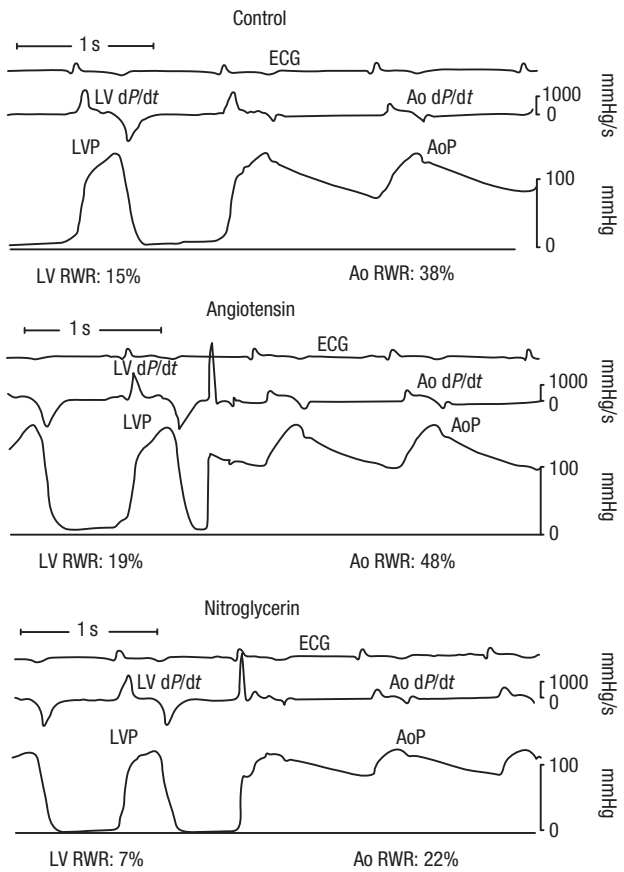
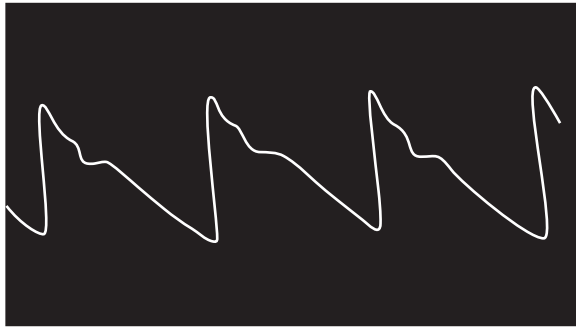


Figure 26.5 Pressure recordings in the left ventricle and ascending aorta of a 44-year-old man under normal conditions (top), after angiotensin infusion (middle), and after sublingual administration of nitroglycerin (bottom). Left ventricular (LV) dP/dt , first derivative of left ventricular pressure; LVP, left ventricular pressure; Ao dP/dt , first derivative of aortic pressure; AoP, ascending aortic pressure. Reflected wave ratios are given below each panel for the left ventricle (LV RWR) and ascending aorta (Ao RWR). Reproduced from Takazawa (1987)

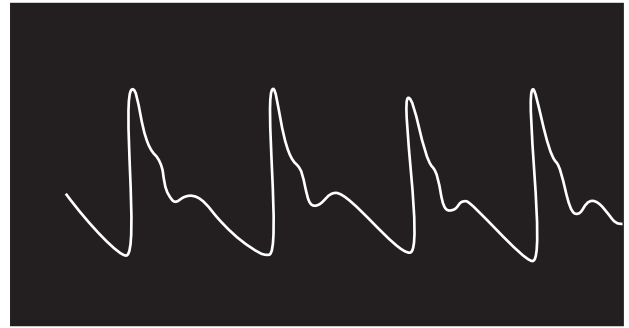
Table 26.1 Overall schema for vasodilation

	Arterial dilation	Arteriolar dilation	Venous dilation
Glyceryl trinitrate	+++	+/0	++
Nitroprusside	+++	++	++
Dipyridamole	0	+++	+
ACEI/ARB	++	+	+
CCB	++	+	0

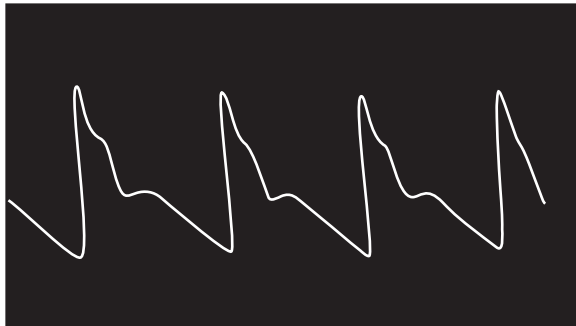
ACEI, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; CCB, calcium channel blockers.



No. 1 – Before dose.



No. 3 – Eight minutes after dose.



No. 2 – Two minutes after dose.



No. 4 – Nine minutes after dose.

Figure 26.6 Radial artery sphygmographs published by Murrell and showing in himself the effects of nitroglycerin as causing a progressive reduction in the late systolic shoulder of the pulse. Reproduced from Murrell (1879)

O'Rourke, 1993; Girerd *et al.*, 1994; Safar and Frohlich, 1995; Nichols and Edwards, 2001).

Carefully conducted studies have shown lesser effects on human carotid arteries where isobaric stiffness appears not to alter, but compliance does increase as a consequence of dilatation only (Boutouyrie *et al.*, 1995). Studies on the effects of vasodilator agents on the human aorta itself have been conflicting (Pepine *et al.*, 1979; Arcaro *et al.*, 1991a; Slørdahl *et al.*, 1991; Stratos *et al.*, 1992; Chen *et al.*, 1995; Stefanadis *et al.*, 1995a; Asmar *et al.*, 2001b; Hirata *et al.*, 2005b). Most human studies have shown no evidence of decreased aortic stiffness with the most powerful arterial dilator (nitroglycerin) beyond that caused by passive decrease in arterial pressure (Yaginuma *et al.*, 1986; Fitchett *et al.*, 1988a; Fitchett *et al.*, 1988b; Carroll *et al.*, 1991). However, in carefully conducted, cross-perfused animal studies where arterial pressure could be maintained constant, Latson *et al.* (1988) showed a very small decrease of aortic pulse wave velocity (PWV), but not of characteristic impedance with nitroglycerin in the dog aorta.

This topic was discussed extensively in *Arterial Vasodilation* (O'Rourke *et al.*, 1993a) and in Safar and O'Rourke (Safar and O'Rourke, 1993; Safar and O'Rourke, 2006). Views have been endorsed by other studies (Stokes *et al.*, 2003; Stewart *et al.*, 2006; Oliver *et al.*, 2010). In these, the point is made that use of vasodilator agents is a legitimate and effective strategy for reducing left ventricular hydraulic load, through an

increase in caliber and a decrease in stiffness of peripheral muscular arteries. However, the major effects of vasodilator agents are not on the stiffened aorta and elastic central arteries, but rather on the peripheral muscular arteries, and are manifest as a reduction in early wave reflection, with a decrease in the lowest-frequency components of impedance modulus (which may easily be overlooked as in Fig. 26.3A) and in left ventricular and aortic systolic pressure augmentation (Figs 26.3 and 26.7). Figure 26.7 shows published data from five studies on the effects of vasodilator agents on ascending aortic impedance moduli spectra. There were consistent decreases in the first two harmonic components of the spectra, but no consistent changes in characteristic impedance (gauged from moduli at higher frequencies). A fall of peripheral resistance (modulus at zero frequency) is not seen with nitroglycerin but is seen with agents such as nitroprusside, nifedipine, and hydralazine, which cause arteriolar dilatation. All, however, do show a reduction in the low-frequency components of the impedance modulus.

These findings indicate variable effects on arteriolar tone, little or no effect on properties of the proximal aorta, but substantial effects on diameter and distensibility of intervening muscular conduit arteries. The reduction in modulus of impedance at low frequencies is explicable on the basis of a reduction in wave reflection. This effect is of great therapeutic importance since it involves impedance values at the frequency of the first and second harmonics of the flow wave, which contain up to 80 percent of the

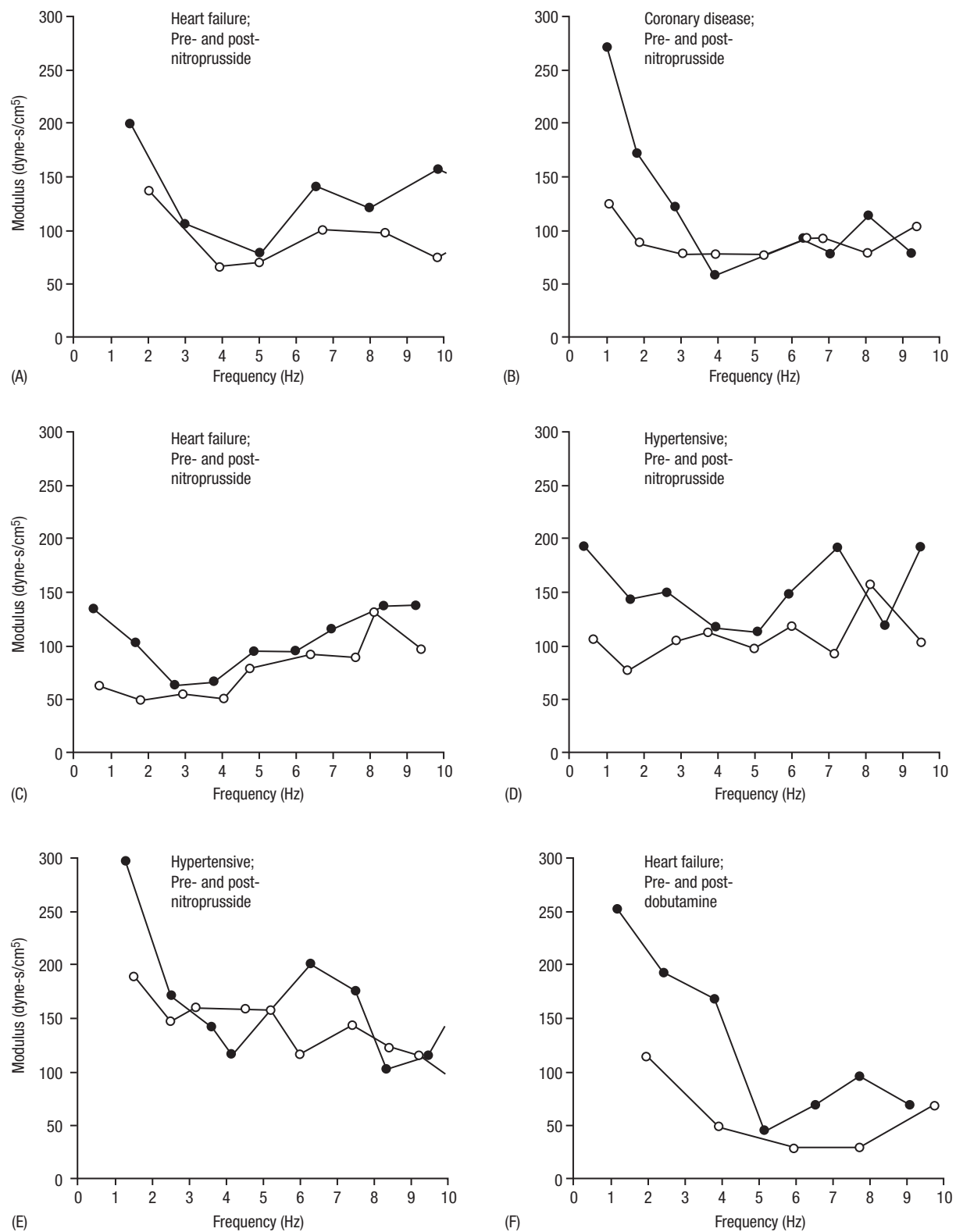


Figure 26.7 The effects of vasodilator therapy on aortic input impedance moduli spectra in humans. The reduction in low-frequency impedance moduli that was invariably seen can be attributed to decrease in wave reflection and is responsible for a reduction in left ventricular afterload. (●-●) Baseline, (○-○) vasodilator. Reproduced from (A) Pepine *et al.* (1979); (B) Gundel *et al.* (1981); (C) and (D) Merillon *et al.* (1984); (E) Ting *et al.* (1986); (F) Binkley *et al.* (1990). See also Figs 26.3A and 12.32

energy of the flow pulse. The impedance changes below 3.0 Hz in Figs 26.3 and 26.7 are therefore of major importance in the reduction of left ventricular load. This is stressed in Chapter 12.

Different-size muscular arteries

It has long been known that nitroglycerin has progressively greater vasodilating actions on the smaller, more distal coronary arteries (Feldman *et al.*, 1981; Sellke *et al.*, 1990), while it has virtually no effect on the aorta itself or peripheral arterioles (see above and Chapter 17). It is now recognized that muscular artery dilation occurs throughout the body and is not an exclusively coronary phenomenon. The mechanism has been described (Kutz *et al.*, 1991; Sellke *et al.*, 1992; Wheatley *et al.*, 1994; Sage *et al.*, 2000) in terms of greater availability of sulfhydryl donors in the smaller arteries and thus greater activation of guanylate cyclase by nitroglycerin leading to greater vasodilatation (see Fig. 17.7). This area warrants closer study in the future and could provide new and different drugs designed to reduce wave reflection when the aorta is stiffened (Oliver and Webb, 2003), while preserving perfusion pressure and arteriolar tone, thus avoiding 'steal'.

MUSCULAR ARTERIES OF DIFFERENT VASCULAR BEDS

It is well known that arteriolar vessels of different organs have different sensitivity to sympathetic nerve stimulation and to catecholamines (O'Rourke *et al.*, 1993a; Daemen and De Mey, 1995). This is the basis for the maintenance of cerebral and myocardial perfusion in hypotensive states when perfusion to the kidneys, splanchnic area, and periphery is markedly reduced. The large muscular arteries share in such regional and organ differences. A notable example relates to serotonin receptors in the coronary and cerebral muscular arteries. In the coronary circulation, these are predominantly of the 5HT₂ type (Connor *et al.*, 1989), whereas in the cerebral circulation they are predominantly 5HT₁ type (Olesen, 1994; Raskin, 1994). Serotonin relieves migraine headaches caused by cerebral vasodilatation, but with an unacceptable incidence of side effects attributable to excessive vasoconstriction elsewhere. The serotonin analog sumatriptan has been 'designed' as a stimulant of 5HT₁ receptors only, with little affinity for other receptors (McIntyre *et al.*, 1992; Bateman, 2000). It is remarkably effective in treatment of migraine headache and only rarely causes symptoms attributable to coronary artery constriction. As discussed earlier, coronary constriction is, however, apt to occur in patients with impaired endothelial function and with known coronary atherosclerosis or predilection to spasm (Golino *et al.*, 1991; McFadden *et al.*, 1991; Vandenbrink *et al.*, 1998).

Differences in blood pressure at different sites

This issue is referred to in many parts of this book, but it is of particular importance in therapy, since evaluation in humans has traditionally been – and still is – from measurement of blood pressure in the brachial artery. Like any peripheral artery, pulse pressure and thus systolic (peak) and diastolic (nadir) values are different from those in the ascending aorta (see Figs 10.26, 10.27, and 26.8). Diastolic pressure is less affected than is systolic by change in pressure pulse contour and pulse pressure with radial diastolic and mean pressure usually within 1 mmHg of aortic (Pauca *et al.*, 1992; Pauca *et al.*, 2001). Elsewhere, it is shown that differences between aortic and brachial systolic pressure are substantial (20 mmHg or more) under control conditions in young healthy adults at rest (see Fig. 10.16), and are extreme (up to 80 mmHg) during exercise (see Fig. 10.14), while in older human subjects they are appreciable at rest but greater still (by 20 mmHg or more) during the use of vasodilator therapy (see Figs 10.26, 10.27, and 26.8) or in shock or heart failure. Newer arterial dilating anti-hypertensive agents (ARBs, ACEIs, calcium channel blockers (CCB)) show greater ability to reduce central compared to peripheral systolic pressure, than older drugs (β -blockers, diuretics) (Miyashita *et al.*, 2010). This can explain their greater benefit in reducing incidence of new diabetes (Elliott and Meyer, 2007), and cardiovascular events (Williams *et al.*, 2006; de Luca *et al.*, 2004). Evaluation of anti-hypertensive drug effect from a reliance on brachial systolic pressure can underestimate the benefit by 5–20 mmHg (see Figs 10.26, 10.27, and 26.8) (Kelly *et al.*, 1990a; Simkus and Fitchett, 1990; Takazawa *et al.*, 1995b). These effects should not be ignored or dismissed, but they are in many current articles in premier cardiological journals often on the authority of a single article (Borow and Newburger, 1982) that showed a higher correlation (0.98) and lower standard deviation between indirect brachial artery pressure and direct aortic pressure than has ever been claimed for indirect and direct brachial pressures. Brachial systolic pressure cannot be considered identical to that in central arteries (see Chapters 28 and 30).

Allowance can be made for pressure wave amplification between central aorta and brachial or radial artery, through consideration of pressure wave contour (see Figs 10.14, 10.16, and 26.8). The method we favor (O'Rourke *et al.*, 2001; Vlachopoulos and O'Rourke, 2001) is based on the relatively constant transfer function for pressure between the ascending aorta and the brachial or radial artery. Other methods are available and are discussed further in Chapter 30. Use of the carotid pulse contour and of other methods are also discussed in Chapter 30. Initially, most attention in the United States and in Europe was directed at analysis of the carotid pulse contour. The major problem is calibration to brachial cuff pressure, and to brachial tonometric waveforms (see Chapter 30) (O'Rourke and Adji, 2010b).

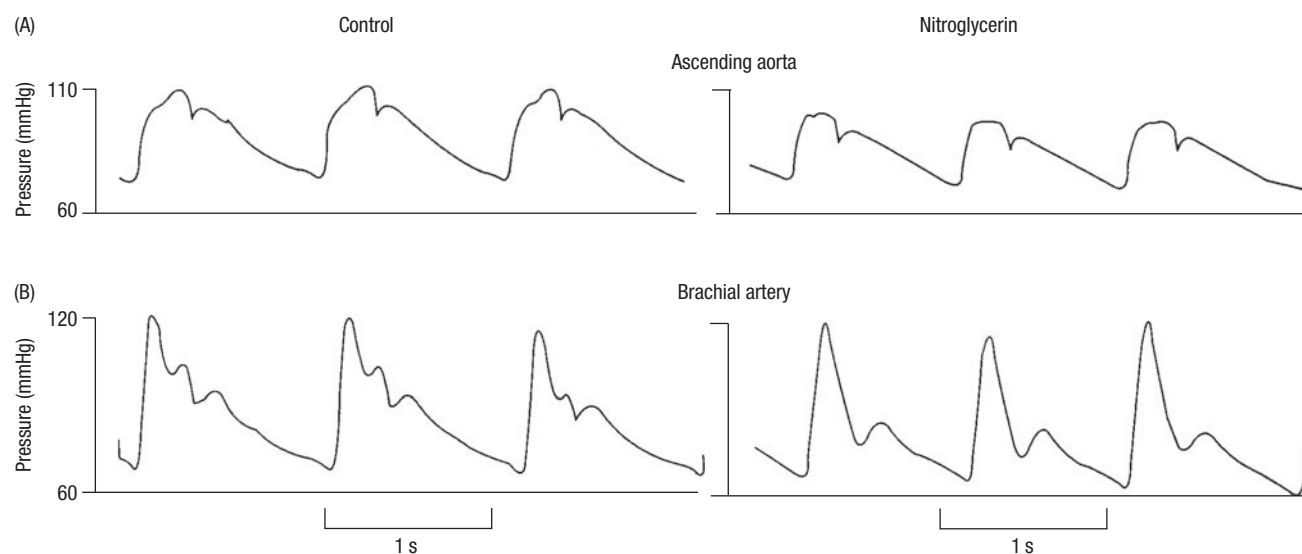


Figure 26.8 High-fidelity pressure waves (Millar micromanometer) recorded directly in the ascending aorta (A) and brachial artery (B) of a 54-year-old man under control conditions (left) and 5.0 minutes after 0.3 mg sublingual nitroglycerin (right). Reproduced from O'Rourke *et al.* (1989a)

While the issue (of pressure wave amplification in peripheral arteries) is of particular importance in humans, it is also relevant to the assessment of anti-hypertensive therapy as trialed in experimental animals. Normal (WKY) rats show considerable amplification of the pulse between the aorta and tail, as do hypertensive (SHR) rats while treated with the calcium antagonist nifedipine, but do not under normal conditions. Hence, the effect of this type of vasodilator drug on aortic pulse pressure is substantially underestimated by the conventional measurement of pressure in the tail (Tsoucaris-Kupfer *et al.*, 1993; Mitchell *et al.*, 1996; Griffin *et al.*, 2003). This is the reason for measuring aortic pressure directly now in acute and chronic mouse experiments.

Conduit function: surgical and procedural

This book cannot serve to describe the many surgical and procedural strategies for improving blood flow in the presence of arterial stenosis. Principles are set out in Chapters 17 and 23. Readers are referred to the many excellent sources that deal with this subject, including the major textbooks (Libby *et al.*, 2008; Libby *et al.*, 2009).

MEDICAL THERAPIES

Hemodynamic principles related to medical therapy of vascular stenoses are dealt with in Chapter 17, with reference to coronary flow. Here attention is directed to the fact that arteries are very low resistance pathways, with high tolerance to occlusive disease. The resistance presented by a stenosis will not provide any limitation to

blood flow until the stenosis approaches 70 percent of the arterial diameter (Feldman *et al.*, 1979a; Feldman *et al.*, 1979b) and then only when the organ is active. The reason for this is that arteriolar resistance is so high as to completely dominate total vascular resistance until a stenosis in a conduit artery reaches this degree of magnitude. Once such a degree of stenosis is attained, however, further narrowing (as by platelet deposition on the intima, swelling within the intima or increased medial tone) will readily restrict flow and bring on symptoms. These points are discussed further in Chapter 17; they explain the long latent period during the development of atherosclerosis and the relatively rapid progression of symptoms once these have commenced. The principles also explain how, in an eccentric organic stenosis, relaxation of smooth muscle in the relatively normal segment of the wall, or modest plaque reduction with lipid-lowering therapy, can relieve symptoms by decreasing arterial resistance, even when only a minor increase in caliber is produced (Brown *et al.*, 1993). The crucial issue is that resistance varies with the inverse of (radius)⁴ (Poiseuille's law; see Chapter 2). Any tiny change in diameter of an already critically narrowed artery can cause a marked change in resistance.

Another important therapeutic issue relates to selection of a vasodilator agent (for the treatment of hypertension or cardiac failure) in the presence of arterial stenoses and occlusions. This is referred to earlier in this chapter and in Chapter 17. When arterial stenoses become critical and restrict flow, collateral vessels develop to bridge arteries beyond major stenoses to other, less-affected vessels. 'Steal' can develop with decrease in collateral flow when agents with predominant arteriolar dilating properties are used. In patients with known cerebral or coronary artery disease, such agents (hydralazine,

prazosin, and terazosin) should be used with caution, and preference given to combined arterial and arteriolar dilating agents, such as ACE inhibitors, angiotensin receptor blockers, calcium channel antagonists or to nitrates (Hankey and Gubbay, 1987; McLaren and Danta, 1987; Jiang *et al.*, 2002).

Cushioning function

Emphasis is directed throughout this book to the disturbances that result in humans with aging and in hypertension as a consequence of arterial stiffening. This is ubiquitous in humans from the first decade on, and is probably the most common cause of cardiac decompensation in the elderly (O'Rourke and Hashimoto, 2007). In youth, arterial distensibility is high and wave reflection ideally timed in diastole to enhance coronary blood flow and cardiac function. In adults, wave reflection returns early to the heart during systole, as a consequence of aortic stiffening and increased pulse wave velocity, and impairs cardiac function. Hence, reduction in wave reflection is an important therapeutic strategy in adult humans with hypertension, angina pectoris, or cardiac failure (see Figs 26.3, 26.4, 26.5, 26.6, 26.7, and 26.8). Therapeutic strategies for treatment of angina pectoris, hypertension, and cardiac failure must also consider slowing of arterial degeneration and thus alleviation of its ill effects on aortic stiffness, measured as PWV, and timing and intensity of wave reflection, measured by aortic AIx (Sakurai *et al.*, 2007; Cockcroft, 2009).

Medical strategies

Dilatation and stiffening with age and hypertension are attributable to fatigue and fracture of elastin fibers in the media of the aorta and other predominantly elastic arteries (see Chapters 4 and 19). Because of a defect in elastin, this process is accelerated in Marfan syndrome (see Chapter 24). Because extent of pulsation is greatest in the ascending aorta, it is affected first. Fatigue is a consequence of cumulative cycles of arterial expansion and is thus related to the extent of dilatation on the one hand, and to age and heart rate on the other. Progression of degeneration has been slowed in patients with Marfan syndrome through the use of β -blocking agents and can be attributed to a reduction in pulse pressure, mean pressure, arterial dP/dt and heart rate (see Chapter 24) (Shores *et al.*, 1994). Progression of arterial degeneration with age and hypertension can similarly be slowed through use of anti-hypertensive therapy combined with β -blockers or other drugs that slow heart rate. Prevention of elastin fracture and medial degeneration appears to be impossible. From an engineering viewpoint, one would expect this to progress as long as the heart continues to beat. One's best hope is to slow the process (see Chapter 19).

A logical goal of medical therapy is to decrease arterial stiffness. When one views microscopically the disorganized structure of the adult aorta (Fig. 19.8), one wonders how this could ever be possible. In peripheral muscular arteries, relaxation induced by nitrates, ACE inhibitors, and calcium antagonists leads to transfer of stress to the elastin fibers so that the dilated artery becomes more distensible (see Chapter 3, and Figs 20.8 and 26.1). When the artery is already degenerate and dilated, further muscular relaxation may not have the same effect, and stiffness may not change appreciably. However, compliance may increase because the artery is dilated – it can accommodate a greater aliquot of blood for a given pressure change (Boutouyrie *et al.*, 1995; Boutouyrie *et al.*, 1999; Boutouyrie *et al.*, 2000), even though stiffness of its wall is unaltered.

There seems no real prospect for direct drug-induced improvement of aortic stiffness in adult humans. Certainly, reports to the contrary have been published. These usually measured distensibility from the unrealistic Windkessel model, did not take full account of the decrease in arterial distension and the passive effects of pressure reduction on aortic stiffness, or used pressure change from a different (usually brachial, sometimes carotid) site. The majority of reports show little or no direct effect of drugs on aortic stiffness in adult humans – when this is measured as elastic modulus, pulse wave velocity, or characteristic impedance. Reports of Mitchell *et al.* (Mitchell *et al.*, 2001; Mitchell *et al.*, 2002; Mitchell *et al.*, 2003) and Mitchell and Izzo (2003) using carotid tonometry do not agree with those studies where aortic pressure and flow were measured directly (Pepine *et al.*, 1979; Gundel *et al.*, 1981; Merillon *et al.*, 1984; Ting *et al.*, 1986; Yaginuma *et al.*, 1986; Binkley *et al.*, 1990; Carroll *et al.*, 1991) nor previous studies in experimental animals that we have reported (see Chapter 12) nor the study of Latson *et al.* (1988) which found no change in aortic characteristic impedance nor aortic diameter with high-dose nitroglycerine. Problems in applying Mitchell's indirect analysis are set out in Table 12.1. Many of these apply to other studies. Our deliberate approach is to keep analyses as simple as possible and assumptions as few as possible. In a methodologic consensus document, Laurent *et al.* (2006) pointed out deficiencies of multiple methods and recommended use of just PWV and AIx; authors noted that favorable outcome studies were available with these two methods, and, surprisingly not for aortic or carotid distensibility calculated from diameter/pressure change.

While active reduction of aortic stiffness may be an impossibility, passive reduction in stiffness always occurs with reduction in arterial pressure (see Nichols and Edwards, 2001). Furthermore, arterial vasodilators do increase the caliber and distensibility of the more peripheral muscular arteries. This provides the main therapeutic effect of such drugs through a decrease in wave reflection returning to the heart from major peripheral

reflecting sites (the arterioles) (see Figs 12.31, 12.32, 14.23, 20.9, 26.3, 26.4, 26.5, 26.6, 26.7, and 26.8), together with branching points upstream.

The effects of vasodilator agents on the properties of large arteries are best illustrated in studies of ascending aortic impedance as left ventricular hydraulic load (see Fig. 26.7). Decrease in wave reflection, and delay of wave reflection, cause a reduction of impedance modulus at those frequencies (up to 3.0 Hz) which normally contain most of the energy of the left ventricular ejection wave (see Fig. 11.13). The consequence of this is a decrease in amplitude of aortic pulse pressure, a decrease in late systolic pressure augmentation (see Figs 26.3, 26.5, and 26.8) and thus a substantial decrease in systolic pressure and in myocardial oxygen demand. These effects add to the benefit derived from the reduction in mean pressure but may occur, as with nitroglycerin, in the absence of any decrease in mean arterial pressure or peripheral resistance (Yaginuma *et al.*, 1986; Fitchett, 1993a; Fitchett, 1993b).

Figure 26.9 from Jiang *et al.* (2002) shows the limitations in assessment of nitroglycerin action from conventional measurement of cuff brachial artery pressure, but the insight provided by pulse waveform analysis. From cuff measures, there were no changes in mean or diastolic pressure with a six-fold increase in dose, but modest reduction in brachial systolic and pulse pressure with the two higher doses. In contrast, waveform analysis showed consistent and dose-dependent changes in aortic systolic, pulse, augmented, and mean pressure, and in LV ejection duration. These findings were interpreted to show evidence of effect on arterial dilation from the lowest dose of 0.1 mg/hour, of venous dilation from the intermediate dose of 0.2 mg/hour and arteriolar dilation at the highest dose of 0.6 mg/hour (Fig. 26.10).

This approach has been used to study the magnitude and time course of interaction between nitroglycerin and sildenafil (O'Rourke and Nichols, 2002), and to show value of the combination in treatment of refractory hypertension (Oliver *et al.*, 2010).

Current guidelines recommend that nitrates should not be used within 24 hours of a PDE5 inhibitor because of the risk of extreme vasodilatation and hypotension (Cheitlin *et al.*, 1999). Such guidelines are based on the use of bolus sublingual doses of nitroglycerin (up to 0.6 mg) and on assessment from measurement of brachial pressure by cuff sphygmomanometer. When appropriately low doses of nitroglycerin are used (by intravenous infusion or by transdermal patch) pulse waveform analysis shows evidence of a slight potentiation of nitrate effect by

sildenafil up to 24 hours after dosing; similar mild potentiation by the longer-acting tadalafil is apparent in Fig. 26.11.

In the presence of heart failure, a decrease in wave reflection may affect flow to a greater degree than pressure, increasing the duration of systole and increasing flow in late systole, thus causing an increase in cardiac stroke volume (see Chapter 14, and Figs 14.10, 14.11, 14.12, and 14.13) (Pepine *et al.*, 1979; Carroll *et al.*, 1991; Maruyama *et al.*, 1993; Miyashita *et al.*, 1993; Sakai *et al.*, 1993; Laskey *et al.*, 1995; Uebaba *et al.*, 2002).

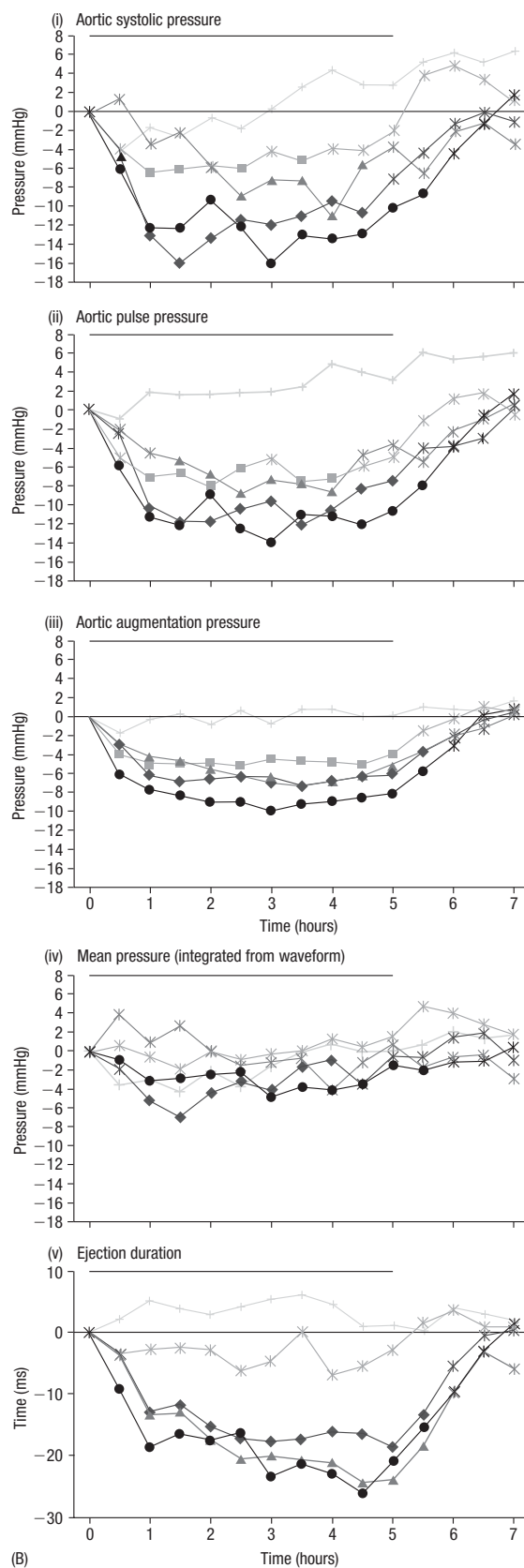
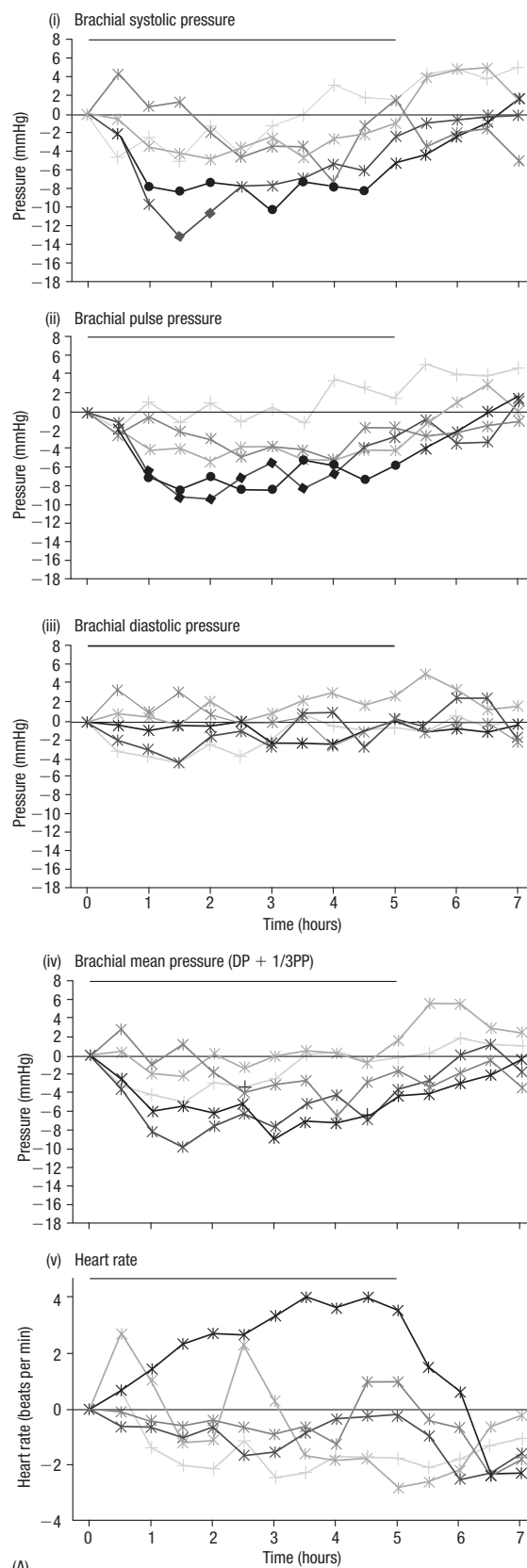
These considerations give insight into and explain the logic of therapies currently used for the treatment of hypertension, angina pectoris, and cardiac failure. They help to separate the effects of change in peripheral resistance and mean pressure from the effects of change in arterial stiffness and pulsatile pressure. They show that benefits may be gained through a change in pulsatile pressure without the necessity for a change in mean pressure.

This whole strategy is based on knowledge of how arterial stiffening distorts the favorable effects of wave reflection so that, with aging or disease, this becomes highly unfavorable. The strategy is based not on direct reduction of aortic stiffness, but on delay in and reduction of wave reflection through the action of drugs on peripheral muscular arteries and arterioles (see Figs 26.2, 26.3, 26.4, 26.5, 26.6, and 26.7).

Figure 26.9 shows the effects of small doses of nitroglycerin on aortic pressure waveform components. Only with the highest dose of 0.6 mg/hour were effects of reduced mean pressure apparent. However, clinically useful effects, attributable to conduit arterial vasodilation (Bank and Kaiser, 1998), were seen with doses as low as 0.1 mg/hour (Jiang *et al.*, 2002).

Views set out here are challenged by findings presented by Mitchell *et al.* (2002) for the vasopeptidase inhibitor omapatrilat; these describe a specific action on the proximal aorta, with increase in distensibility (reduction in characteristic impedance). When one views at surgery or autopsy the degenerate dilated aged proximal aorta, this seems to be a pious hope rather than a realistic prospect. Findings of Mitchell *et al.* (2002) for omapatrilat are out of line for related drugs, including ACE inhibitors, and lead on to highly unusual interpretations on aortic changes with age (Mitchell *et al.*, 2003; O'Rourke *et al.*, 2004a; O'Rourke and Nichols, 2004b). For omapatrilat, minor benefit was found for reduced wave reflection, and all benefit was attributed to improved distensibility

Figure 26.9 Effects of nitroglycerin (GTN) in transdermal patch (Nitrodur®), delivering zero (placebo, –) or (0.104, ■), (0.205, ●), (0.416, *), and (0.625, +) mg/hour over a 5-hour period in 46 patients, with change shown in relation to control values. (A) Conventional measures of brachial systolic, diastolic, and pulse pressure, together with mean pressure calculated as diastolic + 1/3 pulse pressure. (B) Indices determined using SphygmoCor process to calculate aortic systolic, diastolic, integrated mean, and augmented pressure, together with left ventricular (LV) ejection duration. The patch was applied immediately after taking baseline control values (time zero) and removed 5 hours later. Line above each graph indicates the period during which the GTN patch was applied. Closed symbols on a curve indicate times of significant difference from baseline. Reproduced from Jiang *et al.* (2002)



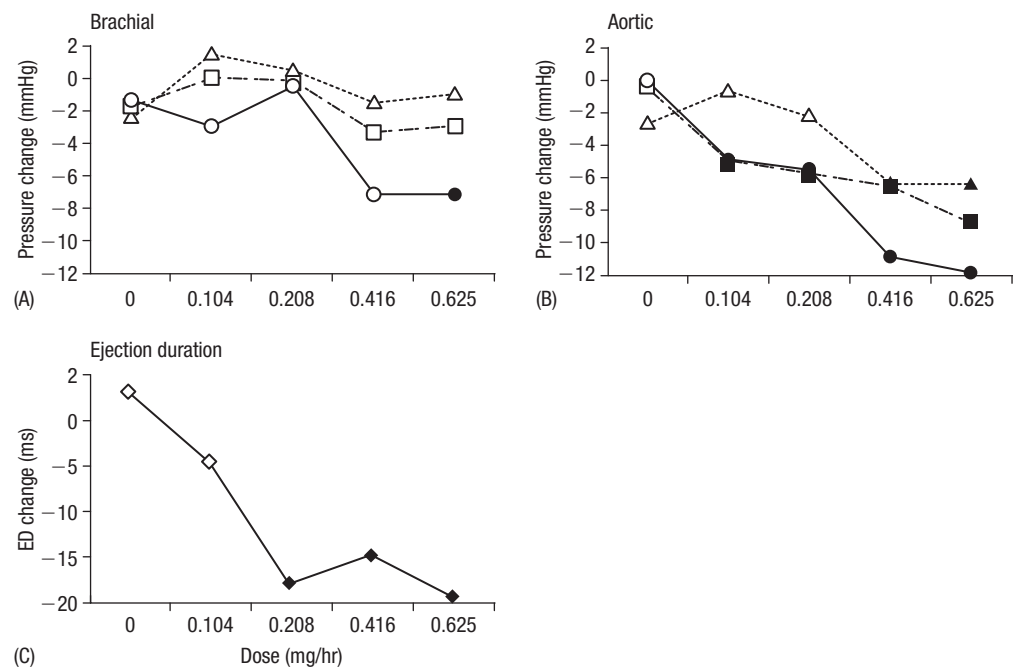


Figure 26.10 Averaged pressure change (between 0.5 and 5.0 hours) as a function of GTN dose for: (A) brachial systolic pressure ($\circ$), brachial diastolic pressure ($\square$), and brachial mean pressure ($\triangle$); (B) estimated ascending aortic systolic pressure ($\circ$), augmented systolic pressure ($\square$), and mean pressure integrated from the pressure waveform ($\triangle$); and (C) ejection duration ($\diamond$). Solid symbols represent values where $P < 0.01$. For brachial systolic pressure ($\bullet$), $P < 0.01$. For aortic systolic ($\bullet$) and aortic augmented pressure ($\blacksquare$), $P < 0.0001$ at all doses. For aortic integrated mean pressure ($\blacktriangle$), $P < 0.001$ at dose > 0.4 mg/h. For left ventricular ejection duration ($\blacklozenge$), $P < 0.001$ at doses > 0.2 mg/h.

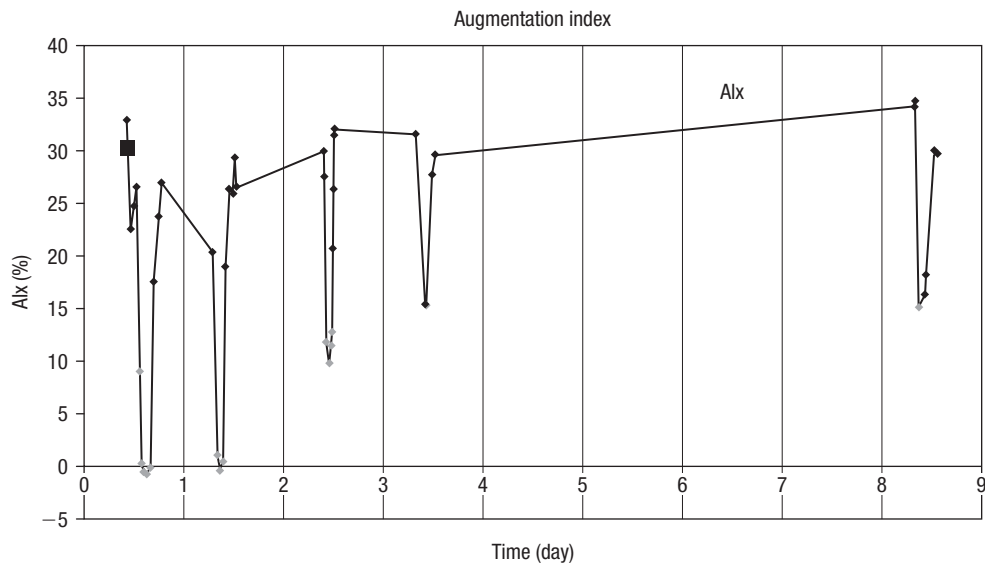


Figure 26.11 Change in aortic augmentation index (Alx), calculated from radial artery tonometry in a 66-year-old man on successive days. Tadalafil (square box) was given initially and caused a small decrease in Alx which promptly returned to control level after 2 hours. Application of a nitroglycerine patch delivering 0.2 mg/hour at this time caused Alx to fall substantially, then revert to baseline when the patch was removed. Application of the patch again 18 hours later caused a similar decrease in Alx. This effect of nitroglycerine was less when the nitroglycerine patch was applied 2, 3, and 8 days after tadalafil. Changes in measured brachial systolic and diastolic pressures were insignificant after tadalafil, or after application of the patch.

(reduced characteristic impedance) of the proximal aorta. Findings may have a simple technical explanation (see Chapter 30 and Fig. 30.44) and relate to capacity to identify the timing of peak aortic flow and the return of wave reflection from the indirectly recorded carotid pressure waveform. Even very experienced researchers can have problems with artifact in the indirectly recorded carotid pressure waveform (Gatzka *et al.*, 2001a; Gatzka *et al.*, 2001b; Kingwell and Gatzka, 2002). It took a very detailed preparation for Latson *et al.* (1988) in Sagawa's laboratory to control for all variables in studying the effects of a powerful arterial dilator (nitroglycerine) in high dose on aortic characteristic impedance. The absence of any reduction in this study contrasts with the 12 percent reported by Mitchell *et al.* (2002) for omapatrilat. The present weight of evidence from multiple sources supports the view that we and others have proposed (O'Rourke *et al.*, 1993a; O'Rourke *et al.*, 2001; Klingbeil *et al.*, 2002; Mahmud and Feely, 2002; Nichols and Singh, 2002; Stokes *et al.*, 2003; Adji *et al.*, 2011).

Vaitkevicius *et al.* (2001) and Kass *et al.* (2001) considered a different method for altering properties of the aorta. They studied the effect of the AGE cross-linked breaker ALT711. This drug cleaves collagen fibers in the degenerate aorta. It remains to be seen if such an effect can stimulate the development of new elastic fibers sufficient to retain integrity of the aorta, as well as to improve distensibility. The data provided to Kass *et al.* (2001) for analysis did not permit separation into local and remote (reflection) effects of the drug. The measure of total arterial compliance was used; this includes both aortic distensibility and wave reflection. Investigations continue into this and related novel drugs.

Surgical strategies

Vascular conduits are widely used for the replacement of diseased aortic segments. Long conduit grafts are also used to bypass diseased segments, as in aortic dissection when it is impracticable to remove the diseased aorta. All these grafts are of stiff Tygon or Dacron material and are designed to act as conduits only. Studies of such conduits have proceeded to examine the effects of altered distensibility. It has been shown that these can favorably affect cardiac performance by reducing aortic systolic pressure and increasing aortic systolic flow (Morita *et al.*, 1991; Kelly and Kass, 1992a; Kelly and Kass, 1992b; Maruyama *et al.*, 1993). It is only a matter of time before surgical (elastic) prostheses are introduced for patients with extensive stiffening and dilation of the aorta and with uncorrectable cardiac problems (angina or heart failure) arising therefrom. Design needs to consider material fatigue and the need to withstand repetitive pulsations of up to 40 million per year over many years. This issue is currently being tackled in Sydney by young cardiovascular surgeons Iliopoulos and colleagues (Iliopoulos, 2006) and

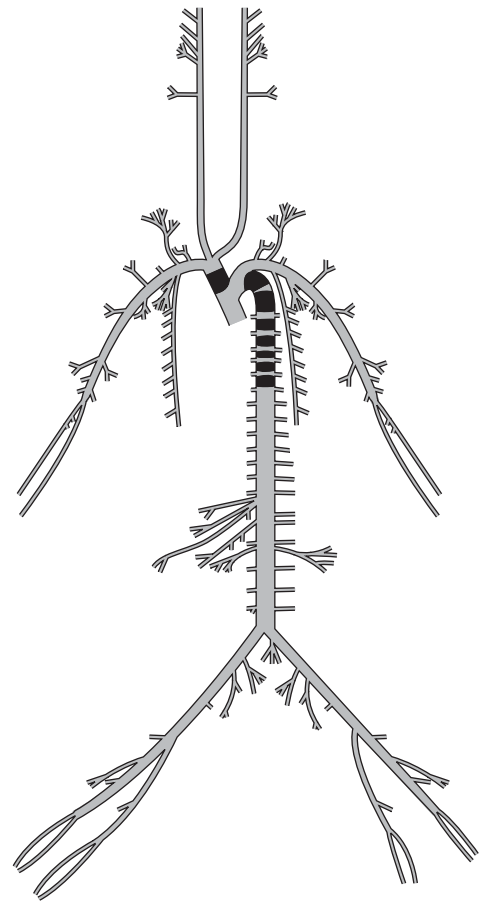


Figure 26.12 Diagram of the arterial system of a dog, showing how aortic stiffness was increased by application of rigid Lucite ferrules to the descending thoracic aorta and brachiocephalic artery. From O'Rourke (1967b)

Byrom (Byrom *et al.*, 2010). The opposite (stiffening of the aorta by application of rigid Lucite ferrules (Fig. 26.12)) was used by O'Rourke (1967b) to simulate aortic stiffening. A similar stenting method is under consideration to achieve aortic 'destiffening' in Sydney and New Haven.

ARTERIAL COUNTERPULSATION

Counterpulsation is the most widely used and effective method of mechanical left ventricular assistance in current practice (see Chapter 17). It is frequently employed for mechanical support following cardiac surgery, in complicated myocardial infarction and in severe heart failure as a bridge to heart transplantation. Also, enhanced external counterpulsation (EECP) has recently been employed in patients with refractory angina (Nichols *et al.*, 2006). Both invasive and non-invasive counterpulsation techniques are based on the same physiologic principles relating to myocardial contraction, ventricular ejection, oxygen demand, and capacity for blood supply,

as discussed in Chapter 17 (Lawson *et al.*, 1992; Kern *et al.*, 1993a; Kern *et al.*, 1993b; Werner *et al.*, 1999; Michaels *et al.*, 2002; Bonetti *et al.*, 2003b). Counterpulsation represents the opposite of an aging effect, since aortic pressure is reduced during systole and increased during diastole. Counterpulsation was originally conceived by the Kantrowitz and Kantrowitz (1953) and used clinically in 1968 (Kantrowitz *et al.*, 1968). The earliest systems pumped blood directly into and out of the proximal aorta. The process was revolutionized by the introduction of the intra-aortic balloon by Mouloupoulos *et al.* (1962). This permitted the same effect to be achieved, but through pumping of gas rather than blood into and out of the body. The procedure has been further simplified by development of a thin, wrapped balloon that can be inserted percutaneously.

With intra-aortic balloon counterpulsation, the balloon is located in the descending thoracic aorta (Fig. 26.13) and with EECP the external cuffs are positioned on the legs. The balloon or cuffs are rhythmically inflated and deflated with each beat of the heart, so that inflation occurs in diastole and deflation in systole. An inviscid, low-density gas (usually helium) is used in the intra-aortic balloon. Timing is set by the QRS complex of the electrocardiogram, so that inflation and deflation occur with the same timing at each cycle of the heart. The intervals from the ECG trigger balloon (or cuff) inflation and deflation;

timing is adjusted to achieve the optimal therapeutic effect on myocardial function.

Ideal timing of counterpulsation follows established physiologic principles and can be set when arterial pressure is measured in the ascending aorta (Sarnoff *et al.*, 1958; Berne and Rubio, 1979; Suga, 1990; Gibbs, 1995). The aortic wave foot and incisura are first identified in normal (unassisted) beats. Then inflation is adjusted such that the diastolic boost to pressure occurs immediately after the incisura, activating the system for one beat out of four. Deflation is then set such that the pressure rise during early systole is minimized. The ideal situation is to have full inflation and deflation creating the greatest possible rise throughout diastole and the greatest possible fall throughout systole (O'Rourke *et al.*, 1975; O'Rourke *et al.*, 1979; O'Rourke, 1986; Michaels *et al.*, 2002). When this is attained at one assisted beat in four with the balloon counterpulsation technique, the machine is switched to one beat in two, the wave contour reviewed, and any further adjustment made. The machine is then switched to assist each beat, and the aortic tracing is reviewed to confirm optimal timing of inflation and deflation.

There has been some controversy as to the ideal timing for balloon deflation. The idea that very early balloon deflation aids left ventricular ejection (Jaron *et al.*, 1983; Welkowitz and Li, 1984; Zelano *et al.*, 1990) is inconsistent with the earlier data of Weber and Janicki (Weber

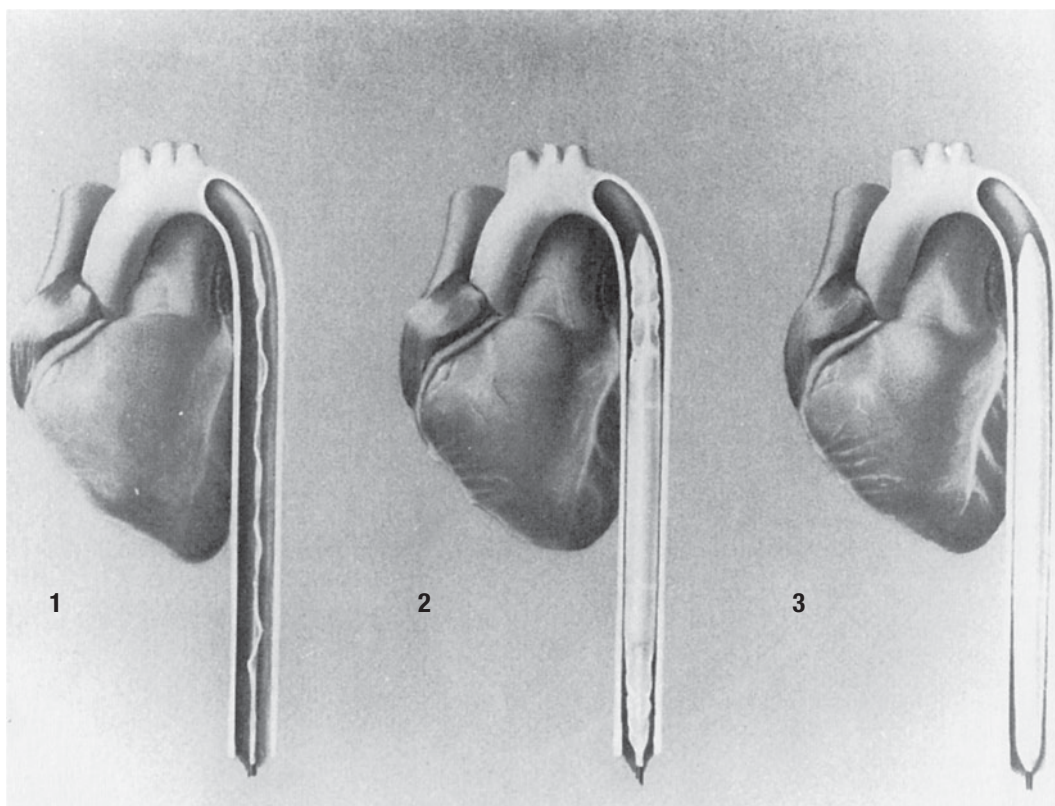


Figure 26.13 Intra-aortic balloon counterpulsation achieved by rhythmic inflation (right) and dilation (left) of a sausage-shaped balloon inserted from the groin into the descending thoracic aorta. Picture courtesy of Avco Corp.

et al., 1972; Weber *et al.*, 1974; Weber and Janicki, 1977) as discussed by Weber in the third edition of this book (see Fig. 14.20). It is inconsistent, too, with physiologic principles related to myocardial oxygen supply and demand (Sarnoff *et al.*, 1958; Berne and Rubio, 1979; Gibbs and Chapman, 1979; Suga, 1990; Gibbs, 1995; Braunwald, 1997a). Ventricular ejection and oxygen demands bear no relationship to the diastolic pressure of the previous beat, but are related to pressure throughout systole (Gibbs and Chapman, 1979; Suga, 1990; Gibbs, 1995; Braunwald, 1997a). Early deflation does not significantly prolong ejection and does not increase ejection (see Fig. 14.22). Early balloon deflation also decreases pressure gradient across the coronary tree and may reduce myocardial blood flow (Berne and Rubio, 1979).

MONITORING AND MEASUREMENT OF THERAPEUTIC EFFECT

An entire chapter (Chapter 30) is now directed at non-invasive assessment of ventricular–vascular interaction

through the use of pulse wave analysis. We see this as an important step forward in assessing the effects of aging, and of hypertension on the heart, and for interpreting the effects of drug therapy, especially vasodilator therapy in the prevention and treatment of cardiac failure. We have already stressed that the beneficial effects of such therapy cannot be determined from conventional measurement of systolic pressure in the brachial artery, since this may substantially underestimate pressure reduction at the heart.

While the process we favor (sphygmocardiography) generates the aortic pressure wave from the radial pressure wave at the wrist, separates the cardiac cycle into systolic and diastolic periods, and provides numerical indices of ventricular–vascular interaction, other approaches are possible, and in some instances may be preferable. All, however, depend on accurate measurement of pressure waveform and on an understanding of pulse wave genesis.

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Exercise

Technical issues related to the recording of pulsatile phenomena	554	Cardiac failure	563
Counterpulsation with running?	556	Aortic coarctation	564
Ascending aortic impedance	558	Angina pectoris	565
What arterial changes are promoted by exercise?	560	Diabetes mellitus	566
Exercise programs	562	Possible ill effects of exercise on the cardiovascular system	566
Aging	562		

The progress of ... degeneration in the arteries is practically the physical history of the individual after the zenith of his strength has been reached, accompanied by an imperceptible but progressive limitation of the field of his heart's response to effort. Although for long he may lead a vigorous life within this limitation, with no evident discomfort, yet during the fourth decade of life the ease with which violent efforts are made is gradually lessening, and violent exertions are as a rule carefully avoided. Running after trains is not to be done in comfort, and the ascent of hills is undertaken with more deliberation. All this proceeds *pari passu* with diminished resiliency of the arterial wall. So insidious is the progress ... that is usually well established before it is recognised.

James Mackenzie, 1902

The first to third editions of this book did not deal specifically with exercise. This was natural. It was hard enough to obtain accurate data on pulsatile phenomena at rest, let alone unscramble motion and other artifacts from waves that were usually recorded invasively in experimental animals and in humans through catheters and implanted devices. The situation has now changed, with the ability to measure pressure, flow, and diameter fluctuations non-invasively in humans or through telemetric techniques in free-ranging animals. Furthermore, with basic principles now established for understanding arterial function and vascular-ventricular interaction at rest, it is important to extend these to real-life situations where the body is physically active. The ability to exercise – to chase prey and escape from predators – determined the ability of species to survive in eons past. The same ability in

humans to exercise determines the ability to enjoy recreational pursuits with comfort. The severity of cardiovascular diseases, such as cardiac failure, angina pectoris, and leg claudication are graded (as in the New York Heart Association (NYHA) criteria) on the basis of the degree to which they limit the ability to exercise.

These basic views on exercise have not changed since the last (2005) edition of this book. A number of things have changed. There is a new view on diastolic left ventricular (LV) dysfunction, and an appreciation that this is responsible for around 50 percent of all patients with cardiac failure (Vasan *et al.*, 1996; Senni and Redfield, 2001) and for a greater percentage of elderly patients (Hundley *et al.*, 2001; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b; Redfield *et al.*, 2003). Diastolic LV dysfunction is attributable to aortic stiffening and early wave reflection from peripheral sites (Hundley *et al.*, 2001; O'Rourke, 2001; Abhayaratna *et al.*, 2008; Weber *et al.*, 2008) and is probably responsible for the decline in exercise capacity with age, as described by Mackenzie in 1902. At a later age, this may become manifest as an episode of cardiac failure with pulmonary edema, precipitated by an arrhythmia, surgical procedure, or intercurrent infection; it is usually preceded by gradual deterioration in exercise capacity in proceeding years. Mackenzie's (1902) view is rarely considered by modern physicians. All mature individuals become aware of their declining capacity to keep up with children and grandchildren, while people of all ages are aware of the toll that aging takes from 18–30 years in competitive sporting prowess. The sole exception is long-distance running – which is discussed later in this chapter, and for which a special unique mechanism can explain the dominance of competitors in their fourth and fifth decades. Mechanistic

issues of aging and diastolic LV dysfunction are discussed in Chapters 14, 15, and 19.

Another issue to be tackled is the relationship between exercise and acute cardiovascular events – the paradox that unaccustomed exercise can precipitate events, whereas regular exercise protects against development of such events (von Klot *et al.*, 2008). Part of the enigma can be explained on the basis that capacity for exercise depends on absence of significant cardiovascular disease, and that people undertaking regular exercise are usually more careful about their health and control of risk factors for cardiovascular disease (Curfman, 1993; Mittleman *et al.*, 1993; Tofler *et al.*, 1996; Maron, 2000; Maron *et al.*, 2004; Pelliccia *et al.*, 2010).

Other new issues are the ability now to take measures of central, as well as peripheral pressure waveforms during rest and physical activity, including exercise (Sharman *et al.*, 2005; Sharman *et al.*, 2006; Brewer *et al.*, 2007; Casey, 2008; Casey *et al.*, 2008b; Duren *et al.*, 2008; Holland *et al.*, 2008; Knez *et al.*, 2008; Dawson *et al.*, 2009; Heffernan *et al.*, 2009; Sharman *et al.*, 2009a; Chirinos *et al.*, 2009c), the value of these and other indices – as of endothelial function – in monitoring exercise programs and prognosis (Sharman *et al.*, 2008; Vogel, 2008; Payvandi *et al.*, 2009; Sakuragi *et al.*, 2009). New recommendations are provided on exercise for normal subjects and patients with cardiovascular disease (Haskell *et al.*, 2007; Nelson *et al.*, 2007; Williams *et al.*, 2007; Mohlenkamp *et al.*, 2008; Scott *et al.*, 2008; Cruickshank *et al.*, 2009; Kenchaiah *et al.*, 2009; Marwick *et al.*, 2009; Myers, 2009; Schulze, 2009; Taylor, 2010a; Taylor, 2010b).

While effect of exercise on arterial stiffness and central hemodynamics has been studied for more than half a century (Kroeker and Wood, 1955), many aspects remain unclarified. It appears that effects depend on the type (aerobic, resistance, combined), the intensity, and the duration of exercise. However, the presence of an association between regular exercise or increased exercise capacity and a well-functioning arterial system does not prove a direct cause-and-effect relationship. While exercise may by itself lead to an improvement in arterial elastic properties, it may also well be that those with already ideal elastic arteries can more easily exercise for this very reason; thus, they are more often engaged in such activities, setting a positive feedback mechanism in operation, and further improving their arterial function.

This book must deal differently from others on the subject of exercise, since its emphasis is on pulsatile phenomena in arteries, and on the interaction between pulsating arteries and beating heart, rather than on the distribution of increased flow from the heart to actively working skeletal muscle. Attention is directed at change in stroke volume and heart rate of the intermittently contracting heart, to the effects of exercise on arterial properties, to arterial integrity and to the load presented to the heart. Emphasis is on humans, with consideration of normal function and the effects of disease. The

objective of exercise will not be considered in any detail, nor will maintenance of blood flow to the brain. Details are given in standard texts and in special reviews, such as the 1994 AHA monograph *Cardiovascular Responses to Exercise* (Fletcher, 1994) (see also Blake, 1991; Störk *et al.*, 1991; Little and Cheng, 1993; Stratton *et al.*, 1994; Belardinelli *et al.*, 1995; Tall, 2002; Myers, 2003; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b), and the most recent AHA statement on exercise in heart failure (Marwick *et al.*, 2009).

Other issues with exercise include the marked change in heart rate with onset of exercise, and the lesser change in stroke volume from the heart. Because of the changes in heart rate and ventricular contractility, there are marked changes in amplification of the pulse in the upper arm, and so in the difference in blood pressure in the arm when pressure is measured and in the aorta where effects are implied (Sharman *et al.*, 2005; Sharman *et al.*, 2006). Further difficulties come from different responses to isometric (resistance) exercise such as weightlifting and 'isotonic' or 'aerobic' exercise such as running (Figueroa *et al.*, 2010) and to acute and chronic exercise. There are still more problems confronted by physicians who wish to interpret changes with exercise as beneficial or disadvantageous to health, in patients with occult or known cardiovascular disease (Mohlenkamp *et al.*, 2008).

TECHNICAL ISSUES RELATED TO THE RECORDING OF PULSATILE PHENOMENA

The most obvious responses to exercise are the increase in heart rate, stroke volume, systolic and pulse pressure. The echocardiogram (ECG) and blood pressure are regularly recorded and confidently reported during diagnostic treadmill stress testing. But how accurate are the recordings? Artifact from motion can be eliminated from the ECG when the skin is properly prepared, electrodes carefully applied, and leads supported so that heart rate should pose no problem. This is not true, however, for pressure recordings.

Arterial pressure is usually measured by sphygmomanometry from the brachial artery using an upper arm cuff. Consistent values are routinely presented in stress test reports for treadmill exercise. However, when pressure waves are measured invasively through a catheter, or non-invasively by tonometry or Finapres, marked fluctuations are usually very obvious (Fig. 27.1) (Rowell *et al.*, 1968; Palatini *et al.*, 1989; O'Rourke and Avolio, 1992; Sharman *et al.*, 2005; Sharman *et al.*, 2006; Casey *et al.*, 2008b; Sharman *et al.*, 2008). These fluctuations have usually been passed over as motion artifact and disregarded. Careful studies, however, have shown that such variations in pressure are very real and correspond to similar, but quantitatively different, variations in proximal aortic and left ventricular pressure (Fig. 10.14) (Rowell *et al.*, 1968; Palatini *et al.*, 1989; O'Rourke and Avolio, 1992). Such

T.P. 26 years

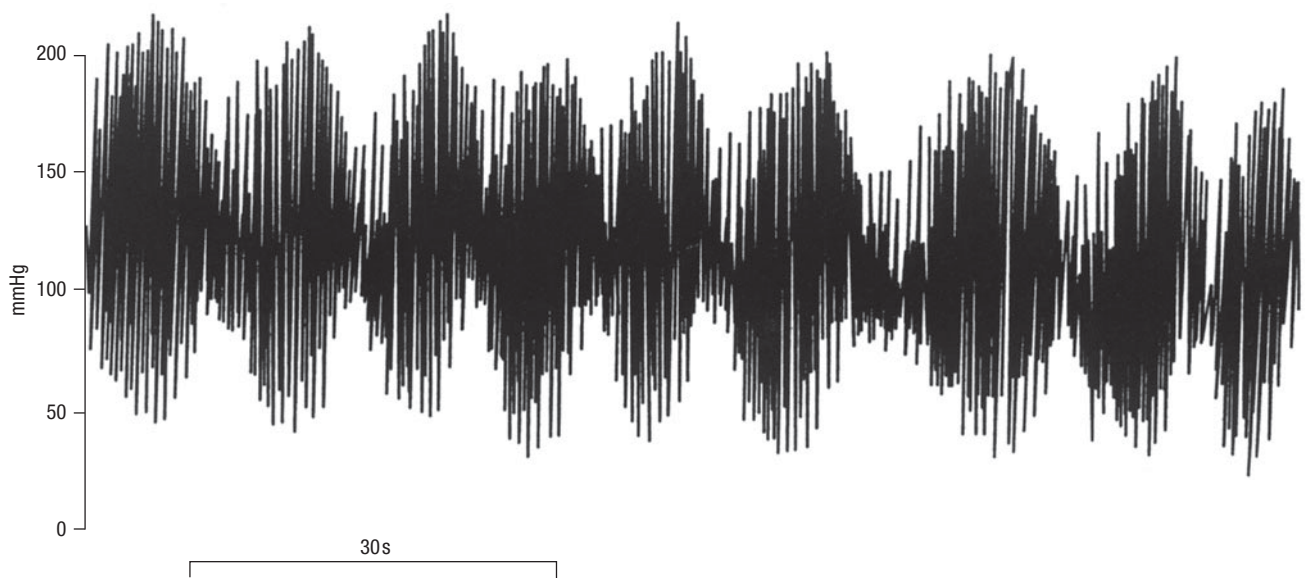


Figure 27.1 Change in arterial pressure with track running, and with heart rate 160/min and step rate 154/min, leading to a 'beat' frequency of 6.0/min. 'Beating' is due to interaction of two waves, one generated by the heart, the other generated by vertical body movement. Reproduced from Palatini *et al.* (1989)

studies cast doubt on the confidence with which blood pressure is conventionally recorded during exercise, and on the indices of cardiac function that are calculated therefrom. There are two problems with recording pressure during exercise – the effect of movement itself, and the change in amplification of the pressure wave between the aorta and the brachial artery when heart rate increases.

Bodily movement

Bodily movement during running and treadmill exercise does generate pressure waves in the arterial system. These are minute when body position remains constant, as with cycling, but the waves are of high amplitude – similar in magnitude to those generated by ventricular ejection itself – when the body moves vertically with running on a treadmill or pathway (Rowell *et al.*, 1968; Palatini *et al.*, 1989; O'Rourke *et al.*, 1992a; O'Rourke *et al.*, 1993b). With running, a stride of 40 inches is accompanied by vertical movement of the torso and head through 4.0–6.0 inches. When pressure waves are recorded continuously, either invasively or non-invasively, fluctuations caused by bodily movement during running can be seen clearly (see Fig. 27.1). At the peak of exercise, stride rate and heart rate are often very close (in the range 140–180 min) so that a type of 'beating' pattern is established, such that pulse pressure rhythmically increases and decreases at a frequency (of around 1.0–10 min) which is determined by the difference between heart rate and stride rate (see Fig. 27.1). Such variations in pulse pressure can be very substantial and equivalent to pulse pressure, i.e. over

60 mmHg. They are due to interaction between the pressure wave generated by the heart and the wave generated by vertical movement of the whole body (Palatini *et al.*, 1989; O'Rourke *et al.*, 1992a; O'Rourke *et al.*, 1993b; Stelliou, 1993). Close inspection of pressure wave contour shows in addition that when pulse amplitude is highest, the wave has a single peak and trough (indicating that the two waves are in phase), whereas when pulse amplitude is lowest, there are two separate waves (indicating a phase delay approximating 180°) (Fig. 27.2) (O'Rourke and Avolio, 1992). Clearly, measurement of arterial pressure by sphygmomanometry under these conditions would be unreliable, since values obtained would depend on whether systolic and diastolic pressure were taken at the peak or nadir of phasic fluctuation. This subject has not been confronted by exercise physiologists, nor by cardiologists who supervise and interpret exercise stress tests in patients with suspected coronary artery disease.

Amplification of the pressure pulse in the upper limb

The second issue related to the accuracy of blood pressure determinations during exercise is the difference between central aortic and brachial (or radial) pressure waves consequent on amplification of the pressure pulse during transmission through the upper limb (see Fig. 10.25) (Kroeker and Wood, 1955; Rowell *et al.*, 1968; O'Rourke, 1970b; Benetos *et al.*, 2010). This subject is discussed in detail elsewhere (Chapters 10 and 30), but it is nowhere more important in absolute magnitude than during

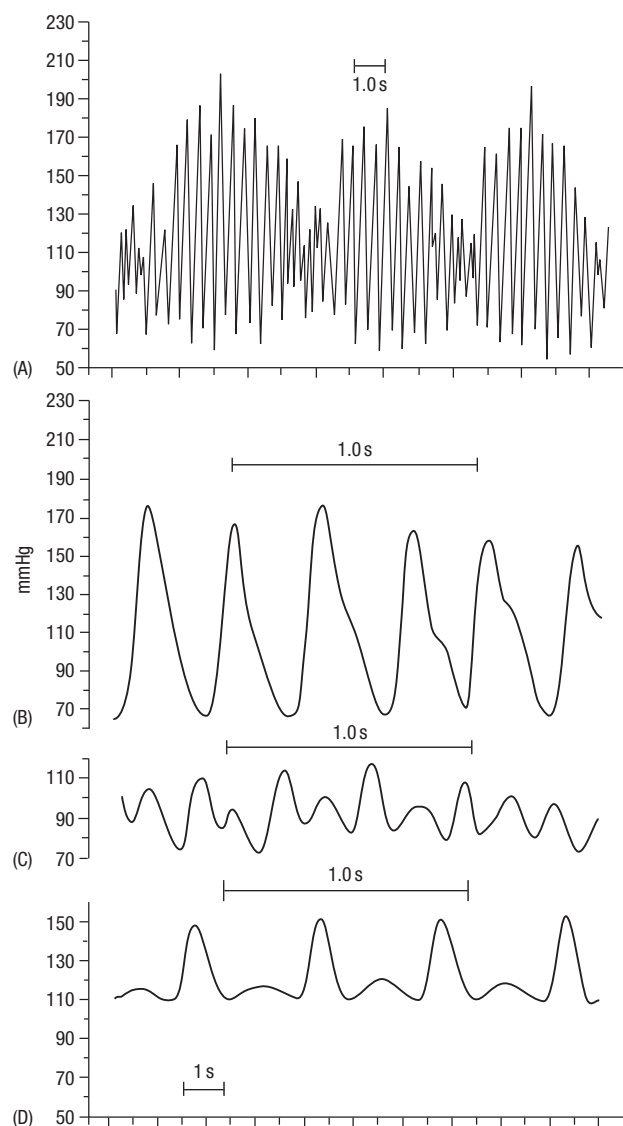


Figure 27.2 Pressure wave recordings during running in a young subject. (A) (Compressed time scale) when heart rate and step rate were not synchronized. (B) When heart rate and step rate were synchronized, with heart rate and body wave in phase. (C) With synchronized heart and step rate, but with heart and body wave out of phase. (D) During rest after exercise. Reproduced from O'Rourke *et al.* (1993b)

exercise when stroke volume increases somewhat, but heart rate increases substantially.

During exercise, differences of greater than 80 mmHg have been recorded between the central aorta and upper limb arteries (see Fig. 10.14). Such differences in systolic pressure must render almost valueless the indices of myocardial oxygen consumption (as 'double product': heart rate $\times$ systolic pressure) that are routinely, confidently, but uncritically reported in stress test reports (see Moffitt *et al.*, 1984). These differences in pulse amplitude between central and peripheral arteries are readily explained on the basis of wave reflection in the upper

limb, as quantified by the transfer function for pressure (O'Rourke, 1970b; Lasance *et al.*, 1976; Karamanoglu *et al.*, 1993; Chen *et al.*, 1997) and which remains unchanged during exercise (Sharman *et al.*, 2006). The transfer function for pressure between the central aorta and brachial artery (see Fig. 30.18) shows a peak value of modulus of about 2.8 at a frequency of 4.0–5.0 Hz. At a slow heart rate and relatively long ejection duration, most of the energy of the aortic pressure wave is at frequencies less than 2.5 Hz where the amplitude of the transfer function is low (O'Rourke, 1970b; Karamanoglu *et al.*, 1993). However, with exercise, heart rate increases and ejection duration decreases, such that all energy is at frequencies greater than 2.0 Hz where amplification is high (O'Rourke, 1970b; Karamanoglu *et al.*, 1993; Chen *et al.*, 1997; Sharman *et al.*, 2005). When wave contour is known for the radial pulse, use of the transfer function enables appropriate corrections to be made so that aortic systolic pressure peak and pressure throughout systole can be calculated accurately (see Chapter 30). This is, however, not done in regular clinical practice. The phenomenon of pressure wave amplification in the upper limb is currently ignored by human exercise physiologists and by clinicians, yet it is clear that such oversight may introduce substantial error and provide misleading results (Mottram *et al.*, 2004; O'Rourke, 2004b; Sharman *et al.*, 2005; Sharman *et al.*, 2006).

COUNTERPULSATION WITH RUNNING?

Arterial counterpulsation is discussed elsewhere (see Chapters 17 and 26). This is a technique of mechanical left ventricular assistance wherein a sausage-shaped balloon is inserted into the thoracic aorta or cuffs positioned on the legs and are rhythmically inflated and deflated in synchrony with the heart and with such timing that pressure during diastole is artificially increased, and pressure during systole, decreased. This has the dual benefit of augmenting coronary perfusion pressure, while decreasing myocardial oxygen demand and decreasing left ventricular hydraulic load. Reduced left ventricular afterload aids ventricular ejection and tends to increase left ventricular ejection fraction and stroke volume.

In the early days, before counterpulsation techniques were perfected, counterpulsation was achieved by rhythmic longitudinal movement of a bed on which the patient was strapped. This, the reverse ballistocardiograph or 'bash bed', was a theoretically correct, but highly uncomfortable, medical procedure. This procedure was abandoned but has recently been resurrected in Japan as treatment of refractory angina (Fukuda *et al.*, 2010). Such 'passive exercise' may perhaps have a place along with external counterpulsation, but this is unlikely to be substantial (see Chapters 17 and 26). However, as described above, the same phenomenon can be achieved during running in humans if heart rate and stride rate can be

entrained, and if appropriate phasing can be achieved. We have shown that this is possible (Fig. 27.3) and have speculated that such a mechanism – utilization of vertical bodily movement to aid cardiac function during long-distance running – may explain the ‘rhythm of running’ to which long-distance and marathon runners aspire (O’Rourke *et al.*, 1992a; O’Rourke *et al.*, 1993b; Heagerty, 1994). We had been impressed by the extent (4.0–5.0 inches) of vertical movement with marathon runners, and the tantalizingly similar heart rate and stride rate recorded at such events. When the data of Rowell *et al.* (1968) and Palatini *et al.* (1989) came to our notice, we began to appreciate the magnitude of pressure waves generated by bodily motion and to wonder if the ‘beating’ phenomenon might not be just part of something more important. Entrainment with appropriate phasing could mimic

arterial counterpulsation and create a favorable situation where heart motion and bodily motion are in complete harmony, with one aiding the other, and with stroke volume and cardiac output enhanced, so that oxygen delivery to the tissues is increased over and above that possible with normal cardiac function alone (Folkow *et al.*, 1970; Folkow *et al.*, 1971; O’Rourke and Avolio, 1992; O’Rourke *et al.*, 1993b; Stelliou, 1993; Heagerty, 1994; Bramble and Lieberman, 2004).

There is another paradox to be explained in long-distance running, which may be answered by the phenomenon of heart action enhanced by bodily movement. Marathon and other long-distance runners reach their peak in the fourth decade of life, in comparison with other athletes who usually achieve peak performance by the end of the second decade (see Chapter 19). It is likely that a

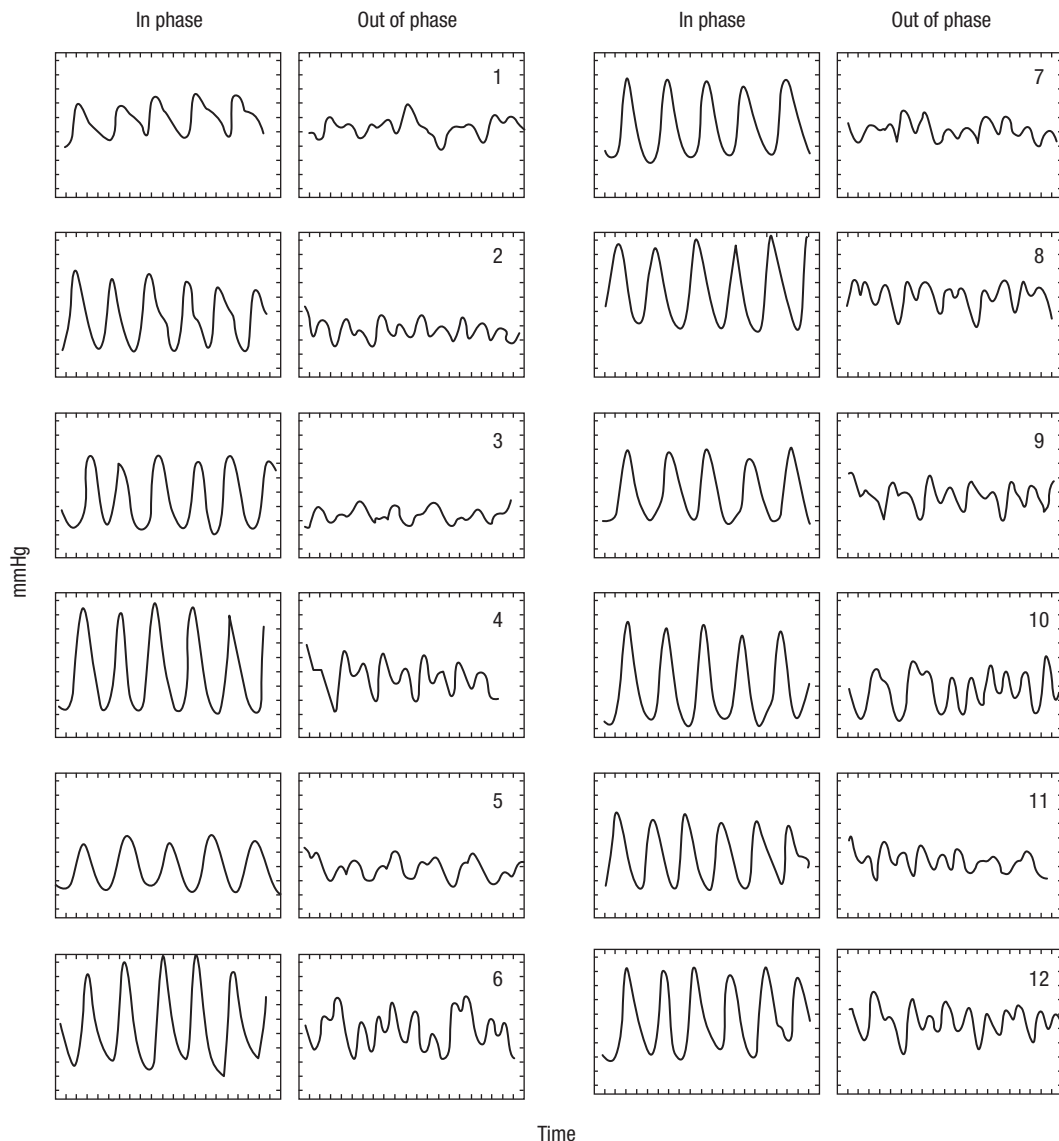


Figure 27.3 Pressure waves in 12 subjects when the heart and step rates were entrained. Left tracing: heart and step in phase. Right tracing: heart and step out of phase. Pressure scale, 30–230 mmHg; time scale, 0–2 s. Average pulse pressure for in-phase entrainment was 108 ± 26 mmHg (s.d.) and for out of phase entrainment 38 ± 17 mmHg. Reproduced from O’Rourke *et al.* (1993b)

degree of arterial stiffening as seen in 40 year olds may be advantageous, rather than disadvantageous, by enabling waves generated by bodily movement to pass more quickly to the heart from the legs (Vlachopoulos *et al.*, 2010d), aiding diastolic myocardial flow, and more quickly from the heart to the legs aiding flow to the ankle, knee, and hip extensors between successive periods of contraction (Folkow *et al.*, 1970; Folkow *et al.*, 1971; O'Rourke and Avolio, 1992; O'Rourke *et al.*, 1993b). Such synchrony between legs and heart (Folkow *et al.*, 1970; Folkow *et al.*, 1971), and the heart and legs (O'Rourke *et al.*, 1993b), with optimal wave speed and phasing would aid flow to each, and peak efficiency of the body. It would, however, need to be attained, then maintained for long periods of distance running. How is this achieved? Training over a long time is clearly important. For such a favorable phenomenon to occur, and to be maintained over a long period of time during a race, one requires not only the appropriate phase of body and heart waves, but also exact entrainment. While this might appear improbable, one has to concede that there are many examples of entrainment in nature, as between movements of the arms in synchrony with the legs of humans during walking and running, as between bodily movement and respiration during running in rabbits and greyhound dogs (Blake, 1991; Lee and Banzett, 1997), and between respiration and heart rate in rabbits. For the latter, Kenner *et al.* (1976) have pointed out that when the rates of two interrelated bodily functions come close, they tend to lock in, with entrainment resulting. For activities other than marathon running, there is good evidence that low arterial stiffness favors the ability to exercise (Vaitkevicius *et al.*, 1993; Nussbacher *et al.*, 1995; Chen *et al.*, 1999).

It must be clear from the above, and from Fig. 10.14, that it is possible to create adverse as well as favorable circumstances for cardiac function when heart rate and stride rate are entrained or close to entrainment. In the adverse situation, pressure during systole is augmented, and during diastole reduced (as can occur during counterpulsation when this is incorrectly timed). Such a situation would have the effect of impeding ventricular ejection and reducing cardiac output, as well as increasing myocardial oxygen demand and decreasing coronary perfusion pressure. The consequence could be a decrease in cardiac output and the development of myocardial ischemia. This would be most marked in older people with stiffened arteries (see Chapter 17) (Ohtsuka *et al.*, 1994), or with some degree of coronary stenosis (Watanabe *et al.*, 1992) or diastolic dysfunction (Masuyama *et al.*, 1994). Sudden cardiac decompensation and death occur not infrequently in mature and older humans during exercise (Siscovick *et al.*, 1984; Curfman, 1993; Mittleman *et al.*, 1993; Willich *et al.*, 1993; Mohlenkamp *et al.*, 2008), especially when unaccustomed. This is not always due to ventricular fibrillation or acute coronary thrombosis, and often occurs in people previously reasonably fit participating in a competitive recreational

event, such as Sydney's City-to-Surf race. A period of extreme myocardial ischemia with cardiac decompensation, such as described above, might explain such an event (Hong *et al.*, 1987). Entrainment is not necessary. If heart rate and stride rate were close, for example 158 and 160, respectively, beat frequency would be 2.0 min and there would be recurrent periods of ischemia lasting 15 seconds at a time (and around 40 beats), sufficient to trigger extreme systolic and diastolic dysfunction, if not a lethal arrhythmia.

There are situations where it is not clear whether exercise is harmful or beneficial. Mohlenkamp *et al.* (2008) describe marathon runners to have high prevalence of atherosclerosis with increased coronary artery calcium, and to be at higher risk of coronary events. In a cross-sectional study, Vlachopoulos *et al.* (2010d) described marathon runners to have high carotid-femoral pulse wave velocity (PWV), indicative of aortic stiffening, but no difference in augmentation index (AIx) compared to controls. Findings are most easily explained (as they were by the authors) on the basis of exercise causing increased PWV as a result of aortic elastin fiber damage. Another explanation, however, is that aortic stiffening (like short body length in the highly successful Ethiopian marathon runners), causes earlier return of waves from the lower body, and so favors development of the rhythm of running, and hence the likelihood of such people participating and succeeding in the sport.

Clearly, more research needs to be done on pressure waves generated by bodily movement during exercise and their effects on the heart (Störk *et al.*, 1991). The subject is relevant to two extremes of human performance – to winning an Olympic marathon at one extreme and to abrupt cardiac decompensation at the other. To date, most clinical studies of exercise on cardiovascular function in humans have required invasive catheterization or cumbersome scanning equipment, and so have been undertaken with the subject recumbent, the body still, and only the legs active. There is one aspect where experts agree, and which is well supported by data on sudden arrhythmic death – that it is the unaccustomed exercise which is dangerous (Curfman, 1993; Mittleman *et al.*, 1993; Tofield, 2010).

ASCENDING AORTIC IMPEDANCE

Limited data only are available on changes in ascending aortic impedance during acute exercise in normal humans (Murgo *et al.*, 1981b; Laskey *et al.*, 1985; Laskey and Kussmaul, 1987). In these published studies, impedance was determined during cardiac catheterization with light leg exercise in the horizontal position. Changes in heart rate were modest (mean 71–105 and 78–122 in the Laskey and Murgo series, respectively), as were changes in cardiac output and mean arterial pressure (91 and 79 percent for cardiac output and 14 and 13 percent for pressure).

Murgo *et al.* (1981b) showed no significant change in modulus or phase of impedance during light exercise, although the small decrease in phase observed at low frequency is consistent with a decrease in peripheral wave reflection associated with a decreased peripheral resistance which fell by 38 percent. Laskey *et al.* (Laskey *et al.*, 1985; Laskey and Kussmaul, 1987) did show evidence of a reduction in wave reflection manifest as a slight reduction in the low-frequency phase, and in fluctuation of modulus and phase at higher frequencies. One would expect that the vasodilatation of more extreme exercise – as attained in racing or on a treadmill – would result in more definite evidence of reduced wave reflection. Both groups showed a shift of impedance modulus minima and of phase crossover to higher frequencies, and attributed this to an earlier return of wave reflection consequent on increased mean arterial pressure and aortic pulse wave velocity. Characteristic impedance (expressed in terms of volume flow) did not change with exercise. Laskey *et al.* also studied patients with cardiomyopathy and controlled cardiac failure. In these patients, there was a decrease in the (previously elevated) peripheral resistance during exercise, but no evidence of decrease in peripheral wave reflection as was seen in the normal subjects. Other studies have shown evidence of a decrease in wave reflection with vasodilator drugs, such as nitroprusside, in patients with heart failure (see Chapter 14) (Pepine *et al.*, 1979;

Binkley *et al.*, 1990; Chiu *et al.*, 1992; Nichols and Pepine, 1992). Difference in exercise and with vasodilator therapy may relate to the localized region of vasodilatation during leg exercise and the more generalized vasodilatation induced by drugs. The improvement in exercise tolerance in cardiac failure patients given nitrates may well be due to a reduction in the ill effects of early wave reflection on the left ventricle (Leier *et al.*, 1983; Cohn, 1994; Westerhof and O'Rourke, 1995; Uebaba *et al.*, 2002).

As previously discussed (Chapter 12) and as seen in Fig. 27.4, the minimum value of impedance modulus occurs at just above 2.5 Hz in healthy young human subjects. This usually corresponds to the frequency of the second harmonic of the aortic flow wave at normal heart frequency. At the frequency of the fundamental harmonic, impedance modulus is very much higher. The tachycardia of exercise results in the first harmonic of flow (the dominant harmonic of the wave) occurring at a frequency where impedance modulus is relatively low. Optimal ventricular–vascular coupling is created when the highest components of the ventricular ejection (flow) wave occur at frequencies where impedance modulus is lowest. Under these circumstances, amplitude of the pressure wave generated by flow will be lowest. The arterial system appears to be designed for optimal function with respect to ventricular performance during exercise, with sub-optimal performance at rest. This is precisely what one

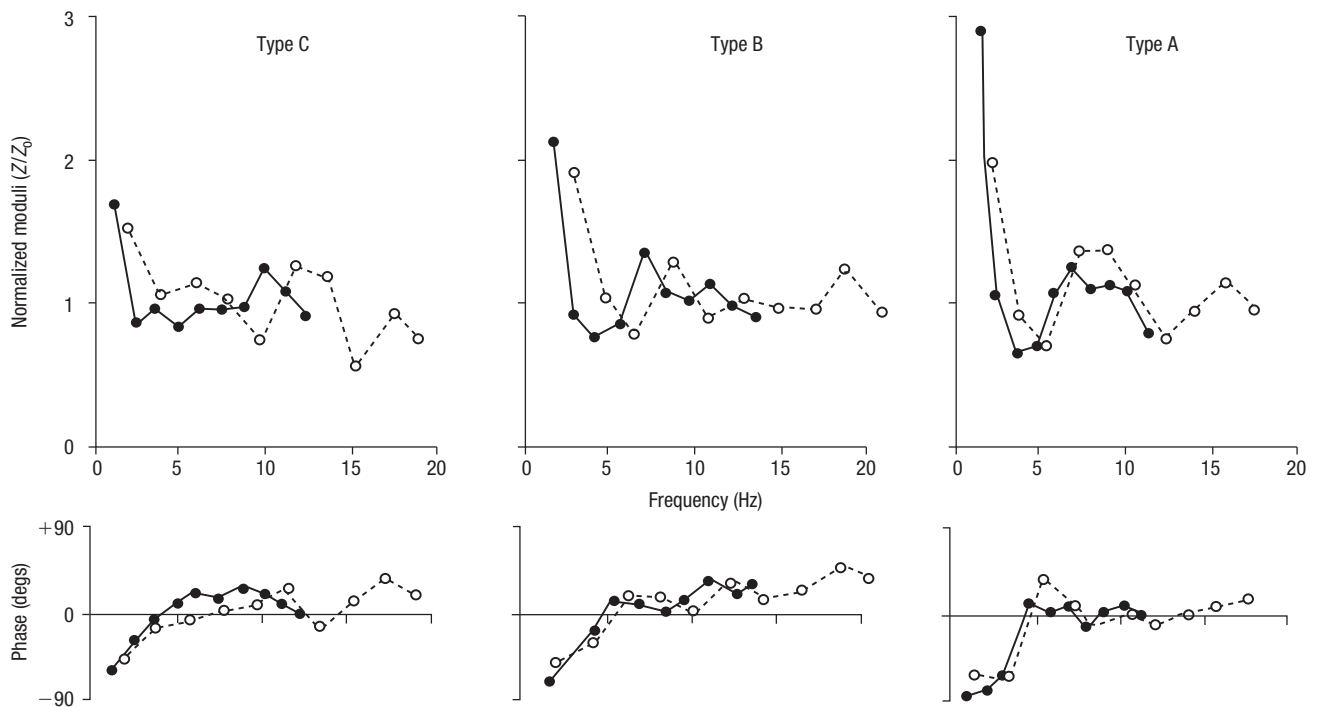


Figure 27.4 Ascending aortic impedance in 13 male subjects without cardiovascular disease at rest (●—●) and during supine bicycle exercise (○—○). Subjects are divided into three groups: right, aortic pressure wave showing type A pattern (four subjects); center, aortic pressure wave showing type B pattern (five subjects); left, aortic pressure wave showing type C pattern (four subjects). Exercise was not associated with any significant change in impedance patterns. Shift of impedance spectra to the right is attributable to increase in mean arterial pressure with exercise. Reproduced from Murgo *et al.* (1981b)

might expect if such had evolved to provide a seminal advantage during 'flight or fight'. This is discussed by Taylor (Taylor, 1964; Taylor, 1994).

WHAT ARTERIAL CHANGES ARE PROMOTED BY EXERCISE?

Multiple studies attest to the benefits of regular aerobic physical exercise (Curfman, 1993; Mittleman *et al.*, 1993; Stratton *et al.*, 1994; Hambrecht *et al.*, 1998; Hambrecht *et al.*, 2000a; Hambrecht *et al.*, 2000b; Tanaka *et al.*, 2000; Vita and Keaney, 2000; Stewart, 2002; Tall, 2002; Myers, 2003) and to the improvement in oxygen extraction from blood, and in cardiovascular function that occur with exercise training (McGuire *et al.*, 2001a; McGuire *et al.*, 2001b; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b; Duren *et al.*, 2008; Heffernan *et al.*, 2009). Data on this subject are colored by a number of factors. People who are ill for any reason, or have subclinical arterial stiffening, coronary arteriosclerosis, or hypertension will not be comfortable with exercise, and are unlikely to persevere. They are thus under-represented in groups of people who engage in regular exercise (Tanaka *et al.*, 2000; Hundley *et al.*, 2001). Furthermore, those who do so engage are more likely than others to take care with other lifestyle measures including diet, weight control, smoking, and drinking (Tall, 2002). Comparisons of groups in the population selected on the basis of physical prowess, especially the fit elderly as in the study by Hundley *et al.* (2001), or in the US National Institution of Aging, may not be representative of their peers in the community at large (Jessup, 2003).

Improved endothelial function with exercise is now established (Hambrecht *et al.*, 1998; Higashi *et al.*, 1999a; Higashi *et al.*, 1999b; DeSouza *et al.*, 2000; Hambrecht *et al.*, 2000a; Hambrecht *et al.*, 2000b; Taddei *et al.*, 2000; Vita and Keaney, 2000; Heffernan *et al.*, 2009). The cause is probably the periodic increased shear stress on the vascular endothelium that accompanies exercise. It is possible that the same mechanism (measured periodic shear stress on endothelium) is responsible for the favorable effects of enhanced external counterpulsation (EECP) (see Chapter 17) (Bonetti *et al.*, 2003b; Nichols *et al.*, 2006; Casey *et al.*, 2008a). The period of training required for persistently improved endothelial function is about 4 weeks; with inactivity, the benefit passes off over a similar period. Improvement in endothelial function (Hayward *et al.*, 2002; Wilkinson *et al.*, 2002a; Edwards *et al.*, 2004a; Mustata *et al.*, 2004) is associated with reduction in peripheral wave reflection. Improvement in endothelial function may explain the benefits of 'passive exercise' and of sessions of external counterpulsation in refractory angina (Fukuda *et al.*, 2010).

Vaitkevicius *et al.* (1993) and Nussbacher *et al.* (1995) found that low arterial stiffness favors the ability to exercise, and attributed reduction in arterial stiffness to

regular exercise. Hundley *et al.* (2001) attributed impaired exercise capacity in older subjects to diastolic LV dysfunction caused by progressive arterial stiffening. Grazi *et al.* (2001) came to the same conclusion in their studies of older hypertensive rodents. Chen *et al.* (1999) have shown that reduction in indices of arterial stiffness with verapamil improved cardiac performance with exercise. In this study, there was a reduction in measured aortic PWV (an index of arterial stiffness), but a much greater reduction in central pressure augmentation (a measure of wave reflection from the peripheral vasculature). This is sufficient to explain most of the favorable effects of physical training on exercise capacity – at least as this influences ventricular–vascular interaction (Chen *et al.*, 1999; McGuire *et al.*, 2001a; McGuire *et al.*, 2001b).

Effects on arterial stiffness and central hemodynamics are better described and explained when divided into three periods: (1) during exercise, (2) acutely post-exercise, and (3) mid- or long-term. Invasive (Murgu *et al.*, 1981b) and non-invasive studies (Sharman *et al.*, 2005) have shown that during exercise there is a disproportionate increase in peripheral compared to central pressures (thus, pulse pressure amplification increases) and in PWV, whereas other wave reflection indices, such as aortic AIx and augmented pressure, decrease. The acute post-exercise effects depend on the type of exercise. In particular, aerobic exercise has no effect (Heffernan *et al.*, 2007) or reduces aortic PWV (Kingwell *et al.*, 1997), while AIx and central diastolic blood pressure are reduced (Heffernan *et al.*, 2007). On the other hand, whole body resistance exercise increases aortic PWV, carotid β -stiffness index, central blood pressure, and AIx (Table 27.1) (Geleris *et al.*, 2004; DeVan *et al.*, 2005).

The explanation that is offered for these acute effects of exercise on arteries is thus similar to that offered for the effects of drugs (see Chapters 26 and 28). We believe that exercise improves endothelial function of arteries and arterioles throughout the body. In older people with stiff dilated proximal elastic arteries, there is minimal effect from resultant smooth muscle relaxation, and the arteries remain stiff and dilated – although perhaps a little less so if reduction in arterial pressure causes decrease in passive distension. However, in these same people, muscular arteries and arterioles actively dilate when smooth muscle relaxes, so that wave reflection is reduced – and hence left ventricular systolic pressure and LV load as well. Central systolic pressure falls to a greater degree than measured in the brachial artery (again, as discussed in Chapter 26). Such an explanation is consistent with most of the published data, including those of Cameron and Dart (1994) whose measure of 'total arterial compliance' (Chapter 30) embraced the effects of exercise on both elastic and muscular arteries, i.e. on aortic stiffness and peripheral wave reflection. It is consistent also with the detailed studies of Seals and colleagues on exercise-induced change in arterial function (Tanaka *et al.*, 1998; DeSouza *et al.*, 2000; Tanaka *et al.*, 2000).

Table 27.1 Overview of effect of exercise on arterial function

	Aortic stiffness	Aortic impedance	Wave reflection indices	Amplification
Acute, during exercise				
Aerobic	↑	–	↓	↑
Resistance	↑	↑	↑	↑
Combined	?	?	?	?
Immediately post-exercise				
Aerobic	↓	–	↓	↑
Resistance	↑	↑	↑	↑
Combined	?	?	?	?
Long-term (non-interventional studies)				
Aerobic	↓	?	↓	↑
Resistance	↑	↑	↑	?
Combined	↓	?	↓	?
Mid- and long-term (interventional studies)				
Aerobic	↓/–	?	↓/–	?
Resistance	↑/–	?	↑/–	↑
Combined	–	?	?	?

↑, increase; ↓, decrease; –, no effect; ?, no definite effect.

Mid- and long-term changes in arterial stiffness and central hemodynamics promoted by exercise have been assessed by two types of studies, non-interventional (cross-sectional and follow up, longitudinal) and interventional ones. The first investigate the relationship of arterial function with physical activity (mainly assessed with self-reported questionnaires on exercise habits) and cardiorespiratory fitness (assessed by maximal oxygen consumption on a treadmill or a cycle ergometer) in various populations and provide valuable insights regarding the effects of exercise over the years. The latter actively engage individuals in training schedules under controlled conditions. They are largely restricted to mid-term effects (ranging from a few weeks to several months); however, they establish a causal relationship.

Despite scarce contrary reports (Schmitz *et al.*, 2001; Knez *et al.*, 2008), overwhelming evidence attests to a beneficial relationship between arterial stiffness or central hemodynamic indices and exercise in various population groups. Effects apply to both sexes, both adults and children, and extend from the general population to subjects with cardiovascular risk factors, such as obese, diabetics, pre-hypertensive, or hypertensive individuals and extend to patients with coronary artery disease (Tanaka *et al.*, 1998; Ferreira *et al.*, 2002; Mackey *et al.*, 2002; Greenfield *et al.*, 2003; Boreham *et al.*, 2004; Binder *et al.*, 2006; Kozakova *et al.*, 2007; Farpour-Lambert *et al.*, 2009; Sakuragi *et al.*, 2009). As stressed earlier in this chapter, it should be kept in mind that the initiation point of the series of events is difficult to define. Exercise may improve arterial function; however, those with already compliant elastic arteries can more easily exercise and are more often engaged in activities for this very reason. Effects on arterial function depend largely on the amount and type of exercise. Although amount is difficult to

quantify, it appears that the effects are dose-dependent (Tanaka *et al.*, 2000; Edwards and Lang, 2005; Otsuki *et al.*, 2007a; Otsuki *et al.*, 2007b); the more one exercises the less stiff (more compliant) their arteries appear to become (Fig. 27.5). Effects are evident mainly in central arterial stiffness (aortic PWV, central elastic artery), while peripheral arterial stiffness may not be altered (arm and leg PWV, peripheral muscular arteries) (Tanaka *et al.*, 1998). Type of exercise is also of the essence. *Aerobic (endurance)* exercise leads to lower aortic and carotid stiffness, while wave reflection indices, such as AIx and

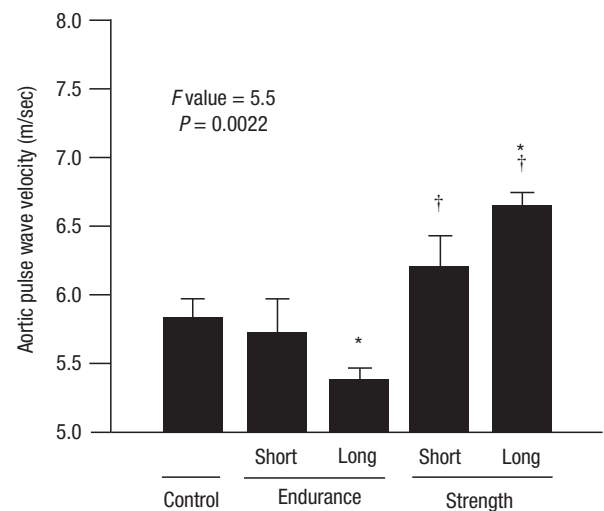


Figure 27.5 Aortic stiffness (pulse wave velocity) in endurance-trained men with short (5 years) and long (12 years) careers and in strength-trained men with short (4 years) and long (7 years). Data are expressed as means $\pm$ SE. * $P < 0.05$ versus sedentary control; † $P < 0.05$ versus sport career-matched endurance-trained men. Reproduced from Otsuki *et al.* (2007a)

augmented pressure, also decrease (Vaitkevicius *et al.*, 1993; Tanaka *et al.*, 2000; Edwards *et al.*, 2004a; Edwards *et al.*, 2004b; Collier *et al.*, 2008; Currie *et al.*, 2009; Madden *et al.*, 2009; Miyaki *et al.*, 2009; Yoshizawa *et al.*, 2009). While typical forms of exercise investigated include jogging or running and cycling, other forms such as soccer, swimming, or basketball could also lead to arterial destiffening (Tanaka *et al.*, 2005). Alternative types of exercise, such as whole body vibration and yoga, may have the same beneficial effects on arterial function as aerobic exercise (Duren *et al.*, 2008; Otsuki *et al.*, 2008). Aerobic exercise of up to three months may not be effective in reducing arterial stiffness in older populations with hypertension as in healthy young adults (Ferrier *et al.*, 2001; Seals *et al.*, 2001). Plausible explanations are that the plasticity of arterial function to adapt to exercise training may diminish with age and that beneficial effects require longer periods of exercise. Recent data suggest that interval, instead of continuous, aerobic training might be more effective in reducing arterial stiffness (Guimaraes *et al.*, 2010). Resistance (strength) training seems to have variable effects ranging from negative to neutral or positive depending on the duration and the way it is performed (Bertovic *et al.*, 1999; Miyachi *et al.*, 2004). Increased stiffness was completely reversed after four months of detraining (Miyachi *et al.*, 2004). Marked and abrupt pressure elevations during resistance training may harm the load-bearing elements of the arterial wall. Eleftheriades and Farkas (2010) have emphasized the marked increase in arterial pressure that accompanies weightlifting, and have linked this to initiation of aortic dissection. Recent studies have shown that when resistance training is executed progressively and with low intensity it has no, or even beneficial effects on arterial stiffness (Okamoto *et al.*, 2008). The combination of aerobic and resistance training, such as in rowing (Cook *et al.*, 2006), seems to have no adverse effect on arterial stiffness (Stewart *et al.*, 2005; Loimaala *et al.*, 2009). We believe that resistance (or strength) training alone has unfavorable effects on arterial stiffness (see Fig. 27.5), but when performed in a progressive manner, and in conjunction with aerobic exercise, these effects are attenuated or even negated.

EXERCISE PROGRAMS

Comments on exercise to this point refer to aerobic activities, such as running, cycling, walking, or swimming. With these types of exercise, increase in heart rate is associated with reduction in diastolic pressure, reduction in aortic PWV, reduction in aortic AIx, increase in central and brachial pulse pressure, and increase in amplification of the pulse between aorta and brachial/radial arteries (Sharman *et al.*, 2005; Heffernan *et al.*, 2007). Changes are explicable on the basis of increased heart rate (with decreased central AIx and increased amplification),

peripheral vasodilatation, and an increase in peak flow ejection from the heart.

However, there is another type of exercise – resistive (or strength) training – as undertaken in weightlifting and rowing. With resistive training, popular for cosmetic reasons among ‘bodybuilders’, activity is associated with increase in peripheral and central systolic and diastolic pressure, increase in aortic PWV and AIx, and often, reduction in amplification of the pulse (Geleris *et al.*, 2004; Miyachi *et al.*, 2004; DeVan *et al.*, 2005; Figueroa *et al.*, 2009).

Resistance training is considered undesirable for people with cardiovascular disease or where improvement in cardiovascular function is desired. A weightlifter is not good at long-distance running. A sumo wrestler is not good at the high jump, and is apt to be disabled early in life by diabetes mellitus or ischemic heart disease.

In practice, some degree of resistance training adds variety and interest to aerobic programs, and aids compliance. If begun after aerobic training, it is unlikely to be deleterious (Cook *et al.*, 2006; Okamoto *et al.*, 2008).

Changes in arterial pressure, PWV and AIx during exercise programs (aerobic and resistance) are apt to be maintained between exercise sessions when these continue, but they wear off over successive weeks or months (Miyachi *et al.*, 2004). Differences in arterial properties with different programs are summarized in Table 27.1.

AGING

The effects of aging on the cardiovascular system are discussed in Chapters 4, 10, and 19 and are such as to distort the favorable ventricular–vascular interaction evident in youth. There is with age a progressive increase in aortic PWV and in aortic characteristic impedance such that wave reflection returns early from the periphery of the body. Ill effects are seen in the time domain as an increase in aortic pressure at the peak of ventricular ejection, with a further rise in pressure to a late systolic peak, and in the frequency domain as an increase in characteristic impedance and in the frequency of modulus minima and phase crossover, such that the impedance modulus is markedly increased at heart rate frequency, for both rest and exercise.

Detailed studies of the effects of aging on ascending aortic impedance have yet to be done, but modeling studies suggest a progressive increase of impedance modulus with age, and with values at any frequency up to 4.0 Hz being some four to six times greater at age 80 than at age 20 (see Chapter 19). This would be expected to have a marked effect on the capacity to exercise (Port *et al.*, 1980; Yin *et al.*, 1981; Vaitkevicius *et al.*, 1993; Fleg *et al.*, 1994; Fleg *et al.*, 1995; Wilson, 1995; O’Rourke, 2001; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b). Certainly, this process can explain progressive left ventricular hypertrophy and cardiac failure in the elderly. We have

pointed out in Chapters 14 and 19 that optimal ventricular–vascular interaction appears to be achieved when full body height is attained, but when aortic degeneration is minimal. Most Olympic gold medals for sustained athletic performance (with the exception of long-distance runners, discussed earlier) are gained by competitors in the 17–23 year age range. We have suggested that cardiac failure from ‘systolic hypertension’ at age 80 is just the end-product of a slow but progressive impairment in ventricular–vascular interaction from age 18, and that this is manifest first as decreased competitive physical performance in the 20s with a further decrease in recreational physical performance in subsequent decades. More data need to be obtained, but those which exist are completely compatible with this hypothesis (see Fig. 19.18) (Murgo *et al.*, 1980a; Murgo *et al.*, 1981b; Holloszy, 1988; Weisfeldt *et al.*, 1992; Vaitkevicius *et al.*, 1993; Fleg *et al.*, 1994; Fleg *et al.*, 1995; Nussbacher *et al.*, 1995; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b). The hypothesis was first put forward by Mackenzie (1902) in his classic textbook.

Conventional studies of cardiovascular function with aging have shown just two consistent alterations in resting parameters: increased brachial arterial systolic and pulse pressure and impaired left ventricular diastolic relaxation (see Chapter 19) (Schulman *et al.*, 1992; Weisfeldt *et al.*, 1992; Folkow and Svanborg, 1993; Lakatta, 1993a; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b); cardiac output and stroke volume decrease to some extent with age (Nichols *et al.*, 1985a; Cheng *et al.*, 2009b), however, this has not been a constant finding (McGuire *et al.*, 2001a; McGuire *et al.*, 2001b). The increase in pressure is referred to above and is due to increased wave reflection amplitude (reflected pressure waves add to the incident pressure wave, see Chapter 9) caused principally by arterial stiffening; the decrease in stroke volume is caused by similar changes (except that reflected waves subtract from the incident flow wave rather than adding to it) (see Chapter 14) (Westerhof and O’Rourke, 1995; Nichols and Edwards, 2001). Since clinically measured systolic pressure is distorted by wave amplification in the upper limb, changes in the ascending aorta are greater than are described in the literature (see Chapter 19 and Figs 19.1 and 20.1). Impaired diastolic relaxation is largely due to myocardial changes of hypertrophy (also referred to as remodeling), but is related also to arterial properties and early wave reflection (Hori *et al.*, 1985; London *et al.*, 1992; Miyashita *et al.*, 1993; Takazawa *et al.*, 1993b; Masuyama *et al.*, 1994; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b; Abhayaratna *et al.*, 2006a; Abhayaratna *et al.*, 2006b; Weber *et al.*, 2006; Weber *et al.*, 2008; Cheng *et al.*, 2010).

In contrast to resting parameters, aging is associated with a considerable deterioration in cardiovascular parameters during exercise, even in carefully screened fit older subjects (Weisfeldt *et al.*, 1992; Fleg *et al.*, 1993; Folkow and Svanborg, 1993; Lakatta, 1993a; Vaitkevicius *et al.*, 1993; Fleg *et al.*, 1994; Fleg *et al.*, 1995; Nussbacher *et al.*, 1995; Hundley *et al.*, 2001). These changes are attributable

to the increased left ventricular hydraulic load described above and to the age-associated decrease in the inotropic, chronotropic, and arterial-vasodilating effects of catecholamine stimulation. With aging, there is lower exercise capacity, measured by maximal oxygen intake, and at any level of exercise, less increase in heart rate, ejection fraction, and cardiac index, no change rather than a fall in left ventricular systolic volume, and a slight rise rather than a fall in left ventricular diastolic volume, together with greater rises in systolic and mean blood pressure (Fig. 27.6). Exercise training had no differential effects on old and young subjects, causing similar increases in maximal oxygen uptake, end diastolic volume, stroke volume, and cardiac output during exercise, and, similar reduction in heart rate and mean blood pressure at rest (Fleg *et al.*, 1993; Vaitkevicius *et al.*, 1993; Fleg *et al.*, 1994; Fleg *et al.*, 1995; McGuire *et al.*, 2001a; McGuire *et al.*, 2001b).

Vaitkevicius *et al.* (1993) and Lakatta (see Lakatta and Levy (2003a); Lakatta and Levy (2003b)) from the Gerontology Research Center, National Institute of Aging in Baltimore have taken this matter further, have confirmed the relationship between systolic blood pressure, aortic PWV, and carotid AIx with age, and have shown an inverse relationship between these variables and maximal oxygen consumption (Fig. 27.7). The relationship between measured oxygen consumption and AIx was independent of age. They then compared a small group of older athletes with their sedentary peers and showed that these (with maximal oxygen uptake similar to younger sedentary subjects) had lower systolic pressure, aortic PWV, and carotid AIx than did their peers. Their conclusion that exercise training improves arterial stiffness needs qualification. This reduction in wave velocity and augmentation may have been due to a reduction in blood pressure as a consequence of training; alternatively, athletic training may have selected out those whose cardiovascular stiffness most comfortably allowed such activity (Ferrier *et al.*, 2001). As stated previously too, change in stiffness may be greater in peripheral muscular than in central predominantly elastic arteries.

There is no doubt that regular exercise is beneficial for health and is associated with lower resting blood pressure and heart rate, less extreme changes in blood pressure and heart rate with a burst of activity, and lower morbidity and mortality from cardiovascular events (Clausen, 1977; Ward *et al.*, 1987; Curfman, 1993; Fletcher, 1994; Rodriguez *et al.*, 1994; Sherman *et al.*, 1994; Stratton *et al.*, 1994; Jennings, 1995; Hambrecht *et al.*, 1998; Hambrecht *et al.*, 2000a; Hambrecht *et al.*, 2000b; Tall, 2002).

CARDIAC FAILURE

The severity of cardiac failure is assessed on the basis of capacity for exercise (New York Heart Association criteria). Treatment of cardiac failure usually improves the capacity for exercise and thus the NYHA class. Judicious aerobic exercise can improve cardiac failure as it can, the

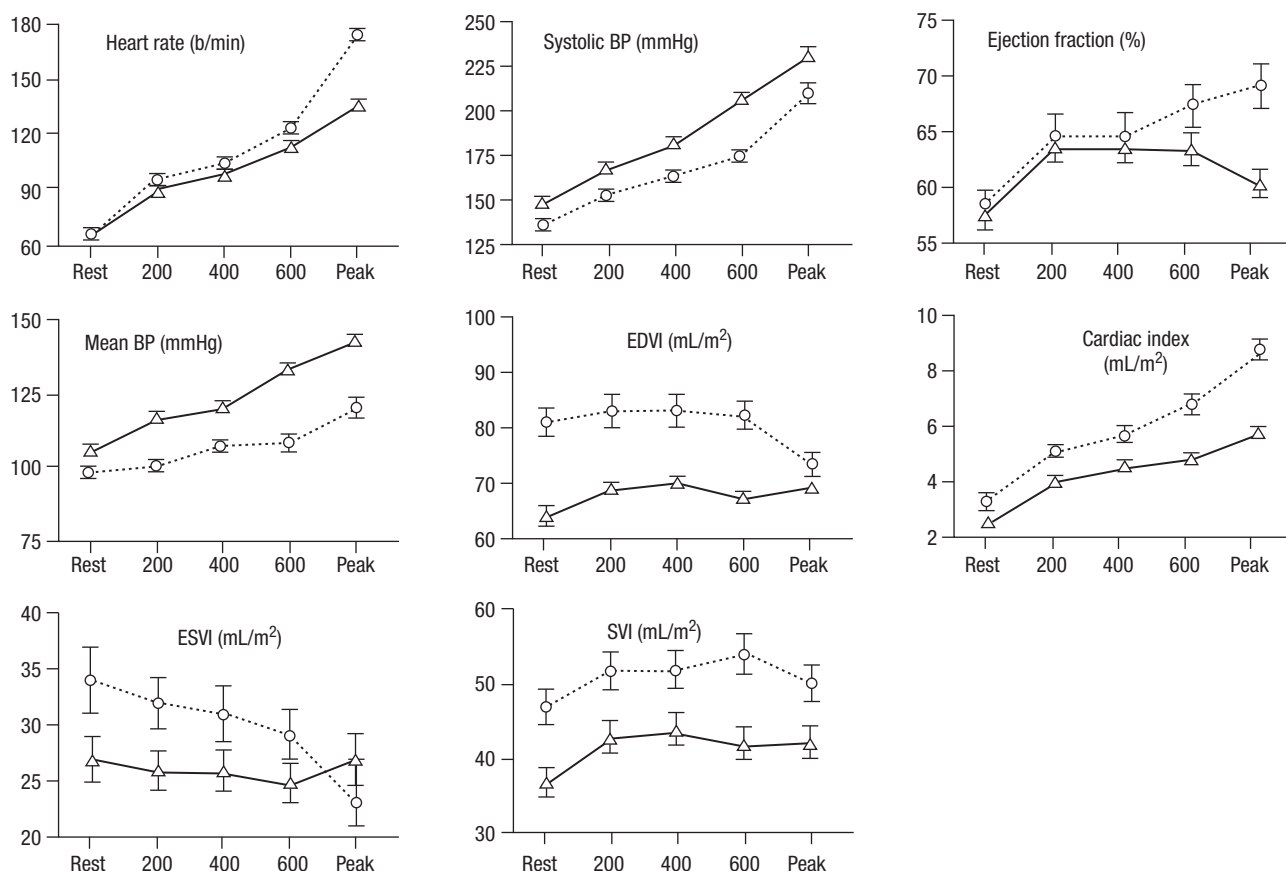


Figure 27.6 Mean response to graded supine exercise in 11 young (age 24–32 years, $\circ-\circ$) and 13 older (age 60–82 years, $\triangle-\triangle$) rigorously screened healthy men. BP, brachial blood pressure; ESVI, end-systolic volume index; EDVI, end-diastolic volume index; SVI, stroke volume index. Reproduced from Stratton *et al.* (1994)

complications of diabetes mellitus (Ward *et al.*, 1987; Belardinelli *et al.*, 1995; Hambrecht *et al.*, 1998) (see AHA Position Statement on Exercise and Cardiac Failure (2003); Marwick *et al.* (2009)). Of all therapies, nitrates appear to be most effective in reducing symptoms and improving exercise capacity (Leier *et al.*, 1983; Cohn *et al.*, 1991; Cohn, 1994). Nitrates reduce venous pressure, as do many other drugs used in heart failure. However, they are most effective in reducing ascending aortic impedance modulus, which remains elevated during exercise in patients with cardiac failure (Laskey *et al.*, 1985; Laskey and Kussmaul, 1987; Nichols and Pepine, 1992), acting through a reduction in peripheral wave reflection (Yaginuma *et al.*, 1986; Fitchett *et al.*, 1988a; Fitchett *et al.*, 1988b; Latson *et al.*, 1988; Jiang *et al.*, 2002; Uebaba *et al.*, 2002). Other vasodilator drugs (Chen *et al.*, 1999) have a similar effect (see Chapter 26), but nitrates appear to be the most powerful and effective during exercise.

AORTIC COARCTATION

In aortic coarctation, the minima of impedance modulus in the ascending aorta occur at a very high frequency as a

consequence of the short distance from the heart to a major reflecting site at the top of the descending aorta (O'Rourke and Cartmill, 1971). Values of impedance modulus are very high at low frequencies, thus accounting for the high amplitude of the arterial pressure pulse. With exercise, one would expect little or no change in ascending aortic impedance, so that the amplitude of the pressure pulse, and of systolic pressure, would be greatly increased (see Chapter 24).

Taylor and Donald (1960) drew attention to the dramatic increases in brachial artery pressure that occur during exercise in patients with aortic coarctation and showed that these are far greater than those seen in patients with severe essential hypertension who would be considered to require urgent medical therapy and restriction of activity (Fig. 24.12). These authors also pointed out that such changes in pressure usually occur without symptoms (although they can induce cardiac failure), with moderate exertion only and in patients otherwise asymptomatic, and who usually have had no advice to reduce their activities. These pressure increases are precisely what one would expect from a reduction in the cushioning capacity of the arterial system. Their potentially damaging effects on the heart and arteries were

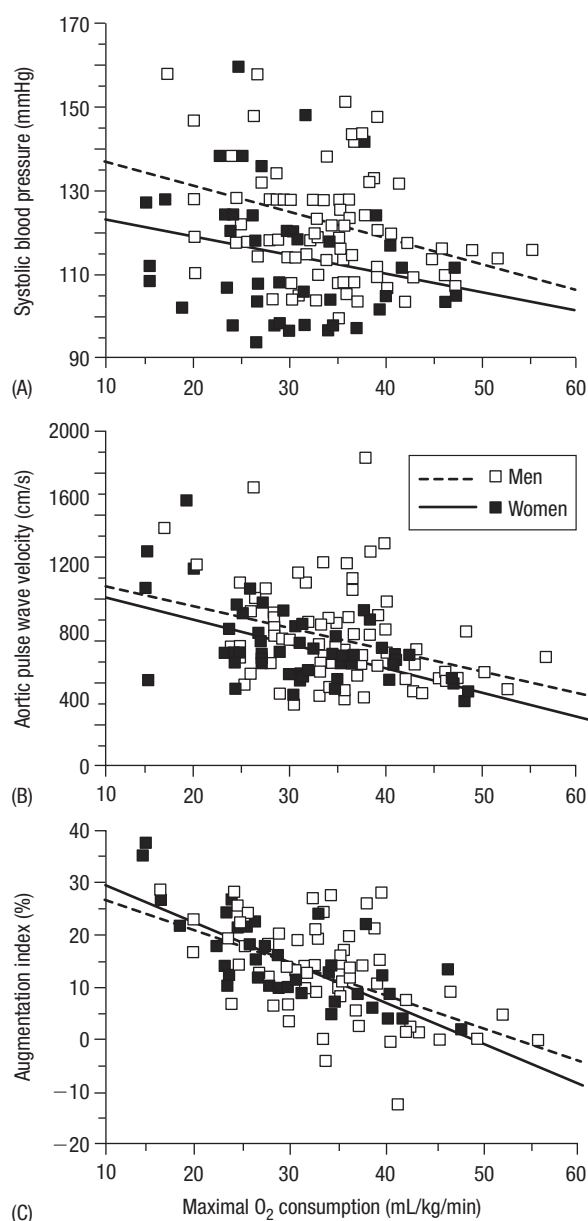


Figure 27.7 Scatterplots of systolic blood pressure (SBP), aortic pulse wave velocity (APWW), and augmentation index (Alx) as a function of maximal oxygen consumption ($\text{VO}_2 \text{ max}$). (A) SBP. Men, $y = 140.2 - 0.55 \text{ VO}_2 \text{ max}$, $r = 0.30$, $P < 0.01$; women, $y = 126.6 - 0.41 \text{ VO}_2 \text{ max}$, $r = 0.22$, $P = \text{NS}$. (B) APWW. Men, $y = 1234 - 13.8 \text{ VO}_2 \text{ max}$, $r = 0.34$, $P < 0.01$; women, $y = 1144 - 4.3 \text{ VO}_2 \text{ max}$, $r = 0.49$, $P < 0.01$. (C) Alx (percent). Men, $y = 34.6 - 0.64 \text{ VO}_2 \text{ max}$, $r = 0.54$, $P < 0.0001$; women, $y = 37.7 - 0.76 \text{ VO}_2 \text{ max}$, $r = 0.74$, $P < 0.0001$. Data from 146 rigidly screened healthy volunteers from the Baltimore Longitudinal Study of Aging. Reproduced from Vaitkevicius *et al.* (1993)

stressed by Taylor and Donald, and by others (O'Rourke and Cartmill, 1971; Maron *et al.*, 1973; Nanton and Olley, 1976; Maron, 2000). Similar changes might be expected with exercise when the aorta is extremely stiff, as in the elderly. This is so (Vaitkevicius *et al.*, 1993; Nussbacher

et al., 1995), but the elderly are less likely to engage in strenuous exertion than are young subjects with aortic coarctation. The values of brachial pressure given in Fig. 24.12 are probably close to those generated in the ascending aorta, unlike the situation shown in Fig. 10.14, on account of the contour of the pressure wave in aortic coarctation, which at rest has far more energy in the first than in the second and higher harmonics (O'Rourke and Cartmill, 1971), and so undergoes less amplification to the periphery.

ANGINA PECTORIS

The diagnosis of angina pectoris is made on the basis of its predictable relationship to physical exercise. Angina is caused by myocardial ischemia, resulting from an imbalance between myocardial oxygen demand and supply. Angina occurs in susceptible people during exercise because myocardial oxygen demands are increased during exercise beyond the level of maximal supply. Thus, angina pectoris is usually caused by obstructive coronary artery disease, and so is discussed in Chapter 17.

One must emphasize, however, that angina does not necessarily require coronary artery disease and can occur when demand is very high, as in severe aortic stenosis or hypertrophic cardiomyopathy. It can also occur when there is an imbalance between the pressure maintained during ventricular systole (which determines oxygen demand by the left ventricle) and the pressure maintained during diastole (which is a major determinant of myocardial perfusion and supply). An increase in the former and a fall in the latter, or an increase in duration of systole relative to diastole, may cause angina pectoris in the presence of minor or no coronary artery disease (see Chapter 17) (Boudoulas, 1991; Ferro *et al.*, 1995; Boudoulas and Leier, 2000; O'Rourke, 2008). This is always a factor contributing to angina during exercise; in some circumstances, as when the ventricle is hypertrophied, or systole is prolonged relative to diastole, or diastolic relaxation is impaired or venous pressure raised, it may be dominant (Schulman *et al.*, 1992; Watanabe *et al.*, 1993; Masuyama *et al.*, 1994; Ohtsuka *et al.*, 1994; Stratton *et al.*, 1994; Ardehali *et al.*, 1995; Gosse and Clementy, 1995; Gould, 1997; Boudoulas and Leier, 2000; Panting *et al.*, 2002; O'Rourke, 2008).

The ability to generate the aortic from the radial pressure pulse and to determine indices of myocardial oxygen supply and demand during exercise, will assist in further studies of this subject, as discussed in Chapters 17 and 30 (Karamanoglu *et al.*, 1993; Takazawa *et al.*, 1995a; Chen *et al.*, 1997). Some cases of 'syndrome X' (angina pectoris with normal coronary arteries at angiography) probably result from such an imbalance in aortic pressure components (see Chapter 17 and Fig. 30.38) (Kemp, 1973; Boudoulas, 1991; Yoshio *et al.*, 1995; Gould, 1997; Panting *et al.*, 2002; Cannon, 2009). Although convincingly

demonstrated by Panting *et al.* (2002), this was not attributed to such hemodynamic factors by the authors nor in subsequent *New England Journal of Medicine* correspondence. The eyes of cardiologists appear to be fixed on narrowed arteries, to the exclusion of perfusion pressure and time.

DIABETES MELLITUS

Regular exercise is very important in the treatment of diabetic patients. Frank type 2 diabetes mellitus may disappear in a person who conforms with an exercise and dietary program (Losordo, personal communication). The American Heart Association has made a scientific statement on this issue (Marwick *et al.*, 2009). Scott *et al.* (2008) from Marwick's department have shown that type 2 diabetic patients are more likely than healthy normals to have a hypertensive brachial pressure response to exercise. They confirmed that this hypertensive response was present in the aorta also in diabetics, and was associated with an increase in relative LV wall thickness, and signified an adverse effect of diabetes on ventricular–vascular coupling.

POSSIBLE ILL EFFECTS OF EXERCISE ON THE CARDIOVASCULAR SYSTEM

Some caveats have already been offered on exercise – in aortic coarctation and in unaccustomed bouts. Anecdotal accounts of harm arising from exercise (Siscovick *et al.*, 1984; Shave *et al.*, 2010; Tofield, 2010) are gaining support from detailed studies (Mittleman *et al.*, 1993; Willich *et al.*, 1993; Tofler *et al.*, 1996). None of these counter the view that regular moderate exercise is beneficial and should be continued through life as long as possible (Ward *et al.*, 1987; Curfman, 1993; Vaitkevicius *et al.*, 1993; Rodriguez *et al.*, 1994; Sherman *et al.*, 1994; Marwick *et al.*, 2009). People who exercise regularly are more likely than those who do not to control their weight and to control other risk factors for coronary and other vascular disease. Their arterial pressure is apt to be lower and resting heart rate slower than their sedentary peers. They are less likely to experience the same surge in blood pressure or in heart rate if a burst of exercise is necessary in normal daily activities.

Recent studies have clarified the association between exercise and coronary thrombosis and myocardial infarction. In sedentary people, coronary thrombosis is more likely to occur during a burst of activity or shortly (within 1 hour) after it than during a comparable period of rest (Mittleman *et al.*, 1993; Willich *et al.*, 1993). In people who take regular exercise, coronary occlusion and myocardial infarction are less frequent (Fig. 27.8), and do not show the increased incidence with exercise. The mechanism of coronary thrombosis is discussed in

Chapter 23. This is usually caused by disruption or rupture of an atherosclerotic plaque by inflammation and weakening at the edge of the plaque or by disruptive forces at the vessel wall (Richardson *et al.*, 1989; Lee *et al.*, 1991; Cheng *et al.*, 1993; Lee *et al.*, 1994; Davies, 1997; Libby, 2001a; Libby *et al.*, 2002; Libby *et al.*, 2009). It is possible that the sudden elevation of blood pressure and/or heart rate that occurs with extreme unaccustomed exercise may be the cause of such disruption and so of coronary thrombosis and the infarction that follows (Elefteriades and Farkas, 2010).

Sudden death during running has been referred to earlier. In a blend of history and legend, Pheidippides (530–490 BC), an Athenian herald, was sent to Sparta to seek help when the Persians landed at Marathon. He ran 240 km (150 miles) in 2 days. He subsequently ran the 40 km (25 miles) from the battlefield near Marathon to Athens to announce the victory of Greeks over Persians in the Battle of Marathon (490 BC) with the word Νενίκηκαμεν (*Nenikékamen*, 'We have won') and promptly after that he collapsed and died on the spot. Sudden death may be caused by an arrhythmia following coronary thrombosis, but may be precipitated by a period of myocardial ischemia brought on by a transient change in aortic pressure waveform. It might be brought on, too, by an arrhythmia

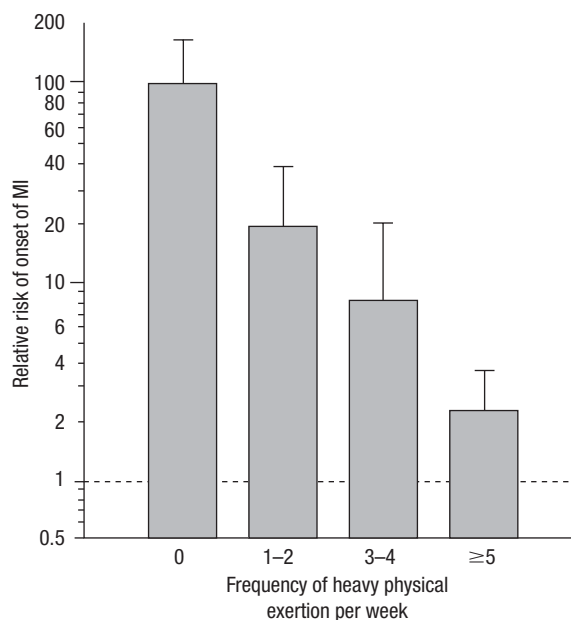


Figure 27.8 Relative risk of myocardial infarction (MI) according to the usual frequency of heavy exertion. Heavy exertion was defined as physical activity at a level of 6 MET or more. The relative risk is shown on a logarithmic scale. Habitually sedentary people had an extreme relative risk, whereas those who reported heavy exertion five or more times per week had a risk only 2.4 times higher than the base-line risk ($P < 0.001$). The T-bars indicate the 95 percent confidence limits. The dotted line indicates the base-line risk. Reproduced from Mittleman *et al.* (1993)

complicating an otherwise minor ischemic episode. Another issue is the possibility of early exercise after myocardial infarction predisposing to infarct expansion and even rupture (Lewis, 1934; White, 1951a; Jugdutt *et al.*, 1988) and to progressive ventricular dilation which can follow acute infarction (White *et al.*, 1987; Pfeffer *et al.*, 1991). In a recent cross-sectional study, we showed that marathon runners had significantly higher aortic PWV, whereas there was no difference in AIx and AIx corrected for heart rate (AIx@75) compared to controls (Vlachopoulos *et al.*, 2010d). While such aortic stiffening, like short

stature in Ethiopian athletes, may assist development and maintenance of ‘the rhythm of running’ as described earlier in this chapter, both may predispose to long-term ill effects, including coronary atherosclerosis (Mohlenkamp *et al.*, 2008). An influential editorial in the *New England Journal of Medicine* (Curfman, 1993) posed the question ‘Is exercise beneficial – or hazardous – to your heart?’ The answer is that it can be both. Hopefully, the discussion in this chapter provides a background for distinguishing why exercise can be good or bad through different mechanisms that affect the heart and arterial tree.

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Central arterial pressure

Introduction	569	Form factor, mean pressure, and the arterial pressure waveform	571
Physiological principles	570		
Physical principles	570		

INTRODUCTION

In undertaking the work described in this book, one of our goals was to develop a better clinical hydraulic measure of left ventricular load than was currently provided. This was available at the end of the twentieth century as arterial blood pressure measured non-invasively from brachial (or radial) cuff sphygmomanometry or from direct cannulation of the radial artery. The cuff is widely applied in clinical practice, in epidemiology, and in drug studies. Radial cannulation is frequently used by intensivists and anesthetists in operating rooms or intensive care wards. The pressure that one wishes to obtain is that in the ascending aorta, since this approximates that in the left ventricle during systole, and that at the coronary ostia during the period of diastole. We expected that such pressures (aortic systolic and pulse pressures) would be more relevant to left ventricular load and central artery circumferential stress than systolic and pulse pressures measured in the upper limb. We also expected that other indices derived from the aortic pressure wave, including its amplification to the upper limb, its augmentation from peripheral wave reflection, its integral during the periods of systole and diastole would provide more information on vascular load, cardiac function, and ventricular-vascular interaction. Results of such application are described later in this chapter and elsewhere, including Chapters 14 and 15. Initially, however, it is desirable to examine theoretic and practical implications which may detract from our ability to apply this information, and the approximations that are made in attempting to do so. The initial approximation was to use the carotid artery pressure waveform as a surrogate of the aortic pressure waveform. The carotid pressure waveform was rarely used by pioneers of sphygmography in the nineteenth century (although Mahomed (1872) published one tracing to show its different contour to the radial – see

Fig. 10.34). Mackenzie and Lewis used carotid waveforms, measured through tambours applied to the neck to identify and understand cardiac arrhythmias prior to availability of the ECG (Snellen, 1883). Carotid tracings were later used by Weissler (1968) and colleagues at Ohio State University for measurement of systolic time intervals as a measure of left ventricular (LV) function. This work was overtaken by introduction of echocardiography in the 1970s, but was reintroduced following development of arterial tonometry (Drzewiecki *et al.*, 1983), and introduction of accurate instruments by Huntly Millar as a modification of his high fidelity catheter manometer (Murgu and Millar, 1972). Ray Kelly (O'Rourke and Kass, 2001) was responsible for the seminal studies with this instrument including calibration and validation (Kelly *et al.*, 1989b; Kelly *et al.*, 1989f) and the first drug study to show benefit of a vasodilating β -blocker (dilevalol) over atenolol on carotid systolic pressure despite similar effects of both drugs on brachial systolic pressure and aortic pulse wave velocity (PWV) (Kelly *et al.*, 1989d). Pressure and diameter waveforms are similar though not identical in the carotid artery and proximal aorta, so that one can be used as a surrogate of the other (van Bortel *et al.*, 2001b). Pressure waves are basically similar in the ascending aorta and carotid artery (Kelly *et al.*, 1989b; Chen *et al.*, 1997), although amplitude of the augmented pressure wave is almost twice as high in the ascending aorta as in the carotid artery (Kelly *et al.*, 1989a; Takazawa *et al.*, 1998). The carotid pressure wave is widely used as a surrogate of ascending aortic pressure and a measure of central pressure (Kelly *et al.*, 1989b; Chen *et al.*, 1997; Kass *et al.*, 1998; Mitchell *et al.*, 2002; Mitchell *et al.*, 2003; Mitchell *et al.*, 2004a; Mitchell *et al.*, 2004b; Mitchell *et al.*, 2005a; Mitchell *et al.*, 2005b; Dart *et al.*, 2006; Mitchell *et al.*, 2006; Dart *et al.*, 2007; Mitchell *et al.*, 2007a; Mitchell *et al.*, 2007b; Segers *et al.*, 2007b; Mitchell *et al.*, 2008; Segers *et al.*, 2009; Mitchell *et al.*, 2010).

PHYSIOLOGICAL PRINCIPLES

Elsewhere, we have shown that the pressure wave is amplified in peripheral arteries, and that the degree of amplification is variable and depends upon arterial stiffness, being close to unity (i.e. no amplification, 0 percent), in older people and subjects with arteriosclerosis, or above 2 (> 100 percent) in young adults, or even in older people in the presence of tachycardia or when ejection duration is short. Figure 28.1 illustrates amplification of the pressure wave between ascending aorta at left and radial artery at right, with carotid and brachial arteries in between. Amplification of the wave has its greatest effect on the wave peak, i.e. on systolic and pulse pressure, and little effect on diastolic pressure. There is little effect either on mean pressure (which falls by < 1 mmHg between aorta and radial artery) (Pauca *et al.*, 1992; Pauca *et al.*, 2001). To a first approximation, one can consider diastolic and mean pressure to be constant between the aorta (or carotid) and radial arteries (Kelly and Fitchett, 1992; Pauca *et al.*, 1992; Pauca *et al.*, 2001).

When one compares the aortic and radial waves as superimposed (Fig. 28.2), one notes the near identity of not only diastolic and mean pressure within two arteries, but also the peak aortic systolic pressure and the beginning of the late systolic pressure wave in the radial artery. Such relationships suggest a method (described below) to calculate aortic systolic and pulse pressure from the radial artery waveform.

Figure 28.3 shows diagrammatically the relationship between the aortic systolic wave and the ventricular pressure wave; these are practically indistinguishable throughout the period of systole, under most conditions, and a change in the aortic pressure wave is always accompanied by an identical change in the left ventricular pressure wave. This relationship holds unless there is an obstruction between the ventricular cavity and ascending aorta as in aortic stenosis or hypertrophic cardiomyopathy (see Figs 6.8 and 6.24) and/or high velocity flow causing a Venturi effect in the aorta (Fig. 6.23) (Takazawa,

1987). Details of simultaneously recorded pressure waves in aorta and LV are discussed on p. 159.

PHYSICAL PRINCIPLES

Figure 28.4 shows the method of applanation as applied to the carotid, brachial, and radial arteries. The theory of applanation tonometry only applies when the sphere (eyeball) or round tube (artery) has its surface flattened (applanated) under the measuring sensor (which in the Millar device is < 1 mm² in area). Applanation can usually be assured in the radial artery, where the artery is sandwiched between sensor and radius bone, and can often be obtained for the carotid artery when the vessel, at a greater depth, is flattened at front and back when compressed gently against the vertebral bodies and ligaments behind. In the brachial artery, applanation cannot be assured for two principal reasons. First, one cannot always flatten the artery's anterior wall against the bone behind, since it tends to move backwards into soft tissue surrounding the humerus. Second, the stiff bicipital aponeurosis intervenes to a varying extent between the

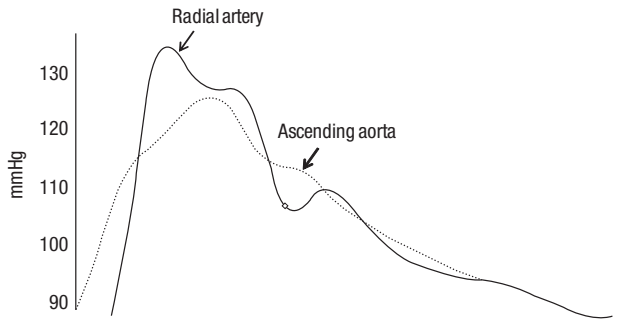


Figure 28.2 Simultaneously recorded ascending aortic (synthesized) and radial artery pressure (measured) waves. Note delay between feet of the pressure waves and similarity of radial artery second shoulder with ascending aorta peak

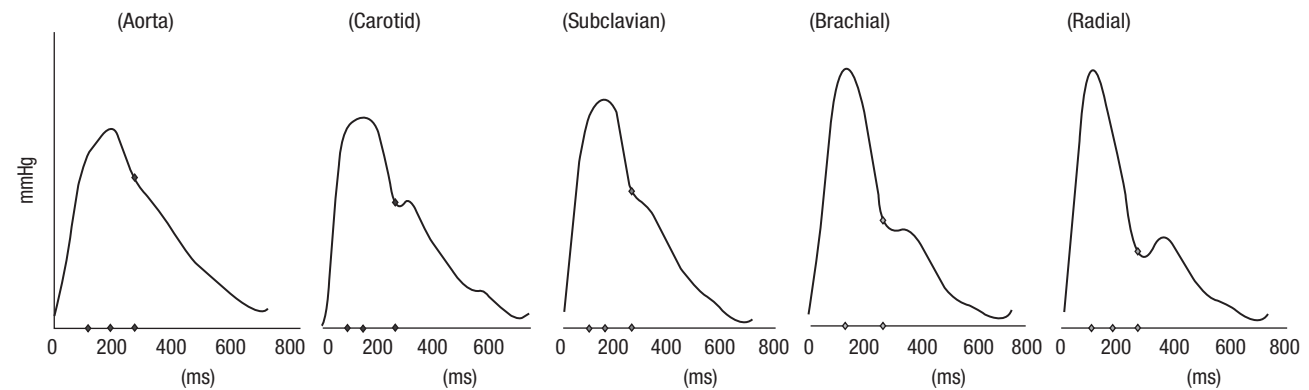


Figure 28.1 Pressure waves measured in a healthy young man between the ascending aorta (at left) and radial artery (at right) with the carotid, subclavian, and brachial arteries in between

sensor and artery and interferes with appplanation of the anterior surface of the artery. Certainly, one can record a waveform at the elbow, but it has not been possible to gain a consistent shape of the wave compared to the invasively recorded brachial wave, whereas this is possible and has been confirmed for the radial and carotid arteries (Kelly *et al.*, 1989a; Kelly *et al.*, 1989f; Chen *et al.*, 1997). The tonometrically recorded brachial wave often has a blunted shape, such that its form factor (height of mean pressure level above diastolic ÷ pulse pressure) lies between the carotid FF and radial FF, and much closer to the carotid (Fig. 28.4) (Segers *et al.*, 2009; O'Rourke and Adji, 2010b).

FORM FACTOR, MEAN PRESSURE, AND THE ARTERIAL PRESSURE WAVEFORM

The term 'form factor' was introduced by Chemla *et al.* (2002) and Bos *et al.* (2007a) to provide a term which describes the difference between $(MP - DP) \div PP$. This expresses where the mean pressure level lies in relation to the top and bottom of the pressure waveform. In the past, it was

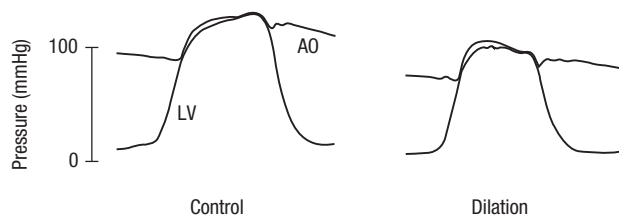


Figure 28.3 Ascending aortic (AO) and left ventricular (LV) pressure waves recorded simultaneously with a multisensor catheter in a young subject before (left, control) and after and sublingual nitroglycerin (right, dilation). Note that aortic and LV pressures are almost identical after opening of the aortic valve until closure of the aortic valve. Reproduced from Murgu and Westerhof (1987)

a tradition to express mean pressure as one-third the difference between systolic and diastolic pressure + diastolic pressure, i.e. mean pressure (MP) = diastolic pressure (DP) + $0.333 \times$ pulse pressure (PP). In this case, the form factor is 33 percent. A number of authors have questioned this on the basis of brachial artery tonometric waveforms which showed form factor varying between 28 and 55 percent with average around 40 percent (O'Rourke and Adji, 2010b). This value is similar to that of the carotid waveform. Form factor for the ascending aortic pressure wave can be as high as 61 percent (O'Rourke and Adji, 2010b).

This issue is illustrated in Fig. 28.5 which assumes a heart beat as consisting of two squares, set side by side, with equal systole and diastole, and with the pressure pulse being confined within the two squares. If the pressure pulse is represented by a triangle made up of half one square, its mean value will be 25 percent of the distance between base and apex, so form factor is 25 percent. If the pressure wave has the appearance of the whole of the first square and zero over the next square, mean pressure will be half way up the wave and form factor is 50 percent. Of course, these are extremes which are not seen in life, but the figure does make the point that form factor depends on the shape of the wave in an artery. If the pulse pressure is different and peaked at the periphery as seen in Fig. 28.1 for the radial artery, form factor will be less than in the artery where pulse pressure is lower. In our cohort of subjects and patients, form factor averaged 33.6 percent in the radial artery, 41 percent for the ascending aorta, and around 40 percent when this was measured in the carotid artery; form factor was similar (around 40 percent) when measured in the brachial artery (O'Rourke and Adji, 2010b). Segers *et al.* (2009) reported similar values of form factor in the carotid and brachial arteries. Form factor in any artery varies with the shape of the pressure waveform and especially on amplification between ascending aorta and radial artery (see Fig. 6.11).

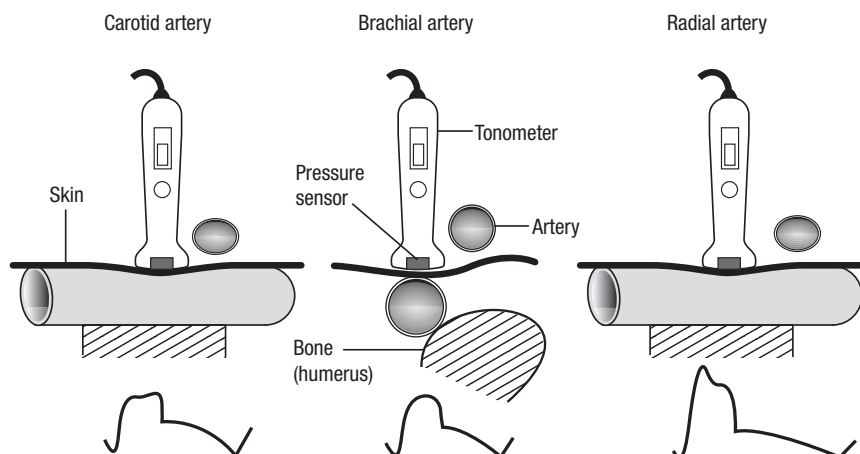


Figure 28.4 Radial artery tonometer applied at three positions – to the carotid artery in the neck, to the brachial artery at the elbow, and to the radial artery at the wrist. Because of the rounded nature of the humerus, and the bicipital aponeurosis under the skin, appplanation (flattening) of the brachial artery cannot be assured

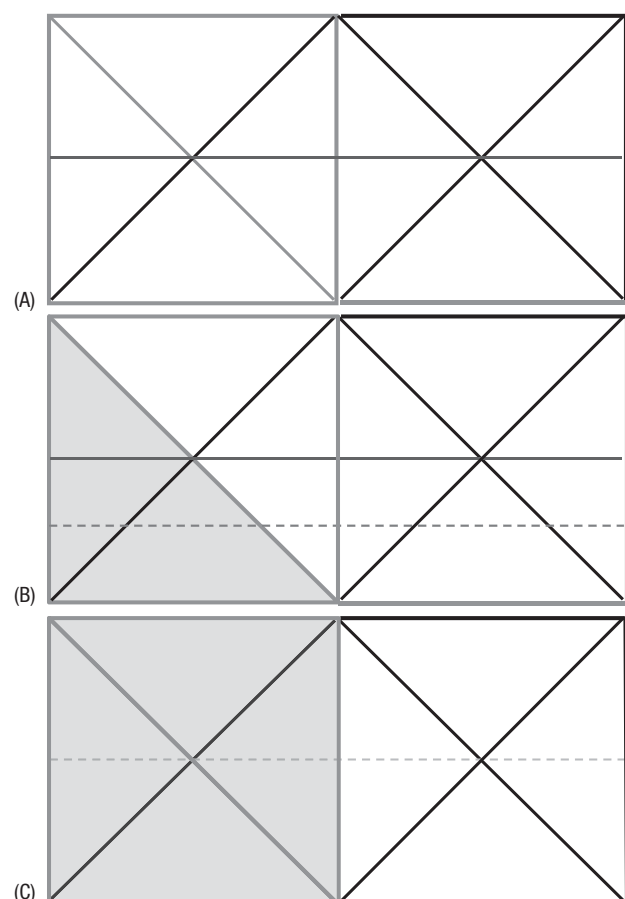


Figure 28.5 Skeleton of the pulse waveform (A) without wave but systolic and diastolic period equal and composed of four triangles. (B) The wave comprises half of systole and one triangle. Form Factor is 25 percent. (C) The wave comprises all of systole and none of diastole. Form Factor is 50 percent. Dotted line denotes mean pressure

Figure 6.11 shows the relationship between radial artery form factor and amplification of the pressure wave from aorta to radial artery.

In this book, we make little of various estimations of form factor, because we know that it varies widely, and that it is unnecessary to consider if the radial artery waveform is integrated arithmetically over the whole cardiac cycle (or better, over eight or so cycles which correspond to slightly different pressure waveforms over one respiratory cycle). It is however, very important when calculating aortic amplification from the peripheral pulse.

Calibration and form factor

When Kelly and colleagues introduced arterial tonometry into clinical practice, they were most concerned about indices provided by the waveforms, including time intervals, amplification of the wave, and augmentation of the pressure wave at different sites and the effect of aging on

the pressure waveform. Their major paper (Kelly *et al.*, 1989a) describing change in arterial pressure waves with age in the radial, carotid, and femoral arteries, described amplitude in millivolts of transducer signal. Clinicians wished the waves to be calibrated to mmHg. In the absence of invasive calibration data, the best estimate of arterial pressure was taken to be brachial cuff pressure, and this was applied to the radial pressure waveform. In so doing, it was recognized that pulse pressure is further amplified in travel between brachial and radial arteries. However, studies of intra-arterial pressure showed that there was little difference in pulse pressure between the brachial and radial arteries (Fig. 28.6, Table 28.1) (Kroeker and Wood, 1955; Rowell *et al.*, 1968; Takazawa *et al.*, unpublished data), so that it seemed reasonable to apply cuff brachial systolic and diastolic pressure to the radial pressure waveform, so taking the magnitude of systolic pressure error as being as small (< 1 mmHg) as is that for mean or diastolic pressure between brachial and radial arteries. An alternative approach was to use a validated radial cuff oscillometry device, such as the Omron HEM-630 as an acceptable substitute for measurement of brachial pressure.

This view was challenged by Verbeke *et al.* (2005) who had concentrated on analysis of the carotid pulse as a measure of central pressure, and calibrated this from the brachial cuff values of systolic and diastolic pressure with mean value obtained by mathematical integration of the tonometrically recorded brachial artery pressure wave at the elbow. From this, they calculated high values of form factor (42.4 percent), with these approaching values in the carotid artery (44 percent). On calculating the value of carotid systolic and pulse pressures from the assumption of identity of mean and diastolic pressure at brachial and carotid sites, they found that there was very little amplification between the carotid and brachial artery, and substantial amplification between the brachial and radial artery (Segers *et al.*, 2005b; Verbeke *et al.*, 2005; Segers *et al.*, 2009). This contradicted our assumptions of trivial amplification between brachial and radial arteries. We repeated the study and obtained similar results – variable amplification between carotid and brachial site and variable, but greater, amplification between brachial and radial arteries (Fig. 28.7). In the process, we became aware of the difficulties in accurately appanating the brachial artery, and found that the tonometric brachial artery waveforms were quite different to those previously recorded invasively, and had a shape and form factor closer to the carotid than the radial artery (Fig. 28.7) (O'Rourke *et al.*, 2005). We could only explain results on the basis of erroneous interpretation of brachial tonometry. We urged authors to check their tonometric brachial artery pressure waves against brachial intra-arterial waves. We had been successful in showing near identity of non-invasive with invasive waves at the radial and carotid sites (Kelly *et al.*, 1989a; Kelly *et al.*, 1989e), but had not been able to find consistent similarity at the brachial site.

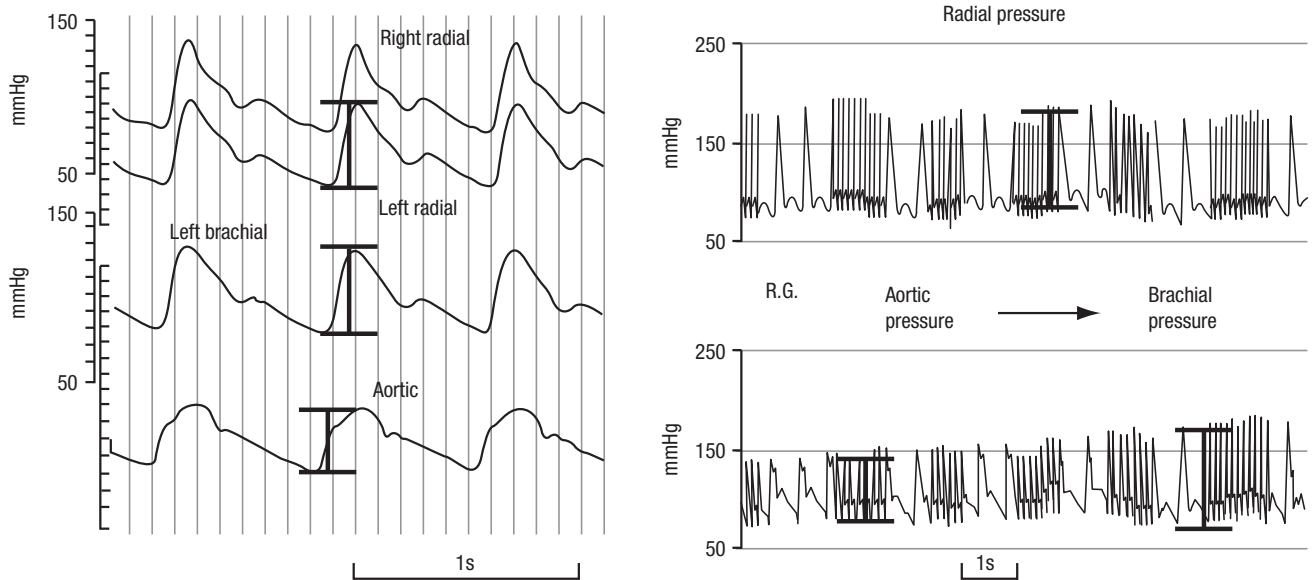


Figure 28.6 Pressure waves recorded simultaneously in ascending aorta, brachial artery, and radial artery. Amplitude of radial and brachial waves are similar, but amplitude of the aortic wave is much smaller. Left: Pressure waves recorded by Kroeker and Wood (1955) in left and right radial artery (top) in left brachial artery and the proximal aorta of a young adult volunteer. Right: Pressure waves recorded in the radial artery (top), simultaneously with pressure waves recorded from catheter inserted into the opposite brachial artery during withdrawal from the ascending aorta (at left) to the brachial artery (at right) in a young adult volunteer. From Rowell *et al.* (1968)

Table 28.1 Comparison of published pressure amplification values

Method	Invasive	Generalized transfer function	Brachial tonometry
Amplification = arm PP/ central PP	1.38 (Kroeker and Wood, 1955)	1.32 (Wilkinson <i>et al.</i> , 2001)	−1.01 (Mitchell <i>et al.</i> , 2002)
	1.34 (Pauca <i>et al.</i> , 1992)	1.27 (Asmare <i>et al.</i> , 2001b)	1.02 (Mitchell <i>et al.</i> , 2003)
	1.40 (Pauca <i>et al.</i> , 2001)	1.28 (Sharman <i>et al.</i> , 2006)	1.01 (Mitchell <i>et al.</i> , 2004a)
		1.26 (Williams <i>et al.</i> , 2006)	−1.01 (Mitchell <i>et al.</i> , 2005a)
		1.25 (Hashimoto <i>et al.</i> , 2007a)	1.01 (Mitchell <i>et al.</i> , 2006)
		1.22 (Hashimoto <i>et al.</i> , 2007b)	1.01 (Mitchell <i>et al.</i> , 2007a)
		1.36 (McEniery <i>et al.</i> , 2008)	1.03 (Mitchell <i>et al.</i> , 2008)
		1.33 (Roman <i>et al.</i> , 2007)	1.04 (Segers <i>et al.</i> , 2009)
		1.26 (O'Rourke and Adji, 2008)	1.14 (Waddell <i>et al.</i> , 2001)
		1.27 (Nichols <i>et al.</i> , 2006)	1.00 (Dart <i>et al.</i> , 2007)
Mean	1.38	1.31	1.03
Amplification = arm SP/central SP (mmHg)	14 (Kroeker and Wood, 1955)	12.3 (Wilkinson <i>et al.</i> , 2001)	−1.0 (Mitchell <i>et al.</i> , 2002)
	12 (Pauca <i>et al.</i> , 1992)	9.2 (Asmare <i>et al.</i> , 2001b)	0.5 (Mitchell <i>et al.</i> , 2003)
	16 (Pauca <i>et al.</i> , 2001)	10.1 (Adji <i>et al.</i> , 2004b)	3.0 (Mitchell <i>et al.</i> , 2004a)
		17 (Sharman <i>et al.</i> , 2006)	−1.0 (Mitchell <i>et al.</i> , 2005a)
		10.2 (Williams <i>et al.</i> , 2006)	1.0 (Mitchell <i>et al.</i> , 2006)
		8.8 (Hashimoto <i>et al.</i> , 2007a)	1.5 (Segers <i>et al.</i> , 2007b)
		12.2 (McEniery <i>et al.</i> , 2008)	5.3 (Waddell <i>et al.</i> , 2001)
		9.0 (Roman <i>et al.</i> , 2007)	1.0 (Dart <i>et al.</i> , 2007)
		12 (O'Rourke and Adji, 2008)	
		13 (Nichols <i>et al.</i> , 2006)	
Mean	14.5	11.4	1.1

After O'Rourke and Adji (2010a).

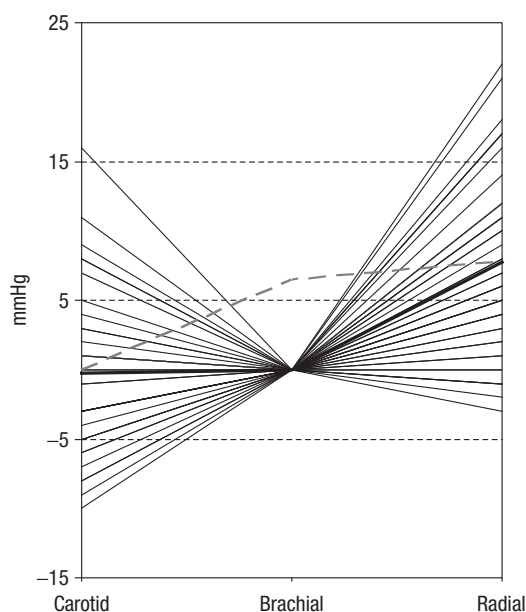


Figure 28.7 Apparent amplification of pulse pressure between carotid and brachial artery, and between brachial and radial artery in 101 patients attending our outpatient clinic, with all waveforms recorded by applanation tonometry and with calibration of the brachial waveform to brachial cuff pressure values. Amplification was very variable with average < 1 mmHg between carotid and brachial arteries, and approximately 7 mmHg between brachial and radial artery (thick lines). Dashed line shows average amplification seen when the radial waveform was calibrated to brachial pressure values. Unusual results (which agree with those of Segers *et al.* (2009)) are explicable on the basis of inability to applanate the brachial artery (see text)

This issue remains unresolved (Avolio *et al.*, 2009b; Segers *et al.*, 2009), although we remain confident that radial and brachial intra-arterial waveforms are practically identical (O'Rourke and Takazawa, 2009; O'Rourke and Adji, 2010a; O'Rourke and Adji, 2010b).

These issues on brachial artery calibration, first raised by Segers *et al.* (2005b) and Verbeke *et al.* (2005) are also relevant to earlier work by Dart, Jennings, and Cameron in Melbourne, and by Mitchell, Izzo, and colleagues in the United States. All had used brachial applanation tonometry and cuff systolic and diastolic pressures for calibration of the peripheral (brachial) pressure wave, and then had extrapolated from the mean and diastolic pressure to calculate central systolic and pulse pressure for the carotid pressure waveform. Just as in the Segers' Asklepios study, papers emanating from these groups show little or no difference between brachial and carotid systolic mean and pulse pressure and concluded that central pressures are no better than brachial pressure for predicting cardiovascular events or explaining benefit of arterial vasodilating drugs (angiotensin-converting enzyme inhibitors (ACEIs), angiotensin receptor blockers (ARBs), calcium channel blockers (CCBs), and nitrates) over β -blockers or

Form Factor: A number which denotes pressure wave shape

Eqn 28.1
Form Factor (FF) = $\frac{\text{Mean pressure} - \text{Diastolic pressure}}{\text{Pulse pressure}} = \frac{\text{MP} - \text{DP}}{\text{SP} - \text{DP}}$

Eqn 28.2
 $\text{FF (carotid)} = \frac{\text{MP(c)} - \text{DP(c)}}{\text{SP(c)} - \text{DP(c)}}$ and $\text{FF (brachial)} = \frac{\text{MP(b)} - \text{DP(b)}}{\text{SP(b)} - \text{DP(b)}}$

Eqn 28.3
 $\frac{\text{FF(c)}}{\text{FF(b)}} = \frac{\text{MP(c)} - \text{DP(c)}}{\text{PP(c)}} \times \frac{\text{PP(b)}}{\text{MP(b)} - \text{DP(b)}}$

Since $\text{MP(c)} = \text{MP(b)}$ and $\text{DP(c)} = \text{DP(b)}$

Eqn 28.4
 $\therefore \frac{\text{FF(c)}}{\text{FF(b)}} = \frac{\text{PP(b)}}{\text{PP(c)}} = \text{Amplification}$

Figure 28.8 Relationship between amplification and ratio of form factors at two sites can be explained mathematically when diastolic and mean pressure are considered to be identical at the two sites. After Bos *et al.* (Bos *et al.*, 2007a; Bos *et al.*, 2007b) and Chemla *et al.* (2002)

diuretics (Dart *et al.*, 2006; Dart *et al.*, 2007). Studies which have shown benefit of central pressures have used the transfer function technique (see below, Asmar *et al.*, 2001b; Williams *et al.*, 2006; Agabiti-Rosei *et al.*, 2007; Roman *et al.*, 2007; Roman *et al.*, 2009) or more recently, the second systolic shoulder technique to calculate central aortic pressure (Hickson *et al.*, 2009b). Systematic differences between peripheral and central pressures expressed as mmHg and percent are set out in Table 28.1 (O'Rourke and Adji, 2010a). Figure 28.8 (including eqns 28.1, 28.2, 28.3, and 28.4) shows how amplification of the pulse between a central artery (aorta or carotid artery) and an upper limb artery (brachial or radial) is inversely related to the ratio of form factors in the central or peripheral artery. If the pressure wave is blunted (as when measured by tonometry in the brachial artery), its form factor will be exaggerated. This explains findings of Verbeke *et al.* (2005), Segers *et al.* (2009), and of O'Rourke and Adji (2010b) in Fig. 28.7.

Methods for measuring central pressure

Different methods for non-invasive measurement of central blood pressures have been alluded to throughout this and other chapters. All are approximations, with value dependent on ability to match invasively recorded pressure waves and their changes with drugs and under different conditions. The biggest approximation for all is the assumption that brachial cuff and brachial intra-arterial pressures are identical; this is discussed in Chapter 20. There are three principal methods:

1. *Use of a generalized transfer function (GTF) method* to generate ascending aortic pressure from the radial artery pressure wave. This method was used in the vast majority of studies set out in the

latter part of this chapter, and is described in detail in Chapter 30. When calibrated to intra-arterial pressure, this method provides values of aortic systolic, diastolic, pulse, and mean pressure which are virtually identical to simultaneously recorded ascending aortic pressure (Pauca *et al.*, 2001) under control conditions and during use of vasoactive drugs, during exercise (Sharman *et al.*, 2006), and with the Valsalva maneuver (Chen *et al.*, 1997). When used with non-invasive tonometry, the vagaries of cuff manometry calibration are the worst problems with precision. Both invasive and non-invasive methods are accepted by the US Food and Drug Administration Agency (FDA) as substantially equivalent to that obtained by aortic catheterization under K002742 and K012487. This method was approved in recent consensus documents (Laurent *et al.*, 2006; Agabiti-Rosei *et al.*, 2007; Mancia *et al.*, 2007a; Mancia *et al.*, 2007b).

2. *Use of the second systolic shoulder of the radial pulse waveform.* From Figs 28.1 and 28.2 of this chapter and in multiple other illustrations in this book, it should be apparent that the beginning of the second systolic shoulder of the brachial or radial waveform approximates systolic peak pressure. Value of this association was recognized at about the same time by Pauca *et al.* (2004), by Nichols (2005), and by Takazawa *et al.* (2007) in Japan and further confirmed by Hickson *et al.* (2009b). The reason for the association is relatively simple. The second systolic peak of the aortic pressure wave is made up of the reflected pressure wave returning from the lower body. This low frequency component of the pulse continues on down the arm with little attenuation or amplification, and is recognized here as the late systolic shoulder of the radial or brachial pulse. This method works well when the second systolic peak is recognizable, but cannot be applied when this localized peak is not present in the radial or brachial pressure wave – as in young adults, with tachycardia, in systolic heart failure, or during use of strong vasodilator drugs. Comparison of the GTF and late systolic shoulder method provide almost identical results of the central pressure wave (Adji and O'Rourke, 2004b). The GTF process works well at all times, the shoulder method works perhaps 90 percent of the time, and only when the late shoulder is apparent.
3. *Use of mean (MP) and diastolic (DP) pressure to calculate carotid pressure waves.* Previous studies in humans (Pauca *et al.*, 1992; Pauca *et al.*, 2001) have shown that diastolic pressure is virtually identical in carotid and radial arteries (carotid > radial by about 2 mmHg), and that

mean pressure is also virtually identical (mean < 1 mmHg less in the radial than carotid waveform). These values can be determined in a peripheral artery, and then applied to the carotid artery. From form factor of the carotid waveform, and if brachial diastolic and mean pressure are known, it is possible to extrapolate to the systolic peak, and so provide values of central systolic and pulse pressure. This method is more subject to sphygmomanometric cuff error since it is dependent on accurately determining diastolic and mean pressure, and then from the difference between (i.e. MP – DP, which is usually relatively small), extrapolating up to the systolic peak. This method magnifies cuff sphygmomanometric errors; often calculated carotid systolic and pulse pressures are equal to or even greater than brachial (Table 28.1). Such is the case in the papers by Dart, Mitchell, and Segers, previously cited. We regard these as flawed, since they are based on the use of applanation tonometry to an artery (the brachial) whose surface cannot reliably be applanated (flattened).

In the following section, data come from central aortic pressure generated from the radial artery and GTF or from invasively recorded aortic pressure, unless specified otherwise.

Views on arterial blood pressure remained relatively stagnant throughout the twentieth century with clinicians, epidemiologists, insurance and military examiners content to rely on brachial cuff systolic and diastolic pressures as the sole expression of left ventricular load and a source of damage to arterial vessels. As pointed out elsewhere in this book, the view improperly attributed to James Mackenzie was unduly simplistic, suggesting that systolic pressure was an indication of LV strength (the higher the better) and diastolic pressure was an indication of peripheral resistance, and so the *sine qua non* of arterial hypertension. Framingham investigators first expressed doubt on this view in 1971 and 1981 (Kannel *et al.*, 1971; Kannel *et al.*, 1981), but it was left to the SHEP trialists to demolish the prevailing view by showing that elevated systolic pressure in the elderly had an adverse prognosis, that was reduced by anti-hypertensive therapy even when diastolic pressure was normal (SHEP Cooperative Research Group, 1991). Up until this time, all other published trials of anti-hypertensive treatment had entered patients on the basis of elevated brachial diastolic pressure.

SHEP, stimulated by Kannel, Dawber, and colleagues began a re-evaluation of arterial pressure, its meaning, and significance. Franklin's analysis of Framingham data followed (Franklin *et al.*, 1999). This showed that in older people, pulse pressure (brachial systolic – brachial diastolic) was more robust than systolic in predicting outcomes. This was promptly confirmed by the European

Syst-Eur study (Staessen *et al.*, 1997), and by other data from China (Liu *et al.*, 1998). It is hardly surprising that the revolution spread to consideration of aortic rather than brachial pressure, especially when Franklin's analysis showed that importance of pulse pressure in predicting outcome only applied to people over age 50, and that there was a disparity in the relationship between aortic and brachial pulse pressure in people (especially males) at different ages (Franklin *et al.*, 1999; Mosley *et al.*, 2007).

Up until publication of Franklin's work on central and brachial pressure, done in association with Cockcroft and Wilkinson in the UK, clinicians had not seriously considered the relevance of aortic and brachial systolic pressure. To clinicians, the blood pressure was the blood pressure, as taken by the cuff method in the arm, and had so been for the past century, with a mass of information on prognosis and treatment available. They resisted change.

In the late twentieth century, there was a small nucleus of scientists, principally of physiological bent and training, who were dissatisfied with the status quo, who considered that the concept of hypertension as practiced by clinicians did not do justice to the enormous leap that had been made in drug therapy by pharmaceutical companies, and the benefits that had flowed on to patients with 'the silent killer'. Our own groups in the United States, Australia, and Greece were among these.

In Sydney, we were intrigued with the information gained at cardiac catheterization on use of nitrates, with the clear demonstration that a tiny dose of nitroglycerine could reduce aortic systolic and pulse pressure without any change in brachial systolic and diastolic values. We were encouraged by McGregor, Fitchett, and others in Canada who had shown marked changes in distensibility of small muscular arteries without corresponding changes in arterial pressure. The beneficial effects of nitrates on angina and heart failure had been known for over a century, and the effect of nitrate on the pulse waveform had been described in Murrell's original article (Murrell, 1879). On the basis of unchanged brachial systolic and diastolic pressure, benefit was ascribed to something else – to an effect 'beyond blood pressure', and probably to venous dilation. This was and still is the favored view of clinicians. Kelly's work showed how nitrates in relatively small dose could reduce central aortic systolic pressure by up to 20 mmHg, without any reduction in brachial systolic pressure. This was promptly confirmed by Fitchett (Fitchett *et al.*, 1986) and by Takazawa (Takazawa *et al.*, 1995b) who also showed the identical reduction in LV systolic pressure, with far less change in venous pressure. These findings led on to consideration of the effects of other drugs on central aortic systolic pressure, including vasodilating β -blockers (Kelly *et al.*, 1989d), ACE inhibitors, ARB agents, and calcium channel blockers. Benefits on central over brachial pressure were attributed to reduction in wave reflection (Yaginuma *et al.*, 1986) and a book on arterial dilation by O'Rourke, Safar, and Dzau was published in 1993 (O'Rourke *et al.*, 1993a).

Two trials, published in 2000 and 2002, had a major influence on prescribing habits in patients with cardiovascular disease, especially hypertension, diabetes mellitus, and left ventricular hypertrophy. Yusuf *et al.* (2000) showed that use of the ACEI ramipril (10 mg daily) compared to placebo in patients at high cardiovascular risk treated with all currently appropriate therapy had reduction in all cardiovascular events over a period of five years. Brachial systolic and diastolic pressure difference between the two groups was trivial, and authors attributed benefit of ramipril to be 'beyond blood pressure reduction'. They were quick to dismiss the suggestion that benefit may have been due to greater reduction in central than brachial systolic pressure (O'Rourke, 2000b). A subsequent acute study conducted in patients showed just such an effect of ramipril compared to placebo or atenolol (Hirata *et al.*, 2005b). The LIFE study (Dahlöf *et al.*, 2002) showed a similar benefit to ramipril for hypertensive patients with left ventricular hypertrophy, treated with the ARB losartan, compared to atenolol. Authors again attributed benefit to 'beyond blood pressure lowering'. They conceded that losartan may have had greater benefit in lowering aortic than brachial pressure, but this possibility had not been considered in design of the trial. Both atenolol and losartan lower brachial cuff blood pressure, however, losartan lowers central systolic and pulse pressure much more than atenolol (Fig. 28.9) (Nichols, 2005). Likewise, CONSENSUS showed greater reduction in cardiovascular events when a CCB (compared to diuretic) was added to an ACEI (Consensus Trial Study Group, 1987). When a similar study (J-CORE) was conducted with similar therapy, it showed greater reduction in central pressure for the CCB/ACEI combination than the diuretic/ACEI combination (Matsui *et al.*, 2009). This has been shown also for benefit of amlodipine/valsartan combination over amlodipine/atenolol in the EXPLOR study (Boutouyrie *et al.*, 2010).

The effects of difference in central pressure was tested in the REASON study which compared an ACEI/diuretic combination against atenolol over a 12-month period in patients with hypertension. The primary end point was to test for difference in carotid-femoral PWV. There was no difference in this. Secondary end points were measures of wave reflection, including aortic augmentation index and difference between carotid and aortic systolic pressure compared to brachial. Substantial and significant differences were seen for reduction in central pressure with the ACEI/diuretic over atenolol (Asmar *et al.*, 2001b; Asmar *et al.*, 2001d; London *et al.*, 2004a), and this was associated with reduction in LV mass, which was not seen for the atenolol-treated group (de Luca *et al.*, 2004).

The most convincing study on difference in therapy for aortic and brachial systolic pressure was seen in the CAFE substudy of ASCOT (see Chapter 21) (Williams *et al.*, 2006). This showed a consistent difference of 4–5 mmHg over the whole period of five years (Fig. 28.10), and was associated with greater reduction of a composite end

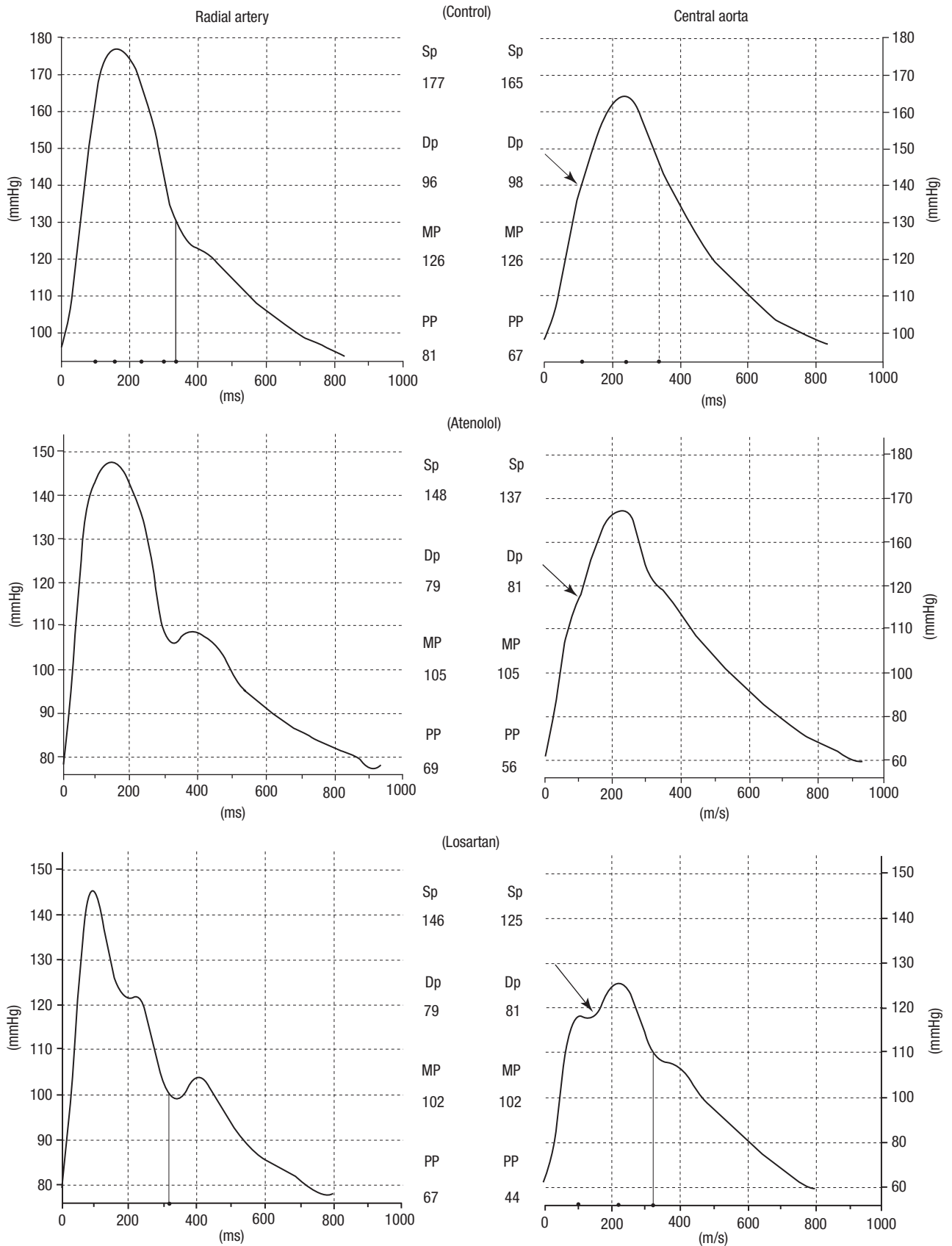


Figure 28.9 Effects of atenolol and losartan on central (aortic) and peripheral (radial artery calibrated from brachial cuff pressure) pressure in a patient with essential hypertension. The reduction in peripheral pressure was essentially the same for the two drugs, however, losartan reduced wave reflection and central systolic and pulse blood pressure much more than atenolol. Reproduced from Nichols (2005)

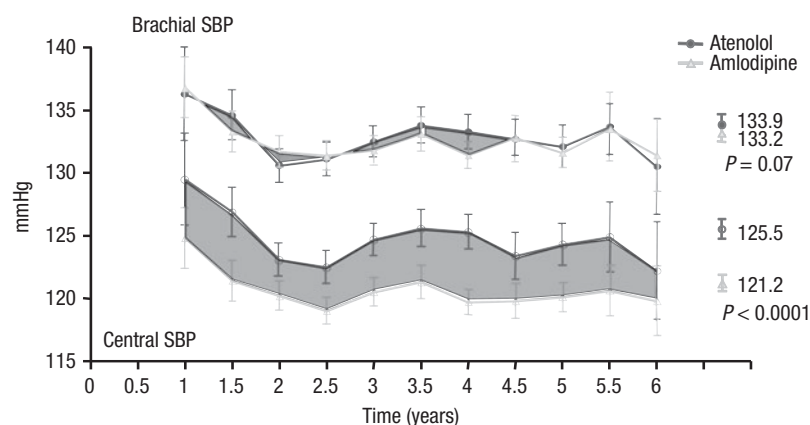


Figure 28.10 In the CAFE study (Williams *et al.*, 2006), a substudy of ASCOT, brachial systolic and diastolic pressures were reduced by an amlodipine- or atenolol-based strategy. There was no difference in brachial systolic pressure over five years, but central pressure was lower by 4–5 mmHg in the amlodipine group

point in CAFE, and explained superiority of amlodipine over atenolol in the treatment of hypertension in the ASCOT study. This study was instrumental in changing guidelines for treatment of hypertension in the UK, with withdrawal of atenolol as first-line therapy.

As it stands in early 2011, all invasive and most of the non-invasive published studies have shown clear superiority of central aortic pressure and indices in determining clinical end points. The non-invasive tonometry publications that showed superiority of central pressures and indices used either radial artery applanation tonometry and a generalized transfer function process to generate ascending aortic pressure waveforms, or tonometry of the carotid artery (without involving tonometry of the brachial artery). Two publications, one by Mitchell *et al.* (2010) and the other by Dart *et al.* (2006) from the ANBP 2 study in Melbourne, Australia report no superiority of central (carotid) systolic or pulse pressure over brachial pressure for predictors of outcome. These studies, however, found central systolic and pulse pressure to differ only by 1.0 mmHg from brachial systolic or pulse pressure (Table 28.1, Fig. 10.25). The method used by these two groups has been criticized in this book (see Chapters 10 and 22), and elsewhere (O'Rourke *et al.*, 2005; O'Rourke and Adji, 2010a; O'Rourke and Adji, 2010b) on the basis of flawed use of applanation tonometry on the brachial artery. Another group (Segers *et al.*, 2005b; Segers *et al.*, 2007b; Segers *et al.*, 2009) in Ghent, Belgium has also used this method, but has not yet been in a position to describe outcome. Data available now (Segers *et al.*, 2005b) show evidence of substantial noise in derived central pressure indices, but little noise in the simpler indices, especially AIx and amplification of the radial wave that do not depend on calibration. Further clarification of the views will surely come from reanalysis where possible of the Framingham, Asklepion, and ANBP2 studies, from further

follow up of Strong Heart and Taiwan (Wang *et al.*, 2010) cohorts, and from the large studies currently being conducted in China where central pressures are being measured in large cohorts with the transfer function method applied to radial applanation tonometry.

We have shown superiority of central aortic pulse pressure over brachial pulse pressure in a meta-analysis of the previously cited studies (Vlachopoulos *et al.*, 2010a, Vlachopoulos *et al.*, 2010b). After criticism was raised by Protogerou *et al.* (2010) on inclusion of the Dart paper which used a different and discredited (O'Rourke and Adji, 2010b) technique, we revised the meta-analysis and confirmed that central pressure provided better information than brachial pressure for prediction of outcome (Vlachopoulos *et al.*, 2010c). This entire subject is dealt with in detail in Chapters 21 and 22.

Biomarkers are usually considered as chemical tests – but they must be considered also to comprise central pressures, features of the aortic pulse waveform and aortic stiffness. A consensus group considered this issue in 2007 (Agabiti-Rosei *et al.*, 2007), and recommended that measures of central aortic pressure are likely to be more useful in making decisions on when and with what intervention to treat arterial hypertension. The ESH/ESC guidelines for management of arterial hypertension were addressed in 2007, reappraised in 2009, and it is anticipated that they will be fully updated in 2011–2012. The JNC guidelines have not been updated since 2003, but a publication is anticipated in 2011. Value of central aortic pressure is likely to be addressed by both, and will depend upon the basis of the material presented in this chapter. It is proper that measures of outcome related to central aortic pressure should be decisive, and that such measures must show incremental value over currently available methods.

Lifestyle and environment

Smoking	581	Air pollution	589
Diet and nutrition	583	Illegal substances	589
Alcohol	587	Gender	589
Obesity and weight loss	588	Race and ethnicity	593
Mental, emotional status, and work stress	588		

... The onset of what may be called physiological arteriosclerosis depends, in the first place, upon the quality of arterial tissue (vital rubber) which the individual has inherited, and secondly upon the amount of wear and tear to which he has subjected it. ... Entire families sometimes show this tendency to early arteriosclerosis – a tendency which cannot be explained in any other way than that in the make-up of the machine, bad material was used for the tubing. More commonly, the arteriosclerosis results from the bad use of good vessels. ...

William Osler, 1905

Atherosclerotic disease comprises four major areas, coronary artery disease (CAD), cerebrovascular disease, peripheral arterial disease (PAD), and thoracic or abdominal aortic aneurysm. The 'Cardiovascular Continuum' was described by Dzau and colleagues in 2006 (Dzau *et al.*, 2006a; Dzau *et al.*, 2006b) to explain the development over many years of coronary disease with its complications, then end-stage heart failure. William Osler described two forms of arteriosclerosis, 'nodular arteriosclerosis' (atherosclerosis) and a disease which may be superimposed on arterial aging (his 'senile arteriosclerosis'). While the cardiovascular continuum put an emphasis on atherosclerosis in prosperous nations, it did not account fully for the problems of aging, which occur in all societies. Aging of the thoracic aorta and elastic arteries causes arterial stiffening and leads to development of cardiac failure and microvascular disease in highly perfused organs, such as the brain and kidneys. We recently introduced the 'Vascular Aging Continuum' which dovetails with the late phases of the cardiovascular continuum and provides a more comprehensive explanation,

especially for vascular diseases in nations with little atherosclerosis (see also Chapter 22) (O'Rourke *et al.*, 2010b).

William Osler visually separated genetic ('vital rubber') and environmental factors ('bad use of good vessels') and pinpointed their interplay. While genetic background has a strong effect on cardiovascular disease (CVD) risk, environmental factors are more important. More than 70 percent of CVD and adult-onset diabetes could be attributed to modifiable, non-genetic factors, which are in turn strongly dependent on environmental factors (Willett, 2002). The uncorrectable factor is repetitive pulsation of arteries, and so, represents arterial aging (Chapter 19). CVD rates differ five- to 100-fold among population groups of similar genetic background and they increase sharply when populations of the same ethnic group migrate from low- to high-risk environments. Among other examples of environmental and social changes, the industrialization in nineteenth-century England and the ban on smoking in public areas, solidly supports the alluring concept that environmental modifications can endanger or promote cardiovascular health (Chow *et al.*, 2009) (Fig. 29.1). Therapeutic lifestyle changes have an important beneficial effect on cardiovascular morbidity and mortality; they are a more 'natural' way towards therapeutic goals and should be an integral component of the prevention or treatment of any CVD. However, the vague and rather immense meaning of the term 'lifestyle' includes many distinct elements of everyday life (Table 29.1). Examples of lifestyle modification are numerous, including apart from what we eat and drink, physical activity, weight reduction, smoking, and stress.

The concept that the beneficial effect of lifestyle modification on CVD is partly mediated through the improvement of arterial elastic properties and wave

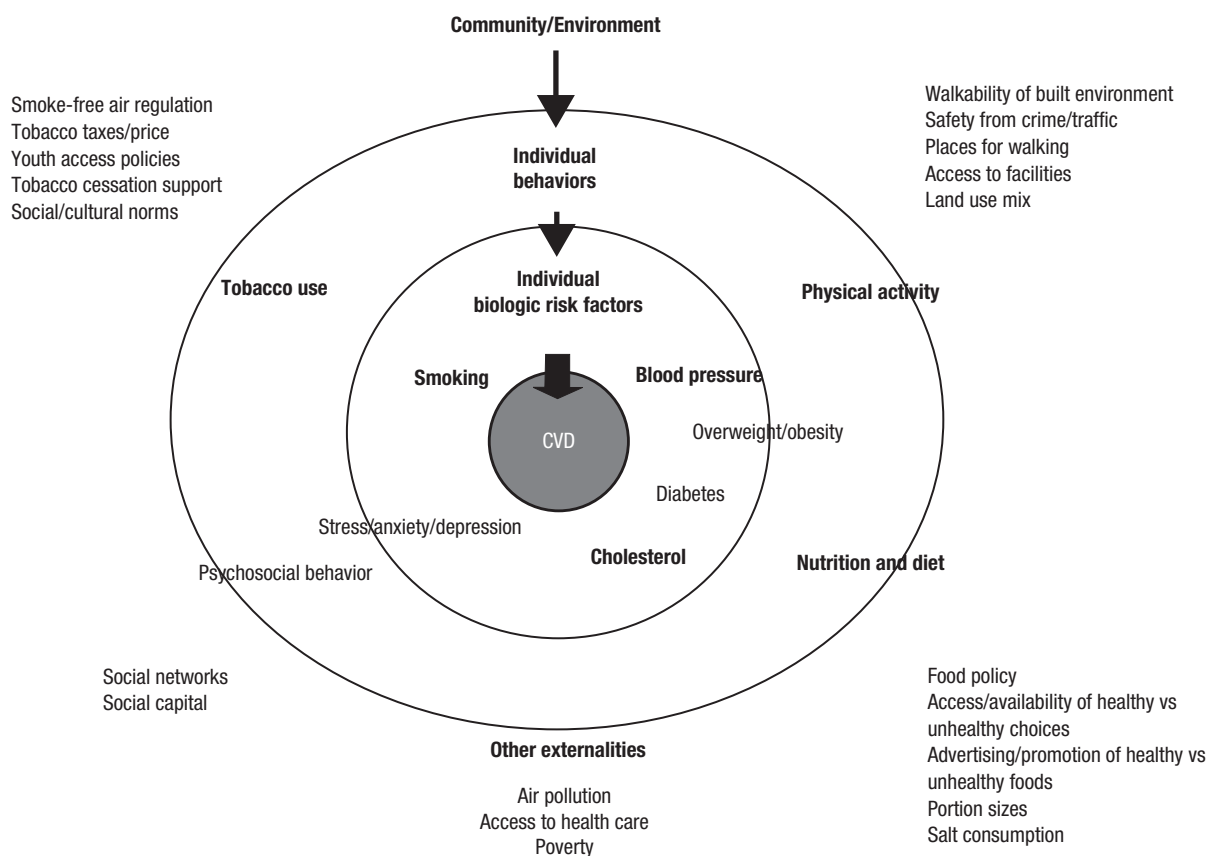


Figure 29.1 Environmental factors that have been linked to human behaviors and established risk factors that cause cardiovascular diseases. Reproduced from Chow *et al.* (2009)

reflections characteristics is reasonable and attractive. Along these lines, numerous studies investigating the link between arterial stiffness (mainly aortic pulse wave velocity (PWV), see Chapters 3 and 4) and wave reflections, as well as central pressures and hemodynamic indices, and environmental influences have been conducted in the last few years. However, most of them are either cross-sectional or pathophysiological studies that clarify mechanisms, consolidate the theoretical background and reveal independent associations, but they do not necessarily extend to clinical associations and causal relationships. Some interventional studies provide evidence of cause-and-effect associations. Unfortunately, longitudinal studies with events in long-term follow up, which provide the more solid evidence are extremely limited regarding some lifestyle elements or are totally lacking regarding others.

The world presently is reeling from the debate on 'global warming' as the most important environmental issue of our age. During the debate, it has become apparent that concepts of science have been compromised by some of the individual experts, and en masse by some expert groups. Issues have become politically sensitive in many if not most nations of the world, and the debate has become more emotional than scientific. There is a possibility of the same human defects entering into effects of

environment on the human body, particularly as these relate to active smoking, passive smoking, dietary supplements, natural therapies, vitamins, common foods and drinks (alcoholic or non-alcoholic), industrial pollutants, as well as recreational drugs and hormones.

Contributors to these areas are more likely than physiologists, physicians, and bioengineers, to underestimate the complexities and artifacts that can arise in taking measurements of arterial stiffness and wave reflections. In what follows therefore, we will adhere to the guidance of the European Consensus Group (O'Rourke *et al.*, 2002a; Laurent *et al.*, 2006) in showing a preference for measuring arterial stiffness as pulse wave velocity along the aorta, and effects of aortic stiffening at controlled heart rate and wave reflections as augmentation index (AIx) (see Chapter 9) or as amplification of the pressure pulse between proximal aorta and radial artery. We will give lesser weight to indices of distensibility or compliance which have not consistently shown benefit in outcome studies (Laurent *et al.*, 2006), especially when the caveats of O'Rourke *et al.* (2002a) have not been followed. We also need note that arterial stiffness varies with arterial pressure and with age (carotid-femoral or aortic PWV changes 60–100 cm/s for a change in mean blood pressure of 10 mmHg or a change in age by ten years (Avolio *et al.*,

Table 29.1 Concise evaluation of available data on lifestyle and aortic stiffness and wave reflections

	Main findings	Areas for further investigation
Smoking	All types (acute, passive, cigarette, cigar) induce increase of AS and indices of WR	More smoking cessation studies needed to assess the reversibility of detrimental effects
<i>Diet/nutrition</i>		
Coffee and caffeine	Increase in AS and indices of WR both acutely and chronically. Synergistic effect with smoking	Need clarification: cut-offs of consumption, effect of other coffee ingredients
Tea	Only acute studies. Neutral effect on AS (green), short-lasting increase in AS (black). Long-lasting increase in indices of WR (both green and black)	Chronic effect to be investigated. Mechanisms to be elucidated (caffeine vs flavonoids)
Cocoa and chocolate	Decrease in AS and indices of WR (chronically)	Ideal dose not known. Effect of milk chocolate? Although effects are flavonoid-related, exact mechanisms to be elucidated
Alcohol	Decrease in AS	Most cross-sectional studies do not discriminate between drinks. Need investigation: alcohol or flavonoids responsible? Differential effect of red wine not established. Mechanisms to be elucidated
Salt	Positive association with AS and indices of WR in cross-sectional studies. Beneficial effect of sodium restriction (interventional studies)	Mechanisms to be elucidated
Obesity/weight loss	In most studies, obesity is associated with increased AS. Central obesity most likely detrimental. Weight loss improves AS and indices of WR (independently of the type of diet)	Mechanisms to be elucidated. Pressure-dependent, or -independent effect?
Mental stress	Increase in AS and indices of WR with mental stress. Laughter reduces AS	The chronic effect to be investigated. The effect of music to be investigated. Mechanisms to be elucidated
Work stress	Increase in AS with work stress	The chronic effect to be investigated. The effects on indices of WR to be investigated. Mechanisms to be elucidated. Better quantification of work-related stress. Assessment of the reversibility of detrimental effects
Environmental factors	Only acute studies. Increase in indices of WR with air pollution. Inconclusive results regarding AS	Assessment of the reversibility of detrimental effects. Assessment of chronic effects

AS, aortic stiffness; WR, wave reflections.

1983; Avolio *et al.*, 1985a; Asmar *et al.*, 2001b)), while change in AIx needs to be interpreted in the light of any change in heart rate, ventricular contractility, mean arterial pressure, and age as well as body height and gender.

SMOKING

Smoking is an independent major risk factor for CVD, and it remains the leading preventable cause of death in the world. While the evidence linking smoking with CVD is clearly present, the exact components and the mechanisms responsible for this association have not been clearly elucidated. Apart from endothelial dysfunction, inflammation, sympathetic activation, and modification of lipid profile, a strong body of evidence indicates that arterial stiffness may be an underlying mechanism by which smoking contributes to CVD (Ambrose and Barua, 2004).

Active smoking has an acute and a chronic detrimental effect on arterial stiffness and wave reflections (Doonan *et al.*, 2010). The effect of **acute** smoking on arterial stiffness and wave reflections has been investigated through interventional studies that lasted for as long as 2.0 hours. Apart from transiently increasing both blood pressure and heart rate, smoking increases arterial stiffness and indices of wave reflections, such as aortic AIx, in habitual smokers (Fig. 29.2) (Kool *et al.*, 1993; Failla *et al.*, 1997; Stefanadis *et al.*, 1997; Mahmud and Feely, 2003a; Vlachopoulos *et al.*, 2004a; Lemogoum *et al.*, 2006), non-smokers (Mahmud and Feely, 2003a), and hypertensives (Rhee *et al.*, 2007). The maximal short-term effect of cigarette smoking seems to take place in the first 5–10 minutes (Failla *et al.*, 1997; Stefanadis *et al.*, 1997; Mahmud and Feely, 2003a; Vlachopoulos *et al.*, 2004a; Lemogoum *et al.*, 2006; Rhee *et al.*, 2007). Cigar smoking is not innocent, since arterial stiffness increases for at least 1.0 hour after smoking one cigar (Vlachopoulos *et al.*,

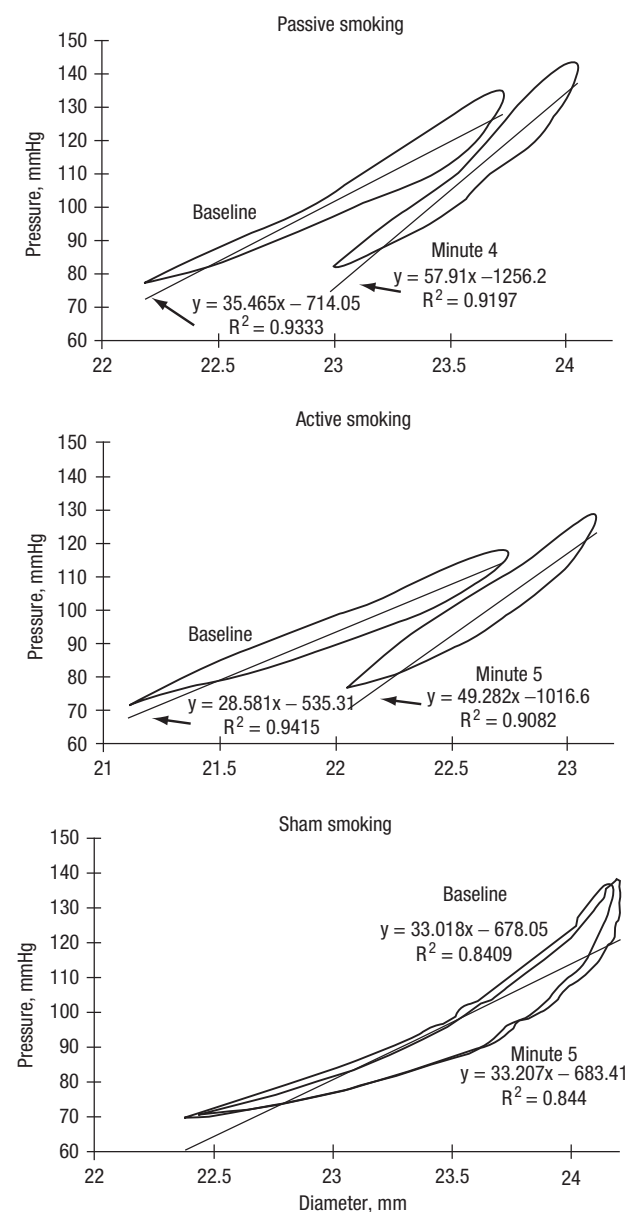


Figure 29.2 Examples of pressure-diameter loops in the thoracic aorta of three subjects (one from the passive, one from the active, and one from the sham smoking group). The loop shifted to different hypothetical line of elasticity during passive and active smoking, indicating contribution of a pressure-independent effect in the deterioration of elastic properties. Reproduced from Stefanadis *et al.* (1998b)

2004a). Interestingly, smoking and coffee consumption have a synergistic effect on arterial function when combined (Vlachopoulos *et al.*, 2004c), while red wine is shown to decrease aortic AIx in uncomplicated habitual smokers (Papamichael *et al.*, 2006).

The results of cross-sectional studies investigating the **chronic** effect of smoking, by comparing current smokers with non-smokers, are in accordance with the studies regarding the acute effect (van Trijp *et al.*, 2004; Vlachopoulos *et al.*, 2004c; Rehill *et al.*, 2006; Jatoi *et al.*,

2007; Polónia *et al.*, 2009). Although there are some contrary reports (van Trijp *et al.*, 2004), the effect on arterial stiffness appears to be unrelated to the intensity and duration of smoking. It seems that even small quantities of smoking are able to produce its deleterious effects. The effect of smoking on arterial stiffness is not only due to increases in blood pressure, but it also has a pressure-independent component (Fig. 29.2) (Stefanadis *et al.*, 1997). Interestingly, several studies have shown that AIx and central blood pressure are higher in current and former smokers as compared to never smokers, while such effects are less evident regarding brachial blood pressure (Mahmud and Feely, 2003a; Jatoi *et al.*, 2007; Minami *et al.*, 2009). The long-term effects of cigarette smoking on hypertension have not been elucidated. In two large population studies, current cigarette smoking was independently, yet modestly, associated with an elevated risk of developing hypertension (Primates *et al.*, 2001; Bowman *et al.*, 2007). However, other studies have reported lower blood pressure levels among smokers (Green *et al.*, 1986) and increases in blood pressure after smoking cessation (Lee *et al.*, 2001). The mildly lower blood pressure in smokers can be attributed to lower body weight (Perkins *et al.*, 1989).

Passive smoking is a well-established risk factor for coronary heart disease (Carnoya and Glantz, 2005) with detrimental effects on the cardiovascular system as grave as those of active smoking. While quitting smoking relies in part on the smokers themselves, smoke-free environments can effectively be achieved through banning of smoking by employers, regulatory authorities, and governments. In an undeclared war lasting for decades, efforts led by public authorities and dedicated scientists such as Stanton A Glantz and William Parmley to persuade public opinion and limit smoking and its health hazards have started to flourish. Outcomes after banning of smoking in public places and workplaces have been set and enforced, have been impressive (Lightwood and Glantz, 2009; Meyers *et al.*, 2009). Rates of acute myocardial infarction can be reduced as much as 17 percent with the greatest effect among younger individuals and non-smokers. While the effect is more prominent when a ban is enforced over several years, short-term bans have equally impressive results. In the Helena study (Sargent *et al.*, 2004), a ban was enforced for six months. Admissions for acute myocardial infarctions were significantly reduced during the six-month period and returned to pre-ban levels after the ban was suspended. Passive smoking has immediate and substantial effects on arterial stiffness in both smokers (Stefanadis *et al.*, 1998b) and non-smokers (Stefanadis *et al.*, 1998b; Mahmud and Feely, 2004; Argacha *et al.*, 2008). These effects are mediated through a nicotine-dependent pathway, impairment of microvascular function and increase of asymmetrical dimethyl-arginine (Argacha *et al.*, 2008).

Effect of smoking *cessation* on arterial stiffness and wave reflections is an important issue. Duration of

smoking cessation in former smokers has a significant linear relationship with improvement in arterial stiffness parameters. Values of aortic PWV, AIx, and transit time equal those of non-smokers after a decade of smoking cessation (Fig. 29.3) (Jatoi *et al.*, 2007). In intervention studies, cessation of smoking for 4 weeks resulted in a decrease in AIx (Rehill *et al.*, 2006), while six months of smoking cessation was associated with clear improvement

of AIx corrected for heart rate, aortic pressure, and aortic PWV, and with a reduction of daytime blood pressure (Polónia *et al.*, 2009). While wave reflections may be rapidly reduced after smoking cessation, PWV may take longer to normalize. In a study that used nicotine replacement therapy, an early (56 days) improvement of AIx, but not of PWV, was observed (Roux *et al.*, 2010).

DIET AND NUTRITION

There is now overwhelming evidence that dietary factors have a significant impact on CVD risk, and their effect on arterial stiffness and wave reflections has been an appealing goal of investigation, with a plethora of emerging evidence supporting this notion. Research has focused especially on caffeine and coffee, tea, cocoa, and other flavonoid-containing foods and beverages, antioxidant vitamins, and sodium.

Although recommendations for accurate arterial stiffness assessment point out that subjects should refrain from eating and drinking beverages containing caffeine for at least 3.0 hours before assessment (Laurent *et al.*, 2006), many studies have not controlled for recent food or drink intake or do not report the relation of measurements with food or drink intake in their protocol. PWV and markers of central hemodynamics, such as aortic AIx, are sensitive to feeding state and therefore should be measured at fasting to avoid variability due to recent (within 3.0 hours) food intake (Ahuja *et al.*, 2009). However, long fasting is not ideal and should be avoided because hunger can cause increase in blood pressure, PWV, and AIx.

Coffee and caffeine

Coffee is a complex mixture of compounds that may have variable (and often unidentified) effects on the cardiovascular system. However, the effects of coffee are often confused with the effects of caffeine, the methylated xanthine derivative. On the other hand, caffeine, consumed in coffee, tea, and soft drinks, is the most widely used pharmacologically active substance in the world and investigation of its effects is worthwhile in its own right. It should be stressed therefore, that coffee and caffeine are not interchangeable terms and results of studies should be interpreted in the context of the particular beverage/substance investigated. Results of studies on coffee consumption are equivocal, varying from a positive to a neutral or to even a J- or a U-shape association with CVD risk (Lopez-Garcia *et al.*, 2008). In a recent large study, based upon self-reported drinking habits, two or more cups of coffee per day seemed to reduce all-cause mortality in women (but not in men), primarily due to a reduction in cardiovascular mortality (Lopez-Garcia *et al.*, 2008). Similar findings were noted with decaffeinated

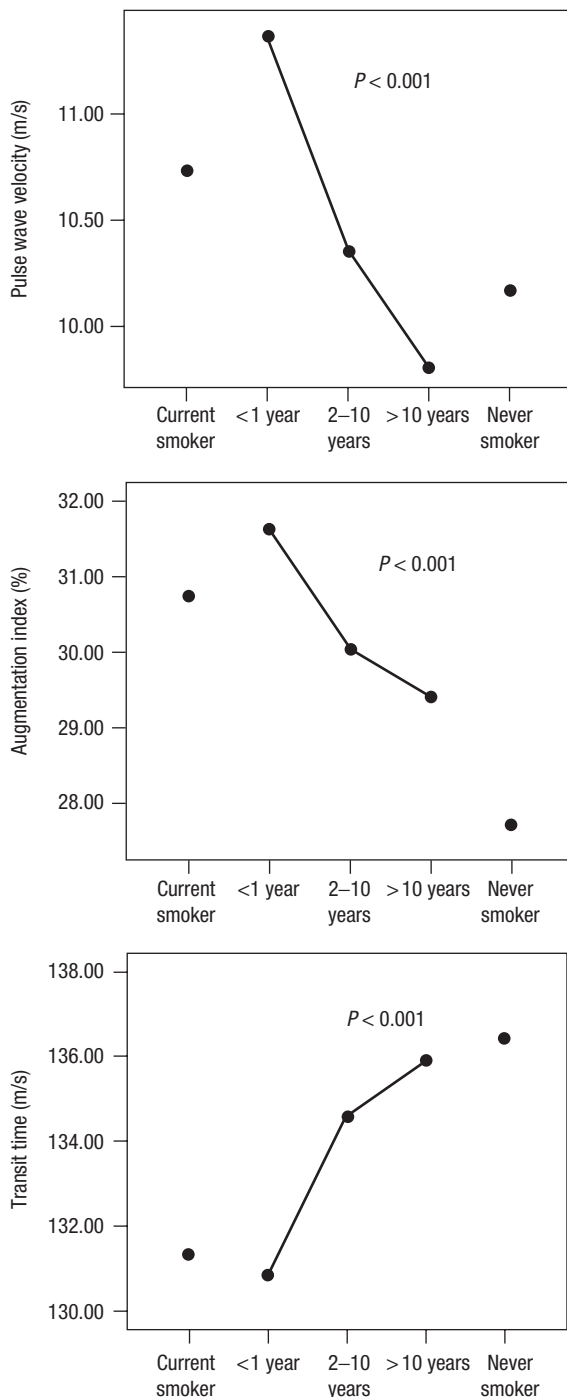


Figure 29.3 Aortic pulse wave velocity, augmentation index, and transit time of current smokers, never smokers, and ex-smokers. Reproduced from Jatoi *et al.* (2007)

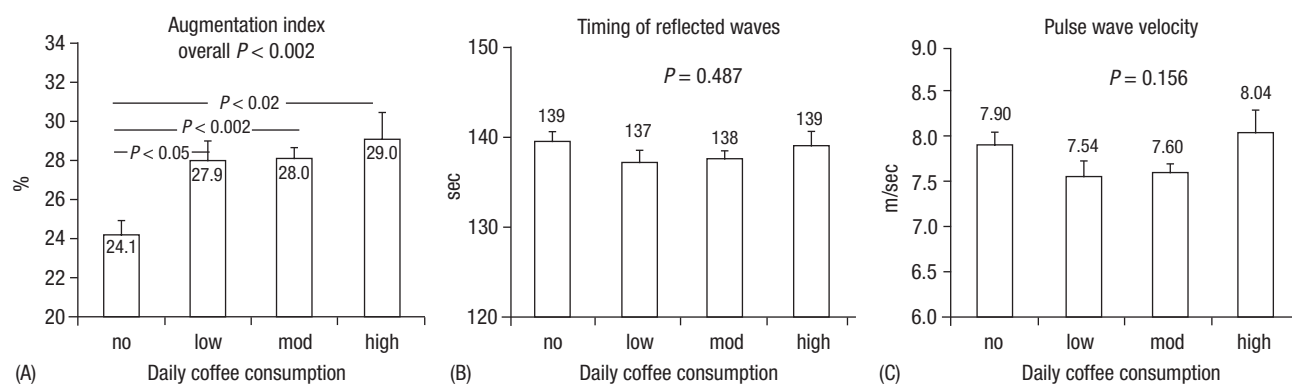


Figure 29.4 Aortic augmentation index, timing of reflected waves and aortic pulse wave velocity in different groups of daily coffee consumption (low: < 200 mL/day, moderate: $200\text{--}450$ mL/day, and high: > 450 mL/day). Reproduced from Vlachopoulos *et al.* (2007c)

coffee, suggesting a beneficial effect from a component other than caffeine. Interestingly, regular caffeine intake increases blood pressure, but when ingested through coffee, the blood pressure effect of caffeine is small (Noordzij *et al.*, 2005). The rate of caffeine metabolism through genetic predisposition may be an important determinant of the effect of coffee intake on coronary heart disease risk and account for these discrepancies in outcomes of studies.

Many studies indicate that caffeine intake, in the form of a tablet or in coffee, exerts an acute detrimental effect on arterial stiffness and indices of wave reflections, both in normal subjects and in hypertensives (Mahmud and Feely, 2001; Vlachopoulos *et al.*, 2001; Vlachopoulos *et al.*, 2003b; Vlachopoulos *et al.*, 2003c; Waring *et al.*, 2003; Vlachopoulos *et al.*, 2004c). The ingestion of caffeine results in a sustained impairment of arterial stiffness, which lasts longer in hypertensives (at least 3.0 hours) compared with normotensives. Interestingly, the unfavorable effect of caffeine is more evident in central hemodynamics than in peripheral pressure (Fig. 29.4) (Vlachopoulos *et al.*, 2007c) and coffee increases arterial stiffness to a lesser degree in habitual drinkers compared to non-habitual ones (Vlachopoulos *et al.*, unpublished data).

Chronic coffee consumption is associated with increased arterial stiffness and wave reflections in healthy normotensives, whereas in hypertensive patients, wave reflections, but not aortic stiffness, are increased (Papaioannou *et al.*, 2005). The latter implies that the stiffening effect of coffee on the aorta is probably less prominent in the already stiff aortas of hypertensive patients, than in normal aortas.

The caffeine-induced effect on arterial stiffness is probably related to vasoconstriction of smooth muscle cells in the muscular arteries due to an increase in sympathetic activity and circulating catecholamine concentrations mediated by central nervous system stimulation and antagonism of endogenous adenosine.

Coffee consumption is very frequently combined with smoking. The combination of acute caffeine intake/chronic

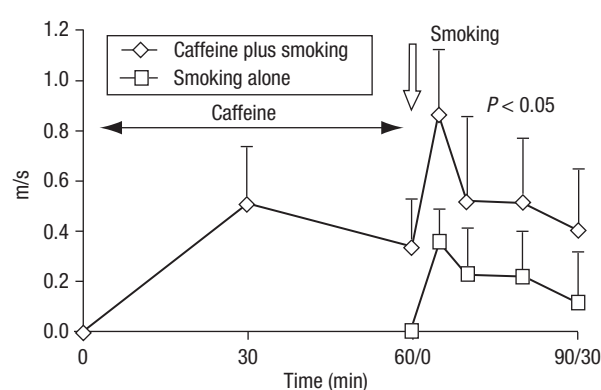


Figure 29.5 Acute pulse wave velocity responses to caffeine, caffeine plus smoking, or smoking alone. Note the synergistic effect when caffeine and smoking are combined. Reproduced from Vlachopoulos *et al.* (2004c)

coffee consumption and smoking has a synergistic detrimental effect on aortic PWV and AIx (Fig. 29.5) (Vlachopoulos *et al.*, 2004c). Clinical evidence shows that coffee drinking is unrelated to coronary artery disease risk in never smokers, but in ex-smokers and current baseline smokers, daily coffee intake is associated with higher coronary artery disease risk (Klatsky *et al.*, 2008).

Tea

The effect of tea consumption on cardiovascular risk has not been clearly defined with results of studies ranging from a neutral effect to a reduction of CVD risk (Kuriyama *et al.*, 2006). Moreover, while tea is reported to have no long-term effect on blood pressure, it might reduce the risk of developing hypertension (Taubert *et al.*, 2007). The possible beneficial effect of tea has been attributed to its flavonoid content. Flavonoids (those in tea are called 'catechins') are polyphenols with antioxidant, anticarcinogenic, and anti-inflammatory properties. The intake of tea flavonoids is associated with a reduction in CVD events (Hooper *et al.*, 2008), possibly as

a result of the improvement in endothelial dysfunction (Alexopoulos *et al.*, 2008). The two types of tea mostly consumed in the Western world are black tea and green tea. Black tea contains more caffeine than green tea, and partly oxidized flavonoids that form dimers and polymers, whereas green tea contains mainly monomers (catechins).

The effect of tea on aortic stiffness and wave reflections has not been so extensively studied. Acutely, both black and green tea increase wave reflections in normal subjects (Vlachopoulos *et al.*, 2006a). Furthermore, both black and green tea improve endothelial function, in the acute phase (Alexopoulos *et al.*, 2008). As regards aortic stiffness, green tea has no significant effect, whereas black tea leads to an increase for less than 2.0 hours. Counterbalancing effects of flavonoids varying in quantity (high in green and low in black tea) and caffeine may account for these findings.

Chocolate (cocoa and components)

Cocoa (*Theobroma*) was named the food (*broma* in ancient Greek) of Gods (*theos* in ancient Greek) by Carl Linnaeus because it had long been considered as a nutritional element with multiple benefits for human health. Recent research revealed that cocoa does indeed exert beneficial cardiovascular effects (Corti *et al.*, 2009), mainly mediated by its polyphenols, a heterogeneous group of molecules found primarily in fruits and vegetables. The beneficial effects of cocoa are most likely due to an increased bioavailability of nitric oxide. This may explain the improvement in endothelial function, the reduction in platelet function, and the potentially

beneficial effects on blood pressure, insulin resistance, and blood lipids (Corti *et al.*, 2009).

As it was recently shown in a cross-sectional study, chronic cocoa consumption is inversely associated with aortic PWV and AIx in healthy subjects. Consumption of at least 5.6 g of cocoa (approximately 12 g of chocolate) per day was associated with a significant decrease in aortic PWV and AIx (Fig. 29.6) (Vlachopoulos *et al.*, 2007a). Furthermore, increasing the intake of chocolate was associated with a decrease of central (aortic) systolic pressure and of central pulse pressure, but not of peripheral (brachial) pressure, denoting that its beneficial effect is not always evident when only peripheral blood pressures are measured.

Most interventional studies have used cocoa and dark chocolate rich in cocoa flavonoids. Both exert a beneficial acute and short-term effect on endothelial function (Corti *et al.*, 2009). Dark chocolate results acutely in a decrease in wave reflections in normal individuals (Fig. 29.7) (Vlachopoulos *et al.*, 2005). Aortic stiffness is not largely affected by acute chocolate consumption, while a tendency to decrease at a later stage (3.0 hours or more) begins to show. These taken together, imply that the beneficial effect of dark chocolate on indices of wave reflections is exerted mainly through vasodilation of the medium- and small-size muscular arteries. Thus, wave reflection amplitude is decreased not because of the delayed return in the elastic aorta (aortic PWV was not decreased), but rather because of the delay (decreased PWV) in the muscular arteries. The effect of the more largely consumed milk chocolate is not known. Effects are anticipated to be similar, although not necessarily identical, because milk chocolate contains much fewer

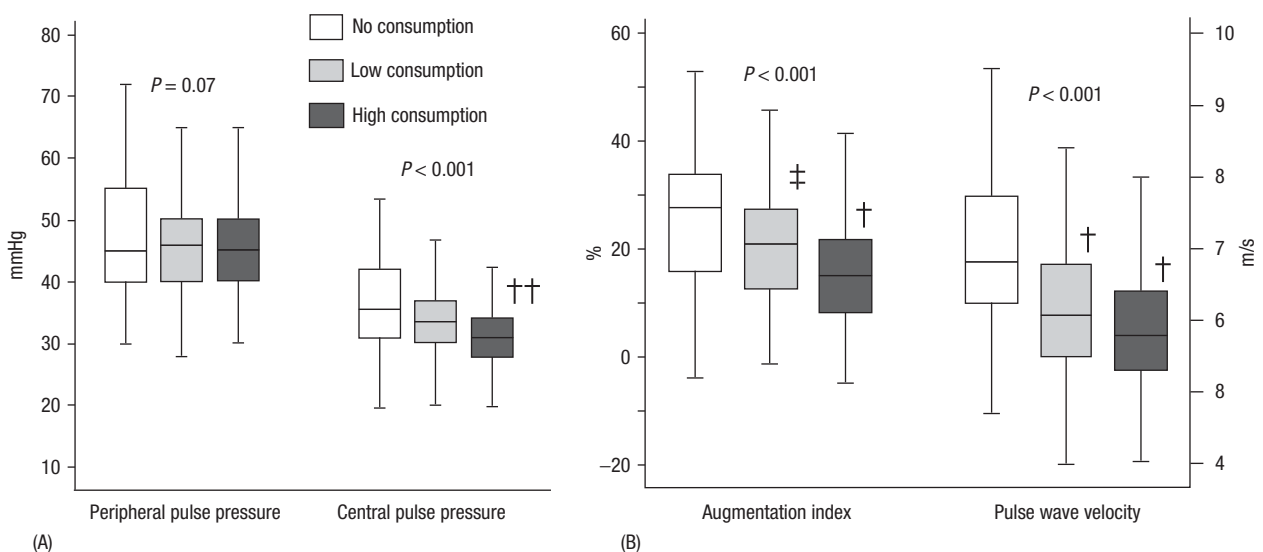


Figure 29.6 Box plots of peripheral and central pulse pressures (A) and aortic augmentation index (AIx) and pulse wave velocity (PWV) (B) in different groups of cocoa consumption, where the low group consumed < 4.63 g/day and the high group consumed > 4.63 g/day. Post hoc (Bonferroni) P values: †P < 0.001, ††P < 0.005, and ‡P < 0.05 compared with non-consumers. Reproduced from Vlachopoulos *et al.* (2007a)

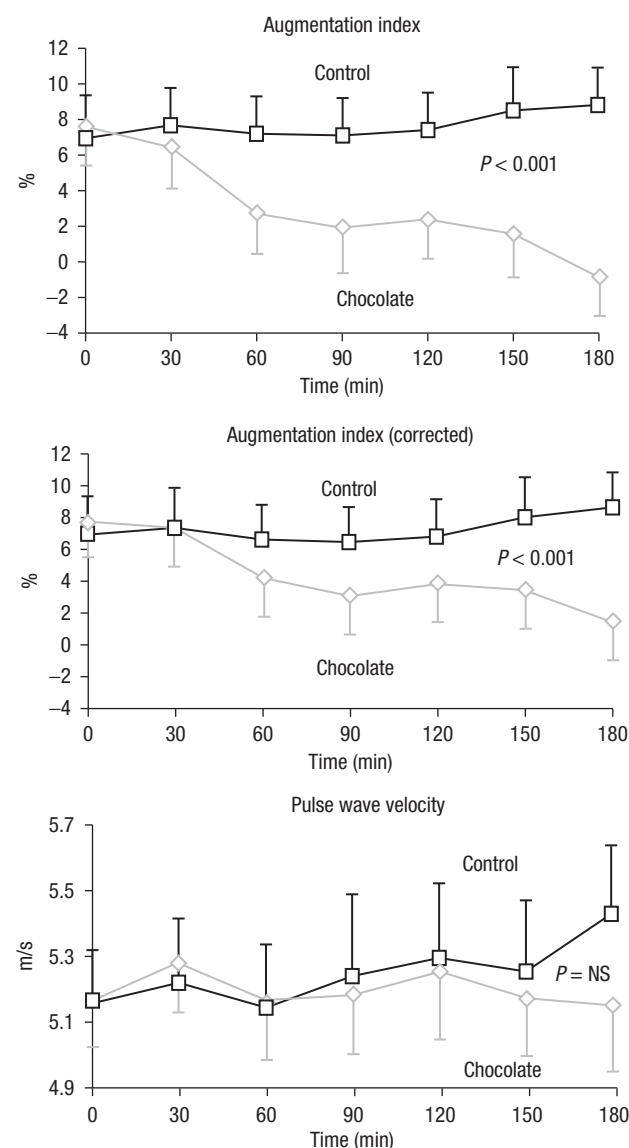


Figure 29.7 Acute effect of chocolate consumption (100 g of a dark chocolate) on indices of wave reflections and aortic stiffness. Reproduced from Vlachopoulos *et al.* (2005)

flavonoids than dark chocolate, and also because flavonoids are absorbed to a lesser degree when milk is present. White chocolate is not considered a real chocolate, because it is made only from cocoa butter after removal of cocoa liquor and it contains no flavonoids.

Foods and vitamins

FRUITS AND VEGETABLES

The Mediterranean diet protects against CVD. Fruits and vegetables are essential components of this diet and their consumption from childhood to adulthood is associated with lower arterial stiffness in young adulthood (Aatola *et al.*, 2010).

SOY PRODUCTS

A possible beneficial effect of soy consumption on cardiovascular risk has long been the target of investigation. Soy contains isoflavones, i.e. flavonoids found in soy and in red clover, have been shown to reduce systemic arterial compliance and aortic PWV in short-term consumption (Nestel *et al.*, 2007), but not in hypertensives (Teede *et al.*, 2006).

ω-3 FATTY ACIDS

The most important finding concerning ω-3 fatty acids intake relates to fish oils. Fish eaters have better aortic elastic properties than non-fish eaters. Oral supplementation with ω-3 fatty acids, like eicosapentaenoic acid and docosahexaenoic acid, both in the acute postprandial phase and on a long-term basis, seems to improve arterial compliance in healthy individuals (Hall *et al.*, 2008).

VITAMINS

Antioxidant vitamins C (ascorbic acid) and E have been shown to reduce AIx and improve arterial compliance both at the immediate post-consumption phase (Wilkinson *et al.*, 1999) and after long-term administration (Mottram *et al.*, 1999; Mullan *et al.*, 2002). However, other studies have shown no effect after oral administration of these vitamins (Eskurza *et al.*, 2004), underscoring the need for further studies.

GARLIC

This nutritional element with possible (but strongly debated) anti-hypertensive properties, has been shown to reduce aortic PWV when consumed chronically (Breithaupt-Grögler *et al.*, 1997).

SALT

In a study of two Chinese populations, the age-associated increase in aortic PWV was blunted in the population with a lower salt intake (Avolio *et al.*, 1985a). Furthermore, salt-sensitive hypertensives have increased arterial stiffness compared to salt-resistant subjects with the same blood pressure levels (Draaijer *et al.*, 1993). In prospective randomized trials, sodium restriction reduces cardiovascular events in prehypertensive subjects. Sodium restriction is accompanied by a reduction in arterial stiffness both in diabetic subjects (Lambert *et al.*, 1997) and in older adults with systolic hypertension (Gates *et al.*, 2004). A modest reduction in salt intake for 6 weeks reduced aortic PWV overall in mild hypertensives, but subgroup analysis indicated that the effect was significant only in blacks, despite the fact that they achieved a smaller

reduction in salt intake and a similar fall in blood pressure, compared with whites and Asians (The China Salt Substitute Collaborative Group, 2007; He *et al.*, 2009; Hu *et al.*, 2009). In addition, the DASH diet is shown to decrease aortic AIx (Al-Solaiman *et al.*, 2009) and PWV (Blumenthal *et al.*, 2010). However, a recent randomized trial in 29 overweight and obese normotensive individuals showed no significant change in aortic PWV when salt intake was reduced from 9.2 to 3.8 g/day for 2 weeks, in spite of a significant improvement in endothelial function (Dickinson *et al.*, 2009).

ALCOHOL

The discovery of late Stone Age beer jugs has established the fact that purposely fermented beverages existed at least as early as the Neolithic period circa 10 000BC. Drinking in moderation may have health benefits, particularly in regard to coronary heart disease and the risk of myocardial infarction. A meta-analysis of 34 studies found that one to two drinks a day in women, and two to four drinks a day in men, are inversely related with total mortality, while greater intake increased mortality (Di Castelnuovo *et al.*, 2006).

Data regarding the relationship between alcohol consumption and arterial stiffness are based on cross-sectional and population-based studies. Regarding the effect of alcohol on arterial function, a J- or U-shaped curve has been demonstrated between its consumption and aortic stiffness and indices of wave reflections both in men and in women (Fig. 29.8) (Mahmud and Feely, 2002a; Sierksma *et al.*, 2004; Hougaku *et al.*, 2005; van den Elzen *et al.*, 2005). The mechanisms involved are not fully clarified; the beneficial effect of moderate consumption

may be mediated through an increase in high density lipoprotein cholesterol and insulin sensitivity, a decrease in inflammatory, oxidative and thrombotic status. The effect of alcohol on endothelial function is not so clear. Experimental data suggest that moderate alcohol consumption increases nitric oxide production, while heavy alcohol drinkers have impaired endothelial function.

Despite the great interest in the long-term effects of alcohol, there are only two studies investigating the longitudinal effects of alcohol on arterial stiffness. Specifically, it was shown that higher daily alcohol consumption (more than 23 g/day of ethanol) for nine years was associated with increased relative risk of arterial stiffness, as measured by aortic PWV, compared with non-drinkers (Nakanishi *et al.*, 2001). In a more recent study with six-year follow up, it was shown that there is a U-shaped relationship between alcohol consumption and brachial-ankle pulse wave velocity, only in patients with normal or optimal blood pressure (Matsumoto *et al.*, 2009). Moreover, subjects with normal or optimal blood pressure and light to moderate consumption show improvement of arterial stiffness, while subjects with high normal blood pressure or higher and heavy alcohol consumption show worsening of arterial stiffness (Matsumoto *et al.*, 2009).

Red wine

The French paradox, that coronary heart disease mortality is lower in France than would be expected from the high national prevalence of smoking and saturated fat intake, has been attributed to frequent red wine consumption. Red wine decreases aortic blood pressures and wave reflections acutely (Fig. 29.9) (Papamichael *et al.*, 2008). Studies with de-alcoholized wine have shown that part of

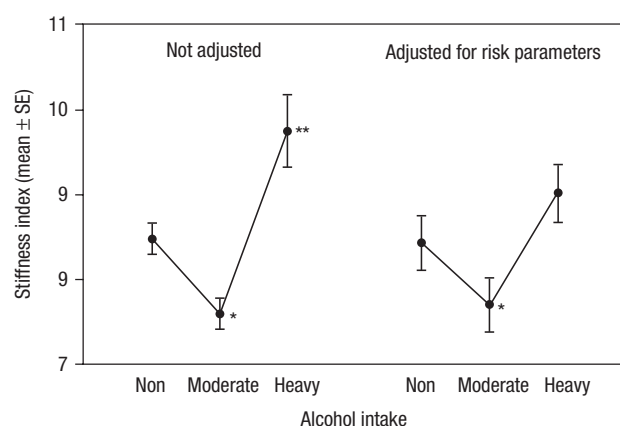


Figure 29.8 Association between stiffness index and categories of alcohol consumption, where non-drinkers consumed <1.0 unit/week, moderate drinkers consumed 1.0 to 9.9 units/week, and heavy drinkers >10 units/week. A unit was defined as 12 oz of beer, 2 oz of spirits, or 4–5 oz (one glass) of wine/day. Reproduced from Hougaku *et al.* (2005)

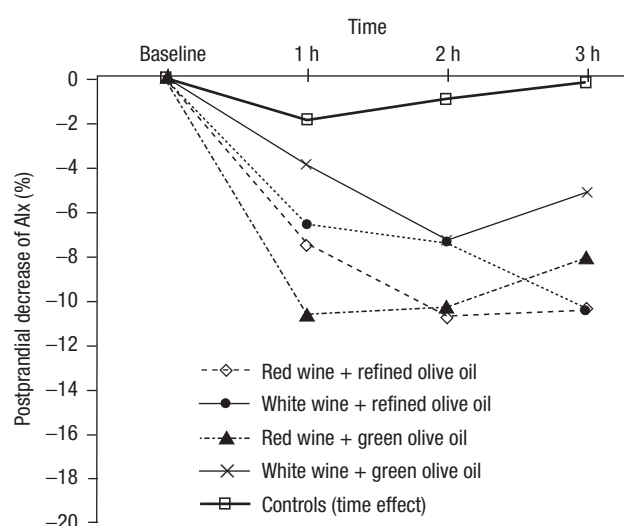


Figure 29.9 Acute changes in augmentation index for various combinations of wine and olive oil consumption. Reproduced from Papamichael *et al.* (2008)

the effect is due to alcohol-induced vasodilation (Mahmud and Feely, 2002a) and part to the high flavonoid content. The short-term (6 weeks) effect of red wine has been studied in postmenopausal women. Neither red wine nor de-alcoholized red wine induced any changes in wave reflections and central hemodynamics (Naissides *et al.*, 2006). Interestingly, red wine counteracts the deleterious effects of smoking on central blood pressures and AIx (Papamichael *et al.*, 2006) and works in synergy with olive oil to decrease central blood pressures and AIx (Papamichael *et al.*, 2008). Despite the extensively studied effects of red wine on central blood pressures and wave reflections, less is known about the effects on aortic PWV. There is only one relevant study that demonstrated that red wine improves aortic PWV and suggested that alcohol might be the most favorable vasoactive substance, since the alternative non-alcoholic drink did not induce any significant alteration in aortic PWV (Mahmud and Feely, 2002a).

OBESITY AND WEIGHT LOSS

Obesity is associated with increased aortic stiffness, as this has been evaluated in various studies (Wildman *et al.*, 2003; Ferreira *et al.*, 2004; Zebekakis *et al.*, 2005; Orr *et al.*, 2008). The negative impact of obesity has also been investigated in children and adolescents (Tounian *et al.*, 2001). Central fat accumulation has gained specific attention. Truncus subcutaneous fat accumulation during adolescence is associated with increased central arterial stiffness during adulthood (Ferreira *et al.*, 2004). From these studies, it is reasonable to conclude that overweight and obesity are also associated with an increase in wave reflections. The main pathophysiological mechanisms linking central obesity and vascular dysfunction include the production of adipocytokines, with detrimental metabolic effects and an inflammatory stimulation.

Weight loss is associated with benefits in multiple metabolic parameters and with a decrease in CVD risk. Its beneficial effect on arterial function has been clearly demonstrated, even in patients with morbid obesity, who underwent bariatric surgery (Ikonmidis *et al.*, 2007). Changes in body weight in a two-year follow-up study were positively associated with changes in aortic PWV. While weight gain increased large-artery stiffness, weight loss had the opposite effect (Wildman *et al.*, 2005). The beneficial effect of weight loss has also been demonstrated in various populations (Schneider *et al.*, 2005; Dengel *et al.*, 2006; Blumenthal *et al.*, 2010) and is independent of the diet used (meat- or plant-based). The improvement in arterial elastic properties observed with weight loss are largely explained by the concomitant reduction in blood pressure; however, the reduction in aortic PWV observed with weight loss was independent of blood pressure changes in one study (Wildman *et al.*, 2005).

MENTAL, EMOTIONAL STATUS, AND WORK STRESS

Every affection of the mind that is attended either with pain or pleasure, hope or fear, is the cause of an agitation whose influence extends to the heart

William Harvey, 1628

According to the evolution theory by Charles Darwin, 'In the struggle for survival, the fittest win out at the expense of their rivals because they succeed in adapting themselves best to their environment.' Somatic responses to stress represent evolutionary adaptive mechanisms that prepare for 'fight or flight' situations. However, excessive or prolonged activation of these mechanisms may have harmful effects. For instance, cutaneous vasoconstriction and activation of clotting cascade protect from bleeding when the organism is injured, but it also increases heart load and risk for inadvertent thrombosis, respectively.

Mental stress evoked in a laboratory setting is linked to increased CVD morbidity and mortality (Chida and Steptoe, 2010). In a clever study conducted during the World Cup soccer tournament in Germany in 2006, the emotional stress from watching a stressful soccer game more than doubled the risk of a cardiovascular event (Wilbert-Lampen *et al.*, 2008). Possible underlying mechanisms include catecholamine release and resultant increase in heart rate and blood pressure. Intense stress evoked by the World Trade Center attacks in New York City on September 11, 2001 resulted in a substantial and sustained (up to two months) increase in blood pressure that was independent of seasonal effects (Gerin *et al.*, 2005). Interestingly, the effects on blood pressure were also evident in remote cities, such as rural Mississippi.

The effect of acute mental stress on arterial function has been studied in various vascular beds. Brief periods of mental stress can have an enduring effect on arterial stiffness that is related to increased blood pressure and heart rate, allowing inferences for the effects of more prolonged or intense stimuli in everyday life. An arithmetic stress results in an increase in aortic stiffness and wave reflections in healthy subjects (Vlachopoulos *et al.*, 2006d; Lydakis *et al.*, 2008). Moreover, a computerized color-word interference task and mirror tracing result in an increase in carotid artery stiffness (Ellins *et al.*, 2008). Watching a stressful movie results in a sustained increase in aortic PWV. Taken from the exactly opposite angle, good mood appears to have a beneficial effect on arterial stiffness. Laughter resulting from watching a comedy reduced aortic PWV (Fig. 29.10) (Vlachopoulos *et al.*, 2009c). Although music lowers both stress and blood pressure and improves endothelial function (Miller *et al.*, 2010), its effect on arterial stiffness has not yet been studied.

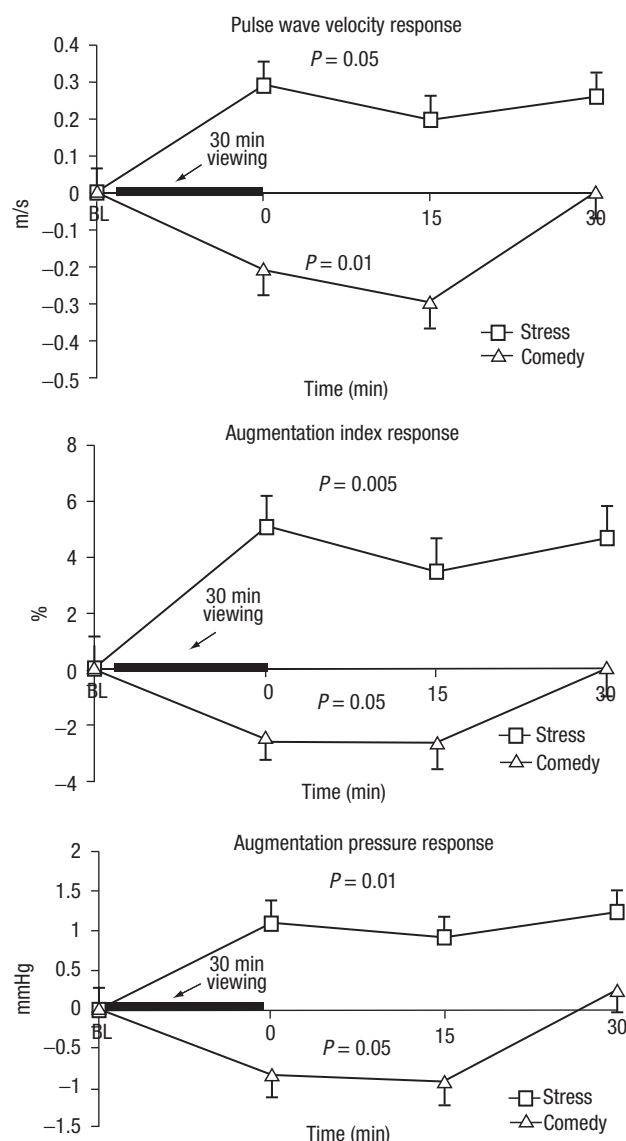


Figure 29.10 Acute responses of pulse wave velocity, augmentation index, and augmented pressure to watching a stressful movie and a comedy. Reproduced from Vlachopoulos *et al.* (2009c)

Data on the effect of subacute or chronic mental stress on arterial function are lacking. Most studies have focused on psychiatric disorders, such as depression, which is associated with endothelial dysfunction (Spieker *et al.*, 2002), even in remission periods; however, central hemodynamics do not seem to be altered in major depression. Anti-depressant therapy improved aortic PWV of newly diagnosed patients with depression and this improvement was related to the amelioration of their clinical status (Oulis *et al.*, 2010). Moreover, psychosocial variables, such as hostility and anxiety, may be important in the development of arterial stiffening in adolescence (Midei and Matthews, 2009). However, larger studies are needed to confirm these associations.

Work stress

Work stress is an independent factor for cardiovascular events. Job strain assessed by self-reported questionnaires is associated with both aortic and brachial-ankle pulse wave velocity (Nomura *et al.*, 2005; Michikawa *et al.*, 2008). Investigation of the role of two different work stress models, the effort reward imbalance (Siegrist model), and demand-control model (Karasek model), showed that a high stress job indicated by high demands and low control is associated with increased arterial stiffness (Utsugi *et al.*, 2009).

AIR POLLUTION

The adverse health effects of air pollution have been studied intensely over the last 50 years. Air pollution is a heterogeneous mixture of gases, such as second-hand smoke, liquids, and particulate matter. Epidemiological studies have identified a strong association between air pollution and increased cardiovascular morbidity and mortality (Brook *et al.*, 2010). Combustion-derived air pollution (acute exposure to diesel exhaust) is associated with an immediate and transient increase in arterial stiffness (Lundbäck *et al.*, 2009). Particle-related health consequences among welders include increased systemic inflammation and poorer vascular function, as measured by AIx and flow-mediated dilatation (Fang *et al.*, 2009). Recently, it was shown that air pollution may accelerate arterial wall stiffening and heighten the amplitude of the reflection wave (Adamopoulos *et al.*, 2010; Lenters *et al.*, 2010). Studies are needed to investigate whether interventions in pollutant environments would reverse these deleterious effects.

ILLEGAL SUBSTANCES

Little is known about the effects of illegal and addictive substances, such as cocaine, heroin, and cannabis, on arterial stiffness. A small cross-sectional study demonstrated that chronic cocaine abuse produced a significant decrease in aortic strain and distensibility and this effect was related to the duration of cocaine abuse (Bigi *et al.*, 2008). Furthermore, indirect evidence of compromised vascular function emerged from a recent study in young habitual cannabis users with erectile dysfunction that demonstrated impaired endothelial vasodilatation in the users with organic erectile dysfunction compared with those with non-organic erectile dysfunction (Aversa *et al.*, 2008). However, further studies are warranted to demonstrate the exact effects of such substances.

GENDER

There are definite differences between males and females for some cardiovascular parameters, and no difference in

others. Some can be explained on the basis of bodily habitus (females as a group are shorter than males, by 5.0–10 cm, and weigh less) and some cannot. Differences are seen for all races (Skurnick *et al.*, 2010).

Peripheral blood pressure, central blood pressures, and indices

Systolic brachial pressure in females during early adulthood is usually lower, but increases in middle life (55–60 years) to match or exceed brachial systolic pressure in males. Between the ages of 17 and 50 years, there is a plateau in systolic pressure in males, then a progressive increase. This plateau is not seen in females and systolic pressure appears to increase progressively with age from

early life. Brachial diastolic blood pressure behaves similarly in both sexes: it increases until approximately 50 years and then declines (Figs 29.11 and 29.12; also, see Chapter 19) (Franklin *et al.*, 1997; McEniery *et al.*, 2005a). Central blood pressures and indices differ between sexes in adult life (Hayward and Kelly, 1997; Smulyan *et al.*, 2001; Mitchell *et al.*, 2004a; Cecelja *et al.*, 2009; Janner *et al.*, 2010). Augmented pressure and AIx are systematically higher in women. In both sexes, augmented pressure increases linearly, while AIx tends to a plateau value above the age of 60 years (see Chapter 9). The marked differences between males and females in contour of the central and peripheral arterial pressure waveforms are retained through later life, although differences become less as aortic augmentation index approaches an asymptomatic value of around 40–45 percent in

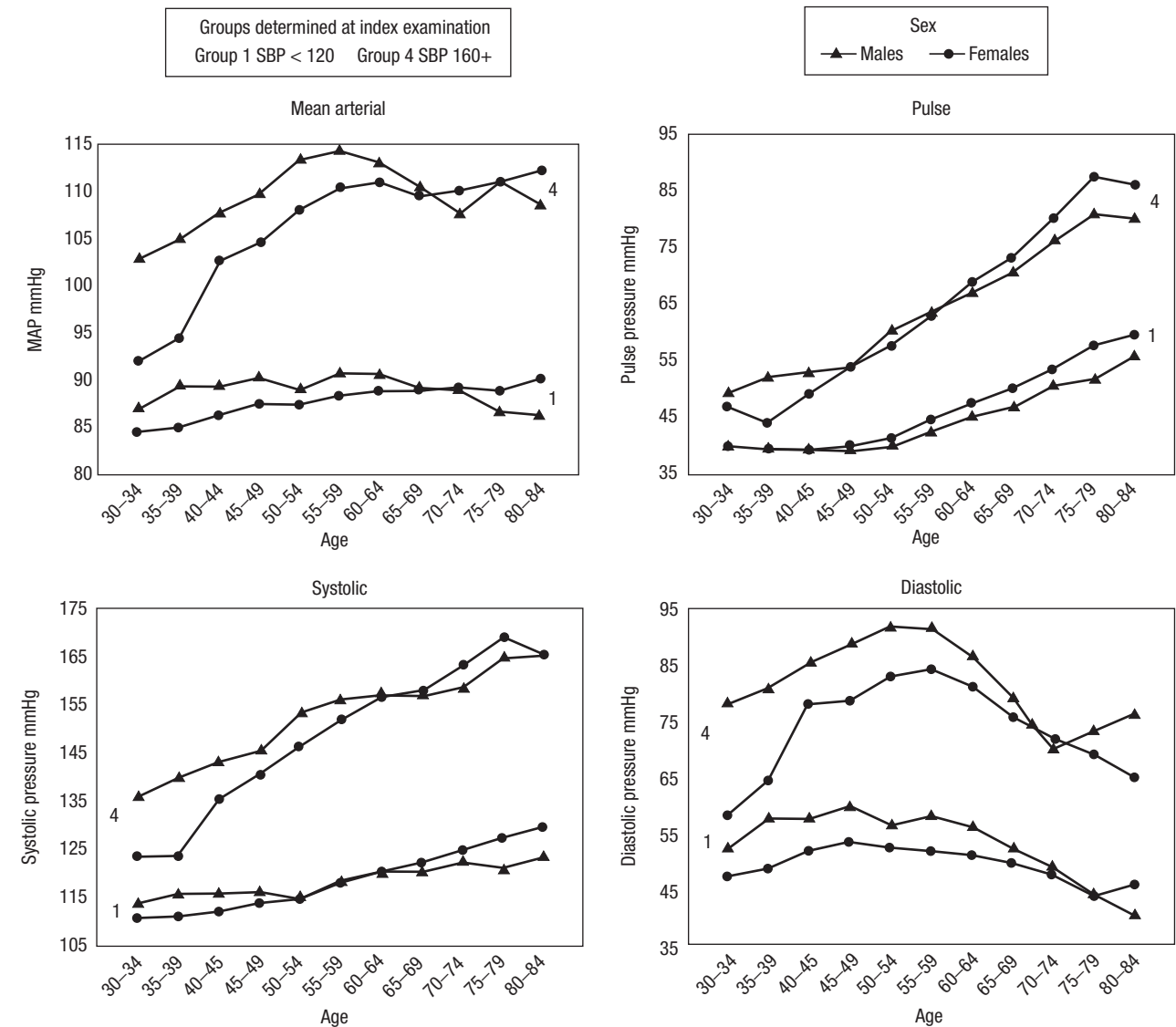


Figure 29.11 Blood pressure components according to age: group averaged data by sex. Averaged blood pressure levels for each subject within five-year age intervals (30–34 to 80–84) according to recorded systolic blood pressure at the index examination (group 1, < 120 mmHg; group 2, 120–139 mmHg; group 3, 140–159 mmHg; group 4, ≥ 160 mmHg). Adapted from Franklin *et al.* (1997)

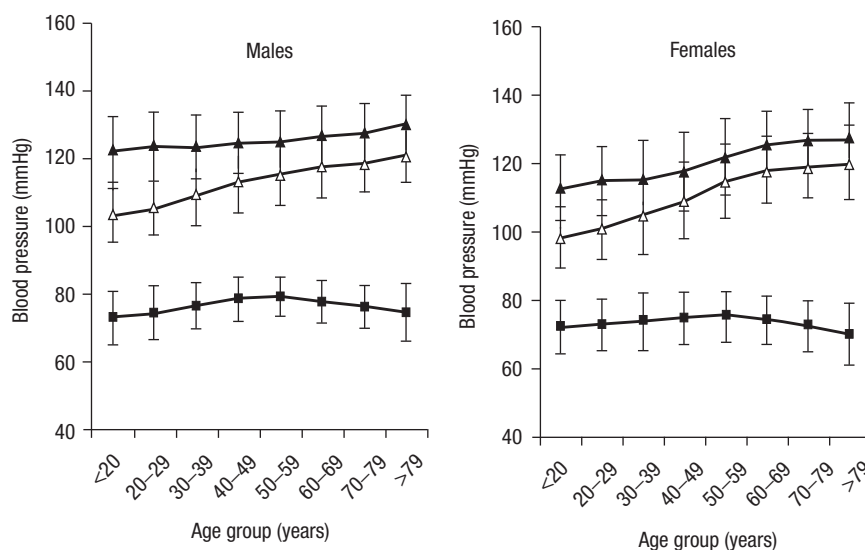


Figure 29.12 Blood pressure averaged for deciles of age for males and females. Symbols represent peripheral systolic (closed triangles), central systolic (open triangles), and diastolic (closed squares) blood pressures. Adapted from McEniery *et al.* (2005a)

octogenarians, and amplification in the upper limb approaches an asymptotic just above 1.0. Curvilinearity of these relationships is described and explained in Chapter 9 (Fig. 9.18) (Kelly *et al.*, 1989a; McEniery *et al.*, 2005a; Avolio *et al.*, 2009b).

The difference in central pressures and indices between females and males appears to be due to smaller amplification of the pressure wave between ascending aorta and upper limb in females, consequent on lower height (Hayward and Kelly, 1997; Smulyan *et al.*, 1998; Smulyan *et al.*, 2001) and not on differences in pulse transit time. Due to shorter body stature, the reflecting site in women is closer to the central circulation than it is in men. Thus, with PWV being equal, the pressure wave, reflected off a shorter arterial tree, returns towards the proximal aorta earlier and tends to amplify the primary wave in systole rather than in diastole. However, wave reflections indices are still elevated in women even after adjustment for height (McEniery *et al.*, 2005a; Lieber *et al.*, 2010) implying contribution of differences in the magnitude of the reflected wave. In the same wavelength, such differences have also been seen between girls and boys at age 8 (Ayer *et al.*, 2010), when there is no definite difference in bodily height. Accordingly, differences in central pressures and indices can also be attributed to greater physical taper of the aorta and large arteries of females, which has the effect of placing the functionally discrete lower body reflecting site closer to the heart, and so accentuating the effect of shorter body height, to cause earlier return of reflected waves to the ascending aorta from the lower body (Cecelja *et al.*, 2009). Heart rate is usually higher in females than in males and tendency towards higher wave reflections indices in women due to the above-mentioned factors may be compensated by a faster heart rate that would make the reflected wave return closer to diastole.

Indeed, significance of difference in such indices increases after adjustment for heart rate (Lieber *et al.*, 2010). All these factors contribute to greater susceptibility to heart failure with preserved systolic function in women (Shim *et al.*, 2011).

The phenomenon of spurious systolic hypertension of youth has only been described in males (who are usually tall, well developed, and physically active and non-smokers (O'Rourke *et al.*, 2000; Mahmud and Feely, 2003b)); this is usually explained as an extreme of pressure wave amplification in the upper limb, due to low aortic distensibility (Mahmud and Feely, 2003b).

Aortic stiffness

The vast majority of studies on 'aortic' pulse wave velocity (the 'gold standard' of aortic stiffness) have shown no difference in carotid-femoral (or aortic) pulse wave velocity with age (see Chapter 19). Similarity between 'aortic' PWV in males and females at any age contrasts with the difference in aortic pressure waveforms. The difference between waveforms in males and females is explicable on the basis of early return of wave reflections (in females), but clearly this is not due to faster travel and return of the wave after reflection. As described above and in Chapters 9 and 19, difference in distance to reflecting sites and difference in taper of large arteries explain earlier wave reflections in the presence of identical aortic stiffness and carotid-femoral PWV. Measures of aortic or arterial compliance do show difference between males and females. This is a scaling issue; since bodily dimensions are greater in males than females, aortic diameter is greater in males, so that the aorta can accommodate a greater amount of blood for a given pressure change. 'Scaling' is

already dealt with when stiffness is described in terms of pulse wave velocity, just as it is when pressure change is related to velocity change in expression of values for aortic impedance (see Chapter 12).

Gender hormones

ANDROGENS

Despite the lack of association in some cross-sectional and prospective studies, evidence originating mostly from the recent large studies supports an essential role of testosterone (and to a lesser degree of dehydroepiandrosterone (DHEA) and DHEA sulfate) in CVD (Basaria and Dobs, 2007; Jackson, 2010). Lower levels of androgen are associated particularly with increased cardiovascular death, adding to other morbidities, such as reduced muscle mass and lower-extremity strength, limitations in physical function, poor mobility, low trauma fractures, and reduced bone mineral density. Although not yet comprehensively investigated, testosterone was recently implicated in CVD in women as well (Laughlin *et al.*, 2010; Sievers *et al.*, 2010).

Aortic or carotid-femoral PWV appears to have a modest negative association between endogenous androgen levels and PWV in men (Ishihara *et al.*, 1992a; Ishihara *et al.*, 1992b; Gyllenberg *et al.*, 2001; Dockery *et al.*, 2003), which becomes stronger in patients with high CVD risk, such as diabetics (Fukui *et al.*, 2007) and erectile dysfunction patients as our own data have shown. Data on the relationship between central hemodynamics and androgen levels in men are presently lacking.

Currently, there is growing enthusiasm on androgen supplementation for patients with reduced testosterone levels stemming from the beneficial effects of testosterone supplementation on cardiovascular risk factors, such as lipids and reduced insulin sensitivity, and the overall safe profile of this therapeutic strategy. In the same line are studies showing that testosterone supplementation administered to men with low testosterone levels can reduce both aortic PWV and AIx (Webb *et al.*, 2008; Yaron *et al.*, 2009). However, a recent large study of testosterone supplementation in older men was discontinued because of increased cardiovascular events in the testosterone-treated group (Basaria *et al.*, 2010). While this study regards a 'fragile' population of elderly men and results cannot be extrapolated easily to patients with different characteristics, it emphasizes that steps should be taken with prudence in order to avoid the precedent of estrogen replacement therapy failure (see below).

FEMALE HORMONES

Female sex steroid hormones include estrogens and progestagens. Endogenous estrogens during premenopause

have shown protective effects on the vasculature and exert both genomic and non-genomic effects on vascular cells. Estrogen deficiency in premenopausal women is associated with an increased risk of coronary atherosclerosis and adverse cardiovascular events (Bailey Merz *et al.*, 2003; Shaw *et al.*, 2009). The possible vasculoprotective effect of endogenous estradiol is also extended to men, with low levels of estradiol being predictive of cardiovascular events; events increase further when low estradiol is combined with low testosterone (Tivesten *et al.*, 2009; Menke *et al.*, 2010).

The relationship between endogenous female hormones and arterial stiffness is not clear. Serving as an indirect evidence of a beneficial effect, in the prepubertal years, when female sex steroids are low, women have stiffer arteries than age-matched men (Ahimastos *et al.*, 2003). During the reproductive years, female sex steroids fluctuate during the menstrual cycle, but PWV remains unchanged (Williams *et al.*, 2001) and no correlation is found between PWV and serum estradiol or progesterone concentrations at any time point of the cycle (Robb *et al.*, 2009). Female sex steroids increase sharply at the first trimester of pregnancy and remain stably increased thereafter. In the first two trimesters, arterial stiffness is reduced (Edouard *et al.*, 1998; Robb *et al.*, 2009), but rises again in the third trimester despite the fact that serum estradiol and progesterone concentrations remain increased with advanced gestation. Again, no correlation is found between PWV and female sex hormones at any time point in pregnancy (Robb *et al.*, 2009) indicating a more complex interrelationship between these female hormones and arterial stiffness. Common variants in estrogen pathway genes implicated in myocardial infarction and elevated blood pressure have been associated with measures of wave reflections in the Framingham cohorts (Peter *et al.*, 2009).

Effects of menstrual cycle on augmentation index are inconsistent, showing either no systematic change during the cycle (Papaioannou *et al.*, 2009) or reduction in the luteal phase of the cycle (Robb *et al.*, 2009). In studies, however, measurements should be made in the same phase of the cycle for optimal data interpretation. During the first two trimesters of pregnancy, wave reflections are decreased and return towards baseline values, thereafter resembling the changes in PWV (Poppas *et al.*, 1997; Robb *et al.*, 2009).

While studies have shown a sharper increase of aortic stiffness with menopause compared to men in respective years (Staessen *et al.*, 2001c), large population studies have shown no gender difference (The Reference Values for Arterial Stiffness' Collaboration, 2010). Accordingly, apart from changes in lipid profile, women at the menopause show no change in arterial parameters or blood pressure that cannot be explained by progressive aging (Matthews *et al.*, 2009; Stramba-Badiale, 2010).

Hormone replacement therapy (HRT) was welcomed as a panacea for cardiovascular protection in the late

1980s, but reality proved disappointing (Grady *et al.*, 2002). Cardiovascular risk was clearly increased in some studies, while effect was neutral in others. Studies on the effect of hormone replacement therapy on arterial stiffness and wave reflections followed the hype of their era. Several studies have measured PWV in postmenopausal women receiving HRT (Rajkumar *et al.*, 1997; Samaras *et al.*, 1999; Stefanadis *et al.*, 1999b; Kawecka-Jaszcz *et al.*, 2002; da Costa *et al.*, 2004; Nichols *et al.*, 2007; Nichols and Epstein, 2009). Most of these studies showed a decrease in PWV in response to HRT, but the beneficial effect was evidenced only if estrogen alone was administered. In an insightful study involving postmenopausal women on long-term estrogen HRT or estrogens and progesterone HRT, improvement in carotid stiffness indices was shown in the estrogen arm, whereas the combination of estrogens and progesterone HRT had an adverse effect (Liang *et al.*, 1997). Apart from the potential antagonistic effect of progestin on estrogen-induced protection, time elapsed between menopause and initiation of therapy may be of essence for the nature of the effect. The KEEPS (Kronos Early Estrogen Prevention Study) trial is in progress to examine whether earlier initiation of estrogen therapy is associated with a decreased risk of CHD.

RACE AND ETHNICITY

There are systematic differences in cardiovascular parameters in different races. These have been best characterized in the United States between Whites and African Americans, and in South Africa between Whites and Blacks. In general, aortic pulse wave velocity is higher in African Americans compared with Whites, despite comparable mean arterial pressure; measurements of PWV in other races or ethnic groups are scarce. These topics are addressed further in Chapter 19.

Central pressures and indices

While European men have no significant differences in aortic AIx compared with Asian men (Sugawara *et al.*, 2010c), aortic AIx, as well as central systolic BP and pulse pressure, are shown to be higher and amplification lower in Blacks in Africa and the United States compared with white males when comparisons were made by age, and by gender (Heffernan *et al.*, 2008). Such differences cannot be explained on the basis of bodily dimensions (height and weight), but can be explained on the basis of different aortic stiffness. With lower amplification of the pressure wave in Blacks through the upper limb, the aortic systolic and pulse pressure will be higher than that in Whites and given brachial pressures will underestimate aortic systolic and pulse pressures if one expects the same difference between central and peripheral pressures as in Whites. Hence, the degree of severity of hypertension will be

underestimated if brachial pressure is taken to be the root cause of damage to heart and arteries. With aortic augmentation and central pressures systematically higher in Blacks than in Whites at the same age, one would expect to gain greater benefit from reduction in wave reflections in Blacks compared to Whites. In this context, Blacks appear to respond better to nitrates and to nitrate combination than do Whites (Carson *et al.*, 1999). On this basis, combination treatment for hypertension with a hydralazine/nitrate combination was accepted by the FDA for use in hypertension in African Americans (Taylor *et al.*, 2004).

Explanation of anti-hypertensive drug effect

The differences between central and peripheral pressure waveforms in males and females, and between Whites and Blacks has led to better understanding of how arterial pressure should be monitored so as to allow for variable amplification of the pressure wave – not just in Blacks and Whites, but also at different ages, heart rates, and pressure levels, and in different diseases and with different drugs.

The value of a generalized transfer function to describe pressure transmission in the upper limb is that it remains relatively stable in different races, in males and females and with different drugs and in different diseases (see Chapter 30). This enables the aortic pressure wave to be determined relatively accurately from the radial pressure waveform. With transfer function for pressure stable, differences in amplification of the pressure wave depend on the harmonic content of the aortic pressure wave, and hence on the pattern of ventricular ejection, and the modulus and phase of ascending aortic impedance (see Chapter 12).

We have described how gender or race can alter the shape of the aortic pressure waveform, and how differences in the waveforms can result in discrepancies in measurement of aortic (and hence LV) systolic pressure. Drugs used for other conditions, such as protease inhibitors for HIV infections (Schillaci *et al.*, 2005) or non-steroidal anti-inflammatory analgesic drugs (Antman *et al.*, 2007), can cause subtle changes in radial and aortic pressure waveforms, and hence ill effects which cannot be measured from cuff brachial pressures alone (O'Rourke, 2005). Such subtle changes in arterial waveforms caused by change in aortic stiffness or wave amplification are attracting attention by regulatory agencies to explain ill effects of non-cardiac drugs.

Figure 19.32 shows the relationship between AIx in the radial artery and AIx calculated for the ascending aorta using the SphygmoCor transfer function process. There is an inverse relationship between both, at least in the range of 1.05 and 1.90 for the radial artery and 0–50 percent for the ascending aorta. Figure 29.13 shows the same data expressed in terms of difference in pulse pressure

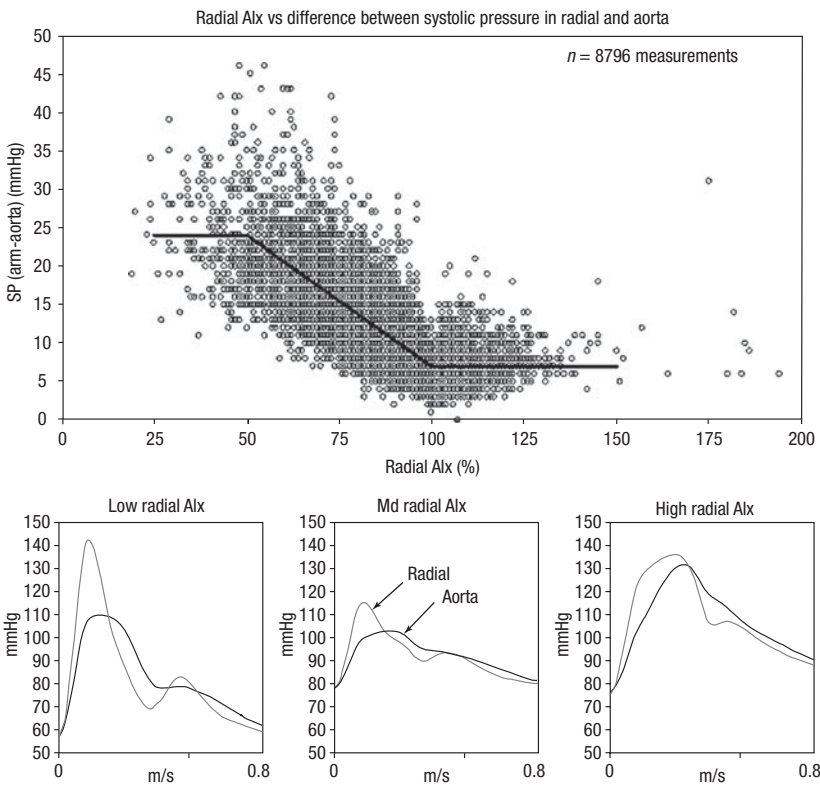


Figure 29.13 (Top) Relationship of amplification of the radial pressure wave (in mmHg) with radial augmentation index (AIx). (Bottom) Central and peripheral pressure waves at different radial AIx (for explanation, see text). Adapted from O'Rourke and Adji (2008)

between aorta and radial artery plotted against radial AIx. Figure 29.13 serves to explain how the difference between systolic and pulse pressure between the two sites can be interpreted from the contour of the pressure wave at the radial and brachial arteries. Between 50 and 100 percent for the radial artery waveform, which represents the vast majority of radial waves, the reciprocal relationship applies, and central pressure can be estimated from the contour of the radial artery pressure wave. However, for radial AIx under 50 percent, or over 100 percent, the relationship does not apply and systolic/pulse pressure difference is stable, < 25 percent, around 25 mmHg, and > 100 percent, around 7.0 mmHg. The implications of these findings are illustrated in the waves below. In the radial wave on the left, radial AIx is around 50 percent and pressure difference is approximately 33 mmHg. In the

middle waves, radial AIx is around 65 percent and pressure difference is 13 mmHg, while for the waves at the right, radial amplification is 125 percent and pressure difference is approximately 5.0 mmHg. The findings can be viewed another way. In the waveform on the left, the peak of the wave is in early systole in aortic and radial tracings, in the waveforms on the right, the peak of the wave is in late systole, whereas in the waves in the center (the most common situation), the radial peak is in early systole and the aortic peak is in late systole. Vasodilator drugs will appear to have the same effect on central and peripheral systolic and peak pressure if the radial augmentation index remains above 100 percent or below 50 percent, but will have differential effects if radial AIx is between 50 and 100 percent before or after administration of the drug.

Pressure pulse waveform analysis

Historical perspective: the sphygmogram	595	Relationship between aortic and radial artery pressure waveforms	606
Applanation tonometry	598	Sphygmocardiography	616
Radial artery waveforms	599	Finger plethysmography	630
Carotid artery waveforms	600	Relationship between indices	631
Relationship between carotid and radial artery pressure waveforms	604	Measurement of arterial properties with arterial models	631
Relationship between carotid and aortic pressure waveforms	606		

The pulse ranks first among our guides; no surgeon can despise its counsel, no physician shut his ears to its appeal. Since, then, the information the pulse affords is of so great importance, and so often consulted, surely it must be to our advantage to appreciate fully all it tells us, and to draw from it all that it is capable of imparting.

FA Mahomed, 1872

However eloquent may be the words of a writer, he cannot in a page convey as clear an idea of the rhythm of the heart as a simple pulse-tracing and if writers had given us more pulse tracings, their works would have been greatly enhanced in value.

Sir James Mackenzie, 1902

It is a supreme paradox that modern physicians, almost without exception, while surrounded by twenty-first century aids to diagnosis and therapy, use the arterial pulse only to assess heart rate and rhythm, and take no notice of the other information that the pulse is capable of imparting. Physicians do, of course, through use of the cuff sphygmomanometer, gauge the peak and nadir of the pulse and record their value as systolic and diastolic blood pressure. We have already discussed how inaccurate these values can be, with standard deviations up to $\pm 10\text{mmHg}$ common in comparison of simultaneously determined indirect to directly recorded values in the brachial arteries. We have also discussed how diastolic brachial blood pressure was over-interpreted as the sole guide to elevated pressure for almost a century, and how, even now, brachial

systolic pressure is misinterpreted as a measure of peak left ventricular pressure (Mullens *et al.*, 2008; Yancy, 2008).

For eons, the presence of the arterial pulse has been recognized as the cardinal sign of life, and its absence the first sign of death. The Fathers of Medicine in Greek, Egyptian, Chinese, and Indian cultures recognized different forms of the pulse in health and disease. These were categorized by the Greek physician Galen, living in Rome, almost 2000 years ago, and the Galenic views on the pulse went unquestioned until the Renaissance. The Royal College of Physicians of London, formed in 1518 to enhance standards of medical practice, took as its crest the physician’s fingers palpating the arterial pulse at the wrist (Fig. 30.1).

In this chapter, we will extend Mahomed’s views of 1872, and show how clinically useful information can be obtained from the arterial pulse waveform through application of the principles set out in previous chapters. We concentrate here on the waveform and explain too how information on arterial stiffness, set out earlier, is obtained from analysis of the waveform.

HISTORICAL PERSPECTIVE: THE SPHYGMOGRAM

If the thermometer and stethoscope are regarded as simple clinic aids, then the sphygmogram (Fig. 30.2) must be acknowledged as the first diagnostic instrument ever introduced into clinical practice. Despite difficulties in its use – being a mechanical device with delicate and complex lever systems, and recording on smoked paper – the



Figure 30.1 Crest of the Royal College of Physicians of London. With permission of the RCP, Regents Park, London, UK

sphygmogram achieved considerable clinical popularity during the years 1860 to 1920, not just for recording pulse waveforms, but for confirming excess risk in life insurance examinations, for describing the effects of disease and drugs, and for identifying the mechanisms of cardiac arrhythmias (Marey, 1860; Marey, 1863; Mahomed, 1872; Mahomed, 1874; Marey, 1875; Mahomed, 1877; Murrell, 1879; Marey, 1881; Broadbent, 1890; Mackenzie, 1902; O'Rourke, 1992; Postel-Vinay, 1996; O'Rourke *et al.*, 2001; Wilkinson *et al.*, 2002a). The great composer Johannes Brahms (1833–97) had his radial pulse waveform recorded as part of a medical examination in Vienna (Fig. 30.2A). Medical journals and textbooks of the period were liberally illustrated with sphygmograms. Users included general practitioners, such as Sir James Mackenzie, hospital clinicians, consultant physicians together with medical examiners for life insurance companies in France. In the United States, Keyt of New Jersey in 1883 introduced the sphygmograph into clinical medicine (Moss, 2006a; Moss, 2006b). An article on the subject was printed in the first edition of the *Journal of the American Medical Association* (Keyt, 1883). The field lapsed with the introduction of the electrocardiogram as a practical clinical device for diagnosing arrhythmias, and with uncritical acceptance of the cuff sphygmomanometer numbers for measuring left ventricular load.

History of the sphygmogram involved the most senior figures in medical practice, especially in England, and the major journals and textbooks. The first instrument was built by Vierordt (1855) in Germany. This was bulky and

could not record any waveform detail. The first practical instrument (Fig. 30.2B) was constructed in Paris by Marey (Marey, 1860; Marey, 1863) (who achieved fame also for his work with movement and rapid sequence photography). The sphygmogram was applied to clinical problems by Burdon-Sanderson in London during the early 1860s. Controversy erupted swiftly with claims of plagiarism in the medical press against an individual who presented similar data, but implied that Marey's device, which he used, had been developed by himself (Ainstie, 1866). By today's standards, where selective quotation is rife, balance often skewed, and where scientists are prone to take an adversarial approach, the offence seems mild. Three points might be made. First, Marey's contribution (see Fig. 30.2) was far greater than many of his English followers were ever prepared to concede; second, difficulties in high-fidelity measurement of pressure waves still lead to conflict in the acceptance of such new views as the usefulness of transfer functions in clinical practice; and third, some national scientific bodies do view selective citation and lack of balance as evidence of scientific misconduct (Pianin, 2003). Some problems with Marey's sphygmogram were overcome by Frederick Mahomed (Mahomed, 1872; Mahomed, 1874; Mahomed, 1877; Mahomed, 1879), initially when a student at Guy's Hospital, London, and whose first presentation on the sphygmogram was made in 1870, at the age of 21. Mahomed's contribution to sphygmography was immense, and is described in a number of reviews (O'Rourke, 1992; Postel-Vinay, 1996; Swales, 1996a), but was terminated prematurely by his death at the age of 35 of the contemporary scourge, typhoid fever. Mahomed characterized the radial pulse of high tension, primarily on the basis of an exaggerated late systolic wave, but showed, as did Marey (Fig. 30.2A), that the same pulse could be seen in normotensive people with senile arteriosclerosis. He traced the natural history of essential hypertension, and separated this from chronic Bright's disease (glomerulonephritis) – all on the basis of pulse waveform analysis, combined with shrewd clinical assessment, and 20 years before Riva Rocci's description of the cuff sphygmomanometer.

Mahomed's work on the arterial pulse was taken up enthusiastically by many, including Sir William Broadbent (Broadbent, 1875; Broadbent, 1890), 14 years his senior, who also worked in London, and attended his colleague in his fatal illness. Broadbent (see below) reconciled sphygmography with clinical palpation of the pulse, and showed that the features demonstrated by Mahomed's tracings could be detected by careful clinical examination. Broadbent continued to champion this viewpoint as he rose to the top of his profession as one of the most successful consultant physicians in London, Censor in Chief of the Royal College of Physicians, and Physician to Queen Victoria and the Prince of Wales.

Mahomed's sphygmogram was superseded by Dudgeon's beautifully compact instrument, which found its way into clinical practice around the world. This instrument was popularized by Sir James Mackenzie

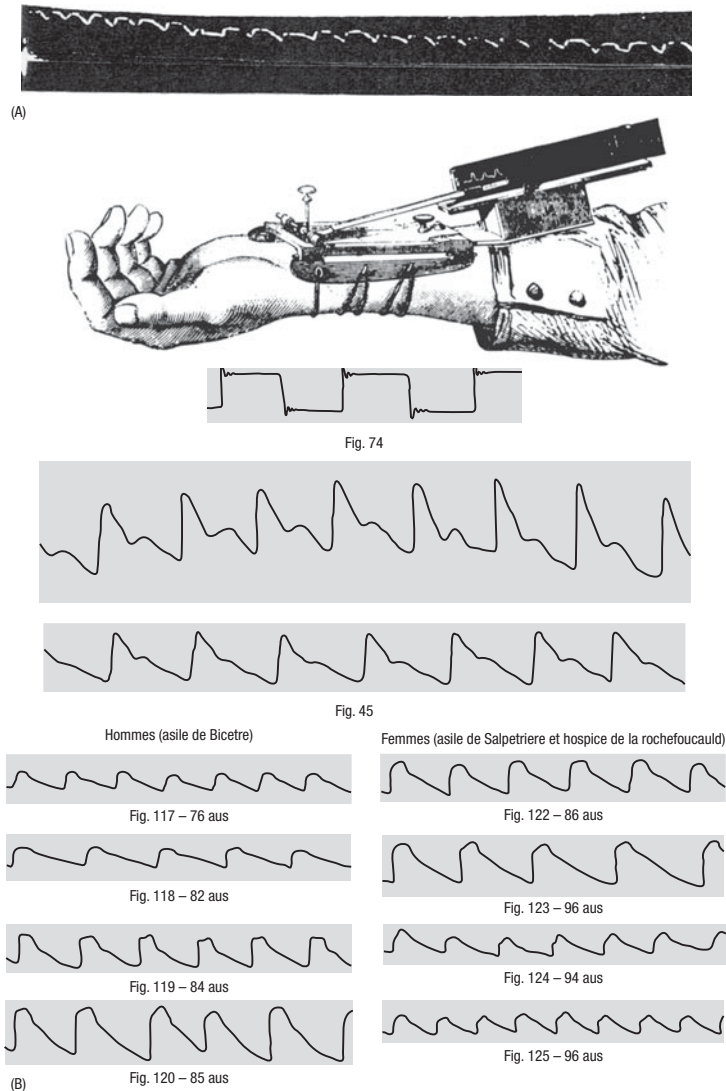


Figure 30.2 (A) Pulse of Brahms. Recorded by Theodor Wilhelm Engelmann, Utrecht University. Courtesy of Dr Gaston Bauer and Professor Alberto Avolio, Sydney, Australia. (B) Tracings from Marey's (1863) original publication on the sphygmogram. Top: The original instrument, with base-plate applied over the radial artery, with long stylus, pivoting near the site of recording, drawing an amplified signal on smoked paper which was moved by a wind-up, clockwork mechanism. Inset below is a 'square wave' test to establish frequency response and instrumental artifact. Center: Sphygmogram tracings in two young adults, showing patterns still regarded as typical for the radial artery. Bottom: Sphygmogram tracings in four old men (left) and women (right), showing typical patterns of senescence with accompanying arteriosclerosis

(1902), who might be considered the father of modern cardiology (at least in the English-speaking world), and whose books on the heart and on the pulse wave were liberally illustrated with sphygmographic tracings. Mackenzie's protégé, Thomas Lewis, became the first editor of *Heart* (the oldest English language cardiac journal) and used the sphygmogram widely in his studies of cardiac arrhythmias, before any practical clinical ECG instrument was available (Lewis, 1906; Lewis, 1933; Snellen, 1983; Snellen, 1984; Krikler, 2000).

The sphygmogram was widely used too in the life insurance industry (Postel-Vinay, 1996). It was 1917 before value of the cuff sphygmomanometer was first established. Prior to this, medical examiners knew the implications of the 'enduring' (Osler, 1906) pulse waveform palpated at the wrist, but sought visual evidence, available with the sphygmogram, for confirming this (Fig. 30.2), and so rejecting an apparently well applicant as being at excessive risk of cardiovascular disease.

The pulse waveform of premature arterial senility had been described in different terms by Marey, Mahomed, Broadbent, Mackenzie, Osler, and others, and comprised a

plateau, broadened, or late systolic peak (Fig. 30.2). This is now known to correspond with enhanced exaggeration of the late systolic peak of the aortic and left ventricular pressure waveform due to wave reflection (Fig. 19.24), and confirmed to confer an ominous prognosis (Fig. 25.9) (London *et al.*, 2001).

Initially, sphygmograms were recorded by mechanical systems and levers on smoked paper by devices made by Marey, Mahomed, and Dudgeon. They were only suitable for recording the radial pressure pulse. It was recognized that distortion occurred, together with delay in transmission of the pulse to the periphery (see Fig. 10.34). Consequently, cup-shaped tambours were developed, connected through air-filled tubes to mechanical levers, again writing on smoked paper. These were particularly suitable for recording the carotid arterial pulse waveform, and were used by Lewis in London and Cannon in Boston (among others) for studies of arrhythmias and cardiovascular reflex control, and with other methods by Wiggers (1928) in his seminal work, which underpinned use of careful manometry, learned from Otto Frank, on human cardiac catheterization following the Second World War.

Human diagnostic cardiac catheterization was introduced in the late 1940s to measure pressure gradients across stenoses, and oxygen saturation gradients across shunts. Imaging techniques were not available as they are now, and careful manometry was essential. Standards were set by pioneers such as Earl Wood at Mayo Clinic, who emphasized the need for adequate frequency response as had Marey and Mackenzie for sphygmograms (Fig. 30.2B), and who confirmed delay and amplification of the pulse between central and peripheral arteries (see Chapter 10, Figs 10.16 and 10.19) (Kroeker and Wood, 1955; Kroeker and Wood, 1956; Remington and Wood, 1956). Regrettably, the modern invasive cardiologist, though skilled in imaging techniques, has little knowledge or concern about manometric frequency response, and uses long tubing with multiple connections, and manometric circuitry which usually provides additional filtering. A 'sloppy' approach (Kaplan, 1998; Messerli *et al.*, 2002; O'Rourke, 2003) seems to have developed towards the measurement of arterial pressure in the catheterization laboratory as it has with the cuff sphygmomanometer in the clinic.

Following introduction of diagnostic cardiac catheterization in the late 1940s, there was renewed interest in recording of pressure pulse waveforms from the carotid artery in the neck. The air-filled tambour systems were replaced by microphones which could be applied directly. These were used by Weissler and colleagues at Ohio State University (Weissler *et al.*, 1963; Weissler *et al.*, 1968; Lewis *et al.*, 1981; Boudoulas, 1990), who drew attention to systolic time intervals (principally pre-ejection period and ejection duration) as indices of myocardial contractility and cardiac function. Attention was directed primarily at timing intervals since these could be measured with acceptable accuracy and reproducibility. Changes in carotid pressure wave contour under different conditions were noted, but there was no method available to calibrate the amplitude of the pulse, and comparison of directly recorded aortic pressure (at cardiac catheterization) and indirectly recorded carotid pressure waveforms gave inconsistent results on amplitude and contour. These studies of carotid tonometry lost favor in the 1970s with introduction of cardiac ultrasound, which could provide the same information, and more, than determination of timing intervals from the carotid sphygmogram.

Present interest in non-invasive analysis of the arterial pulse followed introduction of applanation tonometry, and its application to the cardiovascular system. Applanation tonometry is a specific physical method, and refers to applanation (flattening) and tonometry (measurement of pressure). When a small section of a spherical structure (such as the eyeball) or a circular structure (such as an artery) is flattened, the tangential forces in the wall are eliminated, and the sensor on the flattened surface measures the pressure within (Fig. 30.3). This is the first non-invasive method capable of accurately measuring arterial pressure within the artery; and when a high-fidelity micromanometer is used for sensing, as in the Millar

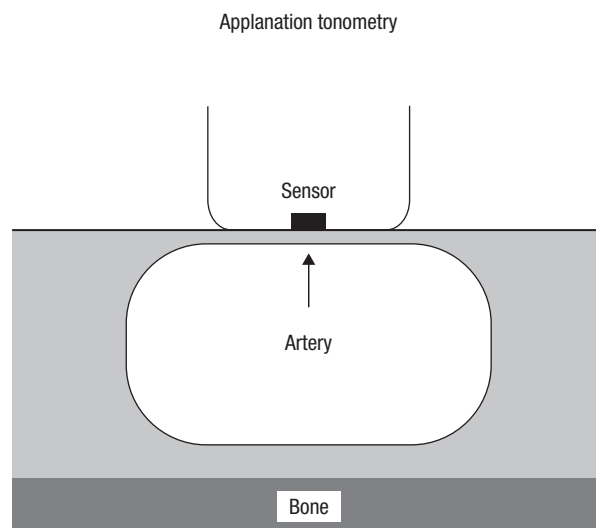


Figure 30.3 The theory of applanation tonometry. When the flat sensor flattens the wall of an artery, tangential pressures are eliminated and the sensor is exposed to the pressure within the artery, which it records accurately. Reproduced from Kelly *et al.* (1989a)

device, the first method capable of recording accurate detail of the waveform as well (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; Kemmotsu *et al.*, 1991a; Kemmotsu *et al.*, 1991b; Chen *et al.*, 1996).

APPLANATION TONOMETRY

Tonometry is widely used in ophthalmology for measuring intraocular pressure, and for the assessment of glaucoma and its response to treatment. Applanation tonometry was first applied to determination of arterial pressure waves by Pressman and Neugard in 1963. This has proved a major advance over all other methods for non-invasive determination of the arterial pressure pulse (Drzewiecki *et al.*, 1983). The single-element transducers introduced by Huntly Millar in 1989 (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b) have been extended into multi-unit arrays (Kemmotsu *et al.*, 1991a; Kemmotsu *et al.*, 1991b; Sato *et al.*, 1993) as in the instrument developed by Nippon-Colin (Komaki, Japan), and subsequently popularized by Omron (Hoofddorp, The Netherlands). The combination of an array that can straddle an artery, together with a servo-controlled placement system, and with the automatic application of pressure from behind, permits automatic operation of the sensor and optimal placement of the operational tonometric element for the recording of radial artery pressure. While this system has definite advantages (Chen *et al.*, 1997), most data have to date been obtained with the single-unit hand-held systems (Kelly *et al.*, 1989c; London *et al.*, 1992; Roman *et al.*, 1992; Hayoz *et al.*, 1994; Chen *et al.*, 1995; Chen *et al.*, 1996; Asmar *et al.*, 2001b; London *et al.*, 2001; O'Rourke *et al.*, 2001; Jiang *et al.*, 2002;

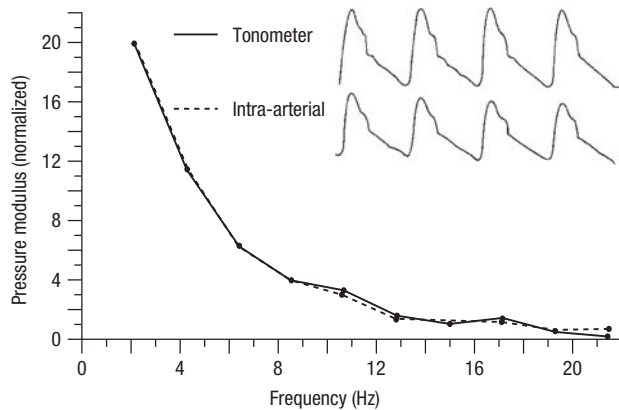


Figure 30.4 Comparison of intra-arterial pressure waves recorded by micromanometer, and tonometric pressure recorded by application of a tonometer to the arterial wall, with expression in the time domain (upper right) and frequency domain (lower left). Reproduced from Kelly *et al.* (1989a)

Nichols and Singh, 2002; O'Rourke, 2002a; Safar *et al.*, 2002b; Schofield *et al.*, 2002a; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b; Stokes *et al.*, 2003; Laurent *et al.*, 2006; Nichols *et al.*, 2006; Williams *et al.*, 2006; Roman *et al.*, 2007; Roman *et al.*, 2009).

Under optimal conditions for applanation (Fig. 30.3), the pressure wave measured non-invasively is virtually identical to that recorded with a high-fidelity transducer within the artery (Fig. 30.4) (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; Chen *et al.*, 1996). In practice, the pressure required to applanate the artery and to compress overlying structures, i.e. the 'hold-down pressure', does vary, so that the absolute values of systolic and diastolic pressure cannot be determined reliably. However, under ideal conditions (which can be achieved with practice in most patients) the contour, harmonic content (Fig. 30.4) and often the amplitude (pulse pressure) (Fig. 30.5) of the pressure pulse can be determined reliably. Several useful indices can be obtained from the pulse wave without calibration.

The theory of applanation tonometry applies when a small part of the arterial wall is flattened (Fig. 30.3). Under these circumstances, accurate waveform recording can be expected. This, of course, will not apply if the sensor is not applied directly over the artery. The theory does not apply either if a large sensor is applied over the artery and adjoining structures. The Millar and Omron devices satisfy requirements, but other sensors may not. All need to be calibrated against intra-arterial pressure to confirm accuracy. Such has been established for the Millar devices in the radial and carotid sites (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; Chen *et al.*, 1996). At these sites, the artery is supported by solid structures (bone, or bone plus ligaments) behind, so that applanation can be ensured, i.e. the artery does not move into soft tissue when external pressure is applied and its surface can be flattened. Most information has been obtained from these two sites. The brachial is not a suitable site for applanation tonometry

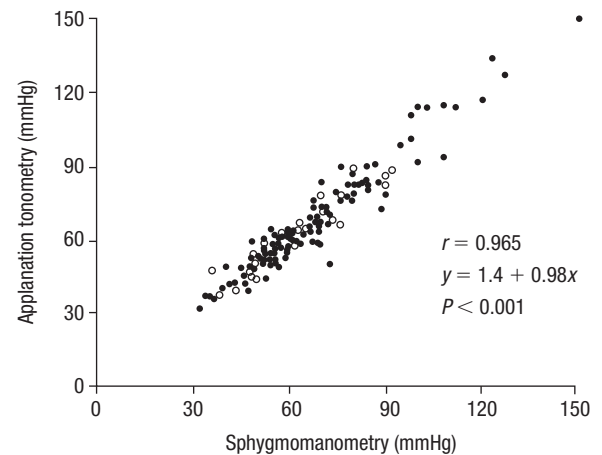


Figure 30.5 Relationship between brachial artery pulse pressure determined by sphygmomanometry and radial artery pulse pressure determined by applanation tonometry. (○) Control subjects; (●) end-stage renal disease patients. Reproduced from London *et al.* (1992)

(O'Rourke *et al.*, 2005; O'Rourke and Adji, 2010b), since the arterial wall cannot reliably be applanated. Implications with respect to central pressure measurement are discussed in Chapters 6 and 28.

RADIAL ARTERY WAVEFORMS

While we have gained considerable experience with carotid artery tonometry, we are most comfortable with the application of tonometry to the radial artery. There are a numbers of reasons for this. First, there is more historical data on the radial pulse wave in health and disease than at any other site. This is the vessel that the physician has been palpating for eons, and virtually the only site at which sphygmography was used in the past to determine the effects of aging, disease, and drugs. Second, accurate, reproducible, and consistent recordings can be taken almost invariably at this site – and compared to those from the contralateral wrist if there is any doubt. Third, recordings can be taken without any concern about baroreceptor activation, which is not uncommon with carotid tonometry. Fourth, use is not associated with the discomfort often felt with carotid tonometry, and there is no hazard from dislodgement of carotid plaque. Fifth, calibration to cuff sphygmomanometric values of systolic and diastolic pressure is reasonable at this upper limb site, whereas calibration of the carotid pressure pulse (see below and Chapter 28) is more difficult and less reliable.

The only disadvantage of using the radial site for applanation tonometry is that the pressure pulse wave is amplified in transmission from proximal aorta, and wave contour is altered. This potential disadvantage can be overcome by exploitation of the (relatively) constant arterial properties in the upper limb; these are little

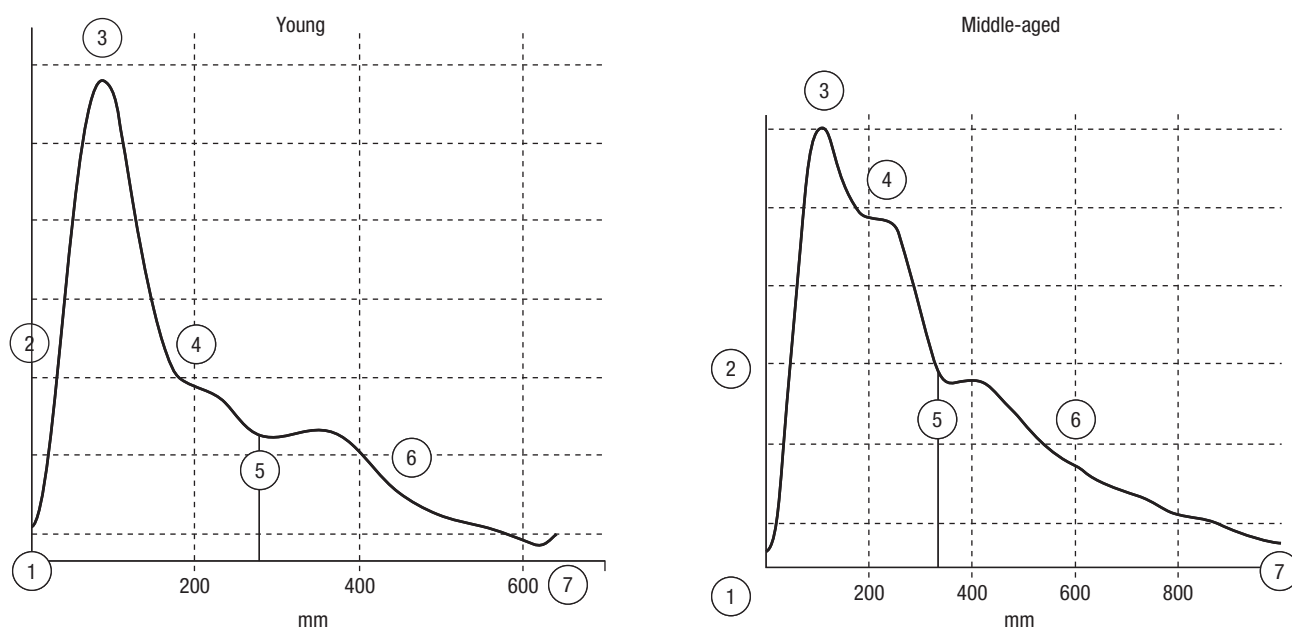


Figure 30.6 Features of a typical radial artery pressure wave in a young (left) and middle-aged human adult (right). For explanation, see text

affected by aging, hypertension, drugs, and disease in adults, so that a generalized transfer function can reasonably be applied to generate the central aortic pressure waveform from the radial waveform (Karamanoglu *et al.*, 1993; O'Rourke *et al.*, 1993a; Chen *et al.*, 1997; Pauca *et al.*, 2001; Sharman *et al.*, 2006) (and FDA K012487 and K002742). This is referred to below under Sphygmocardiography (p. 616).

Typical features of the radial artery pressure waveform (also seen in Fig. 30.2) are shown schematically in Fig. 30.6 for a young adult and a middle-aged person. In recording the radial waveform, the radial artery is palpated at the wrist, then after stabilization of the wrist, the tonometer is applied over its most prominent part. As downward pressure is applied, the pulse waveform will be recorded. When the tonometer is accurately applied, one expects to see:

1. A sharp upstroke of the pulse from its diastolic nadir.
2. A steep, uninterrupted rise of pressure to a peak some 100 ms after the foot of the wave.
3. A second, late systolic shoulder (or rarely a second peak) between the first peak and incisura.
4. A sharp inflection to identify the incisura, which signifies aortic valve closure.
5. A second pressure rise in early diastole (only when the second systolic shoulder is close to the incisura).
6. A near exponential decline in pressure during the latter part of diastole before onset of the next pulse.

Pulses in a given individual should be of the same height and contour, and of maximal amplitude. After application of the tonometer to the skin over the radial artery, it may

be necessary to reduce downward pressure, or to move the sensor slightly sideways for a series of waves to be recorded, all virtually identical and showing the above characteristics.

The radial pulse tracings in Fig. 30.6 show the effects of aging as increase in amplitude of the second systolic shoulder, and decrease in amplitude of the diastolic pressure waveform. Similar features are seen with increase in arterial pressure. The features are illustrated elsewhere, as are the effects of change in heart rate, and of drug therapy.

While we prefer to use the transfer function process in clinical and drug studies, this is a convenience, and all relevant data are present in the radial waveforms themselves (Rietzschel *et al.*, 2001a). Such waveforms, unprocessed, were used in studies of endothelial function (Hayward *et al.*, 2002) and drug action (Millaseau *et al.*, 2002). An alternative way to measure aortic systolic pressure comes from measurement of the beginning of the systolic shoulder (see Chapter 28) (Pauca *et al.*, 2004; Takazawa *et al.*, 2007).

CAROTID ARTERY WAVEFORMS

Many studies have utilized the carotid pulse as a surrogate of the pressure wave in the ascending aorta (Takazawa *et al.*, 1988; Fujii, 1989; Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; Kelly *et al.*, 1989c; Kelly *et al.*, 1989d; Kelly *et al.*, 1989e; Kelly *et al.*, 1989f; Roman *et al.*, 1992; Vaitkevicius *et al.*, 1993; Chen *et al.*, 1995; Ting *et al.*, 1995a; Ting *et al.*, 1995b; Mitchell *et al.*, 2002; Lakatta and Levy, 2003a; Lakatta and Levy, 2003b; Mitchell *et al.*, 2003;

Mitchell *et al.*, 2004a; Laurent and Boutouyrie, 2007; Segers *et al.*, 2007b; Mitchell *et al.*, 2008; Segers *et al.*, 2008; Segers *et al.*, 2009; Mitchell *et al.*, 2010). The proximal subclavian pulse has also been used as a surrogate of the ascending aortic pulse, using air-filled tambours (Wong *et al.*, 1984; Cholley *et al.*, 1995a; Cholley *et al.*, 1995b). There is, however, no doubt that the high-fidelity applanation tonometer is more accurate than an air-filled tambour for recording pressure waves in the neck or at its base (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; Chen *et al.*, 1996). At both sites, calibration is a problem, since one cannot reasonably assume, except in the very elderly, the very young, and the very hypertensive, that systolic and diastolic pressures are similar to those recorded in the brachial or radial artery. A method introduced by Kelly and Fitchett (1992) assumes that the diastolic and mean pressures are identical in the radial and carotid artery (Pauca *et al.*, 1992; Pauca *et al.*, 2001), and determines carotid systolic pressure from appropriate scaling of the carotid wave. Results obtained under different conditions fit reasonably well with those determined through the use of generalized transfer function from ascending aorta to carotid, to radial artery and to femoral artery (Fig. 30.7) (Fitchett, 1993a; Fitchett, 1993b; Gallagher, 1993; Hayoz *et al.*, 1994; Chen *et al.*, 1997). Van Bortel *et al.* (2001b) have applied the same principle for calibrating carotid systolic and diastolic pressure utilizing the carotid diameter waveform as a surrogate of the carotid pressure waveform. Other problems encountered in using the carotid pulse are described above and include difficulty in applanating the carotid artery because of its mobility and depth in the neck, and the changes in heart rate that can occur with mechanical stimulation of carotid

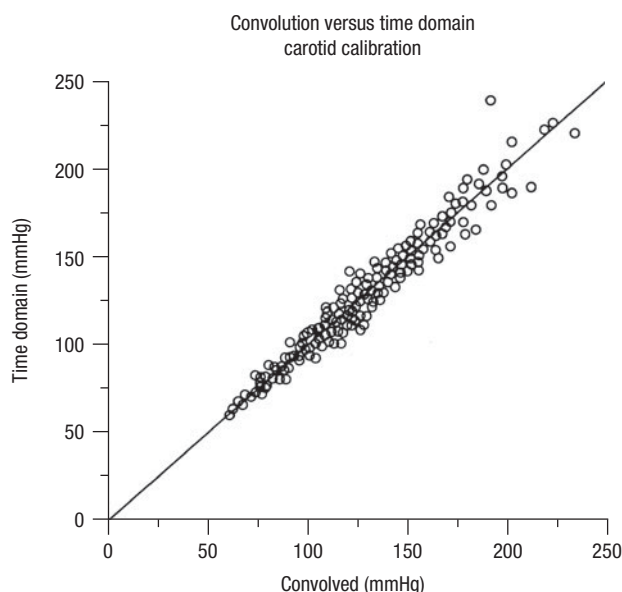


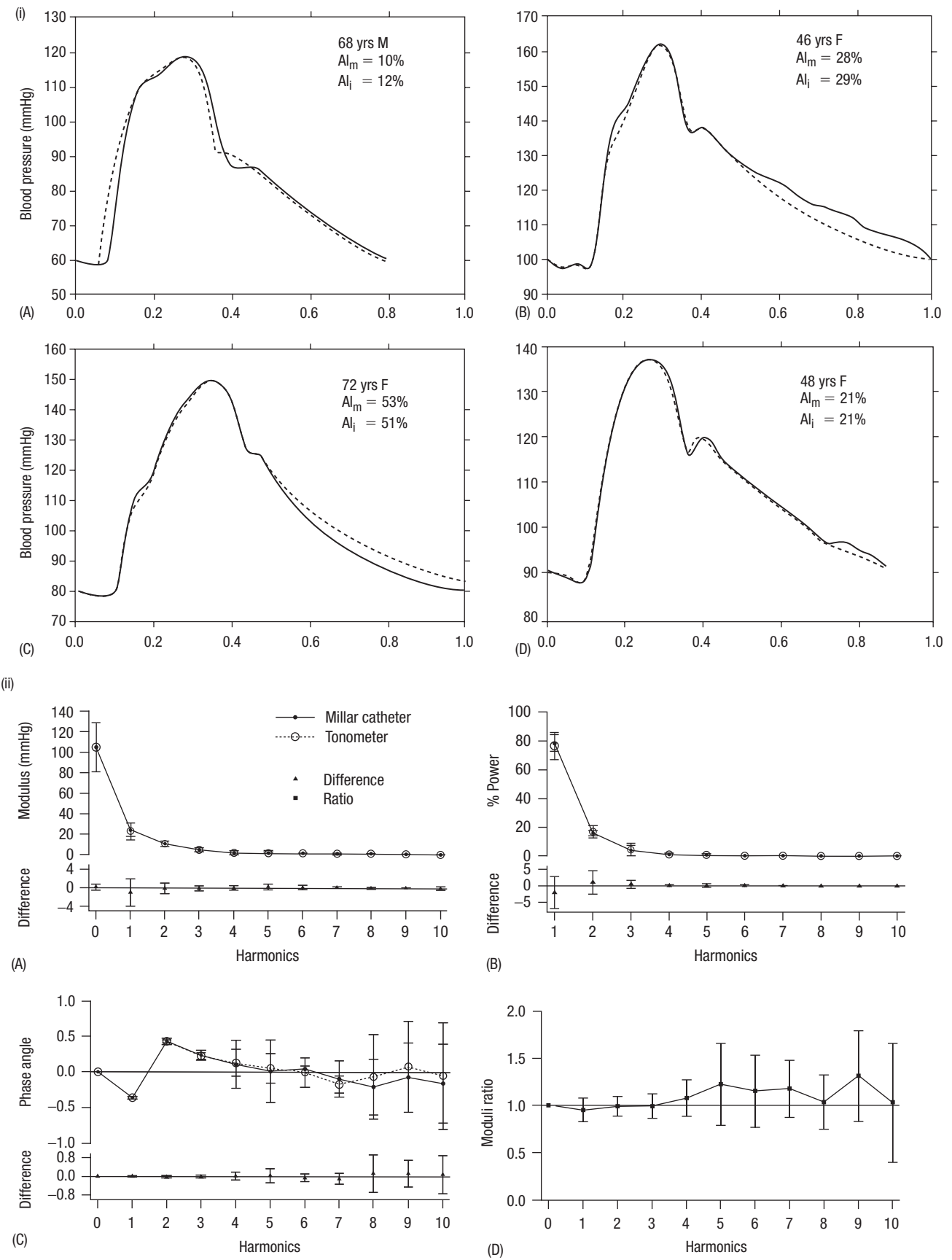
Figure 30.7 Comparison of pressure (systolic and diastolic) in the carotid artery as determined by two different methods. Reproduced from Gallagher (1993)

baroreceptors. Another potential problem is dislodgement of atheromatous material from carotid plaque. A serious problem in calibration of the carotid pulse arises from inappropriate use of applanation to an artery, such as the brachial, whose wall cannot be reliably applanated (Chapter 28) (O'Rourke and Adji, 2010b).

Detailed evaluations of carotid applanation tonometry were performed by Kelly *et al.* (1989b) and by Chen *et al.* (1996). Figure 30.8 shows pressure waves recorded simultaneously at cardiac catheterization from within the carotid artery by Millar micromanometer and from without by a Millar applanation tonometer. There were no significant differences in waveform, contour, nor in harmonic content. Findings are similar to those in Fig. 30.4, and confirm that applanation tonometry can accurately record the intra-arterial pressure waveform. However, both figures have to be regarded as 'best case scenarios', since the intra-arterial waveform was being observed as the tonometer tracings were being obtained. More realistic are data obtained 'blindly' (Fig. 30.9), or in studies of repeatability and reproducibility from different laboratories. Such data from analysis of convolved radial artery pressure waves are given below under Sphygmocardiography (p. 616). We are unaware of such studies of carotid tonometry from its proponents. In our hands, we have not been able to achieve the same reproducibility as for radial tonometry because of problems in achieving stable applanation of the carotid artery, and in avoiding respiratory fluctuations.

Figure 30.8 shows typical carotid artery pressure waveforms as recorded (invasively and non-invasively) in middle-aged or older humans; spectral analysis of these waves are also shown. Figure 10.33 shows such waveforms as recorded in normal subjects at different decades of life, and compares these with corresponding changes in the radial artery. In adults from the fourth decade on (see also Fig. 19.24), the secondary systolic pressure peak is greater than the initial peak in the carotid artery, whereas in the radial artery, the initial pressure peak remains dominant in normal subjects up to the eighth decade of life (Figs 10.33 and 19.24). In the carotid artery, the second systolic pressure peak is always greater than in the radial artery for a given age, but this peak (or shoulder) increases progressively with age at both sites. Aging changes are often expressed in terms of augmentation index, and are shown in Fig. 19.24.

There is now more agreement than in the past on contour of the carotid arterial pressure waveform, but this is not complete. Artifact was common with old air-filled systems and with use of microphones, but still is common with tonometry when the carotid artery is not perfectly applanated, or when the artery is kinked or unstable. Figure 30.10 shows a carotid tonometer tracing presented as typical by Cameron's experienced group (Kingwell and Gatzka, 2002) for the Second Australian High Blood Pressure study. In this, time to the first systolic shoulder was 55 ms and augmentation index was calculated as



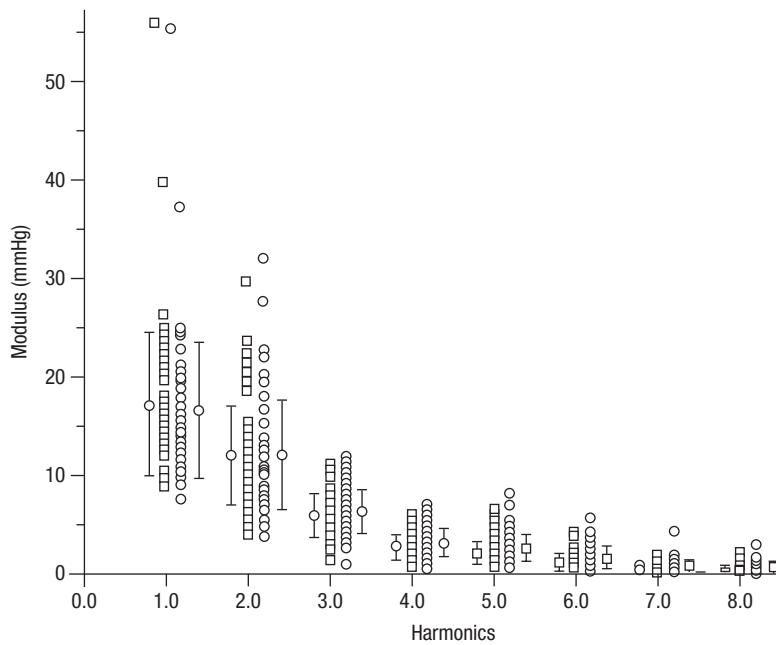


Figure 30.9 Moduli of the first eight harmonics of invasively recorded (open circles) and simultaneously recorded tonometer radial artery pressure waves (squares) in 62 patients. From Kelly *et al.* (1989b)

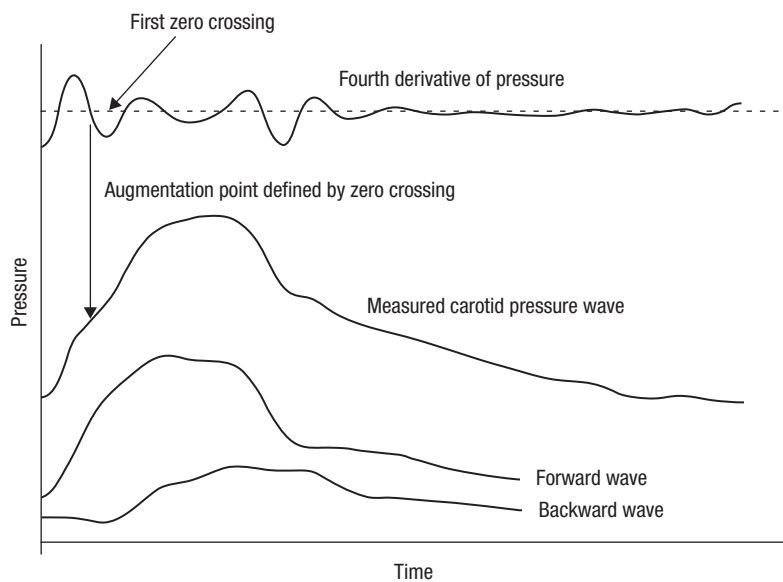


Figure 30.10 Measured carotid pressure waveform and its interpretation. From Kingwell and Gatzka (2002)

62 percent. This is most unusual and probably an artifact (see Fig. 30.44B, p. 637). By way of contrast, Kelly *et al.* (1989c), in a study of 1004 subjects, found time to the first systolic peak to be just over 100 ms, corresponding to the peak of carotid flow, with the average augmentation 30 percent (Figs 10.33 and 19.24). Chen *et al.* (1996) found time to the first systolic peak or shoulder 114 ± 4.0 ms for

the carotid artery and 111 ± 4.0 ms for the aorta, with augmentation less than 50 percent for the carotid in all 66 patients studied. Mitchell *et al.* (2003) calculated T1 as 94–127 ms (from the carotid as surrogate of the aortic pressure wave), whereas Hope *et al.* (2002) from Cameron's group recorded this at 161 ms (mean) from this group for the ascending aorta, using fluid-filled

Figure 30.8 (i) Simultaneously recorded pressure waves in the carotid artery of four different subjects measured by a Millar intra-arterial catheter tip manometer and an externally applied Millar tonometer. Augmentation index for invasive (AI_m) and non-invasive (AI_i) expressed as a percentage of pulse pressure. (ii) Spectral analysis of the simultaneously recorded invasive (micromanometer) and non-invasive (tonometer) carotid wave forms. Error bars represent 95 percent confidence intervals. There were essentially no differences in moduli (A), percentage of power (B) though amplitude of the third and higher harmonics was very small, phase angle (C) and modulus ratio non-invasive/invasive carotid waveforms (D). From Chen *et al.* (1996)

catheter systems. The extreme values (55 and 161 ms) are highly suspect and lead to unusually high, and unusually low, values of pressure augmentation, respectively (see Fig. 30.44B). Figure 30.6 shows the features one needs to see before being confident about the recording of an accurate radial pressure waveform. The same principles apply to the carotid, with the only difference being that the second systolic wave is usually greater than the first in the carotid artery. Need for careful recording of the initial pressure pulse upstroke was stressed by the pioneers of sphygmography for the radial artery; an exaggerated late systolic peak was considered to carry the implication of arteriosclerosis and poor prognosis. Bramwell (1880) wrote forcibly on this, and would have advised that a pulse waveform such as shown in Fig. 30.10 be discarded unless corroborated by repeated measurements, and from the contralateral artery. He wrote (of the radial pulse) 'in some pathological conditions ... aneurysms ... aortic stenosis ... the apex is rounded; but this is so extremely rare that a tracing in which the apices are not pointed should always be regarded as imperfect until verified by repeated and careful readjustments of the instrument'. One can verify by taking recordings from the opposite radial or carotid artery or by checking the radial if the carotid is suspect, or vice versa. Naturally, value of the radial pulse depends on the absence of obstruction upstream; physicians are familiar with this in interpreting brachial cuff pressures.

Kelly *et al.* (1989b) and Chen *et al.* (1996) tested use of carotid tonometry as a surrogate of direct aortic manometry. Chen *et al.* studied 66 patients and showed basic similarities in the waveforms at the two sites under control conditions including time from foot to first systolic shoulder (114 ms for carotid artery; 111 ms for aorta). Maximum rate of rise of pressure was greater for the carotid compared to the aortic tracing, the shoulder was lower on the aortic than the carotid waveform, and carotid augmentation index was accordingly lower ($AI_a = 28$ percent; $AI_c = 14$ percent). Similar results had been shown by Kelly *et al.* (1989b) in this and a later study (Kelly *et al.*, 1989c) which showed carotid augmentation index approximately half of that previously determined for the aorta in their and other institutions (Fig. 19.24). Both Kelly *et al.* (1989b) and Chen *et al.* (1996) examined the effects of nitroglycerin on carotid and aortic augmentation, while Chen *et al.* (1996) examined in addition the effects of handgrip exercise which elevated arterial pressure. Both studies with different interventions showed corresponding changes in carotid and aortic augmentation index.

These studies in Sydney, the United States, Taiwan, Europe, and China, complemented by the extensive studies conducted in Japan by Yaginuma, Takazawa and colleagues, confirm that when appropriately used and with reservations referred to above, carotid artery tonometry is a very useful method for studying arterial hemodynamics. However many, including ourselves, now

prefer radial artery tonometry, with use of the carotid reserved for situations where the radial waveform is suspect or cannot be recorded satisfactorily.

Carotid pressure wave augmentation has been used to determine the favorable effects of vasodilator agents such as nitrates, ACE inhibitors, calcium channel antagonists, and vasodilating β -blocking agents (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; O'Rourke *et al.*, 1989a; Kelly *et al.*, 1992a; Kelly *et al.*, 1992b; O'Rourke *et al.*, 1993a; Chen *et al.*, 1995; Ting *et al.*, 1995a; Ting *et al.*, 1995b; Asmar *et al.*, 2001b; Mitchell *et al.*, 2002; Mitchell *et al.*, 2003; Mitchell *et al.*, 2004a; Yasmin *et al.*, 2004; Mitchell *et al.*, 2005a; Mitchell *et al.*, 2005b; Mitchell *et al.*, 2006; Mitchell *et al.*, 2007a; Mitchell *et al.*, 2007b; Segers *et al.*, 2007b; Mitchell *et al.*, 2008; Mitchell, 2009b; Segers *et al.*, 2009; Mitchell *et al.*, 2010). Carotid pressure augmentation has been related to carotid atherosclerosis, to left ventricular hypertrophy, and to physical performance (Saba *et al.*, 1993; Vaitkevicius *et al.*, 1993; Roman *et al.*, 1995a; Roman *et al.*, 1995b; de Luca *et al.*, 2004). London *et al.* (2001) have related carotid augmentation to outcome in patients with end-stage renal failure.

The most comprehensive studies conducted to verify carotid augmentation (and other indirect indices) have been reported from Japan, notably by Yaginuma and colleagues in Jichi and by Takazawa and colleagues in Tokyo (Yaginuma *et al.*, 1985; Takazawa, 1987; Takazawa *et al.*, 1988; Fujii, 1989; Yabe *et al.*, 1991; Sakai *et al.*, 1992; Sakai *et al.*, 1993; Takazawa *et al.*, 1993a; Takazawa *et al.*, 1993b; Miyashita *et al.*, 1994; Miyashita *et al.*, 1995). Other validations have been reported by Kelly *et al.* (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b) and by Chen *et al.* (Chen *et al.*, 1996). Results are in close agreement, although Takazawa has gone further in comprehensively relating a variety of non-invasive indices to directly recorded aortic and left ventricular pressure waves at cardiac catheterization of patients with different disease conditions, at different ages and with different therapeutic interventions (Yabe *et al.*, 1991; Sakai *et al.*, 1992; Sakai *et al.*, 1993; Takazawa *et al.*, 1993a; Takazawa *et al.*, 1993b; Takazawa *et al.*, 1998).

In individuals greater than 40 years of age, carotid augmentation index underestimates ascending aortic augmentation index by 25–50 percent (see Figs 9.14 and 19.24A), but the relationship is consistent and the latter can reasonably be inferred from the former.

RELATIONSHIP BETWEEN CAROTID AND RADIAL ARTERY PRESSURE WAVEFORMS

The relationship between carotid and radial pressure waveforms has been referred to above, and is apparent in many figures in this book including Figs 10.33, 10.34, 19.24, and 19.30. Amplitude of the pressure pulse is greater in the radial artery than in the carotid artery (Fig. 28.1), and this is due principally to greater absolute

amplitude of the first systolic pressure peak or shoulder (which occurs approximately 110 ms after the wave foot at both sites and corresponds to the peak of flow in the artery). The secondary systolic peak or shoulder is relatively lower in the radial artery than in the carotid, but the actual amplitude of this wave is approximately the same at both sites, as it is also in the ascending aorta (Fig. 28.2). This wave constitutes the reflection of the initial peak from the lower body, and, being of low frequency, it is little amplified in the upper limb. We have taken this as a measure of peak aortic systolic pressure, when measured in the radial artery, and use this as a check against peak aortic pressure determined from sphygmocardiography (see below) (Pauca *et al.*, 2004; Takazawa *et al.*, 2007).

The relationship between radial and carotid pressure waveforms can be expressed in the frequency domain as a transfer function. This (Gallagher, 1993; Gallagher *et al.*, 2004) is shown as determined from 439 normal subjects and patients with a variety of cardiovascular diseases (Fig. 30.11). In Fig. 19.30, the same data are shown and separated into groups above and below 35 years of age. In the whole group, there was considerable scatter, most of which could be attributed to difficulty in accurately recording the carotid pressure waveform by applanation tonometry. For the younger and older subjects, the pattern was similar, with peak of modulus between 1.7 and 2.0 at 4–6 Hz, but there was a definite difference between the two groups, with peak amplification greater by

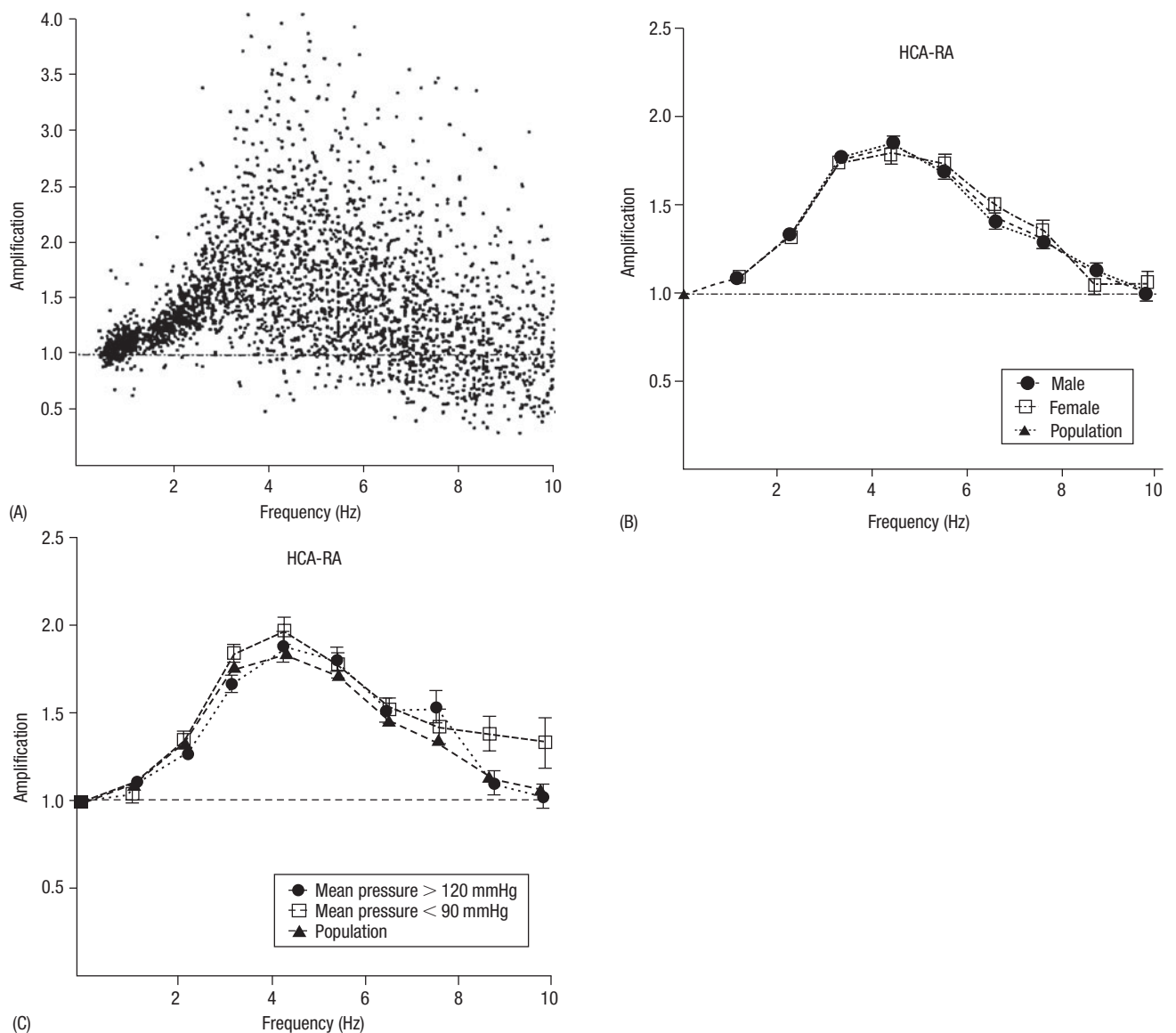


Figure 30.11 (A) Transfer function modulus for carotid-radial artery, determined from 439 simultaneously recorded tonometric carotid and radial artery pressure waves. Age differences in transfer function, and in modulus of the carotid pressure waveforms are shown in Fig. 19.28. (B) and (C) show the same data with mean $\pm$ SEM for males compared to females, and hypertensive people compared to normotensives. From Gallagher (1993) and Gallagher *et al.* (2004)

approximately 17 percent in the younger group (Fig. 19.30). No definite differences were found for gender or arterial pressure (Fig. 30.11B,C). While a difference of 17 percent may seem substantial, it applies to a harmonic component of the waveform which is less than 2 mmHg, and so indicates a difference of around 0.4 mmHg. Potentially more important are lesser differences in modulus of the transfer function at lower frequencies where pressure harmonic components are greater (see Chapter 11).

Systolic time intervals are traditionally recorded from the carotid pressure pulse waveform (Weissler *et al.*, 1963; Weissler *et al.*, 1968; Boudoulas, 1990). We have determined ejection duration measured from the carotid and radial waveforms in 215 patients and normal subjects, using an automatic detection algorithm (Hirata *et al.*, 2001). Correspondence was excellent with mean value 325 ms, difference 4.6, and standard deviation 9.6 ms. Such a finding supports the view, previously proposed, that ejection duration can be determined from the radial as well as from the carotid and aortic pressure pulse. Our only caveat is that the method may be unreliable under conditions (e.g. tachycardia) where the incisura cannot be distinguished from the foot of the reflected diastolic pressure wave. (Similar problems arise in older patients in separating the peak of flow from the foot of the reflected wave in measurement of augmentation pressure.

RELATIONSHIP BETWEEN CAROTID AND AORTIC PRESSURE WAVEFORMS

Differences and similarities between aortic and carotid pressure waves are discussed above. Basic contour is similar, but carotid augmentation is less than in the aorta and amplitude of the pulse is greater. The relationship is best demonstrated in the frequency domain as a transfer function. Figure 30.12 shows results from our group

compared to those of Chen *et al.* (1996). These are basically identical up to the sixth harmonic, beyond which amplitude of pressure harmonics is minuscule. Both show amplification to approximately 1.5 at 4.0 Hz, with apparent fall at higher frequencies.

RELATIONSHIP BETWEEN AORTIC AND RADIAL ARTERY PRESSURE WAVEFORMS

Throughout this book, tracings are shown of pressure waves in the ascending aorta and radial artery at rest (Chapter 10), during exercise (Chapter 27), and with drug therapy (Chapters 15, 20, and 26). Also shown are changes with age in the ascending aorta and radial artery (Chapter 19). We collected these ourselves and from the literature over many years and as we did, became intrigued with the consistent differences that were seen with these under different conditions, when displayed in the time or in the frequency domain (Fig. 30.13). We came to wonder if there was not a consistent relationship between aortic and radial waveforms that could be exploited in order to generate the aortic pressure pulse from the radial pressure pulse. The strongest evidence in this respect came from our studies in the cardiac catheterization laboratory during 1989 (Kelly *et al.*, 1989a; Kelly *et al.*, 1989b; Kelly *et al.*, 1989c; Kelly *et al.*, 1989d; Kelly *et al.*, 1989e; Kelly *et al.*, 1989f).

We were aware of studies which determined transfer functions from pressure waves recorded simultaneously in the aorta and radial artery and elsewhere (O'Rourke, 1970a; O'Rourke, 1970b; Lasance *et al.*, 1976), but appreciated that such transfer functions would not be clinically useful unless they remained stable or were predictable under different conditions. Our enthusiasm was raised by our studies with nitrates and other vasodilators, assessed by applanation tonometry. These showed impressive but consistent and predictable changes in both carotid and radial artery waveforms (Fig. 30.14)

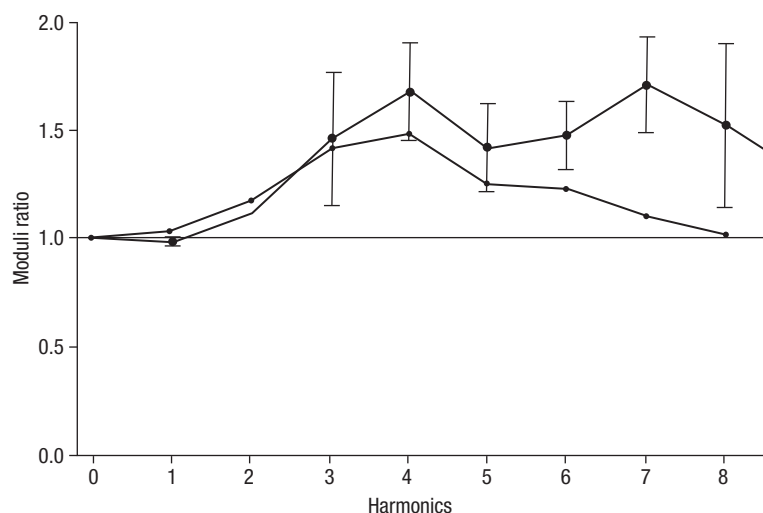


Figure 30.12 Transfer function modulus between ascending aorta and carotid artery as determined by Chen *et al.* (1996); (large circles and error bars which represent 95 percent confidence limits), and by ourselves (Nichols and O'Rourke, 1998)

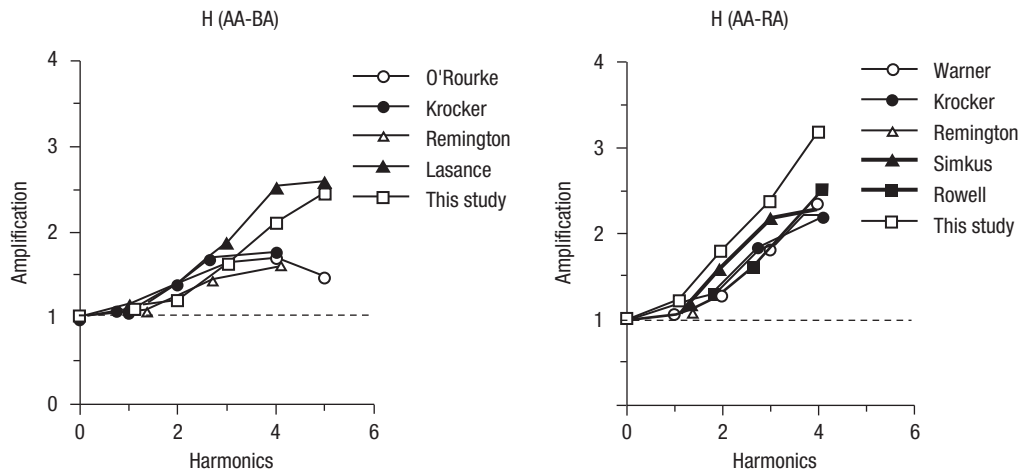


Figure 30.13 Transfer functions calculated for wave pairs recorded in the proximal aorta and brachial artery (left) and between proximal aorta and radial artery (right), from data published in the literature. Wave pairs were subject to Fourier analysis with corresponding harmonics compared. In all cases, transfer function deviated substantially from unity above the first harmonic. Reproduced from Karamanoglu *et al.* (1993)

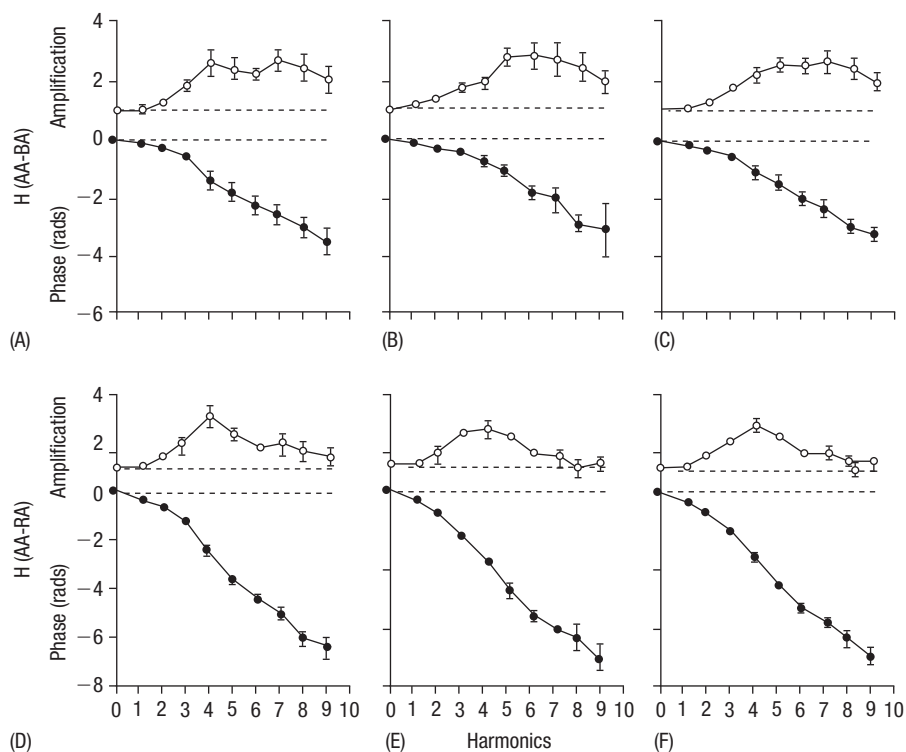


Figure 30.14 Modulus (above) and phase (below) of transfer function for pressure between ascending aorta and brachial artery (upper panels A, B, and C), and between ascending aorta and radial artery (lower panels D, E, and F). Figures A and D relate to transfer function under control conditions, B and E after administration of nitroglycerin 0.3 mg sublingually, and C and F composite data for control conditions and after nitroglycerin administration. Reproduced from Karamanoglu *et al.* (1993)

(O'Rourke *et al.*, 1989a), and suggested that transfer functions may be sufficiently stable under different conditions to be approximated by a generalized transfer function that could be used under all conditions. We were well aware of the huge stiffness changes which occur with aging, hypertension, and disease in the aorta and elastic

arteries (Chapters 19, 20, 24, and 25), but knew that such changes were trivial in the muscular arteries in the upper limbs (Figs 19.13, 19.14, and 19.15).

The catheterization laboratory data we collected in 1989 were published in 1993 (Karamanoglu *et al.*, 1993), and showed transfer functions between the aorta and

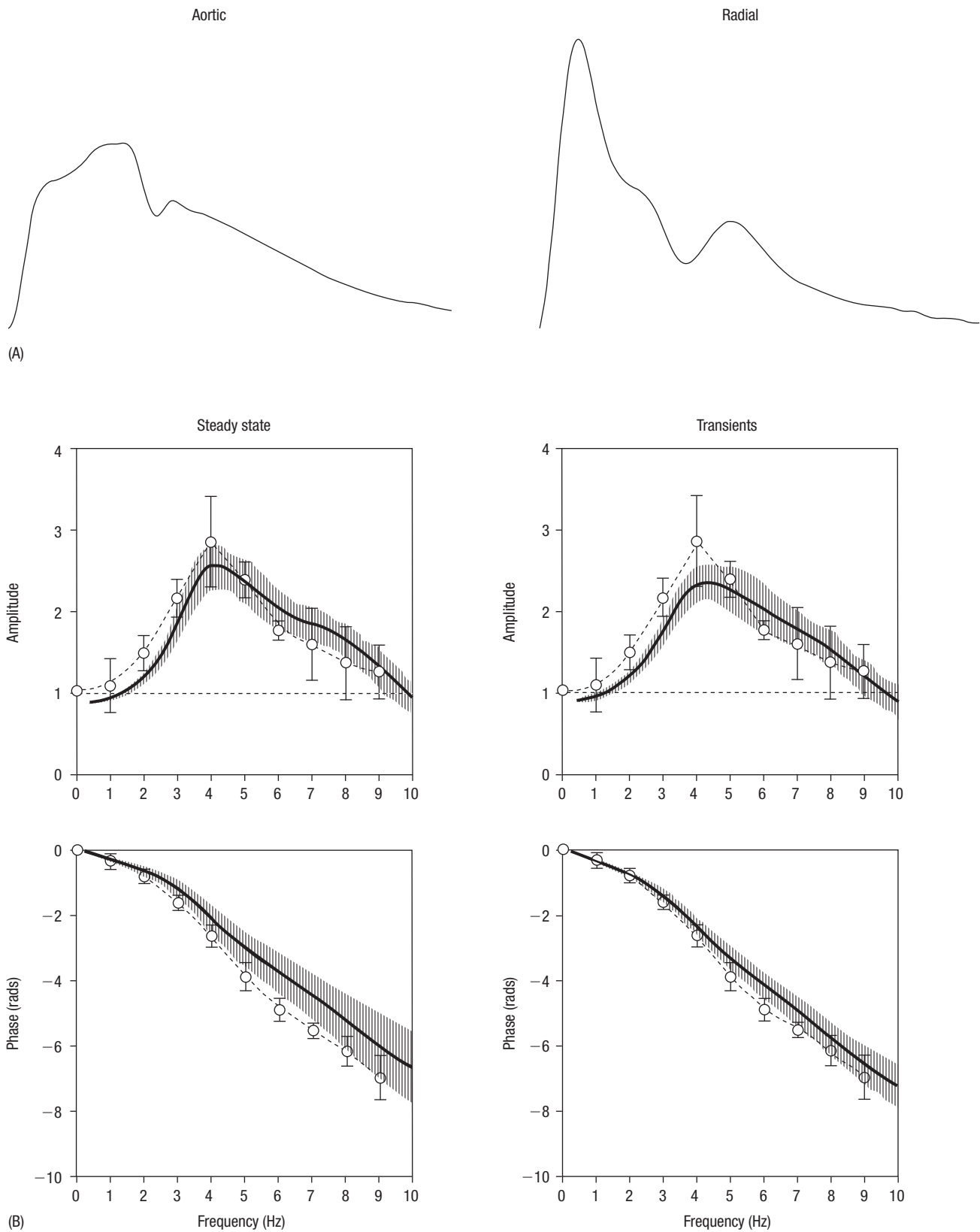


Figure 30.15 (A) Difference between ascending aortic and radial pressure waveforms under control conditions. (B) Spectral plots of generalized transfer functions (shaded area defines 95 percent confidence intervals) generated from the steady state ($n=20$) or hemodynamic transient ($n=14$) data. Generalized transfer function harmonics of Karamanoglu *et al.* (1993) are denoted by open circles with error bars showing 95 percent confidence intervals. Reproduced from Chen *et al.* (1997)

brachial artery and between the aorta and radial artery which were consistent before and after administration of glyceryl trinitrate. The control data were similar to those collected by Lasance *et al.* (1976) in 68 patients and by ourselves in 1970 for 29 patients (O'Rourke, 1970b). We then proposed (O'Rourke *et al.*, 1993a) that this could provide a method for generating the ascending aortic pressure wave non-invasively and for studying drug action. Initial response was extremely skeptical as it should have been, and was only dispelled by two studies from Chen *et al.* (1997) and Fetics *et al.* (1999) in Baltimore (Figs 30.15, page 608 and 30.16). These supported the concept we had proposed, but expressed some concern in identifying features of the aortic waveform if high frequency components of the pulse (above 6.0 Hz) had been attenuated.

Subsequent studies have provided further support for validity of using a generalized transfer function to characterize the relationship between ascending aortic and radial artery pressure waveforms under different conditions. Figure 30.17 (page 610) shows the effects of combining the Gallagher carotid–radial transfer function obtained from 439 subjects, with that we had determined for carotid–ascending aorta. This corresponds almost exactly with that determined by our group and by Chen *et al.* (1997) and Fetics *et al.* (1999) (Figs 30.14 and 30.15). We had also obtained data collected in Sweden by Soderstrom *et al.* (2002) and in Winston-Salem by Pauca *et al.* (2001). All these data are combined in Fig. 30.18 (page 610) together with that from Sydney and Baltimore. Figs 30.19 (page 611), 30.20 (page 611), and 30.21 (page 612) show changes seen with interventions in the different

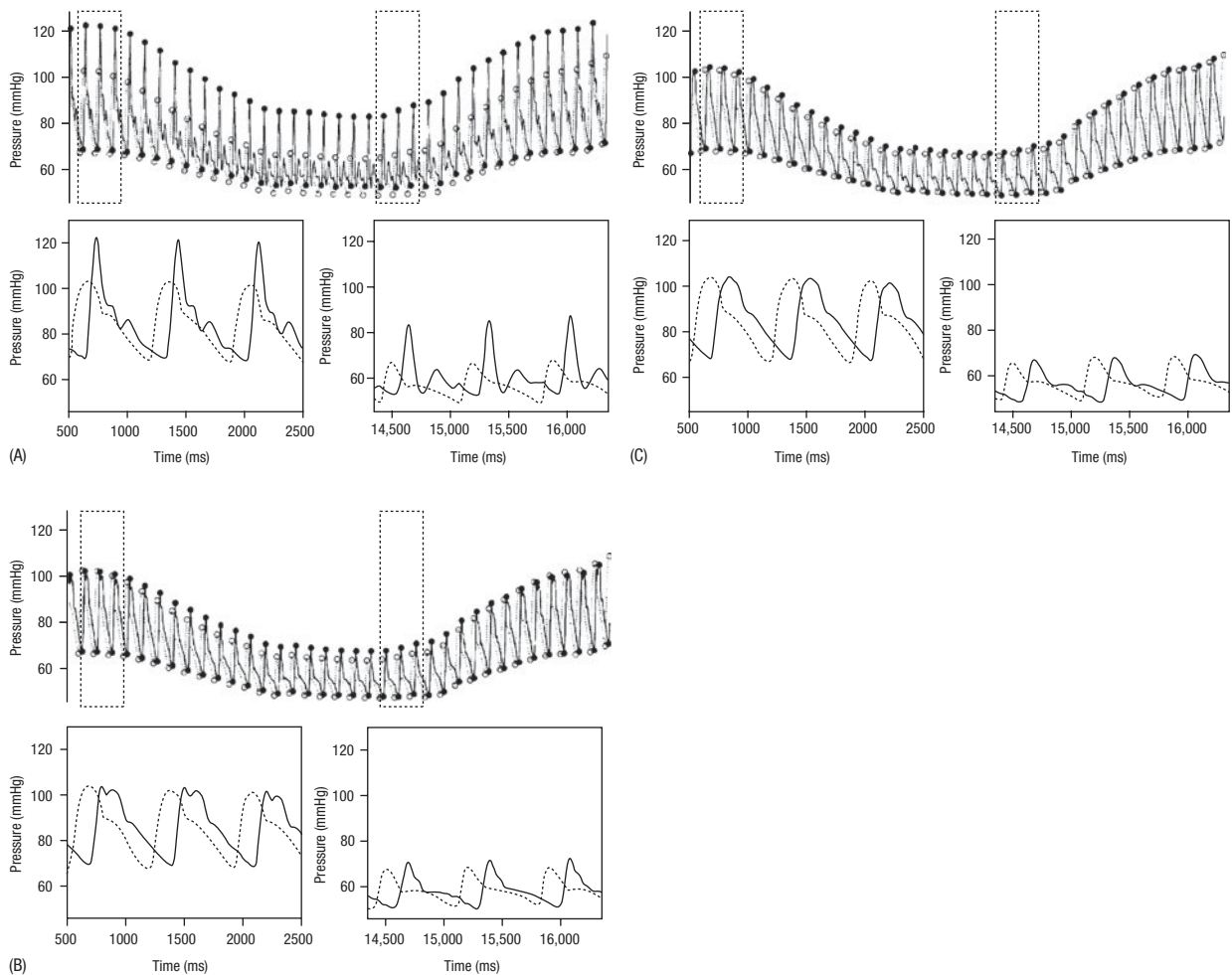


Figure 30.16 (A) Pressure waveforms from the ascending aorta (dotted line) and radial artery (solid line) during the course of a Valsalva maneuver. Systolic and diastolic extremes of the waves are shown in closed circles for radial artery, open circles for ascending aorta. Inset below: control period (left) and peak Valsalva strain (right). (B) Recordings as for (A) but with generalized transfer function used to synthesize the ascending aortic waveform from the radial artery waves. (C) Recordings as for (A) but with individual transfer function used to synthesize the ascending aortic waveform from the radial artery waves. Use of the individual transfer function (determined for that patient) was only marginally superior to use of the generalized transfer function in correcting for the large difference between systolic pressure in the aorta and radial artery before and during the Valsalva maneuver. Reproduced from Chen *et al.* (1997)

studies. Use of vasodilator agents did cause a fall in the peak of transfer function modulus, but little change in its initial rise. Since most of the energy (or variance) of the pressure wave is concentrated in the first two to three harmonics (Nichols, 1970) (Fig. 11.13 and Table 30.1, page 613) such (definite) changes were expected to have little effect on the amplitude or contour of the ascending

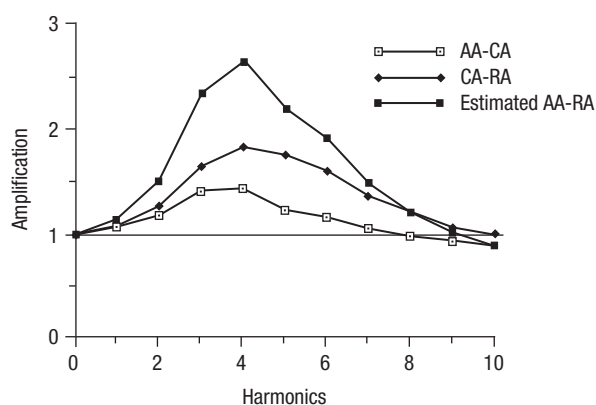


Figure 30.17 Transfer functions of pressure waves recorded by applanation tonometry in the carotid artery and radial artery. The top tracing shows the same data adjusted for transfer function between ascending aorta and carotid artery. Reproduced from Gallagher (1993)

aortic wave synthesized from a generalized transfer function.

Despite the compelling clinical data, we believed that more confirmation was necessary from the type of modeling studies that we had conducted previously with Taylor (Taylor, 1965; Taylor, 1966a; Taylor 1966b), Avolio (1980), and O'Rourke and Avolio (1980) in studies of vascular impedance. Karamanoglu *et al.* (1995), using a modification of Avolio's upper limb vascular model, showed the same transfer function features that we had determined clinically (Fig. 30.22A, page 613). Furthermore, they went on to show that the type of changes described in the literature for aging, hypertension, and drug therapy would have little effect on the transfer function at frequencies up to 5.0 Hz – just the relatively minor changes that are described above (Fig. 30.19, page 611). This modeling work did show that marked changes can occur in a transfer function that would preclude use of a generalized transfer function (Fig. 30.22B, page 613), but these entailed major changes in vascular properties, particularly peripheral resistance such as might follow intra-arterial injection of vasodilator agents or reactive hyperemia. Similar modeling information was published by Stergiopoulos *et al.* (1998).

Validation studies of the transfer function process for synthesizing the aortic from the radial artery pressure waveform have been published by Fetics *et al.* (1999),

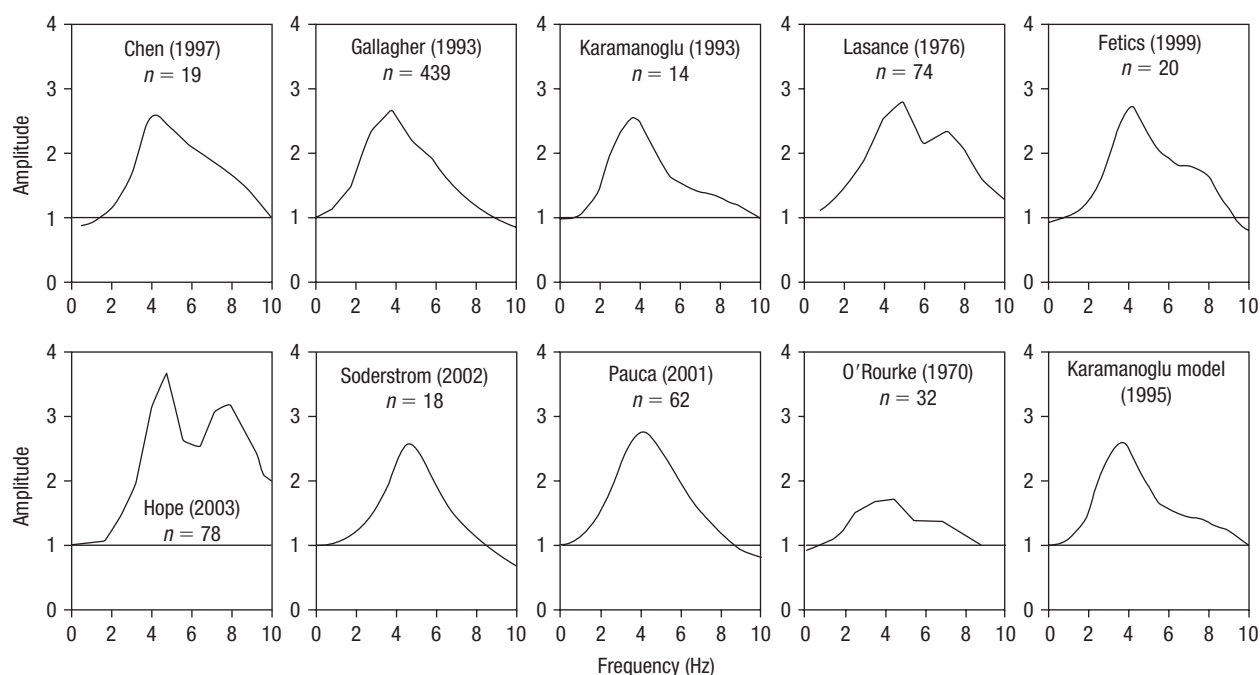


Figure 30.18 Composite figure showing transfer function modulus from Karamanoglu *et al.* (1993), and Gallagher (1993) (in Fig. 30.17), and from data provided by Soderstrom *et al.* (2002) and Pauca *et al.* (2001). Superimposed is transfer function modulus calculated by Karamanoglu *et al.* (1995) in a model of the human upper limb. Low values in O'Rourke (1970b) are attributable to brachial pressure being measured near the axilla, while high and unstable values over 4 Hz in Hope *et al.* (2003) are attributable to artifact created by the uncalibrated, fluid-filled catheter/manometer system (O'Rourke *et al.*, 2004a). After Gallagher *et al.* (2004)

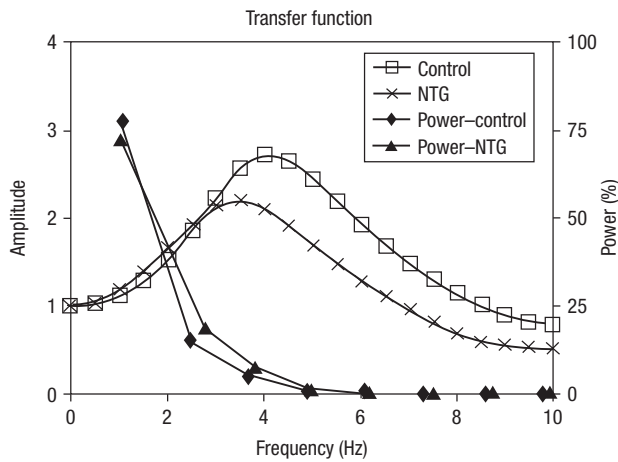


Figure 30.19 Analysis of a data set of 61 simultaneously recorded ascending aortic and radial pressure waveforms before (control) and during intravenous infusion of nitroglycerin (NTG). Transfer function amplitude at baseline (control) and during infusion of NTG is given on the left vertical axis. Energy or power content in (percent) at each harmonic frequency up to 10 Hz is given on the right axis. There was close correspondence of transfer function over the frequency range that contained most of the energy of the pressure waveforms (0–3.3 Hz); more than 98 percent of the energy was contained over this frequency range. Over the frequency range where deviation of transfer function was considerable (3.3–10 Hz), amplitude of pressure components was very low with less than 2.0 percent of the energy contained over these frequencies. Data collected by A Pauca and N Kon, analyzed by A Qasem and M O'Rourke

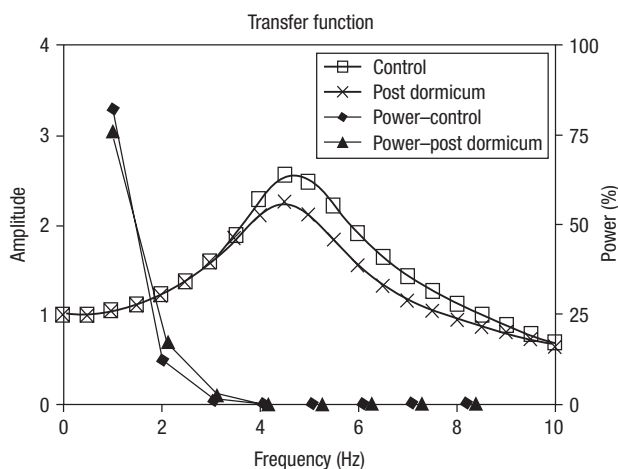


Figure 30.20 Analysis of a data set of 12 simultaneously recorded ascending aortic and radial pressure waveforms before (control) and after (post doricum) a bolus dose of midazolam 1.0–2.0 mg followed by sublingual nitroglycerin 0.25–0.5 mg sublingual. As in Fig. 26.19, there was little difference in transfer function up to 3.0 Hz. Amplitude of pressure harmonics was very low over 3.0 Hz where the transfer function curves diverged. Data collected by S Soderstrom, J Sellgren, J Ponten, G Nyberg, analyzed by A Qasem and M O'Rourke

Takazawa *et al.* (1996), Soderstrom *et al.* (2002), and Pauca *et al.* (2001). The first two used Millar catheter-tip manometer (aorta) and tonometer (radial artery), the third a micromanometer aortic catheter and fluid-filled radial manometer system, and the fourth, two matched fluid-filled systems with frequency response documented at each use. All showed good correspondence between directly measured aortic pressure wave contour and aortic contour calculated from the simultaneously recorded radial pressure waveform, using a generalized transfer function. Correspondence persisted during interventions with drugs. The last two studies (Soderstrom and Pauca) were conducted by anesthetists who are well aware of the need for meticulous manometry. A surprise in the Soderstrom results was that calculated aortic systolic and pulse pressure were consistently lower than actually measured in the aorta, despite similarity of the measured and synthesized aortic waveforms. In this study at baseline, there was however no significant difference between systolic and pulse pressure in aorta and radial artery – rather than the radial artery pressures being slightly higher. A systematic difference in calibration of the two manometer systems was suspected, but could not be confirmed. The Pauca study (Pauca *et al.*, 2001) was the largest (62 patients) and was conducted prospectively. High-fidelity fluid-filled manometer systems were utilized, and these were matched before each use, with tracings from the radial artery confirmed to overlie and with harmonic moduli within 2.0 mmHg (Fig. 30.23, page 614). There was good correspondence between calculated and measured aortic pressures – mean, diastolic, systolic, pulse, and end-diastolic – whereas there were substantial differences between recorded aortic and radial systolic, pulse and end-diastolic pressures (Fig. 30.23, page 614). Correspondence of all calculated aortic pressures was within AAMI SP10 criteria (White *et al.*, 1993) for 'substantial equivalence', with mean offset less than 1.0 mmHg and s.d. less than 5.0 mmHg for all (Table 30.2, page 615). Correspondence was maintained with change in heart rate and rhythm (Fig. 30.24, page 615). The US Food and Drug Administration (FDA) has accepted this method as used in the SphygmoCor Mx and Px systems for measurement of ascending aortic pressure waveforms and indices derived therefrom as K012487 and K002742.

Applicability of a generalized radial–aortic transfer function for generating aortic pressure waveforms strains credibility, and it is appropriate that the concept be challenged and debated. Such has occurred, and considerable skepticism expressed, notably from people who rely on accuracy of fluid-filled manometers with long tubes and multiple taps, as used at diagnostic cardiac catheterization (Hope *et al.*, 2002; Hope *et al.*, 2007). The controversy that arose following Marey's introduction of the sphygmogram has been reignited. William Harvey's words of moderation (Harvey, 1649) are appropriate: these urged his critic Jean Riolan and all observers to steer a path between extreme mistrust

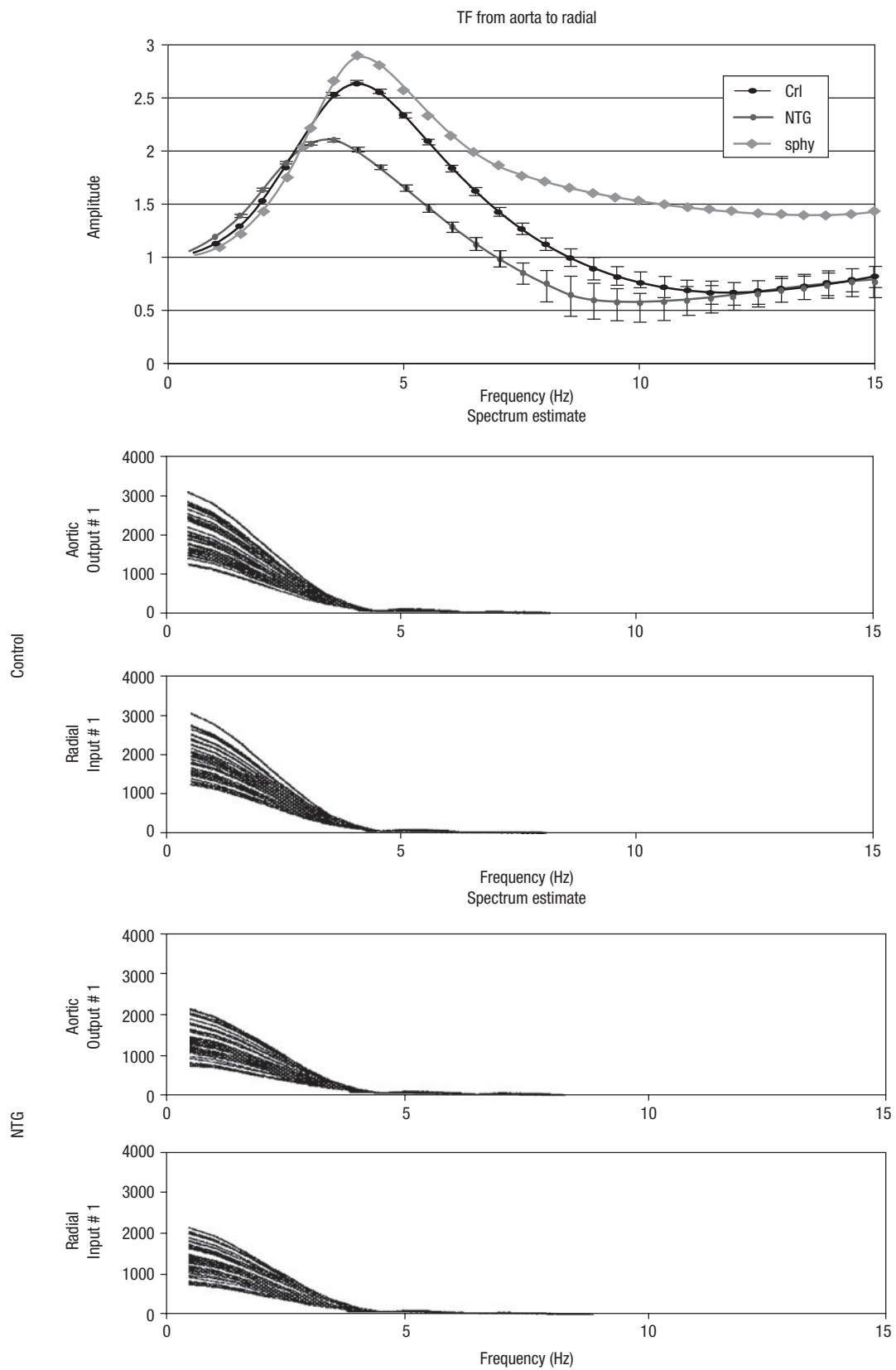


Figure 30.21 Comparison of transfer function as used in SphygmoCor with that determined from the data set shown in Fig. 26.19 under control conditions, and after nitroglycerin with average values and SEM graphed. Below: individual radial and aortic pressure wave power under control conditions and after nitroglycerin. Data collected by A Pauca and N Kon; analyzed by A Qasem and M O'Rourke

Table 30.1 Energy of pressure wave forms

Harmonic	Modulus (mmHg)	Power (%)	Cumulative power (%)
1	17.5 ± 7.3 vs 16.9 ± 7.0 $P=0.02$, $r=0.97$	58.1 ± 15.8 vs 56.0 ± 15.9 $P=0.03$, $r=0.89$	58.1 ± 15.8 vs 56.0 ± 15.9 $P=0.03$, $r=0.89$
2	12.3 ± 5.1 vs 12.4 ± 5.5 $P=0.41$, $r=0.98$	29.2 ± 13.7 vs 29.7 ± 13.4 $P=0.37$, $r=0.95$	87.3 ± 8.5 vs 85.5 ± 8.0 $P=0.01$, $r=0.79$
3	6.1 ± 2.3 vs 6.4 ± 2.3 $P=0.04$, $r=0.90$	8.9 ± 6.7 vs 9.3 ± 6.0 $P=0.37$, $r=0.87$	96.2 ± 3.0 vs 95.0 ± 3.6 $P=0.001$, $r=0.67$
4	2.8 ± 1.2 vs 3.2 ± 1.5 $P=0.002$, $r=0.70$	2.0 ± 1.9 vs 2.6 ± 2.3 $P=0.007$, $r=0.72$	98.2 ± 1.6 vs 97.5 ± 2.0 $P=0.001$, $r=0.68$
5	2.1 ± 1.2 vs 2.5 ± 1.4 $P=0.003$, $r=0.73$	1.1 ± 1.1 vs 1.5 ± 1.3 $P=0.002$, $r=0.74$	99.4 ± 0.7 vs 99.1 ± 0.8 $P=0.001$, $r=0.61$
6	1.2 ± 0.8 vs 1.8 ± 1.0 $P<0.001$, $r=0.77$	0.4 ± 0.4 vs 0.6 ± 0.5 $P<0.001$, $r=0.69$	99.8 ± 0.3 vs 99.7 ± 0.4 $P=0.02$, $r=0.45$
7	0.6 ± 0.4 vs 0.8 ± 0.6 $P=0.005$, $r=0.49$	0.1 ± 0.1 vs 0.2 ± 0.2 $P=0.002$, $r=0.54$	99.9 ± 0.2 vs 99.8 ± 0.2 $P=0.10$, $r=0.42$
8	0.4 ± 0.4 vs 0.6 ± 0.5 $P=0.03$, $r=0.59$	0.10 ± 0.1 vs 0.10 ± 0.1 $P=0.06$, $r=0.53$	99.9 ± 0.1 vs 99.9 ± 0.1 $P=0.5$, $r=0.21$

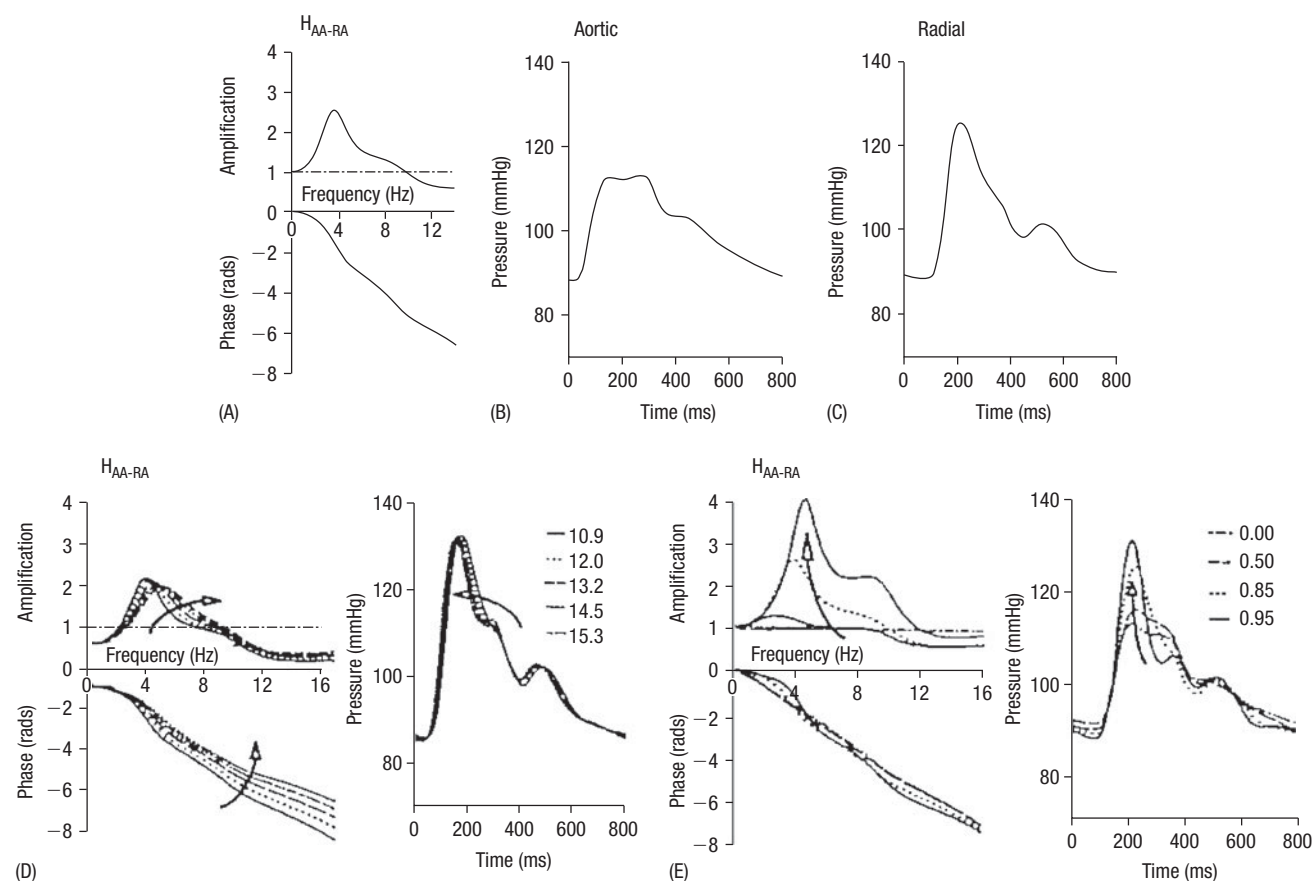


Figure 30.22 (A) Transfer function determined for a realistic multibranched model of the human upper limb. (B) Aortic pressure waveform, and (C) radial artery pressure waveform generated from this waveform using the model. Relatively minor changes in transfer function and synthesized radial waveform result when Young's modulus of the arterial wall is changed over a 40 percent range (D), whereas change in peripheral wave reflection from 0.85 to zero results in substantial change to transfer function and synthesized radial waveform (E). From Karamanoglu *et al.* (1995)

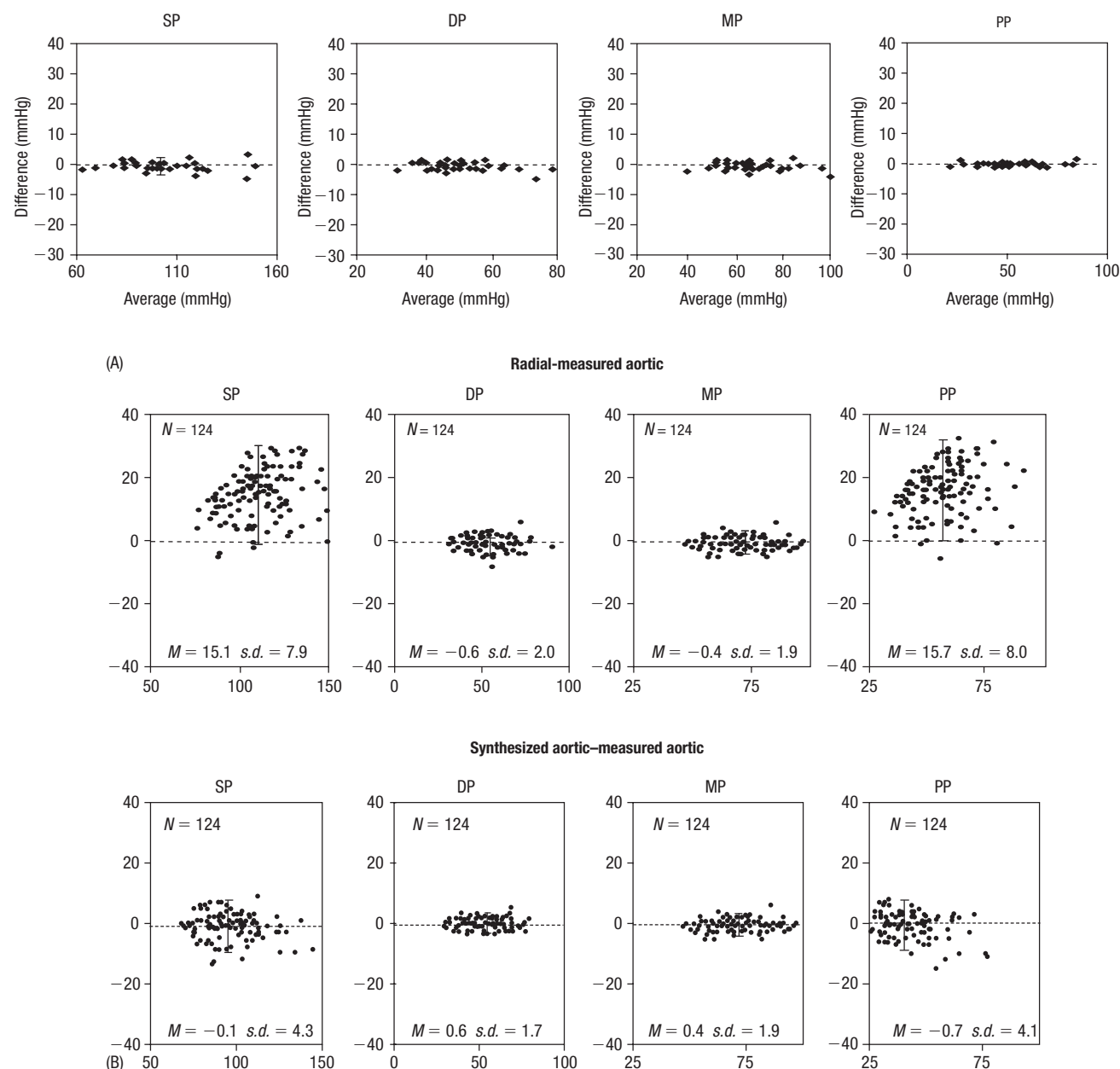


Figure 30.23 Validation of the transfer function process for determining aortic systolic, diastolic, mean and pulse pressure at cardiac surgery in 62 patients with data taken before and after intravenous nitroglycerin infusion. Top: Matching of manometer systems prior to collection of aortic and radial artery data. Center: Relationship between measured radial and measured aortic systolic, diastolic, mean and pulse pressure. Bottom: Relationship between aortic pressure (synthesized from radial) and measured aortic systolic diastolic, mean and pulse pressure. From Pauca *et al.* (2001)

and over-enthusiastic acceptance of any new concept (Chapter 1).

As discussed above, we were led to the possibility of using a generalized upper limb transfer function by the consistent differences in contour of radial and aortic pressure waveforms under different conditions. We had been through a greater controversy in the application of transfer functions to describe pressure/flow relationships as impedance where linearity was assumed but non-linearities were known to be present. This was resolved

by studies from multiple sites, following the proposal by Womersley, McDonald, and Taylor in the 1950s that a linearized approach was reasonable (to a first approximation) (McDonald and Taylor, 1959). We were also impressed by the multiple studies that had shown little change in upper limb and muscular arteries in man with age, hypertension, and drug therapy (see Chapters 19, 20, and 26).

Presented above are data to support our case on generation of the aortic pressure waveform from the radial

Table 30.2 Comparisons of pressure from actual and estimated waveforms

Parameter	Control			Change		
	Mean	Mean diff	s.d.	Mean	Mean diff	s.d.
Synthesized aorta – measured aorta						
SP	105.3	0.0	4.4	84.7	20.2	4.3
DP	59.7	0.6	1.7	50.6	0.6	1.7
MP	78.4	20.5	2.0	63.9	20.4	1.8
PP	45.6	20.7	4.2	34.1	20.8	4.1
ESP	88.1	23.2	3.1	69.6	24.4	4.0
AP	11.0	23.6	7.3	3.1	2.1	7.1
Radial – measured aorta						
SP	121.1	15.7	8.4	99.3	14.5	7.3
DP	58.8	20.5	2.0	49.6	20.7	2.0
MP	78.4	20.5	2.0	63.9	20.4	1.8
PP	62.2	16.3	8.5	49.6	15.1	7.3
ESP	76.7	213.0	2.8	61.4	210.4	2.8
AP	20.7	231.4	14.2	29.5	228.1	17.3

AP, augmentation pressure; DP, diastolic pressure; ESP, end-systolic pressure; MP, mean pressure; PP, pulse pressure; SP, systolic pressure. Results are shown for 62 patients separately for control conditions and during NTG infusion (change).

From Pauca *et al.* (2001).

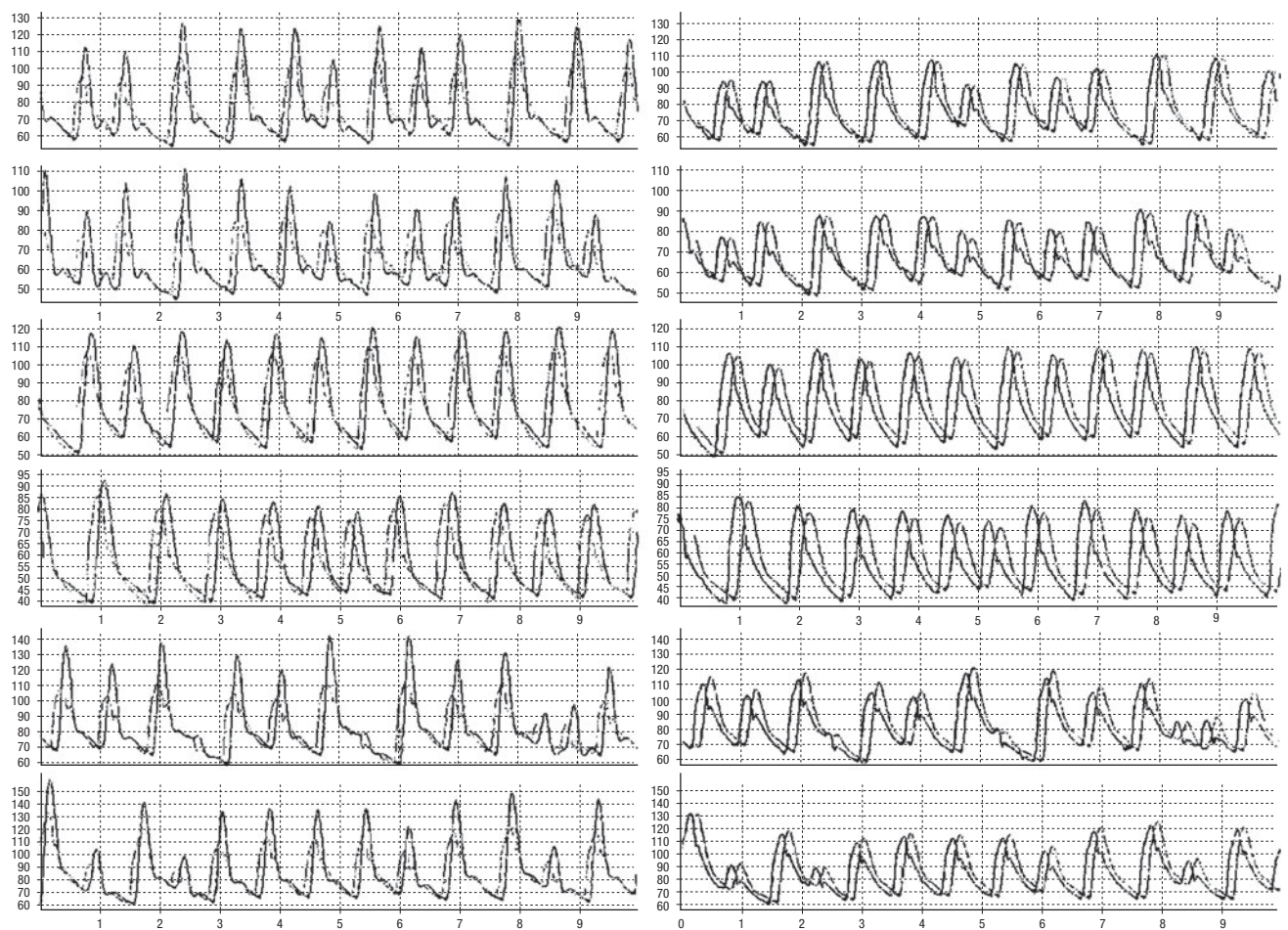


Figure 30.24 Continuous pressure wave tracings in three patients with irregular heart rate. Tracings at left are simultaneously recorded radial and aortic pressures; at right, measured and synthesized aortic pressures (same wave train as at left). From top to bottom; patient 1, control; patient 1, nitroglycerin; patient 2, control; patient 2, nitroglycerin; patient 3, control; and patient 3, nitroglycerin. From Pauca *et al.* (2001)

waveform. There are contrary views. Not all have shown evidence for a generalized transfer function. Many call for more confirmatory data and some suggest that transfer functions be individualized for different patients. This is not a practicable proposition because it would entail direct measurement of aortic pressure in each patient. If the carotid pulse were used as a surrogate of the aorta, one would be at the mercy of all the problems in accurate measurement of the carotid waveform (see above and Fig. 30.10). Chen *et al.* (1997) explored this issue by using a generalized or an individualized transfer function for generation of the aortic pressure waveform from the radial wave. They noted from their series of 20 patients that the generalized transfer function yielded a 96 percent improvement in the fit between measured and estimated central aortic pressure waveform ($P < 0.001$), while the individualized transfer function improved the fit by just another 3.0 percent. This point was emphasized by Kass *et al.* (1998) in a response to criticism on the issue. The data we have presented are from workers who are well versed in the need for meticulous manometry. Even Marey's initial work (Fig. 30.2) showed how his instrument's dynamic accuracy could be calculated. Most modern cardiologists are unaware of instrumental artifact and implicitly trust the accuracy of the pressure waveform data they collect. The modern long fluid-filled manometer systems with multiple connections and entrapped bubbles, and the filtering systems, in all conventional catheter laboratory systems, can cause such marked distortion as to render meaningless comparisons of waveform harmonics whose amplitude may be just 1.0–2.0 mmHg. We are unwilling to consider such data seriously unless frequency response of fluid-filled systems is determined for each system at the beginning and end of the measuring session, as advised in standard texts and reviews (Lambermont *et al.*, 1998; Pepine *et al.*, 1998; Baim and Grossman, 2000).

SPHYGMOCARDIOGRAPHY

The term 'sphygmocardiography' refers to study of the dynamic interaction of the left ventricle and systemic arterial system through analysis of the arterial pressure waveform. The ultimate step involves derivation of the ascending aortic pressure pulse waveform from recordings of peripheral pulse waveforms, with generation of indices which are relevant to left ventricular function. It is an extension of sphygmography (as introduced by Marey in 1860) to cardiac function.

Problems with use of carotid pulse waveform analysis have led to re-evaluation of the radial pulse and to consideration of the possibility that this might be useful for the assessment of central aortic pressure and of indices linking the left ventricle and peripheral circulation. As discussed, such hopes were raised in the late 1980s when we realized that vasodilator drugs, such as nitroglycerin,

appeared to cause similar changes to the late systolic pressure wave in the radial artery as in the simultaneously recorded carotid artery (Fig. 10.27) (O'Rourke *et al.*, 1989a; O'Rourke *et al.*, 1989b). Subsequent invasive studies confirmed that the reduction in wave reflection by nitroglycerin in the ascending aortic pulse tracing could be gauged from a similar change in height of the secondary systolic wave in the brachial artery (see Fig. 14.21) (Kelly *et al.*, 1990a; Takazawa *et al.*, 1995b; Adji and O'Rourke, 2004b; O'Rourke and Pauca, 2004). These findings suggested that changes in central arterial hemodynamics with vasodilator agents could indeed be determined from the contour of the arterial pressure wave in the brachial or radial artery using a generalized transfer function. In pursuing this idea during the late 1980s, we were also buoyed by the knowledge that the advances in sphygmography 100 years ago were based on recordings of the radial pulse in the upper limb (Marey, 1860; Marey, 1863; Gull, 1872; Mahomed, 1872; Mahomed, 1874; Broadbent, 1890; Mackenzie, 1917; Snellen, 1980; O'Rourke, 1992; Swales, 1996a) and by the recognition that the radial artery pulse has been used since antiquity in the diagnosis of disease, as epitomized by Galen's voluminous tomes. We were aware, of course, that clinicians, almost without exception, ignore amplification of the arterial pressure wave between central and peripheral arteries when interpreting brachial arterial sphygmographic recordings, and so take systolic pressure (and diastolic pressure) to be identical throughout the arterial system and in the ascending aorta. We were aware, too, that there are very substantial differences in systolic pressure between peripheral (brachial and radial) and central (aortic and carotid) arteries under different conditions, with systematic differences of 10–25 mmHg noted during the use of vasodilator agents (Kelly *et al.*, 1990a; Simkus and Fitchett, 1990; Takazawa *et al.*, 1993b) and up to 80 mmHg during exercise (see Fig. 10.24) (Rowell *et al.*, 1968). We perceived many advances as flowing from more accurate assessment of brachial and radial artery pressure waves and considered that it was possible to modify the conventional interpretation of upper limb pulses (O'Rourke *et al.*, 1989a; Kelly *et al.*, 1990a).

Consistency of transfer function from ascending aorta to radial artery opens up the possibility of synthesizing the ascending aortic pressure wave from the radial artery pressure wave, determining specific left ventricular properties (such as ejection duration and heart rate), separating the aortic wave into systolic and diastolic periods, and determining indices of ventricular–vascular interaction. We have developed a software process for doing this (O'Rourke *et al.*, 1992c; O'Rourke *et al.*, 1993a). A sample report is given in Fig. 30.25A (page 617). An identical system, but one which uses the ascending aorta to carotid transfer function, can be used to determine the ascending aortic pressure wave from the carotid pressure wave. A sample report is given in Fig. 30.25B (page 617). These reports can be generated in less time than is taken to

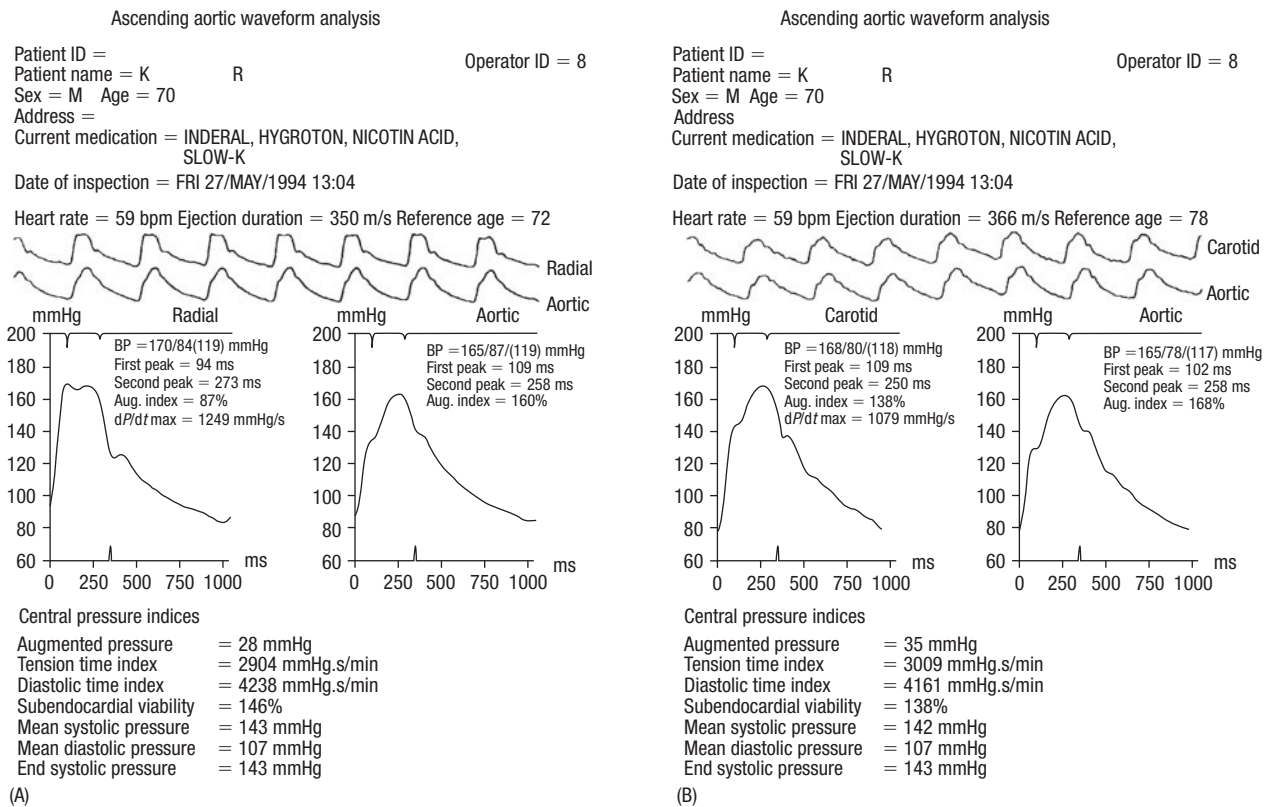


Figure 30.25 (A) Computerized analysis (sphygmocardiograph) of pressure waves recorded in the radial artery of a 70-year-old patient with isolated systolic hypertension. (B) Computerized analysis of pressure waves recorded by applanation tonometry in the carotid artery of the same patient as shown in A. Ascending aortic pressure waves derived from the radial artery (A) or for the carotid artery (B) are virtually identical, as are the indices of ventricular-vascular interaction derived therefrom. For explanation, see text

record an ECG. As would be expected (if the theoretic basis is correct), the calculated ascending aortic pressure wave is virtually identical when determined from either the radial or carotid pulse, as are the indices of ventricular-vascular interaction calculated therefrom (Figs 30.25A and 30.25B, page 617). The process of aortic pressure waveform generated from the radial artery has been accepted by the US FDA for invasive and non-invasively recorded waveforms under K012487 and K002742. For non-invasively recorded radial waveforms, the FDA accepted calibration from the brachial pressure cuff, on the basis that there was no significant difference between brachial and radial pressure recorded invasively or non-invasively. Generation of the aortic from the carotid waveform has not.

These reports show identification data as entered by the operator, together with date and time, then a series of pressure pulses as recorded over 9.0 s in the radial (or carotid) artery. Using a convolutional algorithm derived from the appropriate transfer function, these waves generate the corresponding series of waves predicted for the ascending aorta. This series of waves is shown below the recorded series of waves. Each wave series is then ensemble-averaged to give a single radial (or carotid) wave

at the left and a synthesized ascending aortic pressure wave at the right, using the maximum of the first derivative of the pressure wave for synchronization. The radial wave is calibrated from the systolic and diastolic pressures determined by sphygmomanometer in the brachial artery; the ascending aortic carotid artery calibration is determined by arithmetic integration of the radial waveform from this, assuming that mean pressure is identical at the two sites. Small, but tolerable, errors could result from differences in systolic pressure between radial and brachial artery and in mean pressure between brachial artery and central aorta. Inflection points on the waves (wave foot, incisura, first and second systolic peak) are determined from differentials of the pressure pulse and are flagged. Times to the first and second systolic peak from the wave foot are given, together with augmentation index, which is calculated as the height of the second systolic peak above the wave foot divided either by height of the first systolic peak above the wave foot, or by pulse pressure (Fujii, 1989; Kelly *et al.*, 1989d; O'Rourke *et al.*, 1989a; Kelly *et al.*, 1990a; Nichols and O'Rourke, 1990; Simkus and Fitchett, 1990; Takazawa *et al.*, 1993b; O'Rourke, 2002a; O'Rourke, 2002b). The maximum of the first derivative of recorded pressure is given; this is flagged if low since such

is suggestive of an obstructive lesion at the aortic valve or in the arterial tree proximal to the recording site (Nichols and O'Rourke, 1990). The augmentation index could be used to calculate reference age in normal individuals from the studies of Wilkinson *et al.* (2001), supported by the study of Kelly *et al.* in over 1000 normal subjects (Kelly *et al.*, 1989c); ascending aortic augmentation is given as millimeters of mercury immediately below the ensemble-averaged waves.

Valuable indices of ventricular–vascular interaction can be determined from separation of the aortic wave into systolic and diastolic periods. Tension time index (Sarnoff *et al.*, 1958; Braunwald, 1997a) is the integral of pressure and time during systole and is an index of left ventricular myocardial oxygen and blood demand. The diastolic time index is the integral of pressure and time during diastole, and is an index of capacity for left ventricular perfusion through the coronary arteries (Feigl, 1983; Nichols and O'Rourke, 1990; O'Rourke *et al.*, 1992a). Subendocardial viability ratio (Chapter 17) is the ratio of diastolic time index to tension time index, and is a measure of the propensity for myocardial ischemia on the basis of altered hemodynamic forces (Buckberg *et al.*, 1972a; Buckberg *et al.*, 1972b; Hoffman and Buckberg, 1978; O'Rourke, 1982b; Braunwald, 1997a; Buckberg *et al.*, 2008). Subendocardial ischemia has been shown to occur even when coronary arteries are patent when this index falls below 0.7 (Buckberg *et al.*, 1972a). Mean diastolic pressure is another measure of the capacity for coronary perfusion (Fig. 17.11) (O'Rourke, 1982b; Nichols and O'Rourke, 1990), while both mean aortic systolic pressure and end-systolic pressure have been taken as indices of left ventricular systolic load (Weber and Janicki, 1971; Suga and Sagawa, 1974).

The reports shown (Figs 30.25A and 30.25B, page 617) complement the ECG in non-invasive clinical assessment and give information on mechanical properties of the heart and circulation that are far more precise, more detailed, and more relevant to cardiac function than is the systolic and diastolic blood pressure as recorded conventionally in the brachial artery, or as is now possible by the cuff method in the radial artery.

Examples of clinical application are shown in Figs 30.26 (page 619) and 30.27 (page 620). Changes in the synthesized ascending aortic pressure wave with nitroglycerin (Fig. 30.26, page 619) are virtually identical to those recorded directly and are completely in line with carotid artery recordings. Changes with age are virtually identical to those described for the ascending aorta and inferred from the carotid artery tracings (Kelly *et al.*, 1989c). The effects of vasodilator agents in patients with isolated systolic hypertension are virtually identical to those recorded directly or inferred from the carotid artery tracings (Fig. 30.27, page 620) (Kelly *et al.*, 1989c; Chen *et al.*, 1996). The ill effects of aging are shown in three subjects from one family at age 18, 45, and 97 years (Fig. 30.28, page 620). The ill effects of isolated systolic

hypertension are most apparent in the 97-year-old's calculated aortic tracings. Here the gross increases in aortic systolic pressure, pressure augmentation, and mean systolic pressure and tension time index, and the corresponding decrease in diastolic time index and subendocardial viability ratio, contrast with the relatively small difference in diastolic pressure recorded in the brachial artery. The ability to measure left ventricular ejection duration aids in the assessment of cardiac function at different heart rates. We have found that in elderly people the ejection period is relatively long and does not decrease appropriately with tachycardia, such that there is a relative decrease in diastolic time index and in the subendocardial viability ratio when heart rate increases. These findings draw attention to the likelihood of tachycardia causing myocardial ischemia and impaired left ventricular function in elderly patients (Chapter 17) (Ferro *et al.*, 1995). Recordings are useful in individual patients for tailoring therapy in the management of hypertension, ischemic heart disease, and cardiac failure (O'Rourke *et al.*, 1993a; O'Rourke, 2008).

Studies have been conducted on the capacity to determine central aortic pressure and relevant indices from either the radial or carotid pressure pulse. A total of 593 subjects and patients have been studied on 2224 separate occasions. Synthesized ascending aortic waves usually are quite similar under different conditions, whether recorded from the radial or from the carotid waves (see Fig. 30.25). The original recorded waves (i.e. the carotid compared to the radial artery), however, are usually quite dissimilar, except in the very elderly or hypertensive individuals. Comparisons are given for different data in Figs 30.29 (page 621), 30.30 (page 621), 30.31 (page 622), 30.32 (page 622), and 30.33 (page 623).

The major difference in wave shape is in the degree of augmentation, i.e. in the relative height of the first compared with the second systolic peak. The first peak is caused by the first transit of the impulse generated by the heart, whereas the second systolic peak is generated by the return of this wave, after reflection from the lower part of the body (Nichols and O'Rourke, 1990). In the radial artery, the second peak occurs later than in the carotid artery, and so is lower down on the declining part of the wave (see Fig. 30.25) (Kelly *et al.*, 1989c). Augmentation is consequently less in the radial artery than in the carotid artery (see Fig. 30.29A, page 621). When the aortic wave is synthesized from the radial wave, augmentation is of course much higher and indeed substantially higher than in the raw carotid pressure wave (Fig. 30.29B, page 621). When the aortic wave is synthesized from the carotid wave, however, and augmentation from the two synthesized waves (one from the radial and the other from the carotid) compared, the augmentation index is almost identical and straddles the line of identity (Fig. 30.29C, page 621). The same consistency is seen when the augmentation index is expressed as total augmentation in millimeters of mercury (Fig. 30.29D, page 621). Figure 30.29D (page 621) shows

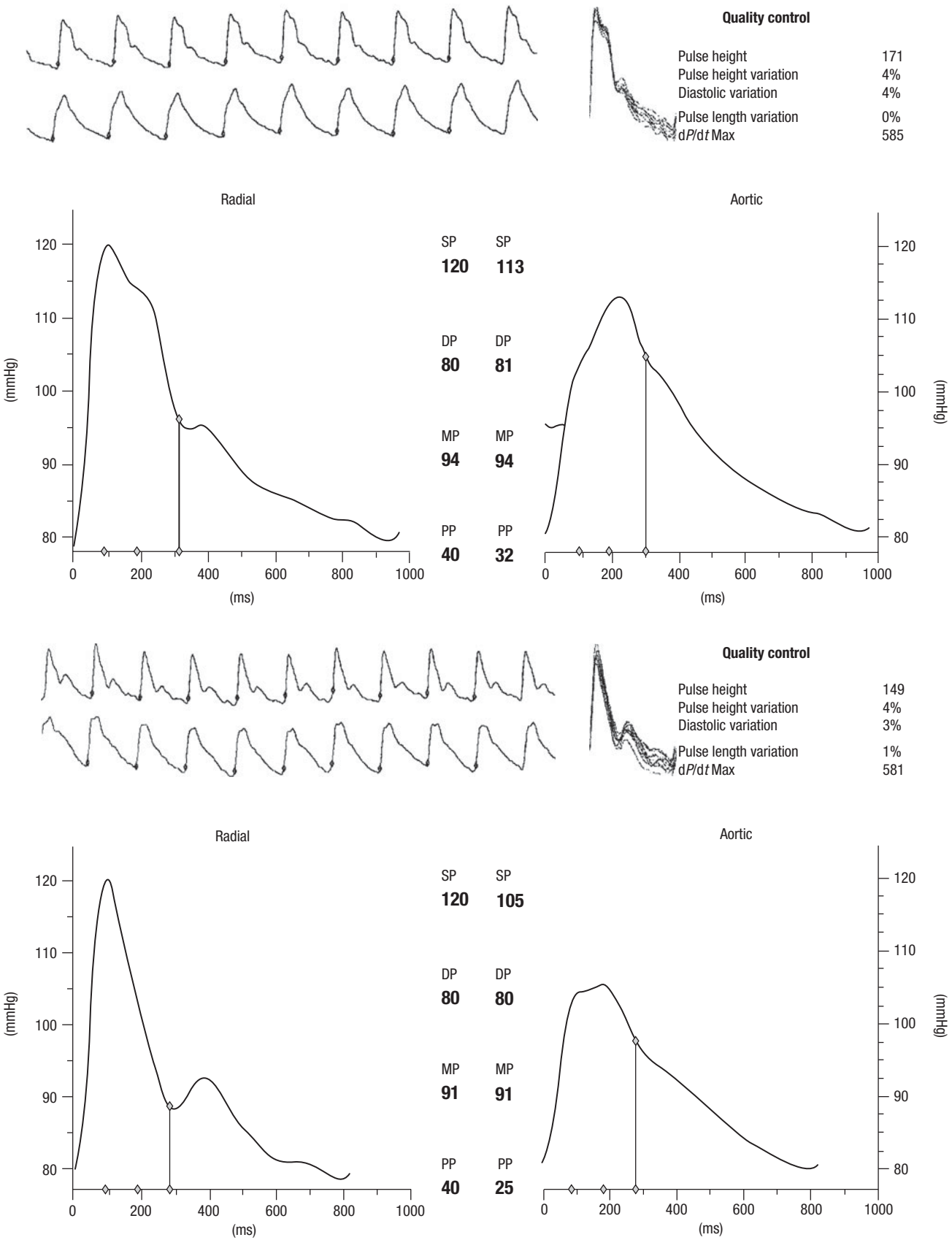


Figure 30.26 Analysis of radial artery pressure waves under control conditions (above) and following administration of nitroglycerin 0.3 mg sublingually (below) in a 54-year-old male. Compare with Figs 10.26, 10.27, and 26.8

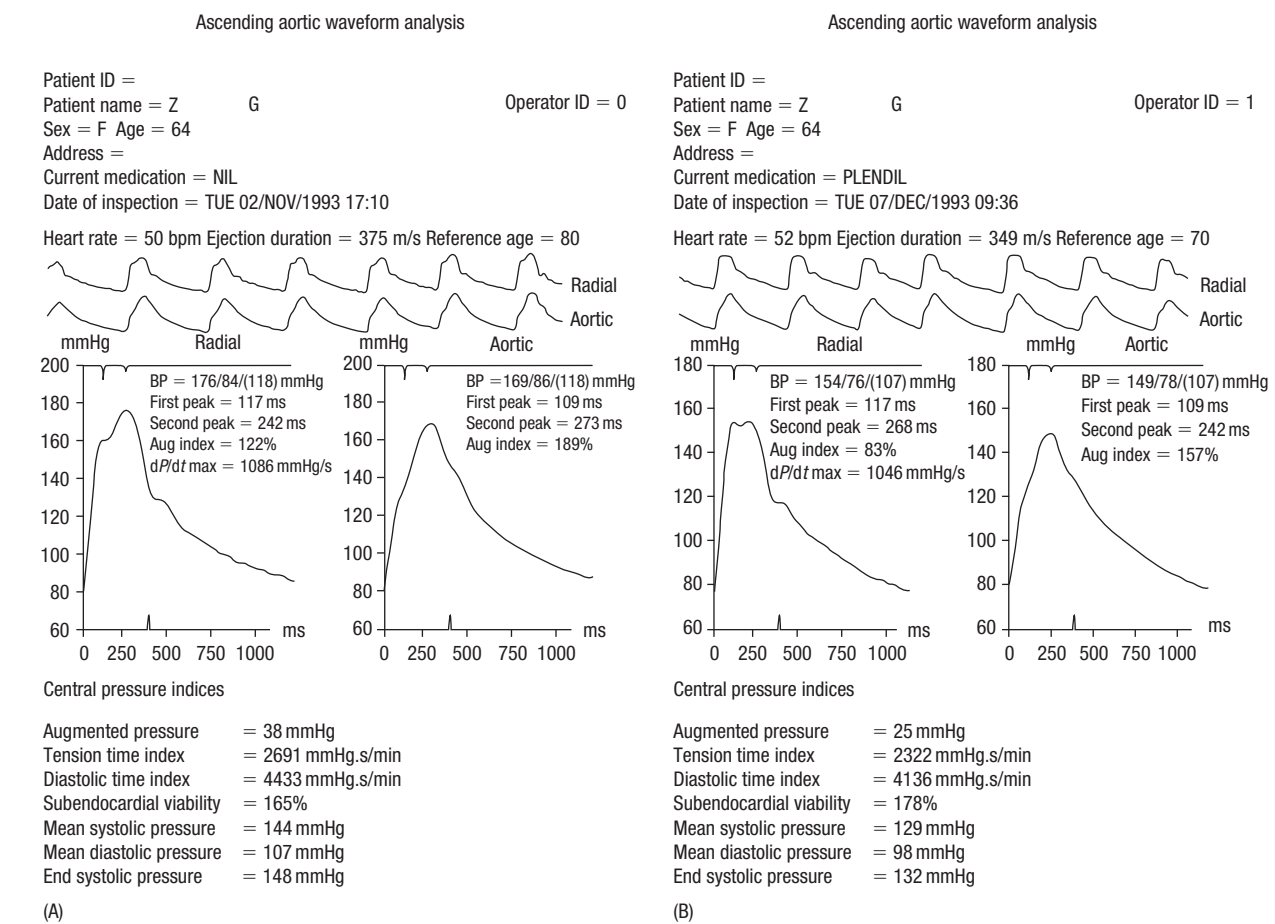


Figure 30.27 Analysis of radial pressure wave contour under control conditions in a patient with isolated systolic hypertension (A), and after 5 days treatment with felodipine (B). Mean pressure fell from 118 to 107 mmHg and calculated ascending aortic augmented pressure from 38 to 25 mmHg

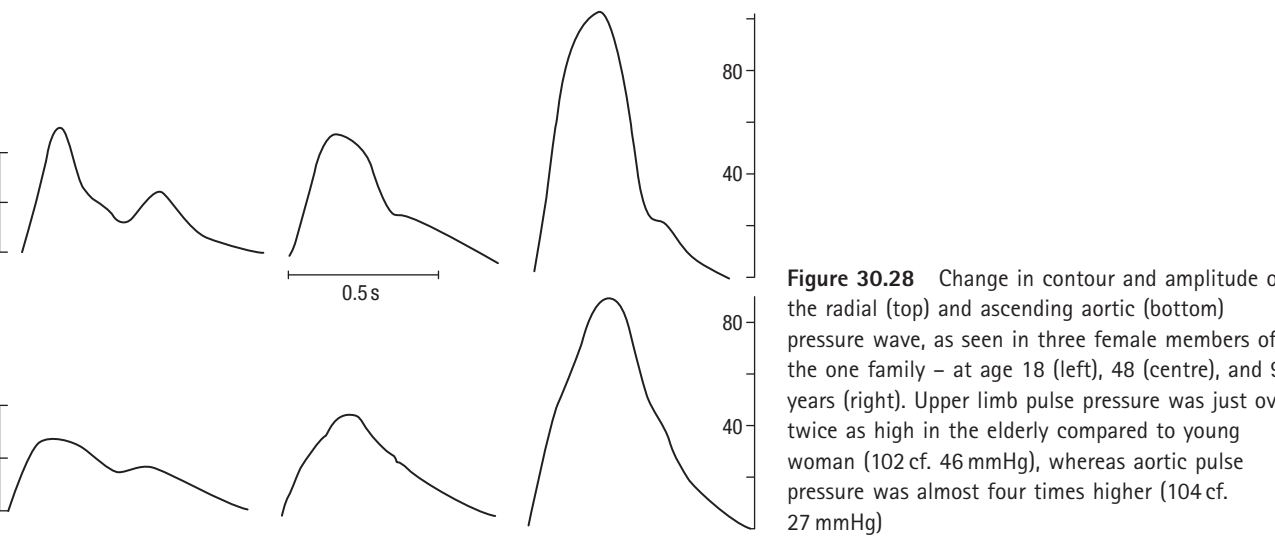


Figure 30.28 Change in contour and amplitude of the radial (top) and ascending aortic (bottom) pressure wave, as seen in three female members of the one family – at age 18 (left), 48 (centre), and 97 years (right). Upper limb pulse pressure was just over twice as high in the elderly compared to young woman (102 cf. 46 mmHg), whereas aortic pulse pressure was almost four times higher (104 cf. 27 mmHg)

results from 871 different observations. There is much overlap close to the line of identity, and the eye is drawn to the outlying points. When expressed in terms of mean and standard error of the mean, results are remarkably

consistent and virtually sit directly on the line of identity (Fig. 30.30, page 621). These results provide strong support for the validity of the method, although they do show considerable scatter for individual results.

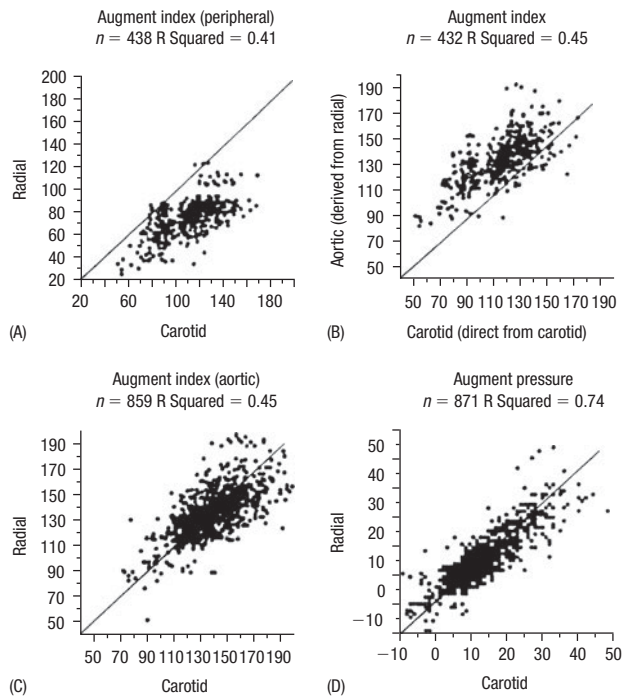


Figure 30.29 Comparison of pressure augmentation as determined from radial and carotid pressure waves in the same subject. (A) Augmentation index in the radial artery (ordinate) against augmentation index in the carotid artery (abscissa) from 438 paired observations in 301 subjects. (B) Comparison of augmentation of ascending aortic pressure waves calculated from the radial wave (ordinate) against augmentation index of the directly recorded carotid pressure wave (abscissa) in 432 paired observations in 295 subjects. (C) Comparison of augmentation index of synthesized ascending aortic pressure waves as determined from the radial artery (ordinate) and carotid artery (abscissa) for 859 observations in 548 subjects. (D) Calculated augmented pressure (mmHg) in the ascending aorta as determined from the radial artery trace (ordinate) and from the carotid artery tracing (abscissa) for 871 observations in 548 subjects

There was little difference in ejection duration determined from the two methods (Fig. 30.31A, page 622), nor in the values of mean systolic pressure (Fig. 30.32A, page 622), mean diastolic pressure (Fig. 30.32B, page 622), tension time index (Fig. 30.32C, page 622), or diastolic time index (Fig. 30.32D, page 622) derived from pressure values and ejection duration. There was more scatter in the derived subendocardial viability ratio, as one might expect (Fig. 30.31D, page 622), but values were close to the line of identity. Maximum dP/dt was higher for the radial than for the carotid pressure tracing, as one might expect (Fig. 30.31B, page 622). End-systolic pressure calculated from either the radial or carotid waves was very closely related (Fig. 30.31C, page 622).

Left ventricular ejection time is closely related to heart rate (Fig. 30.33, page 623). Values obtained by us, using either the carotid or radial waveform, were somewhat

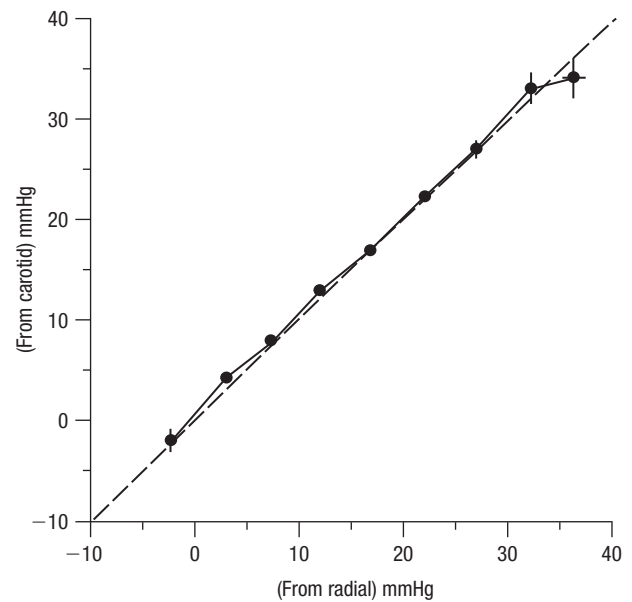


Figure 30.30 Ascending aortic augmented pressure in mmHg as determined from the carotid artery (ordinate) and radial artery (abscissa) and calculated for wave pairs at 5.0 mmHg increments between 25.0 and 135 mmHg, and expressed as mean $\pm$ standard error of mean. Comparison diverges little from the line of identity (dotted line) over this range. Data are the same as in Fig. 30.29

(10–15 ms) greater than those published by Weissler *et al.* (1963), who utilized a low-frequency air-filled tambour over the carotid artery in the neck. There was considerable variability in the value of ejection duration even when expressed as a function of heart rate or total cycle period. Studies on the effects of different disease states are continuing. As previously described (O'Rourke *et al.*, 1992c), patients with cardiac failure and systolic dysfunction tended to have relatively low left ventricular ejection duration at any heart rate. However, patients with apparent diastolic dysfunction appeared to have longer ejection duration (Abhayaratna *et al.*, 2006a; Abhayaratna *et al.*, 2006b; Weber *et al.*, 2006; Abhayaratna *et al.*, 2008; Weber *et al.*, 2008). The use of left ventricular ejection duration measurements may assist in separating patients with diastolic and systolic dysfunction as a cause of cardiac failure (see Fig. 14.25 and below) (Abhayaratna *et al.*, 2006b; Weber *et al.*, 2006; Abhayaratna *et al.*, 2008; Weber *et al.*, 2008; Denardo *et al.*, 2010). Older patients and patients with hypertension appeared to have a relatively longer duration of ejection at any heart rate than younger and normotensive subjects. In some patients, ejection duration appeared inappropriately long and did not shorten with increase in heart rate (Fig. 30.33, page 623). These patients had a low calculated subendocardial viability ratio at normal heart rate, and this decreased further as heart rate increased (Fig. 30.34, page 623) (O'Rourke, 2008).

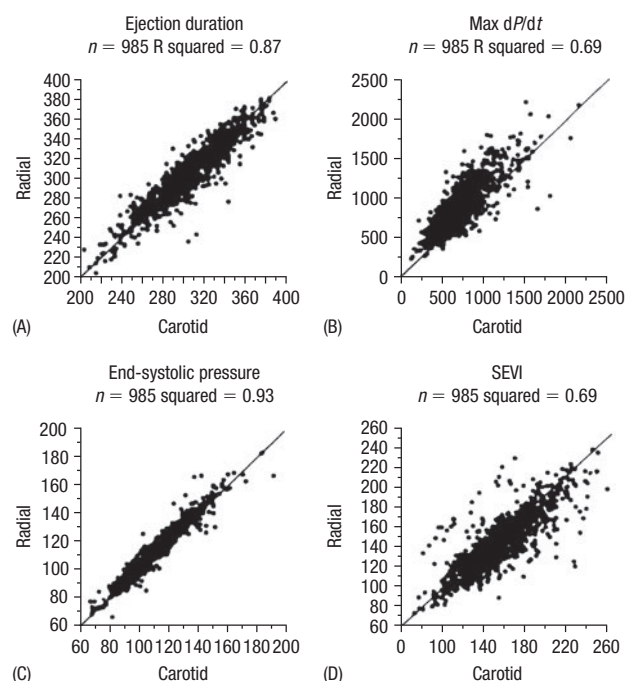


Figure 30.31 (A) Relationship between ejection duration as determined from the radial artery trace (ordinate) and from the carotid artery trace (abscissa) for 985 observations in 548 patients. (B) Relationship between maximum dP/dt as determined from radial artery (ordinate) and from the carotid artery trace (abscissa) from 985 observations in 548 subjects. (C) Comparison of end-systolic pressure in the calculated ascending aortic pressure wave, as determined from the radial artery wave (ordinate) against the carotid artery pressure wave (abscissa) from 985 observations in 548 subjects. (D) Comparison of subendocardial viability ratio in the synthesized ascending aortic pressure wave as determined from the radial artery wave (ordinate) against the carotid artery pressure wave (abscissa) from 985 observations in 548 subjects

Calculated subendocardial viability ratio, like ejection duration, is inversely related to heart rate (Fig. 30.34, page 623). Buckberg *et al.* (Buckberg *et al.*, 1972a; Buckberg *et al.*, 1972b) have shown that, even with normal coronary arteries, coronary blood flow in the subendocardial muscle is compromised when this falls below 70–100 percent. Such values were not seen in people with heart rate under 60/minute, but were relatively common at heart rates of 80/minute (Fig. 30.34, page 623), especially in the elderly and in subjects with hypertension. The subendocardial viability ratio is largely determined by ejection duration, and so was particularly low in people with long ejection (and short diastole), as well as in patients with isolated systolic hypertension where pressure during systole was relatively high and pressure during diastole relatively low.

Maximal dP/dt (see Fig. 30.31B, page 622) was very variable, even when determined from the carotid pressure tracing. Low values were seen as expected in patients with severe left ventricular failure and in patients with stenosis

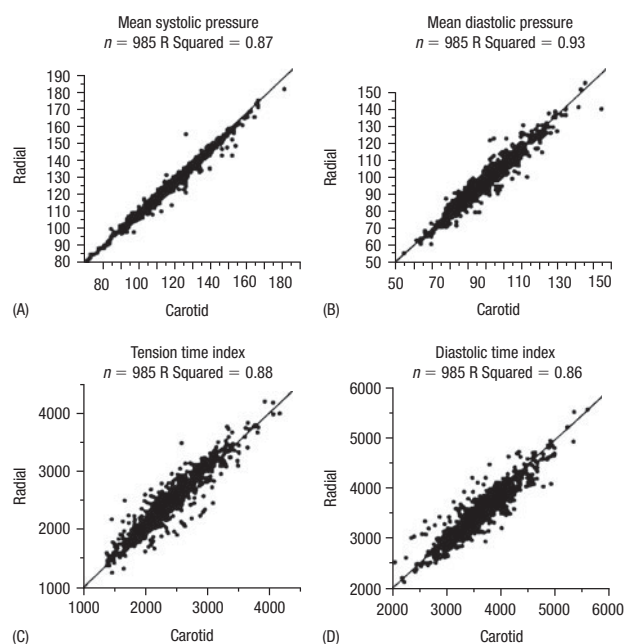


Figure 30.32 (A) Comparison of mean systolic pressure in the calculated ascending aortic wave from the radial (ordinate) and carotid (abscissa) for 985 observations in 548 subjects. (B) Mean diastolic pressure in the ascending aortic synthesized wave as determined from the radial artery pulse (ordinate) and the carotid artery pulse (abscissa) for 985 paired observations in 548 subjects. (C) Tension time index determined for the ascending aortic pressure wave from the radial wave (ordinate) compared to the carotid wave (abscissa) for 985 observations in 548 subjects. (D) Comparison of ascending aortic diastolic time index as determined from the radial artery trace (ordinate) compared with the carotid artery trace (abscissa) for 985 observations in 548 subjects

of the aortic valve. There was, however, considerable variation in other subjects and even in patients with cardiac failure of lesser severity. This measure appeared to be of doubtful utility, although when low it often drew attention to obstructive lesions in the subclavian or brachial artery upstream from the point of measurement of pressure in the radial artery, or to aortic valve stenosis.

The tonometric system as used by us has been tested on Thai and Japanese subjects. Gulprasutdilog and O'Rourke (1996) found similar results for aortic pressure augmentation, ejection duration, and subendocardial viability ratio under different conditions when central pressure was synthesized from the radial and carotid waveforms, using the appropriate generalized transfer functions. Takazawa *et al.* (Takazawa *et al.*, 1996) found similar results for ascending aortic pressure augmentation under control conditions and with nitroglycerin, when the directly recorded ascending aortic pressure wave (at cardiac catheterization) was compared against the aortic wave synthesized from the simultaneously recorded radial pulse (Fig. 30.35, page 624). Basically similar findings have been seen in white people living in the UK (McEniery *et al.*,

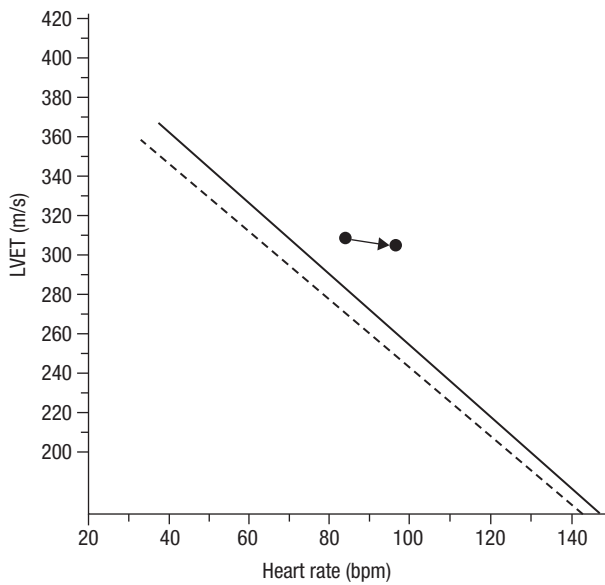


Figure 30.33 Relationship between left ventricular ejection time (LVET) and heart rate for 2224 observations (radial and carotid waves) in 593 subjects. The solid line is the regression line for all data. The dotted line is the regression line for the data of Weissler *et al.* (Weissler *et al.*, 1963; Weissler *et al.*, 1968). The two data points show results in one patient in whom exercise was associated with virtually no change in ejection duration

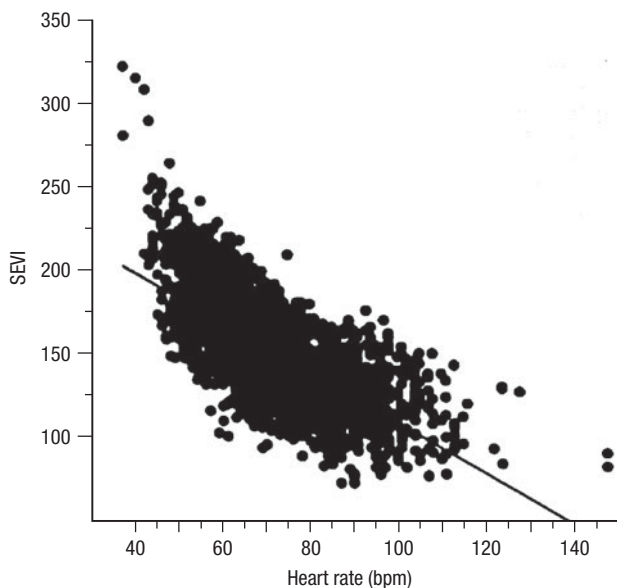


Figure 30.34 Relationship between subendocardial viability ratio (SEVI) and heart rate in 2224 observations on 593 subjects

2005a; McEniery *et al.*, 2008), Europe (Wojciechowska *et al.*, 2004), and the United States (Mitchell *et al.*, 2004a). Black people in the United States (Heffernan *et al.*, 2008) and in South Africa (Shiburi *et al.*, 2006) have higher values of augmentation, and of central systolic pressure

than white people, as they do of aortic pulse wave velocity (PWV) (see Chapter 25). This appears to be due, not to a difference in pressure transfer function in the arm, but to relative increase in modulus of lower frequency components of the aortic pressure wave, consequent on increased aortic impedance (Gallagher *et al.*, 2004).

The most extensive validation study of the transfer function process for determination of aortic systolic and pulse pressure and pulse wave contour is that by Pauca *et al.* (2001) in US subjects undergoing cardiac surgery (Figs 30.23 and 30.24). This study also compared indices as determined by sphygmocardiography, including augmented pressure, end-systolic pressure, augmentation index, and ejection duration (Fig. 30.23, Table 30.3, page 624). All pressure values were within AAMI SP10 criteria, but there was greater scatter with augmented pressure than for systolic and diastolic pressures (O'Rourke *et al.*, 2002b; O'Rourke and Pauca, 2004). This is not unexpected since ability to identify the pressure values depended not just on pressure at its peak or trough, but on pressure where it was changing rapidly, and on timing. Chen *et al.* (1997) raised concerns about the identification of a systolic shoulder for calculation of augmentation; they pointed out that filtering of harmonics higher than the fourth may cause this feature to disappear. This has not been a practical problem with our transfer function process since the transfer function approaches unity after its initial peak at 3.5–4.5 Hz, so that the high frequency components of the pulse are not attenuated. We were not completely satisfied, however, with a comparison of timing intervals with the fluid-filled system used by Pauca *et al.* (Pauca *et al.*, 2001), so we checked independently the correlation between tonometric carotid and radial ejection duration in a further group of 215 patients. Correlation was very good, with mean value for ejection duration 325 ms, mean difference 4.6 ms, and standard deviation 9.6 ms (Hirata *et al.*, 2001). In recent years, we have come to regard amplification of the pulse from aorta to radial artery as a possible substitute for aortic augmentation index. There is a close (inverse) relationship between amplification and aortic AIx over the normal range (Fig. 19.32). Hashimoto *et al.* (2007b) showed greater predictability for amplification than any other measure in regression of left ventricular (LV) mass with drug therapy.

Criticism of the clinical application of transfer functions has been championed by individuals and groups in the UK and Australia. Americans appear generally to accept the FDA endorsement. The valid point has been made that the process of sphygmocardiography has never been validated as it is presently applied, where the peripheral pressure pulse is calibrated to brachial cuff sphygmomanometric systolic and diastolic pressures. Cuff sphygmomanometric pressures are far less accurate than most clinicians are prepared to concede (Fig. 21.5) and such inaccuracies will taint any conventional calibration of peripheral waveforms (Kass *et al.*, 1998). What must be decided by consensus in the future is whether the present

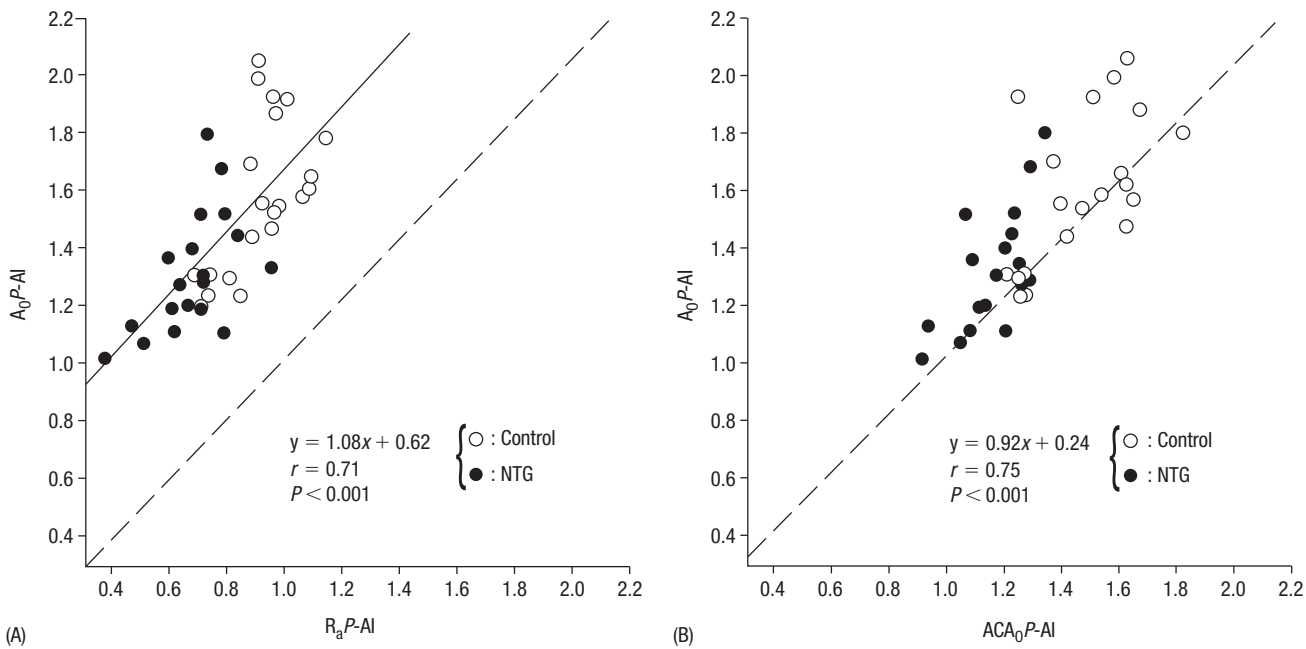


Figure 30.35 (A) Comparison of augmentation index of the directly recorded ascending aortic pressure wave (A_oP -AI, ordinate) and of the simultaneously non-invasively recorded radial pressure wave (R_aP -AI, abscissa) under control conditions (open circles) and after sublingual nitroglycerin (0.3 mg) (closed circles) in 20 patients studied at cardiac catheterization. (B) Comparison of augmentation index of the directly recorded aortic pressure wave (A_oP -AI, ordinate) and of the synthesized wave generated for the aorta through use of a generalized transfer function, from the simultaneously recorded radial pressure wave (ACA_oP -AI, abscissa). Solid lines are regression lines; broken lines are lines of identity. Reproduced from Takazawa *et al.* (1996)

Table 30.3 Coefficient of variation (CV) of each index

	CV among 3 baseline (%)	CV on placebo day (%)
Brachial SBP	1.89	3.87
Brachial DBP	2.42	4.36
Heart rate	3.97	4.15
Aortic SBP	2.13	4.16
Augmentation index	2.03	2.66
Carotid–femoral PWV	3.23	3.32
Leg PWV	3.25	3.25
Reflection time	1.97	1.72
Ejection duration	2.25	2.17

DBP, diastolic blood pressure; PWV, pulse wave velocity; SBP, systolic blood pressure.
From Hirata *et al.* (2002).

process (with brachial or radial cuff calibration) is used, or whether the timing intervals and relative features (such as augmentation index and amplification) are used as a supplement to recordings of conventional cuff pressures. Studies of repeatability and reproducibility of sphygmocardiographic indices have been reported by ourselves (Hirata *et al.*, 2001; Hirata *et al.*, 2002c) and by a number of others (Liang *et al.*, 1998; Wilkinson *et al.*, 1998;

Siebenhofer *et al.*, 1999; Filipovsky *et al.*, 2000; Savage *et al.*, 2002a). All show acceptable concordance, with results at least as good, and usually better than obtained with other methods.

Comparisons of sphygmocardiography with other methods for assessing arterial stiffness and its effects on the heart, likewise are favorable, and better than indices derived from analysis of the diastolic part of the pressure waveform (Izzo *et al.*, 2001; Rietzschel *et al.*, 2001a; Segers *et al.*, 2001; Oliver and Webb, 2003; Laurent *et al.*, 2006).

This clinical system for application of sphygmocardiography (using applanation tonometry) is designed for use in a clinician’s office in association with conventional blood pressure recording by sphygmomanometer, and in association with other clinical aids such as the ECG for patient management. The report is like the ECG, and enables features of the pulse wave to be interpreted, and cardiac function to be assessed with more precision than possible to date. It represents the extension from the sphygmograph of Marey, Mahomed, Mackenzie, and Keyt and shows what is now possible with the combination of high-fidelity manometry, hemodynamic theory, and digital computing. Clearly, it can be improved further. It has other potential for application, as in clinical hemodynamic monitoring during surgery or in intensive care, as well as in research procedures for assessment of ventricular function in association with imaging techniques (Laurent *et al.*, 2006; Mancia *et al.*, 2007a; Mancia

et al., 2007b). (One of the authors – M O'Rourke – has a commercial interest in this process, being founding director of the company which sells the SphygmoCor[®] system.)

Hypertension

As described in Chapter 20, the radial pulse was used by Bright, Mahomed, and their contemporaries in the nineteenth century to gauge elevation of arterial pressure. With his relatively crude mechanical sphygmograph, Mahomed (1874) stressed the diagnostic importance of the late systolic component of the radial artery pulse in people with high arterial tension – 'the tidal wave is prolonged and too much sustained' (Figs 20.1 and 20.2).

With such information in clinical use before the cuff sphygmomanometer was introduced, it is not surprising that sphygmocardiography should find use in modern times as a complement to the cuff sphygmomanometer in the management of hypertension. Examples have already been given (Fig. 19.2) of how systolic pressure elevation in youth may be sufficient for a clinical diagnosis to be made of spurious systolic hypertension – with unnecessary investigation and treatment avoided. The diagnosis of 'spurious systolic hypertension' can be made confidently if elevated systolic pressure is entirely due to a sharp early systolic pressure spike, and if diastolic and integrated mean pressure are normal, and if the measured carotid pressure wave, and derived ascending aortic pressure waves are of normal contour and amplitude for the young person's age. In population studies, this phenomenon of high wave amplification in fully grown adolescents has been used by us to explain the apparent relative plateau of brachial systolic blood pressure (and pulse pressure) between age 18 and 45 years (Fig. 19.1).

Isolated systolic hypertension in elderly people is usually due to arterial stiffening, with early return of wave reflection (Chapter 19). Occasionally, elderly patients are seen with elevated systolic pressure but youthful radial pressure waves, and with relatively low central aortic pressure augmentation (Fig. 30.36, right). These patients may be considered to have 'spurious systolic hypertension' as do the adolescents in Fig. 19.2. They have no stigmata of hypertension and are apt to suffer from hypotensive episodes if treated vigorously. Such a condition may explain some of the poor results seen in trials of very elderly people with hypertension (Beckett *et al.*, 2008).

Treatment of hypertension is more logical if based on correction or targeting of abnormal mechanisms. Use of angiotensin-converting enzymes (ACE) inhibitors, calcium channel antagonists, or nitrates is associated with decrease in central pressure wave augmentation (Figs 30.26 and 30.27) (O'Rourke *et al.*, 1993a; Dahlöf *et al.*, 2005; Hirata *et al.*, 2005b; Williams *et al.*, 2006; Hashimoto *et al.*, 2007a;

Hashimoto *et al.*, 2007b; Jiang *et al.*, 2007; Roman *et al.*, 2007; Matsui *et al.*, 2009), and can be monitored more accurately through sphygmocardiography than when sphygmomanometric blood pressure is recorded alone. As previously emphasized (Figs 10.26 and 10.27) (Kelly *et al.*, 1989f; Kelly *et al.*, 1990a; O'Rourke *et al.*, 1993a; Asmar *et al.*, 2001b; de Luca *et al.*, 2004), reduction in central systolic pressure may be much greater with use of arterial vasodilator drugs, than with apparently equally efficacious therapy using β -blocking agents and diuretics. On this basis, Mitchell *et al.* (1996) (Fig. 20.15) have explained more powerful effects of an ACE inhibitor over hydralazine in rats for regression of left ventricular hypertrophy. It is highly likely that the same principles apply to humans with hypertension and left ventricular hypertrophy, i.e. that regression of left ventricular hypertrophy is attributable to decrease in left ventricular systolic load as can be measured from left ventricular or ascending aortic pressure waves (Hashimoto *et al.*, 2007a; Hashimoto *et al.*, 2007b; Vasan, 2009). Stokes *et al.* (2003) have recently made similar observations to Mitchell *et al.* (1996).

Use of sphygmocardiography may provide insight into adverse effects of anti-hypertensive therapy. Controversy on use of short-acting dihydropyridines in hypertension (Furberg *et al.*, 1995) has highlighted the potential of these drugs to accentuate myocardial ischemia. Short-acting nifedipine preparations often maintain or prolong left ventricular ejection duration, as well as causing tachycardia (Fig. 30.37). Such an effect favors myocardial ischemia, and can be measured as inappropriate reduction of diastolic duration and of subendocardial viability ratio. Importance of diastolic perfusion period in myocardial ischemia has been highlighted by Ferro *et al.* (1995) (Fig. 17.10); sphygmocardiography permits measurement of the relevant indices of myocardial ischemia (Fig. 30.37). Changes seen in this figure are unlikely to be relevant at rest, but could be during tachycardia, when diastolic duration is further decreased (Prichard and Vallance, 2004).

Sphygmocardiography is being used in a number of ongoing epidemiological studies. Previous studies of finger plethysmography ('vasculographs') by the Framingham investigators (Dawber *et al.*, 1973) did show an association between pulse pattern and coronary artery disease, but suffered the disadvantage of great variability and of inability to calibrate or express numerically. New studies of carotid, radial, brachial, and femoral tonometry have been reintroduced into the Framingham study (Mitchell *et al.*, 2004a), and have recently been reported (Mitchell *et al.*, 2010). A serious problem (use of brachial rather than radial tonometry (O'Rourke and Adji, 2010b)) appears to invalidate results. Other studies using validated techniques (Roman *et al.*, 2009; Vlachopoulos *et al.*, 2010a; Vlachopoulos *et al.*, 2010b; Vlachopoulos *et al.*, 2010c; Wang *et al.*, 2010; Weber *et al.*, 2010) confirm benefits of applanation applied to the radial artery. This subject is discussed in Chapter 28.

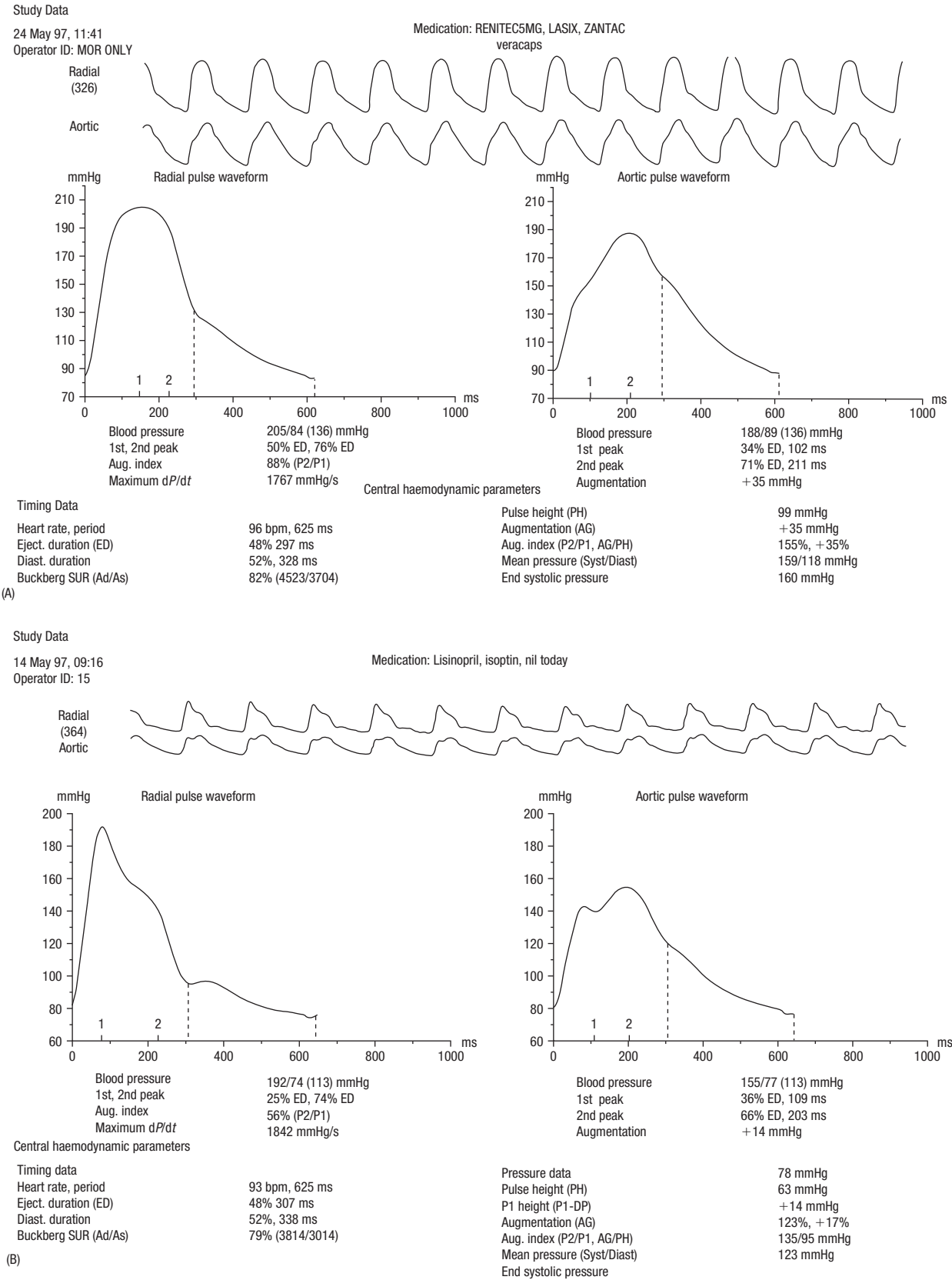


Figure 30.36 'Spurious systolic hypertension'? Sphygmocardiogram reports in two elderly patients with 'isolated systolic hypertension'. The patient in (A) is typical, with early wave reflection, a typical sustained systolic peak of the radial pulse, and high (35 mmHg) augmentation of the derived ascending aortic pressure pulse. The patient in (B) is atypical. Elevated systolic pressure in the upper limb is attributable to amplification of the first systolic peak. Aortic augmentation (14 mmHg) is normal, and calculated aortic systolic pressure at 155 mmHg is 37 mmHg less than recorded in the brachial artery. The patient in (B) warrants far less aggressive therapy than the patient whose waves are shown in (A)

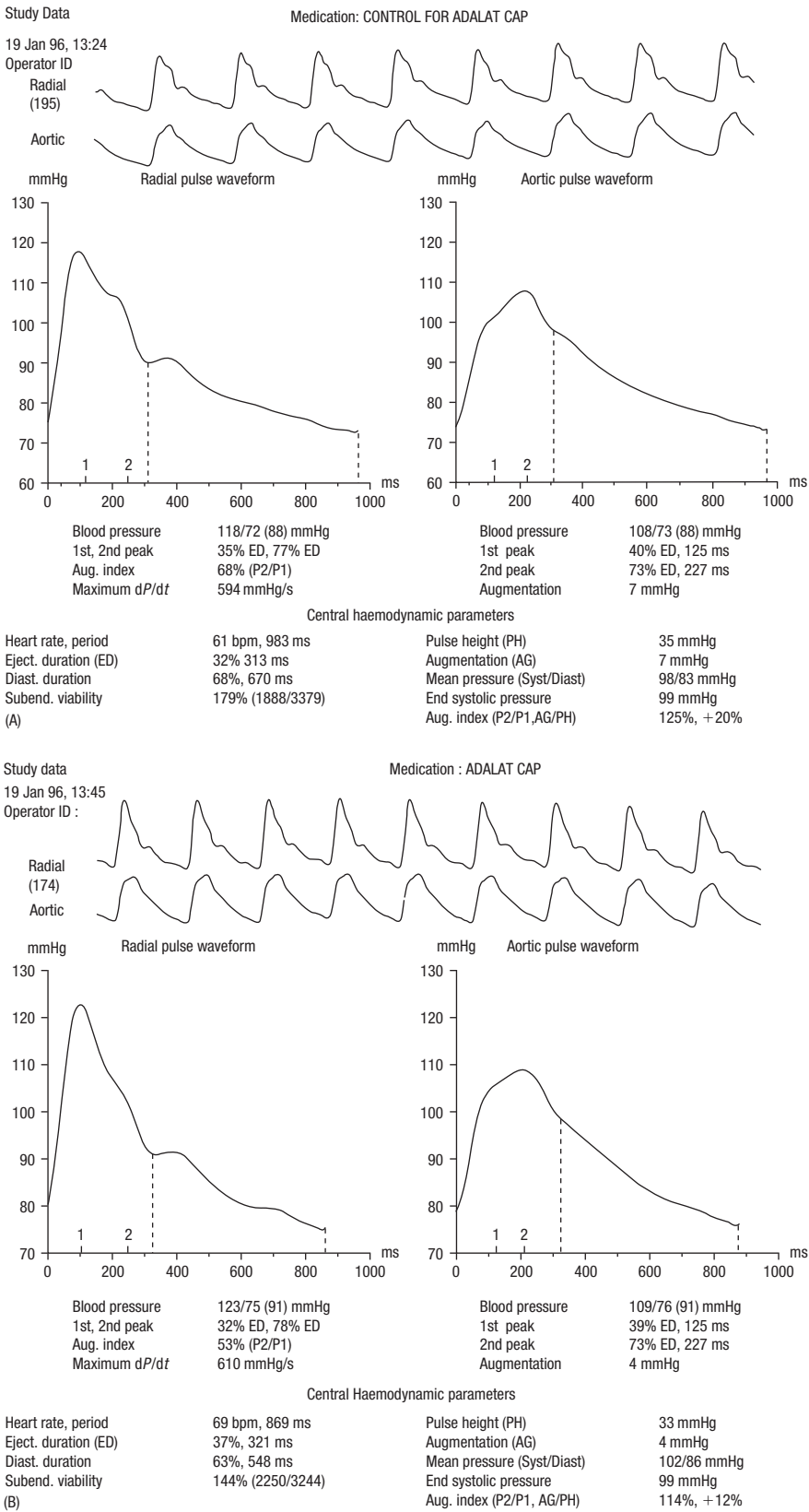


Figure 30.37 Potentially pro-ischemic effects of 'rapid release' nifedipine. Sphygmocardiogram reports in a volunteer (one of the authors) under control conditions (A) and at peak effect of nifedipine in capsule form (B). Despite reduction in heart period from 983 to 869 ms, ejection duration increased from 313 to 321 ms, resulting in a fall of subendocardial viability ratio from 179 percent to 144 percent and diastolic period from 670 ms to 548 ms. Findings were observed consistently on other occasions. A beneficial effect of the drug is apparent as decrease in aortic augmentation from 7.0 to 4.0 mmHg. Adverse effects were not apparent when nifedipine and other dihydropyridines were used in sustained release form, or when combined with β -blocking agents

Myocardial ischemia, angina pectoris

Sphygmocardiography permits greater precision than previously available in medical therapy of angina pectoris through the ability to measure the major hemodynamic determinants of myocardial oxygen requirement (as aortic pressure–time index) and of coronary blood supply (as diastolic pressure–time index). The ratio of the latter to the former is the subendocardial viability ratio (Buckberg *et al.*, 1972a; Buckberg *et al.*, 1972b; Buckberg *et al.*, 2008) and is related to subendocardial perfusion and ischemia (Fig. 17.11). This also is calculated in the sphygmogram report.

The term ‘syndrome X’ has been given as a diagnosis for patients suffering from apparent myocardial ischemia, but showing normal coronary arteries at angiography. This is discussed in Chapter 17. Many mechanisms have been proposed, with emphasis on disease of the micro-circulation (Chilian, 1997), but few consider the hemodynamic determinants of coronary blood flow or the possibility that changes in such determinants may by themselves cause myocardial ischemia and be responsible for this syndrome. This subject has been readdressed by Ferro *et al.* (1995), Julius *et al.* (1967), Gould (1997), Panting *et al.* (2002), Cannon (2009), and O’Rourke and Nichols (2009b). Reduction of diastolic perfusion duration below 45 percent of the cardiac cycle may cause myocardial ischemia, even with lesions generally considered insignificant (Ferro *et al.*, 1995). Figure 30.38 shows the value of sphygmocardiography in patients with angina pectoris and evidence of exercise-induced myocardial ischemia who are considered for coronary angiography. All five patients showed unusual reduction in diastolic duration, sufficient to cause myocardial ischemia according to the findings of Ferro *et al.* (1995), in the absence of ‘significant’ coronary obstruction. The first two patients did undergo coronary angiography without any significant lesion being demonstrated. The remainder were spared this investigation. All were treated with drugs, aimed at optimizing ventricular–vascular interaction, and at minimizing systolic pressure time integral and maximizing diastolic pressure time integral. All responded well to medical therapy.

Sphygmocardiography may prove useful in identifying those patients with ‘syndrome X’ whose symptoms are due to myocardial ischemia and whose ischemia can be attributed to abnormalities in diastolic pressure–time integral (Panting *et al.*, 2002). This is presently under investigation in the WISE study at the University of

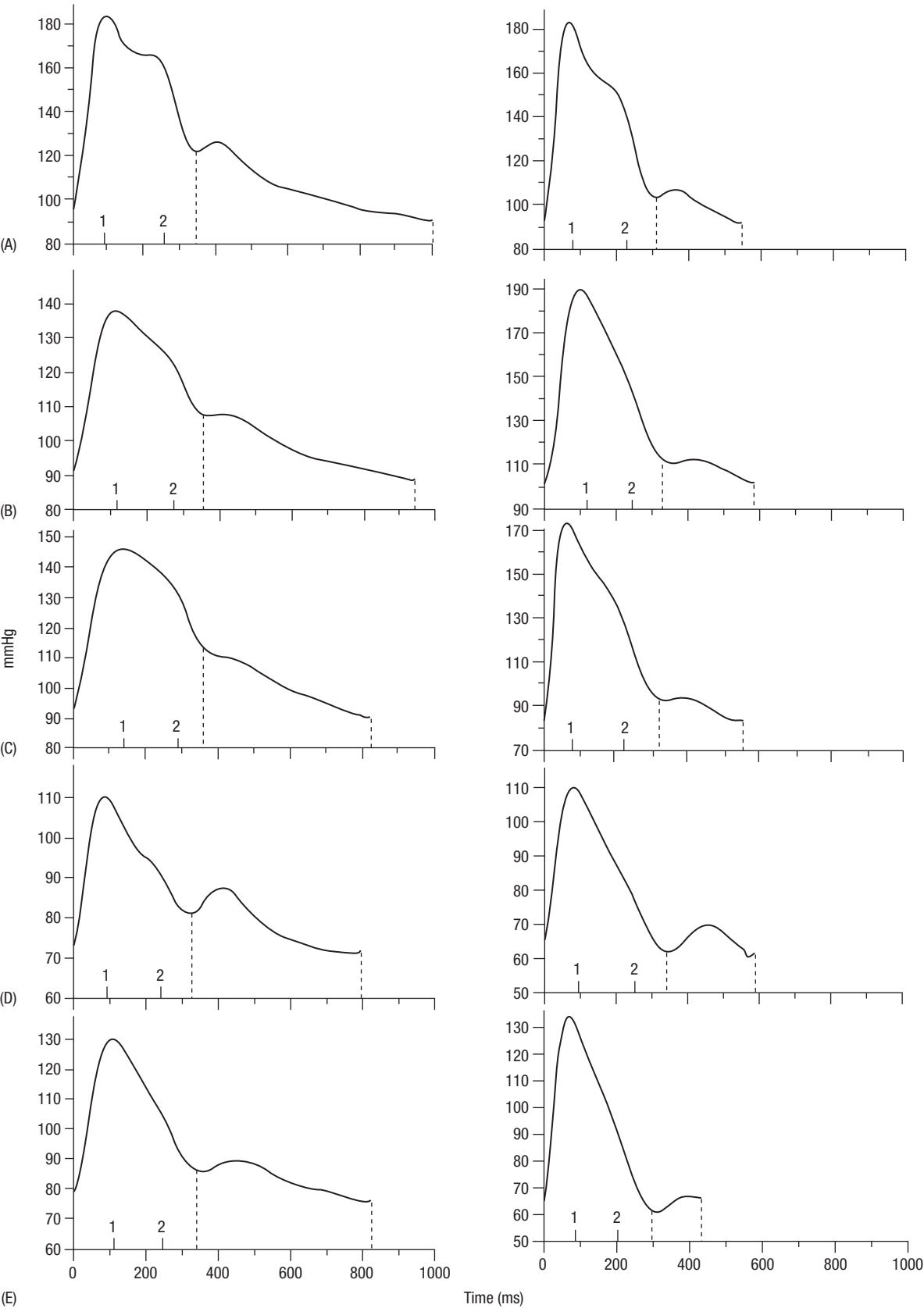
Florida (Pepine *et al.*, 2007; Anderson *et al.*, 2008). Correspondence in the *New England Journal of Medicine* following publication of this paper ignored the impressive results of subendocardial ischemia, and the seminal work of Buckberg, Boudoulas (1990), and Ferro, and confirmed that cardiologists are unwilling to consider the possibility of myocardial ischemia as being caused by anything except coronary artery narrowing (see Panting *et al.* (2002) and O’Rourke (2008)). In patients with angina pectoris, use of sphygmocardiography may permit coronary angiography to be deferred or avoided. This can now be done with more confidence following publication of the COURAGE trial which showed similar benefit for optimal medical therapy as for revascularization in chronic stable angina pectoris (Boden, 2004; Boden *et al.*, 2007).

Use of sphygmocardiography in regular clinical practice enables more intelligent assessment and management of patients with chronic stable angina pectoris. It is likely to promote less referral for angiography, less subjection to the ‘oculostenotic reflex’, ‘oculodilatory reflex’, and ‘auditory dilatory reflex’ (Boden, 2004), and more optimal medical management. Boden’s editorial in the *Journal of American College of Cardiology* commented on yet another paper (Hueb *et al.*, 2004) from the Mayo Clinic, which failed to show benefit for an invasive over conservative strategy in patients with stable angina.

Cardiac failure

The hemodynamic basis of cardiac failure is discussed in Chapter 14. There it was pointed out (Fig. 14.24) that myocardial systolic dysfunction is associated with reduction in ejection duration, whereas diastolic dysfunction is associated with preservation or increase in systolic ejection duration. On this basis, it should be possible to use sphygmocardiography to distinguish between systolic and diastolic dysfunction as the cause of cardiac failure. This was tested in a group of 67 patients with cardiac failure in whom predominant diastolic dysfunction was assumed when left ventricular ejection fraction was estimated at > 35 percent, and systolic dysfunction was diagnosed with ejection fraction ≤ 35 percent (Fig. 15.5) (O’Rourke and Lei, 1997). While there was considerable overlap, suggesting many patients had a mixture of systolic and diastolic dysfunction, the mean values of ejection duration were significantly different at 293 and 259 ms, respectively. Sphygmocardiography may prove useful in making a clinical diagnosis for diastolic dysfunction, and may obviate the

Figure 30.38 An explanation for ‘syndrome X’. Radial artery pressure waves measured by applanation tonometry in five patients with exertional angina pectoris and evidence of myocardial ischemia. Two patients underwent coronary angiography and were shown to have trivial obstructive lesions only. Waves at left were recorded under control conditions, waves at right immediately after exercise. Exercise response was unusual in that the systolic ejection period remained high, and diastolic period was markedly abbreviated to 44, 43, 43, 43, and 32 percent of heart period. Such abbreviation of diastolic period is sufficient from the data of Ferro *et al.* (1995) (Fig. 14.10) to explain myocardial ischemia in the presence of minor (< 50 percent) coronary obstructive lesions. Data obtained with Dr Somprasong Gulprasutdilog



need for echocardiography in some patients. The 'gold standard' of diastolic dysfunction is the invasive measurement of LV pressure/volume relationships during diastole, but this correlates with LV ejection duration, as well as the more frequently used measures of early mitral flow and tissue velocity (Abhayaratna *et al.*, 2006b; Weber *et al.*, 2006; Kasner *et al.*, 2007; Weber *et al.*, 2008; Shim *et al.*, 2011).

In diastolic dysfunction, the main aims of therapy are to reduce late systolic pressure augmentation by decreasing wave reflection – so to improve ventricular relaxation (Yabe *et al.*, 1991; Gillebert *et al.*, 1997; Abhayaratna *et al.*, 2006b; Weber *et al.*, 2006; Weber *et al.*, 2008) – and to increase diastolic time by reducing heart rate. Such therapy can readily be monitored by sphygmocardiography.

In systolic dysfunction, wave reflection has its predominant effect during systole in reducing flow from the heart (Fig. 14.24). When severe, the synthesized aortic pressure wave has little or no late systolic augmentation, the incisura is low down on the wave, and the diastolic pressure wave is prominent. The wave shape is characteristic (Lewis, 1906; Hamilton, 1944; Ewy *et al.*, 1969; Orchard and Craigie, 1980; Tavel, 1985; Vlachopoulos and O'Rourke, 2000a; Vlachopoulos and O'Rourke, 2000b; Denardo *et al.*, 2010), and is diagnostic of severe cardiac failure provided that impaired pulmonary venous return and cardiac tamponade can be excluded. While useful in diagnosis, sphygmocardiography is less helpful in monitoring therapy, since the beneficial effect of decreased wave reflection is apparent in augmentation of late systolic flow from the heart, not on the pressure wave. Progressive change in pressure wave contour (Fig. 30.39A) with increase in ejection duration may, however, be helpful in gauging progress, and following improvement of systolic function after an acute event (Fig. 30.39A) or documenting deterioration with progression of systolic dysfunction (Fig. 30.39B), providing that there has been no decrease in ejection duration (Abhayaratna *et al.*, 2006b; Weber *et al.*, 2006).

The authors are convinced that with modern tonometry, with ready availability of digital computation, and with better understanding of arterial hemodynamics, the old field of sphygmography can now usefully be extended and expanded into 'sphygmocardiography' and in other directions. Some of the approaches are given in this chapter. Others are presented elsewhere, especially in Chapter 26. The approach now is the same as that enunciated in 1872, when FA Mahomed as a medical student wrote the quotation with which this chapter commenced (Mahomed, 1872).

Studies into the clinical value of this new system are continuing. There is every possibility that Pavlov's comments will prove applicable:

It is often said and with good reason that Science advances in leaps, dependent on the introduction of new techniques. In fact, with each such new advance, we rise one step higher and a new vista with previously unforeseen opportunities becomes exposed.

FINGER PLETHYSMOGRAPHY

Photoplethysmography is widely used by anesthetists and intensivists to provide evidence of mechanical cardiac activity and for measurement of oxygen saturation. Little attention is given to detail of the shape of the waveform itself. The recording site is so remote from the heart – at the periphery of a limb – and the wave so different from that recorded in nearby arteries, that one would hardly believe that it may contain information related to ventricular function and ventricular–vascular interaction. Yet it does, as does its second derivative, the so-called 'acceleration plethysmogram'. The Japanese literature contains a great deal on the interpretation of the photoplethysmograph. A medical scientific association directed to the study of the second derivative of the finger plethysmogram has been active in Japan and growing for over two decades. These studies are virtually unknown in the Western world.

Takazawa (1987) in Tokyo recorded the photoplethysmogram and its derivatives concurrently with carotid tonometry and invasive left ventricular and aortic pressure in a series of 49 patients under control conditions and during infusion of angiotensin and administration of nitroglycerin sublingually.

Photoplethysmographic traces are very similar to pressure waves recorded in the carotid artery and bear a like relationship to the ascending aortic pressure wave. Photoplethysmographic waves recorded in the finger, however, are quite different from pressure waves recorded by tonometry at the wrist (see Chapter 10). Augmentation of the finger photoplethysmograph varies in the same way as in the carotid artery with different conditions and interventions, and bears a surprisingly consistent relationship to carotid or ascending aortic pressure wave augmentation (Fig. 20.14).

The photoplethysmogram in Japan is often analyzed in terms of its second differential, through relating the negative 'd' systolic wave that immediately precedes the beginning of diastole to the positive wave 'a' seen at the beginning of systole (Figs 20.14 and 30.40, page 634). A high 'a' wave represents a rapid upslope of the pulse, while a high 'd' wave represents rapid fall of the wave in late systole. The ratio d/a is high in youth, but decreases with aging and in people with hypertension. It decreases, too, after administration of angiotensin, but increases after the administration of nitroglycerin. The ratio d/a of the 'acceleration plethysmograph' is also related to ascending aortic augmentation under different conditions (Takazawa *et al.*, 1993a; Avolio, 2002; Hashimoto *et al.*, 2002).

In Western countries, there had been little interest among clinicians in the analysis of the peripheral arterial pressure pulse, and even less interest in the analysis of the ubiquitous photoplethysmographic pressure wave. Japanese authorities are well ahead of their Western counterparts. This has changed marginally. Chowienzyk *et al.* (1999) and Millaseau *et al.* (2002) in London have

used the photoplethysmographic tracing to examine the effects of vasoactive drugs and endothelial function, and are (re)exploring the relationship between these tracings and the sphygmographic radial and carotid pressure tracings.

Methods for analyzing the finger photoplethysmograph have been introduced and commercialized by Lerman (Bonetti *et al.*, 2004) and by Chowienczyk and colleagues (Chowienczyk *et al.*, 1999), with Lerman using a volume plethysmogram, and Chowienczyk photoplethysmography. Both concentrate on change in the waveform as a measure of endothelial function in the finger, with Lerman's approach depending on change in the waveform after cessation of arterial inflow and generation of reactive hyperemia (see Chapter 5). This method was briefly evaluated by the Framingham investigators (Hamburg *et al.*, 2011). Chowienczyk's approach was to relate body height to the time difference between late systolic peak and early diastolic peak, or amplitude of the volume pulse to this time interval. This approach is rather arbitrary because many waveforms show neither a late systolic peak nor a separate diastolic peak. Furthermore these peaks, when present, are considered by us to represent a single reflected wave, which is interrupted (or 'cut in two') by the incisura caused by aortic valve closure. There is little published with either of the plethysmographic methods, and they appear to be less sophisticated than the Japanese 'acceleration plethysmogram'. Neither has provided diagnostic or prognostic information on top of conventionally measured parameters.

It must be recognized that the finger plethysmogram does not record pressure, but the volume change in multiple small arterial vessels downstream from the digital arteries and upstream from the capillaries. Response to drugs is different also: amplitude can increase markedly after administration of nitroglycerine while amplitude of the arterial pressure waveform remains constant in the radial artery but decreases in the aorta as discussed in Chapter 28.

There is a method through which arterial pressure can be recorded from a fingertip device (Finapres and Portapres). This utilizes the servocontrolled method of Peñáz (Peñáz *et al.*, 1976), whereby a counterpressure is applied to the fingertip in order to keep pressure constant. This applied pressure, registered as a wave, is the arterial pressure in digital arteries supplying the finger. This is utilized in the Finapres and Portapres devices (Wesseling *et al.*, 1974; Wesseling, 1976; Wesseling *et al.*, 1982; Wesseling *et al.*, 1985; Wesseling *et al.*, 1993; Wesseling, 1995; Wesseling *et al.*, 1995).

RELATIONSHIP BETWEEN INDICES

The most comprehensive studies to verify non-invasive with invasive indices of arterial function have been reported from Japan, notably by Yaginuma and colleagues in Jichi and by Takazawa and colleagues in Tokyo, as discussed on p. 606.

Ascending aortic augmentation parallels that in the left ventricle (Fig. 20.12). Expressed in absolute terms the values are similar, although in relative terms (as a percentage) left ventricular augmentation is lower because of the higher amplitude of the left ventricular pulse. Absolute values are not identical (Fig. 20.12), especially when aortic flow velocity is high as in hyperdynamic states or valvular stenoses as a consequence of a Venturi effect in the aorta at peak forward flow, which creates a relative decrease in aortic pressure near that point of the wave which lies close to the foot of the reflected pressure wave. The contribution of this Venturi effect to the wave shoulder is overlooked by most in their measurements of augmentation, but in Takazawa (1987) study it resulted in aortic augmentation being 1–4 mmHg greater than left ventricular pressure augmentation.

Takazawa (1987) showed a close relationship between left ventricular augmentation and the second peak of left ventricular pressure under different conditions (Fig. 30.41). Nitroglycerin tended to cause a greater decrease in augmentation than in the second peak of aortic pressure, suggesting that the decrease in wave reflection from the periphery with nitroglycerin might be associated with an increase in left ventricular ejection and thus preservation of systolic pressure (see Figs 14.12 and 14.13). Increase in left ventricular augmentation by angiotensin caused an increase in left ventricular end-diastolic pressure indicating impairment of cardiac performance when cardiac afterload was increased (Figs 30.42 and 30.43). As predicted (see Chapter 14), the effect was most marked in the presence of impaired ventricular contractility.

MEASUREMENT OF ARTERIAL PROPERTIES WITH ARTERIAL MODELS

Measures of arterial stiffness as utilized clinically are described in Chapters 3 and 4, and elsewhere. Here we describe principles of measurement and how these relate to pressure pulse contour, when using different models of the arterial system.

Windkessel models

In the first edition of *Blood Flow in Arteries*, Donald McDonald (1960) wrote disparagingly about the Windkessel models which had evolved from the simple concept originally set out by Otto Frank (Frank, 1899; Frank, 1927). We would be untrue to the original view if we were to stray from it, but we have to concede that such models are still applied, and that concepts of arterial compliance are still used by responsible colleagues. The original objections of McDonald and Taylor (1959) however still apply – that the arterial system does not behave like a Windkessel, since pressure waves are of different shape and different amplitude at different sites, and that wave

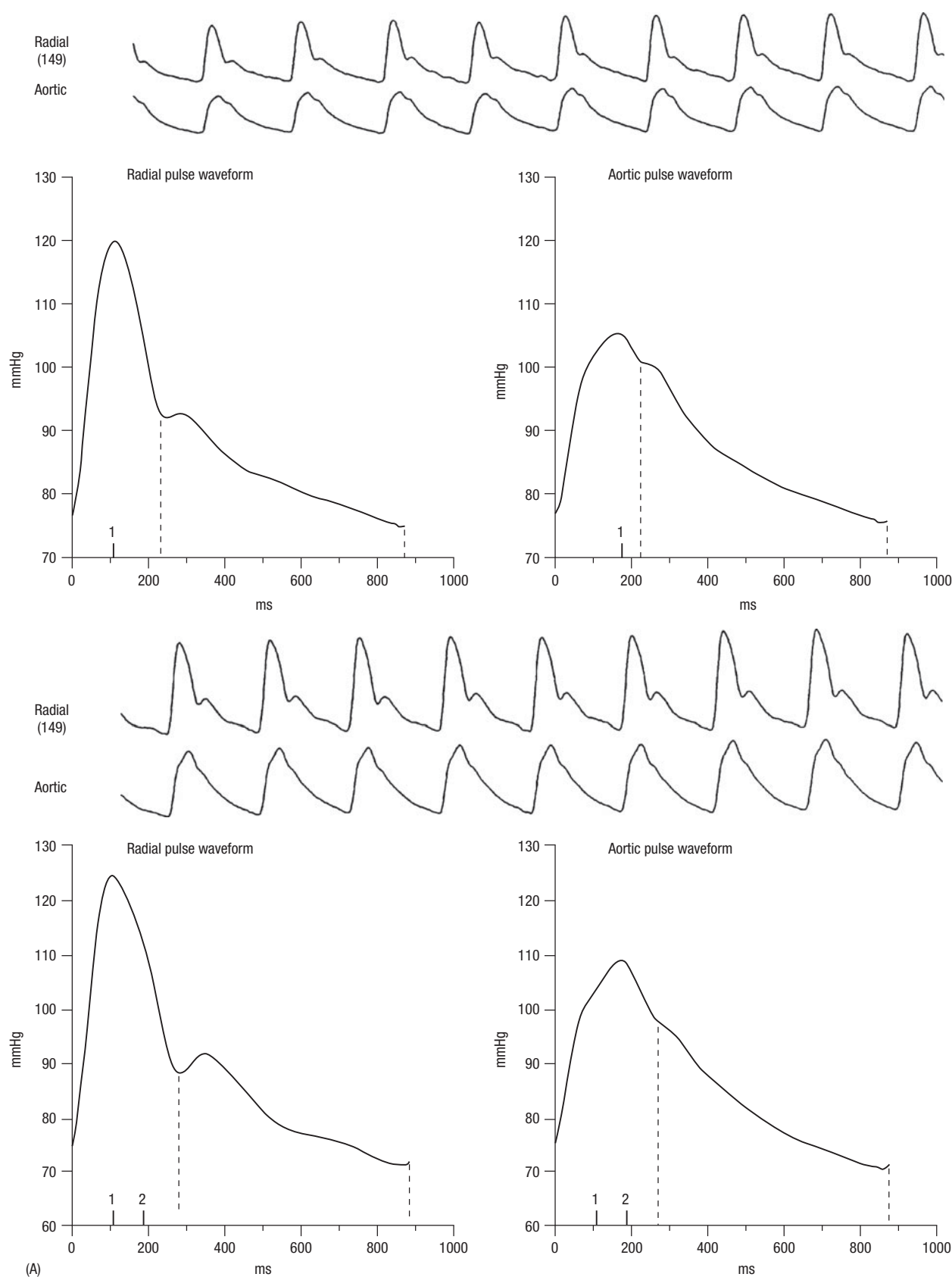


Figure 30.39 (A) Improvement in systolic left ventricular dysfunction. A segment of a sphygmocardiogram report in a 70-year-old male patient with severe left ventricular failure following coronary artery bypass surgery (above) and six months later, when he was stable and active with cardiac failure well-controlled with ACE inhibitor, nitrate, and diuretic (below). (B) Deterioration in systolic left ventricular dysfunction. A segment of a sphygmocardiogram report in a 60-year-old man (above) and five years later, following development of cardiac failure with impaired left ventricular contraction (below)

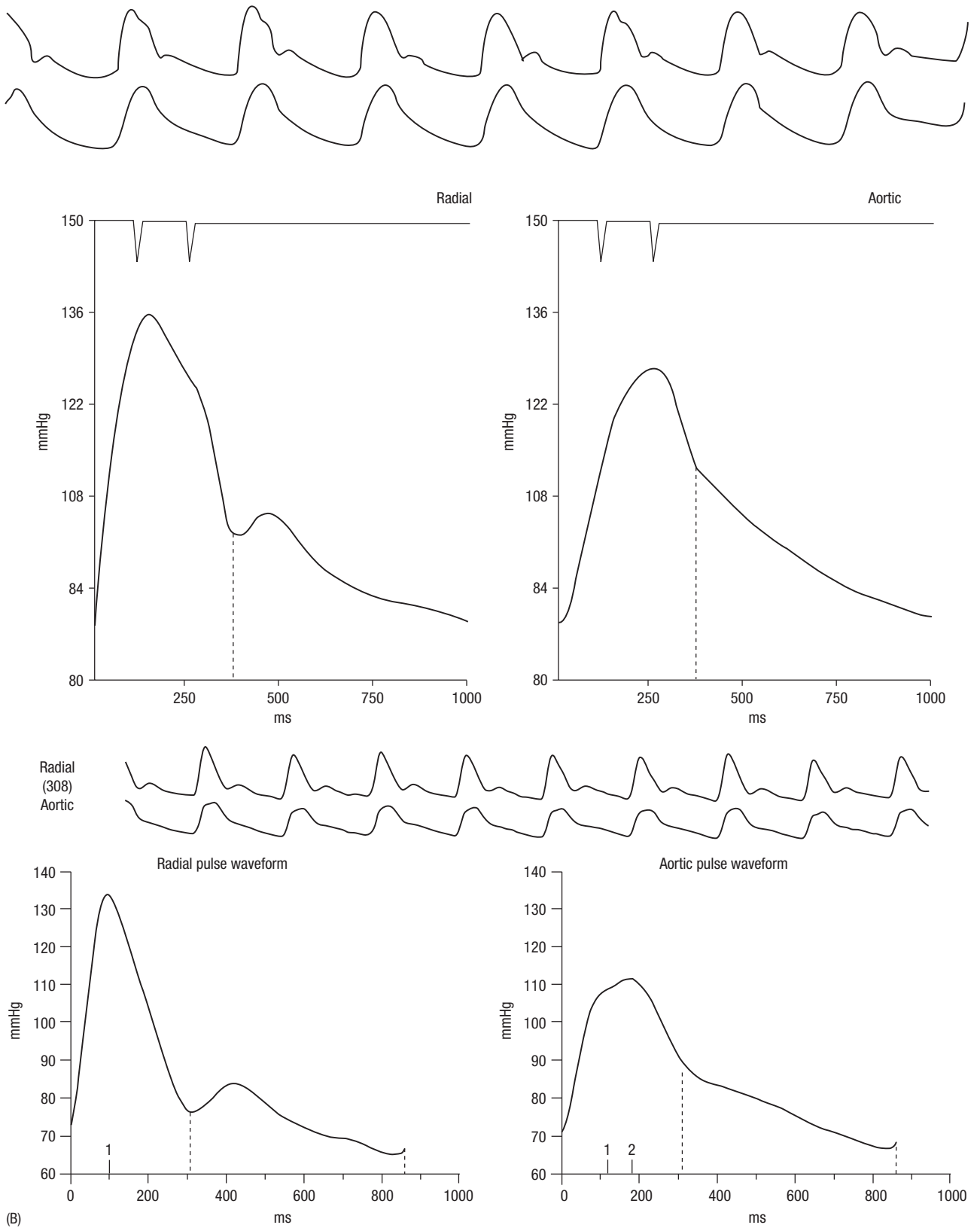


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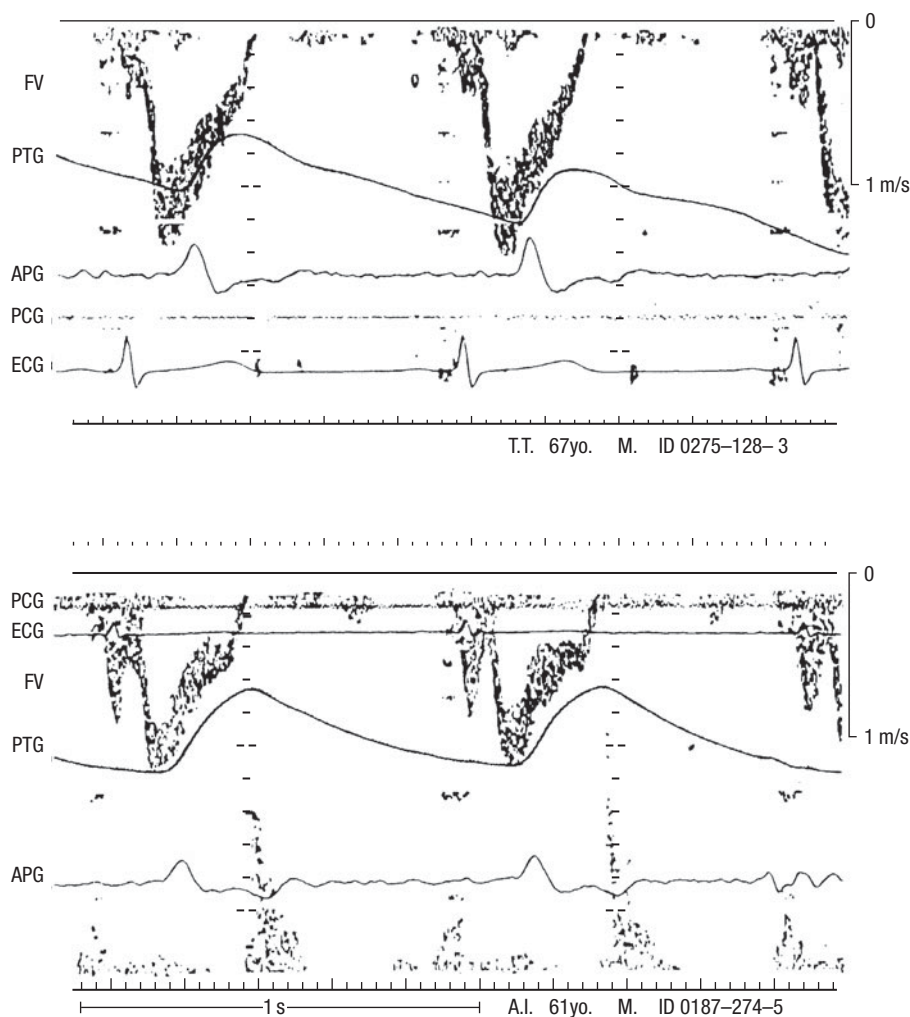


Figure 30.40 Comparison of ascending aortic blood flow velocity, measured non-invasively, in a normotensive subject (above, BP 130/80) with a hypertensive patient of similar age at similar heart rate (below, BP 190/122). Other tracings are finger plethysmograph (PTG), acceleration plethysmogram (APG), phonocardiogram (PCG), and ECG. Reproduced from Sakai *et al.* (1993)

reflection is the reality that cannot be ignored. To these we add a clinical dimension by stating the obvious – that for an approach to be useful in disease states, it must apply comprehensively during changes of heart rate or blood pressure. Nothing is perfect, but the Windkessel theories are built on the least solid ground.

Frank's original studies were on cold-blooded animals with slow heart rate and long periods of systole and diastole. In diastole, pressure fell in a near exponential fashion under different conditions so it was reasonable to equate arterial distensibility to a capacitor and resistance to a resistor, i.e. to use a Windkessel. We also have seen such waveforms in snakes and other cold-blooded animals (see Fig. 12.30) (Avolio *et al.*, 1982). When pressure waves were recorded from dogs, rabbits, humans, and other mammals, the phenomenon of wave reflection became very apparent and needed to be managed realistically. Frank attempted to do so by separating the *Grundform* of the pulse (exponential decline in

diastole) and *Grundswingung* (fluctuation caused by wave reflection).

Yin and Liu (1989) working in association with Milnor in Baltimore conceded the limitations of the Windkessel model, but sought to overcome deficiencies by substituting an exponential decline of pressure in diastole by the average value of pressure throughout diastole, which included the exponential decline and superadded effects of wave reflection. They then characterized behavior of the arterial system (capacitance, inductance, and reflection) as 'large artery compliance'. This required measurement of cardiac output and stroke volume, as well as the central aortic pressure waveform. This parameter was most recently applied by Kass *et al.* (2001) attempting to describe the effects of the advanced glycosylation end-products (AGE) cross-link breaker AT711 in older patients.

Cohn *et al.* (1995) took a related approach, and sought to derive from a peripheral, radial pressure waveform (not the aortic), the fundamental exponential diastolic decline,

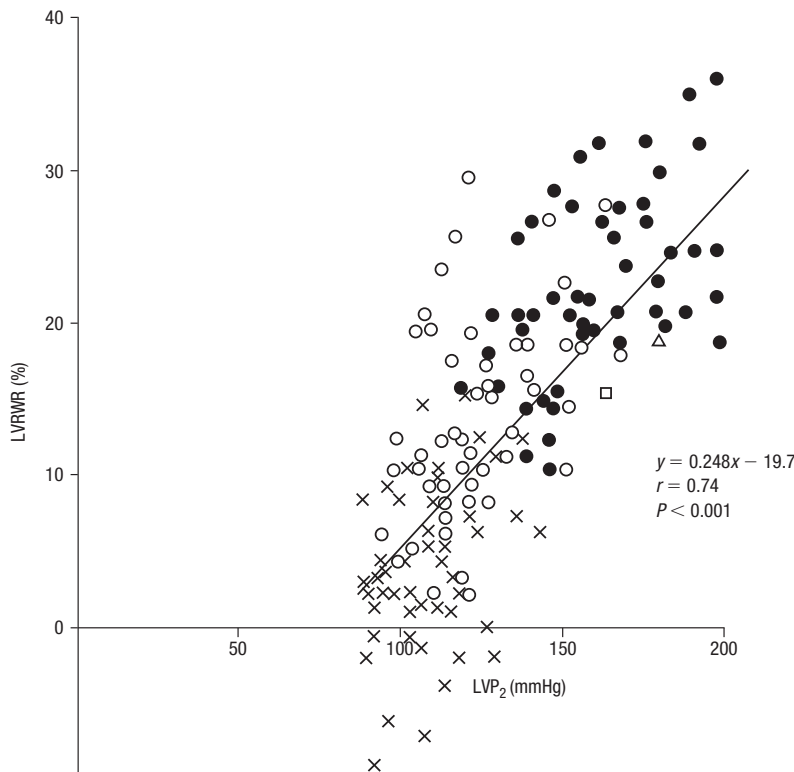


Figure 30.41 Correlation between left ventricular pressure wave augmentation (LVRWR, ordinate) and late peak left ventricular systolic pressure (LVP₂, abscissa) in 49 patients under normal conditions (○), with angiotensin (●), and with nitroglycerin (×). Reproduced from Takazawa (1987)

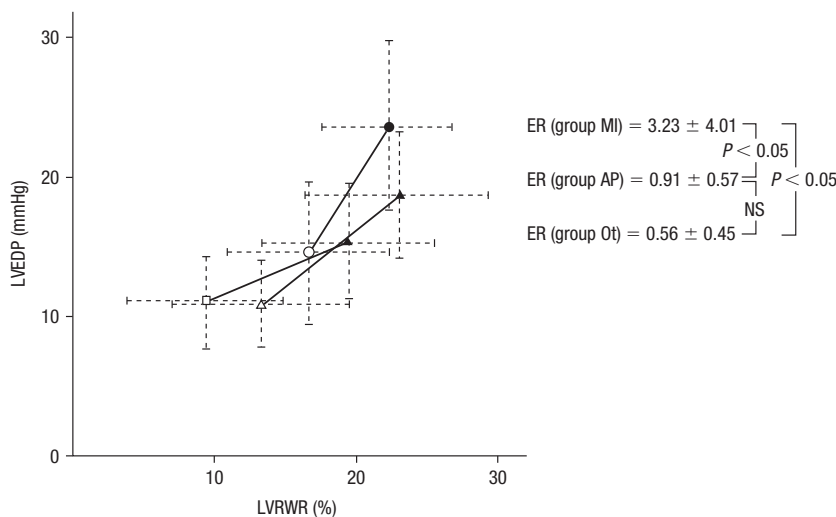


Figure 30.42 Mean changes in LVRWR and left ventricular end-diastolic pressure before and after angiotensin in the 49 patients. LVEDP before (○) and after (●) angiotensin. (○-●) Group MI ($n = 18$); (▲-△) group AP ($n = 20$); (■-□) group Ot ($n = 11$); (ER), LVEDP/LVRWR (mmHg/percent); MI, myocardial infarction; AP, angina pectoris; Ot, other diagnosis. Reproduced from Takazawa (1987)

and the fluctuation which occurred around this as a consequence of wave reflection. He also derived an equation for calculating stroke volume from the pressure pulse, and generated 'large artery compliance' from the exponential decay, and 'oscillatory compliance' from the diastolic pressure fluctuation. This technique has been used in a series of studies, and has shown changes in hypertension and with aging (decreased large artery and oscillatory compliance), but has not been able to characterize effects of therapy and has not provided outcome data. The technique has not escaped criticism (Rietzschel *et al.*, 2001a; Segers *et al.*, 2001; Manning *et al.*, 2002; Oliver and Webb, 2003; Laurent and Boutouyrie, 2007).

There are a host of conceptual, theoretic, and practical problems with this approach. Analysis is confined to the diastolic part of the peripheral pressure wave, and requires simultaneous estimation of cardiac output. It assumes that all relevant wave reflection is in diastole, and that this can be separated from the exponential pressure decay during diastole. As shown elsewhere, wave reflection usually embraces systole and diastole, and can move from diastole into systole or vice versa with changes in blood pressure and heart rate. This can be readily seen with extrasystoles, in atrial fibrillation, with drug therapy, and during the Valsalva maneuver (Figs 10.16 and 30.16). Pressure at the incisura (the beginning of diastole)

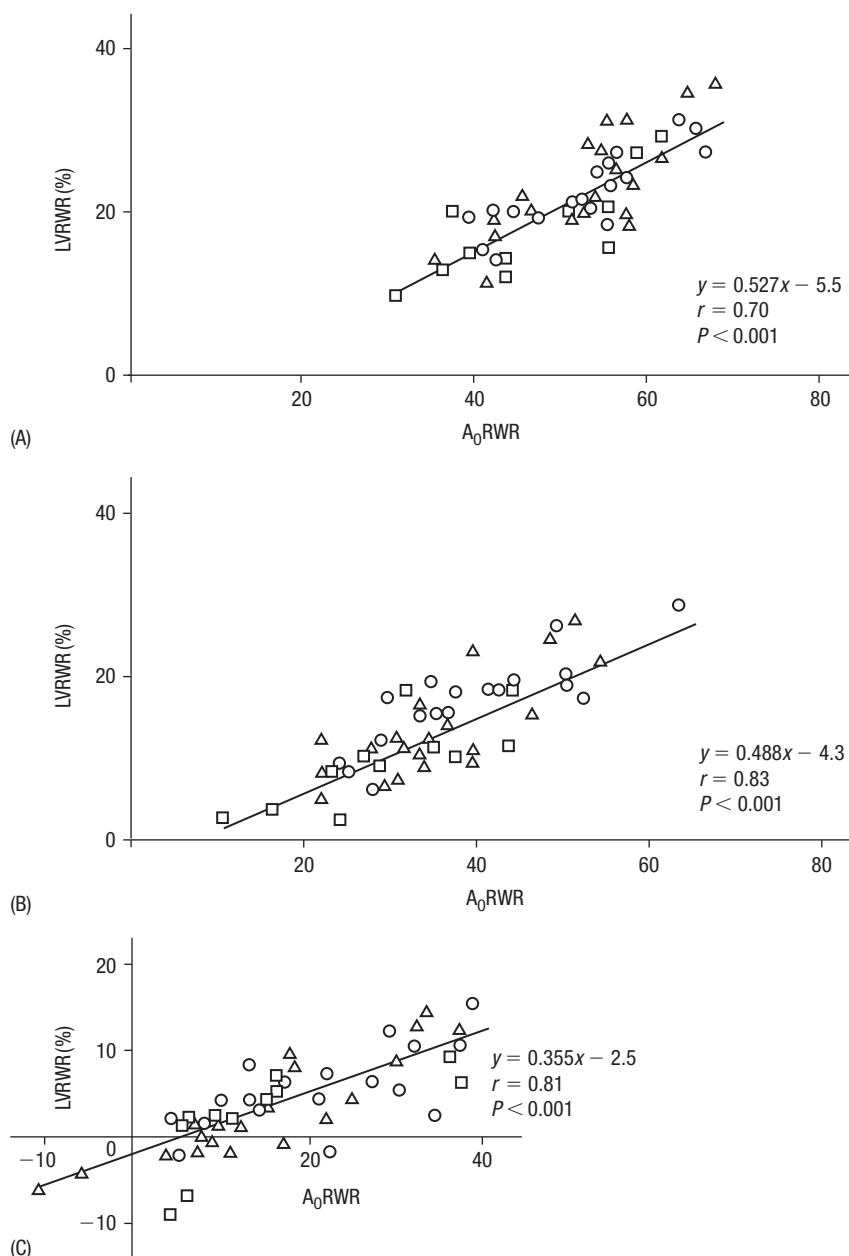


Figure 30.43 Relationship between augmentation of the left ventricular pressure wave (LVRWR, ordinate) and augmentation of the ascending aortic pressure wave (A_0 RWR, abscissa) in 49 patients studied at diagnostic cardiac catheterization under control conditions (B), during infusion of angiotensin (A) and after administration of sublingual nitroglycerin (C). (○) Patients with recent myocardial infarction; (△) patients with angina pectoris; (□) patients with other diagnoses. Reproduced from Takazawa (1987)

depends on timing of the reflected wave, is high when the reflected wave occurs predominantly in systole, and low, even less than end-diastolic pressure, when the reflected wave is strong during diastole (Figs 10.16 and 30.16). End-systolic pressure (pressure at the beginning of diastole) is typically 13 mmHg higher in the aorta than in the radial artery where pressure is recorded for the Cohn process (Table 30.2). Positive results for the ALT711 drug in modification of arterial compliance (Kass *et al.*, 2001) were only seen in data obtained with the HDI (Hypertension Diagnostics Inc., Eagan, MN, USA) transducer after generation of the aortic pressure wave from the radial pulse, using a generalized transfer function process (Chen *et al.*, 1997). Positive results were not apparent in the original (unpublished) analysis. Exponential decay of pressure during diastole can be positive (i.e. upwards) if

the pressure dip at end-systole is lower than end-diastole pressure – as can occur with tachycardia, and with no change in arterial properties (Figs 10.16 and 30.16). Other problems include absence of any data on frequency response and dynamic accuracy of the pressure transducer used with the commercial system (Hypertension Diagnostics Inc.) in contrast to that published for the Millar and Nippon-Colin tonometers (Kelly *et al.*, 1989e; Kemmotsu *et al.*, 1991a; Kemmotsu *et al.*, 1991b; Chen *et al.*, 1997), and the very poor correlation of the cardiac output, generated from pressure, against a validated technique (Cohn *et al.*, 1995). Although championed in preference to other methods (Cohn *et al.*, 2004), this approach is not among those recommended by an international group of authorities in consensus documents for determination of arterial properties (Laurent *et al.*, 2002;

London and Cohn, 2002; O'Rourke *et al.*, 2002a; Pannier *et al.*, 2002; Laurent *et al.*, 2006; Laurent and Boutouyrie, 2007).

Other Windkessel approaches have been used in experimental animals by experienced hemodynamic researchers including Westerhof *et al.* (1969), Quick *et al.* (1998), and Burattini and Campbell (1993), but these have not shown success in application to clinical problems, where comprehensiveness of application is essential.

Transmission line models

Such models were popularized by McDonald, Taylor, and Womersley in consequence of their pivotal interdisciplinary studies in London during the 1950s (McDonald, 1955; Womersley, 1957a; Womersley, 1957b; McDonald and Taylor, 1959; Taylor, 1966b; Taylor, 1973). These are discussed in Chapter 7. Such is the approach we favor, as having the strongest footing in basic science, and is the only approach that we have found useful in physiological and clinical studies (O'Rourke *et al.*, 2001; Nichols and Singh, 2002; O'Rourke *et al.*, 2002a; O'Rourke and Hashimoto, 2007). The approach concentrates on wave travel and wave reflection in a network of tubes which constitutes the systemic circulation, and describes arterial properties and left ventricular load in terms of wave travel and reflection. Disease states have been characterized by changes in wave travel and reflection as have the effects of drugs. We have been able to show that in general, arterial disease and degeneration affects the properties of the large arteries and are characterized by changes in wave velocity, and so in the timing of wave reflection as this affects the pulse waveforms in central arteries. We have also been able to show that in general, the active effects of drugs are manifest as changes in wave reflection from the peripheral arterial tree in consequence of their arterial vasodilator (or constrictor) effects on muscular arteries and arterioles. Such a theme, and evidence in support, is seen throughout this book.

Mitchell and colleagues (Mitchell *et al.*, 2001; Mitchell *et al.*, 2002) have followed this line, but with one particular twist which needs further support before acceptance. These workers have concentrated their attention on the ascending aorta, and on the possibility that new drugs (initially omapatrilat) may have unique effects on the ascending aorta. Attention to this site is completely proper, since the most proximal part of the aorta is known to pulsate most, and to be the major cushion to pulsatile flow from the left ventricle – at least in experimental animals, and in younger adult humans (Merillon *et al.*, 1982a). 'Aortic' pulse wave velocity, as measured non-invasively between the neck and groin, also excludes this important segment of the aorta, since the timing is determined from the carotid to the femoral pulse waveform. Mitchell *et al.* (Mitchell *et al.*, 2001; Mitchell *et al.*, 2002) sought to gauge the properties of the ascending aorta from the timing (foot to inflection) of the central (carotid) pressure waveform, and from

characteristic impedance determined non-invasively from the carotid pressure pulse inflection and the peak of flow (determined non-invasively by Doppler ultrasound) in the ascending aorta. We have serious reservations. This approach also excludes the ascending aorta since it considers the carotid and the ascending aortic pressure pulse to be identical. Differences between carotid and aortic pulses are described on p. 604. The time delay for reflection is usually very short, and difficult to measure accurately. One needs to distinguish between the two components of the early systolic shoulder – the early part which is convex to the left and corresponds to peak aortic flow, and the latter part which is concave to the left and corresponds to the foot of the reflected wave (see Fig. 30.44). With respect to wave reflection we have concentrated on the term Tr , which

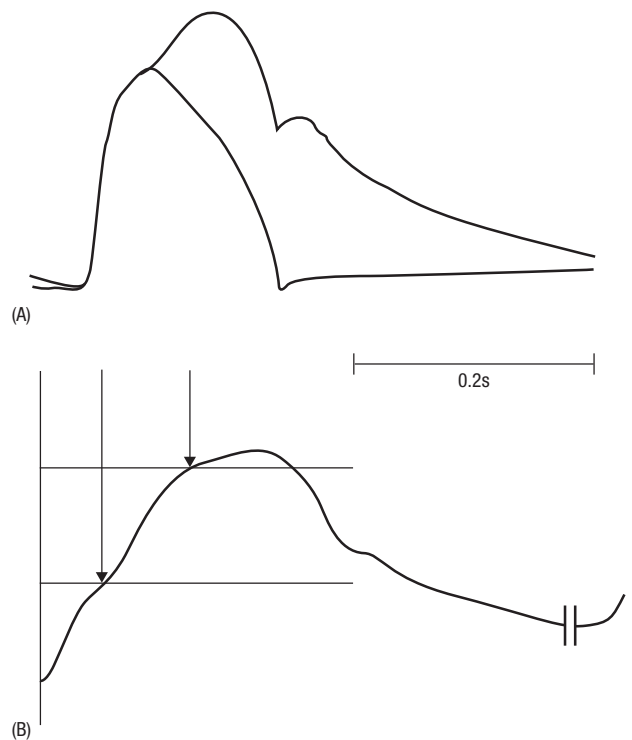


Figure 30.44 (A) Schematic representation of aortic flow and carotid pressure waves, aligned as though recorded simultaneously. The inflection point for pressure corresponds to the return of wave reflection at time T_1 , 94–127 ms after the foot of the waves. Amplitude of pressure to this point is related to the flow peak in calculation of characteristic impedance, while the subsequent rise to wave peak is a manifestation of wave reflection (after Mitchell *et al.* (Mitchell *et al.*, 2002; Mitchell *et al.*, 2003). (B) Carotid pressure waveform redrawn from Gatzka *et al.* (2001b) and Kingwell and Gatzka (2002), as shown in Fig. 26.10. The shoulder of the pressure wave was identified at 55 ms after the foot of the wave, leading to calculation of AI_x as 62 percent (implying that amplitude of reflected wave is greater than amplitude of the incident wave!). A more realistic analysis would identify the shoulder at 133 ms after the foot of the wave, with AI_x calculated as 7 percent. Arrows show T_1 at 55 ms (originally calculated by Gatzka *et al.* 2001b) and at 133 ms (by ourselves)

normally occurs after T1. Cockcroft and Wilkinson (2002) and London *et al.* (1992) have shown that this correlates well with carotid/femoral pulse wave velocity, even when generated from the radial artery pressure waveform. With respect to characteristic impedance, Mitchell *et al.* (2001) have calculated this from the first shoulder of the pressure wave (see above for reservations on accuracy) in association with 95 percent of peak aortic flow. The method is a variant of that originally applied by Dujardin and Stone (1981) for invasively recorded flow. Given the known turbulent flow in the ascending aorta, and the fact that the traced flow envelope describes maximal instantaneous flow, not averaged peak instantaneous flow across the aorta as recordable with MRI (Redheuil *et al.*, 2010), we have concerns about the accuracy and reproducibility of any determinations of characteristic impedance – and so on the parameters (forward-traveling wave and backward-traveling wave) that are calculated from this. Mitchell's approach is refreshing in its application to clinical data and warrants further examination. Our pessimistic (or realistic?) concerns come from a long struggle with a catheter-tipped electromagnetic flow meter, and with high-fidelity measurements of pressure in the left ventricle, ascending aorta and the carotid artery in humans and experimental animals (Nichols *et al.*, 1977a; Nichols *et al.*, 1977b; Merillon *et al.*, 1978; Murgo *et al.*, 1981a; Murgo *et al.*, 1981b; Merillon *et al.*, 1982a; Merillon *et al.*, 1982b; Merillon *et al.*, 1982c; O'Rourke, 1982a; Ting *et al.*, 1986; Ting *et al.*, 1995a; Ting *et al.*, 1995b). Such direct accurate studies encountered extreme difficulties on account of rapid and turbulent flow affecting the recorded aortic flow (actually LV outflow tract) waveform, and through associated Venturi effects in the ascending aorta, on the pressure wave as well. It is hard to conceive that indirect non-invasive studies would be free from such problems, or from the further difficulties of measuring characteristic impedance, then successfully unscrambling the incident and reflected components of the aortic pressure wave.

Hybrid models

Davies and colleagues from Imperial College, London (Davies *et al.*, 2007) have introduced a hybrid model (combined Windkessel and transmission line) in order to apply their wave intensity analysis to explain contour of pressure and flow waves measured non-invasively in the carotid and other arteries.

As did Frank, over a century ago, they attempt to blend the two incompatible concepts, and begin their analysis by subtracting the calculated Windkessel component from the central pressure wave component. They justify this on the basis that the secondary systolic peak of the pressure wave does not represent wave reflection, but influence of a Windkessel, with high pressure achieved because of stiffening of the aorta and proximal elastic artery. They then analyze the remaining pulse wave in terms of wave reflection from anatomically distinct sites in the coronary

arteries and systemic arterial tree. Unsurprisingly, they find that with the Windkessel component excluded, wave reflection is low for low frequency components of the central pressure waveform and high for high frequency waveforms. This is the very opposite to that seen in humans, experimental animals, and explained by Taylor (Taylor, 1966a; Taylor, 1966b) in realistic models of the arterial tree (Table 9.1).

The most recent work from this group (Baksi *et al.*, 2009) (see O'Rourke *et al.* (2010c)) seeks to explain changes in the arterial pulse with aging from a meta-analysis of selected studies where inflections on the pressure wave of patients with various diseases are described as occurring only in the early part of systole, and at the incisura. They do not recognize a late systolic shoulder in younger subjects, nor a secondary reflected wave in diastole for children or adolescents. They argue that all wave reflection occurs in systole, and none in diastole. These views ran contrary to long-standing views of Frank, Wiggers, Hamilton, Remington, and others who studied the pulse in the time domain, of McDonald, Taylor, and others who studied the pulse in the frequency domain, and the vast number since who have used both domains and a variety of other techniques, as set out in this book.

The unusual approach of this Imperial group is apparent from their use of a meta-analysis of patients with different cardiovascular disease, so as to understand an aging change, and to overlook specific studies which are devoted to aging change in normal subjects.

The most recent paper does draw attention to an issue which can be misunderstood – that the apparent return of the reflected wave identified as Tr in the SphygmoCor reports is too high to explain wave reflection when aortic PWV is high (> 12 m/s). As the SphygmoCor process works, Tr is expected to occur after the peak of flow which accounts for T1. Where the two coincide, or when Tr precedes T1, they cannot be identified separately except by very high frequency manometry which is not subject to artifact from movement of heart or catheter. The same applies, as shown in Fig. 10.5, when wave reflection returns late and cannot be separated from the incisura caused by aortic valve closure.

The future of pulse waveform analysis

The future is likely to return to the past (Braunwald, 2010; Nelson *et al.*, 2010), where pulse waveform analysis was widely used in clinical practice to determine type and severity of clinical disease, to assess prognosis and to monitor drug treatment. Instruments now available are more accurate than the old sphygmograms, and easier to use, and their application is based on better understanding of cardiovascular function.

The clinician's aim is the same as in 1872 when FA Mahomed, as a medical student in Guy's Hospital, London wrote, '... surely it must be to our advantage to appreciate fully all it (the pulse) tells us, and to draw from it all that it is capable of imparting.'

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Index

- A-mode ultrasound 180
- AAI *see* arterial stiffness index
- abdominal aorta 37–8
 - aneurysm 511–12
 - atherosclerosis 418, 501
 - elasticity 83–5, 90, 95–6
 - flow waves 64, 233, 239
 - impedance 216, 290, 431
 - length 78
 - pressure waves 102, 229–32
 - pulse wave velocity 104
- ABI *see* ankle-brachial index
- AC *see* alternating current
- acceleration and Poiseuille's equation 17
- ACCT *see* Anglo-Cardiff Collaborative Trial
- ACE *see* angiotensin-converting enzyme
- ACEI *see* angiotensin-converting enzyme inhibitor
- acetylcholine (ACh) 111–13
- adolescence
 - aortic coarctation 519, 521
 - aortic impedance 338, 432
 - cardiovascular function 417
 - emotional stress 588
 - obesity 588
 - pressure wave flow 230
 - pulse wave velocity 427
 - upper limb amplification 439
- advanced glycation end-products (AGE) 482, 525, 529
 - and ALT711 549, 634
- adventitia 56, 82, 511
 - intima-media interface 189–90
 - tunica 82
- afterload
 - left ventricular 11, 210, 354
 - power related to 324, 331–2, 384
 - reduction 322, 336, 391
- aging 105–7, 411–15
 - arterial degeneration 442–6
 - arterial flow waves 433–8
 - arterial pressure waves 438–42
 - arterial stiffness 407–8
 - arterial wall properties 105–7
 - ascending aortic impedance 430–2
 - brachial blood pressure (BP) 469–70
 - cardiac changes 415–16
 - cardiovascular (CV) function 416–17
 - elasticity 411
 - endothelial function 121–2
 - and exercise 562–3
 - functional changes in arteries 423–6
 - and heart disease 417–18
 - intrinsic change 418–23
 - pulse wave velocity 426–30
 - structural changes in arteries 417–23
 - ventricular-vascular interaction 432–3
- agonist intracoronary infusion 114–16
- AHA *see* American Heart Association
- air bubbles 131–2, 140, 178
- air chambers 137
- air pollution 501, 580, 589
- AIx *see* augmentation index
- alcohol 587–8
- aliasing 184–6, 188
 - errors 265–6
- ALLHAT *see* Antihypertensive and Lipid Lowering Heart Attack Prevention Trial
- alpha-blockers 390, 467
- ALT711 482, 549, 636
- alternating current (AC)
 - induction strain gauges 162
 - and vascular impedance 31, 175, 273
- American Heart Association (AHA) 347, 418, 554
- amlodipine 119
 - and atenolol 576, 578
 - trials 354, 466, 480
 - and valsartan 576
- amplification
 - aorta 438–9
 - arterial pressure waves 438–41
 - upper limb 440
- amplitude linearity 255–6
- analog-to-digital converter 70
- androgens 592
- anemometry 51
- aneurysms
 - aorta 53, 141, 453, 502, 505, 507–12
 - Charcot-Bouchard aneurysms 403, 454
 - intracranial 512
 - microaneurysms 373, 405
 - popliteal 195, 502
 - post-stenotic 53–4
 - thoracic 510
 - turbulence 53–4
 - ventricular 354
- angina pectoris 628
 - exercise 565–6
 - Prinzmetal 112, 521–2
 - sphygmocardiography 628
 - see also* syndrome X
- angiography, quantitative coronary (QCA) 393–5
- angiotensin
 - aortic impedance 338
 - aortic pressure 84
 - arterial disease 94
 - elastic modulus 90
 - and nitroglycerin 462, 630, 635–6
 - renin-angiotensin system 123, 353, 451
 - vasoconstriction 509, 532, 540
- angiotensin receptor blockers (ARB)
 - arterial wall 102, 804
 - cardiac failure 340, 352, 354, 356, 390
 - central arterial pressure 574, 576
 - cerebral disease 406
 - diabetes mellitus 527–8
 - endothelial function 123–4
 - in hypertension 465–7
 - in left ventricular hypertrophy 463
 - Marfan syndrome 508–9
 - systolic load 415
 - trials 480–1, 483
 - vasodilatation 456, 538, 540
 - wave reflection 223
- angiotensin-converting enzyme (ACE) 451
- angiotensin-converting enzyme inhibitors (ACEI)
 - aortic distensibility 86
 - aortic impedance 459
 - arterial wall properties 86, 102
 - cardiac failure 337, 340, 348, 352–4, 356, 390
 - cardiovascular mass 86
 - carotid pressure 604
 - central arterial pressure 574–6
 - cerebral disease 406
 - coronary thrombosis 504
 - diabetes mellitus 527–8
 - endothelial function 123
 - hypertension 465–7, 480–4
 - kidney failure 532
 - Marfan syndrome 509–10
 - systolic load 416

- angiotensin-converting enzyme
 - inhibitors (ACEI) (*continued*)
 - trials 465–7, 480–4
 - vasodilatation 456
 - vasodilation 537–8, 540
 - ventricular hypertrophy 463
 - ventricular pumping 86
- angiotensinogen 111
- Anglo-Cardiff Collaborative Trial (ACCT) 105, 218, 222
- Anglo-Scandinavian Cardiac Outcome Trial (ASCOT) 354, 412, 467, 480, 493, 533, 576, 578
- anisotropic material 58, 78, 96
- ankle-brachial index (ABI) 497
- antihypertensive drugs 493
 - and race 593–4
 - therapy 453, 529
 - see also* individual drugs
- Antihypertensive and Lipid Lowering Heart Attack Prevention Trial (ALLHAT) 124, 195, 337, 348, 354, 412–13, 465–6, 480–2, 528
- antinodes 201–2, 298, 438–9
- aorta
 - abdominal *see* abdominal aorta
 - amplification 438–9
 - aneurysms 53, 141, 453, 502, 505, 507–12
 - ascending *see* ascending aorta
 - coarctation *see* aortic coarctation
 - degeneration
 - with aging 11, 297, 563
 - in Marfan syndrome 510
 - descending 430–1, 512–13, 564
 - dimensions 35–7, 533
 - disturbed flow 47
 - flow wave contours 319
 - in hypertension 80, 90, 219, 386, 518
 - impedance patterns 302–3, 305, 313, 339
 - age-related changes 433
 - between peripheral arteries 361, 364
 - inlet length *see* inlet length
 - longitudinal movement 96
 - pressure augmentation 415, 622, 625
 - pressure wave contours 103
 - pressure-strain elastic modulus 90, 106, 455
 - pulsatile diameter 79
 - pulsatile pressure-flow relations 438, 507
 - pulse wave velocity 424, 524, 580
 - thoracic *see* thoracic aorta
 - ventricular afterload 311–13
 - wave attenuation 205, 231
- aortic aneurysms 510–12
- aortic arch, pressure waves 227–9
- aortic coarctation
 - arterial disease 512–21
 - exercise 516–17, 564–5
 - pathophysiology 517–19
 - pressure pulse contours 514–16
 - and surgery 519–20
 - vascular impedance 516
- aortic stiffness 591–2
 - hypertension 455–6
 - increased 389–90
- aortic valvular disease 52
- aortoiliac bifurcation 207–8
- apparent phase velocity 70–1
 - wave reflections 204
- applanation tonometry 598–9, 601
 - aorta 64, 68, 280
 - brachial artery 245–6, 574–5
 - hypertension 413, 625
 - principles of 570–1
 - radial artery 199, 392, 413
 - upper body arteries 239
- ARB *see* angiotensin receptor blocker
- ARIC *see* Atherosclerosis Risk in Communities
- arm *see* upper limb
- arrhythmias 267, 284, 419, 521–2, 588, 596–7
- arterial biomarkers 485–7
 - ankle-brachial index 497
 - arterial stiffness 487–93
 - carotid intima-media thickness 495–7
 - coronary artery calcium score 497
 - endothelial dysfunction 493–5
- arterial blood pressure (BP)
 - measurement 142–5
- arterial changes with exercise 560–2
- arterial counterpulsation
 - coronary circulation 390–1
 - enhanced external (EECP) 380, 391–2, 549–50, 560
 - and exercise 556–8
 - therapy 549–51
- arterial degeneration, aging 442–6
- arterial diameter 161–4
- arterial disease 94–6
 - aortic aneurysms 53, 141, 453, 502, 505, 507–12, 510–12
 - aortic coarctation 512–21
 - arteriovenous shunt 521
 - Marfan syndrome 507–10
 - non-atheromatous coronary artery disease (CAD) 521
 - smooth muscle tone 521–2
 - Takayasu arteritis 521
- arterial distension
 - endothelial function 126
 - pulse wave velocity 73–4
- arterial elastance 340–1
- arterial flow waves, aging 433–8
- arterial models of pressure pulse
 - waveform analysis 631–8
- arterial obstructions 236
- arterial occlusion-reperfusion 114–16
- arterial pressure wave, aging 438–42
- arterial pressure waves
 - aging 433–8, 433–41
 - amplification 438–41
 - central arterial pressure 571–8
 - classification of 210–15
 - growth in children 441–2
- arterial pulse, in disease 246–8
- arterial stenosis 248–9
- arterial stiffness 61
 - aging brain 407–8
- arterial biomarkers 487–93
 - C-reactive protein (CRP) 125, 534
 - embolization 506
 - endothelial function 119–21
 - ethnicity 429–30
 - gender 429
 - pulmonary arteries 360–1
 - race 429–30
- arterial stiffness index (AASI) 69
 - and C-reactive protein (CRP) 534
 - cardiovascular events 479–80
 - and radial pulsation 86–94
- arterial system, steady-state oscillation 8–10
- arterial tonometry 119–21
- arterial tree, branching points (junctions) 205–8, 216, 286, 403, 538
- arterial tree dimensions 35–7
- arterial wall
 - aging 105–7
 - arterial distension and pulse wave velocity 73–4
 - circumferential elastic modulus 83–6
 - compliance 61
 - distending pressure 59–61
 - distensibility 61
 - drugs 107–8
 - elastic dynamics 61–4
 - elastic modulus 81, 96–7
 - elastic parameters 83–97
 - elastic properties 55–9
 - length 78–9
 - phase velocity 70–1
 - pressure wave 69–70
 - propagated wave attenuation 99–109
 - pulse wave velocity 66–9
 - radial pulsation 86–94
 - segment length 78–9
 - smooth muscle cells 97–9
 - stiffness *see* arterial stiffness
 - structure 81–3
 - thickness 79–81
 - transient excitation and pulse wave velocity 71–3
 - wave propagation 64–6
- arterial wall properties, propagated wave attenuation 74–5
- arterial waveform
 - mathematical analysis 260–72
 - recording 255–9, 258–9
- arterial waveform recording
 - amplitude linearity 255–6
 - calibration 258, 268
 - drift 257
 - Fourier analysis 265–7
 - hysteresis 256
 - linearity 270–1
 - noise 256–7
 - power spectrum 267–70
 - resynthesis errors 271–2
 - significant harmonic terms 266–7
 - traveling waves 265
- arteries
 - flow disturbance 52–3
 - flow wave contours in major systemic 239

- flow waves in human upper body
 239–41, 243
 flow waves in pulmonary disease
 367–9
 incremental elastic modulus 81
 inlet length 41–8
 non-laminar flow 48–51
 pressure gradients 171
 pressure waves in human upper
 body 239–43
 pressure waves in pulmonary disease
 367–9
 pulsatile flow velocity 37–41
 pulse wave velocity 104
 structural changes with aging 417–23
 turbulence 51, 53–4
 arterioles
 blood flow 32, 34–5
 capillary exchange 402, 408
 elastic parameters 79, 87
 impedance changes 275
 plasma viscosity 23
 Poiseuille's equation 19, 31
 smooth muscle cells 97
 vasodilator responses 537–8
 see also vasodilator agents
 arteriosclerosis *see* atherosclerosis
 arteriovenous shunt 521
 arteritis, Takayasu 521
 artery size, flow pattern 171
 arthritis, rheumatoid 533–4
 ascending aorta 289–95
 flow waves 226–7
 impedance *see* ascending aortic
 impedance
 left ventricular hydraulic load
 313–15
 and left ventricular pressure 315–16
 pressure and brachial artery pressure
 243–6
 pressure wave contours 637–8
 pulse wave velocity 411
 ascending aortic impedance 289–95
 aging 430–2
 exercise 558–60
 hypertension 457–9
 as left ventricular hydraulic load
 313–15
 systolic heart failure 338–9
 ASCOT *see* Anglo-Scandinavian Cardiac
 Outcomes Trial
 atenolol and amlodipine 576, 578
 atherogenesis 54, 502–3
 atherosclerosis 499–500
 atherogenesis 502–3
 conduit artery narrowing 504
 effects of 503–4
 embolization 504–6
 and inflammation 500–1
 location of 501
 risk factors 501–2
 and thrombosis 504
 Atherosclerosis Risk in Communities
 (ARIC) 89
 atorvastatin 393, 533
 attenuation
 coefficient 75, 177–8
 constant 74
 of propagated waves 74–5
 see also damping
 audio output, continuous wave
 Doppler systems 183
 augmentation
 index (AIx) 210–12
 prediction of cardiovascular events
 93
 of pressure 11
 auscultatory sound technique 143–6
 Korotkoff method 142–3, 470
 automated systems
 auscultatory device 143
 border detection 393
 brachial artery diameter 118
 pressure monitors 144, 472
 quantitative coronary angiography
 394
 axillary artery 37–8, 118, 241
 azelnidipine 466

 B-mode ultrasound 79–81, 87, 160,
 180, 187, 189–91
 BA *see* brachial artery
 Baltimore Longitudinal Study of Aging
 (BLSA) 89
 Bayliss phenomenon 34
 beam formation, ultrasound 179–80
 Bergel's modification of Moens-
 Kortweg equation 66
 Bernoulli effect 18
 Bernoulli principle 6, 26, 235
 Bernoulli's equation 4, 26–7, 53, 260,
 309
 Bessel function 39–40, 71, 168, 173,
 204, 276
 β -blockers
 cardiac failure 354, 356
 central arterial pressure 467, 574,
 576
 hypertension 390, 456, 463, 466
 Marfan syndrome 508–9
 pulse flow magnification 103
 trials 480–1
 BHS *see* Bogalusa Heart Study
 biphasic (bisferiens) pressure pulse
 246–7
 blood flow 382–6, 388–90
 blood flow measurement
 catheter impact artifact 138–9
 catheter-manometer systems 136–7
 electrical cardiometry 150
 electrical impedance cardiography
 149–50
 electromagnetic flowmeters 151–4
 electromagnetic transducers
 158–60
 end-pressure artifact 138
 Fick principle 147
 finapres 145–6
 finometer 145–6
 FloTrac method 149
 forced-oscillation method 134–6
 frequency 132–6
 increased damping 139
 induction strain gauges 162–3
 intravascular velocity transducers
 157–60
 lithium indicator dilution (LIDCO)
 method 148
 magnetic resonance imaging 150
 manometers 129–32, 139–42
 model flow method 150–1
 optical methods 163
 partial CO₂ rebreathing method 147
 PiCCO system 148–9
 'pop' method 132–3
 portapres 145–6
 pressure pulse contour method 148
 pressure recording analytical method
 149
 resistance strain gauges 161–2
 Stewart-Hamilton dilution method
 147
 thermodilution method 148
 transpulmonary thermodilution
 method 148
 ultrasonic flowmeters 154–7
 ultrasonic methods 163–4
 ultrasonic transducers 157–8
 viscosity 24–6
 blood pressure (BP)
 central 590–1
 ESH/ESC guidelines 483–4
 indices 590–1
 measurement 142
 accuracy 470–5
 peripheral 590–1
 therapeutic trials 480–2
 vascular beds 543–4
 vasodilator therapy of cardiac failure
 482–3
 BLSA *see* Baltimore Longitudinal Study
 of Aging
 Bogalusa Heart Study (BHS) 89
 Boyle's law 132
 brachial artery (BA)
 aging 469–70
 ascending aorta pressure 243–6
 cardiovascular events 475–9
 catheterization 114
 NHANES studies 469–70, 473
 pulse wave velocity 106, 426, 455,
 599
 ultrasound with flow-mediated
 vasodilation 116
 brachial blood pressure
 aging 469–70
 and cardiovascular events 475–9
 brachiocephalic artery 37–8
 aging 246, 428, 434
 cardiac failure 354
 elastin 83
 flow waves 228–9, 233–4, 239–44,
 246, 360
 pressure waves 239–44, 360
 stiffness 549
 vascular impedance 282, 288–92,
 294, 298, 306
 Bright, Richard 5, 451, 625
 Broadbent, Sir William 596–7
 bulk modulus 59, 64–5, 132

 C-reactive protein (CRP) 125, 534
 C-spring recorder 6
 CAC *see* coronary artery calcium

- CAD *see* coronary artery disease
 CAFE *see* Conduit Artery Function Evaluation
 caffeine 583–4
 calcification 393
 aortic, age-related 393, 419–20, 495, 497
 carotid intima-media thickness 495
 diabetes mellitus 497
 hyperparathyroidism 533
 kidney disease 529–30
 Monckeberg's sclerosis 419–20
 calcium channel antagonists
 cardiac failure 354
 coronary circulation 390
 diabetes mellitus 525
 hypertension 373, 456, 463–5, 537–8, 625
 left ventricular hypertrophy 390
 smooth muscle tone 522
 calcium channel blockers (CCB) 103, 124
 aging 415–16
 arterial stiffness 480
 azelnidipine 466
 cardiac failure 353–4, 356
 central arterial pressure 574
 diabetes mellitus 527
 hypertension 466–7
 trials 480–2, 528, 576
 vasodilation 540
 wave reflection 223
 capillaries
 aging 416
 cerebral 401–2, 405
 circulation rate 147
 coronary 375
 fluid viscosity 23, 25
 impedance 274, 276, 307, 309, 313
 Poiseuille's equation 4, 14, 18–21
 pulmonary 357, 360, 400
 pulsatile flow 342
 renal 408, 529–30
 resistance 31–3
 size 35–7
 wave reflection 196
 captopril 483
 cardiac disease, and aging 417–18
 cardiac failure
 cause of 347
 concepts of 347–9
 diastolic heart 349–52, 351–2
 drug trials 353–4
 epidemic of 345–7
 exercise 563–4
 left ventricle coupling 334–41
 sphygmocardiography 628–30
 systolic heart 351–2
 vasodilator therapy 482–3
 ventricular unloading treatment 352–3
 wave reflection 351–2
 see also heart
 cardiac output (CO) 6
 aging 415–17, 423, 469
 arterial stiffness 88
 arteriovenous shunt 521
 cerebral circulation 398
 exercise 516, 557–8
 Fick technique 6, 8
 flow 32–6, 52, 339, 341
 hypertension 448–9, 451, 453
 liver circulation 408
 measurement 146–60
 myocardial ischemia 386
 renal circulation 408
 Valsalva maneuver 238
 and vascular resistance 69
 and ventricular pressure 331, 333–5, 337
 cardiac valves 52
 Cardiometrics, Doppler-tipped guidewire 158–9
 cardiomyopathy 161, 347
 hypertrophic 246, 349, 351–3, 565, 570
 impedance 559
 of overload 348
 power-afterload 331–2
 cardiovascular disease (CVD)
 arterial stiffness indices 479–80
 brachial blood pressure (BP) 475–9
 central systolic pressure 479
 diabetes mellitus 592
 prediction of 125–6
 see also arterial disease and specific diseases
 carotid artery
 aging 163, 189–91, 413, 422, 428
 atherosclerosis 504, 506
 augmentation index 437
 elasticity 83, 85–7, 95, 97
 exercise 236
 flow 48, 155
 flow waves 65, 67, 243, 245–6
 form factor 571–2
 hypertension 413, 443
 hypertrophy 461
 imaging 180, 184, 186
 impedance 92, 274, 290–1
 intima-media thickness 520
 peripheral resistance 31
 pressure waves 162, 190–1, 252, 569, 600–4
 pulsatile blood flow 398–401
 pulse wave velocity 191
 radial artery waveforms 604–6
 shear stress 47, 96, 193
 size 127, 189
 stiffness 146, 443
 tonometry 571
 waveforms 600–6
 carotid intima-media thickness 495–7
 Casio wristwatch 455
 catheter impact artifact 138–9
 catheterization
 brachial artery (BA) 115
 cardiac 6, 8
 carotid artery 8, 53
 catheters
 air bubbles 140
 arterial 27
 manometers 103, 105, 129, 136–7
 multisensor 75, 159–61
 transducers 140, 158
 CC *see* compliance coefficient
 CCA *see* common carotid artery
 CCB *see* calcium channel blocker
 CCTA *see* coronary CT angiography
 CDC *see* Center for Disease Control
 central arterial pressure 569–70
 blood pressure (BP) 590–1
 form factor 571–4
 mean pressure 571–2, 574–8
 physical principles 570–1
 physiological principles 570
 race 593
 waveform 571–8
 central elastic arteries 455–6
 central systolic pressure and cardiovascular events 479
 cerebral circulation 398–408
 cerebral edema 407
 cerebral embolism 454, 504
 cerebral infarction 454, 537
 cerebral lesions 405–6
 similarity to pulmonary lesions 405–6
 cerebral microvessels 402–4
 cerebrovascular disease 579
 CFA *see* common femoral artery
 CFR *see* coronary flow reserve
 cGMP *see* cyclic guanosine monophosphate
 characteristic impedance 88, 90
 aorta 93
 arterial stiffness 90, 92
 vascular impedance 275–7, 307–8
 Charcot-Bouchard aneurysms 403, 454
 CHD *see* coronary heart disease
 Chennai Urban Population Study (CUPS) 80
 children 5
 elastic modulus 81
 pressure waves 230
 pulmonary disease 367, 371
 pulse pressure 438, 442
 vascular impedance 300–2
 see also adolescence; infants; neonates
 chlorthalidone 465–6, 480
 chocolate 585–6
 cholesterol
 blood flow 24
 coronary heart disease 123–4, 500–1
 lowering therapy 504, 506, 536
 see also hypercholesterolemia
 chronic pulmonary thromboembolism (CPTE) 368–71
 circulation
 Fourier analysis 9–10
 Poiseuille's equation 17–19
 systemic *see* systemic arterial system/circulation
 see also cardiovascular system
 circumferential elastic modulus 83–6
 clinical trials *see* trials
 CO *see* cardiac output
 cocaine 522
 coefficient of wave reflection 204–6
 coffee 583–4
 coherence 270
 collagen fibers 56, 77, 81–5, 106–7
 color pulsed Doppler systems 187–8

- common carotid artery (CCA) 77, 80, 193
- common femoral artery (CFA) 193
- complex number notation 264
- complex waves 259
velocity 74, 176, 277
- compliance 61
coefficient (CC) 190–1
- Complior device 67, 72–3, 428
- compressive strain 56
- computed tomography (CT) 79, 160–1, 395, 405, 497
- conduit arteries 539–43
- Conduit Artery Function Evaluation (CAFE) 354, 412, 466, 480, 493, 533, 576, 578
- conduit function 544
- connective tissue disease 533–4
- CONSENSUS trial 412–13, 483–4, 576, 580
- continuous wave (CW) Doppler systems
audio output 183
demodulator 182–3
receiver 182–3
spectral analysis 184–5
transducer 182
velocity estimation 183–4
wall Doppler filter 183
- contraction
coefficient 28
smooth muscle 97–100, 387–8
- coronary artery
agonist infusion 115
angiography 393–5
angioplasty 158
atherosclerosis 115
bypass 336, 632
diameter 98, 114–15, 164
epicardial 115, 376, 378
flow 53, 115
lesions 389
narrowing 388–9
occlusion 504
perfusion 199
pressure 142
snare 379
spasm 112, 552
stenosis 378–80
thrombosis 111
- coronary artery calcium (CAC) score 497
- coronary artery disease (CAD) 94–5, 347
blood pressure 477
diagnosis 394–5, 400
and endothelial dysfunction 122
endothelial function 123–4, 126, 493–7
myocardial ischemia 386–7, 390
non-atheromatous 521
pressure-strain elastic modulus 89–90, 95, 106, 360, 455
remodeling 390
stenosis 378–80
see also myocardium, ischemic
- coronary circulation
arterial counterpulsation 390–1
blood flow limitations 388–90
blood flow to normally contracting ventricle 382–6
coronary ‘steal’ 380–2
diagnosis 391–6
flow reserve 376–8
myocardial ischemia 386–8
therapy 390
- coronary computed tomography angiography (CCTA) 395
- coronary flow reserve (CFR) 115, 376–8
invasive (iCFR) 377–8
- coronary heart disease (CHD) 23, 121, 477, 487
- coronary steal 380–2
- CPTC *see* chronic pulmonary thromboembolism
- creep 62
- cross-power spectra 269
- CRP *see* C-reactive protein
- CT *see* computed tomography
- CUPS *see* Chennai Urban Population Study
- Cushing, Harvey 406–7
- CVD *see* cardiovascular disease
- CW *see* continuous wave
- cyclic guanosine monophosphate (cGMP)
- cytokines 113, 500–1, 588
- damping 134–9
- dashpots 61–3
- DC *see* direct current; distensibility coefficient
- death
drug trials 353
rate 346, 489, 494, 496
- dehydroepiandrosterone (DHEA) 592
- dementia 406
- demodulators 182–3
- descending aorta 90–1, 430–1, 455, 510
- diabetes mellitus 524–9, 566
- diastolic heart failure 341, 349–52
early wave reflection 351
- diastolic hypertension 448, 453
- diastolic perfusion pressure 389
- diastolic pressure 26, 69, 72–3, 91
decay 88–9
measurement 143
and systolic pressure 191
- diastolic pressure-time index (DPTI) 385–6, 388–9
- dicrotic notch 69, 92, 243, 479
- diet 583–7
- dihydropyridines *see* calcium channel antagonists
- dilatation, post-stenotic 53–4
- dipyridamole 381–2, 390, 416, 457, 536–40
- direct current (DC) 31, 175
- Dirichlet condition 262
- discharge coefficient 28
- dispersion, spatial 280, 282, 314
- displacement 60–1
detection 188
pulse 67
volume 132, 140
- distending pressure 59–61
- distensibility 61
age-related changes 424
aorta 509
arterial wall 78–9, 85–9, 98, 102, 107
carotid artery 476
coefficient (DC) 190, 476
endothelial function 111, 113, 126
measurements 190–1
pulmonary artery 449
pulse wave velocity 450
- distension measurements, ultrasound 190–1
- disturbed flow 48–53
- diuretics
and b-blockers 103, 390, 464, 466–7, 480
systolic pressure 354, 632
- dobutamine 326, 337, 339, 381, 536, 542
- Doppler ultrasound 181–2
continuous wave 51–5, 181–9
pulsed wave 45, 116–17, 155–6, 185–8
volume flow 191
- Doppler ultrasound for turbulence, pulsed wave 52–3
- Doppler-tipped guidewire 158–9
- doxazosin 465–6
- DPTI *see* diastolic pressure-time index
- drift in recording systems 257
- drugs
arterial wall properties 107–8
hypertension *see* antihypertensive drugs
trials 353–4
and wave reflection 222–3
see also specific (types of) drugs
- dye 13, 45, 47, 49–52, 155, 172
- dynamic elastic modulus 63, 70–1, 85, 101
- dynamic frequency 132–3, 153–4, 160, 258
- dynamics, fluid 4, *see also* hemodynamics; hydrodynamics
- dyslipidemias 125, 419, 533
- echo resolution 178
- echocardiogram (ECG) 351, 554
- EDRF *see* endothelium-derived relaxing factor
- EECP *see* enhanced external counterpulsation
- elastance *see* elastic modulus
- elastic arteries 397, 402–3, 421–9, 442–56
- elastic body 56–8
- elastic limit 56
- elastic materials, dynamic behavior of 61–4
- elastic modulus 58, 60–3, 65–6, 70–1
aging 455
arterial wall 81
circumferential 78–9, 81
collagen vs elastin 81
dynamic 63, 70–1
incremental 58, 64, 81, 423
longitudinal 78–9, 96

- elastic modulus (*continued*)
 - Peterson's 88
 - secant 81
 - tangential 81
 - Young's 5, 88
- elastic tube, wave propagation 64–6
- elastic tube under distending pressure 59–61
- elasticity 5
 - arterial wall 55–9, 83–97
 - creep 62
 - Hooke's law 56, 58–60
 - and wave reflection 208
 - see also* viscoelasticity
- elastin fibers
 - aging 445
 - atherosclerosis 503
 - in Marfan syndrome 453, 545
- elastography 395
- elastomers 56, 63
- electric transmission line 208, 271
- electrical cardiometry 150
- electrical impedance cardiography 149–50
- electrocardiograms (ECG) 67–8, 139, 145, 257
- electrocardiography 381, 387, 519, 522
- electromagnetic flowmeters (EMF) 151–6, 172–4, 256–7
- electromagnetic transducers 158–6
- embolization 504–6
- EMF *see* electromagnetic flowmeters
- enalapril 339, 353, 483, 509
- end-pressure artifact 138
- end-stage renal disease (ESRD) 91, 213, 488–90, 494–5
- end-systolic pressure-volume relationship (ESPVR) 325–8
- endothelial dysfunction
 - arterial biomarkers 493–5
 - and coronary atherosclerosis 122
 - measurement of 114–21
- endothelial function 111–13
 - and age 121–2
 - agonist intracoronary infusion 114–16
 - arterial distensibility 126
 - arterial occlusion-reperfusion 114–16
 - arterial stiffness 119–21
 - arterial tonometry 119–21
 - brachial artery (BA) 114, 116
 - cardiovascular event prediction 125–6
 - and drugs 123–5
 - endothelial progenitor cells (EPC) 121
 - and exercise 123
 - exercise 123
 - glycocalyx 127
 - laser Doppler flowmetry 121
 - mechanotransduction 127
 - microparticles 121
 - nitric oxide 113–14
 - nitroglycerin 118–19
 - peripheral artery reactive hyperemia 116–18
 - pulse wave analysis 119–21
 - shear stress 127
 - venous occlusion plethysmography 114
- endothelial progenitor cells (EPC) 121
- endothelial-derived relaxing factor (EDRF) 97, 112, 535–6
 - see also* nitric oxide
- endothelin 111–12, 126, 502, 505
 - antagonist 359, 373
- endothelium
 - gene expression 126–7
 - nitric oxide production 113–14
 - normal 47, 49
 - shear stress 47, 126–7
 - vascular 2, 111–12
 - vasoactive substances 20
- energy
 - expenditure 323–4
 - kinetic 17–18, 26–7, 53, 138, 309
 - loss 341–2
 - myocardial 334
 - ventricular 321–5
 - see also* power
- enhanced external counterpulsation (EECP) 121, 123, 380, 391–3, 549–50
- entrance length *see* inlet length
- environment, effects of 589
- enzyme nitric oxide synthase (eNOS) 113–15, 120, 123–4, 126
- EPC *see* endothelial progenitor cells
- epicardial coronary artery 115, 376, 378
- epidemiology of hypertension 466–7
- epinephrine 234
- ergot 522
- erythrocyte velocity 21, 151
- ESPVR *see* end-systolic pressure-volume relationship
- ESRD *see* end-stage renal disease
- ethnicity 429–30
- Euler, Leonhart 4
- Euler-Fourier formula 261
- European Societies of Hypertension and of Cardiology (ESH/ESC) guidelines for BP measurement 483–4
- Evaluation and Treatment of Hypertension guidelines, Joint National Committee (JNC7) on Detection 124, 483–4, 532
- exercise 553–4
 - and aging 562–3
 - angina pectoris 565–6
 - aortic coarctation 516–17, 564–5
 - arterial changes 560–2
 - ascending aortic impedance 558–60
 - cardiac failure 563–4
 - counterpulsation with running 556–8
 - diabetes mellitus 566
 - and endothelial function 123
 - endothelial function 123
 - flow waves 236–7
 - ill effects of 566–7
 - pressure waves 236–7
 - programs 562
 - pulsatile phenomena 554–6
- experimental models of velocity profiles 44–8
- external ventricular power 316–20
- Fahraeus, Robin 18
- Fahraeus-Lindquist effect 18–19, 21
- FAI *see* flow augmentation index
- fast Fourier transform (FFT) 181
- female hormones 592–3
- femoral artery 424, 428, 439, 503, 513, 601
 - aging 424, 428, 439
 - analysis of waveforms 269–70
 - arterial wall properties 64, 67, 78, 80–1, 83, 85, 97–104
 - atherosclerosis 503
 - disease 513
 - flow 41–2, 46–7, 52–3
 - flow waves 229–30, 233–4, 238, 241–2, 250–2
 - pressure pulse waveform analysis 601
 - pulsatile pressure flow 165–6, 169–72, 174
 - vascular impedance 277, 283–91, 293, 297
 - wave reflections 192–5, 197, 206–7, 209
- FF *see* form factor
- FFR *see* fractional flow reserve
- Fick, Adolf 6
 - C-spring recorder 6
 - straight-spring manometer 6, 8
- Fick principle 6, 147–8
- Finapres finometer 145–6
- finger plethysmography 630–1
- finometers 145–6
- FloTrac method 149
- flow
 - disturbed *see* non-laminar flow
 - laminar *see* laminar flow
 - and pressure gradient 168–9
 - pulsatile *see* pulsatile flow
 - turbulent *see* turbulent flow
 - ultrasound *see* ultrasound
 - volumetric *see* volume flow
- flow augmentation index (FAI) 13
- flow disturbances
 - cardiac valves 52
 - large arteries 52–3
- flow pressure gradient 169–71
- flow reserve 376–8
- flow waves
 - abdominal aorta 233
 - arterial obstructions 236
 - arteries 239
 - exercise 236–7
 - humans 243
 - lower limb vessels 233
 - mean arterial pressure change 237–8
 - pulmonary arteries 359–60, 367–9
 - thoracic aorta 233
 - tilting 236
 - upper body arteries 239–41, 243
 - Valsalva maneuver 238–9
 - vasoconstriction 234–6
 - vasodilatation 233–4
 - wave reflection 209–10
- flow-mediated vasodilation (FMD) 13

- force-velocity relationships 324
 forced-oscillation method 134–6
 form factor (FF) 571–4
 Fourier (harmonic) analysis 8–10
 arterial wall properties 71, 90–1
 arterial waveform analysis 260–8,
 270–1
 circulation 409
 flow 39–40, 260–8, 270–1
 pressure pulse waveform 607
 pulmonary circulation 362, 373
 pulsatile pressure flow 167, 169
 vascular impedance 277, 284, 292,
 295, 307
 wave reflection 196, 204, 209
 see also fast Fourier transform
 Fourier, Joseph 5, 260
 fractional flow reserve (FFR) 378
 Framingham study
 age-related changes 40, 424, 432–3,
 438
 cardiac failure 346–50
 cardiovascular events 120, 125, 280,
 348–50, 390
 Frank, Otto 5–7, 129, 196, 226
 equation 64, 69, 73, 219, 229–30,
 260
 Frank-Starling mechanism 86, 322–3,
 325–6, 334
 frequency response determination
 132–6
 frequency response and phase linearity
 258–9
 fruit 586

 Galen (Aelius Galenus) 3, 9, 196, 225,
 247, 595, 616
 Galileo Galilei 3
 garlic 586
 gated sine-wave flowmeters 153–4
 gender
 arterial stiffness 429
 hormones 592–3
 risk factor for cardiovascular disease
 589–93
 gene expression, endothelium 126–7
 Girard 14, 19
 glyceryl trinitrate *see* nitroglycerin
 glycocalyx 127
 Goldblatt, Harry 452
 growth, arterial pressure waves
 441–2
 guidelines, World Health Organization/
 International Society of
 Hypertension (WHO/ISH) 483–4
 guinea pig 226–7, 230, 296–8, 315–17

 Hagen, GH 14
 Hagen-Poiseuille equation 14, 48
 Hagenbach, E 4, 14–15
 Hales, Reverend Stephen 3–4, 147
 harmonic analysis *see* Fourier analysis
 harmonic components 40–2, 55, 136
 complex numbers 205
 wave reflection 208–9, 270
 Womersley's equations 169–70,
 173–4
 harmonic wave components 70–1

 Harvey, William 3, 146–7, 195–6, 231,
 357, 478, 588, 611
 head and neck 227–9
 heart, external work 341–2
 heart failure
 CONSENSUS trial 412–13, 483–4,
 576, 580
 Veterans Administration study 483
 Heart Outcomes Prevention Evaluation
 (HOPE) study 124, 246, 354,
 480–4, 527
 heated-film probe *see* hot-film probe
 hematological factors 53
 hemodynamic factors 37, 53
 hemodynamics 4–5, 10, 19
 age-related changes 423, 429–30
 arterial 507, 512–14, 612, 630
 arterial wall 55
 arterial waves 158
 central 389, 429–30, 491, 493, 588,
 592
 and exercise 554, 560–1
 flow 31, 51, 55
 vascular impedance 175
 wave reflection 196
 hemorrhage, cerebral 398, 403, 406,
 453–4, 518
 hemorrhagic shock 247
 high pulse flow 405
 high pulse wave velocity 301
 high-frequency noise 257
 high-resistance
 aortic occlusion 205–6
 arterioles 201, 216, 276, 293
 Hooke, Robert 56
 Hooke's law 56, 58, 60
 non-Hookian 59
 HOPE trial 354, 412, 480–4
 hormones, gender 592–3
 HOT study 481–2, 527–8
 hot-film anemometry 51
 humans
 flow waves 243
 input impedance 300–6
 pressure waves 241–3
 hydralazine
 cardiac failure 301–2, 352–3
 hypertension 456, 464–5, 536
 hydraulic load *see* load
 hydrodynamics
 flow 26, 35, 49, 175
 and Poiseuille's equation 17
 5-hydroxytryptamine (5-HT) *see*
 serotonin
 hypercholesterolemia 533
 and atherosclerosis 501, 503
 hyperemia, reactive 114, 116–18, 120,
 222, 376–80
 hyperparathyroidism 533
 hypertension
 ascending aortic impedance 457–9
 drugs *see* antihypertensive drugs;
 sphygmocardiography
 epidemiology 466–7
 historical viewpoint 451–3
 isolated systolic (ISH) 207, 211, 220,
 453
 large artery stiffness (human) 454–7

 left ventricular hypertrophy 461–3
 mechanisms of 449–51
 pathology 453–4
 peripheral muscular arteries 456–7
 pressure wave augmentation 459–61
 pulmonary *see* pulmonary
 hypertension
 sphygmocardiography 625–7
 systolic *see* systolic hypertension
 therapy 407, 463–6
 Hypertension Optimal Treatment
 (HOT) study 481–2, 527–8
 hypertrophic cardiomyopathy 160–1,
 246, 349, 570
 hypertrophy, left ventricular *see*
 ventricles
 hysteresis 256

 iCFR *see* coronary flow reserve,
 invasive
 ICG *see* impedance cardiography
 iliac artery
 flow wave 38, 44, 53, 84, 438
 impedance 305
 pressure waves 138, 171, 229–31
 see also aortoiliac bifurcation
 illegal substances 589
 iloprost 373
 image generation by ultrasound 180–1
 impact pressure 27
 impedance 273–6
 in aortic coarctation 564
 ascending aorta *see* ascending aorta
 cardiography (ICG) 149–50
 characteristic *see* characteristic
 impedance
 in exercise *see* exercise
 experimental animals, pulmonary
 arteries 361–5
 humans, pulmonary arteries 365–7
 longitudinal 175, 273, 276–7
 modulus 9
 peripheral arteries 283–8
 pulmonary arteries 361–5
 terminal 204–5, 208, 273–6, 278,
 306–7
 theory of 276–8, 282–3
 vascular input 274–83, 308–9
 impedance modulus, units 308–9
 implantable hemodynamic monitor 141
 impulse gradient 167
 IMT *see* intima-media thickness
 incisure 69, 73
 incremental elastic modulus 81
 indapamide 459, 480, 493
 indicator-dilution method 147
 indices
 blood pressure (BP) 590–1
 pressure pulse waveform analysis
 631
 race 593
 indirect measurement of arterial blood
 pressure (BP) 142–5
 inducance 160
 strain gauges 162–3
 inertia 6, 165–6, 171, 273
 infants 211, 251, 367, 441
 neonates 437

- infarction
 - cerebral 454, 537
 - myocardial *see* myocardial infarction
- inflammation 500–1
- inflection point 88, 91, 93
- inlet length 41–8
- input impedance
 - aortic and ventricular afterload 311–13
 - apparent phase velocity spectra 71
 - ascending aorta 289–95, 313–15
 - central aorta 288–9
 - determination of 306–9
 - and external ventricular power 316–20
 - humans 300–6
 - moduli spectra 71
 - peripheral arteries 288–9
 - pulmonary artery 295–6, 316
 - systemic arteries size 296–300
 - ventricular hydraulic load 313–16
- instantaneous pressure and flow 328–31
- International Society of Hypertension (ISH) 88, 473
- interventions 233–9
- intima, tunica 82, 495
- intima-media thickness (IMT) 80, 189–90
- intra-aortic balloon counterpulsation (IABP) 334, 337, 352, 380, 391, 550
- intracerebral capillaries 401–2
- intracerebral veins 401–2
- intracranial aneurysms 512
- intracranial hypertension 406
- intramyocardial coronary artery 389
- intravascular ultrasound (IVUS) 391–3
- intravascular velocity transducers 157–60
- intrinsic changes with aging 418–23
- ischemic myocardium *see* myocardium
- ISH *see* International Society of Hypertension; isolated systolic hypertension
- isolated systolic hypertension (ISH) 453, 480
- isotropic material 57–60, 78
 - anisotropic 58, 78, 96
- IVUS *see* intravascular ultrasound
- Joint National Committee (JNC7) on Detection
 - BP measurement 472
 - Evaluation and Treatment of Hypertension guidelines 124, 483–4, 532
- junctions *see* branching points
- kidney *see* renal
- kidney disease 529–30
- kinematic viscosity 39, 43, 49, 168, 205
- kinetic energy 18, 26–7
 - and end-pressure artefact 138
- Korotkoff auscultatory method 142–3, 470
- L-monomethyl arginine (L-NMMA) 113–14, 126, 376
- lacunar infarcts 243, 403–7, 454
- Lamé's equation 59
- laminar flow 13–14
 - breakdown *see* non-laminar flow
 - disturbance 52
 - Poiseuille's equation 17
 - Poiseuille's equations 14
 - pulsation 37
 - restoration of 50
 - Reynolds' number 50, 174
 - Womersley's equations 174
- Landois 6
- large arteries
 - Reynolds' number 29–30, 35, 52–3
 - stiffness 454–7
- laser Doppler anemometry (LDA) 51
- laser Doppler flowmetry (LDF) 121
- lateral pressure 27
- LBBB *see* left ventricular bundle branch block
- LDA *see* laser Doppler anemometry
- LDF *see* laser Doppler flowmetry
- LDL *see* low-density lipoprotein
- left ventricles 322–3
 - cardiac failure 334–41
- left ventricular bundle branch block (LBBB)
- left ventricular hypertrophy 461–3
 - hypertension 461–3
 - reversal of 463
- left ventricular load, and wave reflections 212–15
- leg *see* lower limb
- length
 - arterial wall 78–9
 - see also* stature
- lercanidipine 539
- Lewis, Thomas 5, 569, 597
- life insurance industry 126, 448, 452, 596
- LIFE (Losartan Intervention For Endpoint reduction in hypertension) study 124, 246, 354, 412, 466, 480–1, 483–4, 527, 576
- lifestyle 581–94
- limbs *see* lower limb; upper limb
- Lindqvist, Johan 18–19, 21
 - Fahraeus-Lindqvist effect 18–19, 21
- linear elastic theory 83
- linearity 270–1
- lipid-lowering drugs *see* statins
- lipids
 - arterial wall 47, 592
 - build-up 112, 122, 499
- lipoproteins 500–3, 533
- liquid-filled catheter-manometer systems 133, 136, 138, 140
- lisinopril 466, 626
- Lissajou loop 154, 256–7
- lithium indicator dilution (LIDCO) method, blood flow measurement 148
- liver circulation 408
- load
 - afterload
 - left ventricular 11, 210, 354
 - power related to 324, 331–2, 384
 - reduction 322, 336, 391
 - ventricular hydraulic 313–16
- local pulse wave velocity 191
- longitudinal elastic modulus 96–7
- longitudinal impedance 273–4, 276–7
- longitudinal strain 56–7, 59
- longitudinal wave 177
- losartan 466, 481, 508–9, 576–7
- low-density lipoprotein (LDL) atherosclerosis 393, 500–1
- immunophoresis 24
- and shear stress 48, 127
- lower limb vessels
 - amplification 438–9
 - flow wave contours 233
 - pressure contours 229–32
- Ludwig, C 6, 165
- LV *see* left ventricular
- M-mode sonography 69, 79, 87, 156, 180
- McDonald, DA 8, 10
- Mackenzie, Sir James 2, 6–7
- magnetic resonance imaging (MRI) 66–7, 150
- Mahomed, Frederick Akbar 5–6
- manometers 2, 6, 129–32, 139–42
 - catheters in *see* catheters
 - fluid filled 611
 - micromanometers 8, 103, 105, 601
 - Poiseuille's equation 6, 14
 - see also* sphygmomanometers
- MAP *see* mean aortic pressure
- Marey, EH 5–6, 129, 144, 435, 596–7, 611, 616, 624
- Marfan syndrome 507–10
- mathematical analysis of arterial waveforms 260–72
- matrix metalloproteinase (MMP) 511
- Maxwell model 61
 - combined Voigt and 62
 - and Voigt model 62
- MCP *see* monocyte chemoattractant protein
- mean aortic pressure (MAP) 32–3, 85, 590
- mean arterial pressure
 - in aortic coarctation 513
 - central 571–2, 574–8
 - flow waves 237–8
 - in hypertension 449, 454, 508
 - interventions causing changes 546
 - pressure waves 14, 237–8
- mechanics 13
 - arterial wall 55–6, 86, 89, 509
 - atherogenesis *see* atherosclerosis
 - fluid 45, 48, 127
- mechanotransduction 127
- media, tunica 82, 495
- medical therapies 544–9
 - see also* drugs
- Medtronic implantable hemodynamic monitor 141
- MERIT-HF study 353
- MESA *see* Multi-Ethnic Study of Atherosclerosis

- metabolic diseases 523–4
 connective tissue disease 533–4
 diabetes mellitus 524–9
 hypercholesterolemia 533
 hyperparathyroidism 533
 kidney failure 529–33
 MI *see* myocardial infarction
 microparticles (MP) 121
 microvascular angina 375, 389
 migraine 522
 Millar catheter-tip catheter-manometer 138–9, 141–6, 158–61, 569–70, 598–603
 Mills catheter-tip probe 159
 MMP *see* matrix metalloproteinase
 model flow method 150–1
 Moens-Kortweg equation 64–6, 68, 74, 83, 176, 206, 208, 277
 Monckeberg's sclerosis 419–20
 monitoring therapeutic strategies 551
 monocyte chemoattractant protein (MCP) 500, 502
 mortality *see* death
 motion
 complex number notation 205, 264–5
see also exercise; Newton
 MP *see* microparticles
 MRFIT 412, 448, 467, 469, 477
 MRI *see* magnetic resonance imaging
 Multi-Ethnic Study of Atherosclerosis (MESA) 87
 multigate pulsed Doppler systems 187
 Multiple Risk Factor Intervention Trial (MRFIT) 412, 448, 467, 469, 477
 muscular arteries 543–4
 myocardial infarction (MI) 115
 myocardial regeneration 635
 risk of 497, 524, 566
 myocardial ischemia
 coronary circulation 386–8
 sphymocardiography 628
 myocardium
 blood flow 20, 210, 383–6
 cells *see* myocytes
 contraction *see* contraction
 oxygen requirements 383
 wave reflection 215–16
 myocytes 348, 353, 355, 453
 myogenic response 34, 97, 527
- National Health and Nutrition Examination Survey (NHANES) studies 469–70, 473
 Navier, M 14
 Navier-Stokes equations 14, 27, 171, 270
 neonates 150, 437
 New York Heart Association (NYHA) 340, 553, 563
 Newton, Sir Isaac 3–4
 second law of motion 130, 135
 third law of motion 198
 velocity of wave propagation 64
 Newtonian fluids 13–14, 17, 20, 22–3, 192
 non-Newtonian fluids 22–5
 viscosity 21–2, 39, 174
- nifedipine 86, 119, 296, 339, 381
 hypertension 541, 625, 627
 wave reflection 330
 nitrates *see* individual drugs
 nitric oxide (NO) 113–14
 nitroglycerin (NTG) 118–19
 nitroprusside 113, 116, 146
 cardiac failure 339, 559
 coronary stenosis 364, 381
 hypertension 458, 508, 510, 540–2
 nodes 202, 208, 229
 antinodes 201–2, 298, 438–9
 noise 256–7
 non-atheromatous coronary artery disease (CAD) 521
 non-invasive pressure recordings 249–53
 non-laminar liquid flow 48–53
 non-linearity 9–10, 22
 detection of 271
 elasticity 74, 83, 86, 88
 Fourier analysis 9
 input impedance 150, 282–3, 309
 phase 259
 pressure-diameter 103, 449
 stress-strain 58, 192
 non-Newtonian fluids 22–5
 norepinephrine
 impedance 280, 287–8, 296, 308
 pressure and flow waves 74, 97–9, 107–8, 234, 285
 NTG *see* nitroglycerin
 Nuvel (Numed) velocity catheter 158
 NYHA *see* New York Heart Association
- occlusion 73
 aortic 205–6, 450
 coronary 504, 522, 566
see also stenosis; venous occlusion
 plethysmography
 occlusion-reperfusion 114–15
 OCT *see* optical coherence tomography
 omapatrilat 482, 537, 546, 549
 optical coherence tomography (OCT) 395
 optical methods for arterial diameter 163
 oscillations
 equations 130
 sinusoidal 204
 steady state 8–10
 oscillatory compliance 88, 635
 oscillatory pressure gradient 165–6 and flow 168
 Osler, William 407–8, 418, 424, 444, 446, 499, 579, 597
 oxygen
 consumption 95
 delivery 25
 myocardial 376–7, 383
 myocardial supply 215
- PA *see* pulmonary arterial
 PAD *see* peripheral arterial disease
 palpography 395
 papillary muscle 312, 324, 339, 342
 parathyroid gland 533
 partial CO₂ rebreathing method 147
- Pascal's law 59
 PAT *see* pulse amplitude tonometry
 patent ductus arteriosus 28, 374, 388, 405
 pathophysiology
 aortic coarctation 517–19
 therapeutic strategies 535–43
 PDE5 *see* phosphodiesterase 5
 perfusion pressure 58, 376–7
 perindopril 353, 480, 509
 peripheral arterial disease (PAD) 497, 579
 peripheral arteries
 input impedance 288–9
 reactive hyperemia 116–18
 resistance *see* resistance
 stiffness 456–7
 peripheral blood pressure 590–1
 peripheral resistance unit (PRU) 33
 peripheral vascular resistance 31–5
 PET *see* positron emission tomography
 phase
 distortion 136, 162, 259, 266
 errors in 136
 linearity 129, 255, 258
 shift 70, 139, 184, 202, 204, 206–7, 265–6, 271
 velocity 70–1, 204
 phonocardiography 634
 phosphodiesterase 5 (PDE5) 111–12, 114, 124–5, 373, 535–6
 photoplethysmography 114, 251–3, 392, 455–62, 630–1
 physical principles
 central arterial pressure 570–1
 ultrasound 177–8
 physiological principles 570
 PI *see* pulsatility index
 PiCCO system 148–9
 Pitot tube 6, 27, 44
 plaque
 atherosclerotic 31, 122, 392–3, 503–4
 development 2, 23
 imaging 395
 rupture 409, 501, 503–4
 plasma, viscosity of 23–4
 plasticity 562
 plethysmography, finger 461–2, 479, 630
 poise 15–16
 Poiseuille, Jean L onard Marie 4
 Poiseuille's equation 58
 circulation 17–19, 30–2, 37, 168–9, 171, 176, 282
 manometer theory 6
 Poisson ratio 57, 59, 63, 65, 78, 97
 popliteal aneurysms 195, 502
 Portapres monitor 145–6
 positron emission tomography (PET) 377, 503
 post-stenotic dilatation, turbulence 53–4
 posture 236, 397
 power
 spectrum 267–70
see also energy
 power-afterload relationship 331
 PP *see* pulse pressure

- PPH *see* primary pulmonary hypertension
- PRAM *see* pressure recording analytical method
- pravastatin 24–5, 393, 533
- prazosin 335, 390, 456, 465, 536, 545
- pressure
- impact 27
 - lateral 27
- pressure gradient
- artery size 171
 - flow 168–71
 - pulse frequency 171
 - time-derivative method 171–2
 - vascular impedance 175–6
 - Womersley equations 174–5
- pressure measurement, vascular
- impedance determination 309
- pressure pulse contour method 148
- pressure pulse waveform analysis 595
- amplification, upper limb 555–6
 - aortic coarctation 514–16
 - applanation tonometry 598–9
 - arterial models 631–8
 - carotid artery waveforms 600–4
 - and aortic pressure waveforms 606
 - and radial artery waveforms 604–6
 - finger plethysmography 630–1
 - indices 631
 - radial artery waveforms 599–600
 - and aortic pressure waveforms 606–16
 - and carotid artery waveforms 604–6
 - sphygmogram 595–8
 - sphygmocardiography 616–30
- pressure recording analytical method (PRAM) 149
- pressure wave, foot velocity 69–70
- pressure waves
- abdominal aorta 229–32
 - amplification *see* amplification
 - arterial obstructions 236
 - arterial wall 69–70
 - augmentation 459–61
 - biphasic (bisferiens) 246–7
 - blood flow measurement 148
 - classification of arterial 210–15
 - in disease 367–9
 - exercise 236–7
 - humans 241–3
 - hypertension 459–61
 - lower limb vessels 229–32
 - in major systemic, arteries 239
 - mean arterial pressure change 237–8
 - non-invasive recordings 249–53
 - pulmonary arteries 359–60, 367–9
 - reflection 209–10
 - thoracic aorta 229–32
 - tilting 236
 - upper body 239–43
 - Valsalva maneuver 238–9
 - vasoconstriction 234–6
 - vasodilatation 233–4
- pressure-flow relationships 3–4, 10, 14–19
- pressure-strain elastic modulus 89–90, 106
- pressure-velocity catheters 159–60
- pressure-volume relationships 325–6, 383
- PRF *see* pulse repetition frequency
- primary pulmonary hypertension (PPH) 372–4
- Prinzmetal angina 112
- smooth muscle tone disturbances 521–2
- propagated wave attenuation 74–5, 99–109
- propranolol 456, 459, 508–9
- prostacyclin 11, 20, 359, 373, 505
- PRU *see* peripheral resistance unit
- pulmonary arteries
- arterial stiffness 360–1
 - disease 367–71
 - flow waves 359–60, 367–9
 - impedance 295–6
 - and right ventricular load 316, 371–2
 - impedance (experimental animals) 361–5
 - impedance (humans) 365–7
 - lesions 405–6
 - pressure waves 359–60
 - in disease 367–9
 - pulse wave velocity 360–1
 - right ventricular load 316, 371–2
 - stenosis 53, 141–2
 - valves 52–3
- pulmonary circulation 357–9
- pulmonary hypertension
- experimental models 369–71
 - primary 159, 370–4, 3669
 - severe 368
- pulsatile flow
- carotid artery 398–402
 - cerebral microvessel 403–4
 - cerebral microvessels 402–3
 - cerebral and pulmonary lesions 405–6
 - exercise 554–6
 - intracerebral capillaries 401–2
 - intracerebral veins 401–2
 - velocity 37–41
- pulsatile pressure-flow relations 168–76
- pulsatility index (PI) 397–402, 406, 408, 494
- pulse
- amplification *see* amplification
 - attenuation/damping *see* attenuation
 - pulse amplitude tonometry (PAT) 120
 - pulse frequency 171
 - pulse pressure (PP) 493–6, 532, 571
 - pulse repetition frequency (PRF) 181, 185–7
 - pulse wave velocity
 - arterial distension 73–4
 - arteries 104
- pulse wave velocity (PWV)
- age-related changes 593
 - aging 426–30
 - aortic 523
 - arterial wall 66–9
 - femoral 525, 576
 - pulmonary arteries 360–1
- pulse waveform analysis (PWA) 119, 196
- endothelial function 119–21
- future of 638–9
- pulsed Doppler systems
- color 187–8
 - multigate 187
 - RF processing 188–9
 - single gate 185–7
- pump function curves 331–3
- pump systems 137
- PWA *see* pulse waveform analysis
- PWV *see* pulse wave velocity
- quantitative coronary angiography (QCA) 393–5
- race
- and antihypertensive drugs 593–4
 - arterial stiffness 429–30
 - indices 593
 - risk factor for cardiovascular disease 593–4
- radial artery 64
- arterial stiffness indices 86–94
 - arterial wall properties 81, 86–94
 - pressure waves in 239, 249
 - tonometry 119, 199, 245
- radial pulsation 86–96
- radiofrequency (RF) 180
- radiofrequency (RF) processing 188–9
- radius *see* diameter
- ramipril 480–1, 576
- Raynaud's phenomenon 522
- RBC *see* red blood cell
- reactance 175–6, 276
- reactive hyperemia 79, 114, 116–18, 233, 236
- REASON (pREterax in regression of Arterial Stiffness in a contrOLled double-bliNd) study 412, 480, 493, 496, 576
- receivers 182–3
- recombinant tissue plasminogen activator (rtPA) 111
- red blood cell (RBC)
- aggregation 20, 25
 - blood viscosity 25
 - deformability 20
 - flow 31
- reflection *see* wave reflection
- reflection coefficient 177, 204–6
- relative risk (RR) 488, 495
- relaxation time 82
- REMATCH study 355
- renal artery 53, 233, 235, 306, 408
- renal circulation 408
- renal failure
- cardiovascular events 529–33
 - end-stage 479–80
 - metabolic diseases 529–33
- renin 408, 451
- angiotensin system 94, 123, 353, 451, 483, 532
 - inhibitors 123
 - suparenin 235
- resistance
- peripheral vessels 31–5
 - and Poiseuille's equation 58, 176
 - strain gauges 161–2

- see also high-resistance arterioles;
 strain gauges
 resonance 219, 239, 241
 rest
 arterial pressure 237
 cardiovascular function 564
 myocardial ischemia 387
 pulmonary vascular impedance 366
 resynthesis errors 271–2
 REVERSAL (Reversal of Atherosclerosis
 with Aggressive Lipid
 Lowering) trial 393
 Reynolds' number 43, 49–51
 and laminar flow 174
 in large arteries 29–30, 35, 52–3
 Reynolds, Osborne 49
 RF see radiofrequency
 rheology 19, 21–2
 rheumatoid arthritis 533–4
 rigidity 23, 59, 106
 risk factors
 for atherosclerosis 501–2
 for cardiovascular disease 581–94
 Riva-Rocci sphygmomanometer 5, 142,
 448, 596
 RR see relative risk
 rtPA see recombinant tissue
 plasminogen activator
 running 555, 566–7
 counterpulsation 556–8
 long-distance and 553–4

 SAC see systemic arterial bed
 St Venant model 62
 salt 586–7
 saphenous artery 83, 171, 206, 217, 226
 SAVE trial 483–4
 sawtooth function 261
 secant elastic modulus 81
 secant viscosity 22
 second law of motion 130, 135
 secondary intracranial hypertension
 (SIH) 401
 secondary medial effects 505
 secondary wave reflection 218–20
 segment length 78–9
 serotonin
 analogs 522
 coronary narrowing 388
 endothelial function 113
 impedance 296, 308
 pulse wave velocity 360–5, 369
 smooth muscle tone 522
 SFA see superficial femoral artery
 shear modulus 59, 97
 shear rate 20–5, 46–7, 192–3
 estimating system (SRES) 47
 shear strain 57, 97
 shear stress 19–23
 aorta 53–4
 in disease 127
 endothelium 126–7
 sheep 90, 226–8, 230
 SHEP see Systolic Hypertension in the
 Elderly Program
 SHM see simple harmonic motion
 shock 247, 337, 352
 signal-to-noise ratio (SNR) 188

 significant harmonic terms, arterial
 waveform recording 266–7
 SIH see secondary intracranial
 hypertension
 sildenafil 111, 113
 nitroglycerin with 535–6, 538, 546
 primary pulmonary hypertension
 373
 simple harmonic motion (SHM) 263
 sine wave meter 153–4
 single gate pulsed Doppler systems
 185–7
 sinus arrhythmias 267, 284, 419, 521–2,
 588, 596–7
 sinusoidal oscillations 204
 size of vascular beds 29–31
 small arteries 389, 457
 smoking 581–3
 smooth muscle cells (SMC) 81–2, 86,
 97–102, 107
 smooth muscle tone
 arterial disease 521–2
 cocaine 522
 embolization 505–6
 ergot 522
 migraine 522
 Prinzmetal angina 521–2
 Raynaud's phenomenon 522
 serotonin 522
 snakes 297, 299
 SNR see signal-to-noise ratio
 SOLVD trial 353, 483–4
 sonography
 ultrasonography 69, 84, 391–2
 see also ultrasound
 soy products 586
 spasm, coronary artery 112, 552
 spectral analysis 184–5
 carotid waveforms 269, 601, 603
 Doppler 184–5, 187, 191
 noise 256
 sphymocardiography 616–30
 SphymoCor 67–8, 211, 220, 246, 476,
 611–12, 638
 aortic augmentation calculation 533,
 546, 593
 sphymography 5–6
 with phonocardiography 249
 pressure waves 249, 252, 595–8
 sphymomanometers 4–5, 472
 blood flow 143–4
 endothelial function 117
 sphymocardiography
 angina pectoris 628
 cardiac failure 628–30
 hypertension 625–7
 pressure pulse waveform analysis
 616–30
 springs 61–2
 SPTI see systolic pressure-time index
 square-wave 132–4, 140, 153, 262, 597
 SRES see shear rate estimating system
 stagnation 27
 standing wave 196, 201–2, 219, 229,
 241
 Starling, Earnest 6–7
 Starling's law of the heart 323, 326
 see also Frank-Starling

 static calibration 153, 162, 257
 statins 124, 533–4
 stature 251, 429, 435, 591
 and disease 507, 533, 567
 steady-state oscillation 8–10
 wave reflection 201–2
 steady-state pressure-flow relations, see
 also Poiseuille's equation
 steady-state pressure-flow relationships
 14–19
 steal, coronary 381, 537
 stenosis
 arterial 248–9
 carotid 52, 398
 coronary 389, 485, 558
 Stewart-Hamilton dilution method 147
 stiffness
 age and see aging
 aorta 455–6
 arterial wall properties 61
 central elastic arteries 455–6
 small arteries 457
 Stokes, G 14
 see also Navier-Stokes
 straight-spring manometer 6, 8
 strain
 elastic body 56–8
 relationship to stress 58–9
 strain gauges 119, 141–2, 146, 161–2,
 167
 stress
 elastic body 58
 relationship to strain 58–9
 risk factor for cardiovascular disease
 588–9
 shear see shear stress
 string, transient wave 198–200
 stroke 5, 402, 405–7, 409, 451–2, 454,
 460–1, 465–6, 473, 480–1
 arterial stiffness 88, 115–16
 hypertension 451–2, 454, 460–1,
 465–6, 480–1
 stroke volume (SV) 148–51, 369, 426,
 428
 index 483, 564
 Studies of left ventricular Dysfunction
 (SOLVD) 353, 483–4
 subclavian artery 37
 arterial wall properties 83
 coronary steal 380
 impedance 282, 287–92, 300
 waveforms 228, 234, 239, 241, 570,
 601
 subendocardium 380–1
 versus subepicardium 382, 387–8
 viability ratio 388, 532, 618, 621–5,
 627–8
 subendothelium 56
 subepicardium 381
 versus subendocardium 382, 387–8
 sumatriptan 522, 536, 543
 superficial femoral artery (SFA) 193
 surgery
 aortic coarctation 519–20
 arterial stiffness 549
 Survival and Ventricular Enlargement
 (SAVE) trial 483–4
 SV see stroke volume

- syndrome X 376, 388–9, 436, 565, 628
 systemic arterial compliance (SAC)
 91–4
 systemic arteries
 effect of body size on impedance
 296–300
 see also large systemic arteries
 systemic lupus erythematosus 533–4
 systemic vasculitis 493, 496
 systolic cardiac failure 351–2
 ascending aortic impedance 338–9
 wave reflection 339–40
 systolic hypertension
 aging 413, 618, 625–6
 aortic stiffness 347, 350, 419
 Systolic Hypertension in the Elderly
 Program (SHEP) 2, 220, 412–13,
 448, 467, 475, 480–2
 systolic pressure 2, 8, 11
 see also end-systolic pressure-
 volume relationships; systolic
 hypertension
 systolic pressure-time index (SPTI)
 383, 388–9
- T-tube, asymmetric 219–20, 232, 294,
 298–9, 303–4
 tadalafil (Cialis) 111, 113–14, 535–6, 538
 nitroglycerin with 546, 548
 Takayasu arteritis 521
 tangential elastic modulus 81
 tea 584–5
 TEE *see* transesophageal
 echocardiography
 tensile stress 57, 502–3
 tension 58
 and elastic modulus 61–2, 78
 elastic tube 59
 law of Laplace 454
 smooth muscle 78
 ventricular energy 323
 tension-time index (TTI) 383, 386, 388
 terazosin 335, 390, 465, 545
 terminal impedance 276, 306–7
 TGF- β *see* transforming growth
 factor- β
 thermodilution method 148
 therapy
 arterial counterpulsation 549–51
 conduit arteries 539–43
 conduit function 544
 coronary circulation 390
 disease conditions 536
 hypertension 407, 463–6
 medical therapies 544–9
 monitoring effect of 551
 muscular arteries 543–4
 pathophysiology 535–43
 trials 480–2
 vessel types 536–9
 see also treatment; trials
 third law of motion 198
 thoracic aorta 37–8, 229–33
 aging 102, 434, 444
 aneurysms 510–11
 aortic coarctation 512–14, 550
 arterial wall properties 80, 83–5,
 96–7, 104
- counterpulsation 556
 elastin 82, 106, 505
 flow 75
 flow waves 64, 233
 impedance 292, 297
 length 78
 pressure waves 69, 229–32
 velocity profile 44–7
 wave attenuation
 waveforms 173
 Womersley equations 174
 thromboembolism 368–70
 thrombosis
 and atherosclerosis 504
 coronary *see* coronary artery
 embolization 505–6
 tilting 236
 time-derivative method, pressure
 gradient 171–2
 tissue plasminogen activator 11
 tonometry
 applanation *see* applanation
 tonometry
 arterial wall properties 92
 brachial artery 246, 272, 438, 573
 carotid artery 438, 598–9, 604, 630
 endothelial function 115, 119–21
 femoral artery 625
 radial artery 599, 601, 625
 Torricelli's equation 6, 28–9
 total peripheral resistance 32–34, 92m
 145, 148–9, 312
 transducers
 continuous wave Doppler systems
 182
 ultrasound 178–9
 transesophageal echocardiography
 (TEE) 80, 87, 160
 transforming growth factor-b (TGF-b)
 507–9
 transient excitation 71–3
 transient wave reflection
 in a string 198–200
 in a tube 200–1
 transit-time method 154–5
 transmission line
 models 637–8
 Windkessel hybrid 638
 wave reflection 202–3
 transpulmonary thermodilution
 method 148
 transverse wave 198
 traveling waves 265
 treadmill exercise testing 376, 381, 387,
 554, 559, 561
 treatment *see* therapy
 trials *see* individual trials
 trinitroglycerin *see* nitroglycerin
 TTI *see* tension-time index
 turbulence, post-stenotic dilatation
 53–4
 turbulence 51–4
 aneurysms 53–4
 atherogenesis 54
- UK Prospective Diabetes Study
 (UKPDS) 481–2
 ultrasonic flowmeters 154–7
- ultrasonic transducers 157–8
 ultrasound
 arterial diameter 163–4
 beam formation 179–80
 distension measurements 190–1
 Doppler *see* Doppler ultrasound
 Doppler principle 181–2
 echo resolution 178
 image generation 180–1
 intima-media thickness 189–90
 intravascular (IVUS) 157–8
 local pulse wave velocity 191
 physical principles of 177–8
 spectral analysis 184–5
 transducers 178–9
 wall diameter estimation 189
 wall shear stress 191–4
 umbilical artery 225, 248–50
 umbilical vein 113
 United Kingdom Prospective Diabetes
 Study (UKPDS) 481–2
 upper limb
 amplification 440
 pressure pulse amplification 555–6
 US *see* ultrasound
- Valsalva maneuver 238–9
 valsartan and amlodipine 576
 valves *see* specific valves
 vardenafil 113–14, 535
 vascular beds
 blood pressure (BP) 543–4
 emotional stress 588
 endothelial function 114, 121
 femoral *see* femoral artery
 flow waves 233, 244
 impedance in 175, 273–4
 peripheral resistance 32, 34
 pulsatile flow 39
 size changes in 29–31
 ultrasound 194, 215–16
 vasoconstriction 234–5
 vasodilatation 233
 wave reflection 222
 vascular endothelium 111–13
 vascular impedance
 aortic coarctation 516
 characteristic impedance 275–7,
 307–8
 determination by pressure
 measurement 309
 impedance modulus 276, 308–9
 input impedance 274–5, 277–83
 longitudinal impedance 273–4,
 276–7
 pressure gradient 175–6
 terminal impedance 276, 306–7
 vascular resistance *see* resistance
 vasculitis
 systemic 493, 496, 534
 see also Takayasu arteritis
 vasoactive drugs 86, 145, 151, 631
 vasoconstriction
 flow waves 234–6
 pressure waves 234–6
 vasoconstrictors
 arterial wall properties 98, 100, 106,
 108

- endothelial function 112, 115, 119
 flow 36
 hypertension 450–1
 impedance 288, 293, 371
 pressure and flow wave effects 248
 and vasodilatation 239
 vasodilatation
 flow waves 233–4
 pressure waves 233–4
 vasodilator therapy, cardiac failure 482–3
 vasopeptidase inhibitor 482, 546
 vegetables 586
 veins
 cerebral 400, 402, 406
 cortical 402
 intrathoracic 17
 penile 538
 Poiseuille's equation 14, 17
 pulmonary 318–19, 358, 360
 vasodilators 538
 vena cava 322, 358
 velocity
 Doppler system estimation 183–4
 profile 44–8
 ventricular contraction 324
 wave propagation 64
 velocity profile in pulsatile flow 37–41
 venous occlusion plethysmography 114, 120
 ventricles
 end-systolic pressure-volume relationship 325–8
 energy expenditure 323–4
 force-velocity relationship 324
 instantaneous pressure to instantaneous flow relationship 328–31
 left 322–3
 power-afterload relationship 331
 pressure to volume relationship 323
 pump function curves 331–3
 ventricular afterload and impedance 311
 ventricular aneurysms 354
 ventricular contraction 6, 167–8, 221, 312, 324
 and ejection duration 341
 intermittent 465
 isometric 327, 382
 left 382
 premature 168
 right 372
 ventricular elastance 340–1
 ventricular hydraulic load, reduction of 334–8
 ventricular pressure, to volume relationship 323
 ventricular septal defect (VSD) 349, 405
 ventricular unloading treatment, cardiac failure 352–3
 ventricular-vascular interaction in aging 432–3
 vessel types, therapeutic strategies 536–9
 Veterans Administration study, heart failure 483
 Vierordt, K 129, 576
 virtual histology 395
 viscoelastic arteries, pulse wave velocity 66–9
 viscoelastic tubes, propagated waves 74–5
 viscoelasticity 20, 23, 163
 viscosity 4
 anomalous 10, 386
 blood 24–6
 c.g.s. unit 16, 49
 coefficient of 20–1, 23
 kinematic 39, 43, 49, 205, 287
 Newtonian 39
 plasma 23–4
 Poiseuille's equation 4, 14–19
 whole blood 19–26
 vitamins 586
 Voigt model 62
 combined Maxwell and 62
 and Maxwell model 62
 volume flow 14, 16, 22
 and perivascular Doppler ultrasound 191
 see also pressure-volume loops;
 pressure-volume relationships
 volume-clamp method 145, 251
 vortex formation 49
 VSD *see* ventricular septal defect
 ω -3 fatty acids 586
 wall Doppler filter 183
 wall shear rate (WSR) 192
 wall shear stress (WSS) 192–4
 and shear rate 21–3
 values, ultrasound 192–4
 in vivo, ultrasound 191–2
 wall track system 68
 water (fluid) hammer equation 68, 83, 91, 206, 222, 308
 wave number 201
 wave propagation
 arterial wall properties 64–6
 elastic tube 64–6
 Young, Thomas 64–5
 wave propagation elastic tube 64–6
 wave reflection
 arterial junctions 206–8
 clinical trials 220–2
 coefficient of 204–6
 dementia 406
 drug modification of 222–3
 and elasticity 208
 flow waves 209–10
 myocardial oxygen 215–16
 phase velocity 204
 pressure waves 209–10
 reduction of 545
 secondary 218–20
 single arterial junction 206–8
 sites of 216–18
 steady-state oscillation 201–2
 systolic heart failure 339–40
 transient waves 198–201
 transmission line 202–3
 wave reflections
 augmentation index 210–12
 clinical trials 220–2
 and left ventricular load 212–15
 magnitude of 196
 secondary 218–20
 waveforms
 pressure *see* pressure waves
 wavefront velocity 70–1
 wavelength 70–1
 quarter-wavelength effect 208, 217, 226
 wave(s)
 attenuation (of propagated waves) *see* attenuation
 flow *see* flow wave
 harmonic components *see* harmonic components
 pressure wave *see* pressure wave
 propagation/transmission, in elastic tube
 reflection *see* reflection
 standing *see* standing wave
 Weber, EH 5, 64
 Weber, WE 5
 weight 588
 Westkessel model 278
 WHO *see* World Health Organization/
 International Society of Hypertension
 whole blood viscosity (WBV) 19–26
 Wiedemann, G 14
 Windkessel chamber 3–4, 7, 77, 148, 198, 219, 278–9, 313, 433, 516
 Windkessel model 4, 6, 88, 105, 148, 165, 176, 220, 260, 402–3, 545, 631–8
 transmission line hybrid 638
 and Westkessel model 278
 wine 587–8
 wires *see* guidewires
 Witzig-Womersley model 63
 Womersley, John 8–10
 equations 18, 40
 arterial blood flow 174–6
 arteries 63, 260, 276, 281–3, 300, 363, 365
 wave reflection 99–101, 192, 196, 206–7
 see also Witzig-Womersley model
 work *see* energy
 World Health Organization/
 International Society of Hypertension (WHO/ISH)
 BP measurement 527
 guidelines 483–4
 WSR *see* wall shear rate
 WSS *see* wall shear stress
 yield point 56
 yield stress 21–2, 25, 47, 403–4
 Young, Thomas 4–5
 compliant aorta 56
 (elastic) modulus 5, 56, 58–60, 63, 81, 88, 97
 wave propagation 64–5
 Young-Helmholtz theory 4
 zero crossing counter 183, 186

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